User Guide

Reverb, dereverberation, spatial audio, analysis and automation

Command-line workflows / Real-time processing

Impulse responses / AUv3 and VST3 architecture

Illustrated DSP and acoustic reference

Colby Leider, PhD

August 5, 2026

Technical reference and musical production handbook,
with educational exercises and applications to Audio AI and Machine Learning

Frontispiece. The nave of Laon Cathedral, built chiefly during the late twelfth and early thirteenth centuries: a long direct axis joined to lateral aisles, repeated bays, and a high vaulted reflection field.

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verbx User Guide

Reverb, Spatial Audio, Dereverberation, Plug-in Design, and Educational Exercises

Documentation and software by Colby Leider, PhD.
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Repository: https://github.com/TheColby/verbx
No part of a cited recording, score, or third-party artwork is reproduced in this book.

Typeset and generated from the repository source.

For Patricia

Preface

[forthcoming]

Acknowledgments

This book and the verbx software stand on a long continuum of work in room acoustics, electroacoustic music, artificial reverberation, digital signal processing, spatial audio, measurement, and software engineering. The research bibliography identifies the authors whose equations, experiments, algorithms, and perceptual models provide the technical foundation; the listening appendix recognizes composers, performers, producers, engineers, architects, and recording traditions that make reverberation a musical material rather than merely an effect.

Particular thanks are due to the maintainers of Python, NumPy, SciPy, SoundFile, Rich, Typer, JUCE, Pandoc, XeLaTeX, TeX Live, MakeIndex, Mermaid, and the many other open-source tools used to build, test, illustrate, typeset, and distribute the project. Museums, libraries, archives, photographers, and score repositories that release public-domain or openly licensed material make the historical plates possible. Every reproduced plate carries its source and license beside the image or in its caption.

Finally, gratitude belongs to the musicians, students, engineers, and patient readers who test long tails, difficult routing, unusual tunings, high channel counts, and the boundary between useful resonance and beautiful excess.

About This Edition

This book combines the verbx command-line manual, plug-in architecture handbook, production and validation cards, illustrated DSP atlas, impulse-response guidance, academic references, and a listening appendix devoted to music in which reverberation is part of the composition or production.

The software is evolving. Statements marked as protocols, foundations, beta behavior, or future work are not compatibility certifications. Commands, schemas, figures, and generated cards are rebuilt from the same repository so that the book can be audited alongside the code.

Part I	Installation, CLI workflows, and sound design
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List of Symbols and Notation

Symbols are italicized in mathematical prose; vectors and matrices are identified by boldface and by context or dimension. Uppercase letters commonly denote transforms, matrices, or totals; lowercase letters commonly denote samples, scalars, vectors, or indices. Unless a section states otherwise, discrete-time indices are integers, frequency is in hertz, time is in seconds, angles are in radians, and level is in decibels. A symbol intentionally reused by different disciplines is defined again in its local equation; subscripts such as “in,” “out,” “FDN,” and “IR” disambiguate dimensions in complexity statements.

Discrete time, transforms, and filters

n, m, k, ℓ	Integer sample, delay, frequency-bin, frame, mode, or summation indices as defined locally.
$x[n], y[n]$	Input and output samples at discrete-time index n.
$\mathbf{x}[n], \mathbf{y}[n]$	Multichannel input and output sample vectors.
$x_c[n], y_o[n]$	Sample on input channel c and output channel o.
$\delta[n]$	Unit impulse: one at $n = 0$ and zero elsewhere.
$u[n]$	Discrete unit-step sequence.
$h[n]$	Impulse response of a linear time-invariant system.
$h_{oc}[n]$	Impulse response from input channel c to output channel o.
$X(z), Y(z)$	One-sided or bilateral z transforms of $x[n]$ and $y[n]$.
$H(z)$	Transfer function $Y(z)/X(z)$ in the z domain.
z	Complex transform variable; $z = e^{j\Omega}$ lies on the unit circle.
z^{-M}	Delay of M samples, typeset with a true negative superscript.
$D_M(z)$	Delay operator, normally equal to z^{-M}.
$X[k], H[k]$	Discrete Fourier-transform bins of a signal and filter.
F_s	Sampling frequency in hertz.
T_s	Sampling period $1/F_s$ in seconds.
f	Ordinary frequency in hertz.
ω	Angular frequency $2\pi f$ in radians per second.
Ω	Normalized digital angular frequency in radians per sample.
N	Signal-frame count, FFT length, transform order, or locally declared problem size.
B	Audio block size or FFT partition length in samples.
L	Signal, filter, delay-line, or impulse-response length in samples.
P, Q	Polynomial orders, partition size and count, or locally declared dimensions.
a_k, b_k	Feedback and feedforward coefficients of a difference equation.
$A(z), B(z)$	Denominator and numerator polynomials of a rational transfer function.
$*$	Linear convolution.
$\odot$	Elementwise multiplication when explicitly declared.
$(\cdot)^*$, $(\cdot)^{\mathsf{H}}$	Complex conjugation and conjugate transpose.
$\Re\{\cdot\}, \lvert\cdot\rvert$	Real part and scalar magnitude.

Reverberation and room-acoustic measures

T_{60}, RT_{60}	Time required for reverberant energy level to decay by 60 dB.
T_{20}, T_{30}	RT estimates extrapolated from 20 dB and 30 dB fitted ranges.

EDT	Early decay time extrapolated from the first 10 dB of decay.
$E(t)$	Remaining or instantaneous acoustic energy at elapsed time t, as defined locally.
$L_E(t)$	Energy-decay level, commonly $10\log_{10}(E(t)/E(0))$ dB.
τ	Exponential time constant.
γ, β	Decay rate, damping factor, or boundary reflection factor as defined locally.
C_{50}, C_{80}	Early-to-late clarity indices using 50 ms and 80 ms boundaries.
D_{50}	Fraction of total energy arriving in the first 50 ms.
T_s^{center}	Center time of an impulse response; the superscript distinguishes it from sampling period.
DRR	Direct-to-reverberant energy ratio, normally reported in decibels.
$IACC$	Interaural cross-correlation coefficient.
G_{room}	Room strength or a context-specific room gain.
V, S	Room volume and total boundary surface area.
S_i, α_i	Area and absorption coefficient of boundary region i.
$\overline{\alpha}$	Area-weighted mean absorption coefficient.
A_{abs}	Equivalent absorption area $\sum_i \alpha_i S_i$.
c	Speed of sound in the propagation medium.
$\bar{\ell}$	Mean free path, approximately $4V/S$ in a diffuse room.
f_{Sch}	Schroeder transition frequency between sparse modal and statistical behavior.
$\mathbf{x}_s$, $\mathbf{x}_r$	Source and receiver position vectors.
r_p, t_p	Length and arrival time r_p/c of propagation path p.
Feedback delay networks and recursive reverberators	
N_{FDN}	Number of delay lines or state channels in an FDN.
m_i	Delay length of FDN line i in samples.
$\mathbf{s}[n]$	FDN delay-state vector at sample n.
$\mathbf{A}$	Feedback, mixing, state-transition, or system matrix defined locally.
$\mathbf{b}$, $\mathbf{c}$	Input and output projection vectors.
$\mathbf{B}$, $\mathbf{C}$	Multichannel input and output projection matrices.
$\mathbf{D}(z)$	Diagonal matrix of delay operators.
$\mathbf{G}(z)$	Diagonal or coupled loop-filter matrix.
g, $g_i(f)$	Scalar feedback gain and frequency-dependent gain of delay line i.
λ_k	Eigenvalue, pole, or modal decay factor at index k.
$\rho(\mathbf{A})$	Spectral radius: maximum magnitude of the eigenvalues of $\mathbf{A}$.
$\mathbf{I}$	Identity matrix of the dimension required by context.
$\mathbf{H}_N$	Hadamard matrix of order N when that matrix family is selected.
$\text{diag}(\cdot)$	Diagonal matrix formed from the listed scalar or filter entries.
Convolution, dereverberation, and statistical estimation	
M_{IR}	Impulse-response length in samples.
C_{in}, C_{out}	Input and output channel counts.
R	Number of active input-output convolution routes.
$H_q[k]$	Frequency-domain filter coefficients for IR partition q.
$X(t,f)$, $Y(t,f)$	Desired and observed short-time Fourier-transform coefficients.
t, f	STFT frame and frequency-bin indices when used as discrete arguments.
K_{FFT}	Number of retained FFT frequency bins.

T_{STFT}	Number of STFT frames in an analyzed signal.
Δ	Prediction delay in frames for WPE or a finite increment elsewhere.
$\mathbf{g}(f)$	Multichannel delayed-prediction coefficient vector at frequency f.
$\lambda(t,f)$	Time-frequency source variance in statistical dereverberation.
$G(t,f)$	Bounded spectral suppression gain.
$\mathbf{R}_{yy}(f)$	Observed-signal spatial covariance matrix.
$\mathbf{w}(f)$	Beamformer or multichannel filter vector.
ε	Small regularization, floor, or numerical-stability constant.
$\hat{x}, \hat{\theta}$	Estimated signal or parameter; a hat denotes an estimate.
$\mathbb{E}[\cdot]$	Expectation under the probability model stated locally.
Spatial audio, geometry, and physical models	
θ, ϕ, r	Azimuth, elevation, and radial distance in a spatial coordinate system.
Ω	Unit direction or solid-angle coordinate; not complexity lower-bound notation.
$Y_l^m(\Omega)$	Spherical harmonic of degree l and order m.
l, m	Spherical-harmonic degree and signed order.
$q = l(l+1) + m$	ACN channel index for a spherical-harmonic component.
$\mathbf{R}$	Rotation matrix or spatial covariance matrix, identified by context.
$H_L(f,\Omega), H_R(f,\Omega)$	Left- and right-ear transfer functions for direction Ω.
N_s, N_r	Numbers of modeled source and receiver positions.
$N_{\mathrm{surf}}, N_{\mathrm{ray}}$	Numbers of geometry surfaces and traced rays.
$\mathbf{M}, \mathbf{C}, \mathbf{K}$	Mass, damping, and stiffness matrices in a finite-element model.
$\mathbf{q}(t), \dot{\mathbf{q}}(t), \ddot{\mathbf{q}}(t)$	Generalized displacement, velocity, and acceleration vectors.
E, ν, ρ_m, h	Young's modulus, Poisson ratio, material density, and plate thickness.
D	Flexural rigidity $Eh^3/[12(1-\nu^2)]$ of an isotropic thin plate.
$\mathbf{L}$	Discrete positive Laplacian used by the structured plate model.
$\Delta x, \Delta y$	Finite-element or finite-difference grid spacing.
$\phi_j, \omega_j, \zeta_j$	Mode shape, natural angular frequency, and damping ratio of mode j.
$N_{\mathrm{mode}}, N_{\mathrm{node}}$	Retained modal count and spatial node count.
Norms, optimization, and complexity	
$\lVert\cdot\rVert_p$	Vector, matrix, or signal p-norm stated by context.
$\langle \mathbf{x}, \mathbf{y} \rangle$	Inner product of compatible vectors or signals.
$\mathscr{L}(\theta)$	Loss or objective function evaluated at parameters θ.
$\nabla_\theta \mathscr{L}$	Gradient of the objective with respect to θ.
$\mathscr{D}_{\mathrm{train}}, \mathscr{D}_{\mathrm{val}}, \mathscr{D}_{\mathrm{test}}$	Training, validation, and test datasets.
$\mathcal{O}(g(n))$	Asymptotic upper bound, normally worst-case unless qualified.
$\Omega(g(n))$	Asymptotic lower bound; context distinguishes it from angular frequency.
$\Theta(g(n))$	Asymptotically tight bound.
$\lceil x \rceil, \lfloor x \rfloor$	Ceiling and floor of a scalar value.

List of Abbreviations

ADM BWF	Audio Definition Model metadata carried in a Broadcast Wave Format file.
AI	Artificial intelligence.
API	Application programming interface.
AUv3	Audio Unit version 3 plug-in format.
BRIR	Binaural room impulse response.
CLI	Command-line interface.
DAW	Digital audio workstation.
DRR	Direct-to-reverberant ratio.
EDC	Energy decay curve.
EDR	Energy decay relief.
EDT	Early decay time.
FDN	Feedback delay network.
FEM	Finite-element method.
FFT	Fast Fourier transform.
FIR	Finite impulse response.
FOA	First-order Ambisonics.
FDTD	Finite-difference time-domain method.
GPU	Graphics processing unit.
HRTF	Head-related transfer function.
IACC	Interaural cross-correlation coefficient.
IR	Impulse response.
LFE	Low-frequency effects channel.
LUFS	Loudness units relative to full scale.
ML	Machine learning.
OLA	Overlap-add convolution.
OLS	Overlap-save convolution.
RT60	Reverberation time extrapolated to a 60 dB decay.
SIMD	Single instruction, multiple data.
SOFA	Spatially Oriented Format for Acoustics.
STFT	Short-time Fourier transform.
VST3	Steinberg Virtual Studio Technology version 3.
WPE	Weighted prediction error dereverberation.

Chronology of Reverberation

This selective chronology is a historical reading map, not a story in which one technology neatly replaces another. Architecture, chambers, plates, springs, magnetic delay, digital networks, convolution, physical modeling, spatial rendering, and learned systems continue to coexist because each stores, shapes, and spatializes acoustic energy differently. Dates identify useful landmarks; broad periods acknowledge practices whose origins are distributed across cultures, institutions, patents, laboratories, studios, compositions, and oral engineering traditions.

Each entry asks three questions. What medium stores the acoustic memory? Which controls become available to a musician or engineer? Which assumptions become hidden? Read the chronology alongside Chapter 2's illustrated history and Appendix E's research bibliography, where primary papers, patents, institutional histories, and surveys supply fuller attribution and technical context.

Prehistory	Caves, rock shelters, canyons, and other resonant sites join sound to place long before acoustics exists as a named science. Repetition, sustained vocalization, percussion, and movement reveal echoes and modes empirically. Modern archaeoacoustics studies these relationships cautiously, combining measurement with archaeological and cultural evidence rather than treating resonance alone as proof of historical intention.
Classical antiquity	Open-air theaters, enclosed ritual spaces, and monumental stone architecture organize projection and reflection through geometry. Builders and performers work with audibility, speech rhythm, ensemble placement, and audience distribution without reducing the result to one decay number. Direct sound, early support, and intelligibility already form a practical design problem.
c. 20 BCE	Vitruvius's *De architectura* describes theater geometry, voice projection, and resonant bronze vessels associated with Greek performance practice. The account mixes observation, inherited theory, and idealized prescription, but it is an early written attempt to connect built form, resonant frequency, and intelligibility. Modern readers should resist treating the vessels as electronic equalizers in disguise: the larger historical fact is that architecture and tuned resonance were already discussed as parts of one system.
Sixth century	Hagia Sophia, completed in 537, demonstrates the acoustic consequences of a vast central dome joined to semidomes, galleries, columns, and reflective masonry. Its field is neither a simple rectangular decay nor a uniform diffuse bath. Long paths, coupled volumes, focusing, and changing occupancy make the building a continuing case study in architectural and liturgical sound.
Medieval sacred architecture	Romanesque and Gothic churches enlarge vertical volume, multiply bays and aisles, and place sung text within long, spectrally evolving decays. Chant and early polyphony are shaped by breath, syllabic pacing, processional movement, and architecture. The room is not an effect added afterward; it is part of performance timing and collective intonation.
c. 800–1200	Maya ceremonial architecture supplies a distinct history of reflection and resonance. At Chichén Itzá, handclaps near the stairway of El Castillo can return as a descending, chirp-like echo because the stepped geometry disperses frequency components in time. Archaeoacoustic study compares measurements, models, architecture, and cultural evidence; the striking sound does not by itself establish that builders intended to imitate a bird. The example remains an unusually clear lesson in how repeated geometry transforms a transient.
c. 1365	Guillaume de Machaut's *Messe de Nostre Dame* (c. 1365) illustrates how sustained lines, staggered syllables, and contrapuntal density can cooperate with a resonant church. Overlap joins phrases, but excessive late energy can erase textual and rhythmic distinctions. The musical problem already resembles modern reverb mixing: continuity must not destroy articulation.

1436	Guillaume Dufay's *Nuper rosarum flores* (1436) was performed for the consecration of Florence Cathedral. Its ceremonial pacing, alternating forces, and sustained sonorities demonstrate how composition, procession, and monumental architecture can be heard as one event. Claims that every musical proportion directly encodes the cathedral remain debated, but the performance history still offers a useful reminder: acoustic space affects formal scale, ensemble deployment, and the listener's perception of arrival.
Sixteenth century	Large ceremonial vocal repertories increasingly use separated groups, registral contrast, and antiphonal exchange. Thomas Tallis's *Spem in alium* (c. 1570) places forty independent parts in a texture whose intelligibility depends on attack spacing, ensemble geometry, and the room's ability to unify without collapsing many lines into one spectral mass.
Late sixteenth–early seventeenth centuries	Venetian polychoral practice makes architectural separation explicitly compositional. Giovanni Gabrieli's *In ecclesiis* (c. 1615) coordinates voices, instruments, balconies, and shared decay. Local early reflections help distinguish groups while the late field connects them, anticipating modern object placement feeding a common immersive reverberator.
1650	Athanasius Kircher's *Musurgia universalis* (1650) gathers contemporary ideas about echo, speaking tubes, resonant spaces, instruments, and the transmission of sound. Its engravings are as important historically as its uneven explanations: they show acoustics becoming a visual and mechanical subject whose paths can be drawn, redirected, and demonstrated. The book belongs to a long prehistory of signal-flow thinking, even though its categories differ sharply from modern linear systems and measured impulse responses.
Seventeenth–eighteenth centuries	Opera houses, court theaters, chapels, and purpose-built concert rooms develop alongside changing ensemble sizes and public listening practices. Designers reason through proportion, materials, ornament, occupancy, and empirical adjustment. Musicians adapt tempo, articulation, scoring, continuo, and dynamics to the response of particular rooms.
Nineteenth century	Industrial materials, larger halls, orchestral expansion, public organs, and new audience expectations intensify questions of power, blend, clarity, and reverberant support. Individual resonances and coupled volumes remain audible, but statistical thinking becomes increasingly attractive as rooms grow and measurement practices mature.
1877–1890s	The phonograph, improved microphones, telephone circuits, and early electrical transmission alter the meaning of acoustic memory. A sound can be stored, replayed, transmitted, and auditioned away from its originating room. Recording initially captures reverberation as an inseparable trace, but repeatability makes comparison possible: performers and engineers can hear how placement, horn geometry, enclosure, and medium change the balance between direct energy and reflected sound.
1890s	Wallace Clement Sabine's work connects room volume, effective absorption, and reverberation time through repeatable experiments. Reverberation becomes a quantity that can be estimated, compared, and designed, while the limits of the diffuse-field assumption remain important. The resulting framework establishes a durable bridge between architectural choices and temporal decay.
1900	Boston Symphony Hall opens after Sabine advises its design. The hall becomes a prominent demonstration that architectural acoustics can combine calculation, material specification, model reasoning, and musical judgment before construction is complete. Its history also cautions against reducing quality to reverberation time: lateral energy, stage support, frequency balance, occupancy, geometry, and performer experience all contribute to the result listeners call a successful room.
Early twentieth century	Telephone, radio, electroacoustic transducers, electrical recording, and amplification separate sound production from immediate acoustic radiation. Once microphones and loudspeakers can connect distant spaces, a room can become a signal-processing component. The conceptual ingredients of an effects send, remote chamber, and returned ambience begin to align.
1920s	Electrical recording improves bandwidth and permits microphones, amplifiers, loudspeakers, and mixers to participate more directly in spatial design. Engineers can favor controlled capture and then restore a chosen sense of space. Reverberation starts to become separable from the original performance location, though editing and routing remain physically cumbersome.

1930	Carl F. Eyring publishes a reverberation relation that improves on the Sabine approximation when average absorption is not small. The distinction is more than a competing formula: it exposes the assumptions behind a decay estimate and the difference between discrete energy loss at boundaries and a continuous exponential model. Later software can compute a number instantly, but responsible interpretation still depends on field diffuseness, source-receiver placement, bandwidth, and fit range.
1930s	Radio, sound film, public-address work, and electrical organs create demand for compact or repeatable ambience. Stairwells, tiled rooms, tanks, pipes, acoustic labyrinths, magnetic paths, and resonant objects are evaluated as practical devices. Noise, isolation, flutter, transducer placement, and bandwidth become engineering constraints as important as nominal decay time.
1939–1941	Laurens Hammond's spring-reverberation patent documents a compact electromechanical route for extending an electrical instrument's decay. Torsional waves, transducers, dispersion, and mechanical damping create a response unlike a miniature room. The spring's audible identity later becomes central to organ, guitar, surf, dub, studio, and experimental practices.
1940s	Studios increasingly treat reflective rooms as outboard processors. A console send drives a loudspeaker in the chamber; microphones capture the response; equalization, compression, placement, and return level shape the result. This topology remains recognizable in every modern auxiliary reverb bus even after the physical room is replaced by code.
1947	Bill Putnam's work on the Harmonicats' *Peg o' My Heart* (1947) is frequently cited as an early artistic landmark for chamber reverberation in record production. The important shift is authorship: reflected sound is deliberately captured, balanced, and printed as a production layer rather than accepted only as an inseparable property of the recording venue.
Late 1940s–1950s	Magnetic tape makes delay, repetition, reversal, varispeed, splicing, and feedback available as compositional operations. Tape echo is not diffuse reverberation, yet recirculating repeats can be filtered, saturated, offset, and layered until they function as an artificial acoustic field. Reversing a printed wet signal creates the rising envelope later associated with reverse reverb, while a gate can turn an otherwise continuous decay into a precisely timed rhythmic event.
1950s	Purpose-built studio chambers turn architecture into repeatable infrastructure. Major facilities develop identifiable spatial signatures, while engineers learn to vary apparent size and brightness through loudspeaker position, microphone pattern, distance, filtering, and return dynamics. Chambers are expensive and fixed, but their physical complexity is difficult to duplicate exactly.
1957	The EMT 140 plate reverberator stores energy in a driven steel sheet and captures vibration with pickups. It supplies dense, bright decay without a room-sized air volume and allows mechanical damping to shorten the tail. Plate sound becomes both a technical solution and a production category, especially effective on voices, drums, and sustained instruments.
Late 1950s–1960s	Springs, plates, chambers, tape delay, magnetic drums, pipes, folded acoustic paths, and electronic feedback coexist. No single device wins; each exposes a different compromise among size, cost, bandwidth, noise, density, portability, controllability, and coloration. Studio practice develops around combining these media rather than pursuing one universal room simulation.
1960	Harry F. Olson and John C. Bleazey describe a synthetic reverberator using loudspeaker, pipe, microphone, and feedback. The system makes the cross-domain abstraction visible: propagation supplies delay, the path supplies frequency-dependent loss, transducers cross between acoustic and electrical domains, and feedback extends effective decay.
1961	Manfred R. Schroeder and Benjamin F. Logan formalize "colorless" artificial reverberation using parallel feedback comb filters and serial allpass sections. Delay lengths, loop gains, mode distribution, echo density, and spectral coloration become analyzable design variables. The topology establishes a vocabulary still heard in software reverberators and teaching diagrams.
1961–1962	Schroeder's related work on natural-sounding artificial reverberation develops the perceptual argument: exact simulation of every reflection is unnecessary when a compact system reproduces the statistics listeners use. This reframes reverberator design as a negotiated relation among physical plausibility, computational economy, coloration, density, and auditory expectation.

1960s measurement practice	Warble tones, interrupted noise, pistol shots, balloon bursts, spark sources, and loudspeaker-driven test signals support increasingly repeatable room measurements. Each excitation reveals a different compromise among bandwidth, directivity, crest factor, portability, repeatability, and signal-to-noise ratio. The shift toward measured decay curves encourages octave-band reporting and separates source behavior, room response, microphone response, and background noise as distinct parts of an experimental chain.
1960s computer music	Digital delay and filter ideas enter computer-music research before economical commercial realtime hardware is widespread. Offline and institutional systems permit experiments with recursion, stochastic structures, spatial projection, and durations impractical in electromechanical devices. Reverb becomes compositional code as well as room simulation.
Late 1960s–early 1970s	Integrated circuits, converters, semiconductor memory, and digital control improve steadily, but high-quality realtime delay remains costly. Designers work under severe memory and bandwidth constraints. Those limits encourage short delays, carefully tuned feedback, restricted bandwidth, and compact control sets that become part of early digital reverb's recognizable sound.
Early 1970s	Feedback delay network concepts emerge from work on coupled digital delay systems and multichannel reverberation. Cross-coupling delay states offers greater richness than independent parallel combs. Matrix structure brings new questions of stability, energy preservation, eigenvalues, mode distribution, and computationally efficient mixing.
1970s spatial audio	Michael Gerzon and collaborators develop Ambisonic theory for representing and reproducing a sound field independently of one fixed loudspeaker feed. Reverberation now has directional order: early reflections, diffuse energy, source rotation, decoding, and listening-area behavior can be expressed within a spatial representation. The work supplies conceptual foundations for later surround, higher-order Ambisonics, binaural conversion, and scene-based reverberation.
1976	The EMT 250 becomes a landmark commercial digital reverberator. Memory-based processing, converters, program selection, and a distinctive tactile control surface turn abstract algorithms into a studio instrument. Digital reverb can now be recalled and adjusted without moving microphones, changing plates, or occupying a purpose-built chamber.
Late 1970s	The Lexicon 224 lineage brings influential modulated digital room and plate programs into major studios. Modulation helps break up fixed resonances and gives tails motion, but excessive depth can reveal chorus or pitch wandering. Remote-console operation anticipates the separation between an engine and its production-facing control surface.
1979	James A. Moorer's "About This Reverberation Business" surveys and extends practical digital reverberator design. The discussion treats early reflections, comb and allpass structures, low-pass feedback, mode density, and the relation between measurable structure and audible character. Its continuing pedagogical value lies in connecting implementation choices to listening consequences: delay topology is never merely a block diagram when every recirculation changes density, spectrum, and time.
Early 1980s	AMS RMX16, Quantec, Yamaha, Lexicon, and other systems expand program-oriented reverberation, nonlinear envelopes, room simulation, and multichannel possibilities. Named programs become a practical vocabulary: hall, room, plate, ambience, reverse, and nonlinear describe production intentions even when the internal structures differ substantially.
Early–mid 1980s	Gated and reverse-style reverbs move the wet envelope into the foreground of arrangement. Noise gates, printed room microphones, digital nonlinear programs, reversed tapes, and transient-aware routing create tails that rise, terminate abruptly, or occupy only selected beats. Reverberation becomes rhythmic orchestration rather than passive environmental realism.
Mid-to-late 1980s	Affordable racks such as Yamaha SPX and Alesis systems bring multiple digital spaces to project studios, live rigs, and individual musicians. MIDI program change and automation improve repeatability. Limited converters, short memory, coarse parameters, and characteristic algorithms become audible aesthetics rather than defects to be forgotten.
1980s sampling and IR experiments	Longer digital storage encourages sampled responses, cross-synthesis, and laboratory convolution even before routine low-latency native processing is practical. An impulse response can be edited like recorded material: trimmed, windowed, equalized, reversed, resampled, or combined. Those freedoms also introduce methodological risks. A transformed IR may be musically useful while no longer documenting the room, device, source position, or microphone geometry named by its filename.

Late 1980s–early 1990s	Multi-effects processors combine reverb with pitch shifting, chorus, delay, gating, dynamics, and modulation. Shimmer-like and impossible-room ideas arise from feedback paths that no physical enclosure would produce. The distinction between "room simulator" and "spatial instrument" becomes increasingly explicit.
1992	Jean-Marc Jot's analysis/synthesis framework connects perceptual controls to realtime artificial reverberation. Frequency-dependent decay, statistical time-frequency behavior, and a more independent treatment of coloration and duration move beyond ad hoc tuning. Reverb design becomes easier to specify in terms meaningful to listeners and production engineers.
1993	Realtime binaural concert-hall simulation research joins room modeling to headphone presentation. A reverberant field must now preserve direction, interaural cues, source distance, and listener interpretation, not merely produce a convincing monophonic decay. This work anticipates virtual acoustics, games, VR, and mobile spatial rendering.
Mid-1990s	Faster processors and native audio systems support longer networks, more modulation, multiband damping, surround outputs, and increasingly detailed parameter automation. Algorithm design and user-interface design begin to separate: many internal coefficients may be controlled through a smaller perceptual parameter set.
1996	Time-frequency analysis and synthesis of room reverberation support restoration of impulse responses corrupted by measurement noise. Decay estimation, noise-floor handling, and model-based reconstruction connect room measurement to synthetic tail generation. The impulse response becomes both evidence and editable source material.
1997	Work by Jot and Chaigne, and by Rocchesso and Smith, sharpens FDN design through diffuseness, matrix structure, and modal analysis. Stability alone is not enough: a network can be mathematically bounded yet ring metallically. Delay selection, feedback eigenstructure, damping, and input-output projection jointly determine audible quality.
Late 1990s	Long convolution becomes more practical as FFT methods, partitioning, memory, and processor performance improve. Measured rooms and devices can travel as impulse-response files. The method preserves one linear response with high specificity, but source position, listener position, level dependence, motion, and nonlinear hardware behavior remain outside an ordinary fixed convolution.
1997 onward	ISO 3382 and related standards consolidate definitions and procedures for measuring room-acoustic parameters in performance spaces and, in later parts and revisions, other room classes. EDT, T20, T30, clarity, definition, strength, and spatial measures describe different evidence rather than interchangeable quality scores. Standardization improves comparison, but a report remains incomplete without source, receiver, occupancy, bandwidth, noise-floor, fitting, and uncertainty details.
2000	The exponential sine-sweep method associated with Angelo Farina provides a robust route from played sweep and recorded room response to an impulse response, while separating harmonic distortion responses in time. Measurement moves closer to an accessible production workflow, and deconvolution becomes standard vocabulary for capturing spaces and devices.
Early 2000s	Native convolution reverbs and growing IR libraries make concert halls, chambers, plates, springs, cabinets, and unusual spaces available as recallable software assets. Fast convolution scheduling must balance direct heads, short early partitions, long late partitions, CPU cost, memory, and monitored latency. Provenance becomes part of good library practice.
2000s fast convolution	Nonuniform partitioned convolution separates an inexpensive low-latency head from progressively larger tail partitions. The scheduling pattern makes long stereo and multichannel responses usable in interactive systems without assigning one giant transform to every audio block. Fast convolution is an algorithmic family, not a promise of zero latency: block size, partition plan, channel count, transform implementation, device transfer, and host buffering still determine end-to-end behavior.
2000s	Plug-in abundance ends the practical scarcity of one shared studio reverberator. Every track can host a different space, but unlimited instances can weaken spatial coherence. Production practice responds with shared buses, early/late separation, return equalization, ducking, automation, and deliberate use of a small family of related rooms.
2000s speech processing	Dereverberation develops through inverse filtering, spectral enhancement, multichannel prediction, source separation, and statistical room models. The goal differs from creative reverb removal: speech systems optimize intelligibility, recognition, localization, or estimation under assumptions about stationarity, microphones, noise, and room-response length.

2010s	Weighted prediction error and related multichannel methods become influential for blind speech dereverberation. Delayed linear prediction estimates late reverberation while attempting to preserve the desired direct and early components. Window length, delay, model order, channel geometry, and source motion define the boundary between suppression and damage.
2012	The survey *Fifty Years of Artificial Reverberation* gathers the field's development from Schroeder structures through FDNs, convolution, modulation, and perceptual approaches. Historical synthesis matters because familiar labels often conceal different mechanisms; the same audible goal may be reached by physical, statistical, recursive, or measured-response methods.
2010s immersive audio	Ambisonics, object audio, binaural rendering, and height-channel formats make reverberation a scene component rather than a stereo insert. Early reflections need directional plausibility; late fields need envelopment and decorrelation; metadata and channel ordering become part of correctness. Fold-down and headphone translation expose errors hidden by a large loudspeaker array.
2012 onward	Cinema and broadcast object-audio workflows, including Dolby Atmos production, increase the practical demand for reverberation that scales across beds, objects, heights, binaural renderers, and changing playback layouts. A convincing mix must survive more than one loudspeaker count. Send design, object width, decorrelated late energy, LFE policy, metadata, renderer behavior, and stereo fold-down all become parts of the reverberation decision.
2015	Scattering delay network research derives efficient room synthesis from geometry-linked delay paths and scattering nodes. First-order reflections can be represented accurately while higher orders become progressively coarser. The approach occupies a useful middle ground between image-source precision and statistical FDN efficiency.
Mid-to-late 2010s	GPU processing, vectorized FFT libraries, and larger memory systems make dense multichannel convolution and corpus-scale rendering more accessible. Acceleration changes throughput but not representation: unnecessary routing paths, oversized transforms, device transfers, and poor partition schedules can still dominate cost or create unacceptable realtime latency.
Late 2010s	Neural networks estimate room parameters, map reverberant speech toward cleaner targets, generate or transform impulse responses, and emulate devices from examples. Data introduces a new acoustic medium: rooms and hardware may be compressed into model weights. Dataset coverage, licensing, determinism, and disclosure become engineering concerns alongside spectral and temporal quality.
Late 2010s–early 2020s	Realtime ray tracing and geometric-acoustics engines become increasingly practical for interactive media. Image sources, stochastic rays, beam tracing, and path validation estimate specular and diffuse arrivals from scene geometry, while a statistical or recursive tail supplies affordable density. CAD ingestion does not solve acoustics automatically: watertight geometry, scale, material absorption and scattering, source directivity, diffraction, and listener motion remain modeling decisions that must be validated against measurement.
Early 2020s	Differentiable DSP and hybrid neural structures retain explicit delays, filters, feedback, convolution, or physical states while learning selected parameters and residual behavior. This can improve interpretability and stability compared with an unconstrained black box, while still permitting data-driven fitting of complex plate, spring, room, or dereverberation behavior.
2020s physical simulation	Finite-element, boundary-element, finite- difference, digital-waveguide, ray-tracing, and hybrid solvers benefit from improved computation and CAD workflows. No solver removes scale limits: wavelength, mesh resolution, boundary models, source directivity, and listener motion determine cost. Practical systems divide early geometry, low modes, and statistical late decay.
2020s Audio AI	Synthetic and measured reverberation become tools for data augmentation, domain randomization, robustness testing, and spatial learning. Training, validation, and test partitions must avoid room leakage and duplicated IR families. Reverb parameters, random seeds, source licenses, model versions, and analysis reports become part of reproducible dataset provenance.
2020s creative practice	Musicians combine convolution onsets, algorithmic tails, freeze, shimmer, reverse envelopes, gates, ducking, resonators, microtonal scale emphasis, and immersive motion. Historical categories become modular layers rather than mutually exclusive products. The artistic question shifts from "Which room?" to "How should acoustic memory evolve through this phrase?"

2026	verbx assembles algorithmic and convolution reverberation, dereverberation, measurement, JSON analysis, long and infinite tails, Scala-tuned resonance, surround and immersive routing, CLI automation, and AUv3/VST3 preparation in one documented system. Its design treats historical vocabulary as testable behavior rather than as preset marketing.
Beyond 2026	The likely direction is hybrid: measured early responses, geometry-aware paths, stable parametric late networks, learned material or residual models, listener-aware spatial rendering, and machine-readable provenance. The durable criteria remain musical and scientific: articulation, stability, controllability, spatial coherence, reproducibility, attribution, and the ability to make a decay serve the form of a sound.

This chronology deliberately ends without declaring a winner. Chambers retain physical complexity; plates and springs retain characteristic matter; recursive networks retain controllability; convolution retains measured specificity; physical models retain causal geometry; learned systems retain interpolation capacity. The history of reverberation is therefore not a march toward invisibility. It is an expanding collection of ways to make acoustic memory audible, editable, spatial, measurable, and musical.

Bibliographical Note

The bibliography is intentionally broader than a software manual's dependency list. It joins foundational papers, standards, books, conference publications, patents, measurement literature, historical documentation, spatial-audio research, and selected production sources. Annotated entries identify why a source matters to verbx; the extended discovery list provides additional paths into the literature without claiming equal depth of review for every record.

Readers new to the field should begin with Sabine and modern room-acoustics texts, then proceed through Schroeder's comb and allpass work, feedback delay network analysis, perceptual and statistical approaches associated with Jean-Marc Jot and collaborators, convolution and impulse-response measurement, and contemporary spatial or learned systems. Citations attached to photographs, scores, quotations, recordings, and web resources serve a different purpose: they document provenance, attribution, date, and license rather than establish a DSP claim.

Titles and author names follow the best available source metadata, normalized for this edition. DOI links are preferred for research papers; stable institutional or author pages are used when no DOI is available. The repository is the correction record for malformed metadata, dead links, duplicate entries, and later bibliographic discoveries.

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Part I

User Manual and Workflows

"Let echo, too, perform her part, prolonging every note with art."

— Joseph Addison, *Ode for St. Cecilia's Day* (1699)

verbx

verbx is a research-grade Python CLI for creating reverb effects that range from subtle room placement to cathedral-scale tails 3600 seconds long. It handles the complete reverb workflow: ingesting and generating impulse responses, processing audio through two independent engines, controlling every parameter with time-varying automation, delivering loudness-targeted multichannel output, reducing late-room smear with deterministic dereverberation, producing reproducible analysis artifacts at every step, and now previewing spaces in realtime from CLI-selectable audio devices.

You can batch reverberate a directory of audio files to create lush Dolby Atmos beds. Or use it as part of your corpus-augmentation workflow for audio AI projects.

This is not a "set RT60 and go" tool. The parameter surface is wide by design, but the first successful render only needs a source, a destination, and a musical intention. Start by choosing whether you need an offline asset or an interactive DAW return, then follow the practical workflow below.

The detailed engine material, including algorithmic architectures, convolution, spring and plate models, room geometry, and finite-element modeling, appears later in the technical reference after the practical workflow has been established.

1.1 Two Production Paths: Command Line and DAW Plug-ins

verbx is **primarily intended for command-line use**. The CLI is the reference workflow and exposes the broadest processing, analysis, automation, batch, multichannel, impulse-response, dereverberation, reporting, and quality-control surface. It is designed for work in which an input file, a fully specified operation, and a verifiable output artifact are the natural units of thought.

AU, AUv3, and VST3 plug-in versions are also provided for music-production work inside a digital audio workstation (DAW). The plug-ins are not command lines placed behind knobs, and the CLI is not a DAW running in a terminal. They are different approaches to the same family of reverb problems. The CLI emphasizes explicit configuration and reproducible rendering; the plug-ins emphasize immediate listening, host automation, low-latency interaction, and session recall. A project can use either path alone, but many serious productions benefit from using both at different stages.

1.1.1 The CLI path: files, experiments, and reproducible renders

Choose the CLI when the task begins with files or data and should end with files, reports, or a repeatable command. A CLI render names its input, output, engine, parameters, channel layout, numerical format, and optional analysis products. That explicitness is valuable when a result must be recreated next week, on another workstation, across hundreds of files, or as part of a research or machine-learning pipeline.

The CLI is the strongest path for:

- offline rendering where sound quality and complete tail calculation matter more than meeting a realtime callback deadline;
- very long or extreme decays whose output duration should be estimated, validated, reported, and written deliberately;
- deterministic batch jobs driven by shell scripts, manifests, seeds, and version-controlled configuration;

- impulse-response synthesis, inspection, conversion, morphing, matrix routing, and convolution across unusual input and output layouts;
- dereverberation, room analysis, loudness measurement, quality control, and machine-readable JSON or CSV evidence;
- corpus augmentation for Audio AI, where split integrity, labels, provenance, and exact repeatability matter;
- rendering wet stems, surround beds, Ambisonic intermediates, and other assets that will be imported into a DAW or renderer later;
- testing parameter sweeps and controlled comparisons without relying on manual knob positions or host-specific session state.

The appropriate CLI habit is **specify, inspect, render, verify**. Begin by making the operation explicit. Use a preset only as a starting point, then record important overrides in the command, config file, or manifest. Inspect the resolved configuration and estimated duration when the output could be large. Render to a new path rather than destructively replacing the source. Finally, analyze the result and retain the command or report alongside the audio.

For example, an offline wet stem can be rendered and documented as one operation:

```
verbx render vocal.wav vocal_hall_wet.wav \
  --engine algo --rt60 3.4 --predelay 32 \
  --wet 1.0 --dry 0.0 --output-subtype FLOAT \
  --json-out vocal_hall_wet.analysis.json
```

The output is an asset rather than a live processor. It can be auditioned, archived, delivered, imported into a DAW, or compared with another render. If a parameter sweep is required, commands can generate ten controlled variants without asking an engineer to move the same plug-in knob ten times and remember the exact states.

CLI automation is similarly explicit. It can be expressed through arguments, automation files, manifests, or generated commands. That is ideal for repeatable experiments and long-form transformations. It is less immediate than drawing a curve against a musical timeline, but it is easier to audit and reproduce.

The CLI's `realtime` command is useful for device testing, low-latency audition, and live command-line workflows. It does not change the product's primary offline and artifact-oriented design. A terminal realtime stream does not know the DAW's bar grid, track routing, region edits, plug-in delay compensation, offline-bounce policy, or project automation. When those relationships are the center of the work, use the plug-in.

1.1.2 The plug-in path: performance, automation, and mix context

Choose AU, AUv3, or VST3 when the reverb must live inside a musical session and respond while the arrangement plays. The host supplies audio buses, project sample rate, block size, transport, tempo, automation, preset and state recall, latency compensation, and final bounce behavior. The plug-in processes the buffers it receives and returns audio before each callback deadline.

The plug-in path is strongest for:

- placing a reverb directly on a track, auxiliary return, subgroup, object, or spatial bus while hearing the complete arrangement;
- adjusting decay, pre-delay, damping, diffusion, width, mix, Freeze, Reverse, and quality controls by ear without rendering a new file after every change;
- writing and editing host automation against bars, beats, regions, markers, picture, or performance gestures;
- changing space at a section boundary, throwing selected words into a tail, or freezing one harmony beneath the next;

- saving processor state inside the DAW session so collaborators can reopen the musical context rather than reconstruct a shell command;
- using the realtime spectrum analyzer and effective-decay display while balancing sources and returns;
- performing the reverb from the full-screen Perform and Expert pages with controls designed for continuous musical interaction.

The appropriate plug-in habit is **insert, route, automate, audition, commit**. First decide whether the processor belongs on an insert or a send. An insert changes one track's complete signal path and may use both dry and wet output. A send normally feeds a 100-percent-wet plug-in return that several tracks can share. Next establish safe gain staging and a useful static sound. Only then add automation, because automation cannot rescue a return whose basic density, bandwidth, or routing is wrong. Audition the result in context and through the required fold-down or binaural path. Commit it through the DAW's freeze, bounce, or export system when the mix is approved.

The DAW, not the plug-in, owns the output audio file. Project sample rate and callback format are host decisions; final PCM subtype, metadata, channel export, and loudness delivery are normally selected in the DAW's bounce dialog. The plug-in may oversample its wet path internally, but it cannot turn a 48 kHz host session into a 192 kHz delivered master by itself. Likewise, an analyzer display inside the editor is immediate feedback, not a substitute for a retained CLI analysis report when formal evidence is required.

Real-time processing also imposes constraints that offline rendering does not. The audio callback cannot wait for a long analysis, allocate unpredictably, open files casually, or calculate an unbounded future tail. A plug-in must honor host buffer deadlines and report latency correctly. The CLI can spend more than one second processing one second of audio when a costly algorithm or large convolution matrix justifies it. The plug-in must either finish on time, use a prepared lower-cost structure, or expose a quality choice with a clear latency and CPU tradeoff.

1.1.3 The surfaces are related, not identical

Do not assume one-to-one parameter or preset parity between the CLI and plug-in. The CLI includes commands and options that do not belong on a realtime effect panel, including file-format selection, batch manifests, report destinations, failure artifacts, offline dereverberation, IR utilities, and corpus tooling. The plug-in includes host-facing state, automation, bus negotiation, callback quality modes, and visual interaction that do not have the same meaning in an offline command.

Even controls with similar names can operate inside different constraints. The plug-in's RT60 range is `0.01` to `360` seconds and is designed for logarithmic coarse and fine adjustment during playback. The CLI can address longer bounded renders and can fail fast when a requested tail would create a surprising output duration. Plug-in Reverse is a realtime reverse-style swell; an offline reverse workflow can use future samples and render noncausal structures unavailable to a live callback. Freeze inside a running host sustains prepared network state; offline processing can inspect, extend, or transform segments without a realtime deadline.

The safest interchange is therefore through documented audio and impulse-response assets, not an assumption that every preset file is portable. If the CLI creates an IR, wet stem, surround bed, or analyzed variant, import that asset into the DAW with its report. If the plug-in creates an important automated effect, print the return and analyze the resulting audio with the CLI. When recreating a sound across paths, transfer the audible design goals and shared parameter values, then verify by listening and analysis rather than assuming numerical identity.

1.1.4 Productive hybrid workflows

Several combined workflows are especially effective:

1. **Design offline, mix interactively.** Use the CLI to synthesize or curate an impulse response, validate it, and save provenance. Load or route the resulting asset in the DAW, then automate level and musical placement with the plug-in or host tools.
2. **Audition interactively, render systematically.** Find the decay family, damping, width, and source treatment in the plug-in. Translate the approved design into explicit CLI renders for batches, alternate versions, datasets, or unusually long tails, verifying differences rather than claiming preset parity.
3. **Render wet stems, retain mix freedom.** Create 100-percent-wet mono, stereo, surround, or Ambisonic stems with the CLI. Import them into the DAW so the mix can edit, automate, compress, and spatialize the return independently of the dry source.
4. **Perform first, document afterward.** Record or bounce plug-in automation in

the DAW, then run `verbx analyze` on the printed return or final mix to produce retained decay, loudness, peak, spectral, and room evidence. 5. **Use the DAW for form and the CLI for scale.** Let the DAW determine which phrases, stems, and transitions excite a space. Export those selections, run large matrix, extreme-tail, or corpus-scale operations through the CLI, and return the rendered assets to the musical timeline.

1.1.5 Choosing quickly

Use the **CLI** when you need repeatability, batch scale, complete files, unusual formats, reports, analysis, dereverberation, IR engineering, extreme offline quality, or a command that another person or machine can rerun. Use the **plug-in** when you need to hear the effect against the arrangement, perform controls, automate against the DAW timeline, share a live return among tracks, or preserve state inside a music-production session. Use **both** when the project needs the CLI's rigor and scale together with the DAW's musical context and immediacy.

Neither path is a lesser edition of the other. The CLI is the primary verbx interface because it can express the complete processing system explicitly. The plug-ins are purpose-built production instruments because a DAW requires a different contract: bounded realtime work, host-owned routing, automatable parameters, visual feedback, and reliable session state.

1.2 AUv3 / VST3 Plug-in Track

The plug-in track brings the same reverb vocabulary into a host-managed realtime setting. The design image establishes the intended information hierarchy; the compiled captures that follow show which parts already exist in native code and which remain targets.

Figure 1-1 presents VERBX full-screen AUv3 and VST3 plug-in design and establishes the visual reference for the discussion that follows.

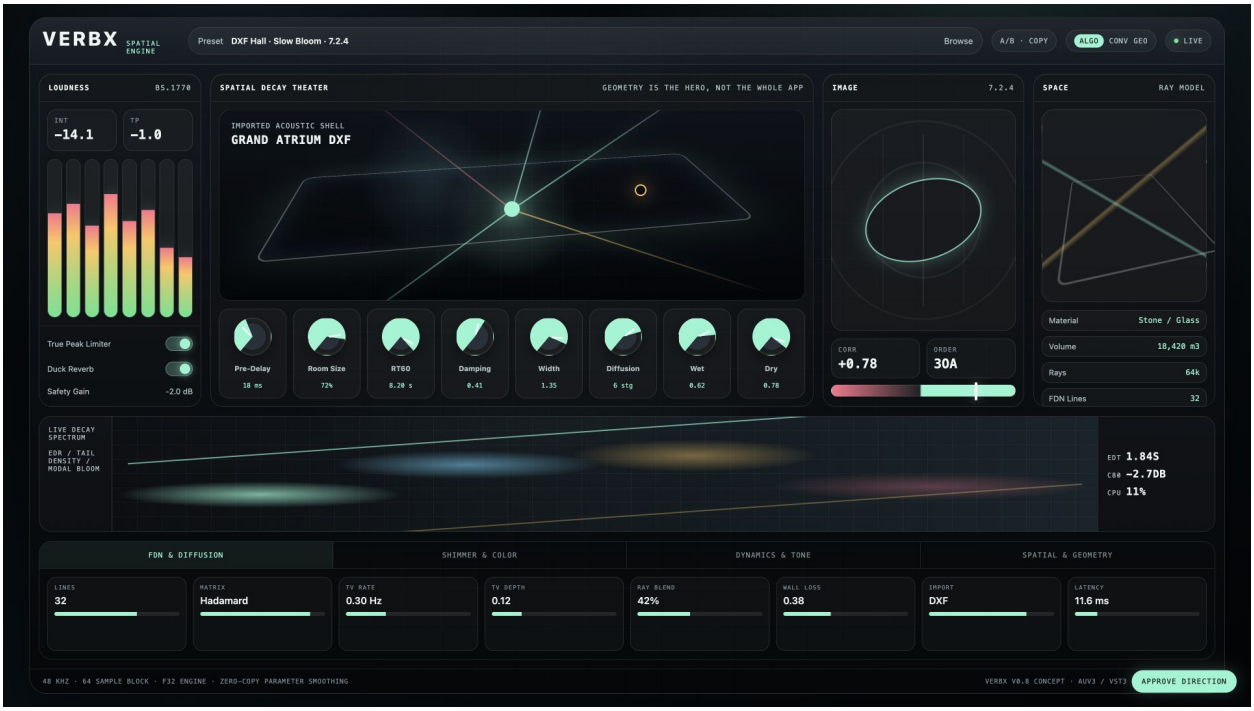

Figure 1-1: VERBX full-screen AUv3 and VST3 plug-in design

The image above is the approved `1920×1080` visual direction for the full-screen spatial console.

The capture below is the currently compiled JUCE editor. It now implements the same spatial-console composition: loudness bank, DXF geometry theater, image and ray-model panels, nine live parameter cards, horizontal decay analyzer, quality/mode controls, and lower expert sections. The host has muted its input, so the live analyzer trace correctly rests at its floor.

Figure 1-2 presents VERBX native realtime spectrum analyzer and establishes the visual reference for the discussion that follows.

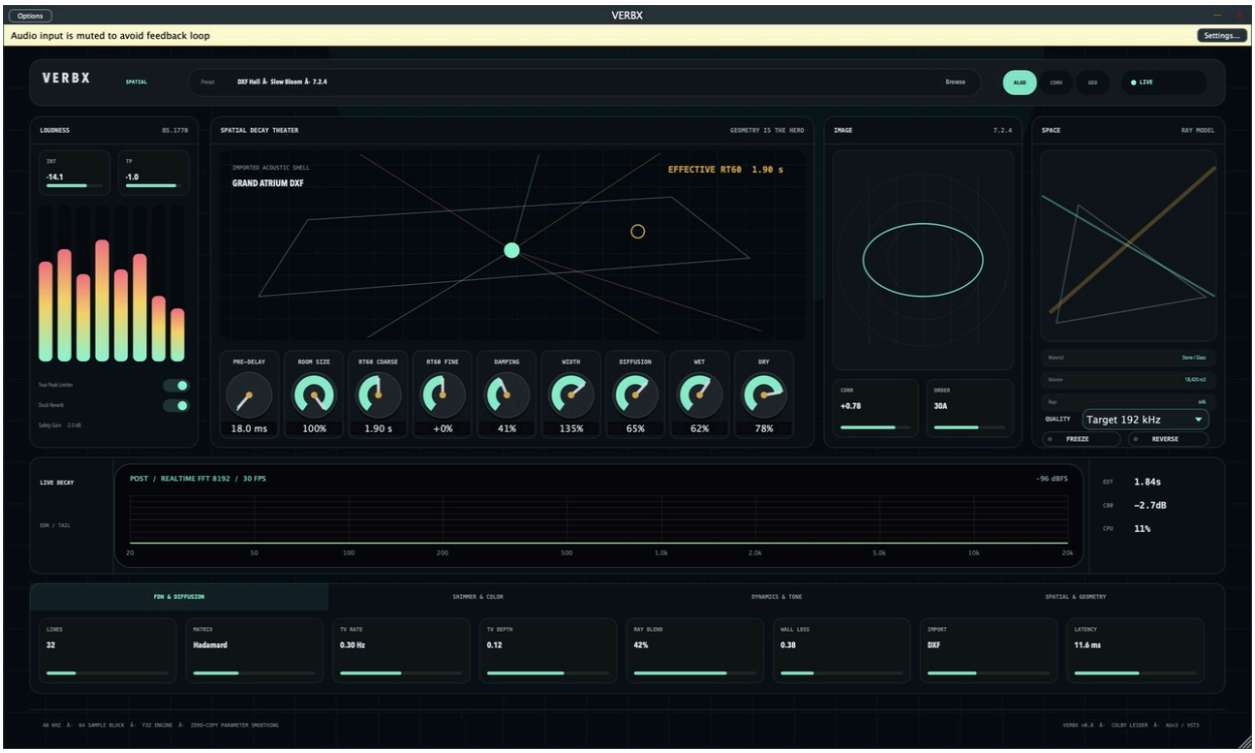

Figure 1-2: VERBX native realtime spectrum analyzer

The compiled **Expert** page mirrors all nine continuous parameters as both rotary controls and high-resolution faders, keeps the realtime spectrum in view, and adds twenty native selector buttons. Its five macro banks write the same host-automatable state for quality, width, logarithmic decay, dry/wet routing, and paired damping/diffusion character settings.

Figure 1-3 presents VERBX compiled Expert control matrix and establishes the visual reference for the discussion that follows.

Figure 1-3: VERBX compiled Expert control matrix

The first native plug-in foundation is implemented under `native/verbx_plugin`:

- C++17/JUCE host shell for AU, AUv3, VST3, and standalone targets
- shared C11 parameter manifest and real-time processing boundary
- logarithmic RT60 mapping from `0.01s` to `360s` with coarse and fine controls
- explicit Freeze and Reverse parameters
- real allocation-free wet-path oversampling for Host, 2x, 4x, and Target 192 kHz quality modes, with a 32-bit-float callback contract
- cached lock-free parameter reads on the audio callback
- overlaid realtime post-DSP spectrum analyzer with an 8192-point Hann FFT, logarithmic frequency grid, smoothed response, and peak trace
- allocation-free mono/stereo Schroeder realtime core with pre-delay, room scaling, RT60, damping, diffusion, width, wet/dry, Freeze, and a zero-lookahead reverse-style swell
- 20 ms realtime parameter smoothing for host automation without zipper noise
- responsive Perform/Expert pages with 18 linked continuous controls and 20 selector buttons; no Expert control is decorative or stored outside host state
- full-screen 16:9 spatial-console editor with the complete initial 12-parameter control surface and effective-RT60 readout

Target mode chooses the smallest integer factor that reaches or exceeds 192 kHz without downsampling. It therefore runs at 4x/192 kHz in a 48 kHz project, 2x/192 kHz at 96 kHz, and 5x/220.5 kHz at 44.1 kHz. The status strip shows the actual host rate, internal rate, factor, block size, and reported latency. Quality changes rebuild the prepared wet-path state off the audio callback.

The complete installer builds and installs the CLI, native executable, man pages, runtime extras, VST3, Audio Unit on macOS, and the standalone app:

```
./install.sh
```

If JUCE is not already available, the installer downloads the pinned JUCE `8.0.6` source release into `build/deps/JUCE`. For an offline installation or an existing checkout, use:

```
./install.sh --juce-source /path/to/JUCE
```

The default macOS plug-in destinations are `~/Library/Audio/Plug-Ins/Components/VERBX.component` and `~/Library/Audio/Plug-Ins/VST3/VERBX.vst3`; the standalone app is installed as `~/Applications/VERBX.app`. That app contains and registers the true AUv3 extension at `Contents/PlugIns/VERBX.appex`. Linux installs VST3 to `~/.vst3` and the standalone binary to `~/.local/bin/verbx-plugin`. Restart or rescan the audio host after

installation. Run `./install.sh --help` for component skips, custom destination directories, offline operation, and build controls.

Repository builds do not require JUCE unless the plug-in target is enabled manually:

```
# Verify the guarded scaffold without JUCE.
cmake -S native/verbx_plugin -B build/native/verbx_plugin

# Configure the real AU/AUv3/VST3/Standalone targets when JUCE is installed.
cmake -S native/verbx_plugin -B build/native/verbx_plugin-juce \
  -DVERBX_ENABLE_JUCE_PLUGIN=ON
cmake --build build/native/verbx_plugin-juce --config Release
```

Use `-DVERBX_JUCE_SOURCE_DIR=/path/to/JUCE` when building from a JUCE source checkout instead of an installed CMake package.

1.3 Instant Sonic Gratification

If you want immediate results with minimal decision-making, run this:

```
git clone https://github.com/TheColby/verbx.git && cd verbx && \
./install.sh --prefix "$HOME/.local" && \
verbx render ../in.wav out.wav --engine algo --rt60 120 --wet 0.99 --dry 0.01
```

That gives you an absurd long-tail render immediately. If `verbx` is not on PATH after install, run:

```
export PATH="$HOME/.local/bin:$PATH"
```

1.4 Quick Start

If `verbx` is already installed, this is the fastest way to start:

```
verbx render input.wav output.wav --engine algo --rt60 2.5 --wet 0.3 --dry 0.7
```

This applies a natural-sounding 2.5-second algorithmic reverb. Output is written to `output.wav`, with analysis at `output.wav.analysis.json`.

Before a first render or a demo, confirm the installed runtime and write a machine-readable readiness report:

```
verbx quickstart --verify --strict --smoke-test --json-out quickstart.json
```

Need a live preview instead of an offline bounce?

```
verbx realtime --engine algo --input-device "Built-in Microphone" \
  --output-device "Headphones" --rt60 20 --freeze --shimmer \
  --fdn-matrix tv-unitary --fdn-tv-rate-hz 0.35 --fdn-tv-depth 0.12 \
  --lowcut 120 --highcut 9000 --tilt 1.5 --duration 10
```

Initial realtime mode runs either direct convolution from `--ir` or an algorithmic proxy IR rendered once and monitored through the streaming convolver. It is meant for auditioning spaces and tails, not yet for the full automation/batch feature set. Realtime `--freeze` is an honest approximation: it renders a long self-sustaining proxy tail for live auditioning rather than reusing the offline segment-freeze operator.

Need live cleanup instead of added space?

```
verbx realtime --live-mode dereverb \
  --input-device "Built-in Microphone" \
  --output-device "Headphones" \
  --sample-rate 48000 --block-size 384 \
  --dereverb-mode wiener --dereverb-strength 0.85 \
  --dereverb-window-ms 12 --dereverb-hop-ms 4 --dereverb-tail-ms 90 \
  --dereverb-pre-emphasis 0.2 --dereverb-max-atten-db 18 \
  --duration 10
```

If you need to install first on macOS:

```
brew tap thecolby/verbx
brew install thecolby/verbx/verbx
verbx quickstart
```

Need starting settings for your file?

```
verbx suggest input.wav
verbx quickstart
```

Need the long-form manual?

```
python3 scripts_generate_docs_pdf.py
```

That writes docs/USERGUIDE.md and USERGUIDE.pdf, combining this README with the user-facing guides and tip-heavy docs in docs/.

1.4.1 Five Runnable Examples

```
# 1. Natural room — voice or piano in a medium hall
verbx render in.wav hall.wav --engine algo --rt60 2.0 --wet 0.25 --dry 0.8 --pre-delay-ms 18

# 2. Convolution with a real IR — character follows the space you measured
verbx render in.wav conv.wav --engine conv --ir hall_ir.wav --partition-size 16384

# 3. Shimmer pad — pitch-shifted ambient wash, good for synths
verbx render in.wav shimmer.wav --engine algo --rt60 12 --wet 0.85 \
  --shimmer --shimmer-semitones 12 --shimmer-mix 0.35 --bloom 2.0

# 4. Broadcast loudness target — -23 LUFS, -1 dBTP true peak
verbx render in.wav broadcast.wav --target-lufs -23 --true-peak --target-peak-dbfs -1

# 5. Extreme ambient — 90-second tail, slow evolution, near-frozen
verbx render in.wav ambient.wav --engine algo --rt60 90 --wet 0.92 \
```

```
  --fdn-matrix tv_unitary --fdn-tv-rate-hz 0.08 --bloom 2.0 --tilt 0.8

# 6. Comb-cloud texture — dense metallic haze before the late field
verbx render in.wav combcloud.wav --engine algo --rt60 6 --wet 0.78 --dry 0.35 \
  --comb-cloud --comb-cloud-count 32 --comb-cloud-feedback 0.42 --comb-cloud-mix 0.30 \
  --fdn-lines 12 --fdn-matrix hadamard
```

```
# Output-definition presets (default is HD)
verbx render in.wav out_hd.wav --engine conv --ir hall_ir.wav
verbx render in.wav out_md.wav --engine conv --ir hall_ir.wav --quality-preset md
verbx render in.wav out_sd.wav --engine conv --ir hall_ir.wav --quality-preset sd
```

1.5 Full Installation Instructions

With uv (fastest):

```
git clone https://github.com/TheColby/verbx.git && cd verbx
uv sync && uv run verbx --help
```

For realtime audio device support:

```
uv sync --extra realtime
```

With pip + venv:

```
python3 -m venv .venv && source .venv/bin/activate
pip install -e . && verbx --help
```

For realtime audio device support:

```
pip install -e ".[realtime]"
```

Complete installation, including native CLI and plug-ins:

```
./install.sh
verbx --help && man verbx-render
man verbx-dereverb
```

This installs the Python package with realtime and SOFA runtime extras by

default. Use `--minimal-python`, `--skip-native`, `--skip-plugins`, `--no-man`,

or custom `--au-dir`, `--vst3-dir`, and `--app-dir` destinations when a smaller or system-managed installation is preferable. Inspect the complete plan without

changing the machine using `./install.sh --dry-run`.

If VERBX does not appear after fully quitting and reopening the DAW, force a signed reinstall and Audio Unit cache rebuild on macOS:

```
./install.sh --reset-plugin-cache
```

VERBX appears under the plug-in vendor **Colby Leider**. Logic and GarageBand can use the AUv2 component or the AUv3 app extension; VST3 hosts use `VERBX.vst3`. The editor opens at a host-safe 1280x720 and remains fully resizable down to 800x450 while preserving the full console layout. The installer now signs the nested AUv3 extension before its containing app, registers it with PlugInKit, and seals and strictly verifies every installed macOS bundle, touches the plug-in paths, and restarts the Audio Component Registrar. The explicit cache-reset option backs up existing Apple Audio Unit cache files beneath `~/.local/share/verbx/cache-backups/` before clearing them. DAW-specific VST3 caches may still require the host's "rescan all plug-ins" command.

An ad-hoc-signed AUv3 can validate but still fail when Logic launches its app extension. To avoid shadowing the reliable AUv2 component, `./install.sh` installs but unregisters ad-hoc AUv3 hosting by default. Use an Apple signing identity for production AUv3 hosting:

```
./install.sh --codesign-identity "Apple Development: Your Name (TEAMID)"
```

`--enable-adhoc-auv3` is available only for local extension debugging.

macOS plug-ins build as universal `arm64+x86_64` binaries by default so they remain visible to native Apple Silicon DAWs and hosts running under Rosetta. The default deployment floor is macOS 12. Override these release defaults only when intentionally producing a narrower local build:

```
./install.sh --macos-architectures arm64 --macos-deployment-target 14.0
```

With Homebrew (macOS):

```
brew tap thecolby/verbx
brew install thecolby/verbx/verbx
verbx version
```

Official tap repository: `TheColby/homebrew-verbx`.

For local maintainer testing, you can also install from the in-repo formula:

```
brew install --build-from-source ./packaging/homebrew/verbx.rb
```

Requirements: Python 3.11+, `libsndfile` on system path. Optional: `numba` (faster algorithmic path), `cupy` (CUDA convolution), `h5py` (SOFA import/extract via `verbx ir sofa-*`), `sounddevice` (realtime input/output via `verbx realtime`).

Homebrew maintainer and release details are consolidated below.

1.6 Homebrew Distribution and Release Maintenance

The installation commands above cover the user path. The tap's release, automation, and compatibility details follow for maintainers and packagers. The published tap formula lives at `Formula/verbx.rb` in `TheColby/homebrew-verbx`.

1.6.1 Maintainer Workflow

A release maintainer refreshes the formula only after the version and release artifact are final. The workflow below keeps the formula pin, source checksum, and published tag aligned.

1. Refresh formula pins for a new release tag:

```
./scripts/refresh_homebrew_formula.sh 0.9.9
```

2. Commit formula changes in this repo:

```
git add packaging/homebrew/verbx.rb scripts/refresh_homebrew_formula.sh docs/HOMEBREW.md
git commit -m "chore(homebrew): refresh formula for v0.9.9"
```

3. Ensure tap repo formula is updated and pushed.

1.6.2 Release Automation

The release workflow can synchronize the tap after a version tag is pushed. Repository secrets and policy variables determine whether a missing tap credential blocks the release.

`.github/workflows/release.yml` includes gated tap sync:

- Trigger: tag push (`v*`)
- Job: `sync-homebrew-tap`

- Requires secret: `HOMEBREW_TAP_TOKEN` (when sync required)
- Optional variable: `HOMEBREW_TAP_REPO` (default `TheColby/homebrew-verbx`)
- Policy variable: `RELEASE_REQUIRE_HOMEBREW` (`true` by default)

Behavior:

- `RELEASE_REQUIRE_HOMEBREW=true` (default): release fails if token is missing.
- `RELEASE_REQUIRE_HOMEBREW=false`: sync is skipped when token is missing.

1.6.3 Compatibility Notes

The formula keeps compiler-sensitive acceleration packages outside its required resource set. This limits installation failures without changing the core command-line DSP contract.

- Formula pins Python resources using `brew update-python-resources`.
- `numba`/`llvmlite`/`scikit-learn` are intentionally excluded from Homebrew resources to avoid fragile compiler-bound builds in common macOS environments.
- Core DSP paths are unaffected; shimmer remains available with librosa and falls back gracefully when acceleration dependencies are absent.

If `verbx` is not found after install, add `~/.local/bin` to your PATH:

```
export PATH="$HOME/.local/bin:$PATH"   # add to ~/.zshrc or ~/.bashrc
```

1.7 Python API (Research Workflows)

Use `verbx` as a library when you need notebook/pipeline integration:

```
from verbx.api import analyze_file, generate_ir, render_file
from verbx.config import RenderConfig
from verbx.ir import IRGenConfig

report = render_file("dry.wav", "wet.wav", RenderConfig(engine="algo", rt60=2.5, wet=0.7))
ir_audio, ir_sr, ir_meta = generate_ir(IRGenConfig(mode="fdn", duration=3.0, sr=48000))
metrics = analyze_file("wet.wav", include_loudness=True)
```

1.8 Audio Examples

Rendered examples are included in `examples/audio/`. The pack is now delivered at 48 kHz, PCM24. Most examples are stereo; the utility click and short hybrid IR files are mono. The shimmer-heavy examples were re-rendered at this higher rate specifically to remove the grit from the older 24 kHz / PCM16 pack.

GitHub repository README pages do not provide reliable inline audio controls. The `Play` links below open each asset directly in the browser's native media player with one click.

1.8.1 Utility and verification files

Use these short files to confirm that playback, convolution, and dry/wet comparison work before judging a complex render.

Table 1-1: Utility and verification files: reference table

File	Play	Description
dry_click.wav	Play	One-shot dry click reference for sanity checks
dry_click_reverbed.wav	Play	Reverberated click for immediate A/B verification
hybrid_ir_short.wav	Play	Short hybrid IR asset used in quick convolution demos

1.8.2 Realistic dry/wet example pairs

These pairs keep the source and processed result adjacent so room character can be judged without searching for a matching dry file.

Table 1-2: Realistic dry/wet example pairs: reference table

File	Play	Description
realistic_speech_dry.wav	Play	Dry speech source used for room and plate examples
realistic_speech_room.wav	Play	Natural speech room render
realistic_music_dry.wav	Play	Dry music source used for ambient and shimmer examples
realistic_music_hall.wav	Play	Natural concert-hall style music render, re-tuned for a cleaner less congested tail
realistic_drums_dry.wav	Play	Dry drum source used for room and cathedral examples
realistic_drums_room.wav	Play	Natural drum room render

1.8.3 Extreme range demos

The extreme examples are stress tests for long decay, dense feedback, and pitch-shifted energy. Match playback level before comparing them with the natural examples.

Table 1-3: Extreme range demos: reference table

File	Play	Description	Key settings
extreme_cathedral_drums.wav	Play	Drums → 8s Hadamard FDN cathedral	`--rt60 8.0 --fdn-lines 16 --fdn-matrix hadamard`
extreme_shimmer_music.wav	Play	Music → 6s reverb with octave shimmer	`--shimmer --shimmer-semitones 12 --shimmer-feedback 0.65`
extreme_plate_speech.wav	Play	Speech → circulant FDN plate simulation	`--rt60 1.8 --fdn-matrix circulant --lowcut 200 --highcut 6000`
extreme_frozen_music.wav	Play	Music → 30s near-infinite tail (32-line FDN)	`--rt60 30.0 --fdn-lines 32 --wet 0.95`

1.8.4 Experimental music tradition demos

Eight examples drawn from the experimental and avant-garde music tradition, each isolating a different reverb behavior or aesthetic.

Table 1-4: Experimental music tradition demos: reference table

File	Play	Inspiration	What to listen for
lucier_sitting_room.wav	Play	Alvin Lucier – *I Am Sitting in a Room* (1969)	Speech run through the room 7× until only resonant frequencies survive
eno_discreet_music.wav	Play	Brian Eno – *Discreet Music* (1975) / Ambient series	12s tail swallowing the source into a continuous wash
oliveros_deep_listening.wav	Play	Pauline Oliveros – *Deep Listening* (1989)	18s cave-scale resonance, very low damping, 32-line FDN
fripp_frippertronics.wav	Play	Robert Fripp – Frippertronics tape-loop	Octave shimmer with 0.78 feedback accumulating over 8s
mbv_shoegaze.wav	Play	My Bloody Valentine – *Loveless* (1991) wall of sound	Dense shimmer wash (mix 0.55) through circulant FDN
reich_phase_drums.wav	Play	Steve Reich – phase minimalism	Tight 0.7s room on percussion, circulant diffusion
radigue_drone.wav	Play	Eliane Radigue – *ADNOS* (1973–1974) / drone electronics	45s near-infinite sustain, 32-line Hadamard, wet 0.97
feldman_sparse_room.wav	Play	Morton Feldman – late period	3.8s room, low wet (0.52), allpass diffusion, contemplative space

Dry source files are in the same directory. See examples/audio/README.md for the full render commands.

1.9 Public Alpha Launch Notes

Current public alpha release: **v0.9.9**.

Current stabilization status:

- v0.9.9 adds full-state A/B comparison to the native plug-in editor for direct reverb decisions without leaving the current session.
- Render-performance budgets and native structural parity are now blocking CI gates. Native source-duration floors and analytic ISM timing references have focused regression coverage.
- Current native-track decision: v0.8 is a hybrid transition release, with verbx-c shipped as an opt-in native render/doctor binary while the Python CLI remains the public alpha default. See docs/ROADMAP_NEXT_4_WEEKS.md.
- verbx is currently research-grade software (public alpha), not production-certified.
- Confirm your environment with verbx quickstart --verify --strict and verbx doctor.
- Verify one algorithmic render and one convolution render before batch usage.
- For live monitoring, verify verbx realtime --list-devices before relying on realtime auditioning.
- For reproducible reports and bug submissions, attach --repro-bundle outputs and verbx doctor --json-out doctor.json.
- For demo-ready outputs, keep --true-peak --target-peak-dbfs -1 enabled when files will be transcoded.
- Public alpha scope, known limitations, and support paths: docs/PUBLIC_ALPHA_NOTES.md
- Launch-week pinned demo commands and expected SHA256 outputs: docs/LAUNCH_WEEK_DEMO_PINS.md
- Canonical launch-example parity check:

```
python scripts/check_launch_examples.py --check
```

- PyPI publish auth setup for maintainers: docs/PYPI_PUBLISH_SETUP.md

---

1.10 v0.8 Native Track

v0.8 is the planned native C executable line. The released tool remains the Python implementation in v0.7.x, but the native rewrite has now started with an executable scaffold under native/verbx_c/README.md. The live feature/gap table is tracked in docs/NATIVE_PARITY.md.

Chosen v0.8 release shape: **hybrid wrapper phase before full replacement**. verbx-c is the opt-in native executable for deterministic offline render, doctor diagnostics, and machine-readable native support bundles. The Python verbx CLI remains the default for realtime, dereverb, convolution, IR tools, batch workflows, immersive utilities, and the full FDN feature surface until native parity is proven by the checked-in contract fixtures.

Current native status:

- standalone verbx-c executable target
- portable C11 build path via scripts/build_verbx_c.sh
- implemented commands: help, version, doctor, render
- mono/stereo WAV input: PCM16/24/32 and float32/float64
- mono/stereo WAV output: pcm16, float32, float64
- deterministic offline render lifecycle in C: read -> process -> tail-finalize -> write
- explicit native process/error contract surfaced in verbx-c doctor
- native tail-stop metric selection: --tail-metric peak|rms
- native tail-padding safety: implicit algorithmic padding over 300 seconds is

rejected unless an explicit --tail-limit bounds the render; native reports the resolved tail_padding_seconds in native-render-report-v1

- native peak-safe output: --peak-safe --peak-ceiling-db DB
- native JSON reports: doctor --json-out and render --json-out
- foundational native algorithmic reverb core with float64 internal processing
- reusable verbx_c_core library with a tested plug-in parameter manifest
- realtime context API with persistent mono/stereo reverb state, quality-target status, zero-latency reporting, Freeze, and reverse-style swell behavior
- guarded C++/JUCE AU, AUv3, VST3, and standalone shell with the complete initial control dock and realtime spectrum overlay
- plug-in RT60 coarse/fine mapping and native render floor aligned at 0.01s

v0.8 in scope:

- verbx-c doctor and verbx-c render
- deterministic mono/stereo WAV offline render
- pcm16, float32, and float64 output
- render/doctor JSON reports for support bundles
- parity comparison against tests/fixtures/native_render_parity_contract.json

Deferred beyond the first v0.8 slice:

- replacing Python verbx as the default command
- full native realtime reverb DSP and device/DAW production validation
- convolution, dereverb, IR synthesis/morphing, batch, immersive, and AI helpers

- full Python FDN parity, automation lanes, shimmer/freeze/repeat, and broad preset coverage

Example native smoke test:

```
./scripts/build_verbx_c.sh --doctor
scripts/install_verbx_c.sh --prefix "$HOME/.local" --doctor
./build/native/verbx_c/verbx-c render in.wav out.wav \
  --rt60 3.5 \
  --peak-safe \
  --out-format float32 \
  --json-out native-render.json
```

1.11 Experimental DXF Room Tracing

`verbx ir trace` is the first bounded physical-acoustics prototype: it reads a simple room-like DXF outline, generates a stereo IR, and writes a `trace-report-v1` support bundle.

```
verbx ir trace room.dxf room_ir.wav \
  --source 2,3,1.5 \
  --listener 6,4,1.5 \
  --height 3 \
  --material studio \
  --layer-material west-wall=concrete \
  --rays 50000 \
  --length 4 \
  --target-sr 48000 \
  --json-out room_trace.json

verbx render dry.wav in_room.wav --engine conv --ir room_ir.wav
```

This MVP supports ASCII DXF `LINE`/`LWPOLYLINE` 2D room outlines and derives an axis-aligned room box with direct path, first-order reflections, and a stochastic late tail. `--material` now validates against octave-band material profiles such as `studio`, `drywall`, `glass`, `concrete`, `carpet`, `acoustic-panel`, and `diffuser`. Repeat `--layer-material LAYER=MATERIAL` to override the material for an existing ASCII-DXF outline layer. `LINE` and `LWPOLYLINE` segments resolve independently to bounding-box walls; a 2D plan cannot assign floor or ceiling materials. The report records the resolved per-surface profiles. See `docs/DXF_TRACE_MVP.md`.

1.12 Announcement Channels

Release notices are published through the project-owned channels below. Homebrew's general blog is included only for ecosystem context, not as the announcement channel for this tap.

- Release announcements: github.com/TheColby/verbx/releases
- Homebrew tap updates: github.com/TheColby/homebrew-verbx
- Homebrew project news/blog: brew.sh/blog

Note: Homebrew blog posts cover Homebrew project releases and ecosystem updates; third-party tap formula launches are announced by the tap/project maintainers.

1.13 Suggested Exercises

1. Render one dry excerpt through a short room, a plate, and a long hall; loudness-match the results and describe the change in distance, width, and spectral decay.
2. Create a fully wet return for one source and a parallel wet/dry insert for the same source. Compare the two routing models in a short mix.
3. Run `verbx analyze` on a dry file and one rendered file, then identify which reported quantities support, complicate, or contradict your listening notes.
4. Make three deterministic variations of one preset by changing only RT60, pre-delay, and damping. Preserve the command lines and JSON reports.
5. Build a one-minute listening test that alternates dry, early-field, late-field, and complete versions without revealing their order to a listener.
6. Create a before-and-after production note that names the source, room role, routing, target loudness, and the one parameter you would revisit first.
7. Render the same preset at two sample rates and compare the tail after level matching; distinguish audible differences from file-format assumptions.
8. Use a supplied preset as a starting point, then make a version appropriate for dialogue, one for drums, and one for sustained harmony.
9. Prepare a one-page handoff for another engineer containing the command, assets, expected output, and a reversible troubleshooting step.
10. Choose one surprising render result and explain whether it arises from source material, parameter interaction, monitoring, or an actual defect.

“Because of the reverberation, there’s always more to the sound than just the sound.”

— Pauline Oliveros (Date unknown)

Chapter 2

What Is Reverb? (and why verbx sounds different)

Reverb begins when a sound should be over but the room keeps it alive. A violinist lifts the bow, a singer closes a consonant, or a drummer strikes a snare. The source may stop almost at once, but the sound does not. It reaches the floor, walls, ceiling, seats, bodies, scenery, and every irregular surface nearby, then returns by paths of different lengths. Each trip changes its level, color, direction, and timing.

We hear those returns as part of the place. Even with our eyes closed, they tell us whether a room feels intimate or immense, bright or muffled, empty or crowded. They help us judge distance and direction, but they also shape the mood of a performance. A dry voice can feel private and immediate. The same voice in a long stone decay can feel ceremonial, remote, or larger than the person who made it.

That is why reverb is more than an effect placed behind a track. It is an acoustic event, a listening cue, and a musical material. We will begin with what happens between the source and the ear, listen to examples, and then take apart the DSP that recreates

or deliberately bends that experience. At ordinary settings, verbx can suggest a recognizable room. At extreme settings, the room gives way to something else: a way of composing with memory and time.

2.1 The Acoustic Event: Direct Sound, Early Reflections, and Late Field

When sound leaves a source, the first arrival is usually the direct path. It carries the clearest localization information and the sharpest transient. Reflections arriving in the next several tens of milliseconds form the early-reflection pattern. Their timing and direction tell the ear whether boundaries are close or far, symmetrical or irregular, hard or absorptive. After enough reflections, individual paths become too dense to follow. The listener hears a late field whose statistical behavior matters more than the identity of any one echo.

The figure below separates those three time regions before they combine at the ear. It is deliberately a perceptual diagram rather than a floor plan: the same room can produce different direct-to-reverberant balances when the source or listener moves.

Figure 2-1 presents Anatomy of a reverberant event from source to perception and establishes the visual reference for the discussion that follows.

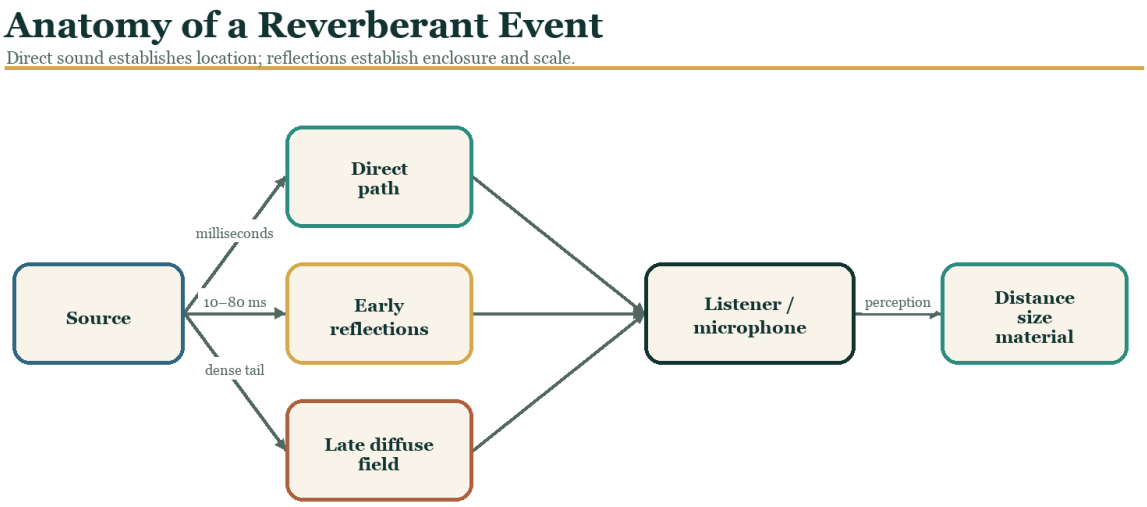

Figure 2-1: Anatomy of a reverberant event from source to perception

How to read this figure. The upper path preserves source identity and position. The middle path supplies discrete boundary information: a strong floor reflection may add weight, while a lateral reflection may add width. The lower path carries the room's integrated memory. A production can alter each path separately with pre-delay, early/late balance, diffusion, damping, and wet/dry mix. "More reverb" is therefore not one operation. It may mean more early energy, a longer late decay, a darker tail, a wider return, or a lower direct-to-reverberant ratio.

The boundaries are not absolute. A reflection at 45 ms may be heard as part of the source on a legato cello phrase and as a distinct slap after a rimshot. Tempo, articulation, register, masking, and playback level all change the perceptual threshold. The useful engineering question is not simply "How many milliseconds is early?" but "What musical role does this arrival play?"

2.1.1 Four Time Scales in One Sound

Reverb becomes easier to design when heard at four nested time scales. The first is the waveform cycle, measured in fractions of a millisecond; phase and comb filtering live here. The second is the onset window, roughly the first 80 ms; localization, pre-delay, and early reflections live here. The third is the phrase window, from a few hundred milliseconds to several seconds; note overlap, clarity, and cadence live here. The fourth is formal time, from tens of seconds to minutes; freeze, recirculation, and extreme RT60 settings live here.

A 20 ms pre-delay is tiny compared with a symphonic phrase but large compared with the period of a 1 kHz tone. A two-second tail is long compared with a sixteenth note at 120 BPM but short compared with the pause after a cathedral cadence. A 120-second tail no

longer describes an ordinary enclosure: it becomes a layer of form. verbx sounds different partly because its controls remain meaningful across all four scales.

2.1.2 A Staircase That Answers Like a Bird: Chichén Itzá and Archaeoacoustics

At El Castillo, the stepped Pyramid of Kukulcán at Chichén Itzá, a handclap made in front of the principal stairway does not return as a neutral copy of the clap. Listeners hear a brief, pitched sweep whose falling contour has often been compared with the call of the resplendent quetzal, *Pharomachrus*

mocinno. The surprising point is not that a bird naturally sounds reverberant. It is that architecture can filter and reorganize an impulsive human sound until the returning echo acquires a birdlike gesture. Garza and colleagues describe this effect as *la cola del quetzal*, or the quetzal's tail, and note that the bird was culturally important to the Maya even though it was not native to the northern Yucatán region.

This is a particularly vivid case of **archaeoacoustics**, the interdisciplinary study of sound and listening in archaeological places. Archaeoacoustics combines acoustic measurement, signal analysis, architectural reconstruction, archaeology, anthropology, and the study of performance or ritual. Its question is not only "What transfer function does this structure have today?" It also asks how audibility, echo, speech projection, musical instruments, movement, and spatial separation may have shaped past human activity. A rigorous study must therefore keep three levels distinct: the physical sound field, its physiological and cognitive perception, and the cultural meanings that people may have attached to it.

The Chichén Itzá effect begins with a broadband transient. A handclap sends energy toward a large, approximately periodic array of stone treads and risers. Contributions scattered from different portions of the staircase travel different path lengths and return at different times. At some frequencies those contributions reinforce one another; at others they partially cancel. The stairway consequently behaves less like a single wall and more like a distributed acoustic filter. The returning energy is both delayed and spectrally reorganized.

In impulse-response language, the received signal is still

$$y(t) = x(t) * h(t),$$

but $h(t)$ is highly structured. It contains a sequence of weighted, frequency-shaped returns associated with the stair geometry rather than a featureless exponential tail. If adjacent scattering paths differed by one fixed delay $\Delta\tau$, their interference would produce an approximate repetition spacing of $f_r \approx 1/\Delta\tau$. The real staircase presents a changing set of path lengths, angles, and amplitudes, so its strongest spectral emphasis moves through time. That motion is heard as a chirp rather than as a stationary comb-filter pitch.

Declercq, Degrieck, Briers, and Leroy (2004) compared diffraction simulations with field recordings and found that the result depends strongly on the excitation. A mathematical impulse and a real handclap do not produce identical echoes because their spectra and temporal envelopes differ. Their model also found that adding the ground reflection changed the calculated effect much less than changing the incident sound. This source dependence is an important reverb lesson: a room response may be linear, yet the part of that response we hear most clearly depends on what the source supplies for the architecture to filter.

The chirp should not be confused with an ordinary diffuse late reverberant field. It is closer to a colored early reflection or a short sequence of coherent echoes. A concert hall tail tends toward dense statistical overlap, while the El Castillo staircase preserves enough periodic structure for the ear to follow a frequency glide. Both phenomena nevertheless belong in the same conceptual chapter. They demonstrate that geometry converts propagation time into timbre and that an impulse response is a record of architecture, source position, receiver position, and boundary structure.

The bird comparison also requires care. The similarity between the chirped echo and a quetzal call is a perceptual and spectrographic comparison, not proof that the monument was designed as a mechanical bird voice. The present structure has a construction, weathering, excavation, and restoration history; modern listeners and handclaps are not direct recordings of ancient practice. The existence of the acoustic effect is measurable. Its intentionality and original cultural function require independent archaeological evidence.

Garza, Medina, Padilla, Ramos, and Zalaquett (2008) therefore argue for systematic Maya archaeoacoustics rather than isolated marvels. Their framework joins calibrated measurements and mathematical models with evidence about architecture, instruments, public events, ritual, and social organization.

It also treats hypotheses as testable: researchers should document source and receiver coordinates, excitation type, weather, crowd conditions, surface state, and uncertainty; compare multiple monuments; and ask whether a proposed effect survives plausible reconstructions of the earlier building.

For a reverb designer, Chichén Itzá offers a useful reversal of the usual goal. Instead of hiding individual delays until they fuse into a smooth room, preserve a controlled periodicity long enough for the reflection pattern to become a recognizable gesture. A short nonuniform multitap network, a bank of dispersive comb filters, or a measured impulse response can produce birdlike sweeps without a long RT60. The musical result is part echo, part filter, and part spatial sign. It reminds us that reverberation can carry identity before it carries duration.

2.1.3 RT60 Is Important, but It Is Not the Sound

RT60 is the time required for reverberant energy to decay by 60 dB. It is a useful summary because it turns a complex response into a number that can be compared across rooms and algorithms. It is not a complete description. Two reverbs can share an RT60 and differ radically in onset, echo density, modal coloration, high-frequency loss, stereo behavior, and how smoothly they reach the noise floor.

For a nearly exponential decay, relative level follows

$$L(t) \approx -60\frac{t}{T_{60}} \text{ dB}.$$

At half the RT60, the idealized tail is 30 dB below its starting level. At one quarter, it is 15 dB down. Music complicates this simple line: the source may continue exciting the room, different frequency bands may decay at different rates, and a time-varying network may exchange energy between modes. The number is still valuable, but the ear hears the complete energy distribution.

The Schroeder frequency gives a second useful boundary:

$$f_s \approx 2000\sqrt{\frac{T_{60}}{V}},$$

where f_s is in hertz, T_{60} is in seconds, and V is room volume in cubic meters. Below this approximate transition, individual modes tend to be perceptually important; above it, a statistical diffuse-field model becomes more plausible. In a virtual reverb, delay lengths and feedback topology create an analogous modal region. Long RT60 values make weakly damped modes easier to hear, so a stable gain formula is not enough: delay distribution, matrix structure, modulation, and spectral correction must also be designed well.

2.1.4 Space, Distance, and the Direct-to-Reverberant Ratio

Listeners often interpret reverb level as distance. A close source normally has a high direct-to-reverberant ratio; a distant source has a lower one. But level alone is ambiguous. A close singer in a highly reflective chamber may have more reverberant energy than a distant singer outdoors. High-frequency attenuation, early-reflection timing, and the ratio of frontal to lateral energy help resolve the scene.

In a mix, preserve these relationships deliberately. Raising only the wet return can move a source backward, but excessive high-frequency content may make the return feel unnaturally close. Increasing pre-delay can preserve a close, intelligible dry onset while retaining a large room. Rolling low frequencies out of the return can reduce masking without making the room seem smaller. The art is to decide which distance cues the music needs and which it can contradict for expressive reasons.

2.2 Open-Source Acoustic Image Portfolio

Reverberation is heard, but it is also designed, built, notated, photographed, and remembered. This chapter gathers twenty-five rights-cleared score pages, manuscript leaves, architectural photographs, and images of physical reverberation devices. Together they connect musical texture to the spaces and mechanisms that prolong it. Every image is stored locally so that the guide remains reproducible offline, every figure has a source and license directly beneath its caption, and the complete provenance ledger follows in `/Users/cleider/dev/verbx/docs/assets/open_source_portfolio/ATTRIBUTIONS.md`.

Two requested twentieth-century examples require a firm rights boundary. Aaron Copland's *Fanfare for the Common Man* (1942) and Olivier Messiaen's *Apparition de l'église éternelle* (1932) remain protected works in relevant jurisdictions, and no permissively licensed opening-score reproductions were located. They are therefore discussed elsewhere in this guide but are not reproduced here. The Dukas figure uses the public-domain fanfare from *La Péri* (1911) as a legally reusable orchestral analogue; the Franck figure uses the public-domain *Choral No. 3* (1890) as an organ-writing analogue. This substitution preserves the listening and production lesson without treating online availability as permission.

The portfolio is arranged in three sequences. The first reads notation as a spatial plan: entrances, rests, registral layers, and antiphonal groups imply an acoustic before a note sounds. The second reads buildings as filters whose volume, materials, and geometry govern early reflections and decay. The third opens the machinery of artificial reverberation, from vibrating metal to folded acoustic paths and deliberately reflective test rooms.

2.2.1 Historical timeline and extended viewing guide

The dates below distinguish the age of a composition, building, or device from the date of its particular photograph or edition. They also prevent an attractive image from standing in for history: each figure represents a specific technology, institution, or musical practice whose acoustic meaning changed over time.

1. Tallis's *Spem in alium* (c. 1570) belongs to the late Tudor Renaissance; the pictured Philip Legge full-score edition dates from 2010 and exposes the forty-part tutti in modern engraving.
2. Machaut's *Messe de Nostre Dame* (c. 1365) belongs to fourteenth-century Ars Nova practice, long before modern concert-hall listening separated musical work from liturgical enclosure.
3. The Waseda chant leaf transmits a medieval notational tradition in which text, breath, ritual pace, and resonant architecture jointly organized time.
4. The Spanish chant manuscript is a hand-produced historical object whose inks, ruling, wear, and irregular script locate music in material culture rather than abstract data.
5. *Ave verum* (traditional; date unknown) uses square notation inherited from medieval chant transmission, making phrase shape more visible than modern bar-based meter.
6. Palestrina's *Missa Papae Marcelli* (c. 1562) is a late-Renaissance polyphonic monument whose sustained contrapuntal entries are highly sensitive to low-mid reverberant buildup.
7. Dukas's *Fanfare pour précéder La Péri* (1912) comes from the early twentieth century and tests whether synchronized brass attacks remain monumental without losing edge definition.
8. Franck's *Choral No. 3* (1890), from the late Romantic organ repertory, treats instrument and church as a coupled sustained-tone system.
9. Gabrieli's *In ecclesiis* (c. 1615) stands at the late-Renaissance and early-Baroque boundary, where separated ensembles and Venetian architecture made placement compositional.
10. The present St. Mark's Basilica was begun in the eleventh century and developed over subsequent centuries; its domes, galleries, and hard ornamented surfaces embody a long-lived coupled-volume acoustic.

11. Notre-Dame de Paris was built principally from the twelfth through fourteenth centuries; its repeated Gothic bays create sequential lateral and overhead reflection zones rather than one simple cavity.
12. Laon Cathedral was built chiefly during the second half of the twelfth and early thirteenth centuries; its nave supplies a clear longitudinal axis with side-aisle and vault paths.
13. Hagia Sophia was completed in 537 during the Byzantine period; its dome and semidomes demonstrate why a visually centralized enclosure can produce acoustically distributed energy.
14. Hamilton Mausoleum was constructed in the mid-nineteenth century; its hard circular interior became known for persistent discrete echoes as well as long decay.
15. The Inchindown oil tanks were built during the Second World War; their vast hard-walled industrial volumes became accidental laboratories for extraordinary reverberation.
16. *Silophone* (2000–2001), by The User, transformed four empty chambers in Montréal's Silo No. 5 into a remotely playable public instrument. Participants sent sound by telephone or web; loudspeakers projected it into the silo, microphones captured the response, and the return was broadcast back to listeners.
17. York Minster's present Gothic fabric developed from the thirteenth through fifteenth centuries, combining a long nave, transepts, tower volume, furnishings, and changing occupancy.
18. The EMT 140, introduced in 1957, converted an electromechanically driven steel plate into a compact studio decay whose density did not depend on a room-sized air volume.
19. Spring reverberators developed through twentieth-century communications, organ, guitar-amplifier, and studio practice; the exposed tank makes wave propagation and transducer coupling physically visible.
20. The Doepfer A-199 belongs to contemporary modular-synthesizer culture, where an external spring tank turns historical electromechanics into a voltage-controlled performance material.
21. The O. C. Electronics folded-line exterior represents a mid-twentieth-century attempt to package a long acoustic propagation path into practical equipment dimensions.
22. The corresponding interior reveals that the folded line was literal geometry: repeated path turns traded compact size against bandwidth, loss, and audible resonant character.
23. The TU Dresden echo chamber photograph dates from 2009, but the room continues the twentieth-century laboratory tradition of deliberately reflective spaces for repeatable acoustic testing.
24. NASA's Reverberant Acoustic Test Facility is a modern aerospace chamber whose horn array and hard enclosure create extreme diffuse acoustic loading for spacecraft qualification, not musical ambience.
25. The IETR chamber is a modern electromagnetic reverberation facility rather than an acoustic room; it is included as a caution that identical terminology can name physically different wave domains.

2.3 Scores, Manuscripts, and Spatial Notation

Thomas Tallis's *Spem in alium* (c. 1570) is an ideal threshold because its forty independent parts make space structurally audible. The following figure replaces the sparse opening with a later climactic page on which all eight five-part choirs are active. Read downward to compare the simultaneous choirs; read horizontally to see sustained tones, imitative entries, and cadential motion overlap in time. In performance, choir placement turns that polyphonic density into motion around the listener. The full-page reproduction deliberately preserves all forty staves because the disposition of active voices is itself a spatial score.

Composed in the late Tudor period, probably around 1570, the motet belongs to a Renaissance culture of large ceremonial vocal polyphony while exceeding ordinary ensemble scale. The page is especially instructive for reverberation design because every new attack enters a field already carrying many decays. A successful long-room treatment must unite the mass without erasing consonants or reducing

forty lines to one broadband pad. Compare the page at the direct sound, early-reflection, and late-tail levels: the notation asks each layer to support a different part of the musical argument.

Figure 2-2 presents Later tutti score page of Thomas Tallis's *Spem in alium* (c. 1570), with all eight five-part choirs active on one page and establishes the visual reference for the discussion that follows.
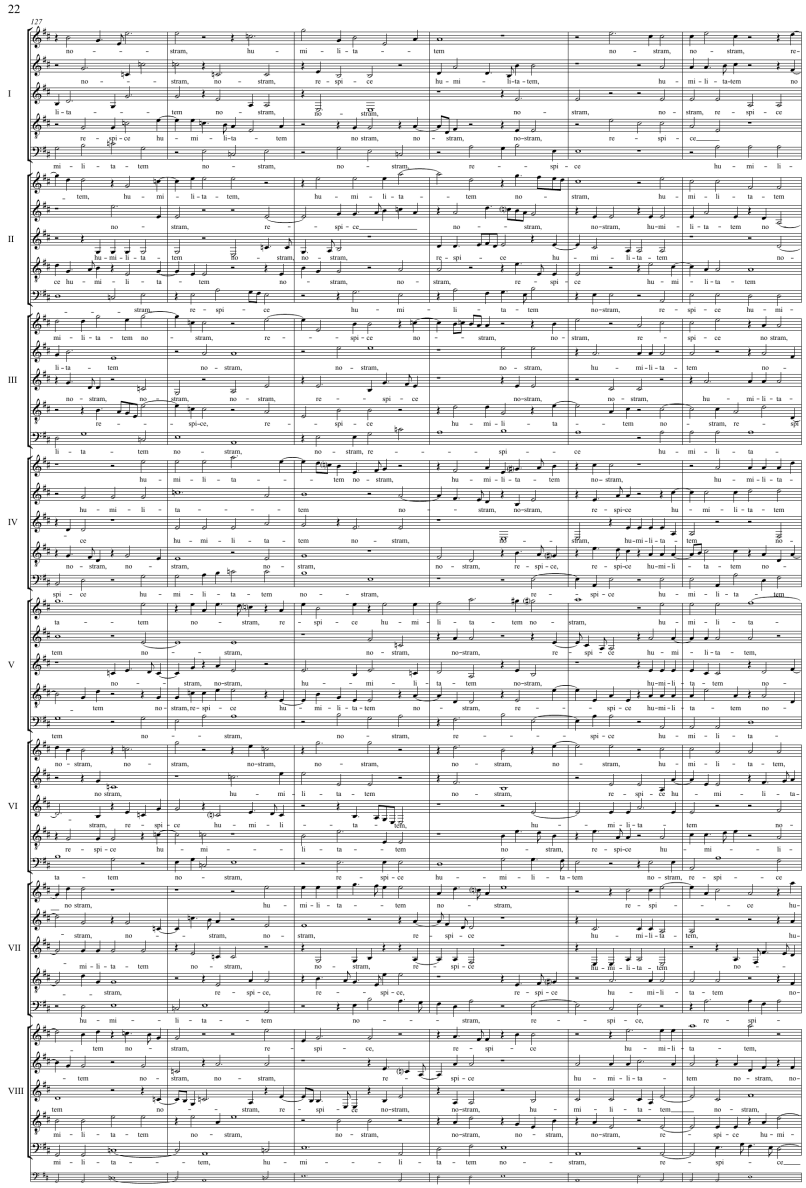
Figure 2-2: Later tutti score page of Thomas Tallis's *Spem in alium* (c. 1570), with all eight five-part choirs active on one page

Source and license: Philip Legge, editor, IMSLP score record, CC BY-SA 2.5 Australia. PDF page 22 rendered without alteration except scaling. The user's supplied IMSLP parts edition remains a useful performer-oriented companion, but the full score is used here so all choirs can be seen together.

Guillaume de Machaut's *Messe de Nostre Dame* (c. 1365) represents a different relationship between line and room. the following figure is a modern public-domain score rather than a manuscript facsimile, making the contrapuntal layers immediately legible. Long values, staggered syllables, and voice crossings leave acoustic room between attacks while allowing sustained energy to overlap. When auditioning this repertoire through verbx, begin with moderate early-reflection density and a long but frequency-shaped tail; excessive broadband sustain can erase the rhythmic distinction between the voices that the notation works so carefully to preserve.

Figure 2-3 presents Opening score page from Guillaume de Machaut's *Messe de Nostre Dame* (c. 1365), showing layered late-medieval polyphony in modern notation and establishes the visual reference for the discussion that follows.
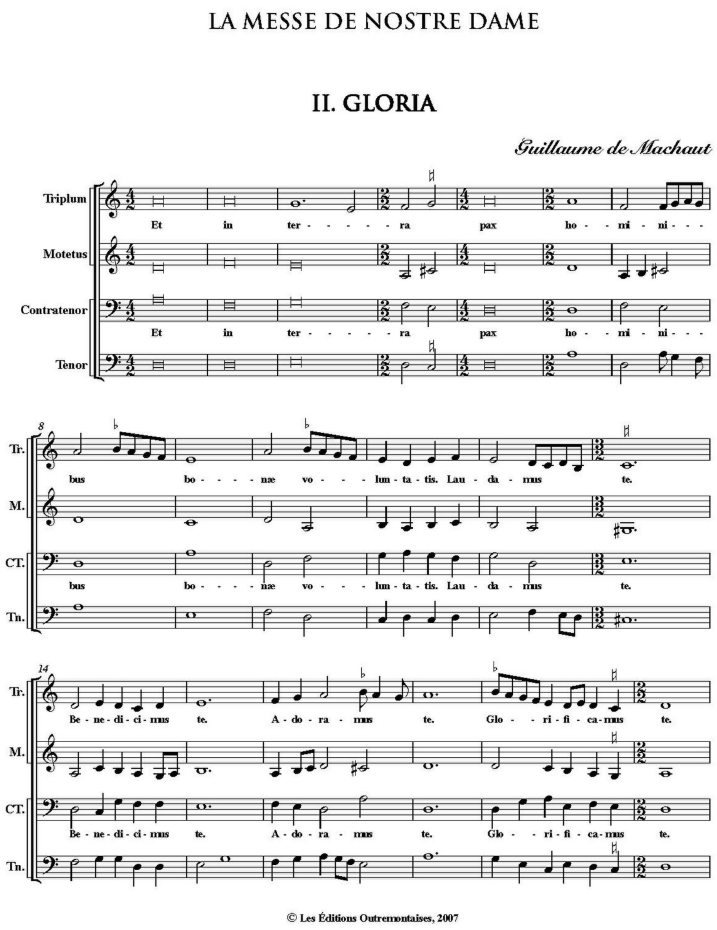
Figure 2-3: Opening score page from Guillaume de Machaut's *Messe de Nostre Dame* (c. 1365), showing layered late-medieval polyphony in modern notation

Source and license: Guillaume de Machaut, Wikimedia Commons scan record, Public Domain Mark 1.0. PDF page 1 rendered without alteration except scaling.

Gregorian chant is often described as a single melodic line, but the physical page and the performance space supply additional dimensions. the following figure shows a chant leaf preserved by Waseda University Library. The compact notation block sits within a broad field of page material, reminding the

reader that pacing is carried by breath, text, liturgy, and architecture rather than by a modern metrical grid. For reverb design, chant is a stringent test: the tail should connect phrases and magnify the contour without smearing consonants into an undifferentiated wash.

Figure 2-4 presents Gregorian chant leaf with staff notation and a wide manuscript field, preserved by Waseda University Library and establishes the visual reference for the discussion that follows.
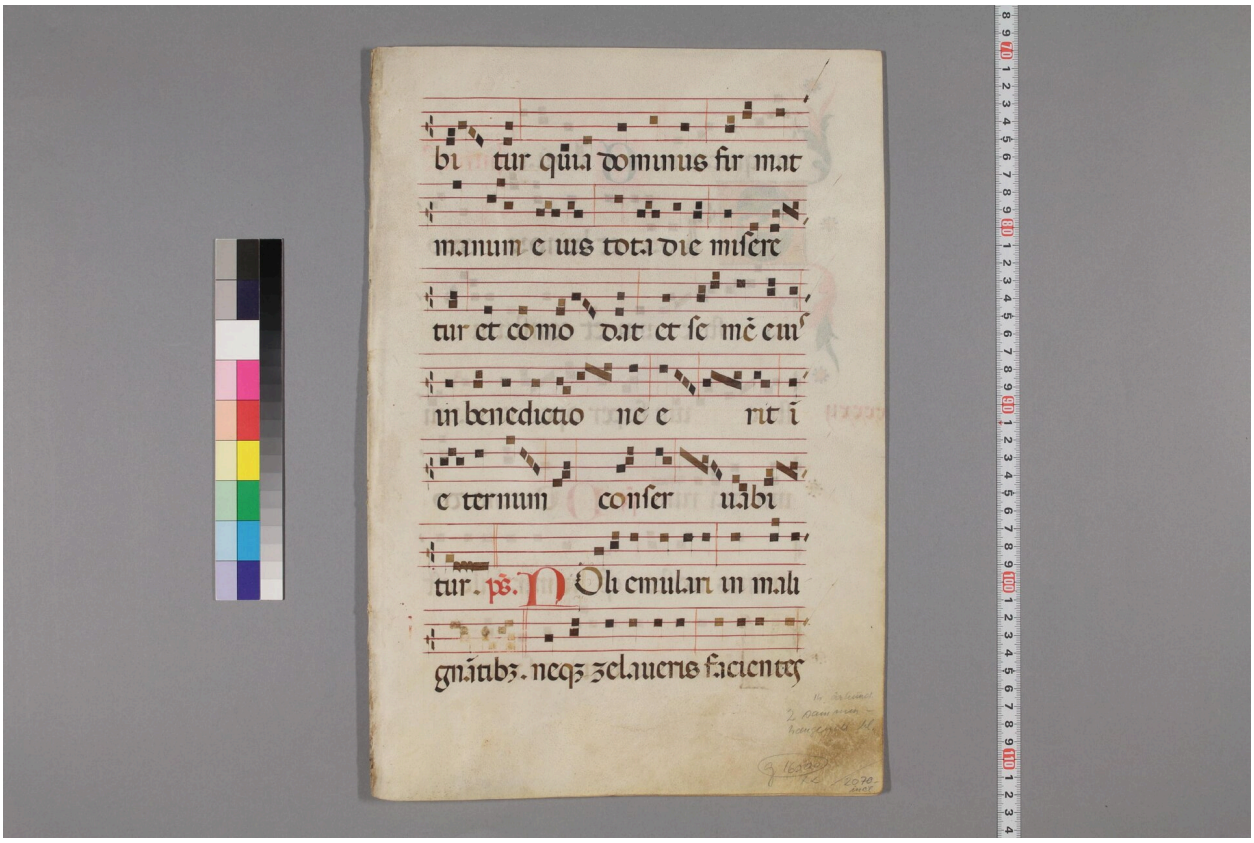
Figure 2-4: Gregorian chant leaf with staff notation and a wide manuscript field, preserved by Waseda University Library

Source and license: Waseda University Library, Wikimedia Commons file record, Public Domain Mark 1.0. PDF page 1 rendered without alteration except scaling.

The manuscript leaf in the following figure brings material history closer. Red and black inks separate structural and textual functions, while the irregular surface and hand-shaped characters resist the visual uniformity of typesetting. A reverberant realization should retain that human-scale articulation. In practical terms, use enough pre-delay to keep each new phrase distinct, attenuate excessive low-mid accumulation, and compare the dry and wet versions at matched loudness. The goal is not to manufacture an imaginary universal cathedral but to hear how a line acquires dimensionality when reflections remain intelligible.

Figure 2-5 presents Spanish chant manuscript leaf, photographed with a color target and scale for archival reference and establishes the visual reference for the discussion that follows.

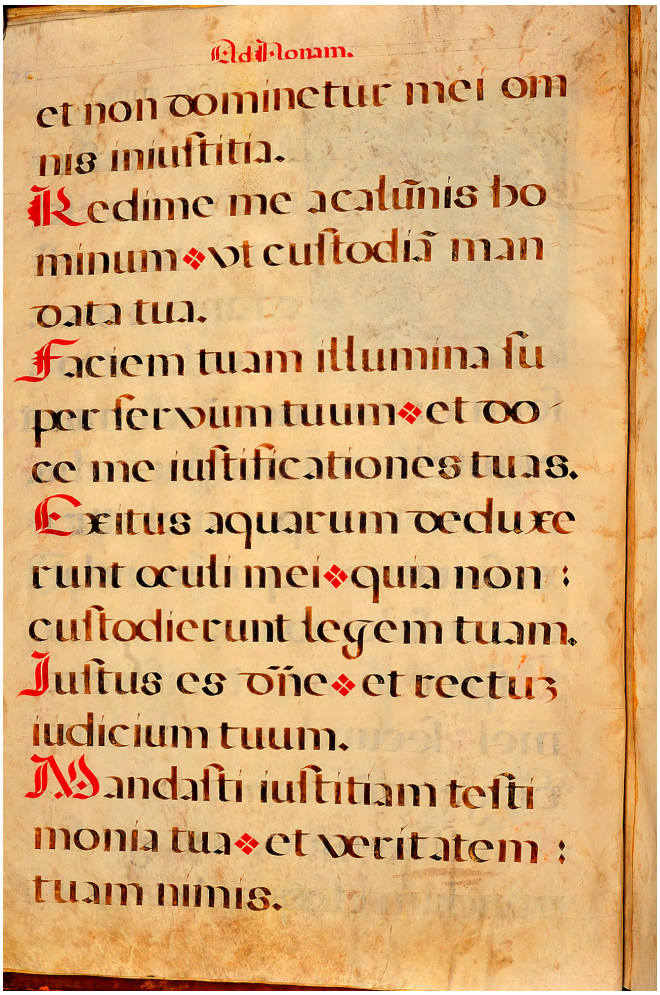

Figure 2-5: Spanish chant manuscript leaf, photographed with a color target and scale for archival reference

Source and license: UBC Library Digitization Centre, Wikimedia Commons file record, no known copyright restrictions. Reproduced without alteration except scaling.

The following figure presents *Ave verum* (traditional; date unknown) in square-note notation. The visual grouping of neumes encourages the performer to think in gestures rather than isolated pitches, an idea that transfers directly to reverb listening. Evaluate whether the acoustic preserves the beginning, trajectory, and release of each gesture. If only the release remains audible, the wet path is too dominant; if no connection survives between groups, the decay or early field may be too short. This page is particularly useful for testing a small chapel preset against a large basilica preset while holding performance and loudness constant.

Figure 2-6 presents Gregorian *Ave verum* (traditional; date unknown) in square-note notation, with grouped neumes that invite phrase-based acoustic listening and establishes the visual reference for the discussion that follows.

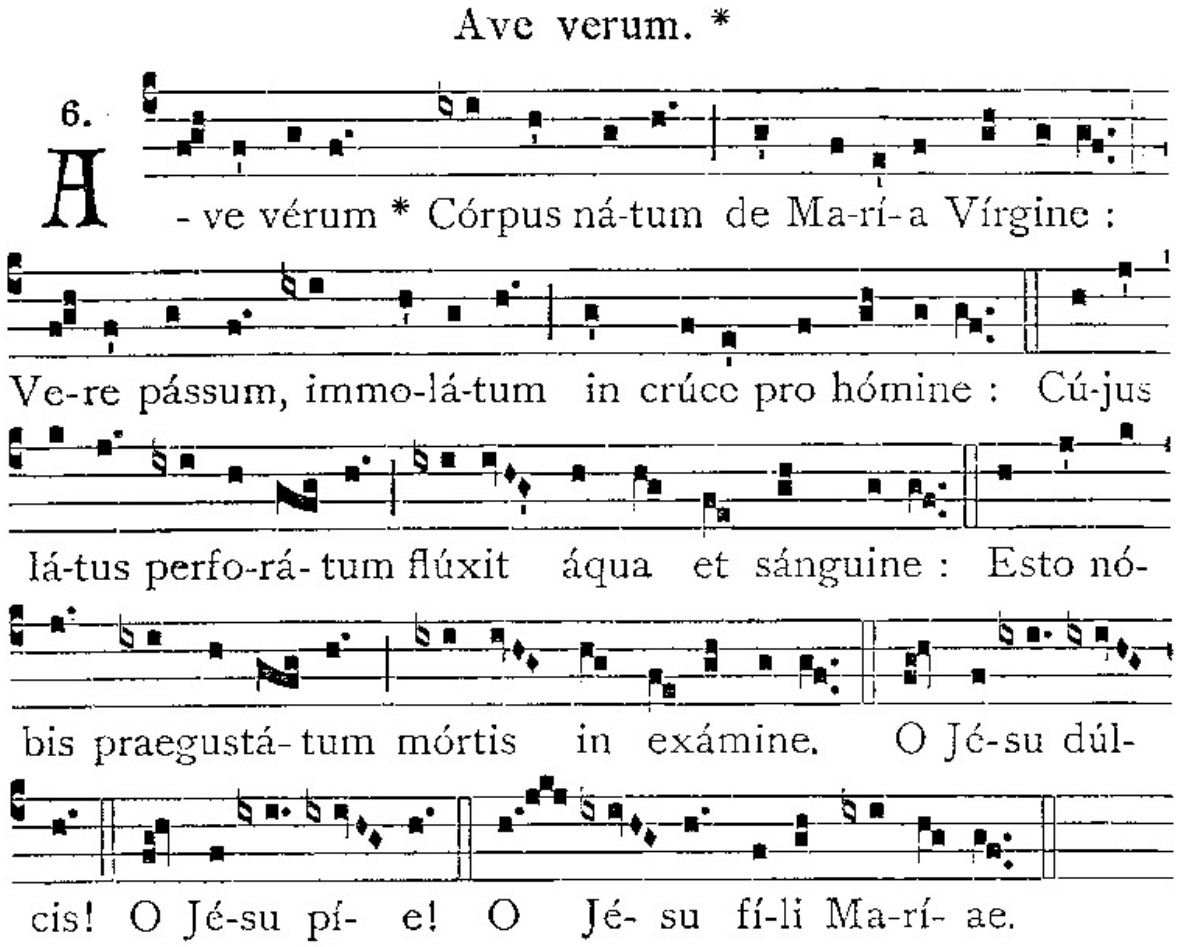

Figure 2-6: Gregorian *Ave verum* (traditional; date unknown) in square-note notation, with grouped neumes that invite phrase-based acoustic listening

Source and license: unknown notator, Wikimedia Commons file record, public domain under the file's rights statement. Reproduced without alteration except scaling.

The opening of Giovanni Pierluigi da Palestrina's *Missa Papae Marcelli* (c. 1562) in the following figure makes imitative density visible. Each voice enters into an already resonating contrapuntal field, so room decay changes not only apparent distance but also the perceived independence of lines. Use this page as a mixing exercise: increase decay until the imitation becomes difficult to follow, then recover clarity with high-frequency damping, low-frequency decay control, and a modest pre-delay rather than simply reducing the wet level. That process separates spectral congestion from reverberant envelopment.

Figure 2-7 presents Mensural opening of Palestrina's *Missa Papae Marcelli* (c. 1562), showing successive polyphonic entries and decorated initials and establishes the visual reference for the discussion that follows.
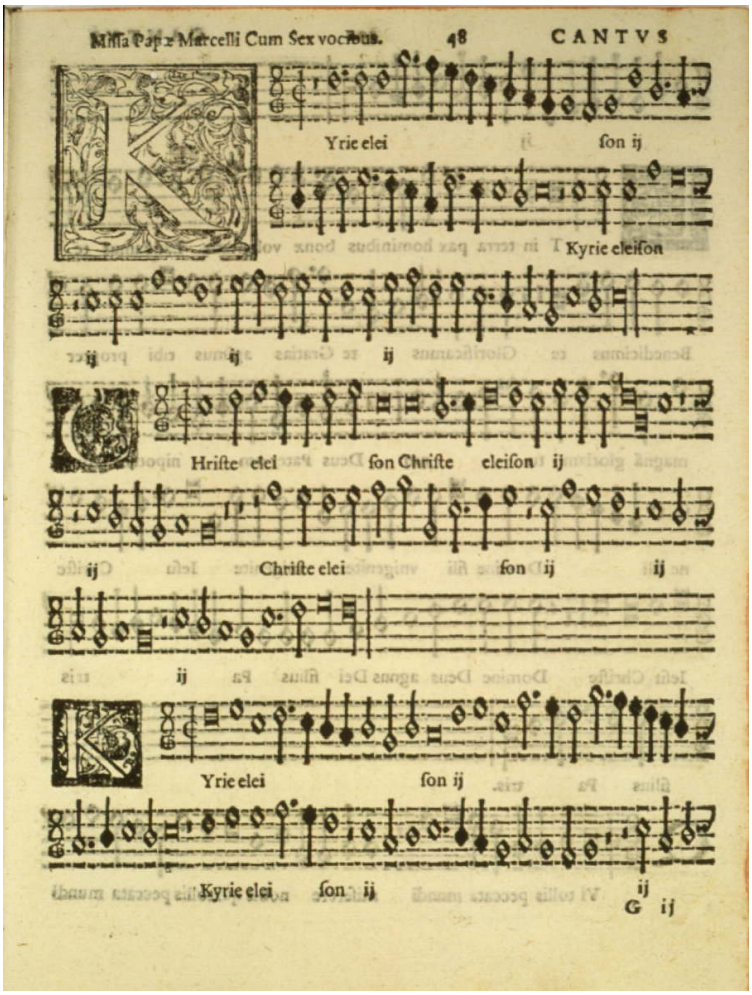
Figure 2-7: Mensural opening of Palestrina's *Missa Papae Marcelli* (c. 1562), showing successive polyphonic entries and decorated initials

Source and license: unknown historical engraver, Wikimedia Commons file record, public domain. Reproduced without alteration except scaling.

Paul Dukas's *Fanfare pour précéder La Péri* (1912) supplies the brass-fanfare comparison in the following figure. The compact excerpt shows brilliant, synchronized attacks followed by held sonorities: exactly the combination that exposes a reverb's transient handling and buildup. A convincing large-space preset should let the attacks remain decisive while the sustained harmony blooms behind them. This rights-cleared excerpt is not a stand-in compositionally for Copland's *Fanfare for the Common Man* (1942); it is a reusable test stimulus for the same production questions of brass onset, low-frequency support, and monumental apparent scale.

Figure 2-8 presents Opening measures of Paul Dukas's *Fanfare pour précéder La Péri* (1912), arranged as a compact brass-attack and sustain test and establishes the visual reference for the discussion that follows.

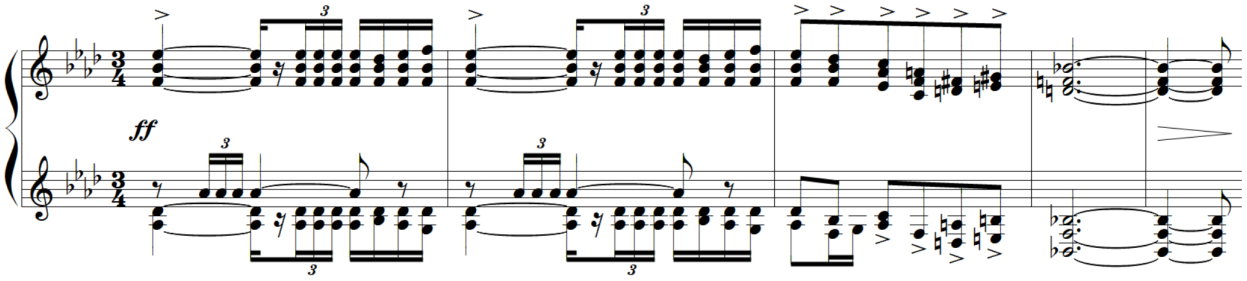

Figure 2-8: Opening measures of Paul Dukas's *Fanfare pour précéder La Péri* (1912), arranged as a compact brass-attack and sustain test

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César Franck's *Choral No. 3* (1890) offers a public-domain organ analogue for studying the acoustic questions raised by Messiaen's *Apparition de l'église éternelle* (1932). the following figure shows dense chordal writing, registral breadth, and pedal-supported mass. Organ and room form a coupled system: held tones continuously feed the acoustic, while release reveals the decay uncontaminated by a new attack. Compare the score against a render and listen specifically at chord changes. If the previous harmony masks the new one, shorten low-band decay or increase modulation decorrelation before reducing the total RT60.

Figure 2-9 presents Opening region of César Franck's *Choral No. 3* (1890) from *Trois chorals pour grand orgue* (1890), a public-domain organ-acoustics study and establishes the visual reference for the discussion that follows.

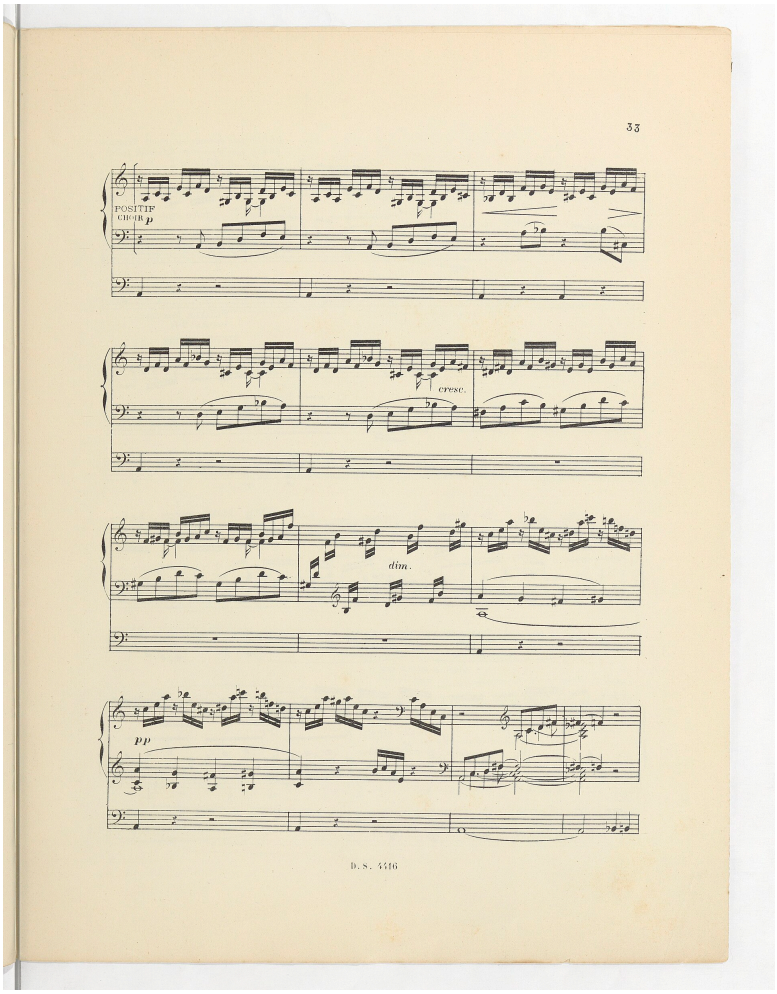

Figure 2-9: Opening region of César Franck's *Choral No. 3* (1890) from *Trois chorals pour grand orgue* (1890), a public-domain organ-acoustics study

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Giovanni Gabrieli's *In ecclesiis* (c. 1615) closes the notation sequence because its antiphonal conception points directly toward architectural placement. the following figure separates vocal and instrumental groups across the page, making alternation, overlap, and tutti arrival visually explicit. A spatial render can honor that design by giving groups distinguishable early-reflection signatures while allowing the late field to unify them. Do not confuse separation with dryness: the most persuasive result often combines localized direct sound, different short reflection patterns, and one shared enveloping decay.

Figure 2-10 presents Score excerpt from Giovanni Gabrieli's *In ecclesiis* (c. 1615), with antiphonal groups organized as a spatial dialogue and establishes the visual reference for the discussion that follows.
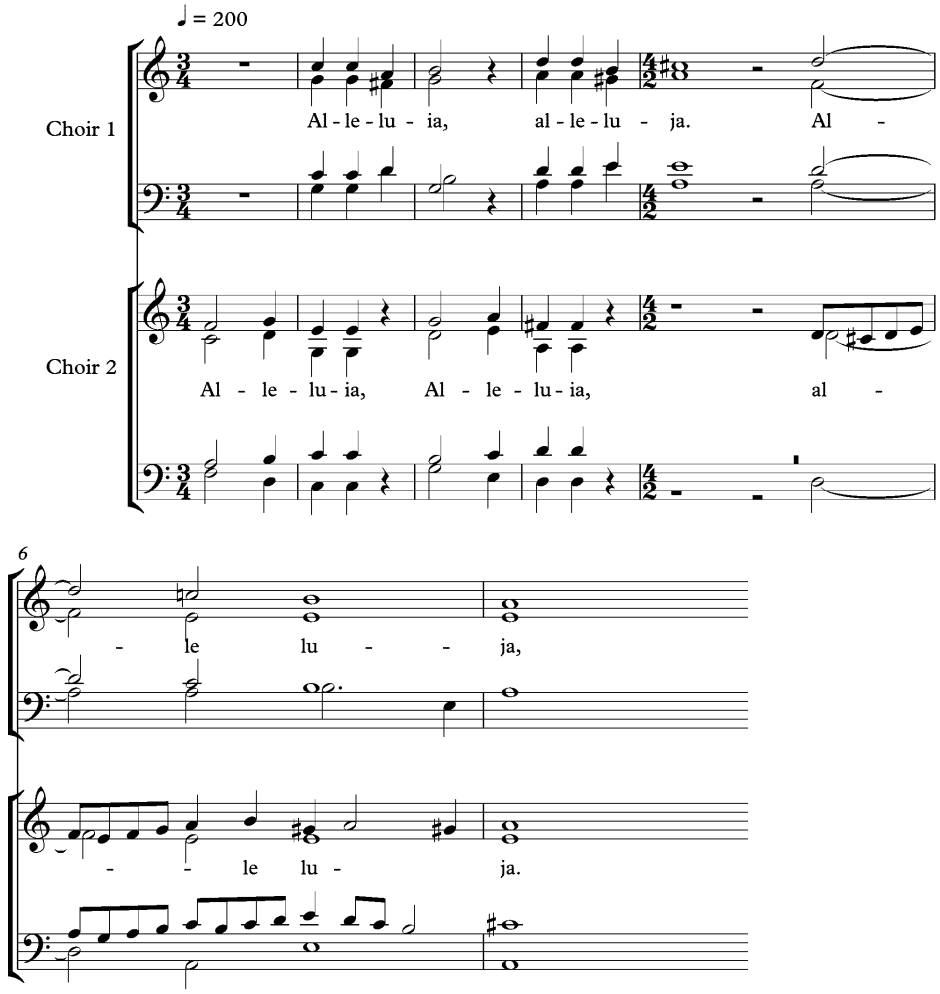
Figure 2-10: Score excerpt from Giovanni Gabrieli's *In ecclesiis* (c. 1615), with antiphonal groups organized as a spatial dialogue

Source and license: Giovanni Gabrieli, Wikimedia Commons file record, public domain. Reproduced without alteration except scaling.

2.4 Resonant Architecture

The interior of St. Mark's Basilica in the following figure supplies an architectural counterpart to Gabrieli's notation. Galleries, domes, stone surfaces, and separated performance positions create multiple paths between source and listener. The photograph should not be read as a complete acoustic model, but it identifies the geometry a model must explain: strong local reflections, coupled volumes, long paths, and a diffuse late field. When building a St. Mark's-inspired preset, avoid representing the entire space with RT60 alone; combine pre-delay, reflection clusters, frequency-dependent decay, and multichannel placement.

Figure 2-11 presents Interior of St. Mark's Basilica in Venice, showing galleries, arches, and coupled domed volumes associated with antiphonal practice and establishes the visual reference for the discussion that follows.

Figure 2-11: Interior of St. Mark's Basilica in Venice, showing galleries, arches, and coupled domed volumes associated with antiphonal practice

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The following figure looks upward through the ribbed vaults of Notre-Dame de Paris. Repeated bays establish a sequence of reflection zones rather than one undifferentiated cavity. Stone boundaries support long decay, but pillars, ribs, furnishings, and occupied surfaces scatter and absorb energy unevenly. This is why a cathedral preset benefits from temporal texture: perfectly smooth exponential decay can sound less plausible than a tail with evolving density and gentle modal variation. The photograph also encourages vertical thinking for immersive mixes, where height returns can carry vault energy without pulling the direct source upward.

Figure 2-12 presents Ribbed vaults and repeated bays inside Notre-Dame de Paris, viewed as a sequence of high reflection zones and establishes the visual reference for the discussion that follows.

Figure 2-12: Ribbed vaults and repeated bays inside Notre-Dame de Paris, viewed as a sequence of high reflection zones

Source and license: Uoaei1, Wikimedia Commons file record, CC BY-SA 4.0. Reproduced without alteration except scaling.

Laon Cathedral's nave in the following figure makes source-listener distance easy to imagine. The central aisle supplies a long direct axis, side aisles create lateral paths, and the high vault supports delayed overhead energy. In a stereo simulation, lateral reflections often contribute more apparent width than an indiscriminate increase in wet level. In surround or Atmos-oriented production, route early energy according to plausible directions, then let the late field broaden across sides, rears, and heights. The visual symmetry is useful, but the acoustic result should still include small asymmetries that prevent a sterile mirror image.

Figure 2-13 presents Nave of Laon Cathedral, with a long central axis, lateral aisles, and a high vaulted reflection field and establishes the visual reference for the discussion that follows.

Figure 2-13: Nave of Laon Cathedral, with a long central axis, lateral aisles, and a high vaulted reflection field

Source and license: David Iliff, Wikimedia Commons file record, CC BY-SA 3.0. Reproduced without alteration except scaling.

The dome of Hagia Sophia in the following figure demonstrates a radically different visual organization: a vast central volume surrounded by layered surfaces and openings. Domes can focus, redistribute, and prolong sound in ways that a rectangular-room model cannot reproduce literally. For creative reverb, the lesson is not to imitate one measured impulse response from appearance alone. Instead, use geometry-informed reflection timing, a dense late network, and restrained modulation to evoke scale while preserving musical control. Compare mono, stereo, and immersive versions to hear how overhead energy changes perceived enclosure.

Figure 2-14 presents Interior panorama of Hagia Sophia, emphasizing its central dome and layered reflecting surfaces and establishes the visual reference for the discussion that follows.

Figure 2-14: Interior panorama of Hagia Sophia, emphasizing its central dome and layered reflecting surfaces

Source and license: Ronan Reinart, Wikimedia Commons file record, CC BY-SA 3.0. Reproduced without alteration except scaling.

Hamilton Mausoleum, shown in the following figure, is famous for an exceptionally persistent echo. Its circular plan, hard surfaces, and domed ceiling create a compact but highly reflective system. The image is a reminder that long audible persistence does not require cathedral-scale floor area; geometry and absorption are decisive. In verbx, compare a smooth long RT60 preset with a sparse repeating-echo design. The latter may better communicate the perceptual identity of a space in which discrete returns remain audible before the response becomes diffuse.

Figure 2-15 presents Interior of Hamilton Mausoleum, a hard circular volume associated with an unusually persistent echo and establishes the visual reference for the discussion that follows.

Figure 2-15: Interior of Hamilton Mausoleum, a hard circular volume associated with an unusually persistent echo

Source and license: Supergolden, Wikimedia Commons file record, CC BY-SA 3.0. Reproduced without alteration except scaling.

The Inchindown oil-storage tunnel in the following figure leads toward enormous underground tanks known for extremely long decay. The narrow passage in the photograph is not itself the full resonant tank, but it conveys the scale, hard boundaries, and isolation of the complex. Long-tail rendering must be bounded operationally: estimate output duration before processing, communicate progress, and distinguish a one-hour RT60 parameter from an infinite or frozen mode. Musically, use such spaces as evolving instruments rather than default rooms; a single impulse can occupy an entire phrase or section.

Figure 2-16 presents Access tunnel in the Inchindown underground oil-storage complex, whose large hard tanks are associated with exceptionally long reverberation and establishes the visual reference for the discussion that follows.

Figure 2-16: Access tunnel in the Inchindown underground oil-storage complex, whose large hard tanks are associated with exceptionally long reverberation

Source and license: Dave Thompson, Wikimedia Commons file record, CC BY-SA 2.0. Reproduced without alteration except scaling.

The following figure documents *Silophone* (2000–2001), by The User, at Montréal's Silo No. 5. The public stations in the foreground make its central idea legible: architecture, sound system, and network interface were one instrument. Submitted sounds were played through loudspeakers in four empty grain-storage chambers, captured by microphones, and returned to listeners. This is more than a grain-silo-inspired preset. Its reverb depended on a specific building, loudspeaker and microphone positions, live noise, and a public routing system.

Figure 2-17 presents *Silophone* public listening and playback stations, with Montréal Silo No. 5 behind them and establishes the visual reference for the discussion that follows.

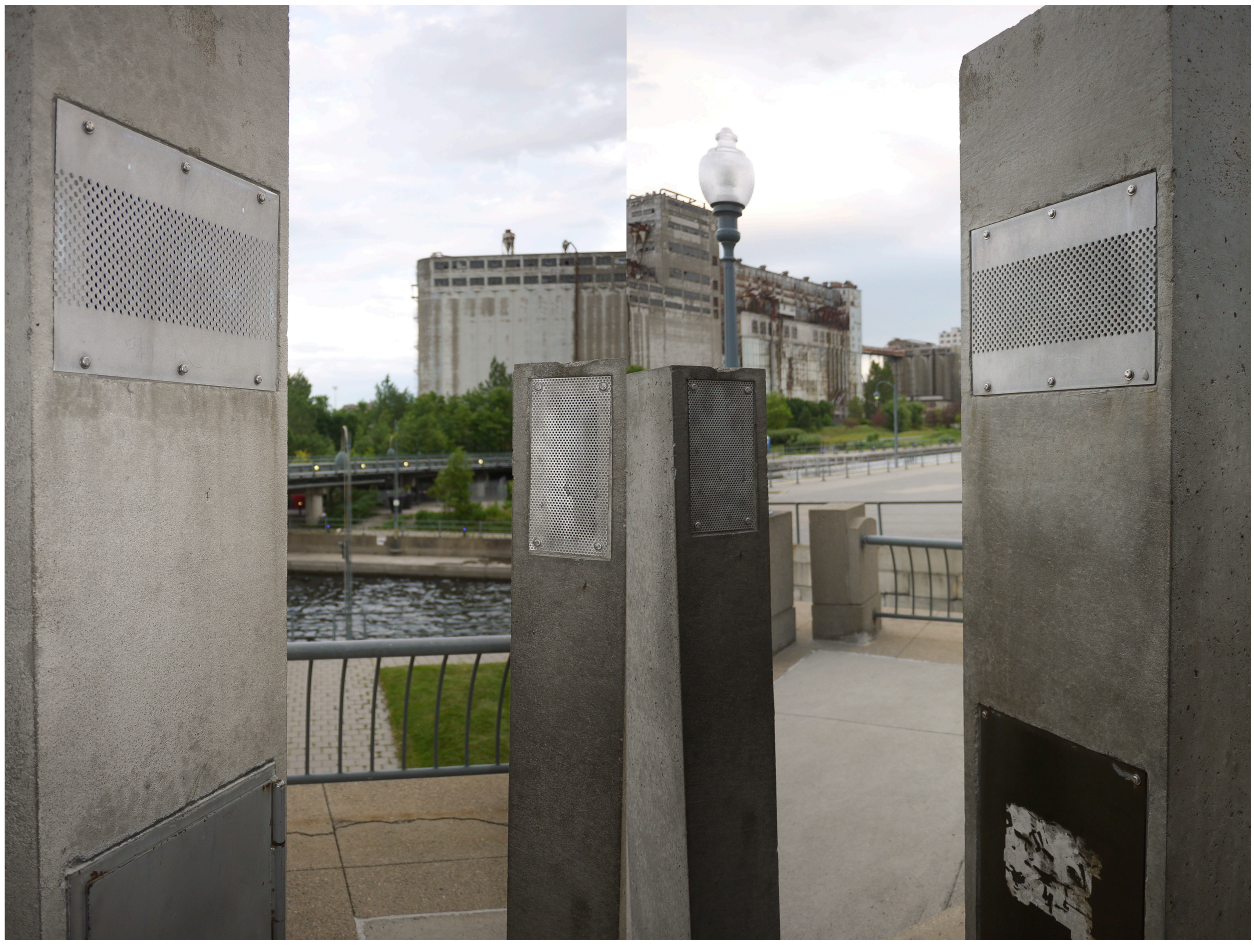

Figure 2-17: *Silophone* public listening and playback stations, with Montréal Silo No. 5 behind them

Source and license: art_inthecity, Wikimedia Commons file record, CC BY 2.0. Reproduced without alteration except scaling.

York Minster's nave in the following figure completes the architectural sequence with a broad frontal view. The image reveals strong axial depth, tall lateral boundaries, and extensive overhead volume while retaining visually complex scattering surfaces. Use it to separate three controls that are often collapsed: apparent source distance, room width, and decay time. More pre-delay can bring a source forward even in a long tail; stronger lateral early energy can widen the room without lengthening it; and multiband decay can preserve warmth without sacrificing consonant clarity.

Figure 2-18 presents Nave of York Minster, viewed along its principal axis with lateral and overhead reflection regions clearly visible and establishes the visual reference for the discussion that follows.

Figure 2-18: Nave of York Minster, viewed along its principal axis with lateral and overhead reflection regions clearly visible

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2.5 Mechanical and Laboratory Reverberation

The EMT 140 in the following figure makes plate reverberation physically legible. An electromechanical driver excites a large suspended steel plate, pickups capture its vibration, and damping changes the decay. The device is not a miniature room; it is a distributed resonant surface whose dense modal response produces a smooth, characteristic tail. When designing a plate-like algorithm, prioritize rapid density growth, controlled low-frequency behavior, and a bright but non-grainy decay. The photograph also explains why a classic plate occupies substantial physical space even though a plug-in reduces it to a few controls.

Figure 2-19 presents EMT 140 plate reverberator, showing the large framed steel-plate assembly used to create dense artificial decay and establishes the visual reference for the discussion that follows.

Figure 2-19: EMT 140 plate reverberator, showing the large framed steel-plate assembly used to create dense artificial decay

Source and license: EMT-Archiv-Lahr, Wikimedia Commons file record, CC BY-SA 4.0. Reproduced without alteration except scaling.

The following figure removes the enclosure from a spring-reverb tank so its transducers and parallel springs are visible. A driver launches torsional waves into the springs; reflections, dispersion, and coupling produce the familiar lively, sometimes “drippy” response. Spring behavior is strongly source-dependent: impulsive guitar or percussion reveals boing and chirp, while sustained material can become a diffuse halo. A digital model should expose enough control over dispersion, damping, and nonlinear drive to move between subtle coloration and obvious effect rather than treating every spring as one fixed preset.

Figure 2-20 presents Open spring-reverb tank with parallel springs, input driver, and output pickup visible and establishes the visual reference for the discussion that follows.

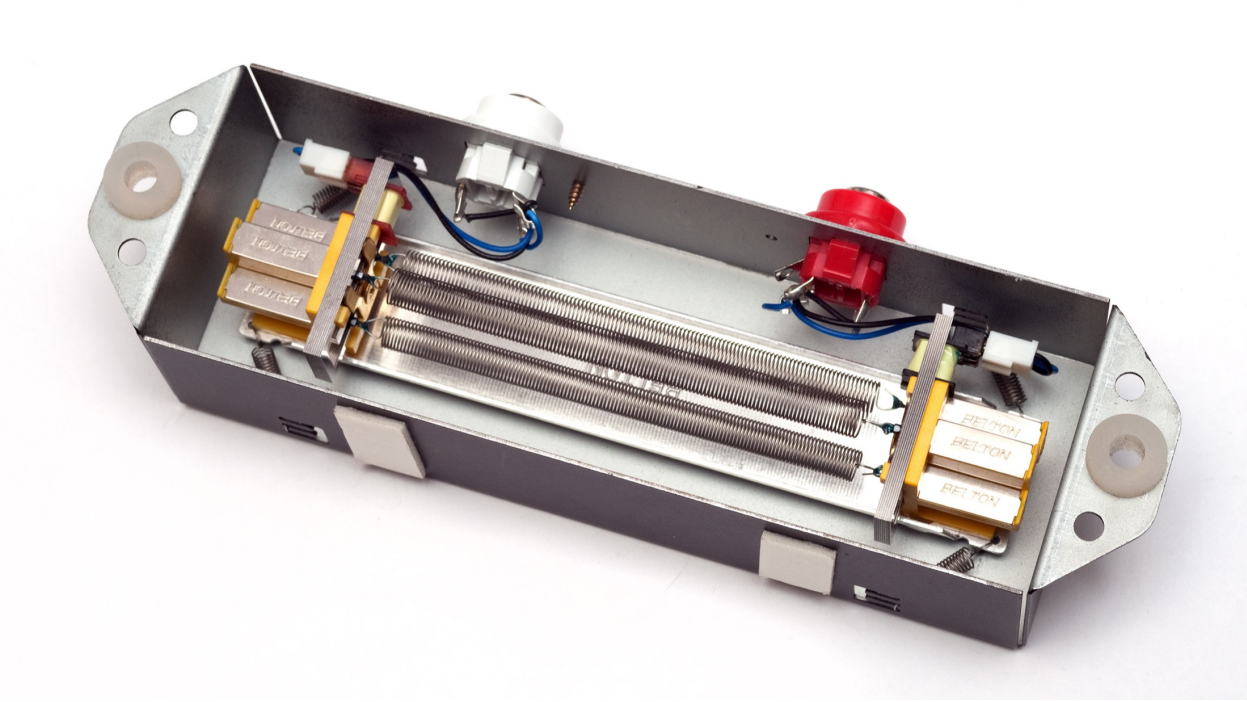

Figure 2-20: Open spring-reverb tank with parallel springs, input driver, and output pickup visible

Source and license: Ashley Pomeroy, Wikimedia Commons file record, CC BY-SA 4.0. Reproduced without alteration except scaling.

The compact Doepfer A-199 spring assembly in the following figure shows how the same principle scales into modular synthesis. The mechanical element is modest, but feedback routing, equalization, and excitation level can turn it into a playable resonator. This suggests a productive plug-in design principle: physical-model parameters should remain available on an Expert page while performance macros expose musically stable combinations. For a spring preset, map one macro to input drive and damping together, another to feedback and tone, and retain a safety limiter because resonant feedback can rise rapidly.

Figure 2-21 presents Compact spring-reverb assembly for a Doepfer A-199 modular-synthesizer module and establishes the visual reference for the discussion that follows.

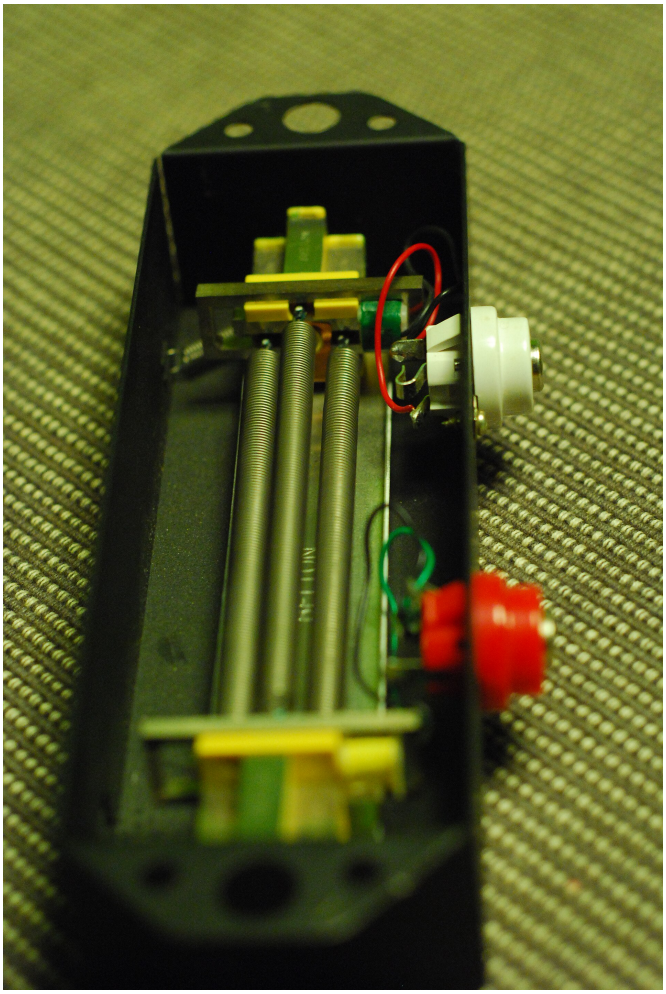

Figure 2-21: Compact spring-reverb assembly for a Doepfer A-199 modular-synthesizer module

Source and license: GeschnittenBrot, Wikimedia Commons file record, CC BY-SA 2.0. Reproduced without alteration except scaling.

The following figure identifies an O. C. Electronics Folded-Line reverberation device from its exterior label. Folded acoustic or mechanical paths pursue a familiar engineering goal: obtain a longer effective propagation path inside a compact enclosure. The label is visually modest, but it introduces an important conceptual bridge from physical delay lines to digital delay networks. In both cases, topology determines recurrence. A model becomes more convincing when delays, feedback gains, filtering, and coupling are treated as a system rather than as independent decorative controls.

Figure 2-22 presents Exterior identification label of an O. C. Electronics Folded-Line reverberation device and establishes the visual reference for the discussion that follows.

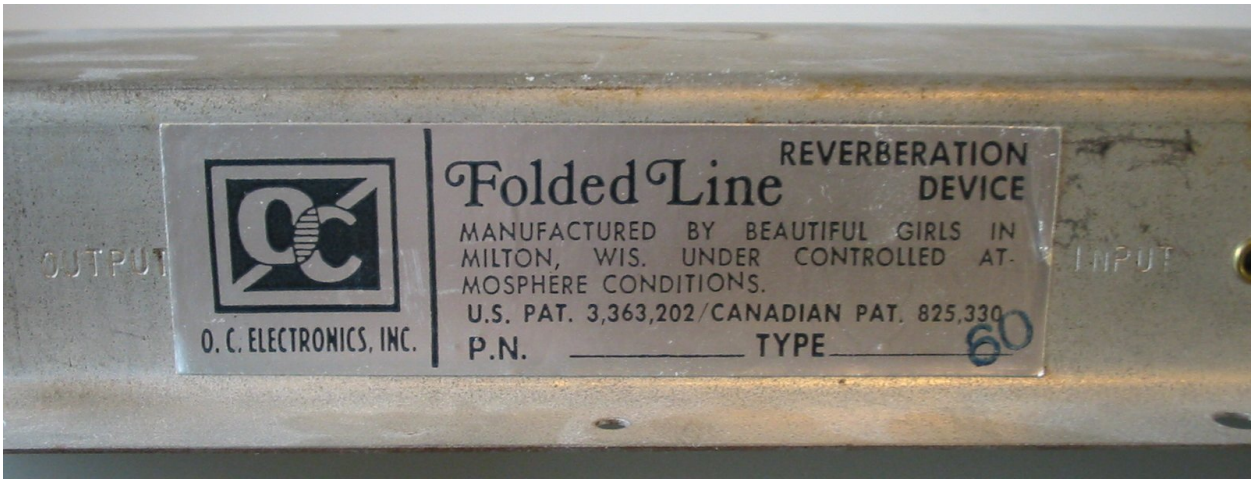

Figure 2-22: Exterior identification label of an O. C. Electronics Folded-Line reverberation device

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The opened device in the following figure reveals the internal folded element alluded to by the label. This paired view is useful because it distinguishes product description from mechanism. In a digital feedback delay network, long physical paths become sample delays, junction behavior becomes a mixing matrix, and losses become filters and gains. Yet the physical object still teaches restraint: a small number of purposeful paths can create a strong identity. More delay lines are not automatically better if their lengths and couplings produce audible regularity or unstable buildup.

Figure 2-23 presents Interior of an O. C. Electronics Folded-Line reverberation device, exposing its compact folded propagation element and establishes the visual reference for the discussion that follows.

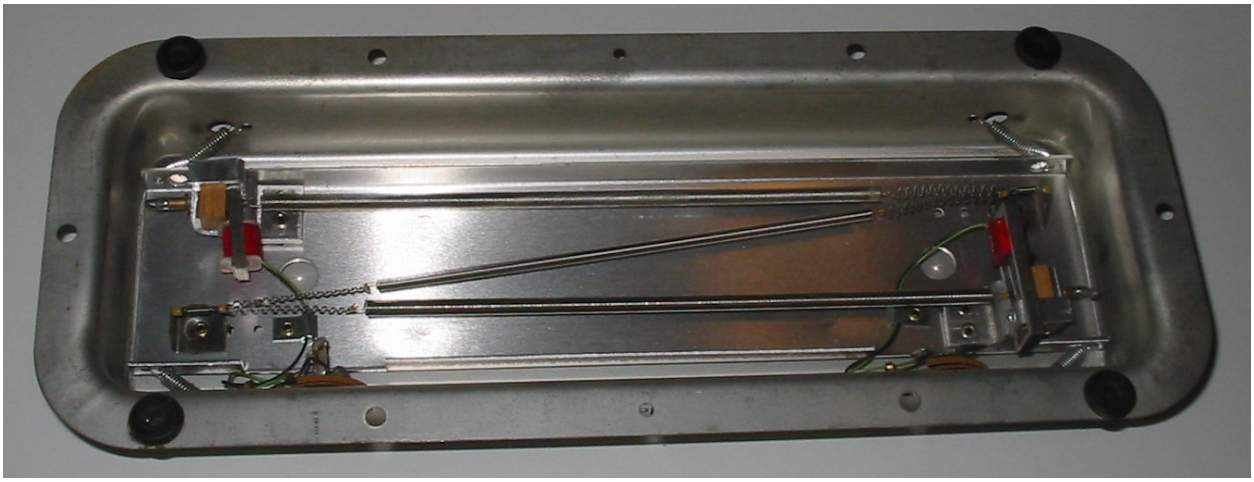

Figure 2-23: Interior of an O. C. Electronics Folded-Line reverberation device, exposing its compact folded propagation element

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The TU Dresden echo chamber in the following figure is an intentionally reflective acoustic room. Irregularly oriented panels disrupt simple parallel-wall paths and help distribute energy through many directions. Such a chamber is useful for material measurements and for generating dense reverberant fields under controlled conditions. The image clarifies why diffusion is not synonymous with decay: scattering changes the direction and density of reflections, while absorption and volume primarily govern energy loss. In verbx, adjust diffusion while holding RT60 constant to hear that distinction directly.

Figure 2-24 presents Acoustic echo chamber at TU Dresden, with hard boundaries and angled scattering panels and establishes the visual reference for the discussion that follows.

Figure 2-24: Acoustic echo chamber at TU Dresden, with hard boundaries and angled scattering panels

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The following figure shows the horn room of a reverberant acoustic test facility. This is not a musical echo chamber: its powerful horns create controlled high-level diffuse sound fields for aerospace testing. The distinction is instructive. Reverb metrics describe the response of a source-room-receiver system, so source directivity, bandwidth, level, and measurement position matter. A room that is excellent for qualification testing may be unusable for music, while its measurement discipline remains valuable. When validating an algorithm, specify stimulus, level, sample rate, channel geometry, and analysis window rather than reporting RT60 alone.

Figure 2-25 presents Horn room in a reverberant acoustic test facility, where high-output sources excite a controlled diffuse field and establishes the visual reference for the discussion that follows.

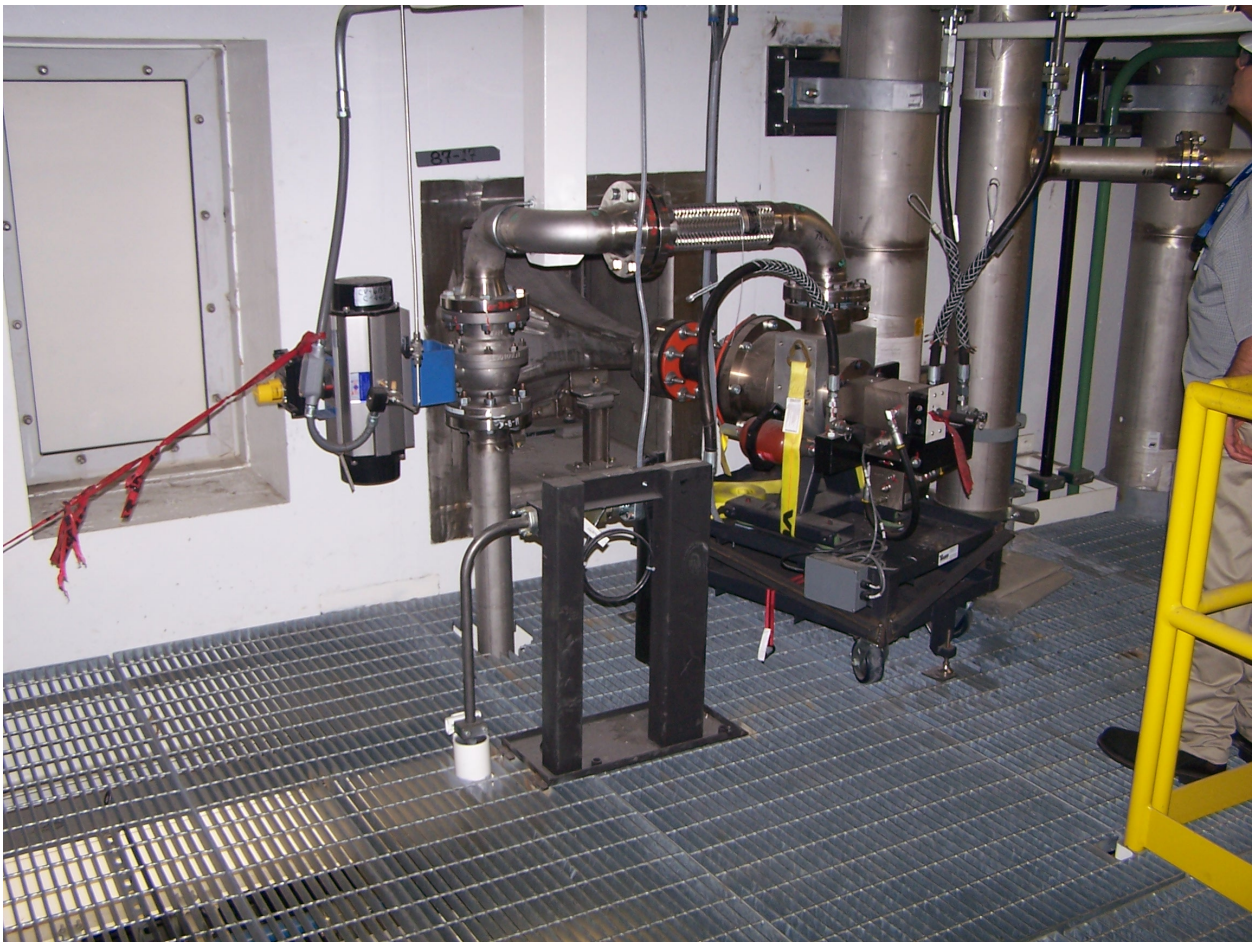

Figure 2-25: Horn room in a reverberant acoustic test facility, where high-output sources excite a controlled diffuse field

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The IETR chamber in the following figure is an electromagnetic reverberation chamber, not an audio room. It belongs here as a carefully labeled comparative image: angled conducting surfaces and stirring structures create a statistically distributed field for radio-frequency measurements. The analogy to acoustic diffusion is conceptually useful, but the wave impedances, boundary conditions, transducers, and frequency scales are different. Keeping that boundary explicit is good scientific practice. The shared lesson is that a “diffuse field” is an experimental objective produced by geometry, source behavior, and averaging, not a visual style or a single reverb knob.

Figure 2-26 presents Electromagnetic reverberation chamber at IETR, included as a labeled cross-domain comparison rather than an acoustic reverb room and establishes the visual reference for the discussion that follows.

Figure 2-26: Electromagnetic reverberation chamber at IETR, included as a labeled cross-domain comparison rather than an acoustic reverb room

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2.6 Reuse and Verification

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2.7 Musical Examples

The musical examples in this section use synthetic study recordings shipped in `examples/audio/`. They are not excerpts from the named historical works discussed in the listening appendix. Each sonogram

plots time in seconds along the horizontal axis, frequency in hertz on a logarithmic vertical axis, and level in decibels relative to the loudest time-frequency cell. Light gold marks stronger energy; dark green marks weaker energy. A sonogram cannot replace listening, but it makes note overlap, spectral decay, and sustained resonances visible. Every paired figure uses one shared time scale from zero to the longer recording's endpoint. When the shorter recording ends, its panel continues at the –80 dB floor instead of stretching the source to fill the plot.

2.7.1 Listening Method: Compare Dry, Wet-Only, and Context

For every example, make three passes. First hear the dry source and mark attacks, releases, rests, and registral changes. Then hear the wet-only return and ask what the space is doing as an independent musical layer. Finally hear the mix and decide whether the return supports or contradicts the source. A tail can be beautiful alone and still damage a phrase by masking its next attack.

Monitor at a stable level. Long tails invite level creep because their peaks are often lower than the dry source while their integrated energy is high. Compare at matched loudness, leave headroom, and listen through the final decay rather than stopping at the last written note. The end of the tail is part of the example.

2.7.2 Example 1: The Click Reveals the Room

An impulse is the most analytical source because it contains no phrase to hide the response. The dry click below occupies a narrow time slice across a broad frequency range. Its reverberated version reveals pre-delay, early clusters, the transition to a dense field, and the final frequency-dependent decay. Listen to the shipped `dry_click.wav` and `dry_click_reverbed.wav` before studying the figure.

The sonogram below makes the room's temporal spreading visible. The dry event is nearly vertical; the wet event extends horizontally as reflected energy persists.

Figure 2-27 presents Dry click compared with its reverberated response and establishes the visual reference for the discussion that follows.

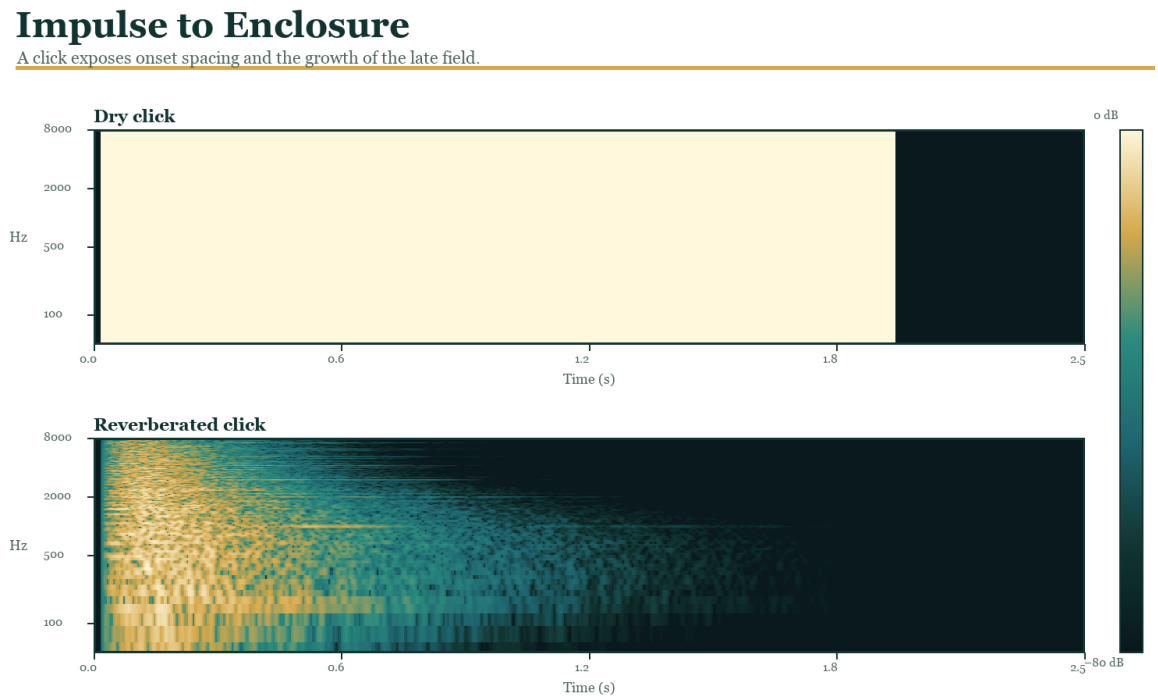


Figure 2-27: Dry click compared with its reverberated response

How to read this figure. The click's broadband stripe is an excitation, not a musical pitch. In the reverberated panel, later vertical traces indicate discrete reflections; the smoother horizontal fade is the late field. If the late field shows isolated horizontal ridges, the algorithm is emphasizing modes. If the entire panel stays uniformly bright, the return may be too dense or too compressed. A good design need not look perfectly smooth, but visible structures should correspond to an intended sonic character rather than accidental ringing.

Use the click when changing matrix family, allpass count, or delay distribution. It is less forgiving than music and quickly exposes flutter, repeated periods, and abrupt gates. Then return to music: a technically smooth click response is only useful if the result serves phrasing.

2.7.3 Example 2: Piano or Harmonic Phrase in a Hall

A chordal phrase tests harmonic continuity. Short room settings add body around each attack; hall settings connect notes across releases and rests. The direct signal must remain intelligible enough that the harmony changes when the player changes it, not one beat later. This tension is central to piano, harp, mallet percussion, and plucked strings.

The paired sonograms below compare the shipped dry musical phrase with its hall render. The same harmonic ridges appear in both panels, but the hall connects them and extends the final cadence beyond the source.

Figure 2-28 presents Harmonic phrase before and after hall reverberation and establishes the visual reference for the discussion that follows.

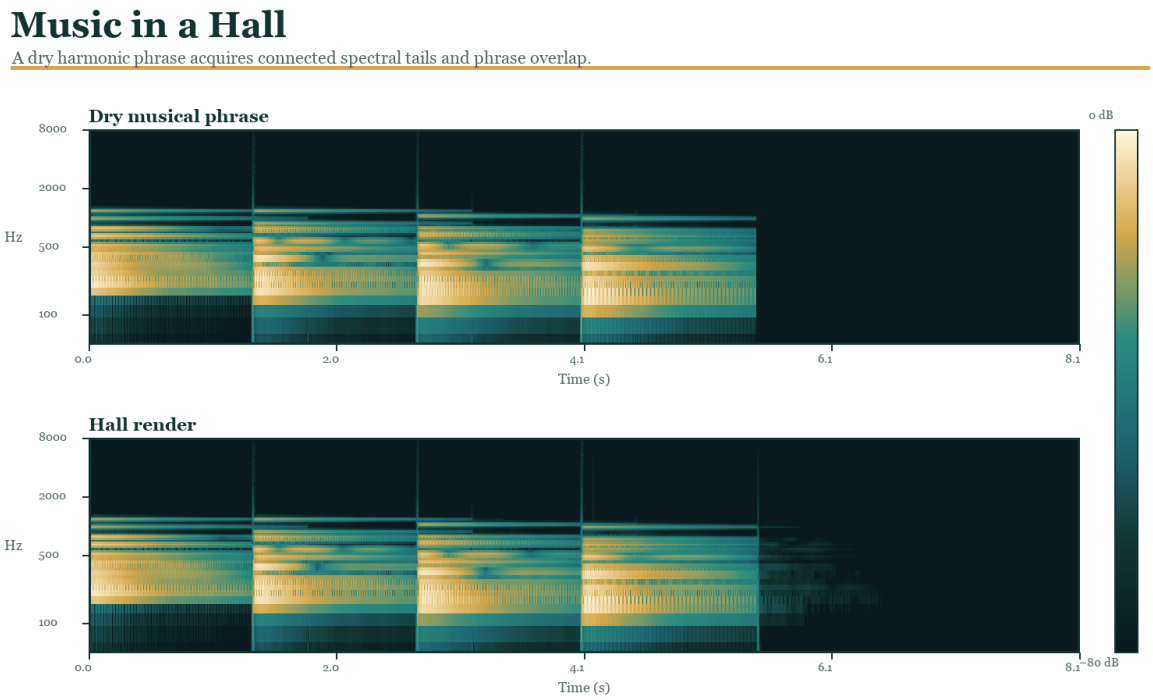


Figure 2-28: Harmonic phrase before and after hall reverberation

How to read this figure. In the dry panel, each chord has a clear left edge and a visible release. In the hall panel, the ridges continue between attacks, especially in the low-middle register. That continuation is musical glue until it obscures harmonic rhythm. The darker high-frequency area near the end shows damping: upper partials leave before lower energy. If every band decayed equally, the hall would tend to sound bright and synthetic; if lows persisted far longer, cadences could become muddy.

Reproduce a controlled version with:

```
verbx render examples/audio/realistic_music_dry.wav /tmp/music_hall.wav \
  --engine algo --rt60 2.4 --pre-delay-ms 24 \
  --wet 0.38 --dry 0.78 --fdn-lines 16 --fdn-matrix hadamard \
  --lowcut 90 --highcut 11000 --target-peak-dbfs -1
```

For a Romantic cadence, increase RT60 before increasing wet level. For a contrapuntal passage, reduce wet level and length, or use ducking so each attack remains legible. Treat pre-delay as phrasing space: 20–35 ms can separate the hammer or pluck from the room without making the response feel like a discrete echo.

2.7.4 Example 3: Drums, Early Reflections, and Groove

Percussion makes the difference between early reflections and late decay obvious. Early energy changes apparent drum size and room boundary; late energy determines how long the groove occupies the mix. A short, bright room can make a snare feel larger without smearing the kick pattern. A cathedral tail can turn the same rhythm into overlapping waves whose accents are defined by the return rather than the source.

The following sonograms compare a dry drum pattern and a room render. The attacks remain vertical and separately readable, while low-level energy fills the gaps.

Figure 2-29 presents Drum transients before and after a short room response and establishes the visual reference for the discussion that follows.

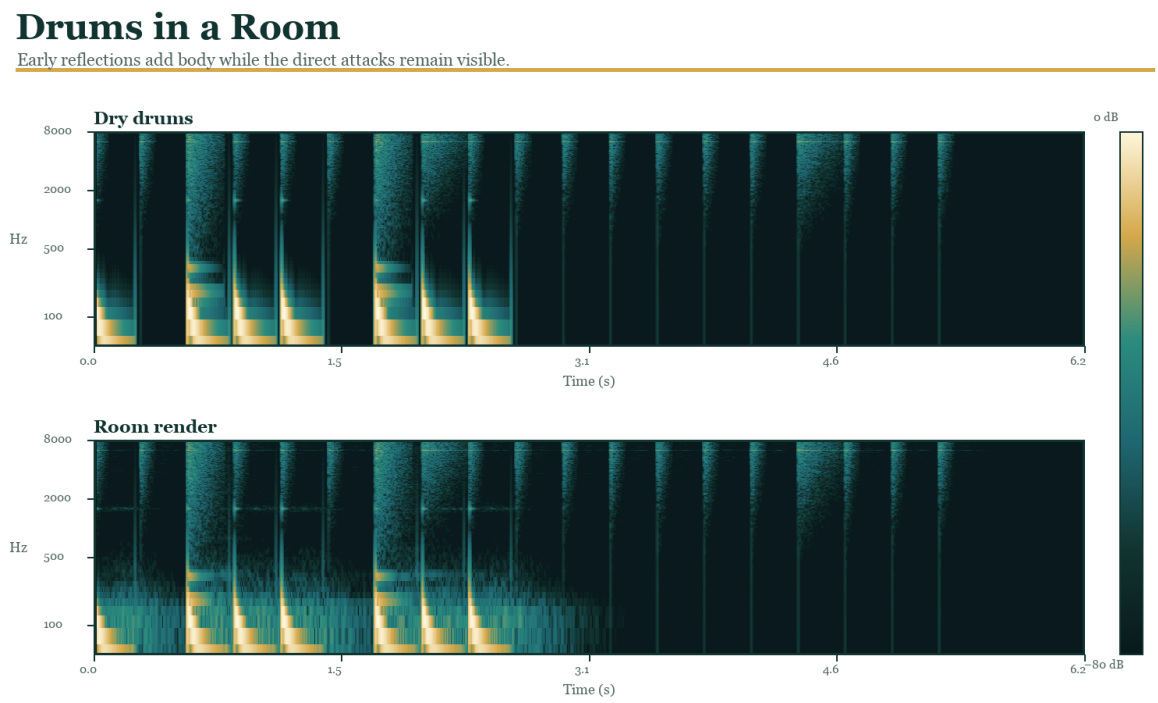


Figure 2-29: Drum transients before and after a short room response

How to read this figure. The dry panel has dark space between attacks. The room panel retains vertical transients but adds short horizontal fans after them. Those fans are long enough to supply body and short enough to preserve subdivision. Watch the low-frequency region: excessive persistence there can make a kick drum appear longer without sounding more spacious. A return high-pass filter often improves groove more effectively than shortening the entire RT60.

Try a compact room and a tempo-related pre-delay:

```
verbx render examples/audio/realistic_drums_dry.wav /tmp/drums_room.wav \
  --engine algo --rt60 0.85 --pre-delay 1/64 --bpm 120 \
  --wet 0.26 --dry 0.88 --fdn-lines 8 --fdn-matrix circulant \
  --lowcut 140 --highcut 9000
```

The note-value pre-delay makes the experiment repeatable at a new tempo. Do not assume that synchronizing always sounds better: a slightly non-metric onset can keep the room from reinforcing every beat in the same way.

2.7.5 Example 4: Cathedral-Scale Percussion

Very long percussion reverb changes the hierarchy of events. The first strike excites a field that is still active when later strikes arrive. Each new attack is therefore both a source event and a modulation of an existing texture. In a physical cathedral, architectural asymmetry and air absorption prevent perfect repetition. In an algorithm, matrix choice and time variation must perform some of that work.

The sonogram below shows the shipped cathedral drum study. Broad attacks enter a field whose low and middle bands remain active for many seconds.

Figure 2-30 presents Percussion rendered through an eight-second Hadamard FDN cathedral and establishes the visual reference for the discussion that follows.

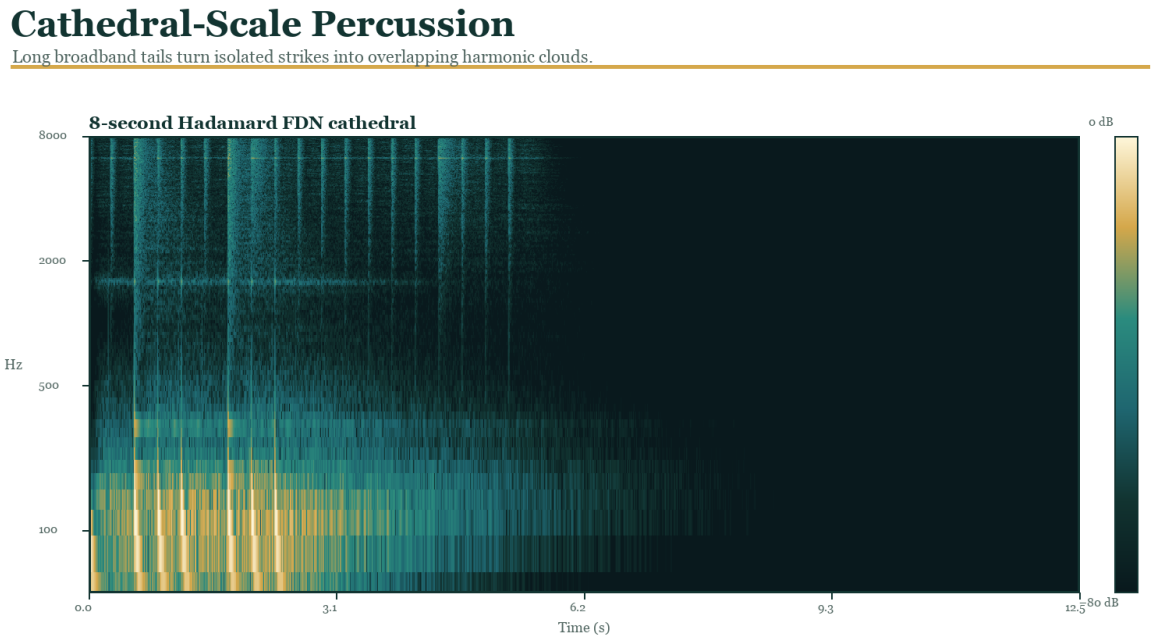


Figure 2-30: Percussion rendered through an eight-second Hadamard FDN cathedral

How to read this figure. Vertical energy identifies new strikes; horizontal clouds show accumulated room memory. Notice that successive attacks do not reset the field. They add energy to frequencies already decaying. The result can create a slow harmonic rhythm unrelated to the written drum rhythm. This is why long percussion reverbs often benefit from lower wet level, greater high-frequency damping, and automation that allows selected accents to enter the long return.

This listening problem appears dramatically in Berlioz's *Grande Messe des morts* (1837) and in antiphonal repertory designed for large architecture: distance and decay become orchestration. A production need not imitate a specific church to learn the lesson. Send only structural attacks to the longest return and let smaller notes articulate the foreground.

2.7.6 Example 5: Voice, Consonants, and Vowels

Voice contains two different reverb tests. Consonants are short, often broadband, and carry intelligibility. Vowels are sustained, harmonic, and carry pitch and identity. A return that flatters vowels can mask consonants; a setting optimized for speech clarity may feel too dry for lyric singing.

The paired sonograms below show dry speech and a room render. Read the short upper-band bursts as consonant activity and the horizontal harmonic stacks as vowels.

Figure 2-31 presents Speech articulation before and after room reverberation and establishes the visual reference for the discussion that follows.

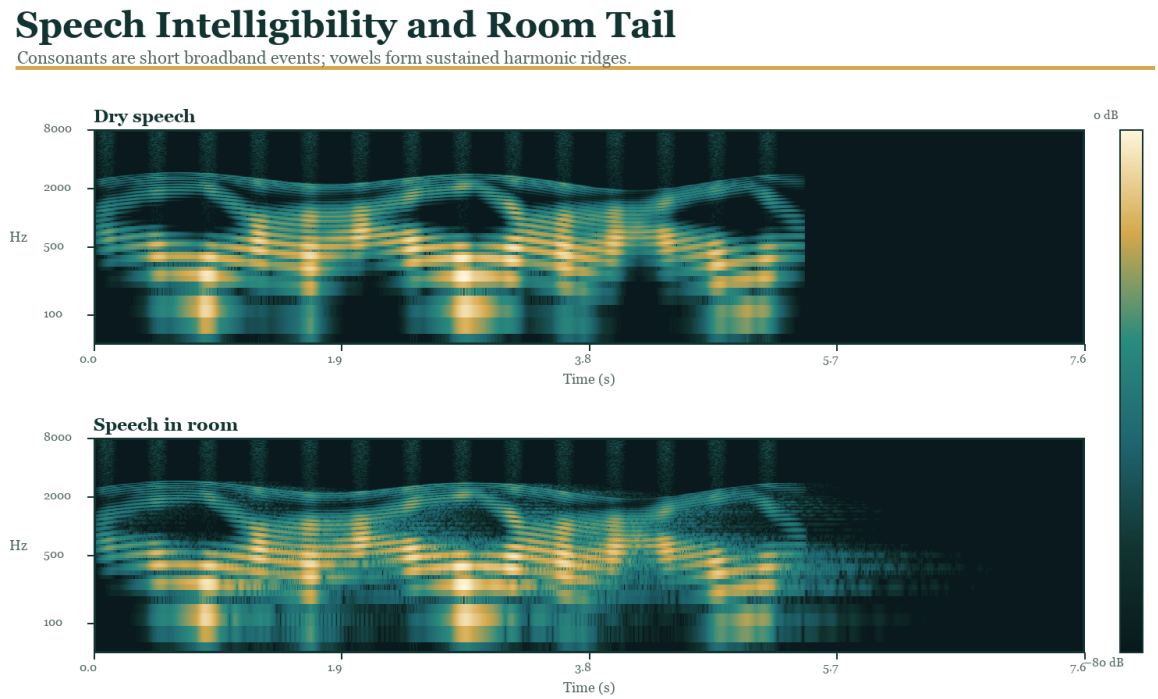


Figure 2-31: Speech articulation before and after room reverberation

How to read this figure. The room extends vowel bands and places a low-level veil after consonant bursts. Intelligibility decreases when that veil is still strong at the next consonant. Pre-delay can protect the initial consonant; ducking can protect every active syllable; high-frequency damping can prevent sibilance from turning into a persistent hiss. None of those changes requires abandoning a long lyrical tail.

For a vocal send, start with:

```
verbx render vocal.wav /tmp/vocal_space.wav \
  --engine algo --rt60 2.8 --pre-delay-ms 32 \
  --wet 0.42 --dry 0.78 --duck --duck-attack 12 --duck-release 260 \
  --duck-strength 0.72 --duck-floor 0.20 --lowcut 160 --highcut 9500
```

The release time is a musical control. A short release lets the tail rise between syllables; a longer release waits for the end of a line. Automate return level at phrase ends rather than forcing one static setting to solve every word.

2.7.7 Example 6: Plate-Like Brightness on Speech and Singing

A plate-like reverb is dense early, smooth, and often bright. It does not need to convince the listener that the singer occupies a literal metal plate. Its musical role is to extend the voice without introducing

obvious room geometry. Circulant and other regular FDN matrices can approach this character when delay distribution, diffusion, and damping are carefully controlled.

The single sonogram below shows the shipped plate speech study. Sibilant energy spreads into a fine upper-band tail while vowels remain stronger lower in the spectrum.

Figure 2-32 presents Speech rendered through a bright circulant-FDN plate study and establishes the visual reference for the discussion that follows.

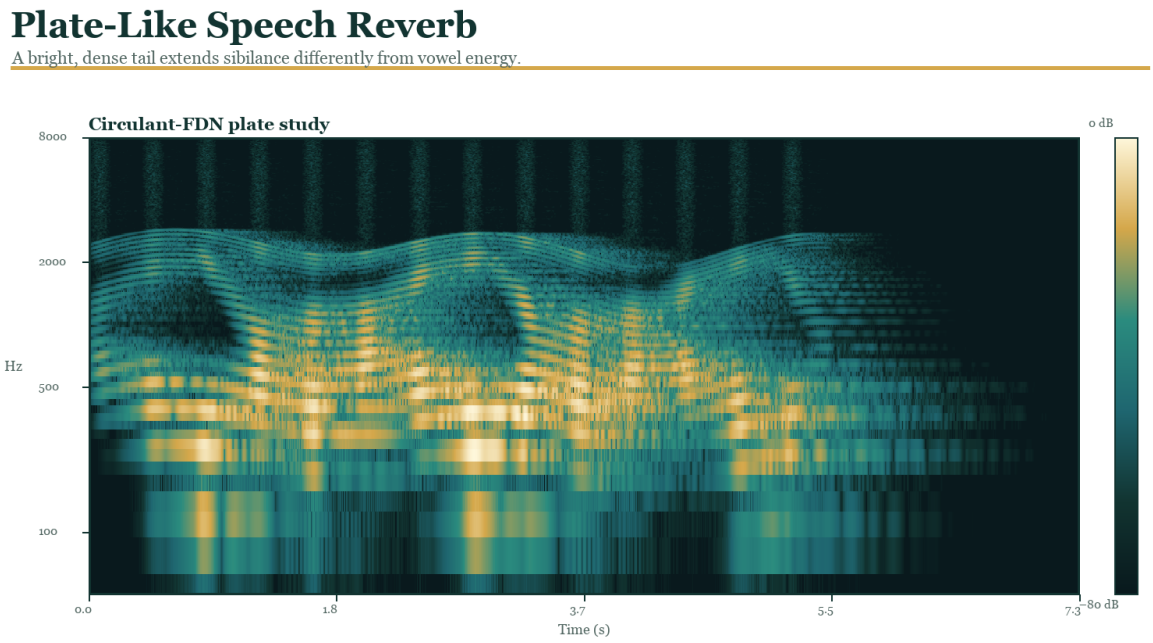


Figure 2-32: Speech rendered through a bright circulant-FDN plate study

How to read this figure. The upper spectrum remains active after several consonants, but it does not dominate the complete tail. A bright return feels polished when that energy is diffuse; it feels brittle when isolated high-frequency modes ring. Compare the visual smoothness with what you hear. A sonogram may show continuous energy while the ear still detects a metallic repetition, especially on headphones.

Plate-like returns work well as a second vocal layer. Keep a darker room or chamber for depth, then add the bright plate at lower level for sustain. The two returns should have different onset and decay profiles; duplicating the same timing merely raises level.

2.7.8 Example 7: Sparse Music and the Composition of Silence

Sparse music reveals the full envelope of every return. A single note is followed by a field, then by the field's disappearance. The listener can hear modulation, modal beating, noise-floor behavior, and the exact moment the processor closes or truncates. In dense music these details hide under new events; in sparse music they become form.

The upper panel below is a sparse-note room study, while the lower panel is a phase-drum study. Together they show how the same idea of "space" must behave differently when silence is structural versus when pulse is continuous.

Figure 2-33 presents Sparse-note room behavior compared with a dense phase-rhythm field and establishes the visual reference for the discussion that follows.

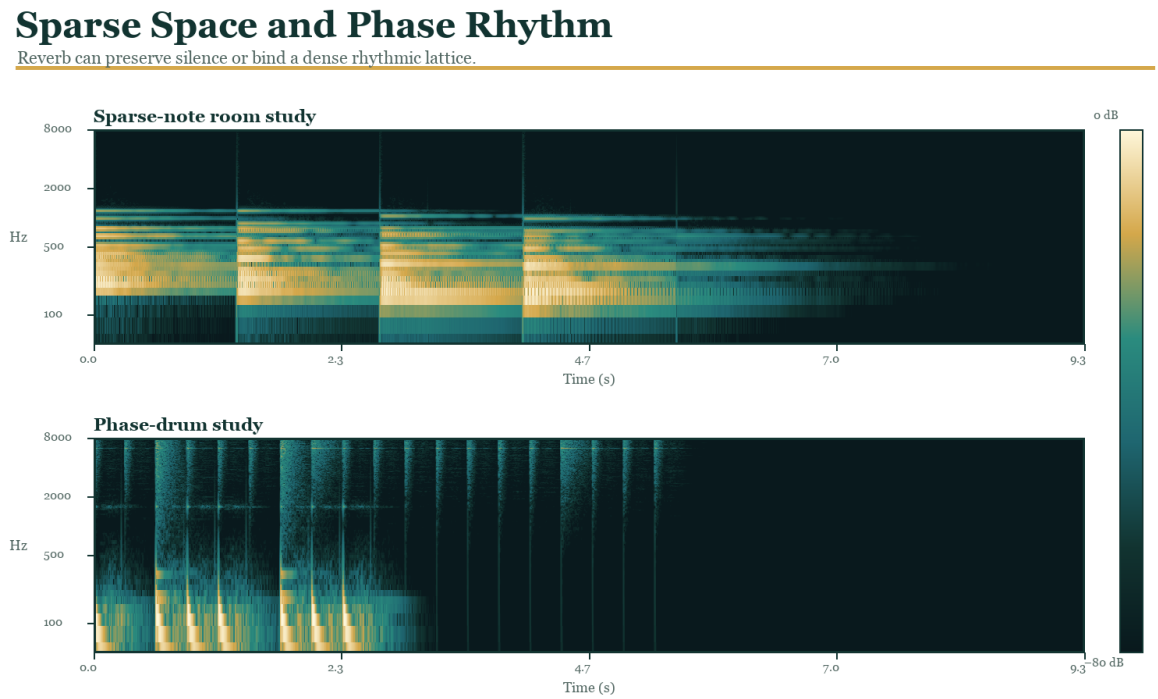


Figure 2-33: Sparse-note room behavior compared with a dense phase-rhythm field

How to read this figure. In the sparse panel, each note owns a visible decay region; the dark gaps are compositional material. An abrupt cutoff would be obvious. In the rhythmic panel, repeated attacks create nearly continuous energy, so the return is judged by groove, masking, and spectral accumulation instead. The same RT60 number has different consequences because the excitation density differs.

The practical lesson is to set reverb while hearing the actual event density. A preset auditioned on sustained pads says little about its behavior on isolated piano harmonics. For music in the orbit of Morton Feldman's spacious pacing, let the tail complete its gesture. For repeating-process textures, decide whether the room should reveal the process or fuse it into a single surface.

2.7.9 Example 8: Antiphony, Choir, and Architectural Counterpoint

Thomas Tallis's *Spem in alium* (c. 1570) and Giovanni Gabrieli's *In ecclesiis* (c. 1615) demonstrate that space can participate in counterpoint. Groups answer across distance; reflections bind the ensemble while directional differences preserve identity. A stereo reverb can suggest some of this behavior, but an immersive or multichannel design can make direction and decay independent compositional parameters.

Begin by separating source groups before adding reverb. Give each group a distinct send level, pre-delay, or early-reflection pattern while sharing a coherent late field. If every group receives an unrelated complete room, the image may become wide but cease to feel like one architecture. If every group receives an identical mono return, the counterpoint may collapse toward the center.

A useful exercise is to route four phrase groups into one long FDN, vary only their input projections or channel positions, and keep the late feedback network common. The shared tail supplies architectural identity; the distinct injections preserve antiphonal placement. At cadences, automate the longest return upward only after the last consonant so the room completes the formal punctuation.

2.7.10 Example 9: Drone, Organ Pedal, and Slow Harmonic Time

Sustained sources do not reveal RT60 directly because they keep feeding the room. They reveal spectral equilibrium: which bands accumulate, which modes beat, and whether the network remains stationary or slowly evolves. Organ pedals, bowed tones, feedback guitar, and electronic drones can expose a one-decibel imbalance that short sources never reveal.

The paired sonograms below compare a layered ambient study and a long-form drone study. The horizontal bands make slow changes visible; the important information is not only how long energy lasts, but how its distribution changes while the source continues.

Figure 2-34 presents Two sustained musical textures viewed across slow harmonic time and establishes the visual reference for the discussion that follows.

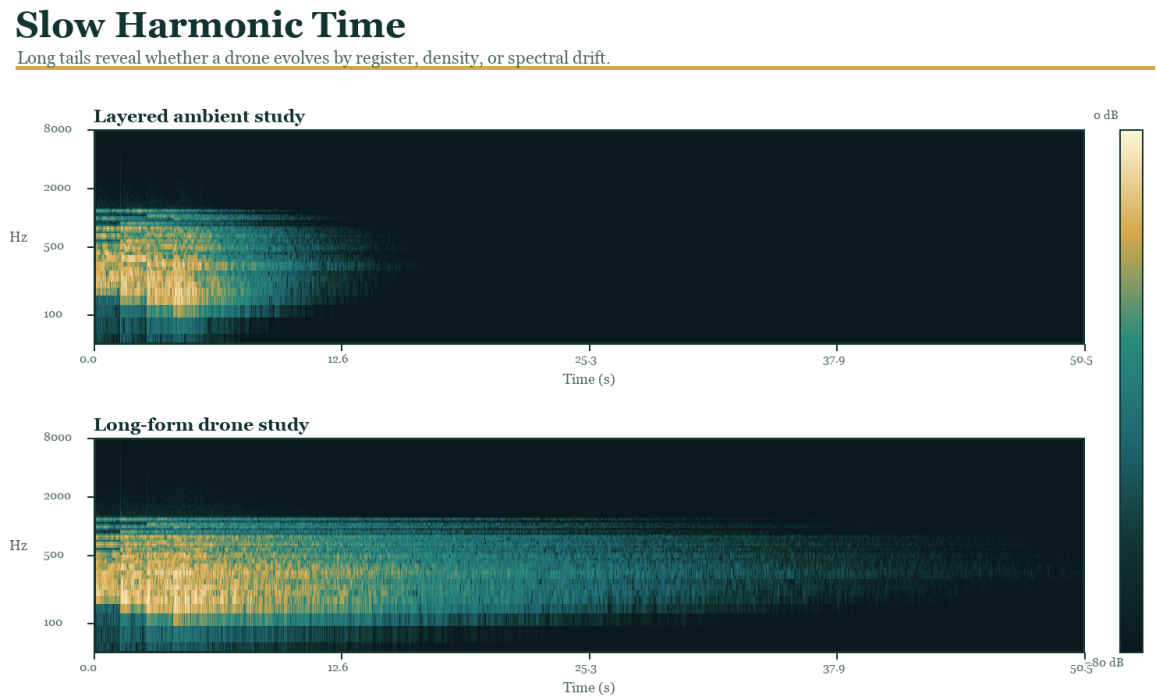


Figure 2-34: Two sustained musical textures viewed across slow harmonic time

How to read this figure. Persistent horizontal ridges represent stable harmonic or modal energy. Changes in brightness reveal entries, exits, or spectral drift. A reverb on this material should avoid pinning the entire texture to a few accidental resonances. Time-varying matrices, slight modulation, or tonal correction can distribute energy without turning the return into an obvious chorus.

In repertoire such as Ligeti's *Lux Aeterna* (1966) or Messiaen's *Et exspecto resurrectionem mortuorum*, spectral mass, register, and architectural decay shape the listener's sense of time. For an electronic analogue, automate matrix motion more slowly than the phrase rate. Fast motion reads as modulation; slow motion reads as a changing room.

2.7.11 Example 10: Recirculation as Musical Form

Repeatedly feeding a recording back through a room or a reverb gradually replaces the source's detailed spectrum with the resonances of the system. Alvin Lucier's *I Am Sitting in a Room* makes this transformation the complete form. Tape-loop and delay practices create related processes: the return is not decoration but a memory that re-enters the next generation.

The sonograms below compare a tape-loop-style study and a room-recirculation study. Both show source detail consolidating into longer spectral bands.

Figure 2-35 presents Recirculation processes that turn a source into a resonant field and establishes the visual reference for the discussion that follows.

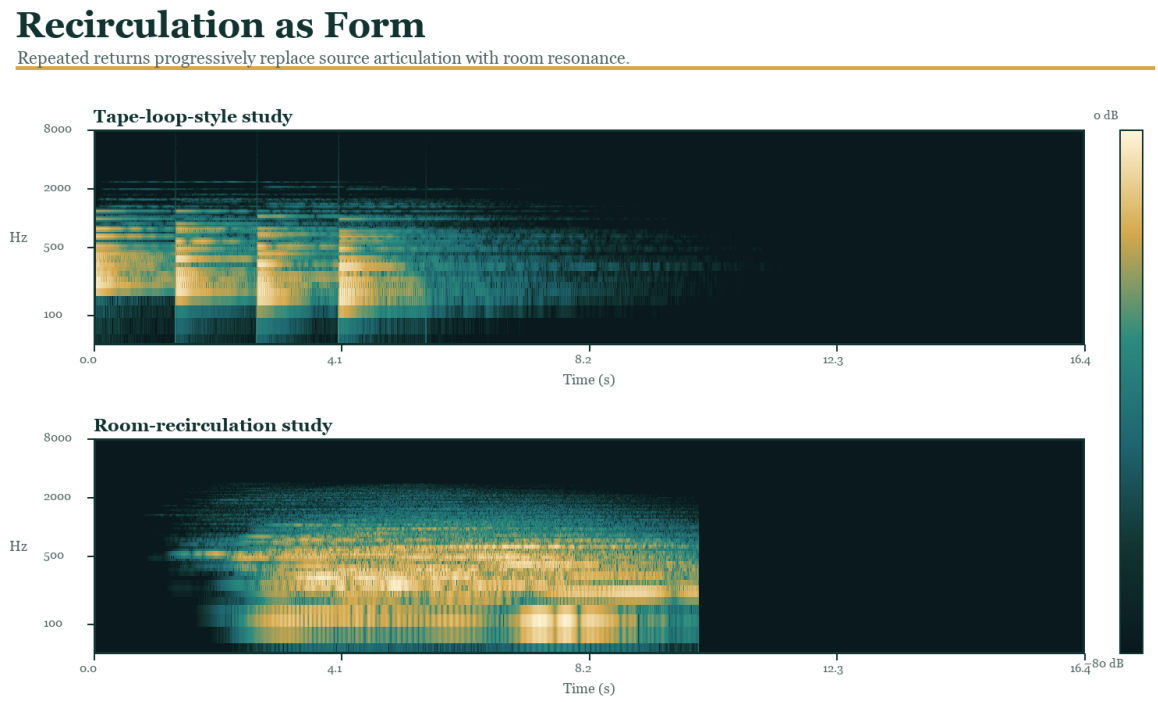


Figure 2-35: Recirculation processes that turn a source into a resonant field

How to read this figure. Early in each panel, attacks and pitch changes remain distinct. Later, selected bands become dominant because every pass reinforces the same system. A perfectly neutral loop would preserve the source indefinitely, but musical recirculation depends on selective memory. The engineering task is to make that selection intentional and bounded.

With verbx, `--repeat` performs sequential reprocessing. Start conservatively, render to floating point, and measure every generation:

```
verbx render source.wav /tmp/room_generation.wav \
  --engine algo --rt60 3.5 --repeat 5 --wet 0.86 --dry 0.14 \
  --fdn-lines 16 --fdn-matrix hadamard --lowcut 80 \
  --out-format float32 --json-out /tmp/room_generation.json
```

The JSON sidecar is part of the composition record. It lets another listener reproduce the transformation rather than relying on a screenshot of controls.

2.7.12 Example 11: Dense Guitar and Deep-Listening Fields

Dense broadband material and slowly unfolding spatial material can both fill a sonogram, but they challenge reverb differently. A dense guitar field may already contain distortion, modulation, and layered delays; a new long return can reduce depth by filling every remaining gap. A deep-listening texture may leave more room for the reverb to reveal low-level spatial changes.

The paired sonograms below show these contrasting fields. The upper panel is dense and continuous; the lower panel contains more visible internal movement and open regions.

Figure 2-36 presents Dense production field compared with a spacious deep-listening study and establishes the visual reference for the discussion that follows.

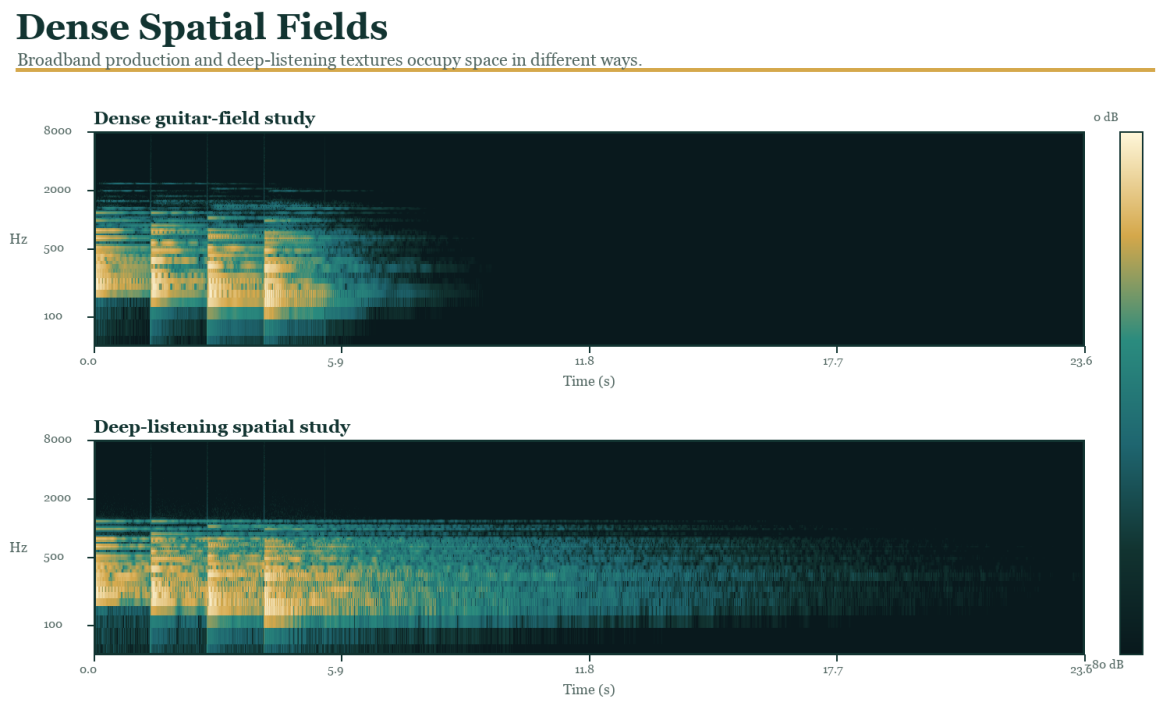


Figure 2-36: Dense production field compared with a spacious deep-listening study

How to read this figure. In the dense panel, added reverb changes envelope and width more readily than it reveals a new room; there is little unoccupied spectrum. In the lower panel, the return can articulate trajectories and decays. Use this comparison to decide whether a source needs more duration or merely a different spatial envelope.

For dense material, automate narrow throws, filter the return aggressively, or place only selected stems in the longest space. For sparse spatial material, preserve dynamic range and inspect the noise floor: the quietest part of the tail may be where motion is most audible.

2.7.13 Example 12: Shimmer as Register, Not Glitter

Shimmer pitch-shifts part of the wet field and feeds or blends it back into the return. The obvious octave-up setting can sound decorative when used continuously, but it becomes compositional when treated as register. A low source can seed a high sustained line; an upper melody can generate a halo that changes harmony at phrase boundaries.

The sonogram below shows the shipped shimmer study. New energy appears above the source bands and persists through the tail.

Figure 2-37 presents Octave-shifted feedback extending a musical phrase into the upper register and establishes the visual reference for the discussion that follows.

Figure 2-37: Octave-shifted feedback extending a musical phrase into the upper register

How to read this figure. Compare the bright upper ridges with the lower source energy that excites them. Shimmer is not a broadband treble boost: it creates new frequency relationships. Excess feedback makes these upper bands accumulate and can change the harmony. A low-cut before the pitch shifter prevents bass energy from producing a crowded octave layer; a high-cut after it can soften alias-like brightness.

```
verbx render music.wav /tmp/music_shimmer.wav \
  --engine algo --rt60 10 --wet 0.78 --dry 0.34 \
  --shimmer --shimmer-semitones 12 --shimmer-mix 0.34 \
  --shimmer-feedback 0.68 --shimmer-lowcut 280 --shimmer-highcut 12000 \
  --target-peak-dbfs -2
```

Write the shimmer entry into the arrangement. Bring it in for one cadence, a bridge, or the final resonance rather than leaving it at one level for the entire piece.

2.7.14 Example 13: Freeze and the Harmonic Field

Freeze captures a segment and sustains it through a near-static or very long loop. The source segment becomes the “instrument body” of the frozen field. A capture containing one clean interval tends to remain legible; a capture spanning a chord change can create beating or harmonic ambiguity. The best capture point is therefore a musical decision, not just a technical one.

The following sonogram shows the shipped frozen-music study. It extends for more than half a minute, making the gradual loss and redistribution of energy easy to see.

Figure 2-38 presents A captured musical interval sustained as a near-infinite reverberant field and establishes the visual reference for the discussion that follows.

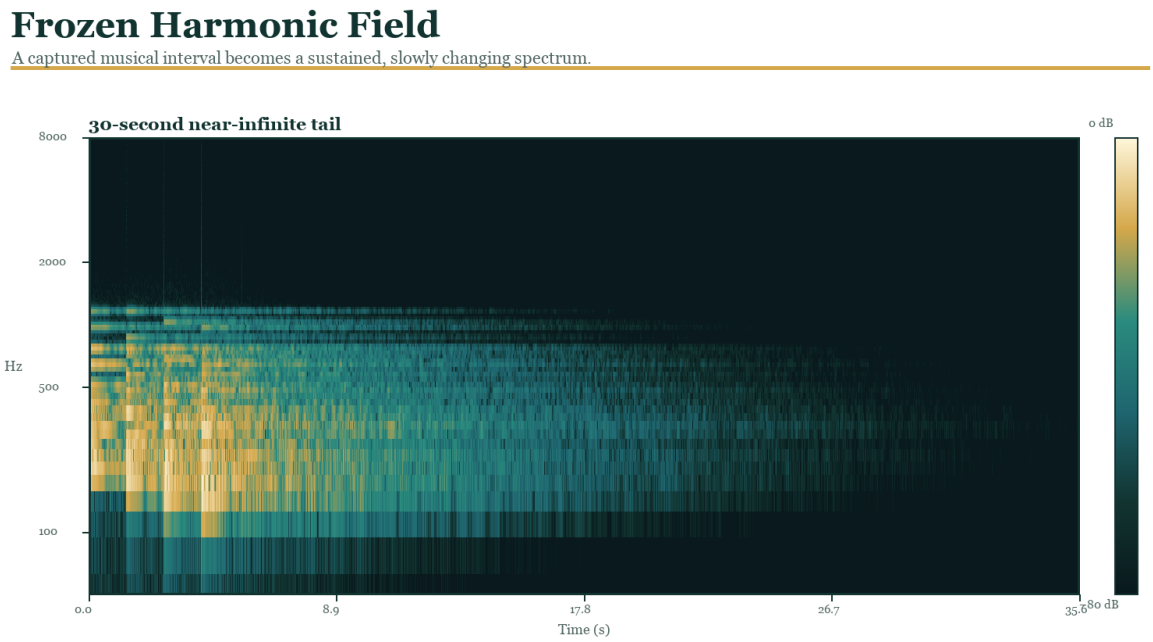


Figure 2-38: A captured musical interval sustained as a near-infinite reverberant field

How to read this figure. The strongest initial excitation sits at the left, but stable harmonic bands continue across much of the panel. Their slow decline distinguishes this bounded freeze from a perfectly lossless loop. A truly infinite mode must still manage DC, denormals, modulation, and output level; "infinite" describes sustain intent, not permission for uncontrolled numerical growth.

Compose a transition by freezing the last harmony of one section, changing the dry material underneath it, and releasing the field only after the new harmony has become stable. The listener hears the previous section as literal acoustic memory.

2.7.15 Example 14: Send/Return Reverb as Counterpoint

Insert reverb and send/return reverb encourage different musical thinking. An insert asks how a processor transforms one track. A send asks what independent spatial layer several sources contribute to. The return can have its own dynamics, automation, equalization, and formal entrances.

The Mermaid signal-flow figure below shows the dry and wet paths as parallel musical voices. The return is intentionally labeled as counterpoint rather than residue.

Figure 2-39 presents Musical send and return treated as parallel articulation and spatial counterpoint and establishes the visual reference for the discussion that follows.

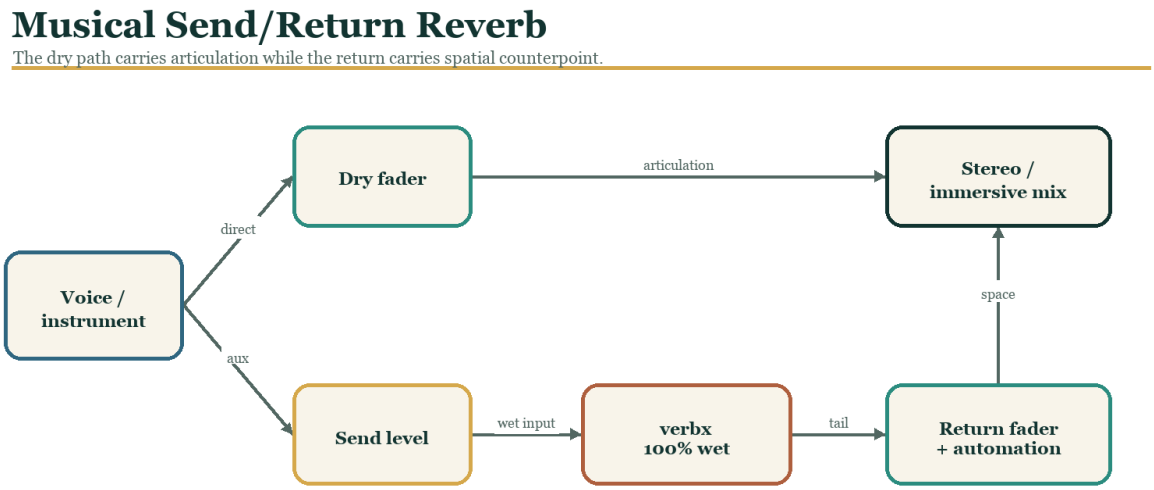


Figure 2-39: Musical send and return treated as parallel articulation and spatial counterpoint

How to read this figure. The dry fader controls foreground articulation. The aux send controls how much each source excites the shared space. The return fader controls the room as a layer after all excitations have combined. Automating the send changes what enters memory; automating the return changes how much of that memory is heard. Those gestures are not equivalent. A post-phrase send throw produces a different tail than raising a return that has been accumulating all along.

2.8 DSP Overview

Digital reverberation begins with an impulse response $h[n]$: the output produced by a unit impulse. For a linear time-invariant system, any input $x[n]$ produces

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Convolution reverb stores or synthesizes $h[n]$ and evaluates that sum efficiently. Algorithmic reverb builds a recursive system whose impulse response emerges from delay, filter, and feedback operations. Both approaches can produce excellent results. Their differences concern control, memory, latency, time variation, and the kinds of spaces they make easy to design.

The detailed diagrams in this section use the implementation-level graph grammar found in Julius O. Smith's Schroeder Reverberators: rectangular transfer blocks, named branch signals, explicit gains and delays, summing

junctions, and output matrices. The figures are original verbx diagrams with independent example values. Read arrows as signal paths, z^{-M} as a delay of M samples, circles as sums, and a returning path as state that will affect a future output sample.

2.8.1 Discrete-Time Foundations: Samples, Delays, and State

A digital reverberator does not manipulate a continuous acoustic field directly. It updates a finite collection of numbers at the sample rate. If the sample rate is f_s hertz, sample n occurs at time

$$t_n = \frac{n}{f_s}.$$

One sample therefore represents $1/f_s$ seconds. At 48 kHz, a 1 ms delay contains 48 samples; at 192 kHz, the same physical delay contains 192 samples. The acoustic time is unchanged, but memory use and the number of arithmetic operations both increase. This distinction matters when a preset is moved between sample rates: delays expressed in samples must be recalculated from their intended duration rather than copied unchanged.

The elementary delay operator is

$$D_M(z) = z^{-M},$$

which means "return the sample written M updates earlier." In code, that operation is usually a circular buffer with one write pointer and one or more read pointers. The buffer is state: its contents summarize enough of the past to determine future output. Recursive reverberators contain many such state variables, so a parameter change can affect audio written before the change as well as audio arriving afterward.

Three implementation scales should remain conceptually separate:

Table 2-1: Discrete-Time Foundations: Samples, Delays, and State: reference table

Scale	Typical unit	What it controls
Sample	samples or microseconds	Delay indexing, interpolation, phase, and causality
Block	frames or milliseconds	Host scheduling, FFT partitions, automation updates, and latency
Musical event	beats or seconds	Pre-delay, buildup, phrase overlap, and audible decay

A robust design converts user-facing time values into sample-domain state once per block or parameter event, then performs the inner audio loop without allocations. It also records which quantities are sample-rate dependent. A 2.4-second RT60 is a physical target; a 1,499-sample comb delay is a topology choice tied to one sample rate.

2.8.2 Difference Equations, Transfer Functions, and Impulse Responses

A digital filter is a rule that maps an input sample sequence to an output sample sequence. A finite impulse response (FIR) filter uses a finite weighted history of the input. An infinite impulse response (IIR) filter also uses previous outputs, so energy can recirculate indefinitely in the mathematical model. Causality requires the present output to depend only on present and past information. Linearity permits superposition, and time invariance means that delaying an input delays its response without changing the response's shape. Those properties make convolution, transfer functions, and pole-zero analysis available as mutually consistent descriptions.

The general scalar, causal, linear difference equation is

$$y[n] = \sum_{k=0}^{P} b_k x[n-k] - \sum_{k=1}^{Q} a_k y[n-k],$$

with feedforward coefficients b_k and feedback coefficients a_k. Its transfer function is the rational polynomial

$$H(z) = \frac{\sum_{k=0}^{P} b_k z^{-k}}{1 + \sum_{k=1}^{Q} a_k z^{-k}} = \frac{B(z)}{A(z)}.$$

Zeros are roots of $B(z)$; poles are roots of $A(z)$. A zero suppresses a complex sinusoidal component at its angle and radius, while a pole describes a natural mode that the system can sustain after excitation. For real-valued coefficients, non-real roots occur in complex-conjugate pairs. Repeated roots have multiplicity, which changes the envelope multiplying the corresponding exponential mode.

A block diagram, a difference equation, and a transfer function describe the same linear time-invariant system from different viewpoints. Consider

$$y[n] = x[n] + g\,y[n - M].$$

The difference equation is closest to code: read one delayed value, multiply by g, add the current input, and write the result. Taking the z transform gives

$$Y(z) = X(z) + gz^{-M}Y(z),$$

and therefore

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - gz^{-M}}.$$

Expanding the denominator as a geometric series gives

$$H(z) = 1 + gz^{-M} + g^2z^{-2M} + g^3z^{-3M} + \cdots,$$

which is the z-domain statement of the audible echo train. This three-way translation is a useful debugging method. If a diagram suggests one sign, the equation another, and the impulse response a third, the implementation is not merely “voiced differently”; one representation is wrong.

2.8.3 Solving Higher-Order Characteristic Equations

Finding the poles of a high-order reverberator is an algebra problem with numerical consequences. Multiplying a denominator expressed in z^{-1} by a sufficient power of z produces an ordinary polynomial. Equivalently, substitute $u = z^{-1}$, solve for u, and invert each nonzero root. Closed-form quadratic formulas remain useful for second-order sections, but production reverberators routinely create orders for which closed forms are unwieldy or do not exist in radicals.

For a monic polynomial

$$p(z) = z^Q + c_{Q-1}z^{Q-1} + \cdots + c_1 z + c_0,$$

the roots are the eigenvalues of its companion matrix

$$K = \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{Q-1} \end{bmatrix}, \qquad p(z) = \det(zI - K).$$

This linear-algebra formulation connects filter analysis to robust eigenvalue algorithms. It also explains why polynomial roots can be ill conditioned: a small coefficient perturbation can move a cluster of nearly repeated roots substantially. Implement long filters as cascaded first- and second-order sections when possible, compute roots in double precision, pair conjugates explicitly, and verify the frequency response after reconstruction. In a feedback network, inspect eigenvalues of the state transition or a suitable polynomial eigenvalue linearization rather than multiplying large sparse delay polynomials by hand.

2.8.4 Poles, Stability, and Decay

The poles of a recursive filter describe the modes that can continue after the input stops. For the feedback comb, the pole condition is

$$z^M = g.$$

There are M roots distributed around the complex plane. Their angles determine modal frequencies, while their common radius is $|g|^{1/M}$. Poles near the unit circle decay slowly; poles outside it grow. The practical stability requirement $|g| < 1$ is therefore not an arbitrary range check but a statement that every circulation must lose energy.

An FDN replaces one scalar pole family with the roots of a matrix-delay characteristic equation. The same geometric intuition survives: delay lengths distribute modal angles, the feedback matrix couples modes, and loop filters pull pole radii inward by frequency-dependent amounts. RT60 is a perceptual summary of those radii, not a complete description of the pole pattern. Two networks can share the same fitted RT60 while one rings at several exposed frequencies and the other decays smoothly.

For extreme decay times, stability margin becomes an audible design parameter. A target of 360 seconds places loop gains extraordinarily close to unity. Parameter interpolation, filter normalization, matrix orthogonality, denormal handling, and limiter placement all matter because a tiny systematic gain error can persist through thousands of loops.

2.8.5 Echo Density, Modal Density, and Mixing Time

Echo density asks how many distinguishable arrivals occur in a time interval. Modal density asks how many resonant modes occupy a frequency interval. They are related but not interchangeable. A signal can have many temporal arrivals yet retain colored modes, or have many modes but a conspicuously sparse onset.

The early response is usually nonstationary: arrivals become more frequent as reflection order increases. After a mixing time, individual paths are no longer the useful mental model and the response behaves statistically. A digital design recreates this transition with tapped delays, nested diffusers, scattering, or an FDN whose recirculations rapidly multiply path combinations. The goal is not maximum density at sample zero. The goal is the right density trajectory for the apparent source distance and enclosure size.

Use a click to hear temporal sparsity and a sustained chord to hear modal sparsity. On the click, listen for flutter and repeated gaps. On the chord, listen for stable tones that outlive neighboring partials. Increasing allpass depth may solve the first problem without solving the second; changing delays or feedback topology may solve the second without producing a convincing onset.

2.8.6 Feedback Comb Filters: Duration and Modes

A feedback comb filter delays its output, scales it, and adds it back to the input. Its transfer function is

$$H_{\text{comb}}(z) = \frac{1}{1 - gz^{-M}},$$

where M is delay in samples and g is feedback gain. An impulse produces repetitions every M samples with amplitudes $1, g, g^2, \ldots$. In frequency, those repetitions form a regularly spaced modal pattern. One comb therefore sounds colored; several combs with carefully chosen delays can create a richer decay.

The diagram below exposes the feedback path that creates both duration and coloration.

Figure 2-40 presents Feedback comb filter with delay length M and loop gain g and establishes the visual reference for the discussion that follows.

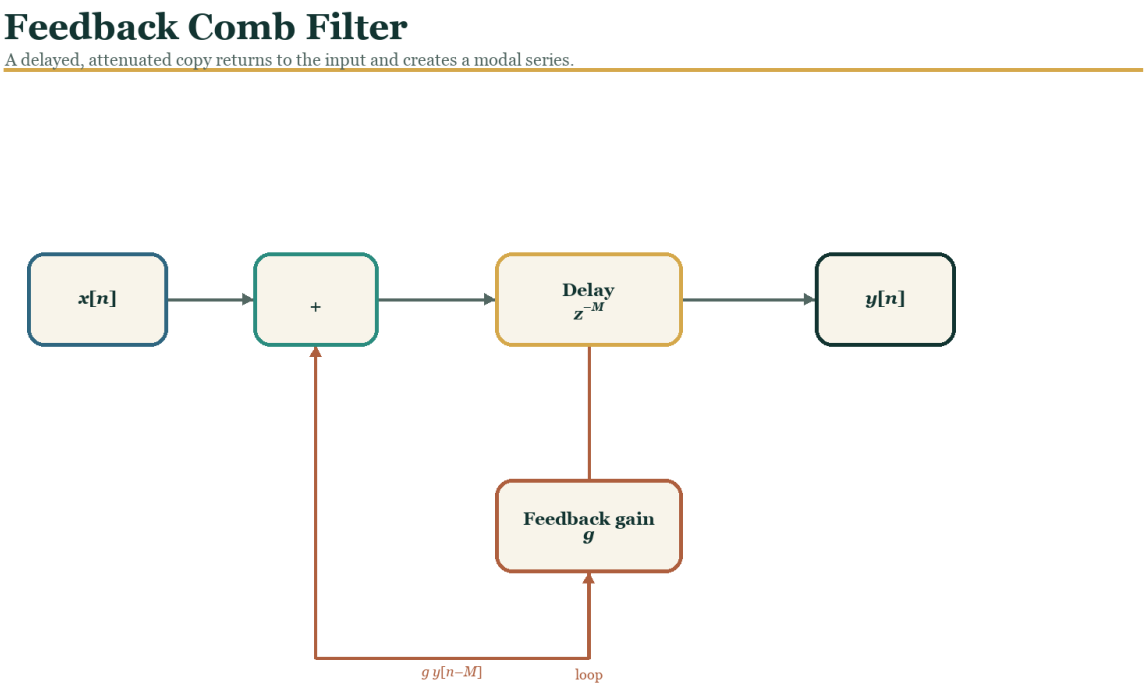
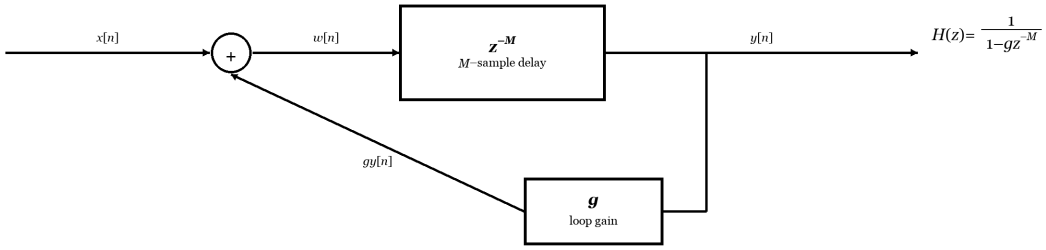


Figure 2-40: Feedback comb filter with delay length M and loop gain g

How to read this figure. Every trip around the loop adds M samples of time and a factor of g. If $|g| < 1$, the loop decays; if $|g| = 1$, it is theoretically lossless; if $|g| > 1$, it grows. Real implementations also require damping, DC control, and protection against finite-precision problems. The spacing f_s/M between modes is as important as nominal RT60. Several delay lengths should avoid obvious common periods unless a pitched or resonant effect is intended.

The expanded flowgraph below labels the internal write signal and places the transfer function beside the loop. It is the form to use when translating the structure into a sample-by-sample difference equation or checking which sample is multiplied by g.

Figure 2-41 presents Implementation-level feedback comb flowgraph with an explicit internal state, M-sample delay, loop gain, and transfer function and establishes the visual reference for the discussion that follows.



Feedback comb: each loop traversal adds M samples and multiplies amplitude by g.

Figure 2-41: Implementation-level feedback comb flowgraph with an explicit internal state, M-sample delay, loop gain, and transfer function

How to read this flowgraph. The summing junction forms $w[n]$ from the new input and the attenuated delayed output. The delay emits $y[n] = w[n-M]$; that value branches to the external output and to gain g before returning to the sum. Following one impulse around the lower return path explains the sequence at $0, M, 2M, \ldots$ samples. The drawing also distinguishes the signal stored in memory, $w[n]$, from the signal currently leaving the delay, $y[n]$, a distinction that prevents common indexing mistakes in implementations.

The following pole-zero diagram turns the comb equation into geometry. The drawing uses $M = 12$ only to keep individual modes legible; a practical delay contains many more samples and therefore many more poles. The same rule survives at every order.

Figure 2-42 presents Unit-circle pole-zero map for a representative feedback comb, showing twelve equal-radius poles and twelve coincident zeros at the origin and establishes the visual reference for the discussion that follows.

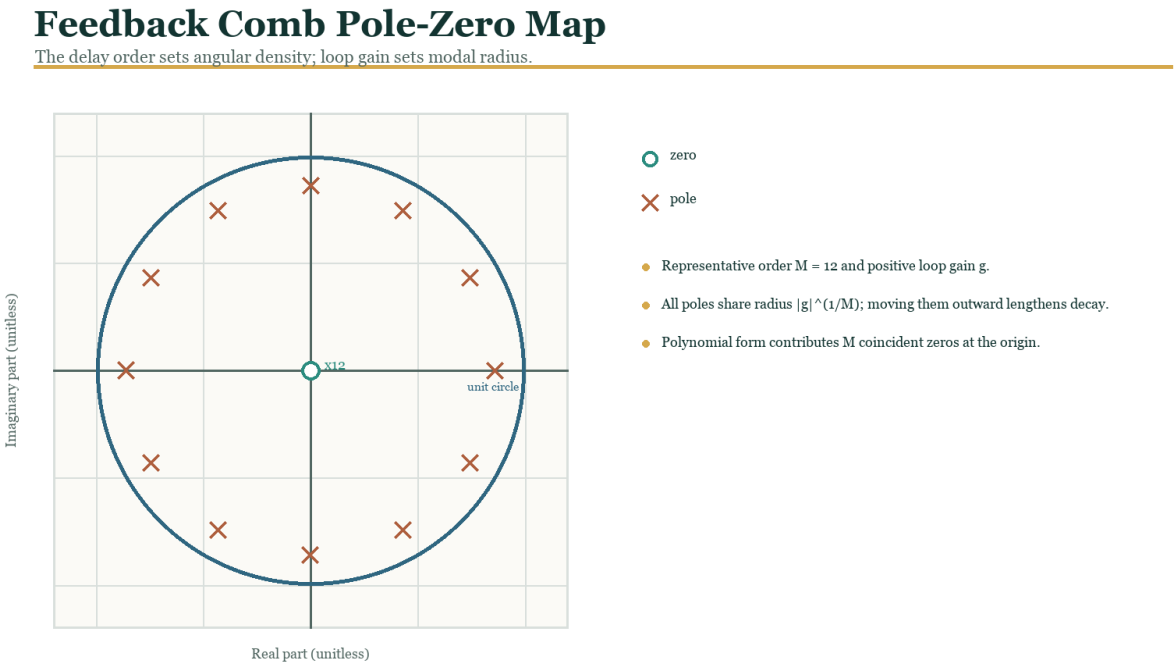


Figure 2-42: Unit-circle pole-zero map for a representative feedback comb, showing twelve equal-radius poles and twelve coincident zeros at the origin

Each solution of $z^M = g$ contributes one pole. The angular spacing is $2\pi/M$, while the radius is $|g|^{1/M}$. Raising g moves every cross toward the unit circle and lengthens all modal decays together. A negative g rotates the pole family by π/M. The zeros at the origin appear after polynomializing the causal transfer function; they do not cancel the resonant pole ring.

For a target T_{60} and a delay duration $d = M/f_s$ seconds, the loop gain is

$$g = 10^{-3d/T_{60}}.$$

At extreme RT60 values, g approaches one. Small numerical or spectral errors then circulate for a long time, which is why internal precision and loop conditioning matter more at 360 seconds than at 1.2 seconds.

The companion magnitude plot turns the pole ring into an audible spectral prediction. Each regular resonance corresponds to one of the equally spaced modal angles above.

Figure 2-43 presents Peak-normalized magnitude response of the representative feedback comb, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

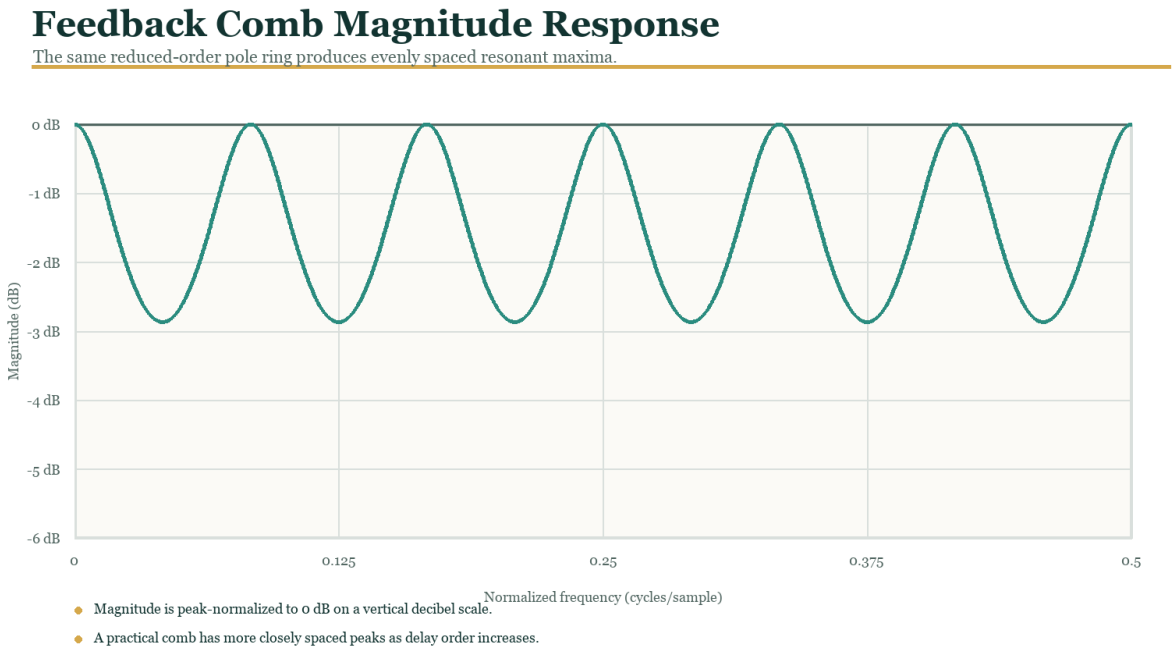


Figure 2-43: Peak-normalized magnitude response of the representative feedback comb, with frequency in cycles per sample and magnitude on a vertical decibel scale

2.8.7 Schroeder Allpass Filters: Density Without Magnitude Coloration

An allpass filter changes phase while maintaining a flat ideal magnitude response. A common Schroeder form is

$$H_{\mathrm{AP}}(z) = \frac{-g + z^{-M}}{1 - gz^{-M}}.$$

Its direct and delayed paths cancel the comb-like magnitude coloration that the feedback loop would otherwise introduce. In time, however, an impulse becomes a pattern of decaying echoes. Serial allpasses can therefore increase echo density before the signal enters a longer late-field network.

The Mermaid diagram below separates feedforward and feedback paths. The signs matter: the topology is not simply a comb with a dry mix.

Figure 2-44 presents Schroeder allpass diffuser with matched feedforward and feedback coefficients and establishes the visual reference for the discussion that follows.

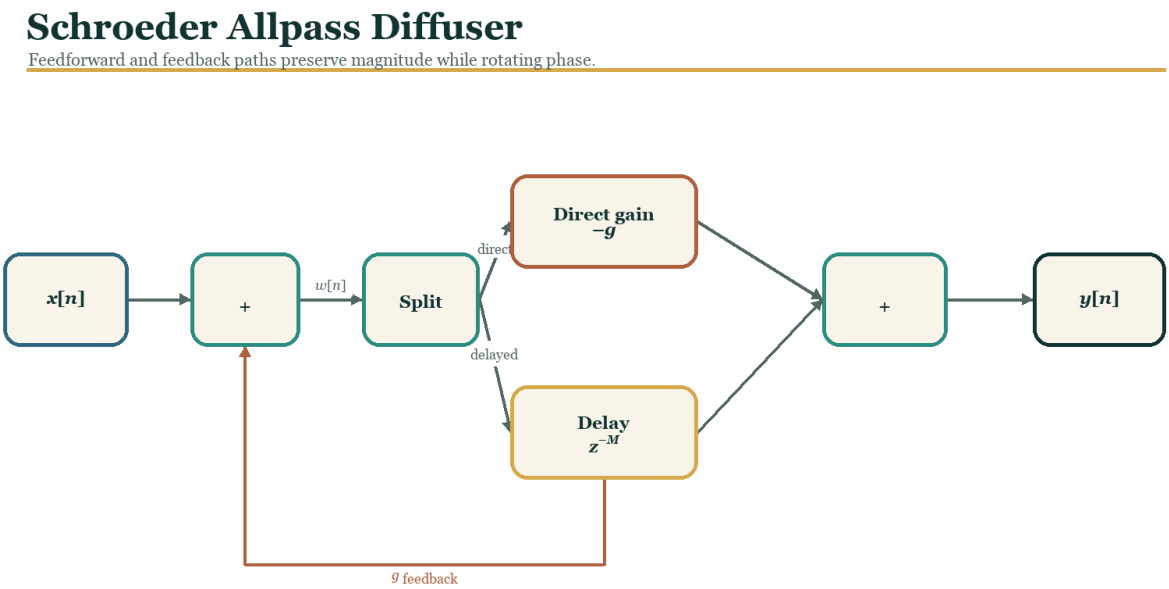


Figure 2-44: Schroeder allpass diffuser with matched feedforward and feedback coefficients

How to read this figure. The first sum forms $w[n] = x[n] + g\,w[n-M]$, and the split sends $w[n]$ to both output paths. The delayed path enters the output sum with unity gain, while the direct path contributes $-g\,w[n]$. Together they produce $y[n] = w[n-M] - g\,w[n]$. The delay and feedback create a decaying echo sequence; the direct coefficient corrects the magnitude response. "Flat magnitude" does not mean "inaudible." Phase structure, transient spreading, delay length, and coefficient still change timbre and spatial impression. Very large g values produce long diffusion patterns that may sound metallic; small values scatter less energy. The allpass chain must be judged as an onset-design tool, not merely by its frequency response.

The implementation graph below separates the allpass's two sums. This makes the signs and the shared delayed state visible, which is essential when comparing the transfer function with code or tracing an impulse by hand.

Figure 2-45 presents Implementation-level Schroeder allpass flowgraph with explicit feedforward and feedback coefficient paths and establishes the visual reference for the discussion that follows.

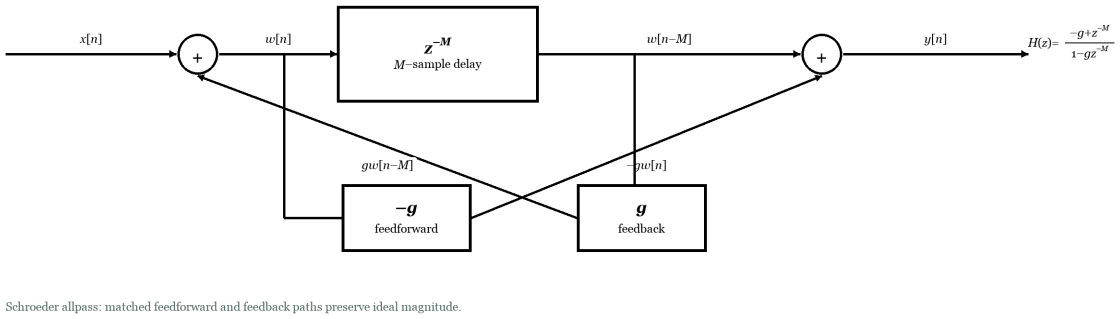


Figure 2-45: Implementation-level Schroeder allpass flowgraph with explicit feedforward and feedback coefficient paths

How to read this flowgraph. The left sum creates the delay-line write value $w[n] = x[n] + g\,w[n-M]$. The right sum combines the delayed state with the negative direct branch, producing $y[n] = w[n-M] - g\,w[n]$. The same coefficient magnitude therefore appears in two different roles: positive feedback sustains the echo sequence, while negative feedforward corrects its magnitude response. Swapping a sign changes the topology and can turn a diffuser into a colored or unstable structure even though every block still looks plausible in isolation.

The reciprocal structure is easiest to see in the following unit-circle plot. It uses a reduced delay order so the individual reciprocal pairs can be inspected rather than merging into a continuous ring.

Figure 2-46 presents Unit-circle pole-zero map for a Schroeder allpass, with stable interior poles and reciprocal exterior zeros and establishes the visual reference for the discussion that follows.

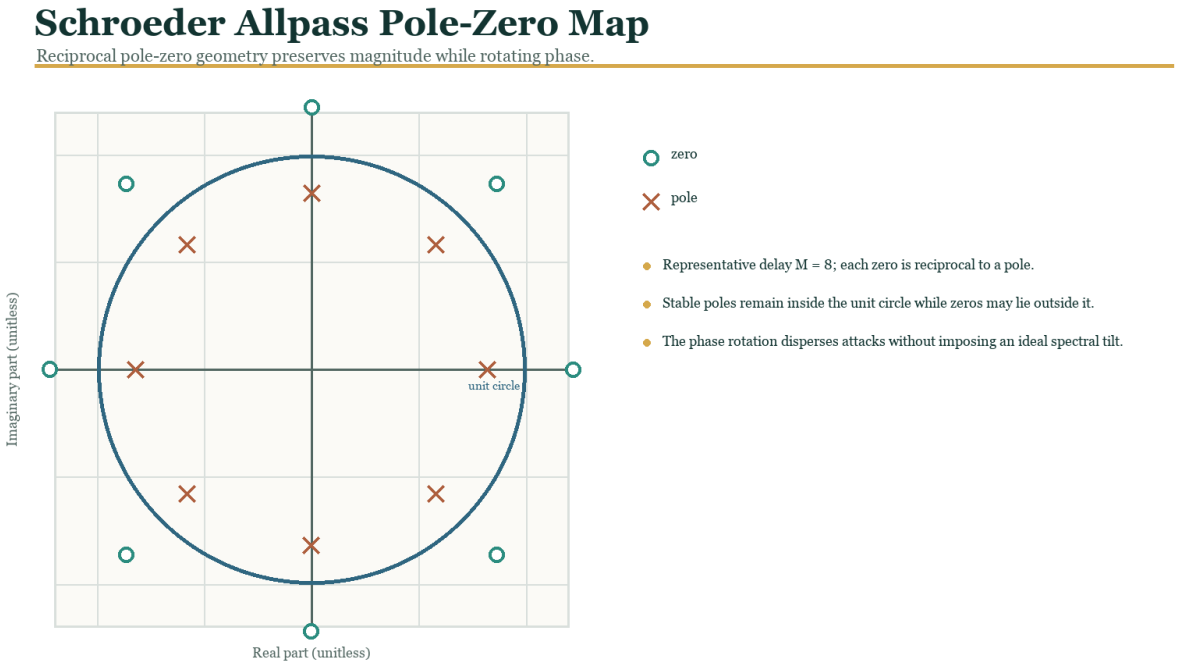


Figure 2-46: Unit-circle pole-zero map for a Schroeder allpass, with stable interior poles and reciprocal exterior zeros

For every pole p_k, the corresponding allpass zero lies at $1/p_k^*$. On the unit circle, numerator and denominator magnitudes therefore match even though their phases do not. The exterior zeros do not make the causal filter unstable because stability is governed by poles. They do make the system non-minimum-phase, which is precisely how the network can redistribute transient energy in time while preserving ideal magnitude.

The following magnitude plot is deliberately flat. It is the essential check on the allpass claim: a non-flat curve would mean the reciprocal-root or gain relationship had been implemented incorrectly.

Figure 2-47 presents Peak-normalized magnitude response of the representative Schroeder allpass, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

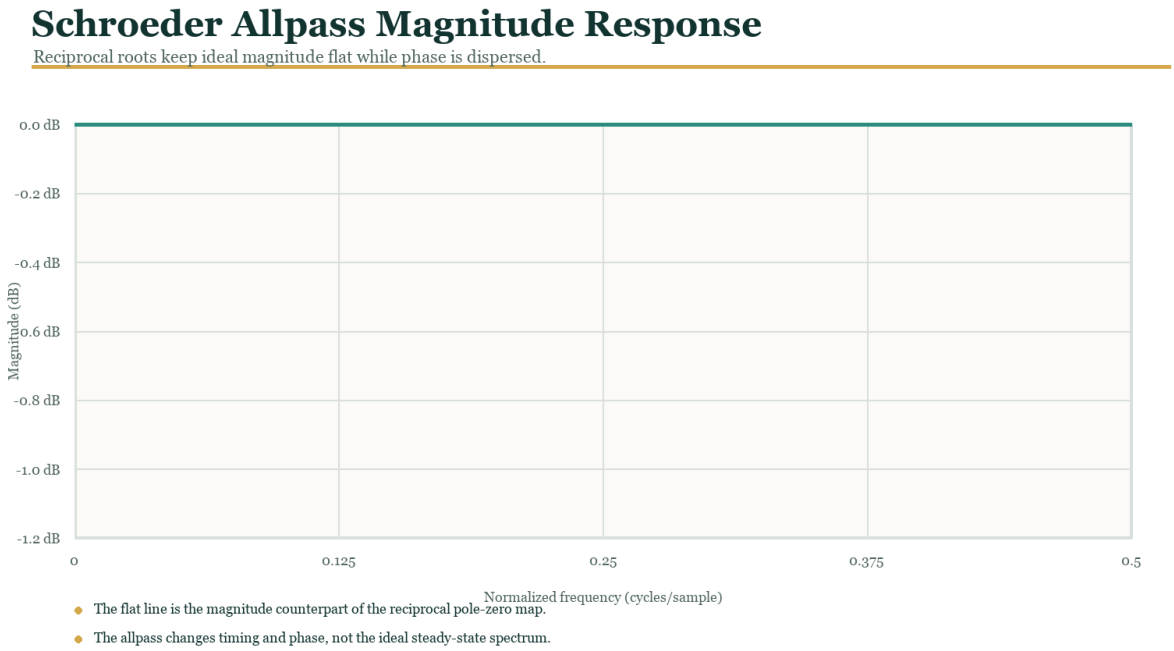


Figure 2-47: Peak-normalized magnitude response of the representative Schroeder allpass, with frequency in cycles per sample and magnitude on a vertical decibel scale

Magnitude alone cannot explain why an allpass diffuser is audible. The unwrapped phase response below rotates rapidly near delay-related frequencies. Its changing slope indicates that nearby frequency neighborhoods accumulate different amounts of phase rotation. For a click, this creates a denser, less obviously periodic cloud of arrivals; for speech or percussion, it can soften the transition from attack to tail without an ideal spectral EQ curve.

Figure 2-48 presents Unwrapped phase in radians for the representative Schroeder allpass, plotted against normalized frequency in cycles per sample and establishes the visual reference for the discussion that follows.

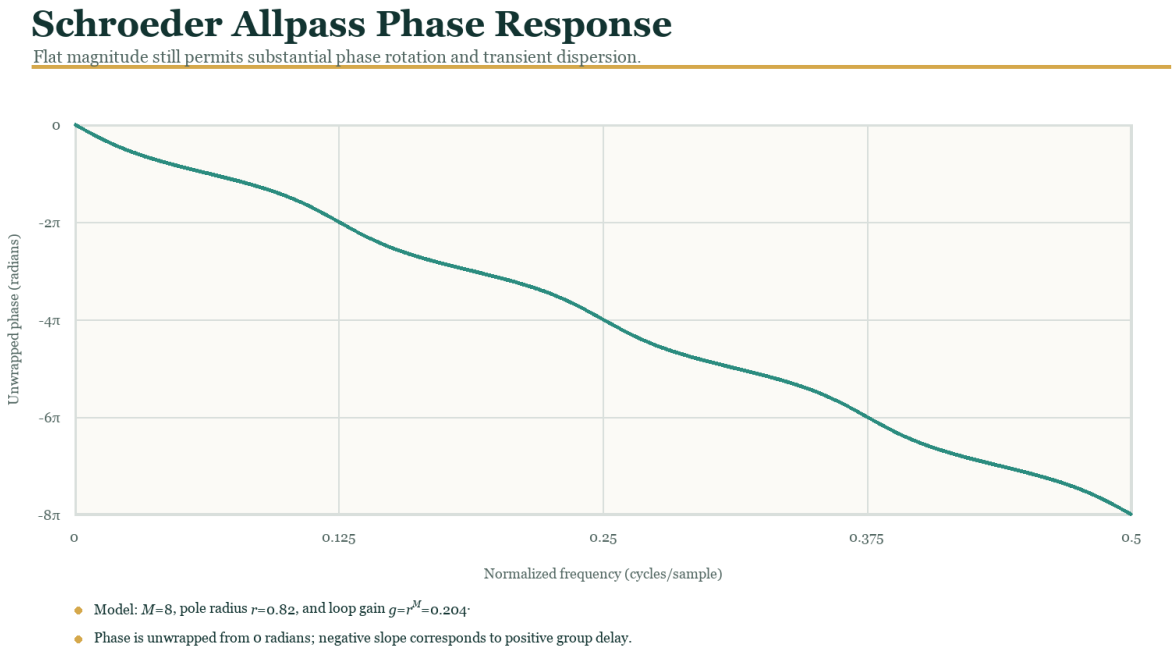


Figure 2-48: Unwrapped phase in radians for the representative Schroeder allpass, plotted against normalized frequency in cycles per sample

This reduced example uses delay order $M = 8$, pole radius $r = 0.82$, and loop gain $g = r^M = 0.82^8 \approx 0.204$. Its unwrapped phase therefore begins at 0 radians and ends at -8π radians at Nyquist.

Read the teal phase trace as accumulated rotation rather than an amplitude measurement. Its slope is the important audible quantity: a steeper slope means larger group delay. Increasing feedback steepens localized phase transitions, lengthening the echo pattern and increasing diffusion, but it can also reveal metallic ringing when delay lengths or stages are too regular. In a serial diffuser, choose mutually incommensurate delays and moderate gains so the individual delay patterns overlap into texture rather than announcing a single repeat rate.

2.8.8 Allpass Networks: From Echoes to a Diffuse Excitation

One allpass creates a recognizable pattern. Several allpasses with mutually awkward delay lengths multiply the number of arrivals. If the stages are too similar, the patterns line up and produce periodicity. If there are too many stages or too much feedback, the attack loses definition before the late field begins.

The serial network below shows the conceptual progression from one impulse to a dense excitation. Each stage is short compared with the final RT60.

Figure 2-49 presents Serial allpass network used to build early echo density and establishes the visual reference for the discussion that follows.

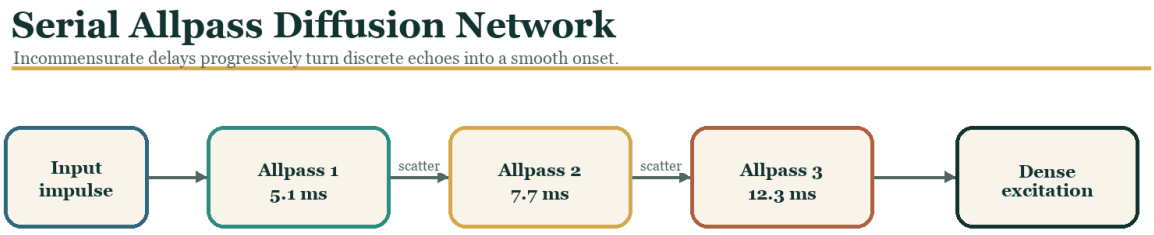


Figure 2-49: Serial allpass network used to build early echo density

How to read this figure. Every box expands the number of time-domain arrivals while approximately preserving spectral magnitude. The listed delays are illustrative, not universal defaults. Percussion may need fewer stages so the attack remains crisp; a synthetic pad can tolerate more diffusion. In verbx, allpass diffusion is a front-end stage. It prepares energy for the FDN rather than carrying the complete decay alone.

2.8.9 The Schroeder Reverberator

The classic Schroeder architecture places several feedback comb filters in parallel, sums them, and follows the sum with serial allpasses. The combs supply a long response; the allpasses increase density. This structure established the vocabulary of digital reverb and remains useful because it is efficient, understandable, and musical.

The signal flow below shows the canonical division of labor. It is a family of designs, not one immutable set of delay values.

Figure 2-50 presents Classic Schroeder reverberator with parallel combs and serial allpasses and establishes the visual reference for the discussion that follows.

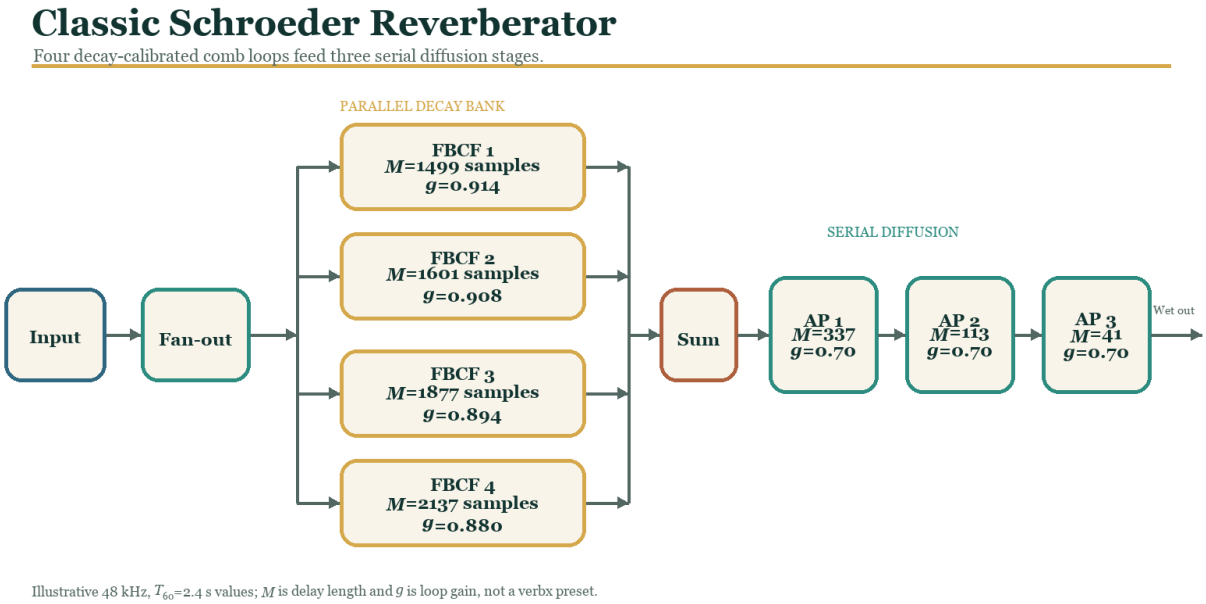


Figure 2-50: Classic Schroeder reverberator with parallel combs and serial allpasses

How to read this figure. The explicit branch labels show one illustrative 48 kHz,

2.4-second design, not verbx defaults. Each comb delay M establishes a different modal spacing, while its loop gain g calibrates how quickly that branch decays. The summing bus recombines all four independent tails. Three short serial allpasses then increase echo density without intentionally changing magnitude response. Coloration remains possible because each comb has a regular modal series and because the branches do not exchange energy. Moorer-style extensions add early reflections and frequency-dependent damping. Modern FDNs generalize the feedback relationship so every delay line can exchange energy with every other line.

The parameterized flowgraph below expands the canonical blocks into a traceable example. Its delays and gains demonstrate the notation rather than prescribe verbx defaults; the four output rows also show how one reverberator state can feed a multichannel matrix.

Figure 2-51 presents Parameterized Schroeder-style reverberator with serial allpasses, four parallel feedback combs, named branch signals, and a four-channel output matrix and establishes the visual reference for the discussion that follows.

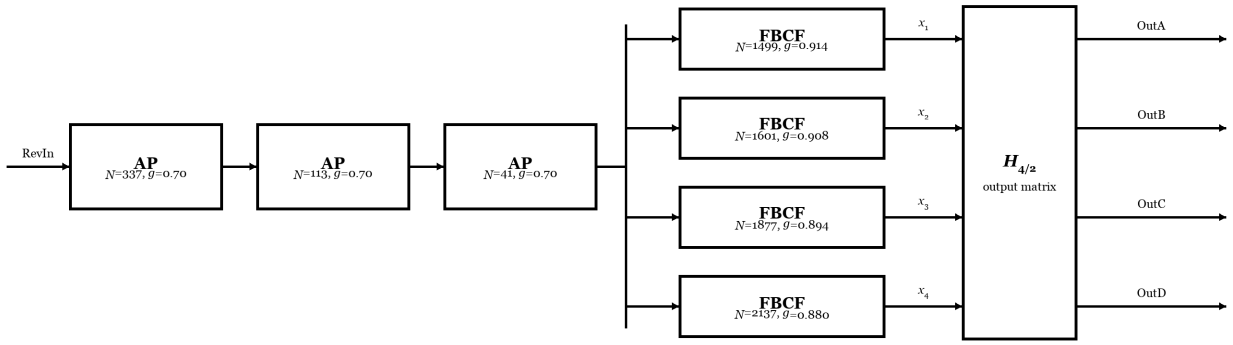


Figure 2-51: Parameterized Schroeder-style reverberator with serial allpasses, four parallel feedback combs, named branch signals, and a four-channel output matrix

How to read this flowgraph. Start at `RevIn` and move through three short allpasses, whose unequal sample delays progressively scatter the onset. The vertical bus then sends the same diffuse excitation to four feedback comb filters. Each comb lists a delay N and a gain calculated for the illustrated 2.4-second target at 48 kHz. Their named outputs x_1 through x_4 enter a normalized Hadamard matrix, whose signed combinations produce four related but decorrelated outputs. The graph therefore exposes three distinct design layers: onset diffusion, late modal duration, and spatial projection.

The following reduced-order modal map combines several unequal comb rings and an illustrative allpass zero family. It is not a root plot of the exact printed delays, whose thousands of points would obscure the structure at book scale.

Figure 2-52 presents Reduced-order unit-circle map of a parameterized Schroeder reverberator, showing interleaved comb-pole rings and reciprocal allpass zeros and establishes the visual reference for the discussion that follows.

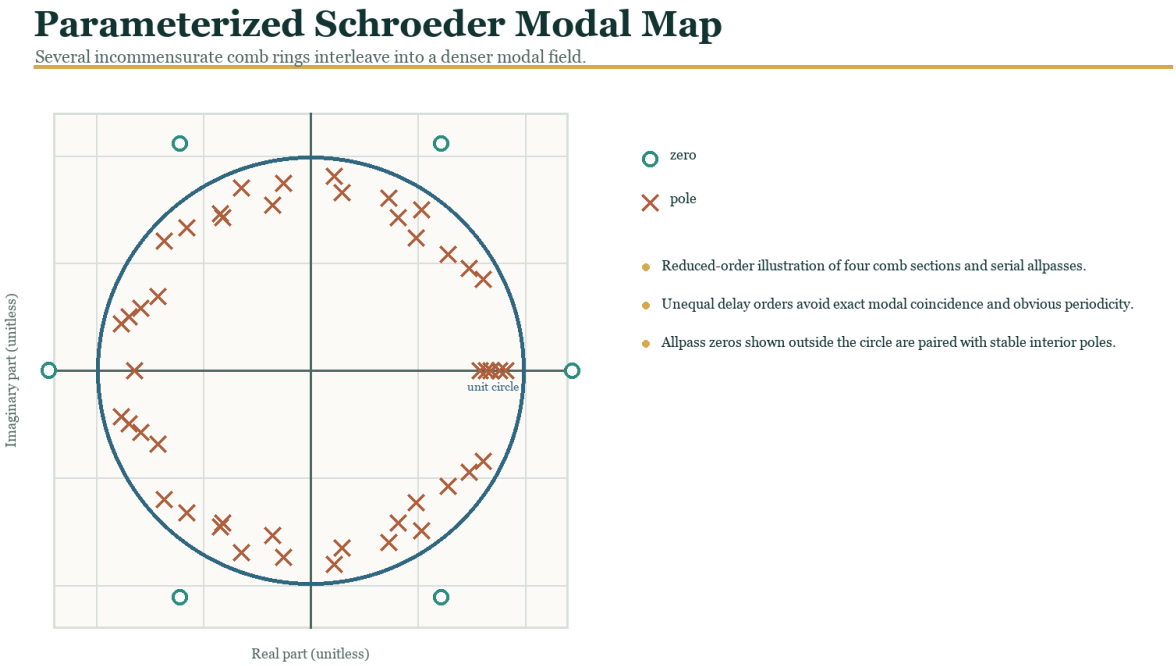


Figure 2-52: Reduced-order unit-circle map of a parameterized Schroeder reverberator, showing interleaved comb-pole rings and reciprocal allpass zeros

Each comb contributes a regular ring, but unequal orders and radii interleave those rings. The allpasses add pole-zero pairs whose magnitude effects cancel ideally while their phase effects accumulate. Coincident or nearly coincident poles predict exposed ringing; a more even angular distribution predicts smoother modal coverage. Because parallel branch transfer functions add, the final transmission zeros also depend on branch gains and summation, not only on the zeros drawn for individual stages.

The Schroeder design remains an excellent teaching instrument. Bypass its allpasses and hear the comb modes. Restore one stage at a time and hear density increase. Change one delay until it shares a common divisor with another and hear periodicity emerge. These experiments turn abstract topology into an audible vocabulary.

The following response emphasizes how unequal comb sections populate the spectrum with many nearby resonances even before the allpass stages redistribute the transient energy.

Figure 2-53 presents Peak-normalized magnitude response for the reduced-order parameterized Schroeder example, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

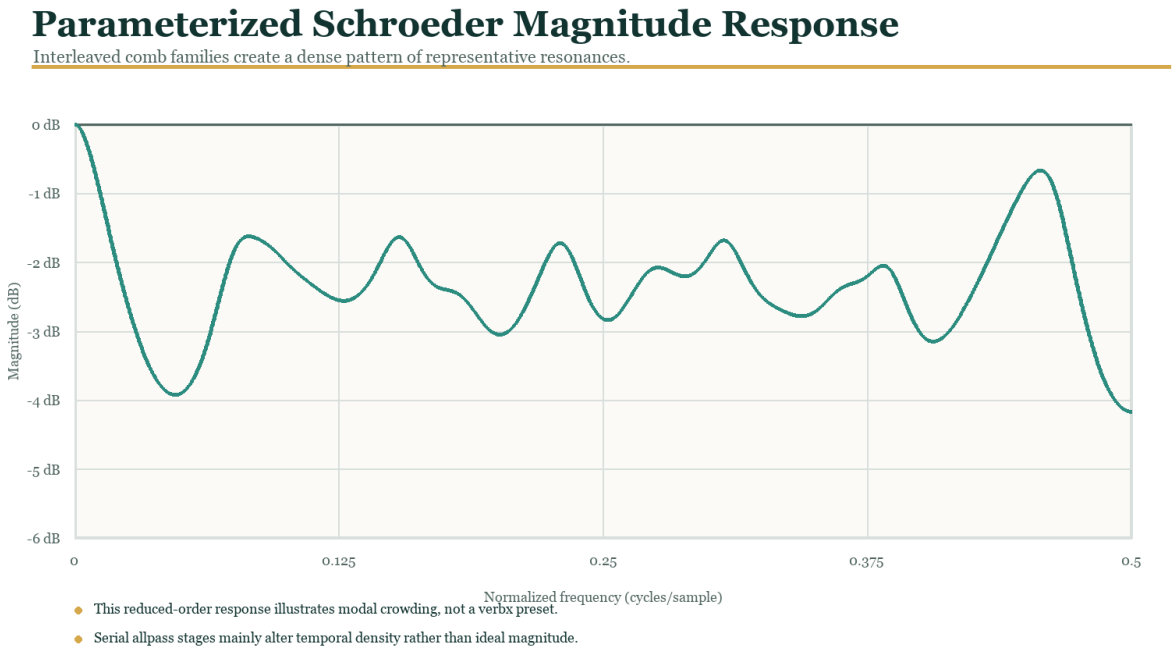


Figure 2-53: Peak-normalized magnitude response for the reduced-order parameterized Schroeder example, with frequency in cycles per sample and magnitude on a vertical decibel scale

2.8.10 Feedback Delay Networks: Coupled Modal Systems

An N-line Feedback Delay Network replaces independent comb feedback with a vector loop. Each delay line produces one state component. A matrix mixes those components before they are written back, allowing energy to move throughout the network. Input and output projections determine how the source excites modes and how the listener receives them.

A simplified state description is

$$s[n+1] = GMD(z)s[n] + Bx[n],$$

$$y[n] = C^{\mathsf{T}}D(z)s[n] + dx[n],$$

where $D(z)$ is the bank of unequal delays, M is the feedback matrix, G contains decay gains or filters, B injects the source, and C projects the state to output channels.

The Mermaid diagram below makes the vector feedback loop explicit.

Figure 2-54 presents Feedback Delay Network with vector delay state and matrix feedback and establishes the visual reference for the discussion that follows.

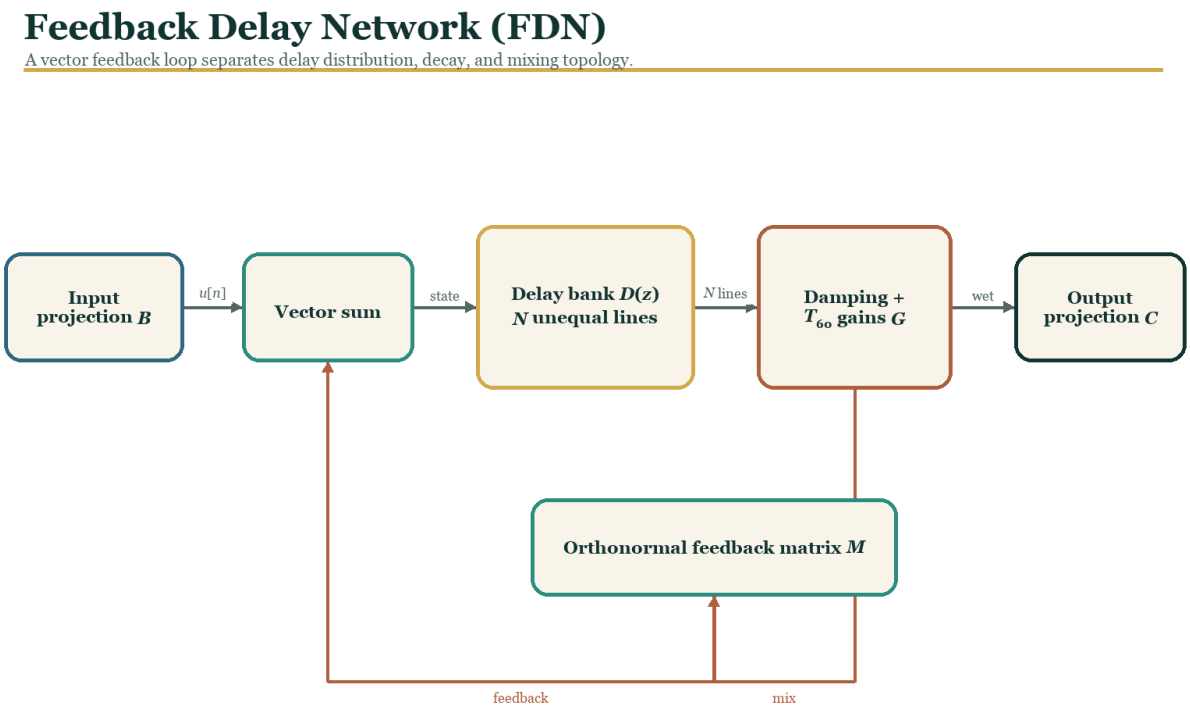


Figure 2-54: Feedback Delay Network with vector delay state and matrix feedback

How to read this figure. The delay bank defines modal timing; the gain and damping stage defines loss; the matrix defines how energy exchanges among lines. Those roles interact but are independently designable. An orthonormal matrix preserves vector energy before damping, making RT60 calibration tractable. A poor input projection may leave some modes weakly excited; a poor output projection may overemphasize others. High line count alone does not guarantee a smooth reverb.

The expanded FDN flowgraph below opens the vector blocks into four representative delay rows and shows both the recirculating state and the separate output tap. The four rows stand for an arbitrary N-line network, not a four-line limit.

Figure 2-55 presents Expanded FDN flowgraph with input projection, unequal delay lines, per-line damping, RT60 gains, unitary feedback matrix, and output projection and establishes the visual reference for the discussion that follows.

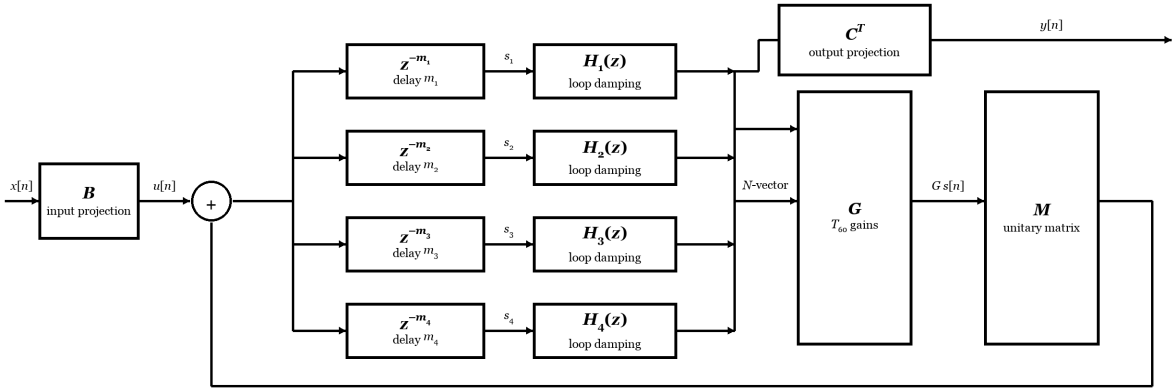


Figure 2-55: Expanded FDN flowgraph with input projection, unequal delay lines, per-line damping, RT60 gains, unitary feedback matrix, and output projection

How to read this flowgraph. Projection B distributes the scalar input into the vector sum. Each row delays one state component by m_i samples and filters it with $H_i(z)$. The gain block G calibrates loss, after which unitary matrix M redistributes energy before the vector returns to the input sum. Output projection C^{T} observes the delayed state without replacing the feedback path. That separation lets a designer change stereo or immersive presentation without changing the poles that govern the late decay.

The following pole-zero figure presents a reduced-order FDN modal projection. Unlike a single comb, an FDN does not generally decompose into one obvious ring per delay line; its coupled characteristic equation mixes delay, matrix, and loop-filter terms.

Figure 2-56 presents Illustrative unit-circle modal projection of an expanded FDN, with coupled poles and projection-dependent transmission zeros and establishes the visual reference for the discussion that follows.

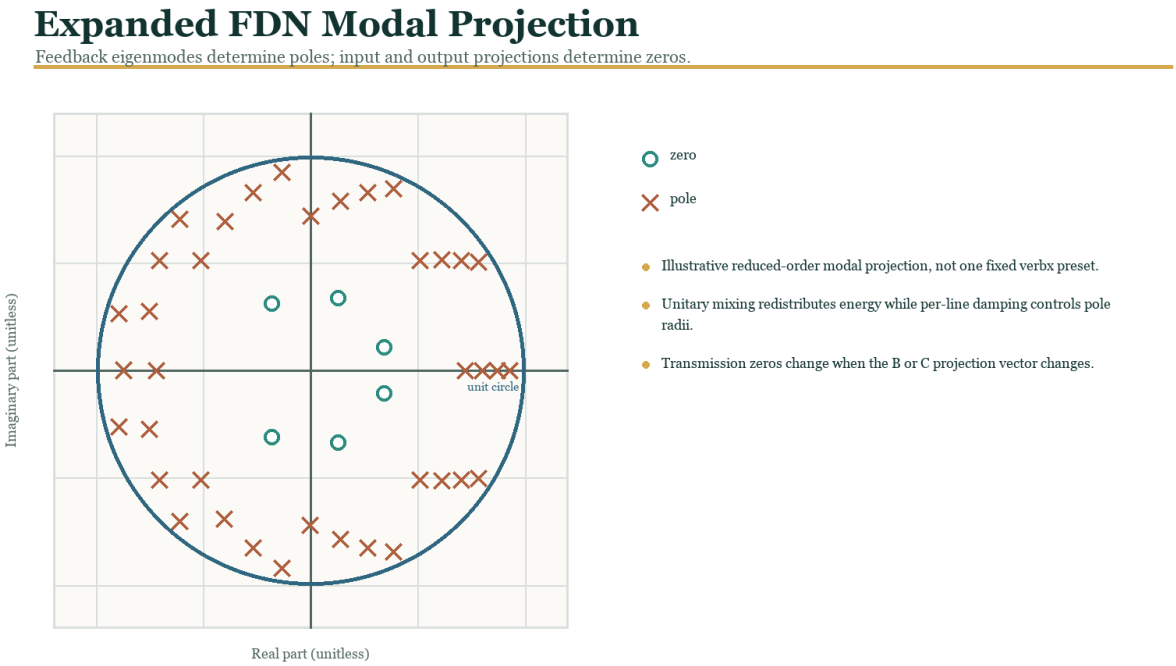


Figure 2-56: Illustrative unit-circle modal projection of an expanded FDN, with coupled poles and projection-dependent transmission zeros

The crosses represent eigenmodes of a stable, damped coupled state. Their unequal radii stand for frequency-dependent and line-dependent loss. The circles are transmission zeros for one illustrative input-output projection. Replacing B or C changes those zeros and the observed color, while leaving the internal state poles unchanged. Replacing M or a loop filter can move the poles themselves, so those changes require renewed stability analysis.

verbx exposes several matrix families because they create different exchange patterns. Hadamard mixing is dense and uniform. Householder mixing is efficient and structured. Circulant and elliptic families have interpretable eigenstructure. Random orthogonal matrices reduce obvious regularity. Graph-derived matrices make connectivity a design parameter. Time-varying unitary matrices alter the modal basis slowly while retaining controlled loop energy.

The companion magnitude response is the exact response of this schematic pole-zero model. Peaks indicate roots that the chosen projection emphasizes; valleys indicate cancellation or weak observation. The regular rings are a reduced modal illustration,

not the transfer function of a specified FDN matrix, delay set, or verbx preset. A real FDN response depends jointly on its delays, feedback matrix, loop filters, and input and output vectors.

Figure 2-57 presents Peak-normalized magnitude response for the illustrative expanded FDN projection, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

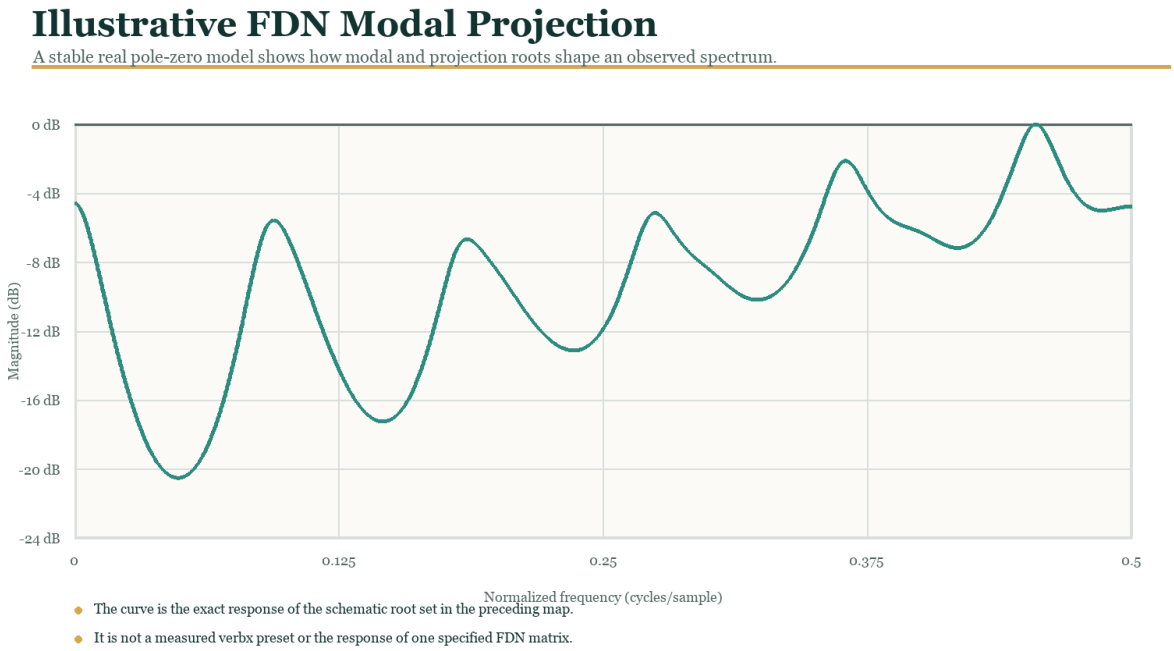


Figure 2-57: Peak-normalized magnitude response for the illustrative expanded FDN projection, with frequency in cycles per sample and magnitude on a vertical decibel scale

2.8.11 The Five Independent Design Coordinates of an FDN

It is tempting to identify an FDN by line count alone, but five coordinate systems act together: delay timing, feedback coupling, loop loss, input excitation, and output observation. Figure below places those choices around the recursive state. Changing any one can alter the result even when the other four are held fixed.

Figure 2-58 presents Five design coordinates that jointly determine an FDN late field and establishes the visual reference for the discussion that follows.

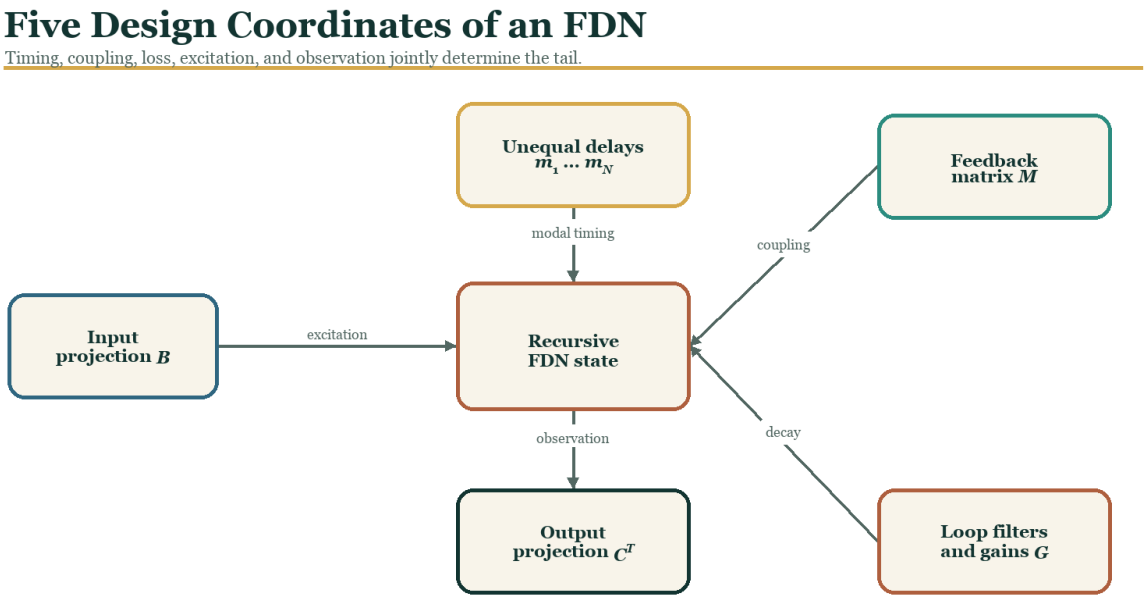


Figure 2-58: Five design coordinates that jointly determine an FDN late field

How to read this figure. The central state is not itself a sound-quality control. The delay set gives the state its temporal memory; the matrix redistributes that memory; the loop filters decide what survives; the input vector decides which modal combinations are excited; and the output vector decides which combinations are observed. This is why changing only C can alter stereo width without moving the network's poles, whereas changing M can alter both modal structure and energy exchange.

The five coordinates suggest a disciplined design order:

1. Choose delay durations for the desired scale and modal distribution.
2. Choose a feedback matrix with known energy behavior.
3. Calibrate broadband or multiband loop loss from the target decay.
4. Choose an input projection that excites the state without privileged lines.
5. Choose output projections that create the required channel image and fold-down.

Randomizing all five at once may produce an interesting preset, but it makes a failure difficult to diagnose. A controlled listening comparison changes one coordinate at a time and uses the same impulse, chord, level, and output projection.

2.8.12 Delay-Set Design and Number-Theoretic Structure

Each delay line contributes a recurrence period and a family of modes. Equal or simply related delays cause recirculating paths to coincide; the network then reveals a common period as flutter, pitched ringing, or cyclic stereo motion. Pairwise coprime lengths are a useful starting heuristic, but primality alone does not guarantee a good network. The complete coupled pole pattern also depends on the matrix, gains, and projections.

For a line of m_i samples, the uncoupled comb spacing is approximately

$$\Delta f_i = \frac{f_s}{m_i}.$$

Longer delays create closer modes and slower recurrence. Shorter delays create wider spacing and faster buildup. A practical set spans a modest range around a size target rather than clustering at one length. Excessive spread can make branches decay with perceptibly different granularity even when their RT60 gains are calibrated correctly.

Delay selection should satisfy several constraints at once:

Table 2-2: Delay-Set Design and Number-Theoretic Structure: reference table

Constraint	Purpose	Failure symptom
No zero or sub-sample line	Preserve causality and valid memory access	Instability or duplicated direct sound
Few shared factors	Reduce coincident recurrences	Flutter or rhythmic ringing
Bounded minimum delay	Limit high modal spacing	Audible isolated modes
Bounded maximum delay	Control buildup and memory	Late, disconnected tail onset
Sample-rate rescaling	Preserve physical durations	Preset changes size with sample rate

When a sample-rate conversion rounds several durations to nearby integers, recheck the set rather than assuming its number-theoretic relationships survived. Deterministic rounding and a stored seed make the resulting topology reproducible.

2.8.13 Feedback Matrices, Energy, and Eigenstructure

If the matrix is orthonormal,

$$M^{\top}M = I,$$

then it preserves Euclidean vector energy before loop loss. This separation is valuable: the matrix controls redistribution while G controls decay. A matrix need not be dense to be lossless, and a dense matrix is not automatically well conditioned.

The spectral radius

$$\rho(A) = \max_{k} |\lambda_k(A)|$$

provides one stability check for a frequency-independent state transition A. With delays and loop filters, the complete condition is frequency dependent, but the intuition remains: no recirculating eigenmode may receive net gain at any frequency. Numerical verification should therefore sweep the loop filters as well as checking the nominal matrix.

Table 2-3: Feedback Matrices, Energy, and Eigenstructure: reference table

Matrix family	Coupling character	Useful reason to choose it
Hadamard	Dense, balanced signed mixing	Predictable diffusion and efficient transforms
Householder	Structured global reflection	Low arithmetic cost with full-state interaction
Circulant	Translation-invariant row structure	Controlled eigenstructure and repeatable color
Random orthogonal	Irregular dense coupling	Reduced visible symmetry with energy preservation
Sparse graph-derived	Local or clustered exchange	Deliberate pathways and unusual spatial textures
Time-varying unitary	Slowly changing modal basis	Reduced stationary ringing in long exposed tails

Orthonormality is a starting condition, not a listening verdict. A lossless matrix can still align poorly with a delay set or projection vector. Conversely, a deliberately structured matrix may be musically useful precisely because it does not erase every pathway at once.

2.8.14 Input and Output Projections

The input projection B distributes source energy among delay lines. A vector with one nonzero entry excites the network from one point; a dense balanced vector begins with broader excitation. Multiple input channels use a matrix whose columns describe distinct injection patterns. Those columns should be normalized so adding channels does not silently increase loop energy.

The output projection C^{T} observes the state. It can be changed without changing the internal recurrence, which makes projection a powerful spatial design layer. Left and right vectors should share the same late history while weighting components differently enough to avoid mono duplication. Immersive outputs extend the same principle to side, rear, and height channels.

Projection quality is measured as well as heard. Inspect channel RMS balance, cross-correlation, interchannel coherence by frequency, and mono fold-down. A return can sound impressively wide in isolation yet collapse unevenly or lose low-frequency energy when summed. Normalized signed projections usually provide a safer starting point than independent random reverbs per channel.

2.8.15 Frequency-Dependent Decay in an FDN

Real materials absorb frequency bands differently, and air attenuates high frequencies more strongly over long paths. A convincing room therefore rarely has one RT60 at every frequency. In an FDN, loop filters can approximate a desired decay curve. The challenge is to shape loss without compromising stability or creating obvious filter resonances inside the feedback loop.

The multiband signal flow below shows low, middle, and high losses before energy returns to the feedback matrix.

Figure 2-59 presents Frequency-dependent FDN loop with independent low, middle, and high decay targets and establishes the visual reference for the discussion that follows.

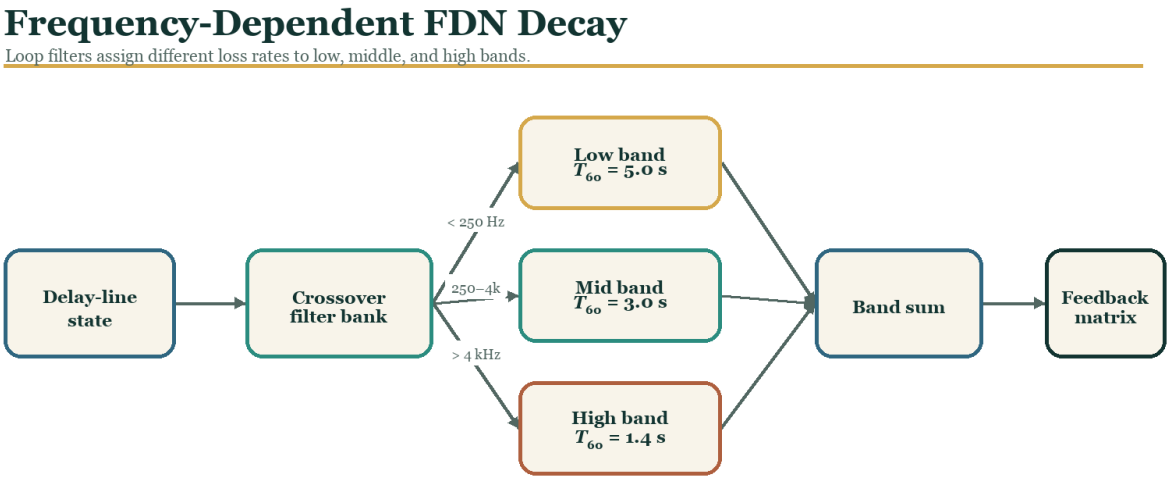
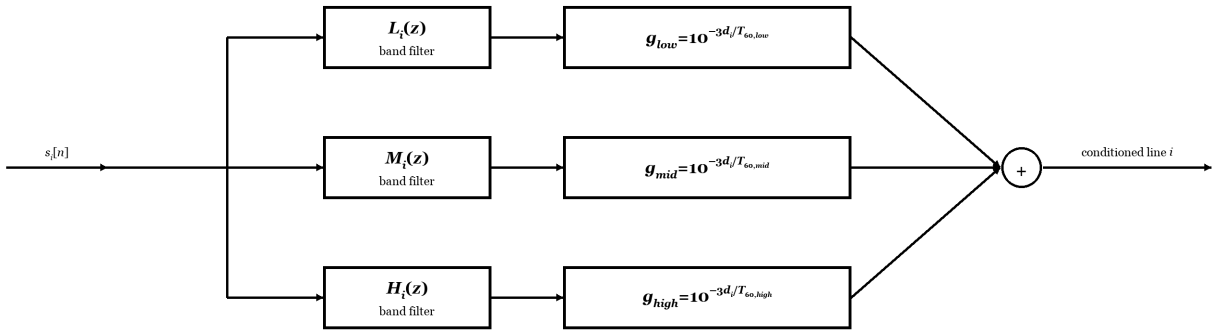


Figure 2-59: Frequency-dependent FDN loop with independent low, middle, and high decay targets

How to read this figure. Every band travels around the same spatial topology but loses a different amount of energy. A longer low-band RT60 adds warmth and weight; too much produces mud or a persistent pitched floor. A short high-band RT60 suggests soft materials or long air paths; too short makes the room dull and detached. Crossovers must be smooth because their errors recirculate.

The line-level graph below zooms into one FDN damping block. It makes the decay-gain calculation explicit for each band before the three paths are recombined.

Figure 2-60 presents One multiband FDN loop-filter row with low-, middle-, and high-band filters and independently calibrated decay gains and establishes the visual reference for the discussion that follows.



Frequency-dependent loop loss: each crossover band receives its own calibrated decay gain.

Figure 2-60: One multiband FDN loop-filter row with low-, middle-, and high-band filters and independently calibrated decay gains

How to read this flowgraph. State component $s_i[n]$ is split among three complementary filters. Each branch receives a gain of the form $10^{-3d_i/T_{60,b}}$, where d_i is the delay-line duration and $T_{60,b}$ is the target for band b. The sum reconstructs one conditioned state component before matrix mixing. Calibrating gain from each line's own duration is important: applying one coefficient to unequal delays gives those lines different decay times and distorts the intended spectral envelope.

The following modal plot shows how three decay targets become families of pole radii. The outer family represents the slowest band and the inner family the fastest.

Figure 2-61 presents Unit-circle pole-zero map for multiband feedback damping, with separate low-, middle-, and high-band pole-radius families and establishes the visual reference for the discussion that follows.

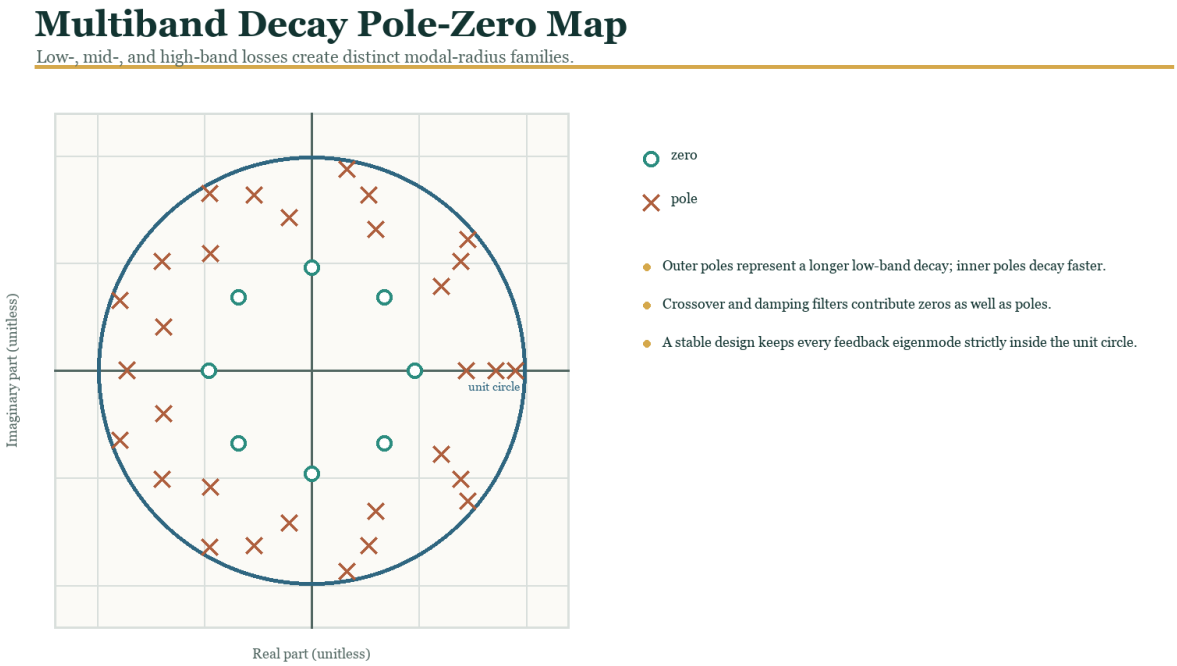


Figure 2-61: Unit-circle pole-zero map for multiband feedback damping, with separate low-, middle-, and high-band pole-radius families

Band filters contribute their own poles and zeros, and the surrounding delay network replicates their influence across many modes. Stability must therefore be checked on the complete loop, not inferred from three scalar gains alone. A smooth target decay curve should produce a smooth radial transition with frequency; abrupt radial clusters can become audible as bands that detach from one another during the tail.

The separate magnitude plot gives that radial family a spectral interpretation. It is a representative loop-filter view, not the transfer function of every possible FDN matrix and delay set that uses these three decay targets.

The plotted loop filter uses two complementary one-pole splits:

$$H_{\text{loop}}(z) = g_L L(z) + g_M (U(z) - L(z)) + g_H (1 - U(z)).$$

where $L(z)$ is the low crossover response, $U(z)$ is the upper crossover lowpass, and g_L, g_M, and g_H are the low-, middle-, and high-band loop gains. The figure uses normalized crossover frequencies of 0.06 and 0.22 cycles per sample and gains of 0.97, 0.90, and 0.78. Because the three paths are complementary, their unweighted sum is unity; because every listed gain is below one, this isolated loop filter does not add energy at DC or Nyquist. The complete FDN still requires a separate stability check.

Figure 2-62 presents Magnitude response of the representative multiband FDN loop filter, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

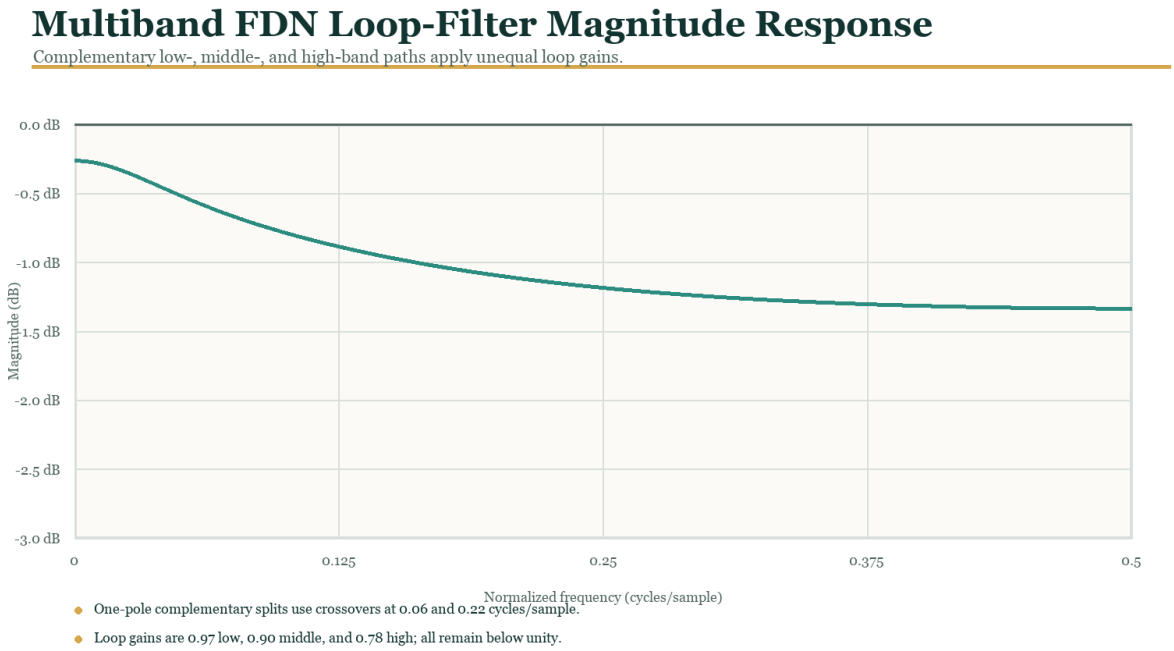


Figure 2-62: Magnitude response of the representative multiband FDN loop filter, with frequency in cycles per sample and magnitude on a vertical decibel scale

```
verbx render music.wav /tmp/multiband_hall.wav \
  --engine algo --rt60 3.0 --fdn-rt60-low 4.2 \
  --fdn-rt60-mid 3.0 --fdn-rt60-high 1.7 \
  --fdn-xover-low-hz 260 --fdn-xover-high-hz 4200 \
  --fdn-lines 24 --fdn-matrix random_orthogonal --wet 0.45 --dry 0.72
```

Listen after the source stops. During the phrase, masking may hide a low-frequency problem that becomes obvious only in the final decay.

2.8.16 Moving Delays, Fractional Reads, and Interpolation

A delay specified in milliseconds rarely lands on an integer sample, and a modulated delay changes continuously. If its desired length is $d[n]$ samples, the processor reads the circular buffer at a fractional position. Writing

$$d[n] = M[n] + \mu[n], \qquad 0 \le \mu[n] < 1,$$

separates the integer offset $M[n]$ from fractional part $\mu[n]$. Linear interpolation uses the two neighboring samples:

$$y[n] = (1 - \mu[n])x[n - M[n]] + \mu[n]x[n - M[n] - 1].$$

It is inexpensive and continuous, but its magnitude response changes with fractional position. Higher-order Lagrange interpolation improves high-frequency magnitude accuracy. A Thiran allpass interpolator prioritizes phase and delay accuracy while maintaining unit magnitude. The right choice depends

on whether the moving read is a creative chorus-like voice, a subtle anti-ringing modulation, or a precision physical delay.

Figure below separates the audio-rate memory path from the slower control path. They meet only at the interpolated read head.

Figure 2-63 presents Separate audio and control paths for a smoothly modulated fractional delay and establishes the visual reference for the discussion that follows.

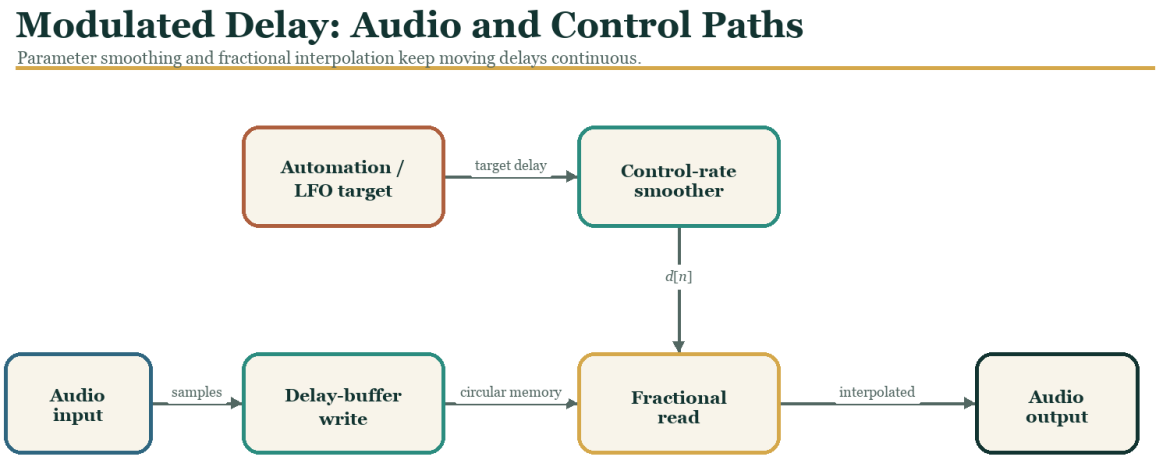


Figure 2-63: Separate audio and control paths for a smoothly modulated fractional delay

How to read this figure. Audio is written into circular memory at the sample rate. The target delay arrives from automation or an oscillator, passes through a smoother, and controls the read position. The smoother prevents a discontinuous pointer jump; the interpolator reconstructs a continuous value between stored samples. Neither block is optional when the delay moves during audible output.

A changing delay introduces pitch shift because the read head moves relative to the write head. Slow shallow motion can decorrelate modes without calling attention to itself. Fast or deep motion becomes chorus, vibrato, or Doppler-like sweep. In an FDN, different lines should not move in lockstep unless coherent pitch motion is intended. The total modulation must also respect minimum and maximum delay bounds so the read head never crosses invalid memory or overtakes the write head.

2.8.17 Parameter Smoothing and Host Automation

Hosts can deliver a new parameter target once per block, at sample offsets within a block, or at irregular GUI rates. Applying a discontinuous target directly to a gain, delay, filter coefficient, or matrix can create clicks or temporarily invalidate a stable design. A one-pole smoother is

$$p[n] = a\,p[n-1] + (1-a)p_{\text{target}}[n],$$

with

$$a = e^{-1/(\tau f_s)},$$

where τ is a smoothing time constant. Linear ramps are also useful when an exact arrival time matters. Gains may be smoothed in decibels or amplitude depending on the desired perceptual trajectory; frequencies are often smoother in logarithmic space; RT60 is best mapped through a bounded logarithmic parameter before loop gains are recalculated.

Recursive parameters need special care. Interpolating directly between two unrelated orthogonal matrices does not generally remain orthogonal. Safer strategies interpolate a constrained rotation, crossfade between complete networks, or update through a factorization whose intermediate states preserve the required energy property. Similar care applies to loop-filter coefficients: every intermediate filter must remain stable, not only the endpoints.

Smoothing has a musical cost. A 500 ms ramp is click-free but can miss a sixteenth-note gesture; a 1 ms ramp may preserve timing but reveal zippering on a 360-second tail. Document smoothing times as part of the DSP contract and test automation at different host block sizes.

2.8.18 Early Reflections and the Early-to-Late Transition

Early reflections carry geometry. Their delays and directions communicate source distance, nearby boundaries, and room shape before the late field becomes statistically dense. A tapped delay line can render a designed pattern; an image-source or ray-tracing model can derive paths from geometry; a measured IR can supply the complete onset.

Each early tap can be represented as

$$y_q[n] = a_q F_q(z) x[n - m_q],$$

where m_q is path delay, a_q is spreading and reflection loss, and $F_q(z)$ models frequency-dependent boundary absorption. Multichannel rendering adds a directional projection for each path. The tap list should not merely be randomized: arrival order, level decay, spectral darkening, and lateral distribution jointly establish a plausible enclosure.

The handoff to the late field is a crossfade in statistical description, not necessarily one literal splice sample. If the FDN begins too early and too densely, the source seems embedded in an abstract wash. If it begins too late, the early response sounds like a cluster of echoes followed by a separate effect. Match energy, spectrum, spatial width, and density across a transition region.

2.8.19 Convolution and Partitioned FFT Processing

Convolution reverb uses an impulse response measured in a real room or designed by a synthesis process. Direct time-domain convolution costs more as the IR grows. FFT convolution turns long convolution into blockwise multiplication in the frequency domain. Partitioning the IR lets a processor use small early blocks for latency and larger later blocks for efficiency.

The Mermaid diagram below shows uniformly or nonuniformly partitioned overlap-save at a conceptual level. Exact scheduling varies by implementation.

Figure 2-64 presents Partitioned FFT convolution path for a long impulse response and establishes the visual reference for the discussion that follows.

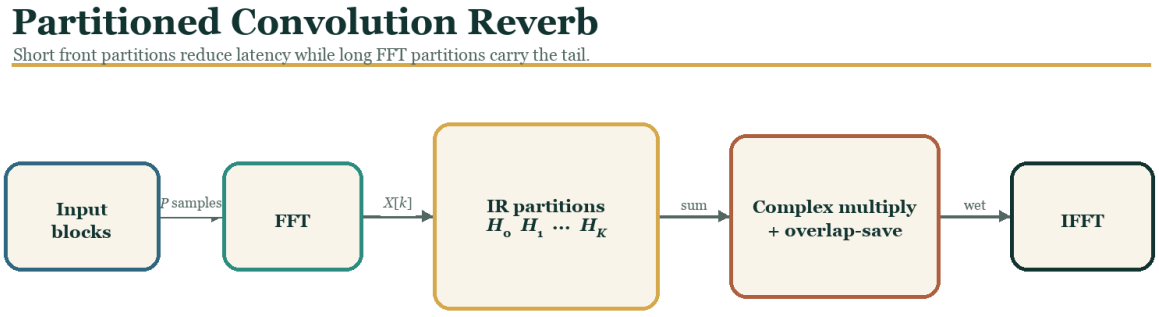


Figure 2-64: Partitioned FFT convolution path for a long impulse response

How to read this figure. The input is transformed once per block. Each IR partition multiplies a correspondingly delayed input spectrum; the products are accumulated and transformed back. The first partition determines a large part of algorithmic latency. Longer partitions improve efficiency but increase buffering. Convolution reproduces the captured linear response exactly within numerical and routing limits, but ordinary convolution cannot continuously change that response without interpolation or a new IR.

For an FIR of length L, direct convolution evaluates

$$y[n] = \sum_{k=0}^{L-1} h[k]x[n-k],$$

which costs L multiply-accumulates per output sample. A two-second mono IR at 192 kHz contains 384,000 taps; direct evaluation is therefore wasteful even before multichannel routing. FFT convolution groups samples into blocks and uses the convolution theorem,

$$Y_r[k] = X_r[k]H[k],$$

where r identifies a processing block and k an FFT bin. The apparent simplicity of that multiplication hides the scheduler that aligns past input spectra with every IR partition.

In overlap-save processing, the FFT length must include enough historical samples to avoid circular-convolution contamination. The invalid prefix is discarded and only the new valid samples are emitted. The implementation must define whether reported latency includes host buffering, input accumulation, FFT scheduling, and output staging; “zero latency convolution” usually means that an initial direct or very small partition is processed before larger tail partitions, not that no buffering exists anywhere.

Uniform and nonuniform partitioning trade simplicity against efficiency:

Table 2-4: Convolution and Partitioned FFT Processing: reference table

Partition plan	Latency behavior	Computational behavior
Uniform small blocks	Low and constant	Many FFTs for the complete tail
Uniform large blocks	Higher and constant	Better throughput for long IRs
Direct head plus FFT tail	Very low onset latency	More scheduler complexity
Geometrically growing blocks	Small early, efficient late	Multiple FFT sizes and deadlines

A practical nonuniform design might process the first few milliseconds directly, the next region in 64- or 128-sample partitions, and the distant tail in progressively larger blocks. Late partitions have more wall-clock time before their contribution is needed, so they can be computed less frequently without delaying the direct onset.

2.8.20 Matrix Convolution and Immersive Routing

A multichannel IR is a matrix of filters. For M inputs and N outputs,

$$Y_j(z) = \sum_{i=1}^{M} H_{ji}(z)X_i(z), \qquad j = 1,\dots,N.$$

The off-diagonal filters are not optional decoration: they encode cross-channel energy that helps a measured room feel coherent. A stereo-to-Atmos bed renderer may require many simultaneous convolution paths, so partition spectra should be shared and batched rather than invoking an unrelated mono convolver for every route.

Channel metadata is part of the DSP. A mathematically correct matrix with incorrect ordering can send left energy to a height channel or swap ambisonic components. Tests should use labeled impulses, one active input at a time, and verify arrival time, level, and polarity at every output. Fold-down and binaural decode tests reveal errors that may be hard to identify while monitoring the full array.

2.8.21 Impulse-Response Conditioning

An IR often needs preparation before it becomes a production filter. Remove unintended leading silence while preserving true propagation delay. Remove DC, but do not erase a legitimate very-low-frequency room mode. Apply fades only where measurement noise or a truncation edge would otherwise circulate. Normalize according to the intended contract: peak normalization preserves headroom, energy normalization supports comparable wet levels, and neither reproduces an absolute acoustic calibration by itself.

Sample-rate conversion must use sufficient stop-band attenuation because any resampling artifact becomes part of every processed signal. Channel lengths should be aligned, and metadata should record sample rate, channel order, direct-arrival position, normalization policy, and provenance. An IR library without these facts may sound useful but is not a reproducible measurement collection.

Choose convolution when the identity of a measured space matters. Choose algorithmic FDN processing when RT60, matrix motion, modulation, or extreme duration must change during the sound. Hybrid workflows often use a short measured early response followed by an algorithmic late field.

2.8.22 Measuring and Capturing an Acoustic Impulse Response

An acoustic impulse response is not simply a recording made in a reverberant room. It is an estimate of the transfer path from a specified source position to a specified receiver position under documented conditions. Move the loudspeaker, microphone, furniture, doors, audience, or temperature and the measured system changes. A useful capture therefore records geometry and calibration alongside audio: room and session identifier, date, source and microphone coordinates, orientation, height, transducers, interface, gain, sample rate, atmosphere, occupancy, and every processing step.

Several excitation methods are practical:

Table 2-5: Measuring and Capturing an Acoustic Impulse Response: reference table

Excitation	Strength	Principal limitation
Acoustic impulse, such as a balloon or starter pistol	Fast, intuitive, portable	Limited repeatability, spectrum, and signal-to-noise ratio
Loudspeaker-driven electrical pulse	Known timing and polarity	High peak demand and possible loudspeaker nonlinearity
Maximum-length sequence or other pseudorandom noise	Efficient averaging for an LTI system	Nonlinear distortion folds into the estimate
Linear sine sweep	Controlled spectrum and level	Harmonic distortion is not separated as cleanly in time
Exponential sine sweep	High energy across the band and nonlinear separation	Requires careful inverse filtering and clock discipline
Program material or ambient excitation	Measures the occupied operating condition	Usually underdetermined and difficult to reproduce

A balloon pop can be excellent for scouting positions, but a calibrated exponential sine sweep is usually the stronger archival measurement because it distributes energy over seconds instead of demanding one extreme peak. Repeat each source-receiver pair at least twice. If the recovered responses disagree, diagnose motion, clipping, clock drift, noise, or time variance before averaging them. Measure the background noise separately and leave headroom in the playback, microphone, preamplifier, and converter stages.

2.8.23 Practical Impulse Sources: Pops, Clappers, and Purpose-Built Devices

An impulsive source is useful because its energy arrives in a short interval, making the first arrival, early reflection pattern, and decay visible directly in the recording. It is also easy to misuse. No handheld source is perfectly omnidirectional, spectrally flat, or repeatable. Treat a pop or strike as a field observation unless the source has been characterized, repeated, and documented. Record several takes without moving the source or microphone, retain every take, and compare them before selecting a representative response.

Table 2-6: Practical Impulse Sources: Pops, Clappers, and Purpose-Built Devices: reference table

Source or device	What it is good for	What can mislead you
Handclap or hand snap	Fast location scouting and audience-perspective listening	Strong performer-to-performer variation; limited bass energy; directional radiation
Balloon pop	Cheap broadband transient for large spaces	Burst position, balloon material, and pop direction vary; latex debris and venue restrictions may apply
Two-by-four clapper or paired hardwood boards	Repeatable mechanical strike with useful midrange energy	It is not an ideal impulse; wood resonances and the operator's geometry color the result
Slate clapper, castanets, or a purpose-built impulse clapper	Compact controlled transient with a recognizable onset	Limited low-frequency output and a characteristic source spectrum
Pistol-style starter device	Very high peak level and a clear onset in a suitable, permitted outdoor or industrial setting	Legal, facility, hearing-safety, fire-safety, and microphone-overload risks; never use it casually or where a firearm-like report is inappropriate
Small loudspeaker playing a needle pulse	Known digital timing and repeatable routing	Loudspeaker bandwidth, excursion, and nonlinear distortion limit the usable impulse
Omnidirectional loudspeaker playing an exponential sweep	Best general-purpose choice for archival room IR work	Requires playback hardware, a matched inverse filter, and time for setup and deconvolution

The two-by-four clapper is worth distinguishing from a random board strike. Two straight, dry hardwood boards with handles can make a practical reusable clapper: bring the broad faces together sharply, keep the operator out of the direct path from source to microphone, and repeat the same stance and height. It provides a more consistent onset than a handclap and avoids the disposal and surprise of balloons. It still produces a device-specific signature, so do not divide a recorded room response by an imaginary flat spectrum. Use it to compare locations, identify obvious flutter or long echoes, and make musically useful rough captures, not to claim laboratory calibration.

A balloon is often the quickest first visit tool. Use one size and material for a session, inflate each balloon similarly, position it at the intended source coordinate, and pop it remotely or with the operator well away from the microphone. A close microphone can clip even when the distant tail looks quiet, so leave generous preamp headroom. Do not use balloons where fragments, noise, accessibility concerns, animals, or fire regulations make them inappropriate. A handclap, wooden clapper, or low-level sweep is usually a better choice for a public venue.

Starter-pistol-style sources appear in older room-acoustics practice because they can provide a high-level, abrupt event outdoors or in large empty structures. Their practical drawbacks are serious: local law and venue rules may treat them as weapons; the report can injure hearing, alarm people, trigger safety systems, overload microphones and converters, and invalidate an otherwise quiet session. Use such a source only under explicit facility authorization, with trained operators and an approved safety plan. In most music-production and research captures, a calibrated loudspeaker sweep is safer, more repeatable, and more informative.

Other useful inventions include a calibrated dodecahedral loudspeaker, an omni source with known response, a small battery-powered clapper for difficult access, and a synchronized electrical loopback box. The loopback does not excite the room; it records a copy of the playback signal beside the microphone capture so later processing can identify latency and clock behavior. For a matrix measurement, label every physical source, microphone, and take before recording. A beautiful IR with uncertain coordinates is not a reusable spatial measurement.

2.8.24 Capturing an IR in Apple Logic Pro

Logic Pro is a capable playback-and-recording environment for an impulse-response session, but it is not a complete measurement application. Its Test Oscillator can generate a user-defined sine sweep, and ordinary audio tracks can record microphone inputs. The missing step is matched deconvolution: export the sweep and capture, then use a validated measurement tool or scientific workflow to apply the inverse sweep and recover the IR. Do not mistake a recorded sweep WAV for an impulse response.

1. **Prepare the project.** Set a documented sample rate and 24-bit recording depth, choose WAV or AIFF, and disable processing on the measurement path. Logic supports common PCM formats and sample rates through 192 kHz; choose a rate that matches the interface, microphone system, and intended IR library rather than changing it mid-session.
2. **Create separate source and capture tracks.** Put the Test Oscillator on a dedicated source track or auxiliary routed only to the measurement loudspeaker. Create a mono audio track for the measurement microphone, select the correct interface input, and record-enable it. If the interface has a spare input, record a loopback from the playback chain on a second mono track.
3. **Set safe routing and levels.** Mute every effect, limiter, automatic gain stage, and spatializer that is not part of the system being measured. Use direct hardware monitoring where appropriate; do not let software-monitoring latency confuse the measurement path. Set the microphone preamp so the direct sound has headroom, then confirm the loudest part of the sweep or impulsive source never clips.
4. **Generate and record.** In the Test Oscillator, select Sine Sweep mode, set the desired start and end frequencies and a duration long enough for the venue and noise floor, then trigger it while recording the microphone and loopback tracks. Start recording before the sweep, allow the full decay to finish, and leave several seconds of post-tail noise for a defensible noise-floor decision. For balloon or clapper work, record the same pre-roll and post-roll, then make at least three takes at each geometry.
5. **Export without alteration.** Keep the raw regions, note source and microphone coordinates in Logic's project or track notes, and bounce or export the raw microphone and loopback files with normalization off. Logic's audio-recording guide covers input assignment and record enable; its bounce documentation explains how output format and tail length affect exported files.
6. **Deconvolve externally, then return to verbx.** Use the exact inverse for the sweep that was played, inspect the recovered direct arrival and harmonic-distortion regions, and save the raw sweep, raw capture, inverse filter, recovered IR, and geometry notes together.

Then run `verbx ir analyze`, `verbx ir process`, and a convolution audition as shown below.

For Logic sessions using a physical impulse rather than a sweep, the microphone track is already a rough IR candidate. Trim only after preserving an untouched original; identify the direct arrival, retain meaningful propagation delay if the capture will represent the real source distance, and apply a gentle terminal fade only after the tail reaches the noise floor. Compare takes before choosing one. A single dramatic pop can be musically compelling but is weak evidence of a room's repeatable transfer function.

The measurement chain is

$$s(t) \longrightarrow \text{D/A} \longrightarrow \text{loudspeaker and room} \longrightarrow \text{microphone} \longrightarrow \text{A/D} \longrightarrow r(t).$$

For an approximately linear, time-invariant room and measurement chain,

$$r(t) = s(t) * h(t) + v(t),$$

where $s(t)$ is the known excitation, $h(t)$ is the desired impulse response, and $v(t)$ contains background and electronic noise. A loopback channel recorded beside the microphone channel reveals playback latency and clock behavior. Calibrated absolute sound pressure is valuable for research, but a production IR may instead retain relative level and document its later normalization.

2.8.25 Deconvolving a Sine Sweep

Deconvolution removes the known excitation from the recorded response. In the frequency domain, the noise-free ideal would be $H(f) = R(f)/S(f)$. Direct division is fragile wherever $S(f)$ is small, so a regularized estimate is safer:

$$\widehat{H}(f) = \frac{R(f)S^*(f)}{|S(f)|^2 + \lambda(f)}.$$

Here $S^*(f)$ is the complex conjugate and $\lambda(f)$ limits noise amplification. The regularizer can follow a measured noise spectrum rather than remaining constant. Transform $\widehat{H}(f)$ back to time, identify the direct arrival, and inspect the pre-arrival region: substantial energy before the causal response often indicates synchronization, windowing, or inverse-filter error.

An exponential sweep from f_1 to f_2 over duration T can be written

$$s(t) = \sin\left[2\pi f_1 \frac{T}{\ln(f_2/f_1)}\left(e^{t\ln(f_2/f_1)/T} - 1\right)\right], \qquad 0 \le t \le T.$$

Apply short fades at both ends to avoid discontinuities, play the sweep through the source, and record enough silence afterward to contain the complete room tail. The inverse filter is a time-reversed sweep with the amplitude correction required by the logarithmic time-to- frequency mapping. Convolving the recording with that inverse compresses the swept energy into the linear room response. Frequency-domain regularization and the time-domain inverse filter are two views of the same system-identification operation.

2.8.26 The Farina Exponential-Sweep Method

Angelo Farina's exponential-sweep method is especially valuable because a weakly nonlinear playback and capture chain does not place every distortion product on top of the linear IR. After deconvolution, the kth harmonic response is displaced from the main linear response by

$$\Delta t_k = T\frac{\ln k}{\ln(f_2/f_1)}.$$

With the usual inverse-filter convention, higher-order harmonic responses appear earlier than the principal linear response. They can be windowed and inspected separately instead of contaminating the room estimate as they do in many pseudorandom measurements. This does not make the room perfectly time invariant or the transducers perfectly separable; it makes nonlinear contamination visible and measurable. Use a sweep long enough for adequate low-frequency energy, but not so long that audience movement, ventilation cycles, wind, or temperature drift violates the stationary-system assumption.

A disciplined Farina session follows this sequence:

1. Calibrate playback level and verify that the complete chain remains below clipping.
2. Record the microphone and electrical loopback channels at one unchanged sample clock.

3. Play an exponentially swept sine with onset and endpoint fades.
4. Continue recording through the required decay and noise-floor interval.
5. Deconvolve with the matched inverse sweep or a regularized frequency-domain equivalent.
6. Separate the earlier harmonic responses from the later linear response before trimming.
7. Repeat the measurement and compare direct arrival, polarity, spectrum, decay, and noise.
8. Preserve the raw sweep, raw capture, inverse filter, recovered IR, and metadata together.

Farina's AES paper provides the canonical derivation and nonlinear-separation framework; the complete citation appears as reference CV6 in Appendix C.

2.8.27 Preparing the Recovered IR for verbx

verbx does not presently control a measurement loudspeaker or record and deconvolve a sweep inside the CLI. Capture and deconvolution should be performed in a measurement application, DAW, or validated scientific script. The recovered WAV can then enter the reproducible

verbx inspection and production path:

```
verbx ir analyze measured_room_raw.wav --json-out measured_room_raw.analysis.json

verbx ir process measured_room_raw.wav measured_room_ready.wav \
  --lowcut 20 --highcut 22000 --normalize peak --peak-dbfs -1

verbx render source.wav source_in_measured_room.wav \
  --engine conv --ir measured_room_ready.wav \
  --json-out source_in_measured_room.analysis.json
```

Do not normalize away information before deciding what the library promises. An absolute, calibrated IR and a peak-normalized production IR serve different purposes. Preserve the raw direct-arrival delay if source distance must remain authentic; remove or shorten it if the IR will be placed behind an independent pre-delay control. Retain enough tail to reach the measured noise floor, then use a gentle terminal fade rather than a hard truncation. For arrays and immersive captures, measure and label every source-receiver route, preserve one common time origin, and test the matrix with one active input channel at a time.

2.8.28 Hybrid Reverberation: Geometry First, Statistics Later

A hybrid processor uses the representation best suited to each time region. A measured, image-source, or ray-traced response supplies direct and early paths. A recursive network supplies the late field at a fraction of the storage cost and remains continuously controllable. Figure below shows both branches rejoining before one spatial projection.

Figure 2-65 presents Hybrid reverberator combining geometrical early energy with an algorithmic late field and establishes the visual reference for the discussion that follows.

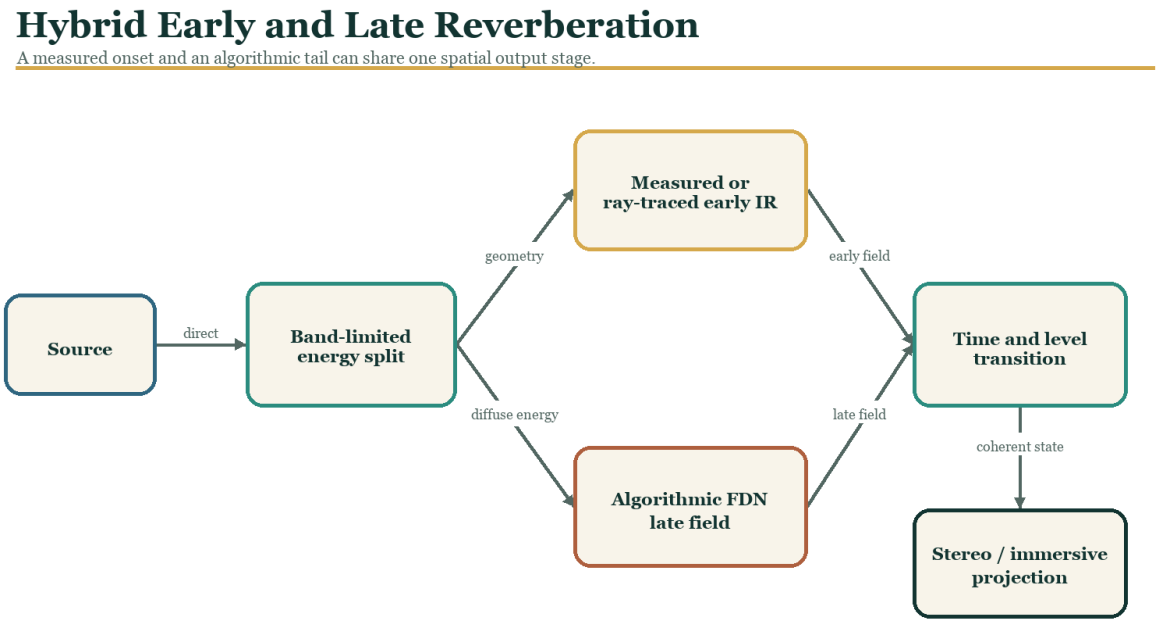


Figure 2-65: Hybrid reverberator combining geometrical early energy with an algorithmic late field

How to read this figure. The upper branch preserves identifiable paths and directional cues. The lower branch turns diffuse energy into a controllable recursive tail. The join is responsible for matching arrival density, spectrum, energy, and spatial character; the final decoder presents one coherent room rather than two stacked effects.

Hybrid calibration begins by selecting a transition interval, then matching the late network to the measured energy-decay slope and spectrum around that interval. Excite the FDN with a decorrelated version of the final early energy rather than an unrelated dry copy when continuity matters. Preserve deterministic seeds so a rebuilt hybrid IR does not change every time documentation, tests, or presets are regenerated.

2.8.29 Measuring Decay with Backward Integration

RT60 should be measured from the response, not assumed from a parameter label. Schroeder backward integration estimates the energy remaining after time index n:

$$E[n] = \sum_{k=n}^{L-1} h^2[k].$$

The normalized energy-decay curve is

$$L_E[n] = 10 \log_{10}\left(\frac{E[n]}{E[0]}\right).$$

A straight line is fitted over a valid decay interval and extrapolated to –60 dB. EDT typically fits 0 to –10 dB; T_{20} uses –5 to –25 dB; T_{30} uses –5 to –35 dB. Those labels describe fitting windows, not the amount of audio that must literally reach –60 dB. Agreement among estimates suggests a reasonably exponential decay. Large disagreement may reveal multiple slopes, a noisy tail, gating, or a source that was not impulsive.

Figure below traces the complete analysis path from response samples to reported metrics. Every transformation should preserve enough metadata to explain a failed fit.

Figure 2-66 presents Measurement pipeline from an impulse response to fitted reverberation metrics and establishes the visual reference for the discussion that follows.

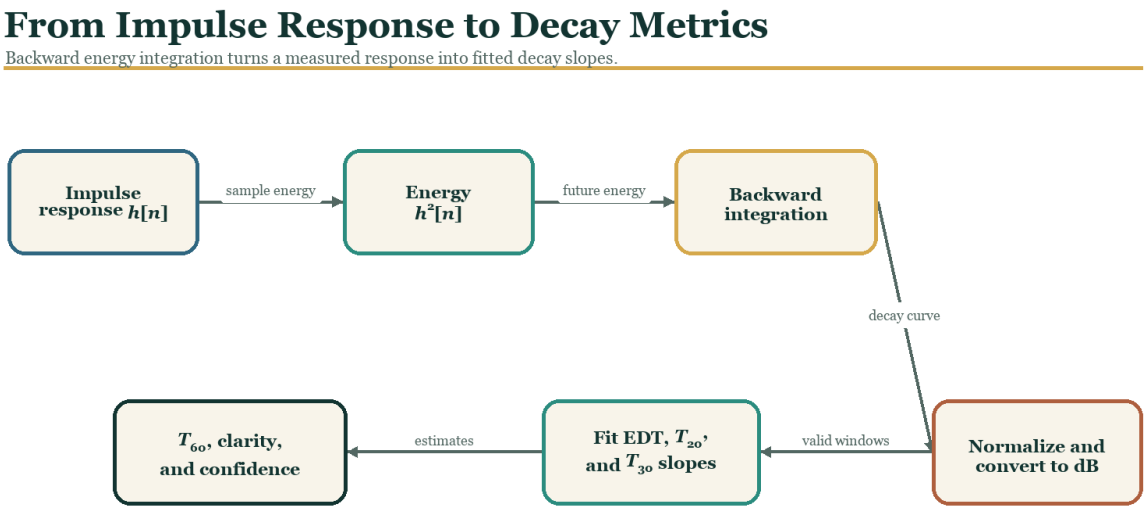


Figure 2-66: Measurement pipeline from an impulse response to fitted reverberation metrics

How to read this figure. Squaring removes polarity and produces instantaneous energy. Backward integration accumulates all future energy at each time. Decibel normalization turns exponential decay into an approximately straight line. The fitter selects only valid dynamic ranges and reports both estimates and confidence instead of forcing one number from an inadequate tail.

Noise-floor handling is essential. Once integrated room energy approaches integrated background noise, the curve bends and a naive regression overestimates decay. A robust analyzer estimates the noise floor, limits the fitting range, reports the achieved dynamic range, and refuses a T_{30} estimate when 30 reliable decibels are unavailable. Band-limited measurements should also report filter center frequencies and bandwidths.

Decay time is only one descriptor. Clarity compares early and late energy:

$$C_t = 10 \log_{10} \left(\frac{\sum_{n=0}^{n_t} h^2[n]}{\sum_{n=n_t+1}^{L-1} h^2[n]} \right),$$

where t is commonly 50 ms for speech or 80 ms for music. Definition D_{50} expresses the first 50 ms as a fraction of total energy, and center time T_s measures the energy- weighted temporal centroid. These metrics can disagree productively: a room may have a long RT60 yet retain useful clarity because its direct and early energy is strong.

2.8.30 Numerical Precision, Denormals, and State Safety

Recursive reverberators magnify small implementation choices because state circulates. Double precision reduces accumulated coefficient and summation error, especially when loop gains approach one, but it does not excuse unstable filters or an energy-increasing matrix. Every processing format still needs explicit bounds and failure behavior.

Subnormal floating-point values can appear near the end of a long decay. Some processors handle them slowly; others flush them to zero. A reverb can avoid pathological tails by using supported flush-to-zero

modes, adding an inaudible terminating rule below a defined threshold, or proving that the target platform handles denormals efficiently. The policy must not create a visible gate at normal listening levels.

State safety includes:

- finite-value checks at control boundaries rather than expensive checks on every sample;
- bounded delay indices and validated circular-buffer lengths;
- stable intermediate coefficients throughout automation ramps;
- deterministic reset behavior for transport starts, sample-rate changes, and preset loads;
- explicit handling of NaN and infinity before they enter recursive memory;
- limiter and loudness stages outside the FDN loop unless nonlinear feedback is intentional.

A limiter inside the recursive loop changes the system into a nonlinear reverberator. That may be a creative instrument, but its RT60 formula and superposition assumptions no longer apply. A safety limiter after wet/dry mixing controls delivery level without rewriting the late-field poles.

2.8.31 Realtime Scheduling and End-to-End Latency

The audio callback has a deadline: one host block must finish before the device needs the next block. If the block contains B samples at sample rate f_s, its wall-clock budget is

$$T_{\text{block}} = \frac{B}{f_s}.$$

At 192 kHz, a 64-sample block allows only 0.333 ms. CPU averages are insufficient; the worst callback matters. File I/O, memory allocation, locks, console output, JSON writing, and device discovery do not belong in that deadline.

End-to-end monitoring latency includes input conversion and safety buffers, host input buffering, plug-in delay, host output buffering, and output conversion. An algorithmic FDN can add zero samples of lookahead while the complete system still has several blocks of latency. Partitioned convolution may add one partition or use a direct head to reduce its reported delay. A limiter with lookahead adds its own explicit samples. Report these components separately so users can distinguish DSP latency from device configuration.

Block-size invariance is a release requirement. Rendering the same deterministic input with 32-, 64-, 256-, and 1,024-sample blocks should not change steady parameters, decay calibration, or automation timing beyond documented interpolation rules. Differences often reveal state reset errors, block-rate smoothing, or FFT partition misalignment.

2.8.32 Verification: Close the Loop Between Math and Listening

No single test establishes reverb quality. An impulse exposes topology, a burst exposes buildup, a sine sweep exposes linear response, a sustained chord exposes modes, speech exposes masking, percussion exposes density, and a long silence exposes numerical decay. Figure below organizes these probes into a repeatable engineering loop.

Figure 2-67 presents Closed DSP validation loop connecting deterministic probes, measurements, and listening and establishes the visual reference for the discussion that follows.

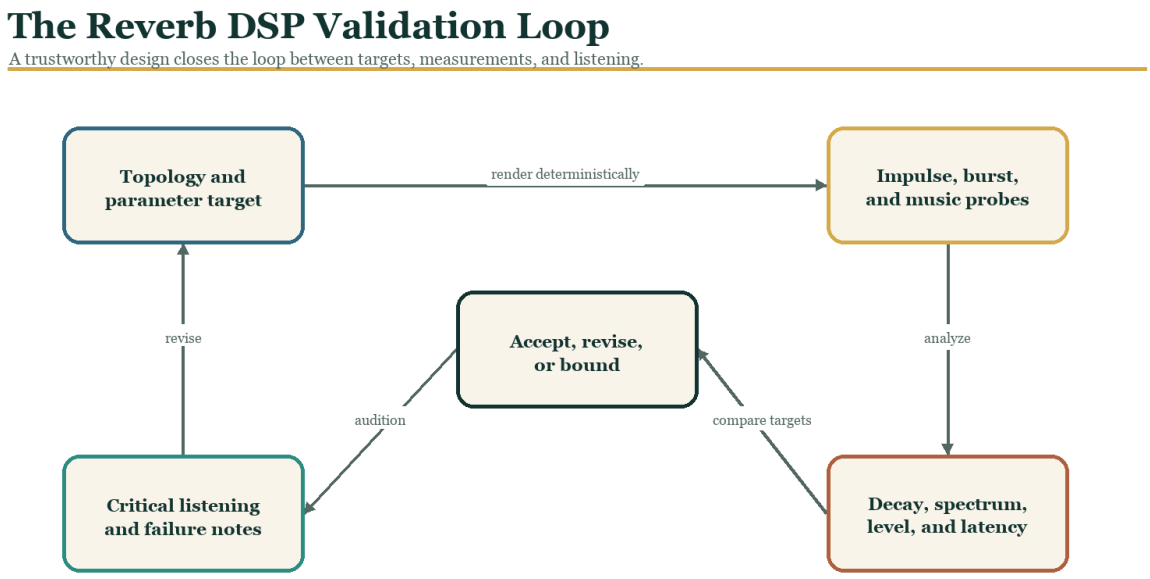


Figure 2-67: Closed DSP validation loop connecting deterministic probes, measurements, and listening

How to read this figure. A design target creates deterministic renders. Analysis compares those renders with numerical bounds. Listening identifies perceptual failures that one metric cannot summarize. The decision either accepts the design, constrains its valid range, or feeds a specific failure back into topology and parameter choices.

A minimal verification matrix includes:

Table 2-7: Verification: Close the Loop Between Math and Listening: reference table

Probe	Measure	Listen for
Unit impulse	Peak, first arrival, RT60, decay linearity	Flutter, isolated echoes, abrupt ending
Log sweep	Magnitude and phase response	Narrow resonances and spectral tilt
Sustained sine or chord	Mode balance and stationarity	Ringing, beating, unintended chorus
Percussion	Echo-density growth and peak headroom	Attack loss, groove smear, pumping
Speech	C50, intelligibility, wet envelope	Consonant masking and sibilant tails
Silence after excitation	DC, denormals, final energy	Noise growth, gate, failure to terminate
Multichannel impulses	Route matrix and correlation	Swaps, polarity errors, weak fold-down

The JSON report is evidence, not decoration. Store sample rate, block size, seed, topology, matrix family, delay set, smoothing policy, output format, measured latency, decay estimates, and warnings. A listening observation becomes actionable when another developer can recreate the exact state that produced it.

2.8.33 The Complete verbx Algorithmic Path

verbx separates onset design, diffusion, late-field recursion, creative transformation, and output safety. That separation is one reason the processor remains controllable at extreme settings. Pre-delay does not need to be hidden inside FDN delays; shimmer does not need to destabilize the core matrix; loudness and limiting can remain outside the reverberation state.

The complete Mermaid overview below shows the major stages. Optional blocks can be bypassed, but their order is intentional.

Figure 2-68 presents Complete verbx algorithmic path from source onset to analyzed output and establishes the visual reference for the discussion that follows.

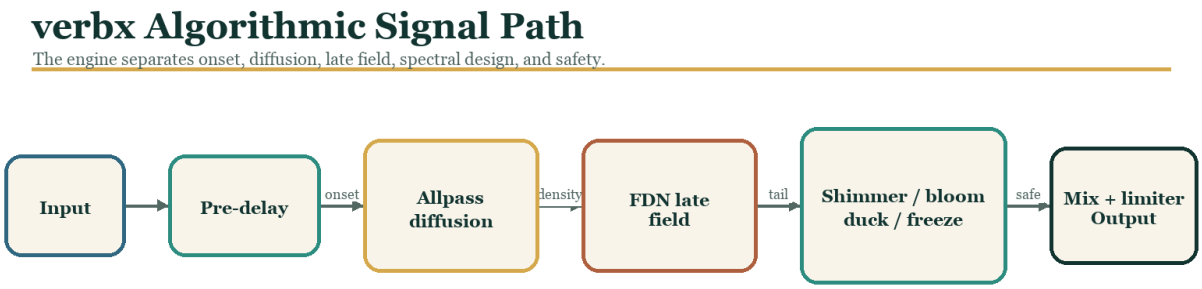


Figure 2-68: Complete verbx algorithmic path from source onset to analyzed output

How to read this figure. The first two stages control when and how the room begins. The FDN controls long-term energy exchange. Creative blocks reinterpret the wet field. Post-processing controls presentation and delivery without changing the internal decay calibration. The analysis sidecar records what happened. Because the stages are separate, a user can design a natural two-second room and a 120-second harmonic cloud with the same conceptual vocabulary.

2.9 A History of Artificial Reverberation: Architectural, Mechanical, Electrical, Electromechanical, and Digital

Artificial reverberation is often narrated as a clean progression from echo chamber to plate, spring, digital rack unit, convolution plug-in, and neural model. The actual history is more complicated and more useful. Old technologies did not disappear when new ones arrived. They became colors, workflows, metaphors, and algorithm names. A twenty-first-century producer may place a sampled room before a modeled plate, route that result into a feedback delay network, and automate it like a dub engineer performing a spring and tape echo. The signal chain contains several eras at once.

This history therefore follows ideas as much as products. Every reverberator must answer four questions. **Where is energy stored?** A room stores it in propagating air waves and boundary interactions; a plate stores it in bending waves; a spring stores it in dispersive torsional and transverse motion; a delay network stores it in memory. **How is the stored energy mixed?** Architecture, transducers, matrices, scattering junctions, and modulation all redistribute it. **How is energy lost?** Air, surfaces, mechanical damping, filters, quantization, and feedback gains determine decay. **How does a user control the result?** Moving microphones, changing damping pads, selecting programs, editing parameters, loading an impulse response, or conditioning a model are historically different interfaces to the same design problem.

The term *artificial* also requires care. A purpose-built chamber produces physically real acoustic reverberation, but it is artificial with respect to the recorded performance because the dry or partly dry signal is sent into the chamber after capture. A plate and spring are physical resonators, but they do not reproduce a room literally. An algorithm can be designed from room statistics without tracing one building. Convolution can reproduce a measured linear response while remaining unable to reproduce source directivity, listener motion, nonlinear vibration, changing occupancy, or the complete three-dimensional field. The relevant distinction is not real versus fake. It is what system is being excited, what response is retained, and what artistic control is gained.

The chronology below is organized into overlapping regimes:

- **architectural and acoustical practice**, in which performers, microphones, loudspeakers, and rooms create the response;
- **mechanical and electromechanical storage**, in which pipes, plates, springs, drums, and other structures replace a full room;
- **electrical and electronic networks**, in which filters, delays, feedback, switching, and modulation synthesize decay;
- **digital computation**, in which memory and arithmetic implement algorithms, measured responses, physical models, and learned systems;
- **software and spatial systems**, in which many reverberators coexist inside a session and can be automated, distributed, rendered, or trained.

These regimes overlap because engineers keep useful limitations. A plate's bright onset, a spring's chirp, an early digital unit's sparse modulation, and a chamber's irregular decay remain desirable precisely because they are not interchangeable. The history of reverberation is the history of turning constraints into recognizable musical behavior.

2.9.1 Architecture Before the Effect Send

Before recording, reverberation could not be separated from the event that produced it. A singer, drum, organ, or ensemble excited a space; performers and listeners heard the direct sound and reflections as one event. Musical practice adapted to that inseparability. Long decays supported sustained chant and large harmonic masses but punished rapid articulation. Antiphonal writing turned distance and delayed response into counterpoint. Organs were voiced for buildings, and orchestral balances developed in relation to halls, stages, shells, and occupied seats.

Architecture supplied all of the operations later reverberators would imitate: delay through propagation, level loss through spreading and absorption, spectral change at boundaries, diffusion through ornament and irregular geometry, and recursive mixing through repeated reflection. Yet architecture did not offer a single global "reverb amount." A listener's experience depended on source position, directivity, receiver position, orientation, occupancy, temperature, and frequency. Moving a choir changed the response. Opening a curtain changed the response. Filling a hall changed the response. The "preset" was a physical arrangement.

The following figure presents St. Mark's Basilica in Venice as an instructive historical model. Its coupled volumes, hard surfaces, galleries, and separated performance positions help explain why spatial exchange and reverberant fusion could become compositional resources long before electronic media.

Figure 2-69 presents St. Mark's Basilica as an architectural reverberator and spatial-performance instrument and establishes the visual reference for the discussion that follows.

Figure 2-69: St. Mark's Basilica as an architectural reverberator and spatial-performance instrument

Source and license: Andrek02, Wikimedia Commons, CC0 1.0.

The photograph shows layered arches, galleries, domes, columns, and reflective finishes. These features create multiple propagation paths rather than one uniform decay. Separated ensembles can remain locally identifiable while sharing a common late field.

The acoustic recording era of the late nineteenth and early twentieth centuries did not preserve such fields transparently. Horns and diaphragms coupled performers mechanically to cutting systems. Recording balance depended on physical distance from the horn, and the capture geometry strongly favored direct energy. Engineers had little opportunity to add a separate response after recording. They could choose a resonant room, place performers strategically, or exploit the recording chain's own resonances, but reverberation remained entangled with capture.

This limitation produced a recurring historical desire: capture intelligibility and balance under controlled conditions, then restore a convincing or artistically useful sense of space afterward. Every artificial-reverb technology answers that desire differently. The chamber externalizes it into another room. The plate and spring compress it into mechanical structures. The digital reverberator abstracts it into delays and coefficients. The plug-in multiplies it across tracks and sessions.

2.9.2 Resonant Spaces Recruited as Devices

Once microphones, electrical amplification, loudspeakers, mixing, and rerecording became practical, a room could become an outboard processor. A feed from a console drove a loudspeaker in a reflective enclosure; one or more microphones captured the room's response; the engineer returned that signal to the mix. The basic send-and-return topology still governs modern reverb buses. What changed was that the "effect unit" occupied architectural space and required transducers at both ends.

Early chambers were not always designed from scratch. Bathrooms, stairwells, hallways, basements, tanks, and storage spaces could be evaluated by ear. A useful chamber needed sufficient isolation, a controllable loudspeaker and microphone arrangement, low noise, and a decay that added complexity without obvious flutter or dominant modes. Surface treatment could be adjusted, but the chamber's fundamental size and geometry remained fixed. Engineers changed apparent character by moving the loudspeaker, moving microphones, changing microphone patterns, equalizing the send or return, and combining more than one pickup.

Accounts of Bill Putnam's work on the Harmonicats' *Peg o' My Heart* in 1947 are often used to mark an early artistic milestone in record-production chamber reverb. The exact priority claim matters less than the workflow it represents: a recorded signal was deliberately sent into a separate reflective room, captured, and blended as an authored layer. By the 1950s, purpose-built chambers became prestige infrastructure in major studios. Capitol's underground chambers, for example, treated architecture as a repeatable part of the recording system rather than an accidental bathroom effect. The AES Reverb Collection places chamber design alongside spring, plate, algorithmic, and perceptual research, emphasizing that chamber history belongs inside audio engineering rather than outside it.

Abbey Road's Studio Two echo chamber offers a compact British example. EMI technician Henry Clark and Abbey Road engineer Stuart Eltham converted a wartime air-raid shelter during the mid-1950s, lining the room with hard reflective surfaces and using glazed sewer-pipe sections as scatterers. A console send drove a loudspeaker in the chamber, while microphones returned the reflected sound to the control room. The room was not a neutral container. Loudspeaker position, microphone distance, boundary reflections, pipe scattering, and the frequency responses of the transducers all became part of its signature. Abbey Road's history of the Studio Two echo chamber describes its use on recordings by the Shadows, Peter Sellers, the Beatles, and Pink Floyd, including John Lennon's vocal in *A Day in the Life* (1967).

The following reconstruction is intended to make that physical signal path visible. It preserves the documented two tall glazed pipes and one shorter pipe, together with a loudspeaker and two microphone positions, but it does not claim to reproduce a particular archival photograph or every historical equipment choice.

Figure 2-70 presents Illustrated reconstruction of the Abbey Road Studio Two echo chamber and establishes the visual reference for the discussion that follows.

Figure 2-70: Illustrated reconstruction of the Abbey Road Studio Two echo chamber

Source and license: Original illustration generated for this guide in 2026 and distributed under the repository's MIT license. Historical details are based on Abbey Road Studios' account of the Studio Two echo chamber.

The loudspeaker excites the reflective room; the microphones capture a return shaped by distance, hard boundaries, and the three glazed pipe scatterers. The image is interpretive rather than an archival photograph.

Unusual spaces demonstrate why chamber selection was an empirical art. The following figure documents a particularly direct example: *Silophone* (2000–2001), a public sound installation by The User, the Montréal collaboration of Thomas McIntosh and Emmanuel Madan. It transformed four empty storage chambers in Montréal's Silo No. 5 into a remotely playable instrument: visitors submitted sounds by telephone or the web, loudspeakers projected them into the silo, microphones captured the resulting resonance, and the return was broadcast back to listeners.

The project treated reverberation as both material and interface. A participant did not simply select a large-room preset; they addressed a specific industrial structure with its own chamber geometry, modes, noise, access constraints, and playback chain. The feedback path between remote sender, loudspeaker, chamber, microphone, and stream made the space audible as a changing instrument rather than a passive effect. This is a useful corrective to the generic "silo reverb" trope: the acoustic result belongs to the actual site and its electroacoustic routing, not merely to a building type.

Figure 2-71 presents *Silophone* public listening and playback stations in Montréal, with Silo No. 5 behind them and establishes the visual reference for the discussion that follows.

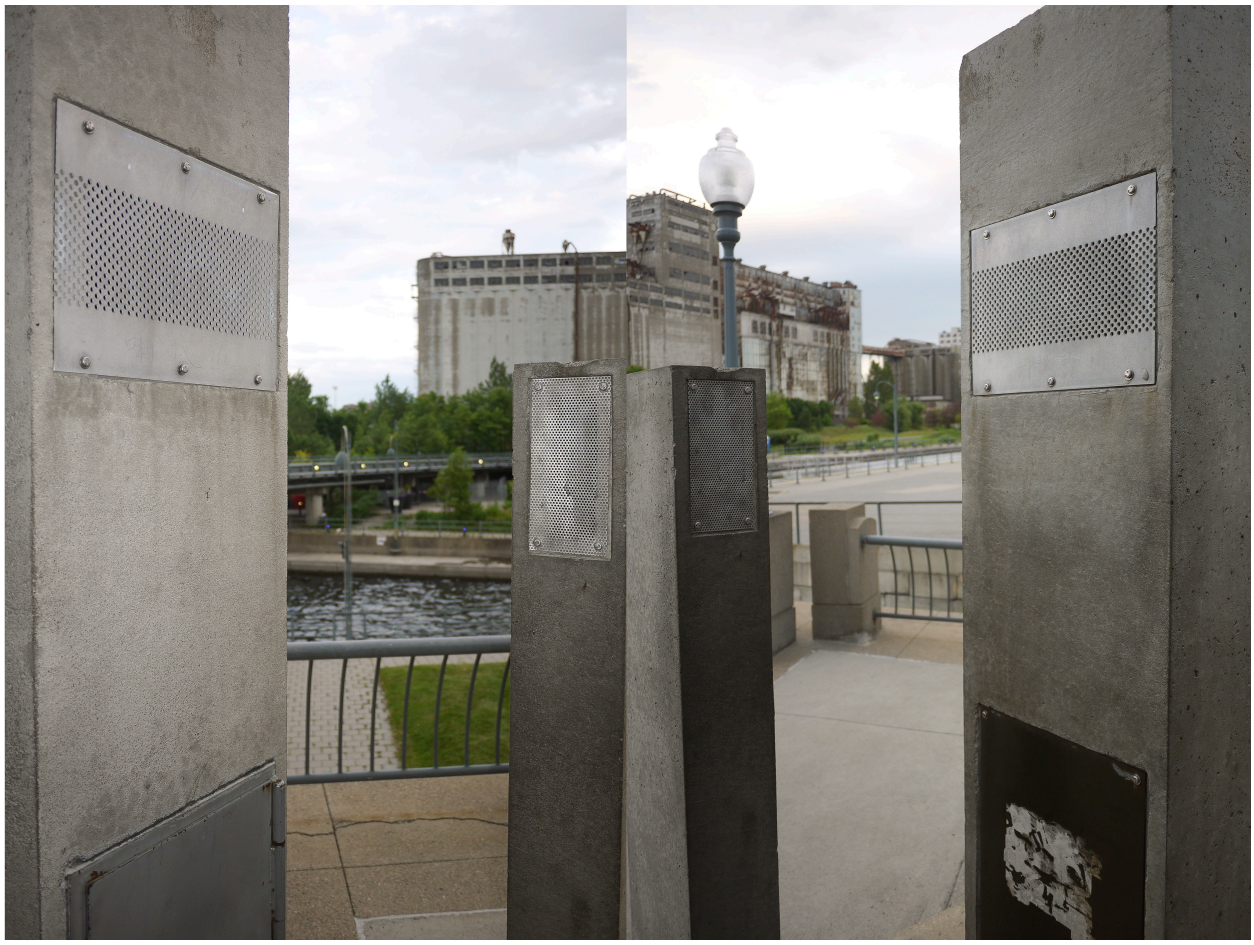


Figure 2-71: *Silophone* public listening and playback stations in Montréal, with Silo No. 5 behind them

Source and license: art_inthecity, Wikimedia Commons, CC BY 2.0.

The installation made the silo's reverberation reachable through a civic interface rather than treating the building as scenery.

The chamber made several production ideas explicit. First, **dry and wet became separable assets**. The dry performance could remain intelligible while the chamber return was filtered, compressed, faded, or muted. Second, **one space could unify many sources**. Vocal, percussion, and instrumental sends could excite the same chamber and acquire shared decay statistics. Third, **space became routable**. A console's auxiliary system determined which events entered the room and when. Fourth, **reverb became recordable independent of performance**. Engineers could print wet stems or rerecord the response.

The cost was operational. A chamber consumed real estate and required isolation from traffic, ventilation, electrical hum, and studio activity. It could not instantly become a different size. Microphone self-noise and amplifier noise accumulated in the return. Low-frequency modes and flutter had to be managed physically or electronically. A chamber could be exceptional, but it was not portable, cheap, or infinitely reproducible.

The following figure shows the TU Dresden echo chamber as a purpose-built acoustic instrument. Its nonparallel elements and reflective surfaces are visible evidence that chamber reverberation is designed through geometry rather than by selecting a room-name label.

Figure 2-72 presents TU Dresden purpose-built acoustic echo chamber and establishes the visual reference for the discussion that follows.

Figure 2-72: TU Dresden purpose-built acoustic echo chamber

Source and license: Henry Muhlpfordt, Wikimedia Commons, CC BY-SA 3.0.

Irregularly oriented panels and hard boundaries promote a complex reflection field and discourage a single dominant flutter path. Loudspeaker and microphone positions turn the enclosure into a send-and-return processor with a physically generated impulse response.

Chambers survived every later technology because they do something difficult to reduce to one algorithm: they combine real transducers, air, boundary vibration, noise, directional radiation, microphone response, and imperfect geometry. Their limitations also create discipline. With one or two chambers, a studio develops a shared spatial identity and engineers learn how send level, equalization, microphone distance, and return compression interact. Unlimited plug-in instances can simulate variety, but they can also remove the productive constraint of a common acoustic world.

2.9.3 Pipes, Tubes, and Folded Acoustic Paths

A full room was not the only way to store acoustic energy. Engineers explored pipes, tubes, and folded paths that could delay and recirculate sound in a smaller footprint. Harry F. Olson and John C. Bleazey's 1960 paper "Synthetic Reverberator" described a loudspeaker, pipe, microphone delay unit, and feedback system. The design is historically important because it sits between architectural and

electronic thinking. Propagation through air provides delay, transducers convert between domains, and electrical feedback extends the response.

An acoustic delay path is conceptually simple. A pressure wave enters a tube, travels at the speed of sound, reflects or exits, and is captured later. In practice, diameter, length, termination, wall loss, dispersion, and higher-order modes color the signal. Folding the path saves space but introduces bends and junctions. Feedback increases effective decay but also reinforces path resonances. The device therefore does not become a miniature neutral room. It becomes a characteristic resonator whose physical dimensions are audible.

These systems foreshadow digital delay networks. A pipe section behaves like a delay with frequency-dependent loss. A reflection or branch behaves like a scattering junction. A microphone and amplifier supply gain. Feedback creates recursion. The later digital abstraction replaces air and transducers with memory and arithmetic, but the block diagram remains recognizable.

The following figure presents the exterior identification of an O. C. Electronics Folded-Line reverberation device. Its enclosure hides the propagation path, encouraging the user to treat a mechanical-acoustic structure as outboard equipment.

Figure 2-73 presents Exterior of a Folded-Line acoustic reverberation device and establishes the visual reference for the discussion that follows.

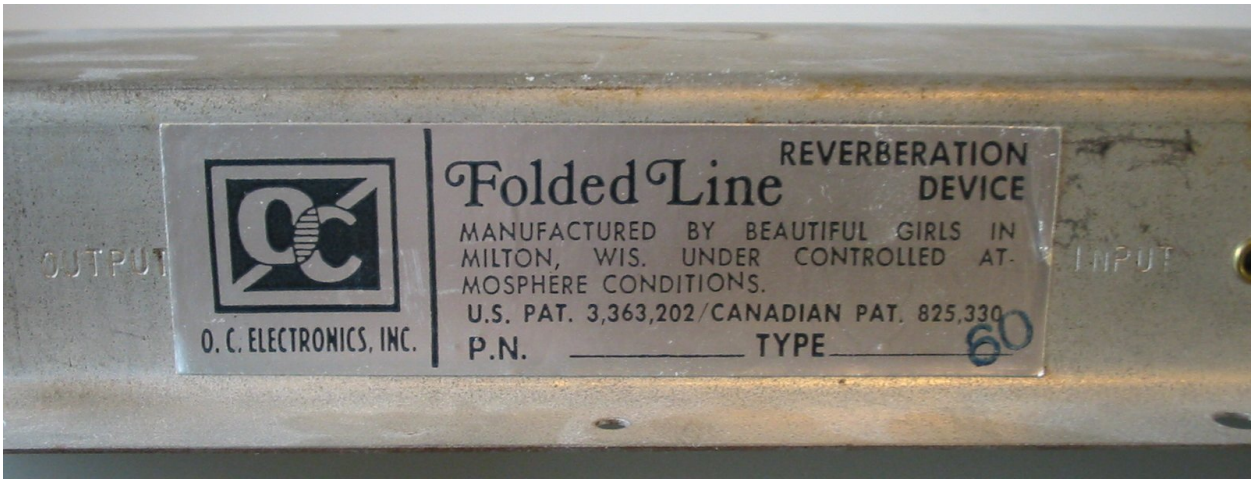


Figure 2-73: Exterior of a Folded-Line acoustic reverberation device

Source and license: Grebe, Wikimedia Commons, CC BY-SA 3.0.

The front label identifies the unit while the sound-producing path remains inside the enclosure. This packaging anticipates later rack processors: a complex energy-storage medium is reduced to input, output, and a small control surface.

The following interior view makes the hidden mechanism visible. The folded structure compresses path length into a cabinet, trading architectural volume for a constrained propagation system.

Figure 2-74 presents Interior folded acoustic path and transducer assembly and establishes the visual reference for the discussion that follows.

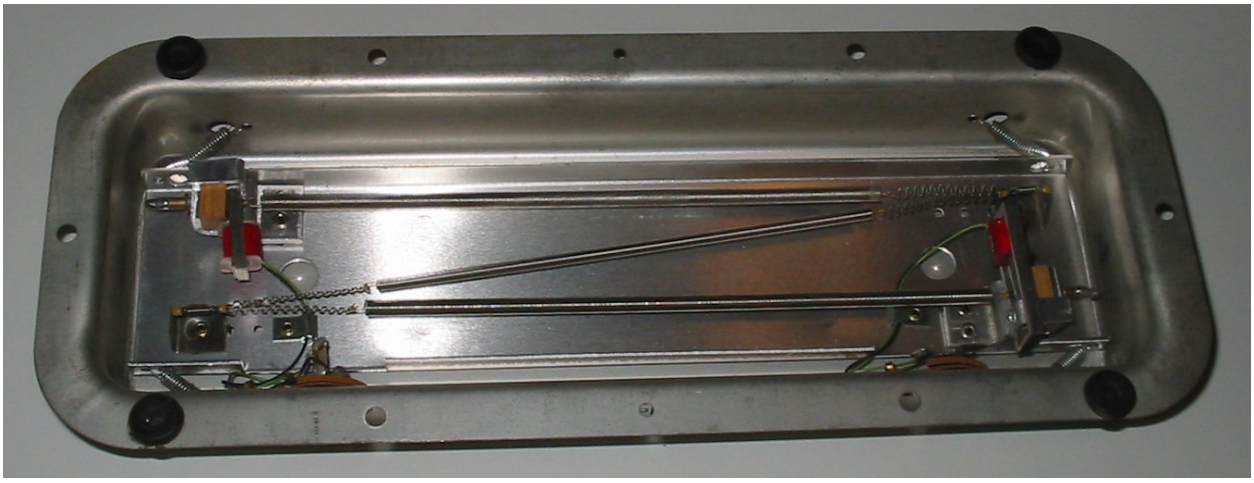


Figure 2-74: Interior folded acoustic path and transducer assembly

Source and license: Grebe, Wikimedia Commons, CC BY-SA 3.0.

The image reveals how physical length can be packed into a smaller enclosure. Bends, walls, and coupling elements contribute delay and coloration; they are not incidental packaging around an otherwise abstract algorithm.

Folded lines did not become the dominant studio standard, but their logic survives in digital waveguides, scattering networks, and physical models. They also remind us that “mechanical,” “acoustical,” “electrical,” and “electronic” are not mutually exclusive labels. Many reverberators cross domains. A loudspeaker launches a wave into air; a microphone recovers it; an amplifier feeds it back. A plate uses electrical transducers and mechanical bending waves. A spring uses electrical drive and pickup with mechanical propagation. Historical categories are best understood by the principal storage medium, not by pretending each device belongs to only one technology.

2.9.4 Plate Reverberation and the EMT 140

The plate reverberator replaced architectural propagation with bending waves in a tensioned sheet of metal. An input transducer excites the plate, pickups sense vibration at other locations, and damping controls shorten or shape decay. Because bending-wave speed depends on frequency, the response is dispersive. The plate’s modal density, boundary conditions, tension, pickup positions, and damping produce a bright, dense character that can suggest space without reproducing one room’s geometry.

EMT introduced the EMT 140 in 1957. It became a studio standard because it offered repeatable high-quality reverberation without a dedicated chamber. “Compact” is relative: the plate and frame are large, heavy, sensitive to vibration, and usually isolated in a machine room. Yet the device could be manufactured, installed, serviced, and used across sessions. Its decay could be changed mechanically by moving damping material toward or away from the plate. Stereo variants used multiple pickups to obtain related but distinct returns.

The plate solved several chamber problems while creating a new aesthetic. It did not capture traffic outside the studio. It required much less architectural volume. Its response was available whenever the studio was open. The onset could be dense and flattering on voice, percussion, and orchestral material. At the same time, the plate did not provide a literal sequence of room reflections. It offered a synthetic field whose credibility came from perceptual sufficiency: enough density, smoothness, bandwidth, and decay to create a stable sense of reverberant extension.

The following figure shows an EMT 140 assembly with its large framed plate, drive and pickup mechanisms, and control hardware. The photograph corrects the modern plug-in user's scale intuition. "Plate" originally meant an industrial electromechanical instrument, not a menu item.

Figure 2-75 presents EMT 140 plate reverberator and establishes the visual reference for the discussion that follows.

Figure 2-75: EMT 140 plate reverberator

Source and license: EMT-Archiv-Lahr, Wikimedia Commons, CC BY-SA 4.0.

A steel plate is suspended under tension inside a rigid frame. An electromechanical driver injects energy, pickups recover the distributed vibration, and a damping mechanism changes decay. The large area supports many interacting modes whose dense response became a canonical studio sound.

The plate's historical importance extends beyond its recordings. It established a durable user category. Engineers learned that a useful reverberator need not claim to be a hall. A plate could be selected because its attack, density, brightness, and decay supported a source. Later digital units named "plate" programs not because their delay networks contained metal, but because the label communicated a family of temporal and spectral expectations.

Digital plate models developed along three broad paths. Algorithmic designs imitate rapid density growth, frequency-dependent decay, and characteristic diffusion without solving a plate equation. Physical models discretize bending-wave behavior or use modal and waveguide approximations. Convolution

captures one setting as an impulse response. Learned models can interpolate or reproduce nonlinear and time-varying details from examples. Each preserves a different part of the instrument. The AES Reverb Collection explicitly connects historical plate hardware with later digital emulation research, while recent DSP-informed neural work models plate and spring behavior as structured rather than generic effects.

The practical lesson is historical and technical. "Plate" is not one transfer function. Individual units differ; pickup combinations differ; damping settings differ; maintenance and mounting differ. A modern plate preset should be evaluated by response, not by name. Listen for how quickly density forms, whether transients acquire a metallic edge, how upper frequencies decay, whether stereo channels feel related without collapsing, and how the return behaves under compression. Those are the inherited design questions that made the EMT 140 influential.

2.9.5 Springs: Portable Reverberation Becomes an Instrument

Spring reverberation stores energy in one or more metal springs rather than a plate or room. A driver converts an electrical input into mechanical motion; waves travel along the spring, reflect at its ends and discontinuities, disperse, and reach one or more pickups. Multiple springs, differing lengths, deliberate imperfections, and mechanical coupling raise echo density and complicate the response. The system is compact enough to fit inside an organ, amplifier, rack unit, or pedal enclosure.

Laurens Hammond's 1939 patent application, published as US 2,230,836, described a system intended to increase reverberation time and simulate the repeated sound of a large enclosure in an electrical musical instrument. The patent is historically useful because it frames the spring not merely as an effect but as compensation for an instrument removed from the architecture associated with its acoustic relatives. An electric organ could acquire sustained environmental response without occupying a cathedral.

The spring's physics ensures that this response is not neutral. Torsional and transverse modes propagate differently; wave speed varies with frequency; reflections at terminations create chirps and repeated structures; transducers impose bandwidth limits; shock can produce the famous crash. Designers use multiple springs and coupling arrangements to increase complexity, but the result retains a recognizable dispersive fingerprint. That fingerprint became a musical language rather than a defect to be eliminated.

The following figure shows the drive and pickup region of a Gibbs Special Products reverberation unit associated with Hammond's electromechanical tradition. The visible coils, magnetic structures, spring terminations, and mounting assembly demonstrate how a patented physical principle became a serviceable component inside electronic instruments.

Figure 2-76 presents Transducer and spring coupling inside a Hammond/Gibbs electromechanical reverberation unit and establishes the visual reference for the discussion that follows.

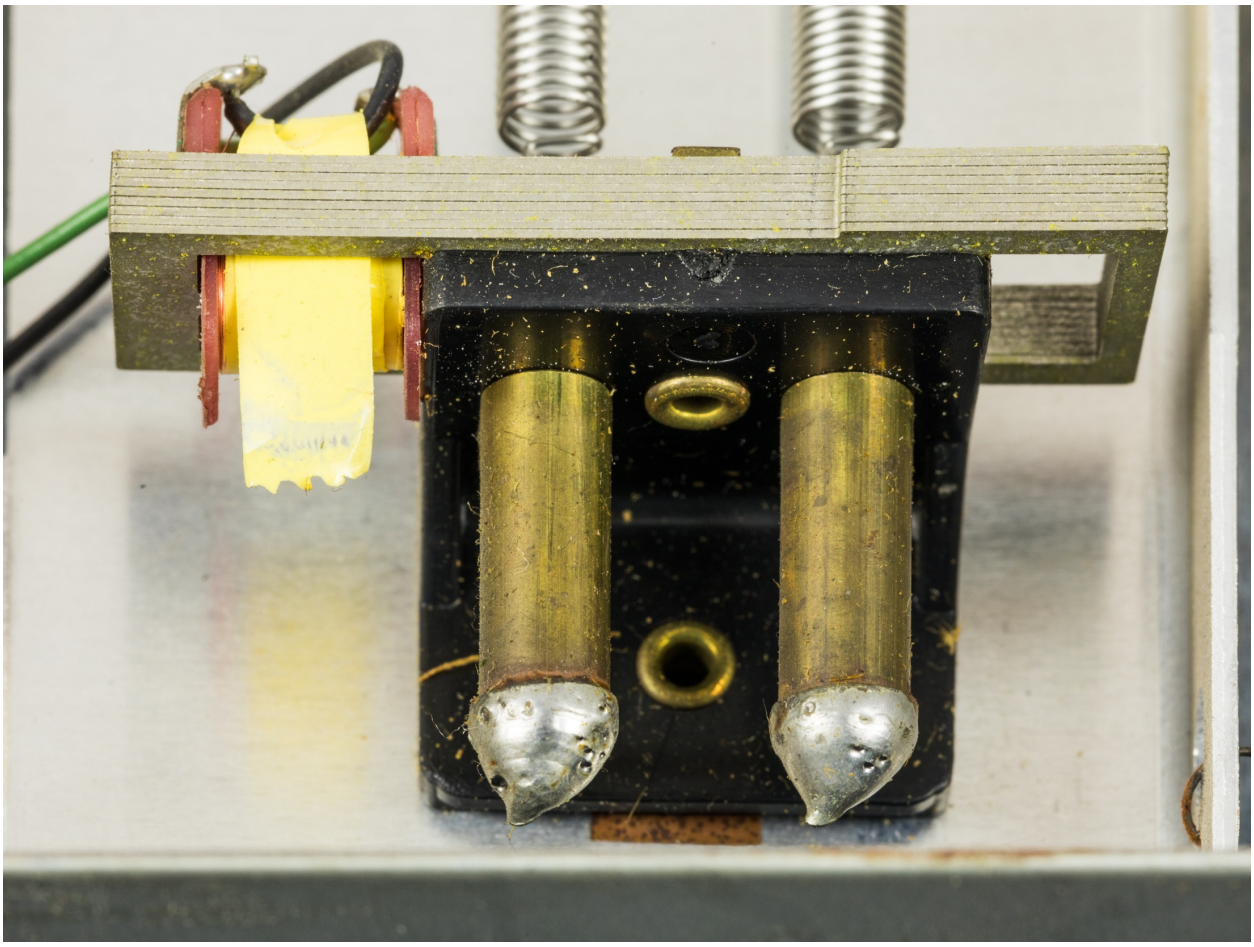


Figure 2-76: Transducer and spring coupling inside a Hammond/Gibbs electromechanical reverberation unit

Source and license: Raimond Spekking, Wikimedia Commons, CC BY-SA 4.0.

The close-up exposes coils, magnetic assemblies, spring terminations, and supporting metalwork. Reverberation is stored mechanically, but electrical drive and pickup make the structure part of the organ's signal path.

Spring reverb entered guitar culture through amplifiers and standalone units, where it interacted with pickups, tubes, loudspeakers, performance dynamics, and stage volume. The Fender Reverb Unit introduced in the early 1960s made "dwell," tone, and mix performable controls. Dwell changed how strongly the tank was excited, which affected both level and apparent behavior. Tone shaped the wet path. Mix balanced direct and returned sound. These controls remain conceptually important because they separate excitation, timbre, and blend instead of reducing the effect to one amount.

The following figure presents the rear chassis of a standalone Fender reverb unit. Its exposed tubes, transformers, connectors, wiring, and cabinet construction show that spring reverberation had become musician-operated amplification gear, not only studio infrastructure.

Figure 2-77 presents Rear chassis view of a Fender standalone spring-reverb unit and establishes the visual reference for the discussion that follows.

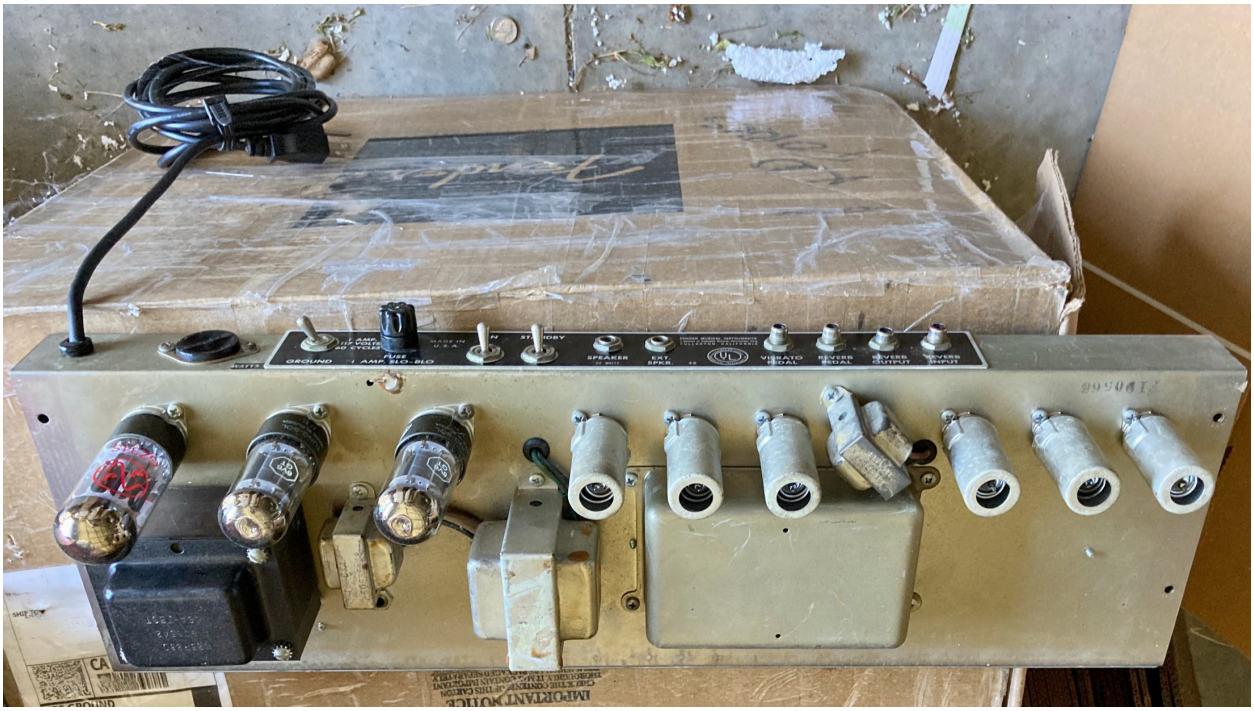


Figure 2-77: Rear chassis view of a Fender standalone spring-reverb unit

Source and license: Bill Abbott, Wikimedia Commons, CC BY-SA 2.0.

Tubes, transformers, connectors, wiring, and the long cabinet are visible from the service side. The unit helped make reverberation part of instrumental articulation: picking dynamics and dwell determine how forcefully the spring system is excited.

Surf guitar foregrounded the spring's splash, brightness, and transient response. Dub engineers later used springs as part of a performative mixing system with tape echo, filtering, feedback, muting, and fader movement. In both cases the device was not judged by how closely it reproduced a concert hall. Its identity came from the relation between attack and mechanical tail. A picked note or snare hit could trigger a burst whose chirp and density occupied rhythmic space.

The following figure removes the surrounding product enclosure so the parallel springs, input driver, and output pickup can be inspected directly. This visible signal path explains why spring emulations must model more than an exponential decay.

Figure 2-78 presents Open spring-reverb tank showing driver, springs, and pickup and establishes the visual reference for the discussion that follows.

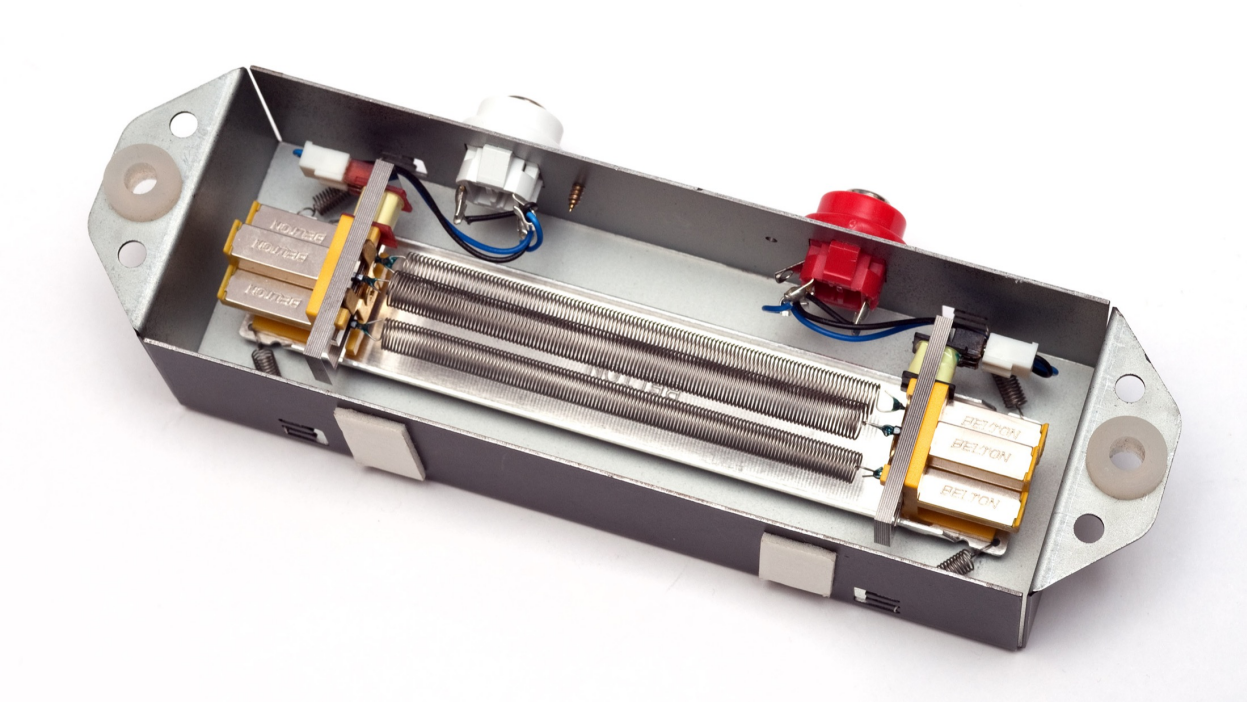


Figure 2-78: Open spring-reverb tank showing driver, springs, and pickup

Source and license: Ashley Pomeroy, Wikimedia Commons, CC BY-SA 4.0.

Several suspended springs provide related propagation paths. Mechanical terminations and transducer coupling generate dispersion, repeated arrivals, limited bandwidth, and sensitivity to physical shock.

The spring also demonstrates how technologies migrate across scale. The large studio chamber became a plate; the plate's principle of distributed mechanical storage became a compact spring assembly; the spring assembly later became a digital algorithm, convolution response, and learned model. Yet physical springs remain in production because no simulation eliminates the attraction of directly exciting a resonant object.

The following figure shows a modern modular-synthesizer spring assembly. The physical tank is still separated from the control electronics, preserving a design lineage in which voltage and code prepare and recover a process that occurs in metal.

Figure 2-79 presents Compact spring assembly for a modular-synthesizer reverberator and establishes the visual reference for the discussion that follows.

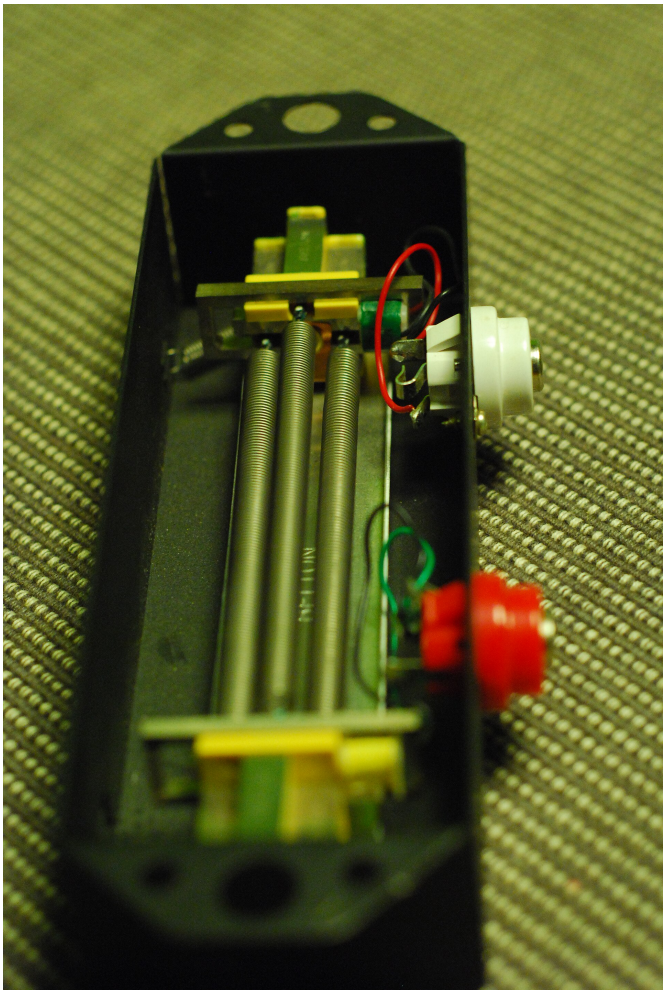


Figure 2-79: Compact spring assembly for a modular-synthesizer reverberator

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The tank remains an external mechanical memory while the module provides drive, recovery, feedback, and level control. Contemporary modular practice can patch the wet signal back into filters, oscillators, or controlled feedback networks.

Other electromechanical devices explored vibrating membranes, foils, drums, wires, rotating magnetic media, and hybrid acoustic paths. Oil-can delays and magnetic drum systems are often discussed with echo rather than reverb, but the distinction becomes blurry when many recirculating repeats, filtering, and diffusion fuse perceptually. Tape delay likewise supplied controllable time and feedback before affordable digital memory. Engineers could build pseudo-reverberant textures by combining several tape heads or machines, changing speed, filtering returns, and feeding one delay into another.

The key transition is conceptual: a reverberant response does not require a literal room if a system can produce enough delayed, attenuated, spectrally evolving, and mutually complicated copies. Mechanical devices proved that alternative storage media could create convincing or musically valuable persistence. Electronic and digital systems would separate those copies from physical propagation and make their timing programmable.

2.9.6 From Electromechanical Media to Delay-and-Feedback Abstraction

The following figure places a Dynacord DRS-78 digital reverberator beside a VRS-23 “vertical reverb.” The pairing is historically valuable because two electronic delay regimes share one photograph. The DRS-78 processes its delay and reverberation digitally, while the VRS-23 uses clocked analog bucket-brigade delay lines arranged for echo or reverberation. Studios did not move from one era to another overnight. They compared, layered, and retained devices according to sound.

Figure 2-80 presents Dynacord digital and analog bucket-brigade reverberators side by side and establishes the visual reference for the discussion that follows.

Figure 2-80: Dynacord digital and analog bucket-brigade reverberators side by side

Source and license: Mikael Altemark, Wikimedia Commons, CC BY 2.0.

The DRS-78 and VRS-23 embody overlapping electronic approaches to artificial decay. Their coexistence shows that digital adoption expanded the palette rather than immediately erasing analog delay technology.

Analog electronic delay was historically difficult. A long audio-band delay requires storage, and pure resistor-capacitor networks change phase without supplying room-scale delay economically. Magnetic tape, rotating media, acoustic paths, and later bucket-brigade devices provided time at the cost of noise, bandwidth, stability, or modulation artifacts. Each imperfection shaped use. Tape delay darkened and compressed repeats. Mechanical paths rang. Bucket-brigade delays traded bandwidth against delay length and clock noise. Reverb designs often embraced these losses because real reverberant energy also becomes darker and less coherent over time.

The abstract building blocks were nevertheless becoming clear: delayed branches, attenuation, filtering, summation, and feedback. Harry Olson and John Bleazey’s pipe-based synthetic reverberator made

that abstraction visible across acoustic and electrical domains. Manfred R. Schroeder and Benjamin F. Logan's 1961 paper "'Colorless' Artificial Reverberation" addressed two central faults of delay-loop systems: comb-filter coloration and insufficient echo density. Cascaded allpass filters could increase density while maintaining an approximately flat magnitude response, and parallel feedback comb filters could establish decaying recirculation.

Schroeder's 1961 convention paper described natural-sounding artificial reverberation produced by electronic means and even discussed an ambiophonic, three-dimensional extension. The historical significance is not that one topology solved reverberation forever. It is that reverberator design became a problem that could be reasoned about using transfer functions, delay statistics, echo density, coloration, and perception. The room was no longer the only explanatory model.

The familiar Schroeder architecture used several parallel feedback comb filters followed by serial allpass diffusers. The comb section built duration through recursion. The allpass section increased temporal complexity. Prime or mutually incommensurate delays reduced obvious common periods. Frequency-dependent feedback shortened high-frequency decay. The topology was computationally economical and theoretically legible, but sparse or poorly chosen settings produced metallic modes, flutter, and audible periodicity.

James A. Moorer's late-1970s work added more explicit early reflections, low-pass behavior in feedback loops, and design detail informed by room responses. Later structures distributed state through feedback matrices, nested allpasses, scattering junctions, velvet-noise sequences, time variation, and multiband decay. Stautner and Puckette's early-1980s work on digital reverberation connected multichannel output with matrix feedback. Jean-Marc Jot and Antoine Chaigne formalized design methods that separated desired decay behavior from a lossless feedback structure. Julius O. Smith's work on digital waveguides and lossless scattering provided a broader physical and mathematical vocabulary.

Barry Blesser's 2001 synthesis, "An Interdisciplinary Synthesis of Reverberation Viewpoints", argues that artificial reverberation is understood best where perceptual metrics, room statistics, musical culture, and studio practice intersect. That observation explains the historical endurance of nonliteral devices. An algorithm need not reproduce every path in a room if it creates the temporal and spectral statistics from which listeners construct a useful sense of space. Conversely, a mathematically elegant network can fail if its modes, density, onset, or spatial behavior violate perceptual expectations.

2.9.7 The First Commercial Digital Reverberators

Digital reverberation required converters, memory, arithmetic, control logic, and enough throughput to update a long recursive system at audio rate. In the 1970s these resources were expensive and limited. Early commercial units therefore encoded design judgment in hardware. Memory size constrained delay lengths. Word length shaped noise and headroom. Processor speed constrained filter count and modulation. Converter quality shaped bandwidth and transient response. Those limits did not merely reduce fidelity; they helped define recognizable products.

The EMT 250, introduced in 1976, is widely recognized as the first commercial digital reverberator. Barry Blesser's historical account identifies his role in developing the system, and EMT's contemporary Courier 26 documents the electronic reverberator in its original period. The freestanding unit used lever-like controls and supported reverberation plus delay, chorus, and related effects. Its furniture-like form made digital computation tactile: engineers could shape time without navigating a general-purpose computer.

The EMT 250 did not imitate the appearance of the EMT 140. This difference symbolizes the change in medium. The plate exposed a large mechanical storage system. The digital unit could hide its state in memory and present only parameters. Decay time could change without moving a damper across steel. Delay and modulation could become separate programs. Recalling a sound increasingly meant restoring numbers rather than recreating microphone positions or mechanical settings.

Lexicon's path began with digital delay technology. Its own Pantheon manual recounts the Delta-T 101 delay and David Griesinger's development work that led to the Lexicon 224. Introduced at the end of the 1970s, the 224 paired a processor with a remote control surface whose sliders and program selection made sophisticated algorithms accessible from the console. The unit's influence came from a combination of sound and interface. Engineers could choose room, hall, plate, and related programs, then alter decay, pre-delay, crossover behavior, and other musically legible dimensions.

The 224's algorithms were not neutral simulations. Limited memory and arithmetic encouraged carefully modulated structures whose density and pitch motion became part of the sound. The result could feel lush because modulation prevented static resonances and because frequency-dependent behavior was tuned by ear and psychoacoustic judgment. "Digital" did not mean transparent. It meant that coloration could be authored in code and repeated across units.

AMS developed digital delay and reverberation as another distinct lineage. The company's official history traces its formation in 1976 and early digital products. The RMX16 documentation describes the original 1982 unit as the first microprocessor-controlled, full-bandwidth digital reverberator. Its programs, including nonlinear and gated responses, became tightly associated with 1980s production because they transformed the envelope of reverberation rather than merely extending decay.

The following figure shows a working studio rack containing a Lexicon 200, AMS RMX16, and Lexicon PCM42 delay. The image is more historically truthful than an isolated museum portrait: landmark processors lived beside delays, compressors, patchbays, and console sends, where engineers combined them into systems.

Figure 2-81 presents Early digital reverberation and delay processors in studio context and establishes the visual reference for the discussion that follows.

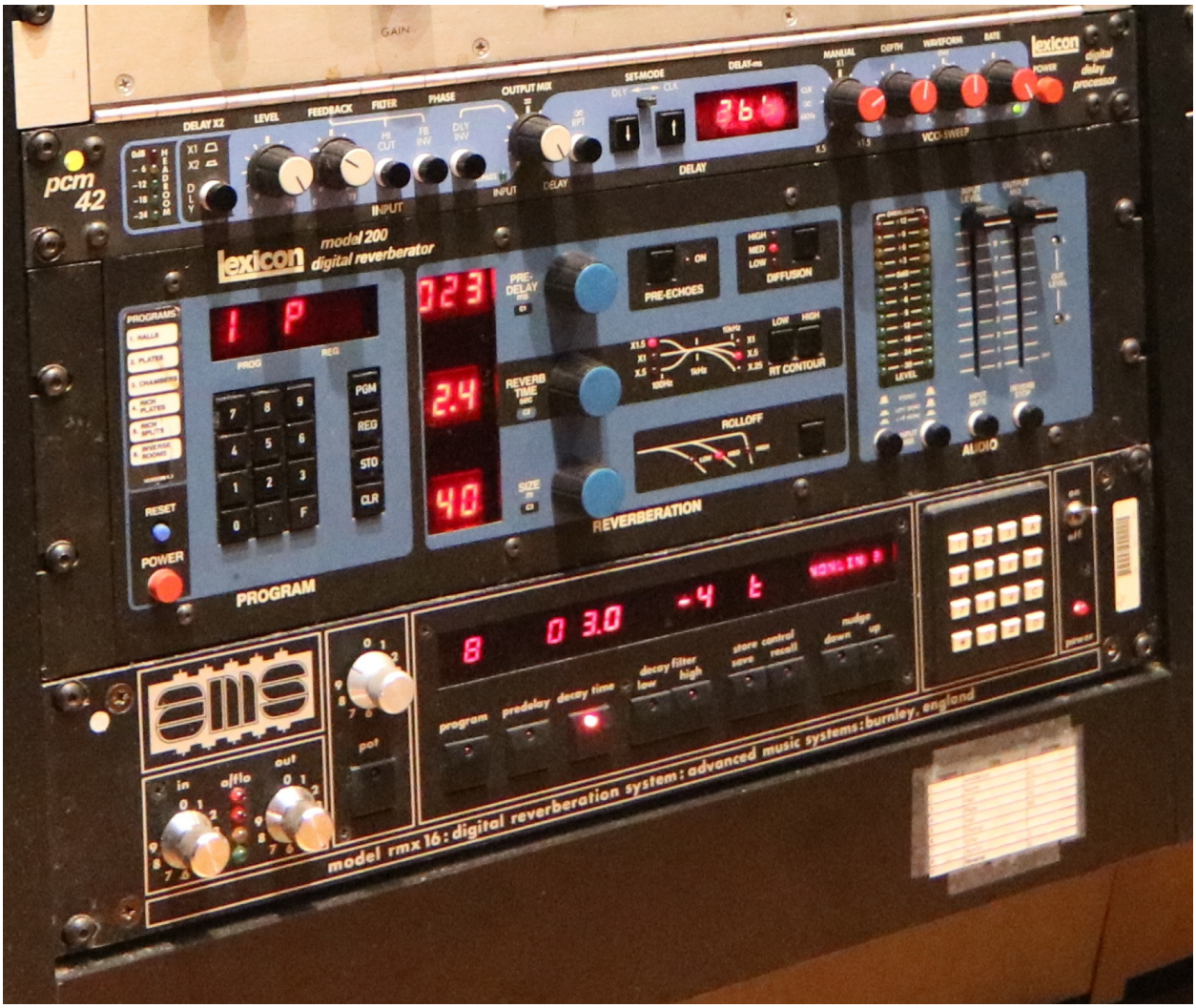

Figure 2-81: Early digital reverberation and delay processors in studio context

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The Lexicon 200 and AMS RMX16 occupy adjacent rack space with a PCM42 delay. Their proximity reflects an era in which delay, pitch change, gated envelopes, and reverberation were patched together as complementary production tools.

Gated and nonlinear programs demonstrate that digital reverberation quickly escaped the goal of invisible realism. A gate could stop a dense response before its natural exponential decay. A nonlinear program could hold or reshape energy before terminating. Used on drums, these envelopes magnified attack and apparent room size while protecting rhythmic space. The effect became part of arrangement and genre. The processor did not ask “what room is this?” so much as “what energy envelope should follow this event?”

The first digital generation also made programmability a studio asset. A chamber could be photographed and documented, but its exact sound depended on physical setup. A digital unit could store

or recall programs, though early recall was not always complete by modern standards. Studios could exchange settings, manufacturers could issue updates, and recognizable algorithms could appear across recordings. Presets became part of production vocabulary.

2.9.8 Affordable Racks and the Democratization of Digital Space

High-end digital reverberators initially belonged to major studios, broadcasters, and institutions. During the 1980s, integrated circuits, memory, converters, and manufacturing improved enough for smaller rack units to reach project studios, touring systems, keyboard rigs, and home recordists. This shift changed production practice as much as sound. A studio no longer needed a chamber, plate, or flagship remote-controlled processor to place several artificial spaces in one arrangement.

Affordable units often exposed fewer editable parameters and used lower-cost converters, restricted bandwidth, short programs, and compact displays. These constraints became aesthetic markers. Grainy tails, audible modulation, narrow bandwidth, and abrupt program envelopes could make a source sit in a mix precisely because the return did not resemble unprocessed high-fidelity audio. A producer could choose a small box for its attack and texture rather than apologize for it as a compromised hall.

Alesis's MIDIVerb family represents this democratization. MIDI control connected effects to increasingly automated studios, while preset-oriented operation let users move quickly among rooms, plates, gates, and special effects. The following figure shows the original MIDIVerb as a compact front-panel device. Its scale contrasts sharply with the EMT plate and freestanding EMT 250.

Figure 2-82 presents Alesis MIDIVerb and the project-studio digital transition and establishes the visual reference for the discussion that follows.

Figure 2-82: Alesis MIDIVerb and the project-studio digital transition

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A compact rack format, program display, and small control set made digital reverberation portable and affordable. Preset identity became more prominent as a large internal algorithm was represented by a number and a few controls.

Yamaha's REV and SPX families likewise helped turn digital effects into routine infrastructure. The SPX90, introduced in the mid-1980s, combined reverberation with delay, pitch change, modulation, gating, and other utilities in a one-rack-space unit. Its gated and reverse-style programs became production signatures, but its broader importance was functional integration. One processor could solve several session problems and could be repatched as the arrangement changed.

The following figure shows a Yamaha SPX90 and Alesis Microverb III inside a working rack. The photograph documents the practical ecosystem of the era: effects shared space with patching, conversion, synthesis, and other studio utilities.

Figure 2-83 presents Yamaha SPX90 and Alesis Microverb III in project-studio use and establishes the visual reference for the discussion that follows.

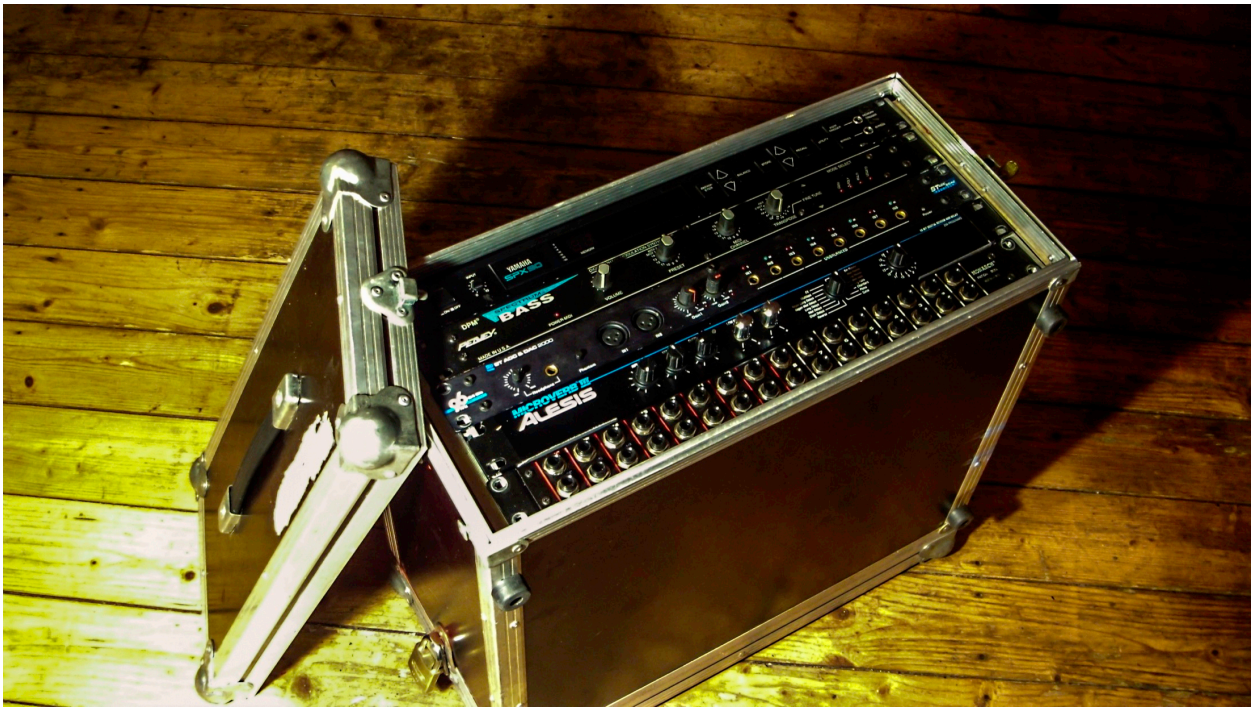

Figure 2-83: Yamaha SPX90 and Alesis Microverb III in project-studio use

Source and license: Klara Kopf, Wikimedia Commons, CC BY 2.0.

The one-rack-space format compresses processing and control into a standardized physical unit. Multiple affordable processors allowed different sources to receive distinct spaces rather than compete for one chamber or plate.

More available processors encouraged layered spatial production. A short ambience could give drums body; a plate could support a vocal; a longer hall could connect strings or synthesizers; a gated program could appear only at structural accents. This was not merely "more reverb." It was orchestration by decay family. Different sends and returns occupied different temporal and spectral roles.

The risk was fragmentation. If every source received an unrelated preset, the mix could lose a common acoustic logic. Skilled engineers often created hierarchy: one principal room or plate established shared space, while secondary devices supplied character or special events. That strategy remains valuable in a DAW with unlimited instances. Historical scarcity taught engineers to distinguish an environmental field from decorative effects.

2.9.9 Multi-Effects, Pitch, Modulation, and Impossible Rooms

Digital memory allowed reverberation to interact with pitch shifting, delay, modulation, filtering, and dynamics inside one programmable system. Eventide's Harmonizer lineage made pitch and delay central studio resources; later multi-effects units such as the H3000 exposed patchable structures and algorithms whose spatial results could be deliberately unreal. A pitch-shifted feedback path could rise or fall indefinitely. Modulated delays could widen and animate a tail. Reverse and multitap patterns could imply spaces that no architecture would produce.

The following figure presents an Eventide H3000 SE. Its dense front panel and program structure represent a move from a dedicated "room simulator" toward a general spatial and timbral computer.

Figure 2-84 presents Eventide H3000 SE as a programmable multi-effects environment and establishes the visual reference for the discussion that follows.

Figure 2-84: Eventide H3000 SE as a programmable multi-effects environment

Source and license: John R. Southern, Wikimedia Commons, CC BY-SA 2.0.

Reverb, delay, pitch transformation, modulation, and feedback can participate in one patch. The device helped normalize the idea that a reverberant tail could change pitch, move, or behave as a synthetic voice.

This branch of history leads directly to shimmer, resonant cloud, freeze, reverse, and granular reverbs. Shimmer commonly feeds a pitch-shifted interval into a long decay so each circulation adds transformed harmonic energy. Freeze raises loop gain toward unity or replaces the decay process with a sustaining spectral state. Granular reverberation distributes short windows across time, pitch, and space rather than relying only on recursive delay lines. These effects inherit the send-and-return concept but no longer claim that the return is a passive environment.

The distinction between reverb and instrument becomes unstable. If feedback is automated, the return has phrasing. If pitch changes on each circulation, it has harmonic motion. If a freeze preserves one chord under the next, it has memory at the level of form. If a spatial trajectory moves only the tail, it has counterpoint independent of the source. The technology's history therefore expands from acoustic simulation to composition with persistent signal state.

2.9.10 Algorithmic Maturity in the 1990s and 2000s

As processing power and memory increased, high-end algorithmic systems could support more delay lines, more elaborate modulation, multiband decay, denser early-reflection patterns, and multichannel output. Lexicon's 480L, PCM series, 960L, and related families represented different scales of that development. Quantec, TC Electronic, Sony, Yamaha, Eventide, and other manufacturers pursued distinct balances of realism, density, spatial behavior, control, and coloration.

Algorithm design also became more systematic. Feedback delay networks used a matrix to couple many delay lines. If the matrix preserved energy and loop filters imposed the desired frequency-dependent loss, decay time could be designed with greater independence from mixing topology. Modulation reduced static modes but had to avoid audible pitch wandering. Early and late fields could be separated. Output matrices and decorrelation could generate stereo or surround fields from shared internal state.

Jean-Marc Jot's work is central to this transition because it treats perceptual decay specification, energy-preserving mixing, and correction of modal response as related design tasks. Rather than tuning every feedback gain by trial, a designer can derive gains from desired decay times. Rather than accept the uncolored ideal as sufficient, correction filters can compensate for irregular mode distribution. The FDN becomes a framework within which different matrices, delays, filters, and modulators create families of reverberators.

The following figure shows a later studio rack containing Lexicon PCM 90, Alex, and Vortex processors with a Sony HR-MP5. The image illustrates diversification within digital effects: dedicated and multi-effects units coexisted, and a studio's "digital reverb" was often a collection of algorithms with different generations and control philosophies.

Figure 2-85 presents Later digital-reverb and multi-effects families in a studio rack and establishes the visual reference for the discussion that follows.

Figure 2-85: Later digital-reverb and multi-effects families in a studio rack

Source and license: fr4dd, Wikimedia Commons, CC BY 2.0.

Several processors provide overlapping but nonidentical rooms, plates, delays, modulation, and special effects. Algorithmic identity increasingly belongs to software and firmware even when it remains housed in dedicated hardware.

Surround production changed the output problem. A stereo reverberator could create width through two decorrelated returns, but cinema and multichannel music needed stable energy around a larger array. Simply copying one tail into every channel raised level and correlation without producing convincing envelopment. Multichannel algorithms used additional output taps, matrices, decorrelation,

and directional early patterns. The internal network could remain shared so the room felt coherent, while outputs sampled that state differently.

This development anticipates today's immersive distinction among source, early field, late bed, and special objects. The late field often benefits from broad statistical continuity. Early reflections may remain connected to source direction. A designed effect may move separately. Modern systems can render many channels, but the fundamental problem remains the historical one: distribute stored energy so listeners hear one intentional environment rather than unrelated echoes.

2.9.11 Convolution and the Impulse-Response Archive

Algorithmic reverberators generate an impulse response from a recursive structure. Convolution reverberators store or load an impulse response and apply it to new audio. If the measured system is linear and time invariant, the response captures its complete input-output behavior for the measured source, receiver, channels, level, and state. This made it possible to bring specific halls, churches, chambers, plates, springs, loudspeakers, and signal chains into software.

Convolution theory is old, but long real-time audio convolution was computationally expensive. Fast Fourier transforms, partitioned convolution, efficient memory, and general-purpose processors made long responses practical. Uniform and nonuniform partitioning allowed a short first partition to preserve low latency while larger later partitions handled the long tail efficiently. Multichannel and matrix convolution extended the idea from one input-output pair to many directional paths.

Impulse-response measurement became its own craft. A pistol shot or balloon burst provides a simple excitation but limited repeatability and signal-to-noise ratio. Maximum-length sequences and swept sinusoids improve energy distribution and allow deconvolution. Exponential sine sweeps can separate harmonic-distortion responses in time. Microphone directivity, source radiation, position, clock stability, ambient noise, and truncation all affect the captured asset.

Convolution changed the status of historical hardware. An EMT plate, spring tank, chamber, or rare digital unit could be sampled and distributed as data. This preservation is powerful but partial. One impulse response captures one setting and one linearized state. It does not necessarily capture level-dependent transducer behavior, mechanical shock, modulation that changes from pass to pass, moving dampers, time-varying converters, or the performative act of driving feedback. Libraries can sample many settings, but a finite collection remains different from a continuously controllable physical system.

Convolution also changed architectural memory. A space could be documented before renovation, closure, or demolition. Researchers could compare positions and layouts. Film and game production could place dry sources into measured environments. Musicians could use a grain silo, mausoleum, tunnel, or cathedral as a portable response. The ethics of that portability matter: an impulse response is both technical data and a representation of a place. Documentation should identify source, receiver, date, equipment, channel convention, license, and processing rather than treating every downloaded WAV as anonymous raw material.

2.9.12 Native Plug-ins and the End of One-Reverb Scarcity

As DAWs and native processors became powerful enough for real-time effects, reverberation moved from shared hardware into software instances. Early plug-ins reproduced familiar categories because users already understood room, hall, plate, spring, chamber, and gated programs. Over time, software interfaces exposed deeper structures: early-reflection editors, multiband decay, modulation matrices, impulse-response browsers, surround routing, freeze, pitch feedback, and graphical decay displays.

The economic change was profound. A hardware studio might own one plate, one chamber, and several digital boxes. A DAW could instantiate dozens of reverbs, automate every parameter, save them with the session, render offline at higher quality, and exchange presets instantly. The effect became abundant. This abundance enabled detailed sound design but weakened the default discipline of a shared return.

Native processing also separated algorithm from dedicated hardware. A classic topology could be reimplemented, modeled, extended, or combined with another method. Some plug-ins emulate a particular unit including converters and bandwidth. Others reproduce only broad behavior. Others use classic names for entirely new networks. Historical literacy helps users distinguish these claims. A “224-style hall,” a sampled 224 program, and a circuit-plus-algorithm model are different products even if their preset browsers use similar language.

Automation turned reverb into a time-varying arrangement layer. Engineers could change decay by section, move early reflections, switch impulse responses, duck returns from source activity, or freeze selected moments. Offline rendering removed realtime constraints for extreme tails and large matrices. At the same time, parameter smoothing, state recall, denormal handling, and plug-in delay compensation became engineering requirements that mechanical devices never faced in the same form.

2.9.13 Reverb Leaves the Rack: Pedals and Modular Systems

Digital signal processors eventually became small and efficient enough for floor pedals and modular-synthesizer modules. This migration changed who performed the reverb and when. A rack return is often controlled by an engineer after capture. A pedal is touched by the performer, placed before or after distortion, and incorporated into technique. A modular reverb can receive control voltage, audio-rate modulation, and feedback from an open patch.

The following figure shows the Make Noise Erbe-Verb in a Eurorack context. Its many continuous controls present reverberation as a voltage-controllable synthesis process rather than a fixed room selector.

Figure 2-86 presents Voltage-controlled digital reverberation in a modular system and establishes the visual reference for the discussion that follows.

Figure 2-86: Voltage-controlled digital reverberation in a modular system

Source and license: Brandon Daniel, Wikimedia Commons, CC BY-SA 2.0.

The Erbe-Verb exposes size, decay, modulation, absorption, tilt, and other dimensions to manual and control-voltage change. The reverb can become an actively modulated voice inside a patch.

Continuous control revives an older physical intuition. Moving a plate damper changed decay mechanically; changing a chamber microphone changed perspective; striking a spring changed its state. Modular control makes analogous changes electronic and repeatable, though rapid modulation can create pitch, zipper noise, or instability if the algorithm is not designed for it. The control surface therefore reveals whether a reverb is a static processor with interpolated parameters or a genuinely time-varying instrument.

The following figure shows the TC Electronic Nova Reverb pedal. Its display, preset architecture, and knobs compress studio-style digital processing into a performance format.

Figure 2-87 presents Programmable digital reverberation as a performance pedal and establishes the visual reference for the discussion that follows.

Figure 2-87: Programmable digital reverberation as a performance pedal

Source and license: Oldangelmidnight, Wikimedia Commons, CC BY-SA 2.0.

Room families and editable parameters are placed at the musician's feet. Reverb can be changed between songs, driven into amplifiers, or positioned within a chain whose distortion and dynamics alter the return.

Pedal and modular culture also encouraged hybrid categories: shimmer, black-hole, bloom, cloud, swell, lofi, granular, and spectral reverbs. These names describe gestures and textures more than architecture. They continue the 1980s move toward program identity, but with far more continuous control. The word *reverb* now covers any system in which sound leaves a temporally extended, spatially suggestive, or recursively transformed memory.

2.9.14 Measurement Rooms, Spatial Audio, and Immersive Reverberation

Artificial reverberation has always depended on measurement, even when design was guided primarily by ear. Decay curves, frequency response, echo density, noise, distortion, and spatial correlation help distinguish a useful field from ringing or instability. Reverberation chambers used in acoustics and aerospace testing push the physical principle toward controlled diffuse excitation. Their purpose may be material testing, source-power measurement, or qualification rather than music, but they reveal the engineering ideal of statistically distributed energy.

The following figure shows a Reverberant Acoustic Test Facility horn room. High-output horns excite a large reflective volume so equipment can be exposed to intense diffuse acoustic energy. The image expands the history beyond studio effects: a reverberant field can be a measurement environment, a stress condition, and a designed laboratory tool.

Figure 2-88 presents High-level reverberant acoustic test facility and establishes the visual reference for the discussion that follows.

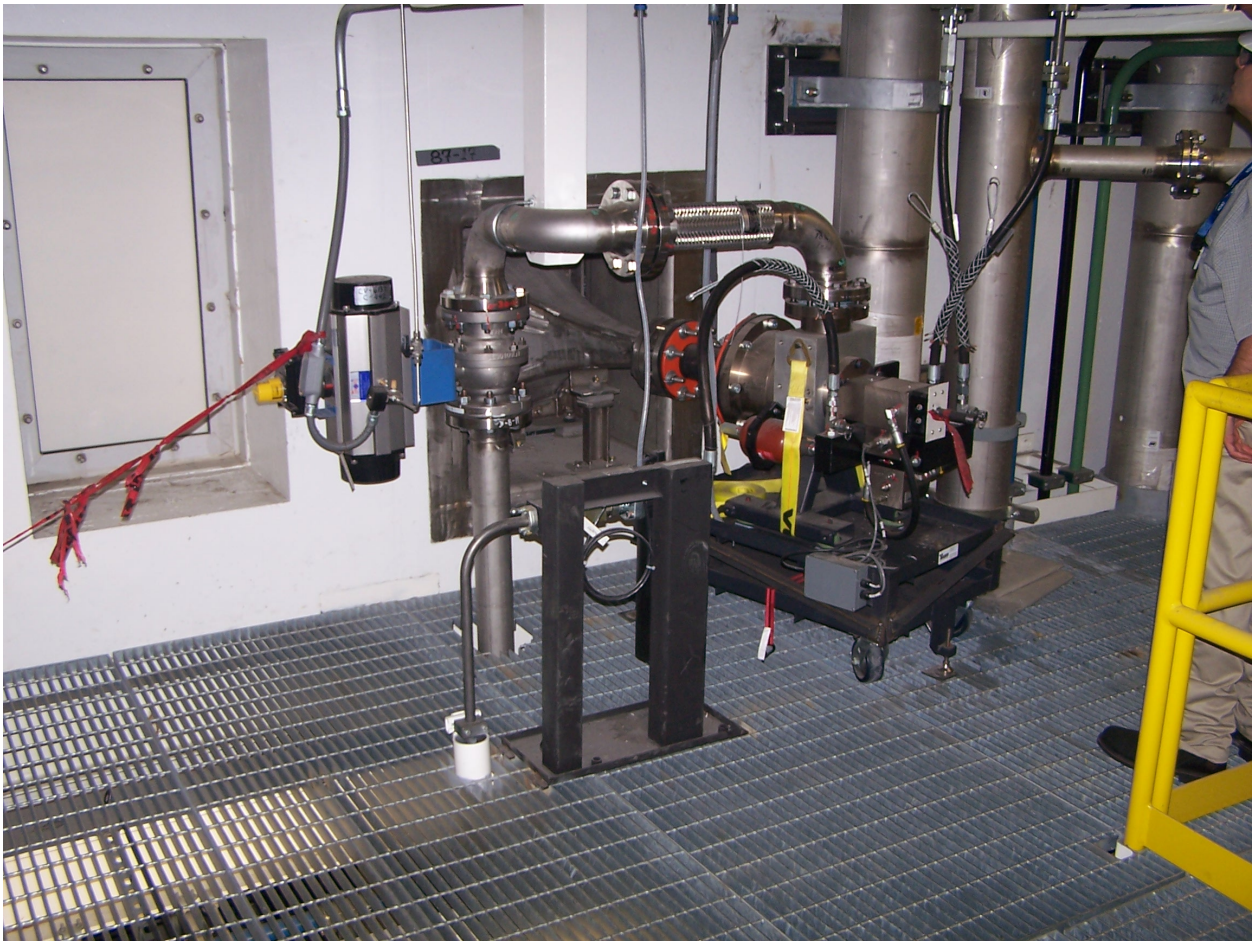

Figure 2-88: High-level reverberant acoustic test facility

Source and license: Illudium Pu-36, Wikimedia Commons, CC BY-SA 3.0.

Horns inject controlled energy into a reflective enclosure intended to build a strong distributed field. The room demonstrates reverberation as an engineered test condition rather than a musical aftereffect.

Spatial audio adds direction to the historical questions of storage, mixing, and loss. A mono chamber return captures one source-to-microphone path. Stereo adds related viewpoints. Surround and Ambisonic measurements preserve more directional information. Matrix impulse responses can map several sources to several outputs. Object-based production may keep direct sources independent while placing diffuse reverberation in a bed and special reflected gestures on objects.

The computational burden grows quickly. An M-input, N-output convolution contains MN impulse responses. A high-order Ambisonic field contains many spherical-harmonic channels. Binaural rendering requires head-related filters and may update with head motion. A dynamic listener or source invalidates the assumption that one fixed impulse response describes the scene. Interpolation among measured responses, geometric simulation, wave methods, and hybrid late-field algorithms address different parts of the problem.

Ray tracing and image-source methods return the history to architecture, but now geometry is data. A CAD model can produce candidate reflection paths; material coefficients attenuate and filter them; source and receiver directivity determine path weights. Geometric acoustics is efficient at frequencies where wavelengths are small relative to surfaces, but it does not naturally capture low-frequency wave behavior or diffraction without extensions. Full wave methods can model those effects at much greater cost. Hybrid room renderers often calculate early paths geometrically and hand the late field to a statistical algorithm.

The most convincing immersive result may therefore combine several historical media: measured early reflections, a ray-traced directional pattern, an FDN late field, modeled plate coloration, and binaural or loudspeaker rendering. The layers are not redundant. Each handles the part of the response for which its assumptions are useful.

2.9.15 Extreme Spaces and the Cultural Memory of Decay

Some spaces become historically important because their decay exceeds ordinary expectations. Hamilton Mausoleum was long associated with an exceptionally extended echo. Tanks, tunnels, silos, cisterns, and industrial chambers attract musicians because their modes and long paths turn isolated events into large-form material. Digital capture allows these spaces to circulate as impulse responses, but the fascination begins with bodily experience: a sound continues after its source has stopped, returning from a structure too large or irregular to apprehend at once.

The following figure presents Hamilton Mausoleum as the final photographic anchor in this history. Its hard circular enclosure illustrates why extraordinary decay is both an acoustic measurement and a cultural story.

Figure 2-89 presents Hamilton Mausoleum and the cultural imagination of extreme reverberation and establishes the visual reference for the discussion that follows.

Figure 2-89: Hamilton Mausoleum and the cultural imagination of extreme reverberation

Source and license: Supergolden, Wikimedia Commons, CC BY-SA 3.0.

Hard curved boundaries and a large enclosed volume support long, conspicuous returns. Such sites motivate measurement, composition, impulse-response capture, and the design of digital “impossible space” presets.

Extreme digital settings extend this cultural idea beyond physical construction. A 360-second or 3,600-second decay does not describe a conventional occupied room. It describes controlled memory. The same stability rules still apply: loop gain must remain bounded, low-frequency accumulation must be managed, modulation must avoid runaway pitch or phase behavior, and the output must preserve headroom. History helps interpret the control. An “infinite” mode descends from chamber recirculation, tape feedback, sustained springs, and digital loop gain, but it becomes a new compositional category when the tail outlives phrases and sections.

2.9.16 Physical Modeling, Machine Learning, and the Current Hybrid Era

Physical modeling attempts to preserve causal structure. Digital waveguides model traveling waves and scattering. Finite-difference and finite-element methods discretize wave equations. Modal synthesis

represents a resonator as decaying modes. Plate and spring models reproduce dispersion and damping rather than only matching one output recording. These methods can expose physically meaningful parameters, but they require computational choices and boundary assumptions that shape the result.

Machine learning introduces another representation: a model estimates a mapping from dry audio and controls to reverberant output, or from a reverberant signal to a less reverberant target. A network may learn a black-box effect from examples, predict room parameters, generate impulse responses, interpolate among spaces, estimate material absorption, or assist dereverberation. DSP-informed models constrain the architecture so delays, filters, convolution, or physical states remain explicit. The research on modeling plate and spring reverberation with DSP-informed neural networks is historically revealing because it does not replace the old mechanisms conceptually; it embeds their structure in a trainable system.

Learned reverberation raises new archival questions. A convolution library can identify the exact impulse response used. A neural model may summarize thousands of responses in weights that are difficult to interpret. Training data may contain rooms, hardware, performances, or licenses that need documentation. A model can generalize between measured conditions, but it can also invent plausible response without preserving one place. Provenance, evaluation, deterministic inference, and disclosure become part of reverb engineering.

The strongest current systems are often hybrid. A geometric model predicts early arrival directions. A learned estimator infers materials or room dimensions. An FDN produces a stable late field. Convolution supplies a measured device onset. Neural residual processing adds time-varying detail. A renderer distributes the result to loudspeakers or headphones. This is not a failure to choose one method. It is the mature recognition that "reverberation" contains several physically and perceptually distinct tasks.

2.9.17 A Condensed Timeline

The following timeline should be read as overlapping milestones rather than replacement dates:

- **Before electrical recording:** architecture, performer placement, and listener position determine reverberation inseparably from the event.
- **Early electrical era:** microphones, amplifiers, loudspeakers, mixers, and rerecording make separate send-and-return rooms possible.
- **1930s–1950s:** improvised and purpose-built echo chambers turn architecture into studio outboard equipment.
- **1939–1941:** Laurens Hammond's spring-reverberation patent establishes a compact electromechanical method for extending an electrical instrument's decay.
- **1940s–1960s:** chambers, springs, magnetic delays, tubes, pipes, and other physical systems coexist in broadcast, organ, studio, and instrument design.
- **1957:** the EMT 140 plate offers dense, controllable electromechanical reverberation without a dedicated room.
- **1960–1961:** Olson and Bleazey describe a synthetic pipe-and-feedback reverberator; Schroeder and Logan formalize comb and allpass approaches to density and coloration.
- **1970s:** digital delay, memory, converters, and real-time arithmetic become sufficient for commercial programmable reverberation.
- **1976:** the EMT 250 establishes a landmark commercial digital reverberator with tactile control.
- **Late 1970s:** Lexicon's 224 lineage brings influential modulated algorithms and remote-console operation into major studios.
- **Early 1980s:** AMS RMX16, Yamaha REV systems, Quantec, and other processors expand full-bandwidth, program-oriented, and multichannel possibilities.
- **Mid-to-late 1980s:** SPX, MIDIVerb, Microverb, PCM, and other compact racks make multiple digital spaces practical outside elite studios.

- **1990s:** higher processing power supports more elaborate modulation, FDNs, multiband decay, surround output, and native software effects.
- **Late 1990s–2000s:** practical long convolution and growing impulse-response libraries make measured spaces and devices portable.
- **2000s–2010s:** native plug-ins, pedals, and modular DSP multiply instances, automation, hybrid algorithms, and performable control.
- **2010s–present:** immersive rendering, large convolution matrices, geometric simulation, neural models, and Audio AI connect reverberation to spatial computing and data-driven production.

2.9.18 How to Listen Historically

A useful history is audible. Compare technologies with the same dry sources at matched loudness and similar nominal decay. Use a click for onset and density, speech for consonant masking, snare for transient envelope, piano for harmonic overlap, and a sustained chord for modulation and modal stability. Do not attempt to force every device to the same bandwidth or exact T_{60} if doing so removes its identity.

Listen in layers. First isolate the wet signal. Identify pre-delay, initial burst, discrete repetitions, density growth, tonal change, pitch motion, noise, and termination. Then restore the dry source and judge distance, blend, intelligibility, and groove. Finally place the return inside a complete arrangement. A plate that sounds bright alone may position a vocal perfectly. A realistic hall may disappear under dense instrumentation. A cheap early digital program may create a superior drum envelope because its limited bandwidth protects the mix.

The historical comparison should ask what each medium makes easy:

- A **chamber** makes transducer, air, geometry, and shared spatial identity unavoidable.
- A **plate** makes dense bright extension and mechanical damping easy.
- A **spring** makes compact dispersive splash and performative excitation easy.
- A **tape or magnetic delay system** makes recirculation, saturation, and progressive bandwidth loss easy.
- A **Schroeder or FDN algorithm** makes continuous parameter control, modulation, multiband decay, and extreme duration easy.
- A **convolution engine** makes one measured linear response repeatable and portable.
- A **physical model** makes causal material or geometry parameters available.
- A **learned model** makes interpolation and data-conditioned behavior possible, while demanding stronger provenance and evaluation.

No category owns realism. A chamber can sound unnatural if badly miked. A plate can feel more appropriate than a sampled hall. A sparse digital algorithm can sound synthetic and still be musically exact. A convolution response can be physically measured yet spatially wrong for a new source position. A neural model can be statistically convincing while obscuring what place or device it represents. The correct question is always: which cues survive, which are transformed, and what role does that transformation play in the music?

2.9.19 Historical Lessons for verbx

verbx inherits this entire history rather than selecting one winner. Its convolution engine belongs to the impulse-response tradition. Its algorithmic engine belongs to Schroeder, Moorer, FDN, scattering, and modulation traditions. Its extreme tails inherit feedback practice. Its reverse, gate, shimmer, freeze, ducking, and automation controls inherit studio techniques that treated the return as an arrangement layer. Its multichannel and Ambisonic features extend the long search for convincing spatial distribution. Its analysis and JSON reporting answer a newer requirement: make complex processing reproducible and machine-readable.

This inheritance creates obligations. Preset names should not substitute for documented behavior. A "plate" should explain what plate-like features are being modeled. A "chamber" should not imply a measured building unless an impulse response and provenance support that claim. Extreme feedback must be bounded. Sample-rate changes must preserve intended delay times. Multichannel routing must declare channel order. Neural augmentation must record model and data provenance. Historical vocabulary is useful only when it clarifies rather than markets.

Artificial reverberation developed by moving acoustic memory from architecture into rooms used as processors, then into vibrating matter, magnetic and electronic delay, digital memory, software, spatial metadata, and learned representations. At every stage, engineers exchanged one set of constraints for another. The chamber offered physical complexity but little portability. The plate offered density but not literal geometry. The spring offered compactness with dispersion. Early digital units offered control under severe memory limits. Convolution offered specificity under fixed-state assumptions. Plug-ins offered abundance at the risk of incoherence. Neural systems offer interpolation at the risk of opacity.

The history matters because those tradeoffs are still present in every reverb decision. Choosing a technology means choosing how energy will be stored, mixed, lost, controlled, documented, and heard. A great reverberator does not erase its medium. It makes the medium musically intelligible.

2.9.20 Sources and Further Historical Reading

The technical and historical claims in this section should be read alongside the book's research bibliography. Particularly useful starting points are Olson and Bleazey's "Synthetic Reverberator", Schroeder and Logan's "'Colorless' Artificial Reverberation", Schroeder's "Natural Sounding Artificial Reverberation", Blesser's "An Interdisciplinary Synthesis of Reverberation Viewpoints", the AES historical Reverb Collection, Laurens Hammond's spring-reverberation patent, EMT's 1976 Courier 26, Lexicon's product-history account, and AMS's company history. The peer-reviewed survey "A History of Audio Effects" provides a broader effects chronology, while the current research bibliography documents FDN, convolution, spatial, physical-modeling, and neural developments in depth.

2.10 Why verbx Sounds Different

No single feature explains a reverb's sound. What matters is how precision, topology, delay distribution, diffusion, spectral loss, time variation, dynamics, and routing

interact. verbx is designed to keep those dimensions explicit rather than compressing them into one "size" macro.

2.10.1 Exact Long-Tail Gain Calibration

For delay line i with duration d_i seconds and target T_{60}, verbx computes

$$g_i = 10^{-3d_i/T_{60}}.$$

Every line receives a gain appropriate to its own delay. A single shared gain would make longer lines decay differently from shorter lines and would distort the modal envelope. At short RT60 values the error may pass unnoticed. At 120 or 360 seconds, it becomes a slow drift in balance that the ear hears as coloration.

The algorithmic path uses 64-bit internal precision for the long-lived state. When loop gains approach one, rounding error persists for many circulations. Double precision does not make an unstable design stable, but it preserves the intended loss and reduces the chance that the tail becomes a numerical artifact before it becomes silence.

2.10.2 Matrix Family as a Musical Texture Control

The feedback matrix is not a cosmetic option. It determines how state components exchange energy. A dense uniform matrix can make the field smooth and neutral; a sparse or graph-derived matrix can preserve more audible pathways; a slowly time-varying matrix can reduce stationary ringing and create subtle motion.

Audition matrix families with a click, then with a sparse chord, then with a dense mix. The click reveals periodicity. The chord reveals modal coloration. The dense mix reveals whether motion becomes blur. Keep RT60, delay count, damping, and wet level fixed so the comparison isolates topology.

```
for matrix in hadamard householder circulant random_orthogonal tv_unitary; do
  verbx render examples/audio/dry_click.wav "/tmp/click_${matrix}.wav" \
    --engine algo --rt60 8 --wet 1 --dry 0 \
    --fdn-lines 16 --fdn-matrix "$matrix" --no-progress
done
```

The best matrix is source-dependent. Uniformity is valuable for exposed classical material; controlled irregularity can be expressive on electronic percussion; slow motion can keep a minute-long tail alive without obvious pitch modulation.

2.10.3 Diffusion Is an Envelope, Not a Quality Score

"More diffusion" is often treated as "better reverb," but diffusion changes the attack of the wet field. A dense immediate onset can support vocals and pads while softening percussion. A slower buildup can preserve transients and imply a larger volume. The correct density is the one that makes the room speak at the right moment in the phrase.

Use allpass stages and bloom deliberately. Allpasses scatter the excitation before it enters the late field. Bloom shapes how the wet envelope rises. They can create related perceptions through different mechanisms. If an attack feels detached, reduce pre-delay or accelerate buildup. If it loses definition, reduce front-end diffusion before shortening the entire tail.

2.10.4 Time Variation Without Obvious Chorus

Stationary delay networks have stationary modes. On a sustained source, those modes can emerge as metallic pitches. Modulation and time-varying unitary matrices distribute energy across changing modal bases. The design goal is often to move slowly enough that the listener hears a living space rather than a pitch effect.

`--fdn-tv-rate-hz` controls update rate and `--fdn-tv-depth` controls how far the matrix moves from its base state. At 0.05–0.3 Hz, motion unfolds over several seconds. That is appropriate for long ambient tails. Faster rates can become audible animation and may be exactly right for sound design.

```
verbx render drone.wav /tmp/drone_moving_room.wav \
  --engine algo --rt60 45 --fdn-lines 32 --fdn-matrix tv_unitary \
  --fdn-tv-rate-hz 0.12 --fdn-tv-depth 0.08 \
  --wet 0.88 --dry 0.18 --target-peak-dbfs -2
```

2.10.5 Extreme RT60 as a New Compositional Dimension

At 0.4 seconds, reverb behaves like surface and room size. At four seconds, it becomes phrase overlap and architectural scale. At forty seconds, it becomes harmonic memory. At 360 seconds, it becomes a slow layer whose beginning and end may belong to different sections of a piece. At 3600 seconds, "room simulation" is no longer the useful metaphor. The processor is a bounded recursive instrument.

Long duration changes orchestration. Low notes occupy more future time because their energy often decays slowly. Dense chords accumulate until the field approaches a spectral average. Silence becomes

the only way to hear internal evolution. A composer must decide what is allowed to enter the field, not merely how much of it is returned.

Start extreme experiments with floating-point output and conservative wet gain:

```
verbx render source.wav /tmp/extreme_field.wav \
  --engine algo --rt60 120 --wet 0.82 --dry 0.12 \
  --fdn-lines 32 --fdn-matrix tv_unitary \
  --fdn-tv-rate-hz 0.08 --fdn-tv-depth 0.06 \
  --lowcut 70 --target-peak-dbfs -3 --out-format float32 \
  --json-out /tmp/extreme_field.json
```

Do not judge the result after ten seconds. Render enough silence to hear the tail's middle and end. Long-tail quality includes how the field leaves, not only how it begins.

2.10.6 Damping, Air, and the Meaning of Warmth

Warmth is not simply a low-pass filter after the reverb. Post-EQ changes what the listener hears; in-loop damping changes what continues to circulate. A high-frequency component removed inside the loop cannot return on the next pass. Over time, this creates a changing spectrum similar to repeated interaction with absorptive surfaces.

Use post-EQ for mix placement and loop damping for decay design. If the return is bright at the onset but should darken naturally, loop filtering is appropriate. If the decay is correct but conflicts with a vocal, post-EQ may solve the mix without changing room behavior. Multiband RT60 targets make this distinction explicit.

2.10.7 Ducking and the Separation of Space from Articulation

Ducking attenuates the wet field while the source is active and releases it into gaps. It allows a long decay to coexist with clear foreground articulation. This is not merely a corrective technique. The release envelope can become rhythmic counterpoint: the room inhales after each phrase.

Set attack fast enough to protect the transient but not so fast that the return clicks or pumps unnaturally. Set release in relation to syllable, beat, or phrase. A 120 ms release may create a rhythmic pulse on drums; 400 ms may wait for a vocal line to open. The floor parameter leaves a minimum ambience so the source does not appear to jump between dry and wet rooms.

2.10.8 Shimmer, Bloom, Freeze, and Reverse as Re-Compositions of the Tail

Shimmer changes register. Bloom changes onset envelope. Freeze changes finite decay into sustain. Reverse changes the direction of the energy envelope so the room anticipates an event rather than remembering it. These modes are most convincing when written into the musical form.

Use shimmer at a harmonic pivot, bloom before an orchestral entrance, freeze across a section boundary, or reverse reverb as an anacrusis into one important word. When every event receives every transformation, the special behavior becomes a static texture. When selected events receive it, reverb becomes orchestration.

2.10.9 Gated Reverse Reverb: A Room That Arrives, Then Vanishes

Gated reverse reverb combines two envelope operations. Reverse reverb makes wet energy rise toward a destination transient; a gate or explicitly drawn gain window limits that rise to a deliberate interval and closes it at, just before, or just after the transient. The result is not simply a backward tail. It is a bounded anticipatory gesture with a composed beginning, trajectory, and ending. Silence around the gesture is part of the effect.

Three related processes should not be confused:

- **Forward gated reverb** begins after the dry event and truncates the normal decay.
- **Ungated reverse reverb** swells into the event but may expose a long, noisy, or harmonically indiscriminate lead-in.
- **Gated reverse reverb** swells toward the event only inside a controlled pre-event window, then closes decisively enough to articulate the arrival.

Gate placement determines the result. A gate before the reverb controls which dry events excite the processor, but the accepted events can still generate complete tails. A gate after the wet processor shapes the audible reverberant field itself and creates the characteristic abrupt boundary. For the canonical effect, key the detector from the dry source while applying gain reduction only to a 100-percent-wet return. Keep the dry transient on a separate path.

For a true offline construction, isolate the destination event, reverse it, render a wet tail, reverse the wet render, trim it to the desired lead time, and align its maximum or chosen cutoff with the original event. Then shape the printed return with clip gain or a linked gate. This two-reversal method has access to future context and can create a genuinely noncausal anticipation. It is the reference against which realtime approximations should be judged.

The current verbx plug-in Reverse button is intentionally different: it is a zero-lookahead, transient-triggered reverse-style swell. It cannot emit sound before a transient that the host has not delivered. To create a true gated anticipation with the current plug-in, print the wet Reverse return, move that stem earlier, and gate or draw its envelope in the DAW. A future bounded-capture implementation could automate this operation by buffering a fixed window, but its latency would be at least the active capture duration and would have to be reported to the host.

Tempo provides a useful first estimate for the reverse window. For tempo B in beats per minute and a lead spanning N beats,

$$T_{\text{lead}} = \frac{60N}{B} \text{ seconds.}$$

At 120 BPM, an eighth-note lead is 250 ms, a quarter-note lead is 500 ms, and a half-note lead is 1 second. These values establish metrical intent rather than final sound. Start the wet window 5 to 30 ms late or early relative to the grid when consonants, bow noise, or drum attacks require a more convincing perceptual landing.

A practical vocal setup uses a 500 to 1,500 ms wet print, high-pass filtering near 120 to 250 Hz, and a gate that closes between 10 ms before and 30 ms after the dry consonant. Drums usually tolerate shorter windows, stronger low-mid energy, and a harder close. Pads, cymbals, and orchestral transitions can use several beats, but the gate should still reveal a clear formal boundary rather than merely hiding the beginning of a long wash. A 5 to 20 ms terminal fade prevents a digital click while retaining the sensation of an abrupt cutoff.

Threshold alone is rarely sufficient. Expose or automate lead duration, detector source, attack, hold, release, hysteresis, range, wet bandwidth, and endpoint offset. Hysteresis prevents a noisy reverse tail from chattering around threshold. Range lets a small room residue survive instead of forcing absolute silence. A sidechain high-pass filter keeps plosives and kick energy from opening the gate prematurely, while a low-pass filter can make the approach dark and leave the destination transient spectrally unobstructed.

For stereo and immersive work, derive one gate envelope from a linked detector and apply it coherently to all wet channels. Independent channel gates can make the image twitch, collapse, or arrive from the wrong direction as thresholds are crossed at different times. The reverse field may widen or move during its approach, but its endpoint should remain stable under stereo fold-down and binaural rendering. Keep LFE empty unless the gesture specifically requires low-frequency impact, and inspect bass-managed monitoring before assuming low-frequency wet energy was written into LFE.

Common failures are diagnostic. If the arrival sounds late, align the envelope endpoint rather than the file boundary. If the dry attack loses force, close the wet gate earlier, shorten the terminal fade, or reduce energy between 2 and 6 kHz. If the swell chatters, increase hysteresis or replace level detection with a drawn window. If every phrase feels predicted, reserve the effect for structural entrances, withheld downbeats, cadences, and selected words. Gated reverse reverb is strongest when it changes musical causality for one event, then returns silence to the foreground.

2.10.10 Multichannel Routing and the Shape of the Late Field

Stereo width is only one spatial dimension. In multichannel work, early energy can localize toward the source while late energy expands to sides, rear, or height. A true $M \times N$ convolution matrix can preserve measured cross-channel relationships. An algorithmic FDN can project one internal state into several output channels with decorrelated but coherent returns.

Avoid creating width by independent random reverbs on every channel. The result may be decorrelated yet lack one shared acoustic identity. Prefer a common feedback state with carefully designed input and output projections. Verify fold-down behavior, channel correlation, and level after decoding.

The stereo projection flowgraph below shows a shared eight-component FDN state feeding two different signed projection vectors. It also keeps dry/wet summing outside the recursive state so mix automation cannot alter decay stability.

Figure 2-90 presents Stereo output projection from a shared eight-line FDN state through normalized signed vectors and independent dry-wet output mixers and establishes the visual reference for the discussion that follows.

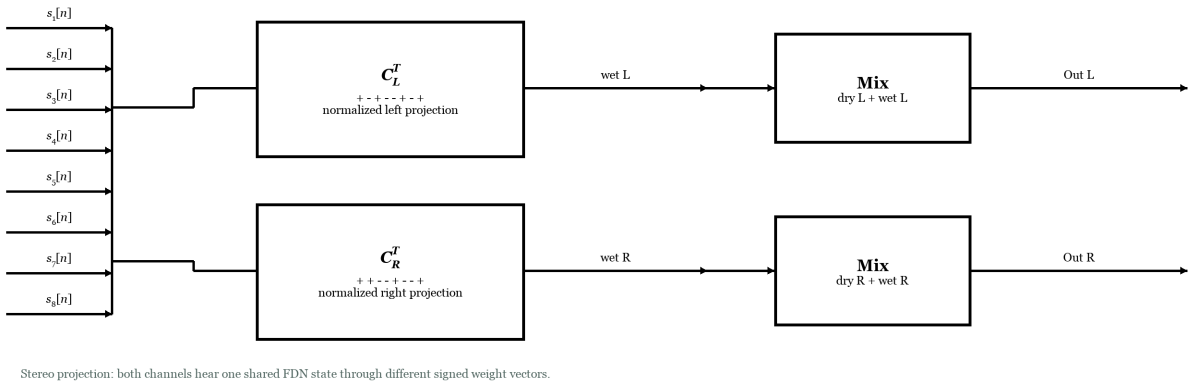

Figure 2-90: Stereo output projection from a shared eight-line FDN state through normalized signed vectors and independent dry-wet output mixers

How to read this flowgraph. Every state component contributes to both wet channels, but the signs in C_L^{T} and C_R^{T} differ. The channels therefore share one decay history while emphasizing different modal combinations. Normalization prevents channel count or coefficient choice from creating an unintended level jump. The final mixers add the projected wet signals to their dry counterparts after recursion, preserving one acoustic identity and predictable fold-down behavior.

The following unit-circle map separates what is shared from what is channel-specific. The internal pole ring belongs to the common FDN state, while the plotted zeros represent one output projection; the other channel generally has a different zero pattern.

Figure 2-91 presents Unit-circle map for stereo FDN projection, with shared modal poles and output-dependent transmission zeros and establishes the visual reference for the discussion that follows.

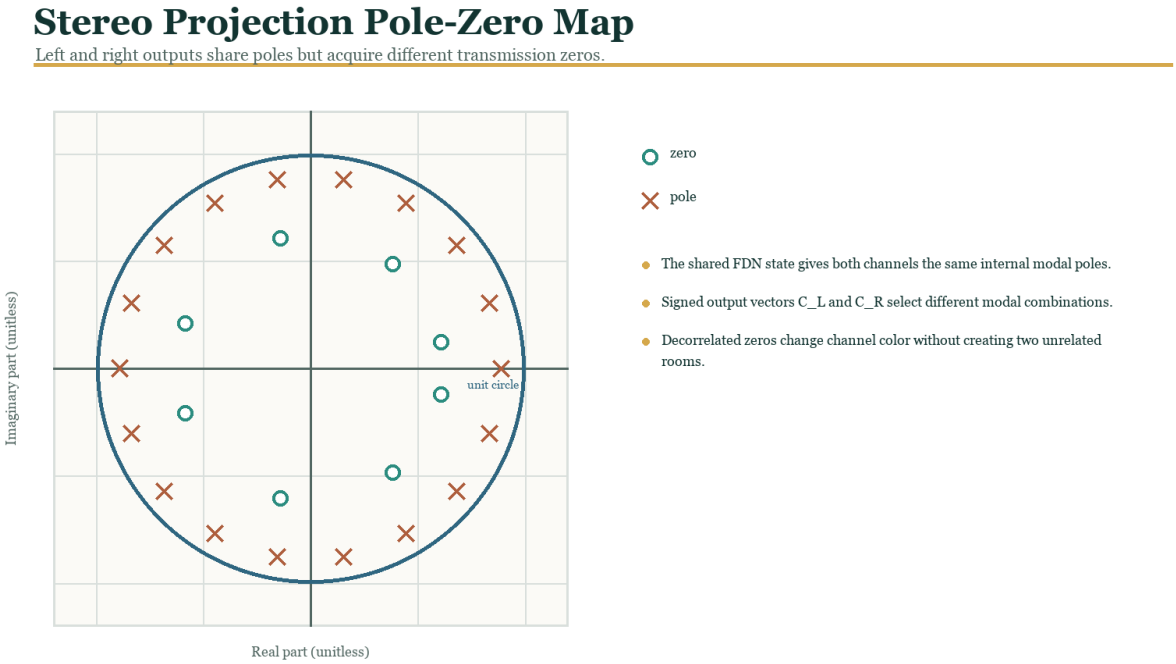

Figure 2-91: Unit-circle map for stereo FDN projection, with shared modal poles and output-dependent transmission zeros

Changing C_L and C_R alters which modes cancel at each output without constructing two unrelated acoustic systems. This is the linear-systems reason signed projections can create width while preserving a coherent room. Mono fold-down combines the two transfer numerators, so it must be checked explicitly: a left-right zero that sounds spacious in stereo can become a broad cancellation after summation.

The final companion plot makes the projection dependency concrete. It is one channel's response; a different signed output vector retains the shared poles while changing the locations and depths of transmission cancellations.

Figure 2-92 presents Peak-normalized magnitude response of one stereo FDN output projection, with frequency in cycles per sample and magnitude on a vertical decibel scale and establishes the visual reference for the discussion that follows.

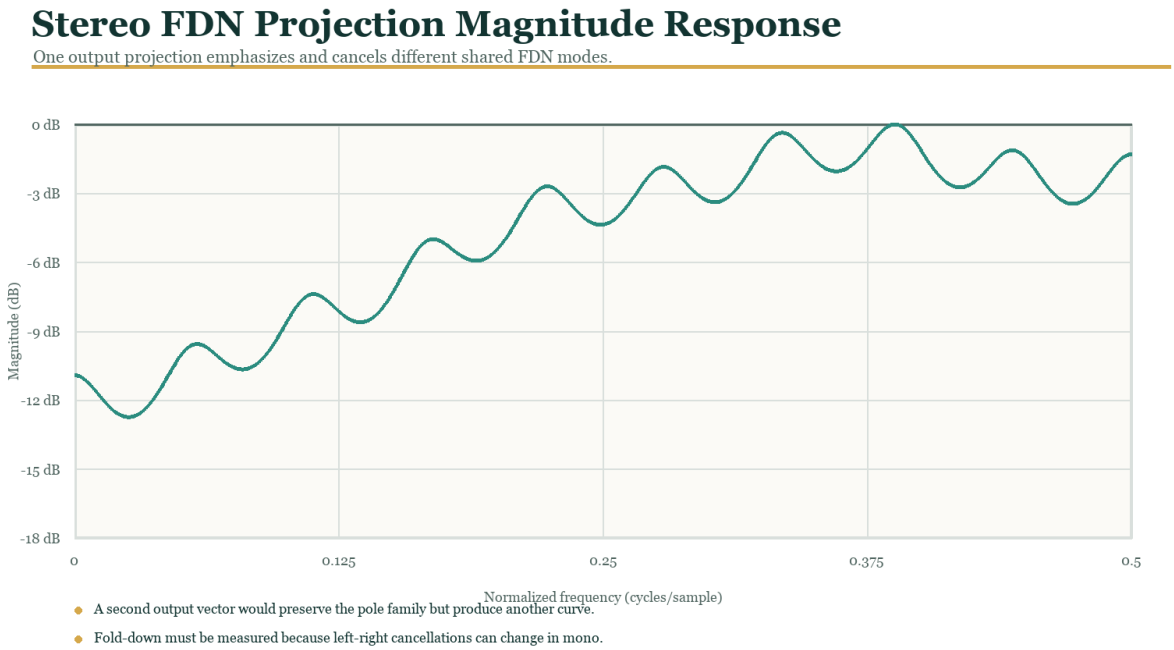

Figure 2-92: Peak-normalized magnitude response of one stereo FDN output projection, with frequency in cycles per sample and magnitude on a vertical decibel scale

2.11 The Science and DSP of Dereverberation

Dereverberation asks a harder question than reverberation synthesis. A reverb processor starts with a comparatively clean signal and a known algorithm or impulse response; dereverberation starts with only the mixture that reached a microphone and tries to infer which part belonged to the source before the room prolonged it. The dry performance, room response, source position, microphone response, background noise, and often even the number of active sources are unknown. Many different combinations of those latent quantities can produce nearly the same recording. Dereverberation is therefore an **ill-posed inverse problem**, not a literal undo button.

That distinction sets the standard for honest use. A successful processor can reduce late energy, restore temporal contrast, improve speech intelligibility, and make a close source feel less distant. It cannot generally recover the exact pressure waveform that would have been recorded in an anechoic chamber. The most useful scientific question is consequently not “Did all reverb disappear?” but “Which acoustic component was estimated, under which assumptions, at what cost in distortion, spatial fidelity, and latency?”

2.11.1 The Forward Acoustic Model

For one stationary source and one microphone, the standard discrete-time model is

$$y[n] = \sum_{q=0}^{L_h-1} h[q]x[n-q] + v[n] = (h * x)[n] + v[n],$$

where $x[n]$ is the unknown dry source, $h[n]$ is the room impulse response, $v[n]$ is additive noise, and $y[n]$ is the observed microphone signal. The convolution is long: a 1.5 s response at 48 kHz contains 72,000 samples before accounting for an even longer noise floor. With S sources and M microphones, the model becomes

$$y_m[n] = \sum_{s=1}^{S} (h_{m,s} * x_s)[n] + v_m[n], \qquad m = 1, \dots, M.$$

Every source-to-microphone path has its own response $h_{m,s}[n]$. If a talker moves, a door opens, or a handheld microphone rotates, that response is time varying rather than a fixed convolution. The familiar linear time-invariant model remains valuable, but its limits should be visible whenever a method is evaluated on real capture.

For dereverberation, the room response is commonly partitioned at an early/late boundary n_e:

$$h[n] = h_{\text{early}}[n] + h_{\text{late}}[n],$$

$$y[n] = x_{\text{early}}[n] + r_{\text{late}}[n] + v[n].$$

Here $x_{\text{early}}[n] = (h_{\text{early}} * x)[n]$ is usually the desired signal, not merely the mathematically dry source. It retains the direct arrival and a short early window because those components support loudness, localization, source width, and naturalness. The interference term $r_{\text{late}}[n]$ contains delayed replicas that smear later phonemes, notes, and transients. The boundary is task dependent: speech systems often preserve tens of milliseconds, while music restoration may preserve more of a room's early signature.

The following figure places the unknown quantities around the one observation actually available to a blind estimator. Its non-linear layout is important: neither the room nor the source lies in a simple known processing chain at restoration time.

Figure 2-93 presents Dereverberation formulated as inference of a dry or early source from a reverberant, noisy observation using acoustic assumptions and regularization and establishes the visual reference for the discussion that follows.

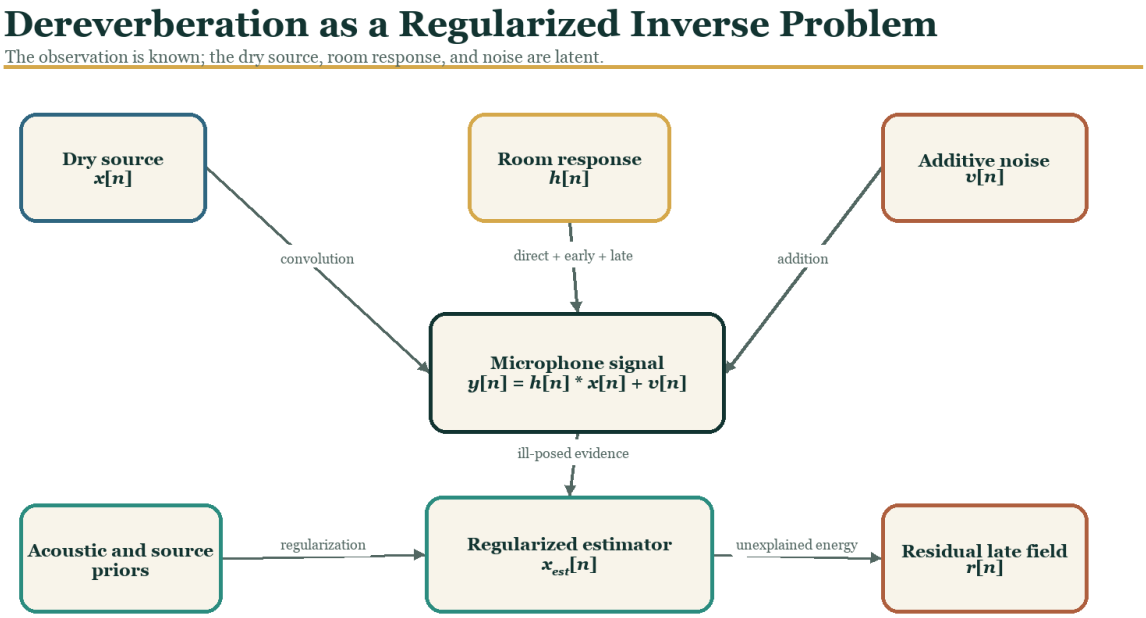

Figure 2-93: Dereverberation formulated as inference of a dry or early source from a reverberant, noisy observation using acoustic assumptions and regularization

How to read this figure. The upper paths describe physical capture; the lower paths describe inference. Only $y[n]$ is observed. A processor must supply assumptions about decay, source statistics, sparsity, spatial covariance, or training data before it can separate an estimate from the residual. Stronger assumptions can produce stronger suppression, but they also create stronger failure modes when the recording violates them.

2.11.2 Why Exact Inverse Filtering Usually Fails

If $h[n]$ were perfectly known, one might propose the inverse filter $G(z) = 1/H(z)$ so that $G(z)H(z) = 1$. Three problems make that expression a poor general solution. First, measured room responses contain deep spectral notches. Dividing by a small $|H(e^{j\omega})|$ gives enormous gain and amplifies microphone noise, quantization, and response-estimation error. Second, a room response is commonly nonminimum phase. Its inverse may be unstable if zeros lie outside the unit circle, or noncausal if a stable inverse is constructed by moving energy before the direct arrival. Third, a room response is position specific. A filter that inverts one source/receiver pair can worsen a signal captured a few centimeters away.

A regularized frequency-domain inverse makes the compromise explicit:

$$G_\lambda(e^{j\omega}) = \frac{H^*(e^{j\omega})}{|H(e^{j\omega})|^2 + \lambda(\omega)}.$$

When $|H|^2$ is large relative to λ, the expression approaches an inverse. Near a notch, the regularizer prevents explosive gain. The function $\lambda(\omega)$ may encode expected noise, uncertainty, perceptual weighting, or a hard robustness floor. This is closely related to Wiener deconvolution, where the regularizer is a ratio of noise and source power spectra. It also exposes the unavoidable bias/variance tradeoff: more regularization leaves more room coloration; less regularization may reduce coloration on one measurement while producing noise and instability elsewhere.

Multichannel inverse filtering can be better conditioned because several microphone paths need not share the same zeros. The multiple-input/output inverse theorem method seeks FIR filters whose combined channel response approximates a delayed impulse. Miyoshi and Kaneda's formulation is foundational here. Yet it still needs measured or estimated room responses, accurate channel synchronization, and a sufficiently stationary geometry. It is best understood as **room-response equalization or shortening with known transfer functions**, not as a universal blind dereverberator.

Blind methods avoid claiming that $h[n]$ is known. They estimate only a useful component: late-reverberant power, a delayed linear prediction, a time-frequency mask, a spatial filter, or a learned mapping. This narrower goal is why practical dereverberation can work despite the impossibility of exact blind inversion.

2.11.3 Statistical Science of the Late Field

After sufficient mixing time, a room's late field is often approximated as diffuse, noise-like, and exponentially decaying. If T_{60} is the reverberation time, an idealized amplitude envelope is

$$a(t) = 10^{-3t/T_{60}},$$

and its energy envelope is

$$a^2(t) = 10^{-6t/T_{60}} = e^{-t/\tau_E}, \qquad \tau_E = \frac{T_{60}}{6 \ln 10}.$$

This does not say that every late sample follows a smooth exponential. Individual samples remain stochastic and interference produces fluctuations. The claim is about expected energy over an ensemble or local time-frequency region. Real rooms also decay at different rates by frequency, and low-frequency modes may violate diffuse-field assumptions entirely. Erkelens and Heusdens model this frequency dependence explicitly; Habets develops the statistical framework into practical late-reverberation estimators.

In an STFT indexed by frequency bin k and frame l, write

$$Y_{k,l} = X_{k,l} + R_{k,l} + V_{k,l}.$$

A simple late-power tracker uses delayed observations as evidence for energy that may still be arriving from earlier source frames:

$$\widehat{\lambda}_{r,k,l} = \alpha_k \widehat{\lambda}_{r,k,l-1} + (1 - \alpha_k) c_k |Y_{k,l-D}|^2.$$

The delay D protects the direct and early portion; the coefficient α_k smooths the estimate; c_k maps delayed energy to an expected residual according to decay rate. A longer T_{60} implies slower forgetting. Frequency-dependent α_k or c_k allows a bright room, dark room, or strongly modal low end to be represented more honestly than one broadband constant.

This model fails predictably. A sustained organ tone looks like persistent source energy and persistent late energy at once. A noise burst can be mistaken for a diffuse tail. A new note arriving while the old note decays violates the simple delayed-cause narrative. Nonstationary background noise biases the tail estimate upward. These are not merely tuning problems; they are ambiguities in the evidence. A robust implementation therefore bounds gain, smooths estimates, preserves a residual floor, and reports its assumptions.

2.11.4 Spectral Suppression and Wiener Estimation

Once desired and late-reverberant power estimates are available, a Wiener-style gain can be formed as

$$G_{k,l}^{\mathrm{W}} = \frac{\widehat{\lambda}_{x,k,l}}{\widehat{\lambda}_{x,k,l} + \widehat{\lambda}_{r,k,l} + \widehat{\lambda}_{v,k,l}} = \frac{\xi_{k,l}}{1 + \xi_{k,l}},$$

where $\xi_{k,l}$ is an a priori desired-to-interference ratio. The estimate is $\widehat{X}_{k,l} = G_{k,l}^{\mathrm{W}} Y_{k,l}$. A spectral-subtraction alternative uses

$$|\widehat{X}_{k,l}|^p = \max\left(|Y_{k,l}|^p - \beta\widehat{\lambda}_{r,k,l}, G_{\min}^p |Y_{k,l}|^p\right),$$

with $p = 1$ for magnitude subtraction or $p = 2$ for power subtraction. The oversubtraction factor β controls aggressiveness and $G_{\min}$ prevents complete spectral holes. Both families normally reuse the observed phase because estimating clean phase is much harder. That phase reuse is one reason a strongly processed signal can remain smeared or phaselike even after its magnitude envelope looks cleaner.

The following figure shows the estimator as two evidence paths meeting at a bounded gain, rather than as a magical “remove room” block.

Figure 2-94 presents Statistical late-reverberation suppression using delayed spectral evidence, desired-signal evidence, a bounded time-frequency gain, and overlap-add resynthesis and establishes the visual reference for the discussion that follows.

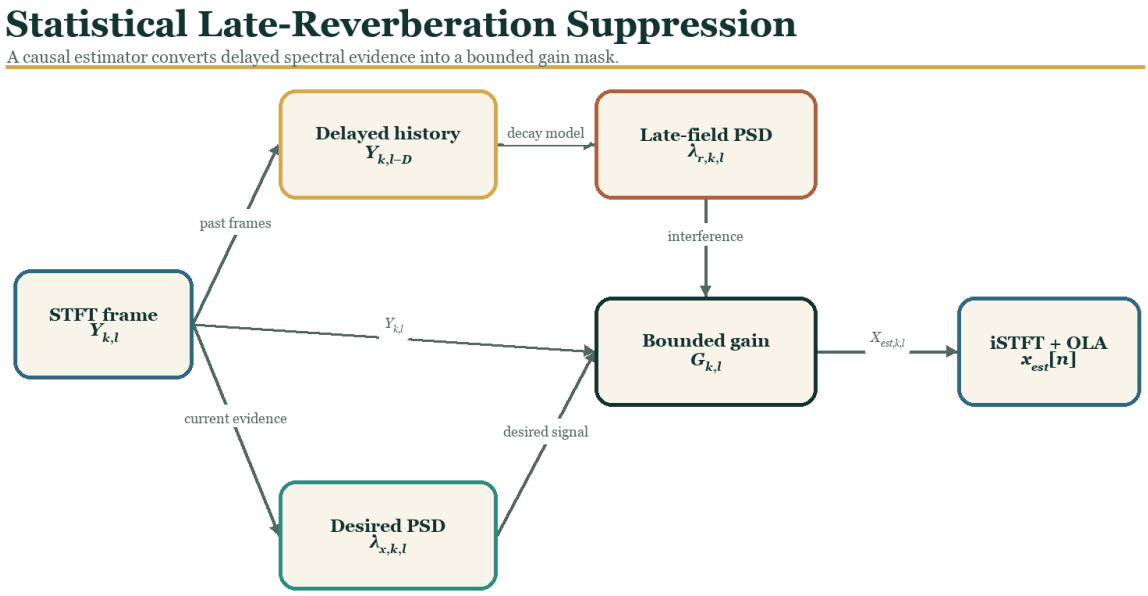

Figure 2-94: Statistical late-reverberation suppression using delayed spectral evidence, desired-signal evidence, a bounded time-frequency gain, and overlap-add resynthesis

How to read this figure. The upper branch estimates interference from past frames; the lower branch estimates what should survive now. Their ratio drives the gain. The unprocessed STFT also reaches the gain because suppression scales observed complex bins rather than synthesizing an unrelated waveform. Resynthesis must use a compatible window and hop so the overlap-add sum remains level and free of modulation.

The engineering details determine whether the equations sound acceptable. Frequency-only smoothing reduces isolated "musical noise" tones but can blur narrow harmonics. Time smoothing reduces pumping but lets late energy leak through attacks. A gain floor protects room tone and consonant texture but limits maximum reduction. Attack/release asymmetry can open quickly for new direct sound while closing slowly on the estimated tail. Stereo-linked gains protect the image; independent gains can remove more channel-specific energy but make a centered source wander.

This is the family implemented by the current `verbx dereverb` command and the live dereverb front end. It is a deterministic STFT late-tail suppressor with `wiener` and `spectral_sub` gain laws. It is **not** currently a WPE solver, measured-RIR inverse, source separator, or neural dry-signal generator. The distinction matters when interpreting its controls:

Table 2-8: Spectral Suppression and Wiener Estimation: reference table

verbx control	Estimation role	Audible risk when pushed
`--strength`	Scales estimated late-field removal	Hollow tone, transient thinning, modulation
`--floor`	Sets $G_{\min}$, the minimum residual gain	Higher values leave tail; lower values expose musical noise
`--window-ms`	Sets time/frequency analysis resolution	Long windows smear attacks; short windows weaken bass resolution
`--hop-ms`	Sets estimator update and overlap cadence	Large hops make gains coarse; incompatible hops fail validation
`--tail-ms`	Sets the late-energy tracking horizon	Too long mistakes sustain for room; too short misses slow decay
`--pre-emphasis`	Changes weighting before estimation	Excess values exaggerate sibilance and sensor noise
`--max-atten-db`	Caps maximum suppression	A high cap increases artifacts; a low cap limits clarity gain
`--stereo-link`	Couples channel gain decisions	Linking preserves image but may retain asymmetric room energy
`--mix`	Blends processed and latency-aligned original	Partial mix restores naturalness and some reverberation

The safest strategy is to begin with moderate strength, a nonzero floor, and a partial mix. Increase reduction while listening to sibilants, cymbal decays, piano attacks, and stereo ambience, not only to steady vowels. A setting that improves a spectrogram may still damage musical phrasing.

2.11.5 Delayed Linear Prediction and WPE

Weighted prediction error (WPE) approaches late reverberation from a different direction. Instead of estimating a scalar late-power spectrum and attenuating bins, it models late reverberation as a delayed linear prediction from prior STFT frames. The key insight is that the direct and early components belong close to the current source event, while the late field remains correlated with older microphone observations.

For M microphones, stack the current observations into $y_{t,f} \in \mathbb{C}^M$. Construct a delayed history vector $\bar{y}_{t-\Delta,f}$ from K prediction taps beginning Δ frames in the past. A selected output channel can then be modeled as

$$Y_{t,f} = X_{t,f} + g_f^{\mathsf{H}} \bar{y}_{t-\Delta,f},$$

so the dereverberated estimate is the prediction error

$$\widehat{X}_{t,f} = Y_{t,f} - g_f^{\mathsf{H}} \bar{y}_{t-\Delta,f}.$$

The prediction delay Δ is not an implementation nuisance. It defines what the algorithm protects. If Δ is too small, the predictor explains and subtracts direct or useful early energy. Attacks become dull and source color changes. If Δ is too large, a portion of harmful late energy lies outside the model and survives. Prediction order K similarly trades model capacity against computation, adaptation speed, and overfitting.

Ordinary least squares is biased toward high-energy frames. WPE assumes that the desired speech or source coefficient is a zero-mean complex Gaussian with time-varying variance $\lambda_{t,f}$ and minimizes

$$J_f(g_f) = \sum_t \frac{\left|Y_{t,f} - g_f^{\mathsf{H}} \bar{y}_{t-\Delta,f}\right|^2}{\lambda_{t,f}}.$$

For fixed variances this is weighted least squares. For fixed predictor coefficients, the residual supplies an updated variance estimate. Batch WPE alternates those steps, usually for several iterations. The inverse-variance weighting prevents high-power source frames from dominating the filter and arises naturally from maximum-likelihood estimation under the time-varying Gaussian model described by Nakatani and colleagues.

Define the weighted covariance and cross-correlation

$$R_f = \sum_t \frac{\bar{y}_{t-\Delta,f} \bar{y}_{t-\Delta,f}^{\mathsf{H}}}{\lambda_{t,f}}, \qquad p_f = \sum_t \frac{\bar{y}_{t-\Delta,f} Y_{t,f}^*}{\lambda_{t,f}}.$$

Then $g_f = R_f^{-1} p_f$, although a numerical implementation should solve the linear system rather than form an explicit inverse. Diagonal loading, condition-number checks, and finite-value guards are essential when the history is poorly excited or channels are nearly redundant.

The following figure separates WPE's undelayed reference, delayed predictor, and iterative variance path. That feedback relation is the reason WPE is more than a fixed comb-cancellation filter.

Figure 2-95 presents WPE delayed-prediction topology in which past multichannel frames estimate the late field, the current reference supplies the desired event, and residual variance reweights the predictor and establishes the visual reference for the discussion that follows.

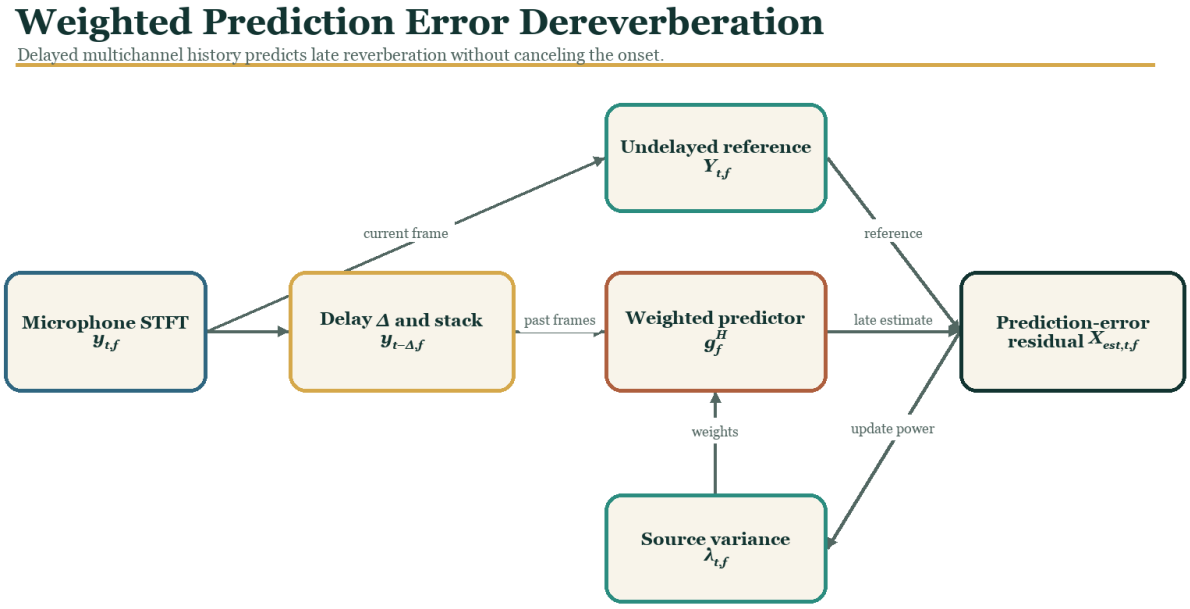

Figure 2-95: WPE delayed-prediction topology in which past multichannel frames estimate the late field, the current reference supplies the desired event, and residual variance reweights the predictor

How to read this figure. The middle horizontal route predicts late energy from frames older than Δ; the upper route protects the undelayed observation; the lower route closes the statistical iteration. Subtraction occurs only after delayed prediction, so the topology explicitly distinguishes a current onset from its older reverberant history.

Single-channel WPE can work because one channel still contains temporal predictability, but multichannel WPE has a richer basis and can exploit spatial diversity. Batch WPE sees the whole utterance and can iterate toward a strong estimate; online WPE must update statistics recursively and accept an adaptation lag. Recursive least squares, recursive covariance updates, or Kalman formulations reduce memory and delay but introduce forgetting factors and tracking tradeoffs. Caroselli and colleagues demonstrate adaptive WPE for large-scale recognition, while Nakatani and Kinoshita formulate frame-by-frame low-latency joint processing.

WPE also has characteristic failure modes. A predictor can suppress sustained harmonic material because periodic music is itself predictable. Rapid movement invalidates its filter before adaptation catches up. Background noise corrupts covariance and variance estimates. Very long filters become computationally expensive: for each frequency, the history dimension grows with MK, and a dense solve scales steeply with that dimension. Practical systems therefore regularize, limit order, update less frequently than every sample, and sometimes estimate source variance with a neural network.

2.11.6 Multichannel Spatial Dereverberation

Multiple microphones do more than provide more samples. They observe different mixtures of the same direct source and reflected field. Direct sound from a compact source has a structured steering vector; a sufficiently diffuse late field has a different spatial covariance; directional interferers differ again. Spatial filtering can exploit those differences even when their spectra overlap.

For STFT observation vector $y_{t,f}$, the spatial covariance is

$$R_{yy,f} = \mathbb{E}\{y_{t,f} y_{t,f}^{\mathsf{H}}\}.$$

If a desired steering vector d_f and interference covariance $R_{vv,f}$ are available, an MVDR beamformer uses

$$w_{\mathrm{MVDR},f} = \frac{R_{vv,f}^{-1} d_f}{d_f^{\mathsf{H}} R_{vv,f}^{-1} d_f}.$$

The denominator enforces unity response in the desired direction while minimizing output power attributed to interference. A multichannel Wiener filter relaxes the distortionless constraint to minimize mean-square error. Weighted power minimization distortionless response (WPD) convolutional beamforming unifies delayed dereverberation and spatial filtering in one spatio-temporal objective.

These methods are complementary rather than interchangeable. WPE attacks temporal predictability in the late tail. Beamforming attacks spatial directions and covariance structure. Applying WPE before a spatial filter can shorten the effective response and improve covariance estimates; joint formulations can avoid a suboptimal fixed cascade. Drude and colleagues report that integrating WPE with neural beamforming outperforms the standalone components in their far-field recognition experiments. That result is evidence for the tested conditions, not a promise that every musical array recording benefits from the same ordering.

The following figure shows the two evidence streams that a serious multichannel system must maintain: calibrated geometry and spatial statistics above, delayed temporal prediction below.

Figure 2-96 presents A multichannel dereverberation stack combining array calibration, delayed late-tail prediction, spatial covariance estimation, and distortion-controlled beamforming and establishes the visual reference for the discussion that follows.

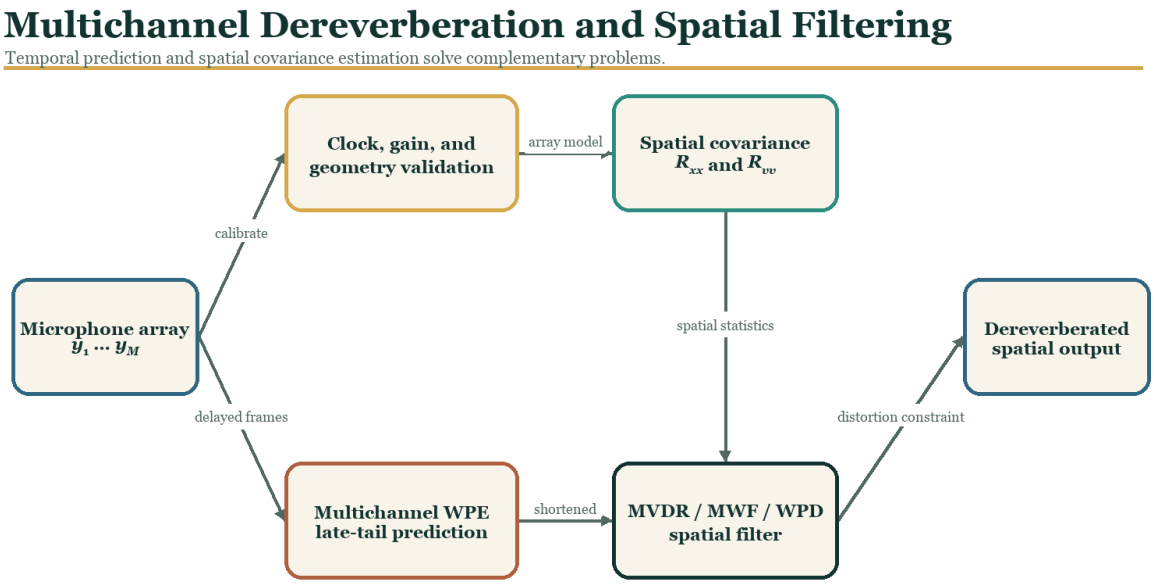

Figure 2-96: A multichannel dereverberation stack combining array calibration, delayed late-tail prediction, spatial covariance estimation, and distortion-controlled beamforming

How to read this figure. The lower path shortens temporal reverberation; the upper path describes array geometry and spatial energy. The spatial filter needs both. A channel-count increase is not automatically useful: unsynchronized clocks, mismatched gains, unknown microphone positions, or poor aperture can make the additional covariance unreliable.

For music and immersive production, the target output also needs definition. Collapsing a concert recording to one beamformed channel may improve speechlike clarity while destroying ensemble width and hall envelopment. A stereo or Ambisonic dereverberator should preserve interchannel level, time, phase, and diffuseness cues intentionally. Gains may be linked, estimated in a mid/side basis, or constrained by a multichannel target covariance. Every choice decides which part of "the room" is unwanted and which part is artistic evidence.

2.11.7 Neural and Hybrid Methods

Neural dereverberation learns a mapping from reverberant observations to a chosen target. The model may estimate a complex ratio mask, desired magnitude, clean waveform, room response, source variance for WPE, or spatial covariance for beamforming. Architectures include convolutional and recurrent networks, temporal convolutional networks, transformers, complex-valued mask estimators, and waveform-domain encoder/decoders.

Supervision makes the target question unavoidable. An anechoic target asks the model to remove both early and late room energy. An "early" target preserves the response up to a chosen boundary. A close-microphone target includes that microphone's coloration and bleed. A studio stem may contain processing absent from the distant observation. These are different inverse problems. Training labels that silently mix them teach inconsistent behavior.

Neural systems can use source regularities unavailable to blind statistical estimators. A speech model learns phonetic and harmonic structure; a music model can learn attacks, sustain, and instrument spectra. That prior can reconstruct plausible detail after severe smearing. It can also hallucinate plausible but incorrect consonants, suppress unfamiliar instruments, or imprint the training corpus's room and microphone biases. For archival, forensic, scientific, and classical-music work, plausibility is not equivalent to fidelity.

Hybrid systems keep an interpretable acoustic core and learn the hard statistics around it. A network may estimate $\lambda_{t,f}$ for WPE, generate masks for covariance estimation, predict T_{60} and direct-to-reverberant ratio, or synthesize a virtual second channel for multichannel WPE. Yang and Chang's virtual acoustic channel expansion is one example; neural-network-assisted Kalman WPE is another. These designs can retain a known prediction topology while improving source/noise discrimination.

Real-time neural deployment adds constraints beyond model accuracy. Look-ahead, receptive field, frame accumulation, accelerator transfer, dynamic allocation, denormal handling, and worst-case execution time all affect audio safety. A model reporting a small arithmetic cost can still miss a callback deadline because of memory traffic or runtime scheduling. Any future neural `verbx` mode should publish model identity, target definition, causal context, algorithmic delay, hardware, confidence, and deterministic fallback behavior in its JSON report.

2.11.8 Psychoacoustics: What Should Be Removed?

The ear does not classify all reflections as damage. Early lateral reflections can increase apparent source width and support intelligibility; a moderate room response can make speech or music sound natural. Late energy is more likely to fill temporal gaps, reduce modulation depth, mask following consonants, and fuse successive notes. But listeners also adapt to a room, and context changes perceived reverberance. The perceptually optimal estimate is often an early response, not an anechoic signal.

Speech illustrates temporal masking clearly. Energy from a vowel extends into the weaker consonant that follows it. The consonant may still exist physically, yet its contrast is reduced by the vowel's late field. Dereverberation restores modulation depth by attenuating that older energy. Music complicates the rule: the same overlap may be legato, resonance, or orchestration rather than interference. A piano pedal and a cathedral tail both prolong energy, but only one is a room response. An estimator driven only by predictability can damage both.

Direct-to-reverberant ratio (DRR), early-to-late ratio, clarity C_{50} or C_{80}, and modulation transfer describe different perceptual axes from T_{60}. Two recordings can share one decay time while one has a strong direct onset and the other sounds distant. Reducing T_{60} alone does not guarantee restored presence. Conversely, increasing the early-to-late ratio by suppressing only late energy may improve clarity without changing the fitted slope of the residual decay very much.

Tsilfidis and Mourjopoulos explicitly connect late suppression to perceptual reverberation modeling. Their work supports a product principle: the goal should be selective reduction of harmful audible late energy, not maximizing numerical attenuation. The best setting is often the lowest strength that reveals articulation without advertising the processor.

2.11.9 Artifacts and the Quality Frontier

Dereverberation sits on a Pareto frontier. More late-tail reduction normally increases at least one cost: coloration, transient damage, background-noise modulation, spatial change, or latency. There is no single strength value that optimizes every source and task.

Table 2-9: Artifacts and the Quality Frontier: reference table

Artifact	DSP cause	Diagnostic source	Conservative remedy
Musical noise	Isolated time-frequency gains fluctuate around estimation errors	Sustained noise, cymbal tail, breath	Raise floor; smooth gains; reduce strength
Phasiness	Observed phase is retained while magnitude changes rapidly	Solo voice, strings, stereo room tone	Increase time smoothing; use partial mix
Transient erosion	Predictor or long window treats onset energy as late field	Castanets, piano, consonants	Shorten window; protect delay; reduce order or strength
Spectral holes	Oversubtraction drives narrow bins near zero	Harmonic sweep, pink noise	Cap attenuation; use Wiener gain; raise floor
Pumping	Tail estimate follows source envelope too directly	Speech pauses, kick pattern	Lengthen release/tracker; reduce strength
Image wobble	Channels receive unrelated gain trajectories	Centered mono source in stereo ambience	Link gains or constrain spatial covariance
Noise breathing	Noise is classified as source in one frame and late field in another	HVAC, preamp hiss, location bed	Estimate noise separately; retain stable floor
Comb coloration	Inaccurate inverse or prediction cancels correlated direct energy	Sweeps, sustained vowels	Increase regularization or prediction delay
Over-dry isolation	Useful early room cues are removed with the tail	Chamber music, location dialogue	Preserve early target; lower mix or strength

Listening tests should include bypass matched for loudness. Dereverberated output often sounds "better" simply because transients become louder or average level changes. Match integrated loudness and peak headroom before judging clarity. Also audition the residual $y[n] - \hat{x}[n]$. A good residual should sound predominantly like diffuse late energy; recognizable dry words, melody, or attacks reveal source cancellation.

2.11.10 Causality, Adaptation, and End-to-End Latency

Offline algorithms may inspect future samples, estimate statistics over a complete file, iterate until convergence, and use zero-phase filtering. A real-time processor must commit to output before that future exists. "Real-time" only means processing keeps pace; it does not mean zero latency. The end-to-end monitoring delay is approximately

$$L_{\text{total}} = L_{\text{ADC}} + L_{\text{input buffer}} + L_{\text{analysis}} + L_{\text{algorithm}} + L_{\text{output buffer}} + L_{\text{DAC}},$$

plus any operating-system safety offsets, resampling, aggregate-device buffering, plugin host compensation, and wireless transport. A dereverberator can control only part of this sum.

For an STFT processor, a causal window of N samples must be filled before its complete spectrum is available. A centered STFT adds future look-ahead and is unsuitable for strict live monitoring unless that delay is accepted. Overlap-add emits updates every hop H, while driver callbacks arrive in blocks of B samples. If B is not compatible with H, the implementation needs an internal adapter or must reject the configuration. `verbx` currently chooses fail-fast validation for the live dereverb path: the block size must be divisible by the resolved hop size.

At 48 kHz, a 12 ms window is approximately 576 samples and a 4 ms hop is approximately 192 samples. A 384-sample callback block contains two complete hops. This does **not** imply that total acoustic round-trip latency is exactly 12 ms or 8 ms. Driver input/output buffers and converter safety offsets remain device dependent, and the analysis/resynthesis implementation can add a window-related delay. The trustworthy measurement is a physical loopback: feed a pulse through the complete input, processor, host, and output chain; record both the reference and return; find their sample offset; repeat under realistic load; and report median plus worst case.

WPE introduces a second time scale: statistical adaptation. A causal WPE output may have a small signal-path delay while its covariance and source variance need many frames to become reliable. After a source moves or speech begins, quality ramps toward steady state. That adaptation lag is not always counted as algorithmic latency, but it affects perceived responsiveness and should be reported. Neural methods similarly separate look-ahead from state warm-up and accelerator scheduling.

Low latency and deep suppression conflict. Short windows react quickly but have broad frequency bins, making low-frequency decay harder to distinguish. Short predictors adapt quickly but model less of the tail. Small smoothers preserve attacks but allow estimate variance to modulate the output. A live preset should be judged under deadline pressure and dropout monitoring, not only by processing a file faster than its duration.

2.11.11 Evaluating Dereverberation Scientifically

No one metric establishes success. Evaluation should combine acoustic quantities, signal-reconstruction metrics, downstream task performance, artifact measures, and controlled listening. Each answers a different question.

Paired intrusive evaluation is possible when a dry or early target is known and aligned. Synthetic tests convolve a clean source with measured or simulated room responses, then compare the estimate with the original target. Scale-invariant signal-to-distortion ratio (SI-SDR) measures reconstruction after removing a scalar gain ambiguity, but it can punish benign spatial or equalization differences and reward solutions that do not sound natural. Short-time objective intelligibility (STOI) and perceptual evaluation of speech quality (PESQ) are speech oriented; neither should be treated as a music-quality score.

Non-intrusive evaluation is needed for ordinary recordings with no dry reference. The speech-to-reverberation modulation energy ratio (SRMR) uses modulation spectra to estimate reverberation-related smearing. Blind T_{60}, DRR, or early-to-late estimators can compare before and after, but processing may violate the statistical model used by the estimator. An algorithm can "game" a proxy by reshaping modulation or spectral energy without restoring the source. Non-intrusive values are evidence, not ground truth.

Room-acoustic evaluation is strongest when an impulse response is available. Backward Schroeder integration yields an energy-decay curve from which EDT, T_{20}, and T_{30} can be fitted. Clarity indices compare early and late energy:

$$C_{t_e} = 10 \log_{10} \frac{\int_0^{t_e} h^2(t)\,dt}{\int_{t_e}^{\infty} h^2(t)\,dt},$$

with $t_e = 50$ ms commonly used for speech and $t_e = 80$ ms for music. A dereverberator aimed at late suppression should normally increase early-to-late clarity, though its output is not itself a pristine room impulse response if the method is nonlinear or signal dependent.

Task evaluation asks whether the processing helps the intended system. For distant speech recognition, report word error rate across rooms, distances, speakers, and noise conditions. The REVERB Challenge established paired simulated and real-room test conditions and made clear that enhancement scores and recognition outcomes need not rank systems in the same order. For dialogue editing, test transcription and listener effort. For source separation, test leakage and source fidelity. For music, use trained listeners and include transient, sustained, sparse, dense, mono, stereo, and immersive material.

Listening evaluation should be randomized, loudness matched, and preferably hidden. A MUSHRA-style panel can include reverberant input, one or more estimates, the known early target when available, and an intentionally poor anchor. Ask separate questions for reverberation reduction, speech clarity, timbral fidelity, spatial stability, transient quality, and overall preference. A single "quality" score hides the tradeoff the experiment is meant to expose.

Every report should preserve the input hash, sample rate, channel layout, algorithm and version, complete parameters, random seed if any, processing mode, machine, wall-clock speed, algorithmic delay, measured loopback delay when relevant, and metric definitions. Without that provenance, small score differences are not reproducible science.

2.11.12 A Reproducible verbx Workflow

Start by preserving the original and measuring what can be measured honestly. For program audio, label blind room estimates as estimates rather than treating them like measured IR parameters:

```
verbx analyze location_dialogue.wav --input-kind program --edr --room \
  --json-out reports/location_dialogue.before.json
```

Render a conservative Wiener pass to 32-bit floating-point WAV and preserve a structured processing report:

```
verbx dereverb location_dialogue.wav location_dialogue.dereverb.wav \
  --mode wiener --strength 0.75 --floor 0.08 \
  --window-ms 20 --hop-ms 5 --tail-ms 120 \
  --pre-emphasis 0.15 --mix 0.9 --out-subtype float32 \
  --json-out reports/location_dialogue.dereverb.json
```

Analyze the result separately so processing metadata and measured output evidence remain distinct records:

```
verbx analyze location_dialogue.dereverb.wav --input-kind program --edr --room \
  --json-out reports/location_dialogue.after.json
```

Then make at least three loudness-matched auditions: bypass, conservative processing, and a deliberately stronger setting. The stronger setting identifies the artifact boundary; it is not automatically the deliverable. Listen to the residual as well as the output. For stereo material, compare linked and unlinked behavior while watching a phase-correlation or vectorscope display.

For live monitoring, begin with a device list and use wired input/output paths:

```
verbx realtime --list-devices
```

```
verbx realtime --live-mode dereverb \
  --input-device "Built-in Microphone" --output-device "Headphones" \
  --sample-rate 48000 --block-size 384 \
  --dereverb-mode wiener --dereverb-strength 0.75 \
  --dereverb-floor 0.08 --dereverb-window-ms 12 \
  --dereverb-hop-ms 4 --dereverb-tail-ms 90 \
  --dereverb-max-atten-db 15 --dereverb-stereo-link
```

Measure physical round-trip delay before performance use. Bluetooth devices can add far more latency than the DSP and should not be used to characterize the processor. If a DAW hosts the plugin path, measure that host and buffer configuration rather than transferring a standalone CLI number to it.

The experiment matrix below is more informative than searching for one universal preset:

Table 2-10: A Reproducible verbx Workflow: reference table

Dimension	Minimum useful conditions	What it reveals
Source	Speech, percussion, piano, sustained strings, dense mix	Source-prior and transient failures
Room	Short dry room, medium room, long diffuse hall, modal small room	Model mismatch and decay tracking
Distance	Near, medium, far	DRR dependence
Noise	Quiet, stationary noise, changing noise	Tail/noise confusion
Channels	Mono, stereo linked, stereo independent	Spatial stability
Strength	Bypass, conservative, nominal, stress	Artifact frontier
Mode	Wiener, spectral subtraction	Gain-law dependence
Metric	Clarity, blind decay, residual, listening, task score	Agreement and disagreement between evidence

2.11.13 Limits, Safety, and Claims

Dereverberation cannot separate every process that prolongs sound. Instrument resonance, sustain pedal, chorus, delay, compression release, distortion, audience noise, and the room may overlap. A single microphone provides no label saying which decay is artistic. Aggressive processing of a mastered recording can remove intentional production ambience and expose edits or noise that the ambience masked.

The processor also cannot repair clipping, recover bandwidth removed before capture, or infer a unique dry waveform from a fully smeared mixture. It may improve intelligibility without making audio studio-dry, or make audio sound drier while decreasing fidelity. Those outcomes are not contradictions because "reverberation amount," "clarity," and "reconstruction accuracy" are different dimensions.

For forensic or evidentiary use, retain the original, process only a copy, preserve hashes and parameters, and describe the method as enhancement rather than recovery of fact. For machine-learning datasets, keep dry/reverberant pairing and train/validation/test room separation intact; otherwise room leakage can make a model appear to generalize. For archival music, prefer reversible conservative passes and document the artistic decision.

The current `verbx` implementation should therefore be described precisely as deterministic spectral late-tail suppression. Its controls are designed to expose the reduction/artifact tradeoff and produce machine-readable reports. WPE, multichannel convolutional beamforming, measured inverse filtering, and neural restoration are discussed here because they define the scientific field and roadmap, not because the CLI silently substitutes those methods.

2.11.14 Selected Primary Literature

The following works are ordered alphabetically by first author. Appendix C contains the guide's larger annotated bibliography and cross-references [PE3], [PE4], [PE9], [SR1], and related room-acoustic sources.

Allen, J. B.; Berkley, D. A. "Image method for efficiently simulating small-room acoustics." *The Journal of the Acoustical Society of America* 65(4): 943–950. DOI: 10.1121/1.382599, 1979.

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Nakatani, T.; Kinoshita, K. "Simultaneous denoising and dereverberation for low-latency applications using frame-by-frame online unified convolutional beamformer." *Proceedings of Interspeech 2019*: 111–115. DOI: 10.21437/Interspeech.2019-1286, 2019.

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Yoshioka, T.; Nakatani, T. "Generalization of multi-channel linear prediction methods for blind MIMO impulse response shortening." *IEEE Transactions on Audio, Speech, and Language Processing* 20(10): 2707–2720. DOI: 10.1109/TASL.2012.2210879, 2012.

2.12 A Thirty-Minute Reverb Laboratory

The following laboratory turns the chapter into an audible sequence. Use headphones and loudspeakers if possible; keep output level fixed; preserve every render and JSON sidecar.

2.12.1 Minute 0–5: Identify the Three Regions

Render the dry click through a one-second room at 100 percent wet. Mark direct onset, the first visible reflections, and the point where individual arrivals become a dense tail. Change only pre-delay and repeat. The late decay should remain similar while the relationship between source and room changes.

2.12.2 Minute 5–10: Isolate Diffusion

Render a rimshot or click with zero, two, four, and eight allpass stages. Keep RT60 and matrix fixed. Listen for loss of attack definition, flutter reduction, and buildup speed. Choose the lowest stage count that supplies the density the source needs.

2.12.3 Minute 10–15: Compare Matrix Families

Use a sparse major seventh chord and an eight-second tail. Compare Hadamard, circulant, random orthogonal, and time-varying unitary matrices. Write three adjectives for each without looking at the option name. Then inspect sonograms and ask whether visible modal ridges agree with what you heard.

2.12.4 Minute 15–20: Design Frequency-Dependent Decay

Set low, middle, and high RT60 values equal. Then shorten only the high band; next, lengthen only the low band. Listen through the complete final tail. The point is not to find a universal curve but to learn how decay spectrum changes perceived material and scale.

2.12.5 Minute 20–25: Make the Return a Musical Voice

Put verbx on a 100 percent wet auxiliary return. Send only the final note of every four-bar phrase. Then leave the send constant and automate the return instead. Compare the two gestures. One controls what the room remembers; the other controls when the memory is revealed.

2.12.6 Minute 25–30: Enter Extreme Time Safely

Render one stable chord into a 60-second time-varying FDN. Leave at least 90 seconds of silence after the source. Write floating-point audio and a JSON report. Inspect peak, integrated loudness, DC offset, and the final 20 seconds. Increase RT60 only after the tail is spectrally and numerically controlled.

2.13 Practical Listening Checklist

Use the checklist after level matching and before changing parameters again. It separates acoustic behavior, musical function, and reproducibility so a striking first impression does not become the only evaluation.

- **Onset:** Does the room begin with the source, behind it, or as a separate echo?
- **Density:** Can individual repetitions be heard after they should have fused?
- **Color:** Which bands survive longest, and is that survival musically useful?
- **Modes:** Do stable pitches emerge that were not structurally important in the source?
- **Width:** Does the return enlarge the image without weakening the center or fold-down?

- **Dynamics:** Does the tail mask attacks, words, or harmonic changes?
- **Motion:** Does time variation feel architectural, chorused, or unstable?
- **Ending:** Does the tail reach silence gracefully, gate abruptly, or expose noise and DC?
- **Form:** Is reverb supporting a phrase, connecting sections, or functioning as its own layer?
- **Evidence:** Can the render be reproduced from its command, preset, seed, and JSON report?

2.14 Summary: Reverb as Memory

Reverb is the memory of an acoustic or synthetic system. Direct sound tells the listener what happened; early reflections tell where it happened; the late field tells how the space continues to respond. In music, that memory can reinforce articulation, connect harmony, enlarge orchestration, preserve a past section, or become a new sustained instrument.

verbx sounds different because it does not reduce this behavior to one size control. It exposes the topology and timing that create density, the matrix that exchanges energy, the filters that shape decay, the projections that create spatial output, and the creative processes that reinterpret the field. At ordinary settings, those controls design rooms, chambers, halls, and plates. At extreme settings, they design musical time.

2.15 Suggested Exercises

1. Measure the direct arrival and first five reflections in a supplied or self-recorded impulse response; sketch the likely boundary geometry without claiming more precision than the evidence permits.
2. Compare a feedback comb, an allpass diffuser, and an FDN using the same impulse. Relate audible density to the corresponding signal-flow diagram.
3. Plot an energy-decay curve for two rooms or two presets, fit EDT and T30 where valid, and explain why the estimates disagree or agree.
4. Make a short percussion study in which early reflections establish one apparent room while the late field establishes another. State the intended perceptual contradiction.
5. Choose one pole-zero plot from the chapter and predict its magnitude response before viewing the accompanying response graph.
6. Record a hand clap or balloon pop in two contrasting spaces and compare their onset-to-late-field transition with the chapter's vocabulary.
7. Derive the feedback gain required for a chosen RT60 and delay length, then test whether the resulting decay is perceptually plausible.
8. Make a sonogram pair with identical time and frequency axes for two reverb designs, then annotate three meaningful differences.
9. Construct a simple allpass or comb filter in any DSP environment and compare its impulse response with the corresponding idealized equation.
10. Write a listening guide for a nontechnical collaborator that distinguishes echo, early reflections, diffusion, and late reverberation.

"Multitudinous echoes awoke and died in the distance."

— Henry Wadsworth Longfellow, *Evangeline* (1847)

Chapter 3

verbx Reference

3.1 Core Concepts

The concepts in this section form the minimum vocabulary for the engines and workflows that follow. Each topic begins with an operational explanation, then adds the implementation detail needed to predict edge cases.

3.1.1 Algorithmic vs. Convolution

For beginners: Algorithmic reverb synthesizes the space from scratch using delay networks and filters. It does not need an external file, responds instantly to parameter changes, and can produce decay times no physical room could sustain. Convolution reverb applies a pre-recorded impulse response – a measurement of what a specific room does to a click – to your audio. The result sounds like the space where the IR was recorded.

For experts: The algorithmic engine in verbx uses a Schroeder allpass diffusion stage feeding a fully coupled N-line FDN with configurable feedback matrix. Convolution uses uniformly-partitioned overlap-save FFT with optional CUDA acceleration via CuPy. The two engines share the same pre-delay, shimmer, freeze, ducking, bloom, tilt, loudness, and spatial stages. Use `--engine auto` and verbx selects based on whether an IR is present.

Choose algorithmic when you want extreme lengths, animated or time-varying decay, spaces that do not exist, low storage overhead. Choose convolution when you want: the character of a specific real or designed space, exact linear reproduction of an IR, or multichannel matrix routing from a measured space.

3.1.2 RT60

For beginners: RT60 is the time required for reverberant sound energy to fall by 60 dB after the source stops. That is a decrease to one millionth of the starting energy, or one thousandth of the starting pressure amplitude. A short RT60 makes events separate quickly and usually improves articulation. A long RT60 joins events into a continuous field and can enlarge, soften, or obscure musical detail. RT60 describes the slope of a decay; it does not by itself describe how loud the reverb is, how soon it begins, whether it is bright or dark, or how wide the return sounds.

A small treated room may have an RT60 below 0.4 s. Many production rooms occupy roughly 0.3–0.8 s, concert halls often lie near 1.5–2.5 s at mid frequencies, and very large stone spaces can sustain substantially longer decays. These are orientation ranges rather than presets or universal targets. Occupancy, frequency, source/receiver position, geometry,

and measurement method all matter. The verbx analyzer can represent estimates from 0.01 s to 3600 s; the current render and IR-synthesis controls accept targets from 0.1 s to 3600 s. That span moves continuously from tight acoustic control to deliberately nonphysical musical sustain without pretending that every extreme is a plausible architectural room.

3.1.2.1 Decibels, Energy, and the Meaning of "60"

The decibel is logarithmic. For pressure or sample amplitude $p(t)$ relative to reference p_0, level is

$$L_p(t) = 20\log_{10}\left(\frac{|p(t)|}{|p_0|}\right) \text{ dB}.$$

For energy $E(t)$ relative to E_0, level is

$$L_E(t) = 10\log_{10}\left(\frac{E(t)}{E_0}\right) \text{ dB}.$$

An amplitude ratio of 10^{-3} and an energy ratio of 10^{-6} therefore both represent –60 dB. This is why RT60 is sometimes described as a thousandfold pressure reduction and sometimes as a millionfold energy reduction. The statements are equivalent, provided the reader knows which physical quantity is being discussed.

An ideal exponential amplitude envelope can be written

$$p_{\text{env}}(t) = p_0 e^{-t/\tau_A},$$

where τ_A is an amplitude time constant. Requiring a –60 dB amplitude change at $t = T_{60}$ gives

$$\tau_A = \frac{T_{60}}{3\ln 10}.$$

The corresponding energy time constant is half as long:

$$\tau_E = \frac{T_{60}}{6\ln 10}.$$

On a decibel-versus-time graph, an ideal exponential is a straight line with slope

$$s = -\frac{60}{T_{60}} \text{ dB/s}.$$

Thus a 2 s decay falls at –30 dB/s, while an 8 s decay falls at –7.5 dB/s. Doubling RT60 halves the magnitude of the decay slope; it does not merely append a fixed amount of audio to the tail.

Figure 3-1 compares four ideal decay families on one time axis. It provides the visual reference for interpreting RT60 as a slope rather than as a cutoff point.

3.1.2.2 From Room Physics to Reverberation Time

In a statistical diffuse-field model, Sabine's relation estimates reverberation time from room volume and equivalent absorption area:

$$T_{60} \approx 0.161\frac{V}{A}, \qquad A = \sum_j \alpha_j S_j,$$

where V is volume in cubic meters, S_j is surface area in square meters, and α_j is the dimensionless absorption coefficient of surface j. The constant 0.161 has units that make the result seconds under ordinary SI assumptions. More volume stores more acoustic energy; more absorption removes a larger fraction at each encounter.

Sabine's approximation is strongest when absorption is modest and distributed, the field is sufficiently diffuse, and source/receiver positions sample the room representatively. For larger average absorption, Eyring's form accounts for the finite fractional loss more directly:

$$T_{60} \approx 0.161 \frac{V}{-S \ln(1 - \overline{\alpha})},$$

where S is total surface area and $\overline{\alpha}$ is area-weighted mean absorption. Neither equation predicts every seat, low-frequency mode, coupled chamber, or strongly directional geometry. They are statistical models of ensemble energy, not substitutes for a measured impulse response or wave simulation.

Absorption is frequency dependent. Carpet, curtains, occupied seating, air, porous panels, wood, and masonry do not remove all frequencies equally. A room consequently has an RT60 curve $T_{60}(f)$ rather than one complete scalar. Published room values should identify octave or one-third-octave bands, occupancy, excitation, microphone arrangement, and the estimator used. A single broadband number is useful only when its compression of that curve is acknowledged.

Figure 3-1 presents Ideal RT60 decay families showing relative decay level in decibels against time after excitation in seconds and establishes the visual reference for the discussion that follows.

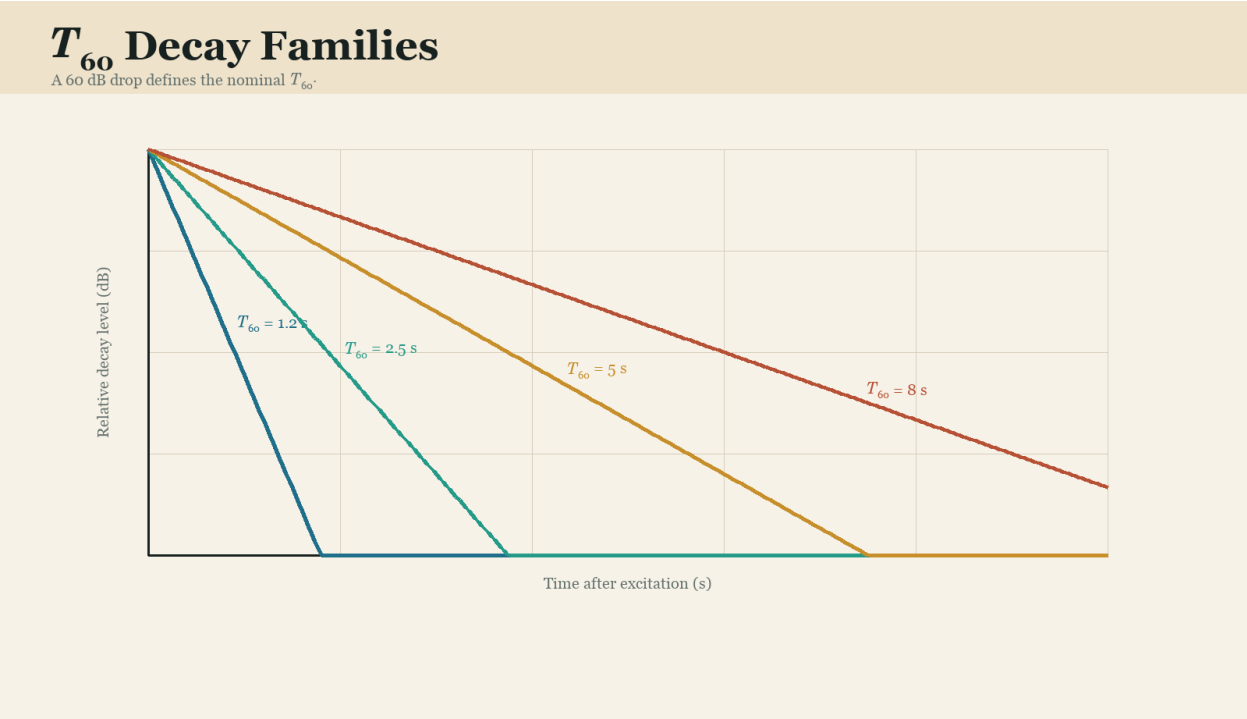


Figure 3-1: Ideal RT60 decay families showing relative decay level in decibels against time after excitation in seconds

How to read Figure 3-1. Every curve begins at the same normalized level. The 1.2 s curve reaches –60 dB first and then remains at the displayed floor; the 8 s curve is still decaying at the right edge. A real response does not become exactly silent at T_{60}. The –60 dB crossing is a reference on an extrapolated or measured slope, while audibility depends on source level, spectrum, masking, and the playback noise floor.

3.1.2.3 RT60 Inside a Feedback Delay Network

For experts, RT60 drives per-line gain calibration in a unitary or approximately energy-preserving feedback delay network. If line i has delay d_i seconds and scalar feedback magnitude g_i, it circulates approximately T_{60}/d_i times during one target decay. Requiring the accumulated amplitude to reach 10^{-3} gives

$$g_i = 10^{-\frac{3d_i}{T_{60}}}$$

or, for a line of m_i samples at sample rate F_s,

$$g_i = 10^{-\frac{3m_i}{F_s T_{60}}}.$$

Shorter delays traverse their loop more often and therefore require gains closer to one for the same total decay. Using one identical gain on unequal delays creates unequal modal decay rates. Calibrating each line by its duration is what turns a set of dissimilar delays into one approximate broadband RT60 target.

This derivation assumes that the feedback matrix preserves energy. For an exactly unitary matrix M, $M^{\mathsf{H}}M = I$, so loss can be assigned primarily by diagonal gains or filters. A nonunitary matrix can contain growing or shrinking eigenmodes even when every individual g_i looks reasonable. Stability depends on the complete loop operator: its spectral radius must remain below one for a strictly decaying linear system.

At very long RT60 values, g_i approaches one. Numerical precision, denormals, DC leakage, nonlinear stages, modulation interpolation, and tiny matrix-normalization errors become audible over time. A network that survives a five-second impulse test may drift during a ten-minute tail. Extreme values therefore require floating-point headroom, DC blocking, bounded modulation, long-duration stress renders, and analysis of the final segment rather than only the onset.

3.1.2.4 Measuring RT60 from an Impulse Response

An acoustician rarely waits for a recorded decay to become exactly 60 dB quieter and then reads a stopwatch. The bottom of a measured response may already be masked by ventilation, electrical noise, audience movement, quantization, or microphone self-noise. Instead, the usual procedure estimates the slope over a clean part of an energy decay curve and extrapolates that line to –60 dB. The reported RT60 is therefore often a model fitted to a smaller observed range, not a literal uninterrupted observation of all 60 dB.

Let $h(t)$ be a measured room impulse response. Schroeder backward integration forms the remaining energy after time t:

$$E(t) = \int_t^\infty h^2(\tau)\, d\tau.$$

For sampled audio with final sample $N-1$, the corresponding cumulative sum is

$$E[n] = \sum_{k=n}^{N-1} h^2[k].$$

Normalizing by the value at the direct arrival and converting to decibels gives the energy decay curve, or EDC:

$$L_E[n] = 10\log_{10}\left(\frac{E[n]}{E[n_0]}\right)\ \text{dB},$$

where n_0 marks the chosen decay onset. Backward integration smooths individual peaks and turns a noisy-looking impulse response into a generally descending energy trajectory. It does not eliminate

measurement noise; because the integral includes every later sample, a stationary noise floor eventually bends the curve away from its true room-decay slope.

Three conventional estimators use different portions of the EDC:

Table 3-1: Measuring RT60 from an Impulse Response: reference table

Estimator	Fitted decay interval	Extrapolation to T_{60}
EDT	0 to -10 dB	fitted time multiplied by 6
T_{20}	-5 to -25 dB	fitted time multiplied by 3
T_{30}	-5 to -35 dB	fitted time multiplied by 2

Early decay time, or EDT, emphasizes the first audible release after excitation and often tracks perceived reverberance more closely than a deep late-tail fit. T20 gives a robust estimate when only about 20 dB of clean decay is available. T30 uses more of the response and is preferable when the measurement has sufficient dynamic range. All three become RT60 estimates by extending the fitted straight line, so a value called "T20" is not a 20-second quantity and does not mean that the room fell by 60 dB during the measured 20 dB interval.

Figure 3-2 places the three regression windows on one idealized EDC. The shared axis makes clear that changing estimators changes the evidence used for the fit, not the definition of the final 60 dB decay target.

A trustworthy implementation must make several decisions before fitting a line:

1. **Locate the onset.** Leading silence must not become part of the decay. For an impulse response, the strongest plausible direct event is a useful anchor; for program audio, peak alignment is only a diagnostic approximation because later source energy may still be arriving.
2. **Estimate the noise floor.** The final portion of the capture provides a noise estimate, but it can also contain a genuine long tail. Automatic methods must distinguish a stable noise plateau from slowly decaying acoustic energy and report uncertainty when they cannot.
3. **Choose a usable range.** A regression boundary below the noise-intersection point makes the EDC look artificially shallow. A boundary too near the onset may instead fit early geometry rather than late statistical decay.
4. **Assess linearity.** A high coefficient of determination, R^2, supports a single-slope model. A low value may indicate poor signal-to-noise ratio, multiple slopes, flutter, source contamination, or a genuinely nonexponential room.
5. **Preserve alternatives.** EDT, T20, T30, fit range, R^2, and noise floor should be retained together. Reporting only the selected RT60 hides the evidence needed to judge whether that selection is acoustically meaningful.

Uncertainty is not an optional footnote to this process. Small changes in onset, filter bandwidth, or noise compensation can move a fitted boundary and therefore change the extrapolated result by more than the visible difference in the measured interval. Repeated captures reveal this sensitivity better than extra decimal places. For a venue survey, report the estimator and spread across positions; for a synthesized IR, hold the random seed, sample rate, output length, and analysis settings constant. For either case, interpret a reported value such as 2.37 s as an estimate produced by a stated procedure, not proof that the underlying acoustic field has one perfectly uniform decay constant.

verbx follows this evidence-preserving approach. `verbx analyze` aligns the strongest event, constructs the backward-integrated EDC, computes EDT, T20, and T30 where their fit windows are usable, and selects the deepest reliable fit as the broadband estimate. Its JSON report also carries fit quality, usable decay range, noise-floor information, input classification, and confidence. This is especially important for machine-learning datasets: a scalar label without its estimator, confidence, and measurement context can turn a physically ambiguous capture into falsely precise training data.

Figure 3-2 presents Schroeder energy decay curve with the EDT, T20, and T30 regression windows marked against time after excitation in seconds and establishes the visual reference for the discussion that follows.

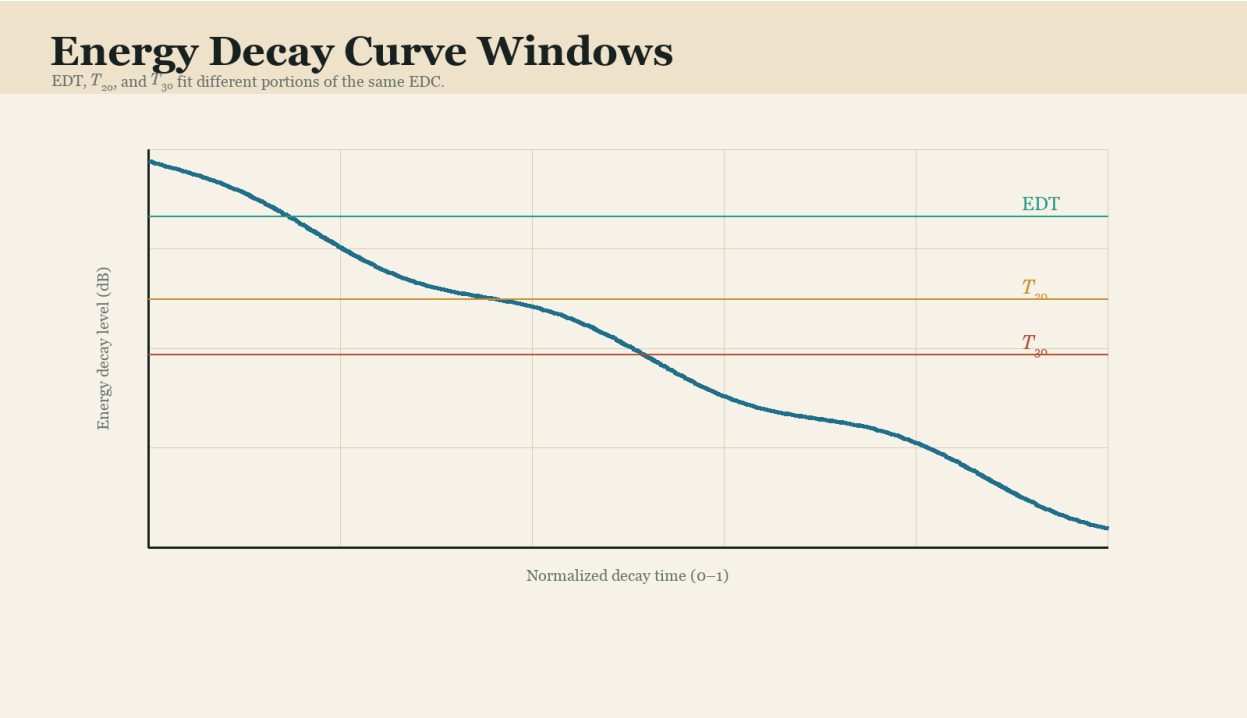


Figure 3-2: Schroeder energy decay curve with the EDT, T20, and T30 regression windows marked against time after excitation in seconds

How to read Figure 3-2. The horizontal axis is elapsed time after the direct event, and the vertical axis is normalized remaining energy in decibels. EDT begins at the top of the decay and is therefore sensitive to early reflections and the transition into the late field. T20 and T30 exclude the first 5 dB so the direct event and the earliest reflection pattern have less leverage. T30 reaches deepest and can be the most representative late estimate, but only if its –35 dB boundary remains safely above the noise-contaminated bend.

3.1.2.5 Energy Decay Relief

A broadband energy decay curve collapses the spectrum before integration. That is useful for one decay estimate, but it can hide a narrow room mode, a plate resonance, or a high band that dies much sooner than the rest of the response. An **energy decay relief** (EDR) retains both time and frequency. It is a time-frequency generalization of the EDC: every frequency bin receives its own backward-integrated decay curve.

Let $H(m,k)$ be STFT bin k in frame m, and let M be the final frame. Following Julius O. Smith III's definition, the remaining energy at frame n and frequency bin k is

$$\mathrm{EDR}(t_n, f_k) = \sum_{m=n}^{M} |H(m,k)|^2 ,$$

with

$$t_n = \frac{nR}{F_s}, \qquad f_k = \frac{kF_s}{N},$$

where R is the STFT hop size in samples, F_s is sample rate in hertz, and N is FFT length. Normalizing each frequency bin at the selected onset gives a decibel surface:

$$L_{\mathrm{EDR}}(n,k) = 10\log_{10}\left(\frac{\mathrm{EDR}(t_n,f_k)}{\mathrm{EDR}(t_0,f_k)}\right)\ \mathrm{dB}.$$

Figure 3-3 shows this surface as a heatmap. Horizontal position is elapsed time, vertical position is logarithmic frequency, and color is normalized remaining energy in decibels. A horizontal ridge indicates a frequency region that stores energy longer than its neighbors. A nearly vertical change shared by all bands is more likely a broadband event, edit, gate, or noise-floor transition. Curved or interrupted ridges can reveal multiple decay regimes that disappear inside one scalar RT60.

Figure 3-3 presents Energy decay relief with elapsed time in seconds, logarithmic frequency in kilohertz, and normalized remaining energy in decibels and establishes the visual reference for the discussion that follows.

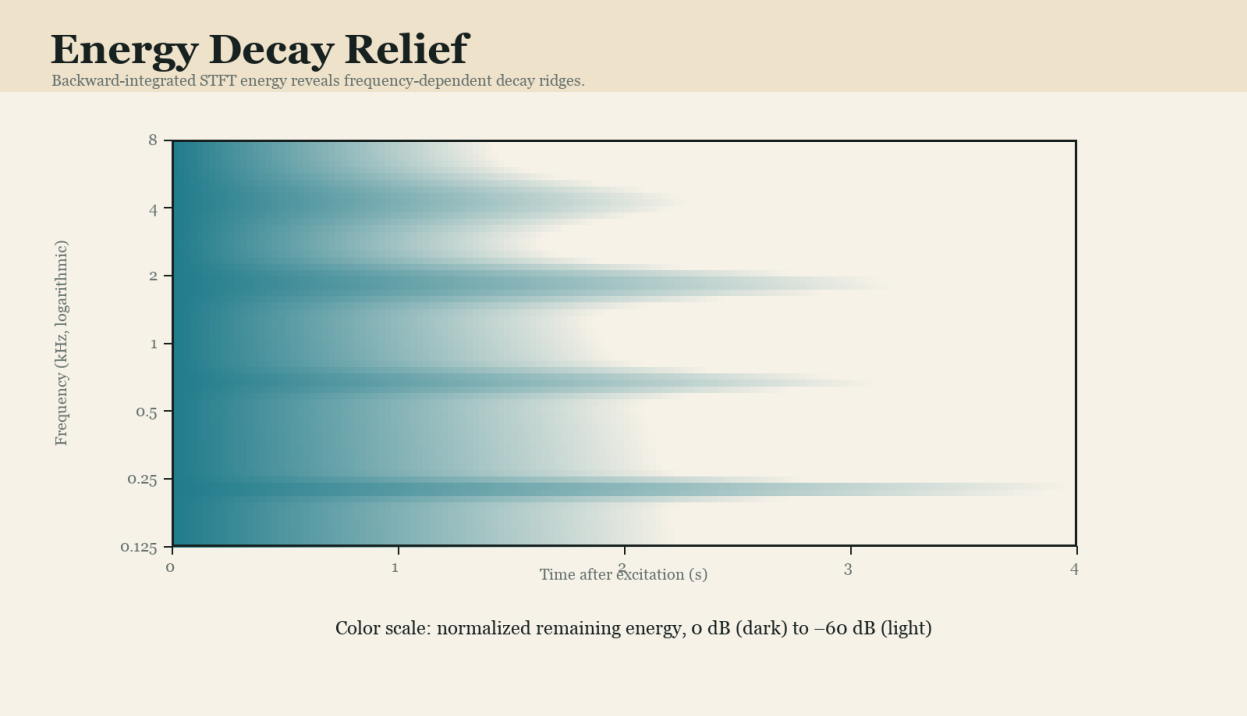


Figure 3-3: Energy decay relief with elapsed time in seconds, logarithmic frequency in kilohertz, and normalized remaining energy in decibels

How to read Figure 3-3. Darker cells contain more remaining energy. Most frequencies fade from left to right, but narrow ridges near 0.22, 0.66, 1.8, and 4.2 kHz persist at different rates. Those ridges model the kind of resonant structure that a broadband EDC averages away. The plotted data are synthetic and explanatory; they are not a measurement of a named room or instrument.

Time-frequency resolution is a design choice. A longer window separates close modes but smears their start and end in time. A shorter window tracks abrupt changes more closely but spreads energy across frequency bins. Smith describes Hann windows around 30–40 ms as a

typical starting range. verbx currently uses a 2,048-sample Hann window and a one-quarter window hop for `--edr`; at 48 kHz those values are approximately 42.7 ms and 10.7 ms. Analysis settings therefore belong beside the result whenever EDR values are compared across files, sample rates, or software.

`verbx analyze --edr` computes this reverse cumulative STFT-energy surface internally, fits a decay slope in each usable frequency bin, and then returns a compact summary: `edr_rt60_median_s`, `edr_rt60_low_s`, `edr_rt60_mid_s`, `edr_rt60_high_s`, and `edr_valid_bins`. The current JSON report stores those summaries, not the complete matrix. This distinction matters: the summary is appropriate for batch comparison and room-size inference, while the full relief is the better diagnostic for isolated modes, plate resonances, coupled slopes, and frequency-selective dereverberation.

Use EDR most confidently on a measured or synthesized impulse response. On music or speech, later source events are included in the backward sum and can resemble stored room energy. The result remains useful as a descriptive recording diagnostic, but it is not a standards-style room measurement. Preserve the input classification, window, hop, sample rate, fit range, valid-bin count, and noise context before interpreting differences as changes in the acoustic system.

The primary derivation is Julius O. Smith III's Energy Decay Relief chapter in *Physical Audio Signal Processing* (2010).

3.1.2.6 Frequency-Dependent and Multiband Decay

A room does not have one decay envelope shared perfectly by every frequency. Air absorption increases toward high frequencies; porous treatment may act mainly above a transition band; panel and membrane absorbers can shorten selected low-frequency resonances; and walls, seating, scenery, people, and openings all have frequency-dependent losses. For that reason, room-acoustic reports commonly estimate $T_{60}(f)$ in octave or one-third-octave bands.

The same idea applies to artificial reverberation. If the desired decay at angular frequency ω is $T_{60}(\omega)$, then a delay line of duration d_i needs an approximate loop filter magnitude

$$\left|G_i\left(e^{j\omega}\right)\right| = 10^{-\frac{3d_i}{T_{60}(\omega)}}.$$

This relation converts a decay-time curve into a loss-per-circulation curve. The loop filter is not merely an equalizer placed after the reverb. A post-EQ changes the level of a band at every instant by the same proportion; a feedback-loop filter changes that band's slope over time. A bright onset followed by a progressively dark tail therefore requires frequency- dependent loop loss, not just a dark output filter.

In practice, a small set of bands is easier to control and often more musically legible than

an unrestricted curve. verbx exposes low-, mid-, and high-band decay targets with two crossovers. The broadband `--rt60` remains the central reference, while `--fdn-rt60-low`, `--fdn-rt60-mid`, and `--fdn-rt60-high` explicitly shape the bands. `--fdn-xover-low-hz` and `--fdn-xover-high-hz` locate their transitions. A compact

`--fdn-rt60-tilt` control skews low and high decay around the middle when a three-number specification would interrupt a production workflow.

Figure 3-4 compares three bandwise decays that share a direct event but diverge as loop loss accumulates. It is the visual distinction between warmth caused by persistent low-frequency energy and simple bass boost.

Different spectral profiles imply different musical spaces. A low band that lasts modestly longer than the middle can suggest occupied halls, timber, or massive architecture. An excessively long low band can mask bass articulation and expose sparse FDN modes as pitched ringing. A short high band usually sounds natural because air and soft materials absorb high frequencies efficiently; an unusually long high band creates gloss, shimmer, or a synthetic metallic halo. The crossover locations matter because

they decide whether a decay change affects fundamental weight, vocal presence, consonant detail, or only air.

Multiband estimates require care. Bandpass filters have finite temporal support and can lengthen an apparent decay, especially in narrow low-frequency bands. Very low bands may contain only a few room modes, violating the diffuse-field assumption behind a straight EDC. A credible report should therefore list band centers, bandwidths, filter design, fit windows, and confidence rather than presenting a smooth colored curve as exact ground truth.

Figure 3-4 presents Low-, mid-, and high-frequency reverberation decay levels in decibels against time after excitation in seconds and establishes the visual reference for the discussion that follows.

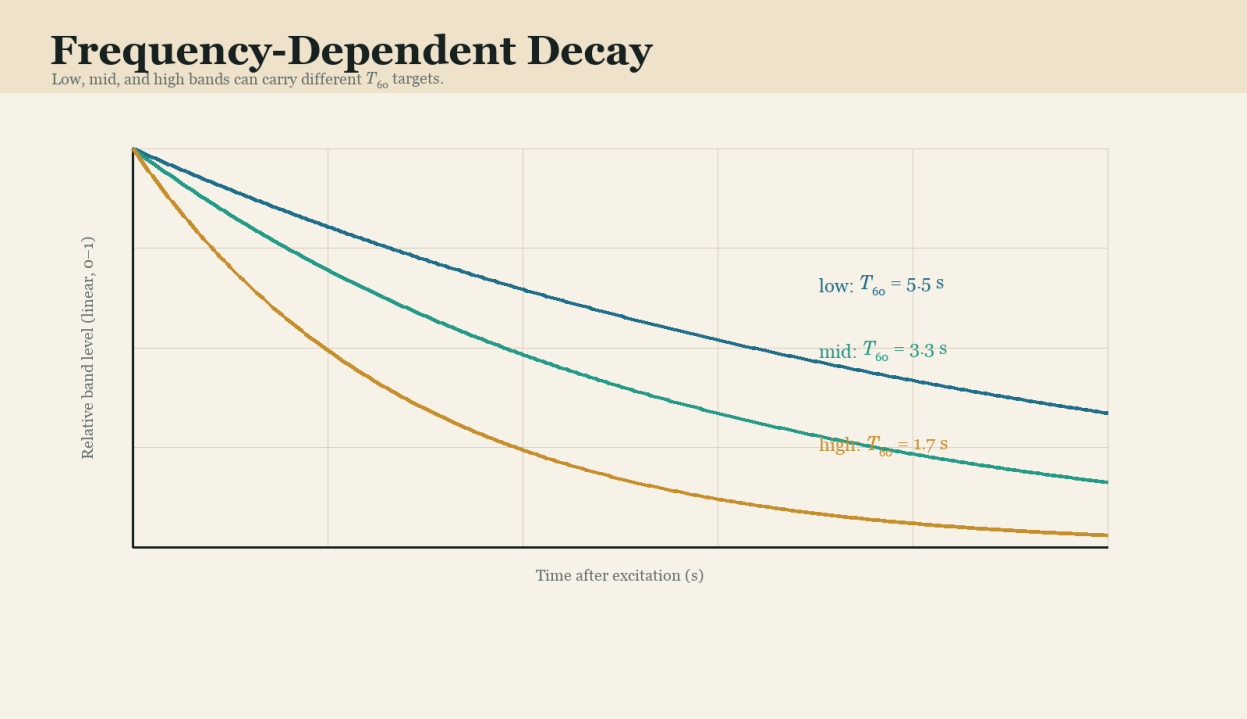


Figure 3-4: Low-, mid-, and high-frequency reverberation decay levels in decibels against time after excitation in seconds

How to read Figure 3-4. At the left edge, all bands are normalized to the same level. The high band then falls fastest, the mid band defines the nominal body of the tail, and the low band persists longest. Their vertical separation grows with time because their slopes, not merely their initial gains, differ. If the high curve were shifted downward but remained parallel to the low curve, the result would be equalization rather than frequency-dependent RT60.

3.1.2.7 Non-Diffuse Rooms, Modes, and Coupled Slopes

The ideal RT60 line assumes a sufficiently mixed field with approximately exponential energy loss. Real spaces can depart from that model in instructive ways. Below a room's transition region, individual modes dominate. Each mode has its own frequency, spatial pattern, and damping rate, so moving a microphone by a fraction of a wavelength can change the apparent low-frequency decay. A broadband average can look acceptable while one note rings for much longer than neighboring notes.

Flutter echo is another departure. Repeated reflections between nearly parallel surfaces create identifiable arrivals rather than a smooth statistical tail. The total energy may decline, but a line fitted through the average does not describe the audible periodicity. Strong focusing, galleries, under-balcony

regions, directional sources, and highly localized absorption likewise make decay depend on position and direction.

Coupled rooms can produce a double-slope EDC. Energy in the primary volume falls quickly at first; energy stored in a secondary chamber then leaks back and supports a slower late tail. In such a response, EDT may be short, T20 intermediate, and T30 long. That disagreement is not necessarily a software defect. It can be evidence that one exponential is an inadequate description of the space. The same behavior is musically useful in artificial reverb: a clear initial release followed by a quiet, persistent halo can preserve articulation while still enlarging phrase endings.

For synthesis, modal density and decay time must be designed together. Raising RT60 does not create more modes; it only lets existing modes remain audible longer. A sparse network with a long target can expose its delay-line frequencies. Increasing line count, choosing incommensurate delays, using an energy-preserving mixing matrix, adding diffusion, and applying subtle modulation all help distribute energy. These operations should not be used to conceal instability. The late field must remain bounded before coloration or motion is judged artistically.

3.1.2.8 RT60 Is Not Wetness, Distance, or Clarity

Several controls can all make a reverb seem "more," but they answer different questions:

Table 3-2: RT60 Is Not Wetness, Distance, or Clarity: reference table

Quantity	Question it answers	Principal audible consequence
RT60	How steeply does late energy decay?	persistence and phrase overlap
Wet level	How loud is the processed return?	effect prominence
Pre-delay	How long before the late field begins?	source separation and apparent scale
DRR	How strong is direct sound relative to reverb?	apparent distance
C_{50} or C_{80}	How much energy arrives early versus late?	speech or musical clarity
EDT	How quickly does the first part release?	perceived immediate reverberance
Spectral RT60	Which frequency bands persist?	warmth, brightness, and masking

A 6 s tail at –30 dB wet level can be subtler than a 1 s room mixed loudly. A long pre-delay can keep a vocal intelligible even when the late field is extensive. Conversely, a short RT60 with dense early reflections and low DRR can push a source far behind the loudspeakers. This is why preset names such as "large," "distant," or "lush" cannot be reduced to one decay-time value.

Clarity metrics also depend on an impulse response's time origin. C_{80} compares energy before and after 80 ms, while C_{50} uses a 50 ms boundary more closely associated with speech. If the direct arrival is misidentified, both ratios become misleading. On complete music rather than an isolated impulse response, ongoing source energy crosses those windows,

so verbx labels such results as program-audio estimates rather than room measurements.

3.1.2.9 Extreme RT60, Freeze, and Infinite-Style Behavior

Finite RT60 and freeze are related but not identical. A finite 3600 s target still specifies a negative slope of –0.0167 dB/s. In exact arithmetic it eventually decays. A freeze mode instead aims for a loop magnitude of one, or introduces recirculation that replaces lost energy, so the state can persist without the usual exponential release. The former is a very long decay; the latter changes the system's operating condition.

As T_{60} grows, the difference between stable and marginal becomes numerically tiny. For a 50 ms delay, the scalar gain is about 0.944 at $T_{60} = 6$ s, but about 0.999904 at $T_{60} = 3600$ s. A normalization error, nonlinear makeup gain, or interpolation overshoot on the order of that remaining margin can change

a slow decay into growth. Extreme settings therefore deserve peak monitoring, output limiting, DC rejection, and long-horizon tests.

Offline rendering introduces a second issue: the requested decay can be much longer than the source. Writing an hour-scale tail for a short test impulse is usually not what a user means, and constructing an equally long proxy impulse response can look like a hung process. Use `--dry-run` to inspect estimated work before an extreme render, choose output duration deliberately, and treat freeze as a performance mode or bounded sound-design operation rather than as permission to allocate an unbounded file.

Figure 3-5 contrasts a normal stable decay, an extremely slow finite decay, an idealized freeze, and an unstable trajectory. The distinction is central to safe long-tail design.

Figure 3-5 presents Stable, near-infinite, frozen, and unstable reverberation-state levels against elapsed time in seconds and establishes the visual reference for the discussion that follows.

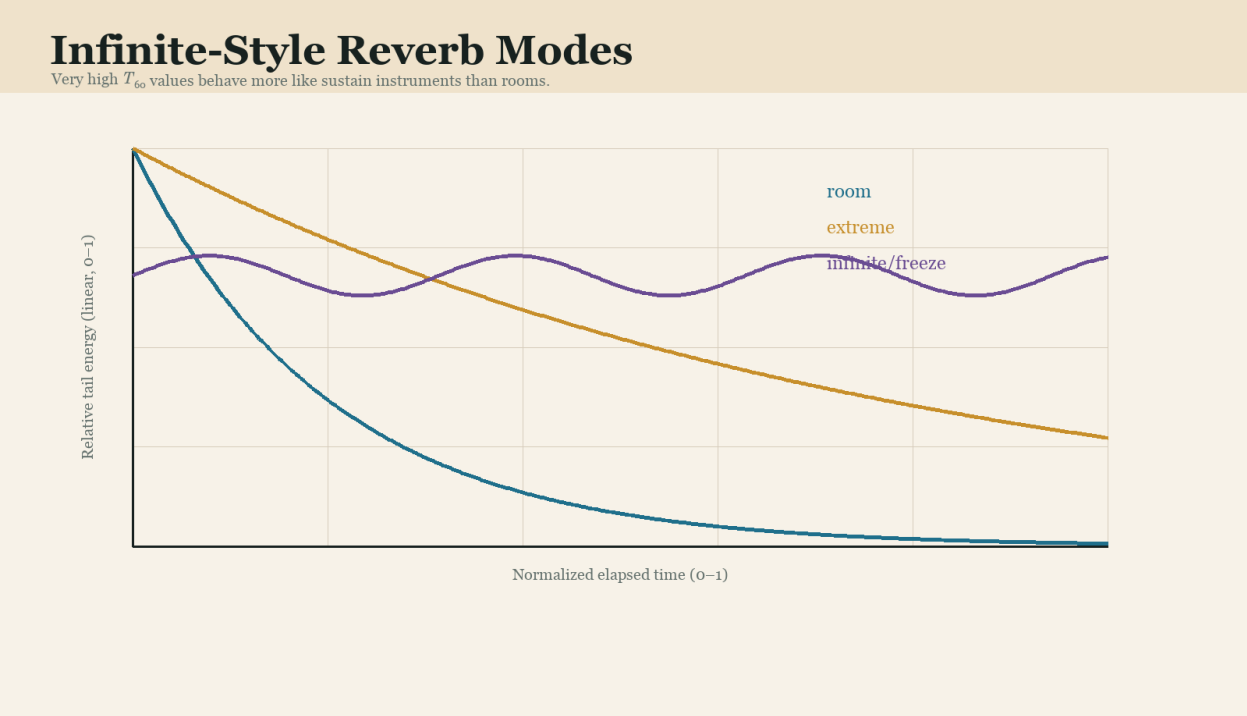


Figure 3-5: Stable, near-infinite, frozen, and unstable reverberation-state levels against elapsed time in seconds

How to read Figure 3-5. The ordinary stable curve slopes visibly downward. The near-infinite curve also descends, but so slowly that a short display can make it look flat. The freeze trajectory remains approximately constant after capture and is intentionally marginal. The unstable trajectory rises; even slow growth is unacceptable because repeated feedback eventually exhausts headroom. The plot should be read as a stability taxonomy, not as a claim that finite-precision freeze can remain mathematically constant forever.

3.1.2.10 Musical Time and RT60

Reverberation time is measured in seconds, but composers and producers hear its interaction with beats, gestures, rests, and harmonic rhythm. At tempo B beats per minute, one beat lasts

$$T_{\text{beat}} = \frac{60}{B} \text{ s}.$$

In 4/4 meter, a four-beat bar lasts

$$T_{\text{bar}} = \frac{240}{B} \text{ s}.$$

These equations do not imply that RT60 must equal a note value. They provide a grid for asking where the tail should sit when the next attack, harmony, or rest arrives. At 120 BPM, a beat is 0.5 s and a bar is 2 s. An RT60 of 2 s reaches –60 dB at the next bar line under the ideal model, but it is already down –15 dB after one beat. Whether that feels connected depends on wet level, spectrum, source density, and masking.

For an ideal decay, the level change after interval t is

$$\Delta L(t) = -60\frac{t}{T_{60}} \text{ dB}.$$

Rearranging gives a useful compositional design equation. If the tail should be D decibels below its onset when the next event arrives after t seconds, choose

$$T_{60} = \frac{60t}{D}.$$

For example, placing the tail about –20 dB below onset at a 1.5 s phrase boundary suggests $T_{60} = 4.5$ s. This is a starting point, not a loudness guarantee, because the source may continue feeding the reverb and frequency bands may decay at different rates.

Short RT60 values can articulate rhythm by giving each transient a compact acoustic frame. Long values can bind separate attacks into a sustained harmonic field, making rests active rather than empty. A long low-frequency decay can retain harmonic roots across chord changes; a long high-frequency decay can carry attacks and noise into the next phrase. Pre-delay can preserve the edge of a note before that field arrives, while ducking can let the tail expand only after the source leaves space.

The most reliable musical method is comparative listening at matched wet loudness. Render three adjacent RT60 values, normalize only for monitoring, and compare the release at exact phrase boundaries. Listen once for articulation, once for harmonic contamination, once for stereo or immersive envelopment, and once at low playback level. A tail that sounds spectacular in solo may occupy every rest in an arrangement; a tail that seems too quiet in solo may be correct in context.

3.1.2.11 Practical verbx Workflows

Begin with a single broadband target and preserve enough dry signal to hear how attacks separate from the room:

```
verbx render in.wav out.wav --engine algo --rt60 2.4 --wet 0.30 --dry 0.80
```

Then shape spectral persistence inside the feedback network rather than relying only on output equalization:

```
verbx render in.wav out_multiband.wav --engine algo --rt60 3.3 \
  --fdn-rt60-low 5.5 --fdn-rt60-mid 3.3 --fdn-rt60-high 1.7 \
  --fdn-xover-low-hz 250 --fdn-xover-high-hz 4000 \
  --wet 0.36 --dry 0.78
```

Generate a deterministic impulse response when the decay itself is the research object. The output length is explicit, so repeated runs remain bounded and comparable:

```
verbx ir gen study_hall.wav --mode fdn --length 8 --rt60 2.4 --seed 42
```

Measure that response and retain both the human-readable summary and machine-readable evidence:

```
verbx analyze study_hall.wav --input-kind ir --edr --room \
  --json-out reports/study_hall.analysis.json
```

For a complete music recording, identify it as program audio so clarity and decay results are not mistaken for standards-style room measurements:

```
verbx analyze wet_mix.wav --input-kind program --lufs \
  --json-out reports/wet_mix.analysis.json
```

A useful validation sequence is to synthesize an 8 s IR with a 2.4 s target, analyze it, and compare EDT, T20, T30, and bandwise estimates. Do not require every value to equal 2.4 s to many decimal places. The early-reflection region, finite file length, frequency shaping, filter transients, stochastic variation, and regression windows all affect measured values. Instead, specify tolerances, hold the seed and sample rate constant, and investigate systematic bias separately from random variation.

When analyzing a physical room, capture multiple source/receiver positions and repeat the measurement. Report the median and spread in each band. One beautifully smooth response can be less representative than several imperfect responses that reveal seat-to-seat variation. For production preset matching, preserve the original IR, its analysis JSON, the verbx version, and the exact command alongside the derived preset.

3.1.2.12 Common Interpretation Failures

"The meter says 2 s, so the room is completely described." A scalar RT60 omits spectral shape, early reflections, direct-to-reverberant ratio, spatial distribution, modulation, and nonexponential behavior. Treat it as one coordinate in a larger acoustic description.

"T30 is always better than T20." T30 uses a deeper range only when that range is clean. If the noise floor intrudes before –35 dB, the apparently more comprehensive estimator can be less trustworthy than T20.

"A longer RT60 should sound louder." RT60 controls slope. Wet gain controls return level. Compare decays at matched loudness before attributing prominence to persistence.

"Post-EQ creates multiband decay." Post-EQ changes spectral balance; loop filtering changes spectral decay slope. Both can be useful, but they are not interchangeable.

"A straight fit proves a diffuse room." A limited window can be straight even when the full response contains modes, double slopes, or directional behavior. Inspect the EDC, the fit interval, bandwise results, and multiple positions.

"A one-hour RT60 is the same as infinite reverb." One has a very small negative slope; the other attempts persistent state. Their stability, rendering, automation, and safety requirements differ.

3.1.2.13 Selected Primary Literature

The following sources provide a compact path from physical reverberation theory through measurement and digital synthesis. Entries are alphabetical by first author.

Eyring, C. F. "Reverberation Time in 'Dead' Rooms." *The Journal of the Acoustical Society of America* 1(2A): 217–241. DOI: 10.1121/1.1915175, 1930.

Jot, J.-M. "An Analysis/Synthesis Approach to Real-Time Artificial Reverberation." *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing*: II-221–II-224. DOI: 10.1109/ICASSP.1992.226080, 1992.

Jot, J.-M.; Chaigne, A. “Maximally Diffusive Yet Efficient Feedback Delay Networks for Artificial Reverberation.” *IEEE Signal Processing Letters* 4(9): 260–263. DOI: 10.1109/97.623041, 1997.

Lundeby, A.; Vigran, T. E.; Bietz, H.; Vorländer, M. “Uncertainties of Measurements in Room Acoustics.” *Acustica* 81: 344–355, 1995.

Schroeder, M. R. “New Method of Measuring Reverberation Time.” *The Journal of the Acoustical Society of America* 37(3): 409–412. DOI: 10.1121/1.1909343, 1965.

Xiang, N. “Generalization of Sabine’s Reverberation Theory.” *The Journal of the Acoustical Society of America* 148(2): R5–R6. DOI: 10.1121/10.0001806, 2020.

3.1.3 Impulse Responses

For beginners: An impulse response (IR) is a recording of what a space does to a single perfect click. When you convolve your audio with an IR, your audio sounds like it was played in that space. verbx can use IR files from external libraries, or generate its own synthetic IRs in four modes. You do not need an IR to use verbx – the algorithmic engine works without one.

For experts: verbx IR synthesis runs in four modes. `fdn` constructs a tail from the same FDN core used in algorithmic rendering, with configurable matrix family and decay parameters. `stochastic` generates exponentially-decayed filtered noise, shaped to match an RT60 curve. `modal` synthesizes a bank of tuned resonators – useful for musically-pitched spaces or physically-inspired objects. `hybrid` combines FDN late field with stochastic early reflections and optional modal resonator coloration. All modes use deterministic content-hash caching so repeated generation with the same parameters retrieves from cache rather than recomputing. The cache is keyed on mode, all synthesis parameters, seed, sample rate, channels, and length.

3.1.4 Wet/Dry Mix

For beginners: `--wet` controls how much reverb you hear; `--dry` controls how much of the original unprocessed signal you keep. Most reverb uses are parallel – you blend the two. Start with `--wet 0.2 --dry 0.8` for subtle room feel and increase wet for more spaciousness. A setting of `--wet 1.0 --dry 0.0` is fully wet with no dry signal – often used in freeze or ambient texture work where you want the reverb itself as the sound.

For experts: verbx allows wet values above 1.0 for deliberate creative overdriving of the wet bus prior to the final mix. This is intentional and distinct from a gain error – it allows the reverb field to dominate with headroom for the loudness and limiter stages downstream to manage levels. Both `--wet` and `--dry` are valid automation targets: you can write time-varying lanes that sweep wet depth over the duration of a render, useful for automating reverb throws or level-responsive gating.

3.2 The Engines

The two engines share routing and post-processing but create the reverberant field differently. Choose by the evidence and behavior required: algorithmic synthesis for controllable or impossible spaces, convolution for a particular measured or designed response.

3.2.1 Algorithmic Engine (`--engine algo`)

The algorithmic engine synthesizes reverb without an impulse response file. It is well suited for extreme tail lengths, evolving or modulated spaces, and creative applications where physical accuracy is not the goal.

What it sounds like: Smooth, dense, fully controllable. At short RT60 values (under 3 seconds) it behaves like a believable room. As RT60 increases past 20–30 seconds, it transitions into something entirely non-physical – a sustained shimmer of harmonic energy that can evolve slowly over minutes. The matrix family is the main texture control: Hadamard produces a more uniform, neutral tail; `tv_unitary` adds slow decorrelation motion; `graph` with ring topology sounds regular and periodic; `random` sounds unpredictable.

Signal flow:

```
input
  └─ pre-delay (z^-Npre)
       └─ allpass diffusion (K stages)
            └─ FDN feedback loop
                 ├─ delay bank (N lines, z^-Ni)
                 ├─ per-line conditioning D_i(z)  [damping + DC block]
                 ├─ RT60 gain G  [diagonal, per-line]
                 ├─ feedback matrix M  [orthonormal family]
                 ├─ optional DFM micro-delays
                 └─ optional link filter
            └─ wet projection
  └─ dry signal
       └─ wet/dry mix → shimmer → bloom/tilt/EQ → loudness → output
```

Delay notation: z^{-N} means an integer-sample delay of N samples.

FDN mechanics: At each sample, the FDN reads from N delay lines, applies per-line damping and DC blocking, multiplies by the gain diagonal G, multiplies by the feedback matrix M, adds the injected excitation from the diffusion stage, and writes back to the delays. The matrix M must be orthonormal (or nearly so) to preserve energy over long tails; verbx orthonormalizes all matrix families before use. The state update is:

$$y[n] = D(x_{\text{fb}}[n])$$

$$x_{\text{fb}}[n+1] = GMy[n] + u[n]$$

where:

- n is the discrete-time sample index.
- $x_{\text{fb}}[n]$ is the feedback-state vector before loop conditioning.
- $y[n]$ is the conditioned state after $D(\cdot)$.
- $D(\cdot)$ is per-line loop conditioning (damping + DC blocking).
- G is the diagonal RT60 gain matrix with entries g_i.
- M is the orthonormal feedback mixing matrix.
- $u[n]$ is the post-diffusion excitation injected into the loop.

FDN gain calibration: For delay line i with period d_i seconds and target decay T_{60}:

$$g_i = 10^{-\frac{3d_i}{T_{60}}}$$

Shorter delay lines require gains closer to 1.0. This is computed per line so different delay lengths in the same network all decay toward the same target RT60.

Matrix families:

Table 3-3: Algorithmic Engine (--engine algo): reference table

Matrix	Sound character	Math note
`hadamard`	Even, neutral density	$N \times N$ Walsh-Hadamard; valid for power-of-2 line counts
`householder`	Similar to Hadamard, slightly more uniform	Householder reflection matrix
`random_orthogonal`	Unpredictable coloration	QR decomposition of random normal matrix
`circulant`	Periodic, regular resonance	Diagonalized by DFT; controlled frequency-domain structure
`elliptic`	Weighted energy distribution	Elliptic rotation-based coupling
`tv_unitary`	Slowly evolving, reduced metallic ringing	Time-varying orthonormal update at `--fdn-tv-rate-hz` Hz
`graph`	Topology-controlled pair mixing	Staged edge interactions over ring/path/star/random graph
`sdn_hybrid`	Geometry-inspired directional scatter	Scattering delay network coupling approach

Key parameters:

Table 3-4: Algorithmic Engine (--engine algo): reference table

Parameter	Range	What it does	Expert note
`--rt60`	0.1–3600	Decay time target (seconds)	Drives per-line gain via $g_i = 10^{-3d_i/T_{60}}$
`--fdn-lines`	2–64	Number of delay lines	Higher line counts increase tail density; above 32 the returns diminish
`--fdn-matrix`	see above	Feedback mixing topology	Controls tail texture and energy diffusion pattern
`--allpass-stages`	0–16	Early diffusion stages	4–10 is typical; 0 disables diffusion entirely
`--allpass-gain`	±0.99	Allpass coefficient	Per-stage or broadcast; must stay inside unit circle
`--comb-cloud`	flag	Optional pre-FDN comb bank	Adds metallic/dense coloration before the late field
`--comb-cloud-mix`	0–1	Comb-cloud blend amount	Start around `0.2` before increasing line count/feedback
`--damping`	0–1	HF rolloff in feedback loop	Higher values darken the tail faster
`--fdn-rt60-tilt`	–1 to 1	Low/high decay skew	Positive = longer lows, shorter highs
`--fdn-link-filter`	none/lowpass/highpass	In-loop spectral shaping	Shapes the spectral flow on feedback edges
`--fdn-tv-rate-hz`	0–5	Time-varying matrix update rate	Active only with `tv_unitary`; slow rates reduce ringing
`--mod-depth-ms`	0–10	Delay modulation depth	Small values suppress metallic resonances
`--width`	0–2	Stereo spread	Increases decorrelation between channels
`--fdn-sparse`	flag	Sparse pair-mixing topology	Higher apparent order at lower compute cost
`--fdn-cascade`	flag	Nested FDN injection	Secondary network feeds early density into primary
`--unsafe-self-oscillate`	flag	UNSAFE above-unity feedback mode	Algorithmic engine only; for intentional self-oscillation
`--unsafe-loop-gain`	0.01–1.25	UNSAFE feedback gain scale	Use `>1.0` to drive oscillation

3.2.2 Spring and Plate Models (`--algo-model`)

`--algo-model spring` and `--algo-model plate` are two constrained voices of the algorithmic engine. They are useful when the musical reference is an electro-mechanical reverb rather than a literal room: guitar tanks, compact combo-amp springs, vocal plates, snare plates, and bright dense returns. They are deterministic signal-processing models by default, not sampled devices.

`--electromechanical-solver modal-fe` additionally provides a bounded offline modal finite-element approximation. Neither mode is a calibration of a particular spring pan or metal plate. That distinction matters. A physical device has transducer nonlinearity, mechanical coupling, mounting loss, noise, pickup placement, and unit-to-unit variation;

verbx deliberately keeps the render reproducible and exposes an auditable parameter set instead.

Select a model explicitly:

```
# Short, splashy guitar-style return.
verbx render guitar.wav guitar_spring.wav \
  --engine algo --algo-model spring --rt60 1.8 \
  --pre-delay-ms 8 --damping 0.70 --wet 0.45 --dry 0.80

# Dense vocal plate. Use dry=0 when this is an auxiliary/send return.
verbx render vocal.wav vocal_plate.wav \
  --engine algo --algo-model plate --rt60 3.2 \
  --pre-delay-ms 14 --damping 0.28 --width 1.35 --wet 1 --dry 0
```

The presets `classic_spring` and `bright_plate` provide conservative starting points. Explicit command-line values still document the intended musical choice, but the model also protects its identity by applying topology bounds. For example, spring keeps short delays, a four-line circulant late field, limited allpass gain, and a minimum amount of motion and damping. Plate uses a larger, denser, brighter diffusion configuration. They should not be treated as arbitrary FDN presets with every underlying topology switch exposed.

3.2.2.1 The Physical Reference

An ideal stretched spring supports longitudinal and torsional wave families. Unlike the ideal string equation, their propagation is visibly dispersive: the group delay depends on frequency, so a transient turns into a succession of frequency-dependent arrivals. A useful conceptual model writes the phase as

$$\phi(\omega) = \omega\tau_0 + \alpha\omega^2,$$

where τ_0 is a nominal transit time and α is a dispersion term. The group delay is the derivative of phase,

$$\tau_g(\omega) = \frac{d\phi}{d\omega} = \tau_0 + 2\alpha\omega.$$

Higher frequencies therefore arrive at a different time from lower frequencies. Real tanks also contain repeated reflections, coupling between multiple springs, and losses at the input/output transducers. A modal view of the response is

$$H_{\text{spring}}(\omega) = \sum_{m=1}^{M} \frac{b_m}{\omega_m^2 - \omega^2 + j\,2\zeta_m\omega_m\omega},$$

with modal frequencies ω_m, damping ratios ζ_m, and pickup/input weights b_m. The irregular mode spacing and frequency-dependent decay are why physical springs can sound "boingy," splashy, or metallic rather than like a small room.

A plate is a thin two-dimensional elastic structure. In an ideal isotropic plate, flexural-wave angular frequency has approximately quadratic dependence on wavenumber:

$$\omega = \sqrt{\frac{D}{\rho h}}\, k^2, \qquad D = \frac{Eh^3}{12(1-\nu^2)},$$

where E is Young's modulus, ρ density, h thickness, ν Poisson's ratio, and D flexural rigidity. A rectangular plate has a mode family often approximated by

$$f_{mn} \propto \left(\frac{m^2}{L_x^2} + \frac{n^2}{L_y^2}\right),$$

for integer mode indices (m, n) and plate dimensions L_x, L_y. Many closely spaced modes, distributed damping, and multiple pickup paths create the characteristically dense, bright, smooth plate tail. The equations describe

the physical reference, not an assertion that verbx solves a plate boundary value problem sample by sample.

3.2.2.2 What verbx Actually Computes

Both models remain inside the stable algorithmic framework. Each delay line uses the same RT60-calibrated feedback gain

$$g_i = 10^{-3d_i/T_{60}},$$

where d_i is the line delay in seconds and T_{60} is `--rt60`. This keeps the decay target interpretable even when the timbre is deliberately electro-mechanical. The feedback network mixes delay states with a matrix M and applies damping inside the loop:

$$x[n+1] = GMD(z)x[n] + u[n].$$

Here $D(z)$ is the frequency-shaping loop filter and $u[n]$ is the diffused input. Because damping is in the feedback path, it changes the decay spectrum rather than merely equalizing the output.

Allpass diffusion supplies echo-density growth without changing the magnitude response of an ideal section. A representative delay-allpass form is

$$A(z) = \frac{a + z^{-N}}{1 + az^{-N}}, \qquad |a| < 1,$$

where N is a delay in samples and a controls the phase/echo pattern. In practice a cascade of sections and short feedback delays creates a dense colored onset before the late field takes over. Small delay modulation changes the effective delay according to

$$d_i[n] = d_i + \Delta d_i \sin(2\pi f_m n / f_s),$$

where Δd_i is `--mod-depth-ms`, f_m is `--mod-rate-hz`, and f_s is sample rate. It is a controlled decorrelation device, not a mechanical model of spring tension changing in real time.

The resulting parameter maps are intentionally practical:

Table 3-5: What verbx Actually Computes: reference table

Model	Fixed character	Main controls to use	Controls deliberately constrained
`spring`	Short, irregular, darkened and moving return	`--rt60`, `--pre-delay-ms`, `--damping`, `--wet`, `--dry`, `--width`	Four-line circulant field; short delay set; allpass stages limited to 3-5; allpass gain no higher than 0.58; modulation no lower than 1.8 ms / 0.35 Hz; damping no lower than 0.58
`plate`	Dense, bright, smooth late field	`--rt60`, `--pre-delay-ms`, `--damping`, `--width`, `--wet`, `--dry`	At least eight FDN lines; Hadamard mixing; at least eight allpass stages; allpass gain at least 0.72; modulation no lower than 0.7 ms / 0.12 Hz; damping no higher than 0.38

`--fdn-lines`, `--fdn-matrix`, and `--comb-delays-ms` are intentionally poor spring/plate controls because the selected model establishes those topology choices. Use `--engine algo --algo-model fdn` when manual FDN topology design is the actual goal.

3.2.2.3 Physical-Proxy CLI Controls

Spring is a multi-element model. `--spring-count` selects how many elements

are active (1 to 8); each repeatable `--spring` option describes one element. The specification is a comma-separated list of `key=value` pairs:

```
verbx render guitar.wav multi_spring.wav --engine algo --algo-model spring \
  --spring-count 3 \
  --spring length_m=0.32,mass_g=20,diameter_mm=0.85,compliance_mm_n=0.45,tension_n=8,damping=0.60 \
  --spring length_m=0.46,mass_g=31,diameter_mm=1.15,compliance_mm_n=0.70,tension_n=5,damping=0.67 \
  --spring length_m=0.64,mass_g=50,diameter_mm=1.55,compliance_mm_n=1.10,tension_n=3,damping=0.76 \
  --rt60 2.1 --wet 0.48 --dry 0.78
```

Each spring accepts `length_m`, `mass_g`, `diameter_mm`, `compliance_mm_n`, `tension_n`, and `damping`. If fewer specifications than `--spring-count` are provided, the remaining elements use the documented defaults. The parser rejects unknown keys, non-positive physical values, and damping outside 0 to 1. Changing length, mass, compliance, or tension changes the derived transit delays; diameter and compliance also influence the bounded decorrelation profile. These values are proxies used to establish a reproducible synthetic topology, not a calibration of a particular commercial spring tank.

Plate controls expose the dimensions and material proxies used to derive its delay-density scale:

```
verbx render vocal.wav large_plate.wav --engine algo --algo-model plate \
  --plate-width-m 2.7 --plate-height-m 1.7 --plate-thickness-mm 0.8 \
  --plate-density-kg-m3 7850 --plate-youngs-gpa 200 \
  --plate-poisson-ratio 0.29 --plate-tension-n 600 \
  --plate-pickup-x 0.18 --plate-pickup-y 0.76 \
  --rt60 3.8 --wet 1 --dry 0
```

`--plate-width-m`, `--plate-height-m`, `--plate-thickness-mm`, `--plate-density-kg-m3`, `--plate-youngs-gpa`, and `--plate-poisson-ratio` form the flexural-rigidity proxy. `--plate-tension-n` adds a bounded tension term. `--plate-pickup-x` and `--plate-pickup-y` are normalized coordinates between 0 and 1 that influence pickup asymmetry. These options are valid for every render for stable configuration files, but they only affect the selected spring or plate model.

3.2.2.4 Modal Finite-Element Solver

For an explicit offline structural response, set `--electromechanical-solver modal-fe`. This replaces the FDN proxy return with a synthesized modal impulse response. It is bounded for repeatable renders: spring chains allow 4 through 128 lumped nodes and plate grids allow 4 through

32 nodes per axis. `--spring-fe-modes` and `--plate-fe-modes` retain up to 128 modes; the render report records the active post-Nyquist mode range.

Figure 3-6 presents Coupled mass-spring-damper tank finite-element model and establishes the visual reference for the discussion that follows.

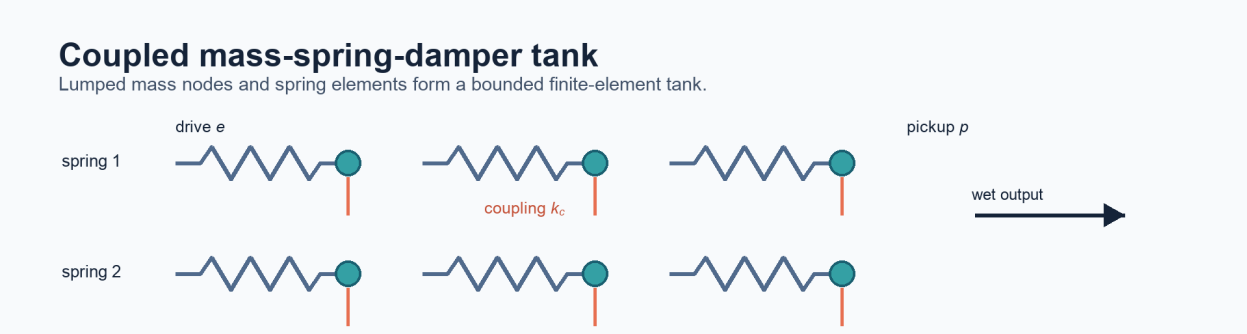


Figure 3-6: Coupled mass-spring-damper tank finite-element model

```
verbx render guitar.wav guitar_tank_fe.wav --engine algo --algo-model spring \
  --electromechanical-solver modal-fe --spring-count 3 \
  --spring-fe-nodes 36 --spring-fe-modes 48 --spring-fe-coupling 0.14 \
  --spring-fe-loss 0.42 --rt60 2.0 --wet 0.55 --dry 0.75

verbx render vocal.wav plate_fe.wav --engine algo --algo-model plate \
  --electromechanical-solver modal-fe --plate-fe-nx 20 --plate-fe-ny 14 \
  --plate-fe-modes 72 --plate-fe-loss 0.18 \
  --plate-pickup-x 0.18 --plate-pickup-y 0.76 --rt60 3.4 --wet 1 --dry 0
```

For a spring, verbx assembles a block mass matrix `M` and a tridiagonal chain stiffness matrix `K`. Multi-spring tanks add a small coupling stiffness between adjacent chains. Normal modes solve:

```
K q[r] = lambda[r] M q[r]
omega[r] = sqrt(lambda[r])
```

The driven/pickup impulse response is a causal sum of damped modes,

```
h(t) = sum over r of [ pickup(r) * drive(r) / omega_d(r) ]
       * exp(-sigma(r) * t) * sin(omega_d(r) * t)
```

`drive` is the input-transducer weighting and `pickup` is the output weighting. RT60 sets the base decay rate; `--spring-fe-loss` increases high-mode loss and `--spring-fe-coupling` controls adjacent-spring coupling.

At element level, the spring displacement vector follows the familiar mass-spring-damper equation

```
M x'' + C x' + K x = e u(t)
```

For a chain with N nodes, verbx assigns node mass $m_i = m_{\text{total}}/N$ and segment stiffness $k_s = (N-1)/C_{\text{total}}$, where C_{total} is the requested metres-per-newton compliance. The tridiagonal stiffness block is assembled from the element contribution

$$k_s \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

plus a driven-end clamp. The solver does not time-step a stiff system sample by sample: it projects the system into normal modes and applies stable exponential modal decay. That retains a mass-spring-damper resonance structure without making the offline render sensitive to an integration step size.

Figure 3-7 presents Clamped plate finite-element grid and establishes the visual reference for the discussion that follows.

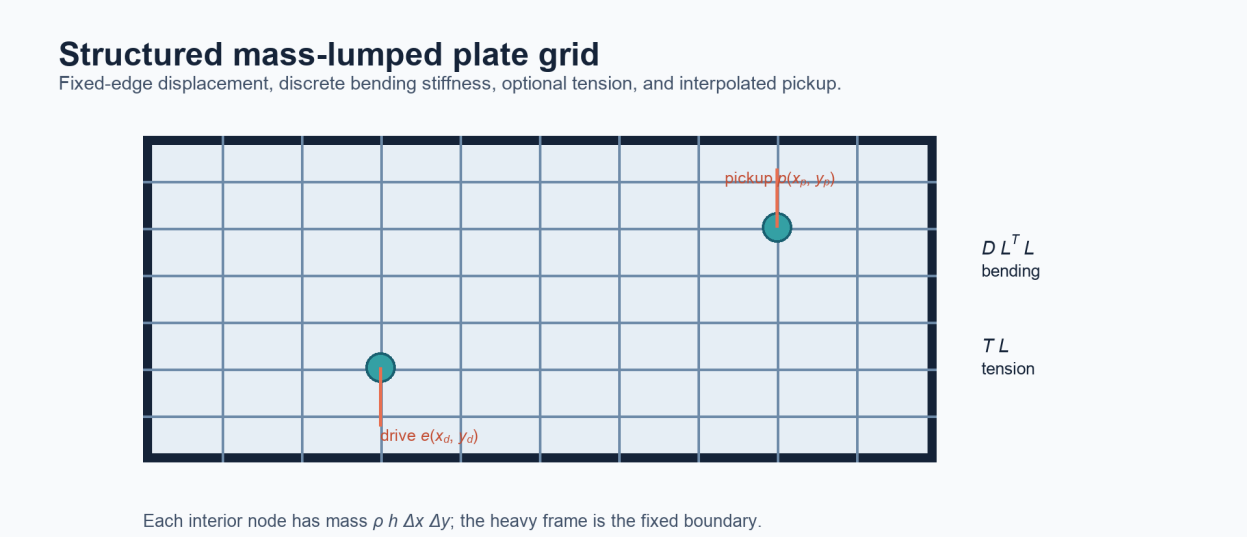


Figure 3-7: Clamped plate finite-element grid

The plate solver uses a structured, mass-lumped clamped grid. Its stiffness is the discrete thin-plate bending term plus optional membrane tension:

$$K = DL^{\mathsf{T}}L + TL,$$
$$M_{ii} = \rho h\,\Delta x\,\Delta y,$$
$$D = \frac{Eh^3}{12\left(1-\nu^2\right)}.$$

Here `L` is the positive finite-difference Laplacian. `--plate-fe-nx`, `--plate-fe-ny`, and `--plate-fe-modes` trade computation for detail; `--plate-fe-loss` gives higher modes stronger decay. The input is fixed off-centre while `--plate-pickup-x/y` select a bilinearly interpolated pickup. This is a deterministic research/sound-design solver, not a fixture, transducer, or hardware-nonlinearity calibration.

For the governing acoustic and structural equations, weak-form discretization, mesh-resolution limits, boundary conditions, modal reduction, room-IR export, hybrid FE/statistical modeling, and validation practice, see the dedicated Finite-Element Modeling chapter.

3.2.2.5 Practical Design and Measurement

For an insert, retain dry signal and keep wet level modest. For a send/return, render 100 percent wet and blend in the DAW or later command stage. Spring usually benefits from a short RT60 and darker damping; plate usually benefits from a longer RT60, low damping, and a pre-delay that lets consonants or drum attacks survive.

```
# Parallel guitar spring: leave the attack clear.
verbx render guitar.wav guitar_spring_parallel.wav \
  --engine algo --algo-model spring --rt60 1.4 --wet 0.32 --dry 0.90 \
  --pre-delay-ms 5 --damping 0.76

# Fully wet snare plate for later bus processing.
verbx render snare.wav snare_plate_return.wav \
  --engine algo --algo-model plate --rt60 1.6 --wet 1 --dry 0 \
  --pre-delay-ms 10 --damping 0.22 --width 1.5
```

Use `verbx analyze` on a rendered wet return to compare EDT, T20/T30, spectral centroid, and clarity across parameter variations. Those measurements describe the generated signal, not a certification that it matches a particular hardware unit. Preserve the command, verbx version, and JSON report when a sound becomes part of a production or a dataset.

3.2.3 Convolution Engine (`--engine conv`)

The convolution engine filters audio through an impulse response. Use it when you want the character of a specific space – measured or synthesized – applied exactly.

What it sounds like: The output has the exact spectral and temporal character of the IR. A measured cathedral IR makes everything sound like it was played in that cathedral. A verbx-generated hybrid IR sounds like a designed space tuned to your specifications. Self-convolution (`--self-convolve`) smears a sound with its own spectral envelope – a different kind of effect.

Partitioned convolution: For long IRs, direct time-domain convolution is impractical. verbx uses uniformly-partitioned overlap-save convolution in the frequency domain:

$$Y_k(\omega) = \sum_{p=0}^{P-1} X_{k-p}(\omega)\, H_p(\omega)$$

where:

- k is the current processing frame index.
- ω is frequency-bin index in the FFT domain.
- P is the number of IR partitions.
- $X_{k-p}(\omega)$ is the stored input spectrum for frame $k-p$.
- $H_p(\omega)$ is the precomputed spectrum of IR partition p.
- $Y_k(\omega)$ is the accumulated output spectrum for frame k.

`--partition-size` controls the partition length: larger partitions reduce per-block FFT overhead but increase latency and peak memory. 16384–65536 samples is a practical range for offline rendering. With CuPy installed and `--device cuda`, the FFT multiply accumulation runs on GPU.

Streaming vs. in-memory: verbx automatically uses streaming convolution (low peak RAM) when the render is simple: engine conv, no repeat, no freeze, no normalization stages, no post-processing effects. All other combinations fall back to full-buffer processing. If RAM is a concern for very long IRs, keep the render chain minimal.

Multichannel routing: For M input channels and N output channels:

$$y_o[n] = \sum_{i=0}^{M-1} \left(x_i * h_{i,o}\right)[n]$$

where:

- M is input-channel count and N is output-channel count.
- $x_i[n]$ is input channel i.
- $h_{i,o}[n]$ is the IR from input channel i to output channel o.
- $y_o[n]$ is output channel o.
- $*$ denotes linear convolution.

The IR file must contain $M \times N$ channels packed in output-major order (channel index $oM + i$) or input-major order ($iN + o$). Set `--ir-matrix-layout output-major` or `input-major` accordingly. Wrong packing order produces valid audio but semantically incorrect routing; verify with `verbx analyze` on the output.

Key parameters:

Table 3-6: Convolution Engine (--engine conv): reference table

Parameter	Range/values	What it does	Expert note
`--ir`	file path	Impulse response to apply	WAV, FLAC, AIFF, OGG, CAF all supported
`--partition-size`	1024–131072	FFT block size	Larger = more throughput, higher latency
`--tail-limit`	seconds	Cap IR tail at this length	Useful to bound compute on long IRs in batch
`--ir-normalize`	peak/rms/none	IR level normalization	`peak` is safest for predictable headroom
`--ir-matrix-layout`	output-major/input-major	Multichannel channel packing	Must match how the IR was created
`--ir-blend`	file path	Second IR for render-time blending	Repeatable; blend alpha automatable
`--ir-blend-mode`	linear/equal-power/spectral/envelope-aware	Blend algorithm	`envelope-aware` preserves early reflection character independently from late tail
`--self-convolve`	flag	Input file is its own IR	Spectral smearing / texture effect
`--device`	auto/cpu/cuda/mps	Compute backend	CUDA requires CuPy; MPS uses Apple Silicon profile

3.3 Impulse Response Synthesis

verbx generates its own IRs in four synthesis modes. The complete parameter reference is in docs/IR_SYNTHESIS.md. The IR toolchain is accessible via `verbx ir gen`, or triggered inline during render with `--ir-gen`.

A larger curated IR set is available in `IRs/library/` with folder-sorted buckets by length (`tiny`, `short`, `medium`, `long`) and mode (`fdn`, `stochastic`, `modal`, `hybrid`), plus deterministic metadata in `IRs/library/manifest.json`.

Synthesis modes:

Table 3-7: Impulse Response Synthesis: reference table

Mode	Character	Best for
`fdn`	Smooth, configurable, FDN-consistent	Spaces that match your algorithmic render topology
`stochastic`	Diffuse, noise-shaped, natural-sounding	Neutral halls, rooms, generic reverbs
`modal`	Resonant, tonal, pitched	Metal objects, tuned rooms, experimental textures
`hybrid`	Stochastic early + FDN late + optional modal resonator	Most general use; strong default

Common use case table:

Table 3-8: Impulse Response Synthesis: reference table

Goal	Mode	Key flags
Neutral hall, natural decay	`stochastic`	`--rt60 3.0 --damping 0.4`
Match your FDN render topology	`fdn`	Same `--fdn-lines`, `--fdn-matrix` as render
Musical, pitched resonances	`modal`	`--f0 "64 Hz" --modal-count 40`
Microtonal or alternate-tuning resonance	any mode	`--scala-file scale.scl --scala-root-hz 220`
General cinematic space	`hybrid`	`--length 120 --seed 42`
Analyze and match audio source	`hybrid`	`--analyze-input source.wav`

3.3.1 Scala and microtonal resonance tuning

`verbx ir gen` can read a standard Scala `.scl` file and emphasize its pitch classes across a bounded frequency range. Scala cents, integer ratios, fractional ratios, comments, non-octave repeat intervals, and a selectable root degree are supported. The same target set tunes modal frequencies and adds a bounded constant-Q emphasis layer to `fdn`, `stochastic`, `modal`, and `hybrid` IRs, so the result can reinforce a scale without turning every tail into a set of isolated sine tones.

```
verbx ir gen irs/19edo_room.wav \
  --mode hybrid --length 12 --rt60 5.5 \
  --scala-file examples/scales/19edo.scl \
  --scala-root-hz 220 --scala-root-degree 0 \
  --scala-low-hz 100 --scala-high-hz 8000 \
  --scala-strength 0.7 --scala-bandwidth-cents 24 --scala-gain-db 5

verbx render in.wav out.wav --engine conv --ir irs/19edo_room.wav
verbx realtime --engine conv --ir irs/19edo_room.wav --block-size 128
```

Scala processing happens while the IR is generated, not inside the realtime callback. This keeps realtime latency unchanged and makes one tuned IR usable from the CLI, a DAW convolution host, or an ML data pipeline. The metadata sidecar records the scale description, SHA-256 content hash, root mapping, resolved target frequencies, strength, bandwidth, gain, and target budget. `--scala-file` is intentionally exclusive with `--analyze-input` and `--f0`; use `--scala-root-hz` when a scale is active.

Generated IRs are cached by content hash plus parameters. Repeated calls with the same settings return from cache immediately; changing the `.scl` contents changes its hash and cache identity even when the filename stays the same.

The complete Microtonal Workflows and Scala Import chapter develops this into a production and composition method. It covers Scala syntax, root mapping, non-octave periods, transposition, tuned-IR libraries, changing harmony, spatial deployment, ML dataset design, and the musical implications of letting a scale remain audible in the decay field.

```
verbx ir gen my_space.wav --mode hybrid --length 120 --rt60 8.0 --seed 42
verbx ir analyze my_space.wav --json-out my_space_analysis.json
verbx ir morph space_A.wav space_B.wav blended.wav --mode equal-power --alpha 0.5
```

3.4 Effects and Post-Processing

These stages act on or around the wet field after its basic room response has been formed. Add one at a time, level-match the result, and preserve enough tail after the source to hear whether the effect remains bounded.

3.4.1 Shimmer

Shimmer pitch-shifts the reverb tail (typically up an octave) and blends it back into the wet signal. The result is a bright, harmonically rich coloration that works well on pads, sustained notes, and ambient textures. The `--shimmer-feedback` parameter is the one most people get wrong: above around 0.85, the feedback loop builds exponentially. This is not a bug – it is the intended mechanism for extreme infinite-rise textures – but it requires either a tail limit, loudness targeting, or deliberate management to avoid runaway gain.

Safe mode clamps `--shimmer-feedback` to `0.98`. For intentional self-oscillation in the algorithmic path, enable `--unsafe-self-oscillate` and use `--unsafe-loop-gain > 1.0` (for example `1.03`).

```
--shimmer --shimmer-semitones 12 --shimmer-mix 0.35 --shimmer-feedback 0.70
--shimmer-lowcut 300 --shimmer-highcut 12000    # control frequency range of shimmer layer
```

3.4.2 Comb Cloud

`--comb-cloud` inserts a separate bank of decorrelated feedback comb filters between the diffusion stage and the main FDN late field. It is an optional color mode for when you want extra density, metallic haze, or exaggerated spatial smear without redefining the core FDN topology. The point is texture, not neutrality.

Use it when the base algorithmic tail feels too smooth or too well-behaved:

- plates and metallic chambers
- sci-fi interiors and synthetic spaces
- frozen, haunted, or intentionally "wrong" ambience
- pre-shimmer thickening before harmonic coloration

Start conservatively. `--comb-cloud-mix 0.15–0.35` is usually enough. Higher `--comb-cloud-feedback` values push the sound toward ringing and resonant buildup.

```
--comb-cloud --comb-cloud-count 24 --comb-cloud-feedback 0.35 --comb-cloud-mix 0.25
--comb-cloud-delays-ms 6,9,13,17,23,29,37,49   # optional custom cloud spacing
```

Practical tip: comb cloud and shimmer solve different problems. Comb cloud thickens and roughens the time structure; shimmer adds pitched harmonic content. If the reverb feels sterile, try comb cloud first. If it feels emotionally flat, try shimmer.

3.4.3 Freeze / Repeat

`--freeze` locks onto a segment of audio (defined by `--start` and `--end` in seconds) and loops it through the reverb engine with an equal-power crossfade at loop boundaries. This produces sustained, near-static textures. `--repeat N` runs the full render chain N times sequentially, each pass using the output of the previous as input – an iterative reprocessing that progressively imprints the room resonance on the source. Classic application: Alvin Lucier's *I Am Sitting in a Room* (1969) technique.

Use `--output-peak-norm input` with repeat chains to keep levels stable across passes.

3.4.4 Ducking

`--duck` is the reverb effect most mix engineers do not use until they hear it. It attenuates the reverb output while the source signal is active, then lets the tail bloom in the gaps. The effect keeps the dry signal clean and articulate while the reverb is still long and spacious. Especially effective on drums, vocals, and anything with rhythmic transients.

```
--duck --duck-attack 15 --duck-release 250 --duck-strength 0.9 --duck-floor 0.18
```

Attack controls how quickly the reverb ducks when signal appears; release controls how quickly it recovers. Shorter release times give a punchier, more gated feel. `--duck-strength` controls how deep the wet attenuation goes, while `--duck-floor` keeps some reverb present even at the deepest point of the pump.

3.4.5 Bloom

`--bloom N` emphasizes the slow build-up phase of the wet field, creating a cinematic swell effect where the reverb tail rises rather than immediately decaying. Values between 1.5 and 3.0 are perceptible as a rise before the decay plateau. Higher values push into dramatic orchestral-swell territory. It operates on the spectral envelope of the wet output and is distinct from simple gain automation.

Use `--bloom-mix` when you want the bloom time constant from `--bloom` but a more restrained or more exaggerated blend than the automatic scaling would choose.

3.4.6 Tilt EQ

`--tilt N` applies a broadband spectral tilt to the wet field. Positive values (try 1.0–3.0) brighten the reverb tail; negative values darken it. This is a post-wet control, so it does not affect the dry signal or the decay mathematics – it only shapes the perceptual tone of the reverb output. Combine with `--lowcut` and `--highcut` for more specific frequency management.

`--tilt-pivot-hz` moves the tonal fulcrum of that tilt, while `--lowcut-order` and `--highcut-order` let you choose gentler or steeper post-wet filter slopes.

3.5 Spatial and Surround

For most uses, stereo output is all you need. Multichannel processing becomes relevant when you are delivering to a surround format, working in Ambisonics, or routing reverb through a spatial bus.

For a complete treatment of channel beds, height layers, Ambisonics, Dolby Atmos beds and objects, binaural monitoring, DAW handoff, deliverables, and immersive QC, read Immersive Reverb, Surround Sound, and Dolby Atmos. The chapter includes signal-flow diagrams, routing recipes, and a precise account of what verbx can and cannot author today.

Wave field synthesis (WFS): verbx can prepare deterministic dry, early-return, and late-return stems plus analysis reports for a WFS installation, but it does not calculate array driving functions, import loudspeaker coordinates, or replace the calibrated WFS renderer. Keep direct source, early images, and late field as separate assets; let the WFS system assign virtual sources and array channels. See Wave Field Synthesis: Virtual Sources, Real Arrays, and verbx Stems for the full workflow.

Channel layouts:

Table 3-9: Spatial and Surround: reference table

Layout	Channels	Use case
`mono`	1	Mono sources or mono IR processing
`stereo`	2	Standard stereo output
`LCR`	3	Left/Center/Right film format
`5.1`	6	Standard surround
`7.1`	8	Expanded surround
`7.1.2`	10	Standard Atmos bed or fixed-channel immersive bus
`7.1.4`	12	Common Atmos monitoring/render layout; not the default Atmos bed
`7.2.4`	13	7-bed + dual-LFE + 4-top layout
`8.0`	8	8-channel bed without dedicated LFE
`16.0`	16	Large-format discrete bed

Layout	Channels	Use case
`64.4`	68	High-density immersive bed + top layer

Use `--input-layout` and `--output-layout` to declare channel semantics explicitly. Without them, verbx uses channel count alone, which can produce ambiguous routing for formats above stereo.

Atmos boundary: verbx writes channel-based WAVE files, Ambisonic material, matrix-routed convolution outputs, JSON analysis, and handoff manifests. It does not currently author Dolby object trajectories, per-object binaural metadata, or a native ADM BWF/DAMF master. Prepare bed and object stems in verbx, then perform object assignment, metadata authoring, endpoint rendering, and master export in an Atmos-capable DAW and the Dolby Atmos Renderer.

The distinction between a bed and a monitoring layout matters. Dolby's standard bed is 7.1.2, while 7.1.4 commonly describes a loudspeaker render with four independently fed height speakers. verbx currently labels channels 9–10 of its symbolic `7.1.2` layout `Ltf/Rtf`; a Dolby bed expects `Ltm/Rtm` in those positions. Verify and explicitly map those channels at handoff rather than relying on channel count.

For large immersive outputs (`16.0`, `64.4`), set `--ir-route-map` explicitly when the IR is mono or channel-matched to the input. Recommended defaults:

- `--ir-route-map broadcast` for mono/channel-matched IRs
- `--ir-route-map full` for matrix-packed $M \times N$ IRs

Other formats are also easy to support: the routing and DSP paths already operate on arbitrary channel counts, and new symbolic layout names are straightforward to add when you need explicit semantics.

Ambisonics: verbx supports First-Order Ambisonics (FOA) with ACN channel ordering and SN3D/N3D/FuMa normalization. Use `--ambi-order 1` to declare FOA mode. `--ambi-encode-from stereo` encodes a stereo input into FOA before processing; `--ambi-decode-to stereo` decodes back out after. `--ambi-rotate-yaw-deg` applies rotation in the Ambisonics domain – useful for spatial orientation of the reverb field relative to a listener position. FUMA is FOA-only; ACN with SN3D is the standard workflow for most Ambisonics toolchains.

IR matrix routing for surround: If your IR file contains $M \times N$ channels (for M input and N output channels), declare the packing order with `--ir-matrix-layout`. Output-major packing stores all inputs for output 0 first, then all inputs for output 1, etc. (channel index $oM + i$). Input-major stores all outputs for input 0 first (channel index $iN + o$). A 5.1 input to 5.1 output full-matrix IR has 36 channels; a diagonal (same IR per channel) has 6. The routing is explicit: verbx does not guess.

3.6 Loudness and Metering

Most audio delivered for broadcast, streaming, or film needs to hit a loudness target. EBU R128 / ITU-R BS.1770 defines integrated loudness in LUFS (Loudness Units relative to Full Scale). The practical difference between targeting –23 LUFS for broadcast and –14 LUFS for streaming can be over 9 dB of apparent level – enough to sound completely wrong in one context if mastered for the other.

verbx has a full loudness pipeline:

- **`--target-lufs N`** measures integrated loudness and scales to the target. Applies after the reverb processing stage.
- **`--target-peak-dbfs N`** enforces a peak ceiling. Use with `--true-peak` for inter-sample peak checking (required for formats that will be transcoded, as codec interpolation can raise peaks above the stored sample values). Sample peak (`--sample-peak`) is sufficient for archival.

- **`--output-peak-norm [input|target|full-scale]`** is a final-stage peak fit applied after all other processing: `input` matches the input file's peak, `target` uses an explicit dBFS value, `full-scale` normalizes to near 0 dBFS.
- **Soft limiter:** enabled by default as a final safety stage. Disable with `--no-limiter` when you want to pass raw dynamics to a downstream limiter in your chain.

The loudness and peak stages are intentionally separate because they serve different goals. Loudness targeting is about program-level normalization. Peak ceiling is about short-term safety. Do not conflate them.

True-peak detection uses oversampled measurement (ITU-R BS.1770). The difference between a sample peak of –0.1 dBFS and a true peak of +0.4 dBFS is invisible in sample-domain inspection but will cause clipping in AAC, MP3, and most streaming codecs. Use `--true-peak --target-peak-dbfs -1` for any output that will be transcoded.

Week 3 delivery sanity checks now fail fast when an explicit limiter threshold is above the limiter ceiling, because that silently collapses the useful gain-reduction range. Explicit container choices also need matching extensions: use `.w64` with `--output-container w64`, `.rf64` with `--output-container rf64`, or leave `--output-container auto` on when you want verbx to infer the container.

Ready-to-run delivery examples:

```
# Broadcast/streaming-safe limiter and peak-normalized output
verbx render in.wav out_limited.wav --preset limiter-broadcast-safe \
  --output-peak-norm target --output-peak-target-dbfs -1

# Bounded long-tail delivery with an explicit W64 container
verbx render in.wav out_long.w64 --preset delivery-long-tail-safe \
  --output-container w64 --tail-limit 12
```

3.7 Automation and Modulation

The automation system lets you change reverb parameters over the duration of a render – wet depth, RT60, room size, decay tilt, IR blend position, and more – without editing the audio manually. This is useful for: reverb throws (sudden wet increase on a vocal), automated room size sweeps during a sound design cue, feature-reactive ducking where loudness in the source drives reverb depth, and batch augmentation where different parameter curves are applied to each variant.

Automation lanes via file or inline points:

```
# Sweep RT60 from 0.8s to 10s over 15 seconds
verbx render in.wav out.wav --engine algo \
  --automation-point "rt60:0.0:0.8:linear" \
  --automation-point "rt60:15.0:10.0:linear"

# From a JSON automation file
verbx render in.wav out.wav --automation-file automation.json
```

Automation targets include: `wet`, `dry`, `gain-db`, `rt60`, `damping`, `room-size`, `room-size-macro`, `clarity-macro`, `warmth-macro`, `envelopment-macro`, `fdn-rt60-tilt`, `fdn-tonal-correction-strength`, `ir-blend-alpha` (requires `--ir-blend`).

Feature vector lanes drive automation from frame-level audio analysis of the source signal. A "feature vector" is a time-series of per-frame descriptors extracted from audio: loudness, transient strength, spectral centroid, spectral flatness, harmonic ratio, MFCCs, formant spread, rhythm pulse, and more. You map these onto render targets with weight, curve shape (linear, smoothstep, power), and hysteresis:

```
# Wet depth tracks loudness and transients in the source
verbx render in.wav out.wav --engine conv --ir hall.wav \
  --feature-vector-lane "target=wet,source=loudness_norm,weight=0.70,curve=smoothstep,combine=
   ↪  replace" \
  --feature-vector-lane "target=wet,source=transient_strength,weight=0.30,curve=power,curve_
   ↪  amount=1.4,combine=add" \
  --feature-vector-trace-out trace.csv
```

Use `--feature-guide GUIDE.wav` to drive feature extraction from a separate audio file rather than the render input – a sidechain-style workflow.

Modulation bus provides LFO and envelope sources for simple periodic or input-reactive variation without needing a full automation file:

```
--mod-target mix --mod-min 0.1 --mod-max 0.9 \
  --mod-source "lfo:sine:0.07:1.0*0.7" \
  --mod-source "env:20:350*0.4"
```

Source syntax: `lfo:<shape>:<rate_hz>[:depth[:phase_deg]][*weight] | env[:attack_ms[:release_ms]][*weight] | audio-env:<path>[:attack_ms[:release_ms]][*weight] | const:<value>[*weight]`

3.8 Python API (stable surface)

For notebook/research pipelines, use `verbx.api` instead of shelling out to the CLI. The stable API surface includes:

- `render_file(infile, outfile, config=RenderConfig(...), **overrides)`
- `generate_ir(out_ir, config=IRGenConfig(...), write_metadata=True)`
- `analyze_file(infile, include_loudness=False, include_edr=False, ...)`

```
from verbx.api import analyze_file, generate_ir, render_file
from verbx.config import RenderConfig
from verbx.ir.generator import IRGenConfig

render_report = render_file(
    "in.wav",
    "out.wav",
    config=RenderConfig(engine="algo", rt60=2.5, progress=False, silent=True),
)

ir_report = generate_ir(
    "ir.wav",
    config=IRGenConfig(mode="stochastic", length=0.5, sr=48_000, channels=2),
)

analysis = analyze_file("out.wav", include_loudness=True, json_out="analysis.json")
```

3.9 CLI Reference

Canonical help snapshots live in docs/CLI_REFERENCE.md. This README section is a curated quick-reference for common switches.

3.9.1 verbx render

```
verbx render INFILE OUTFILE [options]
```

3.9.1.1 Engine and Room Behavior

These switches select the engine and establish the room's time, density, tone, and feedback topology.

Table 3-10: Engine and Room Behavior: reference table

Switch	Range	What it does	Expert note
--engine	algo/conv/auto	Reverb engine	auto picks conv if IR present, else algo
--rt60	0.1–3600	Decay time (seconds)	Per-line gain via $g_i = 10^{-3d_i/T_{60}}$
--wet	0–∞	Wet signal level	Values >1.0 overdrive wet bus intentionally
--dry	0–1	Dry signal level	
--pre-delay-ms	0–500	Reverb onset delay (ms)	
--pre-delay	e.g. 1/8D	Musical note-value pre-delay	Requires --bpm
--bpm	float	Tempo for note-based pre-delay	
--damping	0–1	HF decay rate in feedback	Higher = darker tail
--width	0–2	Stereo decorrelation	
--allpass-stages	0–16	Diffusion stage count	
--allpass-gain	±0.99	Per-stage allpass coefficient	Comma-separated per-stage list accepted
--comb-cloud	flag	Enable optional pre-FDN comb-cloud coloration	Separate color stage; default off
--comb-cloud-count	1–128	Generated comb-cloud line count	Higher = denser, more colored
--comb-cloud-feedback	0–0.95	Comb-cloud feedback amount	Higher = more ringing / metallicity
--comb-cloud-mix	0–1	Blend between diffusion output and comb-cloud output	0.15–0.35 is a strong starting range
--comb-cloud-delays-ms	comma list	Custom comb-cloud delay list in ms	Auto-enables comb cloud
--comb-cloud-seed	int	Deterministic comb-cloud seed	Changes generated spacing/decorrelation
--fdn-lines	2–64	Delay line count	
--fdn-matrix	see table above	Feedback matrix family	
--fdn-tv-rate-hz	0–5	TV-unitary update rate	tv_unitary only
--fdn-tv-depth	0–1	TV-unitary blend depth	tv_unitary only
--fdn-matrix-morph-to	matrix family	Target matrix for gradual morphing	Morphs from --fdn-matrix to target
--fdn-matrix-morph-seconds	seconds	Matrix morph duration	Requires --fdn-matrix-morph-to
--fdn-dfm-delays-ms	float	DFM micro-delay size	One value or one per line
--fdn-sparse	flag	Sparse pair-mixing topology	Exclusive with tv_unitary and graph
--fdn-sparse-degree	1–8	Pair-mixing stages	
--fdn-link-filter	none/lowpass/highpass	In-loop spectral shaping	
--fdn-link-filter-hz	Hz	Link filter cutoff	
--fdn-rt60-tilt	–1 to 1	Low/high RT skew	Positive = longer lows

Switch	Range	What it does	Expert note
`--fdn-tonal-correction-strength`	0–1	Decay-color equalization	Track C control
`--fdn-cascade`	flag	Nested FDN injection	
`--fdn-graph-topology`	ring/path/star/random	Graph topology	`graph` matrix only
`--fdn-spatial-coupling-mode`	none/adjacent/front_rear/bed_top/all_to_all	Channel wet-bus coupling	
`--fdn-nonlinearity`	none/tanh/softclip	In-loop saturation	Keep blend low: 0.05–0.25
`--beast-mode`	1–100	Parameter multiplier	2–5 for heavier ambience, 10+ for extreme

3.9.1.2 RT60 and Multiband Decay

Use the multiband controls when one broadband RT60 hides a musically important difference between low-, mid-, and high-frequency decay.

Table 3-11: RT60 and Multiband Decay: reference table

Switch	Range	What it does
`--fdn-rt60-low`	seconds	Low-band RT60 target
`--fdn-rt60-mid`	seconds	Mid-band RT60 target
`--fdn-rt60-high`	seconds	High-band RT60 target
`--fdn-xover-low-hz`	Hz	Low/mid crossover
`--fdn-xover-high-hz`	Hz	Mid/high crossover

3.9.1.3 Convolution and IR Routing

This group identifies the impulse response, its normalization, channel packing, partitioning, and optional synthesis or blending behavior.

Table 3-12: Convolution and IR Routing: reference table

Switch	Values	What it does
`--ir`	file path	External IR for convolution
`--ir-normalize`	peak/rms/none	IR normalization before convolution
`--ir-matrix-layout`	output-major/input-major	Multichannel IR channel packing
`--ir-route-map`	auto/diagonal/broadcast/full	Channel routing strategy
`--partition-size`	int	FFT partition size
`--tail-limit`	seconds	Cap convolution tail
`--self-convolve`	flag	Use input as its own IR
`--ir-blend`	file path	Blend a second IR at render time (repeatable)
`--ir-blend-mix`	0–1	Blend coefficient(s)
`--ir-blend-mode`	linear/equal-power/spectral/envelope-aware	Blend algorithm
`--ir-gen`	flag	Auto-generate IR before render
`--ir-gen-mode`	fdn/stochastic/modal/hybrid	IR synthesis mode
`--ir-gen-length`	seconds	Generated IR duration
`--ir-gen-seed`	int	Deterministic seed

3.9.1.4 Spatial

Spatial switches declare channel semantics and Ambisonic conventions. A matching channel count without the correct order and normalization is not a valid interchange.

Table 3-13: Spatial: reference table

Switch	Values	What it does
`--input-layout`	auto/mono/stereo/LCR/5.1/7.1/7.1.2/7.1.4/7.2.4/8.0/16.0/64.4	Input channel semantics
`--output-layout`	auto/mono/stereo/LCR/5.1/7.1/7.1.2/7.1.4/7.2.4/8.0/16.0/64.4	Output channel semantics
`--ambi-order`	0–7	Ambisonics order (1 = FOA)
`--ambi-normalization`	auto/sn3d/n3d/fuma	Normalization convention
`--channel-order`	auto/acn/fuma	Channel ordering convention
`--ambi-encode-from`	none/mono/stereo	Encode to FOA before render
`--ambi-decode-to`	none/stereo	Decode from Ambisonics after render
`--ambi-rotate-yaw-deg`	degrees	Yaw rotation in Ambisonics domain

For large immersive buses, declare the target explicitly rather than relying on channel count alone:

```
verbx render in.wav out_7_2_4.wav --output-layout 7.2.4
verbx render in.wav out_8_0.wav --output-layout 8.0
verbx render in.wav out_16_0.wav --output-layout 16.0
verbx render in.wav out_64_4.wav --output-layout 64.4
```

3.9.1.5 Effects

The effect switches reshape the wet return through pitch, dynamics, filtering, repetition, or controlled feedback.

Table 3-14: Effects: reference table

Switch	Values	What it does
`--shimmer`	flag	Enable shimmer (pitch-shifted reverb coloration)
`--shimmer-semitones`	semitones	Pitch shift amount
`--shimmer-mix`	0–1	Shimmer blend
`--shimmer-feedback`	0–1.25	Shimmer feedback (>0.85 = rising; >0.98 requires unsafe mode)
`--unsafe-self-oscillate`	flag	UNSAFE: allow above-unity feedback in algorithmic mode
`--unsafe-loop-gain`	0.01–1.25	UNSAFE algorithmic loop-gain scale (`>1.0` for self-oscillation)
`--shimmer-spatial`	flag	Enable multichannel shimmer decorrelation
`--shimmer-spread-cents`	cents	Per-channel shimmer detune spread
`--shimmer-decorrelation-ms`	ms	Per-channel shimmer delay spread
`--duck`	flag	Enable sidechain ducking
`--duck-attack`	ms	Ducking attack time
`--duck-release`	ms	Ducking release time
`--duck-strength`	0–1	Ducking depth
`--duck-floor`	0–1	Minimum wet gain during ducking
`--bloom`	0–5	Wet field build-up emphasis
`--bloom-mix`	0–1	Bloom blend override
`--lowcut`	Hz	Post-wet high-pass filter
`--lowcut-order`	1–8	High-pass slope order
`--highcut`	Hz	Post-wet low-pass filter
`--highcut-order`	1–8	Low-pass slope order
`--tilt`	dB/oct	Post-wet spectral tilt
`--tilt-pivot-hz`	Hz	Tilt pivot frequency
`--freeze`	flag	Loop a segment through the engine
`--start`	seconds	Freeze segment start
`--end`	seconds	Freeze segment end
`--repeat`	int	Repeat render passes

3.9.1.6 Loudness and Output

These controls define delivery level, peak handling, sample rate, bit depth, container, and tail completion independently of the reverb design.

Table 3-15: Loudness and Output: reference table

Switch	Values	What it does
`--target-lufs`	LUFS	Integrated loudness target
`--target-peak-dbfs`	dBFS	Peak ceiling
`--true-peak` / `--sample-peak`	flag	Peak detection mode
`--limiter` / `--no-limiter`	flag	Final safety limiter
`--normalize-stage`	none/post/per-pass	When normalization applies
`--output-peak-norm`	none/input/target/full-scale	Final peak fit
`--quality-preset`	sd/md/hd	Output-definition preset (`sd`=44.1 kHz PCM16, `md`=48 kHz PCM24, `hd`=192 kHz float32 default)
`--out-subtype`	auto/float32/float64/pcm16/pcm24/pcm32	Output file bit depth (overrides preset subtype)
`--target-sr`	Hz	Render/output sample-rate conversion (overrides preset sample rate)
`--output-container`	auto/wav/w64/rf64	Output container selection
`--tail-stop-threshold-db`	dBFS	Tail detector threshold for write completion
`--tail-stop-hold-ms`	ms	Explicit final zero-hold duration
`--tail-stop-metric`	peak/rms	Tail detector metric

When `--target-sr` differs from the input file rate, `verbx render` performs deterministic internal resampling and writes the output at the requested rate.

3.9.1.7 Execution and Reporting

Execution switches choose compute resources and determine what reproducibility or analysis evidence accompanies the render.

Table 3-16: Execution and Reporting: reference table

Switch	Values	What it does
`--device`	auto/cpu/cuda/mps	Compute backend
`--threads`	int	CPU thread count hint
`--algo-stream`	flag	Algorithmic proxy-IR streaming mode
`--algo-proxy-ir-max-seconds`	seconds	Maximum proxy-IR duration
`--algo-gpu-proxy`	flag	Route algo proxy through CUDA convolution
`--dry-run`	flag	Validate config without writing audio
`--auto-fit`	none/speech/music/drums/ambient	Apply profile-derived starting values
`--preset`	name or `room:WxDxH/material`	Apply named preset or geometry-derived room baseline
`--lucky N`	int	Generate N randomized variants
`--frames-out`	path	Per-frame metrics CSV
`--analysis-out`	path	JSON analysis report path
`--repro-bundle`	flag	Write reproducibility bundle
`--quiet`	flag	Suppress console summary
`--silent`	flag	Suppress all output including analysis JSON

3.9.2 Early Reflection Geometry (Render)

The geometry options place one source and listener inside a rectangular first-order image-source model before the main late-field engine.

Table 3-17: Early Reflection Geometry (Render): reference table

Switch	Values	What it does
`--er-geometry`	flag	Enable first-order image-source early reflections before main engine
`--er-room-dims-m`	`L,W,H`	Room dimensions (meters)
`--er-source-pos-m`	`x,y,z`	Source position (meters)
`--er-listener-pos-m`	`x,y,z`	Listener position (meters)
`--er-absorption`	0.0–0.99	Wall absorption coefficient
`--er-material`	anechoic/dead/studio/hall/stone	Preset absorption profile

---

3.9.3 verbx ir

```
verbx ir gen OUT_IR.wav [options]          # synthesize an IR
verbx ir analyze IR_FILE.wav               # measure RT60, EDT, spectral decay
verbx ir process IN_IR.wav OUT_IR.wav      # shape existing IR (EQ, normalize, tilt)
verbx ir morph IR_A.wav IR_B.wav OUT.wav   # blend two IRs
verbx ir morph-sweep IR_A.wav IR_B.wav OUT_DIR  # alpha-timeline sweep with QA artifacts
verbx ir fit INFILE.wav OUT_IR.wav         # fit an IR to match source audio
verbx ir sofa-info FILE.sofa               # inspect SOFA conventions/dimensions
verbx ir sofa-extract FILE.sofa OUT.wav    # extract FIR matrix for convolution renders
```

`ir gen` key flags: `--mode [fdn|stochastic|modal|hybrid]`, `--length`, `--rt60`, `--damping`, `--seed`, `--sr`, `--channels`, `--er-count`, `--diffusion`, `--fdn-lines`, `--fdn-matrix`, `--resonator`, `--resonator-mix`, `--analyze-input`, `--harmonic-align-strength`, `--f0`, `--scala-file`, `--scala-root-hz`, `--scala-root-degree`, `--scala-low-hz`, `--scala-high-hz`, `--scala-strength`, `--scala-bandwidth-cents`, `--scala-gain-db`, `--scala-max-targets`

`ir morph` key flags: `--mode [linear|equal-power|spectral|envelope-aware]`, `--alpha`, `--early-ms`, `--early-alpha`, `--late-alpha`, `--align-decay`, `--phase-coherence`, `--mismatch-policy [coerce|strict]`

`ir morph-sweep` key flags: Same as morph plus `--alpha-start`, `--alpha-end`, `--alpha-steps`, `--workers`, `--retries`, `--checkpoint-file`, `--resume`, `--qa-json-out`, `--qa-csv-out`

`ir sofa-extract` key flags: `--measurement-index`, `--emitter-index`, `--target-sr`, `--normalize [none|peak|rms]`, `--strict`

3.9.4 verbx realtime

`verbx realtime [options]`

Realtime mode is currently a preview/audition path. `--engine conv` streams a real IR directly; `--engine algo` renders a static proxy IR once, then runs the live monitor through the streaming convolution engine. That means you get live device routing and stable tails today, without pretending the full offline automation surface is callback-safe yet. In other words: convolution settings act live, while algorithmic settings shape the startup proxy IR that the live convolver uses for the session. It now also supports a dedicated low-latency live dereverb path, either standalone or chained in front of the reverb engine.

Transport and device routing

Table 3-18: verbx realtime: reference table

Switch	Values	What it does
`--live-mode`	reverb/dereverb/dereverb-reverb	Choose reverb only, dereverb only, or dereverb feeding the live reverb path
`--engine`	auto/conv/algo	Live engine mode. `auto` chooses convolution when `--ir` is present, else algorithmic proxy
`--ir`	file path	IR source for realtime convolution
`--input-device`	index or substring	Select live input device
`--output-device`	index or substring	Select live output device
`--list-devices`	flag	Print available realtime devices and exit
`--sample-rate`	Hz	Live stream sample rate
`--block-size`	samples	Driver callback block size
`--partition-size`	samples	Convolution partition size used in the live processor
`--input-channels`	int	Processor input width. Defaults to mono/stereo, or to the length of `--input-channel-map`
`--input-channel-map`	comma-separated 1-based ints	Select and reorder hardware input channels, for example `1,3` or `1,3,5,7`
`--output-channels`	int	Processor output width. Defaults to processor width, or to the length of `--output-channel-map`
`--output-channel-map`	comma-separated 1-based ints	Select and reorder hardware output channels that receive processor outputs
`--duration`	seconds	Stop automatically after N seconds; omit for Ctrl-C run
`--quiet`	flag	Reduce console output

Low-latency live dereverb

Table 3-19: verbx realtime: reference table

Switch	Values	What it does
`--dereverb-mode`	wiener/spectral_sub	Realtime dereverb kernel
`--dereverb-strength`	0–2	How aggressively late energy is suppressed
`--dereverb-floor`	0–1	Minimum residual gain floor
`--dereverb-window-ms` / `--dereverb-hop-ms`	ms / ms	Short STFT analysis window and hop for the live dereverb path
`--dereverb-tail-ms`	ms	Exponential late-tail tracking horizon
`--dereverb-pre-emphasis`	0–0.98	Optional pre-emphasis before spectral processing
`--dereverb-mix`	0–1	Blend between dereverbed and latency-aligned dry signal
`--dereverb-max-atten-db`	dB	Clamp the maximum spectral attenuation
`--dereverb-stereo-link`	flag	Link stereo gain decisions to reduce image wobble
`--dereverb-input-gain-db` / `--dereverb-output-gain-db`	dB / dB	Trim into and out of the live dereverb processor

Realtime mix and proxy-room controls

Table 3-20: verbx realtime: reference table

Switch	Values	What it does
`--wet` / `--dry`	0–1	Live wet/dry mix in the convolver
`--rt60`	seconds	Algorithmic proxy decay time

Switch	Values	What it does
`--pre-delay-ms`	ms	Algorithmic proxy pre-delay
`--damping`	0–1	Algorithmic proxy damping
`--width`	0–2	Algorithmic proxy stereo width
`--mod-depth-ms` / `--mod-rate-hz`	ms / Hz	Proxy delay modulation depth and rate
`--freeze`	flag	Realtime algo only: approximate infinite sustain via a long self-sustaining proxy tail
`--algo-proxy-ir-max-seconds`	seconds	Upper bound on startup proxy IR render length
`--lowcut` / `--highcut` / `--tilt`	Hz / dB tilt	Shape the startup proxy IR spectrum before live convolution

FDN topology and feedback options

Table 3-21: verbx realtime: reference table

Switch	Values	What it does
`--fdn-lines`	1–64	Proxy FDN line count
`--fdn-matrix`	hadamard/householder/ random_orthogonal/circulant/elliptic/ tv_unitary/graph/sdn_hybrid	Proxy matrix family
`--fdn-tv-rate-hz` / `--fdn-tv-depth`	Hz / amount	Time-varying matrix motion for supported FDNs
`--fdn-dfm-delays-ms`	comma-separated ms	Delay-feedback modulation taps
`--fdn-sparse` / `--fdn-sparse-degree`	flag / int	Sparse feedback wiring and degree
`--fdn-cascade` and friends	flag / scalars	Enable cascaded/nested FDN behavior
`--fdn-rt60-low` / `--mid` / `--high`	seconds	Multiband RT60 targets
`--fdn-rt60-tilt`	–1 to 1	Tilt the decay profile across bands
`--fdn-link-filter*`	mode / Hz / mix	Filter energy in the feedback links
`--fdn-graph-topology` / `--fdn-graph-degree` / `--fdn-graph-seed`	topology / int / int	Graph-based FDN layout controls
`--fdn-matrix-morph-to` / `--fdn-matrix-morph-seconds`	matrix / seconds	Morph between matrix families during proxy synthesis
`--fdn-spatial-coupling-mode` / `--strength`	mode / 0–1	Immersive cross-cluster coupling
`--fdn-nonlinearity*`	mode / amount / drive	Nonlinear feedback coloration

Diffusion, shimmer, and perceptual macros

Table 3-22: verbx realtime: reference table

Switch	Values	What it does
`--allpass-stages`	0–64	Diffusion depth
`--allpass-gain`	float or comma-separated list	Shared or per-stage diffusion coefficient(s)
`--allpass-delays-ms`	comma-separated ms	Custom allpass delay times
`--comb-delays-ms`	comma-separated ms	Custom FDN/comb delay times
`--shimmer` and `--shimmer-*`	flag / scalars	Startup proxy shimmer block with pitch, mix, feedback, filters, spatial spread
`--room-size-macro` / `--clarity-macro` / `--warmth-macro` / `--envelopment-macro`	–1 to 1	Jot-inspired perceptual macro controls
`--algo-decorrelation-front` / `--rear` / `--top`	0–1	Extra proxy decorrelation for immersive layouts

Switch	Values	What it does
`--unsafe-self-oscillate` / `--unsafe-loop-gain`	flag / scalar	Deliberately allow runaway feedback behavior when you really mean it

Notes:

- `--live-mode dereverb` ignores reverb-engine startup options and runs only the low-latency dereverb processor.
- `--live-mode dereverb-reverb` runs the same dereverb front-end first, then feeds the selected live reverb engine.
- Live dereverb currently supports mono or stereo processor widths.
- `--block-size` must be divisible by the resolved dereverb hop size. At 48 kHz with `--dereverb-hop-ms 4`, safe values include 192, 384, and 576.
- When `--engine conv` is used with `--ir`, algorithmic proxy flags are rejected instead of being silently ignored.
- Realtime `--freeze` is not the offline segment-freeze processor. It is a live-preview approximation built on a long self-sustaining proxy IR.
- Channel maps are 1-based hardware channel numbers. If you pass `--input-channel-map 1,3`, processor input 1 comes from hardware input 1 and processor input 2 comes from hardware input 3.
- Channel-count switches must match the length of the corresponding channel map when both are provided.
- The exhaustive help for every switch lives in `docs/CLI_REFERENCE.md`.

Examples:

```
verbx realtime --engine algo \
  --input-device "Built-in Microphone" \
  --output-device "Headphones" \
  --sample-rate 48000 --block-size 256 \
  --input-channel-map 1,3 --output-channel-map 1,2,5,6 \
  --rt60 24 --freeze --shimmer \
  --fdn-matrix tv-unitary --fdn-tv-rate-hz 0.35 --fdn-tv-depth 0.12 \
  --fdn-graph-topology star --fdn-sparse --fdn-cascade \
  --lowcut 120 --highcut 9000 --tilt 1.5
```

```
verbx realtime --live-mode dereverb \
  --input-device "Built-in Microphone" \
  --output-device "Headphones" \
  --sample-rate 48000 --block-size 384 \
  --dereverb-mode wiener \
  --dereverb-strength 0.85 --dereverb-floor 0.05 \
  --dereverb-window-ms 12 --dereverb-hop-ms 4 --dereverb-tail-ms 90 \
  --dereverb-pre-emphasis 0.2 --dereverb-mix 1.0 \
  --dereverb-max-atten-db 18 --dereverb-stereo-link
```

```
verbx realtime --live-mode dereverb-reverb --engine algo \
  --input-device "Built-in Microphone" \
  --output-device "Headphones" \
  --sample-rate 48000 --block-size 384 \
  --dereverb-mode spectral_sub --dereverb-strength 0.9 \
  --dereverb-window-ms 12 --dereverb-hop-ms 4 \
  --rt60 12 --wet 0.55 --dry 0.45 \
  --fdn-matrix tv-unitary --fdn-tv-rate-hz 0.25 --fdn-tv-depth 0.08
```

3.9.5 verbx analyze

`verbx analyze INFILE [options]`

Extracts reverb and general audio metrics directly from WAV, FLAC, AIFF, OGG, CAF, and other libsndfile-supported files. Reverb analysis is enabled by default. The analyzer aligns the strongest event, builds a backward-integrated Schroeder energy-decay curve, fits EDT/T20/T30 slopes, and selects the deepest reliable fit as its broadband RT60 estimate.

Analyze a captured impulse response and write a machine-readable report:

```
verbx analyze hall_ir.wav --input-kind ir --edr --room \
  --json-out reports/hall_ir.analysis.json
```

Analyze a reverberant music or field recording while explicitly qualifying the result as a program-audio estimate:

```
verbx analyze wet_mix.wav --input-kind program --lufs \
  --direct-window-ms 3.0 --json-out reports/wet_mix.analysis.json
```

The default reverb block reports selected RT60 plus EDT, T20, T30, decay-fit R^2, usable decay range, noise floor, C_{50}, C_{80}, D_{50}, center time, direct-to-reverberant ratio, early IACC, input classification, and a confidence score. `--input-kind auto` distinguishes IR-like signals from program audio; use `ir` or `program` when you know the source type. Clarity, definition, and DRR are conventional room-acoustic quantities for an impulse response. On program audio they are peak-aligned diagnostic estimates, not a standards-compliant room measurement.

Key flags:

- `--reverb` / `--no-reverb`: enable or suppress the default broadband reverb-metric block.
- `--input-kind auto, ir, program`: select automatic classification, an impulse response, or program audio.
- `--direct-window-ms N`: set the direct-sound window used for the DRR estimate; default `2.5` ms.
- `--lufs`: add integrated LUFS, true peak, and LRA.
- `--edr`: add frequency-dependent RT60 estimates via Schroeder backward integration.
- `--room`: add estimated dimensions, volume, absorption, critical distance, class, and confidence.
- `--frames-out path`: write per-frame CSV with time-varying descriptors.
- `--json-out path`: atomically write an `analyze-report-v1` JSON report with source metadata, settings, and all metrics.
- `--ambi-order N`: add Ambisonics spatial metrics for HOA assets.

For a compact legacy feature-only run, use `verbx analyze in.wav --no-reverb`.

3.9.6 verbx room-model

```
verbx room-model --dims-m 6,8,3
verbx room-model --rt60 1.6 --material hall --json-out room.json
```

Use this when you want a physically grounded sanity check before rendering. `verbx room-model` either inspects an explicit rectangular room geometry or infers one from RT60 plus an absorption/material assumption. It reports volume, surface area, direct-path pre-delay, aspect ratios, Bolt-style proportion warnings, and writes JSON when requested.

If you already know the dimensions and want to jump straight to a render, you can skip the inspection step and use the matching render shorthand:

```
verbx render in.wav out.wav --preset room:6x8x3/hall
```

Table 3-23: verbx room-model: reference table

Switch	Values	What it does
`--dims-m`	`width,depth,height`	Inspect an explicit rectangular room
`--rt60`	seconds	Infer room dimensions from RT60 plus absorption
`--absorption`	0.01–0.99	Override the mean absorption used for RT60 inversion
`--material`	preset name	Use a wall material preset when `--absorption` is omitted
`--source-pos-m`	`x,y,z` meters	Source position inside the room
`--listener-pos-m`	`x,y,z` meters	Listener position inside the room
`--json-out`	path	Write the full geometry payload as JSON

3.9.7 verbx dereverb

`verbx dereverb INFILE OUTFILE [options]`

Deterministic spectral late-tail suppression for existing recordings.

Table 3-24: verbx dereverb: reference table

Switch	Values	What it does
`--mode`	wiener/spectral_sub	Suppression algorithm
`--strength`	0–2	Reverberant suppression amount
`--floor`	0–1	Residual floor to reduce musical-noise artifacts
`--window-ms`	ms	STFT analysis window
`--hop-ms`	ms	STFT hop size (must be smaller than window)
`--tail-ms`	ms	Late-field smoothing horizon
`--pre-emphasis`	0–0.98	Optional HF emphasis before suppression
`--mix`	0–1	Blend of processed output
`--out-subtype`	auto/float32/float64/pcm16/pcm24/pcm32	Output encoding
`--json-out`	path	Write structured dereverb report

3.9.8 verbx batch

```
verbx batch template > manifest.json          # generate manifest skeleton
verbx batch render manifest.json --jobs 8     # parallel render
verbx batch augment-template > augment.json   # generate augmentation manifest
verbx batch augment-profiles                  # list built-in profiles
verbx batch augment augment.json --jobs 8     # generate training dataset
```

Batch render flags: `--jobs`, `--schedule [fifo|shortest-first|longest-first]`, `--retries`, `--continue-on-error`, `--checkpoint-file`, `--resume`, `--dry-run`

Batch augment flags: Built-in profiles `asr-reverb-v1`, `music-reverb-v1`, `drums-room-v1`. Key flags: `--copy-dry`, `--dataset-card-out`, `--metrics-csv-out`, `--qa-bundle-out`, `--provenance-hash`, `--verify-split-isolation`

3.9.9 Other Commands

```
verbx suggest INFILE      # analysis-driven starter settings for your specific audio
verbx realtime --list-devices   # list selectable live audio devices
verbx realtime --engine algo --input-device 0 --output-device 3   # live preview
verbx realtime --live-mode dereverb --input-device 0 --output-device 3   # live low-latency dereverb
verbx render in.wav out.wav --preset room:6x8x3/hall   # geometry-derived room baseline
verbx render in.wav out_warm.wav --preset warm-chamber   # one of 280 generated style/space presets
verbx render in.wav out_shimmer.wav --preset shimmer-cathedral   # expansive shimmer preset
verbx render in.wav out_limited.wav --preset limiter-broadcast-safe   # limiter-safe delivery preset
verbx render in.wav out_long.w64 --preset delivery-long-tail-safe --output-container w64   #
↪  bounded long-tail W64 delivery
verbx room-model --rt60 1.8 --material hall   # infer a plausible room geometry
verbx dereverb in.wav out_dry.wav --mode wiener --strength 0.85 --json-out dereverb.json   #
↪  suppress late reverberation from an existing recording
verbx presets             # list built-in presets
verbx presets --show cathedral_extreme   # inspect preset parameters
verbx presets --show warm-chamber   # inspect generated style/space presets
verbx presets --show limiter-broadcast-safe   # inspect limiter/output delivery defaults
verbx quickstart          # copy-paste workflows for first-run scenarios
verbx quickstart --verify --strict       # startup readiness check (useful before demos)
verbx doctor              # platform/acceleration diagnostics
verbx doctor --json-out doctor.json      # machine-readable diagnostics for issue reports
verbx version             # package version string
verbx cache info          # inspect IR cache
verbx cache clear         # clear IR cache
```

3.10 Recipes

The recipes are starting points organized by increasing control depth. Run one unchanged, listen through its complete tail, then alter a single parameter while retaining the original render for comparison.

3.10.1 Beginner Recipes

Subtle room glue – keeps everything sounding like it was recorded together:

```
verbx render mix_bus.wav glued.wav --engine algo --rt60 0.8 --wet 0.15 --dry 0.9 --pre-delay-ms 12
```

Natural vocal hall – spacious without washing the lyrics:

```
verbx render vocals.wav vocals_hall.wav --engine algo \
  --rt60 2.2 --wet 0.28 --dry 0.78 --pre-delay-ms 22 --lowcut 200 --highcut 10000
```

Drums with ducking – tail blooms between hits, never clutters transients:

```
verbx render drums.wav drums_room.wav --engine algo \
  --rt60 1.4 --wet 0.55 --dry 0.6 --duck --duck-attack 10 --duck-release 180
```

Convolution from a free IR library – real space character:

```
verbx render piano.wav piano_conv.wav --engine conv --ir hall_ir.wav --ir-normalize peak --wet 0.5
↪  --dry 0.7
```

Tempo-synced pre-delay – reverb onset lines up with the beat:

```
verbx render snare.wav snare_delay.wav --engine algo --pre-delay 1/8D --bpm 128 --rt60 1.8 --wet 0.
↪  45
```

Loudness-safe delivery – hits –16 LUFS with –1 dBTP ceiling:

```
verbx render master.wav delivered.wav --engine algo --rt60 2.0 --wet 0.2 \
  --target-lufs -16 --true-peak --target-peak-dbfs -1
```

3.10.2 Production Recipes

Broadcast dialogue room – natural placement, EBU R128 compliant:

```
verbx render dialogue.wav dialogue_room.wav --engine conv \
  --ir small_room_ir.wav --wet 0.25 --dry 0.85 --pre-delay-ms 8 \
  --lowcut 150 --highcut 9000 --target-lufs -23 --true-peak --target-peak-dbfs -1
```

Film score hall – wide, clear, cinematic:

```
verbx render strings.wav strings_hall.wav \
  --engine conv --ir large_hall_ir.wav \
  --wet 0.65 --dry 0.55 --pre-delay 1/16 --bpm 72 \
  --width 1.2 --bloom 1.8 --tilt 1.0 \
  --lowcut 80 --target-lufs -20 --target-peak-dbfs -1.5
```

Gated drum space – 1980s aesthetic, punchy tail that cuts off:

```
verbx render drums.wav drums_gated.wav --engine conv \
  --ir plate_short.wav --ir-normalize peak --tail-limit 1.2 \
  --wet 0.75 --dry 0.4 --highcut 9000 --target-peak-dbfs -1
```

Dub chamber send – high-wet parallel texture, bandwidth controlled:

```
verbx render snare_send.wav dub_chamber.wav --engine conv \
  --ir spring_ir.wav --repeat 2 --wet 0.95 --dry 0.05 \
  --lowcut 180 --highcut 4500 --tilt -2.0 --output-peak-norm input
```

Sparse hall for piano or choir – depth without obscuring articulation:

```
verbx render piano.wav piano_hall.wav --engine conv --ir hall_ir.wav \
  --pre-delay 1/16 --bpm 60 --wet 0.55 --dry 0.7 \
  --lowcut 120 --highcut 11000 --target-lufs -20 --target-peak-dbfs -1
```

Cathedral vocal/organ – long, immersive, cinematic:

```
verbx render choir.wav choir_cathedral.wav --engine conv \
  --ir cathedral_ir.wav --wet 0.82 --dry 0.35 --rt60 90 \
  --lowcut 70 --highcut 10000 --target-lufs -21 --true-peak --target-peak-dbfs -1
```

Track D IR blend – morphing between two hall characters during render:

```
verbx render in.wav morphed.wav --engine conv --ir hall_A.wav \
  --ir-blend hall_B.wav --ir-blend-mix 0.6 --ir-blend-mode envelope-aware \
  --ir-blend-early-ms 60 --automation-point "ir-blend-alpha:0.0:0.0" \
  --automation-point "ir-blend-alpha:30.0:1.0"
```

AI dataset batch – augmentation with split isolation and metrics:

```
verbx batch augment augment_manifest.json --profile asr-reverb-v1 \
  --jobs 8 --copy-dry --verify-split-isolation \
  --metrics-csv-out out/metrics.csv --dataset-card-out out/DATASET_CARD.md \
  --qa-bundle-out out/qa_bundle.json --provenance-hash
```

3.10.3 Experimental Recipes

Eight approaches from the experimental music tradition. Rendered demos for all of these are in examples/audio/.

Alvin Lucier – *I Am Sitting in a Room* (1969) (iterative room resonance accumulation)

Each pass imprints the room's modal resonances more deeply. After 12–20 passes, only the resonant frequencies of the virtual room survive – the original speech is gone.

```
mkdir passes && cp voice.wav passes/pass_00.wav && current="passes/pass_00.wav"
for i in $(seq 1 20); do
  next=$(printf "passes/pass_%02d.wav" "$i")
  verbx render "$current" "$next" --engine algo --rt60 4.5 \
    --wet 1.0 --dry 0.0 --fdn-lines 16 --fdn-matrix hadamard \
    --lowcut 60 --no-progress
  current="$next"
done
# Quick single-command version (7 passes baked in):
verbx render voice.wav lucier_7pass.wav --engine algo --rt60 4.5 \
  --wet 1.0 --dry 0.0 --repeat 7 --fdn-lines 16 --fdn-matrix hadamard --lowcut 60
```

Brian Eno – *Discreet Music* (1975) / Ambient series (endless ambient tail)

Decay so long the source dissolves. The wet signal becomes the room's breath.

```
verbx render input.wav eno_ambient.wav --engine algo --rt60 12.0 \
  --wet 0.92 --dry 0.08 --damping 0.25 --pre-delay-ms 35 \
  --fdn-lines 16 --fdn-matrix hadamard --lowcut 50 \
  --target-lufs -22 --target-peak-dbfs -2
```

Pauline Oliveros – *Deep Listening* (1989) (cave-scale resonance)

Inspired by Oliveros's work in underground cisterns. Very low damping lets every frequency sustain; 32-line FDN produces the lateral complexity of stone architecture.

```
verbx render drone.wav deep_listening.wav --engine algo --rt60 18.0 \
  --wet 0.95 --dry 0.10 --fdn-lines 32 --fdn-matrix hadamard \
  --pre-delay-ms 55 --damping 0.15 --lowcut 30 \
  --target-lufs -24 --target-peak-dbfs -2
# For a 240-second synthesized IR version:
verbx render drone.wav deep_ir.wav --ir-gen --ir-gen-mode hybrid \
  --ir-gen-length 240 --ir-gen-seed 108 --engine conv \
  --wet 0.9 --dry 0.15 --target-lufs -24 --target-peak-dbfs -2
```

Robert Fripp / Eno – Frippertronics tape-loop accumulation

Shimmer feedback builds over each block. At 0.78, the octave layer accumulates like a tape recirculation loop growing denser with each pass.

```
verbx render guitar.wav frippertronics.wav --engine algo --rt60 8.0 \
  --wet 0.82 --dry 0.28 --fdn-lines 16 --fdn-matrix hadamard \
  --shimmer --shimmer-semitones 12 --shimmer-mix 0.45 --shimmer-feedback 0.78 \
  --pre-delay-ms 25 --target-peak-dbfs -2
# Iterative version — 12 passes with gradual timbral drift:
mkdir fripp && cp guitar.wav fripp/pass_00.wav && current="fripp/pass_00.wav"
for i in $(seq 1 12); do
```

```
  next=$(printf "fripp/pass_%02d.wav" "$i")
  verbx render "$current" "$next" --engine algo --rt60 8.0 \
    --wet 0.82 --dry 0.12 --shimmer --shimmer-semitones 12 \
    --shimmer-feedback 0.78 --no-progress
  current="$next"
done
```

Shoegaze / My Bloody Valentine – wall of sound (dense shimmer wash)

Freeze a guitar sustain, then bury it in octave shimmer and a circulant FDN. The circulant matrix produces the smeared, tonally undifferentiated density that defines the genre.

```
verbx render guitar.wav shoegaze.wav --engine algo \
  --freeze --start 1.0 --end 2.4 \
  --shimmer --shimmer-semitones 12 --shimmer-mix 0.55 --shimmer-feedback 0.72 \
  --rt60 5.0 --wet 0.88 --dry 0.22 --fdn-matrix circulant --lowcut 80 \
  --width 1.4 --target-peak-dbfs -2
```

Steve Reich – phase minimalism (tight rhythmic room)

Short RT60 with a circulant diffusion matrix keeps individual hits distinct while adding spatial depth. The circulant matrix's circular delay structure creates subtle comb filtering that complements phase-shifted rhythmic material.

```
verbx render percussion.wav reich_room.wav --engine algo --rt60 0.7 \
  --wet 0.55 --dry 0.50 --fdn-lines 8 --fdn-matrix circulant \
  --pre-delay-ms 18 --damping 0.6 --lowcut 60
```

Eliane Radigue – *ADNOS* (1973–1974) / drone electronics (near-infinite sustain)

At RT60=45s with wet=0.97, the dry signal is almost entirely subsumed. Radigue's aesthetic is about sound that has been in the room so long it has become the room.

```
verbx render drone.wav radigue.wav --engine algo --rt60 45.0 \
  --wet 0.97 --dry 0.05 --fdn-lines 32 --fdn-matrix hadamard \
  --damping 0.10 --lowcut 20 --target-lufs -28 --target-peak-dbfs -2
```

Morton Feldman – late period (contemplative sparse space)

Feldman's late works often feature long silences and isolated events in large, reflective spaces. Medium RT60, restrained wet level, allpass diffusion, no shimmer.

```
verbx render piano.wav feldman.wav --engine algo --rt60 3.8 \
  --wet 0.52 --dry 0.52 --fdn-lines 8 --fdn-matrix circulant \
  --pre-delay-ms 30 --damping 0.50 --allpass-stages 4 \
  --target-lufs -26 --target-peak-dbfs -2
```

Self-convolution texture smear – signal convolved with itself:

```
verbx render input.wav self_convolved.wav --self-convolve \
  --beast-mode 12 --partition-size 16384 --normalize-stage none
```

Feature-reactive reverb depth – wet depth tracks source loudness in real time:

```
verbx render in.wav reactive.wav --engine conv --ir hall.wav \
  --feature-vector-lane "target=wet,source=loudness_norm,weight=0.70,curve=smoothstep,combine=
  ↪ replace" \
```

```
  --feature-vector-lane "target=wet,source=transient_strength,weight=0.30,combine=add" \
  --feature-vector-frame-ms 40 --feature-vector-trace-out trace.csv
```

Lucky mode exploration – 12 randomized wild variants from one source:

```
verbx render in.wav out/lucky.wav --lucky 12 --lucky-out-dir out/lucky_set --lucky-seed 2026 --no-
↪  progress
```

3.10.4 Workflow Recipes

Batch parallel render from a manifest:

```
verbx batch template > manifest.json      # edit manifest.json with your jobs
verbx batch render manifest.json --jobs 8 --schedule longest-first --retries 1 \
  --checkpoint-file manifest.checkpoint.json
# interrupted? resume from where it stopped:
verbx batch render manifest.json --jobs 8 --resume --checkpoint-file manifest.checkpoint.json
```

IR sweep QA – morph between two IRs with quality metrics:

```
verbx ir morph-sweep ir_a.wav ir_b.wav out/sweep \
  --alpha-start 0.0 --alpha-end 1.0 --alpha-steps 9 \
  --workers 4 --retries 1 --mismatch-policy strict \
  --checkpoint-file out/sweep.checkpoint.json \
  --qa-json-out out/sweep_summary.json --qa-csv-out out/sweep_metrics.csv
```

Generate a bank of 25 varied IRs:

```
./scripts/generate_ir_bank.sh IRs/bank_25 25 flac
# or with explicit Python control:
./scripts/generate_ir_bank.py --out IRs/bank_25 --count 25 --sr 48000 --channels 2 --format flac
```

Generate a large folder-sorted IR library (varying lengths):

```
uv run python scripts/generate_ir_library.py \
  --out IRs/library --sr 12000 --channels 2 --format flac --seeds-per-shape 1
```

Pre-render validation – catch config errors before a long job:

```
verbx render long_input.wav output.wav --engine algo --rt60 180 --fdn-lines 32 --dry-run
# prints resolved config, estimated output duration, device selection — no audio written
```

3.10.5 Python API (`verbx.api`)

`verbx` now exposes a stable Python surface for file-level workflows:

- `verbx.api.render_file(infile, outfile, **render_options)`
- `verbx.api.generate_ir(outfile, **ir_options)`
- `verbx.api.analyze_file(infile, **analysis_options)`

Minimal notebook examples:

- `docs/notebooks/research_workflow.ipynb`
- `docs/notebooks/dataset_workflow.ipynb`
- Notebook workflow guide

Structured schema docs for machine-generated payloads:

- `docs/SCHEMAS.md` (batch manifest + automation schema references)

```
from verbx.api import analyze_file, generate_ir, render_file

ir = generate_ir("research_ir.wav", mode="hybrid", length=1.5, sr=48000, channels=1)
```

```
metrics = analyze_file("dry.wav")
report = render_file("dry.wav", "wet.wav", engine="conv", ir=ir["outfile"], wet=0.7, dry=0.3)
```

3.11 Performance and Acceleration

Performance in a reverberation system is not one number. A renderer can finish an hour of audio quickly yet still be unsuitable for live use; a plug-in can meet every callback deadline while taking longer than an offline batch renderer to export the same program; and a GPU can increase total throughput while adding enough transfer and scheduling delay to make a small realtime job worse. Before selecting a device, thread count, block size, or partition size, define which constraint actually matters.

Four budgets govern most verbx workloads. **Throughput** asks how much audio can be processed per unit of wall time. **Deadline safety** asks whether every realtime block finishes before the device or host needs it. **Latency** asks how long a signal takes to travel from input to audible output. **Memory** asks whether state, impulse responses, FFT partitions, temporary buffers, and output can remain resident without paging or exhausting the machine. These budgets interact, but they are not interchangeable. Increasing a convolution partition can improve throughput while increasing buffering; adding worker threads can shorten a batch while making callback timing less predictable; and moving an operation to a GPU can reduce arithmetic time while increasing transfer latency and peak memory.

The first useful offline measure is the realtime factor

$$R_{\mathrm{tf}} = \frac{t_{\mathrm{process}}}{t_{\mathrm{audio}}},$$

where t_{process} is elapsed processing time and t_{audio} is the duration of the input plus any deliberately rendered tail. A value below 1 means the job ran faster than realtime; 0.25 means four seconds of audio were produced per second of wall time. This ratio is meaningful only when the test states engine, sample rate, channel count, output duration, precision, IR length, device, and enabled stages. A short mono file at 48 kHz and a twelve-channel file at 192 kHz are not comparable workloads.

Real-time processing requires a stricter measure. For block size B samples and sample rate F_s, the nominal callback interval is

$$T_{\mathrm{block}} = \frac{B}{F_s}.$$

At 48 kHz, a 256-sample block provides about 5.33 ms; at 192 kHz the same block provides about 1.33 ms. The processor must complete every block within that interval after allowing for host work, driver overhead, competing plug-ins, display updates, and operating-system jitter. Average callback time is therefore inadequate evidence. Measure high percentiles and the maximum, count deadline misses, and retain safety headroom. One 8 ms callback in an otherwise fast performance can still produce an audible dropout.

3.11.1 A Cost Model for Algorithmic Reverb

An algorithmic reverberator pays for work per sample. Delay reads and writes, damping filters, allpass stages, modulation, interpolation, and wet projections scale roughly with the number of delay lines and output channels. A dense $L \times L$ feedback matrix has a naive multiplication cost proportional to L^2 per sample, while Hadamard, Householder, circulant, sparse, or otherwise structured transforms can reduce that cost or improve memory locality. Increasing FDN line count usually raises echo density and modal complexity, but it also enlarges state, matrix work, and cache pressure.

RT60 by itself does not make an FDN callback proportionally more expensive. A 3-second and a 300-second decay can use the same delay topology and number of operations per sample; the feedback gains differ. Long RT60 does, however, affect the amount of audio an offline render must write when the requested tail is allowed to continue. It also raises the cost of validation because stability, noise growth, limiting, and end behavior must be observed over a longer duration. Distinguish **cost per sample** from **number of samples emitted**.

Sample rate and channel layout are direct multipliers. Moving from 48 kHz to 192 kHz quadruples the number of sample updates for the same duration before accounting for any larger delay storage, resampling, or oversampling filters. Moving from stereo to 7.1.4 can increase input and output projection work, file bandwidth, analysis cost, and memory even when the internal feedback state remains shared. High sample rate and high channel count should therefore be justified by delivery, measurement, or processing requirements rather than treated as cost-free quality switches.

3.11.2 A Cost Model for Convolution

Direct convolution of an input with an impulse response of M samples requires work proportional to M for every output sample and becomes impractical for long rooms. FFT convolution changes the dominant operation into transforms and complex products over blocks. Partitioned convolution divides the IR into bounded segments so processing can begin before the entire response has been evaluated and so repeated transforms can be reused.

Partition size controls a three-way trade. Small partitions reduce the amount of audio that must be buffered before the first result, but they perform transforms more often and increase scheduling overhead. Large partitions amortize FFT work and often improve offline throughput, but they require larger temporary arrays, produce coarser progress increments, and are unsuitable when low algorithmic latency is required. Hybrid partitioning can use short early partitions and progressively larger late partitions, matching computational effort to the perceptual importance and timing of the response.

Matrix convolution multiplies this problem by routing density. A full $M \times N$ IR bank contains one response for each input-output pair. Diagonal, sparse, symmetric, or shared-tail structures can reduce computation and storage, but only when the omitted cross-channel paths are acoustically and artistically acceptable. “Use the GPU” does not solve an unnecessarily dense routing model; representation remains the first optimization.

3.11.3 Realtime Deadlines and Offline Throughput

Offline rendering can batch work, reorder jobs, use large buffers, wait for JIT compilation, and allow temporary load spikes as long as the final artifact is correct. Real-time processing cannot. Its callback must avoid filesystem access, unbounded allocation, locks, device synchronization, compilation, and any operation whose duration depends unpredictably on external state. Preparation belongs before playback; bounded DSP belongs inside the callback.

This distinction explains why the fastest offline setting is not automatically the best realtime setting. A 65,536-sample FFT partition may process a long IR efficiently in a batch but would imply an unacceptable buffering interval in a live monitor path. Eight workers may saturate a workstation during corpus generation but compete destructively with an audio host. A progress display or spectrum analyzer can be cheap on average yet cause jitter if it performs transforms or allocations on the callback thread.

End-to-end monitored latency includes more than verbx DSP. Input conversion, device buffers, host safety buffers, block adaptation, resampling, lookahead, convolution partitioning, output conversion, and acoustic propagation all contribute. Performance optimization should report which component changed. Reducing render wall time does not necessarily reduce monitored latency, and lowering one internal buffer may expose the system to underruns elsewhere.

3.11.4 Memory Traffic, Locality, and Precision

Modern audio workloads are often limited by movement of data rather than arithmetic. Delay states, FFT arrays, channel matrices, analysis frames, and output buffers compete for cache and memory bandwidth. Contiguous arrays, predictable access, reused workspaces, and structured transforms can matter more than reducing a small number of scalar operations. An algorithm that allocates a fresh array per block may benchmark adequately on a short file and collapse under a long multichannel render because allocation and cache churn dominate.

verbx performs its principal Python DSP in `float64` internally and converts at the output boundary. That choice provides numerical margin for long feedback paths, analysis fits, and repeated transformations, but doubles sample storage relative to `float32` and can reduce SIMD width or GPU occupancy. Output subtype does not retroactively change the cost of internal processing. Writing `float32` instead of `float64` reduces file bandwidth and storage, while the internal engine retains its declared precision.

Peak memory matters as much as final file size. A long IR may be modest on disk yet expand into multiple complex spectra, channel-route copies, overlap buffers, and temporary results. Streaming is the preferred response when stages permit it: process bounded audio chunks, retain only necessary state, and write incrementally. Stages requiring global normalization, complete-file analysis, reversal, or nonlocal transformation may force an additional pass or larger retained state. The pipeline should state that consequence rather than silently abandoning streaming.

3.11.5 Acceleration Is a Hierarchy

Optimization should proceed from semantics outward. First remove unnecessary work: avoid unused channels, redundant resampling, duplicate analysis, excessive tail duration, and dense routes that do not contribute to the deliverable. Next select an algorithm whose complexity matches the task. Then improve data layout and reuse. Only after those steps should implementation accelerators such as vectorized kernels, JIT compilation, multithreading, MPS profiles, or CUDA be evaluated.

CPU vectorization and JIT compilation are especially effective for persistent inner loops with predictable array shapes. Their startup cost can be visible on the first job, so benchmarks should separate cold and warm runs. Multithreading helps when independent channels, files, variants, or partitions provide enough work to amortize scheduling. It hurts when jobs are too small, memory bandwidth is already saturated, or nested libraries create more threads than physical cores can execute efficiently.

GPU acceleration has a break-even point. Data must be transferred, kernels dispatched, and results synchronized. Long convolution, large channel matrices, and repeated batches can provide enough arithmetic intensity to repay that overhead. A small FDN block usually cannot. Keep data resident across multiple operations when possible, avoid transfers per audio block, and benchmark the complete pipeline rather than an isolated kernel. A fast GPU FFT followed by repeated host-device copies may be slower than a well-localized CPU path.

Fallback behavior is part of performance correctness. A requested accelerator that is unavailable should produce a visible status and machine-readable report field. Silent fallback can invalidate a benchmark, surprise a batch deadline, or conceal that a deployment is running a different path than the development workstation. Record resolved device, library availability, thread count, and effective processing mode with every serious measurement.

3.11.6 Benchmarking as Reproducible Evidence

A useful benchmark controls the variables that materially change cost. Record verbx version and commit, operating system, processor and accelerator, memory, Python and library versions, engine, sample rate, channel layout, input duration, rendered-tail duration, IR duration and route count, block or partition size, thread count, precision, and every enabled post-process. Use the same input assets and warm-up policy across runs. Report median and dispersion over repeated trials rather than selecting the fastest run.

For offline work, retain elapsed time, realtime factor, peak resident memory, output size, and if relevant energy consumption. For realtime work, retain callback mean, high percentiles, maximum, deadline misses, host block size, reported algorithmic latency, and the amount of unused deadline headroom. For batch work, retain total makespan, per-job distribution, retry count, checkpoint overhead, and the degree to which short jobs leave workers idle near the end.

Do not optimize against silence or a single impulse alone. Those signals are valuable for correctness but can bypass content-dependent detectors, limiters, modulation, sparse paths, or denormal behavior. Include transient material, sustained low-frequency energy, dense program audio, silence after excitation, and multichannel cases. Confirm output equivalence or declared tolerance after every optimization; a faster render with changed decay, gain, routing, or numerical stability is a different algorithm, not a free speedup.

3.11.7 Musical Quality and Engineering Tradeoffs

Performance settings are audible when they alter topology, partition timing, modulation, precision, channel routing, or the amount of tail rendered. The correct goal is not the highest accelerator utilization. It is the least expensive configuration that preserves the required musical behavior and delivery evidence. A preview may use fewer FDN lines or a shorter IR while composition decisions are fluid; the final render can restore the approved topology. A live performer may prefer moderate quality with large callback headroom over a nominally superior mode that risks interruption.

Treat quality reduction as an authored choice. State which dimension changes: echo density, bandwidth, modulation smoothness, spatial order, IR length, oversampling, or analysis depth. Level-match comparisons and listen to exposed tails, sparse transients, fold-downs, and section boundaries. Some reductions disappear in a dense mix while others damage exactly the silence or sustained decay that gives the music its form.

The practical sequence is therefore: define the delivery and latency requirement, measure a representative baseline, identify the dominant cost, change one layer at a time, verify sonic and numerical equivalence, and retain the benchmark report. Device selection comes near the end of that sequence, not at the beginning.

3.11.8 Current Acceleration Paths

CPU (default): All processing. Algorithmic FDN path benefits from `numba` when installed – install with `pip install numba` and verbx uses JIT-compiled inner loops automatically. Check with `verbx doctor`.

Apple Silicon (MPS): `--device mps` uses the MPS profile for the algorithmic path. The convolution FFT runs on CPU (NumPy/SciPy). Threading helps: `--threads 8` is a good starting point for M-series chips. Apple Silicon is well-suited for the algorithmic engine; the memory bandwidth advantage shows on high line count FDN renders.

CUDA: `--device cuda` enables GPU-accelerated partitioned FFT convolution via CuPy. Install with `pip install cupy-cuda12x` (match your CUDA version). The algorithmic engine does not benefit from CUDA – it runs on CPU regardless. CUDA acceleration is most valuable for long-IR convolution with large files. If CuPy is unavailable, verbx falls back to CPU silently.

Block size and partition size: `--block-size` controls the algorithmic engine's internal block size – larger blocks can improve throughput at the cost of responsiveness per block. `--partition-size` controls convolution FFT partition length – the main tuning knob for convolution throughput. Larger partitions reduce per-block overhead but increase peak memory. For offline rendering, 16384–65536 is a good range. For very long IRs (120s+), larger partition sizes (65536) often give better throughput.

Streaming convolution engages automatically for simple conv renders (no normalization, no post-effects, no freeze, `--repeat 1`). Peak RAM use scales with partition size rather than IR length in this mode.

Numba: When installed, verbx automatically JIT-compiles the FDN inner loop for the algorithmic engine. First render with a new configuration takes a few extra seconds to compile; subsequent renders at the same parameters are significantly faster. To verify it is active: `verbx doctor --json-out doctor.json` and check `numba_available`.

3.11.9 Render Performance Baseline

This directory tracks a small deterministic benchmark suite for CI comparison.

3.11.9.1 What it measures

`scripts/benchmark_render_baseline.py` runs four representative scenarios:

- `algo_short_room`: short algorithmic render (production-style room settings).
- `algo_long_tail_130s`: long-tail algorithmic regression fixture (`RT60=130s`).
- `conv_matrix_stereo`: matrix-routed stereo convolution regression fixture.
- `conv_target_sr_192k`: convolution render with integrated `--target-sr 192000` sample-rate conversion path.

The benchmark emits:

- wall-clock elapsed seconds,
- output duration seconds,
- realtime factor (`output_seconds / elapsed_seconds`),
- comparison against baseline reference/budget values.

3.11.9.2 Baseline file

`ci_baseline.json` defines per-scenario:

- `reference_seconds`: expected reference runtime.
- `max_seconds`: soft budget threshold for obvious regressions.

3.11.9.3 Run locally

```
uv run python scripts/benchmark_render_baseline.py \
  --baseline docs/benchmarks/ci_baseline.json \
  --json-out docs/benchmarks/latest_report.json \
  --compare-threshold 3.0
```

To fail locally on detected regressions:

```
uv run python scripts/benchmark_render_baseline.py \
  --baseline docs/benchmarks/ci_baseline.json \
  --json-out docs/benchmarks/latest_report.json \
  --compare-threshold 3.0 \
  --fail-on-regression
```

3.11.9.4 CI integration

The `perf-baseline` CI job runs this script on `ubuntu-latest` and uploads the JSON report artifact for trend inspection and release triage.

3.12 DSP Architecture

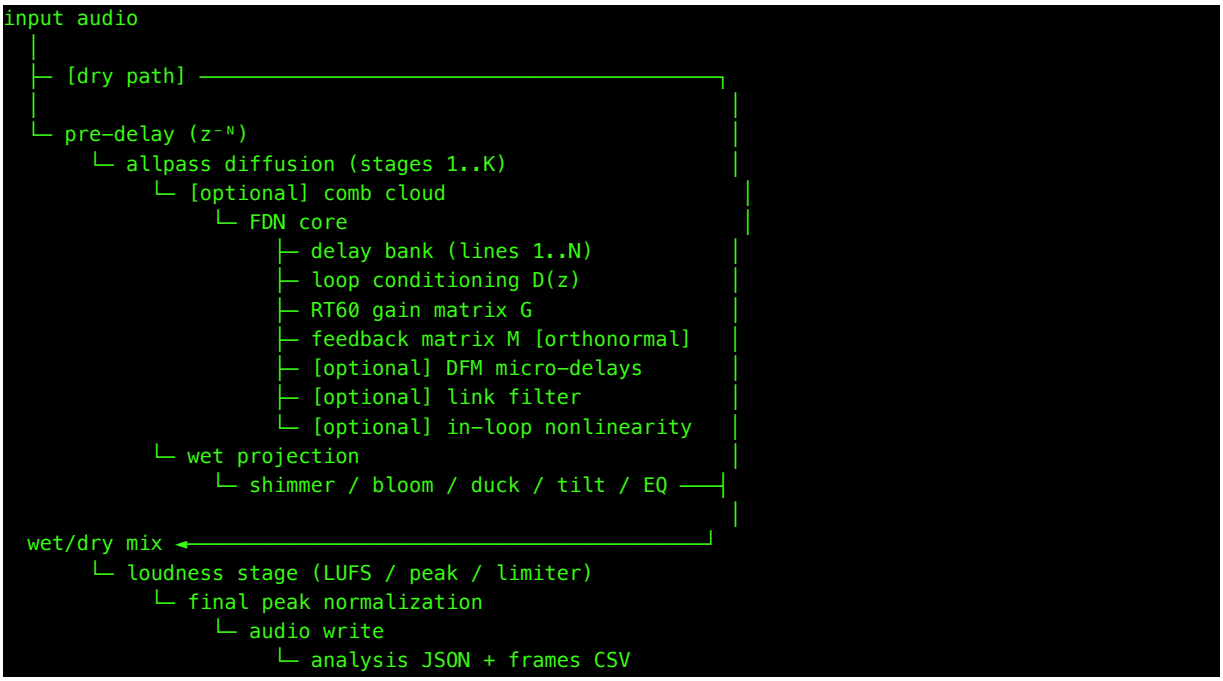
For contributors and people who want to understand the signal chain in code.

Module map:

Table 3-25: DSP Architecture: reference table

Path	Contents
`src/verbx/commands/`	Public command modules and Typer registration surfaces
`src/verbx/cli.py`	Shared CLI validation, config assembly, and helper/report logic
`src/verbx/core/algo_reverb.py`	Algorithmic FDN engine
`src/verbx/core/conv_reverb.py`	Partitioned FFT convolution engine
`src/verbx/core/pipeline.py`	Render orchestration, stage ordering
`src/verbx/core/loudness.py`	LUFS targeting, true-peak, limiter
`src/verbx/core/shimmer.py`	Shimmer, bloom, ducking, tilt
`src/verbx/core/tempo.py`	Note-value pre-delay parsing
`src/verbx/analysis/`	Frame extraction, EDR, Ambisonics metrics
`src/verbx/ir/`	IR synthesis modes, shaping, morphing, fitting, cache
`src/verbx/io/`	Audio I/O, progress reporting

Signal chain (algorithmic engine):



Notation: z^{-N} denotes an integer-sample delay of N samples, K is allpass-stage count, and N (in `lines 1..N`) is FDN delay-line count.

Precision: All DSP – FDN state updates, FFT operations, allpass filters, automation curves, feature vectors, analysis metrics – runs in `float64` internally. Output is downcast at write time according to `--out-subtype`. `verbx render` defaults to HD output (`192000 Hz`, `float32`) unless overridden by `--quality-preset`, `--target-sr`, or `--out-subtype`.

Key design decisions:

- Per-line gain calibration (not global feedback gain) lets all delay lines, regardless of length, track the same RT60 target. This is essential for stable long tails.

- Orthonormalization of all matrix families before use prevents energy accumulation in high-feedback topologies.
- Automation evaluation uses a slew limiter and deadband guard in addition to smoothing to prevent abrupt control jumps and high-frequency control chatter in block-mode evaluation.
- The IR cache uses a content hash (audio samples + metadata) rather than file path, so the same IR content at a different path still hits cache.

3.13 Contributors

The maintained contributor record lives in GitHub; the short list below identifies the creator and links to the complete graph.

- Colby Leider, PhD (creator and maintainer)
- Full contributors graph: github.com/TheColby/verbx/graphs/contributors

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3.14 Contributing

See CONTRIBUTING.md for the full guide. Quick start:

```
hatch env create
hatch run lint      # ruff check
hatch run typecheck # pyright strict
hatch run test      # pytest
```

uv alternative:

```
uv sync --extra dev
uv run ruff check . && uv run pyright && uv run pytest
```

Report security issues via SECURITY.md. See CODE_OF_CONDUCT.md.

3.15 References and Further Reading

Full bibliography: docs/REFERENCES.md

Key book:

- **Pirkle (2019)** – *Designing Audio Effect Plugins in C++: For AAX, AU, and VST3 with DSP Theory*, 2nd ed., Routledge. Recommended companion reading for plug-in anatomy, API-independent DSP cores, host integration, parameter handling, GUI design, and the implementation of delay, reverb, and dynamics processors.

Key papers:

- **Gardner (1998)** – "Reverberation algorithms." Practical implementation guide covering partitioned convolution, early reflections, and late field design.
- **Jot (1992)** – "An analysis/synthesis approach to real-time artificial reverberation." Extends FDN theory to frequency-dependent decay, the basis for multiband RT60 control.
- **Jot & Chaigne (1991)** – "Digital delay networks for designing artificial reverberators." Introduced the Feedback Delay Network in its modern form; directly informs the gain calibration formula used in verbx.

- **Schroeder (1962)** – "Natural sounding artificial reverberation." The foundational work on allpass and comb filter reverb structures that forms the basis for most algorithmic reverb design.
- **Smith (1985)** – "A new approach to digital reverberation using closed waveguide networks." Scattering Delay Networks – a physical wave propagation model distinct from the FDN approach; informs the `sdn_hybrid` matrix family.
- **Valimaki et al. (2012)** – "Fifty years of artificial reverberation." Survey paper; an accessible overview of the full history of algorithmic reverb from Schroeder to modern approaches.

Additional guides in `docs/`:

- Consolidated user guide and `USERGUIDE.pdf` – README plus user-facing docs/tips in one manual
- CLI reference – machine-generated `--help` snapshots for all command groups
- IR synthesis guide – complete parameter reference for all synthesis modes
- AI augmentation guide – dataset generation workflow documentation
- Schema reference – JSON/CSV formats for manifests and automation
- Dataset augmentation notebook – Python API workflow for ML pipelines
- Scala tuning examples – microtonal `.scl` input and convolution workflow
- IR morph QA guide – morph-sweep QA artifacts and CI integration
- Benchmark baseline guide – CI/runtime comparison workflow
- Extreme cookbook – 100 additional workflow examples
- SOFA interoperability note – shipped `sofa-info` / `sofa-extract` workflow and current constraints
- Launch example parity checker – verifies canonical launch commands stay mirrored across docs/man pages

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3.16 License

See LICENSE.

v0.9.9 is the current public alpha. See CHANGELOG.md for version history.

3.17 Suggested Exercises

1. Use a labeled impulse to verify every route in one multichannel or matrix-convolution configuration; document channel order, arrival time, polarity, and level.
2. Capture three candidate impulse responses from one space using unchanged geometry, then compare their noise floors and decay estimates before choosing a production version.
3. Create one convolution render and one algorithmic approximation of the same room role. Identify what the approximation preserves and what it changes.
4. Design a safe long-tail render with explicit duration, container, limiter, and report settings. Explain each guardrail.
5. Create a minimal repeatable render manifest containing input hashes, command line, version, output format, and analysis sidecar location.
6. Use the reference to design a safe batch job, then estimate disk use, render duration, and tail duration before running it.
7. Create a failure-report bundle from a deliberately problematic input and identify the smallest evidence set another developer would need.
8. Compare two output formats and bit depths for a single render, documenting both interoperability and numerical consequences.
9. Validate a channel layout with a single-sample impulse in every channel and archive the result with the render metadata.
10. Write an explicit stop condition for a very long tail, then explain why it is safer than relying on an implicit default.

"Sweetest Echo, sweetest nymph, that liv'st
unseen within thy airy shell."

— John Milton, *Comus* (1637)

Chapter 4

Immersive Reverb, Surround Sound, and Dolby Atmos

Immersive audio is not simply stereo with more loudspeakers. It is the deliberate composition, capture, synthesis, organization, and reproduction of an acoustic world around a listener. That world may be represented as fixed output channels, as a listener-centered sound field, as audio objects accompanied by position metadata, or as a hybrid of all three. Each representation makes a different promise. A channel promises a destination. A scene promises a field that can be decoded. An object promises an element whose position and other rendering instructions can survive until a later stage. None of those promises, by itself, guarantees that the result will feel large, coherent, natural, dramatic, or musically necessary.

Reverberation is where the differences become impossible to ignore. A dry point source can often be moved from one representation to another without immediately exposing every assumption in the signal path. A reverberant field cannot. It contains hundreds or thousands of temporally ordered cues: direct sound, first arrivals, lateral reflections, overhead returns, diffuse buildup, frequency-dependent decay, modulation, directional asymmetry, and noise. It may be compact enough to reinforce a single performer or broad enough to define the apparent architecture of an entire work. A tail can remain tied to a loudspeaker channel, rotate as an Ambisonic scene, spread through a fixed bed, follow an object, or be decomposed into several layers that use different representations. The choice changes not only where the effect is heard, but what kind of space the listener believes is present.

This chapter treats immersive reverb as both an engineering system and a musical material. The engineering problem is to preserve channel meaning, timing, level, phase, metadata, and translation across devices and renderers. The musical problem is to decide what the space is doing. Does it bind an ensemble into one body? Separate simultaneous layers? Reveal a distant source? Extend a cadence? Turn a room into a rhythmic instrument? Make an electronic sound feel architectural? Move a listener between intimacy and monumentality? A technically valid multichannel file can fail every one of those musical tests, while a restrained two-layer design can create a convincing three-dimensional world.

Space is a relationship, not a decoration. A reverb does not sit behind a source as an interchangeable background. It changes the apparent relationship among source, room, listener, and time. Early energy can move a source backward without reducing its fader. Lateral energy can broaden an ensemble without spreading every dry channel. A high, diffuse decay can enlarge a room while a focused direct signal preserves intimacy. A long tail can connect phrases, but it can also erase rhythm and harmony. In immersive production these relationships become directional: the listener can hear a source in front, reflections at the sides, a dome above, and a decaying field behind. The result is not one position plus one effect. It is a coordinated percept assembled from partially independent cues.

Four nested fields. It is useful to imagine every immersive-reverb design as four nested fields. The first is the **source field**: the dry or lightly processed event, including its width, directivity, motion, and distance cues. The second is the **early field**: the first reflections that describe nearby boundaries, reinforce localization, and establish scale. The third is the **late field**: the dense energy that carries reverberance, spectral decay, envelopment, and the memory of the room. The fourth is the **reproduction field**: the loudspeakers, headphones, decoder, renderer, room, and listener through which the other three fields are interpreted. A mix may be carefully authored in the first three fields and still fail

because the fourth field folds channels incorrectly, applies unknown bass management, or produces a binaural image unlike the loudspeaker render.

These fields should not be assumed to share one geometry. A close vocal may remain nearly monophonic while its early reflections occupy the front hemisphere and its late field wraps around the listener. A percussion object may circle overhead while its reverb stays stationary, making the motion readable against a stable environment. An orchestral bed may be broad and fixed while one reverse-reverb gesture crosses the ceiling as an object. Conversely, a sound field captured in Ambisonics may rotate as a coherent whole, preserving the relationship among direct and reverberant energy. The appropriate design depends on whether source motion, room stability, listener orientation, or endpoint translation carries the musical priority.

A long history before loudspeaker formats. Immersive thinking predates electronic audio. Antiphonal practice used separated ensembles and reflective architecture to make location audible as form. Thomas Tallis's *Spem in alium* (c. 1570) distributes forty voices among eight choirs; Giovanni Gabrieli's *In ecclesiis* (c. 1615) turns spatially separated forces and the long decay of a Venetian basilica into compositional structure. Later concert works and electroacoustic practices made placement, trajectory, and environmental response increasingly explicit. Karlheinz Stockhausen's *Gruppen* (1955–1957) and *Kontakte* (1958–1960), Iannis Xenakis's *Persephassa* (1969), and site-specific sound installations do not merely add ambience to otherwise complete musical objects. They organize musical identity through distance, direction, circulation, and the transformation of sound by space.

The lesson for contemporary production is not that every mix should imitate a cathedral or a mid-century multichannel concert. It is that spatial organization can carry counterpoint, orchestration, rhythm, and form. A response moving among groups can become a phrase. A shared decay can turn unrelated sources into one ensemble. A sudden collapse from a large field to a dry center can mark a structural boundary more strongly than an equalizer or level change. A static overhead halo can make a harmony feel suspended, while a moving overhead object can pull attention toward its trajectory. The same controls therefore have acoustic, perceptual, and compositional consequences.

The perceptual foundation. Human spatial hearing does not reconstruct a perfect geometric map from audio. It combines incomplete and sometimes contradictory evidence. Interaural time differences contribute strongly to horizontal localization at lower frequencies; interaural level differences become increasingly important as the head shadows higher-frequency sound. Spectral shaping by the pinnae helps distinguish elevation and front from back. The precedence effect lets an early arrival dominate localization while later related arrivals contribute width and room impression. Direct-to-reverberant ratio, high-frequency loss, and onset structure contribute to distance. Interaural coherence and the direction of early and late energy influence apparent source width and listener envelopment.

Reverberation can reinforce or destabilize every one of those cues. Highly correlated copies sent to many channels may become louder without becoming enveloping. Excessive decorrelation may create width but detach the room from the source or damage a fold-down. Strong overhead transients may read as discrete sources rather than architecture. A bright rear tail may pull attention backward. A center-heavy diffuse return may narrow the front image and mask dialogue. A field that feels continuous on loudspeakers may become inside-the-head or tonally altered through a generic binaural renderer. Immersive mixing is therefore not the pursuit of maximum difference among channels. It is the design of differences that remain perceptually related.

Reverberation occupies time as well as space. A useful spatial decision cannot be separated from the signal's temporal structure. Pre-delay determines whether the source remains distinct before the field begins. Early-reflection timing suggests scale and boundary distance. Echo density determines when isolated events fuse into a late field. Frequency-dependent decay governs how harmony and orchestration accumulate. Modulation can prevent metallic stagnation, but too much movement can destabilize localization and pitch. Ducking can preserve articulation during a phrase and let the environment bloom between gestures. A freeze can turn one instant into a sustained spatial object, but it also changes the system from a decaying room model into a controlled feedback instrument.

In music, the relevant time scale may be a transient, a beat, a phrase, or an entire section. A 2.5-second decay is not intrinsically long or short. It may be transparent around sparse attacks and destructive around dense repeated notes. A 40-millisecond pre-delay can clarify one vocal tempo and create an unwanted flam at another. A rear reflection arriving after a rhythmic subdivision may sound like a response rather than a room. The best immersive settings are therefore evaluated against musical density, tempo, register, articulation, and formal role, not selected from room names alone.

Authoring and rendering are different acts. Authoring describes intent in a representation: channel assignments, object positions, Ambisonic components, automation, routing, and metadata. Rendering turns that representation into signals for an endpoint. In a fixed-channel workflow, much of the rendering decision is committed during the mix. In Ambisonics, a decoder maps the stored scene to loudspeakers or headphones. In Dolby Atmos, the Renderer combines a 7.1.2 bed, objects, and metadata to produce a target-specific loudspeaker or binaural presentation. A 7.1.4 WAV is therefore one possible rendered speaker feed, not a substitute for an editable bed-and-object master.

This distinction matters because two listeners may receive different channel signals from the same authored session. A large Atmos room, a soundbar, stereo loudspeakers, and binaural headphones cannot reproduce identical physical fields. The production goal is not sample-for-sample identity among endpoints. It is preservation of hierarchy and intention: the lead remains legible, the room retains scale, a featured trajectory remains meaningful, low-frequency energy remains controlled, and the work does not collapse when height or rear resolution is reduced.

Beds, objects, and scenes are musical resources. A bed is not a lesser form reserved for material that did not deserve an object. It is often the best representation for a broad, stable, statistically diffuse field. An object is not valuable merely because it can move. A stationary object may preserve a precise location or binaural instruction that a bed cannot express. An Ambisonic scene is not an Atmos bed with different channel labels; it is a basis representation that can rotate and decode as a field. A hybrid production can use all of these deliberately: a front-anchored dry source, a bed-based late field, an Ambisonic environmental stem decoded for the session, and one object carrying a reverse or frozen effect.

The artistic discipline lies in assigning each layer the least complicated representation that preserves its purpose. If a diffuse tail should remain stable around the listener, making twelve animated objects may add routing and translation risk without adding meaning. If a reflected gesture must pass from a side wall to a rear balcony, a fixed bed may erase the event. If head rotation must transform a captured environment coherently, decoding a scene into a fixed print too early may discard useful information. Representation should follow the perceptual job.

Monitoring is part of the instrument. An immersive mix cannot be understood from meters and channel names alone. Loudspeaker geometry, calibration, room response, screen or console reflections, head position, decoder settings, and bass management all affect judgment. Height channels that are too loud can make ordinary reflections feel like effects. Rear loudspeakers that are too close can exaggerate decorrelation. An untreated monitoring room can impose its own decay on every authored room. Headphones remove loudspeaker-room interaction but depend on binaural rendering and individual anatomy.

For that reason, translation checks must begin before the mix is finished. Listen to the discrete target layout, a binaural render, stereo, and at least one reduced surround format. Check at calibrated working level and quietly. Inspect each channel and stem in isolation, but make decisions in the complete render. Verify channel identity with impulses or spoken labels. Compare bass-managed and full-range behavior when possible. A spatial effect that survives only one premium monitoring configuration is not necessarily wrong, but its limitation must be intentional and documented.

The role of verbx. This chapter explains those distinctions and turns them into practical verbx workflows. It covers conventional surround, height layouts, Ambisonics, binaural monitoring, Dolby Atmos beds and objects, reverb routing, DAW handoff, deliverables, and quality control. verbx is particularly useful before and around the final immersive authoring stage: it can design algorithmic fields, apply

matrix impulse responses, create discrete wet stems, process Ambisonic material within documented limits, generate channel-identification and analysis evidence, and preserve reproducible settings in machine-readable reports.

The present product boundary must remain plain: **verbx can generate and process discrete multichannel audio, Ambisonic material, matrix-routed convolution, and machine-readable handoff evidence; it does not currently author Dolby object metadata or create an ADM BWF, DAMF, or IMF IAB master.** Use an Atmos-capable DAW and the Dolby Atmos Renderer for that final authoring stage. A JSON sidecar can describe intended routing and preserve provenance, but it does not become embedded Dolby metadata simply because it accompanies a multichannel file.

Questions this chapter will answer. By the end of the chapter, the reader should be able to:

- distinguish channel-based, scene-based, and object-based representations without treating their channel counts as interchangeable;
- explain why a standard Atmos bed is 7.1.2 even when monitoring uses a 7.1.4 loudspeaker array;
- separate source position, early-reflection geometry, late-field envelopment, and endpoint rendering into controllable layers;
- decide when a reverb belongs in a bed, on an object, in an Ambisonic scene, or across a hybrid routing design;
- map and verify multichannel files by semantic channel label rather than by count alone;
- preserve intelligibility, transient definition, low-frequency headroom, and musical hierarchy while increasing envelopment;
- evaluate interchannel correlation, fold-down behavior, binaural translation, and loudspeaker translation as related but nonidentical tests;
- document the difference between an editable immersive master, a bed or object stem, and a speaker-channel re-render;
- use verbx to create reproducible spatial assets without claiming metadata or mastering capabilities it does not provide.

A disciplined listening method. Begin every immersive-reverb study with the smallest useful source: a spoken phrase, a single percussion hit, a sustained chord, and a short musical excerpt. First listen to the dry source and identify its natural width, onset, register, and phrase rhythm. Next audition early reflections alone and ask whether they change direction, distance, or scale. Then audition the late field alone and listen for density, spectral decay, modulation, and directional balance. Restore the complete mix and decide whether the source and room feel causally connected. Finally, compare the main loudspeaker render, binaural, stereo, and a reduced layout at matched loudness. This sequence makes faults diagnosable. It also prevents the excitement of "more speakers" from concealing a field that is merely louder, less coherent, or less musical.

The goal is not to prescribe one spatial aesthetic. Naturalism, heightened realism, impossible architecture, mobile electronic counterpoint, and deliberately exposed artificial reverb are all valid. The goal is to make the signal path legible enough that a mix decision remains a mix decision, rather than becoming an accidental consequence of channel order, renderer behavior, bass management, or a mistaken assumption about what a file contains. Immersive production becomes tractable when representation, perception, musical function, and delivery are treated as one connected design problem.

4.1 Three Ways to Describe Space

Before choosing a loudspeaker layout or creating a reverb return, a production must decide what kind of spatial information it intends to preserve. Channel-based, scene-based, and object-based audio are often introduced as competing formats, but that framing is misleading. They are three different contracts between authoring and playback. Each contract answers a different question, stores a different kind of information, and postpones a different set of decisions until rendering.

A **channel-based contract** says, in effect, "send this waveform to this named destination." The destination may be Left, Center, Right Surround, or Left Top Middle. The signal can be inspected directly and the route is easy to understand, but the author has committed to an assumed reproduction geometry. A **scene-based contract** says, "this set of components describes a sound field around a reference point." It postpones loudspeaker assignment to a decoder and allows operations such as rotation to act on the field as a whole. An **object-based contract** says, "this waveform is an element with rendering instructions." It postpones endpoint-specific panning so the same object can be interpreted for a large loudspeaker array, a smaller system, stereo, or binaural headphones.

The contracts are not interchangeable because the numbers in their channels do not mean the same thing. Four FOA channels are not four loudspeaker feeds. Ten channels in a 7.1.2 bed are not ten arbitrary tracks. A mono object is not merely a mono channel placed in a larger file. Each representation requires conventions concerning order, normalization, coordinate systems, labels, metadata, and decoding. Losing those conventions can produce a file that opens successfully and sounds plausible while being spatially wrong.

This is especially important for reverberation because a reverberant response contains relationships among directions. Suppose a singer stands in front of a listener in a tall hall. The direct sound arrives from the front. Early reflections may arrive from the front wall, side walls, floor, and ceiling at different times and spectra. The late field may become broadly diffuse but remain brighter above or decay longer behind. There are several valid ways to represent that event:

- A channel-based render can send the direct signal to the front, distribute early reflections among named loudspeakers, and place the late field in side, rear, and height channels.
- A scene-based render can encode the directional field into spherical-harmonic components and allow a decoder to reconstruct it for the available endpoint.
- An object-based session can keep the singer as an object, place a broad late return in a bed, and reserve separate objects for a featured reflection or moving effect.
- A hybrid can combine all three after deliberate conversion and monitoring, provided that each layer's conventions remain explicit.

These approaches can produce similar experiences in one room, but they are not the same asset. The channel render has already committed to feeds. The scene retains a listener-centered field that may be rotated or decoded differently. The object session retains selected source elements and metadata for a renderer. If only the final loudspeaker feed is delivered, the earlier flexibility cannot be recovered reliably from channel count alone.

Representation is not transport. WAV, BWF, CAF, and other containers can carry channel audio, but the container does not establish the complete spatial semantics by itself. A multichannel WAV may contain a standard layout, an Ambisonic scene, an arbitrary stem pack, or a speaker re-render. Labels and metadata may be incomplete, ignored, or interpreted differently by software. A filename such as `mix_7.1.4.wav` communicates intent to a human but does not prove channel order or mastering status. Reliable workflows carry an explicit channel map, format convention, sample rate, reference level, processing history, and intended next stage alongside the audio.

Representation is not perception. A mathematically correct object position does not guarantee stable localization. A high-order scene does not guarantee envelopment. A twelve-channel print does not guarantee height. Perception depends on the rendering method, loudspeaker geometry, binaural filters, room, listener, source spectrum, timing, and competing signals. A diffuse reverb may feel broader from a modest channel bed than from many highly correlated objects. A sparse reflection may localize more clearly as one object than as energy spread through an elaborate scene. Format capability sets the available vocabulary; arrangement and signal design determine whether that vocabulary communicates.

Representation is not musical function. The same source may need different treatment at different moments. A lead voice can begin as a stable front object, merge into a bed-based choral field, and leave behind a rotating Ambisonic texture rendered into the session. A percussion hit can remain fixed while

only its reflected image travels. A room can stay stationary as sources move through it, or the entire scene can rotate with a listener in virtual reality. Choosing a representation is therefore comparable to choosing orchestration: it determines which relationships are easy to articulate, which are expensive, and which are no longer editable after commitment.

When evaluating the three models, ask five questions before asking how many channels are available:

1. What must remain editable at the final authoring stage: loudspeaker feeds, the complete field, or individual source positions?
2. Must the environment rotate with head or camera orientation, or should it remain fixed in the playback room?
3. Is the reverb a stable shared architecture, a source-attached effect, a moving compositional gesture, or several layers with different jobs?
4. Which endpoints matter, and what hierarchy must survive when spatial resolution is reduced?
5. What metadata and channel conventions can the production preserve and verify from render to delivery?

The safest workflow begins with meaning and ends with channel count, not the reverse. Name the source, early field, late field, and special-effect layers. State whether each layer is fixed, listener-centered, or renderer-positioned. Choose a representation for each. Only then choose a bus width, file layout, or object count. This order prevents a common failure in immersive sessions: building an impressive routing graph first and discovering later that no one can explain which parts of the graph carry the room, which carry a speaker print, and which remain editable.

Hybrid workflows are normal because real productions contain more than one spatial job. A bed can provide continuous environmental glue. Objects can preserve important sources and designed gestures. A scene can carry a captured ambience or rotate coherently before being decoded into the authoring environment. The requirement is not ideological purity. It is controlled conversion. Every conversion should state what information is retained, what is committed, and what cannot be reconstructed afterward.

The following three subsections examine the contracts separately. Read them as choices about deferred decisions rather than as a hierarchy of sophistication. Channel-based audio offers directness. Scene-based audio offers field operations and decoder independence. Object-based audio offers endpoint-aware source rendering. The most advanced production is not the one that uses the greatest number of objects or the highest Ambisonic order. It is the one whose representation remains aligned with perception, musical intention, and delivery from the first routing decision to the last listener.

4.1.1 Channel-based audio

A channel-based signal assigns each waveform to a named destination such as Left, Center, Right, Left Surround, or Left Top Front. Stereo, 5.1, 7.1, and a discrete 7.1.4 speaker feed are channel-based formats. Their strengths are directness and predictability: if the playback system honors the labels, the Left Surround signal reaches the left-surround path. Their limitation is that the mix is bound to the assumed loudspeaker geometry. Translating a 7.1.4 print to headphones or a smaller room requires a downmix or a separate renderer.

Channel-based reverb is often the most stable choice for an enveloping late field. A diffuse return can occupy side, rear, and height channels while dry anchors remain in front. The return does not need to carry a moving trajectory; it needs a carefully designed covariance pattern, spectral balance, and decay envelope.

4.1.2 Scene-based audio

Scene-based systems describe a sound field around a listening point. Ambisonics is the most familiar example. Instead of storing one channel per loudspeaker, it stores spherical-harmonic components. A

decoder then maps those components to the available loudspeaker array or to binaural headphones. First-Order Ambisonics (FOA) uses four components; higher orders increase spatial resolution at the cost of more channels and more demanding capture, processing, and decoding.

A scene can be rotated without reauthoring every loudspeaker feed. This makes Ambisonics useful for virtual reality, 360-degree video, game ambience, acoustic documentation, and portable reverberant fields. It is not the same as Dolby Atmos object metadata. Converting an Ambisonic scene into an Atmos session requires a render or a deliberate object/bed authoring step, not a channel-label rename.

4.1.3 Object-based audio

An object combines an audio signal with metadata describing where and how it should render. Position, divergence or size, and binaural behavior may vary over time. The final loudspeaker feeds are computed by a renderer for the target endpoint. That target might be a 7.1.4 room, 5.1, stereo, a soundbar, or binaural headphones.

Object audio is especially valuable when a sound must occupy a precise or moving location that a fixed bed cannot address independently. It is not automatically better for every reverb return. A dense, statistically diffuse late field often belongs in a bed because it should envelop rather than announce a point source. A featured reverse tail, rotating shimmer, or isolated reflection can be an excellent object because its trajectory is part of the composition.

4.2 Reading Surround and Height Layout Names

The notation `7.1.4` counts three groups: seven ear-level full-range channels, one low-frequency-effects channel, and four height channels. It says nothing by itself about whether the signal is a monitoring render, a DAW bus, a codec output, or an authored bed. Context matters.

Table 4-1: Reading Surround and Height Layout Names: reference table

Name	Nominal contents	Typical role
`2.0`	Left, Right	Stereo delivery or monitoring
`5.1`	Five full-range channels plus LFE	Broadcast, film, home-theater surround
`7.1`	Seven full-range channels plus LFE	Expanded side/rear surround
`7.1.2`	7.1 plus two height channels	Standard Dolby Atmos bed; also a fixed-channel immersive bus
`7.1.4`	7.1 plus four height channels	Common Atmos monitoring and speaker-render layout; **not** the default Atmos bed
`7.2.4`	Seven ear-level, two LFE, four height	Installation or custom discrete layout; not a standard Atmos bed name
FOA	Four spherical-harmonic components	Portable first-order scene representation

The distinction is easier to see spatially. Figure 1 compares the ten fixed channels of a 7.1.2 bed with a 7.1.4 monitoring array. The Renderer may distribute bed and object energy among all available loudspeakers; the presence of four physical top loudspeakers does not turn the source bed into 7.1.4.

Figure 4-1 presents Bed channels versus monitor channels and establishes the visual reference for the discussion that follows.

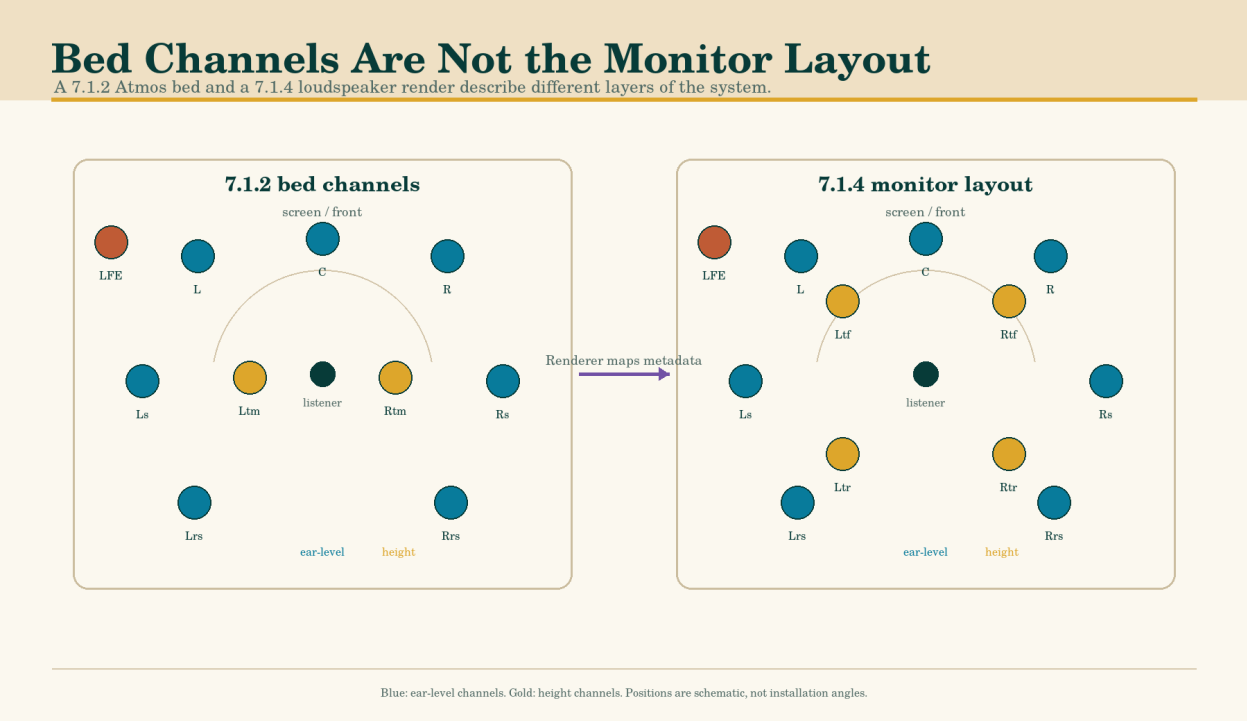


Figure 4-1: Bed channels versus monitor channels

The left plan shows the standard 7.1.2 bed concept: eight channels at or around ear level, including LFE, and two top-middle channels. The right plan shows a common 7.1.4 monitoring arrangement with separate front-top and rear-top loudspeakers. The positions are conceptual rather than installation specifications; use the room-design guidance appropriate to the renderer, room, and delivery contract.

4.2.1 Channel order is part of the format

For a standard Dolby 7.1.2 bed, Dolby documents the SMPTE order as `L, R, C, LFE, Ls, Rs, Lrs, Rrs, Ltm, Rtm`. Channel count alone cannot establish that order. A ten-channel WAV could contain the same signals in a different sequence and sound dramatically wrong while still opening successfully.

verbx currently uses the symbolic `7.1.2` labels `L, R, C, LFE, Ls, Rs, Lrs, Rrs, Ltf, Rtf`. Therefore, when preparing a verbx `7.1.2` file for an Atmos bed, **verify and explicitly map channels 9 and 10 to the Renderer's `Ltm` and `Rtm` inputs**. The labels are not interchangeable merely because the channel count agrees. This is a handoff requirement, not a sonic preference.

Use a spoken channel-identification file or short, non-overlapping impulses before trusting a complex mix. A successful import proves only that the container is readable; it does not prove semantic routing.

4.3 Dolby Atmos Architecture

Dolby Atmos separates audio into beds and objects, then delegates endpoint-specific channel generation to a Renderer. According to Dolby's current professional documentation, a typical music workflow uses one 7.1.2 bed and may use up to 118 objects within a system supporting up to 128 input paths. The exact available count can depend on how stereo objects and beds consume paths.

The bed carries fixed-channel material. It is appropriate for stems whose spatial role is broad and stable: an orchestral room, diffuse crowd, ensemble ambience, or reverberant field. Objects carry audio plus metadata. They are appropriate for a featured source or effect whose authored position, size, motion, or binaural behavior must survive downstream rendering.

Figure 2 shows why an Atmos master cannot be reduced to a 7.1.4 WAV. The bed, object signals, and metadata enter the Renderer separately. It can then derive multiple channel and binaural re-renders from one authored master.

Figure 4-2 presents Dolby Atmos bed/object rendering architecture and establishes the visual reference for the discussion that follows.

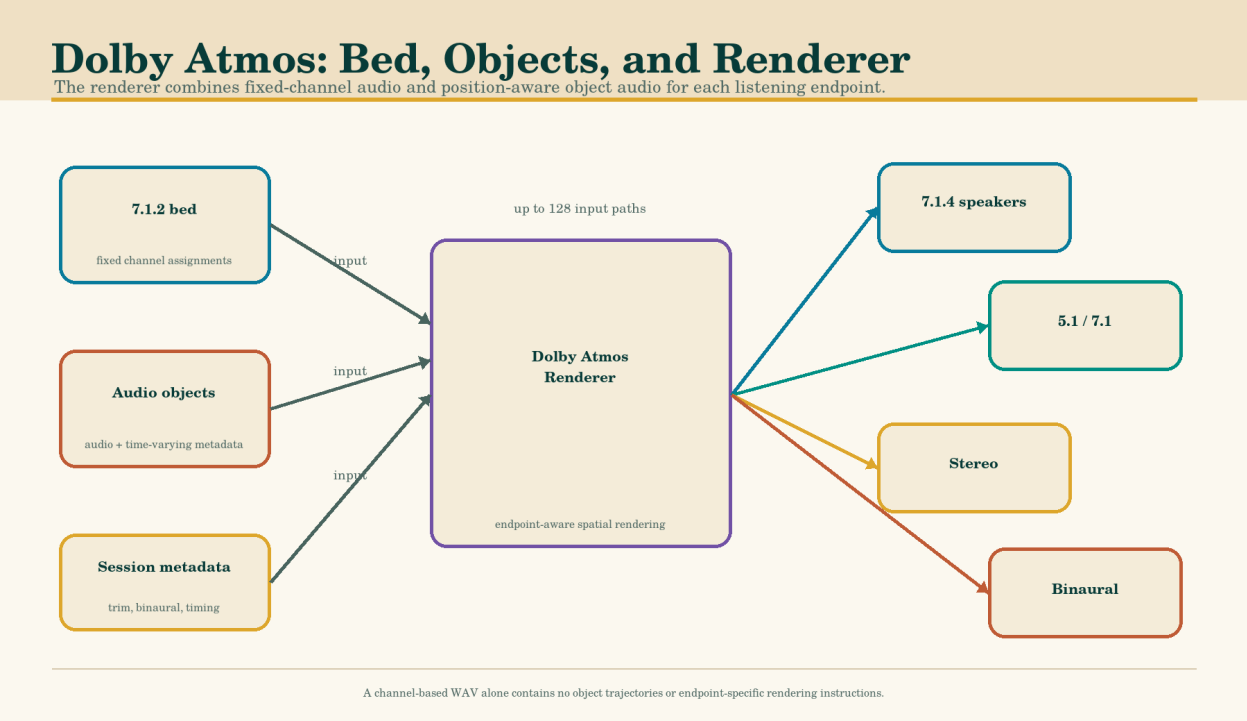


Figure 4-2: Dolby Atmos bed/object rendering architecture

Fixed-channel bed audio, object audio with trajectories, and session metadata remain distinct until the Renderer. The Renderer combines them according to the target endpoint, producing outputs such as 7.1.4, 5.1, stereo, or binaural. A conventional multichannel WAV represents one channel render; it does not carry the complete object-based authoring model shown here.

4.3.1 Why the 7.1.2 bed does not reach every speaker independently

The standard bed provides a stable immersive foundation, but it does not offer separate fixed channels for every possible width or top-front/top-rear location. When a sound needs an independently addressable location outside the bed's fixed destinations, author it as an object. This is one reason a hybrid mix is normal: the bed supplies continuity and envelopment while objects supply location-specific detail.

4.3.2 Objects are not moving tracks by default

An object may remain stationary for an entire piece. The useful property is not motion itself; it is that the Renderer receives a position and can translate it to each endpoint. Excessive motion can shrink perceived scale, distract from musical phrasing, or produce unstable binaural images. Reserve motion for an audible purpose.

4.3.3 The Renderer is part of the instrument

In object-based production, monitoring is already a render. A pan decision heard over 7.1.4 loudspeakers and the corresponding binaural result are two interpretations of the same metadata. Neither should be treated as a disposable preview. The production process must listen through the intended render paths early enough that translation problems can inform the mix.

4.4 Reverb as an Immersive Layer

In stereo production, reverb is often introduced as an effect that places a source farther away, makes it larger, or supplies a pleasing tail. Those descriptions remain useful in immersive production, but they are incomplete. Once sound can arrive from the front, sides, rear, and height plane, reverberation becomes one of the principal means by which the mix establishes a continuous acoustic world among otherwise separate channels and objects. It tells the listener not only how long energy persists, but also where the boundaries of the imagined environment might be, whether sources share that environment, how the environment changes with direction, and whether a spatial event belongs to a source, to a room, or to the form of the music itself.

The central idea of this section is therefore simple: **immersive reverb is spatial orchestration, not speaker filling**. Sending the same stereo return to more loudspeakers does not automatically create a larger or more convincing room. It may merely increase level, duplicate correlation, weaken localization, and produce a fold-down that is louder or more hollow than the immersive monitor. A persuasive field requires differentiated roles. Some energy should preserve the source's direction. Some should describe nearby surfaces. Some should detach from the source and envelop the listener. A small amount may occupy the height plane, while a designed effect may travel independently as an object. The channels become meaningful when those roles are composed rather than populated.

This change in perspective is important because a listener does not perceive twelve independent reverbs when listening to a 7.1.4 presentation. The listener usually perceives sources, distances, enclosing surfaces, openings, reflections, and one or more fields of resonant energy. The production task is to make many electrical paths support a smaller number of coherent perceptual causes. If every channel has unrelated timing, color, and modulation, the result may be wide but cease to sound like one place. If every channel is identical, the result may be stable but collapse toward a loud, correlated image. The useful region lies between those extremes: enough difference to create direction and envelopment, enough shared behavior to imply a common acoustic system.

Three functional categories help organize that region. A **source-bound return** follows the identity and location of a particular voice, instrument, or effect. Its early reflections may move when the source moves, and its pre-delay and direct-to-reverberant ratio help establish the source's apparent distance. An **environment-bound return** belongs to the scene rather than to one emitter. It supplies the persistent late field, room tone, spectral decay, and enveloping continuity that allow several sources to sound as though they inhabit the same architecture. A **gesture-bound return** is explicitly compositional: a reverse swell, frozen harmony, rotating shimmer, isolated echo, or tail that crosses the listener at a formal boundary. These categories can share DSP, but they should not be routed or automated as though they perform the same job.

The distinction also clarifies when to use a bed and when to use an object. A diffuse, stable late field often works well in a bed because its purpose is continuous enclosure, not point localization. A bed gives that field dependable relationships to the front, sides, rear, and available height channels while leaving the Renderer responsible for endpoint translation. A featured reflection or moving tail can work better as an object because its position, size, and trajectory are part of the authored gesture. Treating the entire room as a swarm of objects can make a stable environment difficult to maintain; treating every special effect as part of one fixed bed can make its path vague or impossible to preserve. Hybrid routing is not a compromise here. It is often the clearest expression of the underlying perceptual design.

Early and late energy deserve separate attention within either representation. Early reflections are sparse enough to influence localization, distance, apparent source width, and the inferred position of nearby surfaces. Their timing and direction can tell the listener that a source is near a front wall, deep on a stage, beneath a high ceiling, or close to one side of an enclosure. The late field has a different task. As reflection density rises, individual paths fuse into a statistical field whose decay, spectrum, modulation, and directional balance determine envelopment and room continuity. Routing one undivided reverb signal everywhere sacrifices this distinction. Separating the early and late paths allows the mix to keep a source legible in front while its sustained energy opens around and above the listener.

Interchannel relationship is as important as level. Two channels can have the same RMS level and frequency response while producing very different spatial impressions because their waveforms are correlated differently over time and frequency. Perfect copies tend to reinforce a fixed image and can create large level changes when summed. Completely unrelated returns may enlarge the field but weaken the sense that all channels describe one room. Effective decorrelation varies delay, phase, diffusion state, or modulation while preserving a related envelope and tonal trajectory. In other words, the channels need not contain the same samples, but they should appear to obey the same acoustic history.

Height requires especially deliberate treatment. Overhead loudspeakers are conspicuous resources, and it is tempting to prove that they are active by feeding them a bright, continuous copy of the return. Natural vertical extent is often subtler. Ceiling and upper volume reflections may arrive later, contain less transient definition, or decay with a different spectrum than lateral energy. A restrained height layer can increase apparent volume without pulling the room away from the source. Conversely, a clearly localized overhead tail can be powerful when it is intended as a musical event. The crucial question is not whether the top channels contain reverb; it is whether their energy explains the space or articulates a purposeful gesture.

The most dependable design process begins with hierarchy rather than channel count. First, establish the dry anchors and decide which sources must remain locatable. Second, build a late field that supplies convincing enclosure at a conservative level. Third, add early reflection patterns that establish source distance and architectural orientation without competing with direct attacks. Fourth, tune front, side, rear, and height relationships, listening for one coherent environment rather than maximum activity. Fifth, add objects or automation only where movement, reversal, freezing, or spectral transformation has a specific formal purpose. Finally, print and evaluate every required endpoint. This order keeps decorative motion from masking a weak room model and keeps an impressive loudspeaker demonstration from becoming the only version that works.

Endpoint translation must be considered during design, not after approval. A lateral field that feels smooth over loudspeakers may become diffuse or externalized poorly in binaural. A dramatic rear event may fold into the front image in stereo. Highly anticorrelated side energy may sound expansive discretely but lose body when downmixed. None of these outcomes means that immersive reverb should be reduced to stereo-safe sameness. It means that the production needs a stated hierarchy: which localization, depth, envelopment, spectral, and formal cues are essential, and which may simplify as spatial resolution decreases. The Renderer is part of that instrument, so loudspeaker, binaural, and fold-down monitoring belong inside the creative loop.

Measurement supports this work but does not replace listening. Channel peaks, loudness, correlation, energy decay, spectral balance, and impulse-response plots can reveal routing errors or unstable accumulation. They cannot decide whether a rear bloom completes a phrase, whether the center remains emotionally present, or whether height turns a room into an effect. A disciplined evaluation alternates between isolated paths and the complete mix, between static and automated passages, and between target endpoints. It also checks silence and tail completion: immersive systems can hide low-level buildup across many channels until a long pause, downmix, or limiter exposes it.

With that framework in place, the detailed controls become easier to interpret. Front energy establishes anchor and connection; side and rear energy establish lateral and backward enclosure; height energy establishes vertical extent or a designed overhead gesture; LFE requires a separate editorial decision;

and decorrelation determines whether these regions combine as one field. The following subsections examine those responsibilities

individually, while the routing topology below shows how verbx can keep early definition, diffuse bed energy, and special object returns separate long enough to make each decision audible and reversible.

Reverb contributes several perceptually distinct cues:

- Early reflections communicate apparent room size, source distance, surface placement, and localization.
- The late field communicates envelopment, decay, density, spectral absorption, and the persistence of the acoustic environment.
- Direction-dependent decay communicates architecture: a tall reflective space may retain energy above the listener while side energy damps sooner.
- Interchannel covariance controls whether the field feels broad, stable, phasey, collapsed, or detached.
- Modulation and diffusion determine whether discrete echoes fuse into a field or remain audible as patterns.

Treating all of those cues as one twelve-channel send leaves musical control on the table. Figure 3 presents a more useful topology. It divides the effect into early definition, a diffuse bed return, and an optional designed object return.

Figure 4-3 presents Hybrid immersive-reverb topology and establishes the visual reference for the discussion that follows.

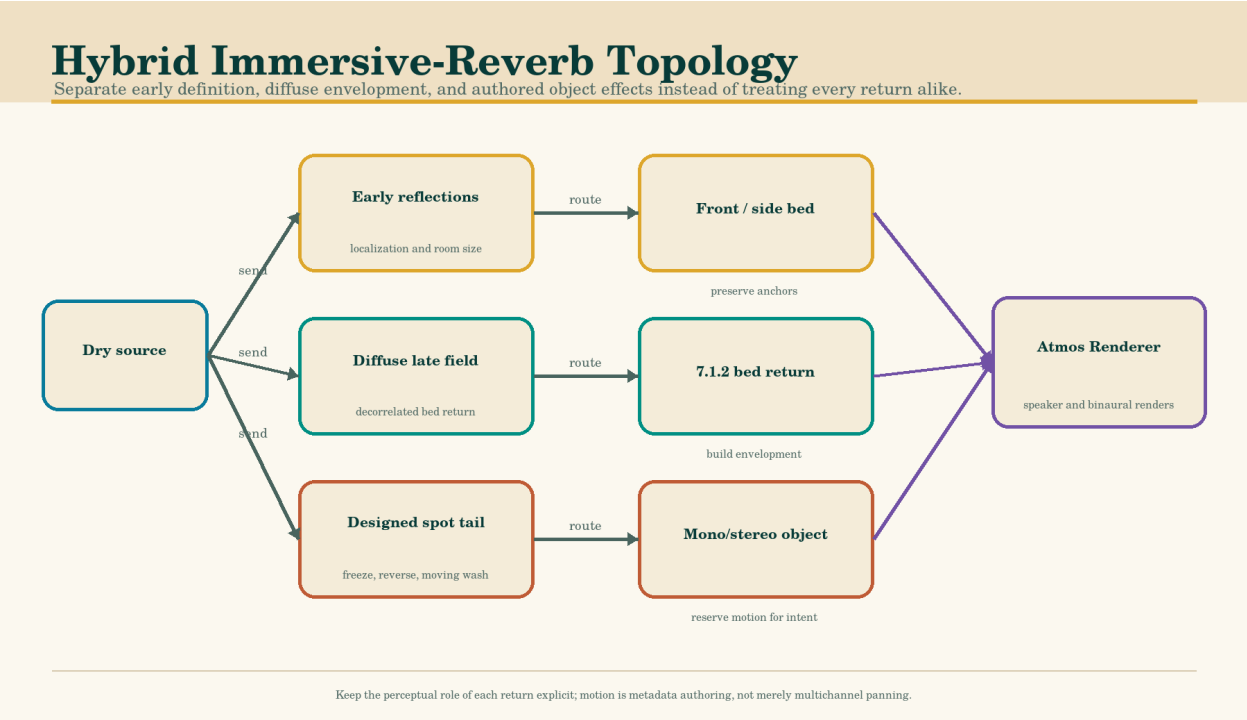


Figure 4-3: Hybrid immersive-reverb topology

A dry source feeds three sends with different perceptual jobs. Early reflections reinforce front and side localization; the diffuse late field enters a stable 7.1.2 bed return; and a special reverse, freeze, or moving tail is returned as a mono or stereo object. The Renderer combines those paths for loudspeaker

and binaural endpoints. This separation prevents decorative motion from destabilizing the entire room impression.

4.4.1 Front channels and the center anchor

The center channel is powerful because it can create a stable screen or stage anchor. It is also easy to overload. Sending a highly correlated reverb return equally to Left, Center, and Right can build a narrow, level-heavy front image. For dialogue and lead vocals, begin with less late-field energy in Center than in Left/Right and surrounds. Preserve intelligibility with front-weighted early reflections, predelay, ducking, or spectral shaping rather than by eliminating all spatial response.

4.4.2 Side and rear channels

Side channels often carry the strongest envelopment cue because they stimulate lateral perception. Rear channels can extend depth, but excessive rear level can pull attention behind the listener or make the room feel detached from the source. A robust late field is usually decorrelated rather than copied. Identical tails in every channel may collapse during downmixing and can produce comb filtering or unstable images.

4.4.3 Height channels

Height should communicate vertical extent, overhead reflections, or a deliberate compositional gesture. Simply duplicating the bed return into top channels raises level without necessarily creating elevation. Consider later or spectrally different overhead energy, independent decorrelation, and a controlled height-to-bed ratio. In many natural rooms, the top field can be smoother and less transient-rich than the direct and early-reflection field.

Four-top monitoring gives a renderer more ways to express front-to-rear height movement, but a standard 7.1.2 bed still supplies only two fixed top-middle channels. Use objects if a return must be explicitly top-front, top-rear, or moving along the ceiling.

4.4.4 LFE is not bass management

The LFE channel is a creative effects channel, not the destination for all low-frequency content. Bass management is a monitoring-system function that redirects low frequencies from main channels according to crossover settings. A large reverb tail can already reach the subwoofer through bass management without being sent to LFE.

As a default, keep diffuse reverb out of LFE unless there is a clear artistic or delivery reason. Uncontrolled low-frequency decay consumes headroom, masks rhythm, and may translate unpredictably across systems. If LFE reverb is intentional, high-pass or band-limit it, monitor the LFE channel directly, and check the full-range re-render separately from a bass-managed room.

4.4.5 Decorrelation without disconnection

Immersive width requires difference among channels, but random difference is not automatically coherent. Useful decorrelation preserves a shared decay envelope and tonal identity while varying delay, modulation phase, diffusion state, or allpass structure. The goal is a single room observed from many directions, not twelve unrelated reverbs.

verbx exposes front, rear, and top decorrelation controls so these regions can be tuned independently. Start conservatively, then compare the discrete layout, stereo fold-down, and binaural render. If the immersive version sounds expansive but the stereo version becomes hollow, the channel relationships are too antagonistic.

4.4.6 Wave field synthesis: virtual sources, real arrays, and verbx stems

Wave field synthesis (WFS) is not a larger surround bed. A conventional channel-based mix assigns waveforms to named loudspeakers. Ambisonics stores coefficients for a sound field and defers decoding. WFS starts from a different practical question: given a dense, measured loudspeaker array, what signals should its individual loudspeakers radiate so that listeners over a useful region encounter the wavefront of a chosen virtual source? The goal is not merely to distribute energy around a room. It is to approximate the propagation of a virtual point source, plane wave, or other source geometry through an extended listening area.

That distinction has musical consequences. In an ideal WFS system, a virtual source can appear in front of, behind, or beyond the loudspeaker array while retaining a meaningful wavefront relationship as listeners move. The apparent source does not depend on one fixed loudspeaker pair or one sweet-spot panning law. In practice, the result is bounded by array geometry, loudspeaker spacing, room reflections, calibration, processing delay, listening distance, and the frequency range over which the array can reproduce the intended field without conspicuous spatial aliasing. WFS is therefore an electroacoustic instrument and an installation discipline, not a file format.

At a conceptual frequency-domain level, an array renderer seeks loudspeaker driving functions $D_\ell(\omega)$ such that the emitted field approximates a target field:

$$p(\mathbf{r}, \omega) \approx \sum_{\ell=1}^{L} D_\ell(\omega)\, G(\mathbf{r} \mid \mathbf{r}_\ell, \omega),$$

where $\mathbf{r}$ is a listening position, $\mathbf{r}_\ell$ is the position of loudspeaker ℓ, G is the propagation model from that loudspeaker to the listener, and L is the number of secondary sources. The approximation is deliberately qualified. A real array is finite, sampled in space, installed in a reflective room, and driven by loudspeakers with nonideal directivity and response. As speaker spacing increases, the usable frequency range for a convincing synthesized wavefront decreases; as the array becomes sparse, the renderer increasingly trades physical precision for practical coverage.

Reverberation in WFS must be designed as a field, not as a stereo send copied around a perimeter. A source-bound early field can be rendered from virtual images that support the direct source's location. A broad late field can be made from many low-level, decorrelated virtual sources or from a diffuse-field strategy chosen by the WFS system. The first approach can articulate architecture and motion; the second can support envelopment without turning every reflection into a locatable event. Neither is automatically better. A sparse musical gesture may benefit from audible moving reflection images, while a dense ensemble often needs a stable, low-attention field whose directionality does not compete with the players.

WFS also clarifies why an apparently impressive reverb can fail in an installation. If every wet component follows the dry virtual source, the room may feel glued to the performer rather than to the venue. If the late field is rendered as many coherent copies, it can become loud and spatially brittle. If the system emits a long bright return from a small subset of array elements, the result can expose aliasing, localization jumps, and room-specific combing. Treat direct sound, early reflections, late field, and special gestures as separate musical layers. Decide whether each layer is source-bound, environment-bound, or intentionally impossible before assigning it to the WFS renderer.

verbx's role is upstream and adjacent to the WFS renderer. verbx can prepare deterministic dry sources, early and late reverb stems, matrix-routed convolution assets, analysis JSON, and repeatable gain and spectral decisions. It can also analyze microphone captures made in the completed array room. It does **not** currently calculate WFS driving functions, ingest an array's loudspeaker-coordinate file, compensate per-loudspeaker latency or transfer functions, select active secondary sources, or directly address the hundreds of channels that a large WFS renderer may use. Those tasks belong to the array renderer, its calibration system, and the installation's routing environment.

This boundary is useful rather than limiting. Keep the reverb design independent of a particular array until the point where array geometry is needed. For example, create a concise early-return candidate and a darker, longer late-return candidate in verbx; retain the dry source unchanged; then let the WFS environment place the dry source, the early images, and the late-field virtual ensemble separately. The verbx reports make the audio decisions auditable, while the WFS session retains the spatial configuration that cannot be represented by a conventional multichannel WAV alone.

4.4.6.1 A practical WFS preparation workflow

Start with three assets rather than one premixed "WFS reverb" file:

1. **Dry source:** preserve the unprocessed event for the WFS renderer's direct virtual source. Do not bake a stereo pan or unrelated room into it.
2. **Early-return candidate:** use a short, controlled reverb or measured early-response asset for virtual reflection images. Keep attacks readable and document the intended source-to-reflection relationship.
3. **Late-return candidate:** render a darker, denser, 100-percent-wet field that the WFS system can distribute as an environmental layer or a cloud of virtual secondary sources.

The following pair creates deliberately separate early and late candidates. They are not WFS driving signals; they are musical source material for the WFS renderer:

```
verbx render source.wav wfs_early_return.wav \
  --engine algo \
  --rt60 0.42 \
  --predelay 4 \
  --damping 0.30 \
  --wet 1.0 --dry 0.0 \
  --output-layout mono \
  --json-out wfs_early_return.analysis.json

verbx render source.wav wfs_late_return.wav \
  --engine algo \
  --rt60 3.4 \
  --predelay 38 \
  --damping 0.62 \
  --wet 1.0 --dry 0.0 \
  --output-layout mono \
  --json-out wfs_late_return.analysis.json
```

In the WFS environment, audition the dry source first. Add only the early-return candidate and test whether its virtual reflection positions support distance rather than blur the source. Add the late return last, initially at a lower level than a stereo instinct might suggest. Walk the intended audience area. Listen for changes in localization, tonal balance, echo density, and stability as the listener leaves the nominal center. A reverb decision that works only at one chair is not necessarily unusable, but it should be described honestly as a local effect rather than as uniform field synthesis.

After the array is tuned, capture representative positions with calibrated microphones and analyze them as a set. Use one report per position, retain the array preset and loudspeaker map with the report, and compare early decay, spectral decay, direct-to-reverberant relationship, and interposition consistency. A minimal evidence pass is:

```
verbx analyze wfs_position_a.wav \
  --edr --room \
  --json-out wfs_position_a.analysis.json
```

Repeat at the center, near an array boundary, and at one musically important off-axis location. The resulting differences are not merely faults. They reveal where the installation produces the intended

perspective, where the venue dominates the renderer, and where a composition may exploit a change in apparent source distance or field texture.

For live WFS, keep latency budgeting explicit. The renderer's delay compensation, audio interface buffers, network transport if present, loudspeaker DSP, and any realtime verbx stage all add to the end-to-end path. A reverb tail can tolerate substantial delay more easily than a direct virtual source, but a source-bound early field cannot be casually delayed without changing precedence and distance cues. If verbx is used live, place it on a dedicated return or pre-render its stems unless its measured realtime latency fits the installation's direct-sound budget.

The compositional opportunity is substantial. WFS can make a dry instrumental line appear to approach from beyond the stage while its late field remains behind the audience; it can let a frozen chord become a stationary architectural surface while solo gestures move through it; it can separate a virtual ensemble into depth layers without equating depth with reverb level. verbx contributes repeatable acoustic materials and analysis to that process. The WFS renderer contributes the physical wavefront strategy. Keeping those roles separate is what makes the workflow both creatively open and technically defensible.

4.5 What verbx Can Produce Today

The following boundary is intentionally explicit.

Table 4-2: What verbx Can Produce Today: reference table

Capability	Current status	Meaning for an Atmos workflow
Discrete multichannel WAVE rendering	Supported	Produce bed candidates, speaker prints, wet stems, and channel-identification files
Named input/output layouts	Supported	Declare routing intent for common layouts rather than relying only on channel count
Matrix convolution	Supported	Route each input independently to one or more output channels with an explicitly packed IR matrix
Algorithmic spatial decorrelation	Supported	Shape front, rear, and top late-field relationships
FOA encode, rotate, and stereo decode	Supported with documented constraints	Prepare or process a scene-based field before a separate Atmos authoring stage
Higher-order Ambisonic validation and channel-order metadata	Supported where documented by the CLI	Keep scene conventions explicit during handoff
JSON analysis and handoff manifests	Supported	Record channels, routing, settings, and QC evidence alongside audio
Dolby object trajectories and size automation	Not currently authored	Create these in an Atmos-capable DAW
Per-object Dolby binaural metadata	Not currently authored	Set and audition this in the DAW/ Renderer
Native ADM BWF, DAMF, or IMF IAB master	Not currently written	Export or record the master through the Dolby Atmos Renderer or an integrated authoring environment
Dolby certification or delivery approval	Not implied	Follow the current distributor, label, broadcaster, studio, and Dolby requirements

A JSON sidecar can be extremely useful, but it is documentation rather than embedded Dolby metadata. Renaming a sidecar or adding the letters "ADM" to a filename does not create the required BWF metadata chunk. Likewise, a 7.1.4 WAV is a speaker-channel render, not an editable object master.

4.6 Practical verbx Recipes

The commands below prepare audio for immersive sessions. They do not bypass the authoring and Renderer stages.

4.6.1 Create a 7.1.2 wet-bed candidate

Render a discrete ten-channel return, then explicitly map the final two channels during DAW import:

```
verbx render source.wav wet_712.wav \
  --engine algo \
  --output-layout 7.1.2 \
  --rt60 2.8 \
  --predelay 28 \
  --algo-decorrelation-front 0.22 \
  --algo-decorrelation-rear 0.48 \
  --algo-decorrelation-top 0.62 \
  --wet 1.0 --dry 0.0 \
  --json-out wet_712.analysis.json
```

On import, confirm `L, R, C, LFE, Ls, Rs, Lrs, Rrs` and remap verbx `Ltf, Rtf` to the intended Atmos bed `Ltm, Rtm` inputs. Do not infer the map from file width alone.

4.6.2 Create a 7.1.4 speaker print

Use this when a twelve-channel file is specifically required for monitoring, installation playback, or comparison. Do not call it an Atmos master:

```
verbx render source.wav wet_714_print.wav \
  --engine algo \
  --output-layout 7.1.4 \
  --rt60 3.6 \
  --algo-decorrelation-front 0.20 \
  --algo-decorrelation-rear 0.52 \
  --algo-decorrelation-top 0.70 \
  --wet 1.0 --dry 0.0 \
  --json-out wet_714_print.analysis.json
```

Name the file with `_print`, `_rerender`, or another unambiguous label so nobody mistakes it for a bed/object master.

4.6.3 Prepare a featured object-return stem

A special tail can remain mono or stereo until Atmos authoring. This preserves the DAW's ability to attach object metadata:

```
verbx render vocal_throw.wav vocal_reverse_object.wav \
  --engine algo \
  --reverse \
  --rt60 6.5 \
  --predelay 90 \
  --wet 1.0 --dry 0.0 \
  --output-layout mono \
  --json-out vocal_reverse_object.analysis.json
```

Import the result, assign it to an object, then author the static position or trajectory while listening through the Renderer. A stereo effect may consume two object paths; use stereo only when its internal width is essential.

4.6.4 Process an FOA field

For a scene-based intermediate, keep ordering and normalization explicit:

```
verbx render stereo_room.wav room_foa.wav \
  --ambi-order 1 \
  --channel-order acn \
  --ambi-encode-from stereo \
  --ambi-rotate-yaw-deg 25 \
  --output-layout auto \
  --json-out room_foa.analysis.json
```

FOA is useful when rotation and portable scene decoding matter more than sharply isolated positions. To use it in Atmos, decode or spatialize it through a compatible plug-in or DAW workflow and audition the resulting bed/object assignment. Do not relabel the four channels as an Atmos bed.

4.6.5 Use a full matrix IR

A true multichannel room response can encode direction-dependent coupling. For M inputs and N outputs, a full matrix contains MN impulse responses. The file packing order must match `--ir-matrix-layout`:

```
verbx render input_51.wav room_matrix_714.wav \
  --input-layout 5.1 \
  --output-layout 7.1.4 \
  --ir concert_hall_6x12.wav \
  --ir-route-map full \
  --ir-matrix-layout output-major \
  --wet 1.0 --dry 0.0 \
  --json-out room_matrix_714.analysis.json
```

Before using a large matrix in a mix, test one input at a time. Confirm that front impulses create the intended early and late energy, rear inputs do not swap, top outputs are truly top outputs, and no LFE row is populated unintentionally.

4.7 Bed, Object, and Hybrid Reverb Strategies

The representation assigned to a return should follow its musical function. The strategies below move from a stable shared environment to a localized event, then combine both approaches in a session architecture that keeps ordinary room energy separate from exceptional gestures.

4.7.1 Bed return

A bed return is the safest default for a shared acoustic environment. Route multiple sources to one immersive reverb, preserve a coherent decay law, and use channel-dependent decorrelation to create width. This approach conserves object paths and keeps the room stable as sources move.

For orchestral or ensemble work, the bed can establish the hall while dry and early-reflection cues retain stage placement. For dialogue, a subtle bed can match production ambience or establish scene scale without attaching the entire room to a moving actor object.

4.7.2 Object return

An object return is useful when the reverb itself is a featured event. Examples include a freeze that rises above the listener, a reverse tail that approaches from the rear, a single reflection that traces a wall, or a transition wash that grows from near to far. Render the audio effect in verbx, then perform the object authoring in the Atmos environment.

Avoid making every source's complete late field a separate moving object. The result can consume paths, complicate metadata, and make the perceived room chase the dry source. In real rooms, direct sound localizes strongly while the late field becomes increasingly diffuse.

4.7.3 Hybrid return

The hybrid strategy uses a shared bed for the ordinary room and one or more objects for exceptional cues. It scales well because the bed carries statistical density while objects remain sparse and meaningful. It also translates more gracefully: if a dramatic object changes character in binaural, the foundational room remains intact.

4.7.4 Stem architecture

A practical immersive session might contain:

- `Room_ER_front`: early reflections emphasizing front and side channels.
- `Room_late_712`: diffuse 7.1.2 bed return.
- `Vocal_throw_mono`: mono special-effect stem assigned to an object.
- `Orchestra_halo_stereo`: stereo height effect assigned as a stereo object or deliberately spread into the bed.
- `LFE_effect`: separately managed, intentional LFE-only content if required.
- `Room_QC`: channel-identification impulses and the associated JSON report, excluded from the final program.

Use names that describe both acoustic function and spatial role. "BigVerb12" is less useful than "Hall-Late_712Bed" when the session is reopened six months later.

4.8 DAW and Renderer Handoff

verbx can prepare audio and evidence, but the authoring environment remains responsible for beds, objects, trajectories, binaural metadata, and the authoritative master. A dependable handoff makes that boundary explicit and gives the receiving engineer enough information to reconstruct channel meaning before making creative changes.

4.8.1 Integrated-renderer workflow

In an integrated environment such as Logic Pro, assign tracks and returns to the bed or to objects, place Atmos-aware processing in the correct part of the signal path, and monitor through the project's Dolby Atmos plug-in. Apple documents an important signal-flow distinction: processing before the Atmos plug-in operates on the bed path, while processing after it affects monitoring and channel-based bounces rather than the exported object master. Verify the current application behavior rather than assuming an ordinary surround-master insert model.

Export an ADM BWF only from the supported spatial-audio export path. Apple describes that export as containing bed audio, object audio, object pan automation, and metadata. A normal multichannel bounce is a channel-based render and serves a different purpose.

4.8.2 External-renderer workflow

With a separate Dolby Atmos Renderer, route each bed channel and object path to its assigned Renderer input. Confirm sample rate, frame rate, synchronization, input assignment, and monitoring return before mixing. Record or export the required master through the Renderer workflow, then generate re-renders and QC material from that same authoritative master.

4.8.3 Handoff package

A strong handoff includes more than audio. The following items prevent a technically valid stem from arriving without the context needed to place or verify it:

- Clearly named mono, stereo, and multichannel WAV files.
- A channel-order document for every file wider than stereo.
- Sample rate, bit depth, start time, frame rate where applicable, and exact duration.
- A verbx JSON report containing processing parameters and measured properties.
- Dry references when the receiving mixer may need to revise the effect.
- A short text note identifying bed candidates, object candidates, and speaker prints.
- Channel-identification impulses or spoken IDs for nonstandard layouts.
- The expected Atmos authoring environment and Renderer version, when contractually relevant.

Do not place a speaker print and an object-master candidate in the same folder with nearly identical names. Make semantic differences visible at a glance.

4.9 Monitoring and Translation

No single playback path proves an immersive master. A 7.1.4 room reveals image placement and envelopment; 5.1 reveals whether height-dependent balances survive; stereo reveals center buildup, phase cancellation, and overall hierarchy; binaural reveals renderer-dependent externalization, elevation, and near/far behavior.

Figure 4 frames monitoring as a loop. The same authoritative master feeds each re-render. Differences that damage the musical intent return to the source session for correction; they are not patched independently into unrelated exports.

Figure 4-4 presents Immersive translation and quality-control loop and establishes the visual reference for the discussion that follows.

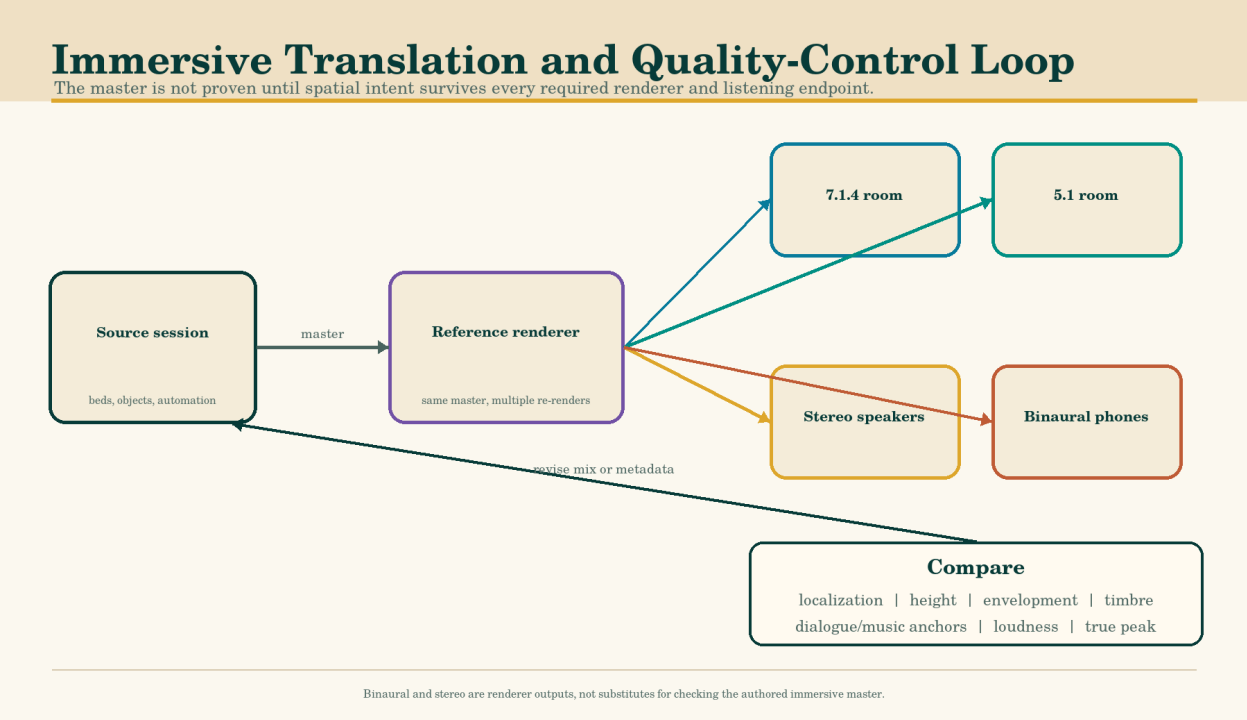


Figure 4-4: Immersive translation and quality-control loop

The authored beds, objects, and automation feed one reference Renderer. Four representative endpoints reveal different failure modes: 7.1.4 tests the intended room image, 5.1 exposes reliance on height, stereo tests hierarchy and phase interaction, and binaural tests headphone localization and metadata behavior. Observations return to the source session so the master remains the single source of truth.

4.9.1 Loudspeaker-room checks

An immersive room should be calibrated, time-aligned, and level-matched according to the applicable standard and facility procedure. Check that monitoring bass management is understood and repeatable. A subwoofer response problem can be misdiagnosed as an LFE-mix problem; a mistimed height speaker can be misdiagnosed as poor elevation metadata.

Walk the room after the reference-position pass. The mix need not be identical in every seat, but catastrophic localization jumps, comb filtering, or disappearing returns reveal fragile channel relationships.

4.9.2 Stereo checks

Listen for excessive front-center buildup, loss of ambience, hollow tonal balance, and transient smearing. A diffuse field can become quieter in stereo because decorrelated channels partially cancel or because the renderer applies conservative downmix coefficients. Correct the spatial design, not merely the stereo file, unless the delivery contract explicitly calls for a separately mastered stereo version.

4.9.3 Binaural checks

Binaural rendering depends on the Renderer's head-related transfer functions and object metadata. A height cue that is obvious over loudspeakers may become subtle over headphones; a large diffuse bed may sound internalized; a near object may become distractingly intimate. Check several good headphones and avoid making decisions from a single consumer spatializer layered on top of the reference Renderer.

When supported, per-object binaural modes such as Off, Near, Mid, and Far affect perceived distance. These modes are metadata decisions, not substitutions for acoustic predelay, direct-to-reverberant ratio, spectral damping, or early-reflection design.

4.9.4 Renderer and consumer-device differences

Soundbars, televisions, headphones, mobile devices, and theatrical systems may use different rendering strategies. The correct response is not to optimize for every device independently. Establish a defensible reference master, test the required official re-renders, and investigate only repeatable failures that compromise intent.

4.10 Ambisonics and Atmos Are Complementary

Ambisonics represents a listener-centered field; Atmos represents beds and objects for endpoint rendering. Both can describe immersion, but their coordinate systems and production assumptions differ.

Table 4-3: Ambisonics and Atmos Are Complementary: reference table

Question	Ambisonics	Dolby Atmos
What is stored?	Spherical-harmonic scene components	Bed channels, object audio, and metadata
Primary strength	Rotation and layout-independent scene decoding	Authored object placement plus broad endpoint ecosystem

Question	Ambisonics	Dolby Atmos
Spatial resolution	Increases with Ambisonic order	Depends on object/bed authoring and endpoint renderer
Typical reverb use	Portable room field or environmental ambience	Bed ambience plus optional object effects
Headphone output	Binaural decode of the scene	Renderer binaural output with object metadata behavior
Conversion	Requires decoding or compatible spatialization	Requires an authoring decision, not relabeling

An Ambisonic room recording can become an excellent Atmos ambience after deliberate decoding and routing. One option is to decode it to a 7.1.2 bed. Another is to use an Ambisonic-capable spatializer that feeds an Atmos-compatible path. A third is to separate salient directional events as objects while retaining the diffuse field in the bed. Choose based on the scene, not on the desire to maximize the number of moving icons.

4.11 Deliverables and File Types

The final file format determines what remains editable and what metadata travels with the audio.

4.11.1 Multichannel WAVE

A conventional multichannel WAVE file carries discrete PCM channels and container metadata. It can be a bed candidate, a loudspeaker print, a re-render, an installation master, or a stem. It does not inherently carry Atmos object trajectories. Channel labels and order must be documented and verified.

4.11.2 ADM BWF

An ADM BWF combines PCM audio with Audio Definition Model metadata in a Broadcast Wave container. Dolby and Apple describe it as an interchange master capable of carrying bed and object audio with the associated metadata. It is not a consumer listening file and should not be confused with a normal multichannel bounce. Dolby notes that changes to an imported ADM BWF are generally made in the originating authoring application rather than edited as if it were a native Renderer session.

4.11.3 DAMF

A Dolby Atmos Master File set is the Renderer's native master representation, commonly involving `.atmos`, `.audio`, and `.metadata` files. It supports continuing work in the Renderer-oriented ecosystem and can be preferable when an editable Renderer master is required.

4.11.4 IMF IAB

IMF Immersive Audio Bitstream packages are associated with interoperable mastering and distribution workflows. Use them only when required by the delivery specification and generated through the appropriate toolchain.

Figure 5 marks the practical boundary. verbx creates audio assets and evidence on the left; the Atmos DAW and Renderer author spatial metadata and master formats on the right.

Figure 4-5 presents verbx-to-Atmos delivery boundary and establishes the visual reference for the discussion that follows.

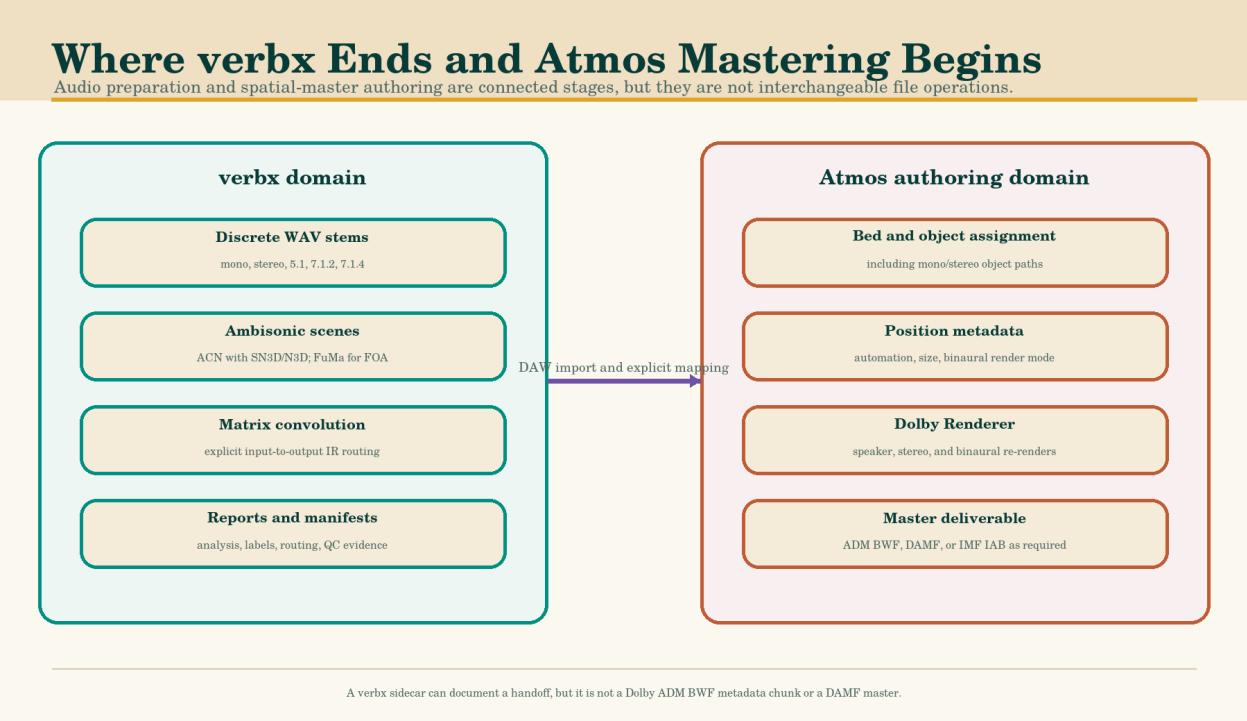


Figure 4-5: verbx-to-Atmos delivery boundary

verbx can create discrete stems, Ambisonic scenes, matrix-convolved outputs, and reports that make a handoff reproducible. The receiving Atmos session assigns beds and objects, writes trajectories and binaural metadata, renders endpoints, and produces the required ADM BWF, DAMF, or IMF IAB master. The arrow represents explicit DAW import and mapping, not an automatic file conversion.

4.12 Immersive Quality-Control Checklist

Quality control is easiest when it follows the same order as production. The three passes below separate source and routing preparation, active authoring checks, and final delivery evidence so that a late discovery can be traced to the stage that introduced it.

4.12.1 Before authoring

Before opening the Atmos session, verify the identity and technical contract of every source. This pass should resolve basic ambiguity before any renderer-specific routing can conceal it.

- Confirm every source file's sample rate, bit depth, duration, start point, and channel count.
- Read the verbx JSON report and compare it with the filename and handoff note.
- Run a channel-identification file through the exact import route.
- Verify whether a wide file is a bed candidate, a speaker print, an Ambisonic scene, or a matrix IR output.
- Confirm that no limiter, normalizer, or loudness stage was applied accidentally to a wet stem.

4.12.2 During Atmos authoring

During authoring, inspect both the authored representation and what the current Renderer produces. The checks below concentrate on bed order, object metadata, endpoint translation, and gain structure.

- Verify 7.1.2 bed order as `L, R, C, LFE, Ls, Rs, Lrs, Rrs, Ltm, Rtm` at the Renderer boundary.
- Solo every bed channel and object input at least once.
- Confirm mono versus stereo object assignments and path consumption.
- Inspect object position, size, motion, and binaural metadata for unintended automation.
- Check whether master-bus processing affects the bed, the monitor render, the exported master, or some combination.
- Monitor both full-range and LFE paths without conflating LFE with bass management.

4.12.3 Before delivery

The delivery pass preserves the authoritative master and the evidence needed to identify it later. Complete these checks only after all required endpoint reviews have returned to the source session.

- Verify or record the final master through the authoritative Renderer workflow.
- Play the master from beginning to end in the Renderer rather than checking only a DAW bounce.
- Check all contractually required re-renders, including stereo and binaural where applicable.
- Measure loudness and true peak on the deliverables required by the recipient; do not assume one target for every platform.
- Confirm sync, frame rate, program start, tail duration, and end-of-file behavior.
- Compare the delivered master's metadata and duration with the source session.
- Preserve the source session, Renderer master, verbx reports, and delivery notes together.

4.13 Creative Spatial-Reverb Recipes

These recipes are starting architectures rather than presets. Each one names a perceptual job, a representation, and an evaluation method so that changes can be judged against musical intention rather than the number of active loudspeakers.

4.13.1 Natural hall around a front ensemble

Keep direct sound and the strongest early reflections in the front stage. Build the late field primarily in side, rear, and top bed channels with moderate decorrelation. Use less Center late energy than Left/Right unless the production calls for a tightly anchored acoustic. Confirm that stereo retains warmth without turning cloudy.

4.13.2 Intimate vocal with an overhead halo

Keep the vocal itself anchored. Send a filtered, predelayed wet stem to a stereo object or to carefully controlled top energy. The halo should respond to phrasing rather than run continuously at the same level. Duck the return during consonants and let it recover into rests. In binaural, compare Near, Mid, and Far metadata choices with the acoustic predelay; they solve different distance cues.

4.13.3 Percussion room with stable transients

Use short front and side early reflections to establish room size. Feed the late field after enough predelay to preserve attacks. Increase rear/top decorrelation more than front decorrelation. Keep low-frequency decay shorter than midrange decay, and avoid LFE unless a specific impact effect requires it.

4.13.4 Reverse transition over the listener

Render a mono or stereo reverse-reverb stem in verbx. Author it as an object that moves only during the transition, then resolves into a stable bed or dry source. Check the movement over 7.1.4 and binaural; simplify the trajectory if headphone localization becomes erratic.

4.13.5 Exterior-to-interior scene change

Automate the acoustic architecture rather than merely raising wet level. Begin with sparse, low-density exterior reflections. Introduce enclosed early reflections as the threshold is crossed, then lengthen and darken the late field. Keep the shared environment in the bed and reserve objects for reflections tied to visible architectural events.

4.13.6 Infinite or frozen field

A frozen field can occupy a bed when it represents an environment, or an object when it behaves as a featured musical entity. Watch low-frequency accumulation and interchannel correlation. A stationary, diffuse freeze often feels larger than a freeze that continuously circles the listener.

4.14 Common Failure Modes

Immersive failures often sound like aesthetic disappointments while originating in routing, correlation, metadata, or monitoring. The symptoms below begin with what a mixer hears and then identify the production assumption most worth checking.

4.14.1 “The height disappears in stereo.”

Height is not a separate axis in two-channel playback. The Renderer must express it through spectral, timing, level, and binaural cues. Strengthen the compositional hierarchy and test the reference binaural render; do not simply add top channels to the stereo mix.

4.14.2 “The 7.1.4 file imported, so the Atmos master is finished.”

It is a channel-based speaker print. It does not contain editable object trajectories or the complete Atmos master metadata. Import stems into an Atmos authoring environment and create the required master there.

4.14.3 “The top-front and top-rear channels are silent in my bed.”

A standard 7.1.2 bed supplies top-middle channels, not four independently addressable top channels. The Renderer maps the bed into the monitoring layout. Use objects for independently authored top-front or top-rear content.

4.14.4 “The room sounds huge over speakers but phasey on headphones.”

Reduce antagonistic decorrelation, inspect copied channels, and compare the Renderer’s binaural output without an additional consumer spatializer. Keep the shared decay envelope coherent even when channel details differ.

4.14.5 “The subwoofer is full of reverb although LFE is empty.”

That can be normal bass management. Full-range bed channels contain low frequencies that the monitor controller redirects below its crossover. Solo LFE and inspect the monitoring configuration before changing the mix.

4.14.6 “Channels 9 and 10 sound too far forward.”

Check the handoff map. verbx’s current symbolic `7.1.2` utility layout labels those channels `Ltf/Rtf`, whereas a Dolby bed expects `Ltm/Rtm`. Map them explicitly and verify with identification signals.

4.14.7 “The ADM BWF does not reflect my last monitor-chain EQ.”

Determine where the processing sits relative to the Atmos authoring and Renderer path. Monitoring or post-render processing may affect a channel bounce without changing the exported object master. Use the DAW’s official signal-flow documentation and re-export from the authoritative session.

4.15 Recommended Official Reading

The following primary documentation is arranged alphabetically by organization:

- **Apple.** Export a spatial audio project as an ADM BWF file in Logic Pro. Explains the difference between an object-based spatial export and a conventional channel bounce.
- **Apple.** Monitor a Spatial Audio mix in Logic Pro. Describes speaker, stereo, and binaural monitoring formats.
- **Apple.** Use beds and objects in Logic Pro. Covers integrated bed/object assignment and path limits.
- **Dolby.** Dolby Atmos Renderer. Product-level description of bed/object rendering, master creation, and re-renders.
- **Dolby.** How do I QC my Dolby Atmos mix?. Emphasizes master verification in the Renderer and delivery-specific requirements.
- **Dolby.** Overview of Dolby Atmos Master File Formats. Compares ADM BWF, DAMF, and IMF IAB roles.
- **Dolby.** What are Beds and Objects in Dolby Atmos?. Defines the default 7.1.2 bed and object workflow.
- **Dolby.** What channel order should be used for assigning bed audio to the Renderer?. Gives the SMPTE 7.1.2 bed order used at the Renderer boundary.
- **International Telecommunication Union.** Recommendation ITU-R BS.2051: Advanced sound system for programme production. Defines reference loudspeaker-layout concepts for advanced sound systems.

Specifications, application behavior, and distributor requirements change. Treat these links and the recipient’s current delivery document as authoritative for a production, and treat this chapter as an engineering and creative workflow guide.

4.16 Suggested Exercises

1. Build a source-bound early field, an environment-bound late field, and one gesture-bound effect from the same dry source. Assign a distinct spatial role to each.
2. Verify a 7.1.2 or 7.1.4 handoff with spoken labels or impulses before auditioning music; record the exact channel map used by the receiving DAW.
3. Render an FOA intermediate, rotate it, decode it, and compare that result with a fixed stereo or surround print at matched loudness.
4. Prepare separate dry, early, and late stems for a WFS renderer. State which spatial decision belongs to verbx and which belongs to the calibrated array system.
5. Audit one immersive mix on its target layout, binaural render, stereo fold-down, and one reduced layout. List the hierarchy that survives each translation.
6. Build a moving source with stationary room energy and a stationary source with moving room energy; compare the two spatial narratives.
7. Create an immersive reverb plan for a one-minute scene, specifying bed channels, objects, height treatment, and fold-down priorities.
8. Measure or listen for interchannel correlation in one return and revise the design to improve envelopment without losing image stability.
9. Use a channel-identification recording to diagnose one incorrect height or surround routing problem in a DAW session.
10. Explain why LFE is normally excluded from a reverb return and identify one exceptional creative use that still preserves mix clarity.

"How sweet the answer Echo makes to music at night."

— Thomas Moore, *Echo* (1821)

Chapter 5

System Orientation Through Block Diagrams

This chapter is a visual map of the signal paths, state boundaries, and evidence loops used throughout verbx. Read each diagram from left to right, then use the paragraph beneath it to identify what may happen in realtime, what must be prepared off-thread, and what should be preserved in a report.

5.1 The complete verbx workflow

The figure below introduces **The complete verbx workflow**. Every workflow begins with an identified source and ends with both an audible artifact and machine-readable evidence. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-1 presents The complete verbx workflow and establishes the visual reference for the discussion that follows.

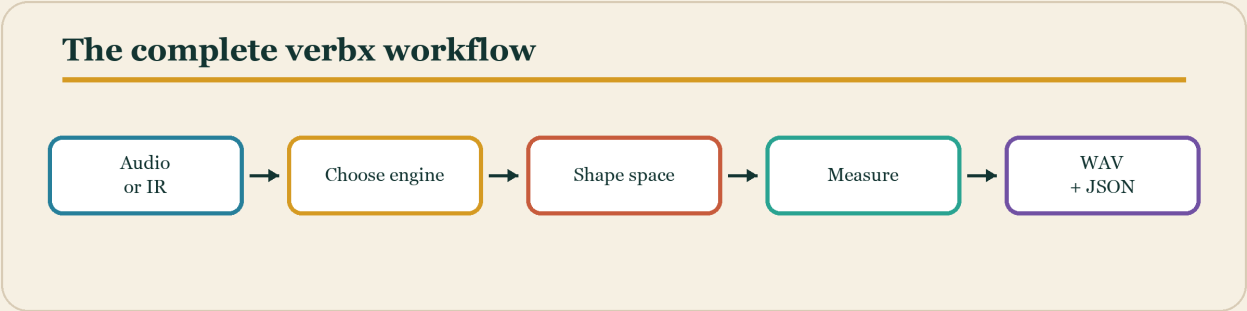


Figure 5-1: The complete verbx workflow

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.2 Algorithmic render path

The figure below introduces **Algorithmic render path**. The algorithmic engine separates arrival cues, density growth, sustained decay, and delivery processing. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-2 presents Algorithmic render path and establishes the visual reference for the discussion that follows.

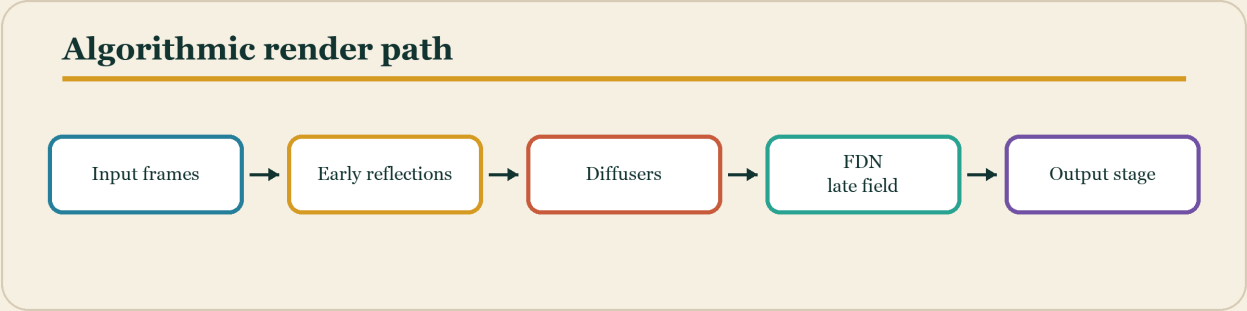


Figure 5-2: Algorithmic render path

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.3 Convolution render path

The figure below introduces **Convolution render path**. Partitioned convolution trades a small fixed scheduling delay for efficient processing of long and multichannel impulse responses. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-3 presents Convolution render path and establishes the visual reference for the discussion that follows.

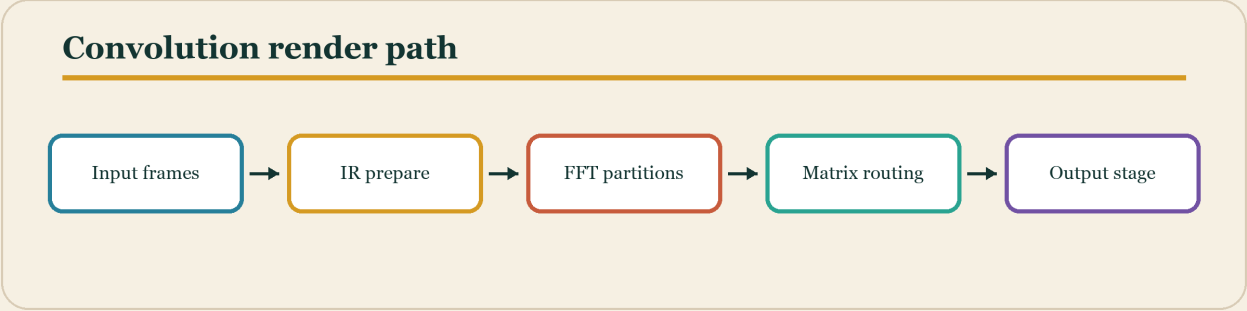


Figure 5-3: Convolution render path

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.4 Realtime callback boundary

The figure below introduces **Realtime callback boundary**. The callback path must remain bounded and allocation-free; preparation, file access, and reporting belong outside it. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-4 presents Realtime callback boundary and establishes the visual reference for the discussion that follows.

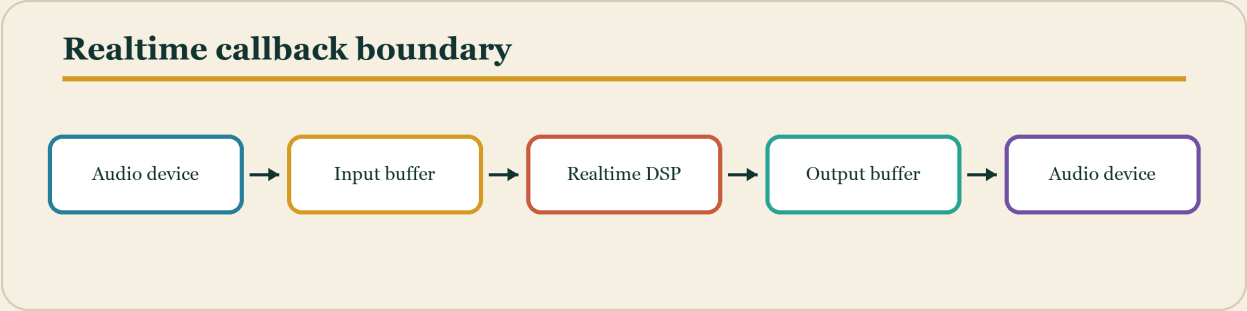


Figure 5-4: Realtime callback boundary

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.5 End-to-end latency stack

The figure below introduces **End-to-end latency stack**. Perceived realtime latency is the sum of device buffers, host scheduling, algorithmic lookahead, and physical propagation. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-5 presents End-to-end latency stack and establishes the visual reference for the discussion that follows.

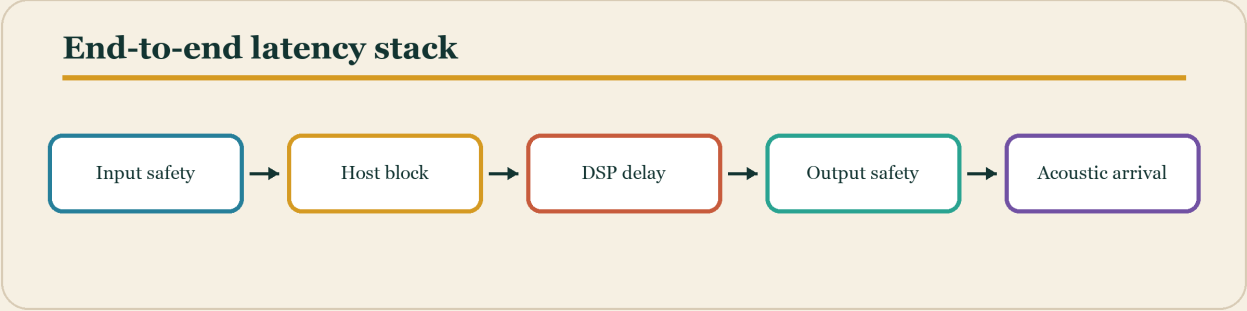


Figure 5-5: End-to-end latency stack

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.6 Reverb event anatomy

The figure below introduces **Reverb event anatomy**. A useful listening model treats reverberation as an ordered event rather than a single wet signal. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-6 presents Reverb event anatomy and establishes the visual reference for the discussion that follows.

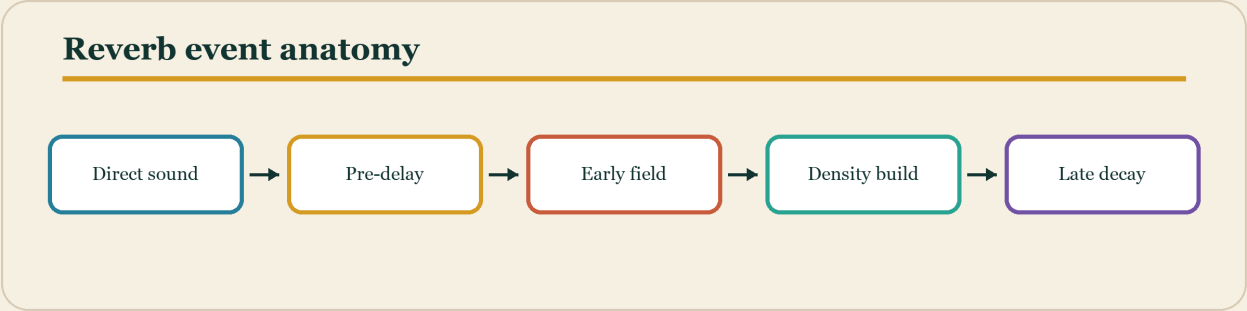


Figure 5-6: Reverb event anatomy

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.7 RT60 control hierarchy

The figure below introduces **RT60 control hierarchy**. Coarse and fine controls establish time scale while frequency-dependent decay and freeze determine how energy persists. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-7 presents RT60 control hierarchy and establishes the visual reference for the discussion that follows.

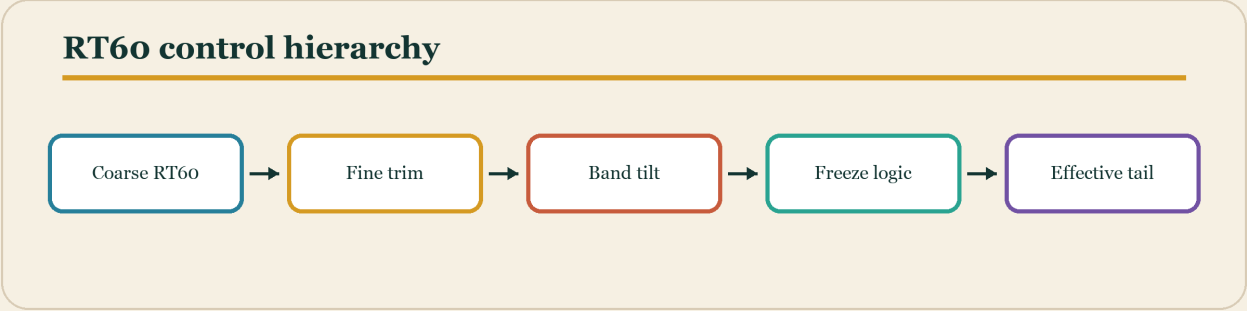


Figure 5-7: RT60 control hierarchy

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.8 FDN feedback loop

The figure below introduces **FDN feedback loop**. Delay lengths create modes; filters set color; the matrix disperses energy; calibrated gain establishes decay. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-8 presents FDN feedback loop and establishes the visual reference for the discussion that follows.

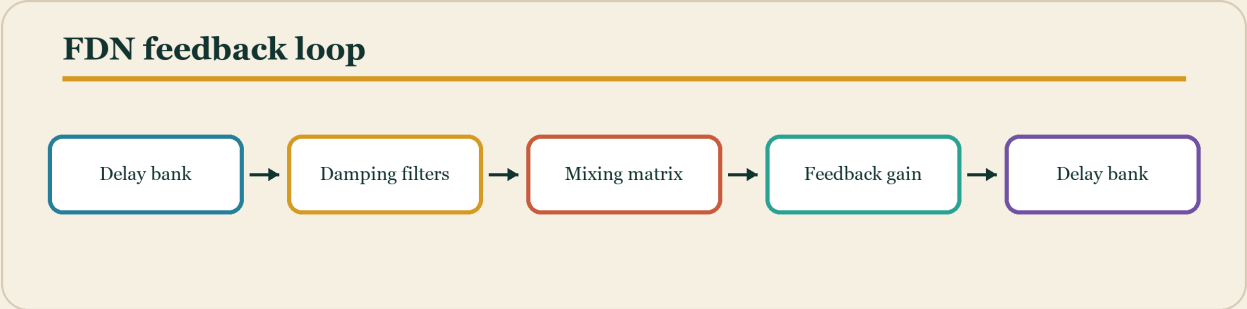


Figure 5-8: FDN feedback loop

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.9 Impulse-response lifecycle

The figure below introduces **Impulse-response lifecycle**. An impulse response is an auditable asset whose provenance and preparation affect every convolution result. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-9 presents Impulse-response lifecycle and establishes the visual reference for the discussion that follows.

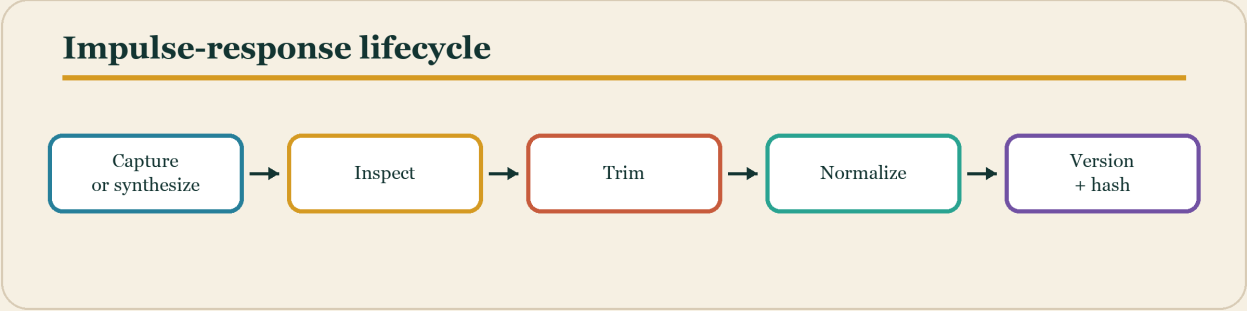


Figure 5-9: Impulse-response lifecycle

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.10 Dereverberation path

The figure below introduces **Dereverberation path**. Dereverberation balances late-energy reduction against speech, transient, and ambience preservation. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-10 presents Dereverberation path and establishes the visual reference for the discussion that follows.

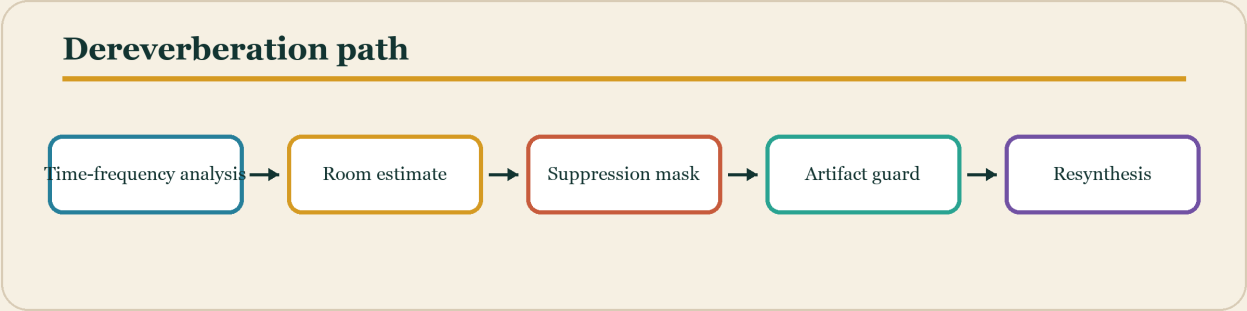


Figure 5-10: Dereverberation path

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.11 Dynamics around reverb

The figure below introduces **Dynamics around reverb**. Ducking creates foreground space before a limiter protects the combined output and meters verify the result. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-11 presents Dynamics around reverb and establishes the visual reference for the discussion that follows.

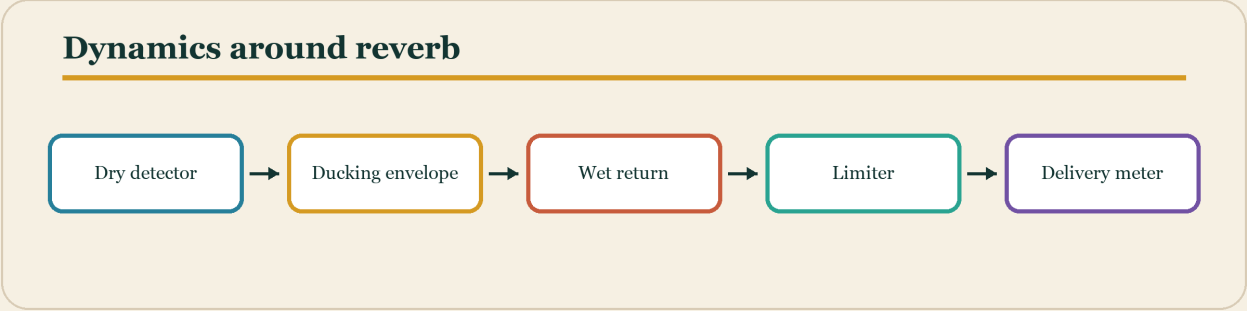


Figure 5-11: Dynamics around reverb

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.12 Spatial routing

The figure below introduces **Spatial routing**. Channel identity must remain explicit from source through room processing and final reproduction. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-12 presents Spatial routing and establishes the visual reference for the discussion that follows.

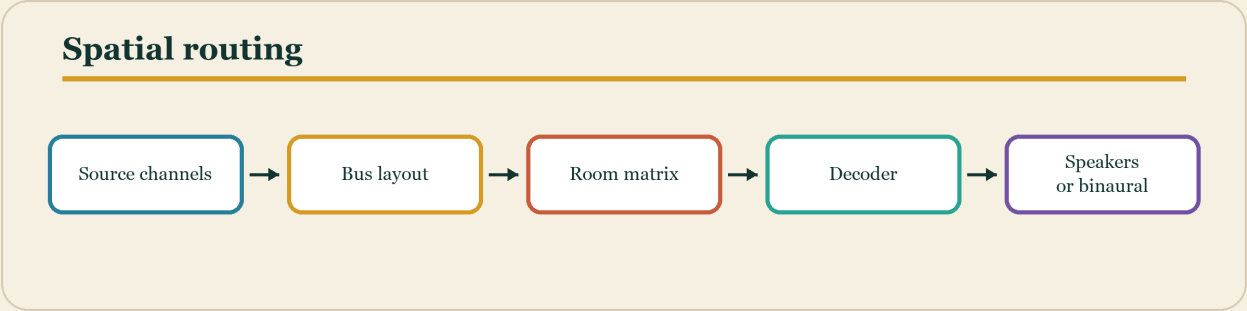


Figure 5-12: Spatial routing

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.13 Automation contract

The figure below introduces **Automation contract**. Automation is safest when parsed and validated off-thread, then applied deterministically at block boundaries. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-13 presents Automation contract and establishes the visual reference for the discussion that follows.

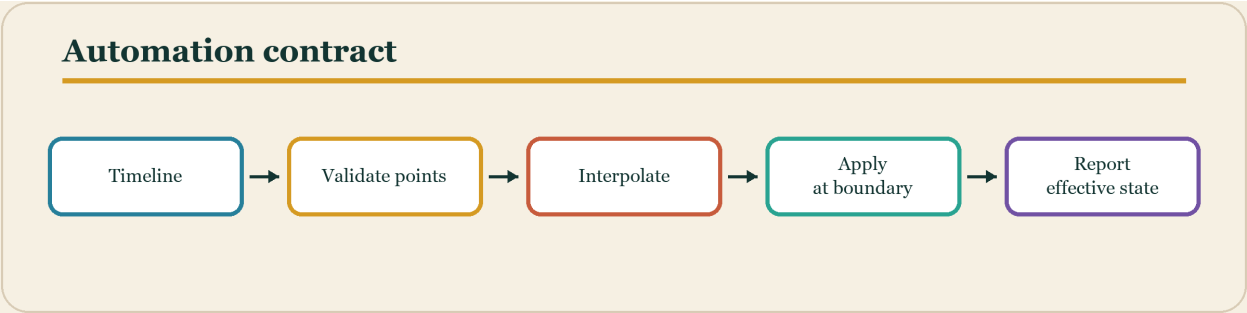


Figure 5-13: Automation contract

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.14 Analysis evidence chain

The figure below introduces **Analysis evidence chain**. Analysis becomes operational evidence when measurements, policy thresholds, warnings, and provenance share one report. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-14 presents Analysis evidence chain and establishes the visual reference for the discussion that follows.

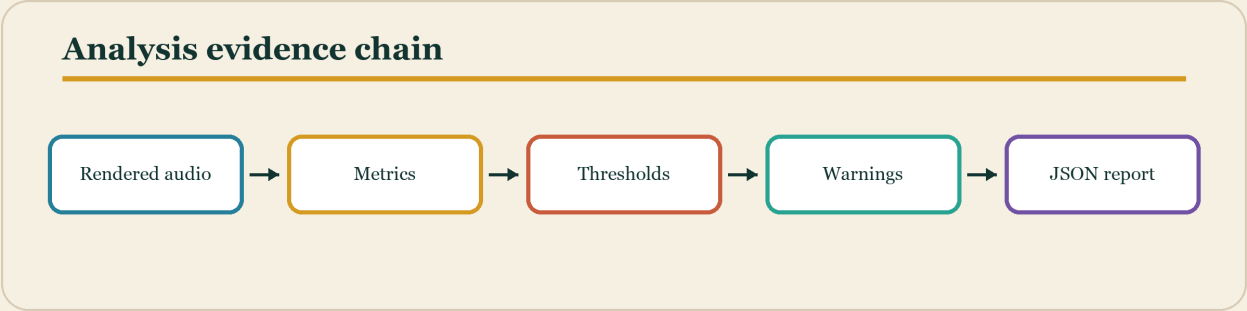


Figure 5-14: Analysis evidence chain

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.15 Preset lifecycle

The figure below introduces **Preset lifecycle**. A preset is more than values: it needs intent, compatibility metadata, calibrated auditioning, and deterministic recall. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-15 presents Preset lifecycle and establishes the visual reference for the discussion that follows.

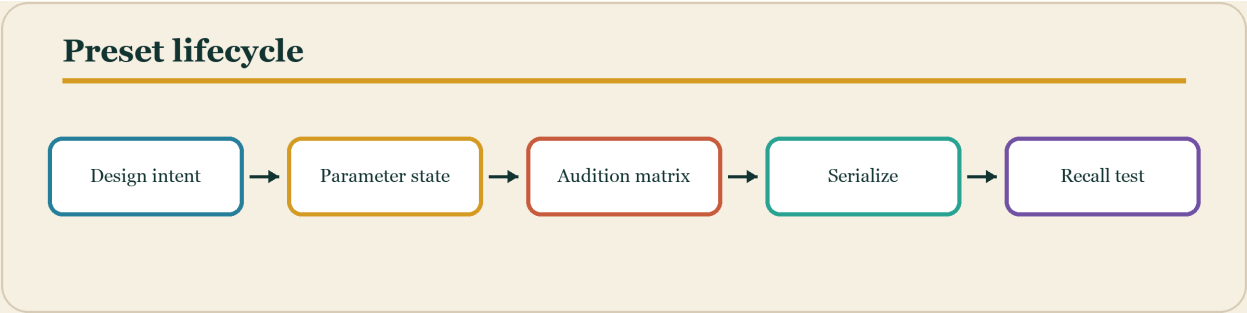


Figure 5-15: Preset lifecycle

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.16 Plug-in host contract

The figure below introduces **Plug-in host contract**. AUv3 and VST3 integration wrap the same DSP in host-specific state, layout, and lifecycle contracts. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-16 presents Plug-in host contract and establishes the visual reference for the discussion that follows.

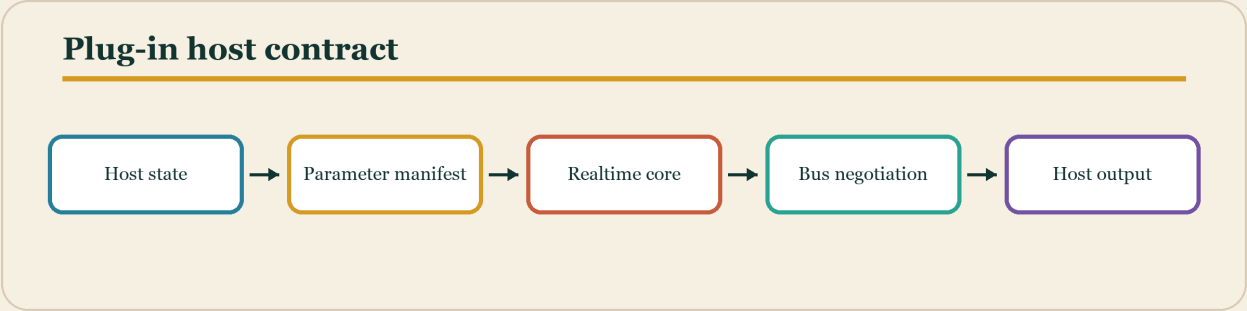


Figure 5-16: Plug-in host contract

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.17 Regression workflow

The figure below introduces **Regression workflow**. Deterministic fixtures make audible and numerical drift discoverable before release. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-17 presents Regression workflow and establishes the visual reference for the discussion that follows.

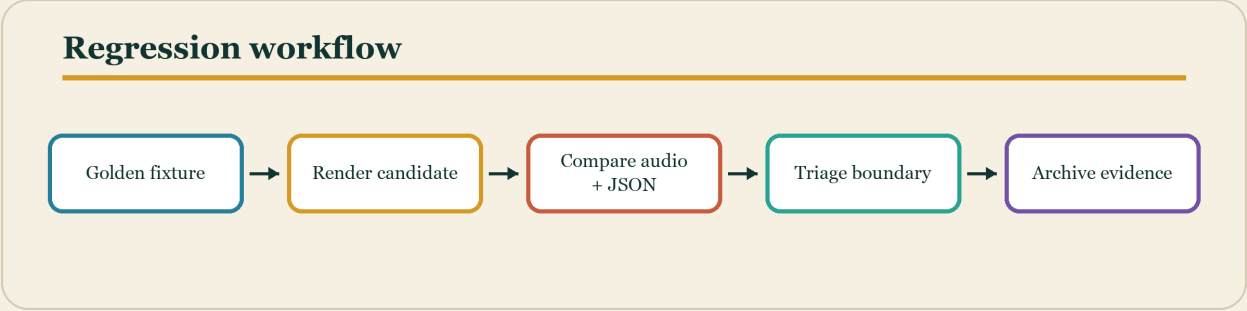


Figure 5-17: Regression workflow

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.18 Learning loop

The figure below introduces **Learning loop**. The textbook exercises use repeated prediction, experiment, measurement, and critical listening to connect controls with perception. Read it from left to right before using the detailed boundary notes beneath the visual.

Figure 5-18 presents Learning loop and establishes the visual reference for the discussion that follows.

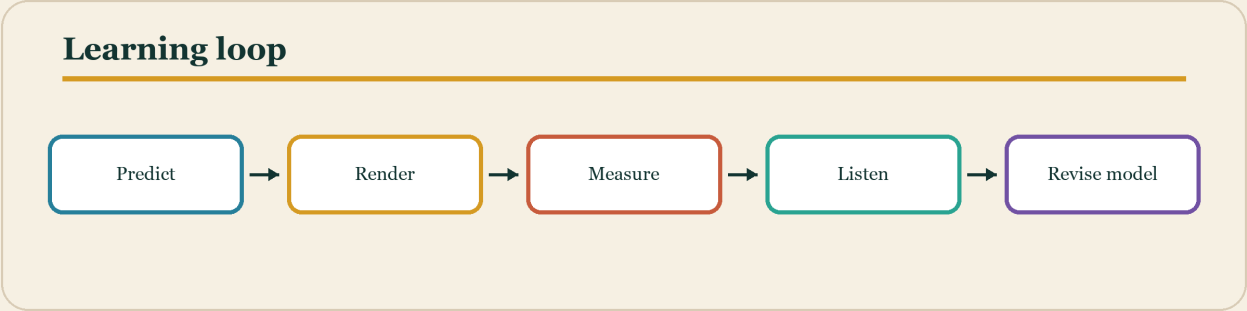


Figure 5-18: Learning loop

The arrows indicate processing or evidence flow, not elapsed-time scale. Every box names a boundary at which parameters, latency, channel identity, or provenance should be checked.

5.19 Suggested Exercises

1. Trace one CLI render from input file to output file and annotate every stage at which gain, channel count, or time alignment can change.
2. Redraw one block diagram for a mono source and one for a multichannel source, naming the data contract at each boundary.
3. Choose one failure mode from the diagrams and design a minimal diagnostic signal that isolates it.
4. Compare offline render and realtime paths for one preset. Identify the state that must be shared and the state that must remain host or device specific.
5. Explain one diagram to a collaborator using only source, early field, late field, safety, and output as your vocabulary.
6. Turn one block diagram into pseudocode, explicitly naming buffers, persistent state, control-rate updates, and output accumulation.
7. Add meters to a diagram at the three most diagnostic points, then justify what each meter can reveal that the others cannot.
8. Identify one feedback loop and explain its stability condition in both signal-flow and musical terms.
9. Make a timing diagram showing audio-block processing, parameter automation, and report generation for one realtime configuration.
10. Review one diagram for an ambiguous arrow, label, or unit; redraw it so a new reader cannot mistake control flow for audio flow.

“And more than echoes talk along the walls.”

— Alexander Pope, *Eloisa to Abelard* (1717)

CLI Reference

This file is generated from live CLI help output via `scripts/generate_cli_reference.py`. Do not edit manually.

6.1 `verbx --help`

```
Usage: root [OPTIONS] COMMAND [ARGS]...

Extreme reverb CLI with scalable DSP architecture.

╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
╭─ Commands ─────────────────────────────────────────────────────────────────╮
│ realtime    Run realtime duplex monitoring with selectable input/output    │
│             devices.                                                       │
│ room-model  Inspect a room geometry or infer one from RT60 and absorption. │
│ version     Print CLI/package version.                                     │
│ quickstart  Print minimal copy/paste commands for first successful renders.│
│ doctor      Print runtime diagnostics for launch-day troubleshooting.      │
│ analyze     Analyze an audio file and print a summary table.               │
│ compare     Side-by-side comparison of two audio files.                    │
│ presets     Print available presets or one preset payload.                 │
│ suggest     Suggest practical render defaults from input analysis.         │
│ render                                                                     │
│ dereverb                                                                   │
│ ir          Impulse response workflows.                                    │
│ cache       IR cache inspection and cleanup.                               │
│ batch       Batch manifest generation and rendering.                       │
│ immersive   Immersive production interoperability workflows.               │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.2 `verbx render --help`

```
Usage: root render [OPTIONS] {infile} {outfile}

╭─ Arguments ────────────────────────────────────────────────────────────────╮
│ *    infile       <path>  [required]                                       │
│ *    outfile      <path>  [required]                                       │
╰────────────────────────────────────────────────────────────────────────────╯
╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --preset                                 <str>            Named preset     │
│                                                           baseline (see    │
│                                                           `verbx presets`) │
│                                                           or dynamic room  │
│                                                           shorthand        │
```

```
|                                                  `room:<width>x<d… |
|                                                  Explicitly       |
|                                                  supplied CLI     |
|                                                  options override  |
|                                                  preset values.   |
| --auto-fit                    <none|speech|mus   Apply            |
|                               ic|drums|ambient   target-oriented  |
|                               >                  heuristic        |
|                                                  profile: none,   |
|                                                  speech, music,   |
|                                                  drums, ambient.  |
|                                                  [default: none]  |
| --engine                      <conv|algo|ism-f   Engine: conv,    |
|                               dn|auto>           algo, ism-fdn, or |
|                                                  auto.            |
|                                                  [default: auto]  |
| --algo-model                  <str>              Algorithmic      |
|                                                  topology: fdn,   |
|                                                  spring, or plate  |
|                                                  (algorithmic     |
|                                                  render path      |
|                                                  only).           |
|                                                  [default: fdn]   |
| --spring-count                <int range>        Number of spring  |
|                               [1<=x<=8]          elements used by  |
|                                                  --algo-model     |
|                                                  spring.          |
|                                                  [default: 1]     |
| --spring                      <str>              Repeatable spring |
|                                                  specification:   |
|                                                  length_m=0.45,ma… |
| --electromechani…             <str>              Spring/plate     |
|                                                  solver: proxy    |
|                                                  (fast FDN voice)  |
|                                                  or modal-fe      |
|                                                  (offline         |
|                                                  structural       |
|                                                  modes).          |
|                                                  [default: proxy]  |
| --spring-fe-nodes             <int range>        [default: 24]     |
|                               [4<=x<=128]                          |
| --spring-fe-modes             <int range>        [default: 24]     |
|                               [1<=x<=128]                          |
| --spring-fe-coup…             <float range>      [default: 0.08]   |
|                               [0.0<=x<=1.0]                        |
| --spring-fe-loss              <float range>      [default: 0.3]    |
|                               [0.0<=x<=2.0]                        |
| --plate-width-m               <float range>      [default: 1.8]    |
|                               [x>=0.1]                             |
| --plate-height-m              <float range>      [default: 1.2]    |
|                               [x>=0.1]                             |
| --plate-thicknes…             <float range>      [default: 0.6]    |
|                               [x>=0.01]                            |
| --plate-density-…             <float range>      [default: 7850.0] |
|                               [x>=1.0]                             |
| --plate-youngs-g…             <float range>      [default: 200.0]  |
|                               [x>=0.1]                             |
| --plate-poisson-…             <float range>      [default: 0.29]   |
|                               [0.0<=x<=0.49]                       |
```

```
| --plate-tension-n                  <float range>     [default: 0.0]    |
|                                    [x>=0.0]                            |
| --plate-pickup-x                   <float range>     [default: 0.72]   |
|                                    [0.0<=x<=1.0]                       |
| --plate-pickup-y                   <float range>     [default: 0.38]   |
|                                    [0.0<=x<=1.0]                       |
| --plate-fe-nx                      <int range>       [default: 12]     |
|                                    [4<=x<=32]                          |
| --plate-fe-ny                      <int range>       [default: 8]      |
|                                    [4<=x<=32]                          |
| --plate-fe-modes                   <int range>       [default: 32]     |
|                                    [1<=x<=128]                         |
| --plate-fe-loss                    <float range>     [default: 0.24]   |
|                                    [0.0<=x<=2.0]                       |
| --rt60                             <float range>     [default: 60.0]   |
|                                    [0.1<=x<=3600.0]                    |
| --wet                              <float range>     [default: 0.8]    |
|                                    [0.0<=x<=1.0]                       |
| --dry                              <float range>     [default: 0.2]    |
|                                    [0.0<=x<=1.0]                       |
| --repeat                           <int range>       [default: 1]      |
|                                    [x>=1]                              |
| --freeze                                             Enable freeze     |
|                                                      segment mode.     |
| --start                            <float range>                       |
|                                    [x>=0.0]                            |
| --end                              <float range>                       |
|                                    [x>=0.0]                            |
| --pre-delay-ms                     <float range>     [default: 20.0]   |
|                                    [x>=0.0]                            |
| --pre-delay                        <str>                               |
| --bpm                              <float range>                       |
|                                    [x>=1.0]                            |
| --damping                          <float range>     [default: 0.45]   |
|                                    [0.0<=x<=1.0]                       |
| --width                            <float range>     [default: 1.0]    |
|                                    [0.0<=x<=2.0]                       |
| --mod-depth-ms                     <float range>     [default: 2.0]    |
|                                    [x>=0.0]                            |
| --mod-rate-hz                      <float range>     [default: 0.1]    |
|                                    [x>=0.0]                            |
| --mod-target                       <none|mix|wet|ga  Dynamic parameter |
|                                    in-db>            target: none,     |
|                                                      mix/wet, or       |
|                                                      gain-db.          |
|                                                      [default: none]   |
| --mod-source                       <str>             Repeatable        |
|                                                      modulation source |
|                                                      spec. Examples:   |
|                                                      lfo:sine:0.08:1.… |
|                                                      env:20:350,       |
|                                                      audio-env:sidech… |
|                                                      const:0.5.        |
| --mod-route                        <str>             Repeatable        |
|                                                      advanced route:   |
|                                                      <target>:<min>:<… |
|                                                      (target:          |
|                                                      mix|wet|gain-db).  |
| --mod-min                          <float>           Minimum mapped    |
```

```
|                                                    value for the    |
|                                                    modulation       |
|                                                    target.          |
|                                                    [default: 0.0]   |
| --mod-max                          <float>         Maximum mapped   |
|                                                    value for the    |
|                                                    modulation       |
|                                                    target.          |
|                                                    [default: 1.0]   |
| --mod-combine                      <sum|avg|max>   How multiple     |
|                                                    sources are      |
|                                                    combined: sum,   |
|                                                    avg, or max.     |
|                                                    [default: sum]   |
| --mod-smooth-ms                    <float range>   One-pole         |
|                                    [x>=0.0]        smoothing time   |
|                                                    for modulation   |
|                                                    control signals. |
|                                                    [default: 20.0]  |
| --allpass-stages                   <int range>     Number of        |
|                                    [0<=x<=64]      Schroeder allpass|
|                                                    diffusion stages |
|                                                    (0 disables      |
|                                                    diffusion).      |
|                                                    [default: 6]     |
| --allpass-gain                     <str>           Allpass gain. Use|
|                                                    one value (e.g.  |
|                                                    0.7) for all     |
|                                                    stages, or a     |
|                                                    comma-separated  |
|                                                    list (e.g.       |
|                                                    0.72,0.70,0.68,0…|
|                                                    for per-stage    |
|                                                    gains.           |
|                                                    [default: 0.7]   |
| --allpass-delays…                  <str>           Optional         |
|                                                    comma-separated  |
|                                                    allpass delay    |
|                                                    list in          |
|                                                    milliseconds.    |
|                                                    Example:         |
|                                                    5,7,11,17,23,29  |
| --comb-delays-ms                   <str>           Optional         |
|                                                    comma-separated  |
|                                                    FDN comb-like    |
|                                                    delay list in    |
|                                                    milliseconds.    |
|                                                    Example:         |
|                                                    31,37,41,43,47,5…|
| --comb-cloud         --no-comb-cloud               Enable an        |
|                                                    optional pre-FDN |
|                                                    cloud of         |
|                                                    decorrelated     |
|                                                    feedback comb    |
|                                                    filters.         |
|                                                    [default:        |
|                                                    no-comb-cloud]   |
| --comb-cloud-cou…                  <int range>     Number of comb   |
|                                    [1<=x<=128]     filters generated|
```

```
|                                                  for the optional  |
|                                                  comb cloud.       |
|                                                  [default: 24]     |
| --comb-cloud-fee…          <float range>         Feedback amount   |
|                            [0.0<=x<=0.95]        used by the       |
|                                                  optional comb     |
|                                                  cloud (0..0.95).  |
|                                                  [default: 0.35]   |
| --comb-cloud-mix           <float range>         Blend from        |
|                            [0.0<=x<=1.0]         diffusion output  |
|                                                  into comb-cloud   |
|                                                  color output      |
|                                                  (0..1).           |
|                                                  [default: 0.25]   |
| --comb-cloud-del…          <str>                 Optional          |
|                                                  comma-separated   |
|                                                  delay list in     |
|                                                  milliseconds for  |
|                                                  the comb cloud.   |
|                                                  Providing this    |
|                                                  auto-enables the  |
|                                                  mode.             |
| --comb-cloud-seed          <int>                 Deterministic     |
|                                                  seed used when    |
|                                                  generating the    |
|                                                  optional comb     |
|                                                  cloud.            |
|                                                  [default: 2026]   |
| --fdn-lines                <int range>           FDN line count    |
|                            [1<=x<=64]            used when         |
|                                                  --comb-delays-ms  |
|                                                  is not provided.  |
|                                                  [default: 8]      |
| --fdn-matrix               <str>                 FDN matrix        |
|                                                  topology:         |
|                                                  hadamard,         |
|                                                  householder,      |
|                                                  random_orthogona… |
|                                                  circulant,        |
|                                                  elliptic,         |
|                                                  tv_unitary,       |
|                                                  graph, or         |
|                                                  sdn_hybrid.       |
|                                                  Default resolves  |
|                                                  to hadamard.      |
|                                                  [default: auto]   |
| --fdn-tv-rate-hz           <float range>         Block-rate update |
|                            [x>=0.0]              speed for         |
|                                                  --fdn-matrix      |
|                                                  tv_unitary (Hz).  |
|                                                  [default: 0.0]    |
| --fdn-tv-depth             <float range>         Blend depth for   |
|                            [0.0<=x<=1.0]         --fdn-matrix      |
|                                                  tv_unitary        |
|                                                  (0..1).           |
|                                                  [default: 0.0]    |
| --fdn-dfm-delays…          <str>                 Optional          |
|                                                  delay-feedback-m… |
|                                                  delays in         |
```

```
|                                                          milliseconds.    |
|                                                          Provide one value |
|                                                          for broadcast or  |
|                                                          one per FDN line. |
| --fdn-sparse          --no-fdn-sparse                    Enable sparse     |
|                                                          high-order FDN    |
|                                                          pair-mixing mode. |
|                                                          [default:         |
|                                                          no-fdn-sparse]    |
| --fdn-sparse-deg…                     <int range>        Number of sparse  |
|                                       [1<=x<=16]         pair-mixing       |
|                                                          stages used when  |
|                                                          --fdn-sparse is   |
|                                                          enabled.          |
|                                                          [default: 2]      |
| --fdn-cascade         --no-fdn-cascade                   Enable            |
|                                                          nested/cascaded   |
|                                                          FDN mode (small   |
|                                                          fast network into |
|                                                          late network).    |
|                                                          [default:         |
|                                                          no-fdn-cascade]   |
| --fdn-cascade-mix                     <float range>      Injection amount  |
|                                       [0.0<=x<=1.0]      from nested FDN   |
|                                                          into the main     |
|                                                          late-field        |
|                                                          network (0..1).   |
|                                                          [default: 0.35]   |
| --fdn-cascade-de…                     <float range>      Delay scaling for |
|                                       [0.2<=x<=1.0]      nested FDN        |
|                                                          relative to       |
|                                                          primary FDN       |
|                                                          delays            |
|                                                          (0.2..1.0).       |
|                                                          [default: 0.5]    |
| --fdn-cascade-rt…                     <float range>      RT60 ratio for    |
|                                       [0.1<=x<=1.0]      nested FDN        |
|                                                          relative to       |
|                                                          --rt60            |
|                                                          (0.1..1.0).       |
|                                                          [default: 0.55]   |
| --fdn-rt60-low                        <float range>      Low-band RT60     |
|                                       [0.1<=x<=3600.0]   target for        |
|                                                          multiband FDN     |
|                                                          decay shaping     |
|                                                          (seconds).        |
| --fdn-rt60-mid                        <float range>      Mid-band RT60     |
|                                       [0.1<=x<=3600.0]   target for        |
|                                                          multiband FDN     |
|                                                          decay shaping     |
|                                                          (seconds).        |
| --fdn-rt60-high                       <float range>      High-band RT60    |
|                                       [0.1<=x<=3600.0]   target for        |
|                                                          multiband FDN     |
|                                                          decay shaping     |
|                                                          (seconds).        |
| --fdn-rt60-tilt                       <float range>      Jot-style         |
|                                       [-1.0<=x<=1.0]     low/high RT skew  |
|                                                          around mid band   |
```

```
|                                                  (-1..1). Positive |
|                                                  extends low-band  |
|                                                  decay and         |
|                                                  shortens highs.   |
|                                                  [default: 0.0]    |
| --fdn-tonal-corr…          <float range>         Track C tonal     |
|                            [0.0<=x<=1.0]         correction        |
|                                                  strength for      |
|                                                  multiband/tilted  |
|                                                  FDN response      |
|                                                  (0..1). Higher    |
|                                                  values apply      |
|                                                  stronger          |
|                                                  decay-color       |
|                                                  equalization.     |
|                                                  [default: 0.0]    |
| --fdn-xover-low-…          <float range>         Low/mid crossover |
|                            [x>=20.0]             frequency used by |
|                                                  multiband FDN     |
|                                                  decay shaping.    |
|                                                  [default: 250.0]  |
| --fdn-xover-high…          <float range>         Mid/high          |
|                            [x>=100.0]            crossover         |
|                                                  frequency used by |
|                                                  multiband FDN     |
|                                                  decay shaping.    |
|                                                  [default: 4000.0] |
| --fdn-link-filter          <str>                 Feedback-link     |
|                                                  filter mode       |
|                                                  inside the FDN    |
|                                                  matrix path:      |
|                                                  none, lowpass, or |
|                                                  highpass.         |
|                                                  [default: none]   |
| --fdn-link-filte…          <float range>         Cutoff frequency  |
|                            [x>=20.0]             used by           |
|                                                  --fdn-link-filter |
|                                                  (Hz).             |
|                                                  [default: 2500.0] |
| --fdn-link-filte…          <float range>         Wet mix of        |
|                            [0.0<=x<=1.0]         feedback-link     |
|                                                  filter processing |
|                                                  (0..1).           |
|                                                  [default: 1.0]    |
| --fdn-graph-topo…          <str>                 Graph topology    |
|                                                  for --fdn-matrix  |
|                                                  graph: ring,      |
|                                                  path, star, or    |
|                                                  random.           |
|                                                  [default: ring]   |
| --fdn-graph-degr…          <int range>           Graph             |
|                            [1<=x<=32]            neighborhood/con… |
|                                                  degree for        |
|                                                  --fdn-matrix      |
|                                                  graph.            |
|                                                  [default: 2]      |
| --fdn-graph-seed           <int>                 Deterministic     |
|                                                  seed used to      |
|                                                  build             |
```

```
|                                                  graph-structured  |
|                                                  FDN pairings.     |
|                                                  [default: 2026]   |
| --fdn-matrix-mor…          <str>                 Optional target   |
|                                                  matrix family for |
|                                                  gradual           |
|                                                  feedback-matrix   |
|                                                  morphing.         |
| --fdn-matrix-mor…          <float range>         Duration          |
|                            [x>=0.0]              (seconds) for     |
|                                                  matrix morph from |
|                                                  --fdn-matrix to   |
|                                                  --fdn-matrix-mor… |
|                                                  [default: 0.0]    |
| --fdn-spatial-co…          <none|adjacent|f      Directional       |
|                            ront_rear|bed_to      wet-bus coupling  |
|                            p|all_to_all>         mode: none,       |
|                                                  adjacent,         |
|                                                  front_rear,       |
|                                                  bed_top,          |
|                                                  all_to_all.       |
|                                                  [default: none]   |
| --fdn-spatial-co…          <float range>         Wet-bus           |
|                            [0.0<=x<=1.0]         directional       |
|                                                  coupling amount   |
|                                                  (0..1).           |
|                                                  [default: 0.0]    |
| --fdn-nonlineari…          <none|tanh|softc      Optional in-loop  |
|                            lip>                  nonlinearity:     |
|                                                  none, tanh, or    |
|                                                  softclip.         |
|                                                  [default: none]   |
| --fdn-nonlineari…          <float range>         Blend amount for  |
|                            [0.0<=x<=1.0]         in-loop           |
|                                                  nonlinearity      |
|                                                  shaping (0..1).   |
|                                                  [default: 0.0]    |
| --fdn-nonlineari…          <float range>         Drive multiplier  |
|                            [0.1<=x<=8.0]         for in-loop       |
|                                                  nonlinearity      |
|                                                  shaping.          |
|                                                  [default: 1.0]    |
| --room-size-macro          <float range>         Perceptual        |
|                            [-1.0<=x<=1.0]        room-size macro   |
|                                                  (-1..1) mapped to |
|                                                  decay-time and    |
|                                                  spacing behavior. |
|                                                  [default: 0.0]    |
| --clarity-macro            <float range>         Perceptual        |
|                            [-1.0<=x<=1.0]        clarity macro     |
|                                                  (-1..1) mapped to |
|                                                  decay, damping,   |
|                                                  and wet balance.  |
|                                                  [default: 0.0]    |
| --warmth-macro             <float range>         Perceptual warmth |
|                            [-1.0<=x<=1.0]        macro (-1..1)     |
|                                                  mapped to damping |
|                                                  and spectral      |
|                                                  decay tilt.       |
```

```
|                                                                  [default: 0.0]  |
| --envelopment-ma…          <float range>     Perceptual                        |
|                            [-1.0<=x<=1.0]    envelopment macro                 |
|                                              (-1..1) mapped to                 |
|                                              width/decorrelat…                 |
|                                              emphasis.                         |
|                                              [default: 0.0]                    |
| --beast-mode               <int range>       Scales core                       |
|                            [1<=x<=100]       reverb parameters                 |
|                                              by an intensity                   |
|                                              multiplier                        |
|                                              (1-100) to push                   |
|                                              denser, longer,                   |
|                                              freeze-like                       |
|                                              tails.                            |
|                                              [default: 1]                      |
| --ir                       <path>                                              |
| --ir-blend                 <path>            Repeatable                        |
|                                              additional IR                     |
|                                              path for                          |
|                                              render-time                       |
|                                              convolution                       |
|                                              blending.                         |
|                                              Requires                          |
|                                              convolution                       |
|                                              render path.                      |
| --ir-blend-mix             <float range>     Repeatable blend                  |
|                            [0.0<=x<=1.0]     coefficient for                   |
|                                              each --ir-blend                   |
|                                              IR (0..1).                        |
|                                              Provide one value                 |
|                                              to broadcast to                   |
|                                              all blend IRs.                    |
| --ir-blend-mode            <str>             IR blend morph                    |
|                                              mode: linear,                     |
|                                              equal-power,                      |
|                                              spectral, or                      |
|                                              envelope-aware.                   |
|                                              [default:                         |
|                                              equal-power]                      |
| --ir-blend-early…          <float range>     Early/late split                  |
|                            [x>=0.0]          time (ms) used by                 |
|                                              envelope-aware                    |
|                                              and split                         |
|                                              blending modes.                   |
|                                              [default: 80.0]                   |
| --ir-blend-early…          <float range>     Optional override                 |
|                            [0.0<=x<=1.0]     alpha for                         |
|                                              early-reflection                  |
|                                              blend region.                     |
| --ir-blend-late-…          <float range>     Optional override                 |
|                            [0.0<=x<=1.0]     alpha for                         |
|                                              late-tail blend                   |
|                                              region.                           |
| --ir-blend-align…  --no-ir-blend-a…          Enable RT60                       |
|                                              alignment before                  |
|                                              morphing to                       |
|                                              stabilize blend                   |
|                                              trajectories.                     |
```

```
|                                                  |                                  | [default:                                  |
|                                                  |                                  | ir-blend-align-d…                          |
| --ir-blend-phase…                                | <float range>                    | Phase-coherence                            |
|                                                  | [0.0<=x<=1.0]                    | safeguard                                  |
|                                                  |                                  | strength for                               |
|                                                  |                                  | spectral/envelop…                          |
|                                                  |                                  | blending.                                  |
|                                                  |                                  | [default: 0.75]                            |
| --ir-blend-spect…                                | <int range>                      | Frequency                                  |
|                                                  | [0<=x<=128]                      | smoothing radius                           |
|                                                  |                                  | (FFT bins) used                            |
|                                                  |                                  | by spectral blend                          |
|                                                  |                                  | modes.                                     |
|                                                  |                                  | [default: 3]                               |
| --ir-blend-misma…                                | <coerce|strict>                  | Mismatch behavior                          |
|                                                  |                                  | for blend-source                           |
|                                                  |                                  | sample-rate/chan…                          |
|                                                  |                                  | differences:                               |
|                                                  |                                  | coerce                                     |
|                                                  |                                  | (resample/align)                           |
|                                                  |                                  | or strict (fail).                          |
|                                                  |                                  | [default: coerce]                          |
| --ir-blend-cache…                                | <str>                            | Cache directory                            |
|                                                  |                                  | for                                        |
|                                                  |                                  | blended/morphed                            |
|                                                  |                                  | IR artifacts used                          |
|                                                  |                                  | by render                                  |
|                                                  |                                  | workflow.                                  |
|                                                  |                                  | [default:                                  |
|                                                  |                                  | .verbx_cache/ir_…                          |
| --self-convolve                                  |                                  | Use INFILE as its                          |
|                                                  |                                  | own IR and force                           |
|                                                  |                                  | fast partitioned                           |
|                                                  |                                  | convolution                                |
|                                                  |                                  | (equivalent to                             |
|                                                  |                                  | --engine conv                              |
|                                                  |                                  | --ir INFILE).                              |
| --ir-route-map                                   | <str>                            | Convolution                                |
|                                                  |                                  | route-map mode:                            |
|                                                  |                                  | auto, diagonal,                            |
|                                                  |                                  | broadcast, or                              |
|                                                  |                                  | full.                                      |
|                                                  |                                  | [default: auto]                            |
| --input-layout                                   | <auto|mono|stere                 | Input signal                               |
|                                                  | o|LCR|5.1|7.1|7.                 | channel layout:                            |
|                                                  | 1.2|7.1.4|7.2.4|                 | auto, mono,                                |
|                                                  | 8.0|16.0|64.4>                 | stereo, LCR, 5.1,                          |
|                                                  |                                  | 7.1, 7.1.2,                                |
|                                                  |                                  | 7.1.4, 7.2.4,                              |
|                                                  |                                  | 8.0, 16.0, 64.4                            |
|                                                  |                                  | [default: auto]                            |
| --output-layout                                  | <auto|mono|stere                 | Output signal                              |
|                                                  | o|LCR|5.1|7.1|7.                 | channel layout:                            |
|                                                  | 1.2|7.1.4|7.2.4|                 | auto, mono,                                |
|                                                  | 8.0|16.0|64.4>                 | stereo, LCR, 5.1,                          |
|                                                  |                                  | 7.1, 7.1.2,                                |
|                                                  |                                  | 7.1.4, 7.2.4,                              |
|                                                  |                                  | 8.0, 16.0, 64.4                            |
|                                                  |                                  | [default: auto]                            |
```

--ir-normalize	<peak|rms|none>	[default: peak]
--ir-matrix-layo…	<output-major|input-major>	[default: output-major]
--conv-route-sta…	<str>	Convolution trajectory start position (index or alias, e.g. left, rear-left).
--conv-route-end	<str>	Convolution trajectory end position (index or alias).
--conv-route-cur…	<str>	Convolution trajectory curve: linear or equal-power. [default: equal-power]
--ambi-order	<int range> [0<=x<=7]	Ambisonics order (0 disables Ambisonics-speci… processing). [default: 0]
--ambi-normaliza…	<auto|sn3d|n3d|fuma>	Ambisonics normalization convention: auto, sn3d, n3d, or fuma. [default: auto]
--channel-order	<auto|acn|fuma>	Ambisonics channel order convention: auto, acn, or fuma. [default: auto]
--ambi-encode-fr…	<none|mono|stereo>	Encode input bus into FOA before render: none, mono, or stereo. [default: none]
--ambi-decode-to	<none|stereo>	Decode Ambisonics output after render: none or stereo. [default: none]
--ambi-rotate-ya…	<float>	Listener yaw rotation in degrees applied in Ambisonic domain. [default: 0.0]
--algo-front-var…	<float range> [0.0<=x<=1.0]	Algorithmic surround decorrelation variance for front channels. [default: 0.0]
--algo-rear-vari…	<float range> [0.0<=x<=1.0]	Algorithmic surround decorrelation

```
|                                                    variance for rear |
|                                                    channels.         |
|                                                    [default: 0.0]    |
| --algo-top-varia…          <float range>           Algorithmic       |
|                            [0.0<=x<=1.0]           surround          |
|                                                    decorrelation     |
|                                                    variance for top  |
|                                                    channels.         |
|                                                    [default: 0.0]    |
| --tail-limit               <float range>                             |
|                            [x>=0.0]                                  |
| --tail-stop-thre…          <float range>           Tail completion   |
|                            [-240.0<=x<=0.0]        threshold in dBFS |
|                                                    used for final    |
|                                                    zero-tail         |
|                                                    writeout.         |
|                                                    [default: -120.0] |
| --tail-stop-hold…          <float range>           Explicit          |
|                            [x>=0.0]                zero-hold         |
|                                                    duration appended |
|                                                    after tail        |
|                                                    completion.       |
|                                                    [default: 10.0]   |
| --tail-stop-metr…          <peak|rms>              Tail stop         |
|                                                    detector metric:  |
|                                                    peak or rms.      |
|                                                    [default: peak]   |
| --threads                  <int range>                               |
|                            [x>=1]                                    |
| --device                   <auto|cpu|cuda|m        Compute device    |
|                            ps>                     preference: auto, |
|                                                    cpu, cuda, or mps |
|                                                    (Apple Silicon).  |
|                                                    [default: auto]   |
| --algo-stream        --no-algo-stream              Use               |
|                                                    algorithmic-to-c… |
|                                                    proxy streaming   |
|                                                    path for long     |
|                                                    algorithmic       |
|                                                    renders.          |
|                                                    [default:         |
|                                                    no-algo-stream]   |
| --algo-proxy-ir-…          <float range>           Maximum proxy-IR  |
|                            [x>=1.0]                duration used by  |
|                                                    --algo-stream.    |
|                                                    [default: 120.0]  |
| --algo-gpu-proxy     --no-algo-gpu-p…              Route algorithmic |
|                                                    render through    |
|                                                    proxy convolution |
|                                                    path to leverage  |
|                                                    CUDA convolution. |
|                                                    [default:         |
|                                                    no-algo-gpu-prox… |
| --partition-size           <int range>             [default: 16384]  |
|                            [x>=256]                                  |
| --quality-preset           <str>                   Output-definition |
|                                                    preset: sd=44.1   |
|                                                    kHz PCM16, md=48  |
|                                                    kHz PCM24, hd=192 |
```

```
|                                                          kHz float32      |
|                                                          (default).       |
|                                                          Explicit         |
|                                                          --target-sr/--ou… |
|                                                          override.        |
|                                                          [default: hd]    |
| --target-sr                          <int range>         Optional         |
|                                      [x>=1]              output/render    |
|                                                          sample rate (Hz). |
|                                                          Input is         |
|                                                          resampled        |
|                                                          internally if    |
|                                                          needed.          |
| --ir-gen                                                                  |
| --ir-gen-mode                        <fdn|stochastic|    [default: hybrid] |
|                                      modal|hybrid>                        |
| --ir-gen-length                      <float range>       [default: 60.0]  |
|                                      [x>=0.1]                             |
| --ir-gen-seed                        <int>               [default: 0]     |
| --ir-gen-cache-d…                    <str>               [default:        |
|                                                          .verbx_cache/irs] |
| --block-size                         <int range>         [default: 4096]  |
|                                      [x>=256]                             |
| --target-lufs                        <float>                              |
| --target-peak-db…                    <float>                              |
| --true-peak          --sample-peak                       [default:        |
|                                                          true-peak]       |
| --limiter            --no-limiter                        [default:        |
|                                                          limiter]         |
| --limiter-mode                       <tanh|arctan|sof    Limiter transfer |
|                                      tsign|hard>         curve: tanh,     |
|                                                          arctan, softsign, |
|                                                          or hard.         |
|                                                          [default: tanh]  |
| --limiter-detect                     <peak|rms>          Limiter detector |
|                                                          mode: peak or    |
|                                                          rms.             |
|                                                          [default: peak]  |
| --limiter-thresh…                    <float>             Limiter onset    |
|                                                          threshold in     |
|                                                          dBFS. Defaults to |
|                                                          the active peak  |
|                                                          target/ceiling.  |
| --limiter-ceilin…                    <float>             Limiter output   |
|                                                          ceiling in dBFS. |
|                                                          Defaults to the  |
|                                                          active peak      |
|                                                          target or -1     |
|                                                          dBFS.            |
| --limiter-knee-db                    <float range>       [default: 6.0]   |
|                                      [x>=0.0]                             |
| --limiter-drive                      <float range>       [default: 1.0]   |
|                                      [x>=1e-06]                           |
| --limiter-mix                        <float range>       [default: 1.0]   |
|                                      [0.0<=x<=1.0]                        |
| --limiter-attack…                    <float range>       [default: 0.5]   |
|                                      [x>=0.0]                             |
| --limiter-releas…                    <float range>       [default: 80.0]  |
|                                      [x>=0.0]                             |
```

--limiter-lookah…		<float range> [x>=0.0]	[default: 1.5]
--limiter-stereo…	--no-limiter-st…		Link channels in the limiter detector to preserve stereo image. [default: limiter-stereo-l…
--limiter-oversa…		<int range> [1<=x<=16]	[default: 2]
--limiter-pre-ga…		<float range> [-48.0<=x<=48.0]	[default: 0.0]
--limiter-post-g…		<float range> [-48.0<=x<=48.0]	[default: 0.0]
--limiter-dc-blo…	--no-limiter-dc…		Apply a gentle DC blocker before limiter detection. [default: no-limiter-dc-bl…
--normalize-stage		<none|post|per-pass>	[default: post]
--repeat-target-…		<float>	
--repeat-target-…		<float>	
--out-subtype		<auto|float32|float64|pcm16|pcm24|pcm32>	Output file subtype. Internal DSP runs in float64 regardless of container subtype; use float64/float32/… per delivery needs. [default: auto]
--output-contain…		<auto|wav|w64|rf64>	Output container mode: auto, wav, w64, or rf64. [default: auto]
--output-peak-no…		<none|input|target|full-scale>	Final peak normalization mode: none, input peak match, explicit target, or full-scale. [default: none]
--output-peak-ta…		<float>	Target dBFS used when --output-peak-no… target is selected.
--shimmer			
--shimmer-semito…		<float>	[default: 12.0]
--shimmer-mix		<float range> [0.0<=x<=1.0]	[default: 0.25]
--shimmer-feedba…		<float range> [0.0<=x<=1.25]	[default: 0.35]
--shimmer-highcut		<float range>	[default:

```
|                                          [x>=10.0]          10000.0]          |
| --shimmer-lowcut                         <float range>      [default: 300.0]  |
|                                          [x>=10.0]                            |
| --shimmer-spatial    --no-shimmer-sp…                       Enable            |
|                                                             multichannel      |
|                                                             shimmer spatial   |
|                                                             decorrelation.    |
|                                                             [default:         |
|                                                             no-shimmer-spati… |
| --shimmer-spread…                        <float range>      Per-channel       |
|                                          [x>=0.0]           shimmer detune    |
|                                                             spread in cents   |
|                                                             (multichannel).   |
|                                                             [default: 8.0]    |
| --shimmer-decorr…                        <float range>      Per-channel       |
|                                          [x>=0.0]           shimmer delay     |
|                                                             spread in         |
|                                                             milliseconds.     |
|                                                             [default: 1.5]    |
| --er-geometry        --no-er-geometry                       Enable            |
|                                                             first-order       |
|                                                             image-source      |
|                                                             early-reflection  |
|                                                             pre-stage.        |
|                                                             [default:         |
|                                                             no-er-geometry]   |
| --ism-order                              <int range>        Maximum           |
|                                          [0<=x<=6]          image-source      |
|                                                             reflection order  |
|                                                             for --engine      |
|                                                             ism-fdn or        |
|                                                             --er-geometry.    |
|                                                             [default: 1]      |
| --er-room-dims-m                         <str>              Room dimensions   |
|                                                             in meters: L,W,H  |
|                                                             [default: 10,7,3] |
| --er-source-pos-m                        <str>              Source position   |
|                                                             in meters: x,y,z  |
|                                                             [default:         |
|                                                             2,2,1.5]          |
| --er-listener-po…                        <str>              Listener position |
|                                                             in meters: x,y,z  |
|                                                             [default:         |
|                                                             5,3.5,1.5]        |
| --er-absorption                          <float range>      Wall absorption   |
|                                          [0.0<=x<=0.99]     coefficient for   |
|                                                             early-reflection  |
|                                                             stage.            |
|                                                             [default: 0.35]   |
| --er-material                            <str>              Early-reflection  |
|                                                             material preset:  |
|                                                             anechoic, dead,   |
|                                                             studio, hall,     |
|                                                             stone, or custom. |
|                                                             [default: studio] |
| --unsafe-self-os…    --safe-no-self-…                       UNSAFE: permit    |
|                                                             feedback-path     |
|                                                             gains above unity |
|                                                             in algorithmic    |
```

```
|                                                  mode for          |
|                                                  self-oscillating  |
|                                                  tails.            |
|                                                  [default:         |
|                                                  safe-no-self-osc… |
| --unsafe-loop-ga…          <float range>         UNSAFE loop-gain  |
|                            [0.01<=x<=1.25]       scale used with   |
|                                                  --unsafe-self-os… |
|                                                  Values >1.0       |
|                                                  encourage         |
|                                                  self-oscillation. |
|                                                  [default: 1.02]   |
| --duck                                                             |
| --duck-attack              <float range>         [default: 20.0]   |
|                            [x>=0.1]                                |
| --duck-release             <float range>         [default: 350.0]  |
|                            [x>=0.1]                                |
| --duck-strength            <float range>         How strongly the  |
|                            [0.0<=x<=1.0]         wet field is      |
|                                                  attenuated when   |
|                                                  the sidechain     |
|                                                  rises.            |
|                                                  [default: 0.75]   |
| --duck-floor               <float range>         Minimum wet gain  |
|                            [0.0<=x<=1.0]         held during       |
|                                                  ducking; useful   |
|                                                  for softer        |
|                                                  pumping.          |
|                                                  [default: 0.0]    |
| --bloom                    <float range>         [default: 0.0]    |
|                            [x>=0.0]                                |
| --bloom-mix                <float range>         Override bloom    |
|                            [0.0<=x<=1.0]         blend amount.     |
|                                                  Default           |
|                                                  auto-scales from  |
|                                                  --bloom.          |
| --lowcut                   <float range>                           |
|                            [x>=10.0]                               |
| --lowcut-order             <int range>           Butterworth order |
|                            [1<=x<=8]             used by the       |
|                                                  post-wet          |
|                                                  high-pass filter. |
|                                                  [default: 2]      |
| --highcut                  <float range>                           |
|                            [x>=10.0]                               |
| --highcut-order            <int range>           Butterworth order |
|                            [1<=x<=8]             used by the       |
|                                                  post-wet low-pass |
|                                                  filter.           |
|                                                  [default: 2]      |
| --tilt                     <float>               [default: 0.0]    |
| --tilt-pivot-hz            <float range>         Pivot frequency   |
|                            [x>=20.0]             used by the       |
|                                                  post-wet tilt EQ. |
|                                                  [default: 1000.0] |
| --automation-file          <path>                JSON/CSV          |
|                                                  automation lanes  |
|                                                  used for          |
|                                                  time-varying      |
```

```
|                                                    |                                  | render control.                  |
| --automation-mode                                  | <auto|sample|blo ck>             | Automation evaluation mode: auto, sample, or block. [default: auto] |
| --automation-blo…                                  | <float range> [x>=0.1]           | Control block size in milliseconds when automation mode is block. [default: 20.0] |
| --automation-smo…                                  | <float range> [x>=0.0]           | Default smoothing time (ms) applied to automation lanes. [default: 20.0] |
| --automation-sle…                                  | <float range> [x>=0.0]           | Optional max control slew as target-range fraction per second; 0/None disables slew guard. |
| --automation-dea…                                  | <float range> [0.0<=x<=1.0]      | Optional control deadband as target-range fraction; small changes below threshold are suppressed. [default: 0.0] |
| --automation-cla…                                  | <str>                            | Clamp override in target:min:max format (repeatable). |
| --automation-poi…                                  | <str>                            | Inline automation control point in target:time_s:va… format (repeatable). |
| --automation-tra…                                  | <str>                            | Optional CSV path for resolved sample-level automation curves. |
| --feature-vector…                                  | <str>                            | Feature-vector mapping lane (repeatable). Format: target=<target>,… |
| --feature-vector…                                  | <float range> [x>=1.0]           | Frame size used for feature-vector extraction (ms). [default: 40.0] |
| --feature-vector…                                  | <float range> [x>=1.0]           | Hop size used for feature-vector extraction (ms). |
```

```
|                                                    [default: 20.0]   |
| --feature-guide                  <path>            Optional external |
|                                                    guide audio used  |
|                                                    for               |
|                                                    feature-vector    |
|                                                    extraction        |
|                                                    instead of INFILE |
|                                                    (Track B external |
|                                                    feature-guide     |
|                                                    ingest).          |
| --feature-guide-…                <align|strict>    Mismatch policy   |
|                                                    for               |
|                                                    --feature-guide   |
|                                                    relative to       |
|                                                    render context:   |
|                                                    align             |
|                                                    (deterministic    |
|                                                    resample +        |
|                                                    hold/trim +       |
|                                                    mixdown) or       |
|                                                    strict.           |
|                                                    [default: align]  |
| --feature-vector…                <str>             Optional CSV path |
|                                                    for               |
|                                                    feature+parameter |
|                                                    trace exports.    |
| --frames-out                     <str>                               |
| --analysis-out                   <str>                               |
| --lucky                          <int range>       Generate N wild   |
|                                  [1<=x<=500]       random renders    |
|                                                    from one input    |
|                                                    using randomized  |
|                                                    parameters.       |
|                                                    Outputs are       |
|                                                    written to        |
|                                                    --lucky-out-dir   |
|                                                    (or OUTFILE       |
|                                                    parent by         |
|                                                    default).         |
| --lucky-out-dir                  <path>            Output directory  |
|                                                    used when --lucky |
|                                                    is enabled.       |
| --lucky-seed                     <int>             Optional          |
|                                                    deterministic     |
|                                                    seed for --lucky  |
|                                                    render            |
|                                                    generation.       |
| --quiet                                            Suppress console  |
|                                                    summary tables    |
|                                                    while still       |
|                                                    writing output    |
|                                                    and analysis      |
|                                                    artifacts.        |
| --verbosity                      <int range>       Console detail    |
|                                  [0<=x<=2]         level: 0=minimal  |
|                                                    summary,          |
|                                                    1=summary +       |
|                                                    output features   |
|                                                    (default), 2=also |
```

```
                                                    include input
                                                    feature table.
                                                    [default: 1]
 --silent                                           Disable analysis
                                                    JSON generation
                                                    and console
                                                    output.
 --dry-run                                          Validate inputs
                                                    and print
                                                    resolved render
                                                    plan without
                                                    writing audio.
 --repro-bundle                                     Write a
                                                    reproducibility/…
                                                    JSON bundle next
                                                    to OUTFILE.
 --repro-bundle-o…                  <path>          Optional explicit
                                                    path for
                                                    reproducibility/…
                                                    JSON bundle.
 --failure-report…                  <path>          Optional JSON
                                                    report path
                                                    populated when
                                                    render execution
                                                    fails.
 --progress          --no-progress                  [default:
                                                    progress]
 --json-out                         <path>          Optional path to
                                                    write the full
                                                    render report as
                                                    JSON.
 --help                                             Show this message
                                                    and exit.
```

6.3 `verbx realtime --help`

```
Usage: root realtime [OPTIONS]

Run realtime duplex monitoring with selectable input/output devices.

─ Options ──────────────────────────────────────────────────────────────
 --live-mode                        <reverb|dereverb  Real-time
                                    |dereverb-reverb  processing mode:
                                    >                 reverb only,
                                                      dereverb only, or
                                                      dereverb feeding
                                                      the live reverb
                                                      path.
                                                      [default: reverb]
 --engine                           <auto|conv|algo>  Realtime engine:
                                                      convolution IR,
                                                      or algorithmic
                                                      proxy rendered
                                                      into a live
                                                      convolver.
                                                      [default: auto]
```

```
| --ir                                <path>              Impulse response  |
|                                                         path for realtime |
|                                                         convolution mode. |
| --input-device                      <str>               Input device      |
|                                                         index or          |
|                                                         case-insensitive  |
|                                                         name substring.   |
| --output-device                     <str>               Output device     |
|                                                         index or          |
|                                                         case-insensitive  |
|                                                         name substring.   |
| --list-devices                                          List available    |
|                                                         realtime audio    |
|                                                         devices and exit. |
| --sample-rate                       <int range>         [default: 48000]  |
|                                     [x>=8000]           |
| --block-size                        <int range>         [default: 512]    |
|                                     [x>=64]             |
| --partition-size                    <int range>         [default: 2048]   |
|                                     [x>=256]            |
| --input-channels                    <int range>         Requested live    |
|                                     [x>=1]              input channel     |
|                                                         count. Defaults   |
|                                                         to mono or stereo |
|                                                         depending on      |
|                                                         device.           |
| --input-channel-…                   <str>               Comma-separated   |
|                                                         1-based hardware  |
|                                                         input channels to |
|                                                         feed the          |
|                                                         processor, for    |
|                                                         example 1,3 or    |
|                                                         1,3,5,7.          |
| --output-channels                   <int range>         Requested live    |
|                                     [x>=1]              output channel    |
|                                                         count. Defaults   |
|                                                         to the            |
|                                                         processor's       |
|                                                         natural output    |
|                                                         width.            |
| --output-channel…                   <str>               Comma-separated   |
|                                                         1-based hardware  |
|                                                         output channels   |
|                                                         that receive      |
|                                                         processor         |
|                                                         outputs, in       |
|                                                         order.            |
| --duration                          <float range>       Optional duration |
|                                     [x>=0.0]            in seconds. Omit  |
|                                                         to run until      |
|                                                         Ctrl-C.           |
| --dereverb-mode                     <wiener|spectral    Low-latency       |
|                                     _sub>               dereverb kernel   |
|                                                         used by           |
|                                                         --live-mode       |
|                                                         dereverb*.        |
|                                                         [default: wiener] |
| --dereverb-stren…                   <float range>       [default: 0.7]    |
|                                     [0.0<=x<=2.0]       |
```

--dereverb-floor	<float range> [1e-06<=x<=1.0]	[default: 0.08]	
--dereverb-windo…	<float range> [x>=2.0]	[default: 16.0]	
--dereverb-hop-ms	<float range> [x>=1.0]	[default: 8.0]	
--dereverb-tail-…	<float range> [x>=10.0]	[default: 120.0]	
--dereverb-pre-e…	<float range> [0.0<=x<=0.98]	[default: 0.0]	
--dereverb-mix	<float range> [0.0<=x<=1.0]	[default: 1.0]	
--dereverb-max-a…	<float range> [0.0<=x<=48.0]	[default: 18.0]	
--dereverb-stere…	--no-dereverb-s…		Link stereo gain decisions to reduce image wobble. [default: dereverb-stereo-…
--dereverb-input…	<float range> [-24.0<=x<=24.0]	[default: 0.0]	
--dereverb-outpu…	<float range> [-24.0<=x<=24.0]	[default: 0.0]	
--dereverb-windo…	<str>		Live dereverb analysis window family (hann, hamming, blackman, kaiser, dpss, tukey, chebwin, and many more). [default: hann]
--dereverb-synth…	<str>		Optional live dereverb synthesis window family. Defaults to the analysis window.
--dereverb-windo…	--dereverb-wind…		Use symmetric instead of periodic live dereverb windows. [default: dereverb-window-…
--dereverb-windo…	<float range> [x>=0.0]	[default: 0.5]	
--dereverb-windo…	<float range> [x>=0.0]	[default: 14.0]	
--dereverb-windo…	<float range> [x>=1e-06]	[default: 2.5]	
--dereverb-windo…	<float range> [x>=1e-06]	[default: 1.5]	
--dereverb-windo…	<float range> [x>=0.001]	[default: 100.0]	
--dereverb-windo…	<int range> [x>=2]	[default: 4]	
--dereverb-windo…	<float range> [x>=0.001]	[default: 2.5]	

```
| --dereverb-windo…  <float range>  [default: 3.0]  |
|                    [x>=1e-06]                     |
| --dereverb-windo…  <str>          Optional       |
|                                   comma-separated |
|                                   weights for     |
|                                   general_cosine  |
|                                   live dereverb   |
|                                   windows.        |
| --wet              <float range>  [default: 0.8]  |
|                    [0.0<=x<=1.0]                  |
| --dry              <float range>  [default: 0.2]  |
|                    [0.0<=x<=1.0]                  |
| --rt60             <float range>  [default: 6.0]  |
|                    [x>=0.1]                       |
| --pre-delay-ms     <float range>  [default: 20.0] |
|                    [x>=0.0]                       |
| --damping          <float range>  [default: 0.45] |
|                    [0.0<=x<=1.0]                  |
| --width            <float range>  [default: 1.0]  |
|                    [0.0<=x<=2.0]                  |
| --mod-depth-ms     <float range>  [default: 2.0]  |
|                    [x>=0.0]                       |
| --mod-rate-hz      <float range>  [default: 0.1]  |
|                    [x>=0.0]                       |
| --fdn-lines        <int range>    [default: 8]    |
|                    [1<=x<=64]                     |
| --fdn-matrix       <str>          Algorithmic proxy |
|                                   matrix for      |
|                                   realtime --engine |
|                                   algo.           |
|                                   [default:       |
|                                   hadamard]       |
| --fdn-tv-rate-hz   <float range>  [default: 0.0]  |
|                    [x>=0.0]                       |
| --fdn-tv-depth     <float range>  [default: 0.0]  |
|                    [x>=0.0]                       |
| --fdn-dfm-delays…  <str>          Comma-separated |
|                                   delay-feedback  |
|                                   modulation taps |
|                                   in milliseconds. |
| --fdn-sparse                                      |
| --fdn-sparse-deg…  <int range>    [default: 2]    |
|                    [1<=x<=16]                     |
| --fdn-cascade                                     |
| --fdn-cascade-mix  <float range>  [default: 0.35] |
|                    [0.0<=x<=1.0]                  |
| --fdn-cascade-de…  <float range>  [default: 0.5]  |
|                    [0.2<=x<=1.0]                  |
| --fdn-cascade-rt…  <float range>  [default: 0.55] |
|                    [0.1<=x<=1.0]                  |
| --fdn-rt60-low     <float range>                  |
|                    [x>=0.1]                       |
| --fdn-rt60-mid     <float range>                  |
|                    [x>=0.1]                       |
| --fdn-rt60-high    <float range>                  |
|                    [x>=0.1]                       |
| --fdn-rt60-tilt    <float range>  [default: 0.0]  |
|                    [-1.0<=x<=1.0]                 |
| --fdn-tonal-corr…  <float range>  [default: 0.0]  |
```

	[0.0<=x<=1.0]		
--fdn-xover-low-…	<float range> [x>=10.0]	[default: 250.0]	
--fdn-xover-high…	<float range> [x>=10.0]	[default: 4000.0]	
--fdn-link-filter	<none|lowpass|highpass>	[default: none]	
--fdn-link-filte…	<float range> [x>=10.0]	[default: 2500.0]	
--fdn-link-filte…	<float range> [0.0<=x<=1.0]	[default: 1.0]	
--fdn-graph-topo…	<ring|path|star|random>	[default: ring]	
--fdn-graph-degr…	<int range> [1<=x<=16]	[default: 2]	
--fdn-graph-seed	<int>	[default: 2026]	
--fdn-matrix-mor…	<str>	Optional second FDN matrix family for gradual morphing.	
--fdn-matrix-mor…	<float range> [x>=0.0]	[default: 0.0]	
--fdn-spatial-co…	<none|adjacent|front_rear|bed_top|all_to_all>	[default: none]	
--fdn-spatial-co…	<float range> [0.0<=x<=1.0]	[default: 0.0]	
--fdn-nonlineari…	<none|tanh|softclip>	[default: none]	
--fdn-nonlineari…	<float range> [0.0<=x<=1.0]	[default: 0.0]	
--fdn-nonlineari…	<float range> [0.1<=x<=8.0]	[default: 1.0]	
--room-size-macro	<float range> [-1.0<=x<=1.0]	[default: 0.0]	
--clarity-macro	<float range> [-1.0<=x<=1.0]	[default: 0.0]	
--warmth-macro	<float range> [-1.0<=x<=1.0]	[default: 0.0]	
--envelopment-ma…	<float range> [-1.0<=x<=1.0]	[default: 0.0]	
--algo-decorrela…	<float range> [0.0<=x<=1.0]	[default: 0.0]	
--algo-decorrela…	<float range> [0.0<=x<=1.0]	[default: 0.0]	
--algo-decorrela…	<float range> [0.0<=x<=1.0]	[default: 0.0]	
--allpass-stages	<int range> [0<=x<=64]	[default: 6]	
--allpass-gain	<str>	Single allpass gain or comma-separated per-stage gains. [default: 0.7]	
--allpass-delays…	<str>	Comma-separated diffusion delays in milliseconds.	
--comb-delays-ms	<str>	Comma-separated FDN/comb delay	

```
                                                    taps in
                                                    milliseconds.
 --freeze                                           Realtime algo
                                                    only: approximate
                                                    a frozen-space
                                                    sustain by
                                                    forcing a long
                                                    near-infinite
                                                    proxy tail.
 --shimmer
 --shimmer-semito…          <float>                 [default: 12.0]
 --shimmer-mix              <float range>           [default: 0.25]
                            [0.0<=x<=1.0]
 --shimmer-feedba…          <float range>           [default: 0.35]
                            [0.0<=x<=1.25]
 --shimmer-highcut          <float range>
                            [x>=10.0]
 --shimmer-lowcut           <float range>
                            [x>=10.0]
 --shimmer-spatial
 --shimmer-spread…          <float range>           [default: 8.0]
                            [x>=0.0]
 --shimmer-decorr…          <float range>           [default: 1.5]
                            [x>=0.0]
 --unsafe-self-os…
 --unsafe-loop-ga…          <float range>           [default: 1.02]
                            [x>=0.001]
 --algo-proxy-ir-…          <float range>           Maximum rendered
                            [x>=0.1]                proxy IR duration
                                                    used by realtime
                                                    --engine algo.
                                                    [default: 120.0]
 --lowcut                   <float range>
                            [x>=10.0]
 --highcut                  <float range>
                            [x>=10.0]
 --tilt                     <float range>           [default: 0.0]
                            [-18.0<=x<=18.0]
 --json-out                 <path>                  Optional path for
                                                    a
                                                    machine-readable
                                                    realtime session
                                                    report JSON.
 --quiet                                            Reduce console
                                                    output.
 --help                                             Show this message
                                                    and exit.
```

6.4 `verbx room-model --help`

```
Usage: root room-model [OPTIONS]

Inspect a room geometry or infer one from RT60 and absorption.

╭─ Options ──────────────────────────────────────────────────────────────╮
 --dims-m               <str>                    Explicit room dimensions
```

```
                                                as width,depth,height in
                                                meters.
 --rt60                  <float range> [x>=0.05] Infer a rectangular room
                                                from RT60 plus
                                                absorption/material
                                                assumptions.
 --absorption            <float range>          Mean absorption
                         [0.01<=x<=0.99]        coefficient used with
                                                --rt60 inference.
 --material              <str>                  Material preset for wall
                                                absorption when
                                                --absorption is not
                                                given.
                                                [default: studio]
 --source-pos-m          <str>                  Source position as x,y,z
                                                in meters.
                                                [default: 2.0,2.0,1.5]
 --listener-pos-m        <str>                  Listener position as
                                                x,y,z in meters.
                                                [default: 5.0,3.5,1.5]
 --json-out              <path>                 Optional path to write
                                                the full room-model
                                                payload as JSON.
 --help                                         Show this message and
                                                exit.
```

6.5 `verbx analyze --help`

```
Usage: root analyze [OPTIONS] {infile}

Analyze an audio file and print a summary table.

┌─ Arguments ──────────────────────────────────────────────────────────────┐
│ *    infile      <path>  [required]                                      │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────┐
 --json-out                          <path>
 --reverb              --no-reverb                      Include
                                                        peak-aligned
                                                        RT60/EDT/T20/T30,
                                                        clarity,
                                                        definition,
                                                        center-time, DRR,
                                                        confidence, and
                                                        early-IACC metrics.
                                                        [default: reverb]
 --input-kind                        <str>              Reverb-analysis
                                                        source model: auto,
                                                        ir, or program.
                                                        [default: auto]
 --direct-window-ms                  <float range>      Direct-sound
                                     [0.1<=x<=100.0]    integration window
                                                        used for the DRR
                                                        estimate.
                                                        [default: 2.5]
 --lufs                                                 Include
```

```
│                                                    LUFS/true-peak/LRA  │
│                                                    metrics.            │
│ --edr                                              Include EDR (Energy │
│                                                    Decay Relief)       │
│                                                    summary metrics.    │
│ --frames-out                       <path>                              │
│ --ambi-order                       <int range>     Enable Ambisonics   │
│                                    [0<=x<=7]       spatial metrics for │
│                                                    the given order.    │
│                                                    [default: 0]        │
│ --ambi-normalizati…                <auto|sn3d|n3d|fu Ambisonics        │
│                                    ma>             normalization       │
│                                                    convention for      │
│                                                    analysis mode.      │
│                                                    [default: auto]     │
│ --channel-order                    <auto|acn|fuma> Ambisonics channel  │
│                                                    order convention    │
│                                                    for analysis mode.  │
│                                                    [default: auto]     │
│ --room                                             Estimate room size, │
│                                                    dimensions,         │
│                                                    absorption,         │
│                                                    critical distance,  │
│                                                    and class from the  │
│                                                    signal's            │
│                                                    reverberant decay   │
│                                                    characteristics.    │
│                                                    Works best on       │
│                                                    reverberant         │
│                                                    recordings or       │
│                                                    rendered impulse    │
│                                                    responses.          │
│ --help                                             Show this message   │
│                                                    and exit.           │
└────────────────────────────────────────────────────────────────────────┘
```

6.6 verbx suggest --help

```
Usage: root suggest [OPTIONS] {infile}

Suggest practical render defaults from input analysis.

┌─ Arguments ────────────────────────────────────────────────────────────┐
│ *    infile      <path>  [required]                                    │
└────────────────────────────────────────────────────────────────────────┘
┌─ Options ──────────────────────────────────────────────────────────────┐
│ --pin         <path>  Write suggested parameters as a JSON preset file.│
│ --help                Show this message and exit.                      │
└────────────────────────────────────────────────────────────────────────┘
```

6.7 verbx quickstart --help

```
Usage: root quickstart [OPTIONS]
```

```
Print minimal copy/paste commands for first successful renders.

┌─ Options ──────────────────────────────────────────────────────────────┐
│ --verify                      Run startup readiness checks for first-run │
│                               confidence.                              │
│ --strict                      Exit non-zero when --verify finds one or more │
│                               failed checks.                           │
│ --json-out            <path>  Optional path to write quickstart        │
│                               verification/smoke JSON.                 │
│ --smoke-test                  Run a tiny end-to-end render smoke test with │
│                               synthetic input audio.                   │
│ --smoke-out-dir       <path>  Optional output directory for smoke-test │
│                               artifacts.                               │
│ --help                        Show this message and exit.              │
└────────────────────────────────────────────────────────────────────────┘
```

6.8 verbx doctor --help

```
Usage: root doctor [OPTIONS]

Print runtime diagnostics for launch-day troubleshooting.

┌─ Options ──────────────────────────────────────────────────────────────┐
│ --json-out                <path>  Optional path to write machine-readable │
│                                   diagnostics JSON.                    │
│ --strict                          Exit non-zero when startup checks fail. │
│ --render-smoke-test               Run a tiny end-to-end render smoke test │
│                                   after diagnostics.                   │
│ --smoke-out-dir           <path>  Optional output directory for doctor │
│                                   smoke-test artifacts.                │
│ --help                            Show this message and exit.          │
└────────────────────────────────────────────────────────────────────────┘
```

6.9 verbx presets --help

```
Usage: root presets [OPTIONS]

Print available presets or one preset payload.

┌─ Options ──────────────────────────────────────────────────────────────┐
│ --show            <str>  Show resolved values for one preset.          │
│ --validate        <str>  Validate a preset's fields against RenderConfig and │
│                          report any errors.                            │
│ --help                   Show this message and exit.                   │
└────────────────────────────────────────────────────────────────────────┘
```

6.10 verbx version --help

```
Usage: root version [OPTIONS]

Print CLI/package version.
```

```
╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.11 verbx ir --help

```
Usage: root ir [OPTIONS] COMMAND [ARGS]...

Impulse response workflows.

╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
╭─ Commands ─────────────────────────────────────────────────────────────────╮
│ gen                                                                        │
│ analyze                                                                    │
│ sofa-info                                                                  │
│ sofa-extract                                                               │
│ trace                                                                      │
│ process                                                                    │
│ morph                                                                      │
│ morph-sweep                                                                │
│ fit                                                                        │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.12 verbx ir gen --help

```
Usage: root ir gen [OPTIONS] {out_ir}

╭─ Arguments ────────────────────────────────────────────────────────────────╮
│ *    out_ir      <path>  [required]                                        │
╰────────────────────────────────────────────────────────────────────────────╯
╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --format                                <auto|wav|flac|a  [default: auto]  │
│                                         iff|aif|ogg|caf>                   │
│ --mode                                  <fdn|stochastic|  [default: hybrid] │
│                                         modal|hybrid>                      │
│ --length                                <float range>     [default: 60.0]  │
│                                         [x>=0.1]                           │
│ --sr                                    <int range>       [default: 48000] │
│                                         [x>=8000]                          │
│ --channels                              <int range>       [default: 2]     │
│                                         [x>=1]                             │
│ --seed                                  <int>             [default: 0]     │
│ --rt60                                  <float range>                      │
│                                         [0.1<=x<=3600.0]                   │
│ --rt60-low                              <float range>                      │
│                                         [0.1<=x<=3600.0]                   │
│ --rt60-high                             <float range>                      │
│                                         [0.1<=x<=3600.0]                   │
│ --damping                               <float range>     [default: 0.4]   │
│                                         [0.0<=x<=1.0]                      │
│ --lowcut                                <float range>                      │
│                                         [x>=10.0]                          │
```

--highcut		<float range> [x>=10.0]	
--tilt		<float>	[default: 0.0]
--normalize		<none|peak|rms>	[default: peak]
--peak-dbfs		<float>	[default: -1.0]
--target-lufs		<float>	
--true-peak	--sample-peak		[default: true-peak]
--er-count		<int range> [x>=0]	[default: 24]
--er-max-delay-ms		<float range> [x>=1.0]	[default: 90.0]
--er-decay-shape		<str>	[default: exp]
--er-stereo-width		<float range> [0.0<=x<=2.0]	[default: 1.0]
--er-room		<float range> [x>=0.1]	[default: 1.0]
--diffusion		<float range> [0.0<=x<=1.0]	[default: 0.5]
--mod-depth-ms		<float range> [x>=0.0]	[default: 1.5]
--mod-rate-hz		<float range> [x>=0.0]	[default: 0.12]
--density		<float range> [x>=0.01]	[default: 1.0]
--tuning		<str>	[default: A4=440]
--scala-file		<path>	Scala .scl scale used to tune and emphasize synthetic IR resonances.
--scala-root-hz		<float range> [x>=1.0]	[default: 440.0]
--scala-root-deg…		<int range> [x>=0]	[default: 0]
--scala-low-hz		<float range> [x>=20.0]	
--scala-high-hz		<float range> [x>=30.0]	
--scala-strength		<float range> [0.0<=x<=1.0]	[default: 1.0]
--scala-bandwidt…		<float range> [1.0<=x<=1200.0]	[default: 25.0]
--scala-gain-db		<float range> [0.0<=x<=24.0]	[default: 4.0]
--scala-max-targ…		<int range> [1<=x<=512]	[default: 128]
--modal-count		<int range> [x>=1]	[default: 48]
--modal-q-min		<float range> [x>=0.5]	[default: 5.0]
--modal-q-max		<float range> [x>=0.5]	[default: 60.0]
--modal-spread-c…		<float range> [x>=0.0]	[default: 5.0]
--modal-low-hz		<float range> [x>=20.0]	[default: 80.0]
--modal-high-hz		<float range> [x>=50.0]	[default: 12000.0]

```
| --fdn-lines                          <int range>       [default: 8]      |
|                                      [x>=1]                              |
| --fdn-matrix                         <str>             FDN matrix        |
|                                                        topology:         |
|                                                        hadamard,         |
|                                                        householder,      |
|                                                        random_orthogona… |
|                                                        circulant,        |
|                                                        elliptic,         |
|                                                        tv_unitary,       |
|                                                        graph, or         |
|                                                        sdn_hybrid.       |
|                                                        [default:         |
|                                                        hadamard]         |
| --fdn-tv-rate-hz                     <float range>     Block-rate update |
|                                      [x>=0.0]          speed for         |
|                                                        --fdn-matrix      |
|                                                        tv_unitary (Hz).  |
|                                                        [default: 0.0]    |
| --fdn-tv-depth                       <float range>     Blend depth for   |
|                                      [0.0<=x<=1.0]     --fdn-matrix      |
|                                                        tv_unitary        |
|                                                        (0..1).           |
|                                                        [default: 0.0]    |
| --fdn-dfm-delays…                    <str>             Optional          |
|                                                        delay-feedback-m… |
|                                                        delays in         |
|                                                        milliseconds.     |
|                                                        Provide one value |
|                                                        for broadcast or  |
|                                                        one per FDN line. |
| --fdn-sparse         --no-fdn-sparse                   Enable sparse     |
|                                                        high-order FDN    |
|                                                        pair-mixing mode. |
|                                                        [default:         |
|                                                        no-fdn-sparse]    |
| --fdn-sparse-deg…                    <int range>       Number of sparse  |
|                                      [1<=x<=16]        pair-mixing       |
|                                                        stages used when  |
|                                                        --fdn-sparse is   |
|                                                        enabled.          |
|                                                        [default: 2]      |
| --fdn-cascade        --no-fdn-cascade                  Enable            |
|                                                        nested/cascaded   |
|                                                        FDN mode (small   |
|                                                        fast network into |
|                                                        late network).    |
|                                                        [default:         |
|                                                        no-fdn-cascade]   |
| --fdn-cascade-mix                    <float range>     Injection amount  |
|                                      [0.0<=x<=1.0]     from nested FDN   |
|                                                        into the main     |
|                                                        late-field        |
|                                                        network (0..1).   |
|                                                        [default: 0.35]   |
| --fdn-cascade-de…                    <float range>     Delay scaling for |
|                                      [0.2<=x<=1.0]     nested FDN        |
|                                                        relative to       |
|                                                        primary FDN       |
```

```
|                                                  delays                |
|                                                  (0.2..1.0).           |
|                                                  [default: 0.5]        |
| --fdn-cascade-rt…          <float range>         RT60 ratio for        |
|                            [0.1<=x<=1.0]         nested FDN            |
|                                                  relative to           |
|                                                  --rt60                |
|                                                  (0.1..1.0).           |
|                                                  [default: 0.55]       |
| --fdn-rt60-low             <float range>         Low-band RT60         |
|                            [0.1<=x<=3600.0]      target for            |
|                                                  multiband FDN         |
|                                                  decay shaping         |
|                                                  (seconds).            |
| --fdn-rt60-mid             <float range>         Mid-band RT60         |
|                            [0.1<=x<=3600.0]      target for            |
|                                                  multiband FDN         |
|                                                  decay shaping         |
|                                                  (seconds).            |
| --fdn-rt60-high            <float range>         High-band RT60        |
|                            [0.1<=x<=3600.0]      target for            |
|                                                  multiband FDN         |
|                                                  decay shaping         |
|                                                  (seconds).            |
| --fdn-rt60-tilt            <float range>         Jot-style             |
|                            [-1.0<=x<=1.0]        low/high RT skew      |
|                                                  around mid band       |
|                                                  (-1..1). Positive     |
|                                                  extends low-band      |
|                                                  decay and             |
|                                                  shortens highs.       |
|                                                  [default: 0.0]        |
| --fdn-tonal-corr…          <float range>         Track C tonal         |
|                            [0.0<=x<=1.0]         correction            |
|                                                  strength for          |
|                                                  multiband/tilted      |
|                                                  FDN response          |
|                                                  (0..1). Higher        |
|                                                  values apply          |
|                                                  stronger              |
|                                                  decay-color           |
|                                                  equalization.         |
|                                                  [default: 0.0]        |
| --fdn-xover-low-…          <float range>         Low/mid crossover     |
|                            [x>=20.0]             frequency used by     |
|                                                  multiband FDN         |
|                                                  decay shaping.        |
|                                                  [default: 250.0]      |
| --fdn-xover-high…          <float range>         Mid/high              |
|                            [x>=100.0]            crossover             |
|                                                  frequency used by     |
|                                                  multiband FDN         |
|                                                  decay shaping.        |
|                                                  [default: 4000.0]     |
| --fdn-link-filter          <str>                 Feedback-link         |
|                                                  filter mode           |
|                                                  inside the FDN        |
|                                                  matrix path:          |
|                                                  none, lowpass, or     |
```

```
|                                                  highpass.          |
|                                                  [default: none]    |
| --fdn-link-filte…  <float range>                 Cutoff frequency   |
|                    [x>=20.0]                     used by            |
|                                                  --fdn-link-filter  |
|                                                  (Hz).              |
|                                                  [default: 2500.0]  |
| --fdn-link-filte…  <float range>                 Wet mix of         |
|                    [0.0<=x<=1.0]                 feedback-link      |
|                                                  filter processing  |
|                                                  (0..1).            |
|                                                  [default: 1.0]     |
| --fdn-graph-topo…  <str>                         Graph topology     |
|                                                  for --fdn-matrix   |
|                                                  graph: ring,       |
|                                                  path, star, or     |
|                                                  random.            |
|                                                  [default: ring]    |
| --fdn-graph-degr…  <int range>                   Graph              |
|                    [1<=x<=32]                    neighborhood/con…  |
|                                                  degree for         |
|                                                  --fdn-matrix       |
|                                                  graph.             |
|                                                  [default: 2]       |
| --fdn-graph-seed   <int>                         Deterministic      |
|                                                  seed used to       |
|                                                  build              |
|                                                  graph-structured   |
|                                                  FDN pairings.      |
|                                                  [default: 2026]    |
| --fdn-spatial-co…  <none|adjacent|f              Directional        |
|                    ront_rear|bed_to              wet-bus coupling   |
|                    p|all_to_all>                  mode: none,        |
|                                                  adjacent,          |
|                                                  front_rear,        |
|                                                  bed_top,           |
|                                                  all_to_all.        |
|                                                  [default: none]    |
| --fdn-spatial-co…  <float range>                 Wet-bus            |
|                    [0.0<=x<=1.0]                 directional        |
|                                                  coupling amount    |
|                                                  (0..1).            |
|                                                  [default: 0.0]     |
| --fdn-nonlineari…  <none|tanh|softc              Optional in-loop   |
|                    lip>                          nonlinearity:      |
|                                                  none, tanh, or     |
|                                                  softclip.          |
|                                                  [default: none]    |
| --fdn-nonlineari…  <float range>                 Blend amount for   |
|                    [0.0<=x<=1.0]                 in-loop            |
|                                                  nonlinearity       |
|                                                  shaping (0..1).    |
|                                                  [default: 0.0]     |
| --fdn-nonlineari…  <float range>                 Drive multiplier   |
|                    [0.1<=x<=8.0]                 for in-loop        |
|                                                  nonlinearity       |
|                                                  shaping.           |
|                                                  [default: 1.0]     |
| --room-size-macro  <float range>                 Perceptual         |
```

```
|                                        [-1.0<=x<=1.0]    room-size macro   |
|                                                          (-1..1) mapped to |
|                                                          decay-time and    |
|                                                          spacing behavior. |
|                                                          [default: 0.0]    |
| --clarity-macro                        <float range>     Perceptual        |
|                                        [-1.0<=x<=1.0]    clarity macro     |
|                                                          (-1..1) mapped to |
|                                                          decay, damping,   |
|                                                          and wet balance.  |
|                                                          [default: 0.0]    |
| --warmth-macro                         <float range>     Perceptual warmth |
|                                        [-1.0<=x<=1.0]    macro (-1..1)     |
|                                                          mapped to damping |
|                                                          and spectral      |
|                                                          decay tilt.       |
|                                                          [default: 0.0]    |
| --envelopment-ma…                      <float range>     Perceptual        |
|                                        [-1.0<=x<=1.0]    envelopment macro |
|                                                          (-1..1) mapped to |
|                                                          width/decorrelat… |
|                                                          emphasis.         |
|                                                          [default: 0.0]    |
| --fdn-stereo-inj…                      <float range>     [default: 1.0]    |
|                                        [0.0<=x<=1.0]                       |
| --f0                                   <str>             e.g. 64, 64Hz, or |
|                                                          64 Hz             |
| --analyze-input                        <path>            Input audio to    |
|                                                          estimate          |
|                                                          fundamentals/har… |
|                                                          for IR tuning     |
| --harmonic-align…                      <float range>     [default: 0.75]   |
|                                        [0.0<=x<=1.0]                       |
| --resonator          --no-resonator                      Enable            |
|                                                          Modalys-inspired  |
|                                                          physical          |
|                                                          modal-bank        |
|                                                          late-tail         |
|                                                          coloration.       |
|                                                          [default:         |
|                                                          no-resonator]     |
| --resonator-mix                        <float range>     [default: 0.35]   |
|                                        [0.0<=x<=1.0]                       |
| --resonator-modes                      <int range>       [default: 32]     |
|                                        [x>=1]                              |
| --resonator-q-min                      <float range>     [default: 8.0]    |
|                                        [x>=0.5]                            |
| --resonator-q-max                      <float range>     [default: 90.0]   |
|                                        [x>=0.5]                            |
| --resonator-low-…                      <float range>     [default: 50.0]   |
|                                        [x>=20.0]                           |
| --resonator-high…                      <float range>     [default: 9000.0] |
|                                        [x>=30.0]                           |
| --resonator-late…                      <float range>     [default: 80.0]   |
|                                        [x>=0.0]                            |
| --cache-dir                            <str>             [default:         |
|                                                          .verbx_cache/irs] |
| --lucky                                <int range>       Generate N        |
|                                        [1<=x<=500]       randomized IR     |
```

```
                                                          files from one
                                                          base setup.
                                                          Outputs are
                                                          written to
                                                          --lucky-out-dir
                                                          (or OUT_IR parent
                                                          by default).
 --lucky-out-dir                          <path>          Output directory
                                                          used when --lucky
                                                          is enabled.
 --lucky-seed                             <int>           Optional
                                                          deterministic
                                                          seed for --lucky
                                                          IR generation.

 --silent
 --help                                                   Show this message
                                                          and exit.
```

6.13 `verbx ir analyze --help`

```
Usage: root ir analyze [OPTIONS] {ir_file}

┌─ Arguments ──────────────────────────────────────────────────────────────┐
│ *    ir_file      <path>  [required]                                     │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────┐
│ --json-out        <path>                                                 │
│ --help                    Show this message and exit.                    │
└──────────────────────────────────────────────────────────────────────────┘
```

6.14 `verbx ir process --help`

```
Usage: root ir process [OPTIONS] {in_ir} {out_ir}

┌─ Arguments ──────────────────────────────────────────────────────────────┐
│ *    in_ir       <path>  [required]                                      │
│ *    out_ir      <path>  [required]                                      │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────┐
│ --damping                           <float range>        [default: 0.4]  │
│                                     [0.0<=x<=1.0]                        │
│ --lowcut                            <float range>                        │
│                                     [x>=10.0]                            │
│ --highcut                           <float range>                        │
│                                     [x>=10.0]                            │
│ --tilt                              <float>              [default: 0.0]  │
│ --normalize                         <none|peak|rms>      [default: peak] │
│ --peak-dbfs                         <float>              [default: -1.0] │
│ --target-lufs                       <float>                              │
│ --true-peak        --sample-peak                         [default:       │
│                                                          true-peak]      │
│ --lucky                             <int range>          Generate N      │
│                                     [1<=x<=500]          randomized      │
```

```
                                                    processed IR files
                                                    from one input IR.
                                                    Outputs are written
                                                    to --lucky-out-dir
                                                    (or OUT_IR parent
                                                    by default).
 --lucky-out-dir                  <path>            Output directory
                                                    used when --lucky
                                                    is enabled.
 --lucky-seed                     <int>             Optional
                                                    deterministic seed
                                                    for --lucky IR
                                                    processing.
 --silent
 --help                                             Show this message
                                                    and exit.
```

6.15 `verbx ir morph --help`

```
Usage: root ir morph [OPTIONS] {ir_a} {ir_b} {out_ir}

┌─ Arguments ──────────────────────────────────────────────────────────┐
│ *    ir_a        <path>  [required]                                  │
│ *    ir_b        <path>  [required]                                  │
│ *    out_ir      <path>  [required]                                  │
└──────────────────────────────────────────────────────────────────────┘

┌─ Options ────────────────────────────────────────────────────────────┐
 --mode                              <str>            Morph mode:
                                                      linear,
                                                      equal-power,
                                                      spectral, or
                                                      envelope-aware.
                                                      [default:
                                                      equal-power]
 --alpha                             <float range>    [default: 0.5]
                                     [0.0<=x<=1.0]
 --early-ms                          <float range>    Early/late split
                                     [x>=0.0]         used by
                                                      split/envelope-a…
                                                      morphing (ms).
                                                      [default: 80.0]
 --early-alpha                       <float range>    Optional alpha
                                     [0.0<=x<=1.0]    override for
                                                      early-reflection
                                                      region.
 --late-alpha                        <float range>    Optional alpha
                                     [0.0<=x<=1.0]    override for
                                                      late-tail region.
 --align-decay       --no-align-decay                 Align decay
                                                      profiles before
                                                      morphing for
                                                      stable RT
                                                      trajectories.
                                                      [default:
                                                      align-decay]
 --phase-coherence                   <float range>    Phase-coherence
```

```
                                        [0.0<=x<=1.0]     safeguard
                                                          strength for
                                                          spectral
                                                          morphing.
                                                          [default: 0.75]
 --spectral-smoot…                      <int range>       Frequency
                                        [0<=x<=128]       smoothing radius
                                                          (FFT bins) used
                                                          by spectral
                                                          modes.
                                                          [default: 3]
 --mismatch-policy                      <coerce|strict>   Mismatch behavior
                                                          for
                                                          sample-rate/chan…
                                                          differences:
                                                          coerce (align) or
                                                          strict (fail).
                                                          [default: coerce]
 --target-sr                            <int range>       Optional target
                                        [x>=1]            sample rate for
                                                          morph processing
                                                          and output.
 --cache-dir                            <str>             [default:
                                                          .verbx_cache/ir_…
 --silent
 --help                                                   Show this message
                                                          and exit.
```

6.16 `verbx ir morph-sweep --help`

```
Usage: root ir morph-sweep [OPTIONS] {ir_a} {ir_b} {out_dir}

┌─ Arguments ──────────────────────────────────────────────────────────────┐
│ *    ir_a         <path>  [required]                                     │
│ *    ir_b         <path>  [required]                                     │
│ *    out_dir      <path>  [required]                                     │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────┐
 --mode                                 <str>             Morph mode:
                                                          linear,
                                                          equal-power,
                                                          spectral, or
                                                          envelope-aware.
                                                          [default:
                                                          equal-power]
 --alpha                                <float range>     Explicit alpha
                                        [0.0<=x<=1.0]     point. Repeat to
                                                          define custom
                                                          sweep timeline.
 --alpha-start                          <float range>     [default: 0.0]
                                        [0.0<=x<=1.0]
 --alpha-end                            <float range>     [default: 1.0]
                                        [0.0<=x<=1.0]
 --alpha-steps                          <int range>       [default: 9]
                                        [2<=x<=257]
 --out-prefix                           <str>             Output filename
```

```
|                                                          prefix for      |
|                                                          generated sweep |
|                                                          IR files.       |
|                                                          [default: morph]|
| --early-ms                          <float range>        [default: 80.0] |
|                                     [x>=0.0]                             |
| --early-alpha                       <float range>                        |
|                                     [0.0<=x<=1.0]                        |
| --late-alpha                        <float range>                        |
|                                     [0.0<=x<=1.0]                        |
| --align-decay       --no-align-decay                 Align decay         |
|                                                      profiles before     |
|                                                      morphing for        |
|                                                      stable RT           |
|                                                      trajectories.       |
|                                                      [default:           |
|                                                      align-decay]        |
| --phase-coherence                   <float range>    [default: 0.75]     |
|                                     [0.0<=x<=1.0]                        |
| --spectral-smoot…                   <int range>      [default: 3]        |
|                                     [0<=x<=128]                          |
| --mismatch-policy                   <coerce|strict>  Mismatch behavior   |
|                                                      for                 |
|                                                      sample-rate/chan…   |
|                                                      differences:        |
|                                                      coerce (align) or   |
|                                                      strict (fail).      |
|                                                      [default: coerce]   |
| --target-sr                         <int range>                          |
|                                     [x>=1]                               |
| --cache-dir                         <str>            [default:           |
|                                                      .verbx_cache/ir_…   |
| --workers                           <int range>      0 = auto            |
|                                     [x>=0]           [default: 0]        |
| --schedule                          <fifo|shortest-f [default:           |
|                                     irst|longest-fir  longest-first]     |
|                                     st>                                  |
| --retries                           <int range>      [default: 0]        |
|                                     [x>=0]                               |
| --continue-on-er…   --fail-fast                      [default:           |
|                                                      fail-fast]          |
| --fail-if-any-fa…   --allow-failed                   Exit non-zero       |
|                                                      when any sweep      |
|                                                      step fails.         |
|                                                      [default:           |
|                                                      fail-if-any-fail…   |
| --checkpoint-file                   <path>           Optional            |
|                                                      checkpoint JSON     |
|                                                      path for            |
|                                                      resume-safe sweep   |
|                                                      execution.          |
| --resume                                             Resume from         |
|                                                      --checkpoint-fil…   |
| --qa-json-out                       <path>           Summary JSON        |
|                                                      output path         |
|                                                      (default:           |
|                                                      <out_dir>/morph_…   |
| --qa-csv-out                        <path>           Per-step QA         |
|                                                      metrics CSV path    |
```

```
│                                                          (default:          │
│                                                          <out_dir>/morph_…  │
│ --silent                                                                    │
│ --help                                                   Show this message  │
│                                                          and exit.          │
└─────────────────────────────────────────────────────────────────────────────┘
```

6.17 verbx ir fit --help

```
Usage: root ir fit [OPTIONS] {infile} {out_ir}

┌─ Arguments ─────────────────────────────────────────────────────────────────┐
│ *    infile      <path>  [required]                                         │
│ *    out_ir      <path>  [required]                                         │
└─────────────────────────────────────────────────────────────────────────────┘
┌─ Options ───────────────────────────────────────────────────────────────────┐
│ --top-k                                 <int range>        [default: 3]     │
│                                         [x>=1]                              │
│ --base-mode                             <fdn|stochastic|m  [default: hybrid] │
│                                         odal|hybrid>                        │
│ --length                                <float range>      [default: 60.0]  │
│                                         [x>=0.1]                            │
│ --seed                                  <int>              [default: 0]     │
│ --candidate-pool                        <int range>        [default: 12]    │
│                                         [x>=1]                              │
│ --fit-workers                           <int range>        0 = auto         │
│                                         [x>=0]             [default: 0]     │
│ --analyze-tuning    --no-analyze-tu…                       [default:        │
│                                                            analyze-tuning]  │
│ --cache-dir                             <str>              [default:        │
│                                                            .verbx_cache/irs] │
│ --help                                                     Show this message │
│                                                            and exit.        │
└─────────────────────────────────────────────────────────────────────────────┘
```

6.18 verbx batch --help

```
Usage: root batch [OPTIONS] COMMAND [ARGS]...

Batch manifest generation and rendering.

┌─ Options ───────────────────────────────────────────────────────────────────┐
│ --help          Show this message and exit.                                 │
└─────────────────────────────────────────────────────────────────────────────┘
┌─ Commands ──────────────────────────────────────────────────────────────────┐
│ template                                                                    │
│ augment-template                                                            │
│ augment-profiles                                                            │
│ augment                                                                     │
│ render                                                                      │
└─────────────────────────────────────────────────────────────────────────────┘
```

6.19 verbx batch template --help

```
Usage: root batch template [OPTIONS]

╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.20 verbx batch augment-template --help

```
Usage: root batch augment-template [OPTIONS]

╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.21 verbx batch augment-profiles --help

```
Usage: root batch augment-profiles [OPTIONS]

╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --json          Emit profile definitions as JSON instead of a table.       │
│ --help          Show this message and exit.                                │
╰────────────────────────────────────────────────────────────────────────────╯
```

6.22 verbx batch augment --help

```
Usage: root batch augment [OPTIONS] {manifest}

╭─ Arguments ────────────────────────────────────────────────────────────────╮
│ *    manifest      <path>  [required]                                      │
╰────────────────────────────────────────────────────────────────────────────╯
╭─ Options ──────────────────────────────────────────────────────────────────╮
│ --output-root                          <path>           Override output    │
│                                                         root directory     │
│                                                         from manifest.     │
│ --profile                              <str>            Optional profile   │
│                                                         override (for      │
│                                                         quick profile A/B  │
│                                                         against one        │
│                                                         manifest).         │
│ --seed                                 <int>            Optional           │
│                                                         deterministic      │
│                                                         seed override.     │
│ --variants-per-i…                      <int range>      Optional           │
│                                        [1<=x<=500]      variants-per-sou…  │
│                                                         override.          │
│ --write-analysis     --no-write-anal…                   Override manifest  │
│                                                         write_analysis     │
│                                                         behavior.          │
│ --copy-dry           --no-copy-dry                      Copy clean source  │
```

```
|                                                    files into output |
|                                                    tree (paired      |
|                                                    dry/wet dataset   |
|                                                    layout).          |
|                                                    [default:         |
|                                                    no-copy-dry]      |
| --verify-split-i…  --allow-split-o…                Require one       |
|                                                    source id/input   |
|                                                    file to belong to |
|                                                    exactly one split |
|                                                    (prevents         |
|                                                    train/val/test    |
|                                                    leakage).         |
|                                                    [default:         |
|                                                    verify-split-iso… |
| --jobs                           <int range>       0 = auto          |
|                                  [x>=0]            [default: 0]      |
| --schedule                       <fifo|shortest-f  [default:         |
|                                  irst|longest-fir  longest-first]    |
|                                  st>               |
| --retries                        <int range>       [default: 0]      |
|                                  [x>=0]            |
| --continue-on-er…  --fail-fast                     [default:         |
|                                                    fail-fast]        |
| --fail-if-any-fa…  --allow-failed                  Exit non-zero     |
|                                                    when any          |
|                                                    augmentation      |
|                                                    render fails.     |
|                                                    [default:         |
|                                                    fail-if-any-fail… |
| --dry-run                                          |
| --jsonl-out                      <path>            Path for dataset  |
|                                                    metadata JSONL    |
|                                                    (default:         |
|                                                    <output_root>/au… |
| --summary-out                    <path>            Path for run      |
|                                                    summary JSON      |
|                                                    (default:         |
|                                                    <output_root>/au… |
| --dataset-card-o…                <path>            Optional Markdown |
|                                                    dataset-card path |
|                                                    for ML dataset    |
|                                                    documentation.    |
| --metrics-csv-out                <path>            Optional CSV path |
|                                                    for per-output    |
|                                                    acoustic          |
|                                                    features.         |
| --metrics-includ…  --metrics-fast                  Include           |
|                                                    LUFS/true-peak/L… |
|                                                    in metrics CSV    |
|                                                    export (slower).  |
|                                                    [default:         |
|                                                    metrics-fast]     |
| --qa-bundle-out                  <path>            Optional QA       |
|                                                    bundle JSON path  |
|                                                    (default:         |
|                                                    <output_root>/au… |
| --baseline-summa…                <path>            Optional prior    |
|                                                    augmentation      |
```

```
│                                                    summary/QA bundle │
│                                                    JSON used for     │
│                                                    regeneration      │
│                                                    deltas.           │
│ --provenance-hash    --no-provenance…              Emit              │
│                                                    deterministic     │
│                                                    provenance hash   │
│                                                    over manifest     │
│                                                    payload + source  │
│                                                    signatures.       │
│                                                    [default:         │
│                                                    no-provenance-ha… │
│ --help                                             Show this message │
│                                                    and exit.         │
└──────────────────────────────────────────────────────────────────────┘
```

6.23 `verbx batch render --help`

```
Usage: root batch render [OPTIONS] {manifest}

┌─ Arguments ──────────────────────────────────────────────────────────┐
│ *    manifest      <path>  [required]                                │
└──────────────────────────────────────────────────────────────────────┘

┌─ Options ────────────────────────────────────────────────────────────┐
│ --jobs                              <int range> [x>=0]  0 = auto     │
│                                                         [default: 0] │
│ --schedule                          <fifo|shortest-fir  [default:    │
│                                     st|longest-first>   longest-first]
│ --retries                           <int range> [x>=0]  [default: 0] │
│ --continue-on-error    --fail-fast                      [default:    │
│                                                         fail-fast]   │
│ --checkpoint-file                   <path>              Optional     │
│                                                         checkpoint file
│                                                         used to persist
│                                                         per-job completion
│                                                         state.       │
│ --resume                                                Resume from  │
│                                                         --checkpoint-file
│                                                         and skip already
│                                                         completed jobs.
│ --dry-run                                                            │
│ --lucky                             <int range>         For each manifest
│                                     [1<=x<=500]         job, generate N
│                                                         wild random render
│                                                         variants. Outputs
│                                                         are written to
│                                                         --lucky-out-dir
│                                                         (or each job
│                                                         OUTFILE parent by
│                                                         default).    │
│ --lucky-out-dir                     <path>              Output directory
│                                                         used when --lucky
│                                                         is enabled.  │
│ --lucky-seed                        <int>               Optional     │
│                                                         deterministic seed
│                                                         for --lucky batch
│                                                         generation.  │
```

```
| --progress-json                      <path>            Append one JSONL   |
|                                                        line per completed |
|                                                        job to this file.  |
|                                                        Each line contains |
|                                                        index, outfile,    |
|                                                        success,           |
|                                                        duration_seconds,  |
|                                                        and error.         |
| --help                                                 Show this message  |
|                                                        and exit.          |
└─────────────────────────────────────────────────────────────────────────────┘
```

6.24 verbx immersive --help

```
Usage: root immersive [OPTIONS] COMMAND [ARGS]...

Immersive production interoperability workflows.

┌─ Options ──────────────────────────────────────────────────────────────────┐
| --help          Show this message and exit.                                |
└────────────────────────────────────────────────────────────────────────────┘
┌─ Commands ─────────────────────────────────────────────────────────────────┐
| template                                                                   |
| handoff                                                                    |
| qc                                                                         |
| queue     Distributed immersive queue workflows.                           |
└────────────────────────────────────────────────────────────────────────────┘
```

6.25 verbx immersive template --help

```
Usage: root immersive template [OPTIONS]

┌─ Options ──────────────────────────────────────────────────────────────────┐
| --help          Show this message and exit.                                |
└────────────────────────────────────────────────────────────────────────────┘
```

6.26 verbx immersive handoff --help

```
Usage: root immersive handoff [OPTIONS] {scene_file} {out_dir}

┌─ Arguments ────────────────────────────────────────────────────────────────┐
| *    scene_file      <path>  [required]                                    |
| *    out_dir         <path>  [required]                                    |
└────────────────────────────────────────────────────────────────────────────┘
┌─ Options ──────────────────────────────────────────────────────────────────┐
| --strict    --warn-only      Fail if policy/QC errors are detected.        |
|                              [default: strict]                             |
| --help                       Show this message and exit.                   |
└────────────────────────────────────────────────────────────────────────────┘
```

6.27 verbx immersive qc --help

```
Usage: root immersive qc [OPTIONS] {infile}

┌─ Arguments ──────────────────────────────────────────────────────────────┐
│ *    infile      <path>  [required]                                      │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────┐
│ --layout                    <str>                Channel layout hint:    │
│                                                  auto, mono, stereo,     │
│                                                  lcr, 5.1, 7.1, 7.1.2,   │
│                                                  7.1.4, 7.2.4, 8.0,      │
│                                                  16.0, 64.4              │
│                                                  [default: auto]         │
│ --target-lufs               <float>              [default: -18.0]        │
│ --lufs-tolerance            <float range>        [default: 3.0]          │
│                             [x>=0.0]                                     │
│ --max-true-peak-dbfs        <float>              [default: -1.0]         │
│ --max-fold-down-delta…      <float range>        [default: 4.0]          │
│                             [x>=0.0]                                     │
│ --min-channel-occupan…      <float range>        [default: 0.34]         │
│                             [0.0<=x<=1.0]                                │
│ --occupancy-threshold…      <float>              [default: -45.0]        │
│ --json-out                  <path>               Optional output path    │
│                                                  for QC JSON payload.    │
│ --fail-on-violation                              Exit with code 2 when   │
│                                                  any QC gate fails.      │
│ --help                                           Show this message and   │
│                                                  exit.                   │
└──────────────────────────────────────────────────────────────────────────┘
```

6.28 verbx immersive queue --help

```
Usage: root immersive queue [OPTIONS] COMMAND [ARGS]...

Distributed immersive queue workflows.

┌─ Options ────────────────────────────────────────────────────────────────┐
│ --help          Show this message and exit.                              │
└──────────────────────────────────────────────────────────────────────────┘
┌─ Commands ───────────────────────────────────────────────────────────────┐
│ template                                                                 │
│ status                                                                   │
│ worker                                                                   │
└──────────────────────────────────────────────────────────────────────────┘
```

6.29 verbx immersive queue template --help

```
Usage: root immersive queue template [OPTIONS]

┌─ Options ────────────────────────────────────────────────────────────────┐
│ --help          Show this message and exit.                              │
└──────────────────────────────────────────────────────────────────────────┘
```

6.30 verbx immersive queue status --help

```
Usage: root immersive queue status [OPTIONS] {queue_file}

┌─ Arguments ──────────────────────────────────────────────────────────────────┐
│ *    queue_file      <path>  [required]                                      │
└──────────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────────┐
│ --help          Show this message and exit.                                  │
└──────────────────────────────────────────────────────────────────────────────┘
```

6.31 verbx immersive queue worker --help

```
Usage: root immersive queue worker [OPTIONS] {queue_file}

┌─ Arguments ──────────────────────────────────────────────────────────────────┐
│ *    queue_file      <path>  [required]                                      │
└──────────────────────────────────────────────────────────────────────────────┘
┌─ Options ────────────────────────────────────────────────────────────────────┐
│ --worker-id                          <str>              Worker identifier.   │
│                                                         Defaults to host     │
│                                                         PID-based value.     │
│ --heartbeat-dir                      <path>             Directory for        │
│                                                         per-worker           │
│                                                         heartbeat JSON       │
│                                                         files.               │
│                                                         [default:            │
│                                                         .verbx_queue_hear…   │
│ --poll-ms                            <int range>        [default: 800]       │
│                                      [x>=50]                                 │
│ --max-jobs                           <int range> [x>=0] 0 = run until        │
│                                                         queue drain          │
│                                                         [default: 0]         │
│ --stale-claim-seco…                  <float range>      [default: 120.0]     │
│                                      [x>=1.0]                                │
│ --continue-on-error    --fail-fast                      [default:            │
│                                                         continue-on-error]   │
│ --fail-if-any-fail…                                     Exit with code 2     │
│                                                         if any queue jobs    │
│                                                         end in failed        │
│                                                         state.               │
│ --help                                                  Show this message    │
│                                                         and exit.            │
└──────────────────────────────────────────────────────────────────────────────┘
```

6.32 Suggested Exercises

1. Write a command that creates a 100-percent-wet stereo return with an analysis JSON, then explain why each output option is explicit.
2. Use `--dry-run` or the equivalent validation path on one expensive render and record which estimated resources or warnings change after one parameter edit.
3. Create a preset, render it twice with a fixed seed, and verify reproducibility using hashes and analysis reports.
4. Deliberately submit one invalid layout or incompatible option combination, then rewrite the command so it fails fast for the right reason.
5. Make a concise shell script that renders three controlled variants and preserves their reports beside the audio.
6. Use shell variables to make a parameter sweep readable, then save the expanded commands with their output names and reports.
7. Write one command that intentionally uses a nondefault sample rate, floating-point output, and explicit channel layout; verify every choice in the report.
8. Build a command-line recipe for reverse reverb and identify the source duration and tail conditions that make it practical.
9. Compare two limiter configurations on the same extreme tail and document the tradeoff between peak safety and transient character.
10. Create a JSON-only analysis workflow that can run in CI without auditioning audio, then list its blind spots.

“The reverb was so long that a trainwreck would sound good in there.”

— Frank Speller (1938–2017), American organist and composer,
on his recital in Westminster Abbey (1992)

Chapter 7

verbx Extreme Workflow Cookbook (with 100 Recipes)

A high-intensity command library for experimental and large-space reverb design in `verbx`. This is not a beginner tutorial – it is a working reference for sound designers, composers, and engineers who want to push the tool into territory where things get genuinely strange. Some of these sounds are useful. Some are purely destructive. All of them are instructive.

7.1 Before You Start

What should `in.wav` be?

Almost anything works, but the most interesting results come from material with transient information – percussive hits, short melodic phrases, spoken words, field recordings with attack. Sustained drones and pads tend to disappear into long tails. Heavily compressed material loses dynamic articulation in the freeze and bloom passes. A single piano note, a snare hit, a short spoken syllable, a plucked string – these are ideal starting points.

For multichannel recipes (sections 8), replace `in.wav` with `in_5p1.wav` or `in_7p1.wav` as noted. These should be properly interleaved multichannel files, not summed stereo.

How to read the commands

Every recipe is a single shell command. Arguments with `--` are named flags. Numeric values follow directly after the flag name. File paths are relative to wherever you run the command from. If you want to run these from a different directory, adjust the paths accordingly.

The output folder must exist before you run anything:

```
mkdir -p out
```

What do the output files mean?

Each output file is named with a three-digit index and a short descriptor. The index matches the recipe number. The descriptor is a rough hint at what parameter dominated the result. `verbx render` defaults to HD output (`192000 Hz`, `float32` WAV) unless overridden. Use `--quality-preset sd|md|hd`, `--target-sr`, and/or `--out-subtype` to set delivery format explicitly. Loudness normalization is applied at the output stage unless you explicitly disable it.

A note on headphones vs. speakers

Wide-field and shimmer recipes will behave very differently on headphones vs. speakers. Multichannel outputs need a properly configured monitoring chain. Several of the self-convolution recipes produce subsonic energy that is inaudible on small speakers but will stress subwoofers and some headphone amplifiers. You have been warned.

7.2 Building Blocks

Before diving into the extremes, here are the six commands worth understanding deeply. Everything else in this cookbook is a variation or combination of these.

```
# 1. Simplest possible algo reverb — 10 seconds of decay, mostly wet
verbx render in.wav out/block_01_algo.wav --engine algo --rt60 10 --wet 0.9 --dry 0.1

# 2. Convolution with a real IR — the cleanest path to a convincing space
verbx render in.wav out/block_02_conv.wav --engine conv --ir hall.wav

# 3. Freeze — grabs a slice of the audio and holds it as an infinite pad
verbx render in.wav out/block_03_freeze.wav --freeze --start 1 --end 2

# 4. Shimmer — pitch-shifted feedback that creates the classic ethereal octave shimmer
verbx render in.wav out/block_04_shimmer.wav --engine algo --shimmer --shimmer-semitones 12

# 5. Self-convolution — uses the audio as its own impulse response (things escalate fast)
verbx render in.wav out/block_05_self.wav --self-convolve

# 6. Lucky mode — randomizes parameters and generates multiple variations at once
verbx render in.wav out/block_06_lucky.wav --lucky 5 --lucky-out-dir out/lucky_block
```

These six commands are the grammar. The 100 recipes below are the sentences.

7.3 Public Alpha Musical Reference Workflows

These five recipes are the canonical musical examples used in the public alpha launch narrative. They are maintained in `README.md` and mirrored here so they do not drift.

1) Alvin Lucier / *I Am Sitting in a Room* (1969) (iterative room resonance)

```
verbx render voice.wav lucier_7pass.wav --engine algo --rt60 4.5 \
  --wet 1.0 --dry 0.0 --repeat 7 --fdn-lines 16 --fdn-matrix hadamard --lowcut 60
```

2) Brian Eno / *Discreet Music* (1975) (ambient loopbed)

```
verbx render input.wav eno_ambient.wav --engine algo --rt60 12.0 \
  --wet 0.92 --dry 0.08 --damping 0.25 --pre-delay-ms 35 \
  --fdn-lines 16 --fdn-matrix hadamard --lowcut 50 \
  --target-lufs -22 --target-peak-dbfs -2
```

3) Pauline Oliveros / *Deep Listening* (1989) (extended drone-space)

```
verbx render drone.wav deep_listening.wav --engine algo --rt60 18.0 \
  --wet 0.95 --dry 0.10 --fdn-lines 32 --fdn-matrix hadamard \
  --pre-delay-ms 55 --damping 0.15 --lowcut 30 \
  --target-lufs -24 --target-peak-dbfs -2
```

4) Frippertronics-style tape-loop accumulation

```
verbx render guitar.wav frippertronics.wav --engine algo --rt60 8.0 \
  --wet 0.82 --dry 0.28 --fdn-lines 16 --fdn-matrix hadamard \
  --shimmer --shimmer-semitones 12 --shimmer-mix 0.45 --shimmer-feedback 0.78 \
  --pre-delay-ms 25 --target-peak-dbfs -2
```

5) Shoegaze reverse-wash (freeze + shimmer)

```
verbx render guitar.wav shoegaze.wav --engine algo \
  --freeze --start 1.0 --end 2.4 \
```

```
--shimmer --shimmer-semitones 12 --shimmer-mix 0.55 --shimmer-feedback 0.72 \
--rt60 5.0 --wet 0.88 --dry 0.22 --fdn-matrix circulant --lowcut 80 \
--width 1.4 --target-peak-dbfs -2
```

7.4 How to Listen Critically

This section sits between the building blocks and the recipes because the way you listen matters as much as the settings you use.

For algorithmic reverbs (section 1): Listen for the density of the diffusion network. A well-tuned algo reverb should feel like a continuous fog rather than a series of distinct echoes. Listen for colorations in the decay – most algo engines have a characteristic tone that emerges around 3–6 seconds of tail. High `--beast-mode` values will push the feedback networks toward instability; you will hear this as a shift from smooth decay to a kind of churning or fluttering texture.

For freeze and repeat chains (section 2): The transition point between the frozen segment and the incoming signal is the interesting moment. Listen for phase artifacts at loop boundaries, for build-up of low-mid energy across repeats, and for how the freeze interacts with the original signal's wet/dry balance. Repeat chains tend to accumulate energy – listen for how normalization (or its absence) shapes the perceived loudness arc.

For convolution (section 3): The first 50ms of a convolution reverb contains the early reflections, which define the perceived size of the space. The tail defines the character. Listen to them separately by shortening and extending `--tail-limit`. A good IR should feel like moving the source into a real space; a bad IR will sound like filtering. Partition size affects latency but not sound quality in offline rendering – in these recipes it primarily affects CPU allocation.

For self-convolution (section 4): This is mathematically interesting and sonically extreme. The audio is convolved with itself, which has the effect of squaring the spectral envelope. Peaks in the original spectrum become dramatically exaggerated. Quiet passages become very quiet. Loud passages become overwhelming. The result is often described as "the audio eating itself." Listen for the spectral imbalance this creates, and for how `--tilt` can be used to correct or exaggerate it.

For shimmer, duck, and bloom (section 5): Shimmer is additive – it adds pitched content that was not there before. Duck is subtractive – it removes the reverb tail while the dry signal is present. Bloom is temporal – it delays the onset of the reverb buildup. Listen to each in isolation before combining them. When all three are active simultaneously (recipe 50), the result is layered enough that you should listen to it at least three times: once for the dry signal, once for the reverb tail, once for the shimmer layer.

For self-generated IRs (section 7): The most useful skill when working with synthetic IRs is learning to distinguish the IR's character from the reverb engine's character. Process the same input with different IRs and compare. A stochastic IR will produce a different kind of density than an FDN IR even if the RT60 is identical.

7.5 Section 1: Algorithmic Extremes (Recipes 1–10)

Algorithmic reverb builds synthetic spaces using networks of delay lines, all-pass filters, and feedback matrices – no recorded impulse required. This makes it fast to iterate and infinitely malleable, but it also means the "space" it simulates is fundamentally fictional. That is its power. In film post, algo reverbs are used for sci-fi environments, supernatural spaces, and any situation where the reverb needs to be emotionally correct rather than acoustically accurate. In electronic music, they are the texture underneath ambient pads, the wash behind a snare, the infinite hallway a note disappears into.

The `--beast-mode` flag increases the density and complexity of the internal diffusion network. Low values (1–4) are broadly useful. Values above 8 begin to produce audible coloration. Values above 12 are genuinely unstable on some inputs and will produce results that range from magnificent to unusable.

7.5.1 Recipe 1: Two-Minute Diffusion Field

```
verbx render in.wav out/001_algo_long.wav --engine algo --rt60 120 --wet 0.95 --dry 0.1 --beast-
↪  mode 6
```

What it sounds like: A two-minute reverb tail that utterly swallows the source. The original signal is barely audible.

DSP note: RT60 of 120 seconds means the energy takes two full minutes to decay by 60dB. Combined with `--beast-mode 6`, the diffusion network is running at higher complexity, producing a very smooth, dense tail. At `--wet 0.95`, you are hearing almost exclusively the reverberant field.

7.5.2 Recipe 2: Lightless Cavern

```
verbx render in.wav out/002_algo_dark.wav --engine algo --rt60 90 --damping 0.9 --wet 0.9 --dry 0.15
```

What it sounds like: A very long, very dark reverb – like shouting into a large cave and hearing the sound transform into low-frequency mush.

DSP note: `--damping 0.9` applies frequency-dependent attenuation to the high frequencies on each feedback pass through the reverb network. High damping values simulate absorptive surfaces (heavy drapes, acoustic foam, water). The result is a reverb that loses its high-frequency content exponentially faster than its low-frequency content.

7.5.3 Recipe 3: Beyond-the-Speakers Width

```
verbx render in.wav out/003_algo_wide.wav --engine algo --rt60 75 --width 2.0 --wet 0.85 --dry 0.2
```

What it sounds like: An expansive stereo reverb where the tail feels wider than the speakers. Sounds placed at center appear to spread outward as they decay.

DSP note: `--width 2.0` applies a Mid-Side matrix transform to the stereo output of the reverb, scaling the Side component. Values above 1.0 exaggerate the stereo difference signal. This can produce phantom imaging that extends beyond the speaker baseline. Listen on headphones for the full effect – and be aware that mono compatibility degrades as width increases.

7.5.4 Recipe 4: Twenty-Second Breathing Tail

```
verbx render in.wav out/004_algo_mod_slow.wav --engine algo --rt60 80 --mod-depth-ms 12 --mod-rate-
↪  hz 0.05
```

What it sounds like: A subtly breathing reverb, almost alive. The tail slowly warps over time. Works well for sustained material.

DSP note: The modulation LFO at 0.05 Hz has a period of 20 seconds. It is modulating the delay lengths in the reverb network by up to 12ms, which creates gentle pitch and timing variations in the late reverb field. This is the mechanism behind the characteristic "swimming" quality of classic hardware reverbs like the Lexicon 224.

7.5.5 Recipe 5: Seasick Modulation

```
verbx render in.wav out/005_algo_mod_fast.wav --engine algo --rt60 65 --mod-depth-ms 8 --mod-rate-
↪ hz 1.2
```

What it sounds like: A chorusing, slightly seasick reverb. The tail has a metallic shimmer that is different from the shimmer flag – this one is caused by interference patterns in the modulated delays.

DSP note: At 1.2 Hz with 8ms depth, the modulation is fast enough to produce audible sideband frequencies. The interaction between the modulation rate and the delay network produces beating artifacts that some ears read as pleasant chorus and others read as instability. At higher depths it crosses into vibrato territory.

7.5.6 Recipe 6: Delayed Reverb Cloud

```
verbx render in.wav out/006_algo_predelay_cloud.wav --engine algo --rt60 70 --pre-delay-ms 220 --
↪ wet 0.9
```

What it sounds like: The source plays, there is a noticeable gap of silence, then a massive reverb cloud arrives. Very dramatic. Works extremely well with vocals and solo instruments.

DSP note: 220ms pre-delay is well above the Haas threshold (~30ms), meaning the brain perceives the reverb tail as a distinct echo event rather than as spatial information. This is a widely used production technique for creating "space" without burying the source in reverb. The gap also allows transient information from the dry signal to remain fully articulate.

7.5.7 Recipe 7: Double-Pass Diffusion Splash

```
verbx render in.wav out/007_algo_splash.wav --engine algo --rt60 45 --beast-mode 10 --repeat 2
```

What it sounds like: A dense, complex reverb splash that repeats twice. The beast-mode density combined with repetition creates a layered wash.

DSP note: `--repeat 2` runs the render pipeline twice and stacks the outputs. At `--beast-mode 10`, the diffusion network is at high complexity, so each pass through adds a distinct but related layer of dense reflection patterns.

7.5.8 Recipe 8: Three-Minute Wet Abyss

```
verbx render in.wav out/008_algo_massive.wav --engine algo --rt60 180 --wet 1.0 --dry 0.0 --beast-
↪ mode 12
```

What it sounds like: The source disappears completely. What remains is three minutes of dense, evolving reverb that has almost no relationship to what went in. Use headphones. This one is genuinely terrifying at volume.

DSP note: `--wet 1.0 --dry 0.0` eliminates the direct signal path entirely. Combined with a 180-second RT60 and beast-mode 12, the reverb network is operating at high feedback complexity with an enormous decay envelope. The output is essentially a synthetic texture derived from the input's early moments. On spectrally complex input, this is often beautiful. On input with strong low-frequency content, it will rumble and build in unexpected ways.

Expert note: At RT60 values this extreme, the reverb network is operating at the edge of stability. The internal feedback matrix is nominally convergent, but floating-point accumulation over very long tails can still produce low-level artifacts that surface in the noise floor of spectral analysis tools. In verbx this occurs in 64-bit (`float64`) internal DSP, which substantially lowers numerical drift versus single-precision paths.

7.5.9 Recipe 9: Airy Bright Expanse

```
verbx render in.wav out/009_algo_air.wav --engine algo --rt60 50 --damping 0.2 --highcut 18000 --
↪  tilt 3
```

What it sounds like: A bright, airy reverb. The tail sparkles. Good on acoustic instruments and vocals where you want presence, not darkness.

DSP note: Low damping (0.2) preserves high-frequency content across feedback passes, producing a reverb tail that retains treble energy. `--tilt 3` applies a gentle high-shelf tilt to the output, further emphasizing the air frequencies above 4kHz.

7.5.10 Recipe 10: Low-Frequency Mud Chamber

```
verbx render in.wav out/010_algo_mud.wav --engine algo --rt60 110 --lowcut 30 --highcut 1800 --tilt
↪  -5
```

What it sounds like: A thick, dark, low-mid reverb that eliminates almost all high-frequency content. Like being inside a concrete room full of water.

DSP note: The combined effect of `--highcut 1800` and `--tilt -5` aggressively attenuates frequencies above 1.8kHz. The result is a reverb that exists almost entirely in the 30–1800Hz band. Long RT60 in this band produces the characteristic "mud" of untreated room reverb. Useful for horror sound design, industrial textures, and sub-bass reinforcement.

7.6 How to Listen: Freeze and Repeat

Freeze is distinct from reverb – it is not a space simulation, it is a temporal manipulation. When you freeze a segment, you are extracting a window of audio and playing it back as a loop. The quality of the loop (smoothness, phase coherence, the presence or absence of click artifacts) depends on the nature of the source material at the freeze point. Sustained tones freeze cleanly. Transients at the freeze boundary create artifacts that can be used deliberately.

Repeat chains stack the entire render output and re-process it. This accumulates artifacts, saturates the frequency response, and gradually transforms the material. Listen to how each repeat changes the spectral balance, and notice whether normalization is fighting the accumulation or allowing it.

7.7 Section 2: Freeze and Repeat Chains (Recipes 11–20)

Freeze workflows are central to ambient music production and sound design for picture. The ability to hold a moment of audio indefinitely – creating a synthetic drone from a single phrase, or sustaining a room ambience between cuts – is one of the more practically useful things `verbx` does. Repeat chains go further, iterating the reverb process multiple times and creating layered, progressively transformed

textures. In scoring, this is used to build tension by accumulating reverberant energy. In experimental music, it is used to watch a source material destroy itself through repeated convolution.

7.7.1 Recipe 11: Triple Short Freeze

```
verbx render in.wav out/011_freeze_short.wav --freeze --start 2 --end 3 --repeat 3 --engine algo
```

What it sounds like: A one-second window of the source audio frozen and repeated three times through an algo reverb. You will hear the source phrase briefly, then a looped, reverberant version of it.

DSP note: `--start 2 --end 3` extracts the audio between the 2-second and 3-second marks. This window is looped internally and then passed through the reverb engine. The algo engine processes the frozen loop, so the reverb is applied to the looped content rather than to the original recording.

7.7.2 Recipe 12: Wide Four-Pass Freeze

```
verbx render in.wav out/012_freeze_wide.wav --freeze --start 4 --end 6 --repeat 4 --width 1.8
```

What it sounds like: A two-second freeze, widened dramatically, repeated four times. The stereo image expands with each repeat.

DSP note: Width processing at 1.8 applied to a frozen loop will amplify any stereo differences present in the extracted window. If the source is mono, the width processing will apply decorrelation to create artificial stereo spread, which accumulates across the four repeat passes.

7.7.3 Recipe 13: High-Density Freeze Cascade

```
verbx render in.wav out/013_freeze_beast.wav --freeze --start 1 --end 2.2 --repeat 5 --beast-mode 14
```

What it sounds like: A beast-mode freeze – dense, complex, with five passes of increasingly saturated reverb. The original source becomes unrecognizable by pass three.

DSP note: Beast-mode 14 pushes the diffusion network into high-complexity territory. Applied to a freeze loop, each repeat pass processes an already-reverberant signal through a high-density network, compounding the diffusion and creating what is effectively reverb-of-reverb.

7.7.4 Recipe 14: Dark Frozen Loop

```
verbx render in.wav out/014_freeze_dark.wav --freeze --start 3 --end 5 --repeat 3 --damping 0.85 --
↪  tilt -4
```

What it sounds like: A dark, muffled freeze. The repeated passes become progressively more low-frequency dominated. Good for horror, industrial, or ominous ambient textures.

DSP note: High damping combined with negative tilt stacks two separate high-frequency attenuation mechanisms. Damping reduces HF within the reverb feedback loop; tilt applies a static EQ shelf to the output. Over three repeat passes, the cumulative HF loss is significant.

7.7.5 Recipe 15: Frozen Shimmer Cloud

```
verbx render in.wav out/015_freeze_shimmer.wav --freeze --start 2.5 --end 4 --repeat 2 --shimmer
```

What it sounds like: A frozen segment with shimmering pitched overtones that build across two repeat passes. The shimmer adds an octave-up harmonic that makes the loop feel larger and more cinematic.

DSP note: The shimmer algorithm pitch-shifts the reverb signal up by 12 semitones (default) and feeds it back into the reverb input with a configurable mix level. Applied to a freeze loop, this creates a self-sustaining harmonic layer that re-enters the loop on each repeat.

7.7.6 Recipe 16: Six-Pass Algorithmic Accumulation

```
verbx render in.wav out/016_repeat_algo.wav --engine algo --rt60 55 --repeat 6 --normalize-stage
↪  per-pass
```

What it sounds like: Six passes of algo reverb, each normalized independently. The loudness stays consistent but the texture becomes progressively denser and more complex.

DSP note: `--normalize-stage per-pass` applies normalization at the output of each repeat iteration before passing it into the next. This prevents the accumulation from causing clipping, but it also means the normalization gain changes between passes, which can introduce subtle level-stepping artifacts on very short repeat segments.

7.7.7 Recipe 17: Four-Pass Convolution Hall

```
verbx render in.wav out/017_repeat_conv.wav --engine conv --ir hall.wav --repeat 4 --normalize-
↪  stage per-pass
```

What it sounds like: Four convolution passes with a hall IR. The space expands with each pass – by pass four, the room sounds enormous.

DSP note: Convolving the output of a convolution with the same IR is mathematically equivalent to squaring the transfer function of the impulse response (approximately). This causes the early reflections to multiply and the tail to extend non-linearly. The result is a reverb that grows beyond what the original IR's RT60 would suggest.

7.7.8 Recipe 18: Repeated Ducked Tail

```
verbx render in.wav out/018_repeat_duck.wav --engine algo --repeat 5 --duck --duck-attack 5 --duck-
↪  release 800
```

What it sounds like: The dry signal punches through cleanly, then the reverb blooms afterward. Across five repeats, the duck-and-bloom pattern becomes a rhythmic feature of the texture.

DSP note: Duck is a sidechain-style gain reduction applied to the wet signal when the dry signal is above a threshold. `--duck-attack 5` means the attenuation engages very quickly (5ms), protecting the transient. `--duck-release 800` means the attenuation releases slowly (800ms), giving the reverb a long fade-in after each transient. Combined with repeat, this creates a forward-moving texture that maintains rhythmic articulation.

7.7.9 Recipe 19: Four-Pass Bloom

```
verbx render in.wav out/019_repeat_bloom.wav --engine algo --repeat 4 --bloom 4.5 --wet 0.95
```

What it sounds like: The reverb builds slowly rather than appearing instantly. Over four repeats, the bloom tail becomes the dominant sound – a slow, rising wash.

DSP note: Bloom delays the onset of the reverb density curve. A bloom value of 4.5 means the reverb takes 4.5 seconds to reach its full diffusion density. On repeat pass one, you hear a slow build. On subsequent passes, the input to each pass already contains reverberant energy, which the bloom curve then delays further – producing increasingly stretched decay envelopes.

7.7.10 Recipe 20: Deep Noise-Floor Cathedral

```
verbx render in.wav out/020_repeat_floor.wav --engine algo --rt60 140 --repeat 3 --target-lufs -26
```

What it sounds like: Three passes of a very long reverb, normalized to a broadcast-appropriate loudness level. Useful as a bed or texture layer.

DSP note: `--target-lufs -26` is roughly the integrated loudness target for film broadcast (–24 LUFS) plus a 2dB headroom buffer. Applying this to a three-pass, 140-second reverb tail ensures the output is at a known loudness level suitable for mixing into a larger context without additional gain staging.

7.8 How to Listen: Convolution Modes

Convolution reverb is multiplication in the frequency domain – the spectrum of the input is multiplied by the spectrum of the impulse response. This makes it inherently linear and phase-coherent in a way that algorithmic reverbs are not. What you hear in a convolution reverb is literally what was captured in the IR, faithfully reproduced. The quality of the output is bounded by the quality of the IR. A noisy or poorly captured IR will produce a noisy reverb. A pristine IR from a great acoustic space will produce something that sounds like the real thing.

The interesting edge cases come from using IRs that are not captured from real spaces – synthetic IRs, self-convolution, matrix IRs for multichannel formats. In these cases, convolution stops being a space simulator and becomes a spectral transformation tool.

7.9 Section 3: Convolution Heavy Modes (Recipes 21–30)

Convolution is the workhorse of post-production reverb. It is what music supervisors use when they need a score to sound like it was performed in a specific church. It is what sound designers use when they need a gunshot to sound like it happened in a parking garage rather than a foley stage. The goal is accuracy and plausibility, and when it works well, it is invisible.

The recipes in this section push convolution into less conventional territory – matrix IRs for multichannel, extremely large or small partition sizes, truncated tails, and combination with other processing.

7.9.1 Recipe 21: Large-Partition Hall

```
verbx render in.wav out/021_conv_hall.wav --engine conv --ir hall.wav --partition-size 32768
```

What it sounds like: A clean hall reverb with a large FFT partition. Sounds like a real concert hall.

DSP note: Partition-based convolution divides the IR into blocks that are convolved separately and summed. Larger partition sizes reduce CPU load at the cost of increased latency (in real-time contexts). In offline rendering, the partition size primarily affects memory allocation and processing chunk size – a partition of 32768 samples at 48kHz represents about 682ms per block.

7.9.2 Recipe 22: Peak-Normalized Plate

```
verbx render in.wav out/022_conv_plate.wav --engine conv --ir plate.wav --ir-normalize peak
```

What it sounds like: A plate reverb character – dense, bright, fast-building. Classic on vocals and snare.

DSP note: `--ir-normalize peak` normalizes the impulse response to its peak sample value before convolution. This ensures consistent output levels regardless of how the IR was captured. Without normalization, the loudness of the convolution output depends entirely on the amplitude of the IR, which varies between recordings.

7.9.3 Recipe 23: Extended Church Tail

```
verbx render in.wav out/023_conv_church.wav --engine conv --ir church.wav --tail-limit 150
```

What it sounds like: A church reverb capped at 150 seconds. Long, but not infinite.

DSP note: `--tail-limit 150` truncates the IR at 150 seconds and applies a fade-out at the truncation point. Church IRs can run very long (some cathedrals have RT60s exceeding 10 seconds), and limiting the tail reduces both file size and processing time without significantly affecting the perceptual character of the early reverb.

7.9.4 Recipe 24: Mono IR Across the Field

```
verbx render in.wav out/024_conv_mono_to_all.wav --engine conv --ir mono_ir.wav --wet 0.9 --dry 0.2
```

What it sounds like: A stereo input processed with a mono IR. The reverb will be mono-width, which can sound narrower than expected.

DSP note: When a mono IR is used on a stereo input, `verbx` applies the IR to each channel independently. The resulting reverb is correlated (identical in both channels) rather than decorrelated. This is technically correct but may sound narrower than a stereo IR. Some engineers prefer this for dialogue processing where reverb width should match the source.

7.9.5 Recipe 25: Output-Major Surround Matrix

```
verbx render in.wav out/025_conv_surround.wav --engine conv --ir matrix_5p1.wav --ir-matrix-layout
↪  output-major
```

What it sounds like: A multichannel convolution where each output channel gets its own IR. Requires a multichannel IR file.

DSP note: Output-major matrix layout means the IR file is organized so that the first group of channels represents the IR for output channel 1 from all inputs, the second group for output channel 2, and so on. This is the format used when you want each output speaker to have a distinct reverb tail derived from all inputs.

7.9.6 Recipe 26: Input-Major Surround Matrix

```
verbx render in.wav out/026_conv_input_major.wav --engine conv --ir matrix_5p1.wav --ir-matrix-
↪  layout input-major
```

What it sounds like: The same multichannel convolution, but with the IR channels interpreted differently. The output will be different from recipe 25 even with the same IR file.

DSP note: Input-major layout means the IR file is organized so that the first group of channels represents the IR for input channel 1 to all outputs, the second group for input channel 2, etc. The two layout modes are equivalent for symmetric IRs but produce different results for asymmetric multichannel IRs. Which is "correct" depends on how the IR was captured and exported.

7.9.7 Recipe 27: High-Throughput Hall Convolution

```
verbx render in.wav out/027_conv_fast.wav --engine conv --ir hall.wav --partition-size 65536 --
↪  normalize-stage none
```

What it sounds like: A hall reverb with the largest partition size and no output normalization. Results in faster offline processing but potentially louder or quieter output depending on the IR.

DSP note: Disabling normalization with `--normalize-stage none` passes the raw convolution output to the file. If the IR has a high amplitude and the input is also loud, the output can clip. This is intentional in certain mastering and sound design contexts where you want the raw convolution result without any gain adjustment.

7.9.8 Recipe 28: Triple Dense Hall

```
verbx render in.wav out/028_conv_dense.wav --engine conv --ir hall.wav --repeat 3 --beast-mode 5
```

What it sounds like: Three convolution passes with beast-mode density. The hall becomes impossibly dense – more reflection than room.

DSP note: Beast-mode applied to a convolution engine increases the density of the convolution kernel by internal upsampling and interpolation of the IR. Combined with three repeat passes, the result is a reverb that has far more dense reflections than the original hall IR would suggest.

7.9.9 Recipe 29: Twenty-Second Trimmed Hall

```
verbx render in.wav out/029_conv_trimmed.wav --engine conv --ir hall.wav --tail-limit 20
```

What it sounds like: A very short hall reverb – you get the early reflections and room character but almost none of the tail. Sounds like a large room with very absorptive surfaces.

DSP note: Truncating the IR at 20 seconds cuts most of the reverberation tail while preserving the crucial early reflection structure that defines the perceived size and shape of the space. This is useful when you want spatial information but not acoustic congestion in a mix.

7.9.10 Recipe 30: Full-Tail Input-Peak Match

```
verbx render in.wav out/030_conv_fulltail.wav --engine conv --ir hall.wav --output-peak-norm input
```

What it sounds like: A full hall reverb normalized so that its peak matches the peak of the input file. This ensures a predictable output level.

DSP note: `--output-peak-norm input` measures the peak amplitude of the input file and applies gain to the output so that both share the same peak level. This is a reference-based normalization approach, useful for comparative listening where you want multiple output files at matched levels.

7.10 Section 4: Self-Convolution and Feedback Smear (Recipes 31–40)

Self-convolution is where things get genuinely experimental. Using the audio as its own impulse response creates a spectral feedback loop – the signal's frequency peaks are amplified and its nulls are deepened, producing a transformation that is related to but distinct from the source. It is not a reverb in any conventional sense. It is more like the audio reflecting itself into a new shape.

This section is primarily useful for experimental sound design, academic study, and extreme texture generation. In production contexts, the output is rarely usable without subsequent processing, but as a raw material generator it is remarkable. The sounds here have appeared in horror films, industrial music, and electroacoustic composition.

7.10.1 Recipe 31: Raw Self-Convolution

```
verbx render in.wav out/031_self_base.wav --self-convolve --normalize-stage none
```

What it sounds like: The source convolved with itself. Spectral peaks become dominant. The result sounds like an exaggerated, smeared version of the original.

DSP note: Self-convolution in the frequency domain squares the magnitude spectrum and doubles the phase. The squared magnitude means that any spectral peak becomes a more prominent peak, and any spectral trough becomes a deeper trough. The phase doubling causes phase alignment differences that create comb filtering when compared with the original. No normalization means the output level will vary dramatically depending on the input spectrum.

7.10.2 Recipe 32: High-Density Self-Convolution

```
verbx render in.wav out/032_self_beast.wav --self-convolve --beast-mode 20 --normalize-stage none
```

What it sounds like: Self-convolution at extreme beast-mode. Do not play this loud without listening at low volume first.

DSP note: Beast-mode 20 is at the upper end of the useful range. Applied to self-convolution, it adds multiple additional feedback paths through the diffusion network, each of which processes the already-self-convolved signal. The cumulative effect can produce output levels orders of magnitude above the input. `--normalize-stage none` means these levels are uncontrolled.

> **Expert note:** Self-convolution at high beast-mode values is computationally similar to running a feedback network with gain slightly above unity. In a real-time context this would cause unbounded growth (infinite clipping). In offline rendering, the process terminates

when the tail reaches the end of the computed buffer, so the output is finite, but the energy distribution within that buffer may be extremely non-uniform. Spectral analysis of the output is informative – you will see the input's spectral peaks raised to extreme prominence.

7.10.3 Recipe 33: Two-Hundred-Second Self Tail

```
verbx render in.wav out/033_self_longtail.wav --self-convolve --tail-limit 200 --partition-size
↪  32768
```

What it sounds like: A very long self-convolution tail. The smeared version of the source continues for over three minutes.

DSP note: Self-convolution produces a signal whose duration is approximately twice the duration of the input. `--tail-limit 200` extends the computed output to 200 seconds, allowing the self-convolved tail to fully develop. The large partition size handles the long convolution kernel efficiently.

7.10.4 Recipe 34: Bright Self-Convolution

```
verbx render in.wav out/034_self_bright.wav --self-convolve --tilt 5 --highcut 18000
```

What it sounds like: A self-convolution biased toward high frequencies. Sounds metallic, glassy, sometimes resembling granular synthesis.

DSP note: Self-convolution tends to bias toward whichever spectral region has the highest energy in the source. `--tilt 5` pre-emphasizes the high frequencies before the convolution operation, which means the squared spectrum will have an even stronger high-frequency component in the output. The hard cut at 18kHz prevents aliased content from contaminating the output.

7.10.5 Recipe 35: Dark Self-Convolution

```
verbx render in.wav out/035_self_dark.wav --self-convolve --tilt -6 --highcut 2500
```

What it sounds like: A dark self-convolution. The result is dominated by low-to-mid frequencies. Sounds like a monster or machine dreaming.

DSP note: `--tilt -6` and `--highcut 2500` together severely limit the bandwidth before and after the self-convolution. The squared low-frequency spectrum produces a dense, rumbling texture with very little high-frequency content. Apply EQ and saturation downstream to shape this into something usable.

7.10.6 Recipe 36: Ducked Self-Convolution

```
verbx render in.wav out/036_self_duck.wav --self-convolve --duck --duck-attack 3 --duck-release 600
```

What it sounds like: Self-convolution with the dry signal punching through the self-convolved wash. The articulation of the original is preserved within the smeared texture.

DSP note: Ducking applied to self-convolution uses the dry signal as the sidechain to attenuate the self-convolved output. This is unusual because the "wet" signal being ducked is derived from the same source as the sidechain. The result depends heavily on the timing relationship between the dry transients and the smeared output. Fast attack (3ms) means the duck engages almost immediately with each transient.

7.10.7 Recipe 37: Shimmering Self-Convolution

```
verbx render in.wav out/037_self_shimmer.wav --self-convolve --shimmer --shimmer-mix 0.55
```

What it sounds like: Self-convolution with an octave-up harmonic layer. Sounds enormous and harmonically complex.

DSP note: Shimmer applied to the output of self-convolution adds a pitch-shifted copy of the already-spectrally-extreme self-convolved signal. At 0.55 mix, the shimmer layer is almost as loud as the self-convolved base. The harmonic content of the shimmer is a pitch-shifted version of the squared spectrum, which can create unusual harmonic relationships.

7.10.8 Recipe 38: Triple Self-Convolution

```
verbx render in.wav out/038_self_repeat.wav --self-convolve --repeat 3 --normalize-stage per-pass
```

What it sounds like: Three passes of self-convolution. The first pass is already extreme; the second and third are increasingly abstract.

DSP note: Each repeat pass uses the output of the previous pass as the input to the next self-convolution. Mathematically, this raises the original spectrum to the power of 2^n (where n is the pass number). By pass three, the spectral peaks are at $2^3 = 8$ times their original prominence (in log scale, +18dB relative to the original spectral balance). Per-pass normalization prevents clipping between passes.

7.10.9 Recipe 39: Loudness-Targeted Self-Convolution

```
verbx render in.wav out/039_self_loud.wav --self-convolve --target-lufs -16 --target-peak-dbfs -1
```

What it sounds like: Self-convolution normalized to streaming-loudness standards. Useful when the self-convolve output needs to be delivered at a calibrated level.

DSP note: –16 LUFS is approximately the target for music streaming platforms. Combined with –1 dBFS true peak headroom, this configuration takes the unpredictable amplitude of self-convolution output and brings it to a usable, deliverable level. The nature of the sound is unchanged; only the gain is adjusted.

7.10.10 Recipe 40: Beast-Mode Self-Consumption

```
verbx render in.wav out/040_self_huge.wav --self-convolve --beast-mode 40 --repeat 2
```

What it sounds like: Beast-mode 40, two repeat passes of self-convolution. This is at the theoretical limit of what the engine can compute in a reasonable time. The output is unpredictable. Back up your input first.

DSP note: Beast-mode 40 is at the edge of numerical stability for most input signals. Combined with two repeat passes of self-convolution, the internal state of the engine accumulates floating-point error at a rate that can produce sub-audible noise floors that are louder than the intended output. This is extreme use and is documented here for completeness, not as a recommended workflow.

7.11 Section 5: Shimmer, Comb Cloud, Ducking, Bloom, and Tilt (Recipes 41–50)

These five tools are the texture controls. Shimmer adds pitched content. Comb cloud adds time-domain metallic density. Duck subtracts. Bloom delays and builds. Tilt tilts. Each has a clear sonic function that can be grasped immediately, but the interesting work happens at the intersections – when comb cloud roughens a pristine FDN tail, when shimmer feeds into a bloomed reverb, or when duck is combined with slow attack times and long tails.

In film scoring, shimmer is used to signal memory, transcendence, or altered states. Comb cloud is good for synthetic chambers, haunted-metal interiors, and any space that should feel a little unstable before the late field smooths out. Duck is ubiquitous in dialogue-heavy mixes where reverb would otherwise obscure speech. Bloom simulates the acoustic behavior of large spaces where diffuse reflections arrive after the direct sound. Tilt shapes the perceived brightness of a space to match the scene.

Quick comb-cloud starter:

```
verbx render in.wav out/comb_cloud_plate.wav --engine algo --rt60 4.5 \
  --comb-cloud --comb-cloud-count 28 --comb-cloud-feedback 0.38 --comb-cloud-mix 0.24 \
  --fdn-lines 12 --fdn-matrix hadamard
```

What it sounds like: A slightly metallic, plate-adjacent haze ahead of the main tail.

DSP note: Comb cloud sits before the late FDN. It changes the excitation entering the network rather than replacing the FDN itself, so you get extra grain and density without throwing away the core RT60 calibration and matrix behavior.

7.11.1 Recipe 41: Octave Shimmer

```
verbx render in.wav out/041_shimmer_oct.wav --engine algo --shimmer --shimmer-semitones 12 --
↪  shimmer-mix 0.35
```

What it sounds like: The classic shimmer reverb – an octave above the source, feeding back into the reverb tail. Recognizable from ambient music, film scores, and certain synthesizer patches.

DSP note: Shimmer works by pitch-shifting the reverb output up by the specified number of semitones and feeding it back into the reverb input with the specified mix level. At 12 semitones (one octave), the feedback loop creates a self-sustaining harmonic layer that grows proportionally to the reverb's RT60. At 0.35 mix, it is present but not dominant.

7.11.2 Recipe 42: Two-Octave Feedback Shimmer

```
verbx render in.wav out/042_shimmer_double.wav --engine algo --shimmer --shimmer-semitones 24 --
↪  shimmer-feedback 0.8
```

What it sounds like: Two octaves up, with very high feedback. The shimmer dominates the reverb completely. Sounds angelic or catastrophic depending on the source material.

DSP note: `--shimmer-feedback 0.8` means the pitch-shifted signal is re-injected at 80% of its output level on each pass through the feedback loop. At two octaves, this produces an extremely bright, self-sustaining shimmer that can outlast the reverb tail itself. On input with strong low-frequency content, the two-octave shift brings bass frequencies into the midrange, which can produce unexpectedly clear harmonic content in the shimmer.

7.11.3 Recipe 43: Perfect-Fifth Shimmer

```
verbx render in.wav out/043_shimmer_fifth.wav --engine algo --shimmer --shimmer-semitones 7 --
↪  shimmer-mix 0.5
```

What it sounds like: A shimmer a perfect fifth above the source. Less obviously "shimmer-y" than the octave version, more harmonically complex. Try this on minor-key source material for interesting results.

DSP note: Seven semitones is a perfect fifth in equal temperament. The pitch ratio is $2^{7/12} \approx 1.498$, very close to the just-intonation ratio of 3:2 (1.5). The slight deviation from pure tuning creates slow beating between the shimmer and source harmonics, which contributes to the characteristic "movement" of shimmer reverb.

7.11.4 Recipe 44: Hard Reverb Ducking

```
verbx render in.wav out/044_duck_hard.wav --engine algo --duck --duck-attack 2 --duck-release 900
```

What it sounds like: The reverb ducks hard and fast when the source is present, then blooms slowly afterward. Classic dialogue-reverb behavior.

DSP note: 2ms attack means the sidechain engages almost instantaneously – essentially every transient in the source will cause the reverb to duck. 900ms release means the reverb takes nearly a second to fully return after the source falls below the threshold. This is a deliberate choice to keep the reverb audible only in pauses and phrase endings, which is standard practice in narrative dialogue mixing.

7.11.5 Recipe 45: Soft Reverb Ducking

```
verbx render in.wav out/045_duck_soft.wav --engine algo --duck --duck-attack 60 --duck-release 180
```

What it sounds like: A gentle, subtle duck. The reverb barely moves. Good for music where you want slight movement without obvious pumping.

DSP note: 60ms attack is slow enough that transients pass through before the duck engages, which means percussive events do not cause sudden reverb dips. 180ms release is fast enough that the reverb returns before the next transient arrives (assuming material above ~160 BPM). The result is a slow, gentle breathing motion in the reverb rather than a clearly audible pumping effect.

7.11.6 Recipe 46: Five-Second Bloom

```
verbx render in.wav out/046_bloom_long.wav --engine algo --bloom 5 --rt60 90
```

What it sounds like: A reverb that takes five seconds to reach full density. For the first few seconds after the source, you hear a thin, distant reflection; then the full reverb gradually fills in.

DSP note: Bloom shapes the density envelope of the early reverb diffusion. A bloom value of 5 seconds means the all-pass diffuser network is fed input progressively over the first five seconds rather than all at once. The mathematical model is an exponential ramp of the diffusion feedback coefficient from 0 to its nominal value over the bloom duration.

7.11.7 Recipe 47: Blooming Shimmer

```
verbx render in.wav out/047_bloom_shimmer.wav --engine algo --bloom 3 --shimmer --shimmer-mix 0.4
```

What it sounds like: A reverb that builds slowly into a shimmer cloud. The shimmer layer arrives as the reverb density fills in. Very cinematic.

DSP note: Because shimmer feeds into the reverb input, the bloom envelope affects how the shimmer builds as well. In the first three seconds, there is very little shimmer because there is very little reverb density to pitch-shift. As the density grows, the shimmer grows with it. This creates a natural swell that feels less mechanical than fixed shimmer.

7.11.8 Recipe 48: Upward Spectral Tilt

```
verbx render in.wav out/048_tilt_up.wav --engine algo --tilt 6 --lowcut 120
```

What it sounds like: A bright, clear reverb with the low frequencies removed. Useful on source material where the reverb tail would otherwise accumulate low-mid energy.

DSP note: `--tilt 6` applies a +6dB/octave high-shelf tilt to the reverb output, progressively boosting frequencies as they increase. Combined with `--lowcut 120`, which removes everything below 120Hz, the result is a reverb that exists almost entirely in the mid-to-high frequency range.

7.11.9 Recipe 49: Downward Spectral Tilt

```
verbx render in.wav out/049_tilt_down.wav --engine algo --tilt -6 --highcut 4500
```

What it sounds like: A dark reverb with a strong low-frequency bias. The tail rumbles and warms. Good for thunder, drums, and low-frequency sound design.

DSP note: `--tilt -6` applies a –6 dB/octave downward slope, attenuating frequencies as they increase. Combined with `--highcut 4500`, which removes everything above 4.5kHz, the result is a reverb confined to the bass and low-mid range.

7.11.10 Recipe 50: Duck-Bloom-Shimmer Combination

```
verbx render in.wav out/050_combo_extreme.wav --engine algo --duck --bloom 4 --tilt 4 --shimmer --
↪  beast-mode 8
```

What it sounds like: Everything at once. Duck, bloom, shimmer, tilt, and high beast-mode. This is not a subtle reverb. It is a feature.

DSP note: The signal chain here is complex: the source is passed into the algo engine with beast-mode 8 diffusion, then bloom shapes the density buildup over 4 seconds, then shimmer adds a pitch-shifted layer into the reverb feedback, then tilt tilts the output toward brightness, then duck dynamically attenuates the entire wet signal when the dry source is present. Each of these processes interacts with the others. You will likely need to A/B this against a simpler configuration to hear what each element is contributing.

> **Expert note:** When combining duck and bloom, be aware that the duck sidechain is typically keyed from the dry input, not the wet output. This means the duck engages based on the source dynamics, while the bloom is shaping the reverb density based on elapsed time. In cases where the source has a slow attack (strings, pads), the duck may not engage strongly enough to prevent reverb buildup during the bloom phase, resulting in a more forward reverb than expected. Adjust duck threshold to compensate.

7.12 Section 6: Loudness and Output Format (Recipes 51–60)

Loudness normalization is one of those topics that sounds boring until you work in a context where it matters – and then it is everything. Streaming platforms, broadcast, cinema, and immersive audio all have different integrated loudness targets, true peak limits, and gating rules. This section covers the practical configurations for common delivery requirements, as well as the edge cases where you want no normalization at all.

Output format matters more than most engineers think. Float32 preserves more dynamic range than PCM24 in intermediate processing stages. Float64 is belt-and-suspenders overkill for most work but useful when accumulating multiple processing passes. These recipes cover the full range.

7.12.1 Recipe 51: Quiet Streaming Loudness

```
verbx render in.wav out/051_lufs_24.wav --target-lufs -24 --target-peak-dbfs -2
```

What it sounds like: Output normalized to –24 LUFS integrated loudness with –2 dBFS peak limit. This is a conservative broadcast target.

DSP note: Integrated LUFS measurement gates silence and weights frequencies according to the ITU-R BS.1770 loudness model (which de-emphasizes very low and very high frequencies relative to mid frequencies). The measurement reflects perceived loudness rather than signal amplitude. –24 LUFS is the US broadcast standard; the European standard is –23 LUFS.

7.12.2 Recipe 52: Loud True-Peak Master

```
verbx render in.wav out/052_lufs_18.wav --target-lufs -18 --target-peak-dbfs -1 --true-peak
```

What it sounds like: Music streaming target loudness with true peak limiting. Appropriate for Spotify, Apple Music, and similar platforms.

DSP note: True peak measurement uses oversampled analysis (typically 4x) to detect inter-sample peaks that sample-level measurement would miss. An inter-sample peak of –1 dBFS can produce a true peak well above 0 dBFS after digital-to-analog conversion, causing clipping in the playback chain. `--true-peak` engages a true peak limiter that accounts for this.

7.12.3 Recipe 53: Sample-Peak Ceiling

```
verbx render in.wav out/053_sample_peak.wav --target-peak-dbfs -0.5 --sample-peak
```

What it sounds like: Peak normalized to –0.5 dBFS at the sample level. This is a basic peak normalization without loudness weighting.

DSP note: Sample peak normalization finds the single loudest sample in the file and applies uniform gain to bring it to the specified level. It ignores loudness perception entirely – a file that is –0.5 dBFS peak could be at –35 LUFS (very quiet) or –8 LUFS (very loud) depending on its dynamic range. Use this when you need predictable amplitude, not predictable loudness.

7.12.4 Recipe 54: Per-Pass Loudness Control

```
verbx render in.wav out/054_per_pass.wav --repeat 4 --normalize-stage per-pass --repeat-target-lufs
↪  -22
```

What it sounds like: Four repeat passes, each normalized to –22 LUFS before the next pass begins. The loudness remains consistent across passes but the texture accumulates.

DSP note: `--repeat-target-lufs -22` sets the per-pass loudness target independently from any output-stage loudness target. This is useful for repeat chains where you want to prevent energy accumulation from causing clipping in intermediate passes while still having the final output normalized differently at the output stage.

7.12.5 Recipe 55: Unrestricted Tail Without Limiting

```
verbx render in.wav out/055_no_limiter.wav --engine algo --rt60 70 --no-limiter
```

What it sounds like: Reverb output with no peak limiting. The output can exceed 0 dBFS. You need a float-format output file or downstream clipping control for this to be useful.

DSP note: The `--no-limiter` flag bypasses the output-stage limiter and leaves the raw reverb output at its natural amplitude. On inputs with strong transients, the reverb energy can accumulate to levels well above 0 dBFS. This is only appropriate for intermediate processing stages where the output will be further processed before final delivery.

7.12.6 Recipe 56: 192 kHz Float32 Convolution

```
verbx render in.wav out/056_float32.wav --engine conv --ir hall.wav --target-sr 192000 --out-
↪  subtype float32
```

What it sounds like: Hall convolution rendered to 32-bit float WAV. Transparent and suitable for further processing.

DSP note: Float32 encoding provides approximately 24 bits of dynamic range at any given magnitude, with the advantage that the encoding range is not fixed – levels above 0 dBFS are representable without clipping. This makes float32 ideal for intermediate processing files. The cost is slightly larger files than PCM24 for equivalent audible quality. Here, `--target-sr 192000` performs deterministic internal sample-rate conversion during render.

7.12.7 Recipe 57: Float64 Convolution Master

```
verbx render in.wav out/057_float64.wav --engine conv --ir hall.wav --out-subtype float64
```

What it sounds like: Hall convolution rendered to 64-bit double float WAV. Identical in audible quality to float32 for most material.

DSP note: Float64 provides approximately 52 bits of mantissa precision. The audible difference from float32 is essentially nonexistent for any material that remains in a normal dynamic range. Float64 is primarily useful for accumulation-heavy operations (many repeat passes, self-convolution chains) where floating-point rounding errors in float32 could accumulate to audible levels. For single-pass renders, float32 is sufficient.

7.12.8 Recipe 58: 24-Bit PCM Delivery

```
verbx render in.wav out/058_pcm24.wav --engine conv --ir hall.wav --out-subtype pcm24
```

What it sounds like: Hall convolution rendered to 24-bit integer PCM. The standard delivery format for most professional audio workflows.

DSP note: PCM24 provides a fixed dynamic range of approximately 144dB theoretical, 120dB practical. It cannot represent values above 0 dBFS without clipping. Unlike float formats, clipping in PCM24 is hard – the value wraps or clips at the encoding limit. For final delivery files, PCM24 is usually appropriate. For intermediate processing, prefer float32 or float64.

7.12.9 Recipe 59: Input-Referenced Peak Normalization

```
verbx render in.wav out/059_peak_input.wav --engine algo --output-peak-norm input
```

What it sounds like: The reverb output is gain-adjusted so its peak matches the peak of the input file. Useful for comparative listening.

DSP note: `--output-peak-norm input` measures the peak sample amplitude of the input file (`in.wav`) and applies the gain necessary to bring the output to the same peak level. This is reference-relative normalization – the output level is defined by the input level rather than by an absolute target. Useful when batch-processing variations and wanting all outputs at consistent relative levels.

7.12.10 Recipe 60: Fixed Target Peak Normalization

```
verbx render in.wav out/060_peak_target.wav --engine algo --output-peak-norm target --output-peak-
↪  target-dbfs -9
```

What it sounds like: The reverb output is peak-normalized to a specific absolute level of –9 dBFS.

DSP note: `--output-peak-norm target` enables absolute peak normalization, and `--output-peak-target-dbfs -9` sets the target. –9 dBFS is a common bus-level target for mixing – it provides substantial headroom for subsequent processing while keeping the signal well above the noise floor. This is different from LUFS normalization: –9 dBFS peak does not guarantee any particular loudness.

7.13 Section 7: Synthetic IR Generation (Recipes 61–70)

Generating synthetic impulse responses is where `verbx` overlaps with acoustic modeling. Rather than measuring a real space, these commands synthesize IRs from mathematical models – Feedback Delay Networks, stochastic noise models, and modal synthesis. The results range from plausible-but-not-real-sounding rooms to completely abstract spectral transformers.

The practical value of synthetic IRs is flexibility: you can tune the RT60, the modal density, the fundamental resonant frequency, and the reflection pattern, all without access to a real space. For science fiction and fantasy sound design, this is invaluable. For electronic music, it allows creating reverbs that are tuned to the key of the track. For theatrical foley, it allows matching a reverb exactly to the visual environment.

The workflow in recipes 61–70 forms a complete IR generation and processing pipeline: generate, process, analyze, then use.

7.13.1 Recipe 61: Hybrid Synthetic Space

```
verbx ir gen out/061_ir_hybrid.wav --mode hybrid --length 120 --seed 61
```

What it sounds like: An IR for a synthetic room with hybrid early-reflection and statistical-tail modeling. Sounds plausible but not identifiably real.

DSP note: Hybrid mode combines a deterministic early-reflection model (which generates discrete room reflections based on a shoe-box room approximation) with a statistical late-reverberation model (which uses a filtered noise model for the tail). The seed ensures reproducibility – the same seed and parameters will always produce the same IR.

7.13.2 Recipe 62: Twelve-Line FDN Impulse

```
verbx ir gen out/062_ir_fdn.wav --mode fdn --length 180 --fdn-lines 12 --seed 62
```

What it sounds like: An FDN-derived IR. Dense, smooth, mathematically perfect in its diffusion. Does not sound like a real room but sounds like a very convincing synthetic space.

DSP note: Feedback Delay Networks use a matrix of delay lines with a feedback matrix (usually unitary, to preserve energy). 12 FDN lines produces a dense, highly decorrelated late-reverb tail. The Hadamard or Householder matrix typically used for the feedback connections ensures maximum decorrelation with minimal computation. The resulting IR captures the FDN's impulse response rather than running the FDN in real time.

7.13.3 Recipe 63: Dense Stochastic Tail

```
verbx ir gen out/063_ir_stochastic.wav --mode stochastic --length 240 --density 1.8 --seed 63
```

What it sounds like: A stochastic IR – the late reverb tail is synthesized from a filtered and shaped noise burst. Density 1.8 produces a very dense, continuous tail.

DSP note: Stochastic IR synthesis models the late-field reverberation as an exponentially decaying noise process, shaped by a target spectral envelope. Density 1.8 controls the number of reflections per unit time in the early part of the stochastic model. Higher density values produce smoother, more "diffuse" reverb, while lower values produce grainier, more textured tails.

7.13.4 Recipe 64: Ninety-Six-Mode Resonator Bank

```
verbx ir gen out/064_ir_modal.wav --mode modal --length 90 --modal-count 96 --seed 64
```

What it sounds like: A modal IR – synthesized from 96 resonant modes. Depending on the source material, you will hear clear resonant frequencies in the reverb tail.

DSP note: Modal synthesis generates an IR as a sum of decaying sinusoids, each at a different frequency (mode) with a different amplitude and decay rate. 96 modes is relatively dense – individual modes may not be distinguishable. At lower mode counts (8–16), the individual resonances are clearly audible and the "room" sounds like a resonator rather than a space.

7.13.5 Recipe 65: Tuned 64 Hz Modal Space

```
verbx ir gen out/065_ir_tuned.wav --mode modal --length 120 --f0 64Hz --seed 65
```

What it sounds like: A modal IR tuned to a fundamental frequency of 64Hz (roughly a C2). The reverb will reinforce pitches harmonically related to C2.

DSP note: Setting `--f0 64Hz` anchors the modal synthesis so that the lowest-frequency mode is at 64Hz, and subsequent modes are placed at harmonically related frequencies (or near-harmonic distributions, depending on the mode generation algorithm). This creates a reverb that colors the source material based on its harmonic relationship to C2. For music in C, this is complementary; for music in other keys, it can create tension or harmonic muddiness.

7.13.6 Recipe 66: Input-Informed Hybrid IR

```
verbx ir gen out/066_ir_from_input.wav --mode hybrid --analyze-input in.wav --seed 66
```

What it sounds like: A synthetic IR derived from the spectral and dynamic characteristics of the input file itself. The reverb will have a character that is somehow "related" to the source.

DSP note: `--analyze-input` extracts the spectral envelope, dynamic range, and tempo/transient information from the input file and uses this to shape the IR synthesis parameters. The result is an IR that resonates sympathetically with the source material. This is useful for creating reverbs that feel "organic" to a specific piece of music or audio.

7.13.7 Recipe 67: Hybrid Resonator IR

```
verbx ir gen out/067_ir_resonator.wav --mode hybrid --resonator --resonator-mix 0.6 --seed 67
```

What it sounds like: A hybrid IR with additional resonator modes mixed in at 60%. The tail has clear resonant peaks that color whatever is processed through it.

DSP note: The resonator adds a set of high-Q bandpass filters to the IR synthesis chain. `--resonator-mix 0.6` blends the resonator output with the base hybrid IR at 60%. The resonant frequencies are determined by the mode generation algorithm and the seed. This is closer to a comb filter or spring reverb character than a room reverb.

7.13.8 Recipe 68: Darkened Peak-Normalized IR

```
verbx ir process out/067_ir_resonator.wav out/068_ir_processed.wav --tilt -4 --normalize peak
```

What it sounds like: The resonator IR processed with a dark tilt and peak normalized. Ready to use as a dark, resonant convolution IR.

DSP note: IR processing applies DSP to the impulse response itself rather than to the output of the reverb. Tilting an IR by –4 dB/octave means the convolution will produce a dark, bass-heavy reverb regardless of what the source material sounds like. Peak normalizing the IR ensures consistent output levels when the IR is used in recipe 70.

7.13.9 Recipe 69: Machine-Readable IR Analysis

```
verbx ir analyze out/068_ir_processed.wav --json-out out/069_ir_analysis.json
```

What it sounds like: This produces no audio – it generates a JSON analysis report of the processed IR.

DSP note: IR analysis typically measures RT20, RT30, RT60 (derived from the decay curve), early decay time, clarity (C50, C80), and spectral centroid as a function of time. This data is useful for predicting how the IR will behave as a reverb and for comparing multiple IRs quantitatively. The JSON output can be imported into analysis or visualization tools.

7.13.10 Recipe 70: Custom-IR Double Render

```
verbx render in.wav out/070_ir_render.wav --engine conv --ir out/068_ir_processed.wav --repeat 2
```

What it sounds like: The input processed with the custom IR we generated and processed in recipes 67–68, repeated twice. This completes the synthetic IR pipeline.

DSP note: This is the payoff of the IR generation workflow. The convolution engine uses the processed synthetic IR as if it were a measured room IR. Two repeat passes mean the input is convolved twice with the resonant, dark-tilted synthetic IR, producing a dense, colored reverb with the resonator's character amplified.

7.14 How to Listen: Spatial and Multichannel

Multichannel listening requires a properly calibrated monitoring chain. Without it, you cannot judge the spatial balance of a 5.1 or 7.1 render – you are listening to a subset of the output. If you are working on multichannel material without a multichannel monitor system, use a downmix for reference, but always verify critical decisions on proper playback.

For the algorithmic multichannel recipes, pay attention to the relationship between the center channel and the surrounds. In ambience and reverb for picture, the surrounds are typically at a lower level than the fronts. Over-reverberant surrounds are one of the most common mixing errors in immersive audio.

7.15 Section 8: Multichannel and Spatial Processing (Recipes 71–80)

Multichannel reverb is a workflow unto itself. The spatial distribution of reverb energy – which sounds arrive from which directions, how the early reflections establish the size and shape of the space, how the late field wraps around the listener – is as important as the RT60 or the dry/wet balance. Bad multichannel reverb sounds like stereo reverb playing from four speakers. Good multichannel reverb sounds like being inside a space.

In cinema mixing, the surround channels carry ambient reverb and environmental information; the front channels carry the action and dialogue. The LFE carries bass and sub-bass reverb energy. The workflow for managing these independently in convolution is through matrix IRs – a different impulse response for each output channel.

7.15.1 Recipe 71: 5.1 Matrix Convolution

```
verbx render in_5p1.wav out/071_5p1_conv.wav --engine conv --ir ir_5p1_matrix.wav --ir-matrix-
↪ layout output-major
```

What it sounds like: A 5.1 convolution reverb with a properly configured matrix IR. Each output channel gets its own reverb character derived from the appropriate IR channel.

DSP note: Output-major layout: the IR file contains IR channels ordered as (IR for output L from all inputs), (IR for output R from all inputs), (IR for output C from all inputs), and so on. This is the standard layout for most multichannel IR libraries captured with separate microphone positions.

7.15.2 Recipe 72: 7.1 Matrix Convolution

```
verbx render in_7p1.wav out/072_7p1_conv.wav --engine conv --ir ir_7p1_matrix.wav --ir-matrix-
↪ layout output-major
```

What it sounds like: The same as recipe 71 but for 7.1 – eight output channels. The additional side channels extend the envelopment around the listener.

DSP note: 7.1 requires an IR with at least 8 channels (or a matrix of 8 output IRs). The side-surround channels in 7.1 typically receive IRs with slightly later direct energy and more diffuse early reflections than the rear surrounds, creating a front-to-back sense of depth.

7.15.3 Recipe 73: Wide 5.1 Algorithmic Hall

```
verbx render in_5p1.wav out/073_5p1_algo.wav --engine algo --rt60 85 --width 1.6
```

What it sounds like: A 5.1 algorithmic reverb. Faster to compute than convolution and more flexible for creative work.

DSP note: Algorithmic multichannel reverb applies the reverb engine independently or jointly across the output channels, depending on the engine's internal architecture. Width at 1.6 in a 5.1 context affects the spread between left and right channels but does not directly control the front-to-back depth, which is managed by the engine's channel assignment logic.

7.15.4 Recipe 74: Three-Pass 7.1 Beast Field

```
verbx render in_7p1.wav out/074_7p1_algo_beast.wav --engine algo --beast-mode 12 --repeat 3
```

What it sounds like: High-density algo reverb on a 7.1 input, with three repeat passes. Very dense, enveloping. Not for dialogue – this is for music or texture.

DSP note: Beast-mode 12 in multichannel increases the diffusion network complexity proportionally across channels. Three repeat passes in 7.1 produce a very large output file. Ensure you have sufficient storage before running this recipe.

7.15.5 Recipe 75: Frozen 5.1 Moment

```
verbx render in_5p1.wav out/075_5p1_freeze.wav --engine algo --freeze --start 3 --end 4.5 --repeat 2
```

What it sounds like: A frozen 5.1 pad extracted from the 3–4.5 second range, repeated twice. Creates a multichannel ambient drone.

DSP note: Freeze in multichannel mode extracts the specified window from all channels simultaneously, maintaining the spatial coherence of the frozen segment. The frozen loop, when processed through the algo reverb, produces a reverberant pad that preserves the original spatial distribution of the source material.

7.15.6 Recipe 76: 7.1 Shimmer Field

```
verbx render in_7p1.wav out/076_7p1_shimmer.wav --engine algo --shimmer --shimmer-mix 0.45
```

What it sounds like: Shimmer on 7.1. The pitch-shifted layer fills the entire surround field. Very immersive, slightly overwhelming on material with strong low-frequency content.

DSP note: The shimmer feedback path in multichannel wraps around all channels. The pitch-shifted output from any given channel feeds back into all channels based on the internal routing matrix. This means shimmer in multichannel is not simply the stereo shimmer experience multiplied – it has a distinct spatial character because the feedback paths are longer and more complex.

7.15.7 Recipe 77: Broadcast-Loudness 5.1 Render

```
verbx render in_5p1.wav out/077_5p1_target.wav --target-lufs -23 --target-peak-dbfs -2
```

What it sounds like: 5.1 loudness normalization to the European broadcast standard (–23 LUFS) with –2 dBFS peak limit.

DSP note: Integrated LUFS for multichannel uses the channel-weighting defined in ITU-R BS.1770, which applies –1.5 dB attenuation to surround channels (Ls, Rs) in the loudness measurement. This means a 5.1 mix at –23 LUFS measured will have higher actual SPL in the surrounds than a stereo mix at the same measured loudness value. Be aware of this when comparing multichannel and stereo versions.

7.15.8 Recipe 78: Float32 Full-Scale 7.1 Render

```
verbx render in_7p1.wav out/078_7p1_fullscale.wav --output-peak-norm full-scale --out-subtype
↪ float32
```

What it sounds like: 7.1 output normalized to full scale in float32 format. Every bit of dynamic range used.

DSP note: `--output-peak-norm full-scale` applies gain to bring the loudest sample in the entire multichannel file to 0 dBFS. In float32, this is representable without clipping. This is an intermediate processing format – do not deliver float32 full-scale as a final output without confirming the downstream system can handle it.

7.15.9 Recipe 79: Twelve-Second 5.1 Tail

```
verbx render in_5p1.wav out/079_5p1_tailcap.wav --engine conv --ir ir_5p1_matrix.wav --tail-limit
↪ 12
```

What it sounds like: A 5.1 convolution reverb with a 12-second tail cap. Short enough that the surround field does not overwhelm the dry signal.

DSP note: 12 seconds is generous for most indoor spaces but short for cathedrals and large outdoors captures. Truncating at 12 seconds in multichannel reduces processing time significantly for long IRs. The fade-out applied at the truncation point should be long enough to avoid a click but short enough not to audibly shorten the perceived tail.

7.15.10 Recipe 80: Four-Minute 7.1 Tail

```
verbx render in_7p1.wav out/080_7p1_longtail.wav --engine conv --ir ir_7p1_matrix.wav --tail-limit
↪  240
```

What it sounds like: A 7.1 convolution with a four-minute tail cap. This is a large file and a long render. The surround field will sustain for an extremely long time.

DSP note: 240-second tail-limit on an 8-channel 48kHz float32 file is approximately 7.5GB of output. Monitor disk space. The long tail is primarily useful for theatrical contexts where the reverb needs to sustain through a long pause or for ambisonic processing where the tail contributes to a sense of space even at very low levels.

7.16 Section 9: Analysis, Suggestion, and Batch Tools (Recipes 81–90)

The analysis and batch tools are where `verbx` starts to function as a production pipeline rather than a single-render tool. These commands are less dramatic than the extremes in sections 1–8, but they are often more useful in practice. Understanding what your audio contains before applying reverb is good engineering. Batching a large set of renders with known parameters is how you scale up to feature-length projects.

The `suggest` command is worth studying – it uses analysis of the input to generate parameter recommendations, which can be a useful starting point for creative decisions or a sanity check on aggressive settings.

7.16.1 Recipe 81: Dotted-Eighth Pre-Delay at 96 BPM

```
verbx render in.wav out/081_predelay_8d.wav --pre-delay 1/8D --bpm 96 --engine algo
```

What it sounds like: The reverb is delayed by a dotted eighth note at 96 BPM (approximately 469ms). The reverb arrives on the off-beat, which can create a rhythmic push-pull effect.

DSP note: Tempo-synced pre-delay uses the BPM value to calculate the delay in milliseconds. A dotted eighth at 96 BPM = (60000 / 96) * 1.5 / 2 ≈ 469ms. This pre-delay duration means the reverb onset aligns with the dotted eighth note position. At 96 BPM, this falls between beats in a way that complements the rhythm without fighting it.

7.16.2 Recipe 82: Sixteenth-Triplet Pre-Delay at 132 BPM

```
verbx render in.wav out/082_predelay_triplet.wav --pre-delay 1/16T --bpm 132 --engine algo
```

What it sounds like: Pre-delay synced to a sixteenth-note triplet at 132 BPM. The reverb arrives in an unusual rhythmic position – syncopated and slightly off-center.

DSP note: A sixteenth-note triplet at 132 BPM = (60000 / 132) / 4 * (2/3) ≈ 75.8ms. This is within the Haas fusion zone (above 30ms), so it may or may not be perceived as a distinct echo – the perception depends on the source material and the reverb level.

7.16.3 Recipe 83: LUFS and EDR JSON Analysis

```
verbx analyze in.wav --lufs --edr --json-out out/083_analysis.json
```

What it sounds like: No audio output. Produces a JSON file with integrated loudness (LUFS) and early decay rate (EDR) measurements of the input file.

DSP note: EDR (Energy Decay Relief) is a time-frequency representation of how energy decays across the frequency spectrum. It is calculated from the short-time power spectral density as a function of time. High EDR at a specific frequency means that frequency decays more slowly than others – this identifies resonant problems in room recordings and helps predict where synthetic reverbs will be tonally uneven.

7.16.4 Recipe 84: Framewise EDR Export

```
verbx analyze in.wav --frames-out out/084_frames.csv --edr
```

What it sounds like: No audio output. Produces a CSV file with per-frame EDR measurements. Suitable for import into data analysis or visualization tools.

DSP note: Per-frame analysis provides time-resolved measurement at the frame rate defined by the analysis window size (typically 512 or 1024 samples). The CSV format is useful for plotting or further analysis in Python, R, or MATLAB. The EDR frames show how the energy decay distribution changes over time – useful for identifying time-varying resonances in long recordings.

7.16.5 Recipe 85: Input-Aware Parameter Suggestions

```
verbx suggest in.wav
```

What it sounds like: No audio output. Prints parameter suggestions to the console based on analysis of the input file.

DSP note: The suggestion engine analyzes the input's dynamic range, spectral centroid, transient density, and estimated RT60 (if the input contains room ambience), then recommends `--rt60`, `--damping`, `--pre-delay-ms`, and `--wet`/`--dry` values appropriate to the input's character. These are starting points, not definitive recommendations. Use them to anchor your creative decisions, then adjust from there.

7.16.6 Recipe 86: Batch Manifest Template

```
verbx batch template > out/086_manifest.json
```

What it sounds like: No audio output. Generates a template batch manifest JSON file.

DSP note: The manifest file defines a list of render jobs with all parameters specified. The template contains example entries for each engine type and common parameter configurations. Edit the manifest to define your actual batch before running recipes 87–88.

7.16.7 Recipe 87: Longest-First Parallel Batch

```
verbx batch render manifest.json --jobs 8 --schedule longest-first --retries 1
```

What it sounds like: Executes all renders defined in manifest.json using 8 parallel workers, processing the longest renders first, with one retry on failure.

DSP note: `--schedule longest-first` is a work-stealing heuristic that prioritizes long renders early so that parallel workers remain active throughout the batch. If you schedule shortest-first, you may complete many small renders quickly but then have only one long render running at the end, leaving 7 workers idle. Longest-first minimizes total wall-clock time for heterogeneous batches.

7.16.8 Recipe 88: Shortest-First Batch Dry Run

```
verbx batch render manifest.json --jobs 4 --schedule shortest-first --dry-run
```

What it sounds like: No audio output. Dry run – prints what would be rendered without executing anything.

DSP note: Always run `--dry-run` first on a new manifest to verify that all paths are correct and all parameters are valid before committing to a long batch render. The dry-run output shows the resolved parameters for each job, including any defaults that will be applied, which makes it easy to spot configuration errors.

7.16.9 Recipe 89: Cache Inventory

```
verbx cache info
```

What it sounds like: No audio output. Prints information about the convolution cache – what is cached, how large, when last accessed.

DSP note: `verbx` caches partitioned IR FFTs to avoid recomputing them on repeated renders with the same IR. The cache is particularly valuable for batch renders where the same IR is used across many jobs. Cache entries include the IR file path, partition size, sample rate, and the cached FFT data. Stale cache entries (from modified IRs) are detected by file modification time.

7.16.10 Recipe 90: Cache Cleanup

```
verbx cache clear
```

What it sounds like: No audio output. Clears the convolution cache.

DSP note: Clearing the cache forces recomputation of IR FFTs on the next render. Do this after modifying IR files that you have previously used, or when disk space is constrained. The cache is stored in a platform-specific location (typically in the user's application data directory). Cache clearing does not affect output files or manifests.

7.17 Section 10: Lucky-Mode Wildcards (Recipes 91–100)

Lucky mode is what happens when you hand the parameter space to the engine and say "find something interesting." It randomizes settings within configured bounds and generates a specified number of variations in a single run. It is not random in the degenerate sense – the parameter space is constrained to avoid unlistenable results – but it covers territory you would not explore by hand.

The practical value of lucky mode is speed of discovery. When you do not know what you want, running lucky with a good seed can surface directions you would not have chosen deliberately. The seed is important: fix it for reproducibility, vary it for exploration. Documenting seeds that produce interesting results is worth doing – a single seed number represents an entire family of related variations.

Lucky mode is also useful at the end of a project when you think you are done. Run a lucky set on your primary source material. One of the variations will be better than what you have. This is not always true. But it is true often enough to make the practice worthwhile.

7.17.1 Recipe 91: Five Lucky Variations

```
verbx render in.wav out/lucky.wav --lucky 5 --lucky-out-dir out/lucky_01
```

What it sounds like: Five random variations of the input with randomized parameters, written to the `out/lucky_01` directory.

DSP note: Without `--lucky-seed`, each run produces different results. Lucky mode randomizes within safe parameter ranges – RT60 is bounded to prevent extremely short or extremely long tails, beast-mode is bounded to avoid instability, wet/dry ratios are biased toward useful mixes. The five files are numbered and can be compared directly.

7.17.2 Recipe 92: Ten Seeded Lucky Variations

```
verbx render in.wav out/lucky.wav --lucky 10 --lucky-out-dir out/lucky_02 --lucky-seed 2026
```

What it sounds like: Ten variations with a fixed seed. Running this command again will produce identical outputs. Use this when you want to share a specific lucky set with collaborators.

DSP note: The seed initializes the pseudo-random number generator used for parameter selection. A fixed seed ensures deterministic parameter sampling, meaning the same `--lucky-seed` on the same input file will always produce the same output files. Vary the seed to explore different families of variations.

7.17.3 Recipe 93: Twenty-Five Device-Aware Variations

```
verbx render in.wav out/lucky.wav --lucky 25 --lucky-out-dir out/lucky_03 --device auto
```

What it sounds like: Twenty-five variations with automatic GPU/CPU device selection for rendering. If a GPU is available, it will be used; otherwise falls back to CPU.

DSP note: `--device auto` allows `verbx` to select the most efficient compute device. For convolution-heavy lucky sets, GPU acceleration can significantly reduce render time. For algo-only sets, the difference is less pronounced. The device selection is made once at the start of the batch and applied to all 25 renders.

7.17.4 Recipe 94: Headless Fifty-Variation Run

```
verbx render in.wav out/lucky.wav --lucky 50 --lucky-out-dir out/lucky_04 --no-progress
```

What it sounds like: Fifty variations with progress output suppressed. This is designed for batch environments where terminal output is logged.

DSP note: `--no-progress` disables the progress bar and per-file status output. In CI/CD pipelines, Docker containers, or other environments where terminal control codes cause issues in log output, suppressing progress is important for log readability. Output errors and warnings are still printed.

7.17.5 Recipe 95: Eight Algorithmic Wildcards

```
verbx render in.wav out/lucky.wav --lucky 8 --lucky-out-dir out/lucky_05 --engine algo
```

What it sounds like: Eight algorithmic reverb variations. Lucky mode constrained to the algo engine – no convolution.

DSP note: Constraining lucky mode to a specific engine narrows the parameter space and produces a more consistent set of variations. All eight files will share the same engine character (algo diffusion) but vary in RT60, modulation, beast-mode, pre-delay, and other algo-specific parameters.

7.17.6 Recipe 96: Eight Convolution Wildcards

```
verbx render in.wav out/lucky.wav --lucky 8 --lucky-out-dir out/lucky_06 --engine conv --ir hall.wav
```

What it sounds like: Eight convolution variations using the hall IR. Lucky mode varies the convolution parameters – partition size, tail limit, repeat, normalization – rather than the IR itself.

DSP note: When lucky mode is constrained to the conv engine with a specific IR, the randomization applies to the processing around the convolution: pre-delay, wet/dry balance, tail truncation, repeat count, and loudness. The hall IR's character is consistent across all eight; the variations represent different ways of using that IR.

7.17.7 Recipe 97: Twelve Self-Convolution Wildcards

```
verbx render in.wav out/lucky.wav --lucky 12 --lucky-out-dir out/lucky_07 --self-convolve
```

What it sounds like: Twelve self-convolution variations. Wild. Some will be unusable; some will be remarkable. The variance is higher in self-convolve lucky mode than in any other configuration.

DSP note: Self-convolve lucky mode varies the beast-mode level, tilt, tail limit, shimmer parameters, and repeat count within ranges that produce interesting results rather than simply clipped noise. That said, the signal path for self-convolution is inherently less stable than algo or conv modes, so the variance between "good" and "bad" outputs is higher. Plan to audition all 12.

7.17.8 Recipe 98: Loudness-Matched Lucky Set

```
verbx render in.wav out/lucky.wav --lucky 15 --lucky-out-dir out/lucky_08 --target-lufs -20
```

What it sounds like: Fifteen variations, each normalized to –20 LUFS. All outputs are at a consistent loudness, making A/B comparison easier.

DSP note: Applying `--target-lufs` to lucky mode normalizes each output independently after rendering. This is useful for comparative listening because the loudness differences between variations are removed, allowing you to evaluate the texture and character rather than the level. –20 LUFS is a comfortable listening level for extended auditioning.

7.17.9 Recipe 99: Thirty Float32 Variations

```
verbx render in.wav out/lucky.wav --lucky 30 --lucky-out-dir out/lucky_09 --out-subtype float32
```

What it sounds like: Thirty float32 variations. The float format preserves headroom for downstream processing – useful if you plan to layer or process the lucky outputs further.

DSP note: Using float32 for lucky output means you do not need to worry about clipping during the render – the format accommodates values above 0 dBFS. This is particularly useful with self-convolve and beast-mode combinations in lucky mode, where the amplitude is less predictable. Process and normalize at the final stage of your signal chain rather than at the lucky render stage.

7.17.10 Recipe 100: Hundred-Variation Reproducibility Stress Test

```
verbx render in.wav out/lucky.wav --lucky 100 --lucky-out-dir out/lucky_10 --lucky-seed 404
```

What it sounds like: One hundred variations, fixed seed 404. The largest single lucky run in this cookbook. Expect a long render time and a large directory.

DSP note: A 100-variation lucky set with a fixed seed is a complete exploration of one corner of the parameter space. With seed 404, the parameter sampling will explore the same 100 points in the parameter space every time. This is useful as a reference set – a known, reproducible collection of 100 reverb treatments for a given input. At 48kHz stereo float32, even 60-second outputs will total several gigabytes. Plan accordingly.

> **Expert note:** The seed 404 is not special. But it is documented here as a reference: if you run recipe 100 on the same input file and do not get the same output, something has changed in the parameter sampling algorithm, the random number generator implementation, or the engine version. Treat the output of seed 404 as a regression test for reproducibility.

7.18 Notes for Large Runs

Start with a small lucky count (3–5) to confirm runtime and output size before committing to a 50 or 100-variation run.

Use `--no-progress` in log-friendly batch environments. The progress bar uses terminal control codes that corrupt plain-text logs.

Fix `--lucky-seed` whenever reproducibility matters. Document the seed alongside the source file and any post-processing applied.

Monitor storage. Long tails, high repeat counts, and multichannel outputs in float format accumulate quickly. A batch of 100 stereo float32 files at 3 minutes each is approximately 34GB at 48kHz.

The `verbx cache info` and `verbx cache clear` commands (recipes 89–90) are useful before and after large batch runs. Clear the cache if disk space is constrained; leave it in place if you will be reusing the same IRs across multiple batches.

When a run produces something you want to keep, immediately rename the output file out of the numbered sequence. The output directory will be overwritten if you rerun the batch with the same output path.

Part II

Plug-in Architecture and Operational Cards

7.19 Suggested Exercises

1. Select five recipes that target the same source class and arrange them from most transparent to most transformed; explain the progression.
2. Take one recipe and create a conservative, moderate, and extreme version while changing only the parameters essential to its stated effect.
3. Convert one recipe into a reusable preset and add a machine-readable analysis report to the output directory.
4. Test one recipe on speech, percussion, and sustained harmony. Identify which source property determines whether it succeeds.
5. Invent a recipe title and provide a runnable command, a listening goal, a safety check, and one likely failure mode.
6. Combine two compatible recipes in serial and in parallel, then explain which result preserves source intelligibility better.
7. Translate one recipe to a different tempo and meter while retaining its musical timing relationship.
8. Create a troubleshooting variant of one recipe with deliberately conservative settings and compare it to the published extreme version.
9. Organize ten recipes into a preset browser taxonomy based on musical function rather than algorithm name.
10. Document a recipe failure on unsuitable material and propose a revised source-selection rule for future users.

“Repeating your ultimate word.”

— John Godfrey Saxe, *The Story of Echo* (1865)

Chapter 8

VERBX AUv3/VST3 Plug-in Handbook

This handbook describes the VERBX plug-in direction, the native foundation that already exists, and the operational practices required to turn that foundation into a dependable AU, AUv3, VST3, and standalone product. It is deliberately honest about maturity. The repository contains a tested parameter manifest, a realtime context boundary, a guarded C++17/JUCE shell, state serialization, and a realtime spectrum-overlay component, a complete initial control dock, and an allocation-free mono/stereo Schroeder reverb with real wet-path oversampling. It is a usable native engine, but not yet the final multichannel architecture. The full-screen image below is the approved visual target and a live design-prototype capture, not a screenshot of a shipping binary.

Figure 8-1 presents VERBX full-screen plug-in design and establishes the visual reference for the discussion that follows.

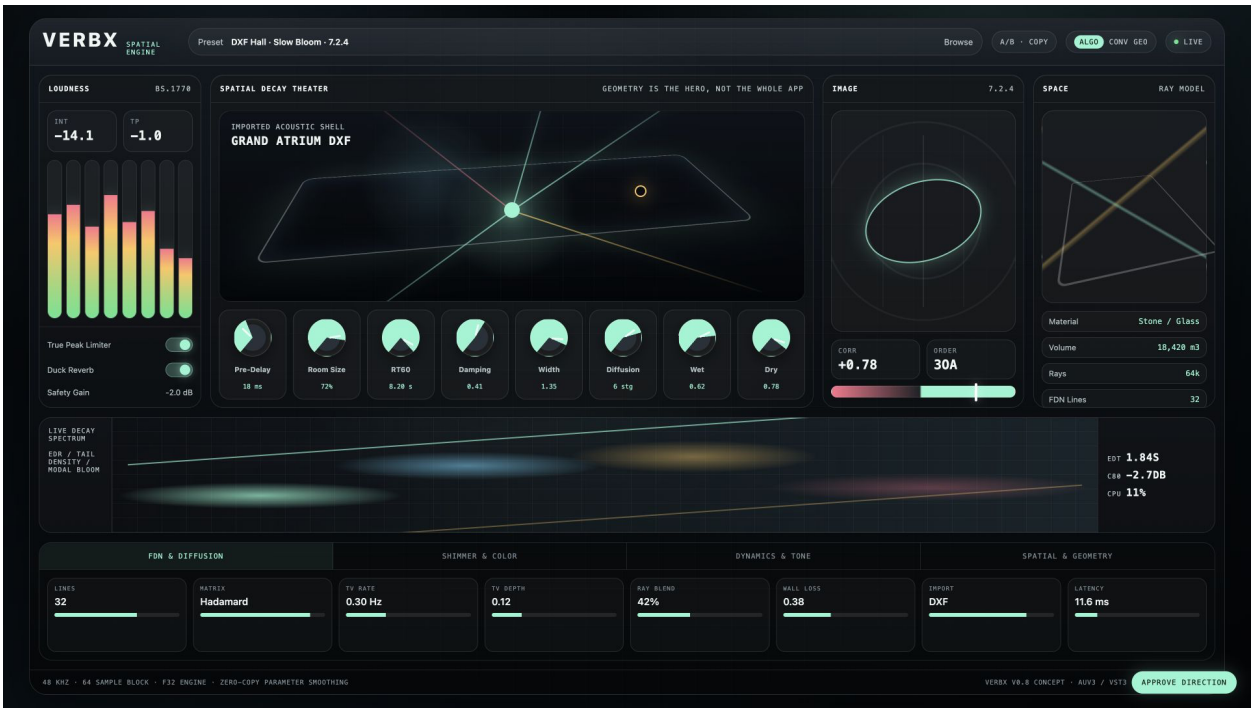


Figure 8-1: VERBX full-screen plug-in design

The next image is an actual capture of the compiled JUCE standalone editor. Its input is muted by the host, so the post-DSP trace sits at the analyzer floor; the logarithmic frequency grid and overlaid display are production C++ rather than browser-prototype artwork.

Figure 8-2 presents VERBX native realtime spectrum analyzer and establishes the visual reference for the discussion that follows.

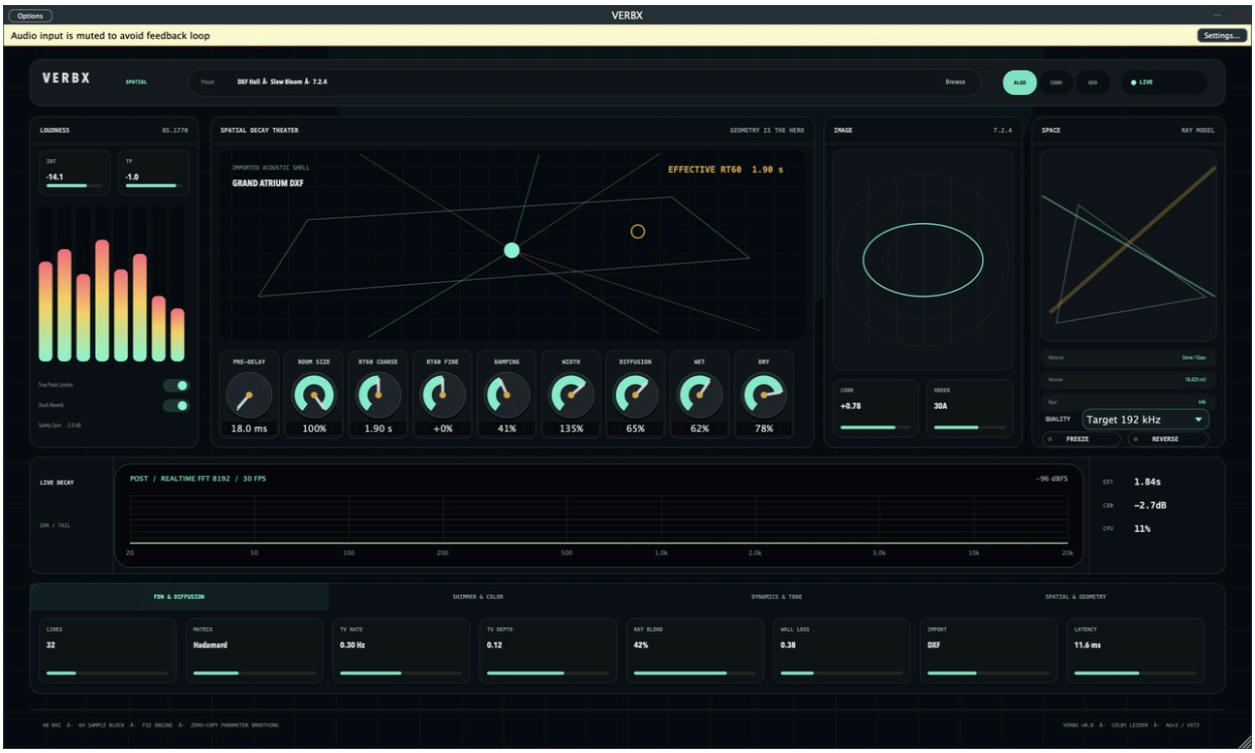


Figure 8-2: VERBX native realtime spectrum analyzer

The next image is generated directly from the compiled editor interaction smoke test. Expert mode keeps every continuous parameter available as both a dial and a precision fader, retains the realtime analyzer, and adds five four-way selector banks. These selectors write existing host state rather than creating hidden or decorative settings.

Figure 8-3 presents VERBX compiled Expert control matrix and establishes the visual reference for the discussion that follows.

Figure 8-3: VERBX compiled Expert control matrix

8.1 Product Intent

VERBX is designed as a spatial architecture instrument rather than a generic reverb with a decorative room picture. Geometry, decay, imaging, dynamics, and quality status should be readable at a glance. The front panel must remain playable even though the underlying engine has a deep parameter surface. That leads to three control layers:

- A performance layer for the controls a musician or mixer reaches for during playback: pre-delay, room size, RT60 coarse and fine, damping, width, diffusion, wet, dry, Freeze, Reverse, and quality.
- An expert layer that currently provides linked precision control and safe macros for quality, width, decay, mix routing, and tail character; future FDN topology, modulation, dynamics, and geometry controls land only after those behaviors are realtime-safe and stable.
- An automation layer for parameters that a host can recall and automate even when they are not continuously visible on the main page.

The plug-in must never imply that visual complexity is equivalent to acoustic accuracy. A geometry display is useful only when it corresponds to validated room metadata, an early-reflection model, or a deterministic imported asset. The display should state whether it is showing a parametric room, a DXF-derived profile, a measured impulse response, or an illustrative preview.

8.1.1 Recommended Companion Text

Will C. Pirkle's *Designing Audio Effect Plugins in C++: For AAX, AU, and VST3 with DSP Theory*, 2nd edition (Routledge, 2019), is the recommended companion text for this handbook. Its treatment of plug-in anatomy, an API-independent processing core, host wrappers, parameters, GUI design, delay structures, reverberation, and dynamics provides a useful engineering vocabulary for the boundaries

described here. VERBX does not adopt Pirkle's example framework wholesale: its C realtime contract, JUCE adapter, parameter manifest, state format, analyzer telemetry, and validation matrix remain repository-specific designs. Readers should use the book for the broader implementation discipline and this handbook for the exact VERBX contracts.

8.2 What Exists In The Repository

The first foundation slice is intentionally narrow and testable:

- `verbx_c_core` is a reusable C11 static library target.
- `plugin_params.h` defines twelve stable initial parameters and the quality choices.
- `plugin_params.c` implements deterministic clamping and logarithmic RT60 coarse/fine mapping from 0.01 seconds to 360 seconds.
- `plugin_realtime.h` defines host configuration, realtime parameters, status, context lifecycle, latency accessors, and processing entry points.
- `plugin_realtime.c` validates host configuration, allocates persistent state only during preparation, and provides bounded mono/stereo Schroeder processing with pre-delay, room scale, RT60, damping, diffusion, width, wet/dry, Freeze, and a zero-lookahead reverse-style swell. Continuous controls use 20 ms smoothing inside the native state so host automation does not zipper. Host, 2x, 4x, and Target quality modes execute that wet network at the reported internal rate without callback allocation.
- `native/verbx_plugin` contains the guarded JUCE shell for AU, AUv3, VST3, and standalone targets.
- The processor caches atomic parameter pointers during construction so the callback reads values without rebuilding strings or searching parameter maps.
- The resizable editor includes a post-DSP spectrum overlay. A fixed SPSC ring carries mono output snapshots off the callback; the message thread performs an 8192-point Hann FFT at 30 visual frames per second, with logarithmic frequency spacing, release smoothing, and a decaying peak trace.
- Perform and Expert are native JUCE pages. Expert contains nine linked rotary controls, nine precision faders, the live analyzer, and twenty selector buttons; all write the existing APVTS host state.
- The complete initial twelve-parameter surface is attached to host automation, with compact musical units and a live effective-RT60 readout.

This boundary is valuable even before the reverb DSP is connected. Host code, parameter identity, state recall, bus negotiation, callback constraints, and latency reporting can be stabilized independently of the sound engine.

8.3 Building The Plug-in Foundation

Default repository configuration does not require JUCE:

```
cmake -S native/verbx_plugin -B build/native/verbx_plugin
```

The configure output should say that `VERBX_ENABLE_JUCE_PLUGIN` is off. This is the supported path for contributors who only need the C core or Python CLI.

When JUCE is installed as a discoverable CMake package, enable the actual host targets:

```
cmake -S native/verbx_plugin -B build/native/verbx_plugin-juce \
  -DVERBX_ENABLE_JUCE_PLUGIN=ON
cmake --build build/native/verbx_plugin-juce --config Release
```

For a JUCE source checkout, add `-DVERBX_JUCE_SOURCE_DIR=/path/to/JUCE` to the configure command.

The enabled target requests AU, AUv3, VST3, and Standalone formats. A successful compile is only the beginning of validation. The resulting formats must be scanned, instantiated, automated, saved, reopened, and stress-tested in their actual hosts before compatibility is claimed.

8.4 Signal And Ownership Boundaries

The plug-in has three architectural layers. The host shell owns format-specific lifecycle, buses, state, parameters, editor creation, and host latency notification. The adapter maps host blocks and normalized automation into the native realtime contract. The DSP core owns audio behavior and must remain testable without a DAW.

Figure 8-4 presents Algorithmic signal topology and establishes the visual reference for the discussion that follows.

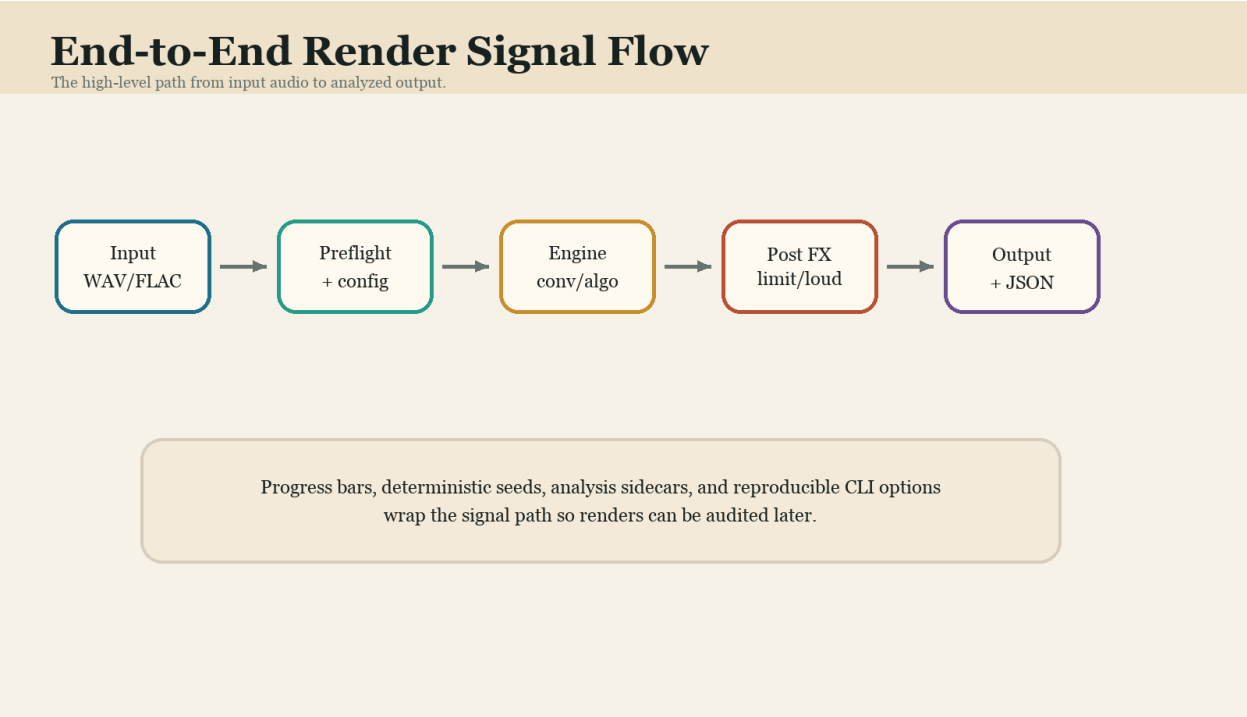


Figure 8-4: Algorithmic signal topology

The audio callback is a hard boundary. It must not parse DXF, open files, update the preset browser, allocate variable-sized containers, rebuild parameter IDs, take UI locks, or wait for background work. Large assets are prepared away from the callback and swapped through bounded, versioned handles. Telemetry flows in the opposite direction through compact snapshots that the editor may poll.

8.5 Precision And Sample Rate

The DAW controls project sample rate. VERBX cannot force a host session to 192 kHz. The default quality choice, Target 192 kHz, selects the smallest integer factor whose internal rate reaches or exceeds 192 kHz. At a 48 kHz host rate this is 4x/192 kHz; at 96 kHz it is 2x/192 kHz; at 44.1 kHz it is 5x/220.5 kHz; and at 192 kHz or above it does not increase the rate. Host, 2x, and 4x select their exact factors. The wet network uses causal linear interpolation and box-filter decimation, while the dry path retains the original host samples. Quality changes allocate and prepare away from the callback, then cross the processor boundary through a nonblocking atomic guard.

The initial callback contract uses 32-bit float because that is the common native exchange type for AU/VST processing and keeps bandwidth bounded. This does not forbid double-precision accumulators, offline float64 rendering, or a future double-precision host path. Precision claims should always distinguish host buffer format, internal state precision, oversampled processing rate, and file-render output format.

Figure 8-5 presents Sample-rate cost and establishes the visual reference for the discussion that follows.

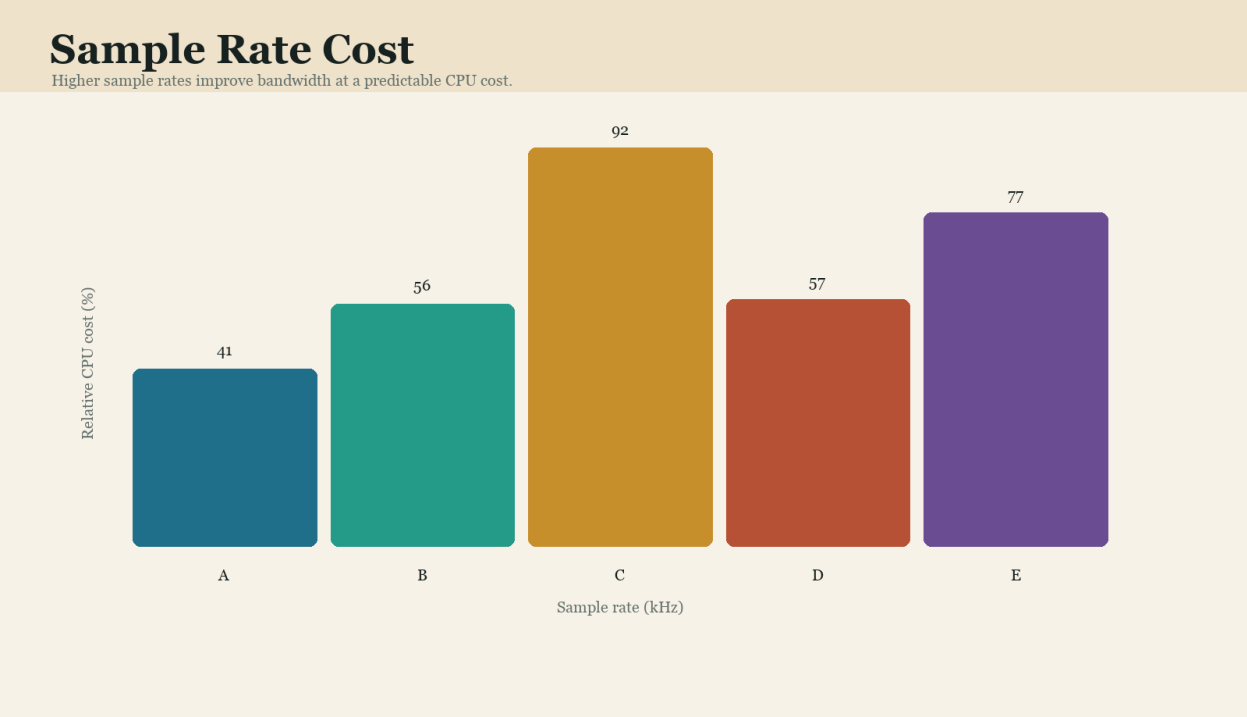


Figure 8-5: Sample-rate cost

8.6 RT60 Coarse And Fine

One linear knob cannot offer useful control from 0.01 seconds to 360 seconds. The coarse control therefore maps normalized automation logarithmically:

```
coarse = exp(log(0.01) + normalized * (log(360.0) - log(0.01)))
fine_ratio = exp(log(1.20) * bipolar_fine)
effective = clamp(coarse * fine_ratio, 0.01, 360.0)
```

The fine control provides a plus-or-minus 20 percent log-space trim. It remains musically useful around a short booth, a medium hall, and an enormous ambient tail. The effective value shown by the editor is authoritative; the coarse knob position alone is not.

Freeze is separate from maximum RT60. Freeze changes the energy behavior of the network and needs its own smoothing and safety semantics. Reverse is also a separate mode because it introduces a fundamentally different envelope and may require buffering and reported latency.

Figure 8-6 presents RT60 decay families and establishes the visual reference for the discussion that follows.

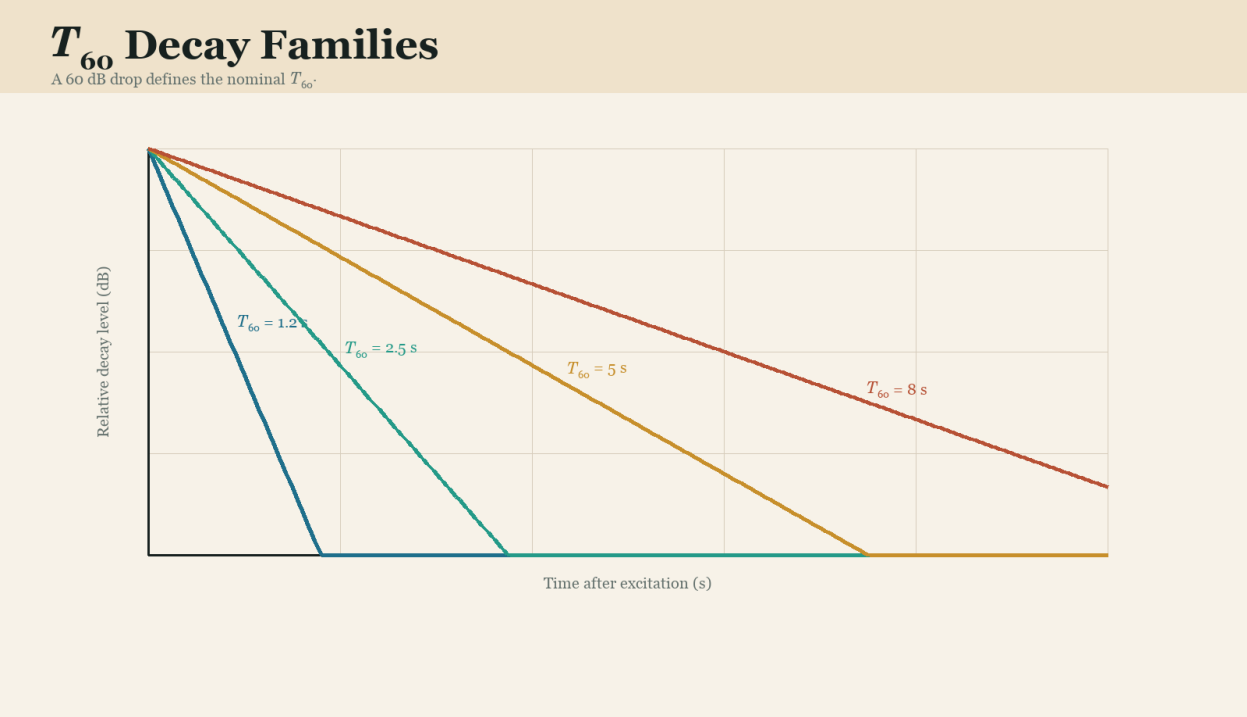


Figure 8-6: RT60 decay families

8.7 Initial Parameter Reference

The first manifest contains twelve parameters. Their IDs are intended to remain stable once released because DAW automation and saved sessions depend on them.

Table 8-1: Initial Parameter Reference: reference table

Parameter ID	User meaning	Initial range	Default	Realtime note
`pre_delay_ms`	Gap before the reverberant field	0 to 1000 ms	18 ms	Delay changes require smoothing or crossfade
`room_size`	Macro geometry/scale control	0 to 1	0.72	Must not resize unbounded memory in callback
`rt60_coarse`	Logarithmic decay position	0 to 1	0.50	Maps to 0.01 to 360 seconds
`rt60_fine`	Bipolar log trim	–1 to 1	0	Applies about plus/minus 20 percent
`damping`	High-frequency decay loss	0 to 0.98	0.41	Coefficients need stable interpolation
`width`	Stereo/spatial spread	0 to 2	1.35	Check mono and correlation behavior

Parameter ID	User meaning	Initial range	Default	Realtime note
`diffusion`	Echo-density macro	0 to 1	0.65	Structural changes may need a safe transition
`wet`	Processed contribution	0 to 1	0.62	Use a deliberate mix law
`dry`	Direct contribution	0 to 1	0.78	Preserve gain staging and bypass behavior
`freeze`	Infinite/sustaining mode	off/on	off	Separate energy-state transition
`reverse`	Reverse-envelope mode	off/on	off	Buffering and latency must be explicit
`quality_mode`	Internal rate policy	Host/2x/4x/ Target	Target	Reprepare outside callback when needed

8.8 State, Presets, And Session Recall

The JUCE shell serializes its parameter tree into host state. A production version also needs an explicit schema version, migration rules, asset identity, and diagnostics for partial restoration. A preset should not silently become a different sound because a geometry file moved or an IR cache was regenerated.

Small deterministic data belongs in host state. Large DXF files, measured IRs, and generated spatial assets should normally be referenced by a stable identity plus a content hash. The state can embed a compact fallback profile when that is small enough. Missing assets should produce a visible warning and a safe known sound, not silence, noise, or an unexplained default room.

8.9 Geometry And DXF

DXF import is an offline preparation workflow. Parsing, topology repair, ray generation, acceleration-structure construction, and IR synthesis must not run inside the audio callback. The realtime side consumes a validated bounded profile: room dimensions, material identifiers, source/listener transforms, early-reflection taps, spatial metadata, and optional prepared IR partitions.

The visual theater should show provenance. Labels such as `Parametric Room`, `Imported DXF Profile`, `Measured IR`, or `Illustrative View` prevent the user from confusing a decorative image with a physical simulation. Geometry edits can be staged in the editor and committed through a background preparation job; the audio thread receives the finished immutable result only after the preparation job completes.

Figure 8-7 presents Material absorption map and establishes the visual reference for the discussion that follows.

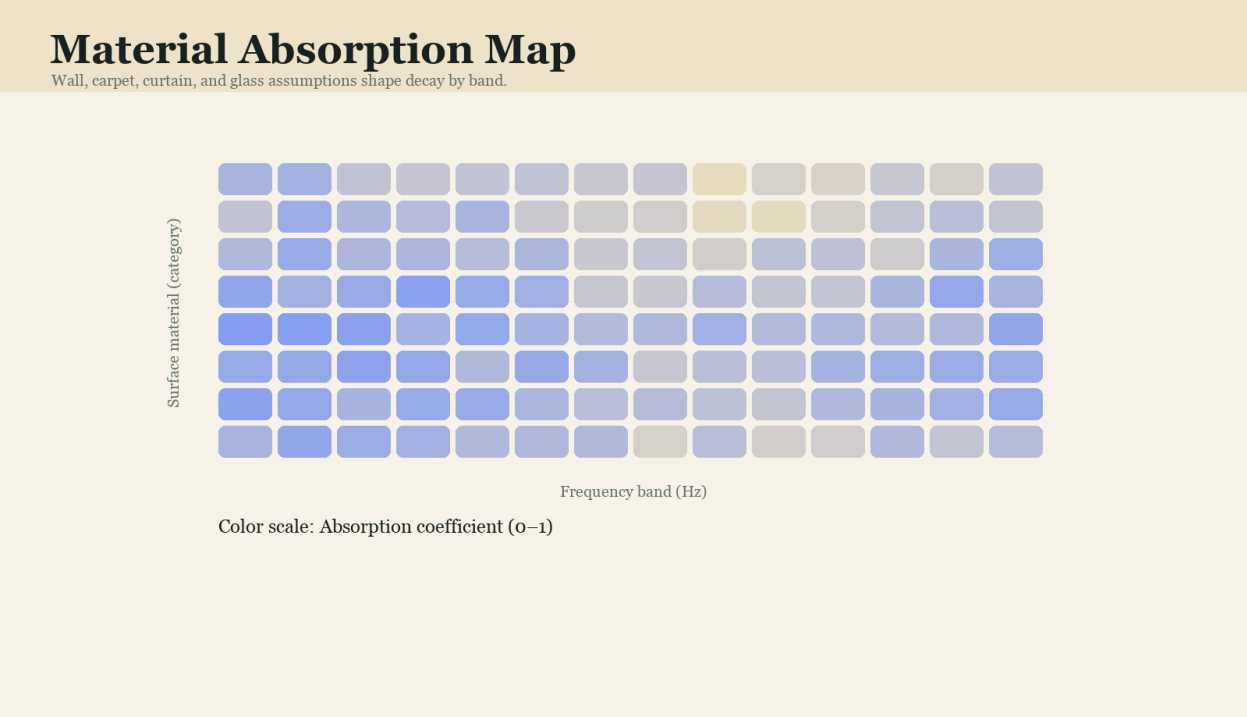


Figure 8-7: Material absorption map

8.10 Latency

Total plug-in latency may include block adaptation, oversampling filters, lookahead dynamics, convolution partitions, and reverse-reverb buffering. The processor must report the exact stable value to the host whenever a mode changes the processing graph. If latency cannot change safely during playback, the UI should state that the change will apply on transport stop or reprepare.

Figure 8-8 presents Realtime latency components and establishes the visual reference for the discussion that follows.

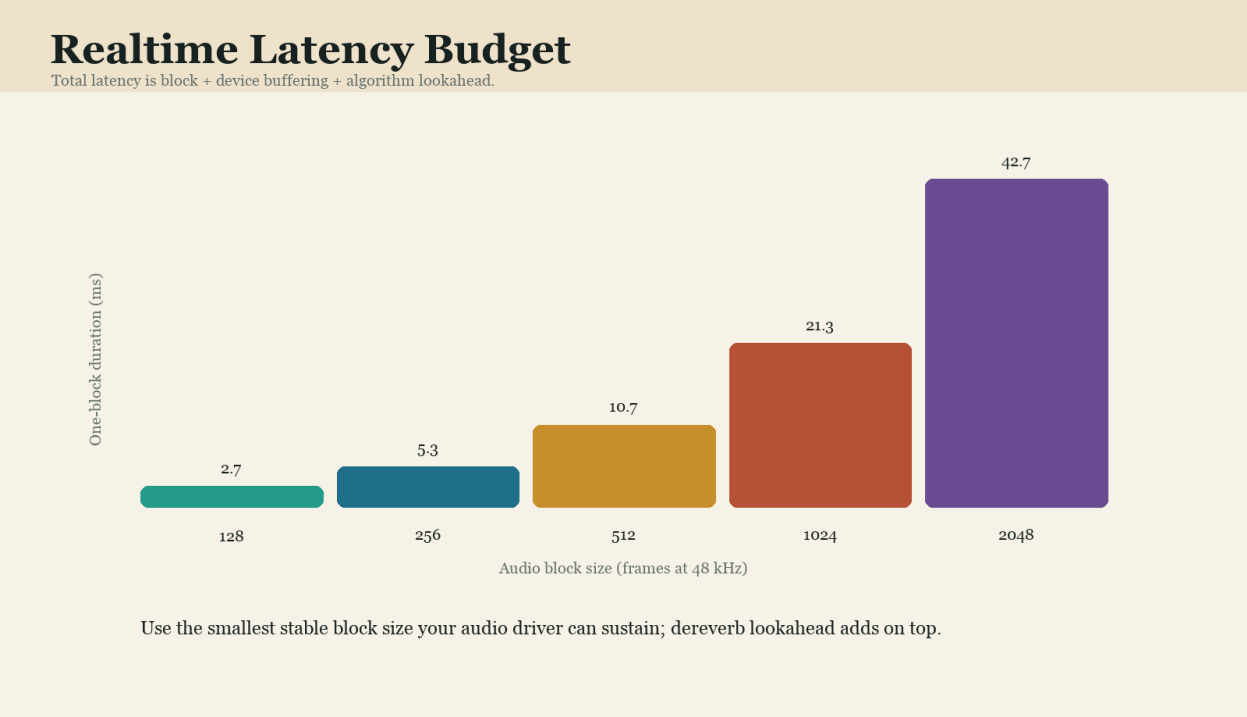


Figure 8-8: Realtime latency components

The current realtime reverb and causal oversampling path report zero frames: they do not buffer future host samples. That value must change when a latency-producing filter, convolution partition, or bounded reverse-lookahead stage lands. The live status display distinguishes host and internal rates, factor, block size, and algorithmic latency. Device I/O and end-to-end monitored latency remain separate measurements.

8.11 Realtime Safety

The callback should perform bounded arithmetic, pointer traversal, atomic loads, and DSP over memory prepared in advance. It should avoid filesystem access, logging, UI calls, mutex acquisition, heap allocation, and operations whose runtime grows unpredictably with external assets. Denormal handling, NaN/Inf containment, and explicit channel/block bounds are part of the audio contract.

Figure 8-9 presents Realtime callback budget and establishes the visual reference for the discussion that follows.

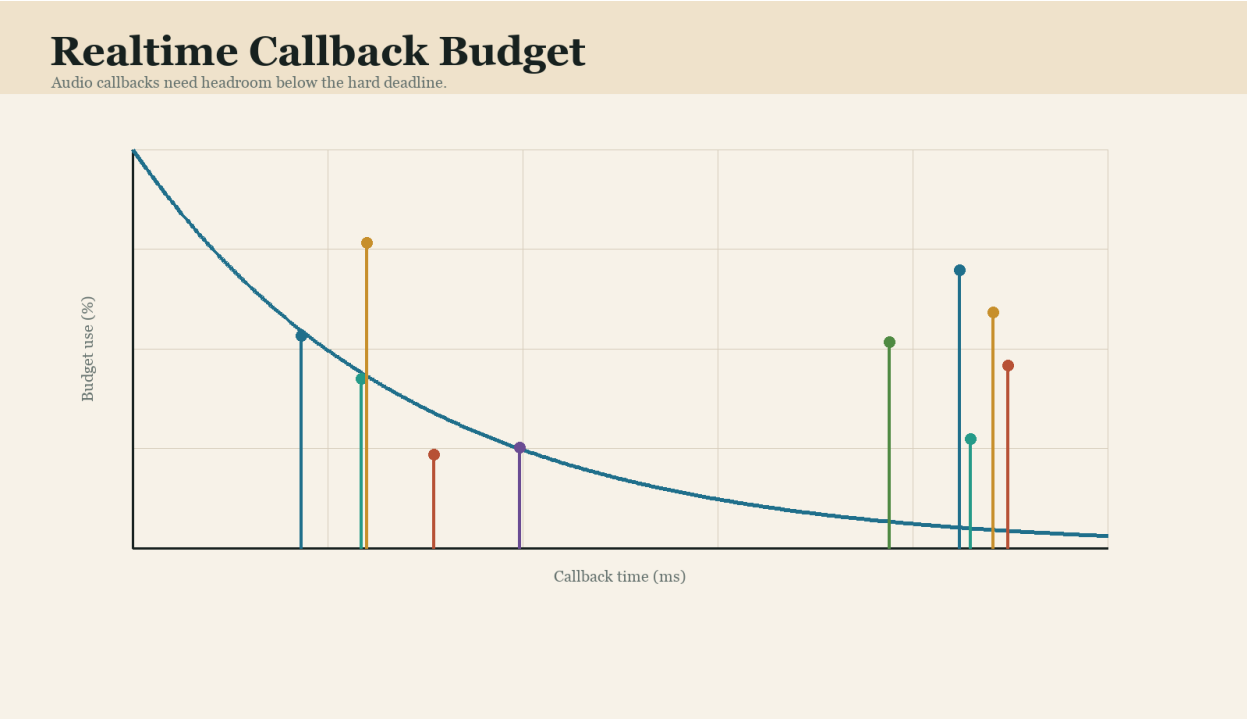


Figure 8-9: Realtime callback budget

Parameter smoothing is not cosmetic. Abrupt pre-delay, damping, mix, and matrix changes can click or destabilize feedback. Each parameter needs a declared transition strategy: sample ramp, coefficient interpolation, dual-engine crossfade, transport-gated reprepare, or intentionally discrete switch.

8.12 Freeze And Reverse

Freeze should enter and leave with controlled energy. The production algorithm must define whether input injection stops, decay gains approach unity, damping continues, modulation remains active, and limiter protection stays engaged. A freeze button is not permission for unbounded gain.

Figure 8-10 presents Infinite reverb behavior and establishes the visual reference for the discussion that follows.

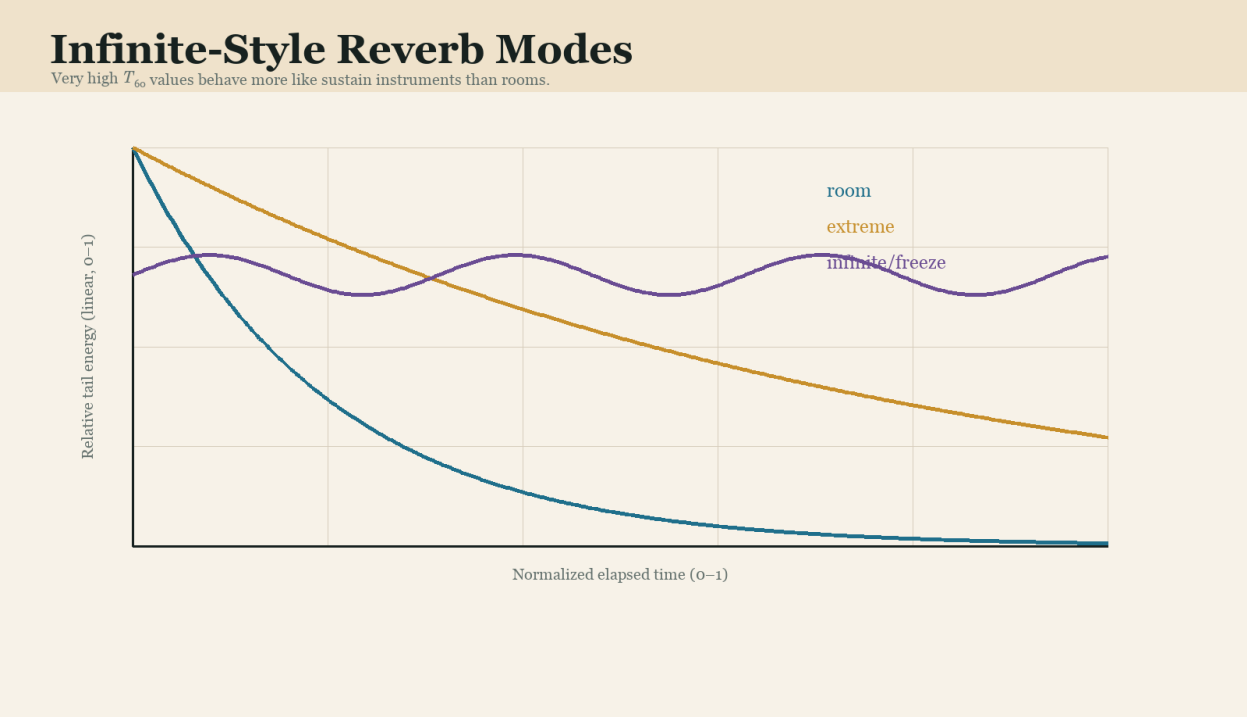


Figure 8-10: Infinite reverb behavior

Reverse needs an explicit latency model. A true reverse response requires future context, pre-rendered material, or a bounded capture window. The UI should show the active window and reported latency. If a low-latency approximation is offered, it must be named as an approximation rather than presented as identical to offline reverse rendering.

Figure 8-11 presents Reverse reverb envelope and establishes the visual reference for the discussion that follows.

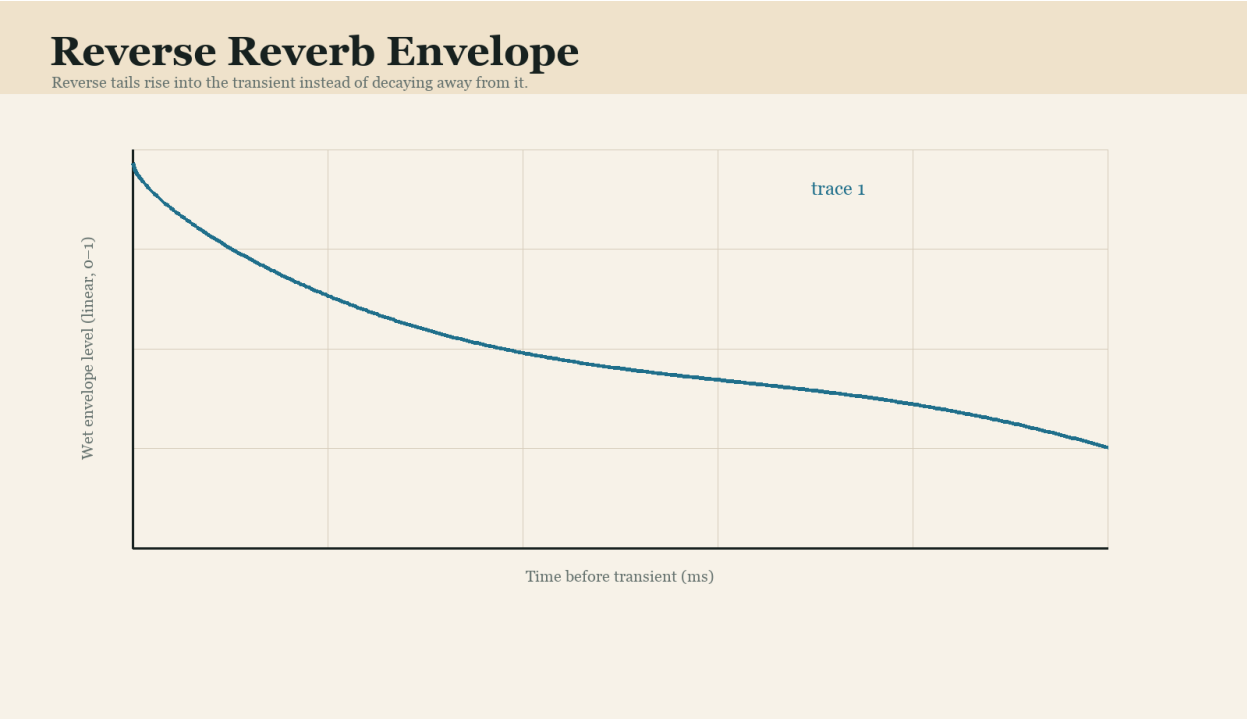


Figure 8-11: Reverse reverb envelope

8.12.1 Gated Reverse Reverb

Gated reverse reverb is a reverse or reverse-style wet field constrained by a gate or authored gain window. The gate belongs after the wet processor when the goal is to shape the audible tail. Placing it before the reverb only decides which dry events excite the network; it does not truncate the resulting field. Key the post-wet gate from the dry source, keep the plug-in return 100 percent wet, and leave the destination transient on its uncompromised dry path.

The current Reverse parameter is a zero-lookahead musical approximation. It detects a transient and raises the wet envelope after that event; it cannot anticipate audio that has not reached the callback. For a true pre-event gated reverse effect, print the wet return, move it earlier so its endpoint meets the destination transient, then apply a host gate or clip-gain window. A future capture-window mode must report at least the window duration as algorithmic latency and must never allocate, seek, or resize its buffer in the callback.

Useful controls for a production gated mode include lead duration, endpoint offset, detector source, detector filtering, threshold, hysteresis, attack, hold, release, range, and channel linking. A 5 to 20 ms terminal fade prevents a click without erasing the abrupt boundary. Tempo synchronization should derive the lead duration from $T_{\text{lead}} = 60N/B$, where N is the selected beat span and B is tempo in beats per minute.

Stereo and multichannel gates should normally share one detector and one gain envelope. Independent thresholds can make the image jump as channels close at different samples. Verify endpoint localization, stereo fold-down, binaural rendering, and host delay compensation. The UI must label the current zero-lookahead behavior separately from any future buffered or offline mode so users never mistake a bloom for a physically reversed response.

8.13 Loudness, Ducking, And Limiting

The limiter is a safety layer, not a substitute for stable feedback design. Meters should distinguish input, wet return, output, gain reduction, true peak, and any safety attenuation. Ducking should expose detector source, attack, release, range, and whether the dry path participates.

Host bypass semantics matter. A hard host bypass may skip internal tails, while an effect bypass parameter can preserve or drain them. The product must define both behaviors and test them in every supported format.

8.14 Bus Layouts And Spatial Formats

The foundation accepts matched mono or stereo main buses. Production expansion should proceed through explicit layouts rather than accepting arbitrary channel counts. Stereo, 5.1, 7.1, 7.1.4, and ambisonic modes have different routing, normalization, and host metadata requirements. A layout is supported only after processing, state recall, metering, and host scanning have been validated.

Figure 8-12 presents Speaker layout coverage and establishes the visual reference for the discussion that follows.

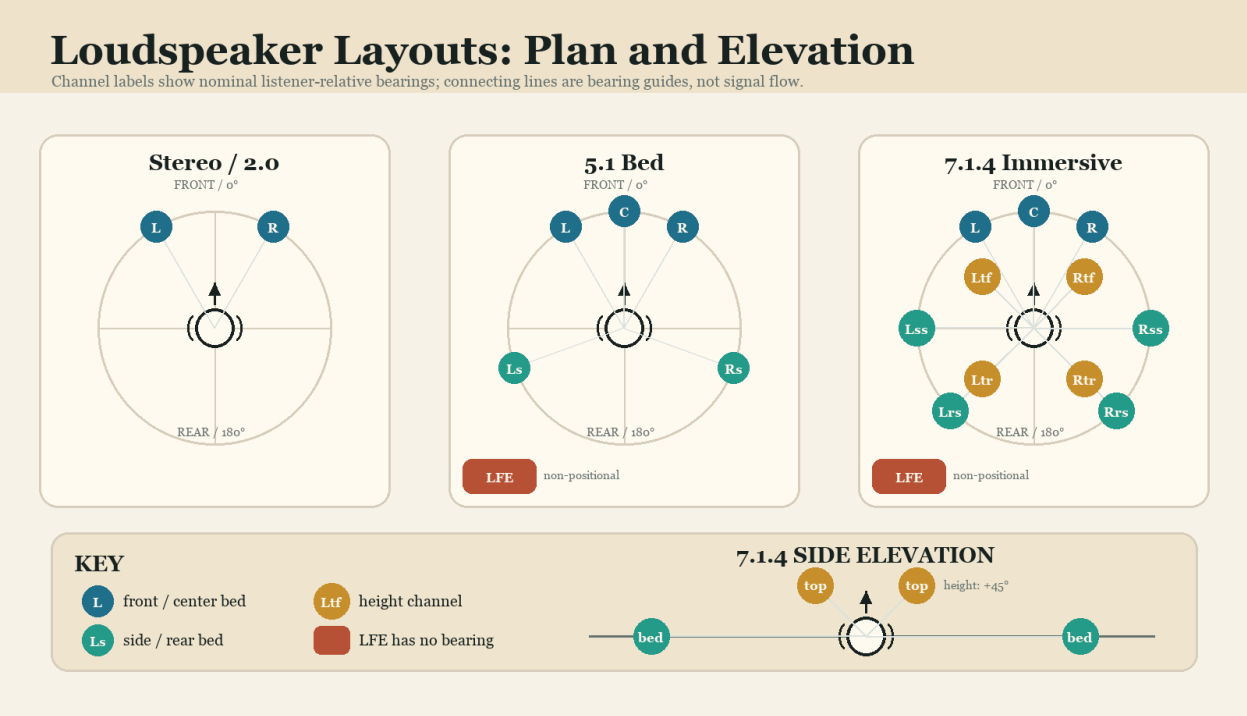


Figure 8-12: Speaker layout coverage

8.15 Editor And Accessibility

The full-screen design is dense, so hierarchy is essential. Keyboard focus, screen-reader labels, scalable text, high-contrast status colors, and a compact window mode should be designed alongside custom drawing. A parameter remains usable when its decorative visualization is disabled.

Telemetry should be rate-limited and decoupled from audio. The editor reads snapshots; it does not interrogate mutable DSP structures. Closing the editor must not change the sound or CPU behavior of the processor.

The implemented spectrum overlay follows that rule: the callback only writes post-DSP mono samples into a fixed lock-free ring and drops new analyzer samples if the display falls behind. The editor drains that ring, windows and transforms 8192 samples, smooths the dB response, and paints the fill and peak paths at 30 Hz. No FFT, path allocation, repaint, or UI lock occurs on the audio thread.

8.15.1 Expert Control Matrix

Select **Expert** in the editor header to replace the visual performance console with a dense precision workspace. The top row contains nine rotary controls and the center matrix contains nine linked horizontal faders. Each dial/fader pair is attached to the same APVTS parameter, so moving either control updates host automation, the other control, saved state, and the DSP. The spectrum analyzer remains visible while editing.

The five selector banks each provide four native buttons:

- **Quality** writes Host, 2x, 4x, or Target 192 kHz policy.
- **Width Matrix** writes calibrated Mono, Natural, Wide, or Ultra width.
- **Decay Range** writes logarithmic Tight, Room, or Hall RT60 values; Freeze also writes the separate Freeze state.
- **Mix Routing** writes matched Dry, Insert, Parallel, or Send dry/wet pairs.
- **Tail Character** writes matched damping/diffusion pairs for Clean, Warm, Dark, or Air behavior.

Selector highlighting follows current host parameter values. If automation or manual editing creates a value that does not exactly match a macro, the bank clears its highlight rather than claiming a preset that is no longer active. This makes the selector state descriptive, not authoritative.

Numeric entry uses the units shown by the control. Enter pre-delay in milliseconds, RT60 directly in seconds, and Room Size, RT60 Fine, Damping, Width, Diffusion, Wet, and Dry as percentages. RT60 seconds are inverted through the logarithmic 0.01-to-360-second mapping; for example, typing `4.8 s` produces an effective 4.8-second decay rather than treating 4.8 as a normalized value. RT60 Fine uses its displayed plus-or-minus 20 percent scale. Click a dial arc for immediate positioning, drag vertically for precision, use the mouse wheel for increments, or double-click to restore the declared parameter default.

8.16 Compatibility Claims

The repository currently provides build scaffolding, not a published host compatibility matrix. A format name in CMake means that JUCE can generate that target when dependencies and platform tools are available. It does not mean the binary has passed scanning and session-recall tests in every DAW.

Compatibility statements should name plug-in format, CPU architecture, operating-system version, host/version, sample rate, block size, bus layout, and validation date. Results from a standalone build do not substitute for an AUv3 sandbox test or VST3 host scan.

8.17 Validation Commands

The native foundation can be checked without JUCE:

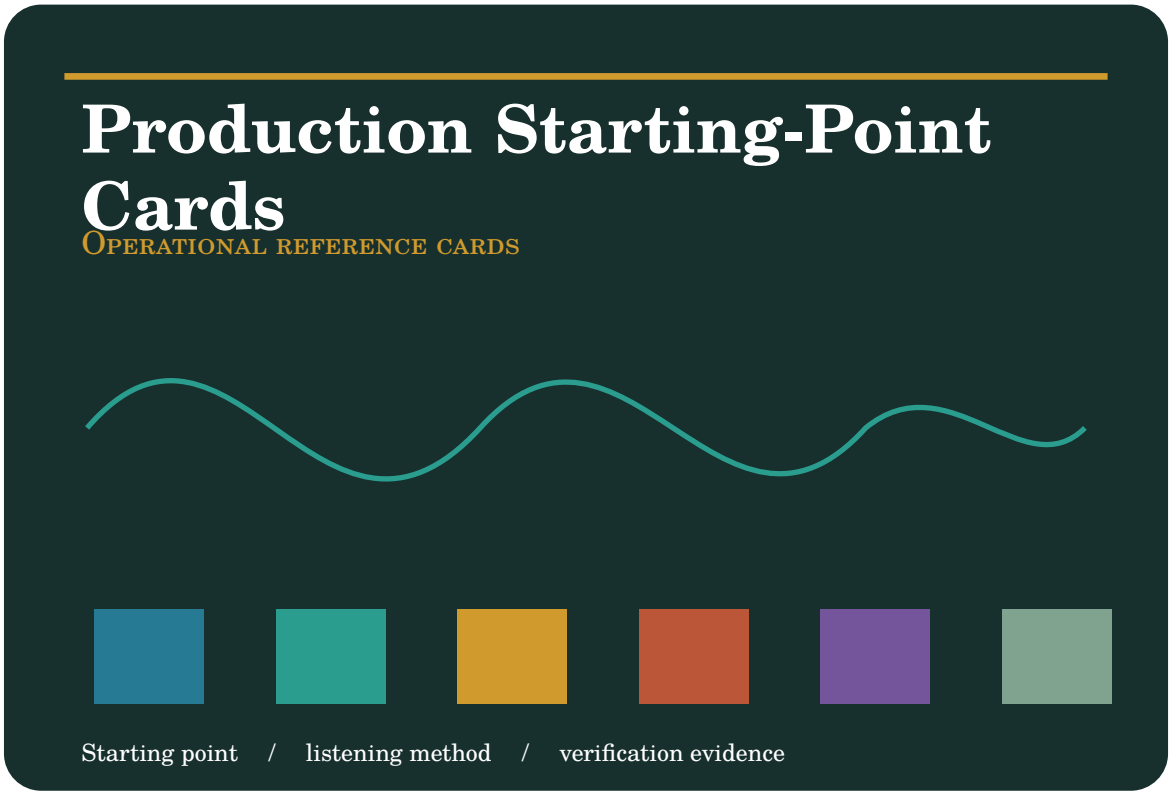
```
cmake -S native/verbx_c -B build/native/verbx_c-plan
cmake --build build/native/verbx_c-plan
ctest --test-dir build/native/verbx_c-plan --output-on-failure
uv run pytest tests/test_native_scaffold.py -q
./scripts/build_verbx_c.sh --clean --doctor
cmake -S native/verbx_plugin -B build/native/verbx_plugin-plan
```

These checks verify the C boundary, direct-build alignment, native render regressions, and JUCE-disabled configure path. They do not compile or validate a JUCE-enabled plug-in when JUCE is absent.

8.18 Reading The Operational Cards

The remainder of this handbook is organized as one-page cards. They are not compatibility certifications or fixed presets. They are repeatable starting points, automation studies, validation procedures, and troubleshooting drills. Record the plug-in build, host, sample rate, block size, channel layout, and asset hashes whenever a card is used for a formal test.

8.19 Production Starting-Point Cards



These cards are coordinates for listening, not finished presets. Calibrate the dry source first, raise the return at matched loudness, and distinguish the direct onset, early reflections, and late field before changing several controls together. Read effective RT60 from the status display rather than estimating it from the coarse knob.

For every candidate, compare stereo and mono, reopen the session, and record host rate, quality mode, block size, layout, latency, peak reduction, and any asset hash. Save a preset only after its parameters, modes, and external assets recall identically.

Production card: Lead vocal in Tight room

For lead vocal in tight room, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Studio chamber

For lead vocal in studio chamber, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Scoring stage

For lead vocal in scoring stage, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Concert hall

For lead vocal in concert hall, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Stone cathedral

For lead vocal in stone cathedral, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Plate-like field

For lead vocal in plate-like field, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze off.

Production card: Lead vocal in Reverse chamber

For lead vocal in reverse chamber, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse on; Freeze off.

Production card: Lead vocal in Frozen architecture

For lead vocal in frozen architecture, aim to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: preserve consonants and front-of-mix intelligibility.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 18 to 45 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Spoken word in Tight room

For spoken word in tight room, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Studio chamber

For spoken word in studio chamber, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Scoring stage

For spoken word in scoring stage, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Concert hall

For spoken word in concert hall, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Stone cathedral

For spoken word in stone cathedral, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Plate-like field

For spoken word in plate-like field, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze off.

Production card: Spoken word in Reverse chamber

For spoken word in reverse chamber, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse on; Freeze off.

Production card: Spoken word in Frozen architecture

For spoken word in frozen architecture, aim to add believable room cues without masking language; check breaths, plosives, and noise-floor lift. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: add believable room cues without masking language.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 8 to 28 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Drum kit in Tight room

For drum kit in tight room, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Studio chamber

For drum kit in studio chamber, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Scoring stage

For drum kit in scoring stage, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Concert hall

For drum kit in concert hall, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Stone cathedral

For drum kit in stone cathedral, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Plate-like field

For drum kit in plate-like field, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze off.

Production card: Drum kit in Reverse chamber

For drum kit in reverse chamber, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse on; Freeze off.

Production card: Drum kit in Frozen architecture

For drum kit in frozen architecture, aim to build size while preserving transient geometry; check kick definition and snare tail density. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: build size while preserving transient geometry.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 4 to 24 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Piano in Tight room

For piano in tight room, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Studio chamber

For piano in studio chamber, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Scoring stage

For piano in scoring stage, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Concert hall

For piano in concert hall, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Stone cathedral

For piano in stone cathedral, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Plate-like field

For piano in plate-like field, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Piano in Reverse chamber

For piano in reverse chamber, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse on; Freeze off.

Production card: Piano in Frozen architecture

For piano in frozen architecture, aim to support sustain without blurring note attacks; listen for low-mid modal buildup. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: support sustain without blurring note attacks.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 12 to 40 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Acoustic guitar in Tight room

For acoustic guitar in tight room, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Studio chamber

For acoustic guitar in studio chamber, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Scoring stage

For acoustic guitar in scoring stage, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Concert hall

For acoustic guitar in concert hall, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Stone cathedral

For acoustic guitar in stone cathedral, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Plate-like field

For acoustic guitar in plate-like field, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze off.

Production card: Acoustic guitar in Reverse chamber

For acoustic guitar in reverse chamber, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse on; Freeze off.

Production card: Acoustic guitar in Frozen architecture

For acoustic guitar in frozen architecture, aim to add depth without combing the direct image; check pick articulation and mono fold-down. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: add depth without combing the direct image.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 10 to 32 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Electric guitar in Tight room

For electric guitar in tight room, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Studio chamber

For electric guitar in studio chamber, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Scoring stage

For electric guitar in scoring stage, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Concert hall

For electric guitar in concert hall, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Stone cathedral

For electric guitar in stone cathedral, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Plate-like field

For electric guitar in plate-like field, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze off.

Production card: Electric guitar in Reverse chamber

For electric guitar in reverse chamber, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse on; Freeze off.

Production card: Electric guitar in Frozen architecture

For electric guitar in frozen architecture, aim to place the cabinet in a designed environment; watch upper-mid glare in the return. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: place the cabinet in a designed environment.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 6 to 30 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Strings in Tight room

For strings in tight room, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Studio chamber

For strings in studio chamber, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Scoring stage

For strings in scoring stage, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Concert hall

For strings in concert hall, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Stone cathedral

For strings in stone cathedral, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Plate-like field

For strings in plate-like field, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze off.

Production card: Strings in Reverse chamber

For strings in reverse chamber, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse on; Freeze off.

Production card: Strings in Frozen architecture

For strings in frozen architecture, aim to extend bow sustain and ensemble width; check section localization and high decay. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: extend bow sustain and ensemble width.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 16 to 55 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Synth pad in Tight room

For synth pad in tight room, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Studio chamber

For synth pad in studio chamber, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Scoring stage

For synth pad in scoring stage, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Concert hall

For synth pad in concert hall, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Stone cathedral

For synth pad in stone cathedral, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Plate-like field

For synth pad in plate-like field, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze off.

Production card: Synth pad in Reverse chamber

For synth pad in reverse chamber, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse on; Freeze off.

Production card: Synth pad in Frozen architecture

For synth pad in frozen architecture, aim to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: turn sustained harmony into an evolving field.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 0 to 40 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Percussion in Tight room

For percussion in tight room, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Studio chamber

For percussion in studio chamber, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Scoring stage

For percussion in scoring stage, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Concert hall

For percussion in concert hall, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Stone cathedral

For percussion in stone cathedral, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Plate-like field

For percussion in plate-like field, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze off.

Production card: Percussion in Reverse chamber

For percussion in reverse chamber, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse on; Freeze off.

Production card: Percussion in Frozen architecture

For percussion in frozen architecture, aim to create rhythmic depth around short impulses; check early reflections against tempo. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: create rhythmic depth around short impulses.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 2 to 22 ms.
- Modes: Reverse off; Freeze prepared/on.

Production card: Field recording in Tight room

For field recording in tight room, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is dense early cues and a controlled short tail.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: dense early cues and a controlled short tail.
- Starting macros: room size 0.28, damping 0.20, diffusion 0.52, wet 0.28.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Studio chamber

For field recording in studio chamber, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is a smooth useful chamber with moderate width.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: a smooth useful chamber with moderate width.
- Starting macros: room size 0.42, damping 0.34, diffusion 0.64, wet 0.38.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Scoring stage

For field recording in scoring stage, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is clear source distance and a broad late field.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: clear source distance and a broad late field.
- Starting macros: room size 0.58, damping 0.46, diffusion 0.72, wet 0.44.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Concert hall

For field recording in concert hall, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is a long integrated decay with stable imaging.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: a long integrated decay with stable imaging.
- Starting macros: room size 0.68, damping 0.52, diffusion 0.78, wet 0.50.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Stone cathedral

For field recording in stone cathedral, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is slow spectral decay and monumental scale.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: slow spectral decay and monumental scale.
- Starting macros: room size 0.78, damping 0.38, diffusion 0.84, wet 0.58.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Plate-like field

For field recording in plate-like field, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is fast diffusion with less geometric localization.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: fast diffusion with less geometric localization.
- Starting macros: room size 0.48, damping 0.62, diffusion 0.88, wet 0.46.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze off.

Production card: Field recording in Reverse chamber

For field recording in reverse chamber, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is a bounded reverse envelope with explicit latency.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: a bounded reverse envelope with explicit latency.
- Starting macros: room size 0.50, damping 0.45, diffusion 0.70, wet 0.52.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse on; Freeze off.

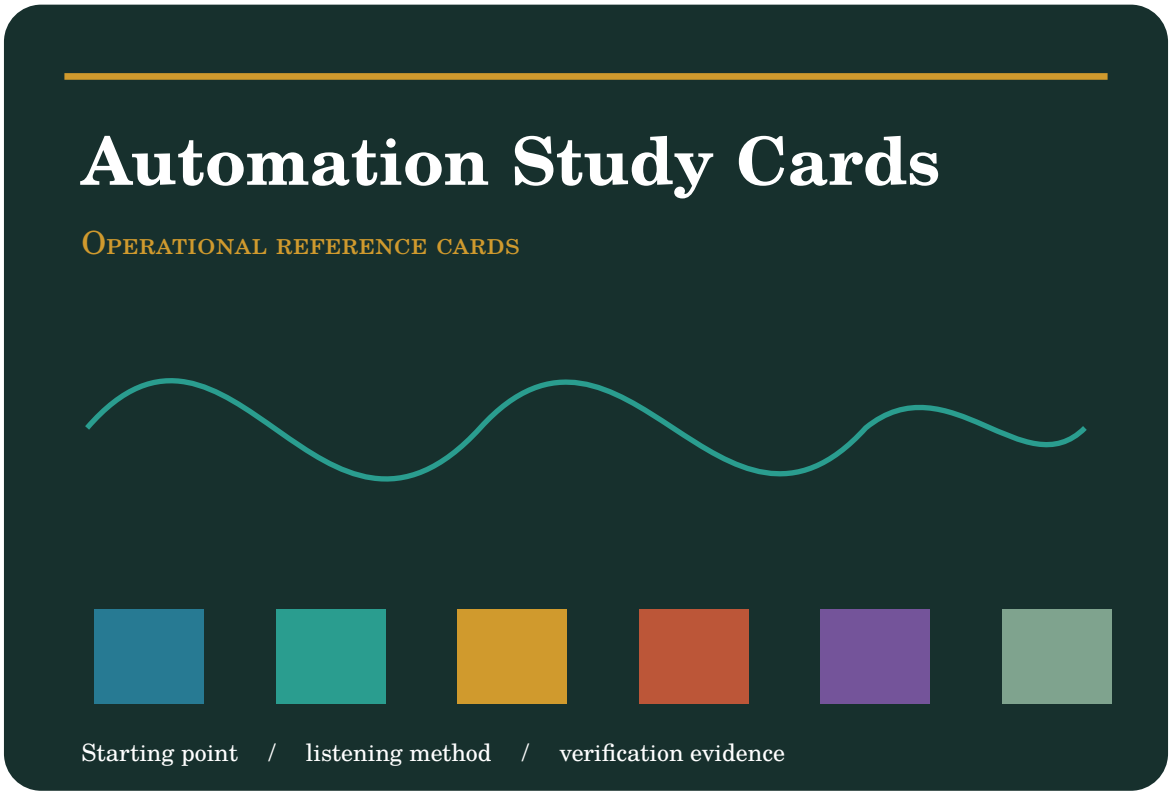
Production card: Field recording in Frozen architecture

For field recording in frozen architecture, aim to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility. The intended room contribution is a sustained field entered and exited safely.

Record these card-specific values:

- Intent: recontextualize a scene without losing its anchors.
- Space character: a sustained field entered and exited safely.
- Starting macros: room size 0.72, damping 0.44, diffusion 0.82, wet 0.66.
- Pre-delay working range: 0 to 60 ms.
- Modes: Reverse off; Freeze prepared/on.

8.20 Automation Study Cards



Write each move in the host and audition it with the editor closed before checking whether the control display follows correctly. Repeat the pass after reopening the project. An automation result belongs to the processor, not to the visible editor state.

Use an impulse, a sustained tone, and representative music. Inspect the transition for clicks, non-finite samples, gain jumps, channel asymmetry, and undeclared latency changes. Record whether the implementation smooths, crossfades, switches discretely, or waits for a preparation boundary.

Motion grammar: Move from the lower setting to the upper setting over eight or more bars. **Primary listening question:** This pattern reveals zipper noise and coefficient discontinuities.

Automation card: Pre-Delay: Slow rise

This slow rise study asks how pre-delay changes separates the direct event from the room onset; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: use a ramp or delay-line crossfade.

Record these card-specific values:

- Host parameter: `pre_delay_ms`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: separates the direct event from the room onset.
- Transition requirement: use a ramp or delay-line crossfade.

Automation card: Room Size: Slow rise

This slow rise study asks how room size changes changes perceived scale and reflection spacing; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: stage structural changes outside the callback.

Record these card-specific values:

- Host parameter: `room_size`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes perceived scale and reflection spacing.
- Transition requirement: stage structural changes outside the callback.

Automation card: RT60 Coarse: Slow rise

This slow rise study asks how rt60 coarse changes moves through the full logarithmic decay range; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: display the effective seconds value.

Record these card-specific values:

- Host parameter: `rt60_coarse`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: moves through the full logarithmic decay range.
- Transition requirement: display the effective seconds value.

Automation card: RT60 Fine: Slow rise

This slow rise study asks how rt60 fine changes trims decay proportionally around the coarse value; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: keep zero as the exact neutral point.

Record these card-specific values:

- Host parameter: `rt60_fine`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: trims decay proportionally around the coarse value.
- Transition requirement: keep zero as the exact neutral point.

Automation card: Damping: Slow rise

This slow rise study asks how damping changes changes high-frequency persistence; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: interpolate stable filter coefficients.

Record these card-specific values:

- Host parameter: `damping`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes high-frequency persistence.
- Transition requirement: interpolate stable filter coefficients.

Automation card: Width: Slow rise

This slow rise study asks how width changes changes lateral energy and correlation; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: monitor mono compatibility during movement.

Record these card-specific values:

- Host parameter: `width`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes lateral energy and correlation.
- Transition requirement: monitor mono compatibility during movement.

Automation card: Diffusion: Slow rise

This slow rise study asks how diffusion changes changes echo-density buildup; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: crossfade when topology must change.

Record these card-specific values:

- Host parameter: `diffusion`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes echo-density buildup.
- Transition requirement: crossfade when topology must change.

Automation card: Wet: Slow rise

This slow rise study asks how wet changes sets processed contribution; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: choose and document the mix law.

Record these card-specific values:

- Host parameter: `wet`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: sets processed contribution.
- Transition requirement: choose and document the mix law.

Automation card: Dry: Slow rise

This slow rise study asks how dry changes sets direct contribution; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: preserve bypass and gain staging.

Record these card-specific values:

- Host parameter: `dry`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: sets direct contribution.
- Transition requirement: preserve bypass and gain staging.

Automation card: Freeze: Slow rise

This slow rise study asks how freeze changes changes network energy behavior; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: use a debounced, smoothed mode transition.

Record these card-specific values:

- Host parameter: `freeze`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes network energy behavior.
- Transition requirement: use a debounced, smoothed mode transition.

Automation card: Reverse: Slow rise

This slow rise study asks how reverse changes changes the envelope and buffering model; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: report added latency before activation.

Record these card-specific values:

- Host parameter: `reverse`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: changes the envelope and buffering model.
- Transition requirement: report added latency before activation.

Automation card: Quality: Slow rise

This slow rise study asks how quality changes selects the internal rate policy; the principal risk is that it reveals zipper noise and coefficient discontinuities. Expected handling: apply through a safe reprepare boundary.

Record these card-specific values:

- Host parameter: `quality_mode`.
- Motion: move from the lower setting to the upper setting over eight or more bars.
- Primary observation: selects the internal rate policy.
- Transition requirement: apply through a safe reprepare boundary.

Motion grammar: Return gradually toward the dry or compact state. **Primary listening question:** This pattern tests whether stored energy decays naturally.

Automation card: Pre-Delay: Slow fall

This slow fall study asks how pre-delay changes separates the direct event from the room onset; the principal risk is that it tests whether stored energy decays naturally. Expected handling: use a ramp or delay-line crossfade.

Record these card-specific values:

- Host parameter: `pre_delay_ms`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: separates the direct event from the room onset.
- Transition requirement: use a ramp or delay-line crossfade.

Automation card: Room Size: Slow fall

This slow fall study asks how room size changes changes perceived scale and reflection spacing; the principal risk is that it tests whether stored energy decays naturally. Expected handling: stage structural changes outside the callback.

Record these card-specific values:

- Host parameter: `room_size`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes perceived scale and reflection spacing.
- Transition requirement: stage structural changes outside the callback.

Automation card: RT60 Coarse: Slow fall

This slow fall study asks how rt60 coarse changes moves through the full logarithmic decay range; the principal risk is that it tests whether stored energy decays naturally. Expected handling: display the effective seconds value.

Record these card-specific values:

- Host parameter: `rt60_coarse`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: moves through the full logarithmic decay range.
- Transition requirement: display the effective seconds value.

Automation card: RT60 Fine: Slow fall

This slow fall study asks how rt60 fine changes trims decay proportionally around the coarse value; the principal risk is that it tests whether stored energy decays naturally. Expected handling: keep zero as the exact neutral point.

Record these card-specific values:

- Host parameter: `rt60_fine`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: trims decay proportionally around the coarse value.
- Transition requirement: keep zero as the exact neutral point.

Automation card: Damping: Slow fall

This slow fall study asks how damping changes changes high-frequency persistence; the principal risk is that it tests whether stored energy decays naturally. Expected handling: interpolate stable filter coefficients.

Record these card-specific values:

- Host parameter: `damping`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes high-frequency persistence.
- Transition requirement: interpolate stable filter coefficients.

Automation card: Width: Slow fall

This slow fall study asks how width changes changes lateral energy and correlation; the principal risk is that it tests whether stored energy decays naturally. Expected handling: monitor mono compatibility during movement.

Record these card-specific values:

- Host parameter: `width`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes lateral energy and correlation.
- Transition requirement: monitor mono compatibility during movement.

Automation card: Diffusion: Slow fall

This slow fall study asks how diffusion changes changes echo-density buildup; the principal risk is that it tests whether stored energy decays naturally. Expected handling: crossfade when topology must change.

Record these card-specific values:

- Host parameter: `diffusion`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes echo-density buildup.
- Transition requirement: crossfade when topology must change.

Automation card: Wet: Slow fall

This slow fall study asks how wet changes sets processed contribution; the principal risk is that it tests whether stored energy decays naturally. Expected handling: choose and document the mix law.

Record these card-specific values:

- Host parameter: `wet`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: sets processed contribution.
- Transition requirement: choose and document the mix law.

Automation card: Dry: Slow fall

This slow fall study asks how dry changes sets direct contribution; the principal risk is that it tests whether stored energy decays naturally. Expected handling: preserve bypass and gain staging.

Record these card-specific values:

- Host parameter: `dry`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: sets direct contribution.
- Transition requirement: preserve bypass and gain staging.

Automation card: Freeze: Slow fall

This slow fall study asks how freeze changes changes network energy behavior; the principal risk is that it tests whether stored energy decays naturally. Expected handling: use a debounced, smoothed mode transition.

Record these card-specific values:

- Host parameter: `freeze`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes network energy behavior.
- Transition requirement: use a debounced, smoothed mode transition.

Automation card: Reverse: Slow fall

This slow fall study asks how reverse changes changes the envelope and buffering model; the principal risk is that it tests whether stored energy decays naturally. Expected handling: report added latency before activation.

Record these card-specific values:

- Host parameter: `reverse`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: changes the envelope and buffering model.
- Transition requirement: report added latency before activation.

Automation card: Quality: Slow fall

This slow fall study asks how quality changes selects the internal rate policy; the principal risk is that it tests whether stored energy decays naturally. Expected handling: apply through a safe reprepare boundary.

Record these card-specific values:

- Host parameter: `quality_mode`.
- Motion: return gradually toward the dry or compact state.
- Primary observation: selects the internal rate policy.
- Transition requirement: apply through a safe reprepare boundary.

Motion grammar: Alternate two musically useful values on a bar or phrase boundary. **Primary listening question:** This pattern tests repeatability and transition timing.

Automation card: Pre-Delay: Tempo pulse

This tempo pulse study asks how pre-delay changes separates the direct event from the room onset; the principal risk is that it tests repeatability and transition timing. Expected handling: use a ramp or delay-line crossfade.

Record these card-specific values:

- Host parameter: `pre_delay_ms`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: separates the direct event from the room onset.
- Transition requirement: use a ramp or delay-line crossfade.

Automation card: Room Size: Tempo pulse

This tempo pulse study asks how room size changes changes perceived scale and reflection spacing; the principal risk is that it tests repeatability and transition timing. Expected handling: stage structural changes outside the callback.

Record these card-specific values:

- Host parameter: `room_size`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes perceived scale and reflection spacing.
- Transition requirement: stage structural changes outside the callback.

Automation card: RT60 Coarse: Tempo pulse

This tempo pulse study asks how rt60 coarse changes moves through the full logarithmic decay range; the principal risk is that it tests repeatability and transition timing. Expected handling: display the effective seconds value.

Record these card-specific values:

- Host parameter: `rt60_coarse`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: moves through the full logarithmic decay range.
- Transition requirement: display the effective seconds value.

Automation card: RT60 Fine: Tempo pulse

This tempo pulse study asks how rt60 fine changes trims decay proportionally around the coarse value; the principal risk is that it tests repeatability and transition timing. Expected handling: keep zero as the exact neutral point.

Record these card-specific values:

- Host parameter: `rt60_fine`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: trims decay proportionally around the coarse value.
- Transition requirement: keep zero as the exact neutral point.

Automation card: Damping: Tempo pulse

This tempo pulse study asks how damping changes changes high-frequency persistence; the principal risk is that it tests repeatability and transition timing. Expected handling: interpolate stable filter coefficients.

Record these card-specific values:

- Host parameter: `damping`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes high-frequency persistence.
- Transition requirement: interpolate stable filter coefficients.

Automation card: Width: Tempo pulse

This tempo pulse study asks how width changes changes lateral energy and correlation; the principal risk is that it tests repeatability and transition timing. Expected handling: monitor mono compatibility during movement.

Record these card-specific values:

- Host parameter: `width`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes lateral energy and correlation.
- Transition requirement: monitor mono compatibility during movement.

Automation card: Diffusion: Tempo pulse

This tempo pulse study asks how diffusion changes changes echo-density buildup; the principal risk is that it tests repeatability and transition timing. Expected handling: crossfade when topology must change.

Record these card-specific values:

- Host parameter: `diffusion`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes echo-density buildup.
- Transition requirement: crossfade when topology must change.

Automation card: Wet: Tempo pulse

This tempo pulse study asks how wet changes sets processed contribution; the principal risk is that it tests repeatability and transition timing. Expected handling: choose and document the mix law.

Record these card-specific values:

- Host parameter: `wet`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: sets processed contribution.
- Transition requirement: choose and document the mix law.

Automation card: Dry: Tempo pulse

This tempo pulse study asks how dry changes sets direct contribution; the principal risk is that it tests repeatability and transition timing. Expected handling: preserve bypass and gain staging.

Record these card-specific values:

- Host parameter: `dry`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: sets direct contribution.
- Transition requirement: preserve bypass and gain staging.

Automation card: Freeze: Tempo pulse

This tempo pulse study asks how freeze changes changes network energy behavior; the principal risk is that it tests repeatability and transition timing. Expected handling: use a debounced, smoothed mode transition.

Record these card-specific values:

- Host parameter: `freeze`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes network energy behavior.
- Transition requirement: use a debounced, smoothed mode transition.

Automation card: Reverse: Tempo pulse

This tempo pulse study asks how reverse changes changes the envelope and buffering model; the principal risk is that it tests repeatability and transition timing. Expected handling: report added latency before activation.

Record these card-specific values:

- Host parameter: `reverse`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: changes the envelope and buffering model.
- Transition requirement: report added latency before activation.

Automation card: Quality: Tempo pulse

This tempo pulse study asks how quality changes selects the internal rate policy; the principal risk is that it tests repeatability and transition timing. Expected handling: apply through a safe reprepare boundary.

Record these card-specific values:

- Host parameter: `quality_mode`.
- Motion: alternate two musically useful values on a bar or phrase boundary.
- Primary observation: selects the internal rate policy.
- Transition requirement: apply through a safe reprepare boundary.

Motion grammar: Change once at an arrangement boundary and hold. **Primary listening question:** This pattern tests state recall and discrete transition behavior.

Automation card: Pre-Delay: Scene switch

This scene switch study asks how pre-delay changes separates the direct event from the room onset; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: use a ramp or delay-line crossfade.

Record these card-specific values:

- Host parameter: `pre_delay_ms`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: separates the direct event from the room onset.
- Transition requirement: use a ramp or delay-line crossfade.

Automation card: Room Size: Scene switch

This scene switch study asks how room size changes changes perceived scale and reflection spacing; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: stage structural changes outside the callback.

Record these card-specific values:

- Host parameter: `room_size`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes perceived scale and reflection spacing.
- Transition requirement: stage structural changes outside the callback.

Automation card: RT60 Coarse: Scene switch

This scene switch study asks how rt60 coarse changes moves through the full logarithmic decay range; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: display the effective seconds value.

Record these card-specific values:

- Host parameter: `rt60_coarse`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: moves through the full logarithmic decay range.
- Transition requirement: display the effective seconds value.

Automation card: RT60 Fine: Scene switch

This scene switch study asks how rt60 fine changes trims decay proportionally around the coarse value; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: keep zero as the exact neutral point.

Record these card-specific values:

- Host parameter: `rt60_fine`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: trims decay proportionally around the coarse value.
- Transition requirement: keep zero as the exact neutral point.

Automation card: Damping: Scene switch

This scene switch study asks how damping changes changes high-frequency persistence; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: interpolate stable filter coefficients.

Record these card-specific values:

- Host parameter: `damping`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes high-frequency persistence.
- Transition requirement: interpolate stable filter coefficients.

Automation card: Width: Scene switch

This scene switch study asks how width changes changes lateral energy and correlation; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: monitor mono compatibility during movement.

Record these card-specific values:

- Host parameter: `width`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes lateral energy and correlation.
- Transition requirement: monitor mono compatibility during movement.

Automation card: Diffusion: Scene switch

This scene switch study asks how diffusion changes changes echo-density buildup; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: crossfade when topology must change.

Record these card-specific values:

- Host parameter: `diffusion`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes echo-density buildup.
- Transition requirement: crossfade when topology must change.

Automation card: Wet: Scene switch

This scene switch study asks how wet changes sets processed contribution; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: choose and document the mix law.

Record these card-specific values:

- Host parameter: `wet`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: sets processed contribution.
- Transition requirement: choose and document the mix law.

Automation card: Dry: Scene switch

This scene switch study asks how dry changes sets direct contribution; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: preserve bypass and gain staging.

Record these card-specific values:

- Host parameter: `dry`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: sets direct contribution.
- Transition requirement: preserve bypass and gain staging.

Automation card: Freeze: Scene switch

This scene switch study asks how freeze changes changes network energy behavior; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: use a debounced, smoothed mode transition.

Record these card-specific values:

- Host parameter: `freeze`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes network energy behavior.
- Transition requirement: use a debounced, smoothed mode transition.

Automation card: Reverse: Scene switch

This scene switch study asks how reverse changes changes the envelope and buffering model; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: report added latency before activation.

Record these card-specific values:

- Host parameter: `reverse`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: changes the envelope and buffering model.
- Transition requirement: report added latency before activation.

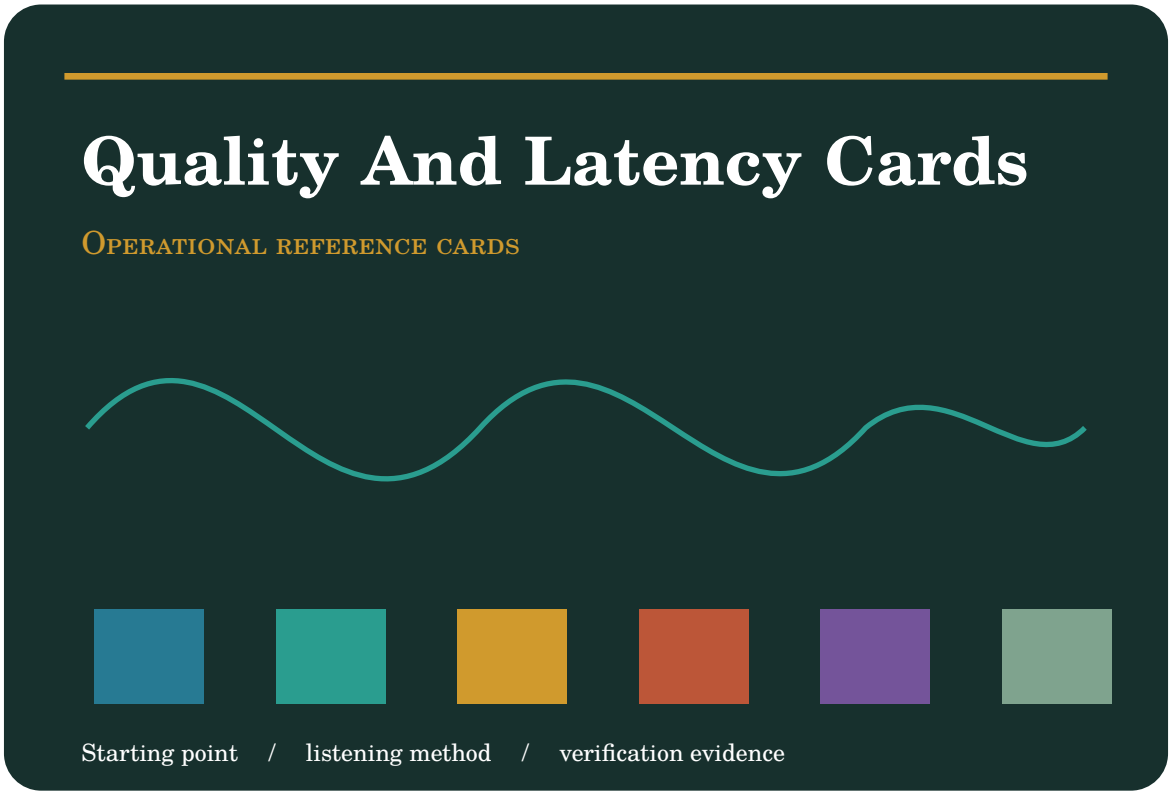
Automation card: Quality: Scene switch

This scene switch study asks how quality changes selects the internal rate policy; the principal risk is that it tests state recall and discrete transition behavior. Expected handling: apply through a safe reprepare boundary.

Record these card-specific values:

- Host parameter: `quality_mode`.
- Motion: change once at an arrangement boundary and hold.
- Primary observation: selects the internal rate policy.
- Transition requirement: apply through a safe reprepare boundary.

8.21 Quality And Latency Cards



Prepare each configuration at the declared rate and maximum block size, then process zero, one, nominal, maximum, and irregular final blocks. Verify the internal-rate accessor, measure CPU with the editor open and closed, and distinguish resampling cost from DSP and drawing cost.

Measure algorithmic latency with an impulse and compare it with the value reported to the host. Device and safety-buffer latency belongs in a separate end-to-end estimate. Retain architecture, operating system, host version, build commit, and warning state with every result.

Quality card 1: 44100 Hz, Host, 64 frames

This case combines a 1.451 ms host block with a 44100 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 44100 Hz.
- Host block duration: 1.451 ms before device and plug-in latency.

Quality card 2: 44100 Hz, Host, 512 frames

This case combines a 11.610 ms host block with a 44100 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 44100 Hz.
- Host block duration: 11.610 ms before device and plug-in latency.

Quality card 3: 44100 Hz, 2x, 64 frames

This case combines a 1.451 ms host block with a 88200 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 88200 Hz.
- Host block duration: 1.451 ms before device and plug-in latency.

Quality card 4: 44100 Hz, 2x, 512 frames

This case combines a 11.610 ms host block with a 88200 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 88200 Hz.
- Host block duration: 11.610 ms before device and plug-in latency.

Quality card 5: 44100 Hz, 4x, 64 frames

This case combines a 1.451 ms host block with a 176400 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 176400 Hz.
- Host block duration: 1.451 ms before device and plug-in latency.

Quality card 6: 44100 Hz, 4x, 512 frames

This case combines a 11.610 ms host block with a 176400 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 176400 Hz.
- Host block duration: 11.610 ms before device and plug-in latency.

Quality card 7: 44100 Hz, Target 192 kHz, 64 frames

This case combines a 1.451 ms host block with a 220500 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 220500 Hz.
- Host block duration: 1.451 ms before device and plug-in latency.

Quality card 8: 44100 Hz, Target 192 kHz, 512 frames

This case combines a 11.610 ms host block with a 220500 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 44100 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 220500 Hz.
- Host block duration: 11.610 ms before device and plug-in latency.

Quality card 9: 48000 Hz, Host, 64 frames

This case combines a 1.333 ms host block with a 48000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 48000 Hz.
- Host block duration: 1.333 ms before device and plug-in latency.

Quality card 10: 48000 Hz, Host, 512 frames

This case combines a 10.667 ms host block with a 48000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 48000 Hz.
- Host block duration: 10.667 ms before device and plug-in latency.

Quality card 11: 48000 Hz, 2x, 64 frames

This case combines a 1.333 ms host block with a 96000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 96000 Hz.
- Host block duration: 1.333 ms before device and plug-in latency.

Quality card 12: 48000 Hz, 2x, 512 frames

This case combines a 10.667 ms host block with a 96000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 96000 Hz.
- Host block duration: 10.667 ms before device and plug-in latency.

Quality card 13: 48000 Hz, 4x, 64 frames

This case combines a 1.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 1.333 ms before device and plug-in latency.

Quality card 14: 48000 Hz, 4x, 512 frames

This case combines a 10.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 10.667 ms before device and plug-in latency.

Quality card 15: 48000 Hz, Target 192 kHz, 64 frames

This case combines a 1.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 1.333 ms before device and plug-in latency.

Quality card 16: 48000 Hz, Target 192 kHz, 512 frames

This case combines a 10.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 48000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 10.667 ms before device and plug-in latency.

Quality card 17: 96000 Hz, Host, 64 frames

This case combines a 0.667 ms host block with a 96000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 96000 Hz.
- Host block duration: 0.667 ms before device and plug-in latency.

Quality card 18: 96000 Hz, Host, 512 frames

This case combines a 5.333 ms host block with a 96000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 96000 Hz.
- Host block duration: 5.333 ms before device and plug-in latency.

Quality card 19: 96000 Hz, 2x, 64 frames

This case combines a 0.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 0.667 ms before device and plug-in latency.

Quality card 20: 96000 Hz, 2x, 512 frames

This case combines a 5.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 5.333 ms before device and plug-in latency.

Quality card 21: 96000 Hz, 4x, 64 frames

This case combines a 0.667 ms host block with a 384000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 384000 Hz.
- Host block duration: 0.667 ms before device and plug-in latency.

Quality card 22: 96000 Hz, 4x, 512 frames

This case combines a 5.333 ms host block with a 384000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 384000 Hz.
- Host block duration: 5.333 ms before device and plug-in latency.

Quality card 23: 96000 Hz, Target 192 kHz, 64 frames

This case combines a 0.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 0.667 ms before device and plug-in latency.

Quality card 24: 96000 Hz, Target 192 kHz, 512 frames

This case combines a 5.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 96000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 5.333 ms before device and plug-in latency.

Quality card 25: 192000 Hz, Host, 64 frames

This case combines a 0.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 0.333 ms before device and plug-in latency.

Quality card 26: 192000 Hz, Host, 512 frames

This case combines a 2.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: Host: no intentional internal rate multiplication.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 2.667 ms before device and plug-in latency.

Quality card 27: 192000 Hz, 2x, 64 frames

This case combines a 0.333 ms host block with a 384000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 384000 Hz.
- Host block duration: 0.333 ms before device and plug-in latency.

Quality card 28: 192000 Hz, 2x, 512 frames

This case combines a 2.667 ms host block with a 384000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: 2x: twice the host rate.
- Expected internal-rate contract: 384000 Hz.
- Host block duration: 2.667 ms before device and plug-in latency.

Quality card 29: 192000 Hz, 4x, 64 frames

This case combines a 0.333 ms host block with a 768000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 768000 Hz.
- Host block duration: 0.333 ms before device and plug-in latency.

Quality card 30: 192000 Hz, 4x, 512 frames

This case combines a 2.667 ms host block with a 768000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: 4x: four times the host rate.
- Expected internal-rate contract: 768000 Hz.
- Host block duration: 2.667 ms before device and plug-in latency.

Quality card 31: 192000 Hz, Target 192 kHz, 64 frames

This case combines a 0.333 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 0.333 ms before device and plug-in latency.

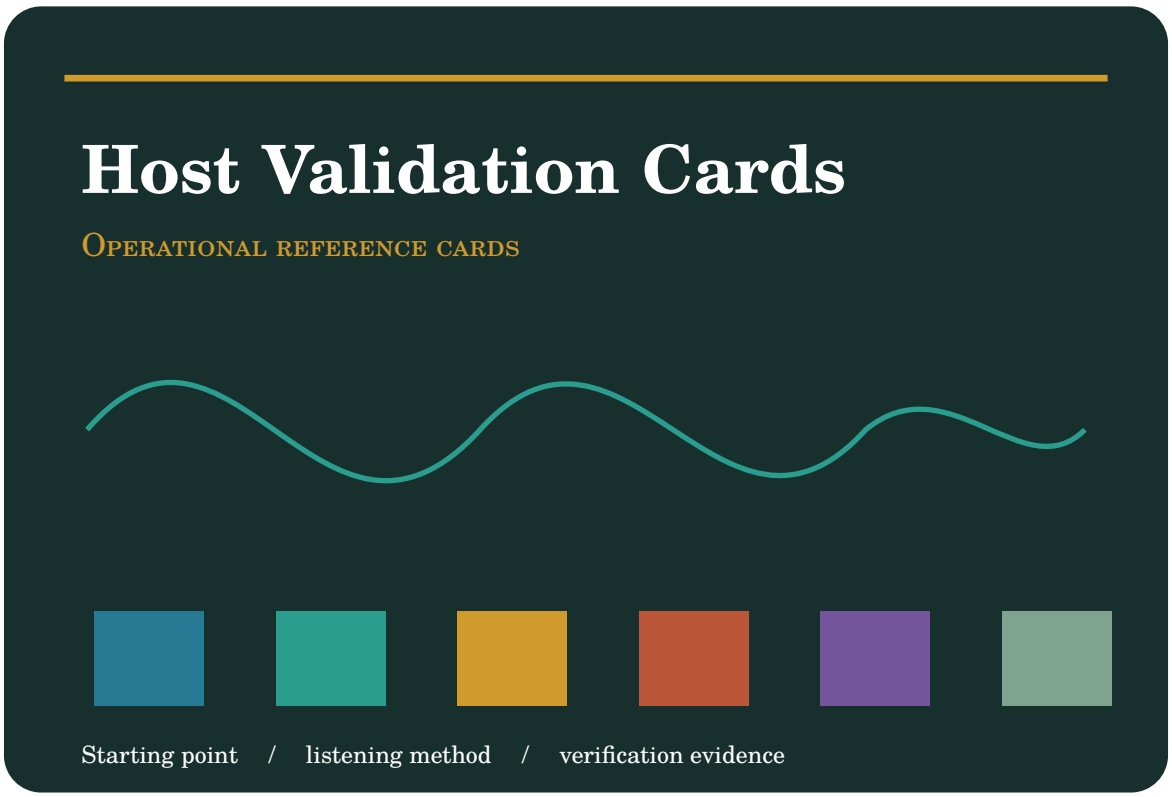
Quality card 32: 192000 Hz, Target 192 kHz, 512 frames

This case combines a 2.667 ms host block with a 192000 Hz internal-rate contract. Confirm that the displayed policy and measured behavior agree.

Record these card-specific values:

- Host rate: 192000 Hz.
- Quality policy: Target 192 kHz: the smallest integer factor reaching at least 192 kHz.
- Expected internal-rate contract: 192000 Hz.
- Host block duration: 2.667 ms before device and plug-in latency.

8.22 Host Validation Cards



Begin from a clean release build and a new host project. Record the binary path, architecture, signature state, format, host version, and commit; test first at 48 kHz, then at the lowest-latency and highest-quality supported settings.

Use deterministic audio, save and reopen the project in a fresh process, and compare state, buses, latency, and assets. Mark the result pass, fail, or blocked. A pass names the dated environment; a failure includes the smallest reproduction and observed fallback; a blocked result names the missing tool or entitlement.

Validation card: Standalone: Scan and instantiate

In Standalone, test scan and instantiate with particular attention to device setup and callback behavior without DAW compensation. The criterion is confirm the format scans, loads, and creates a stable processor/editor pair.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: confirm the format scans, loads, and creates a stable processor/editor pair.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Scan and instantiate

In Desktop AU, test scan and instantiate with particular attention to Apple scanning, state, buses, and latency notification. The criterion is confirm the format scans, loads, and creates a stable processor/editor pair.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: confirm the format scans, loads, and creates a stable processor/editor pair.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Scan and instantiate

In AUv3, test scan and instantiate with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is confirm the format scans, loads, and creates a stable processor/editor pair.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: confirm the format scans, loads, and creates a stable processor/editor pair.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Scan and instantiate

In VST3, test scan and instantiate with particular attention to component/controller state, scanning, and automation identity. The criterion is confirm the format scans, loads, and creates a stable processor/editor pair.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: confirm the format scans, loads, and creates a stable processor/editor pair.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Parameter automation

In Standalone, test parameter automation with particular attention to device setup and callback behavior without DAW compensation. The criterion is write, read, trim, suspend, and replay every exposed parameter.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: write, read, trim, suspend, and replay every exposed parameter.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Parameter automation

In Desktop AU, test parameter automation with particular attention to Apple scanning, state, buses, and latency notification. The criterion is write, read, trim, suspend, and replay every exposed parameter.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: write, read, trim, suspend, and replay every exposed parameter.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Parameter automation

In AUv3, test parameter automation with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is write, read, trim, suspend, and replay every exposed parameter.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: write, read, trim, suspend, and replay every exposed parameter.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Parameter automation

In VST3, test parameter automation with particular attention to component/controller state, scanning, and automation identity. The criterion is write, read, trim, suspend, and replay every exposed parameter.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: write, read, trim, suspend, and replay every exposed parameter.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: State recall

In Standalone, test state recall with particular attention to device setup and callback behavior without DAW compensation. The criterion is save, close, reopen, and compare all parameters and asset identities.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: save, close, reopen, and compare all parameters and asset identities.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: State recall

In Desktop AU, test state recall with particular attention to Apple scanning, state, buses, and latency notification. The criterion is save, close, reopen, and compare all parameters and asset identities.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: save, close, reopen, and compare all parameters and asset identities.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: State recall

In AUv3, test state recall with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is save, close, reopen, and compare all parameters and asset identities.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: save, close, reopen, and compare all parameters and asset identities.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: State recall

In VST3, test state recall with particular attention to component/controller state, scanning, and automation identity. The criterion is save, close, reopen, and compare all parameters and asset identities.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: save, close, reopen, and compare all parameters and asset identities.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Latency compensation

In Standalone, test latency compensation with particular attention to device setup and callback behavior without DAW compensation. The criterion is measure impulse alignment and compare it with the reported frame count.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: measure impulse alignment and compare it with the reported frame count.
- Context emphasis: device setup and callback behavior without DAW compensation.

- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Latency compensation

In Desktop AU, test latency compensation with particular attention to Apple scanning, state, buses, and latency notification. The criterion is measure impulse alignment and compare it with the reported frame count.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: measure impulse alignment and compare it with the reported frame count.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Latency compensation

In AUv3, test latency compensation with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is measure impulse alignment and compare it with the reported frame count.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: measure impulse alignment and compare it with the reported frame count.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Latency compensation

In VST3, test latency compensation with particular attention to component/controller state, scanning, and automation identity. The criterion is measure impulse alignment and compare it with the reported frame count.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: measure impulse alignment and compare it with the reported frame count.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Bus negotiation

In Standalone, test bus negotiation with particular attention to device setup and callback behavior without DAW compensation. The criterion is exercise supported mono/stereo layouts and reject unsupported layouts clearly.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: exercise supported mono/stereo layouts and reject unsupported layouts clearly.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Bus negotiation

In Desktop AU, test bus negotiation with particular attention to Apple scanning, state, buses, and latency notification. The criterion is exercise supported mono/stereo layouts and reject unsupported layouts clearly.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: exercise supported mono/stereo layouts and reject unsupported layouts clearly.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Bus negotiation

In AUv3, test bus negotiation with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is exercise supported mono/stereo layouts and reject unsupported layouts clearly.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: exercise supported mono/stereo layouts and reject unsupported layouts clearly.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Bus negotiation

In VST3, test bus negotiation with particular attention to component/controller state, scanning, and automation identity. The criterion is exercise supported mono/stereo layouts and reject unsupported layouts clearly.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: exercise supported mono/stereo layouts and reject unsupported layouts clearly.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Transport changes

In Standalone, test transport changes with particular attention to device setup and callback behavior without DAW compensation. The criterion is start, stop, loop, seek, and change tempo without corrupting the tail.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: start, stop, loop, seek, and change tempo without corrupting the tail.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Transport changes

In Desktop AU, test transport changes with particular attention to Apple scanning, state, buses, and latency notification. The criterion is start, stop, loop, seek, and change tempo without corrupting the tail.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: start, stop, loop, seek, and change tempo without corrupting the tail.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Transport changes

In AUv3, test transport changes with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is start, stop, loop, seek, and change tempo without corrupting the tail.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: start, stop, loop, seek, and change tempo without corrupting the tail.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Transport changes

In VST3, test transport changes with particular attention to component/controller state, scanning, and automation identity. The criterion is start, stop, loop, seek, and change tempo without corrupting the tail.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: start, stop, loop, seek, and change tempo without corrupting the tail.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Sample-rate changes

In Standalone, test sample-rate changes with particular attention to device setup and callback behavior without DAW compensation. The criterion is reprepare at each supported host rate without stale buffers or status.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: reprepare at each supported host rate without stale buffers or status.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Sample-rate changes

In Desktop AU, test sample-rate changes with particular attention to Apple scanning, state, buses, and latency notification. The criterion is reprepare at each supported host rate without stale buffers or status.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: reprepare at each supported host rate without stale buffers or status.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Sample-rate changes

In AUv3, test sample-rate changes with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is reprepare at each supported host rate without stale buffers or status.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: reprepare at each supported host rate without stale buffers or status.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

Validation card: VST3: Sample-rate changes

In VST3, test sample-rate changes with particular attention to component/controller state, scanning, and automation identity. The criterion is reprepare at each supported host rate without stale buffers or status.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: reprepare at each supported host rate without stale buffers or status.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

Validation card: Standalone: Editor lifecycle

In Standalone, test editor lifecycle with particular attention to device setup and callback behavior without DAW compensation. The criterion is open, resize, close, and reopen the editor while audio remains unchanged.

Record these card-specific values:

- Surface: the JUCE standalone wrapper.
- Goal: open, resize, close, and reopen the editor while audio remains unchanged.
- Context emphasis: device setup and callback behavior without DAW compensation.
- Status: protocol only until a dated result is recorded.

Validation card: Desktop AU: Editor lifecycle

In Desktop AU, test editor lifecycle with particular attention to Apple scanning, state, buses, and latency notification. The criterion is open, resize, close, and reopen the editor while audio remains unchanged.

Record these card-specific values:

- Surface: an Audio Unit host.
- Goal: open, resize, close, and reopen the editor while audio remains unchanged.
- Context emphasis: Apple scanning, state, buses, and latency notification.
- Status: protocol only until a dated result is recorded.

Validation card: AUv3: Editor lifecycle

In AUv3, test editor lifecycle with particular attention to sandbox lifecycle, resources, and compact-window behavior. The criterion is open, resize, close, and reopen the editor while audio remains unchanged.

Record these card-specific values:

- Surface: an AUv3-capable sandboxed host.
- Goal: open, resize, close, and reopen the editor while audio remains unchanged.
- Context emphasis: sandbox lifecycle, resources, and compact-window behavior.
- Status: protocol only until a dated result is recorded.

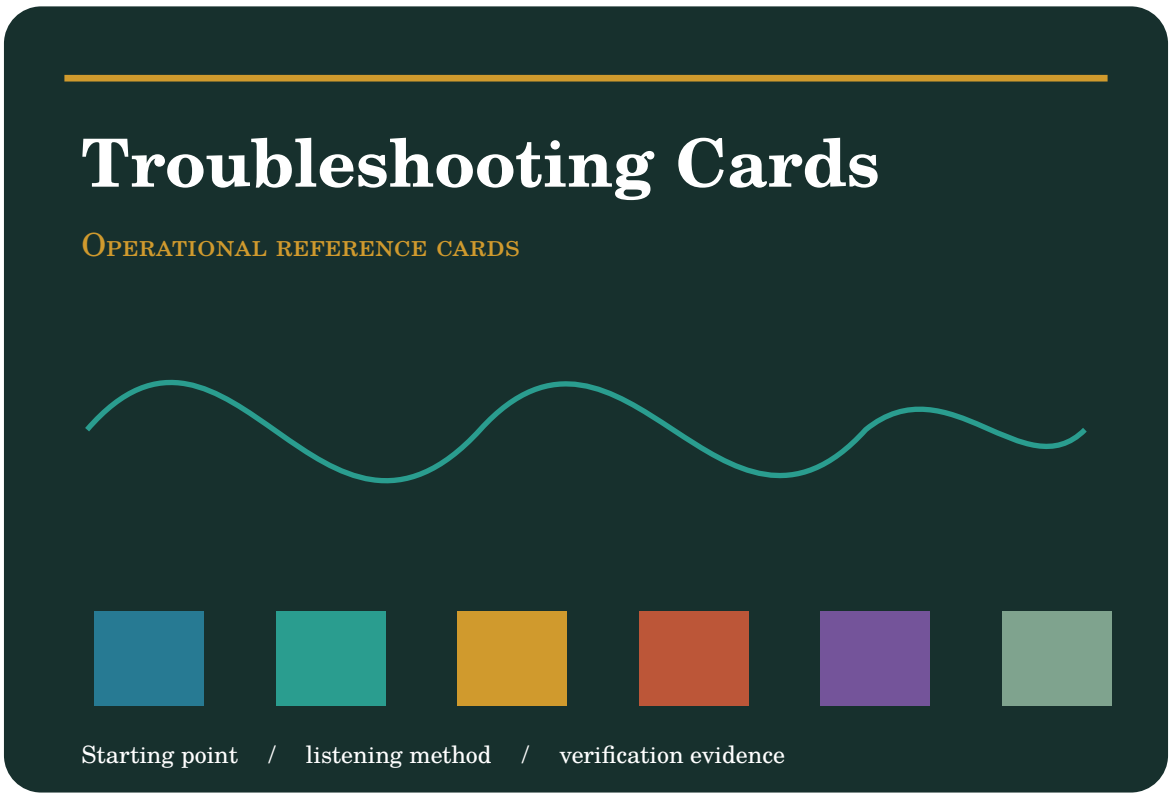
Validation card: VST3: Editor lifecycle

In VST3, test editor lifecycle with particular attention to component/controller state, scanning, and automation identity. The criterion is open, resize, close, and reopen the editor while audio remains unchanged.

Record these card-specific values:

- Surface: a VST3 host.
- Goal: open, resize, close, and reopen the editor while audio remains unchanged.
- Context emphasis: component/controller state, scanning, and automation identity.
- Status: protocol only until a dated result is recorded.

8.23 Troubleshooting Cards



Preserve evidence before changing the system: build, format, host, rate, block size, layout, quality, effective RT60, asset hashes, logs, and the triggering action. Reduce one condition at a time, beginning with static mono or stereo, Host quality, short decay, conservative gain, and no imported assets.

After correction, add the smallest native regression available and retain the host project as a format smoke test. Improve the visible diagnostic when the same failure would otherwise require a debugger.

Troubleshooting card 1: The plug-in does not appear after scanning

Begin with this recovery: inspect the host scan log, confirm architecture and format, then rescan a clean build. The evidence should distinguish the binary is in the wrong format location, failed validation, or was built for the wrong architecture before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The plug-in does not appear after scanning.
- Likely causes: the binary is in the wrong format location, failed validation, or was built for the wrong architecture.
- First recovery: inspect the host scan log, confirm architecture and format, then rescan a clean build.

Troubleshooting card 2: The editor opens but audio is dry

Begin with this recovery: confirm Wet/Dry values, live internal-rate status, and host logs before treating this as a scanner fault. The evidence should distinguish wet gain is down, routing state is stale, or the prepared DSP context failed before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The editor opens but audio is dry.
- Likely causes: wet gain is down, routing state is stale, or the prepared DSP context failed.
- First recovery: confirm Wet/Dry values, live internal-rate status, and host logs before treating this as a scanner fault.

Troubleshooting card 3: Automation recalls the wrong control

Begin with this recovery: compare manifest IDs and restore stable identifiers; never repair this by reordering blindly. The evidence should distinguish a parameter ID or version changed after a session was saved before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Automation recalls the wrong control.
- Likely causes: a parameter ID or version changed after a session was saved.
- First recovery: compare manifest IDs and restore stable identifiers; never repair this by reordering blindly.

Troubleshooting card 4: A preset opens with missing geometry

Begin with this recovery: locate the exact asset, verify its hash, or use the stored bounded fallback profile. The evidence should distinguish the referenced DXF/profile asset moved or its hash changed before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: A preset opens with missing geometry.
- Likely causes: the referenced DXF/profile asset moved or its hash changed.
- First recovery: locate the exact asset, verify its hash, or use the stored bounded fallback profile.

Troubleshooting card 5: CPU rises sharply at 48 kHz

Begin with this recovery: compare Host and 2x modes, increase block size, and record the quality tradeoff. The evidence should distinguish Target 192 kHz implies a 4x internal-rate goal before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: CPU rises sharply at 48 kHz.
- Likely causes: Target 192 kHz implies a 4x internal-rate goal.
- First recovery: compare Host and 2x modes, increase block size, and record the quality tradeoff.

Troubleshooting card 6: CPU rises at 192 kHz

Begin with this recovery: use Host mode and reduce expensive topology before changing musical controls. The evidence should distinguish the host is already processing a very high sample rate before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: CPU rises at 192 kHz.
- Likely causes: the host is already processing a very high sample rate.
- First recovery: use Host mode and reduce expensive topology before changing musical controls.

Troubleshooting card 7: The host reports no latency

Begin with this recovery: measure with an impulse and compare the result with the status accessor. The evidence should distinguish the current causal reverb/oversampling graph is intentionally zero-lookahead or a later buffering graph did not notify the host before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The host reports no latency.
- Likely causes: the current causal reverb/oversampling graph is intentionally zero-lookahead or a later buffering graph did not notify the host.
- First recovery: measure with an impulse and compare the result with the status accessor.

Troubleshooting card 8: Reverse feels late

Begin with this recovery: verify the declared reverse window and host delay compensation. The evidence should distinguish reverse processing requires a capture or lookahead window before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Reverse feels late.
- Likely causes: reverse processing requires a capture or lookahead window.
- First recovery: verify the declared reverse window and host delay compensation.

Troubleshooting card 9: Freeze gets louder over time

Begin with this recovery: disengage safely, lower the stored energy, and inspect freeze gain and limiter reduction. The evidence should distinguish feedback energy is above a stable bound or input injection remains active before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Freeze gets louder over time.
- Likely causes: feedback energy is above a stable bound or input injection remains active.
- First recovery: disengage safely, lower the stored energy, and inspect freeze gain and limiter reduction.

Troubleshooting card 10: Freeze clicks when toggled

Begin with this recovery: add a state transition ramp or dual-path crossfade and retest at full-scale impulses. The evidence should distinguish the mode changes coefficients or injection abruptly before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Freeze clicks when toggled.
- Likely causes: the mode changes coefficients or injection abruptly.
- First recovery: add a state transition ramp or dual-path crossfade and retest at full-scale impulses.

Troubleshooting card 11: The tail disappears on bypass

Begin with this recovery: offer a tail-preserving effect bypass parameter and document the host bypass behavior. The evidence should distinguish the host uses hard bypass and skips processing before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The tail disappears on bypass.
- Likely causes: the host uses hard bypass and skips processing.
- First recovery: offer a tail-preserving effect bypass parameter and document the host bypass behavior.

Troubleshooting card 12: The image collapses in mono

Begin with this recovery: reduce width, check early/late balance, and validate the mono sum. The evidence should distinguish width or decorrelation has produced excessive anti-correlation before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The image collapses in mono.
- Likely causes: width or decorrelation has produced excessive anti-correlation.
- First recovery: reduce width, check early/late balance, and validate the mono sum.

Troubleshooting card 13: The output contains denormal CPU spikes

Begin with this recovery: enable denormal protection and test long decays into digital silence. The evidence should distinguish very small feedback values are entering slow floating-point paths before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The output contains denormal CPU spikes.
- Likely causes: very small feedback values are entering slow floating-point paths.
- First recovery: enable denormal protection and test long decays into digital silence.

Troubleshooting card 14: The output contains NaN or Inf

Begin with this recovery: mute safely, capture diagnostics, clamp inputs, and fix the originating invariant. The evidence should distinguish unstable feedback, invalid coefficients, or corrupted state reached the DSP before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The output contains NaN or Inf.
- Likely causes: unstable feedback, invalid coefficients, or corrupted state reached the DSP.
- First recovery: mute safely, capture diagnostics, clamp inputs, and fix the originating invariant.

Troubleshooting card 15: Changing room size causes a click

Begin with this recovery: prepare the new network off-thread and crossfade bounded states. The evidence should distinguish delay topology changed without a transition before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Changing room size causes a click.
- Likely causes: delay topology changed without a transition.
- First recovery: prepare the new network off-thread and crossfade bounded states.

Troubleshooting card 16: Changing quality interrupts playback

Begin with this recovery: apply quality changes at a declared safe boundary and show pending status. The evidence should distinguish the internal processing graph requires reprepare before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Changing quality interrupts playback.
- Likely causes: the internal processing graph requires reprepare.
- First recovery: apply quality changes at a declared safe boundary and show pending status.

Troubleshooting card 17: Meters freeze when the editor closes

Begin with this recovery: keep DSP telemetry independent and let the editor subscribe only while visible. The evidence should distinguish telemetry ownership is incorrectly coupled to the editor before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Meters freeze when the editor closes.
- Likely causes: telemetry ownership is incorrectly coupled to the editor.
- First recovery: keep DSP telemetry independent and let the editor subscribe only while visible.

Troubleshooting card 18: Closing the editor changes CPU

Begin with this recovery: separate processor state from the editor and repeat the lifecycle test. The evidence should distinguish visualization work or DSP ownership is attached to editor lifetime before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Closing the editor changes CPU.
- Likely causes: visualization work or DSP ownership is attached to editor lifetime.
- First recovery: separate processor state from the editor and repeat the lifecycle test.

Troubleshooting card 19: A DAW project reopens silently

Begin with this recovery: load a safe default audibly, show a blocking status, and preserve diagnostic metadata. The evidence should distinguish state restoration failed or a required asset was unavailable before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: A DAW project reopens silently.
- Likely causes: state restoration failed or a required asset was unavailable.
- First recovery: load a safe default audibly, show a blocking status, and preserve diagnostic metadata.

Troubleshooting card 20: Wet and dry at unity clip

Begin with this recovery: choose equal-power or gain-compensated behavior and expose output safety metering. The evidence should distinguish the mix law sums correlated paths above full scale before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Wet and dry at unity clip.
- Likely causes: the mix law sums correlated paths above full scale.
- First recovery: choose equal-power or gain-compensated behavior and expose output safety metering.

Troubleshooting card 21: Pre-delay automation flanges

Begin with this recovery: crossfade read heads or constrain automation to safe transitions. The evidence should distinguish a moving delay is being read without an intentional interpolation strategy before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: Pre-delay automation flanges.
- Likely causes: a moving delay is being read without an intentional interpolation strategy.
- First recovery: crossfade read heads or constrain automation to safe transitions.

Troubleshooting card 22: High damping sounds unstable

Begin with this recovery: bound the coefficient domain and test every quality mode. The evidence should distinguish filter coefficients approach an unsafe limit at the active internal rate before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: High damping sounds unstable.
- Likely causes: filter coefficients approach an unsafe limit at the active internal rate.
- First recovery: bound the coefficient domain and test every quality mode.

Troubleshooting card 23: The host rejects the channel layout

Begin with this recovery: request matched mono/stereo for the foundation and log the rejected layout clearly. The evidence should distinguish input and output buses are mismatched or unsupported before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The host rejects the channel layout.
- Likely causes: input and output buses are mismatched or unsupported.
- First recovery: request matched mono/stereo for the foundation and log the rejected layout clearly.

Troubleshooting card 24: The screenshot and editor differ

Begin with this recovery: use the maturity statement and track UI implementation separately from DSP readiness. The evidence should distinguish the screenshot is the visual design target while the JUCE editor remains a scaffold before broader DSP changes are attempted.

Record these card-specific values:

- Symptom: The screenshot and editor differ.
- Likely causes: the screenshot is the visual design target while the JUCE editor remains a scaffold.
- First recovery: use the maturity statement and track UI implementation separately from DSP readiness.

8.24 Preset Design Cards

Design every preset against speech, drums, harmonic music, and an impulse at matched loudness. Set family scale with coarse RT60 and place the decay with fine RT60. Decide explicitly whether quality mode, Freeze, Reverse, geometry, and IR identity belong to the preset.

Reload in standalone and each supported format, verify stable parameter IDs, and document schema, build, author, purpose, tags, layout, CPU implications, assets, and fallback behavior. Publish only claims supported by that metadata and the listening tests.

Preset card: Rooms / Intimate

The intimate member should preserve the rooms identity, natural early cues and controlled short decay, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: natural early cues and controlled short decay.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.20, damping 0.40, diffusion 0.55.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Rooms / Open

The open member should preserve the rooms identity, natural early cues and controlled short decay, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: natural early cues and controlled short decay.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.20, damping 0.40, diffusion 0.55.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Rooms / Dark

The dark member should preserve the rooms identity, natural early cues and controlled short decay, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: natural early cues and controlled short decay.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.20, damping 0.40, diffusion 0.55.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Rooms / Infinite

The infinite member should preserve the rooms identity, natural early cues and controlled short decay, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: natural early cues and controlled short decay.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.20, damping 0.40, diffusion 0.55.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Chambers / Intimate

The intimate member should preserve the chambers identity, dense useful depth around voices and instruments, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: dense useful depth around voices and instruments.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.38, damping 0.52, diffusion 0.68.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Chambers / Open

The open member should preserve the chambers identity, dense useful depth around voices and instruments, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: dense useful depth around voices and instruments.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.38, damping 0.52, diffusion 0.68.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Chambers / Dark

The dark member should preserve the chambers identity, dense useful depth around voices and instruments, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: dense useful depth around voices and instruments.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.38, damping 0.52, diffusion 0.68.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Chambers / Infinite

The infinite member should preserve the chambers identity, dense useful depth around voices and instruments, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: dense useful depth around voices and instruments.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.38, damping 0.52, diffusion 0.68.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Halls / Intimate

The intimate member should preserve the halls identity, integrated long decay with stable source localization, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: integrated long decay with stable source localization.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.58, damping 0.64, diffusion 0.76.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Halls / Open

The open member should preserve the halls identity, integrated long decay with stable source localization, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: integrated long decay with stable source localization.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.58, damping 0.64, diffusion 0.76.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Halls / Dark

The dark member should preserve the halls identity, integrated long decay with stable source localization, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: integrated long decay with stable source localization.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.58, damping 0.64, diffusion 0.76.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Halls / Infinite

The infinite member should preserve the halls identity, integrated long decay with stable source localization, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: integrated long decay with stable source localization.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.58, damping 0.64, diffusion 0.76.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Plates / Intimate

The intimate member should preserve the plates identity, fast diffusion and bright sustained density, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: fast diffusion and bright sustained density.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.44, damping 0.72, diffusion 0.86.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Plates / Open

The open member should preserve the plates identity, fast diffusion and bright sustained density, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: fast diffusion and bright sustained density.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.44, damping 0.72, diffusion 0.86.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Plates / Dark

The dark member should preserve the plates identity, fast diffusion and bright sustained density, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: fast diffusion and bright sustained density.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.44, damping 0.72, diffusion 0.86.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Plates / Infinite

The infinite member should preserve the plates identity, fast diffusion and bright sustained density, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: fast diffusion and bright sustained density.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.44, damping 0.72, diffusion 0.86.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Architectures / Intimate

The intimate member should preserve the architectures identity, geometry-led spaces with explicit source/listener context, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: geometry-led spaces with explicit source/listener context.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.66, damping 0.56, diffusion 0.78.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Architectures / Open

The open member should preserve the architectures identity, geometry-led spaces with explicit source/listener context, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: geometry-led spaces with explicit source/listener context.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.66, damping 0.56, diffusion 0.78.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Architectures / Dark

The dark member should preserve the architectures identity, geometry-led spaces with explicit source/listener context, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: geometry-led spaces with explicit source/listener context.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.66, damping 0.56, diffusion 0.78.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Architectures / Infinite

The infinite member should preserve the architectures identity, geometry-led spaces with explicit source/listener context, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: geometry-led spaces with explicit source/listener context.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.66, damping 0.56, diffusion 0.78.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Experimental / Intimate

The intimate member should preserve the experimental identity, reverse, freeze, and exaggerated spatial behavior, while moving the sound toward keep pre-delay and width restrained; prioritize direct connection.

Record these card-specific values:

- Family identity: reverse, freeze, and exaggerated spatial behavior.
- Variant direction: keep pre-delay and width restrained; prioritize direct connection.
- Macro anchors: room size 0.76, damping 0.48, diffusion 0.82.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Experimental / Open

The open member should preserve the experimental identity, reverse, freeze, and exaggerated spatial behavior, while moving the sound toward increase width and early/late separation while preserving center focus.

Record these card-specific values:

- Family identity: reverse, freeze, and exaggerated spatial behavior.
- Variant direction: increase width and early/late separation while preserving center focus.
- Macro anchors: room size 0.76, damping 0.48, diffusion 0.82.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

Preset card: Experimental / Dark

The dark member should preserve the experimental identity, reverse, freeze, and exaggerated spatial behavior, while moving the sound toward increase damping and reduce high-frequency persistence.

Record these card-specific values:

- Family identity: reverse, freeze, and exaggerated spatial behavior.
- Variant direction: increase damping and reduce high-frequency persistence.
- Macro anchors: room size 0.76, damping 0.48, diffusion 0.82.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

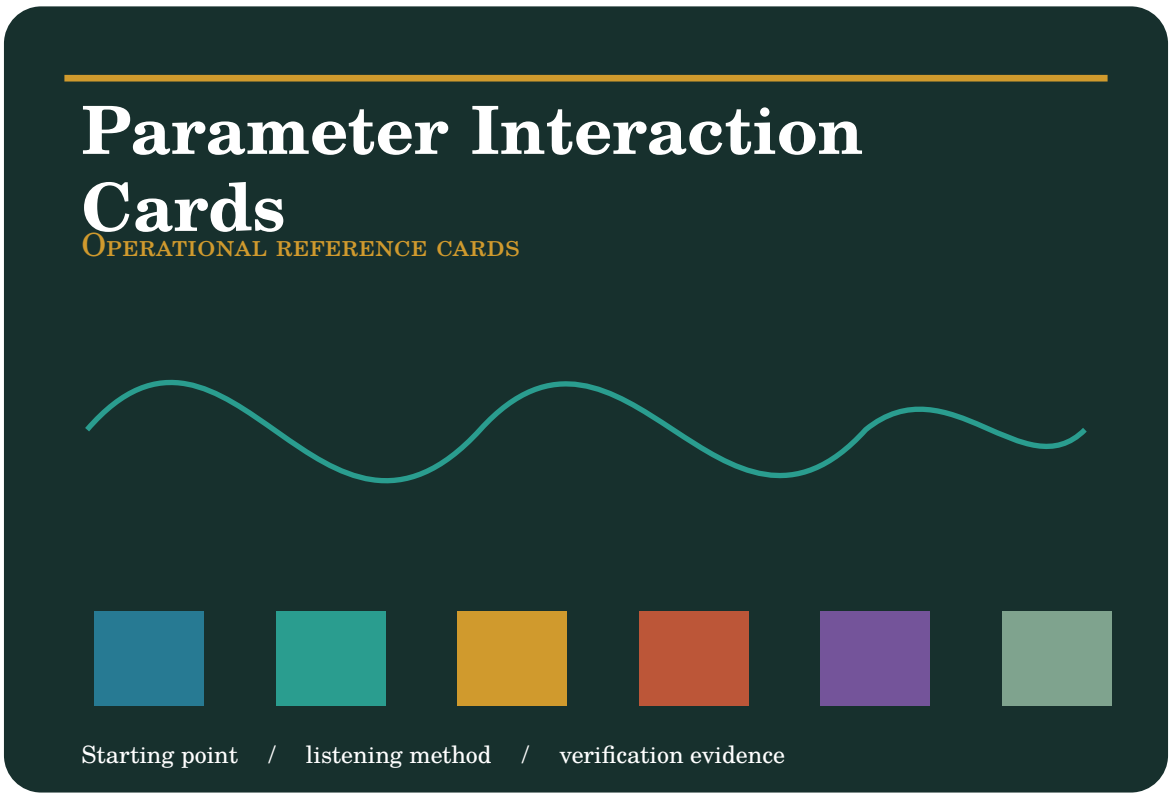
Preset card: Experimental / Infinite

The infinite member should preserve the experimental identity, reverse, freeze, and exaggerated spatial behavior, while moving the sound toward prepare a safe Freeze transition and conservative output protection.

Record these card-specific values:

- Family identity: reverse, freeze, and exaggerated spatial behavior.
- Variant direction: prepare a safe Freeze transition and conservative output protection.
- Macro anchors: room size 0.76, damping 0.48, diffusion 0.82.
- Required metadata: schema, build, author, description, tags, quality policy, layout, and asset identity.

8.25 Parameter Interaction Cards



For each pair, hold one control at default while sweeping the other, reverse the roles, then test parallel and opposing moves. Use an impulse, sustained noise, speech, and music at matched loudness; observe correlation, peak, limiter reduction, effective RT60, latency, CPU, and any structural preparation.

Save the four corners and center as host states. Reopen them in a fresh process, verify stable parameter identity, and make discrete or pending behavior explicit in value text and status rather than implying smooth intermediate states that do not exist.

Interaction card 1: Pre-Delay with Room Size

This pair combines pre-delay, which separates the direct event from the room onset, with room size, which changes perceived scale and reflection spacing. Expected transitions are use a ramp or delay-line crossfade and stage structural changes outside the callback.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `room_size`.
- First role: separates the direct event from the room onset.
- Second role: changes perceived scale and reflection spacing.
- Transition rules: use a ramp or delay-line crossfade; stage structural changes outside the callback.

Interaction card 2: Pre-Delay with RT60 Coarse

This pair combines pre-delay, which separates the direct event from the room onset, with rt60 coarse, which moves through the full logarithmic decay range. Expected transitions are use a ramp or delay-line crossfade and display the effective seconds value.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `rt60_coarse`.
- First role: separates the direct event from the room onset.
- Second role: moves through the full logarithmic decay range.
- Transition rules: use a ramp or delay-line crossfade; display the effective seconds value.

Interaction card 3: Pre-Delay with RT60 Fine

This pair combines pre-delay, which separates the direct event from the room onset, with rt60 fine, which trims decay proportionally around the coarse value. Expected transitions are use a ramp or delay-line crossfade and keep zero as the exact neutral point.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `rt60_fine`.
- First role: separates the direct event from the room onset.
- Second role: trims decay proportionally around the coarse value.
- Transition rules: use a ramp or delay-line crossfade; keep zero as the exact neutral point.

Interaction card 4: Pre-Delay with Damping

This pair combines pre-delay, which separates the direct event from the room onset, with damping, which changes high-frequency persistence. Expected transitions are use a ramp or delay-line crossfade and interpolate stable filter coefficients.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `damping`.
- First role: separates the direct event from the room onset.
- Second role: changes high-frequency persistence.
- Transition rules: use a ramp or delay-line crossfade; interpolate stable filter coefficients.

Interaction card 5: Pre-Delay with Width

This pair combines pre-delay, which separates the direct event from the room onset, with width, which changes lateral energy and correlation. Expected transitions are use a ramp or delay-line crossfade and monitor mono compatibility during movement.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `width`.
- First role: separates the direct event from the room onset.
- Second role: changes lateral energy and correlation.
- Transition rules: use a ramp or delay-line crossfade; monitor mono compatibility during movement.

Interaction card 6: Pre-Delay with Diffusion

This pair combines pre-delay, which separates the direct event from the room onset, with diffusion, which changes echo-density buildup. Expected transitions are use a ramp or delay-line crossfade and crossfade when topology must change.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `diffusion`.
- First role: separates the direct event from the room onset.
- Second role: changes echo-density buildup.
- Transition rules: use a ramp or delay-line crossfade; crossfade when topology must change.

Interaction card 7: Pre-Delay with Wet

This pair combines pre-delay, which separates the direct event from the room onset, with wet, which sets processed contribution. Expected transitions are use a ramp or delay-line crossfade and choose and document the mix law.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `wet`.
- First role: separates the direct event from the room onset.
- Second role: sets processed contribution.
- Transition rules: use a ramp or delay-line crossfade; choose and document the mix law.

Interaction card 8: Pre-Delay with Dry

This pair combines pre-delay, which separates the direct event from the room onset, with dry, which sets direct contribution. Expected transitions are use a ramp or delay-line crossfade and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `dry`.
- First role: separates the direct event from the room onset.
- Second role: sets direct contribution.
- Transition rules: use a ramp or delay-line crossfade; preserve bypass and gain staging.

Interaction card 9: Pre-Delay with Freeze

This pair combines pre-delay, which separates the direct event from the room onset, with freeze, which changes network energy behavior. Expected transitions are use a ramp or delay-line crossfade and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `freeze`.
- First role: separates the direct event from the room onset.
- Second role: changes network energy behavior.
- Transition rules: use a ramp or delay-line crossfade; use a debounced, smoothed mode transition.

Interaction card 10: Pre-Delay with Reverse

This pair combines pre-delay, which separates the direct event from the room onset, with reverse, which changes the envelope and buffering model. Expected transitions are use a ramp or delay-line crossfade and report added latency before activation.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `reverse`.
- First role: separates the direct event from the room onset.
- Second role: changes the envelope and buffering model.
- Transition rules: use a ramp or delay-line crossfade; report added latency before activation.

Interaction card 11: Pre-Delay with Quality

This pair combines pre-delay, which separates the direct event from the room onset, with quality, which selects the internal rate policy. Expected transitions are use a ramp or delay-line crossfade and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `pre_delay_ms` and `quality_mode`.
- First role: separates the direct event from the room onset.
- Second role: selects the internal rate policy.
- Transition rules: use a ramp or delay-line crossfade; apply through a safe reprepare boundary.

Interaction card 12: Room Size with RT60 Coarse

This pair combines room size, which changes perceived scale and reflection spacing, with rt60 coarse, which moves through the full logarithmic decay range. Expected transitions are stage structural changes outside the callback and display the effective seconds value.

Record these card-specific values:

- Parameters: `room_size` and `rt60_coarse`.
- First role: changes perceived scale and reflection spacing.
- Second role: moves through the full logarithmic decay range.
- Transition rules: stage structural changes outside the callback; display the effective seconds value.

Interaction card 13: Room Size with RT60 Fine

This pair combines room size, which changes perceived scale and reflection spacing, with rt60 fine, which trims decay proportionally around the coarse value. Expected transitions are stage structural changes outside the callback and keep zero as the exact neutral point.

Record these card-specific values:

- Parameters: `room_size` and `rt60_fine`.
- First role: changes perceived scale and reflection spacing.
- Second role: trims decay proportionally around the coarse value.
- Transition rules: stage structural changes outside the callback; keep zero as the exact neutral point.

Interaction card 14: Room Size with Damping

This pair combines room size, which changes perceived scale and reflection spacing, with damping, which changes high-frequency persistence. Expected transitions are stage structural changes outside the callback and interpolate stable filter coefficients.

Record these card-specific values:

- Parameters: `room_size` and `damping`.
- First role: changes perceived scale and reflection spacing.
- Second role: changes high-frequency persistence.
- Transition rules: stage structural changes outside the callback; interpolate stable filter coefficients.

Interaction card 15: Room Size with Width

This pair combines room size, which changes perceived scale and reflection spacing, with width, which changes lateral energy and correlation. Expected transitions are stage structural changes outside the callback and monitor mono compatibility during movement.

Record these card-specific values:

- Parameters: `room_size` and `width`.
- First role: changes perceived scale and reflection spacing.
- Second role: changes lateral energy and correlation.
- Transition rules: stage structural changes outside the callback; monitor mono compatibility during movement.

Interaction card 16: Room Size with Diffusion

This pair combines room size, which changes perceived scale and reflection spacing, with diffusion, which changes echo-density buildup. Expected transitions are stage structural changes outside the callback and crossfade when topology must change.

Record these card-specific values:

- Parameters: `room_size` and `diffusion`.
- First role: changes perceived scale and reflection spacing.
- Second role: changes echo-density buildup.
- Transition rules: stage structural changes outside the callback; crossfade when topology must change.

Interaction card 17: Room Size with Wet

This pair combines room size, which changes perceived scale and reflection spacing, with wet, which sets processed contribution. Expected transitions are stage structural changes outside the callback and choose and document the mix law.

Record these card-specific values:

- Parameters: `room_size` and `wet`.
- First role: changes perceived scale and reflection spacing.
- Second role: sets processed contribution.
- Transition rules: stage structural changes outside the callback; choose and document the mix law.

Interaction card 18: Room Size with Dry

This pair combines room size, which changes perceived scale and reflection spacing, with dry, which sets direct contribution. Expected transitions are stage structural changes outside the callback and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `room_size` and `dry`.
- First role: changes perceived scale and reflection spacing.
- Second role: sets direct contribution.
- Transition rules: stage structural changes outside the callback; preserve bypass and gain staging.

Interaction card 19: Room Size with Freeze

This pair combines room size, which changes perceived scale and reflection spacing, with freeze, which changes network energy behavior. Expected transitions are stage structural changes outside the callback and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `room_size` and `freeze`.
- First role: changes perceived scale and reflection spacing.
- Second role: changes network energy behavior.
- Transition rules: stage structural changes outside the callback; use a debounced, smoothed mode transition.

Interaction card 20: Room Size with Reverse

This pair combines room size, which changes perceived scale and reflection spacing, with reverse, which changes the envelope and buffering model. Expected transitions are stage structural changes outside the callback and report added latency before activation.

Record these card-specific values:

- Parameters: `room_size` and `reverse`.
- First role: changes perceived scale and reflection spacing.
- Second role: changes the envelope and buffering model.
- Transition rules: stage structural changes outside the callback; report added latency before activation.

Interaction card 21: Room Size with Quality

This pair combines room size, which changes perceived scale and reflection spacing, with quality, which selects the internal rate policy. Expected transitions are stage structural changes outside the callback and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `room_size` and `quality_mode`.
- First role: changes perceived scale and reflection spacing.
- Second role: selects the internal rate policy.
- Transition rules: stage structural changes outside the callback; apply through a safe reprepare boundary.

Interaction card 22: RT60 Coarse with RT60 Fine

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with rt60 fine, which trims decay proportionally around the coarse value. Expected transitions are display the effective seconds value and keep zero as the exact neutral point.

Record these card-specific values:

- Parameters: `rt60_coarse` and `rt60_fine`.
- First role: moves through the full logarithmic decay range.
- Second role: trims decay proportionally around the coarse value.
- Transition rules: display the effective seconds value; keep zero as the exact neutral point.

Interaction card 23: RT60 Coarse with Damping

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with damping, which changes high-frequency persistence. Expected transitions are display the effective seconds value and interpolate stable filter coefficients.

Record these card-specific values:

- Parameters: `rt60_coarse` and `damping`.
- First role: moves through the full logarithmic decay range.
- Second role: changes high-frequency persistence.
- Transition rules: display the effective seconds value; interpolate stable filter coefficients.

Interaction card 24: RT60 Coarse with Width

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with width, which changes lateral energy and correlation. Expected transitions are display the effective seconds value and monitor mono compatibility during movement.

Record these card-specific values:

- Parameters: `rt60_coarse` and `width`.
- First role: moves through the full logarithmic decay range.
- Second role: changes lateral energy and correlation.
- Transition rules: display the effective seconds value; monitor mono compatibility during movement.

Interaction card 25: RT60 Coarse with Diffusion

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with diffusion, which changes echo-density buildup. Expected transitions are display the effective seconds value and crossfade when topology must change.

Record these card-specific values:

- Parameters: `rt60_coarse` and `diffusion`.
- First role: moves through the full logarithmic decay range.
- Second role: changes echo-density buildup.
- Transition rules: display the effective seconds value; crossfade when topology must change.

Interaction card 26: RT60 Coarse with Wet

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with wet, which sets processed contribution. Expected transitions are display the effective seconds value and choose and document the mix law.

Record these card-specific values:

- Parameters: `rt60_coarse` and `wet`.
- First role: moves through the full logarithmic decay range.
- Second role: sets processed contribution.
- Transition rules: display the effective seconds value; choose and document the mix law.

Interaction card 27: RT60 Coarse with Dry

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with dry, which sets direct contribution. Expected transitions are display the effective seconds value and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `rt60_coarse` and `dry`.
- First role: moves through the full logarithmic decay range.
- Second role: sets direct contribution.
- Transition rules: display the effective seconds value; preserve bypass and gain staging.

Interaction card 28: RT60 Coarse with Freeze

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with freeze, which changes network energy behavior. Expected transitions are display the effective seconds value and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `rt60_coarse` and `freeze`.
- First role: moves through the full logarithmic decay range.
- Second role: changes network energy behavior.
- Transition rules: display the effective seconds value; use a debounced, smoothed mode transition.

Interaction card 29: RT60 Coarse with Reverse

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with reverse, which changes the envelope and buffering model. Expected transitions are display the effective seconds value and report added latency before activation.

Record these card-specific values:

- Parameters: `rt60_coarse` and `reverse`.
- First role: moves through the full logarithmic decay range.
- Second role: changes the envelope and buffering model.
- Transition rules: display the effective seconds value; report added latency before activation.

Interaction card 30: RT60 Coarse with Quality

This pair combines rt60 coarse, which moves through the full logarithmic decay range, with quality, which selects the internal rate policy. Expected transitions are display the effective seconds value and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `rt60_coarse` and `quality_mode`.
- First role: moves through the full logarithmic decay range.
- Second role: selects the internal rate policy.
- Transition rules: display the effective seconds value; apply through a safe reprepare boundary.

Interaction card 31: RT60 Fine with Damping

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with damping, which changes high-frequency persistence. Expected transitions are keep zero as the exact neutral point and interpolate stable filter coefficients.

Record these card-specific values:

- Parameters: `rt60_fine` and `damping`.
- First role: trims decay proportionally around the coarse value.
- Second role: changes high-frequency persistence.
- Transition rules: keep zero as the exact neutral point; interpolate stable filter coefficients.

Interaction card 32: RT60 Fine with Width

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with width, which changes lateral energy and correlation. Expected transitions are keep zero as the exact neutral point and monitor mono compatibility during movement.

Record these card-specific values:

- Parameters: `rt60_fine` and `width`.
- First role: trims decay proportionally around the coarse value.
- Second role: changes lateral energy and correlation.
- Transition rules: keep zero as the exact neutral point; monitor mono compatibility during movement.

Interaction card 33: RT60 Fine with Diffusion

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with diffusion, which changes echo-density buildup. Expected transitions are keep zero as the exact neutral point and crossfade when topology must change.

Record these card-specific values:

- Parameters: `rt60_fine` and `diffusion`.
- First role: trims decay proportionally around the coarse value.
- Second role: changes echo-density buildup.
- Transition rules: keep zero as the exact neutral point; crossfade when topology must change.

Interaction card 34: RT60 Fine with Wet

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with wet, which sets processed contribution. Expected transitions are keep zero as the exact neutral point and choose and document the mix law.

Record these card-specific values:

- Parameters: `rt60_fine` and `wet`.
- First role: trims decay proportionally around the coarse value.
- Second role: sets processed contribution.
- Transition rules: keep zero as the exact neutral point; choose and document the mix law.

Interaction card 35: RT60 Fine with Dry

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with dry, which sets direct contribution. Expected transitions are keep zero as the exact neutral point and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `rt60_fine` and `dry`.
- First role: trims decay proportionally around the coarse value.
- Second role: sets direct contribution.
- Transition rules: keep zero as the exact neutral point; preserve bypass and gain staging.

Interaction card 36: RT60 Fine with Freeze

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with freeze, which changes network energy behavior. Expected transitions are keep zero as the exact neutral point and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `rt60_fine` and `freeze`.
- First role: trims decay proportionally around the coarse value.
- Second role: changes network energy behavior.
- Transition rules: keep zero as the exact neutral point; use a debounced, smoothed mode transition.

Interaction card 37: RT60 Fine with Reverse

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with reverse, which changes the envelope and buffering model. Expected transitions are keep zero as the exact neutral point and report added latency before activation.

Record these card-specific values:

- Parameters: `rt60_fine` and `reverse`.
- First role: trims decay proportionally around the coarse value.
- Second role: changes the envelope and buffering model.
- Transition rules: keep zero as the exact neutral point; report added latency before activation.

Interaction card 38: RT60 Fine with Quality

This pair combines rt60 fine, which trims decay proportionally around the coarse value, with quality, which selects the internal rate policy. Expected transitions are keep zero as the exact neutral point and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `rt60_fine` and `quality_mode`.
- First role: trims decay proportionally around the coarse value.
- Second role: selects the internal rate policy.
- Transition rules: keep zero as the exact neutral point; apply through a safe reprepare boundary.

Interaction card 39: Damping with Width

This pair combines damping, which changes high-frequency persistence, with width, which changes lateral energy and correlation. Expected transitions are interpolate stable filter coefficients and monitor mono compatibility during movement.

Record these card-specific values:

- Parameters: `damping` and `width`.
- First role: changes high-frequency persistence.
- Second role: changes lateral energy and correlation.
- Transition rules: interpolate stable filter coefficients; monitor mono compatibility during movement.

Interaction card 40: Damping with Diffusion

This pair combines damping, which changes high-frequency persistence, with diffusion, which changes echo-density buildup. Expected transitions are interpolate stable filter coefficients and crossfade when topology must change.

Record these card-specific values:

- Parameters: `damping` and `diffusion`.
- First role: changes high-frequency persistence.
- Second role: changes echo-density buildup.
- Transition rules: interpolate stable filter coefficients; crossfade when topology must change.

Interaction card 41: Damping with Wet

This pair combines damping, which changes high-frequency persistence, with wet, which sets processed contribution. Expected transitions are interpolate stable filter coefficients and choose and document the mix law.

Record these card-specific values:

- Parameters: `damping` and `wet`.
- First role: changes high-frequency persistence.
- Second role: sets processed contribution.
- Transition rules: interpolate stable filter coefficients; choose and document the mix law.

Interaction card 42: Damping with Dry

This pair combines damping, which changes high-frequency persistence, with dry, which sets direct contribution. Expected transitions are interpolate stable filter coefficients and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `damping` and `dry`.
- First role: changes high-frequency persistence.
- Second role: sets direct contribution.
- Transition rules: interpolate stable filter coefficients; preserve bypass and gain staging.

Interaction card 43: Damping with Freeze

This pair combines damping, which changes high-frequency persistence, with freeze, which changes network energy behavior. Expected transitions are interpolate stable filter coefficients and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `damping` and `freeze`.
- First role: changes high-frequency persistence.
- Second role: changes network energy behavior.
- Transition rules: interpolate stable filter coefficients; use a debounced, smoothed mode transition.

Interaction card 44: Damping with Reverse

This pair combines damping, which changes high-frequency persistence, with reverse, which changes the envelope and buffering model. Expected transitions are interpolate stable filter coefficients and report added latency before activation.

Record these card-specific values:

- Parameters: `damping` and `reverse`.
- First role: changes high-frequency persistence.
- Second role: changes the envelope and buffering model.
- Transition rules: interpolate stable filter coefficients; report added latency before activation.

Interaction card 45: Damping with Quality

This pair combines damping, which changes high-frequency persistence, with quality, which selects the internal rate policy. Expected transitions are interpolate stable filter coefficients and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `damping` and `quality_mode`.
- First role: changes high-frequency persistence.
- Second role: selects the internal rate policy.
- Transition rules: interpolate stable filter coefficients; apply through a safe reprepare boundary.

Interaction card 46: Width with Diffusion

This pair combines width, which changes lateral energy and correlation, with diffusion, which changes echo-density buildup. Expected transitions are monitor mono compatibility during movement and crossfade when topology must change.

Record these card-specific values:

- Parameters: `width` and `diffusion`.
- First role: changes lateral energy and correlation.
- Second role: changes echo-density buildup.
- Transition rules: monitor mono compatibility during movement; crossfade when topology must change.

Interaction card 47: Width with Wet

This pair combines width, which changes lateral energy and correlation, with wet, which sets processed contribution. Expected transitions are monitor mono compatibility during movement and choose and document the mix law.

Record these card-specific values:

- Parameters: `width` and `wet`.
- First role: changes lateral energy and correlation.
- Second role: sets processed contribution.
- Transition rules: monitor mono compatibility during movement; choose and document the mix law.

Interaction card 48: Width with Dry

This pair combines width, which changes lateral energy and correlation, with dry, which sets direct contribution. Expected transitions are monitor mono compatibility during movement and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `width` and `dry`.
- First role: changes lateral energy and correlation.
- Second role: sets direct contribution.
- Transition rules: monitor mono compatibility during movement; preserve bypass and gain staging.

Interaction card 49: Width with Freeze

This pair combines width, which changes lateral energy and correlation, with freeze, which changes network energy behavior. Expected transitions are monitor mono compatibility during movement and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `width` and `freeze`.
- First role: changes lateral energy and correlation.
- Second role: changes network energy behavior.
- Transition rules: monitor mono compatibility during movement; use a debounced, smoothed mode transition.

Interaction card 50: Width with Reverse

This pair combines width, which changes lateral energy and correlation, with reverse, which changes the envelope and buffering model. Expected transitions are monitor mono compatibility during movement and report added latency before activation.

Record these card-specific values:

- Parameters: `width` and `reverse`.
- First role: changes lateral energy and correlation.
- Second role: changes the envelope and buffering model.
- Transition rules: monitor mono compatibility during movement; report added latency before activation.

Interaction card 51: Width with Quality

This pair combines width, which changes lateral energy and correlation, with quality, which selects the internal rate policy. Expected transitions are monitor mono compatibility during movement and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `width` and `quality_mode`.
- First role: changes lateral energy and correlation.
- Second role: selects the internal rate policy.
- Transition rules: monitor mono compatibility during movement; apply through a safe reprepare boundary.

Interaction card 52: Diffusion with Wet

This pair combines diffusion, which changes echo-density buildup, with wet, which sets processed contribution. Expected transitions are crossfade when topology must change and choose and document the mix law.

Record these card-specific values:

- Parameters: `diffusion` and `wet`.
- First role: changes echo-density buildup.
- Second role: sets processed contribution.
- Transition rules: crossfade when topology must change; choose and document the mix law.

Interaction card 53: Diffusion with Dry

This pair combines diffusion, which changes echo-density buildup, with dry, which sets direct contribution. Expected transitions are crossfade when topology must change and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `diffusion` and `dry`.
- First role: changes echo-density buildup.
- Second role: sets direct contribution.
- Transition rules: crossfade when topology must change; preserve bypass and gain staging.

Interaction card 54: Diffusion with Freeze

This pair combines diffusion, which changes echo-density buildup, with freeze, which changes network energy behavior. Expected transitions are crossfade when topology must change and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `diffusion` and `freeze`.
- First role: changes echo-density buildup.
- Second role: changes network energy behavior.
- Transition rules: crossfade when topology must change; use a debounced, smoothed mode transition.

Interaction card 55: Diffusion with Reverse

This pair combines diffusion, which changes echo-density buildup, with reverse, which changes the envelope and buffering model. Expected transitions are crossfade when topology must change and report added latency before activation.

Record these card-specific values:

- Parameters: `diffusion` and `reverse`.
- First role: changes echo-density buildup.
- Second role: changes the envelope and buffering model.
- Transition rules: crossfade when topology must change; report added latency before activation.

Interaction card 56: Diffusion with Quality

This pair combines diffusion, which changes echo-density buildup, with quality, which selects the internal rate policy. Expected transitions are crossfade when topology must change and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `diffusion` and `quality_mode`.
- First role: changes echo-density buildup.
- Second role: selects the internal rate policy.
- Transition rules: crossfade when topology must change; apply through a safe reprepare boundary.

Interaction card 57: Wet with Dry

This pair combines wet, which sets processed contribution, with dry, which sets direct contribution. Expected transitions are choose and document the mix law and preserve bypass and gain staging.

Record these card-specific values:

- Parameters: `wet` and `dry`.
- First role: sets processed contribution.
- Second role: sets direct contribution.
- Transition rules: choose and document the mix law; preserve bypass and gain staging.

Interaction card 58: Wet with Freeze

This pair combines wet, which sets processed contribution, with freeze, which changes network energy behavior. Expected transitions are choose and document the mix law and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `wet` and `freeze`.
- First role: sets processed contribution.
- Second role: changes network energy behavior.
- Transition rules: choose and document the mix law; use a debounced, smoothed mode transition.

Interaction card 59: Wet with Reverse

This pair combines wet, which sets processed contribution, with reverse, which changes the envelope and buffering model. Expected transitions are choose and document the mix law and report added latency before activation.

Record these card-specific values:

- Parameters: `wet` and `reverse`.
- First role: sets processed contribution.
- Second role: changes the envelope and buffering model.
- Transition rules: choose and document the mix law; report added latency before activation.

Interaction card 60: Wet with Quality

This pair combines wet, which sets processed contribution, with quality, which selects the internal rate policy. Expected transitions are choose and document the mix law and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `wet` and `quality_mode`.
- First role: sets processed contribution.
- Second role: selects the internal rate policy.
- Transition rules: choose and document the mix law; apply through a safe reprepare boundary.

Interaction card 61: Dry with Freeze

This pair combines dry, which sets direct contribution, with freeze, which changes network energy behavior. Expected transitions are preserve bypass and gain staging and use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameters: `dry` and `freeze`.
- First role: sets direct contribution.
- Second role: changes network energy behavior.
- Transition rules: preserve bypass and gain staging; use a debounced, smoothed mode transition.

Interaction card 62: Dry with Reverse

This pair combines dry, which sets direct contribution, with reverse, which changes the envelope and buffering model. Expected transitions are preserve bypass and gain staging and report added latency before activation.

Record these card-specific values:

- Parameters: `dry` and `reverse`.
- First role: sets direct contribution.
- Second role: changes the envelope and buffering model.
- Transition rules: preserve bypass and gain staging; report added latency before activation.

Interaction card 63: Dry with Quality

This pair combines dry, which sets direct contribution, with quality, which selects the internal rate policy. Expected transitions are preserve bypass and gain staging and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `dry` and `quality_mode`.
- First role: sets direct contribution.
- Second role: selects the internal rate policy.
- Transition rules: preserve bypass and gain staging; apply through a safe reprepare boundary.

Interaction card 64: Freeze with Reverse

This pair combines freeze, which changes network energy behavior, with reverse, which changes the envelope and buffering model. Expected transitions are use a debounced, smoothed mode transition and report added latency before activation.

Record these card-specific values:

- Parameters: `freeze` and `reverse`.
- First role: changes network energy behavior.
- Second role: changes the envelope and buffering model.
- Transition rules: use a debounced, smoothed mode transition; report added latency before activation.

Interaction card 65: Freeze with Quality

This pair combines freeze, which changes network energy behavior, with quality, which selects the internal rate policy. Expected transitions are use a debounced, smoothed mode transition and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `freeze` and `quality_mode`.
- First role: changes network energy behavior.
- Second role: selects the internal rate policy.
- Transition rules: use a debounced, smoothed mode transition; apply through a safe reprepare boundary.

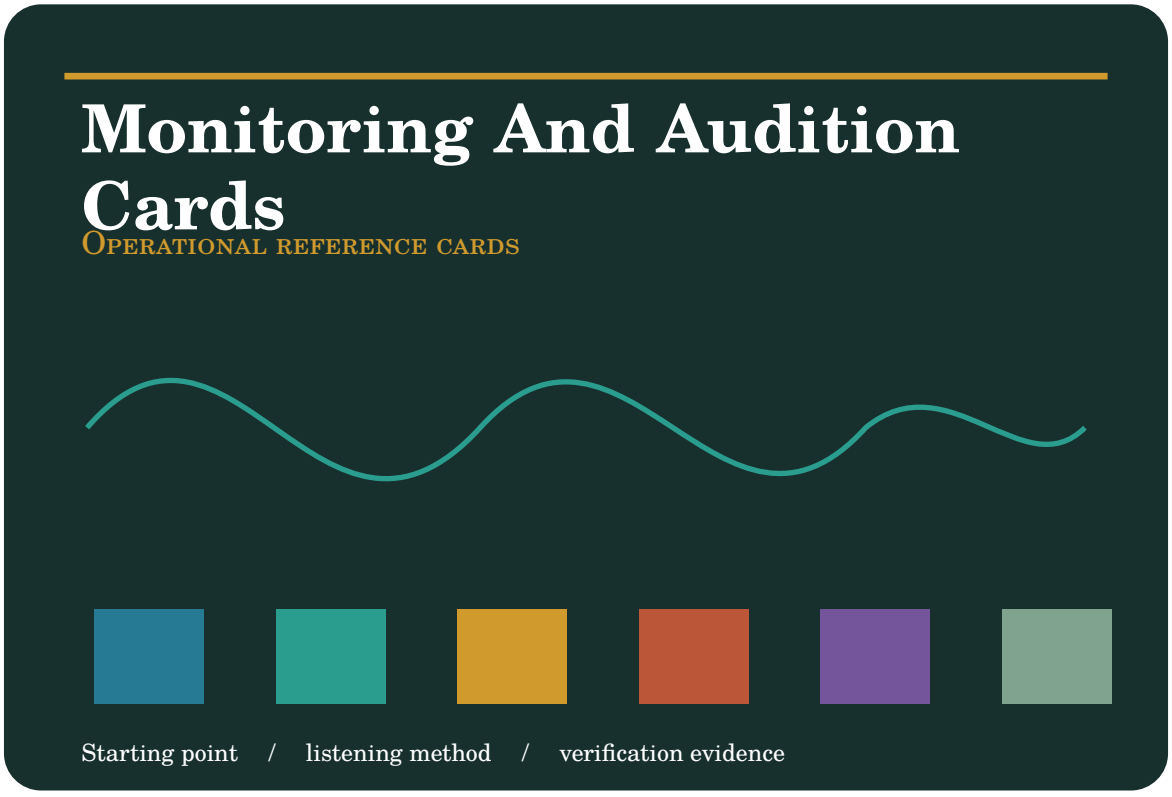
Interaction card 66: Reverse with Quality

This pair combines reverse, which changes the envelope and buffering model, with quality, which selects the internal rate policy. Expected transitions are report added latency before activation and apply through a safe reprepare boundary.

Record these card-specific values:

- Parameters: `reverse` and `quality_mode`.
- First role: changes the envelope and buffering model.
- Second role: selects the internal rate policy.
- Transition rules: report added latency before activation; apply through a safe reprepare boundary.

8.26 Monitoring And Audition Cards



Choose a loop with sparse and dense moments, calibrate the dry path, and compare at unity host gain with enough time for the tail to complete. Judge onset, early reflections, modal buildup, late decay, and noise floor separately, and repeat with the editor closed.

Record the monitor chain, playback level, correction, host rate, block size, quality, layout, and effective RT60. No single context proves translation: pair headphones with speakers, stereo with mono, and channel checks with the intended immersive renderer.

Audition card: Lead vocal on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For lead vocal, the artistic aim is to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability.

Record these card-specific values:

- Source: Lead vocal.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 18 to 45 ms.

Audition card: Lead vocal on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For lead vocal, the artistic aim is to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability.

Record these card-specific values:

- Source: Lead vocal.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 18 to 45 ms.

Audition card: Lead vocal on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For lead vocal, the artistic aim is to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability.

Record these card-specific values:

- Source: Lead vocal.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 18 to 45 ms.

Audition card: Lead vocal on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For lead vocal, the artistic aim is to preserve consonants and front-of-mix intelligibility; watch sibilance and center stability.

Record these card-specific values:

- Source: Lead vocal.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 18 to 45 ms.

Audition card: Spoken word on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For spoken word, the artistic aim is to add believable room cues without masking language; check breaths, plosives, and noise-floor lift.

Record these card-specific values:

- Source: Spoken word.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 8 to 28 ms.

Audition card: Spoken word on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For spoken word, the artistic aim is to add believable room cues without masking language; check breaths, plosives, and noise-floor lift.

Record these card-specific values:

- Source: Spoken word.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 8 to 28 ms.

Audition card: Spoken word on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For spoken word, the artistic aim is to add believable room cues without masking language; check breaths, plosives, and noise-floor lift.

Record these card-specific values:

- Source: Spoken word.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 8 to 28 ms.

Audition card: Spoken word on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For spoken word, the artistic aim is to add believable room cues without masking language; check breaths, plosives, and noise-floor lift.

Record these card-specific values:

- Source: Spoken word.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 8 to 28 ms.

Audition card: Drum kit on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For drum kit, the artistic aim is to build size while preserving transient geometry; check kick definition and snare tail density.

Record these card-specific values:

- Source: Drum kit.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 4 to 24 ms.

Audition card: Drum kit on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For drum kit, the artistic aim is to build size while preserving transient geometry; check kick definition and snare tail density.

Record these card-specific values:

- Source: Drum kit.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 4 to 24 ms.

Audition card: Drum kit on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For drum kit, the artistic aim is to build size while preserving transient geometry; check kick definition and snare tail density.

Record these card-specific values:

- Source: Drum kit.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 4 to 24 ms.

Audition card: Drum kit on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For drum kit, the artistic aim is to build size while preserving transient geometry; check kick definition and snare tail density.

Record these card-specific values:

- Source: Drum kit.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 4 to 24 ms.

Audition card: Piano on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For piano, the artistic aim is to support sustain without blurring note attacks; listen for low-mid modal buildup.

Record these card-specific values:

- Source: Piano.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 12 to 40 ms.

Audition card: Piano on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For piano, the artistic aim is to support sustain without blurring note attacks; listen for low-mid modal buildup.

Record these card-specific values:

- Source: Piano.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 12 to 40 ms.

Audition card: Piano on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For piano, the artistic aim is to support sustain without blurring note attacks; listen for low-mid modal buildup.

Record these card-specific values:

- Source: Piano.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 12 to 40 ms.

Audition card: Piano on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For piano, the artistic aim is to support sustain without blurring note attacks; listen for low-mid modal buildup.

Record these card-specific values:

- Source: Piano.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 12 to 40 ms.

Audition card: Acoustic guitar on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For acoustic guitar, the artistic aim is to add depth without combing the direct image; check pick articulation and mono fold-down.

Record these card-specific values:

- Source: Acoustic guitar.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 10 to 32 ms.

Audition card: Acoustic guitar on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For acoustic guitar, the artistic aim is to add depth without combing the direct image; check pick articulation and mono fold-down.

Record these card-specific values:

- Source: Acoustic guitar.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 10 to 32 ms.

Audition card: Acoustic guitar on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For acoustic guitar, the artistic aim is to add depth without combing the direct image; check pick articulation and mono fold-down.

Record these card-specific values:

- Source: Acoustic guitar.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 10 to 32 ms.

Audition card: Acoustic guitar on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For acoustic guitar, the artistic aim is to add depth without combing the direct image; check pick articulation and mono fold-down.

Record these card-specific values:

- Source: Acoustic guitar.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 10 to 32 ms.

Audition card: Electric guitar on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For electric guitar, the artistic aim is to place the cabinet in a designed environment; watch upper-mid glare in the return.

Record these card-specific values:

- Source: Electric guitar.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 6 to 30 ms.

Audition card: Electric guitar on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For electric guitar, the artistic aim is to place the cabinet in a designed environment; watch upper-mid glare in the return.

Record these card-specific values:

- Source: Electric guitar.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 6 to 30 ms.

Audition card: Electric guitar on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For electric guitar, the artistic aim is to place the cabinet in a designed environment; watch upper-mid glare in the return.

Record these card-specific values:

- Source: Electric guitar.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 6 to 30 ms.

Audition card: Electric guitar on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For electric guitar, the artistic aim is to place the cabinet in a designed environment; watch upper-mid glare in the return.

Record these card-specific values:

- Source: Electric guitar.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 6 to 30 ms.

Audition card: Strings on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For strings, the artistic aim is to extend bow sustain and ensemble width; check section localization and high decay.

Record these card-specific values:

- Source: Strings.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 16 to 55 ms.

Audition card: Strings on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For strings, the artistic aim is to extend bow sustain and ensemble width; check section localization and high decay.

Record these card-specific values:

- Source: Strings.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 16 to 55 ms.

Audition card: Strings on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For strings, the artistic aim is to extend bow sustain and ensemble width; check section localization and high decay.

Record these card-specific values:

- Source: Strings.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 16 to 55 ms.

Audition card: Strings on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For strings, the artistic aim is to extend bow sustain and ensemble width; check section localization and high decay.

Record these card-specific values:

- Source: Strings.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 16 to 55 ms.

Audition card: Synth pad on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For synth pad, the artistic aim is to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation.

Record these card-specific values:

- Source: Synth pad.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 0 to 40 ms.

Audition card: Synth pad on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For synth pad, the artistic aim is to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation.

Record these card-specific values:

- Source: Synth pad.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 0 to 40 ms.

Audition card: Synth pad on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For synth pad, the artistic aim is to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation.

Record these card-specific values:

- Source: Synth pad.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 0 to 40 ms.

Audition card: Synth pad on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For synth pad, the artistic aim is to turn sustained harmony into an evolving field; watch feedback energy and stereo correlation.

Record these card-specific values:

- Source: Synth pad.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 0 to 40 ms.

Audition card: Percussion on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For percussion, the artistic aim is to create rhythmic depth around short impulses; check early reflections against tempo.

Record these card-specific values:

- Source: Percussion.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 2 to 22 ms.

Audition card: Percussion on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For percussion, the artistic aim is to create rhythmic depth around short impulses; check early reflections against tempo.

Record these card-specific values:

- Source: Percussion.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 2 to 22 ms.

Audition card: Percussion on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For percussion, the artistic aim is to create rhythmic depth around short impulses; check early reflections against tempo.

Record these card-specific values:

- Source: Percussion.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 2 to 22 ms.

Audition card: Percussion on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For percussion, the artistic aim is to create rhythmic depth around short impulses; check early reflections against tempo.

Record these card-specific values:

- Source: Percussion.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 2 to 22 ms.

Audition card: Field recording on Nearfield monitors

Use nearfield monitors to judge center focus, depth layers, and low-mid buildup at a calibrated position. For field recording, the artistic aim is to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility.

Record these card-specific values:

- Source: Field recording.
- Monitoring context: Nearfield monitors.
- Judgment goal: judge center focus, depth layers, and low-mid buildup at a calibrated position.
- Useful pre-delay range: 0 to 60 ms.

Audition card: Field recording on Headphones

Use headphones to inspect modulation, tail texture, and left/right discontinuities without room masking. For field recording, the artistic aim is to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility.

Record these card-specific values:

- Source: Field recording.
- Monitoring context: Headphones.
- Judgment goal: inspect modulation, tail texture, and left/right discontinuities without room masking.
- Useful pre-delay range: 0 to 60 ms.

Audition card: Field recording on Mono sum

Use mono sum to expose anti-correlation, combing, and source loss caused by excessive width. For field recording, the artistic aim is to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility.

Record these card-specific values:

- Source: Field recording.
- Monitoring context: Mono sum.
- Judgment goal: expose anti-correlation, combing, and source loss caused by excessive width.
- Useful pre-delay range: 0 to 60 ms.

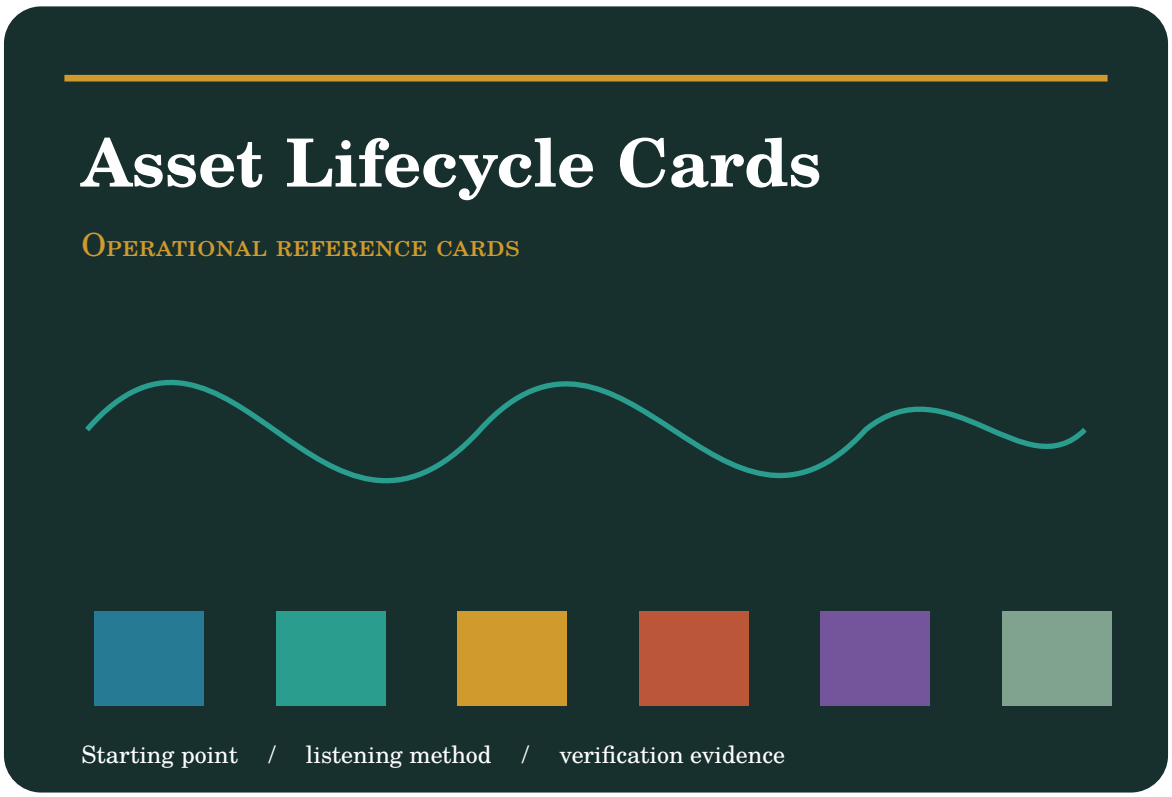
Audition card: Field recording on Low-level playback

Use low-level playback to test whether the space remains legible without relying on loudness. For field recording, the artistic aim is to recontextualize a scene without losing its anchors; compare spectral floor and spatial plausibility.

Record these card-specific values:

- Source: Field recording.
- Monitoring context: Low-level playback.
- Judgment goal: test whether the space remains legible without relying on loudness.
- Useful pre-delay range: 0 to 60 ms.

8.27 Asset Lifecycle Cards



Perform parsing, validation, preparation, caching, and retirement outside the realtime callback. Bound size, count, channels, duration, and memory before allocation; retain provenance while normalizing paths and units; and report the precise invalid field.

Hash canonical content with a schema identifier and every output-affecting input. Test missing, moved, corrupt, oversized, unsupported, and version-mismatched assets. Host state must remain loadable and must disclose any embedded fallback or substituted default.

Asset card: DXF room shell: Import

For the dxf room shell import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on geometry topology, units, transforms, and source/listener coordinates.

Record these card-specific values:

- Asset: DXF room shell.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: geometry topology, units, transforms, and source/listener coordinates.

Asset card: DXF room shell: Validate

For the dxf room shell validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on geometry topology, units, transforms, and source/listener coordinates.

Record these card-specific values:

- Asset: DXF room shell.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: geometry topology, units, transforms, and source/listener coordinates.

Asset card: DXF room shell: Prepare and cache

For the dxf room shell prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on geometry topology, units, transforms, and source/listener coordinates.

Record these card-specific values:

- Asset: DXF room shell.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: geometry topology, units, transforms, and source/listener coordinates.

Asset card: DXF room shell: Recall

For the dxf room shell recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on geometry topology, units, transforms, and source/listener coordinates.

Record these card-specific values:

- Asset: DXF room shell.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: geometry topology, units, transforms, and source/listener coordinates.

Asset card: Early-reflection profile: Import

For the early-reflection profile import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on bounded tap times, gains, directions, and profile version.

Record these card-specific values:

- Asset: Early-reflection profile.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: bounded tap times, gains, directions, and profile version.

Asset card: Early-reflection profile: Validate

For the early-reflection profile validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on bounded tap times, gains, directions, and profile version.

Record these card-specific values:

- Asset: Early-reflection profile.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: bounded tap times, gains, directions, and profile version.

Asset card: Early-reflection profile: Prepare and cache

For the early-reflection profile prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on bounded tap times, gains, directions, and profile version.

Record these card-specific values:

- Asset: Early-reflection profile.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: bounded tap times, gains, directions, and profile version.

Asset card: Early-reflection profile: Recall

For the early-reflection profile recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on bounded tap times, gains, directions, and profile version.

Record these card-specific values:

- Asset: Early-reflection profile.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: bounded tap times, gains, directions, and profile version.

Asset card: Measured impulse response: Import

For the measured impulse response import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on sample rate, channels, trim, normalization, provenance, and checksum.

Record these card-specific values:

- Asset: Measured impulse response.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: sample rate, channels, trim, normalization, provenance, and checksum.

Asset card: Measured impulse response: Validate

For the measured impulse response validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on sample rate, channels, trim, normalization, provenance, and checksum.

Record these card-specific values:

- Asset: Measured impulse response.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: sample rate, channels, trim, normalization, provenance, and checksum.

Asset card: Measured impulse response: Prepare and cache

For the measured impulse response prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on sample rate, channels, trim, normalization, provenance, and checksum.

Record these card-specific values:

- Asset: Measured impulse response.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: sample rate, channels, trim, normalization, provenance, and checksum.

Asset card: Measured impulse response: Recall

For the measured impulse response recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on sample rate, channels, trim, normalization, provenance, and checksum.

Record these card-specific values:

- Asset: Measured impulse response.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: sample rate, channels, trim, normalization, provenance, and checksum.

Asset card: Generated impulse response: Import

For the generated impulse response import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on generator version, seed, parameters, output format, and checksum.

Record these card-specific values:

- Asset: Generated impulse response.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: generator version, seed, parameters, output format, and checksum.

Asset card: Generated impulse response: Validate

For the generated impulse response validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on generator version, seed, parameters, output format, and checksum.

Record these card-specific values:

- Asset: Generated impulse response.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: generator version, seed, parameters, output format, and checksum.

Asset card: Generated impulse response: Prepare and cache

For the generated impulse response prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on generator version, seed, parameters, output format, and checksum.

Record these card-specific values:

- Asset: Generated impulse response.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: generator version, seed, parameters, output format, and checksum.

Asset card: Generated impulse response: Recall

For the generated impulse response recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on generator version, seed, parameters, output format, and checksum.

Record these card-specific values:

- Asset: Generated impulse response.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: generator version, seed, parameters, output format, and checksum.

Asset card: Material library: Import

For the material library import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on stable material IDs, absorption bands, scattering values, and revision.

Record these card-specific values:

- Asset: Material library.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: stable material IDs, absorption bands, scattering values, and revision.

Asset card: Material library: Validate

For the material library validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on stable material IDs, absorption bands, scattering values, and revision.

Record these card-specific values:

- Asset: Material library.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: stable material IDs, absorption bands, scattering values, and revision.

Asset card: Material library: Prepare and cache

For the material library prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on stable material IDs, absorption bands, scattering values, and revision.

Record these card-specific values:

- Asset: Material library.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: stable material IDs, absorption bands, scattering values, and revision.

Asset card: Material library: Recall

For the material library recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on stable material IDs, absorption bands, scattering values, and revision.

Record these card-specific values:

- Asset: Material library.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: stable material IDs, absorption bands, scattering values, and revision.

Asset card: Preset bank: Import

For the preset bank import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on parameter schema, author metadata, tags, asset references, and migration version.

Record these card-specific values:

- Asset: Preset bank.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: parameter schema, author metadata, tags, asset references, and migration version.

Asset card: Preset bank: Validate

For the preset bank validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on parameter schema, author metadata, tags, asset references, and migration version.

Record these card-specific values:

- Asset: Preset bank.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: parameter schema, author metadata, tags, asset references, and migration version.

Asset card: Preset bank: Prepare and cache

For the preset bank prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on parameter schema, author metadata, tags, asset references, and migration version.

Record these card-specific values:

- Asset: Preset bank.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: parameter schema, author metadata, tags, asset references, and migration version.

Asset card: Preset bank: Recall

For the preset bank recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on parameter schema, author metadata, tags, asset references, and migration version.

Record these card-specific values:

- Asset: Preset bank.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: parameter schema, author metadata, tags, asset references, and migration version.

Asset card: Telemetry configuration: Import

For the telemetry configuration import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on meter rates, history lengths, visualization channels, and safety bounds.

Record these card-specific values:

- Asset: Telemetry configuration.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: meter rates, history lengths, visualization channels, and safety bounds.

Asset card: Telemetry configuration: Validate

For the telemetry configuration validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on meter rates, history lengths, visualization channels, and safety bounds.

Record these card-specific values:

- Asset: Telemetry configuration.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: meter rates, history lengths, visualization channels, and safety bounds.

Asset card: Telemetry configuration: Prepare and cache

For the telemetry configuration prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on meter rates, history lengths, visualization channels, and safety bounds.

Record these card-specific values:

- Asset: Telemetry configuration.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: meter rates, history lengths, visualization channels, and safety bounds.

Asset card: Telemetry configuration: Recall

For the telemetry configuration recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on meter rates, history lengths, visualization channels, and safety bounds.

Record these card-specific values:

- Asset: Telemetry configuration.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: meter rates, history lengths, visualization channels, and safety bounds.

Asset card: HRTF or SOFA set: Import

For the hrtf or sofa set import stage, the objective is to read and normalize the external representation away from the audio callback. Identity depends on convention, receiver/emitter indices, coordinate system, sample rate, and license.

Record these card-specific values:

- Asset: HRTF or SOFA set.
- Lifecycle stage: Import.
- Stage objective: read and normalize the external representation away from the audio callback.
- Identity fields: convention, receiver/emitter indices, coordinate system, sample rate, and license.

Asset card: HRTF or SOFA set: Validate

For the hrtf or sofa set validate stage, the objective is to reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors. Identity depends on convention, receiver/emitter indices, coordinate system, sample rate, and license.

Record these card-specific values:

- Asset: HRTF or SOFA set.
- Lifecycle stage: Validate.
- Stage objective: reject malformed, unbounded, unsupported, or ambiguous content with field-specific errors.
- Identity fields: convention, receiver/emitter indices, coordinate system, sample rate, and license.

Asset card: HRTF or SOFA set: Prepare and cache

For the hrtf or sofa set prepare and cache stage, the objective is to produce an immutable realtime-ready representation with a deterministic key. Identity depends on convention, receiver/emitter indices, coordinate system, sample rate, and license.

Record these card-specific values:

- Asset: HRTF or SOFA set.
- Lifecycle stage: Prepare and cache.
- Stage objective: produce an immutable realtime-ready representation with a deterministic key.
- Identity fields: convention, receiver/emitter indices, coordinate system, sample rate, and license.

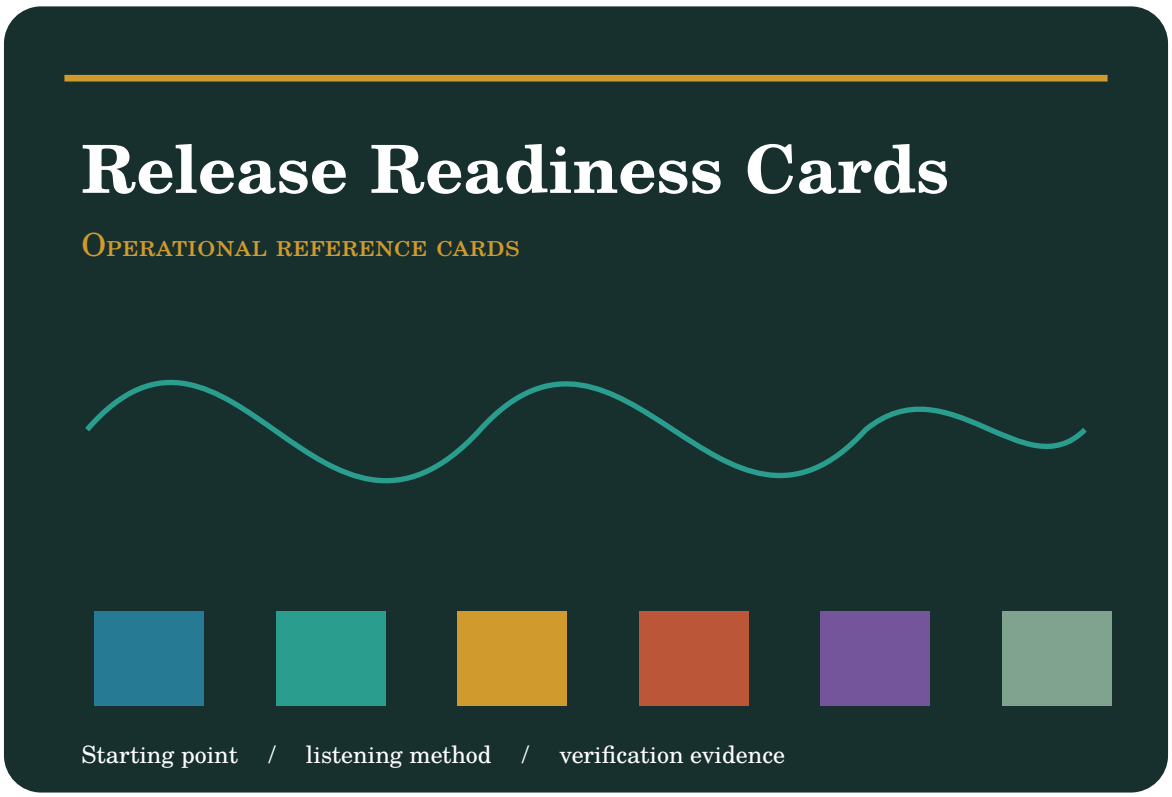
Asset card: HRTF or SOFA set: Recall

For the hrtf or sofa set recall stage, the objective is to resolve the exact asset by identity and degrade safely when it is unavailable. Identity depends on convention, receiver/emitter indices, coordinate system, sample rate, and license.

Record these card-specific values:

- Asset: HRTF or SOFA set.
- Lifecycle stage: Recall.
- Stage objective: resolve the exact asset by identity and degrade safely when it is unavailable.
- Identity fields: convention, receiver/emitter indices, coordinate system, sample rate, and license.

8.28 Release Readiness Cards



Run every gate from a clean release candidate. Confirm architecture, optimization, symbols, identity, version, parameter schema, a new host project, and a reopened representative project. Evidence may be a scanner log, render comparison, state round trip, latency measurement, performance trace, accessibility result, installer receipt, or documentation review.

A warning counts only when explicitly accepted and documented. After a fix, repeat the focused reproduction and neighboring gates, archive the dated matrix with the release tag, and publish only the environments actually tested.

Release card: macOS AU: Scanning

For macOS AU, apply the scanning gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover clean install, discovery, validation logs, duplicate IDs, and rescan behavior.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: clean install, discovery, validation logs, duplicate IDs, and rescan behavior.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Scanning

For macOS AUv3, apply the scanning gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover clean install, discovery, validation logs, duplicate IDs, and rescan behavior.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: clean install, discovery, validation logs, duplicate IDs, and rescan behavior.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Scanning

For VST3, apply the scanning gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover clean install, discovery, validation logs, duplicate IDs, and rescan behavior.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: clean install, discovery, validation logs, duplicate IDs, and rescan behavior.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: State

For macOS AU, apply the state gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover project recall, preset migration, asset identity, defaults, and corrupted-state recovery.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: project recall, preset migration, asset identity, defaults, and corrupted-state recovery.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: State

For macOS AUv3, apply the state gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover project recall, preset migration, asset identity, defaults, and corrupted-state recovery.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: project recall, preset migration, asset identity, defaults, and corrupted-state recovery.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: State

For VST3, apply the state gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover project recall, preset migration, asset identity, defaults, and corrupted-state recovery.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: project recall, preset migration, asset identity, defaults, and corrupted-state recovery.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Audio

For macOS AU, apply the audio gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Audio

For macOS AUv3, apply the audio gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Audio

For VST3, apply the audio gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: silence, impulses, full-scale signals, long tails, NaN/Inf containment, and channel integrity.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Automation

For macOS AU, apply the automation gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover all write/read modes, undo, copy, parameter identity, and editor synchronization.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: all write/read modes, undo, copy, parameter identity, and editor synchronization.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Automation

For macOS AUv3, apply the automation gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover all write/read modes, undo, copy, parameter identity, and editor synchronization.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: all write/read modes, undo, copy, parameter identity, and editor synchronization.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Automation

For VST3, apply the automation gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover all write/read modes, undo, copy, parameter identity, and editor synchronization.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: all write/read modes, undo, copy, parameter identity, and editor synchronization.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Latency

For macOS AU, apply the latency gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover reported frames, impulse measurement, mode changes, compensation, and transport boundaries.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: reported frames, impulse measurement, mode changes, compensation, and transport boundaries.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Latency

For macOS AUv3, apply the latency gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover reported frames, impulse measurement, mode changes, compensation, and transport boundaries.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: reported frames, impulse measurement, mode changes, compensation, and transport boundaries.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Latency

For VST3, apply the latency gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover reported frames, impulse measurement, mode changes, compensation, and transport boundaries.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: reported frames, impulse measurement, mode changes, compensation, and transport boundaries.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Performance

For macOS AU, apply the performance gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover CPU, memory, denormals, editor cost, quality modes, and long-session stability.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: CPU, memory, denormals, editor cost, quality modes, and long-session stability.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Performance

For macOS AUv3, apply the performance gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover CPU, memory, denormals, editor cost, quality modes, and long-session stability.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: CPU, memory, denormals, editor cost, quality modes, and long-session stability.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Performance

For VST3, apply the performance gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover CPU, memory, denormals, editor cost, quality modes, and long-session stability.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: CPU, memory, denormals, editor cost, quality modes, and long-session stability.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Editor

For macOS AU, apply the editor gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Editor

For macOS AUv3, apply the editor gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Editor

For VST3, apply the editor gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: resize, scale, accessibility, keyboard focus, reopen, telemetry, and headless processing.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Distribution

For macOS AU, apply the distribution gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover bundle contents, architecture slices, signing, notarization, installer, and uninstall.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: bundle contents, architecture slices, signing, notarization, installer, and uninstall.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Distribution

For macOS AUv3, apply the distribution gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover bundle contents, architecture slices, signing, notarization, installer, and uninstall.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: bundle contents, architecture slices, signing, notarization, installer, and uninstall.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Distribution

For VST3, apply the distribution gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover bundle contents, architecture slices, signing, notarization, installer, and uninstall.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: bundle contents, architecture slices, signing, notarization, installer, and uninstall.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Diagnostics

For macOS AU, apply the diagnostics gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover status messages, support bundle, crash context, asset hashes, and privacy review.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: status messages, support bundle, crash context, asset hashes, and privacy review.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Diagnostics

For macOS AUv3, apply the diagnostics gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover status messages, support bundle, crash context, asset hashes, and privacy review.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: status messages, support bundle, crash context, asset hashes, and privacy review.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: VST3: Diagnostics

For VST3, apply the diagnostics gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover status messages, support bundle, crash context, asset hashes, and privacy review.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: status messages, support bundle, crash context, asset hashes, and privacy review.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AU: Documentation

For macOS AU, apply the documentation gate to desktop Audio Unit hosts and Apple validation tooling; the evidence must cover build status, supported hosts, limitations, examples, screenshots, and release notes.

Record these card-specific values:

- Target: macOS AU.
- Target scope: desktop Audio Unit hosts and Apple validation tooling.
- Readiness area: build status, supported hosts, limitations, examples, screenshots, and release notes.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

Release card: macOS AUv3: Documentation

For macOS AUv3, apply the documentation gate to sandboxed extension lifecycle and AUv3-capable hosts; the evidence must cover build status, supported hosts, limitations, examples, screenshots, and release notes.

Record these card-specific values:

- Target: macOS AUv3.
- Target scope: sandboxed extension lifecycle and AUv3-capable hosts.
- Readiness area: build status, supported hosts, limitations, examples, screenshots, and release notes.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

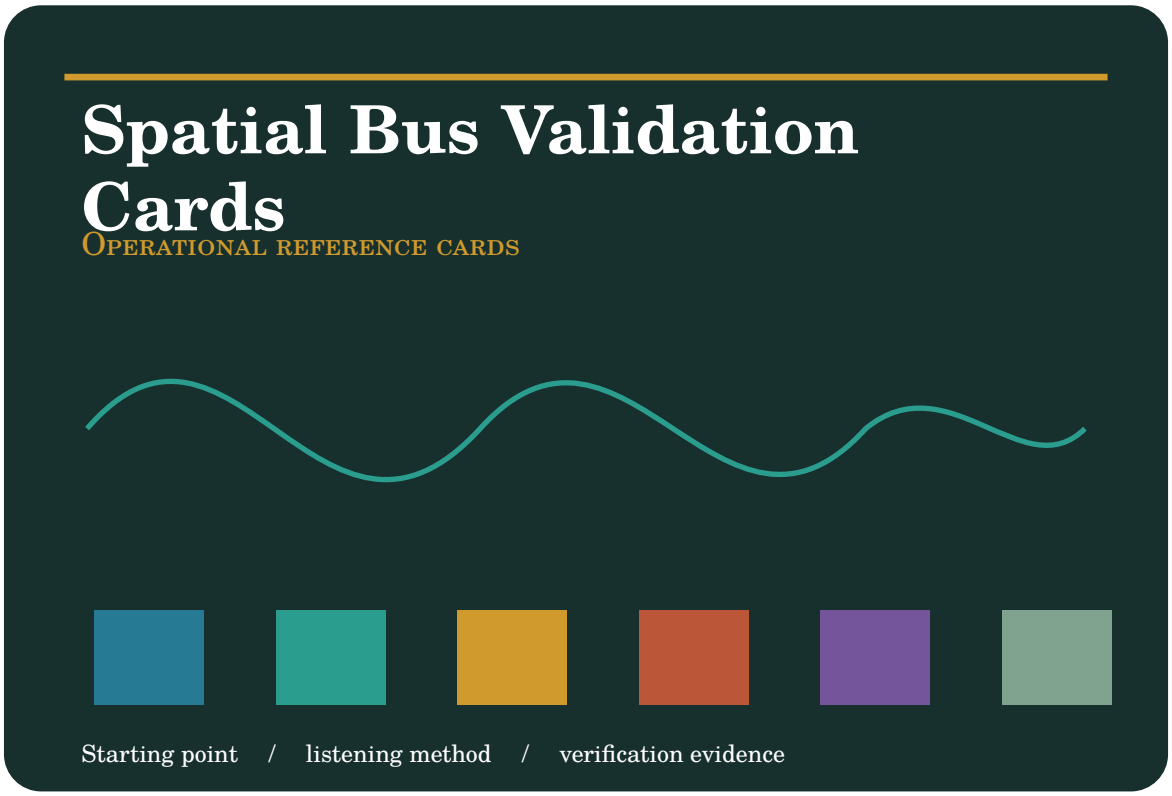
Release card: VST3: Documentation

For VST3, apply the documentation gate to VST3 scanning, component/controller state, and supported desktop hosts; the evidence must cover build status, supported hosts, limitations, examples, screenshots, and release notes.

Record these card-specific values:

- Target: VST3.
- Target scope: VST3 scanning, component/controller state, and supported desktop hosts.
- Readiness area: build status, supported hosts, limitations, examples, screenshots, and release notes.
- Evidence required: dated environment, build commit, result, and retained logs/artifacts.

8.29 Spatial Bus Validation Cards



Negotiate the exact layout before processing, then probe each input with a labeled impulse and inspect every output, meter, channel label, gain, and latency. Test wet/dry, bypass, Freeze, Reverse, limiting, and telemetry against a declared routing policy rather than inherited stereo assumptions.

Save and reopen the project, change to another supported layout, and return. Channel-dependent preparation occurs outside the callback. Fold-downs are translation checks, not substitutes for correct native routing; LFE, height decorrelation, and Ambisonic normalization each require an explicit contract.

Bus card: Mono: Algorithmic

This card checks Mono under algorithmic processing: one matched input and output channel with no hidden stereo assumptions; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: Mono: one matched input and output channel with no hidden stereo assumptions.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Mono: Reverse

This card checks Mono under reverse processing: one matched input and output channel with no hidden stereo assumptions; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: Mono: one matched input and output channel with no hidden stereo assumptions.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Mono: Freeze

This card checks Mono under freeze processing: one matched input and output channel with no hidden stereo assumptions; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: Mono: one matched input and output channel with no hidden stereo assumptions.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Mono: Geometry or IR

This card checks Mono under geometry or ir processing: one matched input and output channel with no hidden stereo assumptions; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: Mono: one matched input and output channel with no hidden stereo assumptions.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.

- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Stereo: Algorithmic

This card checks Stereo under algorithmic processing: matched left/right buses, stable center, width, correlation, and mono fold-down; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: Stereo: matched left/right buses, stable center, width, correlation, and mono fold-down.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Stereo: Reverse

This card checks Stereo under reverse processing: matched left/right buses, stable center, width, correlation, and mono fold-down; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: Stereo: matched left/right buses, stable center, width, correlation, and mono fold-down.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Stereo: Freeze

This card checks Stereo under freeze processing: matched left/right buses, stable center, width, correlation, and mono fold-down; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: Stereo: matched left/right buses, stable center, width, correlation, and mono fold-down.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: Stereo: Geometry or IR

This card checks Stereo under geometry or ir processing: matched left/right buses, stable center, width, correlation, and mono fold-down; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: Stereo: matched left/right buses, stable center, width, correlation, and mono fold-down.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 5.1: Algorithmic

This card checks 5.1 under algorithmic processing: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: 5.1: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 5.1: Reverse

This card checks 5.1 under reverse processing: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: 5.1: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 5.1: Freeze

This card checks 5.1 under freeze processing: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: 5.1: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 5.1: Geometry or IR

This card checks 5.1 under geometry or ir processing: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: 5.1: explicit L/R/C/LFE/Ls/Rs routing and a declared LFE policy.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1: Algorithmic

This card checks 7.1 under algorithmic processing: side/rear separation, normalization, and host channel-order verification; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: 7.1: side/rear separation, normalization, and host channel-order verification.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1: Reverse

This card checks 7.1 under reverse processing: side/rear separation, normalization, and host channel-order verification; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: 7.1: side/rear separation, normalization, and host channel-order verification.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1: Freeze

This card checks 7.1 under freeze processing: side/rear separation, normalization, and host channel-order verification; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: 7.1: side/rear separation, normalization, and host channel-order verification.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1: Geometry or IR

This card checks 7.1 under geometry or ir processing: side/rear separation, normalization, and host channel-order verification; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: 7.1: side/rear separation, normalization, and host channel-order verification.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1.4: Algorithmic

This card checks 7.1.4 under algorithmic processing: bed plus height routing, elevation behavior, and immersive meter coverage; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: 7.1.4: bed plus height routing, elevation behavior, and immersive meter coverage.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1.4: Reverse

This card checks 7.1.4 under reverse processing: bed plus height routing, elevation behavior, and immersive meter coverage; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: 7.1.4: bed plus height routing, elevation behavior, and immersive meter coverage.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1.4: Freeze

This card checks 7.1.4 under freeze processing: bed plus height routing, elevation behavior, and immersive meter coverage; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: 7.1.4: bed plus height routing, elevation behavior, and immersive meter coverage.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: 7.1.4: Geometry or IR

This card checks 7.1.4 under geometry or ir processing: bed plus height routing, elevation behavior, and immersive meter coverage; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: 7.1.4: bed plus height routing, elevation behavior, and immersive meter coverage.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: First-order ambisonics: Algorithmic

This card checks First-order ambisonics under algorithmic processing: ACN/SN3D ordering, rotation behavior, and decoder-independent energy; the core FDN/diffuser path with matched input/output layout.

Record these card-specific values:

- Layout: First-order ambisonics: ACN/SN3D ordering, rotation behavior, and decoder-independent energy.
- Processing mode: Algorithmic: the core FDN/diffuser path with matched input/output layout.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: First-order ambisonics: Reverse

This card checks First-order ambisonics under reverse processing: ACN/SN3D ordering, rotation behavior, and decoder-independent energy; the bounded reverse window, channel alignment, and reported latency.

Record these card-specific values:

- Layout: First-order ambisonics: ACN/SN3D ordering, rotation behavior, and decoder-independent energy.
- Processing mode: Reverse: the bounded reverse window, channel alignment, and reported latency.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: First-order ambisonics: Freeze

This card checks First-order ambisonics under freeze processing: ACN/SN3D ordering, rotation behavior, and decoder-independent energy; sustained energy, channel stability, exit behavior, and safety limiting.

Record these card-specific values:

- Layout: First-order ambisonics: ACN/SN3D ordering, rotation behavior, and decoder-independent energy.
- Processing mode: Freeze: sustained energy, channel stability, exit behavior, and safety limiting.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.
- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

Bus card: First-order ambisonics: Geometry or IR

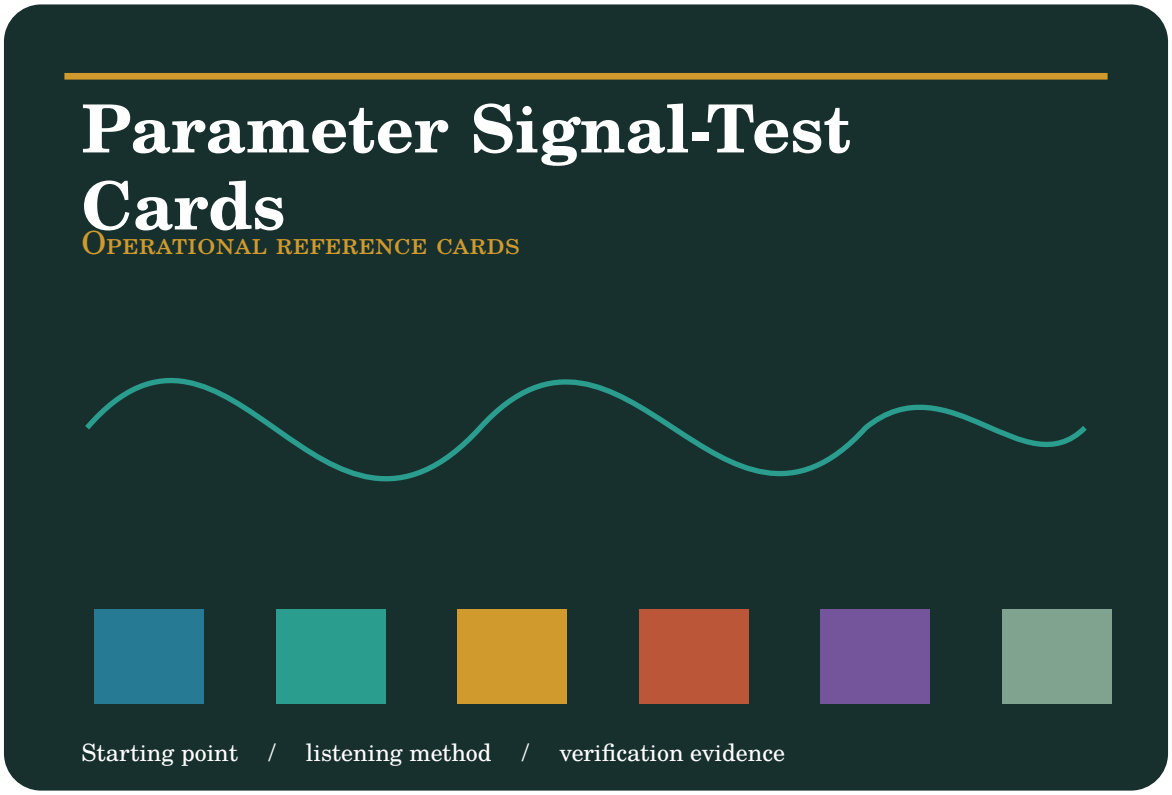
This card checks First-order ambisonics under geometry or ir processing: ACN/SN3D ordering, rotation behavior, and decoder-independent energy; prepared spatial assets, channel metadata, and deterministic fallback.

Record these card-specific values:

- Layout: First-order ambisonics: ACN/SN3D ordering, rotation behavior, and decoder-independent energy.
- Processing mode: Geometry or IR: prepared spatial assets, channel metadata, and deterministic fallback.
- Foundation status: mono/stereo only; larger layouts are future validation protocols.

- Required probes: per-channel impulses, correlated program, decorrelated program, silence, and full-scale safety input.

8.30 Parameter Signal-Test Cards



Prepare matched mono and stereo contexts at 48 kHz, then repeat failures at 44.1, 96, and 192 kHz. Capture default, endpoints, midpoint or discrete choices, a slow move, and an abrupt host change together with output, state, latency, peak, and channel count.

The same input and state must reproduce within the declared tolerance. A click, non-finite sample, unexplained latency jump, channel mismatch, or value-dependent allocation is a defect. Stabilize the native result before comparing formats; keep small golden fixtures or metrics and hashes for long renders.

Signal-test card: Pre-Delay with Digital silence

With digital silence, vary Pre-Delay to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Single-sample impulse

With single-sample impulse, vary Pre-Delay to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Full-scale alternating impulses

With full-scale alternating impulses, vary Pre-Delay to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with 80 Hz sine

With 80 hz sine, vary Pre-Delay to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with 8 kHz sine

With 8 khz sine, vary Pre-Delay to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Pink noise burst

With pink noise burst, vary Pre-Delay to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Dry speech phrase

With dry speech phrase, vary Pre-Delay to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Transient drum loop

With transient drum loop, vary Pre-Delay to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Sustained harmonic pad

With sustained harmonic pad, vary Pre-Delay to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Pre-Delay with Sixty-second tail capture

With sixty-second tail capture, vary Pre-Delay to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is use a ramp or delay-line crossfade.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: use a ramp or delay-line crossfade.

Signal-test card: Room Size with Digital silence

With digital silence, vary Room Size to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Single-sample impulse

With single-sample impulse, vary Room Size to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Full-scale alternating impulses

With full-scale alternating impulses, vary Room Size to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with 80 Hz sine

With 80 hz sine, vary Room Size to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with 8 kHz sine

With 8 khz sine, vary Room Size to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Pink noise burst

With pink noise burst, vary Room Size to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Dry speech phrase

With dry speech phrase, vary Room Size to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Transient drum loop

With transient drum loop, vary Room Size to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Sustained harmonic pad

With sustained harmonic pad, vary Room Size to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: Room Size with Sixty-second tail capture

With sixty-second tail capture, vary Room Size to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is stage structural changes outside the callback.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: stage structural changes outside the callback.

Signal-test card: RT60 Coarse with Digital silence

With digital silence, vary RT60 Coarse to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Single-sample impulse

With single-sample impulse, vary RT60 Coarse to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Full-scale alternating impulses

With full-scale alternating impulses, vary RT60 Coarse to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with 80 Hz sine

With 80 hz sine, vary RT60 Coarse to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with 8 kHz sine

With 8 khz sine, vary RT60 Coarse to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Pink noise burst

With pink noise burst, vary RT60 Coarse to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Dry speech phrase

With dry speech phrase, vary RT60 Coarse to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Transient drum loop

With transient drum loop, vary RT60 Coarse to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Sustained harmonic pad

With sustained harmonic pad, vary RT60 Coarse to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Coarse with Sixty-second tail capture

With sixty-second tail capture, vary RT60 Coarse to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is display the effective seconds value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: display the effective seconds value.

Signal-test card: RT60 Fine with Digital silence

With digital silence, vary RT60 Fine to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Single-sample impulse

With single-sample impulse, vary RT60 Fine to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Full-scale alternating impulses

With full-scale alternating impulses, vary RT60 Fine to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with 80 Hz sine

With 80 hz sine, vary RT60 Fine to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with 8 kHz sine

With 8 khz sine, vary RT60 Fine to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Pink noise burst

With pink noise burst, vary RT60 Fine to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Dry speech phrase

With dry speech phrase, vary RT60 Fine to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Transient drum loop

With transient drum loop, vary RT60 Fine to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Sustained harmonic pad

With sustained harmonic pad, vary RT60 Fine to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: RT60 Fine with Sixty-second tail capture

With sixty-second tail capture, vary RT60 Fine to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is keep zero as the exact neutral point.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: keep zero as the exact neutral point.

Signal-test card: Damping with Digital silence

With digital silence, vary Damping to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Single-sample impulse

With single-sample impulse, vary Damping to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Full-scale alternating impulses

With full-scale alternating impulses, vary Damping to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with 80 Hz sine

With 80 hz sine, vary Damping to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with 8 kHz sine

With 8 khz sine, vary Damping to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Pink noise burst

With pink noise burst, vary Damping to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Dry speech phrase

With dry speech phrase, vary Damping to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Transient drum loop

With transient drum loop, vary Damping to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Sustained harmonic pad

With sustained harmonic pad, vary Damping to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Damping with Sixty-second tail capture

With sixty-second tail capture, vary Damping to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is interpolate stable filter coefficients.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: interpolate stable filter coefficients.

Signal-test card: Width with Digital silence

With digital silence, vary Width to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Single-sample impulse

With single-sample impulse, vary Width to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Full-scale alternating impulses

With full-scale alternating impulses, vary Width to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with 80 Hz sine

With 80 hz sine, vary Width to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with 8 kHz sine

With 8 khz sine, vary Width to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Pink noise burst

With pink noise burst, vary Width to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Dry speech phrase

With dry speech phrase, vary Width to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Transient drum loop

With transient drum loop, vary Width to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Sustained harmonic pad

With sustained harmonic pad, vary Width to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Width with Sixty-second tail capture

With sixty-second tail capture, vary Width to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is monitor mono compatibility during movement.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: monitor mono compatibility during movement.

Signal-test card: Diffusion with Digital silence

With digital silence, vary Diffusion to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Single-sample impulse

With single-sample impulse, vary Diffusion to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Full-scale alternating impulses

With full-scale alternating impulses, vary Diffusion to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with 80 Hz sine

With 80 hz sine, vary Diffusion to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with 8 kHz sine

With 8 khz sine, vary Diffusion to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Pink noise burst

With pink noise burst, vary Diffusion to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Dry speech phrase

With dry speech phrase, vary Diffusion to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Transient drum loop

With transient drum loop, vary Diffusion to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Sustained harmonic pad

With sustained harmonic pad, vary Diffusion to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Diffusion with Sixty-second tail capture

With sixty-second tail capture, vary Diffusion to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is crossfade when topology must change.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: crossfade when topology must change.

Signal-test card: Wet with Digital silence

With digital silence, vary Wet to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Single-sample impulse

With single-sample impulse, vary Wet to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Full-scale alternating impulses

With full-scale alternating impulses, vary Wet to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with 80 Hz sine

With 80 hz sine, vary Wet to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with 8 kHz sine

With 8 khz sine, vary Wet to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Pink noise burst

With pink noise burst, vary Wet to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Dry speech phrase

With dry speech phrase, vary Wet to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Transient drum loop

With transient drum loop, vary Wet to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Sustained harmonic pad

With sustained harmonic pad, vary Wet to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: choose and document the mix law.

Signal-test card: Wet with Sixty-second tail capture

With sixty-second tail capture, vary Wet to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is choose and document the mix law.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: choose and document the mix law.

Signal-test card: Dry with Digital silence

With digital silence, vary Dry to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Single-sample impulse

With single-sample impulse, vary Dry to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Full-scale alternating impulses

With full-scale alternating impulses, vary Dry to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with 80 Hz sine

With 80 hz sine, vary Dry to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with 8 kHz sine

With 8 khz sine, vary Dry to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Pink noise burst

With pink noise burst, vary Dry to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Dry speech phrase

With dry speech phrase, vary Dry to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Transient drum loop

With transient drum loop, vary Dry to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Sustained harmonic pad

With sustained harmonic pad, vary Dry to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Dry with Sixty-second tail capture

With sixty-second tail capture, vary Dry to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is preserve bypass and gain staging.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: preserve bypass and gain staging.

Signal-test card: Freeze with Digital silence

With digital silence, vary Freeze to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Single-sample impulse

With single-sample impulse, vary Freeze to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Full-scale alternating impulses

With full-scale alternating impulses, vary Freeze to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with 80 Hz sine

With 80 hz sine, vary Freeze to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with 8 kHz sine

With 8 khz sine, vary Freeze to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Pink noise burst

With pink noise burst, vary Freeze to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Dry speech phrase

With dry speech phrase, vary Freeze to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Transient drum loop

With transient drum loop, vary Freeze to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Sustained harmonic pad

With sustained harmonic pad, vary Freeze to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Freeze with Sixty-second tail capture

With sixty-second tail capture, vary Freeze to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is use a debounced, smoothed mode transition.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: use a debounced, smoothed mode transition.

Signal-test card: Reverse with Digital silence

With digital silence, vary Reverse to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Single-sample impulse

With single-sample impulse, vary Reverse to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Full-scale alternating impulses

With full-scale alternating impulses, vary Reverse to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with 80 Hz sine

With 80 hz sine, vary Reverse to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with 8 kHz sine

With 8 khz sine, vary Reverse to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Pink noise burst

With pink noise burst, vary Reverse to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Dry speech phrase

With dry speech phrase, vary Reverse to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Transient drum loop

With transient drum loop, vary Reverse to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Sustained harmonic pad

With sustained harmonic pad, vary Reverse to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: report added latency before activation.

Signal-test card: Reverse with Sixty-second tail capture

With sixty-second tail capture, vary Reverse to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is report added latency before activation.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: report added latency before activation.

Signal-test card: Quality with Digital silence

With digital silence, vary Quality to detect denormals, stale buffers, uninitialized state, and noise-floor growth. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Digital silence.
- Observation goal: detect denormals, stale buffers, uninitialized state, and noise-floor growth.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Single-sample impulse

With single-sample impulse, vary Quality to reveal latency, early reflections, channel routing, and deterministic tail shape. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Single-sample impulse.
- Observation goal: reveal latency, early reflections, channel routing, and deterministic tail shape.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Full-scale alternating impulses

With full-scale alternating impulses, vary Quality to stress headroom, limiter response, mode transitions, and sign symmetry. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Full-scale alternating impulses.
- Observation goal: stress headroom, limiter response, mode transitions, and sign symmetry.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with 80 Hz sine

With 80 hz sine, vary Quality to expose low-frequency decay, modulation, modal buildup, and channel phase differences. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: 80 Hz sine.
- Observation goal: expose low-frequency decay, modulation, modal buildup, and channel phase differences.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with 8 kHz sine

With 8 khz sine, vary Quality to expose damping, interpolation, aliasing, and high-frequency stability. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: 8 kHz sine.
- Observation goal: expose damping, interpolation, aliasing, and high-frequency stability.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Pink noise burst

With pink noise burst, vary Quality to show broadband spectral decay, gain behavior, and early/late balance. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Pink noise burst.
- Observation goal: show broadband spectral decay, gain behavior, and early/late balance.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Dry speech phrase

With dry speech phrase, vary Quality to test intelligibility, sibilance, plosives, pre-delay, and ducking behavior. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Dry speech phrase.
- Observation goal: test intelligibility, sibilance, plosives, pre-delay, and ducking behavior.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Transient drum loop

With transient drum loop, vary Quality to test attack preservation, density buildup, tempo interaction, and peak safety. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Transient drum loop.
- Observation goal: test attack preservation, density buildup, tempo interaction, and peak safety.
- Required transition behavior: apply through a safe reprepare boundary.

Signal-test card: Quality with Sustained harmonic pad

With sustained harmonic pad, vary Quality to test modulation smoothness, correlation, freeze energy, and long-term stability. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Sustained harmonic pad.
- Observation goal: test modulation smoothness, correlation, freeze energy, and long-term stability.
- Required transition behavior: apply through a safe reprepare boundary.

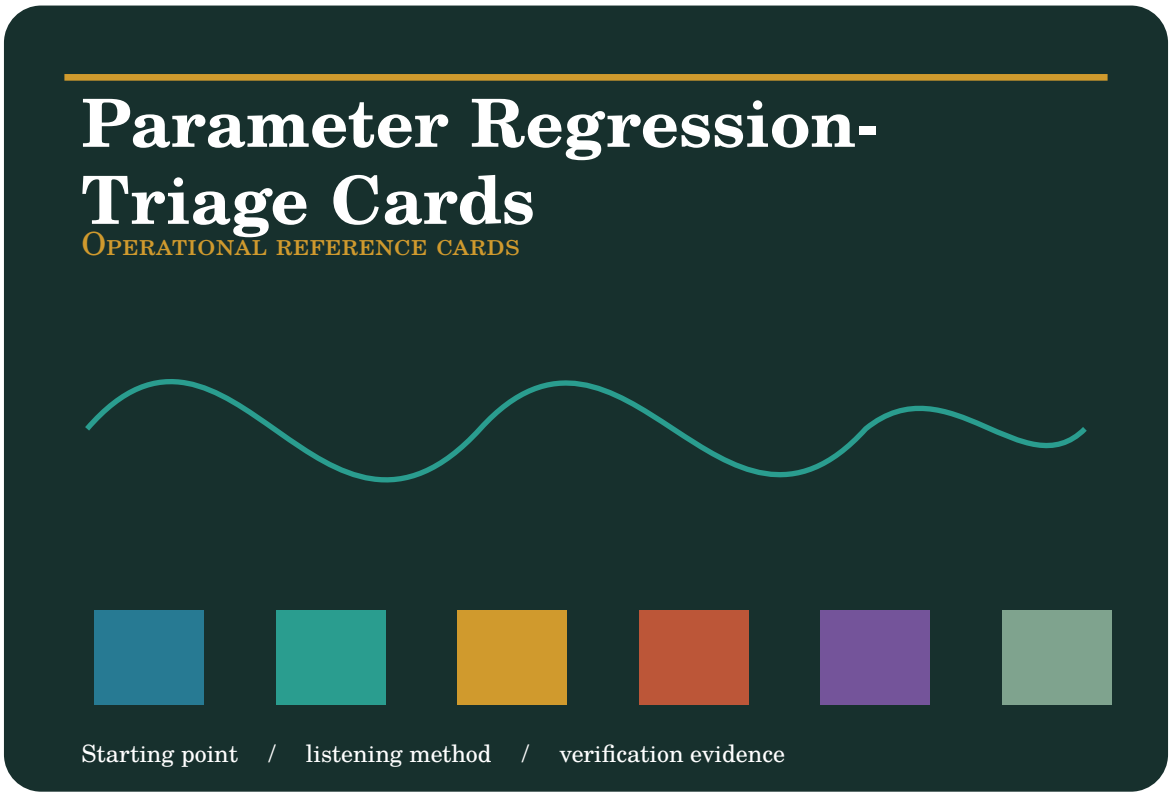
Signal-test card: Quality with Sixty-second tail capture

With sixty-second tail capture, vary Quality to test memory stability, decay completion, noise floor, and repeatability. The required transition behavior is apply through a safe reprepare boundary.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Signal: Sixty-second tail capture.
- Observation goal: test memory stability, decay completion, noise floor, and repeatability.
- Required transition behavior: apply through a safe reprepare boundary.

8.31 Parameter Regression-Triage Cards



Freeze the failing build and preserve the smallest audio fixture, state blob, automation lane, or screenshot that reproduces the outcome. Record format, host, architecture, rate, block size, layout, quality, parameter values, transport, latency, and editor state.

Classify severity by user consequence, find the first contract boundary that diverges, and fix the lowest shared owner. Add a native regression, then repeat format smoke tests and the original recovery. Close triage only when the project is recovered or a tested migration path is documented.

Triage card: Pre-Delay: Audio safety

For Pre-Delay, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is use a ramp or delay-line crossfade; trace `pre_delay_ms` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: use a ramp or delay-line crossfade.

Triage card: Pre-Delay: State and automation

For Pre-Delay, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is use a ramp or delay-line crossfade; trace `pre_delay_ms` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: use a ramp or delay-line crossfade.

Triage card: Pre-Delay: User-facing behavior

For Pre-Delay, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is use a ramp or delay-line crossfade; trace `pre_delay_ms` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `pre_delay_ms`: separates the direct event from the room onset.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: use a ramp or delay-line crossfade.

Triage card: Room Size: Audio safety

For Room Size, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is stage structural changes outside the callback; trace `room_size` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: stage structural changes outside the callback.

Triage card: Room Size: State and automation

For Room Size, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is stage structural changes outside the callback; trace `room_size` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: stage structural changes outside the callback.

Triage card: Room Size: User-facing behavior

For Room Size, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is stage structural changes outside the callback; trace `room_size` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `room_size`: changes perceived scale and reflection spacing.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: stage structural changes outside the callback.

Triage card: RT60 Coarse: Audio safety

For RT60 Coarse, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is display the effective seconds value; trace `rt60_coarse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: display the effective seconds value.

Triage card: RT60 Coarse: State and automation

For RT60 Coarse, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is display the effective seconds value; trace `rt60_coarse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: display the effective seconds value.

Triage card: RT60 Coarse: User-facing behavior

For RT60 Coarse, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is display the effective seconds value; trace `rt60_coarse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_coarse`: moves through the full logarithmic decay range.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: display the effective seconds value.

Triage card: RT60 Fine: Audio safety

For RT60 Fine, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is keep zero as the exact neutral point; trace `rt60_fine` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: keep zero as the exact neutral point.

Triage card: RT60 Fine: State and automation

For RT60 Fine, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is keep zero as the exact neutral point; trace `rt60_fine` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: keep zero as the exact neutral point.

Triage card: RT60 Fine: User-facing behavior

For RT60 Fine, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is keep zero as the exact neutral point; trace `rt60_fine` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `rt60_fine`: trims decay proportionally around the coarse value.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: keep zero as the exact neutral point.

Triage card: Damping: Audio safety

For Damping, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is interpolate stable filter coefficients; trace `damping` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: interpolate stable filter coefficients.

Triage card: Damping: State and automation

For Damping, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is interpolate stable filter coefficients; trace `damping` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: interpolate stable filter coefficients.

Triage card: Damping: User-facing behavior

For Damping, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is interpolate stable filter coefficients; trace `damping` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `damping`: changes high-frequency persistence.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: interpolate stable filter coefficients.

Triage card: Width: Audio safety

For Width, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is monitor mono compatibility during movement; trace `width` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: monitor mono compatibility during movement.

Triage card: Width: State and automation

For Width, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is monitor mono compatibility during movement; trace `width` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: monitor mono compatibility during movement.

Triage card: Width: User-facing behavior

For Width, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is monitor mono compatibility during movement; trace `width` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `width`: changes lateral energy and correlation.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: monitor mono compatibility during movement.

Triage card: Diffusion: Audio safety

For Diffusion, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is crossfade when topology must change; trace `diffusion` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: crossfade when topology must change.

Triage card: Diffusion: State and automation

For Diffusion, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is crossfade when topology must change; trace `diffusion` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: crossfade when topology must change.

Triage card: Diffusion: User-facing behavior

For Diffusion, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is crossfade when topology must change; trace `diffusion` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `diffusion`: changes echo-density buildup.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: crossfade when topology must change.

Triage card: Wet: Audio safety

For Wet, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is choose and document the mix law; trace `wet` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: choose and document the mix law.

Triage card: Wet: State and automation

For Wet, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is choose and document the mix law; trace `wet` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: choose and document the mix law.

Triage card: Wet: User-facing behavior

For Wet, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is choose and document the mix law; trace `wet` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `wet`: sets processed contribution.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: choose and document the mix law.

Triage card: Dry: Audio safety

For Dry, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is preserve bypass and gain staging; trace `dry` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: preserve bypass and gain staging.

Triage card: Dry: State and automation

For Dry, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is preserve bypass and gain staging; trace `dry` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: preserve bypass and gain staging.

Triage card: Dry: User-facing behavior

For Dry, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is preserve bypass and gain staging; trace `dry` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `dry`: sets direct contribution.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: preserve bypass and gain staging.

Triage card: Freeze: Audio safety

For Freeze, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is use a debounced, smoothed mode transition; trace `freeze` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: use a debounced, smoothed mode transition.

Triage card: Freeze: State and automation

For Freeze, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is use a debounced, smoothed mode transition; trace `freeze` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: use a debounced, smoothed mode transition.

Triage card: Freeze: User-facing behavior

For Freeze, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is use a debounced, smoothed mode transition; trace `freeze` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `freeze`: changes network energy behavior.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: use a debounced, smoothed mode transition.

Triage card: Reverse: Audio safety

For Reverse, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is report added latency before activation; trace `reverse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: report added latency before activation.

Triage card: Reverse: State and automation

For Reverse, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is report added latency before activation; trace `reverse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: report added latency before activation.

Triage card: Reverse: User-facing behavior

For Reverse, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is report added latency before activation; trace `reverse` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `reverse`: changes the envelope and buffering model.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: report added latency before activation.

Triage card: Quality: Audio safety

For Quality, classify audio safety against clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency. The expected transition is apply through a safe reprepare boundary; trace `quality_mode` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Failure class: Audio safety.
- Scope: clicks, instability, non-finite output, runaway gain, channel corruption, or unreported latency.
- Expected transition: apply through a safe reprepare boundary.

Triage card: Quality: State and automation

For Quality, classify state and automation against wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement. The expected transition is apply through a safe reprepare boundary; trace `quality_mode` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Failure class: State and automation.
- Scope: wrong recall, parameter-ID drift, host-write mismatch, migration loss, or format disagreement.
- Expected transition: apply through a safe reprepare boundary.

Triage card: Quality: User-facing behavior

For Quality, classify user-facing behavior against misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition. The expected transition is apply through a safe reprepare boundary; trace `quality_mode` from manifest through state, processor, and displayed value.

Record these card-specific values:

- Parameter: `quality_mode`: selects the internal rate policy.
- Failure class: User-facing behavior.
- Scope: misleading value text, stale UI, unclear status, inaccessible control, or undocumented transition.
- Expected transition: apply through a safe reprepare boundary.

8.32 Closing Checklist

Before calling a VERBX plug-in build production-ready, confirm:

- the JUCE-enabled targets compile for every claimed architecture and format;
- parameter IDs and state migrations are frozen and tested;
- the realtime DSP replaces pass-through and has deterministic golden tests;
- callback code is allocation-free, lock-free, bounded, and NaN-safe;
- latency is measured and reported for every processing mode;
- sample-rate, block-size, layout, bypass, and transport changes are tested;
- Freeze and Reverse have explicit energy, transition, and latency semantics;
- geometry and IR assets are prepared outside the callback and recalled by hash;
- the production editor implements the visual target accessibly and without affecting sound;
- every host compatibility claim names a dated tested environment; and
- installers, signatures, notarization, scanning, crash recovery, and support bundles are complete.

The current stateful oversampled reverb makes these obligations visible and testable. The next major milestone is bounded-lookahead reverse processing with exact host latency notification, followed by multichannel layouts and a dated compatibility matrix.

Part III

DSP, Impulse Responses, and Technical Reference

8.33 Suggested Exercises

1. Recreate one CLI reverb design in the plug-in, then print the return and analyze it with the CLI. Document intentional differences.
2. Automate RT60 coarse and fine controls separately over a phrase. Describe the musical reason to use each control's range.
3. Use the Expert page to make a stable wide late field without changing the dry source's localization; compare stereo and mono fold-downs.
4. Use the spectrum analyzer to identify a spectral buildup, then correct it with one reverb parameter before reaching for unrelated channel EQ.
5. Create a DAW preset with a reverse or freeze gesture and record the host automation so another session can reproduce the form.
6. Build an A/B automation lane that compares two room sizes without changing gain, then blind-test the result with a collaborator.
7. Use the analyzer during a full mix and make one intervention that reduces masking without lowering the reverb return fader.
8. Design a performance-safe freeze patch with a clear entry, sustain, exit, and emergency-reset gesture.
9. Compare host tempo-synced pre-delay with a fixed-millisecond pre-delay across two tempi and describe the musical consequence.
10. Create a preset naming scheme that distinguishes source role, room role, algorithm, and expected automation behavior.

"As if a double hunt were heard at once."

— William Shakespeare, *Titus Andronicus* (1594)

Chapter 9

IR Synthesis – A Dual-Layer Reference

Version: v0.7.7 (public alpha)

This document is written for two readers at once. The plain-English sections give you enough to use the tool effectively without reading source code. The technical sections go deep – modal decay math, FDN topology, stochastic density shaping, morph blend modes – and assume you know what a Schroeder allpass is and roughly why it matters. Skip what you don't need.

9.1 What Is an Impulse Response?

9.1.1 Plain English

Clap your hands once in a cathedral. The sound you hear after that single clap — all those echoes, the wash, the way it slowly dies away – that is the room's impulse response. It is the room's acoustic fingerprint.

Convolution reverb works by capturing that fingerprint as a digital audio file and mathematically "stamping" it onto whatever audio you feed in. The result sounds like the audio was recorded in that room. It is not simulation, it is multiplication in the frequency domain – and it sounds convincing because it uses real (or carefully synthesized) physics.

verbx does not require you to record a real room. It synthesizes IRs from scratch, with deterministic parameters, so you get the same result every time from the same seed.

9.1.2 For DSP Engineers

An impulse response $h[n]$ completely characterizes a linear time-invariant (LTI) system. In the context of room acoustics, convolution reverb computes:

$$y[n] = (x * h)[n] = \sum_{k} x[k]\, h[n-k]$$

which in practice is implemented as OLA or OLS partitioned convolution to keep block latency manageable. verbx's IR synthesis produces $h[n]$ directly, bypassing any physical measurement chain. The goal is parametric control rather than physical accuracy.

All internal DSP runs in float64. Output subtype (float32, float64, PCM-16/24) is a container choice at export time and does not affect synthesis precision. The output is always normalized (peak or RMS/LUFS depending on config) before writing.

RT60 is defined in the Sabine sense: the time for the sound pressure level to

decay by 60 dB. Internally verbx resolves T_{60} into per-mode envelope time constants via:

$$\tau = \frac{T_{60}}{\ln(10^6)} = \frac{T_{60}}{13.816}$$

which gives the exponential decay envelope $e^{-t/\tau}$ used across modes.

9.2 The Four Synthesis Modes

9.2.1 2a. `fdn` – Feedback Delay Network

9.2.1.1 What it sounds like

Dense, smooth, algorithmic reverb. Think classic hardware reverb units from the 1980s – less "room" character, more "sound in space." Works well for clean ambience that sits behind a mix without calling attention to itself. The modulation options add subtle pitch shimmer that keeps it from sounding static.

9.2.1.2 How it works

The FDN is the workhorse of algorithmic reverb. The architecture is a set of N delay lines connected in a feedback loop through a mixing matrix M:

$$s[n] = M\,s[n-d] + b\,x[n]$$
$$y[n] = c^{\mathsf{T}} s[n]$$

where d is the vector of delay lengths (in samples), b is the input gains, and c is the output tap weights. For the loop to be stable and lossless, M must be unitary or near-unitary – verbx supports Hadamard matrices (default) and random orthogonal matrices. The Hadamard construction at order N is efficient ($\mathcal{O}(N \log N)$ multiply) and guarantees perfect echo density growth.

Delay line lengths are chosen to be mutually coprime (no shared factors) to maximize echo density. The default `fdn_lines = 8` gives 8 delay lines; going to 16 significantly increases diffusion at the cost of memory and compute.

The FDN includes optional time-varying modulation (`fdn_tv_rate_hz`, `fdn_tv_depth`): each delay line length is sinusoidally modulated at an individually offset rate derived from the master rate. This breaks up metallic resonances at the cost of LTI purity – the output is no longer strictly an IR of a fixed system, but the artifact is perceptually benign at low modulation depths.

Three-band RT60 control (`fdn_rt60_low`, `fdn_rt60_mid`, `fdn_rt60_high`) is implemented as per-band absorption shelving filters inserted in each feedback loop – similar in spirit to the Jot-Chaigne absorptive FDN. Low-frequency RT60 is typically longer than mid in real rooms; setting `fdn_rt60_low` slightly above `fdn_rt60_mid` makes the tail feel more physical.

Graph topology options (`ring`, `random`) control how the N delay lines interconnect before the full mixing matrix is applied. Ring is the classic single-loop structure; random graph with degree k introduces k cross-connections per node and increases early diffusion at the cost of slightly less predictable echo pattern.

The cascade FDN option (`fdn_cascade`) chains two FDN stages at different delay scales, with the second stage fed by the first at a ratio of `fdn_cascade_delay_scale`. This is loosely analogous to Griesinger's nested FDN and significantly increases low-frequency density without proportionally increasing the delay buffer size.

Known limitation: The FDN IR is rendered by impulse-exciting the network and reading $N \times L$ output samples, where L is the requested IR length. For very long IRs (120s at 48kHz) this is about 5.76e9

operations – non-trivial. Sparse mode (`fdn_sparse`) reduces this by zeroing low-magnitude delay connections, with controllable degree.

9.2.2 2b. `stochastic` – Noise-Shaped Diffuse Tail

9.2.2.1 What it sounds like

Pure reverb wash with no discrete echoes and no tonal coloration. Sounds like the late tail of a large space – a long, smooth decay. Useful as a bed when you want pure texture without any pitched character. Can sound a bit thin on its own (there is no direct sound energy), but as a component in hybrid mode it provides exactly what it promises: texture.

9.2.2.2 How it works

Stochastic synthesis generates band-limited noise shaped by a smooth exponential envelope. The core idea is that a diffuse reverb tail, at sufficient echo density, is perceptually indistinguishable from exponentially decaying noise filtered to match the room's frequency-dependent absorption. This is the statistical basis for Gardner's observation that above roughly 2000 echoes/second the ear cannot resolve individual reflections.

The implementation generates a Gaussian noise vector of the full IR length, applies a per-channel decorrelation (different seed offsets per channel), then multiplies by an RT60 envelope:

$$h[n] = \text{noise}[n]\, e^{-n/(f_s \tau)}$$

where $\tau = \frac{T_{60}}{\ln(10^6)}$. The `diffusion` parameter controls a frequency- domain shaping pass: values above 0.5 increasingly smooth the spectral envelope toward white (flat), while lower values allow more spectral irregularity to persist, giving a slightly colored tail that can sound more like a real room.

The `density` parameter scales the RMS of the noise before the envelope is applied. It does not change the decay shape, only the apparent "thickness" of the tail. At density > 1.0 the tail can clip before normalization, which is intentional – normalization re-scales – but very high density values will pull the effective noise floor up and reduce dynamic range in the output.

Band control (`rt60_low`, `rt60_high`): when these differ, the stochastic tail interpolates between two exponential envelopes across frequency using a crossover filter. This approximates frequency-dependent absorption without a full filter-per-sample approach.

9.2.3 2c. `modal` – Decaying Modal Resonator Bank

9.2.3.1 What it sounds like

Pitched, tonal, resonant – like a metal plate, a wine glass, or a very live tiled room. Modes ring at specific frequencies, giving reverb a musical quality that can either blend into pitched content or clash with it depending on tuning. Excellent for designed reverbs on melodic material when you want the space to have harmonic relationship to the source.

9.2.3.2 How it works

Each mode is an exponentially decaying sinusoid:

$$h_k[n] = A_k \sin\left(2\pi f_k n / f_s + \phi_k\right) e^{-n/\tau_k}$$

with amplitude A_k, frequency f_k, phase ϕ_k, and time constant τ_k. The bank sums N such modes:

$$h[n] = \frac{1}{\sqrt{N}} \sum_{k=1}^{N} h_k[n] + \epsilon \text{ noise_bed}[n]$$

The $1/\sqrt{N}$ normalization keeps total energy approximately constant regardless of modal count, which is the right thing to do (amplitude does not add linearly for uncorrelated modes – RMS does).

Frequencies are sampled log-uniformly between `modal_low_hz` and `modal_high_hz`:

$$f_k = \exp\Big(U[\log(f_{\text{low}}), \log(f_{\text{high}})]\Big)$$

This is the correct distribution for musical pitch perception – equal intervals on a log scale correspond to equal musical intervals. Linear uniform sampling would cluster modes near `modal_high_hz` perceptually.

The Q-to-τ relationship follows standard resonator theory:

$$\tau_q = \frac{Q}{\pi f_k}$$

but verbx blends this with the global RT60 target:

$$\tau_{\text{RT}} = \frac{T_{60}}{6.91}$$
$$\tau_k = \min\big(\max(\tau_q, 0.01), \max(2\tau_{\text{RT}}, 0.03)\big)$$

This clipping prevents individual modes from ringing dramatically longer than the RT60 budget, which would otherwise create audible individual tones at the end of the tail – a problem that sounds terrible and is hard to fix after the fact. In practice you can push `modal_q_max` to 200 or higher if you want those long-ringing tones intentionally (bells, chimes), just know the IR will have frequency-domain spikes that may interact poorly with the source material unless tuned deliberately.

Harmonic alignment (`--f0`, `--analyze-input`) nudges mode frequencies toward integer multiples of a fundamental using `tune_frequency_to_targets()`. The `align_strength` parameter (0–1) linearly interpolates each mode's sampled frequency toward the nearest harmonic target. At strength 1.0, all modes collapse to exact harmonics – a fully pitched comb. At 0.0, no alignment. Values around 0.5–0.7 give a nice blend of harmonic coloration without the artificiality of a pure comb filter.

9.2.3.3 Scala scales and microtonal emphasis

`--scala-file SCALE.scl` replaces the ordinary harmonic target list with pitch

classes parsed from a Scala scale. verbx accepts cents entries such as `63.1579`, integer ratios such as `2`, and fractional ratios such as `7/4`. Lines beginning with `!`, and text following `!` on a data line, are comments. The final declared pitch defines the repeat interval, so octave, tritave, and other non-octave scales use the same path.

Let r_d be the ratio of scale degree d, r_p the repeat interval, f_r the root frequency, and d_r the selected root degree. The resolved frequency for degree d in register k is

$$f_{d,k} = f_r \frac{r_d}{r_{d_r}} r_p^k.$$

verbx expands this lattice only between `--scala-low-hz` and `--scala-high-hz`, then removes targets above 0.49 times the sample rate. If a high-division scale exceeds `--scala-max-targets`, targets are selected at even positions on the logarithmically ordered list so low and high registers remain represented. This is a deterministic DSP budget, not a random pitch omission.

The resolved frequencies serve two related purposes. First, modal frequencies are pulled toward the nearest scale target by `--scala-strength`. Second, after the selected synthesis mode is complete, a parallel constant-Q filter bank reinforces those bands in every mode. For a center frequency f_c and bandwidth b cents, the approximate filter quality is

$$Q = \frac{f_c}{f_c \left(2^{b/2400} - 2^{-b/2400}\right)}.$$

The filtered bank is RMS-normalized before `--scala-gain-db` and `--scala-strength` are applied. Final IR normalization still runs afterward, which bounds export headroom but does not erase the relative spectral emphasis. Narrow widths produce clearly pitched resonances; broader widths produce a gentler spectral affinity. Start at 20 to 35 cents, 3 to 6 dB, and strength 0.5 to 0.75 for musical material.

Table 9-1: Scala scales and microtonal emphasis: reference table

Option	Default	Meaning
`--scala-file`	none	UTF-8 Scala `.scl` file
`--scala-root-hz`	440 Hz	Frequency assigned to the selected root degree
`--scala-root-degree`	0	Zero-based degree treated as the root
`--scala-low-hz`	modal low limit	Lowest generated target
`--scala-high-hz`	modal high limit	Highest target before Nyquist clamping
`--scala-strength`	1.0	Modal attraction and emphasis blend, 0 to 1
`--scala-bandwidth-cents`	25 cents	Constant-Q emphasis width
`--scala-gain-db`	4 dB	Emphasis-bank gain before final normalization
`--scala-max-targets`	128	Maximum filter and tuning target count

The parser rejects malformed counts, nonpositive ratios, unsorted degrees, invalid root degrees, and empty post-Nyquist ranges before synthesis. Scala tuning cannot be combined with `--analyze-input` or `--f0` because those options would define a competing frequency target set. Use `--scala-root-hz` instead.

For complete import examples and a musical treatment of consonance through time, root mapping, register, scale cardinality, non-octave periods, transposition, harmonic rhythm, orchestration, spatialization, and changing harmony, continue with Microtonal Workflows, Scala Import, and Scale-Tuned Reverberation.

An air noise bed at amplitude $0.02\, r_{\mathrm{RMS}}$ is added to prevent the IR from being spectrally empty between modal peaks. Without it, convolution with broadband content can reveal the gaps as spectral dips.

Stereo: each mode is panned independently using equal-power panning: $L = \sqrt{0.5(1-p)}$ and $R = \sqrt{0.5(1+p)}$, where p is pan position. For $C > 2$ channels, additional channels receive the mono sum scaled by $1/C$. Not ideal for surround, but functional for stereo and tolerable for LCR.

9.2.4 2d. `hybrid` – The Mode You Will Actually Use

9.2.4.1 What it sounds like

Combines discrete early echoes (the "slap" you hear in the first 50–120ms) with a blended late tail. The early reflections give spatial cues – your brain uses them to estimate room size and shape. The late tail does the wash. Together they sound more like a real room than any single mode alone.

In practice, hybrid is what you actually want for most work. The other modes are useful for sound design extremes (very pure algorithmic color, pitched modal effects) or as components you mix yourself. Hybrid handles general-purpose reverb production well.

9.2.4.2 How it works

Hybrid generation follows a four-step pipeline:

1. Generate early reflections (see Section 3)
2. Generate three late-tail components independently: stochastic (seed + 11), modal (seed + 17, half the modal count), FDN (seed + 23)
3. Blend the three tails with fixed weights:

$$h_{\text{tail}} = 0.55\, h_{\text{stoch}} + 0.25\, h_{\text{modal}} + 0.20\, h_{\text{FDN}}$$

4. Add early reflections by overlap-add into the first er_max_delay_ms samples

The seed offsets (+11, +17, +23) are not magic numbers – they just ensure the three RNGs produce independent sequences from the same user seed. The blending weights (0.55/0.25/0.20) were chosen empirically: stochastic dominates because it produces the smoothest late tail, modal adds tonal character, FDN adds algorithmic density structure. These weights are fixed in the current release; future versions may expose them as parameters.

The post-blend pipeline applies globally: harmonic alignment, Scala constant-Q emphasis, optional Modalys resonator layer, then IR shaping (filters, normalization). Tuning flags such as `--f0`, `--scala-file`, and `--resonator` therefore work across all modes, not just modal.

9.3 Early Reflections

9.3.1 Perceptual Role

Early reflections – the first 5–80ms or so after the direct sound – are the most important part of reverb for spatial perception. They are what lets you tell a bathroom from a concert hall even before the tail arrives. They carry IACC (interaural cross-correlation) cues that localize you in space, and their density and timing pattern gives a strong sense of room size.

The direct sound is always present at sample 0 (amplitude 1.0 per channel) as a convolution anchor – this is the Kronecker delta that initiates the impulse response and ensures the convolution engine sees a properly calibrated gain reference.

9.3.2 Stochastic Cloud Implementation

Early reflections in verbx are a sparse, deterministic stochastic cloud rather than a geometric ray-tracing approximation. This is a deliberate choice: geometric computation would require room dimensions and surface materials as inputs, adding complexity with limited benefit for non-realistic spaces.

The generator (`generate_early_reflections`) places `er_count` taps into a buffer of length `er_max_delay_ms` samples. Each tap's delay is sampled uniformly over [1, max_delay_samples]:

$$d_k \sim U\left[1, \left\lfloor \frac{t_{\max} f_s}{1000} \right\rfloor\right]$$

Amplitude follows one of three decay laws indexed by `er_decay_shape`:

$$\begin{aligned}
\text{linear:} \quad & amp = 1 - \left(\frac{\text{delay}}{\text{max_delay}}\right) \\
\text{sqrt:} \quad & amp = \sqrt{\max\left(10^{-6}, 1 - \frac{\text{delay}}{\text{max_delay}}\right)} \\
\text{exp:} \quad & amp = \exp\left(-3.2 \frac{\text{delay}}{\text{max_delay}}\right)
\end{aligned}$$

Here d_k is tap k's delay in samples, $t_{\max}$ is the `er_max_delay_ms` control, and f_s is sample rate in hertz. The exponential law (default) corresponds to a room with absorption coefficient around 0.2 uniformly distributed, which is plausible for a moderately live space. The sqrt law gives a slower initial roll-off and is useful for very live spaces like stairwells. Linear is rarely what you want but is there for completeness.

The raw decay amplitude is then multiplied by a uniform random jitter in [0.35, 1.0] and by `er_room` (clipped to [0.1, 3.0]):

$$a_k = q(d_k)\, r_{\text{ER}}\, U[0.35, 1.0]$$

Here a_k is tap amplitude, $q(d_k)$ is the selected decay law, and r_{ER} is the `er_room` value. The `er_room` parameter is loosely a room-size proxy: values > 1.0 increase reflection amplitude, making the space feel more reverberant; values < 1.0 create a drier early field.

Stereo width is applied as pan spread per tap. Pan p_k is sampled from $U[-1, 1]$ and scaled by `er_stereo_width`. Left/right amplitudes use a simple linear pan law (not equal-power) because the taps are sparse enough that the distinction matters less than in dense diffuse material:

$$\begin{aligned}
L_k &= 0.5\, a_k\, (2 - \max(0, p_k)) \\
R_k &= 0.5\, a_k\, (2 + \min(0, p_k))
\end{aligned}$$

For er_stereo_width = 0.0, all taps are center-panned (mono ER). For 1.0, full ±1.0 pan range. For 2.0, panning can exceed the unit range, intentionally widening beyond the speaker base – useful for envelopment effects but beware mono compatibility.

One thing I would change in a future version: the tap delays should ideally be constrained to avoid clustering (e.g., via Poisson process sampling or minimum inter-tap spacing). Uniform sampling occasionally produces bunches of taps at similar delays that create audible comb-filter artifacts on transient material. For now, setting `er_count` to 16–24 rather than 64 reduces this risk.

9.4 IR Morphing

9.4.1 What It Does

Morphing takes two existing IRs – say, a small room and a large hall – and interpolates between them to create a new IR that has characteristics of both. You control the blend position with `--alpha` (0.0 = first IR, 1.0 = second IR, 0.5 = midpoint). The cache means repeated morphs at the same parameters are instantaneous.

This is useful in production when you want to automate a sense of space expanding or contracting across a section of a piece, or when you have two reference IRs and want something "between" them tonally.

9.4.2 Blend Modes

Three blend modes are available via `--mode`:

`linear`

Simple time-domain weighted sum:

$$h_{\text{out}}[n] = (1 - \alpha)h_A[n] + \alpha h_B[n]$$

Fast but acoustically naive. If A and B have different RT60 values, the crossfaded IR will have a decay shape that is neither A's nor B's – it will be a linear interpolation of amplitudes, which for exponential decays means the instantaneous decay rate changes non-monotonically with α. At $\alpha = 0.5$ between a 1s and 4s RT60, you get approximately a 2s decay, but the shape has a kink where the two envelopes cross. For subtle blends ($\alpha < 0.2$ or $\alpha > 0.8$) this is usually fine.

`spectral`

Blend in the frequency domain:

$$H_{\text{out}}[k] = (1 - \alpha)H_A[k] + \alpha H_B[k]$$

where H_A and H_B are the FFTs of the respective IRs (zero-padded to avoid circular convolution artifacts on the way back). This preserves the spectral shape of each source more faithfully than time-domain blending. Particularly useful when the two IRs have different frequency-dependent coloration (one bright, one dark) and you want to smoothly interpolate that color.

Phase handling: the implementation blends complex spectra, not magnitude-only. If A and B have substantially different phase responses (which real IRs almost always do), the interpolated phase can produce notches in the output. The `--align-decay` flag pre-aligns the energy onset of both IRs before blending, which partially mitigates this.

`envelope-aware`

The most sophisticated option and the one I would recommend as a starting point for serious work. It estimates the temporal energy envelope of each IR, blends the envelopes separately, then re-modulates the blended spectral content. This is loosely analogous to the approach in Cross-Synthesis of exponential decays.

Specifically:

1. Compute short-time energy envelopes $E_A[t]$ and $E_B[t]$ via squared Hilbert analytic signal or RMS-over-window (current implementation uses the latter at 10ms windows for speed)
2. Interpolate envelopes:

$$E_{\text{out}}[t] = (1 - \alpha)E_A[t] + \alpha E_B[t]$$

3. Generate late tail by blending at the spectral level with envelope correction: scale $H_{\text{out}}[k]$ such that the resulting time-domain signal matches $E_{\text{out}}[t]$

Morph quality diagnostics are reported in the metadata: RT60 drift ($|T_{60,\text{out}} - T_{60,\text{expected}}|$ in seconds) and spectral distance (log-spectral deviation in dB between H_{out} and the target interpolated spectrum). Use these to detect degenerate morphs before committing them to a batch render.

9.5 The Cache

9.5.1 Why It Matters

Generating a 120-second hybrid IR at 48kHz involves several million floating- point operations across three independently synthesized late tails plus early reflections. That takes a couple of seconds on a modern CPU. If you are running a batch of 50 renders that all use the same IR parameters, regenerating it every time is wasteful and slows iteration.

The cache stores both the audio and a metadata sidecar, so a cache hit is just a file read – negligible latency.

9.5.2 How the Hash Key Works

The cache key is a 16-character hex prefix of SHA-256 applied to the JSON-serialized `IRGenConfig` dataclass:

```
payload = asdict(config)
payload["_schema"] = "verbx-ir-v0.5"
text = json.dumps(payload, sort_keys=True, separators=(",", ":"))
key = sha256(text.encode("utf-8")).hexdigest()[:16]
```

Every field of `IRGenConfig` contributes to the hash – `mode`, `length`, `sr`, `channels`, `seed`, `rt60`, all FDN topology flags, all modal parameters, all resonator settings, tuning, normalization target, everything. Changing any single field produces a different hash and therefore a different cache entry.

The schema version string `"verbx-ir-v0.5"` is a namespace guard. When the IRGenConfig schema changes in a breaking way (field added, removed, or renamed), bump this version string. All existing cache entries will become unreachable (not deleted, just miss) and will be regenerated on next access. This is intentional: it avoids silent stale-cache bugs at the cost of some wasted disk space during version transitions.

Cache files live at:

```
.verbx_cache/irs/<hash>.wav
.verbx_cache/irs/<hash>.meta.json
```

The metadata JSON contains the full config, version, seed, and IR metrics (RT60 measured, spectral centroid, etc.) as computed by `analyze_ir()`. For a Scala-tuned IR it also preserves the source filename, description, SHA-256 content hash, root frequency and degree, all resolved targets, filter width, gain, strength, and target budget. The source hash is part of the cache identity, so editing a scale in place cannot silently reuse the previous IR.

Management:

```
hatch run verbx cache info    # show size, entry count, oldest/newest
hatch run verbx cache clear   # remove all cached IR files
```

Note: `cache clear` is non-reversible. If you have long IRs cached that took significant time to generate, consider archiving them before clearing.

9.6 Quick Start Recipes

9.6.1 Wash / Ambient Bed (120s)

For pad layers, atmospheric beds, tape-delay washout. Long RT60, high diffusion, gentle early reflections. Peak normalization at –1 dBFS leaves headroom for downstream summing.

```
hatch run verbx ir gen irs/wash_120.wav \
  --mode hybrid --length 120 --rt60 95 --damping 0.45 \
  --diffusion 0.7 --density 1.2 --er-count 32 --er-max-delay-ms 120 \
  --normalize peak --peak-dbfs -1
```

9.6.2 Cinematic Hybrid

Good starting point for score and sound design work. Moderate length, gentle low-frequency roll-off, subtle modulation to avoid static digital quality. `--tilt 1.5` tilts the spectrum slightly darker (positive tilt = more low end).

```
hatch run verbx ir gen irs/cinematic_hybrid.wav \
  --mode hybrid --length 75 --rt60 60 --tilt 1.5 --lowcut 80 --highcut 12000 \
  --er-room 1.3 --er-stereo-width 1.2 --mod-depth-ms 2.0 --mod-rate-hz 0.1
```

9.6.3 Pitched Modal – Tuned Resonance

For pitched instruments where you want the room to harmonize with the source. Set `--f0` to the root note of your key. High Q range (7–90) gives a mix of short and long-ringing modes. Increase `--modal-count` for denser resonance.

```
hatch run verbx ir gen irs/pitched_modal.wav \
  --mode modal --length 45 --rt60 35 --seed 99 --tuning A4=432 \
  --modal-count 64 --modal-q-min 7 --modal-q-max 90 \
  --modal-low-hz 60 --modal-high-hz 9000 --modal-spread-cents 8
```

9.6.4 Microtonal Hybrid from a Scala Scale

Scale-tuned reverberation can range from a nearly invisible spectral affinity to an exposed bank of sympathetic resonances. The examples below keep Scala import offline and deterministic: each command generates an ordinary WAV IR that can be auditioned through convolution, reused in a DAW, or archived with its metadata sidecar. They also demonstrate why scale, root, degree mapping, bandwidth, register, synthesis mode, and source material must be considered as one musical design.

The included scale library provides three contrasting starting points:

- `19edo.scl` divides the octave into 19 equal steps.
- `5_limit_major.scl` uses a seven-degree, 5-limit just-intonation collection.
- `bohlen_pierce_13edo.scl` divides the 3/1 tritave into 13 equal steps rather than repeating at the octave.

9.6.4.1 Example 1: A subtle 19-EDO harmonic halo

This first recipe maps degree 0 of 19-EDO to 220 Hz and expands the lattice through the useful audio range. Moderate strength, 36-cent bandwidth, and 3 dB of emphasis make the tuning more likely to be heard as room color than as a separate resonator. The hybrid mode retains diffuse energy between scale bands, which lets chromatic and continuously pitched source material move without feeling quantized.

```
mkdir -p irs renders

hatch run verbx ir gen irs/19edo_halo.wav \
```

```
  --mode hybrid --length 18 --rt60 7 --seed 19 \
  --scala-file examples/scales/19edo.scl \
  --scala-root-hz 220 --scala-root-degree 0 \
  --scala-low-hz 140 --scala-high-hz 9000 \
  --scala-strength 0.48 --scala-bandwidth-cents 36 \
  --scala-gain-db 3 --scala-max-targets 128

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/19edo_halo.wav --engine conv --ir irs/19edo_halo.wav \
  --wet 0.28 --dry 1
```

Listen first at the stated wet level, then solo the wet return. In context, the tail should feel unusually coherent without producing obvious isolated notes. Soloed, the 19-step lattice should be easier to hear in sustained harmonics and broadband attacks. Raising `--scala-low-hz` keeps the reverb from inventing a strong bass pedal beneath harmonically mobile music.

9.6.4.2 Example 2: Exposed 5-limit just-intonation resonance

The next design uses a sparse 5-limit scale and modal synthesis. Ratios such as 5/4, 3/2, and 5/3 generate exact frequency relationships to the mapped root. Narrower 14-cent bands, stronger attraction, and a lower target ceiling turn the IR into an audible resonant object. Dry clicks and percussion are good test sources because their broad attacks excite many modes at once.

```
hatch run verbx ir gen irs/just_modal_A3.wav \
  --mode modal --length 12 --rt60 8 --seed 51 \
  --modal-count 72 --modal-q-min 18 --modal-q-max 110 \
  --scala-file examples/scales/5_limit_major.scl \
  --scala-root-hz 220 --scala-root-degree 0 \
  --scala-low-hz 80 --scala-high-hz 5000 \
  --scala-strength 0.88 --scala-bandwidth-cents 14 \
  --scala-gain-db 8 --scala-max-targets 96

hatch run verbx render examples/audio/dry_click.wav \
  renders/just_modal_click.wav --engine conv \
  --ir irs/just_modal_A3.wav --wet 1 --dry 0

hatch run verbx render examples/audio/realistic_drums_dry.wav \
  renders/just_modal_drums.wav --engine conv \
  --ir irs/just_modal_A3.wav --wet 0.42 --dry 1
```

On the click, identify the lowest stable pitch and then follow upper modes as they decay at different rates. On drums, listen for a pitched residue after the transient. If the result sounds like an added chord rather than reverberation, reduce strength toward 0.65, increase bandwidth toward 24 cents, or change from `modal` to `hybrid`. If the kick drum pulls the perceived root too strongly, raise `--scala-low-hz` instead of removing low frequencies from the source.

9.6.4.3 Example 3: A non-octave Bohlen–Pierce field

Bohlen–Pierce tuning repeats at the 3/1 tritave. Its scale degrees therefore do not return to equivalent pitch classes at each octave. This example uses a hybrid tail, a restrained lower limit, and a 7.5-second RT60 so the non-octave relationships can overlap without creating an indefinitely accumulating drone.

```
hatch run verbx ir gen irs/bohlen_pierce_space.wav \
  --mode hybrid --length 16 --rt60 7.5 --seed 313 \
  --scala-file examples/scales/bohlen_pierce_13edo.scl \
  --scala-root-hz 130.8128 --scala-root-degree 0 \
  --scala-low-hz 90 --scala-high-hz 7800 \
  --scala-strength 0.72 --scala-bandwidth-cents 26 \
  --scala-gain-db 5 --scala-max-targets 104
```

```
hatch run verbx render examples/audio/realistic_speech_dry.wav \
  renders/bohlen_pierce_speech.wav --engine conv \
  --ir irs/bohlen_pierce_space.wav --wet 0.34 --dry 1

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/bohlen_pierce_music.wav --engine conv \
  --ir irs/bohlen_pierce_space.wav --wet 0.34 --dry 1
```

Speech reveals the scale as a changing coloration because vowels excite different subsets of the lattice. Harmonic music reveals disagreement between octave-repeating source partials and tritave-repeating resonances. Compare the two renders at matched level: the same IR may read as timbre on speech and as a second harmonic system on sustained music.

9.6.4.4 Example 4: Rotate the root degree without changing the reference pitch

`--scala-root-degree` decides which scale degree receives `--scala-root-hz`. The following matched-seed pair assigns 220 Hz first to the implicit 1/1 and then to degree 5 of 19-EDO. Every non-tuning parameter is held constant, so the comparison isolates degree rotation rather than random-tail variation.

```
hatch run verbx ir gen irs/19edo_degree_0.wav \
  --mode hybrid --length 14 --rt60 6 --seed 1905 \
  --scala-file examples/scales/19edo.scl \
  --scala-root-hz 220 --scala-root-degree 0 \
  --scala-low-hz 100 --scala-high-hz 8000 \
  --scala-strength 0.68 --scala-bandwidth-cents 22 --scala-gain-db 5

hatch run verbx ir gen irs/19edo_degree_5.wav \
  --mode hybrid --length 14 --rt60 6 --seed 1905 \
  --scala-file examples/scales/19edo.scl \
  --scala-root-hz 220 --scala-root-degree 5 \
  --scala-low-hz 100 --scala-high-hz 8000 \
  --scala-strength 0.68 --scala-bandwidth-cents 22 --scala-gain-db 5

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/19edo_degree_0.wav --engine conv --ir irs/19edo_degree_0.wav \
  --wet 0.38 --dry 1

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/19edo_degree_5.wav --engine conv --ir irs/19edo_degree_5.wav \
  --wet 0.38 --dry 1
```

The second IR does not merely transpose the first by five ordinary semitones. It maps a different degree ratio to the same physical reference frequency, then reconstructs the complete lattice around that mapping. Compare sustained notes, cadences, and silence after phrase endings. The changed relationship is often most audible after the dry source stops.

9.6.4.5 Example 5: Separate tuning from random topology in an A/B test

A rigorous comparison needs an untuned control. Use the same mode, seed, length, RT60, filters, and channel count, then change only the Scala options. The first IR below establishes the control; the second adds a broad, moderate 5-limit affinity.

```
hatch run verbx ir gen irs/control_seed_808.wav \
  --mode hybrid --length 20 --rt60 9 --seed 808 \
  --lowcut 70 --highcut 11000

hatch run verbx ir gen irs/just_seed_808.wav \
```

```
  --mode hybrid --length 20 --rt60 9 --seed 808 \
  --lowcut 70 --highcut 11000 \
  --scala-file examples/scales/5_limit_major.scl \
  --scala-root-hz 196 --scala-root-degree 0 \
  --scala-low-hz 100 --scala-high-hz 9000 \
  --scala-strength 0.55 --scala-bandwidth-cents 32 --scala-gain-db 4

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/control_seed_808.wav --engine conv --ir irs/control_seed_808.wav \
  --wet 0.4 --dry 1

hatch run verbx render examples/audio/realistic_music_dry.wav \
  renders/just_seed_808.wav --engine conv --ir irs/just_seed_808.wav \
  --wet 0.4 --dry 1

hatch run verbx analyze renders/control_seed_808.wav \
  --json-out renders/control_seed_808.analysis.json

hatch run verbx analyze renders/just_seed_808.wav \
  --json-out renders/just_seed_808.analysis.json
```

Level-match before judging the pair. Use the JSON reports to check that a level or decay difference is not masquerading as a tuning preference. Then describe whether the scale-conditioned version changes perceived consonance, roughness, brightness, pitch stability, or the boundary between source and room.

9.6.4.6 Example 6: Prepare a rooted bank for realtime or DAW use

A static convolution IR cannot follow chord symbols automatically, but a small rooted library can be generated before a performance or mix. The command below shows one member of such a bank. Repeat it with the same seed and different root frequencies, preserving a filename that states the mapping.

```
hatch run verbx ir gen irs/ji_root_C3_130.8128.wav \
  --mode hybrid --length 10 --rt60 4.8 --seed 5150 \
  --scala-file examples/scales/5_limit_major.scl \
  --scala-root-hz 130.8128 --scala-root-degree 0 \
  --scala-low-hz 90 --scala-high-hz 7500 \
  --scala-strength 0.6 --scala-bandwidth-cents 28 --scala-gain-db 4

hatch run verbx realtime --engine conv \
  --ir irs/ji_root_C3_130.8128.wav --block-size 128
```

For section changes, overlap two wet returns and use an equal-power crossfade long enough to preserve the old IR's early field. An abrupt replacement cuts off the previous convolution state and can sound like a gate. In a DAW, place each rooted IR on a separate return and automate sends or return gains. In an offline CLI workflow, render sections with complete tails and assemble them afterward.

The realtime latency is the same as for any other IR with the same partition and block settings. Scala parsing, target expansion, and filter-bank construction do not run on the callback thread. The metadata sidecar records the scale hash and resolved targets, so an IR bank remains auditable even if two source files happen to share a filename.

9.6.5 Resonator-Colored Hybrid (Modalys-Inspired)

The Modalys resonator layer sits on top of the hybrid tail starting at

`--resonator-late-start-ms`. Good for adding physical resonance character to an otherwise smooth hybrid – think the body resonance of a large wooden instrument or a tuned metal plate. `--resonator-mix 0.38` is a conservative setting; go to 0.6+ for obviously colored effects.

```
hatch run verbx ir gen irs/resonator_hybrid.wav \
  --mode hybrid --length 90 --rt60 70 --seed 33 \
  --resonator --resonator-mix 0.38 --resonator-modes 24 \
  --resonator-q-min 10 --resonator-q-max 120 \
  --resonator-low-hz 80 --resonator-high-hz 7000 \
  --resonator-late-start-ms 90 --f0 "64 Hz"
```

9.6.6 Batch Render

```
hatch run verbx batch template > manifest.json
hatch run verbx batch render manifest.json --jobs 4
```

9.6.7 Tempo-Synced Pre-Delay

```
hatch run verbx render in.wav out.wav --pre-delay 1/8D --bpm 120
```

Supported note forms: `1/4`, `1/8D` (dotted), `1/8T` (triplet), or raw seconds (`0.125`).

9.6.8 Morph Two IRs

```
hatch run verbx ir morph irs/hall_A.wav irs/hall_B.wav irs/hall_AB.wav \
  --mode envelope-aware --alpha 0.6 --early-ms 80 --align-decay
```

9.6.9 Auto-Tune IR to Source Analysis

Analyzes the source file's detected fundamentals and harmonic content, then aligns modal frequencies accordingly. Useful when you want the reverb to reinforce the harmonic content of a specific instrument recording.

```
hatch run verbx ir gen irs/tuned_from_input.wav \
  --mode hybrid --analyze-input source.wav
```

9.7 Choosing Parameters for Specific Acoustic Spaces

This section is opinionated. These are starting points, not prescriptions.

9.7.1 Small Live Room (drum room, bathroom)

Characteristics: short RT60 (0.3–0.8s), dense early reflections with hard flutter risk, bright spectral character, strong IACC asymmetry.

```
--mode hybrid --length 3 --rt60 0.5 \
--er-count 48 --er-max-delay-ms 30 --er-decay-shape exp \
--er-room 0.8 --damping 0.6 --diffusion 0.4 \
--highcut 14000 --lowcut 100
```

Keep `er_count` high and `er_max_delay_ms` short. The dense early cloud in a small room is the defining character – do not cut corners here. Low `--rt60` with moderate `--damping` will give you that tight, controlled sound.

9.7.2 Medium Hall (recital hall, 300–800 seats)

The “bread and butter” reverb space. RT60 around 1.5–2.0s mid, slightly longer at low frequencies.

```
--mode hybrid --length 6 --rt60 1.8 \
--fdn-rt60-low 2.2 --fdn-rt60-mid 1.8 --fdn-rt60-high 1.2 \
--er-count 28 --er-max-delay-ms 60 --er-room 1.0 \
--damping 0.35 --diffusion 0.6 --tilt 0.5
```

The three-band RT60 control is worth using here. Real halls have longer bass decay – it is the signature of mass and volume. Skipping this makes algorithmic reverb sound synthetic quickly.

9.7.3 Cathedral / Large Church

Long RT60, extreme low-frequency prolongation, relatively sparse early reflections (the room geometry produces long inter-reflection paths), heavy high-frequency absorption.

```
--mode hybrid --length 30 --rt60 7 \
--fdn-rt60-low 9.5 --fdn-rt60-high 4.5 \
--er-count 18 --er-max-delay-ms 110 --er-room 1.2 \
--damping 0.5 --diffusion 0.75 --density 0.9 \
--lowcut 40 --highcut 10000 --tilt 2.0
```

High `--tilt` darkens the spectrum to match the heavy air absorption over long distances. With a 30-second IR the cache becomes practically essential.

9.7.4 Metal Plate / Physical Resonator

Not a room at all – a physical resonator with well-defined modal structure. The modal mode is the right tool here, not hybrid.

```
--mode modal --length 8 --rt60 5 \
--modal-count 96 --modal-q-min 40 --modal-q-max 400 \
--modal-low-hz 200 --modal-high-hz 16000 \
--modal-spread-cents 3 --f0 "220 Hz"
```

High Q values (40–400) give the long, singing ring of plate reverb. The very narrow `--modal-spread-cents 3` keeps modes tightly clustered around harmonic targets for a more pitched character. Lower Q (5–15) at wider spread approaches spring reverb.

9.7.5 Infinite / Frozen Pad

Not physically realistic, obviously, but useful as a creative tool – the IR never really decays. Very long length with high RT60 relative to length.

```
--mode stochastic --length 120 --rt60 180 \
--diffusion 0.85 --density 0.8 --damping 0.15
```

The RT60 exceeding the IR length means the decay envelope never reaches –60 dB within the buffer – the tail is essentially flat. Combined with high diffusion you get a dense, white-ish wash. Works well convolved with a long pad or a heavily sustained note.

9.8 Expert Reference: Implementation Notes, Limitations, Future Work

9.8.1 Precision

All synthesis arrays are `np.float64` throughout. The `AudioArray` type alias is `npt.NDArray[np.float64]`. Normalization and shaping are also f64. Conversion to f32 or integer PCM happens only at the `sf.write()` call in `write_ir_artifacts()` and `generate_or_load_cached_ir()`. If you are loading cached IRs programmatically, always request `dtype="float64"` in `sf.read()` — this is what the library does internally, but third-party code may not.

9.8.2 Determinism Guarantees

Given identical `IRGenConfig`, `generate_ir()` will produce bit-identical output across runs on the same platform and NumPy version. The RNG throughout is `numpy.random.default_rng(seed)` (PCG64), not the legacy `np.random` global state. Cross-platform determinism (x86 vs. ARM) is not formally guaranteed because of potential FPU differences in transcendental function evaluation (specifically `np.exp` and `np.sin` on long arrays). In practice, for seeds used in production it has not been an issue, but do not rely on cross-platform bit-identity for regression tests.

9.8.3 Hybrid Blend Weights

The fixed weights `(0.55 stoch, 0.25 modal, 0.20 fdn)` in hybrid mode are the result of listening tests with speech and music material. They are hard-coded. If you need different weights, the cleanest approach is to generate the three components separately (using the three individual modes with appropriate seeds) and mix them externally. This is more verbose but gives full control.

9.8.4 The Resonator Layer

`apply_modalys_resonator_layer()` is not a faithful Modalys physical simulation — the name is aspirational. It is a second modal bank applied as a late-tail coloration on top of the synthesized IR, with a soft gate that only activates after `resonator_late_start_ms`. The gate is not hard-gated but uses a smooth fade-in envelope over approximately 20ms to avoid clicks. The resonator bank shares the same harmonic alignment infrastructure as the main modal mode, so `--f0` and `--analyze-input` targets propagate through to it.

9.8.5 Known Limitations

No perceptual diffusion modeling. The FDN diffusion is purely algorithmic; there is no Schroeder allpass diffuser chain in the current release. Adding one (e.g., a series of 4–6 allpass sections before the FDN input) would improve early echo density at short RT60 settings. This is on the roadmap.

Uniform reflection time sampling. As noted in Section 3, uniform tap-time sampling in the early reflections module occasionally produces clusters. A Poisson-disk or minimum-interval constraint would give more physically plausible tap spacing.

Spectral morph phase interpolation. The `spectral` blend mode blends complex spectra, meaning phase interpolation is linear in the complex plane, which can produce magnitude notches at the midpoint. A magnitude-only blend with phase selection (or phase interpolation via circular statistics) would be more robust.

FDN IR length. Very long FDN IRs (60s+) are slow to generate because the FDN must be simulated sample-by-sample (or block-by-block). There is no shortcut for a time-varying FDN – you cannot analytically compute its IR without running the network. The cache makes this practical for production use, but the first generation of a long FDN IR takes wall-clock time proportional to IR length.

Channel count > 2. The modal panner for channels > 2 distributes all extra channels as mono sum / ch, which is not proper ambisonic or surround panning. Do not use verbx-synthesized IRs for surround production without verifying the channel assignments match your downstream decoder expectations.

9.8.6 Roadmap Alignment (v0.7.5)

From the R3 milestones:

- `R3.1 cache determinism`: cross-sample-rate cache lookup with canonical resampling. An IR generated at 96kHz should be retrievable (with a quality warning) by a 48kHz render job without full regeneration.
- `R3.2 operational QA`: morph diagnostic artifacts – small reference renders that can be compared against golden files in CI. The morph metadata already includes RT60 drift and spectral distance; R3.2 formalizes the acceptance thresholds.
- `R3.3 failure safety`: retry/resume logic for long morph batches that fail mid-job. Currently a failed morph job leaves partial output; R3.3 will add a checkpoint file and graceful resume.

When updating `ir morph` or render-time `--ir-blend` behavior, keep this document, the CLI switch tables in `README.md`, and `docs/REFERENCES.md` source links in sync.

9.8.7 References

For the DSP foundations underlying this implementation:

- Gardner, W. G. (1992). A realtime multichannel room simulation system. JASA.
- Griesinger, D. (1996). Spaciousness and envelopment in musical acoustics. AES 101st Convention.
- Jot, J.-M. & Chaigne, A. (1991). Digital delay networks for designing artificial reverberators. AES 90th Convention.
- Schlecht, S. J. & Habets, E. A. P. (2017). On lossless feedback delay networks. IEEE Transactions on Signal Processing.
- Valimaki, V. et al. (2012). Fifty years of artificial reverberation. IEEE Transactions on Audio, Speech, and Language Processing.

See `docs/REFERENCES.md` for full citation list including Modalys and convolution engine references.

9.9 Suggested Exercises

1. Synthesize three IRs with equal nominal RT60 but different early-reflection structures. Compare direct-to-reverberant ratio and perceived distance.
2. Morph two IRs while retaining a fixed source and output level; identify the moment at which the result stops reading as one coherent room.
3. Create a stereo or multichannel IR matrix and verify every route with isolated impulses before processing musical audio.
4. Analyze a synthesized IR for decay, spectrum, and channel correlation, then revise only the parameter most directly related to the weakest metric.
5. Design an IR library naming convention that preserves geometry, source/receiver roles, normalization, sample rate, and licensing evidence.
6. Capture or synthesize an IR with a clearly audible early reflection, then attenuate that reflection and compare localization.
7. Create an IR with the same magnitude spectrum but a different temporal distribution, then explain why convolution results still differ.
8. Test IR normalization methods on the same source and decide which supports fair listening comparison.
9. Use a logarithmic sine sweep and deconvolution workflow to make an IR, documenting latency, trimming, and noise-reduction decisions.
10. Build a hybrid IR whose early part comes from measurement and whose late part is synthesized; state the crossfade criterion.

“Lost Echo sits amid the voiceless mountains, and feeds her grief.”

— Percy Bysshe Shelley, *Adonais* (1821)

Chapter 10

AI Research and Data Augmentation

`verbx` is both a production reverb system and a deterministic acoustic-data generator. The same controls used to design a vocal plate, a drum room, or a long diffuse field can create reproducible variations for Audio AI and machine-learning research. Chapter 11 explains how to turn that capability into defensible datasets, how to avoid leakage, how to select labels, how to audit a regeneration, and how to evaluate whether augmentation improves a model rather than merely increasing the number of files.

The implementation boundary is important. `verbx batch augment` renders audio and emits lineage, summary, quality, and provenance artifacts. It does **not** train a neural network, choose a loss function, license source material, or prove that synthetic rooms transfer to recorded rooms. Those responsibilities remain with the researcher or production team. The goal is a reliable acoustic layer that can be connected to PyTorch, JAX, TensorFlow, scikit-learn, or a custom training system without hiding how each example was made.

10.1 The acoustic dataset as an evidence chain

An audio file alone is not a research dataset. A useful example also needs a stable source identity, a split assignment, a label policy, a resolved render configuration, a generation status, and enough provenance to reconstruct the plan. Figure 11-1 shows the full chain. The audio corpus and evidence bundle travel together into the model experiment; the held-out evaluation then tests whether the learned behavior survives new sources and rooms.

Figure 10-1 presents The Chapter 11 evidence chain. A source file enters only after licensing and split assignment. The manifest records profile, seed, labels, and overrides; verbx expands that description into deterministic render plans. Audio outputs and machine-readable evidence are separate products of one generation event, and both are required to interpret a model result and establishes the visual reference for the discussion that follows.

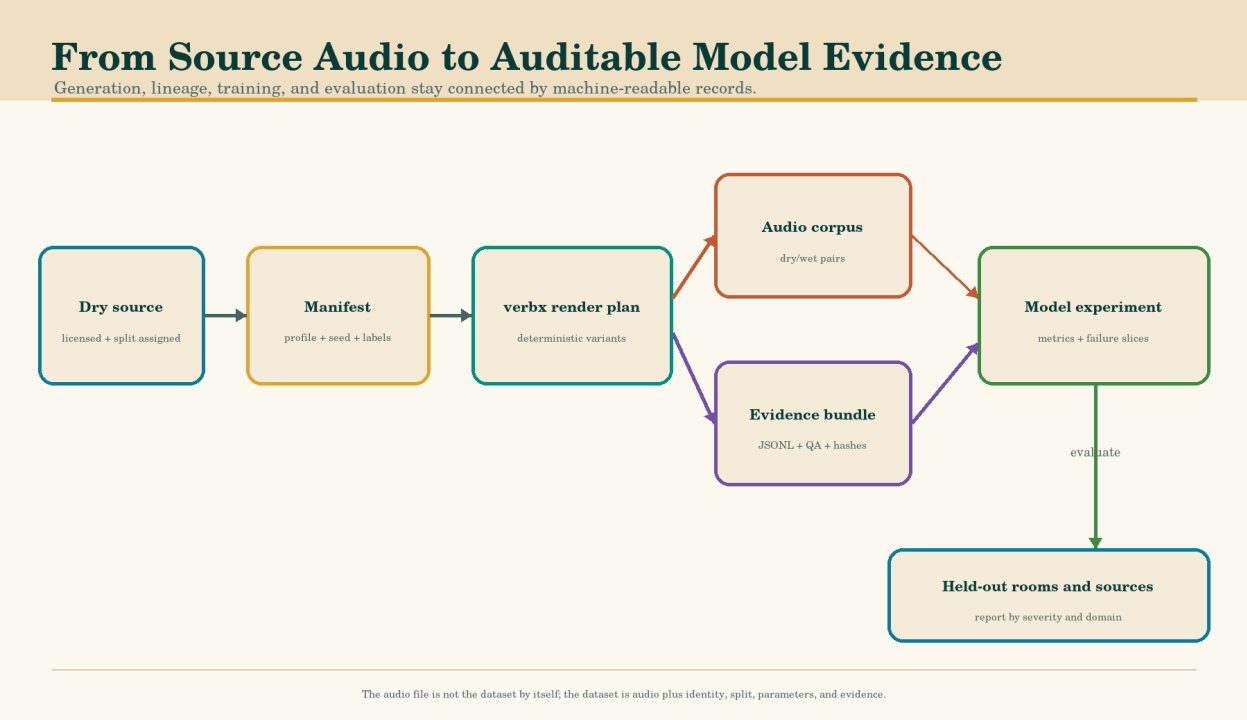


Figure 10-1: The Chapter 11 evidence chain. A source file enters only after licensing and split assignment. The manifest records profile, seed, labels, and overrides; verbx expands that description into deterministic render plans. Audio outputs and machine-readable evidence are separate products of one generation event, and both are required to interpret a model result

This chain supports four distinct claims, which should not be confused:

- **Generation reproducibility:** the same manifest, seed, source bytes, and software behavior expand to the same plan.
- **Dataset identity:** a provenance digest identifies the canonical manifest and source-file signatures used for one corpus generation.
- **Experiment reproducibility:** model code, dependencies, initialization, optimizer state, hardware behavior, and training order are also recorded outside verbx.
- **Scientific validity:** the evaluation design demonstrates that the synthetic variation represents the deployment problem closely enough to improve a relevant outcome.

The first two claims are directly supported by verbx. The latter two require an experiment protocol. A hash can prove that two teams started from the same bytes; it cannot prove that their room distribution was realistic or that their metric captures musical quality.

10.2 Signal model and the unit of augmentation

For a time-invariant room model, a reverberant observation may be written as

$$y_c[n] = \sum_{q=1}^{Q} \sum_{m=0}^{M-1} h_{cq}[m] x_q[n-m] + v_c[n],$$

where $x_q[n]$ is source channel q, $h_{cq}[m]$ is the room or effect response from source channel q to output channel c, and $v_c[n]$ is additive interference. An algorithmic reverberator does not need to materialize

one fixed $h_{cq}[m]$ internally; it may generate the same class of response with delay lines, filters, modulation, and feedback state. For dataset design, the important object is the **resolved transformation**: the source identity, parameter vector, seed, output format, and success state that produced $y_c[n]$.

A single manifest job is a source-level declaration. If a job requests K variants, verbx creates K descendant plans. For source index i and zero-based variant index k, the current seed rule is deterministic:

$$s_{i,k} = s_0 + 100003i + k,$$

where s_0 is the manifest seed. This rule keeps variants reproducible and separates the pseudorandom stream of one source from its neighbors. The per-variant JSONL record stores the resolved seed, profile, archetype, output path, labels, tags, source metadata, render configuration, attempts, duration, and error state.

10.2.1 Labels generated by control and labels measured from audio

Chapter 11 distinguishes three label classes:

1. **Control labels** are parameters requested from the renderer, such as configured RT60, wet gain, damping, pre-delay, FDN matrix, and line count.
2. **Measured labels** are quantities estimated from the rendered waveform, such as an RT60 estimate, early-to-late ratio, spectral centroid, stereo coherence, or integrated loudness.
3. **Semantic labels** describe the source or task, such as speaker, instrument, genre, room class, language, recording session, or production role.

A configured RT60 is not automatically the same as an RT60 estimated from program material. The source may stop before a full decay is observable, the direct signal may dominate, and a frequency-dependent or modulated tail may not fit one broadband exponential. Use control labels for known-generation supervision and measured labels for waveform evidence. If a model predicts physical room quantities, validate those targets against measured impulse responses rather than assuming that a renderer control is a calibrated architectural measurement.

10.3 Split first, augment second

The most damaging augmentation error is often not a bad parameter range; it is leakage. If an utterance, singer, song, stem, session, room, or impulse response contributes descendants to both training and test sets, the model can recognize shared content instead of learning robust acoustics. The score may improve while real-world generalization becomes worse.

Figure 11-2 illustrates the safe order. Assign a source group to train, validation, or test once, then keep every dry copy and every reverberant descendant inside that boundary. The default `--verify-split-isolation` guard rejects a repeated source ID or resolved input path that appears in more than one split.

Figure 10-2 presents Split-before-augmentation design. Each source group produces several variants, but no arrow crosses a split boundary. In real corpora, the grouping key may need to represent a performer, song, session, room, microphone setup, or impulse response rather than merely a file path. The

built-in guard catches repeated IDs and paths; the research manifest must encode the larger identity correctly and establishes the visual reference for the discussion that follows.

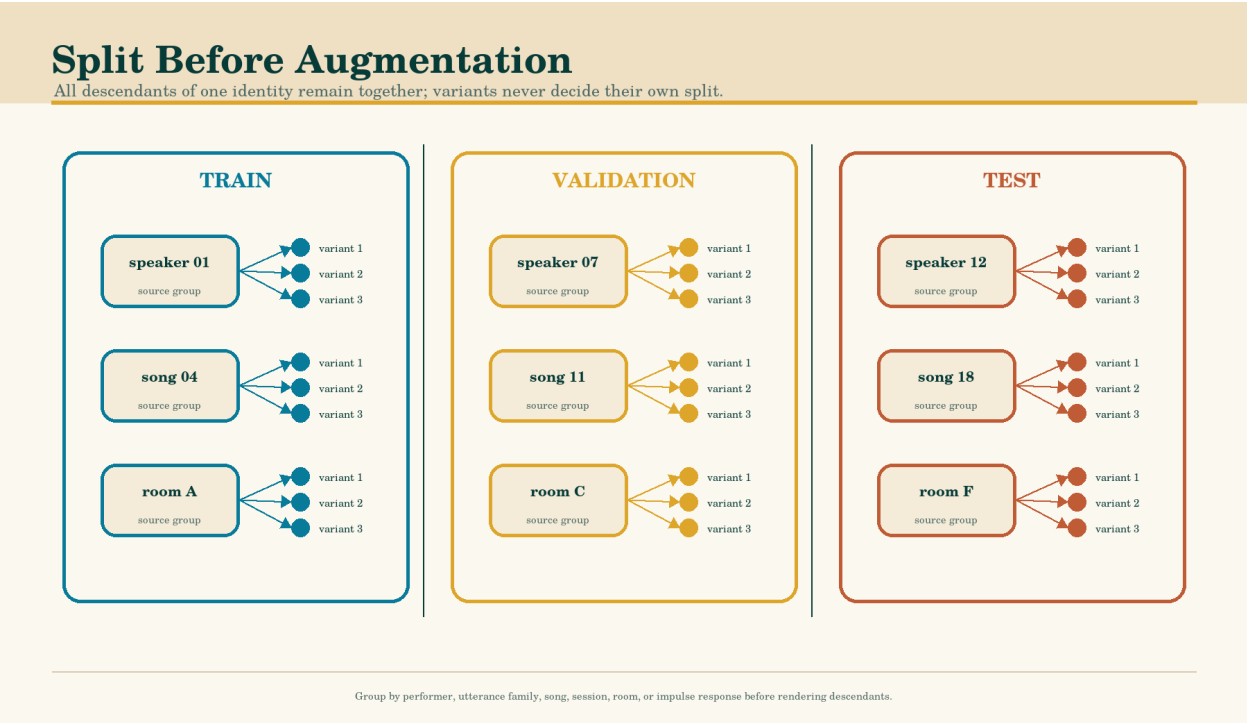


Figure 10-2: Split-before-augmentation design. Each source group produces several variants, but no arrow crosses a split boundary. In real corpora, the grouping key may need to represent a performer, song, session, room, microphone setup, or impulse response rather than merely a file path. The built-in guard catches repeated IDs and paths; the research manifest must encode the larger identity correctly

10.3.1 Choose the leakage unit for the task

The correct grouping boundary depends on the scientific claim. This table separates a minimum defensible grouping unit from a stronger held-out unit and names the hidden identity most likely to cross a naive file-level split.

Table 10-1: Choose the leakage unit for the task: reference table

Task	Minimum grouping unit	Stronger held-out unit	Typical hidden leakage
ASR or keyword spotting	utterance family	speaker and recording session	adjacent cuts from one take
Speaker recognition	speaker	speaker plus room/session	the same room noise in every split
Music tagging	full recording	artist, album, or production session	excerpts from one master
Source separation	song and stem set	artist or multitrack session	mixtures sharing the same stems
Drum transcription	performance take	drummer, kit, and room	duplicated one-shots
Room estimation	source recording	physical room and microphone geometry	multiple positions from one room
Dereverberation	clean source	source identity plus room response	one dry target convolved many ways
Acoustic retrieval	recording event	site and capture session	repeated ambience beds

`--allow-split-overlap` exists for intentional experiments, such as measuring room transfer while holding linguistic content constant. It should be an explicit exception documented in the dataset card, not a convenient way around a validation error. When overlap is allowed, report exactly which identity is shared and which factor remains held out.

10.4 Supervision geometries

The same rendered output can serve different learning problems, but each problem needs a different target. Figure 11-3 maps one dry-to-wet render onto five common supervision choices. Design the target before generation; otherwise a large corpus may be unusable because the necessary dry pair, parameter field, or grouping key was not preserved.

Figure 10-3 presents Supervision geometry for Audio AI. Paired enhancement uses the dry waveform as the target. Estimation uses known controls or measured metrics. Classification uses a defensible room archetype or semantic category. Contrastive learning treats related acoustic views as positives. Neural control prediction maps audio back to a renderer parameter vector. One file may participate in several tasks, but the validity of each target must be argued separately and establishes the visual reference for the discussion that follows.

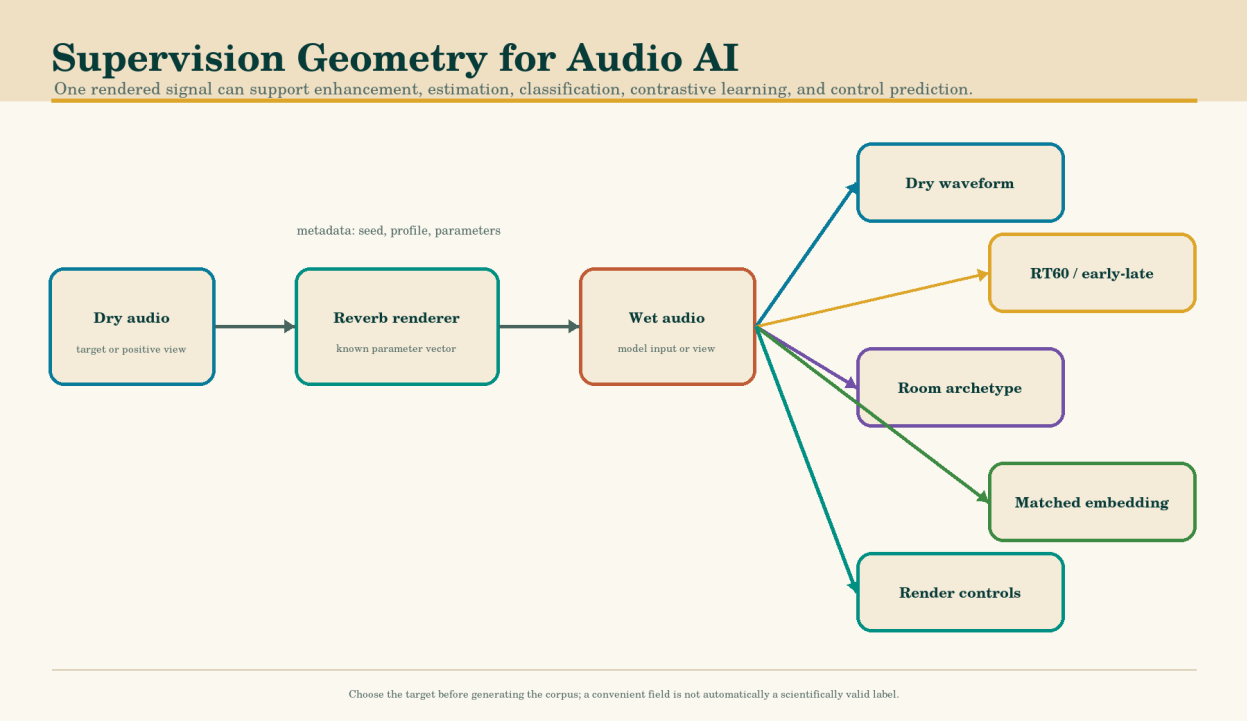


Figure 10-3: Supervision geometry for Audio AI. Paired enhancement uses the dry waveform as the target. Estimation uses known controls or measured metrics. Classification uses a defensible room archetype or semantic category. Contrastive learning treats related acoustic views as positives. Neural control prediction maps audio back to a renderer parameter vector. One file may participate in several tasks, but the validity of each target must be argued separately

10.4.1 Paired waveform supervision

Use `--copy-dry` when every wet output needs an accessible clean reference. This geometry suits dereverberation, enhancement, room transfer, and objective waveform comparisons. Preserve exact alignment: a model trained with sample losses or short-time Fourier transform losses should not be forced

to learn an accidental trim, delay, or resampling difference. For tails that extend beyond the dry signal, define whether the target is padded dry audio, a fixed analysis window, or a loss mask that excludes unsupported time regions.

10.4.2 Scalar and vector regression

Known render controls can supervise RT60, pre-delay, wet depth, damping, width, FDN topology, or a composite parameter vector. Use bounded or transformed targets where the geometry warrants it. RT60 covers orders of magnitude, so $\log T_{60}$ often produces a better-conditioned regression target than raw seconds. Circular angles need periodic losses; categorical matrices need classification or embeddings rather than pretending their names form a numeric scale.

Scala-tuned synthetic IRs add controlled pitch-lattice supervision for music, source-separation, transcription, and robust acoustic-retrieval experiments. Generate them with `verbx ir gen --scala-file SCALE.scl`, preserve the emitted scale hash and resolved frequency list, and treat scale family as a grouping identity during dataset splitting. If variants derived from one `.scl` file appear in both training and test sets, an apparently strong model may only have memorized the same resonance lattice. Useful ablations vary scale, root degree, root frequency, emphasis strength, bandwidth, and RT60 independently while holding the dry source and random seed fixed.

10.4.3 Framewise targets

Time-varying room estimation, adaptive dereverberation, and online control models require targets aligned to frames rather than one label per file. The current augmentation manifest records one resolved render configuration per variant. If parameters are automated within a render, archive the automation file and derive the target timeline with the same frame origin, hop size, and sample rate used by the model. Never interpolate categorical topology changes as if they were continuous values.

10.4.4 Contrastive and metric learning

Two descendants of one source can be a positive pair when the goal is content invariance, while descendants from different sources can be negatives. That choice is reversed for room retrieval, where recordings sharing an acoustic transformation may be positives even when their content differs. Declare whether the embedding should preserve content, performer identity, room, production style, or some factorized combination. A contrastive pipeline without an explicit invariance statement tends to learn the easiest shared artifact.

10.4.5 Multi-task learning

Multi-task models can jointly predict acoustic parameters, source labels, and a restored waveform. This may regularize a representation, but it also couples label quality. A noisy broadband RT60 estimate can damage an otherwise useful enhancement objective. Record loss weights, target normalization, missing-label masks, and task-specific validation metrics so that one aggregate loss does not conceal a failed head.

10.5 Audio AI task map

Figure 11-4 summarizes the main applications supported by a deterministic acoustic generator. The map is intentionally broader than dereverberation: reverberation changes intelligibility, source separation, instrument recognition, scene identity, retrieval, and creative control.

Figure 10-4 presents Applications to Audio AI and machine learning. The central renderer supplies controlled acoustic variants. The surrounding tasks use those variants differently: some seek invariance to room, some estimate the room, some remove it, and some learn to reproduce or control it. A

corpus designed for one arrow should not be assumed valid for all others and establishes the visual reference for the discussion that follows.

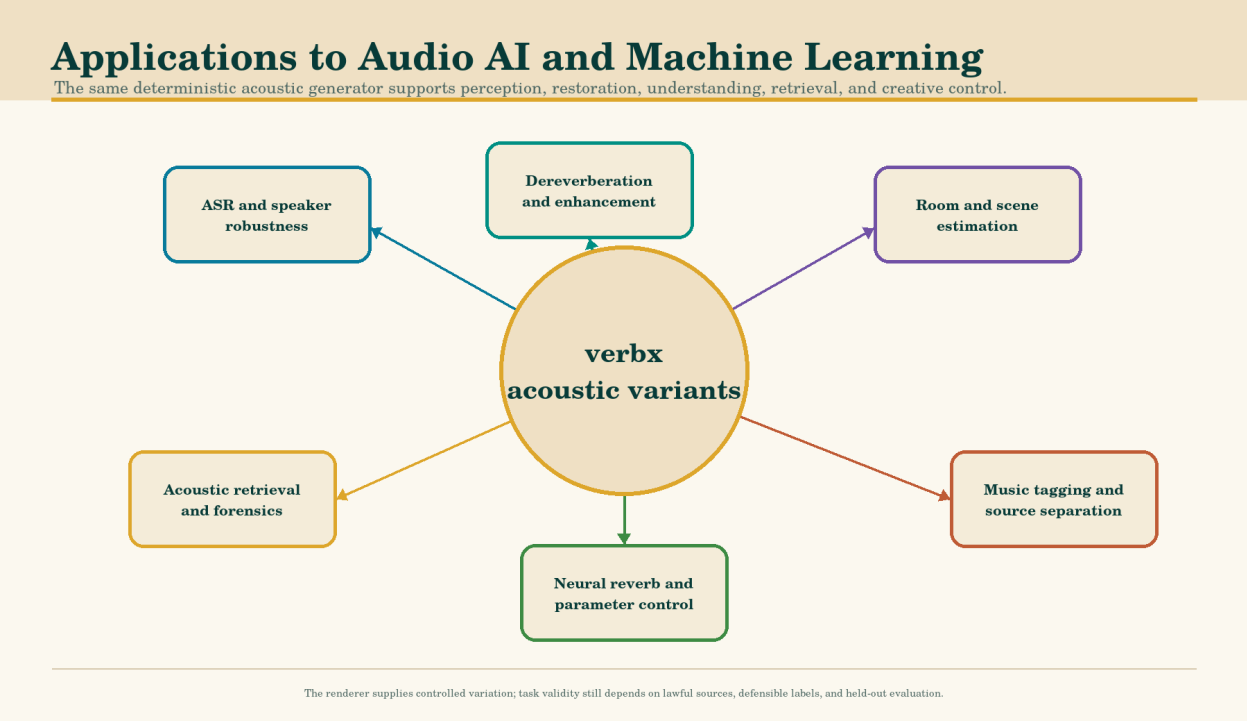


Figure 10-4: Applications to Audio AI and machine learning. The central renderer supplies controlled acoustic variants. The surrounding tasks use those variants differently: some seek invariance to room, some estimate the room, some remove it, and some learn to reproduce or control it. A corpus designed for one arrow should not be assumed valid for all others

The task-to-evidence matrix in Figure 11-5 makes those differences operational. A held-out room dimension is important for every task, but dry pairs, analysis metrics, and listening tests have different priority depending on the model objective.

Figure 10-5 presents Task-to-evidence matrix. Dark cells indicate evidence that is usually required, pale cells indicate useful support, and cream cells are task-dependent. Neural reverb design requires both technical and perceptual evidence; ASR robustness may be validated primarily by recognition

performance, while dereverberation also needs paired targets and artifact review and establishes the visual reference for the discussion that follows.

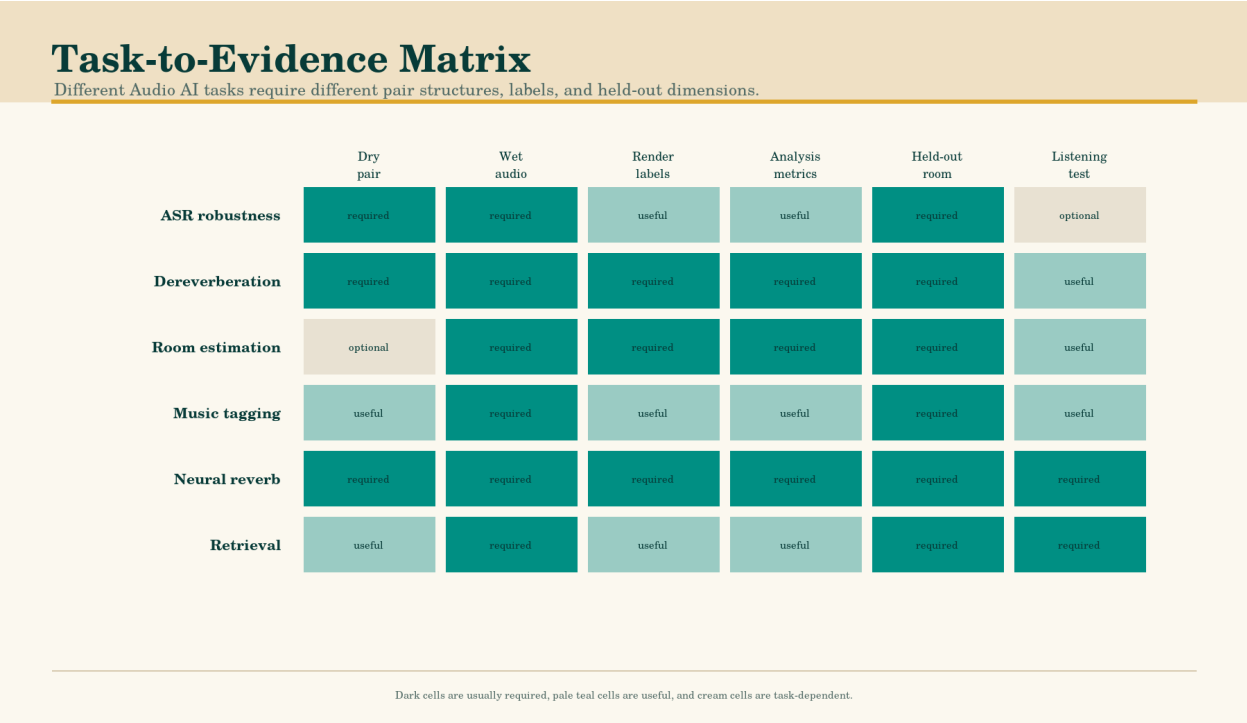


Figure 10-5: Task-to-evidence matrix. Dark cells indicate evidence that is usually required, pale cells indicate useful support, and cream cells are task-dependent. Neural reverb design requires both technical and perceptual evidence; ASR robustness may be validated primarily by recognition performance, while dereverberation also needs paired targets and artifact review

10.6 Built-in profile families

`verbx` currently ships three profile families. A profile first samples an archetype according to its weight, then samples numeric parameters uniformly inside that archetype's bounds and categorical parameters uniformly from its allowed choices. Integer ranges, including FDN line counts, include both endpoints. Job-level options override sampled values, and manifest defaults override the profile before job-specific overrides are applied.

Figure 11-6 plots the outer RT60 and pre-delay envelopes. These rectangles show the union of a profile's archetype ranges, not a probability density. The archetype weights and internal bounds determine the actual distribution.

Figure 10-6 presents Outer parameter envelopes of the three built-in profiles. The ASR family spans approximately 0.18–3.40 seconds and 0–52 milliseconds; the drum family emphasizes 0.22–3.20 seconds

and 0–34 milliseconds; the music family extends to 6.00 seconds and 80 milliseconds. Other dimensions such as wet/dry balance, damping, width, matrix family, and FDN line count are sampled jointly through the selected archetype and establishes the visual reference for the discussion that follows.

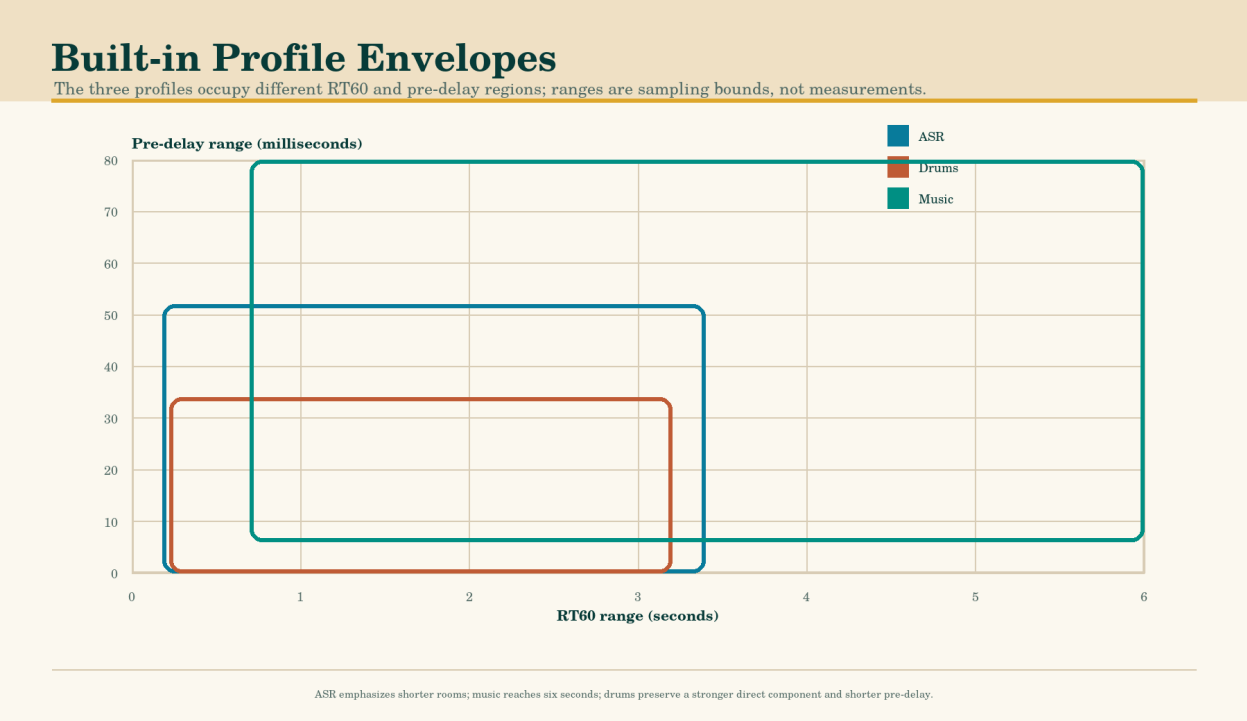


Figure 10-6: Outer parameter envelopes of the three built-in profiles. The ASR family spans approximately 0.18–3.40 seconds and 0–52 milliseconds; the drum family emphasizes 0.22–3.20 seconds and 0–34 milliseconds; the music family extends to 6.00 seconds and 80 milliseconds. Other dimensions such as wet/dry balance, damping, width, matrix family, and FDN line count are sampled jointly through the selected archetype

10.6.1 `asr-reverb-v1`

This profile favors speech robustness with three archetypes:

Table 10-2: asr-reverb-v1: reference table

Archetype	Weight	RT60 (s)	Pre-delay (ms)	Wet	Dry	FDN lines
booth	0.34	0.18–0.65	0–12	0.10–0.28	0.82–1.00	6–10
room	0.43	0.45–1.65	4–24	0.18–0.42	0.72–1.00	8–14
hall	0.23	1.40–3.40	18–52	0.28–0.58	0.62–0.92	10–18

Use it for ASR, keyword spotting, speaker models, speech embeddings, and speech-oriented room estimation. It does not add background noise by itself. If deployment includes HVAC, street, crowd, or device noise, add a separately seeded noise stage and record signal-to-noise ratio as another factor. Keep the room and noise seeds independent so they can be ablated.

10.6.2 `music-reverb-v1`

This profile allows broader stereo width, longer tails, and wetter balances:

Table 10-3: music-reverb-v1: reference table

Archetype	Weight	RT60 (s)	Pre-delay (ms)	Wet	Dry	FDN lines
plate	0.32	0.70–2.20	6–22	0.24–0.50	0.62–0.92	8–16
chamber	0.38	1.30–3.10	12–42	0.28–0.60	0.54–0.88	10–20
long hall	0.30	2.60–6.00	24–80	0.34–0.72	0.40–0.78	12–24

Use it for music tagging, instrument recognition, source separation, acoustic-style retrieval, mix translation, and neural effect control. The profile describes production-like algorithmic spaces, not an unbiased survey of physical rooms. A serious transfer study should reserve measured room impulse responses, real studio returns, or independent plug-ins for evaluation.

10.6.3 `drums-room-v1`

This profile retains a stronger direct component and shorter pre-delay for transient-rich material:

Table 10-4: drums-room-v1: reference table

Archetype	Weight	RT60 (s)	Pre-delay (ms)	Wet	Dry	FDN lines
tight room	0.52	0.22–0.95	0–10	0.12–0.34	0.84–1.00	6–12
studio room	0.33	0.62–1.70	4–18	0.22–0.46	0.68–0.96	8–16
arena	0.15	1.50–3.20	12–34	0.28–0.58	0.52–0.86	10–20

Use it for onset detection, beat tracking, drum transcription, drum replacement, groove classification, and transient-aware separation. If the target task depends on close-mic bleed, overhead geometry, or kit-piece directionality, those factors require multitrack or spatial data beyond a single rendered mix.

10.6.4 Profile selection is a hypothesis

A built-in profile is a documented starting distribution, not a universal truth. Before a large run, compare its generated metrics with a small deployment sample. If real speech rooms cluster around shorter RT60 values than the profile, or production music contains brighter plate tails, change the manifest overrides or create a balanced mixture of runs. Record the difference as a dataset version rather than silently editing a prior manifest.

10.7 Manifest design

Generate a current template rather than copying an old blog post or help transcript:

```
verbx batch augment-template > augment_manifest.json
verbx batch augment-profiles --json > augmentation_profiles.json
```

A practical manifest can mix source-level metadata, job-specific labels, and controlled overrides while leaving most acoustic variation to the profile:

```
{
  "version": "0.7",
  "dataset_name": "studio_voice_rooms_v1",
  "profile": "asr-reverb-v1",
  "seed": 20260716,
  "variants_per_input": 6,
  "output_root": "out/studio_voice_rooms_v1",
  "write_analysis": true,
  "default_options": {
    "engine": "algo",
```

```
    "target_sr": 48000,
    "output_subtype": "float32",
    "normalize_stage": "none",
    "output_peak_norm": "input"
  },
  "jobs": [
    {
      "id": "voice_session_001_take_04",
      "infile": "data/clean/session_001/take_04.wav",
      "split": "train",
      "label": "soprano",
      "tags": ["voice", "sustained", "studio"],
      "variants": 8,
      "metadata": {
        "performer_id": "performer_014",
        "session_id": "session_001",
        "language": "it"
      }
    },
    {
      "id": "voice_session_019_take_02",
      "infile": "data/clean/session_019/take_02.wav",
      "split": "test",
      "label": "baritone",
      "tags": ["voice", "transient", "studio"],
      "variants": 8,
      "options": {
        "rt60": 2.4,
        "pre_delay_ms": 36.0
      },
      "metadata": {
        "performer_id": "performer_031",
        "session_id": "session_019",
        "language": "en"
      }
    }
  ]
}
```

The second job fixes RT60 and pre-delay, but other profile dimensions still vary. This is useful for a controlled slice, although it changes the joint distribution. If the intent is a fully fixed acoustic condition, override every relevant parameter or generate an explicit batch render manifest instead of assuming that two controls determine the room.

10.7.1 Manifest field responsibilities

Not every field belongs at the same level of the evidence chain. The table below distinguishes values inherited from a source, choices made for one transformation, and results observed after rendering.

Table 10-5: Manifest field responsibilities: reference table

Field	Scope	Research use
`dataset_name`	corpus	stable human-readable dataset identity
`profile`	corpus or CLI override	named acoustic sampling family
`seed`	corpus or CLI override	root of deterministic variant seeds
`variants_per_input`	corpus, job, or CLI	default descendant count
`output_root`	corpus or CLI	destination tree
`write_analysis`	corpus or CLI	per-output analysis sidecars

Field	Scope	Research use
`default_options`	corpus	shared render configuration
`jobs[].id`	source	leakage guard and lineage key
`jobs[].split`	source	train, validation, test, or custom split
`jobs[].label`	source	primary class label
`jobs[].tags`	source	searchable secondary categories
`jobs[].metadata`	source	task-specific structured context
`jobs[].options`	source	acoustic overrides for one source group

Paths relative to the manifest resolve from the manifest directory. Output extensions may be WAV, FLAC, AIFF, OGG, or CAF. For training pipelines that depend on exact floating-point samples, prefer WAV float32 and preserve the generated file rather than transcoding it later.

10.8 Plan, smoke test, then render

Large augmentation should be a staged operation. First inspect the expanded plan without writing audio:

```
verbx batch augment augment_manifest.json \
  --dry-run \
  --copy-dry \
  --jobs 8 \
  --schedule longest-first
```

The dry run validates paths, split isolation, profile selection, source identities, output collisions, and descendant counts. It also makes the scale visible before storage is consumed. Next, run a small manifest containing at least one source from every split and label family. Listen to dry/wet pairs, inspect the JSONL rows, and verify that a training loader can parse the output tree. Only then expand to the complete corpus.

A production run with the full evidence set is:

```
verbx batch augment augment_manifest.json \
  --jobs 8 \
  --schedule longest-first \
  --retries 1 \
  --copy-dry \
  --write-analysis \
  --dataset-card-out out/studio_voice_rooms_v1/DATASET_CARD.md \
  --metrics-csv-out out/studio_voice_rooms_v1/metrics.csv \
  --metrics-include-loudness \
  --qa-bundle-out out/studio_voice_rooms_v1/qa_bundle.json \
  --provenance-hash \
  --summary-out out/studio_voice_rooms_v1/summary.json \
  --jsonl-out out/studio_voice_rooms_v1/records.jsonl
```

`--metrics-include-loudness` adds slower loudness, true-peak, and loudness-range analysis. Use `--metrics-fast` for exploratory runs. Keep fail-fast behavior while establishing a pipeline. For unattended generation, `--continue-on-error` can preserve successful work, but leave

`--fail-if-any-failed` enabled so orchestration still receives a nonzero result when any example failed.

10.9 Output artifacts and directory contract

Rendered audio is written under `<output_root>/<split>/`. The filename includes a source index, sanitized source ID, sampled archetype, and one-based variant number. With `--copy-dry`, the dry source

is copied once per source group into the same split tree. Analysis sidecars, when enabled, are stored under `<output_root>/analysis/`.

The main machine-readable artifacts are:

- **`augmentation_manifest.jsonl`:** one record per planned variant, including paths, labels, tags, source metadata, profile, archetype, seed, resolved render configuration, success, attempts, duration, and error.
- **`augmentation_summary.json`:** aggregate planned, successful, failed, source, split, label, archetype, and tag counts plus scheduling context and output paths.
- **`augmentation_qa_bundle.json`:** split-level acoustic summaries, split-by-label counts, and optional baseline deltas.
- **Dataset card:** Markdown identity, generation settings, split distributions, parameter envelope, and links to evidence artifacts.
- **Metrics CSV:** per-output acoustic features suitable for plotting, filtering, or joining to experiment results.
- **Provenance entry:** canonical manifest hash, source signatures, source count, and combined SHA-256 dataset-generation identity.

Treat JSONL as the authoritative join table. Training code should not infer labels by parsing a filename when the same information exists as a typed field. Filenames are convenient for humans; records are safer for schema evolution.

10.10 Corpus size and storage planning

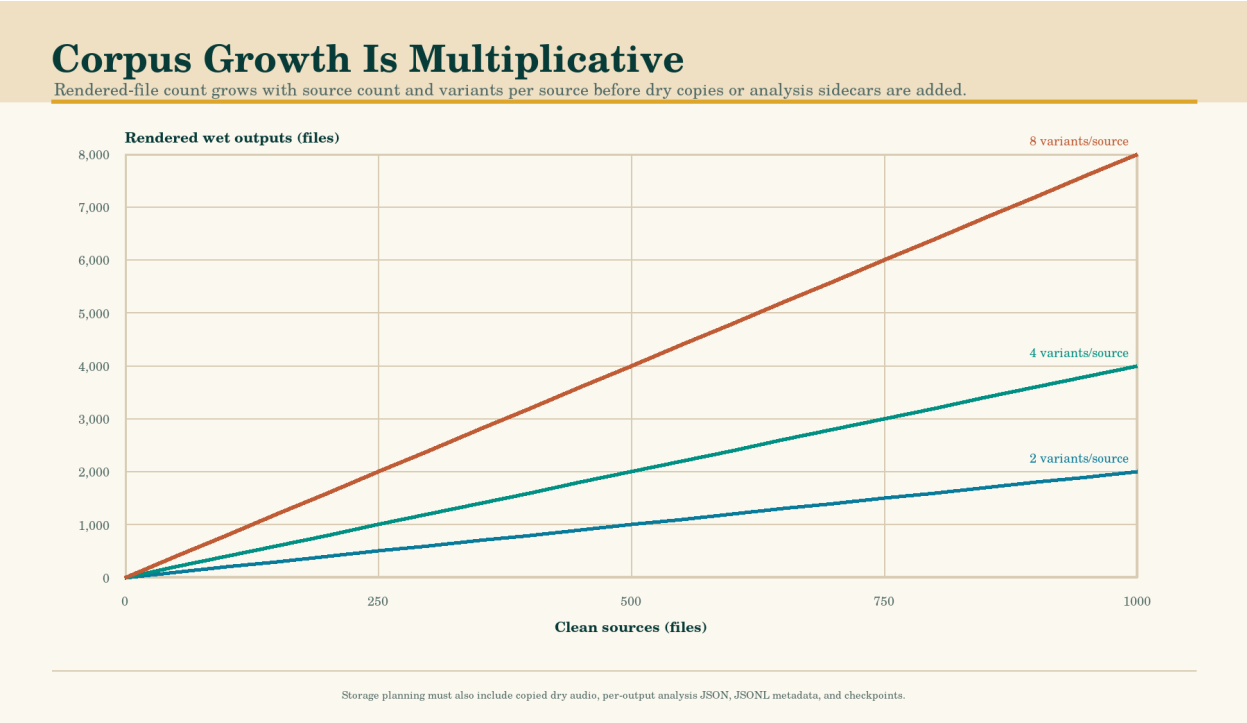
If N source jobs each request K variants, the wet-output count is

$$N_{\text{wet}} = \sum_{i=1}^{N} K_i,$$

which reduces to NK when every job uses the same count. Figure 11-7 shows this multiplicative growth for two, four, and eight variants per source. Dry copies add at most one file per source; analysis sidecars can add one file per wet output.

Figure 10-7 presents Corpus growth in files. One thousand sources produce 2,000, 4,000, or 8,000 wet outputs at two, four, or eight variants per source. The chart excludes dry copies, analysis JSON, metadata, checkpoints, and model-ready feature caches. Audio duration and sample format determine bytes,

so file count is only the first storage estimate and establishes the visual reference for the discussion that follows.



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For uncompressed floating-point audio, a rough payload estimate is

$$B \approx f_s \cdot T \cdot C \cdot \frac{b}{8},$$

where f_s is sample rate in hertz, T is duration in seconds, C is channel count, and b is bits per sample. Add filesystem overhead and account for tails that extend the output duration. At 192 kHz, stereo float32 audio consumes four times the sample payload of 48 kHz stereo float32 audio. Use the sample rate required by the model and target deployment; the book's 192 kHz production default is not automatically the right dataset default.

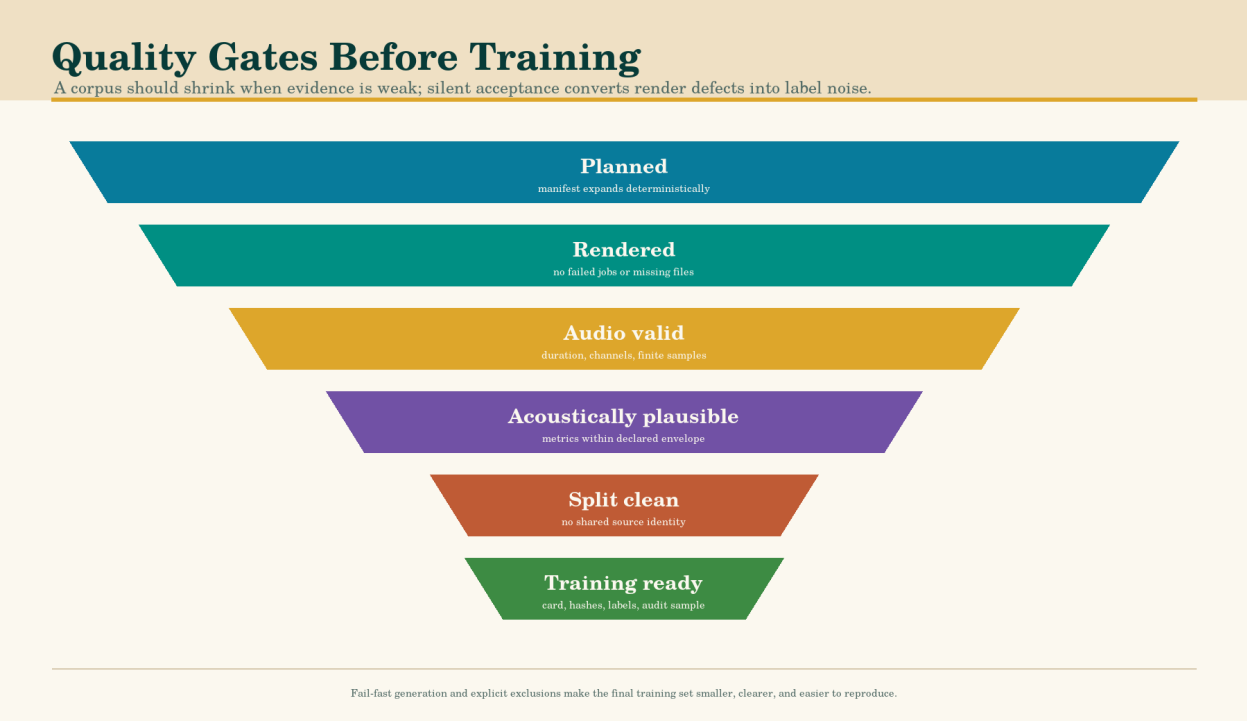
10.10.1 Shard by evidence boundary, not arbitrary file count

When a corpus must be sharded, keep all descendants of a source together and store a shard-level index containing source IDs and record hashes. This prevents a downstream loader from assigning related variants to different distributed workers in ways that accidentally alter sampling balance. Preserve a global JSONL manifest even when audio is packed into tar archives, object storage, or a feature database.

10.11 Quality gates

A successful process exit means the renderer completed; it does not mean every output is suitable for training. Figure 11-8 presents a quality funnel. Each stage may exclude examples, and every exclusion should be recorded rather than silently deleting a file.

Figure 10-8 presents Quality-gate funnel. The widest stage is the deterministic plan. Render failures, invalid audio, implausible metrics, split contamination, or incomplete evidence reduce the set. The final training-ready corpus is intentionally narrower and carries its dataset card, hashes, labels, and an audited listening sample and establishes the visual reference for the discussion that follows.



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10.11.1 Gate 1: plan integrity

Plan integrity confirms that the requested experiment is internally consistent before expensive rendering begins. Check identity, split assignment, and parameter bounds at this stage.

- Every input exists and resolves to the intended file.
- Every source ID is stable, unique at the chosen grouping level, and isolated by split.
- No output path equals an input path.
- Variant counts, output format, target sample rate, and profile match the experiment plan.
- Dataset and manifest versions are immutable once a run begins.

10.11.2 Gate 2: render integrity

Render integrity checks that execution produced the jobs the plan promised. It treats retries and exclusions as recorded events rather than silently accepting a smaller corpus.

- Planned count equals successful count unless a documented exclusion policy applies.
- Failed rows include an actionable error and are not fed to training.
- Retries are reported; repeated transient failures may indicate storage or resource pressure.
- Output duration is compatible with the model loader and tail policy.

10.11.3 Gate 3: audio integrity

Audio integrity examines the resulting containers and samples before any model consumes them. These checks catch corruption, invalid values, and duration or channel surprises that a successful process exit cannot rule out.

- Samples are finite and channel counts are expected.
- Files are not silent, truncated, unexpectedly clipped, or encoded in a surprise subtype.
- Sample rate and frame count match the JSONL or analysis metadata.
- Dry/wet pairs share the intended alignment and source identity.

10.11.4 Gate 4: acoustic plausibility

The metrics CSV and QA bundle can summarize `rt60_estimate_seconds`, `early_late_ratio_db`, `stereo_width`, `stereo_coherence`, `spectral_centroid`, `spectral_flatness`, `dynamic_range`, and `lufs_integrated` by split. These are diagnostic features, not universal pass criteria. Build thresholds from the declared profile and task. A wide stereo field may be correct for music and invalid for a mono ASR experiment.

10.11.5 Gate 5: human audit

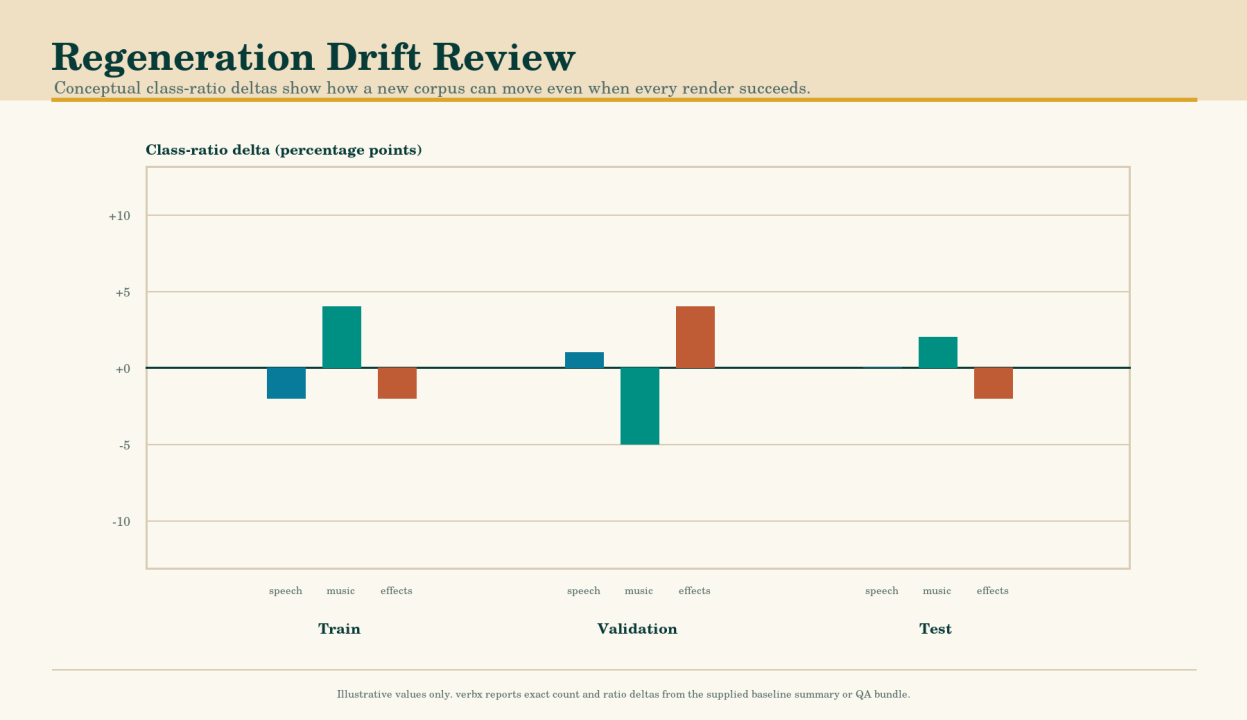
Listen to a stratified sample from every split, archetype, severity bucket, and source class. Include the shortest and longest RT60, driest and wettest mix, narrowest and widest image, and every categorical FDN matrix. Record audible defects with the JSONL `augment_index` so they can be traced to the exact plan.

10.12 Regeneration and distribution drift

Re-running a corpus with a new source inventory or profile can preserve technical success while

changing class balance. Pass a prior summary or QA bundle with `--baseline-summary` to compute split-level and global count and ratio deltas. Figure 11-9 shows how those deltas should be read.

Figure 10-9 presents Conceptual regeneration-drift review. Bars above zero indicate a larger class share than the baseline; bars below zero indicate a smaller share. The plotted values are illustrative, not verbx measurements. The actual QA bundle reports exact current, baseline, count-delta, ratio, and ratio-delta fields for every label represented in each split and establishes the visual reference for the discussion that follows.



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Use the baseline comparison for more than class labels. Preserve a compact external report for parameter histograms, duration, source language, performer, instrument, and room identity. The current QA bundle gives min, max, and mean for supported acoustic metrics; downstream notebooks can join the metrics CSV for quantiles, histograms, and two-sample tests.

Do not compare only global totals. A stable overall soprano/baritone ratio can conceal a test set that lost most baritone examples. Review every split independently, then review interaction slices such as label by archetype or source type by severity.

10.13 Application protocols

Reverberation changes the observation differently for speech recognition, music analysis, spatial models, and generative systems. The protocols below adapt the shared evidence contract to each task instead of assuming one room distribution or metric is universal.

10.13.1 ASR, keyword spotting, and speech embeddings

Begin with `asr-reverb-v1`, preserve transcripts and speaker IDs in `metadata`, and group by speaker or session before augmentation. Report word error rate or task error separately for dry, short, medium, and long reverberation. Include a clean-only training baseline, an augmented model, and a matched-size baseline that repeats clean examples so gains cannot be attributed only to more optimizer steps.

For streaming ASR, test the exact chunk and state policy used at deployment. Long tails cross chunk boundaries and interact with endpointing. A model that performs well on isolated files may fail when reverb state continues across speech segments.

10.13.2 Speaker recognition and diarization

The desired representation should preserve speaker identity while discarding room. Put all audio from one speaker in one split, and check whether microphones or sessions form shortcuts. Evaluate same-speaker trials across different rooms and different-speaker trials inside the same room. If the model improves only when room and speaker remain correlated, augmentation has reinforced the wrong cue.

10.13.3 Dereverberation and speech enhancement

Use copied dry targets and record whether the wet input includes only reverb or also noise. Keep reverb-only, noise-only, and combined conditions as separate ablations. Evaluate waveform or spectral error, intelligibility or recognition performance, and artifact rate. A model can reduce late energy while introducing musical noise, transient holes, phase distortion, or an unnaturally close voice. Human listening remains necessary.

For causal models, restrict the receptive field and algorithmic latency in the experiment report. An offline network with future context is not a valid baseline for a low-latency realtime product.

10.13.4 Blind RT60, DRR, and early-to-late estimation

Use log-scaled RT60 targets and report errors by phonetic or musical content, duration, and severity. Joint prediction of RT60 and early-to-late ratio can expose contradictory estimates. Compare control labels with waveform-derived metrics, but do not train and evaluate on the same estimator's output without acknowledging circularity.

The blind-estimation literature in Appendix C includes work by Deng, Mack, Habets, Xiong, Goetze, Kollmeier, Lollmann, Brendel, Kellermann, and others. Use those methods to define non-neural and neural baselines rather than comparing only against a mean predictor.

10.13.5 Music tagging and instrument recognition

Use `music-reverb-v1`, but split by recording or artist before cutting excerpts. Preserve genre, instrument, artist, album, session, and production metadata when lawful. Report whether the model becomes invariant to room while retaining production-relevant style. For tasks where reverb is part of the label, complete invariance may be undesirable.

Musical examples require special caution because algorithmic reverb can change onset density, harmonic masking, stereo width, and perceived tempo. Test percussive, sustained, sparse, and dense material separately. A tagging model may improve on sustained instruments while losing transient classification accuracy.

10.13.6 Music source separation

Apply reverb at the correct point in the mixture graph. Reverberating the final mix teaches a different problem from reverberating stems independently before summation. Real productions may share one room across sources, use separate sends, or combine close and room microphones. Encode the intended geometry and keep stems, mixtures, dry targets, and render records synchronized.

Evaluate scale-invariant distortion metrics only alongside perceptual and spatial checks. A separator can achieve a better scalar score by narrowing stereo ambience or discarding a late tail that belongs musically to the source.

10.13.7 Drum transcription and onset detection

Use `drums-room-v1` and preserve onset labels at the dry-signal timeline. Confirm whether pre-delay or rendering changes effective alignment for the feature extractor. Report performance by kit piece,

velocity, density, room archetype, and RT60. Long arena variants are a useful stress slice, but they should not dominate training unless deployment contains similar spaces.

10.13.8 Acoustic scene classification and room retrieval

For scene classification, room is evidence rather than nuisance. Use different source content in the same acoustic class so the model cannot identify a scene from one repeated signal. For room retrieval, define positives by room or parameter neighborhood and negatives by acoustically distinct rooms. Reserve real measured rooms as a transfer test whenever possible.

10.13.9 Neural reverb, matching, and parameter inference

A neural reverb may predict audio directly, predict verbx controls, or learn a residual around a deterministic renderer. Parameter prediction preserves an interpretable interface but inherits the renderer's model class. Audio prediction can represent more detail but is harder to constrain and validate. A hybrid model can predict early reflections or spectral corrections while verbx provides a stable late field.

Evaluate parameter error, waveform behavior, stability under automation, and blinded musical preference. Two parameter vectors can sound similar, and a low parameter error can sound worse than a different vector that matches the perceptual target.

10.13.10 Generative audio and controllable music systems

Reverb augmentation can teach a generator to separate musical content from production space or to condition on spatial intent. Preserve dry and wet versions when rights permit, and use clear conditioning fields such as `room_archetype`, `rt60_seconds`, `pre_delay_ms`, and `wet_depth`. Do not label a synthetic long hall as a specific real venue. If text prompts are generated, state that they describe a renderer condition rather than a measured site.

10.13.11 Audio forensics and recording provenance

Reverberation carries information about recording environment, but synthetic augmentation can also create shortcuts. A forensic model may learn verbx's matrix families instead of general room structure. Hold out the complete generator family, test on measured recordings, and avoid claims about authenticity based only on synthetic validation. The provenance hash documents the augmentation corpus; it is not a forensic watermark in the rendered audio.

10.14 Evaluation as a grid, not one score

Model performance should be conditioned on both source novelty and room novelty. Figure 11-10 defines a practical evaluation grid. Every cell should report the task metric and sample count; the unseen-source, unseen-room cells are usually the most informative transfer test.

Figure 10-10 presents Domain-by-severity evaluation grid. Columns progress from dry through increasingly severe reverberation, while rows separate seen and unseen source/room combinations. The colors indicate increasing challenge, not measured performance. Report individual cells before any weighted

mean so failures in unseen long or extreme rooms remain visible and establishes the visual reference for the discussion that follows.

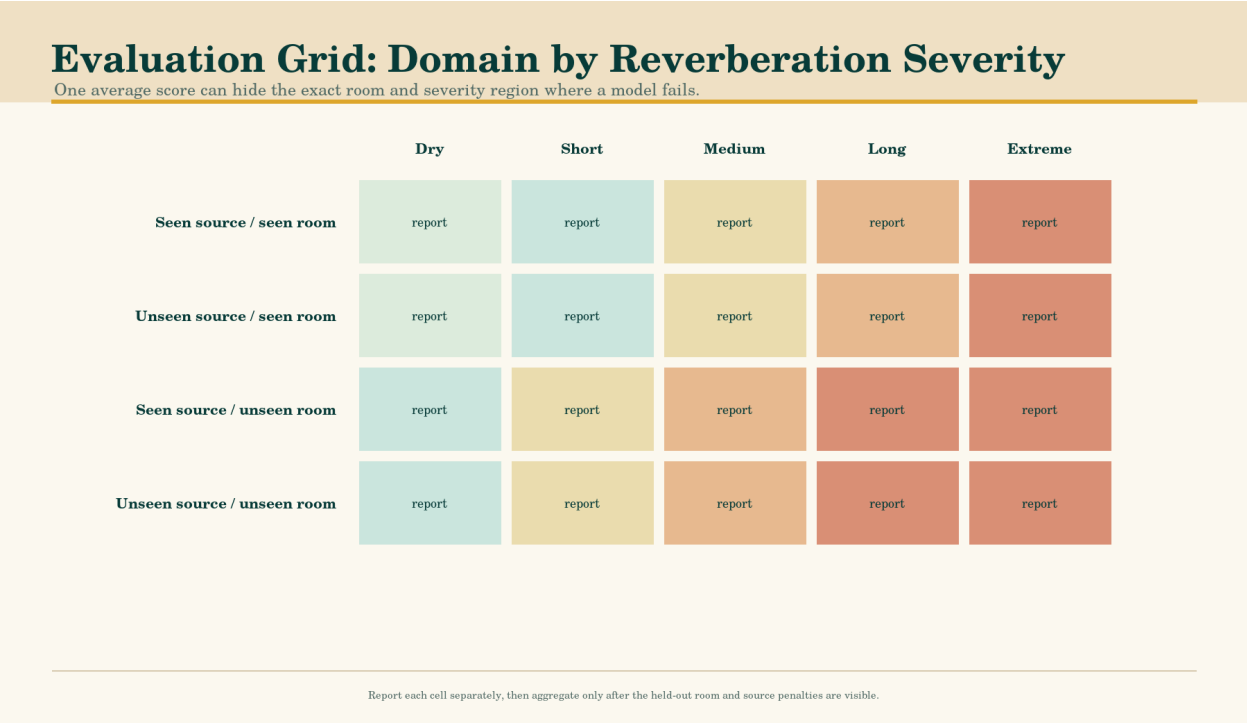


Figure 10-10: Domain-by-severity evaluation grid. Columns progress from dry through increasingly severe reverberation, while rows separate seen and unseen source/room combinations. The colors indicate increasing challenge, not measured performance. Report individual cells before any weighted mean so failures in unseen long or extreme rooms remain visible

10.14.1 Severity buckets

Define severity before examining model results. A speech study might use dry, short below 0.6 seconds, medium 0.6–1.5 seconds, long 1.5–3.0 seconds, and extreme above 3.0 seconds. A music study may need different boundaries and additional wet-depth or early-to-late bins. Store the bucket rule in experiment configuration, not only in plotting code.

10.14.2 Task metrics

Every task needs a primary outcome and at least one companion analysis that explains where the outcome changed. The examples below are starting points for a preregistered evaluation plan, not interchangeable scores.

Table 10-6: Task metrics: reference table

Task	Primary metric examples	Required companion analysis
ASR	word or character error rate	error by speaker, room, duration, and severity
Keyword spotting	false accept/reject, area under curve	threshold stability by room
Dereverberation	spectral/waveform distance, recognition score	artifact listening and latency
Room estimation	log-RT60 error, DRR error, class accuracy	calibration and confidence intervals
Separation	source distortion or perceptual score	stereo image and tail preservation
Music tagging	macro F1 or mean average precision	artist/session leakage audit

Task	Primary metric examples	Required companion analysis
Retrieval	recall at rank, mean reciprocal rank	room-held-out and generator-held-out tests
Neural reverb	control error, spectral/decay distance	blinded preference and stability

Use paired statistical tests when the same evaluation examples are processed by two systems. Bootstrap over the correct independent unit: speakers for speaker-held-out ASR, songs for music, or rooms for room-transfer studies. Bootstrapping thousands of correlated variants as though they were independent produces confidence intervals that are too narrow.

10.15 Ablation studies

An augmentation study should isolate which acoustic variation helps. Figure 11-11 gives a five-run ladder: no augmentation, decay only, decay plus distance cue, the full algorithmic profile, and a full profile evaluated with measured-room holdout.

Figure 10-11 presents Reverb-augmentation ablation ladder. Each row adds one evidence-bearing family while source splits, training budget, model architecture, and evaluation grid remain fixed. The final row introduces measured room impulse responses as a transfer condition rather than treating synthetic success as the endpoint and establishes the visual reference for the discussion that follows.

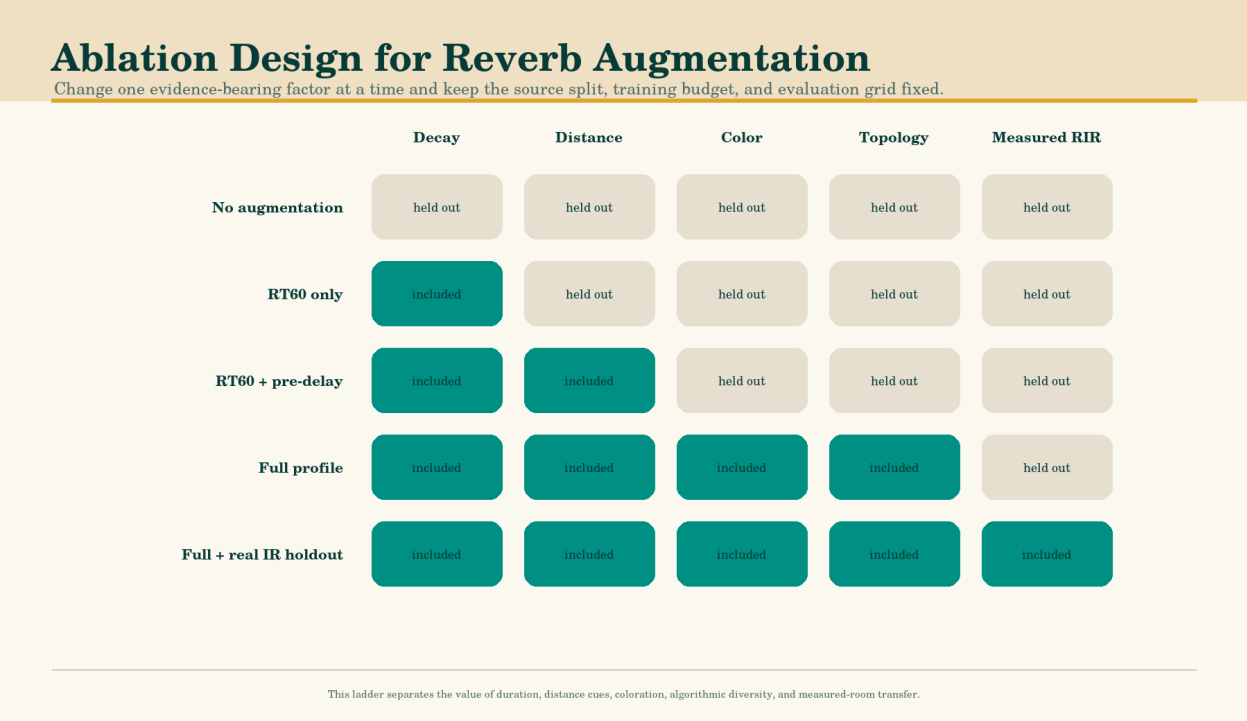


Figure 10-11: Reverb-augmentation ablation ladder. Each row adds one evidence-bearing family while source splits, training budget, model architecture, and evaluation grid remain fixed. The final row introduces measured room impulse responses as a transfer condition rather than treating synthetic success as the endpoint

A complete ablation table should include:

- clean-only training with matched optimizer steps;
- duplicated-clean training with matched example count;

- one-factor RT60 augmentation;
- RT60 plus pre-delay or early/late control;
- full profile variation;
- profile variation without matrix diversity;
- profile variation without stereo-width variation;
- an independent convolution or measured-RIR condition;
- an unseen generator or plug-in condition;
- deployment recordings when rights and privacy permit.

Keep seeds fixed across comparable runs so the source and archetype sequence changes only where the ablation requires it. If removing a factor changes the number of files, match training steps or sampling weights. Report both the best checkpoint and a predefined selection rule to avoid choosing a different favorable epoch for every condition.

10.16 Reproducibility and provenance

`--provenance-hash` canonicalizes the JSON manifest when possible, computes signatures for unique source files, includes their resolved paths and audio metadata, and combines those records with the planned record count. Figure 11-12 shows how that dataset-generation identity relates to the downstream experiment report.

Figure 10-12 presents Reproducibility lineage. Source SHA-256 signatures and canonical manifest content form the dataset-generation identity. Rendered audio and JSONL records feed an experiment report that must additionally preserve code, environment, model configuration, and metrics. The verbx hash establishes generation inputs; it does not by itself capture every training variable and establishes the visual reference for the discussion that follows.

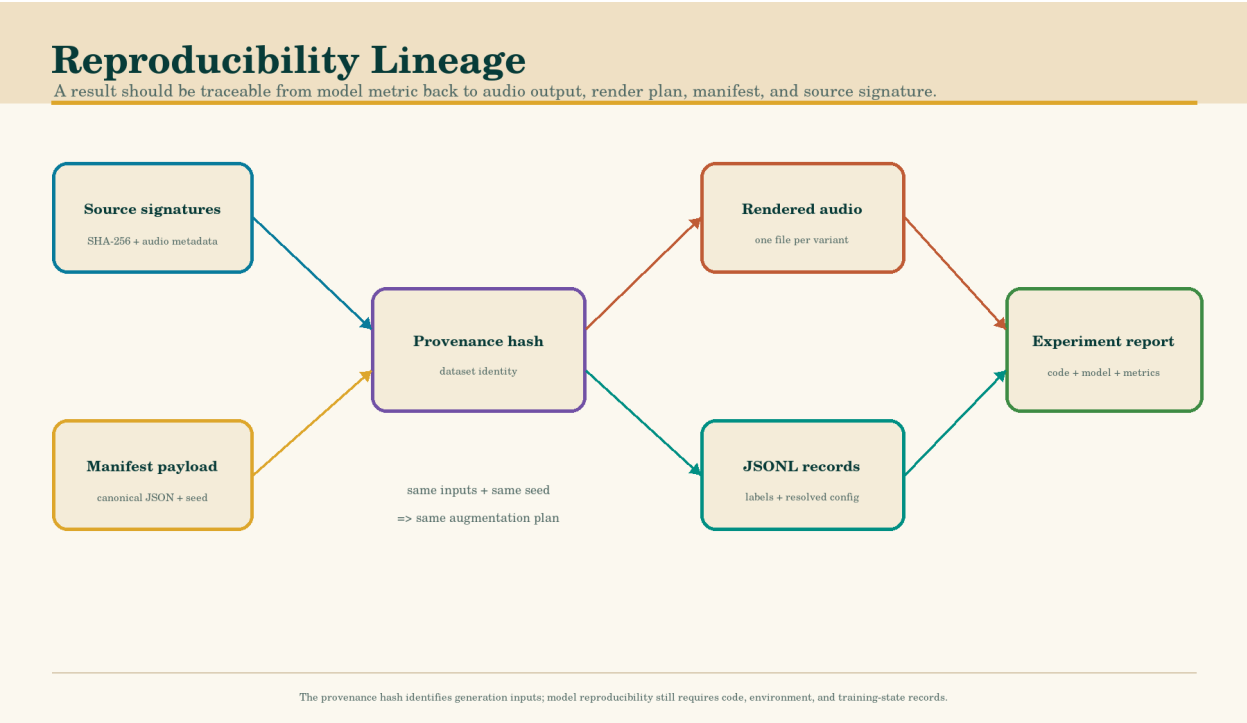


Figure 10-12: Reproducibility lineage. Source SHA-256 signatures and canonical manifest content form the dataset-generation identity. Rendered audio and JSONL records feed an experiment report that must additionally preserve code, environment, model configuration, and metrics. The verbx hash establishes generation inputs; it does not by itself capture every training variable

Archive the following together:

1. the exact source manifest;
2. the exported profile definitions;
3. JSONL records and summary JSON;
4. QA bundle, metrics CSV, and dataset card;
5. provenance payload and source-access policy;
6. verbx version and installation lockfile;
7. training configuration, code commit, environment, and random seeds;
8. evaluation inventory and exclusion list;
9. plots generated from immutable evidence tables;
10. a short listening audit with stable variant IDs.

Absolute source paths are included in current signatures for local reproducibility. When sharing a dataset externally, publish a path-remapping policy or content-addressed source registry rather than assuming another machine has the same directory layout. Do not publish private paths or source bytes that the dataset license forbids.

10.17 TensorFlow and PyTorch integration

verbx should sit at a clearly defined boundary in a machine-learning input pipeline. There are two valid patterns. A **materialized pipeline** renders WAV files and manifests before training. An **on-the-fly pipeline** renders or convolves inside a dataset worker. Materialization costs storage but makes every waveform inspectable, cacheable, and exactly repeatable. On-the-fly augmentation saves storage and can expose a model to more variants, but it complicates worker safety, distributed determinism, performance, and postmortem analysis. Use materialized data for validation and test even when training is dynamic.

The governing rule is

$$\mathcal{D}_{\text{train}}^{(e)} = A(\mathcal{D}_{\text{train}}, e, s), \qquad \mathcal{D}'_{\text{val}} = A(\mathcal{D}_{\text{val}}, s_{\text{val}}), \qquad \mathcal{D}'_{\text{test}} = A(\mathcal{D}_{\text{test}}, s_{\text{test}}),$$

where epoch e may change training variants, while validation and test seeds remain fixed. The clean source split must be created before A is applied. No recording, speaker, session, musical stem family, room capture, or generated IR family designated as a leakage group may cross that boundary.

10.17.1 A split-safe materialized workflow

Materializing audio before training makes the transformation set inspectable and repeatable, but only if split identity is fixed first. The following order prevents augmentation from turning one source family into nominally independent examples across partitions.

1. Assign immutable `source_id`, `group_id`, and `split` fields to the clean corpus.
2. Freeze the split ledger in version control or content-addressed object storage.
3. Assign disjoint room or IR families when room generalization is an evaluation target.
4. Plan augmentation from the frozen ledger and write the planned JSONL manifest.

5. Smoke-render a small stratified sample and run audio, label, and tail checks.
6. Render training variants, then render one fixed validation and test matrix.
7. Hash every output and join acoustic analysis fields back into the manifest.
8. Train from the manifest rather than rediscovering labels from directory names.
9. Report clean, seen-room, unseen-room, and severity-bucket metrics separately.

For a dereverberation model, each row should retain the dry target, reverberant input, IR identifier, direct-arrival convention, wet/dry policy, output gain, and seed. For ASR, the transcript remains attached to the dry source identity while room metadata describes the transformed observation. For music tasks, stems from one multitrack session usually belong to one leakage group even when their filenames differ.

10.17.2 PyTorch `Dataset` and `DataLoader`

The simplest robust PyTorch design reads a verbx JSONL manifest and returns already materialized paths. The dataset object does not invoke a subprocess in `__getitem__`, so worker restarts cannot create partial files or race for one output path.

```python
import json
from pathlib import Path

import soundfile as sf
import torch
from torch.utils.data import Dataset


class VerbxManifestDataset(Dataset):
    def __init__(self, manifest_path: str, split: str):
        rows = [json.loads(line) for line in Path(manifest_path).read_text().splitlines()]
        self.rows = [row for row in rows if row["split"] == split]

    def __len__(self):
        return len(self.rows)

    def __getitem__(self, index):
        row = self.rows[index]
        audio, sample_rate = sf.read(row["output_path"], dtype="float32", always_2d=True)
        waveform = torch.from_numpy(audio.T.copy())
        target = torch.tensor(row["target"], dtype=torch.long)
        return waveform, sample_rate, target, row["source_id"], row["room_id"]
```

Use a collate function that pads or crops waveforms deliberately and returns a valid- sample mask. Do not silently truncate long reverb tails if the task depends on decay. With `DistributedSampler`, shard rows after loading the fixed manifest, call `sampler.set_epoch(e)` for training order, and keep validation order fixed. Seed each worker from a run seed, rank, worker identifier, and epoch when any residual online randomness remains.

For on-the-fly training, prefer an in-process convolution or Python API over spawning the CLI for every item. Load IRs once per worker, bound the cache, derive the variant seed from `hash(source_id, epoch, global_seed)`, and return the chosen parameters with the waveform. Never overwrite the source manifest with epoch-specific values. Record enough seed state to regenerate a failed or unusually informative example later.

10.17.3 TensorFlow `tf.data`

TensorFlow follows the same evidence contract. Construct the dataset from manifest columns, shard deterministically in distributed jobs, decode audio, and batch with an explicit shape policy.

```
import json
from pathlib import Path

import tensorflow as tf

rows = [json.loads(line) for line in Path("manifests/augmented.jsonl").read_text().splitlines()]
train_rows = [row for row in rows if row["split"] == "train"]

paths = [row["output_path"] for row in train_rows]
labels = [row["target"] for row in train_rows]
dataset = tf.data.Dataset.from_tensor_slices((paths, labels))

options = tf.data.Options()
options.experimental_deterministic = True
dataset = dataset.with_options(options)

def decode(path, label):
    audio_bytes = tf.io.read_file(path)
    waveform, sample_rate = tf.audio.decode_wav(audio_bytes, desired_channels=-1)
    return waveform, sample_rate, label

dataset = dataset.map(decode, num_parallel_calls=tf.data.AUTOTUNE, deterministic=True)
dataset = dataset.padded_batch(16, padded_shapes=([None, None], [], []))
dataset = dataset.prefetch(tf.data.AUTOTUNE)
```

`tf.py_function` can wrap Python DSP, but it weakens graph portability, shape inference, and accelerator input optimization. Use it only for training-time experiments whose reproducibility and throughput have been measured. A stronger production path renders

with verbx ahead of time or exports tensors into sharded TFRecord files while retaining the original JSONL provenance ledger.

10.17.4 Validation, test, and challenge matrices

A single random reverberant test set cannot answer whether a model generalizes. Freeze a matrix with meaningful axes:

Table 10-7: Validation, test, and challenge matrices: reference table

Axis	Example buckets
Room identity	seen room, unseen measured room, unseen synthesized family
Decay	dry, short, medium, long, extreme
Direct-to-reverberant ratio	foreground, balanced, distant
Noise	clean, stationary, transient, competing speech
Device	matched microphone, unseen microphone, codec path
Spatial layout	mono, stereo, array, binaural or immersive fold-down

Keep a clean test condition in the same report. For ASR, report word error rate by each cell and the relative degradation from clean. For dereverberation, combine signal metrics with acoustic estimates such as residual RT60, DRR change, spectral distortion, and listening tests. For tagging or separation, inspect class or source-specific failures; reverberation can alter instruments with sparse attacks more than sustained ones.

10.17.5 Caching, distributed workers, and failure recovery

Hash the transformation specification, source content, verbx version, and IR content into one cache key. A filename alone is not sufficient because a file may be replaced in place. Write to a temporary path, validate the container and expected duration, then atomically publish the completed artifact. In a multi-worker render farm, acquire work from immutable manifest rows and make retries idempotent.

For PyTorch Distributed Data Parallel, JAX, TensorFlow MultiWorkerMirroredStrategy, or TPU input services, avoid letting each rank invent a different interpretation of the split. Distribute the same frozen manifest and shard by stable row index or source group. Measure input-stall time, cache hit rate, augmentation CPU time, storage throughput, and accelerator utilization together. A theoretically rich online augmenter that starves the model can reduce effective experimental coverage compared with a simpler pre-rendered corpus.

10.17.6 Preventing augmentation leakage

Augmentation leakage has several forms. A dry utterance and its reverberant copy in different splits is direct leakage. Two excerpts from one recording session can leak the same speaker, noise floor, and microphone. Two IRs measured in the same room can leak room identity even when filenames differ. Two synthesized IRs created from one parameter seed family can leak generator fingerprints. The split ledger must therefore name the unit of independence appropriate to the scientific claim.

When evaluating unseen-room robustness, hold out room families before generating any variants. When evaluating a neural reverb estimator, hold out both source recordings and IR families. When training generative audio, audit whether wet tails reveal copyrighted or sensitive source material beyond the intended crop. A defensible result states not only the framework and model but also exactly what was isolated between training, validation, and test.

10.18 Scaling and operational behavior

Large augmentation studies are systems experiments as well as signal-processing experiments. Throughput, storage, worker failures, cache identity, and deterministic sharding all affect which examples a model actually sees.

10.18.1 Parallel workers

Parallelism should be chosen from measured end-to-end throughput rather than core count alone. More workers can reduce performance when IR state, memory bandwidth, or output storage becomes the limiting resource.

`--jobs 0` selects an automatic worker count. Explicit worker counts are better for controlled benchmarks. Rendering can become limited by CPU, memory bandwidth, storage throughput, or file creation rather than arithmetic alone. Measure end-to-end examples per second and error rate, not only processor utilization.

10.18.2 Scheduling

`longest-first` is the default and often reduces the tail of mixed-duration runs by starting expensive jobs early. `shortest-first` provides quick early outputs and can be useful for smoke tests. `fifo` preserves manifest order. Scheduling should not change variant content, but it can change completion order, resource pressure, and which files exist when an interrupted run stops.

10.18.3 Retries and failures

Retries are appropriate for transient filesystem or resource failures, not deterministic invalid configuration. A repeated success after retry should remain visible in `attempts`. If a render fails because the input is corrupt, fix or exclude the source with a recorded reason rather than raising retry count.

10.18.4 Analysis cost

Per-output waveform analysis can be a substantial second pass. Use fast metrics during parameter development, then enable loudness metrics for the release corpus. If a downstream feature extractor already computes equivalent quantities, decide which implementation is authoritative

and compare a sample before discarding verbx metrics.

10.18.5 Versioning

Use immutable dataset names such as `speech_rooms_v1` and `speech_rooms_v2`, not a mutable `latest` directory. A regeneration with different sources, profile bounds, verbx version, sample rate, or label policy is a new dataset version even if filenames are similar. Preserve baseline summaries so drift is reviewable.

10.19 Lawful and ethical dataset practice

Synthetic acoustics do not remove the obligations attached to source audio. Confirm rights for reproduction, transformation, model training, and distribution separately. A license that permits listening or research access may not permit publishing augmented copies. Dataset cards should name the source license category, access restrictions, retention policy, and whether dry audio is redistributed or must be reconstructed by an authorized user.

For speech, protect identity and sensitive content. Avoid embedding personal names or private paths in public metadata. For music, preserve attribution and do not imply that a synthetic room is part of the original recording. For cultural or site-specific recordings, document consent and context rather than treating acoustic variation as detached raw material.

Bias can enter through both content and acoustics. Speaker demographics, languages, genres, instruments, recording devices, rooms, and production conventions may be unequally represented. An apparently balanced label table can still put one population mostly in long rooms or noisy conditions. Audit intersections, not only marginal counts.

10.20 Failure modes and corrective actions

The validation score is excellent, but deployment is poor. Check performer, song, session, room, and generator leakage. Add an unseen-room and unseen-generator test. Verify that excerpts from the same master did not cross splits.

The model predicts the archetype but not real room behavior. It may recognize matrix color or profile-specific parameter correlations. Hold out matrix families, use measured RIRs, and compare against another renderer.

Dereverberation sounds phasey despite better metrics. Add spatial, transient, and listening criteria. Confirm dry/wet alignment. Reduce reliance on a pointwise waveform loss that punishes perceptually acceptable phase differences.

RT60 regression fails on short clips. The clip may contain insufficient decay evidence. Stratify by duration and source activity, use confidence targets, or exclude windows without an observable decay interval.

Class balance changed after regeneration. Compare the QA bundle to the baseline, inspect split-specific ratio deltas, and version the corpus. Do not repair only the global total.

The corpus is too large to iterate. Build a miniature manifest that preserves every split, label, archetype, and severity bucket. Validate loaders and metrics there before the release run.

Two machines create different model results from the same corpus. Confirm audio and JSONL hashes first, then inspect training dependencies, nondeterministic kernels, data-loader order, precision, and checkpoint selection. Dataset determinism and optimizer determinism are separate.

A profile override unexpectedly narrows variation. Remember the precedence order: sampled profile values are followed by manifest defaults and then job-specific options. An override fixes that field for all affected descendants.

10.21 A reproducible research protocol

The following protocol turns Chapter 11 into an experiment that can be reviewed by another team.

1. **State the deployment question.** Name the source domain, acoustic domain, realtime or offline constraint, and primary outcome.
2. **Declare the independent unit.** Choose speaker, song, session, room, or another grouping key.
3. **Freeze source splits.** Save the source inventory before rendering any variant.
4. **Choose supervision.** Specify dry pairs, control labels, measured labels, semantic labels, and any framewise targets.
5. **Export profile definitions.** Archive `verbx batch augment-profiles --json` with the study.
6. **Write the manifest.** Include stable IDs, lawful metadata, tags, and immutable dataset name.
7. **Dry-run the plan.** Review counts, paths, split isolation, and parameter overrides.
8. **Render a stratified smoke set.** Exercise every split, class, archetype, format, and loader.
9. **Run the full corpus.** Emit JSONL, summary, QA, metrics, dataset card, and provenance.
10. **Apply quality gates.** Record exclusions by `augment_index` and preserve the pre-filter table.
11. **Train matched baselines.** Match source split, optimizer budget, and checkpoint selection.
12. **Evaluate the full grid.** Report unseen sources, unseen rooms, and severity buckets.
13. **Perform ablations.** Isolate decay, distance, color, topology, and real-room transfer.
14. **Listen.** Audit restoration and creative tasks with randomized, level-matched examples.
15. **Publish evidence.** Include code commit, environment, manifest, hashes, and limitations.

10.22 Chapter 11 checklist

Use this checklist after the chapter's design, rendering, and evaluation decisions have been written down. An unchecked item should remain visible in the dataset card or release notes rather than disappearing behind a successful training run.

- ☐ Source rights cover transformation, training, and intended distribution.
- ☐ The split unit matches the scientific claim.
- ☐ Every descendant remains in its parent source split.
- ☐ Control, measured, and semantic labels are distinguished.
- ☐ A dry-run count and storage estimate were reviewed.
- ☐ A stratified smoke corpus passed loader and listening checks.
- ☐ JSONL, summary, QA, metrics, card, and provenance outputs are archived.
- ☐ Failed or excluded examples retain stable IDs and reasons.

- ☐ Baselines match training budget and source inventory.
- ☐ Evaluation includes unseen sources and unseen rooms.
- ☐ Metrics are reported by severity and important intersection slices.
- ☐ At least one independent renderer or measured-room condition tests transfer.
- ☐ Creative or restorative outputs receive a blinded listening audit.
- ☐ Dataset and experiment versions are immutable and citable.

10.23 Literature bridge

Appendix C, Section 7 collects machine-learning and blind-estimation literature directly relevant to this chapter. It includes CRNN-based online RT60 estimation, joint RT60 and early-to-late estimation, late-reverberation PSD estimation with denoising autoencoders, room-aware speech models, reverberation-controllable voice conversion, and non-neural maximum-likelihood baselines. Appendix C, Section 8 adds speech intelligibility and room-acoustic context. Those papers should

guide baseline selection and failure analysis; the verbx profile families are generation tools, not substitutes for the literature or for measured-room evaluation.

The central principle is simple: augmentation is useful when it changes the training evidence in a controlled way and the improvement survives a genuinely new acoustic domain. More files are not the objective. Better, auditable generalization is.

10.24 Suggested Exercises

1. Create a small train, validation, and test augmentation plan that prevents room-identity leakage across splits.
2. Render a controlled augmentation set in which RT60 changes while source, seed, level, and channel layout remain fixed. Produce a CSV or JSON comparison.
3. Choose one task, such as ASR or source separation, and state which acoustic variations should be invariant versus predictive for the model.
4. Run quality checks on a small batch and reject at least one intentionally implausible output using an explicit metric rule.
5. Write a compact dataset card that records provenance, render parameters, split policy, and known limitations of the augmented audio.
6. Create a manifest that makes augmentation seed, room identity, parameter range, and source identity queryable for every output.
7. Design an ablation in which only early reflections vary, then predict the downstream task effect before training a model.
8. Define a perceptual and a machine-oriented acceptance criterion for augmented files, and explain why neither alone is sufficient.
9. Create one realistic and one adversarial reverberant example for a target model, documenting why each belongs in the dataset.
10. Write a reproducibility checklist for rerunning an augmentation corpus six months later on another machine.

“I used the bathtub for reverberation.”

— Pauline Oliveros (January 2003)

verbx Schema Reference

This reference defines the JSON and CSV contracts used by batch manifests, automation files, generated-IR metadata, and analysis reports. Read the prose before each table: a field's scope and provenance are as important as its serialized type.

Current as of v0.9.9.

The current boundary is:

- Batch manifests map to the offline `verbx render` surface.
- Realtime device selection (`verbx realtime`, `--input-device`, `--output-device`) is CLI/session state and is not represented in these schemas.
- Render reports are produced internally by a typed `RenderReport` object, but the written JSON remains backward-compatible dictionary-style output.

11.1 Batch Manifest (`batch.json` / `batch.jsonl`)

Batch manifests describe offline render jobs for `verbx batch render --manifest <file>`. A JSON document groups jobs under one versioned object, while JSONL stores one independent job per line for streaming and append-oriented workflows.

11.1.1 JSON format

Use the object form when a complete batch is authored and reviewed as one document. Relative paths are interpreted in the command's execution context, so production manifests should be run from a documented working directory or use resolved paths.

```
{
  "version": "0.5",
  "jobs": [
    {
      "infile": "path/to/dry.wav",
      "outfile": "path/to/wet.wav",
      "options": {
        "engine": "algo",
        "rt60": 2.5,
        "wet": 0.7,
        "dry": 0.3,
        "pre_delay_ms": 20,
        "fdn_lines": 16,
        "target_sr": 192000,
        "output_subtype": "float32"
      }
    }
  ]
}
```

11.1.2 JSONL format (newline-delimited)

Use JSONL when tooling needs to process or generate records incrementally. Each line contains one job object with no wrapping object and no `"jobs"` key:

```
{"infile": "dry/kick.wav", "outfile": "wet/kick.wav", "options": {"engine": "algo", "rt60": 0.4}}
{"infile": "dry/snare.wav", "outfile": "wet/snare.wav", "options": {"engine": "algo", "rt60": 0.8}}
```

11.1.3 Job object fields

Every job names an input and output. The optional `options` object overrides render defaults for that job and uses `RenderConfig` field names rather than shell spellings.

Table 11-1: Job object fields: reference table

Field	Type	Required	Description
`infile`	string	yes	Path to the dry input audio file
`outfile`	string	yes	Path for the processed output file
`options`	object	no	Any `verbx render` option as a key/value pair

`options` keys map directly to `RenderConfig` fields. All keys are optional – unset fields use defaults. Boolean flags use JSON booleans (`true`/`false`).

The following table covers common options rather than every experimental control:

Table 11-2: Job object fields: reference table

Option	Type	Default	Range
`engine`	string	`"auto"`	`"algo"`, `"conv"`, `"ism-fdn"`, `"auto"`
`algo_model`	string	`"fdn"`	`"fdn"`, `"spring"`, `"plate"`
`electromechanical_solver`	string	`"proxy"`	`"proxy"` or offline structural `"modal-fe"`
`spring_fe_nodes` / `spring_fe_modes`	integer	24 / 24	Nodes `4..128`; modes `1..128`
`spring_fe_coupling` / `spring_fe_loss`	number	`0.08` / `0.30`	`0..1` coupling; `0..2` high-mode loss
`plate_fe_nx` / `plate_fe_ny` / `plate_fe_modes`	integer	12 / 8 / 32	Grid dimensions `4..32`; modes `1..128`
`plate_fe_loss`	number	`0.24`	Frequency-dependent modal loss `0..2`
`rt60`	number	`60.0`	`0.1` – `3600.0`
`wet`	number	`0.8`	`0.0` – `1.0`
`dry`	number	`0.2`	`0.0` – `1.0`
`pre_delay_ms`	number	`20.0`	`0` – `500`
`fdn_lines`	integer	8	`1` – `64`
`fdn_matrix`	string	`"hadamard"`	`"hadamard"`, `"householder"`, `"random_orthogonal"`, `"circulant"`, `"elliptic"`, `"tv_unitary"`, `"graph"`, `"sdn_hybrid"`
`shimmer`	boolean	`false`	–
`shimmer_semitones`	number	12	`-24` – `24`
`shimmer_mix`	number	`0.25`	`0.0` – `1.0`
`shimmer_feedback`	number	`0.35`	`0.0` – `0.98` (safe), up to `1.25` with `unsafe_self_oscillate=true`
`unsafe_self_oscillate`	boolean	`false`	Enables unsafe above-unity feedback path in algorithmic mode
`unsafe_loop_gain`	number	`1.02`	`> 1.0` for intentional self-oscillation (`<= 1.25`)
`auto_fit`	string	`"none"`	`"none"`, `"speech"`, `"music"`, `"drums"`, `"ambient"`

Option	Type	Default	Range
`fdn_matrix_morph_to`	string/null	`null`	Optional matrix morph target family
`fdn_matrix_morph_seconds`	number	`0.0`	`>= 0.0`
`tail_stop_threshold_db`	number	`-120.0`	`-240.0` – `0.0`
`tail_stop_hold_ms`	number	`10.0`	`>= 0.0`
`tail_stop_metric`	string	`"peak"`	`"peak"`, `"rms"`
`algo_stream`	boolean	`false`	Enable algorithmic proxy streaming path
`algo_proxy_ir_max_seconds`	number	`120.0`	`> 0.0`
`algo_gpu_proxy`	boolean	`false`	Use CUDA convolution backend for proxy streaming
`output_subtype`	string	`"auto"`	`"auto"`, `"float32"`, `"float64"`, `"pcm16"`, `"pcm24"`, `"pcm32"`
`output_container`	string	`"auto"`	`"auto"`, `"wav"`, `"w64"`, `"rf64"`
`target_sr`	integer	`null`	`>= 1` (Hz; `null` uses input sample rate)
`lowcut`	number	`null`	Hz; `null` = disabled
`highcut`	number	`null`	Hz; `null` = disabled
`shimmer_spatial`	boolean	`false`	Enable multichannel shimmer decorrelation
`shimmer_spread_cents`	number	`8.0`	`>= 0.0`
`shimmer_decorrelation_ms`	number	`1.5`	`>= 0.0`
`er_geometry`	boolean	`false`	Enable first-order image-source early reflections
`er_room_dims_m`	array[3]	`[10.0, 7.0, 3.0]`	Positive room dimensions (meters)
`er_source_pos_m`	array[3]	`[2.0, 2.0, 1.5]`	Source position (meters)
`er_listener_pos_m`	array[3]	`[5.0, 3.5, 1.5]`	Listener position (meters)
`er_absorption`	number	`0.35`	`0.0` – `0.99`
`er_material`	string	`"studio"`	Material preset name
`repeat`	integer	`1`	`1` – `100`

Generate a template with: `verbx batch template`

11.2 Automation File (`.json` or `.csv`)

Automation files describe time-varying controls for `verbx render --automation-file <file>`. JSON supports several lane types and combination rules; CSV provides a compact breakpoint format for tools that naturally emit rows.

11.2.1 JSON format

The JSON object selects a timing mode and contains one or more lanes. This example combines a wet-mix envelope with a slow RT60 oscillator:

```
{
  "mode": "block",
  "block_ms": 20.0,
  "lanes": [
    {
      "target": "wet",
      "type": "breakpoints",
      "interp": "linear",
```

```
      "points": [
        {"time": 0.0, "value": 0.2},
        {"time": 5.0, "value": 0.9},
        {"time": 10.0, "value": 0.5}
      ]
    },
    {
      "target": "rt60",
      "type": "lfo",
      "shape": "sine",
      "rate_hz": 0.1,
      "depth": 0.4,
      "center": 2.0
    }
  ]
}
```

11.2.2 Top-level fields

Top-level fields establish the evaluation rate and collect the lanes. A lane may operate at control-block or sample rate according to the selected mode and target implementation.

Table 11-3: Top-level fields: reference table

Field	Type	Default	Description
`mode`	string	`"block"`	`"block"` (control-rate) or `"sample"` (sample-rate)
`block_ms`	number	`20.0`	Control block size in ms (when `mode = "block"`)
`lanes`	array	required	List of automation lane objects

11.2.3 Lane types

Lane type determines how values are generated between or within the times supplied by the author. Several lanes may address one target when their combination rules are intentional.

11.2.3.1 `breakpoints` – interpolated envelope

A breakpoint lane is the clearest choice for authored cues and gradual transitions. Points are expressed in seconds and evaluated using the selected interpolation rule.

```
{
  "target": "wet",
  "type": "breakpoints",
  "interp": "linear",
  "combine": "replace",
  "points": [
    {"time": 0.0, "value": 0.3},
    {"time": 4.0, "value": 0.9}
  ]
}
```

Table 11-4: breakpoints – interpolated envelope: reference table

Field	Type	Default	Description
`target`	string	required	Automation target name (see below)
`type`	string	required	`"breakpoints"`

Field	Type	Default	Description
`interp`	string	`"linear"`	Interpolation mode (see below)
`combine`	string	`"replace"`	How to merge with prior lanes for same target
`points`	array	required	Array of `{"time": seconds, "value": float}`

Available interpolation modes are `linear`, `hold`, `step`, `smooth`, `smoothstep`, `exp`, and `exponential`.

Available combination modes are `replace`, `add`, and `multiply`.

11.2.3.2 `lfo` – low-frequency oscillator

An LFO lane generates periodic or seeded low-rate variation around a center value. Its `depth` uses the target's native unit, so the same numeric depth has different meaning for damping and RT60.

```
{
  "target": "damping",
  "type": "lfo",
  "shape": "sine",
  "rate_hz": 0.25,
  "depth": 0.3,
  "center": 0.5,
  "phase_deg": 0.0
}
```

Table 11-5: lfo – low-frequency oscillator: reference table

Field	Type	Default	Description
`target`	string	required	Automation target name
`type`	string	required	`"lfo"`
`shape`	string	`"sine"`	`"sine"`, `"triangle"`, `"square"`, `"sawtooth"`, `"random"`
`rate_hz`	number	`1.0`	Oscillation frequency in Hz
`depth`	number	`0.5`	Peak deviation from center (±depth)
`center`	number	`0.5`	Center value
`phase_deg`	number	`0.0`	Starting phase in degrees

11.2.3.3 `segments` – piecewise constant/linear segments

A segment lane gives each interval an explicit start, end, value, and interpolation policy. It is useful when a control timeline comes from an edit decision list or cue system rather than isolated points.

```
{
  "target": "gain-db",
  "type": "segments",
  "segments": [
    {"start": 0.0, "end": 2.0, "value": -6.0, "interp": "hold"},
    {"start": 2.0, "end": 5.0, "value": 0.0,  "interp": "linear"},
    {"start": 5.0, "end": 8.0, "value": -12.0, "interp": "smooth"}
  ]
}
```

11.2.4 CSV format

CSV automation represents breakpoint rows only. It is convenient for spreadsheets and measurement software, but it cannot express every JSON lane property.

```
target,time_s,value,interp
wet,0.0,0.2,linear
wet,5.0,0.8,linear
wet,10.0,0.5,linear
rt60,0.0,1.0,smooth
rt60,8.0,4.0,smooth
```

The required columns are `target`, `time_s`, and `value`; `interp` is optional.

11.2.5 Automation targets

Targets are grouped by the stage at which the value acts. Ranges below are serialization limits, not promises that every extreme will be musically useful for every source.

Table 11-6: Automation targets: reference table

Target	Domain	Range	Description
`wet`	post	0.0 – 1.0	Wet mix level
`dry`	post	0.0 – 1.0	Dry mix level
`gain-db`	post	–48.0 – 24.0	Output gain in dB
`rt60`	engine	0.1 – 3600.0	Reverberation time in seconds
`damping`	engine	0.0 – 1.0	High-frequency damping
`room-size`	engine	0.25 – 4.0	Room size scalar
`room-size-macro`	engine	–1.0 – 1.0	Room size macro (normalized)
`clarity-macro`	engine	–1.0 – 1.0	Clarity macro
`warmth-macro`	engine	–1.0 – 1.0	Warmth macro
`envelopment-macro`	engine	–1.0 – 1.0	Envelopment macro
`fdn-rt60-tilt`	engine	–1.0 – 1.0	FDN RT60 spectral tilt
`fdn-rt60-low`	engine	0.1 – 3600.0	Low-band RT60 target
`fdn-rt60-mid`	engine	0.1 – 3600.0	Mid-band RT60 target
`fdn-rt60-high`	engine	0.1 – 3600.0	High-band RT60 target
`fdn-xover-low-hz`	engine	20.0 – 20000.0	Low/mid crossover
`fdn-xover-high-hz`	engine	20.0 – 20000.0	Mid/high crossover
`fdn-tonal-correction-strength`	engine	0.0 – 1.0	Tonal correction amount
`ir-blend-alpha`	conv	0.0 – 1.0	IR morph blend (convolution engine)

Aliases: Most targets accept common shorthand – e.g. `t60` → `rt60`, `gain` → `gain-db`, `room` → `room-size`, `blend-alpha` → `ir-blend-alpha`.

11.3 Generated IR Metadata (`*.ir.meta.json`)

`verbx ir gen OUT.wav` writes `OUT.wav.ir.meta.json` unless `--silent` is used. The top-level `version`, `mode`, and `seed` identify the generator; the `params` object contains the complete resolved `IRGenConfig`; and `metrics` contains measurements of the rendered IR. Every parameter contributes to the deterministic cache key.

Scala-tuned IRs add the following fields inside `params`:

Table 11-7: Generated IR Metadata (*.ir.meta.json): reference table

Field	Type	Meaning
`scala_file_name`	string or null	Source `.scl` basename

Field	Type	Meaning
`scala_description`	string or null	First non-comment Scala line
`scala_sha256`	string or null	SHA-256 of the exact source bytes
`scala_root_hz`	number or null	Frequency assigned to the root degree
`scala_root_degree`	integer	Zero-based selected degree
`scala_targets_hz`	array of numbers	Resolved in-range frequency targets
`scala_strength`	number	Tuning and emphasis blend from 0 to 1
`scala_bandwidth_cents`	number	Constant-Q band width in cents
`scala_gain_db`	number	Pre-normalization emphasis gain in dB
`scala_max_targets`	integer	Deterministic DSP target budget

The hash, resolved target list, and root mapping are the reproducibility contract. A consumer does not need the original Scala file to audit which frequencies shaped an existing IR. The synthetic IR cache namespace is `verbx-ir-v0.5`; older entries miss once and regenerate under this schema.

11.4 Analysis Output (`verbx analyze --json-out`)

JSON produced by `verbx analyze --json-out <file>`.

```json
{
  "schema": "analyze-report-v1",
  "source": {
    "path": "/absolute/path/hall_ir.wav",
    "sample_rate_hz": 48000,
    "channels": 2,
    "frames": 241536,
    "duration_seconds": 5.032
  },
  "analysis": {
    "reverb": true,
    "input_kind": "ir",
    "direct_window_ms": 2.5,
    "loudness": false,
    "edr": true,
    "room": false,
    "ambi_order": 0,
    "ambi_normalization": "auto",
    "channel_order": "auto"
  },
  "sample_rate": 48000,
  "channels": 2,
  "metrics": {
    "duration": 5.032,
    "reverb_rt60_seconds": 1.842,
    "reverb_rt60_fit": "t30",
    "reverb_decay_fit_r2": 0.994,
    "reverb_c80_db": 1.73,
    "reverb_confidence": "high",
    "...": "..."
  }
}
```

`sample_rate` and `channels` remain at the top level for compatibility with the pre-v1 shape. New readers should use `source`. Metric values are floats except for classification, fit-method, interpretation, and confidence fields.

11.4.1 Default reverb fields

Default reverb analysis fields combine decay estimates, energy ratios, source classification, and confidence. Units are part of the key contract and should be retained when values are moved into a table or plotting system.

Table 11-8: Default reverb fields: reference table

Key	Unit/values	Description
`reverb_rt60_seconds`	s	Selected broadband RT60 estimate; T30 preferred, then T20, then EDT
`reverb_rt60_fit`	`t30`, `t20`, `edt`, `none`	Regression window selected for RT60
`reverb_edt_seconds`	s	Early Decay Time extrapolated from 0 to –10 dB
`reverb_t20_seconds`	s	RT60 extrapolated from the –5 to –25 dB fit
`reverb_t30_seconds`	s	RT60 extrapolated from the –5 to –35 dB fit
`reverb_decay_fit_r2`	0–1	Coefficient of determination for the selected decay fit
`reverb_decay_range_db`	dB	Peak-to-tail-noise range available in the file
`reverb_noise_floor_dbfs`	dBFS	RMS level of the final ten percent of the peak-aligned tail
`reverb_c50_db`	dB	C_{50} early/late energy ratio at 50 ms
`reverb_c80_db`	dB	C_{80} early/late energy ratio at 80 ms
`reverb_d50_percent`	%	D_{50} energy arriving in the first 50 ms
`reverb_center_time_ms`	ms	Energy-weighted temporal center of the response
`reverb_direct_to_reverberant_db`	dB	Direct-window energy versus the remaining tail
`reverb_iacc_early`	0–1	Maximum early L/R cross-correlation within ±1 ms
`reverb_decay_start_seconds`	s	Strongest-event location used to align the decay
`reverb_tail_duration_seconds`	s	Available signal duration after decay alignment
`reverb_input_kind`	`impulse_response`, `program_audio`	Resolved source classification
`reverb_interpretation`	string	Indicates room-acoustic or program-audio interpretation
`reverb_confidence_score`	0–1	Composite fit, range, duration, and source-suitability score
`reverb_confidence`	`high`, `medium`, `low`	Human-readable confidence label

These fields are present by default. Pass `--no-reverb` to omit them. For program audio, the analyzer reports peak-aligned estimates; standards-oriented clarity, definition, DRR, and RT measurements require a suitable impulse response and controlled measurement procedure.

11.5 Room Size Estimate (`--room` flag on `verbx analyze` / `verbx compare`)

When `--room` is passed the analysis output includes the following additional keys (all prefixed `room_`):

11.5.1 Numeric fields (float)

Numeric room fields describe the fitted acoustic model and its uncertainty. They are estimates inferred from the analyzed response, not surveyed architectural dimensions.

Table 11-9: Numeric fields (float): reference table

Key	Unit	Description
`room_rt60_s`	s	Best RT60 estimate used as sizing input (mid-band preferred)
`room_rt60_low_s`	s	EDR RT60 estimate for the low band (20–250 Hz)
`room_rt60_mid_s`	s	EDR RT60 estimate for the mid band (250–2 000 Hz)
`room_rt60_high_s`	s	EDR RT60 estimate for the high band (2 000 Hz+)
`room_volume_m3`	m^3	Primary volume estimate (Sabine/Eyring blend)
`room_volume_m3_sabine`	m^3	Sabine-only volume estimate
`room_volume_m3_eyring`	m^3	Eyring-only volume estimate
`room_volume_m3_low`	m^3	Conservative lower bound (–30 %)
`room_volume_m3_high`	m^3	Conservative upper bound (+30 %)
`room_dim_width_m`	m	Estimated room width (shortest horizontal dimension)
`room_dim_depth_m`	m	Estimated room depth (1.25 × width)
`room_dim_height_m`	m	Estimated room height (0.62 × width)
`room_surface_area_m2`	m^2	Total surface area of the estimated rectangular box
`room_mean_absorption`	–	Estimated mean absorption coefficient [0, 1]
`room_critical_distance_m`	m	Schroeder critical distance (direct = reverberant field)
`room_confidence_score`	–	Numeric confidence rating [0, 1]

11.5.2 String fields

String fields summarize the selected estimation method and qualitative classifications. Preserve the underlying numeric fields when using these labels in reports.

Table 11-10: String fields: reference table

Key	Values	Description
`room_class`	`"closet"` `"small"` `"medium"` `"large"` `"very_large"` `"cathedral"` `"unknown"`	Qualitative room size label
`room_estimation_method`	`"sabine"` `"eyring"` `"none"`	Formula used for primary volume
`room_confidence`	`"high"` `"medium"` `"low"`	Qualitative confidence rating

11.5.3 Room class thresholds

Room classes are coarse interpretive bins derived from approximate volume and decay. They help organize results but do not replace the volume interval or confidence score.

Table 11-11: Room class thresholds: reference table

Class	Approx. volume	Approx. RT60
`closet`	< 8 m^3	< 0.12 s
`small`	8–50 m^3	0.12–0.35 s
`medium`	50–250 m^3	0.35–0.80 s

Class	Approx. volume	Approx. RT60
`large`	250–1 500 m³	0.80–2.0 s
`very_large`	1 500–10 000 m³	2.0–5.0 s
`cathedral`	> 10 000 m³	> 5.0 s

11.5.4 Example

```
verbx analyze my_reverb.wav --room --json-out room_report.json
```

```
{
  "sample_rate": 48000,
  "channels": 2,
  "metrics": {
    "room_rt60_s": 1.42,
    "room_volume_m3": 320.5,
    "room_volume_m3_low": 224.4,
    "room_volume_m3_high": 416.7,
    "room_dim_width_m": 7.38,
    "room_dim_depth_m": 9.22,
    "room_dim_height_m": 4.58,
    "room_surface_area_m2": 271.2,
    "room_mean_absorption": 0.175,
    "room_critical_distance_m": 1.94,
    "room_class": "large",
    "room_estimation_method": "sabine",
    "room_confidence": "high",
    "room_confidence_score": 0.8,
    "...": "..."
  }
}
```

11.6 Compare Report (`verbx compare --json-out`)

JSON produced by `verbx compare FILE_A FILE_B --json-out <file>`.

```
{
  "schema": "compare-report-v1",
  "file_a": "/path/to/dry.wav",
  "file_b": "/path/to/wet.wav",
  "sample_rate_a": 48000,
  "sample_rate_b": 48000,
  "channels_a": 2,
  "channels_b": 2,
  "metrics_a": { "rms": 0.12, "..." : "..." },
  "metrics_b": { "rms": 0.18, "...": "..." },
  "delta": { "rms": 0.06, "...": "..." }
}
```

`delta` only includes keys whose values are numeric (`float`) in both files — string-valued fields like `room_class` are present in `metrics_a`/`metrics_b` but omitted from `delta`.

11.7 Inline automation (`--automation-point`)

For simple one-off control points without a file:

```
verbx render in.wav out.wav    --automation-point wet:0.0:0.2:linear    --automation-point
↪  wet:5.0:0.9:linear    --automation-point rt60:0.0:1.5
```

Format: `target:time_s:value[:interp]`

- `target` – automation target name
- `time_s` – time in seconds (float)
- `value` – parameter value (float)
- `interp` – interpolation mode (optional, default `linear`)

"The only technical things I know are treble, volume and reverb, that's all."

— Johnny Thunders (Date unknown)

Chapter 12

IR Morph Quality Assurance

`verbx ir morph-sweep` renders a repeatable sequence between two impulse responses and records enough evidence to compare that sequence across revisions. It is intended for regression work, parameter studies, and review sessions in which one isolated morph is not sufficient. The command does not pronounce a render perceptually correct. It produces audio, measurements, and provenance that make an informed decision easier to repeat.

12.1 What the Sweep Produces

A sweep evaluates an ordered alpha timeline. Alpha zero corresponds to the first IR, alpha one corresponds to the second, and intermediate values describe the selected morph algorithm's path between them. For each value, verbx writes a morphed WAV and one row of quality measurements. It also writes a CSV for detailed comparison and a JSON summary for automation.

The command supports parallel workers, bounded retries, checkpointing, and resume. Those controls matter when the endpoint IRs are long or highly multichannel: an interrupted study can continue without discarding successful results, and repeated inputs can reuse deterministic cache artifacts.

12.2 A Uniform Timeline

The shortest useful experiment samples the interval uniformly. Nine steps include both endpoints and seven intermediate states:

```
verbx ir morph-sweep ir_a.wav ir_b.wav out/morph_sweep \
  --alpha-start 0.0 \
  --alpha-end 1.0 \
  --alpha-steps 9 \
  --workers 4 \
  --retries 1 \
  --checkpoint-file out/morph_sweep.checkpoint.json
```

Listen to the endpoints first. A smooth numerical trajectory cannot rescue an incorrect source IR, channel mismatch, or unintended normalization. Then audition the sequence in order and at matched monitoring level. Discontinuities are often easier to hear as a progression than as unrelated files.

12.3 A Deliberate Timeline

Uniform sampling is convenient, but it can miss a narrow transition. Repeat `--alpha` to concentrate renders near a region where timbre, decay, or spatial image changes quickly:

```
verbx ir morph-sweep ir_a.wav ir_b.wav out/morph_sweep \
  --alpha 0.0 \
  --alpha 0.1 \
  --alpha 0.25 \
  --alpha 0.5 \
  --alpha 0.75 \
  --alpha 1.0 \
```

```
  --qa-json-out out/custom_summary.json \
  --qa-csv-out out/custom_metrics.csv
```

An explicit timeline is also useful when alpha values correspond to scene cues, edit points, or a control curve used elsewhere. Preserve the list with the test assets so that a later run compares the same states rather than merely the same endpoints.

12.4 Checkpoint and Resume

Long studies should write a checkpoint outside the output filenames themselves. The first command below records each completed job and continues after an isolated failure. The second reads that checkpoint and skips completed outputs:

```
# Initial pass
verbx ir morph-sweep ir_a.wav ir_b.wav out/morph_sweep \
  --alpha-start 0 --alpha-end 1 --alpha-steps 17 \
  --checkpoint-file out/morph.checkpoint.json \
  --retries 2 \
  --continue-on-error \
  --allow-failed

# Resume after correcting the failure or restoring the worker
verbx ir morph-sweep ir_a.wav ir_b.wav out/morph_sweep \
  --alpha-start 0 --alpha-end 1 --alpha-steps 17 \
  --checkpoint-file out/morph.checkpoint.json \
  --resume
```

`--allow-failed` is appropriate for exploratory batches in which partial evidence is useful. It should not quietly weaken a release gate. In CI, prefer a nonzero exit when any required render fails, and archive the checkpoint only as diagnostic evidence.

12.5 Output Artifacts

The output directory is a small experiment record, not merely a folder of audio. Its contents have separate jobs:

- `<out_dir>/<prefix>_<index>_aNNN.wav` contains the rendered IR for one alpha value.
- `<out_dir>/morph_sweep_metrics.csv` contains one row per planned render, including status, timing, cache state, audio properties, and quality measurements.
- `<out_dir>/morph_sweep_summary.json` records the command mode, endpoints, alpha list, scheduling choices, completion counts, artifact paths, and aggregate statistics.
- The optional checkpoint JSON at `--checkpoint-file` records recoverable progress for a later `--resume` run.

Keep the CSV and summary beside any listening notes. A WAV sequence without its alpha values and mode is difficult to reproduce, while measurements without the corresponding sound invite conclusions detached from audibility.

12.6 Reading the Measurements

Each metric compares the output with a target interpolated from measurements of the two endpoints. Smaller drift is generally desirable, but none of the quantities is a universal acceptance threshold.

- `rt60_target_s` and `rt60_out_s` report target and measured reverberation time in seconds. `rt60_drift_s` is their absolute difference in seconds.
- `early_late_target_db` and `early_late_out_db` report the target and measured early-to-late energy ratio in decibels. `early_late_drift_db` is the absolute difference in decibels.
- `spectral_distance_db` measures distance from the interpolated endpoint spectrum in decibels. Inspect frequency-localized problems with a separate spectral plot rather than treating this scalar as a complete tonal diagnosis.
- `interchannel_coherence_target` and `interchannel_coherence_out` are unitless coherence scores. `interchannel_coherence_delta` is their absolute difference. A small delta preserves the target statistic; it does not prove correct localization or a convincing immersive field.

A missing value can be meaningful. Very short, silent, or pathological IRs may not support a stable decay estimate. Do not convert missing measurements to zero, because zero drift and unavailable evidence are different conditions.

12.7 Building a Regression Gate

A useful gate begins with a known fixture pair and limits chosen from repeated good renders, not arbitrary round numbers. Measure several accepted runs, study normal variation, then set tolerances wide enough to ignore harmless numerical noise and narrow enough to catch the artifact under investigation. Keep separate gates for channel count, sample rate, completion, and objective drift so that a routing failure is not hidden inside an aggregate score.

After the automated gate passes, audition at least the endpoints, midpoint, and the states nearest the largest reported drift. For multichannel material, include the intended layout and at least one required fold-down. The final question is not whether every curve is smooth. It is whether the morph behaves continuously, preserves the intended spatial and decay relationships, and remains usable in the production context that motivated the test.

"Let echo, too, perform her part, prolonging every note with art."

— Joseph Addison, *Ode for St. Cecilia's Day* (1699)

Chapter 13

SOFA Import and Extraction

Reviewed for verbx v0.9.9 on August 5, 2026.

The Spatially Oriented Format for Acoustics (SOFA) stores measured spatial-audio data in HDF5 containers governed by named conventions. A SOFA file is not merely a multichannel impulse response with a different extension. Its arrays may describe many source positions, receivers, listeners, and emitters, and the coordinate metadata determines what those samples mean. `verbx` therefore treats SOFA as an explicit import step rather than accepting it silently wherever a WAV impulse response is expected.

13.1 Supported Workflow

The current command-line surface is deliberately narrow. It lets an engineer inspect a container, select one FIR measurement, and export that selection as an ordinary WAV matrix for the existing convolution tools. The two commands are:

```
verbx ir sofa-info measurements.sofa
verbx ir sofa-extract measurements.sofa selected_ir.wav
```

`sofa-info` reports the convention and version, sample rate in hertz, `Data/IR` shape and dimension labels, and the available source, listener, receiver, and emitter position shapes. Inspect this report before extraction. A plausible channel count does not establish receiver order, coordinate units, or intended loudspeaker routing.

`sofa-extract` writes three related artifacts: the extracted WAV matrix, an IR metadata sidecar, and extraction metadata recording the selection and conversion choices. The WAV is then suitable for the same explicit matrix-routing workflow used by other verbx impulse responses.

13.2 Selecting a Measurement

SOFA datasets often contain several measurements of the same acoustic system at different positions. The default extracts measurement zero and, for a rank-four array, emitter zero. Select other entries by index:

```
verbx ir sofa-extract hall.sofa hall_m012_e01.wav \
  --measurement-index 12 \
  --emitter-index 1
```

In strict mode, verbx accepts `Data/IR` rank three with dimensions `(M, R, N)` or rank four with dimensions `(M, R, E, N)`. Here, `M` is measurement index, `R` is receiver channel, `E` is emitter index, and `N` is sample index. Best-effort mode, which is the default, can accommodate less regular containers, but its output still requires listening and routing verification.

13.3 Sample Rate and Normalization

An extracted IR retains the source sample rate unless `--target-sr` requests deterministic resampling. Normalization is explicit because changing IR gain changes the gain of every later convolution. The accepted modes are `none`, `peak`, and `rms`; the default is `peak`.

```
verbx ir sofa-extract room.sofa room_48k.wav \
  --target-sr 48000 \
  --normalize peak \
  --strict
```

Use `none` when calibrated level relationships must survive extraction. Use `peak` for a convenient bounded IR when absolute calibration is not part of the dataset contract. Use `rms` only when equalizing average IR energy is the intended comparison. Record the choice alongside a production master rather than relying on a filename to preserve it.

13.4 Routing the Exported Matrix

The receiver dimension becomes the channel dimension of the exported WAV. That operation preserves samples, but it does not invent a loudspeaker layout. Before rendering program material, make a short identification render or inspect isolated impulses so that every receiver channel is mapped intentionally through `--ir-matrix-layout` or `--ir-route-map`.

A useful handoff record includes the source file hash, SOFA convention, measurement and emitter indices, sample-rate conversion, normalization mode, receiver order, and intended output layout. The extraction metadata records the computational choices; the engineer must still document the production meaning of the channels.

13.5 Dependency and Failure Modes

SOFA support uses the optional `h5py` dependency so that the baseline verbx installation does not require an HDF5 stack. When the dependency is unavailable, the commands stop with an installation message rather than interpreting the file partially.

The most important failures are semantic rather than syntactic. A container can open successfully while using a convention, coordinate system, orientation, or receiver order inappropriate for the intended render. Strict mode catches unsupported array rank, but it cannot decide whether a chosen viewpoint is musically or geometrically correct. Treat the info report, metadata sidecars, channel-identification render, and listening test as one validation sequence.

13.6 Current Boundary

The shipped feature extracts indexed FIR data. It does not provide coordinate queries such as nearest azimuth and elevation, automatic Ambisonic decoding, arbitrary convention conversion, or direct object-audio authoring. Those operations require convention-specific decisions that should remain visible rather than being guessed during a render.

This boundary keeps SOFA interoperability reproducible: first select and document a view, then convert it to an explicit matrix, and finally route that matrix through the established convolution engine.

13.7 References

The formal and project references below describe the container family and its conventions. Consult the convention named by a dataset in addition to the general standard.

- AES69, *AES Standard for File Exchange: Spatial Acoustic Data File Format*: https://www.aes.org/publications/standards/search.cfm?docID=99
- SOFA conventions project: https://www.sofaconventions.org/

“Because of the reverberation, there’s always more to the sound than just the sound.”

— Pauline Oliveros (Date unknown)

Illustrated Guide

This chapter collects the visual reference material used throughout the guide: signal-flow diagrams, graphs of key DSP tradeoffs, analysis dashboards, and topology sketches. Every quantitative axis is labeled with a unit; conceptual scales are explicitly marked as normalized or categorical. The figures are generated with `python3 scripts/generate_userguide_figures.py`, and this chapter is generated with `python3 scripts/generate_figure_guide.py`, so the PDF can be rebuilt reproducibly.

14.1 How to Read This Chapter

Read quantitative plots from the labeled independent axis toward the reported response, and compare traces only when they share a scale. Boxes and arrows describe topology, not physical size. Layout drawings identify nominal positions rather than installation tolerances. Unless a figure says that it contains measured data, treat it as an explanatory model; use `verbx analyze` and its JSON report for claims about a particular render, device, or room.

14.2 System Flow

These diagrams follow audio, control, and evidence through the system. Read arrows as dependencies or signal direction; box size is not a measure of cost or importance.

Figure 14-1 presents End-to-end render signal flow and establishes the visual reference for the discussion that follows.

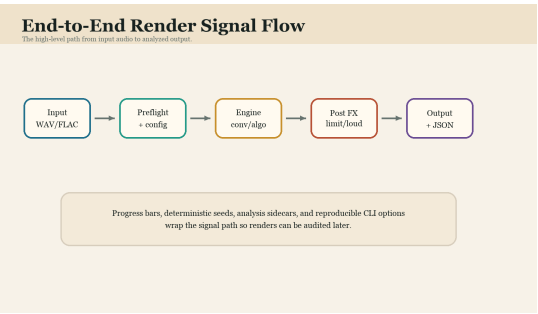


Figure 14-1: End-to-end render signal flow

The figure below introduces **End-to-end render signal flow**. The complete render path from source file through validation, DSP, post-processing, and report output. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-2 presents Realtime latency budget by block size and establishes the visual reference for the discussion that follows.

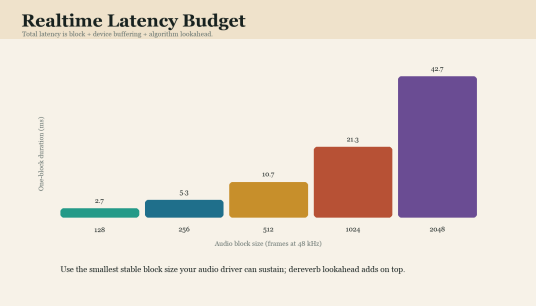


Figure 14-2: Realtime latency budget by block size

The figure below introduces **Realtime latency budget by block size**. The duration contributed by one audio block at 48 kHz, before driver and algorithmic buffering are added. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Audio block size (frames at 48 kHz)** and the vertical axis is **One-block duration (ms)**.

Figure 14-3 presents CLI command map and establishes the visual reference for the discussion that follows.

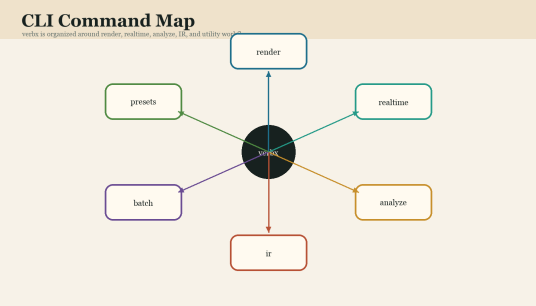


Figure 14-3: CLI command map

The figure below introduces **CLI command map**. The primary command families arranged around the verbx executable. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-4 presents Analysis JSON structure and establishes the visual reference for the discussion that follows.

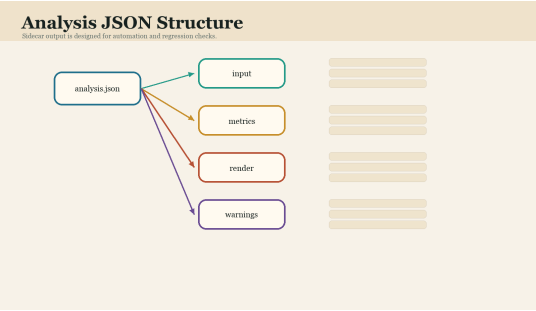


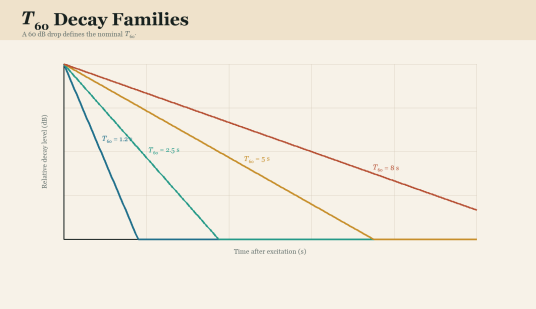
Figure 14-4: Analysis JSON structure

The figure below introduces **Analysis JSON structure**. The top-level input, metrics, render, and warning groups in a machine-readable sidecar. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

14.3 Reverb Physics and Analysis

The plots in this group connect audible decay to quantities that can be measured. Compare units before comparing slopes: seconds, decibels, linear amplitude, and inferred volume answer different questions.

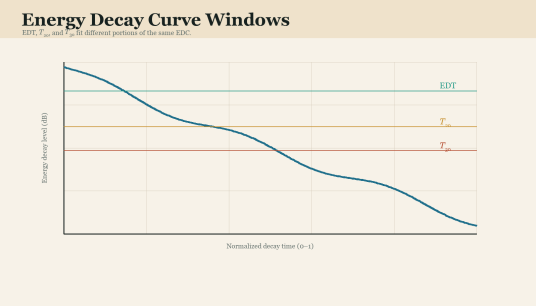
Figure 14-5 presents T_{60} decay families and establishes the visual reference for the discussion that follows.



The figure below introduces T_{60} **decay families**. Idealized decay slopes for several nominal reverberation times, each reaching a 60 dB loss at its labeled T_{60}. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Time after excitation (s)** and the vertical axis is **Relative decay level (dB)**.

Figure 14-5: T_{60} decay families

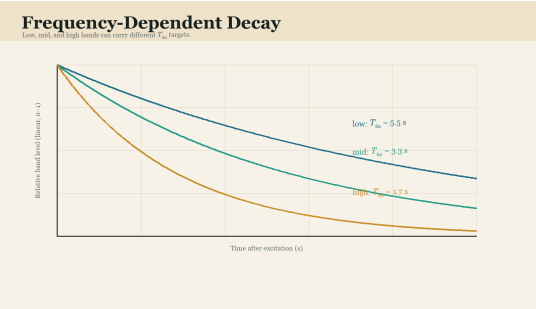
Figure 14-6 presents Energy decay curve fitting windows and establishes the visual reference for the discussion that follows.



The figure below introduces **Energy decay curve fitting windows**. The EDT, T_{20}, and T_{30} regression regions used to estimate decay from different portions of an energy decay curve. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Normalized decay time (0–1)** and the vertical axis is **Energy decay level (dB)**.

Figure 14-6: Energy decay curve fitting windows

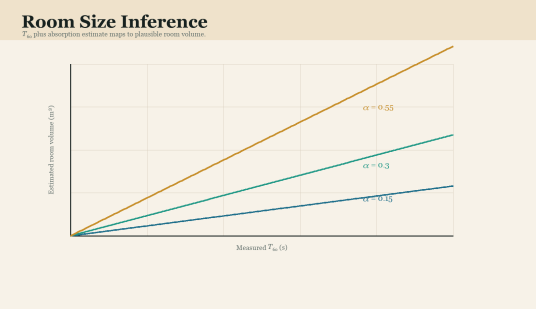
Figure 14-7 presents Frequency-dependent decay bands and establishes the visual reference for the discussion that follows.



The figure below introduces **Frequency-dependent decay bands**. Low-, mid-, and high-frequency tails with different nominal decay constants. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Time after excitation (s)** and the vertical axis is **Relative band level (linear, 0–1)**.

Figure 14-7: Frequency-dependent decay bands

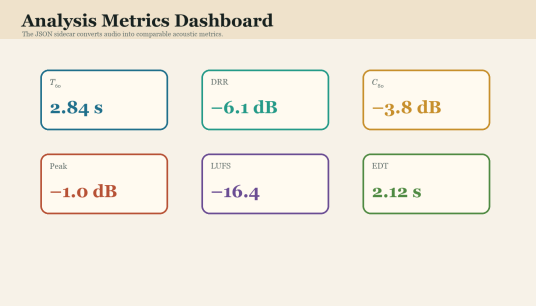
Figure 14-8 presents Room size inference curves and establishes the visual reference for the discussion that follows.



The figure below introduces **Room size inference curves**. Sabine-style volume estimates across T_{60} for three assumed mean absorption coefficients. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Measured T_{60} (s)** and the vertical axis is **Estimated room volume (m³)**.

Figure 14-8: Room size inference curves

Figure 14-9 presents Analysis metrics dashboard and establishes the visual reference for the discussion that follows.



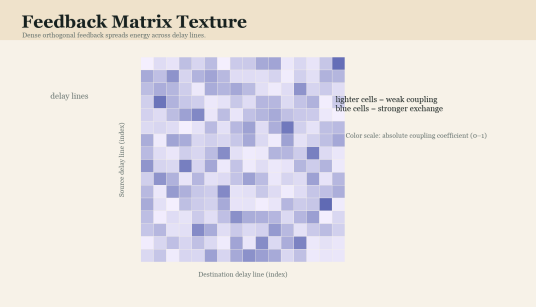
The figure below introduces **Analysis metrics dashboard**. A compact view of representative T_{60}, DRR, C_{80}, peak, LUFS, and EDT fields emitted to JSON. Each card reports a separate metric with its own printed unit, so card size and position do not encode magnitude. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-9: Analysis metrics dashboard

14.4 Algorithms and Processing

This group moves from network topology to implementation. Structural diagrams explain order and feedback; plotted responses explain what those structures do over time or frequency.

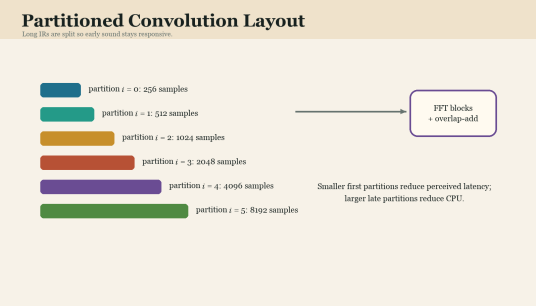
Figure 14-10 presents Feedback matrix texture heatmap and establishes the visual reference for the discussion that follows.



The figure below introduces **Feedback matrix texture heatmap**. A 16 by 16 orthogonal feedback matrix illustrating weak and strong coupling among FDN delay lines. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Destination delay line (index)** and the vertical axis is **Source delay line (index)**. The color or radial scale reports **Absolute coupling coefficient (0–1)**.

Figure 14-10: Feedback matrix texture heatmap

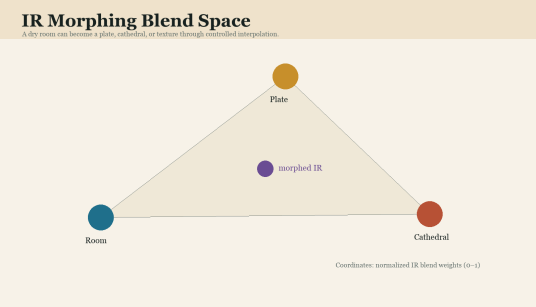
Figure 14-11 presents Partitioned convolution layout and establishes the visual reference for the discussion that follows.



The figure below introduces **Partitioned convolution layout**. An impulse response divided into progressively larger FFT partitions to balance latency and throughput. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-11: Partitioned convolution layout

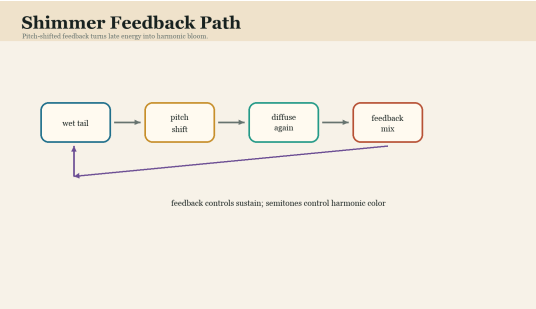
Figure 14-12 presents IR morphing blend space and establishes the visual reference for the discussion that follows.



The figure below introduces **IR morphing blend space**. A conceptual interpolation triangle connecting room, plate, and cathedral impulse-response families. Points and paths occupy a two-dimensional design space; proximity indicates similar states, not physical distance. The horizontal axis is **Blend coordinate *A* (normalized, 0–1)** and the vertical axis is **Blend coordinate *B* (normalized, 0–1)**.

Figure 14-12: IR morphing blend space

Figure 14-13 presents Shimmer feedback path and establishes the visual reference for the discussion that follows.



The figure below introduces **Shimmer feedback path**. The pitch-shift, diffusion, and feedback loop that turns late energy into a sustained harmonic layer. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-13: Shimmer feedback path

Figure 14-14 presents Infinite-style reverb tail behavior and establishes the visual reference for the discussion that follows.

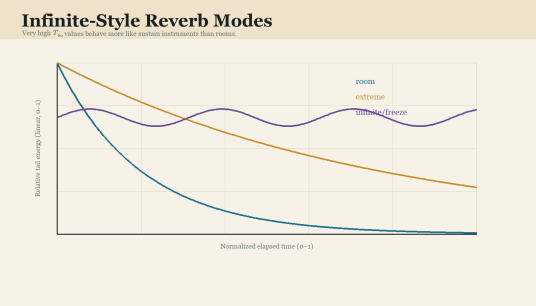


Figure 14-14: Infinite-style reverb tail behavior

The figure below introduces **Infinite-style reverb tail behavior**. Ordinary, extreme, and freeze-like tail energy compared over normalized time. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Normalized elapsed time (0–1)** and the vertical axis is **Relative tail energy (linear, 0–1)**.

14.5 Controls and Tradeoffs

The control plots are decision aids rather than performance benchmarks. They show where a parameter becomes sensitive, where two objectives conflict, and where additional adjustment produces little change.

Figure 14-15 presents Analysis window function shapes and establishes the visual reference for the discussion that follows.

Figure 14-15: Analysis window function shapes

The figure below introduces **Analysis window function shapes**. Hann, Blackman, Kaiser, and Tukey tapers plotted over a common normalized frame. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Normalized sample position (0–1)** and the vertical axis is **Window amplitude (linear, 0–1)**.

Figure 14-16 presents Limiter transfer curves and establishes the visual reference for the discussion that follows.

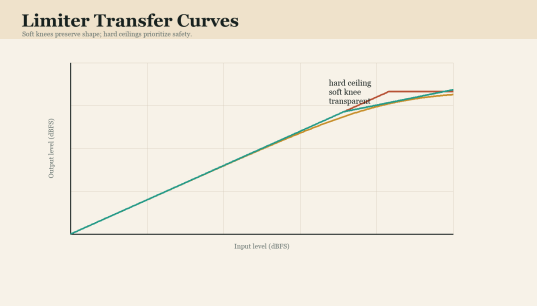


Figure 14-16: Limiter transfer curves

The figure below introduces **Limiter transfer curves**. Hard, soft-knee, and transparent limiting laws compared around the ceiling region. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Input level (dBFS)** and the vertical axis is **Output level (dBFS)**.

Figure 14-17 presents Reverb ducking envelope and establishes the visual reference for the discussion that follows.

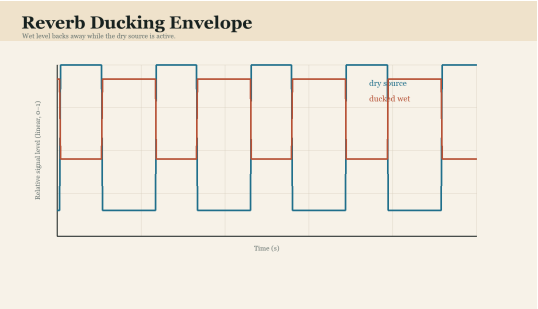


Figure 14-17: Reverb ducking envelope

The figure below introduces **Reverb ducking envelope**. Dry-source activity and the resulting attenuation of the wet return over an eight-second example. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Time (s)** and the vertical axis is **Relative signal level (linear, 0–1)**.

Figure 14-18 presents Dereverb strength versus artifact tradeoff and establishes the visual reference for the discussion that follows.

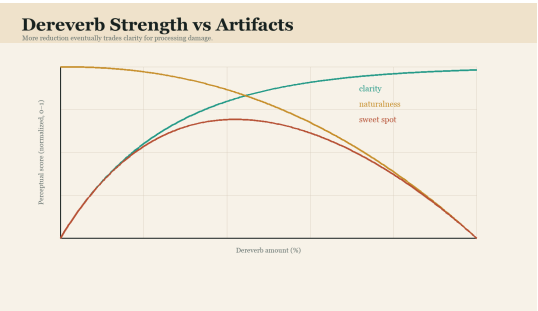
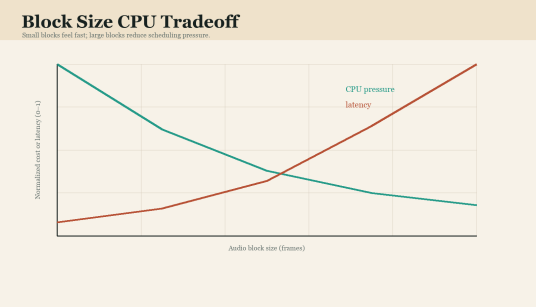


Figure 14-18: Dereverb strength versus artifact tradeoff

The figure below introduces **Dereverb strength versus artifact tradeoff**. The competing trends of clarity, naturalness, and aggregate usefulness as reduction strength increases. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Dereverb amount (%)** and the vertical axis is **Perceptual score (normalized, 0–1)**.

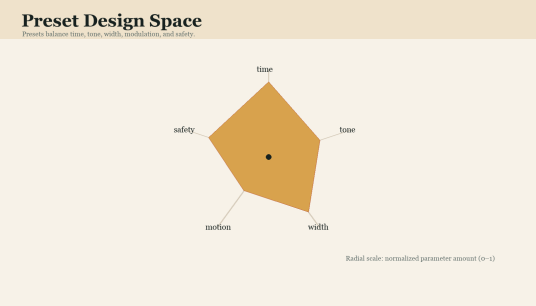
Figure 14-19 presents Block size CPU and latency tradeoff and establishes the visual reference for the discussion that follows.



The figure below introduces **Block size CPU and latency tradeoff**. Normalized scheduling pressure and block latency plotted against common audio buffer sizes. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Audio block size (frames)** and the vertical axis is **Normalized cost or latency (0–1)**.

Figure 14-19: Block size CPU and latency tradeoff

Figure 14-20 presents Preset design radar and establishes the visual reference for the discussion that follows.



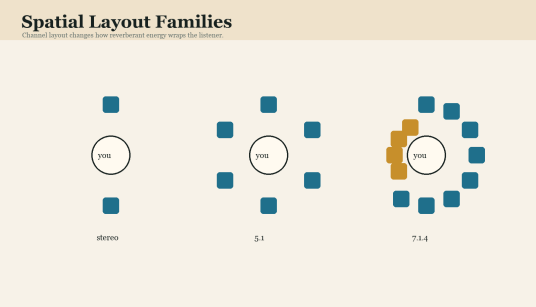
The figure below introduces **Preset design radar**. A five-axis profile for time, tone, width, motion, and safety. Each spoke is a named category and distance from the center is the normalized radial score. The color or radial scale reports **Normalized parameter amount (0–1)**.

Figure 14-20: Preset design radar

14.6 Spatial and Library Views

Spatial figures use plan, elevation, category, and grid views. Their labels identify channels and data organization; geometric distance is meaningful only when an axis or angle is printed.

Figure 14-21 presents Spatial layout families and establishes the visual reference for the discussion that follows.



The figure below introduces **Spatial layout families**. Listener-centered stereo, 5.1, and 7.1.4 speaker arrangements. Boxes, arrows, and spatial placement communicate topology and sequence; their dimensions are schematic and not measurements. It has no numeric axes because it is a structural diagram rather than a measurement plot.

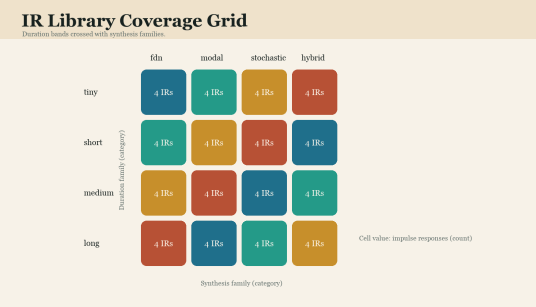
Figure 14-21: Spatial layout families

Figure 14-22 presents Ambisonics order channel growth and establishes the visual reference for the discussion that follows.

The figure below introduces **Ambisonics order channel growth**. The quadratic channel-count growth produced by the three-dimensional Ambisonics relation $(N+1)^2$. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Ambisonics order N (integer)** and the vertical axis is **Channel count (channels)**.

Figure 14-22: Ambisonics order channel growth

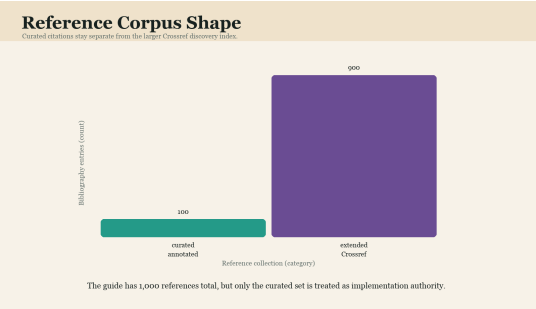
Figure 14-23 presents IR library coverage grid and establishes the visual reference for the discussion that follows.



The figure below introduces **IR library coverage grid**. Duration families crossed with synthesis methods, with four impulse responses represented per cell. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Synthesis family (category)** and the vertical axis is **Duration family (category)**. The color or radial scale reports **Impulse responses (count/cell)**.

Figure 14-23: IR library coverage grid

Figure 14-24 presents Reference corpus shape and establishes the visual reference for the discussion that follows.



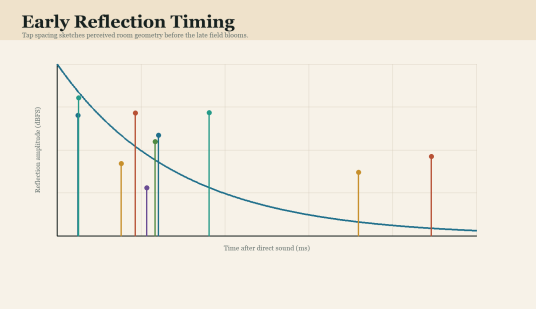
The figure below introduces **Reference corpus shape**. The relative sizes of the curated implementation bibliography and extended discovery index. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Reference collection (category)** and the vertical axis is **Bibliography entries (count)**.

Figure 14-24: Reference corpus shape

14.7 Additional Diagnostics and Design Graphs

These figures extend the core set with diagnostic responses and design comparisons. Each introduction names the independent quantity, the response, and the unit before the graphic appears.

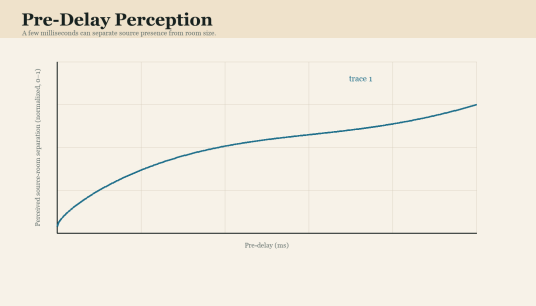
Figure 14-25 presents Early Reflection Timing and establishes the visual reference for the discussion that follows.



The figure below introduces **Early Reflection Timing**. Tap spacing sketches perceived room geometry before the late field blooms. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time after direct sound (ms)** and the vertical axis is **Reflection amplitude (dBFS)**.

Figure 14-25: Early Reflection Timing

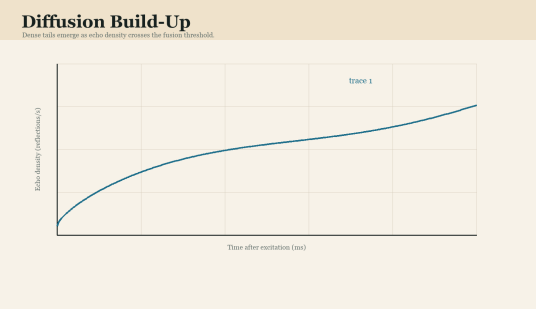
Figure 14-26 presents Pre-Delay Perception and establishes the visual reference for the discussion that follows.



The figure below introduces **Pre-Delay Perception**. A few milliseconds can separate source presence from room size. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Pre-delay (ms)** and the vertical axis is **Perceived source-room separation (normalized, 0–1)**.

Figure 14-26: Pre-Delay Perception

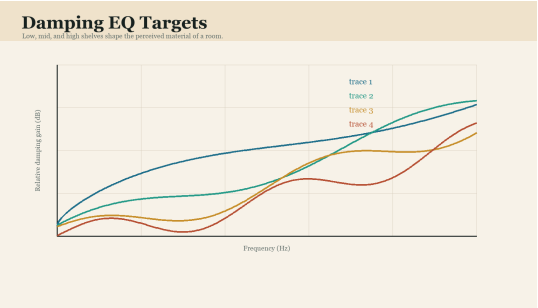
Figure 14-27 presents Diffusion Build-Up and establishes the visual reference for the discussion that follows.



The figure below introduces **Diffusion Build-Up**. Dense tails emerge as echo density crosses the fusion threshold. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Time after excitation (ms)** and the vertical axis is **Echo density (reflections/s)**.

Figure 14-27: Diffusion Build-Up

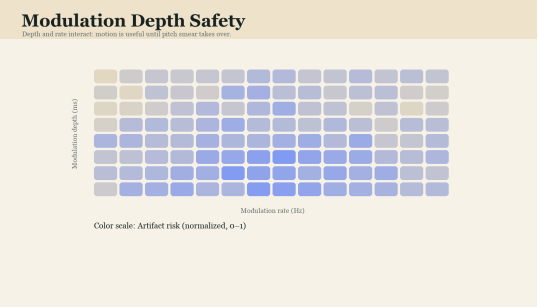
Figure 14-28 presents Damping EQ Targets and establishes the visual reference for the discussion that follows.



The figure below introduces **Damping EQ Targets**. Low, mid, and high shelves shape the perceived material of a room. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Relative damping gain (dB)**.

Figure 14-28: Damping EQ Targets

Figure 14-29 presents Modulation Depth Safety and establishes the visual reference for the discussion that follows.



The figure below introduces **Modulation Depth Safety**. Depth and rate interact: motion is useful until pitch smear takes over. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Modulation rate (Hz)** and the vertical axis is **Modulation depth (ms)**. The color or radial scale reports **Artifact risk (normalized, 0–1)**.

Figure 14-29: Modulation Depth Safety

Figure 14-30 presents Stereo Width Correlation and establishes the visual reference for the discussion that follows.

The figure below introduces **Stereo Width Correlation**. Width controls should preserve mono safety while expanding ambience. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Stereo width (%)** and the vertical axis is **Inter-channel correlation (unitless, –1 to +1)**.

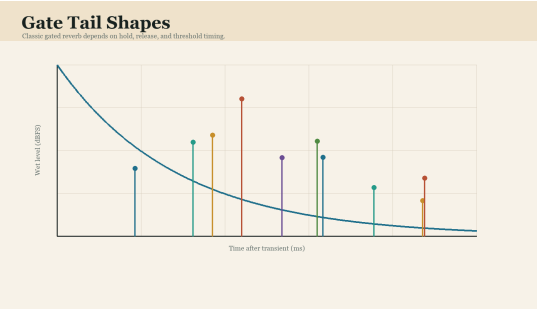
Figure 14-30: Stereo Width Correlation

Figure 14-31 presents Haas Zone and establishes the visual reference for the discussion that follows.

The figure below introduces **Haas Zone**. Tiny left/right delays can widen image before becoming discrete echoes. Colored regions divide the horizontal quantity into operational or perceptual regimes rather than implying a continuous measured response. The horizontal axis is **Inter-channel delay (ms)** and the vertical axis is **Perceptual region (category)**.

Figure 14-31: Haas Zone

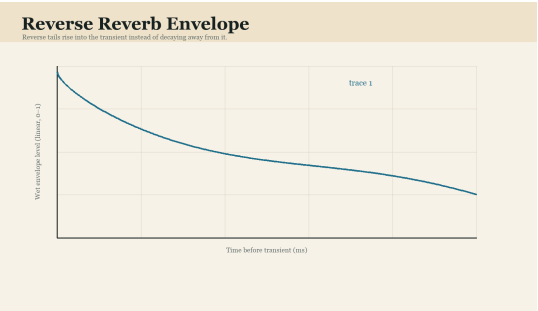
Figure 14-32 presents Gate Tail Shapes and establishes the visual reference for the discussion that follows.



The figure below introduces **Gate Tail Shapes**. Classic gated reverb depends on hold, release, and threshold timing. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time after transient (ms)** and the vertical axis is **Wet level (dBFS)**.

Figure 14-32: Gate Tail Shapes

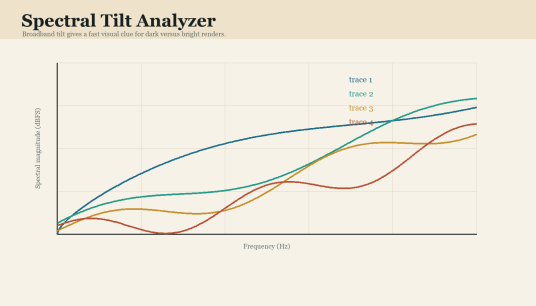
Figure 14-33 presents Reverse Reverb Envelope and establishes the visual reference for the discussion that follows.



The figure below introduces **Reverse Reverb Envelope**. Reverse tails rise into the transient instead of decaying away from it. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Time before transient (ms)** and the vertical axis is **Wet envelope level (linear, 0–1)**.

Figure 14-33: Reverse Reverb Envelope

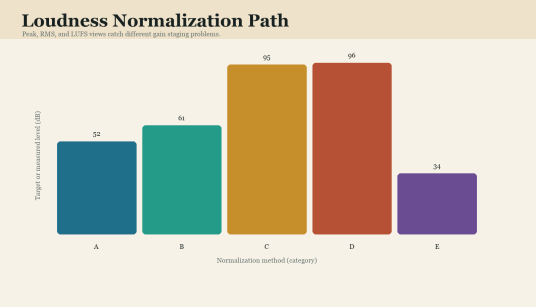
Figure 14-34 presents Spectral Tilt Analyzer and establishes the visual reference for the discussion that follows.



The figure below introduces **Spectral Tilt Analyzer**. Broadband tilt gives a fast visual clue for dark versus bright renders. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Spectral magnitude (dBFS)**.

Figure 14-34: Spectral Tilt Analyzer

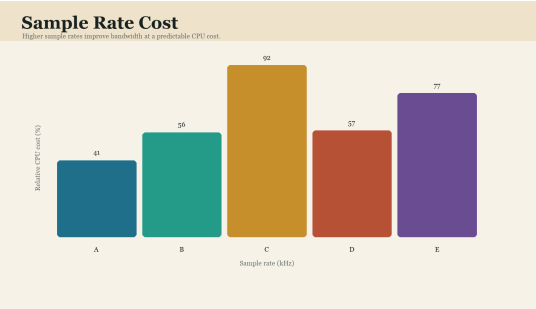
Figure 14-35 presents Loudness Normalization Path and establishes the visual reference for the discussion that follows.



The figure below introduces **Loudness Normalization Path**. Peak, RMS, and LUFS views catch different gain staging problems. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Normalization method (category)** and the vertical axis is **Target or measured level (dB)**.

Figure 14-35: Loudness Normalization Path

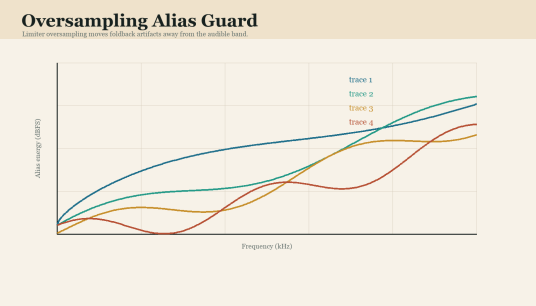
Figure 14-36 presents Sample Rate Cost and establishes the visual reference for the discussion that follows.



The figure below introduces **Sample Rate Cost**. Higher sample rates improve bandwidth at a predictable CPU cost. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Sample rate (kHz)** and the vertical axis is **Relative CPU cost (%)**.

Figure 14-36: Sample Rate Cost

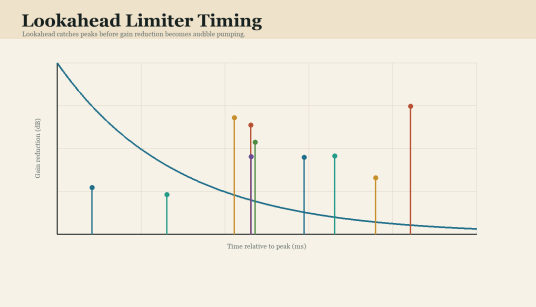
Figure 14-37 presents Oversampling Alias Guard and establishes the visual reference for the discussion that follows.



The figure below introduces **Oversampling Alias Guard**. Limiter oversampling moves foldback artifacts away from the audible band. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Frequency (kHz)** and the vertical axis is **Alias energy (dBFS)**.

Figure 14-37: Oversampling Alias Guard

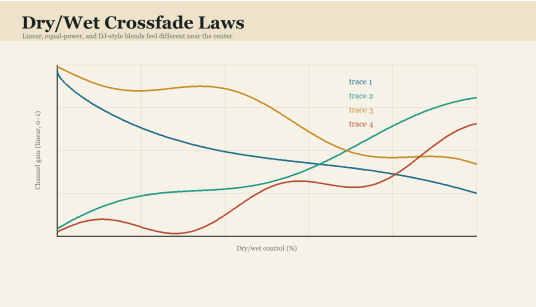
Figure 14-38 presents Lookahead Limiter Timing and establishes the visual reference for the discussion that follows.



The figure below introduces **Lookahead Limiter Timing**. Lookahead catches peaks before gain reduction becomes audible pumping. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time relative to peak (ms)** and the vertical axis is **Gain reduction (dB)**.

Figure 14-38: Lookahead Limiter Timing

Figure 14-39 presents Dry/Wet Crossfade Laws and establishes the visual reference for the discussion that follows.



The figure below introduces **Dry/Wet Crossfade Laws**. Linear, equal-power, and DJ-style blends feel different near the center. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Dry/wet control (%)** and the vertical axis is **Channel gain (linear, 0–1)**.

Figure 14-39: Dry/Wet Crossfade Laws

Figure 14-40 presents IR Trim Finder and establishes the visual reference for the discussion that follows.

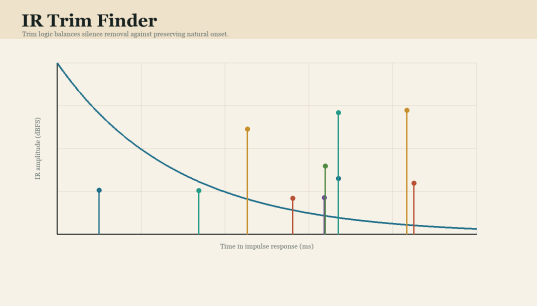


Figure 14-40: IR Trim Finder

The figure below introduces **IR Trim Finder**. Trim logic balances silence removal against preserving natural onset. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time in impulse response (ms)** and the vertical axis is **IR amplitude (dBFS)**.

Figure 14-41 presents Silence Detector Thresholds and establishes the visual reference for the discussion that follows.

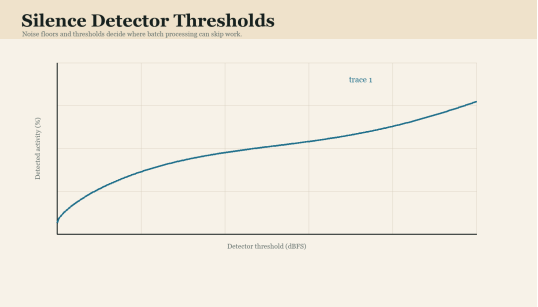


Figure 14-41: Silence Detector Thresholds

The figure below introduces **Silence Detector Thresholds**. Noise floors and thresholds decide where batch processing can skip work. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Detector threshold (dBFS)** and the vertical axis is **Detected activity (%)**.

Figure 14-42 presents Batch Render Throughput and establishes the visual reference for the discussion that follows.

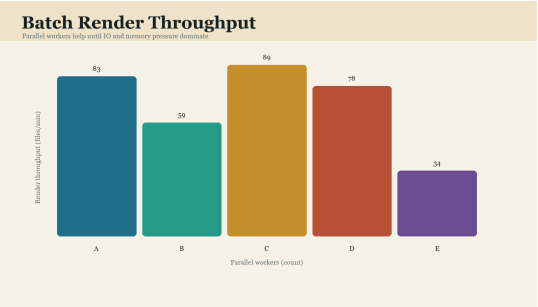


Figure 14-42: Batch Render Throughput

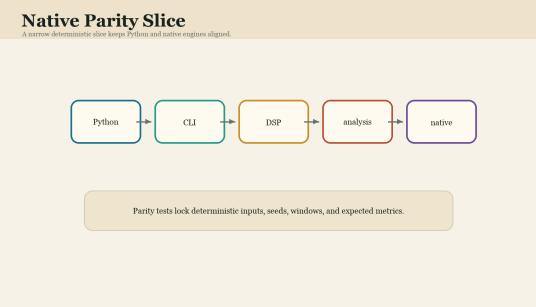
The figure below introduces **Batch Render Throughput**. Parallel workers help until IO and memory pressure dominate. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Parallel workers (count)** and the vertical axis is **Render throughput (files/min)**.

Figure 14-43 presents Cache Hit Savings and establishes the visual reference for the discussion that follows.

The figure below introduces **Cache Hit Savings**. Reusable analysis and IR material turn repeated renders into fast paths. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Cache condition (category)** and the vertical axis is **Elapsed processing time (%)**.

Figure 14-43: Cache Hit Savings

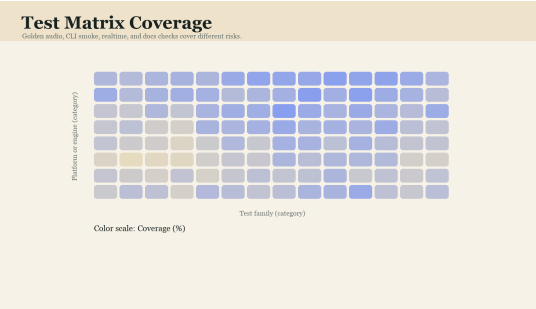
Figure 14-44 presents Native Parity Slice and establishes the visual reference for the discussion that follows.



The figure below introduces **Native Parity Slice**. A narrow deterministic slice keeps Python and native engines aligned. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-44: Native Parity Slice

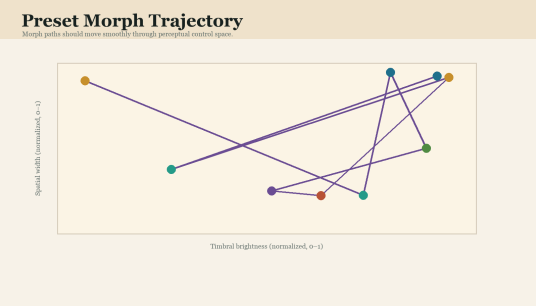
Figure 14-45 presents Test Matrix Coverage and establishes the visual reference for the discussion that follows.



The figure below introduces **Test Matrix Coverage**. Golden audio, CLI smoke, realtime, and docs checks cover different risks. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Test family (category)** and the vertical axis is **Platform or engine (category)**. The color or radial scale reports **Coverage (%)**.

Figure 14-45: Test Matrix Coverage

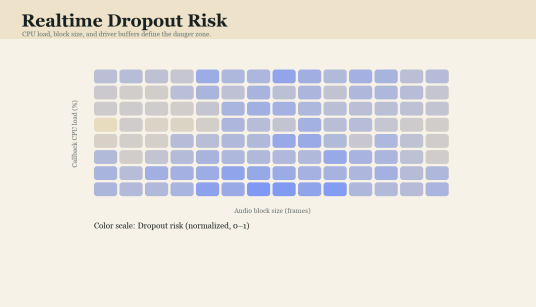
Figure 14-46 presents Preset Morph Trajectory and establishes the visual reference for the discussion that follows.



The figure below introduces **Preset Morph Trajectory**. Morph paths should move smoothly through perceptual control space. Points and paths occupy a two-dimensional design space; proximity indicates similar states, not physical distance. The horizontal axis is **Timbral brightness (normalized, 0–1)** and the vertical axis is **Spatial width (normalized, 0–1)**.

Figure 14-46: Preset Morph Trajectory

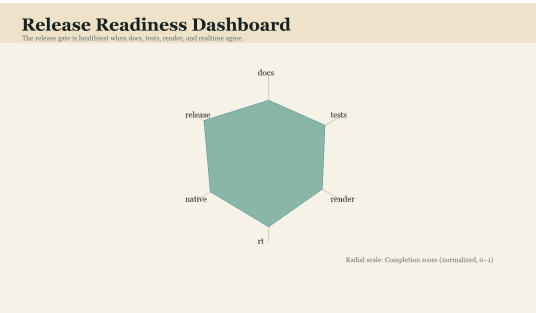
Figure 14-47 presents Realtime Dropout Risk and establishes the visual reference for the discussion that follows.



The figure below introduces **Realtime Dropout Risk**. CPU load, block size, and driver buffers define the danger zone. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Audio block size (frames)** and the vertical axis is **Callback CPU load (%)**. The color or radial scale reports **Dropout risk (normalized, 0–1)**.

Figure 14-47: Realtime Dropout Risk

Figure 14-48 presents Release Readiness Dashboard and establishes the visual reference for the discussion that follows.



The figure below introduces **Release Readiness Dashboard**. The release gate is healthiest when docs, tests, render, and realtime agree. Each spoke is a named category and distance from the center is the normalized radial score. The color or radial scale reports **Completion score (normalized, 0–1)**.

Figure 14-48: Release Readiness Dashboard

14.8 Extended Figure Atlas

The final atlas gathers specialized transfer functions, automation traces, spatial views, and engineering diagrams. Use it as a lookup section, then return to the surrounding chapter for assumptions and listening context.

Figure 14-49 presents Comb Filter Notches and establishes the visual reference for the discussion that follows.

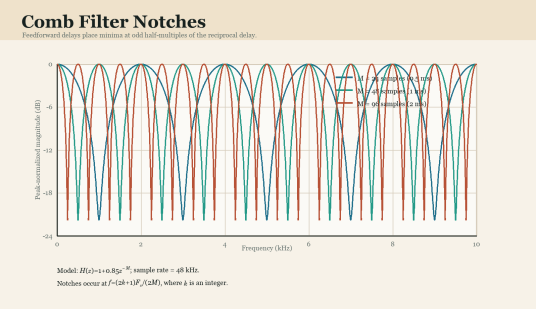


Figure 14-49: Comb Filter Notches

The figure below introduces **Comb Filter Notches** by comparing three feedforward comb filters at a 48 kHz sample rate. The horizontal axis is **Frequency (kHz)** and the vertical axis is **Magnitude response (dB)**, peak-normalized to 0 dB.
Read the figure as a direct evaluation of $H(z) = 1 + 0.85z^{-M}$. Its notches occur at $f = (2k + 1)F_s/(2M)$, where k is an integer, F_s is the sample rate, and M is the delay in samples. Increasing M decreases the spacing between notches. The finite delayed gain keeps each minimum finite rather than drawing an unbounded negative-decibel singularity.

Figure 14-50 presents Allpass Diffuser Group Delay and establishes the visual reference for the discussion that follows.

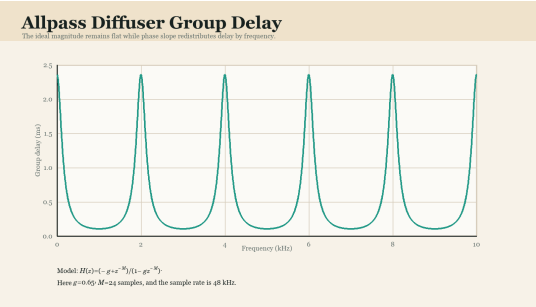


Figure 14-50: Allpass Diffuser Group Delay

The figure below introduces **Allpass Diffuser Group Delay** by plotting the exact group delay of a Schroeder allpass section. The horizontal axis is **Frequency (kHz)** and the vertical axis is **Group delay (ms)**.
Read the figure as an evaluation of $H(z) = (-g + z^{-M})/(1 - gz^{-M})$ with $g = 0.65$, $M = 24$ samples, and $F_s = 48$ kHz. Its ideal magnitude is unity at every frequency, while the phase slope produces the plotted delay variation. Here g is the feedback coefficient, M is the delay order, and F_s is the sample rate.

Figure 14-51 presents FDN Delay Distribution and establishes the visual reference for the discussion that follows.

Figure 14-51: FDN Delay Distribution

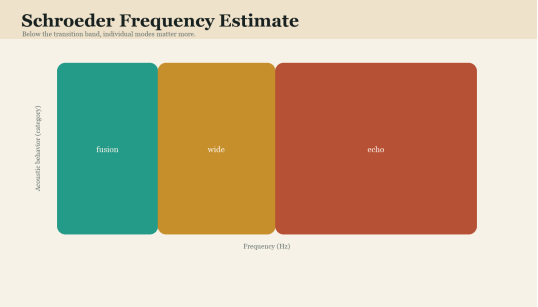
The figure below introduces **FDN Delay Distribution**. Prime-ish delay spacing avoids obvious repeating echo patterns. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **FDN delay line (index)** and the vertical axis is **Delay length (samples)**.

Figure 14-52 presents Modal Density Growth and establishes the visual reference for the discussion that follows.

The figure below introduces **Modal Density Growth**. Large rooms pack more resonances into each octave. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Modes per octave (count)**.

Figure 14-52: Modal Density Growth

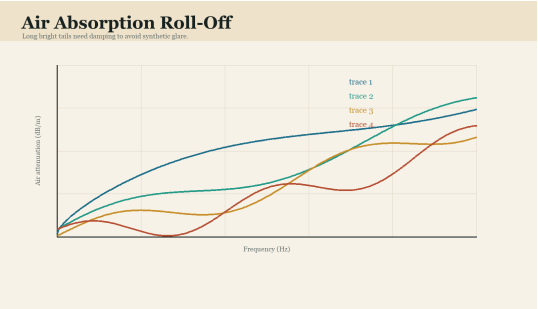
Figure 14-53 presents Schroeder Frequency Estimate and establishes the visual reference for the discussion that follows.



The figure below introduces **Schroeder Frequency Estimate**. Below the transition band, individual modes matter more. Colored regions divide the horizontal quantity into operational or perceptual regimes rather than implying a continuous measured response. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Acoustic behavior (category)**.

Figure 14-53: Schroeder Frequency Estimate

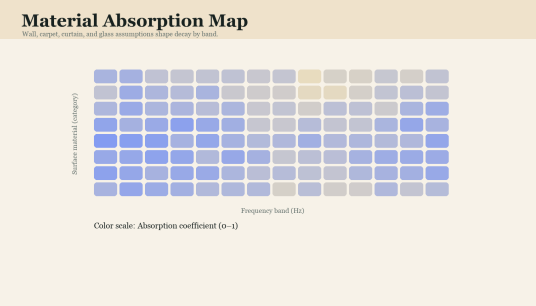
Figure 14-54 presents Air Absorption Roll-Off and establishes the visual reference for the discussion that follows.



The figure below introduces **Air Absorption Roll-Off**. Long bright tails need damping to avoid synthetic glare. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Air attenuation (dB/m)**.

Figure 14-54: Air Absorption Roll-Off

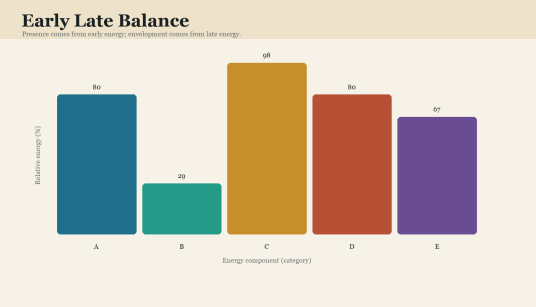
Figure 14-55 presents Material Absorption Map and establishes the visual reference for the discussion that follows.



The figure below introduces **Material Absorption Map**. Wall, carpet, curtain, and glass assumptions shape decay by band. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Frequency band (Hz)** and the vertical axis is **Surface material (category)**. The color or radial scale reports **Absorption coefficient (0–1)**.

Figure 14-55: Material Absorption Map

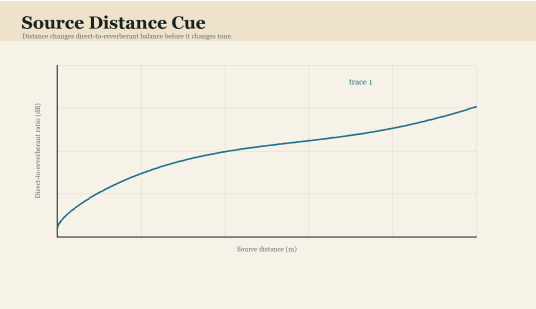
Figure 14-56 presents Early Late Balance and establishes the visual reference for the discussion that follows.



The figure below introduces **Early Late Balance**. Presence comes from early energy; envelopment comes from late energy. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Energy component (category)** and the vertical axis is **Relative energy (%)**.

Figure 14-56: Early Late Balance

Figure 14-57 presents Source Distance Cue and establishes the visual reference for the discussion that follows.



The figure below introduces **Source Distance Cue**. Distance changes direct-to-reverberant balance before it changes tone. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Source distance (m)** and the vertical axis is **Direct-to-reverberant ratio (dB)**.

Figure 14-57: Source Distance Cue

Figure 14-58 presents Mic Pattern Pickup and establishes the visual reference for the discussion that follows.

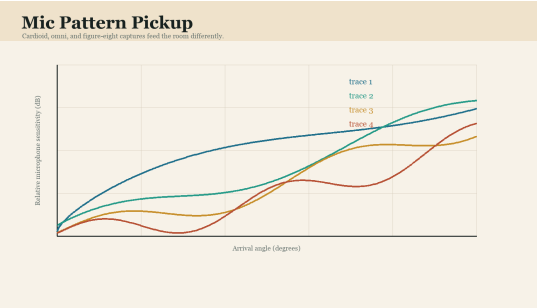


Figure 14-58: Mic Pattern Pickup

The figure below introduces **Mic Pattern Pickup**. Cardioid, omni, and figure-eight captures feed the room differently. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Arrival angle (degrees)** and the vertical axis is **Relative microphone sensitivity (dB)**.

Figure 14-59 presents Sidechain Detector Modes and establishes the visual reference for the discussion that follows.

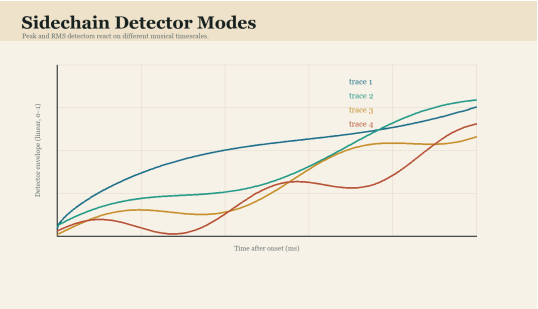


Figure 14-59: Sidechain Detector Modes

The figure below introduces **Sidechain Detector Modes**. Peak and RMS detectors react on different musical timescales. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Time after onset (ms)** and the vertical axis is **Detector envelope (linear, 0–1)**.

Figure 14-60 presents Ducking Release Families and establishes the visual reference for the discussion that follows.

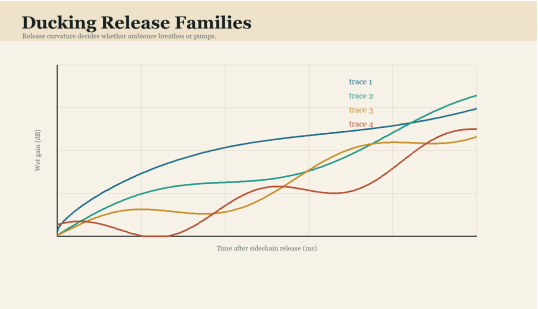
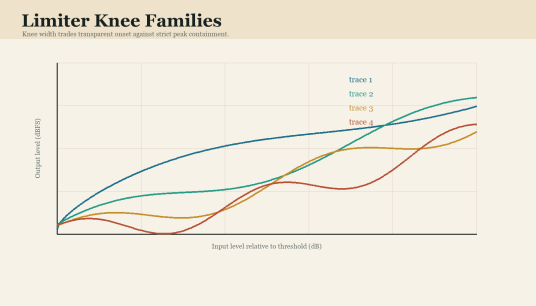


Figure 14-60: Ducking Release Families

The figure below introduces **Ducking Release Families**. Release curvature decides whether ambience breathes or pumps. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Time after sidechain release (ms)** and the vertical axis is **Wet gain (dB)**.

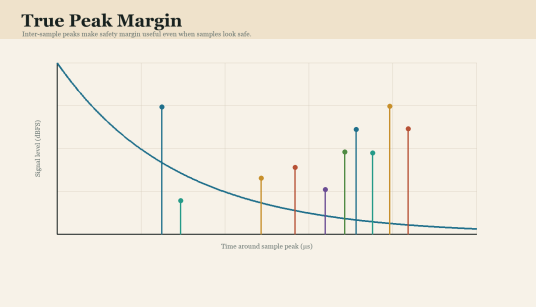
Figure 14-61 presents Limiter Knee Families and establishes the visual reference for the discussion that follows.



The figure below introduces **Limiter Knee Families**. Knee width trades transparent onset against strict peak containment. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Input level relative to threshold (dB)** and the vertical axis is **Output level (dBFS)**.

Figure 14-61: Limiter Knee Families

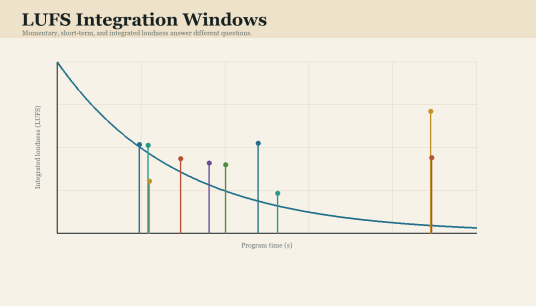
Figure 14-62 presents True Peak Margin and establishes the visual reference for the discussion that follows.



The figure below introduces **True Peak Margin**. Inter-sample peaks make safety margin useful even when samples look safe. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time around sample peak (µs)** and the vertical axis is **Signal level (dBFS)**.

Figure 14-62: True Peak Margin

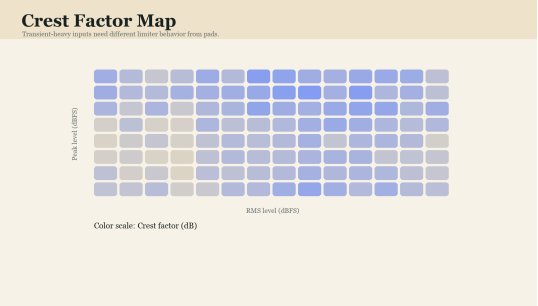
Figure 14-63 presents LUFS Integration Windows and establishes the visual reference for the discussion that follows.



The figure below introduces **LUFS Integration Windows**. Momentary, short-term, and integrated loudness answer different questions. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Program time (s)** and the vertical axis is **Integrated loudness (LUFS)**.

Figure 14-63: LUFS Integration Windows

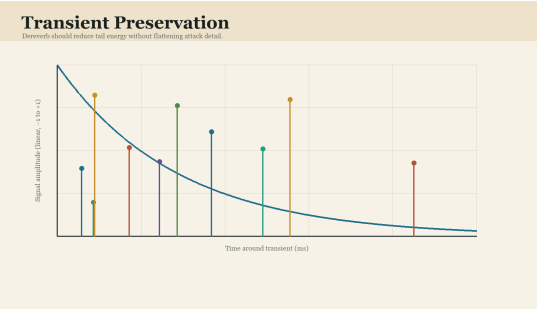
Figure 14-64 presents Crest Factor Map and establishes the visual reference for the discussion that follows.



The figure below introduces **Crest Factor Map**. Transient-heavy inputs need different limiter behavior from pads. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **RMS level (dBFS)** and the vertical axis is **Peak level (dBFS)**. The color or radial scale reports **Crest factor (dB)**.

Figure 14-64: Crest Factor Map

Figure 14-65 presents Transient Preservation and establishes the visual reference for the discussion that follows.



The figure below introduces **Transient Preservation**. Dereverb should reduce tail energy without flattening attack detail. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time around transient (ms)** and the vertical axis is **Signal amplitude (linear, –1 to +1)**.

Figure 14-65: Transient Preservation

Figure 14-66 presents Dereverb Mask Strength and establishes the visual reference for the discussion that follows.

The figure below introduces **Dereverb Mask Strength**. Mask aggressiveness governs the speech-cleanup versus artifact tradeoff. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Dereverb mask strength (%)** and the vertical axis is **Artifact or suppression score (normalized, 0–1)**.

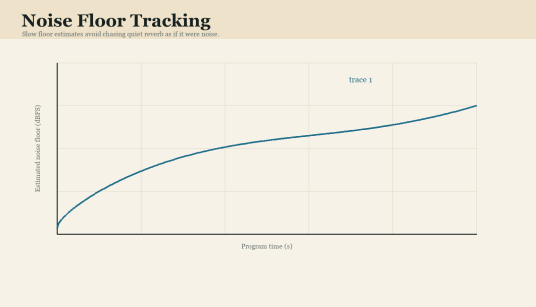
Figure 14-66: Dereverb Mask Strength

Figure 14-67 presents Spectral Gate Residuals and establishes the visual reference for the discussion that follows.

The figure below introduces **Spectral Gate Residuals**. Residual maps reveal where dereverb is leaving flutter or musical noise. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Time (s)** and the vertical axis is **Frequency (Hz)**. The color or radial scale reports **Residual magnitude (dBFS)**.

Figure 14-67: Spectral Gate Residuals

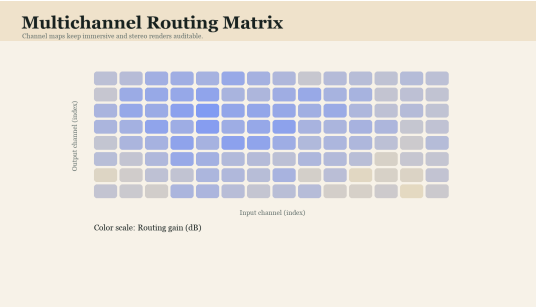
Figure 14-68 presents Noise Floor Tracking and establishes the visual reference for the discussion that follows.



The figure below introduces **Noise Floor Tracking**. Slow floor estimates avoid chasing quiet reverb as if it were noise. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Program time (s)** and the vertical axis is **Estimated noise floor (dBFS)**.

Figure 14-68: Noise Floor Tracking

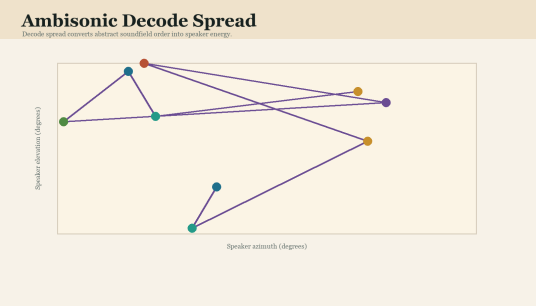
Figure 14-69 presents Multichannel Routing Matrix and establishes the visual reference for the discussion that follows.



The figure below introduces **Multichannel Routing Matrix**. Channel maps keep immersive and stereo renders auditable. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Input channel (index)** and the vertical axis is **Output channel (index)**. The color or radial scale reports **Routing gain (dB)**.

Figure 14-69: Multichannel Routing Matrix

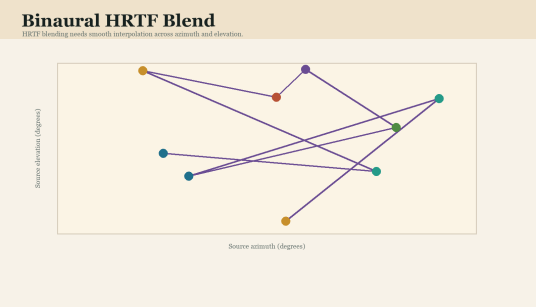
Figure 14-70 presents Ambisonic Decode Spread and establishes the visual reference for the discussion that follows.



The figure below introduces **Ambisonic Decode Spread**. Decode spread converts abstract soundfield order into speaker energy. Points and paths occupy a two-dimensional design space; proximity indicates similar states, not physical distance. The horizontal axis is **Speaker azimuth (degrees)** and the vertical axis is **Speaker elevation (degrees)**. The color or radial scale reports **Relative decode energy (0–1)**.

Figure 14-70: Ambisonic Decode Spread

Figure 14-71 presents Binaural HRTF Blend and establishes the visual reference for the discussion that follows.



The figure below introduces **Binaural HRTF Blend**. HRTF blending needs smooth interpolation across azimuth and elevation. Points and paths occupy a two-dimensional design space; proximity indicates similar states, not physical distance. The horizontal axis is **Source azimuth (degrees)** and the vertical axis is **Source elevation (degrees)**. The color or radial scale reports **HRTF blend weight (0–1)**.

Figure 14-71: Binaural HRTF Blend

The figure below introduces **Loudspeaker Layouts: Plan and Elevation**. Nominal channel bearings for stereo, 5.1, and 7.1.4, with the immersive height layer shown separately. Listener-centered plan views encode nominal azimuth, while the separate side elevation distinguishes bed and height layers. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-72 presents Loudspeaker Layouts: Plan and Elevation and establishes the visual reference for the discussion that follows.

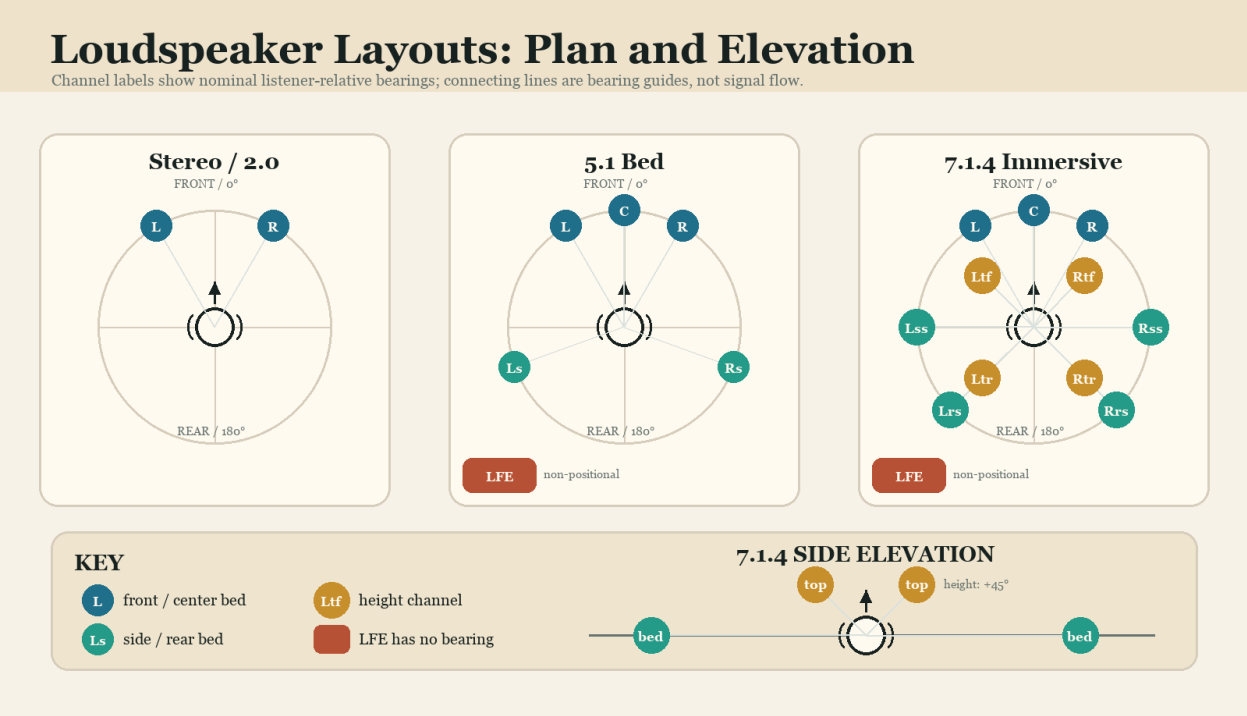


Figure 14-72: Loudspeaker Layouts: Plan and Elevation

Read each plan with front at the top and the listener at the center. Blue marks identify front and center bed channels, teal marks identify side and rear bed channels, and gold marks identify overhead channels. Radial lines indicate nominal bearing only; they are not cables or signal-flow paths. The elevation inset shows why the four height channels cannot be understood from azimuth alone, while the separate LFE key emphasizes that the subwoofer channel has no prescribed bearing. These angles are explanatory nominal placements; use the applicable monitoring standard and room-calibration procedure for installation.

Figure 14-73 presents IR Capture Checklist and establishes the visual reference for the discussion that follows.

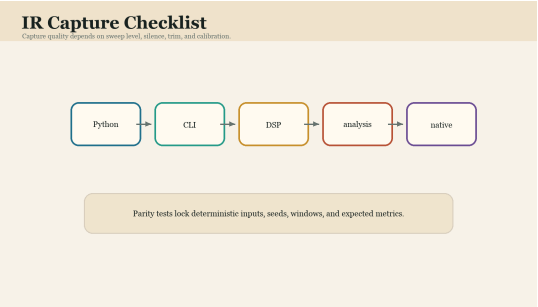
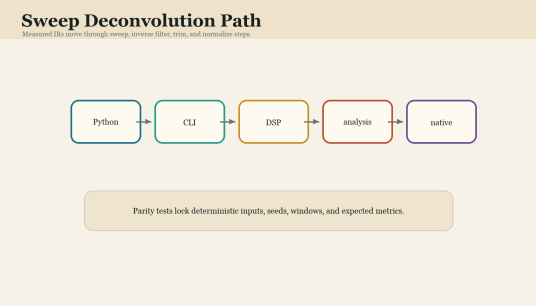


Figure 14-73: IR Capture Checklist

The figure below introduces **IR Capture Checklist**. Capture quality depends on sweep level, silence, trim, and calibration. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

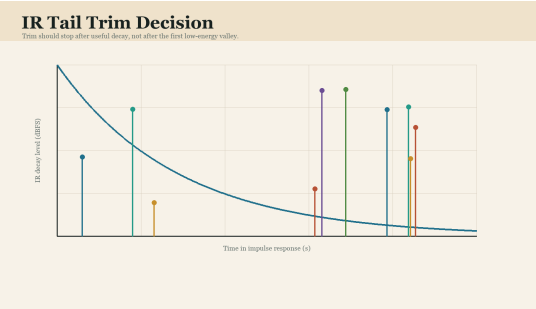
Figure 14-74 presents Sweep Deconvolution Path and establishes the visual reference for the discussion that follows.



The figure below introduces **Sweep Deconvolution Path**. Measured IRs move through sweep, inverse filter, trim, and normalize steps. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-74: Sweep Deconvolution Path

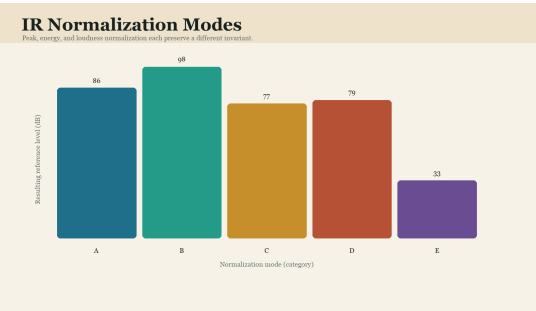
Figure 14-75 presents IR Tail Trim Decision and establishes the visual reference for the discussion that follows.



The figure below introduces **IR Tail Trim Decision**. Trim should stop after useful decay, not after the first low-energy valley. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Time in impulse response (s)** and the vertical axis is **IR decay level (dBFS)**.

Figure 14-75: IR Tail Trim Decision

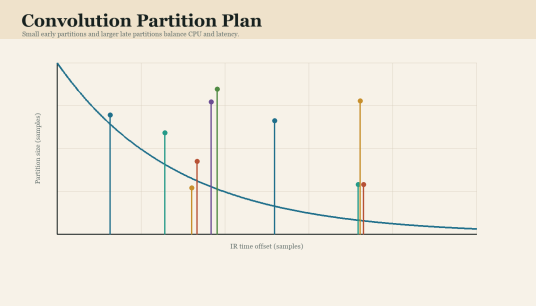
Figure 14-76 presents IR Normalization Modes and establishes the visual reference for the discussion that follows.



The figure below introduces **IR Normalization Modes**. Peak, energy, and loudness normalization each preserve a different invariant. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Normalization mode (category)** and the vertical axis is **Resulting reference level (dB)**.

Figure 14-76: IR Normalization Modes

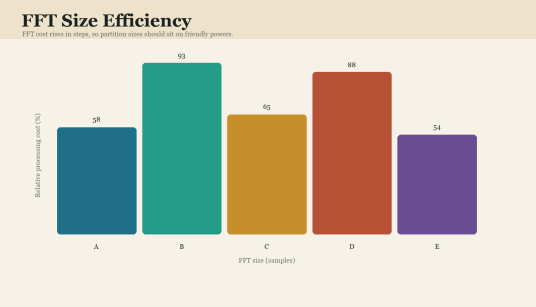
Figure 14-77 presents Convolution Partition Plan and establishes the visual reference for the discussion that follows.



The figure below introduces **Convolution Partition Plan**. Small early partitions and larger late partitions balance CPU and latency. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **IR time offset (samples)** and the vertical axis is **Partition size (samples)**.

Figure 14-77: Convolution Partition Plan

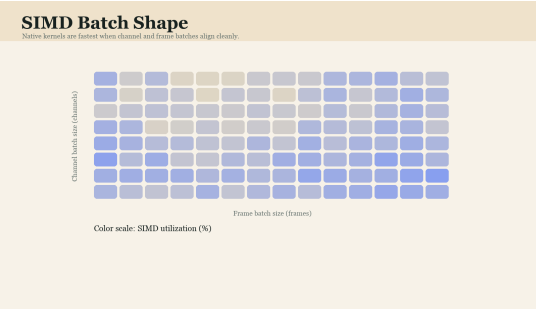
Figure 14-78 presents FFT Size Efficiency and establishes the visual reference for the discussion that follows.



The figure below introduces **FFT Size Efficiency**. FFT cost rises in steps, so partition sizes should sit on friendly powers. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **FFT size (samples)** and the vertical axis is **Relative processing cost (%)**.

Figure 14-78: FFT Size Efficiency

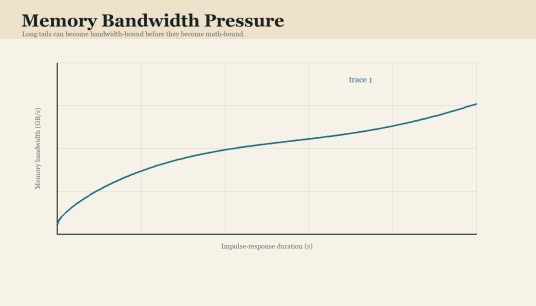
Figure 14-79 presents SIMD Batch Shape and establishes the visual reference for the discussion that follows.



The figure below introduces **SIMD Batch Shape**. Native kernels are fastest when channel and frame batches align cleanly. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Frame batch size (frames)** and the vertical axis is **Channel batch size (channels)**. The color or radial scale reports **SIMD utilization (%)**.

Figure 14-79: SIMD Batch Shape

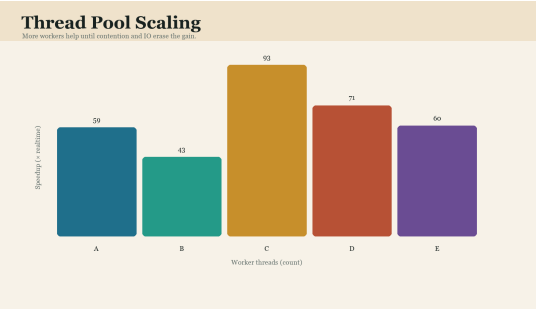
Figure 14-80 presents Memory Bandwidth Pressure and establishes the visual reference for the discussion that follows.



The figure below introduces **Memory Bandwidth Pressure**. Long tails can become bandwidth-bound before they become math-bound. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Impulse-response duration (s)** and the vertical axis is **Memory bandwidth (GB/s)**.

Figure 14-80: Memory Bandwidth Pressure

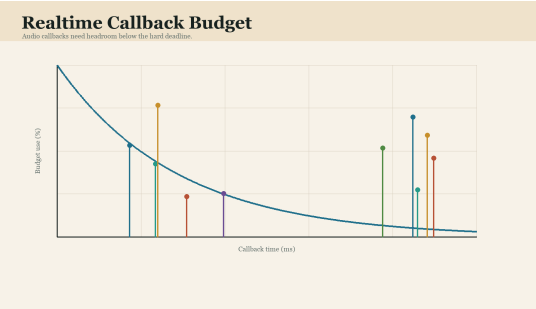
Figure 14-81 presents Thread Pool Scaling and establishes the visual reference for the discussion that follows.



The figure below introduces **Thread Pool Scaling**. More workers help until contention and IO erase the gain. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Worker threads (count)** and the vertical axis is **Speedup (× realtime)**.

Figure 14-81: Thread Pool Scaling

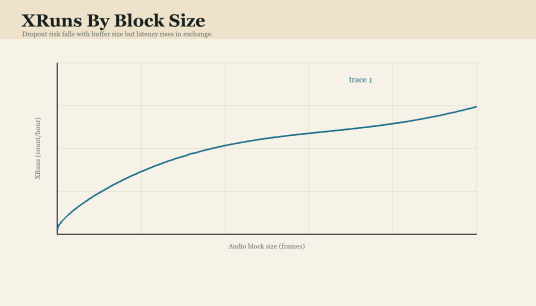
Figure 14-82 presents Realtime Callback Budget and establishes the visual reference for the discussion that follows.



The figure below introduces **Realtime Callback Budget**. Audio callbacks need headroom below the hard deadline. Vertical events and the envelope are positioned against a common time base, making onset, hold, and decay relationships visible. The horizontal axis is **Callback time (ms)** and the vertical axis is **Budget use (%)**.

Figure 14-82: Realtime Callback Budget

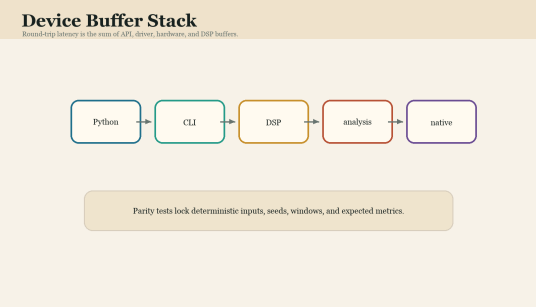
Figure 14-83 presents XRuns By Block Size and establishes the visual reference for the discussion that follows.



The figure below introduces **XRuns By Block Size**. Dropout risk falls with buffer size but latency rises in exchange. The trace shows how the vertical response changes as the horizontal control or measurement advances. The horizontal axis is **Audio block size (frames)** and the vertical axis is **XRuns (count/hour)**.

Figure 14-83: XRuns By Block Size

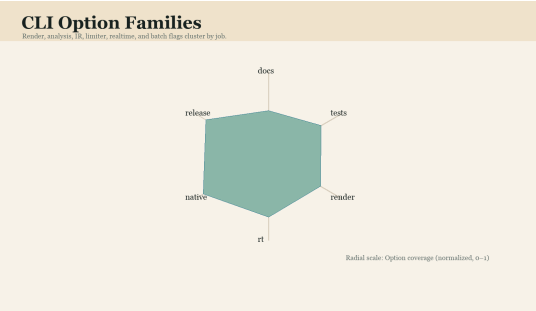
Figure 14-84 presents Device Buffer Stack and establishes the visual reference for the discussion that follows.



The figure below introduces **Device Buffer Stack**. Round-trip latency is the sum of API, driver, hardware, and DSP buffers. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-84: Device Buffer Stack

Figure 14-85 presents CLI Option Families and establishes the visual reference for the discussion that follows.



The figure below introduces **CLI Option Families**. Render, analysis, IR, limiter, realtime, and batch flags cluster by job. Each spoke is a named category and distance from the center is the normalized radial score. The color or radial scale reports **Option coverage (normalized, 0–1)**.

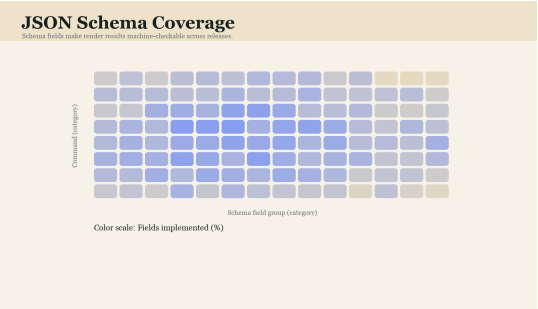
Figure 14-85: CLI Option Families

Figure 14-86 presents Preset Taxonomy and establishes the visual reference for the discussion that follows.

The figure below introduces **Preset Taxonomy**. Presets should be searchable by space, tone, motion, and safety intent. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Preset attribute (category)** and the vertical axis is **Preset family (category)**. The color or radial scale reports **Library density (presets/cell)**.

Figure 14-86: Preset Taxonomy

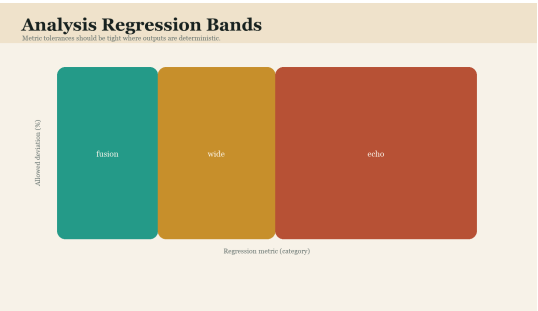
Figure 14-87 presents JSON Schema Coverage and establishes the visual reference for the discussion that follows.



The figure below introduces **JSON Schema Coverage**. Schema fields make render results machine-checkable across releases. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Schema field group (category)** and the vertical axis is **Command (category)**. The color or radial scale reports **Fields implemented (%)**.

Figure 14-87: JSON Schema Coverage

Figure 14-88 presents Analysis Regression Bands and establishes the visual reference for the discussion that follows.



The figure below introduces **Analysis Regression Bands**. Metric tolerances should be tight where outputs are deterministic. Colored regions divide the horizontal quantity into operational or perceptual regimes rather than implying a continuous measured response. The horizontal axis is **Regression metric (category)** and the vertical axis is **Allowed deviation (%)**.

Figure 14-88: Analysis Regression Bands

Figure 14-89 presents Golden Audio Drift and establishes the visual reference for the discussion that follows.

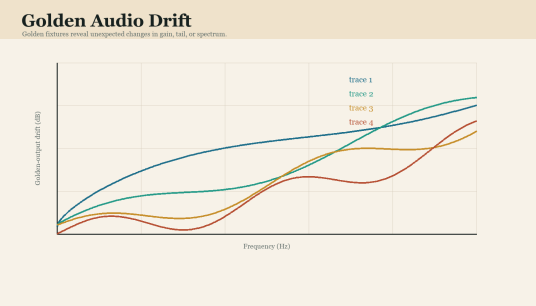


Figure 14-89: Golden Audio Drift

The figure below introduces **Golden Audio Drift**. Golden fixtures reveal unexpected changes in gain, tail, or spectrum. The colored traces share one coordinate system so their slopes, crossings, and endpoints can be compared directly. The horizontal axis is **Frequency (Hz)** and the vertical axis is **Golden-output drift (dB)**.

Figure 14-90 presents Documentation Build Pipeline and establishes the visual reference for the discussion that follows.

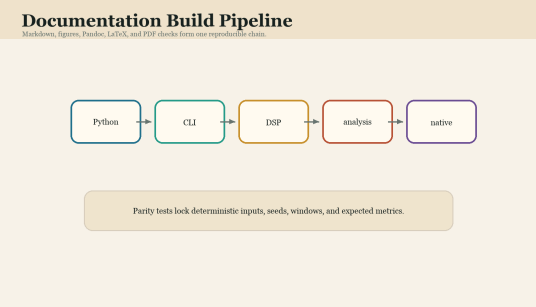


Figure 14-90: Documentation Build Pipeline

The figure below introduces **Documentation Build Pipeline**. Markdown, figures, Pandoc, LaTeX, and PDF checks form one reproducible chain. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-91 presents Table Wrap Stress Test and establishes the visual reference for the discussion that follows.

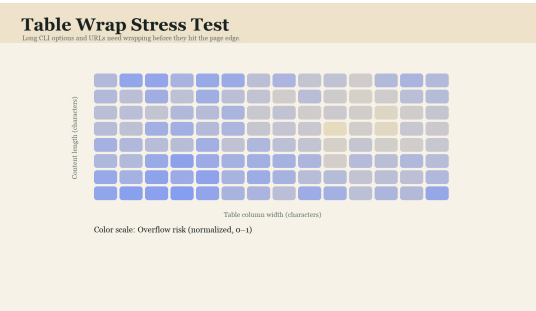
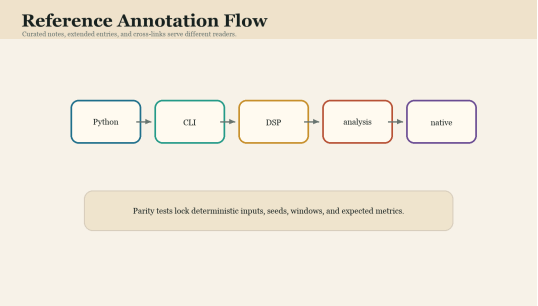


Figure 14-91: Table Wrap Stress Test

The figure below introduces **Table Wrap Stress Test**. Long CLI options and URLs need wrapping before they hit the page edge. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Table column width (characters)** and the vertical axis is **Content length (characters)**. The color or radial scale reports **Overflow risk (normalized, 0–1)**.

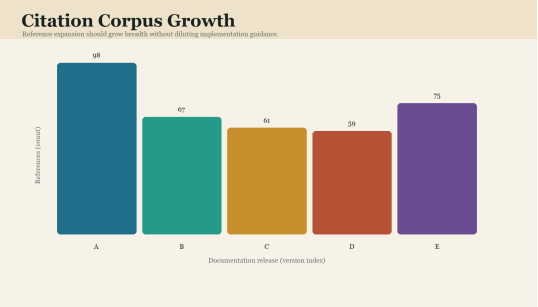
Figure 14-92 presents Reference Annotation Flow and establishes the visual reference for the discussion that follows.



The figure below introduces **Reference Annotation Flow**. Curated notes, extended entries, and cross-links serve different readers. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-92: Reference Annotation Flow

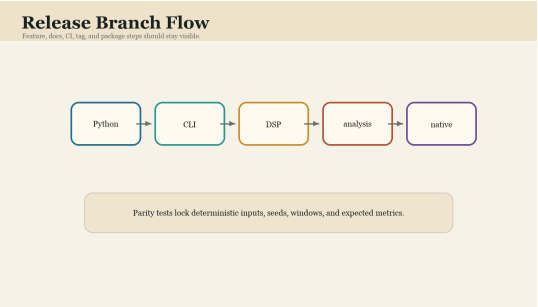
Figure 14-93 presents Citation Corpus Growth and establishes the visual reference for the discussion that follows.



The figure below introduces **Citation Corpus Growth**. Reference expansion should grow breadth without diluting implementation guidance. Bar height encodes the quantity on the vertical axis, while each horizontal category identifies a tested or illustrative condition. The horizontal axis is **Documentation release (version index)** and the vertical axis is **References (count)**.

Figure 14-93: Citation Corpus Growth

Figure 14-94 presents Release Branch Flow and establishes the visual reference for the discussion that follows.



The figure below introduces **Release Branch Flow**. Feature, docs, CI, tag, and package steps should stay visible. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-94: Release Branch Flow

Figure 14-95 presents Homebrew Formula Refresh and establishes the visual reference for the discussion that follows.

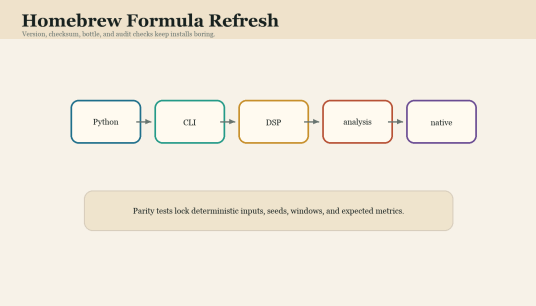


Figure 14-95: Homebrew Formula Refresh

The figure below introduces **Homebrew Formula Refresh**. Version, checksum, bottle, and audit checks keep installs boring. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-96 presents Platform Support Grid and establishes the visual reference for the discussion that follows.

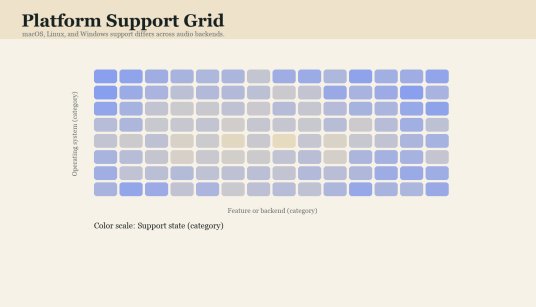


Figure 14-96: Platform Support Grid

The figure below introduces **Platform Support Grid**. macOS, Linux, and Windows support differs across audio backends. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Feature or backend (category)** and the vertical axis is **Operating system (category)**. The color or radial scale reports **Support state (category)**.

Figure 14-97 presents Error Message Quality and establishes the visual reference for the discussion that follows.

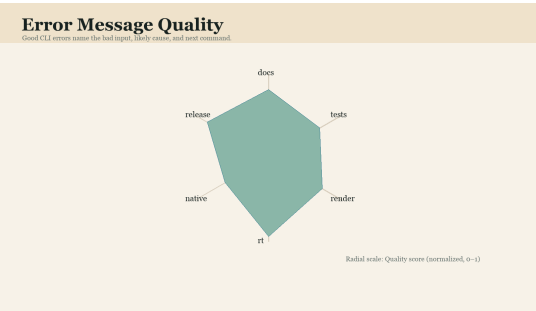


Figure 14-97: Error Message Quality

The figure below introduces **Error Message Quality**. Good CLI errors name the bad input, likely cause, and next command. Each spoke is a named category and distance from the center is the normalized radial score. The color or radial scale reports **Quality score (normalized, 0–1)**.

Figure 14-98 presents User Workflow Map and establishes the visual reference for the discussion that follows.

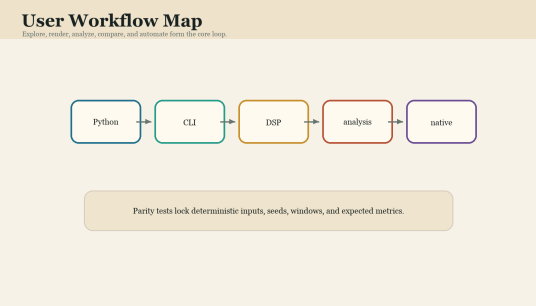


Figure 14-98: User Workflow Map

The figure below introduces **User Workflow Map**. Explore, render, analyze, compare, and automate form the core loop. Boxes and arrows show order and dependency. Their position and size are schematic and carry no numeric scale. It has no numeric axes because it is a structural diagram rather than a measurement plot.

Figure 14-99 presents Feature Maturity Radar and establishes the visual reference for the discussion that follows.

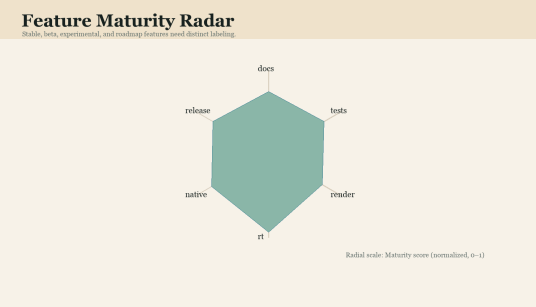


Figure 14-99: Feature Maturity Radar

The figure below introduces **Feature Maturity Radar**. Stable, beta, experimental, and roadmap features need distinct labeling. Each spoke is a named category and distance from the center is the normalized radial score. The color or radial scale reports **Maturity score (normalized, 0–1)**.

Figure 14-100 presents End-to-End Confidence Map and establishes the visual reference for the discussion that follows.

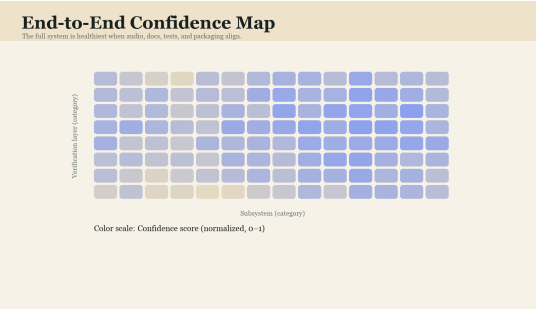


Figure 14-100: End-to-End Confidence Map

The figure below introduces **End-to-End Confidence Map**. The full system is healthiest when audio, docs, tests, and packaging align. Each cell combines one horizontal and one vertical condition; color encodes the third quantity named in the scale label. The horizontal axis is **Subsystem (category)** and the vertical axis is **Verification layer (category)**. The color or radial scale reports **Confidence score (normalized, 0–1)**.

14.9 Suggested Exercises

1. Choose five figures from different chapters and write one prediction before reading each caption, then compare prediction with the diagram's actual claim.
2. Redraw one signal-flow diagram with different parameter values while preserving its causal structure.
3. Use one sonogram pair to identify a time-scale and frequency-scale invariant, then explain why equal axes matter for the comparison.
4. Pair one pole-zero plot with its magnitude response and identify a feature that cannot be inferred safely from one plot alone.
5. Make one original labeled figure from a verbx render, including units, a caption, source data, and a one-paragraph interpretation.
6. Compare two figures that use different visual encodings and identify which question each encoding answers most directly.
7. Audit one graph for axis units, scale choice, and caption completeness; revise its caption to stand alone outside the chapter.
8. Create a small figure sequence that shows input, processing state, and output rather than a single left-to-right block chain.
9. Use one figure as evidence in a short technical argument, clearly separating what it shows from what you infer.
10. Design an accessible text description for a diagram so a reader can understand its causal structure without seeing it.

"Multitudinous echoes awoke and died in the distance."

— Henry Wadsworth Longfellow, *Evangeline* (1847)

Important Musical Pieces

This appendix is a practical listening canon for engineers, musicians, researchers, and plug-in designers working with reverberation. It is not a ranking and it is not a general history of Western music. Every selection uses architecture, resonance, echo, spatial placement, room tone, feedback, or artificial ambience as audible musical material: the reverberation changes rhythm, form, harmony, orchestration, drama, or the listener's sense of place.

Dates identify the composition, completion, premiere, or commercially released recording most relevant to the discussion; sources sometimes differ when a work evolved across several versions. Listen to more than one performance where possible. A dry studio transfer, a reverberant concert recording, a remaster, and an immersive remix may present substantially different evidence.

A.1 How to Use This Appendix

For each work, listen once without analysis. On the second pass, mark the direct sound, first audible reflection, apparent room size, decay color, stereo or multichannel motion, and moments when ambience becomes an instrument rather than a background. On a third pass, build a short verbx study using your own source material. The aim is not to counterfeit a copyrighted recording; it is to isolate and understand an acoustic or production principle.

When a named recording is central to the entry, use that recording or a documented reissue. When the composition itself is central, compare performances made in contrasting rooms. Level-match comparisons, protect your monitoring, and remember that streaming normalization can conceal large differences in crest factor and perceived depth.

A.2 Architecture as an Instrument

Thomas Tallis, *Spem in alium* (c. 1570). Forty independent voices arranged in eight choirs make location part of the counterpoint. Listen for waves of activity passing between groups and for the moment when the full ensemble turns many local sources into one enveloping field. In verbx, route eight stem groups to slightly different early-reflection, pre-delay, and width settings while keeping one coherent late decay.

Giovanni Gabrieli, *In ecclesiis* (c. 1615). Gabrieli's separated vocal and instrumental groups exploit the delayed answers and blended sustain of a large Venetian church. Listen for the alternation between intelligible local gestures and the collective halo that joins them. Recreate the principle with several buses: distinct pre-delays and positions for each ensemble, shared low-frequency decay, and conservative high-frequency damping.

Hector Berlioz, *Grande Messe des morts* (1837). The work's multiple brass groups and very large forces turn distance and direction into orchestration. Listen to how remote attacks retain identity while their decays merge with the central ensemble. A useful verbx study places four brass-like sources around a listener, varies direct-to-reverberant ratio rather than level alone, and checks whether the combined low end remains controlled.

Charles Ives, *The Unanswered Question* (1906; revised later). Ives separates sustained strings, a questioning trumpet, and increasingly agitated winds into different physical and conceptual layers.

Listen for how distance stabilizes the strings while the foreground groups remain sharply profiled. Model the three ensembles with different room sends and early-reflection densities instead of putting one global reverb across the mix.

Ralph Vaughan Williams, *Fantasia on a Theme by Thomas Tallis* (1910). A large string orchestra, a smaller antiphonal group, and a quartet create depth through changing ensemble scale. Listen for transitions where apparent distance changes without a conventional fade. In verbx, compare level automation with DRR automation: move a string stem backward by reducing direct energy and lengthening its early-reflection pattern while leaving musical dynamics intact.

Olivier Messiaen, *Et exspecto resurrectionem mortuorum* (1964). Monumental wind and percussion sonorities are written to activate a large resonant space, with silence functioning as part of the decay. Listen through the rests rather than treating them as empty time. Build a study with sparse gong or tam-tam attacks, long multiband decay, and enough headroom that each tail can be heard to disappear naturally.

A.3 Marcel Pérès and Ensemble Organum: Ritual, Place, and Vocal Grain

Marcel Pérès founded Ensemble Organum in 1982. Its work is especially useful here because it treats medieval repertory as embodied, situated sound rather than as neutral notation placed in a generic "church reverb." The ensemble's work moves among European and Mediterranean sacred traditions, and its research-oriented approach makes vocal color, declamation, drone, ornament, tuning, and architectural resonance mutually dependent listening variables. The ensemble's own history and discography document that breadth, from early medieval repertories to later regional chant traditions. Ensemble Organum history and discography.

For verbx work, do not imitate a recording's room by immediately choosing a huge, bright cathedral preset. First preserve the direct vocal event: narrow the early field, use enough pre-delay for consonants and melismas to remain legible, darken the return gradually, and compare a close, dryer perspective against a more distant one. The artistic question is not simply "how long is the tail?" but how breath, unison, drone, ornament, and reverberant memory establish ritual time. A useful analysis compares energy decay with the persistence of vowels and with the moment at which individual singers stop reading as separate sources.

Ensemble Organum and Marcel Pérès, *École de Notre-Dame de Paris: Mass for the Nativity of the Virgin* (1995). The twelfth- and thirteenth-century Notre-Dame repertory invites a close comparison between written polyphony, vocal blend, and the imagined or recorded architectural field. Listen for how sustained tones establish a shared acoustic ground while rhythmic and textual detail remains locally articulated. Build two versions of an original vocal study: a controlled early-reflection field with a modest late tail, and a more diffuse late field. Match loudness, then identify the point at which the second version begins to obscure the line. The Harmonia Mundi release page confirms the program and artists.

Ensemble Organum and Marcel Pérès, *Guillaume de Machaut: Messe de Notre-Dame* (1996). Machaut's *Messe de Nostre Dame* (c. 1365) is a powerful reverb test because independent vocal entries must remain intelligible while the whole mass acquires ceremonial continuity. Listen for the way dense contrapuntal activity changes the maximum usable decay more than a single RT60 number predicts. In verbx, send four to eight original vocal layers through one coherent late field but vary their early-reflection timing and direct-to-reverberant ratio; compare the result with a single shared stereo return. The album record identifies the Pérès and Ensemble Organum recording.

Ensemble Organum and Marcel Pérès, *Chant Corse: Manuscrits franciscains des XVIIe–XVIIIe siècles* (1994). This Corsican chant program is a reminder that resonant sacred sound is not one homogeneous northern-European cathedral style. Attend to the relation of solo and collective vocal weight, drone-like support, irregular phrase length, and the acoustic space between entries. A

productive verbx exercise is to make three alternate responses for an original chant-like phrase: intimate and nearly dry, a short stone enclosure, and a dark extended field. Keep the source performance identical, log the render parameters, and judge which spatial treatment clarifies rather than exoticizes the material. The release listing identifies the 1994 recording and repertory.

A.4 Recorded Space and Tape-Era Studio Craft

Les Paul and Mary Ford, *How High the Moon* (1951). Layering, close recording, speed manipulation, and repeated parts create an impossible studio ensemble whose space is assembled rather than documented. Listen for the contrast between intimate vocal presence and the depth implied by overdubs. Use a short room, narrow slap, and different pre-delays per layer; avoid one wash that flattens the arrangement.

Elvis Presley, *Mystery Train* (1955). The Sun recording's slapback makes a compact performance feel rhythmically larger without suggesting a concert hall. Listen to the repeat as a second attack that locks to the groove. In verbx, begin with one short repeat, little diffusion, limited bandwidth, and no long tail; move the delay by a few milliseconds to hear how quickly pocket becomes clutter.

The Ronettes, *Be My Baby* (1963). The production combines dense orchestration and echo-chamber return into a unified wall while preserving the lead vocal's authority. Listen for ambience that thickens sustained energy more than transient detail. Test parallel chamber returns with strong filtering and compression, then compare a single long algorithmic preset against several shorter, differently colored returns.

The Beach Boys, *Good Vibrations* (1966). Contrasting modules, instruments, and recording spaces are edited into a coherent form. Listen for changes in depth as part of the arrangement, not merely changes in instrumentation. Build several scene presets with matched loudness but different pre-delay, width, and decay color, then automate scene changes at structural boundaries rather than continuously morphing everything.

The Beatles, *Tomorrow Never Knows* (1966). Tape loops, compression, rotating-speaker color, and studio effects dissolve the boundary between performance and environment. Listen for moving layers whose apparent spaces are not mutually consistent yet remain compositionally clear. A verbx exercise uses multiple effect returns with different bandwidths and modulation rates, each ducked by a different source family.

Led Zeppelin, *When the Levee Breaks* (1971). The drum sound demonstrates how a stairwell, distant microphones, compression, and delay can turn a performance into an architectural event. Listen to the separation between the initial hit and the enormous room response. Approximate the relationship with a transient-preserving dry path, a dark long room, modest pre-delay, and aggressive return compression rather than excessive wet percentage.

A.5 Electroacoustic Space and Feedback

Karlheinz Stockhausen, *Gesang der Jünglinge* (1955–1956). Electronic sound and a recorded boy's voice move through a multichannel field, making trajectory and location structural parameters. Listen for timbral transitions that are reinforced by spatial transitions. In verbx, distribute several related source transformations around an ambisonic or surround layout and preserve common spectral landmarks so motion reads as continuity rather than random panning.

Karlheinz Stockhausen, *Gruppen* (1955–1957). Three orchestras with three conductors surround the audience and exchange blocks of tempo, density, and color. Listen for mass moving between ensembles and for events that only make sense as spatial simultaneities. Use three buses with independent early fields and a shared late field; automate source-group routing rather than relying only on a master-width control.

Edgard Varèse, *Poème électronique* (1958). Conceived for the Philips Pavilion, the work binds electronic montage to a purpose-built architectural and loudspeaker environment. Listen for abrupt scale changes, trajectories, and sound masses that imply spaces larger or stranger than the playback room. Create a sequence of short rendered scenes using sharply contrasting IRs, then crossfade only where the composition calls for continuity.

Karlheinz Stockhausen, *Kontakte* (1958–1960). Electronic gestures rotate, accelerate, and transform while percussion and piano connect synthetic and acoustic time. Listen to spatial movement as a form of articulation. A useful study maps one modulation source to position, brightness, and pre-delay with different depths, producing related motion without making every parameter trace the same curve.

Iannis Xenakis, *Persephassa* (1969). Six percussionists surround the audience, allowing rhythmic patterns to become rotations and distributed masses. Listen for speed inferred from sequential attacks rather than from a continuously moving source. Recreate the principle with six discrete positions, very short directional early reflections, and a common late field that does not blur the attack order.

Alvin Lucier, *I am sitting in a room* (1969). Repeated recording and playback progressively reinforce room modes until speech becomes resonant pattern. The room is the filter and the iterative process is the form; the definitive concept is documented in the Lovely Music album notes. Reproduce the experiment safely with a short spoken phrase, bounded feedback or repeated offline renders, peak normalization between generations, and a JSON record of spectral change.

A.6 Dub, Echo, and the Mixing Desk as Performance Instrument

Augustus Pablo and King Tubby, *King Tubby Meets Rockers Uptown* (1974). Dropouts, spring-like reverberation, filtered echo, and sudden restoration of the rhythm section turn the mix into a live arrangement. Listen for negative space: the effect is powerful because elements disappear. Build a verbx automation pass that mutes sends and sources decisively, then introduces short colored tails only on selected phrases.

Lee Scratch Perry and the Upsetters, *Dread Lion* (1976). The production treats percussion, bass, noise, and reverberant fragments as inhabitants of a mutable environment. Listen for returns that change tone and apparent distance over time. Use feedback filtering, modulation, and return saturation, but cap feedback and peak level so instability remains expressive rather than destructive.

Augustus Pablo, *East of the River Nile* (1977). Melodica tone, bass, and spacious rhythm demonstrate how a narrow lead can float above a deep, economical foundation. Listen for the lead's delayed shadows and the restraint of the low-frequency ambience. In verbx, high-pass the long return, keep bass largely direct, and use a darker short echo to support the lead without widening every element.

Scientist, *Dance of the Vampires* (1981). Extreme dropouts and animated effect throws make the studio feel like a changing stage. Listen for the way echo feedback is performed in phrases rather than left at one setting. Record automation for send level, feedback, and damping as separate gestures; use limiter telemetry to identify throws that approach unsafe accumulation.

Grace Jones, *Walking in the Rain* (1981). The Compass Point production uses disciplined space around bass, drums, guitar, voice, and atmospheric detail. Listen to how depth comes from selective ambience and arrangement gaps rather than from a uniformly wet mix. Rebuild the hierarchy with different decay times for voice and percussion, leaving the bass center dry and stable.

The Orb, *Little Fluffy Clouds* (1990). Ambient collage, speech, loops, and dub-derived delays create a large but rhythmically legible environment. Listen for repeated fragments that become landmarks inside a slowly changing field. Use tempo-related echoes feeding a long dark reverb, then automate return EQ so repetition recedes without merely becoming quieter.

A.7 Minimalism, Ambient Music, and Slow Decay

Terry Riley, *Poppy Nogood and the Phantom Band* (1967–1969). Repetition and tape-delay accumulation create lines that remember their recent past. Listen for the point where echo stops reading as accompaniment and becomes counterpoint. Build a bounded delay-reverb loop with gradual filtering, then compare a fixed feedback value with a carefully performed feedback envelope.

Brian Eno, *Discreet Music* (1975). Long tape-delay systems and sparse source material establish a process that can unfold without conventional foreground development. Listen for change at the scale of minutes and for the musical value of near-stasis. In verbx, use deterministic modulation, a long but finite RT60, conservative high-frequency content, and a one-minute analysis window rather than judging from a five-second excerpt.

Steve Reich, *Music for 18 Musicians* (1976). Pulsation, resonance, and gradual harmonic change make ensemble blend and room support essential. Listen to how mallet attacks remain distinct while voices and sustained instruments connect the harmony. Test medium decay with substantial pre-delay, then shorten only the high-band RT60 until rhythmic detail survives without making the ensemble feel dry.

Brian Eno, *1/1* (1978), from *Ambient 1: Music for Airports* (1978). Sparse piano events leave enough time for decay to become part of the phrase. Listen for the threshold between identifiable note, reflected note, and continuous atmosphere. Render the same piano pattern through several RT60 values at matched loudness; compare how note density changes the maximum usable decay.

Harold Budd and Brian Eno, *First Light* (1980). Soft piano attacks and a luminous sustained field demonstrate how damping and modulation can imply depth without drawing attention to an effect. Listen to the stability of the center and the gentle motion at the edges. Use a dark early field, slow low-depth modulation, and equal-power wet/dry control; check mono compatibility before increasing width.

Aphex Twin, *#3 (Rhubarb)* (1994). A simple harmonic surface becomes emotionally specific through blur, detuning, noise, and very long ambience. Listen for pitch identity surviving inside a nearly continuous tail. In verbx, start with a finite extreme RT60, reduce high-frequency damping gradually, and use a limiter with slow release to contain rare build-ups without making the pad pump.

A.8 Gated, Digital, and Alternative Production Spaces

Peter Gabriel, *Intruder* (1980). The famous drum environment uses a gated ambience that makes room sound terminate as an arranged event. Listen for the explosive onset and unnaturally decisive end. In verbx, tune gate hold and release before increasing wet level, and compare a hard gate with a shaped downward expander so the tail remains forceful without clicking.

Phil Collins, *In the Air Tonight* (1981). The late drum entrance turns a production technique into a structural revelation. Listen to the dry, restrained opening as preparation for the gated impact. Build two complete scenes rather than one automated preset: an intimate vocal environment and a protected high-headroom drum environment, then switch at the arrangement boundary.

Kate Bush, *Running Up That Hill* (1985). Layered drums, synthesizers, voice, and controlled digital space create scale without losing rhythmic insistence. Listen for how ambience supports sustained material while percussion remains forward. Use multiband decay and ducking keyed from the main pulse; release the ducking slowly enough that the field blooms between attacks.

Cocteau Twins, *Carolyn's Fingers* (1988). Guitar processing and vocal ambience produce a dense, luminous surface in which source identity is suggestive rather than literal. Listen for several depths coexisting inside the same wide image. Use parallel shimmer, chorus-like modulation, and dark reverb returns at low individual levels; bypass each return to verify that no single layer carries the entire illusion.

My Bloody Valentine, *To Here Knows When* (1991). Pitch motion, reverse-like envelopes, distortion, and diffuse ambience collapse guitar and room into one unstable texture. Listen for attack ambiguity and slow spectral movement. Experiment with reverse reverb before distortion, then reverse the order; preserve a dry or lightly processed anchor so the texture remains intelligible at high density.

Talk Talk, *After the Flood* (1991). Performance dynamics, sustained organ, amplified noise, and studio space produce a huge field that still feels physically inhabited. Listen to the contrast between near-silence and saturation. Use automation to change send topology and return color across sections, retaining wide dynamic range instead of normalizing every passage to a constant loudness.

A.9 Advanced Concert Spatialization

Jonathan Harvey, *Mortuos Plango, Vivos Voco* (1980). Bell spectra and a boy's voice are analyzed, transformed, and projected through an eight-channel electronic space. Listen for movement that follows spectral identity rather than mimicking a traveling point source. Build related IR or resonator states from one spectral family and move between them with smooth, phase-conscious interpolation.

Pierre Boulez, *Répons* (1981–1984). Soloists, ensemble, live electronics, and loudspeaker placement create call-and-response across physical and transformed spaces. Listen to the distinction between a player's local sound and its electronic continuation around the hall. Design separate early and late spatial paths, report their latency explicitly, and test whether transformed responses remain rhythmically attached to the initiating gesture.

Luigi Nono, *Prometeo: Tragedia dell'ascolto* (1981–1984). Nono's "tragedy of listening" distributes performers and live electronics through an acoustic structure conceived with the venue; IRCAM documents the 1984 San Lorenzo premiere and four instrumental groups in its work record. Listen for fragile sounds whose location and audibility demand attention. Use low-noise sources, subtle movement, and calibrated monitoring rather than exaggerated effects.

Kaija Saariaho, *Lichtbogen* (1986). Instrumental color and live electronics evolve as a continuous spectral and spatial fabric. Listen for transitions where electronic extension is difficult to separate from instrumental resonance. In verbx, derive damping, modulation, and spatial-width automation from one slow control curve, but scale each destination differently to avoid obvious lockstep motion.

Georg Friedrich Haas, *in vain* (2000). Spectral harmony, tuning, darkness, and performance space alter the listener's orientation even without a conventional electronic reverb concept. Listen for beating patterns and apparent depth created by frequency relationships. Use this as a warning against treating every spatial sensation as an effect: compare true room processing with dry microtonal clusters before adding modulation.

Olga Neuwirth, *Le Encantadas o le avventure nel mare delle meraviglie* (2015). The work imports a modeled acoustic memory of Venice's San Lorenzo into another venue and distributes performers as an archipelago; IRCAM describes the virtual acoustic and spatial conception in its work presentation. Compare physical and simulated place as compositional material. A verbx study can morph between measured or synthetic IRs while logging the exact interpolation path.

A.10 Contemporary Depth, Texture, and Immersion

Oneohtrix Point Never, *Chrome Country* (2013). Synthetic voices, keyboard sonorities, and expanding ambience make scale feel unstable and cinematic. Listen for the transition from recognizable source to overwhelming field. Automate decay and spectral tilt together, then use loudness matching to determine whether the perceived expansion comes from space or simply from level.

Holly Herndon, *Chorus* (2014). Processed voices and digitally fragmented space make the studio feel both intimate and distributed. Listen for micro-edits that retain a shared acoustic identity. Use short

convolution states on individual fragments and a common algorithmic tail on the group, balancing local discontinuity against global coherence.

Björk, *Family* (2015). Strings, voice, electronics, and deep spatial perspective turn emotional distance into arrangement. Listen for the voice moving between exposed presence and a larger enveloping environment. Build a dual-engine vocal path with a precise early field and a separately automated late field; avoid using width as a substitute for depth.

Burial, *Archangel* (2007). Vocal fragments, vinyl-like noise, filtered percussion, and cavernous returns create an imagined nocturnal city. Listen for ambience that carries environmental identity while transients remain deliberately incomplete. Use band-limited reverbs, irregular pre-delay, and low-level noise as separate layers, then verify that the rhythm still reads when the long return is muted.

Grouper, *Heavy Water* / *I'd Rather Be Sleeping* (2008). Voice and guitar sit inside a dense veil where noise, sustain, and room-like blur are inseparable. Listen for intelligibility arriving in flashes rather than remaining constant. In verbx, combine moderate reverb with source-side filtering and saturation; an extreme RT60 alone will usually sound cleaner and less integrated than the reference principle.

FKA twigs, *cellophane* (2019). Exposed vocal detail, piano, silence, and carefully scaled ambience make every decay psychologically legible. Listen for how little processing is required when arrangement leaves genuine space. Start with an almost dry vocal, add a quiet long return with substantial pre-delay, and automate only selected phrase endings rather than maintaining a constant wet bed.

A.11 Recording and YouTube Citations

Links below identify the exact performance, master, or reference upload used for the listening note. **Official** means an artist, label, catalog, or automated Topic channel; **institutional** means an orchestra, broadcaster, ensemble, festival, or university channel. **Reference upload** means no stable first-party full recording was located on YouTube during this edition's link audit. Availability and territorial access can change.

A.11.1 Architecture as an Instrument

- Tallis, Thomas. *Spem in alium* (c. 1570). Harry Christophers, live concert. AVROTROS Klassiek, institutional video. YouTube.
- Gabrieli, Giovanni. *In ecclesiis* (c. 1615). Camerata Nova. Concert-performance reference upload. YouTube.
- Berlioz, Hector. *Grande Messe des morts* (1837). WDR Radio Choir and WDR Symphony Orchestra. ARD Klassik, institutional video. YouTube.
- Ives, Charles. *The Unanswered Question* (1906; revised later). Frankfurt Radio Symphony; Andrés Orozco-Estrada. Institutional video. YouTube.
- Vaughan Williams, Ralph. *Fantasia on a Theme by Thomas Tallis* (1910). Kristiansand Symphony Orchestra; Tabita Berglund. Institutional video. YouTube.
- Messiaen, Olivier. *Et exspecto resurrectionem mortuorum* (1964). North Texas Wind Symphony; Eugene Migliaro Corporon, 2013. Institutional video. YouTube.

A.11.2 Recorded Space and Tape-Era Studio Craft

- Les Paul and Mary Ford. *How High the Moon* (1951). Remastered catalog recording. Mary Ford Topic channel, official audio. YouTube.
- Presley, Elvis. *Mystery Train* (1955). Elvis Presley channel, official audio. YouTube.
- The Ronettes. *Be My Baby* (1963). The Ronettes channel, official audio. YouTube.

- The Beach Boys. *Good Vibrations* (1966). The Beach Boys channel, official catalog audio. YouTube.
- The Beatles. *Tomorrow Never Knows* (1966). The Beatles channel, official audio. YouTube.
- Led Zeppelin. *When the Levee Breaks* (1971). Led Zeppelin channel, official audio. YouTube.

A.11.3 Electroacoustic Space and Feedback

- Stockhausen, Karlheinz. *Gesang der Jünglinge* (1955–1956). Fonurgia Nuova, reference transfer. YouTube. See also the Stockhausen complete-edition catalog for the authoritative edition.
- Stockhausen, Karlheinz. *Gruppen* (1955–1957). Ensemble intercontemporain, institutional performance. YouTube.
- Varèse, Edgard. *Poème électronique* (1958). Sebastian Ars Acoustica, reference transfer. YouTube.
- Stockhausen, Karlheinz. *Kontakte* (1958–1960). Ensemble Musikfabrik, institutional performance. YouTube.
- Xenakis, Iannis. *Persephassa* (1969). University of Toronto Percussion, institutional performance. YouTube.
- Lucier, Alvin. *I am sitting in a room* (1969). Reference upload of the recorded work. YouTube. Recording history and procedure: Lovely Music album notes.

A.11.4 Dub, Echo, and the Mixing Desk

- Augustus Pablo and King Tubby. *King Tubby Meets Rockers Uptown* (1974). King Tubby Topic channel, official catalog audio. YouTube.
- Lee “Scratch” Perry and the Upsetters. *Dread Lion* (1976). Lee “Scratch” Perry Topic channel, official catalog audio. YouTube.
- Augustus Pablo. *East of the River Nile* (1977). Augustus Pablo Topic channel, official catalog audio. YouTube.
- Scientist. *Dance of the Vampires* (1981). Reference transfer. YouTube.
- Jones, Grace. *Walking in the Rain* (1981). Grace Jones Topic channel, official catalog audio. YouTube.
- The Orb. *Little Fluffy Clouds* (1990). The Orb channel, official catalog audio. YouTube.

A.11.5 Minimalism, Ambient Music, and Slow Decay

- Riley, Terry. *Poppy Nogood and the Phantom Band* (1967–1969). Reference transfer. YouTube.
- Eno, Brian. *Discreet Music* (1975). 2004 remaster; Brian Eno channel, official audio. YouTube.
- Reich, Steve. *Music for 18 Musicians* (1976). Eighth Blackbird and guests; Vic Firth institutional performance. YouTube.
- Eno, Brian. *1/1* (1978), from *Ambient 1: Music for Airports*. 2004 remaster; Brian Eno channel, official audio. YouTube.
- Budd, Harold, and Brian Eno. *First Light* (1980). 2004 remaster; Harold Budd Topic channel, official audio. YouTube.
- Aphex Twin. *#3 (Rhubarb)* (1994), from *Selected Ambient Works Volume II*. Aphex Twin channel, official audio. YouTube.

A.11.6 Gated, Digital, and Alternative Production Spaces

- Gabriel, Peter. *Intruder* (1980). Peter Gabriel channel, official audio. YouTube.
- Collins, Phil. *In the Air Tonight* (1981). Phil Collins channel, official music video. YouTube.

- Bush, Kate. *Running Up That Hill (A Deal with God)* (1985). 2018 remaster; Kate Bush channel, official audio. YouTube.
- Cocteau Twins. *Carolyn's Fingers* (1988). Cocteau Twins channel, official video. YouTube.
- My Bloody Valentine. *To Here Knows When* (1991). Official music video distributed by UPROXX Indie Mixtape. YouTube.
- Talk Talk. *After the Flood* (1991). Talk Talk band channel, catalog audio. YouTube.

A.11.7 Advanced Concert Spatialization

- Harvey, Jonathan. *Mortuos Plango, Vivos Voco* (1980). Jonathan Harvey Topic channel, official catalog audio. YouTube.
- Boulez, Pierre. *Répons* (1981–1984). Ensemble intercontemporain; Matthias Pintscher. Institutional performance. YouTube.
- Nono, Luigi. *Prometeo: Tragedia dell'ascolto* (1981–1984). New Music Ostrava, reference performance. YouTube. Work and premiere record: IRCAM.
- Saariaho, Kaija. *Lichtbogen* (1986). Wellesz Theatre, reference performance. YouTube.
- Haas, Georg Friedrich. *in vain* (2000). Argento Chamber Ensemble; Michel Galante. Ensemble performance. YouTube.
- Neuwirth, Olga. *Le Encantadas o le avventure nel mare delle meraviglie* (2015). Ensemble intercontemporain; Holland Festival 2016, institutional performance. YouTube. Virtual-acoustic documentation: IRCAM.

A.11.8 Contemporary Depth, Texture, and Immersion

- Oneohtrix Point Never. *Chrome Country* (2013). Oneohtrix Point Never channel, official audio. YouTube.
- Herndon, Holly. *Chorus* (2014). RVNG Intl., official video. YouTube.
- Björk. *Family* (2015). One Little Independent/One Little Indian Jukebox, label audio. YouTube.
- Burial. *Archangel* (2007). Hyperdub, label upload. YouTube.
- Grouper. *Heavy Water* / *I'd Rather Be Sleeping* (2008). Grouper Topic channel, official catalog audio. YouTube.
- FKA twigs. *cellophane* (2019). FKA twigs channel, official video. YouTube.

A.12 Comparative Listening Exercises

1. **One source, six eras.** Render the same eight-bar vocal or percussion phrase using slapback, chamber, dub throw, gated room, ambient tail, and contemporary sparse vocal settings. Loudness-match all versions and record the parameters in JSON.
2. **Architecture versus algorithm.** Compare a measured church or hall IR with an algorithmic preset at the same broadband RT60. Note differences in early-reflection pattern, modal coloration, stereo stability, and high-band decay.
3. **Depth without width.** Keep the output mono while changing pre-delay, DRR, damping, and early density. Restore stereo only after the front-to-back hierarchy works.
4. **Silence as orchestration.** Choose a sparse work from this appendix and reproduce its spacing with your own material. Do not add ambience until the dry arrangement leaves enough time for a decay to be heard.
5. **Finite versus frozen.** Compare a 30-second finite decay, a 360-second extreme decay, and freeze. Measure peak growth and spectral accumulation over at least one minute.

6. **Version audit.** Compare two recordings or mixes of the same composition. Document which differences arise from performance venue, microphone distance, mix decisions, mastering, or playback format.

A.13 Documentation Sources

The listening notes are interpretive and intentionally avoid reproducing lyrics or score text. Work dates and technical framing were checked against composer, publisher, label, and institutional documentation where available. Particularly useful primary or specialist records include the Stockhausen complete-edition catalog, the Lovely Music notes for Lucier's *I am sitting in a room* (1969), IRCAM's *Prometeo: Tragedia dell'ascolto* (1981–1984) record, and IRCAM's account of the virtual San Lorenzo acoustic in *Le Encantadas o le avventure nel mare delle meraviglie* (2015).

A.14 Expanded Listening Canon

The following 192 additions broaden the appendix across sacred, orchestral, jazz, popular, ambient, theater, installation, environmental, game, and contemporary immersive practices. Piece and album titles are italicized throughout. Each entry pairs a listening cue with a practical study prompt. Links open a stable YouTube catalog search so readers can select an authorized or territorially available recording; recording-specific citations in the preceding section and the primary documentation below remain the preferred references where supplied.

A.14.1 Curated Listening Routes

- **Architecture and choir.** Begin with *Spem in alium* (c. 1570), *Miserere mei, Deus* (c. 1638), *Passio* (1982), *The Forty Part Motet* (2001), and *Kits Beach Soundwalk* (1989) to compare written spatial design, recorded architecture, installation, and narrated place.
- **Rhythm and short space.** Move from *Mystery Train* (1955) through *King Tubby Meets Rockers Uptown* (1974), *Quadrant Dub I* (1994), *Welcome to the Terrordome* (1990), and *Dream House* (2013) to hear when reflection becomes pulse, syncopation, or masking.
- **Long decay and near-stasis.** Compare *Apparition de l'église éternelle* (1932), *Schlingen-Bläangen* (1974), *Deep Listening* (1989), *Jetsun Mila* (1986), *The Sacrificial Code* (2019), and *#3 (Rhubarb)* (1994) at matched monitoring level.
- **Voice and psychological distance.** Trace the changing foreground in *A Change Is Gonna Come* (1964), *Amazing Grace* (1972), *Where Will I Be* (1995), *Roads* (1994), *cellophane* (2019), and *when the party's over* (2018).
- **Movement and immersion.** Study *Kontakte* (1958–1960), *Persephassa* (1969), *Répons* (1981–1984), *Le Encantadas o le avventure nel mare delle meraviglie* (2015), *Movement 6* (2021), and *Billions* (2022) with explicit attention to trajectory and envelopment.
- **Environment as form.** Hear *Presque rien No. 1* (1970), *Music for Wilderness Lake* (1979), *A Sound Map of the Hudson River* (1982), *Kits Beach Soundwalk* (1989), *Vatnajökull* (2003), and *Aquaculture* (2010) as composed relationships among listener, place, microphone, and playback room.

A.14.2 Sacred Resonance And Ritual

Guillaume de Machaut, *Messe de Nostre Dame* (c. 1365). from *polyphonic mass*. Track how long stone-room sustain joins successive voices without erasing their independent entries. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Messe de Nostre Dame* (c. 1365) with original material, then preserve the exact render command and JSON report.

Gregorio Allegri, *Miserere mei, Deus* (c. 1638). from *choral work*. Compare distant solo-group height with the grounded main choir and listen for architecture acting as an antiphonal mixer. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Miserere mei, Deus* (c. 1638) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

J. S. Bach, *St Matthew Passion* (1727). from *oratorio*. Use contrasting recordings to separate compositional double-choir space from venue and microphone perspective. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *St Matthew Passion* (1727); identify which path carries its spatial identity.

Anton Bruckner, *Locus iste* (1869). from *motet*. Notice how rests expose the room and how harmonic arrivals briefly convert decay into part of the chord. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Locus iste* (1869) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Arvo Pärt, *Passio* (1982). from *ECM New Series recording*. Study how narrow dynamics, sustained tones, and silence make the recording acoustic an active formal layer. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Passio* (1982) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

John Tavener, *The Protecting Veil* (1988). from *cello and strings*. Listen for solo-cello presence floating ahead of a slowly changing orchestral halo. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *The Protecting Veil* (1988); test whether the form remains legible without level automation.

Giovanni Gabrieli, *In ecclesiis* (c. 1615). from *Sacrae symphoniae II*. Separated vocal and instrumental choirs turn architectural distance into contrapuntal punctuation, with rests leaving the basilica-sized response fully exposed. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *In ecclesiis* (c. 1615) with original material, then preserve the exact render command and JSON report.

Wolfgang Amadeus Mozart, *Ave verum corpus, K. 618* (1791). from *motet*. Compact phrases and soft orchestration reveal how a modest sacred work can acquire scale from blended choral onset and restrained decay. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Ave verum corpus, K. 618* (1791) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Hector Berlioz, *Grande Messe des morts* (1837). from *Requiem*. Widely separated brass groups and massed forces make direction, propagation, and long decay part of the written ceremonial drama. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Grande Messe des morts* (1837); identify which path carries its spatial identity.

Sergei Rachmaninoff, *All-Night Vigil* (1915). from *choral cycle*. Low bass fundamentals, close semitone motion, and sustained vowels test whether a room supports harmonic fusion without masking inner voices. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *All-Night Vigil* (1915) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Igor Stravinsky, *Symphony of Psalms* (1930). from *choral symphony*. Dry rhythmic blocks alternate with resonant choral planes, making articulation and collective bloom equally important to the work's ritual character. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Symphony of Psalms* (1930) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Olivier Messiaen, *Et exspecto resurrectionem mortuorum* (1964). from *orchestral work*. Gongs, winds, and silence exploit long architectural decay as measured continuation, especially when attacks are allowed to vanish completely before the next event. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Et exspecto resurrectionem mortuorum* (1964); test whether the form remains legible without level automation.

A.14.3 Orchestral Depth And Concert-Hall Scale

Gustav Mahler, *Symphony No. 2* (1894). from *symphony*. Map offstage and distant events against close orchestral attacks, especially during the final movement's expanding scale. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Symphony No. 2* (1894) with original material, then preserve the exact render command and JSON report.

Claude Debussy, *La mer* (1905). from *orchestral work*. Follow how orchestral color, hall bloom, and dynamic shaping create depth without fixed foreground/background roles. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *La mer* (1905) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Gustav Holst, *Neptune, the Mystic* (1916). from *The Planets*. The disappearing offstage chorus is a model for level, distance, filtering, and decay acting as one continuous gesture. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Neptune, the Mystic* (1916); identify which path carries its spatial identity.

Maurice Ravel, *Daphnis et Chloé, Suite No. 2* (1913). from *ballet suite*. Compare transient woodwind detail with the broad chorus and string field at the sunrise climax. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Daphnis et Chloé, Suite No. 2* (1913) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Benjamin Britten, *War Requiem* (1962). from *requiem*. Separate chamber, orchestral, and distant ceremonial layers by apparent location as well as orchestration. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *War Requiem* (1962) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

György Ligeti, *Lontano* (1967). from *orchestral work*. Dense micropolyphony provides a reference for diffusion build-up whose source identities remain barely perceptible. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Lontano* (1967); test whether the form remains legible without level automation.

Hector Berlioz, *Symphonie fantastique* (1830). from *symphony*. Extreme orchestral contrasts expose how hall return can enlarge distant bells and brass while preserving the nervous detail of strings and percussion. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Symphonie fantastique* (1830) with original material, then preserve the exact render command and JSON report.

Charles Ives, *The Unanswered Question* (1906; revised later). from *orchestral work*. Spatially separated strings, trumpet, and winds demonstrate that distance can distinguish simultaneous musical roles more clearly than timbre alone. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *The Unanswered Question* (1906; revised later) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Ralph Vaughan Williams, *Fantasia on a Theme by Thomas Tallis* (1910). from *string orchestra*. Divided string groups and antiphonal placement create nested depth planes whose apparent boundaries expand and contract with register and dynamics. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Fantasia on a Theme by Thomas Tallis* (1910); identify which path carries its spatial identity.

Ottorino Respighi, *Pines of Rome* (1924). from *symphonic poem*. Offstage instruments, organ pedal, recorded birds, and monumental brass make the final spatial expansion a model of cumulative depth orchestration. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Pines of Rome* (1924) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Witold Lutosławski, *Livre pour orchestre* (1968). from *orchestral work*. Controlled aleatory distributes transient detail across the ensemble, offering a demanding test for clarity during rapid changes in density and apparent width. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Livre pour orchestre* (1968) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Kaija Saariaho, *Orion* (2002). from *orchestral work*. Spectral orchestration and sustained resonance make instrumental color appear to move through foreground, haze, and distant radiance. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Orion* (2002); test whether the form remains legible without level automation.

A.14.4 Jazz Rooms And Engineered Intimacy

Miles Davis, *So What* (1959). from *Kind of Blue*. Use the studio's restrained ambience to study depth that does not announce itself as an effect. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *So What* (1959) with original material, then preserve the exact render command and JSON report.

Bill Evans Trio, *Waltz for Debby* (1961). on *Waltz for Debby* (1961). Audience, room, and trio occupy one acoustic scene; note how venue sound supplies scale without obscuring touch. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Waltz for Debby* (1961) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

John Coltrane, *A Love Supreme, Part I: Acknowledgement* (1965). from *A Love Supreme*. Listen for cohesive room tone around forceful close detail and a rhythm section with stable depth. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *A Love Supreme, Part I: Acknowledgement* (1965); identify which path carries its spatial identity.

Miles Davis, *In a Silent Way / It's About That Time* (1969). from *In a Silent Way*. Edits, electric timbres, and sustained studio space turn ambience into continuity across assembled performances. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *In a Silent Way / It's About That Time* (1969) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Alice Coltrane, *Journey in Satchidananda* (1971). on *Journey in Satchidananda* (1971). Harp resonance, drone, percussion, and studio ambience create layered sustain with distinct spectral zones. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Journey in Satchidananda* (1971) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Jan Garbarek and The Hilliard Ensemble, *Parce mihi Domine* (1994). from *Officium*. The saxophone and vocal ensemble demonstrate how one large church can preserve separate source distances. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Parce mihi Domine* (1994); test whether the form remains legible without level automation.

Duke Ellington, *Diminuendo and Crescendo in Blue* (1937). from *orchestral jazz work*. Compare studio and live versions to hear how audience, stage leakage, and room excitation transform the same long-form crescendo. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Diminuendo and Crescendo in Blue* (1937) with original material, then preserve the exact render command and JSON report.

Thelonious Monk, *'Round Midnight* (1944). from *jazz standard*. Sparse attacks and deliberate rests reveal the recording room around piano resonance while leaving harmonic ambiguity intact. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *'Round Midnight* (1944) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Charles Mingus, *Goodbye Pork Pie Hat* (1959). from *Mingus Ah Um*. Closely voiced horns, bass, and drums share a warm studio field whose cohesion depends on decay color more than conspicuous wetness. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Goodbye Pork Pie Hat* (1959); identify which path carries its spatial identity.

Keith Jarrett, *The Köln Concert, Part I* (1975). from *The Köln Concert*. Piano, performer sounds, audience, and hall response form one continuous document in which resonant sustain guides improvisational pacing. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *The Köln Concert, Part I* (1975) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Pat Metheny Group, *Are You Going with Me?* (1982). from *Offramp*. Sustained guitar, synth, percussion, and long returns demonstrate how electronic jazz can maintain a stable center inside a panoramic field. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Are You Going with Me?* (1982) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Nils Petter Molvær, *Khmer* (1997). on *Khmer* (1997). Muted trumpet remains physically present against electronically extended drums and atmosphere, creating deliberate tension between intimacy and landscape. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Khmer* (1997); test whether the form remains legible without level automation.

A.14.5 Studio Architecture And Iconic Production

Roy Orbison, *In Dreams* (1963). on *In Dreams* (1963). Orchestral build and chamber-like return increase scale while Orbison's voice remains the fixed perceptual anchor. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *In Dreams* (1963) with original material, then preserve the exact render command and JSON report.

Patsy Cline, *Crazy* (1961). from *Showcase*. A controlled plate-or-chamber halo supports vocal sustain without moving the singer behind the ensemble. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Crazy* (1961) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Pink Floyd, *Echoes* (1971). from *Meddle*. Treat the work as a catalog of synthetic distance, feedback, panning, and long-form transitions between acoustic worlds. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Echoes* (1971); identify which path carries its spatial identity.

David Bowie, *Heroes* (1977). on *Heroes* (1977). The vocal microphone-gate arrangement links performance intensity to apparent room size and should be studied dynamically. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Heroes* (1977) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Prince and the Revolution, *Purple Rain* (1984). on *Purple Rain* (1984). Arena scale comes from performance, audience, delays, and long returns whose energy is carefully limited around the lead. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Purple Rain* (1984) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

U2, *A Sort of Homecoming* (1984). from *The Unforgettable Fire*. Diffuse guitar and drum spaces show how modulation and ambience can bind a mix while preserving propulsion. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *A Sort of Homecoming* (1984); test whether the form remains legible without level automation.

The Ronettes, *Be My Baby* (1963). from *Presenting the Fabulous Ronettes Featuring Veronica*. Chamber reverberation, layered percussion, and dense orchestration show how a mono production can imply enormous depth through spectral and dynamic hierarchy. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Be My Baby* (1963) with original material, then preserve the exact render command and JSON report.

The Beatles, *A Day in the Life* (1967). from *Sgt. Pepper's Lonely Hearts Club Band*. Contrasting vocal spaces and orchestral transitions make studio acoustics an explicit formal boundary between otherwise discontinuous sections. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *A Day in the Life* (1967) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Led Zeppelin, *When the Levee Breaks* (1971). from *Led Zeppelin IV*. The stairwell drum sound is a canonical lesson in distant microphones, compression, pre-delay, and a room response that becomes the groove's principal instrument. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *When the Levee Breaks* (1971); identify which path carries its spatial identity.

Kate Bush, *Running Up That Hill (A Deal with God)* (1985). from *Hounds of Love*. Gated and layered drum ambience supports a driving pulse while voice and synthesizers occupy independently controlled depth planes. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Running Up That Hill (A Deal with God)* (1985) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

My Bloody Valentine, *To Here Knows When* (1991). from *Loveless* (1991). Reverse envelopes, diffuse guitars, and obscured attacks replace ordinary room cues with a continuously suspended synthetic space. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *To Here Knows When* (1991) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Björk, *Jóga* (1997). from *Homogenic*. Close voice, electronic percussion, strings, and volcanic-scale ambience demonstrate how abrupt depth contrast can intensify a song without weakening diction. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Jóga* (1997); test whether the form remains legible without level automation.

A.14.6 Ambient Continuums And Decaying Form

Tangerine Dream, *Phaedra* (1974). on *Phaedra* (1974). Sequenced pulses and evolving electronic clouds offer a test for separating repeated attacks from a continuously moving field. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Phaedra* (1974) with original material, then preserve the exact render command and JSON report.

Laraaji, *Meditation No. 1* (1980). from *Ambient 3: Day of Radiance*. Bright struck transients accumulate into a harmonically dense tail that tests damping and feedback stability. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Meditation No. 1* (1980) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Biosphere, *Poa Alpina* (1997). from *Substrata*. Environmental scale is implied through filtered layers and quiet returns rather than a single identifiable room. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Poa Alpina* (1997); identify which path carries its spatial identity.

Stars of the Lid, *Requiem for Dying Mothers, Part 2* (2001). from *The Tired Sounds of Stars of the Lid*. Slow orchestral layers demonstrate how several long envelopes can overlap without uncontrolled low-mid buildup. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Requiem for Dying Mothers, Part 2* (2001) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

William Basinski, *d|p 1.1* (2002). from *The Disintegration Loops*. Loop decay makes spectral loss and accumulating noise part of form, useful for comparing finite tail and degradation models. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *d|p 1.1* (2002) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Tim Hecker, *In the Fog II* (2011). from *Ravedeath, 1972*. Distortion, organ-like sustain, and unstable depth show why reverb color cannot be separated from source processing. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *In the Fog II* (2011); test whether the form remains legible without level automation.

Brian Eno, *Discreet Music* (1975). on *Discreet Music* (1975). Slow tape-system evolution makes feedback, delay, and gradual spectral change audible as compositional process rather than decorative ambience. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Discreet Music* (1975) with original material, then preserve the exact render command and JSON report.

Harold Budd and Brian Eno, *First Light* (1980). from *The Plateaux of Mirror*. Soft piano attacks enter a diffuse electronic field whose decay blurs clock time while preserving each note's harmonic consequence. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *First Light* (1980) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Aphex Twin, *#3 (Rhubarb)* (1994). from *Selected Ambient Works Volume II*. A nearly static harmonic loop reveals minute changes in tail color, noise, and beating that become perceptually large over repeated listening. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *#3 (Rhubarb)* (1994); identify which path carries its spatial identity.

Gas, *Königsforst 1* (1998). from *Königsforst*. Buried orchestral loops and a distant pulse make the virtual environment feel deep even when no discrete source has a stable location. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Königsforst 1* (1998) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Loscil, *Endless Falls* (2010). on *Endless Falls* (2010). Rain, low-frequency beds, and restrained tonal events offer a study in separating environmental texture from synthesized late-field energy. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Endless Falls* (2010) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Sarah Davachi, *For Voice* (2020). from *Cantus, Descant*. Organ-like sustain and narrow-band beating expose how playback-room modes interact with slowly changing electronic spectra. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *For Voice* (2020); test whether the form remains legible without level automation.

A.14.7 Dub, Trip-Hop, And Vocal Depth

Public Enemy, *Welcome to the Terrordome* (1990). from *Fear of a Black Planet*. Dense sampling demonstrates selective short-space placement where a global long tail would destroy rhythmic hierarchy. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Welcome to the Terrordome* (1990) with original material, then preserve the exact render command and JSON report.

Massive Attack, *Teardrop* (1998). from *Mezzanine*. The vocal occupies an intimate center while percussion and tonal beds imply a larger dark enclosure. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Teardrop* (1998) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Portishead, *Roads* (1994). from *Dummy*. Sparse arrangement makes the scale and automation of vocal and string ambience unusually exposed. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Roads* (1994); identify which path carries its spatial identity.

Radiohead, *Pyramid Song* (2001). from *Amnesiac*. Piano resonance, voice, strings, and delayed percussion establish multiple depth planes that converge near climaxes. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Pyramid Song* (2001) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

James Blake, *Limit to Your Love* (2010). from *James Blake*. Extreme contrast between dry intimacy, sub-bass energy, and expanding vocal space is a lesson in spectral headroom. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Limit to Your Love* (2010) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Frank Ocean, *Seigfried* (2016). from *Blonde*. Shifting voice layers and quiet environmental tails create psychological distance with very low effect visibility. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Seigfried* (2016); test whether the form remains legible without level automation.

Lee "Scratch" Perry and the Upsetters, *Dub Revolution* (1973). from *Blackboard Jungle Dub*. Mutes, spring-like splashes, filtering, and feedback demonstrate that the wet return can function as an improvising answer rather than a fixed background. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Dub Revolution* (1973) with original material, then preserve the exact render command and JSON report.

Augustus Pablo, *King Tubby Meets Rockers Uptown* (1974). from *King Tubbys Meets Rockers Uptown*. Melodica phrases and rhythm fragments trigger sharply timed echoes whose decay becomes counterpoint inside the groove. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *King Tubby Meets Rockers Uptown* (1974) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Grace Jones, *Nightclubbing* (1981). on *Nightclubbing* (1981). Dry rhythmic authority coexists with selective vocal and instrumental depth, illustrating how restraint makes occasional spatial expansion more powerful. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Nightclubbing* (1981); identify which path carries its spatial identity.

Cocteau Twins, *Heaven or Las Vegas* (1990). on *Heaven or Las Vegas* (1990). Layered guitar and voice use modulation and bright diffusion to create a wide field that remains buoyant instead of collapsing into masking. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Heaven or Las Vegas* (1990) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Tricky, *Hell Is Round the Corner* (1995). from *Maxinquaye*. Whisper-close voices sit against a darker sampled environment, making low-level ambience and spectral contrast carry psychological distance. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Hell Is Round the Corner* (1995) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

FKA twigs, *cellophane* (2019). from *Magdalene*. An exposed vocal, piano resonance, and carefully delayed blooms demonstrate how a nearly empty arrangement can support radical changes in emotional scale. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *cellophane* (2019); test whether the form remains legible without level automation.

A.14.8 Screen, Installation, And Game Worlds

Bernard Herrmann, *Scene d'Amour* (1958). from *Vertigo*. Orchestral surges and scoring-stage depth link harmonic recurrence with a powerful sense of spatial return. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Scene d'Amour* (1958) with original material, then preserve the exact render command and JSON report.

György Ligeti, *Lux Aeterna* (1966). from *2001: A Space Odyssey soundtrack context*. The choral field demonstrates how dense voices and long room decay can become an apparently source-less texture. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Lux Aeterna* (1966) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Vangelis, *Blade Runner Blues* (1982). from *Blade Runner*. Synth sustain, saxophone, city-like ambience, and delay construct a space larger than any visible room. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Blade Runner Blues* (1982); identify which path carries its spatial identity.

Angelo Badalamenti, *Laura Palmer's Theme* (1990). from *Twin Peaks*. Foreground keyboard attacks and a dark orchestral halo show how reverberation can signal emotional scale changes. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Laura Palmer's Theme* (1990) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

John Williams, *Theme from Jurassic Park* (1993). from *Jurassic Park*. The scoring-stage image balances monumental brass and strings with intelligible inner detail at very high density. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Theme from Jurassic Park* (1993) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Martin O'Donnell and Michael Salvatori, *A Walk in the Woods* (2001). from *Halo: Original Soundtrack*. Choral and synthetic layers demonstrate a game-space aesthetic that remains readable under repeated playback and combat effects. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *A Walk in the Woods* (2001); test whether the form remains legible without level automation.

Austin Wintory, *Nascence* (2012). from *Journey*. The score's gradual spatial expansion offers a model for automating envelopment as a narrative parameter. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Nascence* (2012) with original material, then preserve the exact render command and JSON report.

Jerry Goldsmith, *Main Title from Alien* (1979). from *Alien*. Orchestral timbre and scoring-stage depth suggest a vast hostile environment before image-specific effects define the fictional space. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Main Title from Alien* (1979) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Wendy Carlos and Rachel Elkind, *Main Title from The Shining* (1980). from *The Shining*. Electronic sustain, chant-like material, and low-frequency bloom create architectural dread through slow spectral motion rather than realistic room simulation. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Main Title from The Shining* (1980); identify which path carries its spatial identity.

Akira Yamaoka, *Theme of Laura* (2001). from *Silent Hill 2*. Guitar, drums, and grainy atmosphere balance song-like immediacy with the unstable environmental identity of a psychological game world. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Theme of Laura* (2001) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Jeremy Soule, *Secunda* (2011). from *The Elder Scrolls V: Skyrim*. Quiet piano and orchestral haze are designed to coexist with changing gameplay ambience, offering a model for depth that tolerates interruption and repetition. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Secunda* (2011) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Hildur Guðnadóttir, *Bridge of Death* (2019). from *Chernobyl*. Cello-derived textures, noise, and industrial resonance blur the line between score, environment, and imagined physical danger. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Bridge of Death* (2019); test whether the form remains legible without level automation.

A.14.9 Contemporary Immersive Production

SOPHIE, *Is It Cold in the Water?* (2018). from *Oil of Every Pearl's Un-Insides*. Hyper-detailed dry events and enormous synthetic depth shifts reveal the expressive value of abrupt spatial discontinuity. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Is It Cold in the Water?* (2018) with original material, then preserve the exact render command and JSON report.

Billie Eilish, *when the party's over* (2018). from *WHEN WE ALL FALL ASLEEP, WHERE DO WE GO?*. Close vocal layers and restrained long returns show how arrangement can make subtle reverb feel immense. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *when the party's over* (2018) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Jon Hopkins, *Luminous Beings* (2018). from *Singularity*. Pulsing detail and expanding harmonic clouds invite analysis of modulation, stereo correlation, and long-tail density. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Luminous Beings* (2018); identify which path carries its spatial identity.

Floating Points, Pharoah Sanders, *and the London Symphony Orchestra,* Movement 6 (2021). from *Promises*. A recurring electronic figure, saxophone, and orchestra share a slowly changing field without losing their independent depths. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Movement 6* (2021) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Caroline Polachek, *Billions* (2022). from *Desire, I Want to Turn Into You*. Layered voice, percussion, and choral scale demonstrate precise contrast between intimate articulation and spectacular bloom. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Billions* (2022) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Hans Zimmer, *Paul's Dream* (2021). from *Dune*. Voice, percussion, synthetic low end, and immense multichannel scale demonstrate how immersive reverberation can imply geography and ritual simultaneously. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Paul's Dream* (2021); test whether the form remains legible without level automation.

Nicolas Jaar, *Space Is Only Noise If You Can See* (2011). from *Space Is Only Noise*. Speech, pulses, close detail, and oblique room cues make negative space and changing perspective central to the track's identity. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Space Is Only Noise If You Can See* (2011) with original material, then preserve the exact render command and JSON report.

Oneohtrix Point Never, *Chrome Country* (2013). from *R Plus Seven*. Synthetic choir, bright transients, and sudden depth changes turn familiar reverberant signs into unstable digital architecture. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Chrome Country* (2013) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Holly Herndon, *Chorus* (2014). from *Platform*. Distributed vocal fragments and machine-cut ambience expose the expressive boundary between spatial continuity and frame-level interruption. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Chorus* (2014); identify which path carries its spatial identity.

Björk, *Black Lake* (2015). from *Vulnicura*. Voice, strings, electronics, and long-form spatial expansion create a dramatic landscape whose scale changes with the emotional argument. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Black Lake* (2015) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Arooj Aftab, *Mohabbat* (2021). from *Vulture Prince*. The voice remains poised and intimate while harp, strings, and subtle returns construct a deep, slow-moving nocturnal field. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Mohabbat* (2021) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Beyoncé, *VIRGO'S GROOVE* (2022). from *Renaissance*. Dense vocal arrangement, groove, and immersive release formats invite comparison of front-focused clarity with enveloping production detail. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *VIRGO'S GROOVE* (2022); test whether the form remains legible without level automation.

A.14.10 Early Music, Organ, And Monumental Acoustics

Hildegard von Bingen, *O vis aeternitatis* (c. 1151). from *A Feather on the Breath of God*. Follow the monophonic line into the room and distinguish melodic continuation from architectural sustain. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *O vis aeternitatis* (c. 1151) with original material, then preserve the exact render command and JSON report.

Pérotin, *Viderunt omnes* (c. 1198). from *Perotin*. Hear how sustained lower voices and measured upper parts use cathedral decay as a harmonic joining mechanism. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Viderunt omnes* (c. 1198) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Heinrich Schütz, *Saul, Saul, was verfolgst du mich?* (1650). from *Symphoniae sacrae III*. Antiphonal calls, rests, and sharply profiled consonants reveal how a large room can become part of rhetorical timing. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Saul, Saul, was verfolgst du mich?* (1650); identify which path carries its spatial identity.

J. S. Bach, *Passacaglia and Fugue in C minor, BWV 582* (c. 1710). Track the repeated bass through registration changes and ask when the building reinforces structure rather than merely adding grandeur. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Passacaglia and Fugue in C minor, BWV 582* (c. 1710) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Olivier Messiaen, *Apparition de l'église éternelle* (1932). The slow crescendo and decrescendo turn organ, low-frequency room modes, and silence into one monumental envelope. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Apparition de l'église éternelle* (1932) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Charlemagne Palestine, *Schlingen-Bläangen* (1974). on *Schlingen-Bläangen* (1974). Sustained organ overtones expose beating, modal reinforcement, and the point at which the room seems to continue the instrument. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Schlingen-Bläangen* (1974); test whether the form remains legible without level automation.

Guillaume Dufay, *Nuper rosarum flores* (1436). from *motet*. Proportional form and ceremonial placement invite listeners to separate claims about architectural symbolism from directly audible spatial organization. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Nuper rosarum flores* (1436) with original material, then preserve the exact render command and JSON report.

Johannes Ockeghem, *Missa prolationum* (c. 1470). from *cyclic mass*. Canonic voices at different mensural relationships require a resonant acoustic that blends sonority without erasing independent temporal layers. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Missa prolationum* (c. 1470) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Thomas Tallis, *If ye love me* (c. 1540). from *motet*. Clear syllabic writing provides a controlled reference for consonant intelligibility and harmonic warmth in contrasting church acoustics. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *If ye love me* (c. 1540); identify which path carries its spatial identity.

Charles-Marie Widor, *Symphony for Organ No. 5: Toccata* (1879). from *organ symphony*. Repeated manual figures and sustained pedal energy expose low-frequency decay, registration-dependent masking, and the building's contribution to apparent power. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Symphony for Organ No. 5: Toccata* (1879) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

César Franck, *Choral No. 3 in A minor* (1890). from *Trois chorals*. Dense chromatic harmony and organ sustain reveal when reverberation reinforces phrase architecture and when it obscures harmonic rhythm. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Choral No. 3 in A minor* (1890) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

György Ligeti, *Volumina* (1962). Clusters, changing wind pressure, and massive spectral blocks activate the organ and room as one nonlinear, slowly decaying instrument. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Volumina* (1962); test whether the form remains legible without level automation.

A.14.11 Opera, Theater, And Staged Distance

Claudio Monteverdi, *L'Orfeo: Toccata and Prologue* (1607). from *L'Orfeo*. Compare ceremonial brass projection with the prologue's vocal presence and note how contrasting spaces establish the drama before the plot advances. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *L'Orfeo: Toccata and Prologue* (1607) with original material, then preserve the exact render command and JSON report.

Wolfgang Amadeus Mozart, *Don Giovanni: Commendatore Scene* (1787). from *Don Giovanni*. The confrontation uses register, orchestration, stage position, and hall scale to turn apparent distance into supernatural authority. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Don Giovanni: Commendatore Scene* (1787) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Richard Wagner, *Parsifal: Transformation Music* (1882). from *Parsifal*. Listen for orchestral continuity making the imagined room appear to change size while the musical pulse remains suspended. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Parsifal: Transformation Music* (1882); identify which path carries its spatial identity.

Luciano Berio, *Coro* (1975–1976). on *Coro* (1975–1976). Singers paired with instruments create a distributed ensemble in which local timbres and collective resonance continually exchange roles. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Coro* (1975–1976) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Unsuk Chin, *Alice in Wonderland* (2007). on *Alice in Wonderland* (2007). Rapid changes of scale and orchestral perspective demonstrate how abrupt spatial contrast can support theatrical surrealism. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Alice in Wonderland* (2007) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Michel van der Aa, *Sunken Garden* (2013). on *Sunken Garden* (2013). Live voices, orchestra, electronics, and film-derived space invite a comparison between visible location and heard location. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Sunken Garden* (2013); test whether the form remains legible without level automation.

Henry Purcell, *Dido and Aeneas: Dido's Lament* (1689). from *Dido and Aeneas*. A grounded bass and exposed vocal line let hall decay extend grief without displacing the singer from the dramatic foreground. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Dido and Aeneas: Dido's Lament* (1689) with original material, then preserve the exact render command and JSON report.

Giuseppe Verdi, *Messa da Requiem: Dies irae* (1874). from *Messa da Requiem*. Explosive chorus and bass drum alternate with near-silence, making room recovery time and dynamic headroom essential to the theatrical effect. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Messa da Requiem: Dies irae* (1874) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Giacomo Puccini, *Tosca: Te Deum* (1900). from *Tosca*. Processional layers, bells, chorus, organ, and solo voice build a convincing vertical hierarchy within a single stage and pit perspective. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Tosca: Te Deum* (1900); identify which path carries its spatial identity.

Benjamin Britten, *Peter Grimes: Four Sea Interludes* (1945). from *Peter Grimes*. Orchestral depth and changing spectral weather translate an unseen environment into the psychological space surrounding the drama. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Peter Grimes: Four Sea Interludes* (1945) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Kaija Saariaho, *L'Amour de loin* (2000). on *L'Amour de loin* (2000). Orchestra and electronics mediate literal and emotional distance, making unattainable proximity the opera's central spatial condition. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *L'Amour de loin* (2000) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

George Benjamin, *Written on Skin* (2012). on *Written on Skin* (2012). Transparent orchestration and sharply controlled resonance permit sudden shifts between narrated distance, immediate violence, and suspended time. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Written on Skin* (2012); test whether the form remains legible without level automation.

A.14.12 Resonant Instruments And Chamber Scale

Claude Debussy, *La cathédrale engloutie* (1910). from *Préludes, Book I*. Pedal, register, and dynamics make a piano imply architecture; identify which depth cues exist before any recorded-room contribution. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *La cathédrale engloutie* (1910) with original material, then preserve the exact render command and JSON report.

John Cage, *In a Landscape* (1948). on *In a Landscape* (1948). A limited pitch field and gentle attack make pedal resonance and room decay audible as part of the piece's continuity. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *In a Landscape* (1948) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Morton Feldman, *Rothko Chapel* (1971). on *Rothko Chapel* (1971). Sparse attacks, voices, viola, and percussion leave the listener enough time to hear each resonance change the color of the next event. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Rothko Chapel* (1971); identify which path carries its spatial identity.

George Crumb, *The Phantom Gondolier* (1972). from *Makrokosmos, Volume I*. Extended piano techniques and vocalized sound create several apparent distances inside one instrument. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *The Phantom Gondolier* (1972) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Ryuichi Sakamoto, *andata* (2017). from *async*. Piano, mechanical detail, and electronic atmosphere balance tactile proximity against a broad, unstable background field. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *andata* (2017) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Kali Malone, *Spectacle of Ritual* (2019). from *The Sacrificial Code* (2019). Organ tuning and patient voice leading expose slow interference patterns that can be mistaken for modulation or moving reflections. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Spectacle of Ritual* (2019); test whether the form remains legible without level automation.

Ludwig van Beethoven, *Piano Sonata No. 14 in C-sharp minor, Op. 27 No. 2* (1801). from *piano sonata*. Repeated accompaniment and long pedal invite comparison between notated resonance, instrument decay, and the room's added harmonic blur. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Piano Sonata No. 14 in C-sharp minor, Op. 27 No. 2* (1801) with original material, then preserve the exact render command and JSON report.

Frédéric Chopin, *Prelude in E minor, Op. 28 No. 4* (1839). from *prelude*. Sparse attacks and changing inner voices make even a short room response consequential to harmonic pacing and cadential silence. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Prelude in E minor, Op. 28 No. 4* (1839) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Maurice Ravel, *Ondine* (1908). from *Gaspard de la nuit*. Rapid figuration, pedal, and register create a liquid halo whose clarity depends on balancing instrumental resonance against room decay. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Ondine* (1908); identify which path carries its spatial identity.

Tōru Takemitsu, *Rain Tree Sketch II* (1992). from *piano work*. Isolated sonorities and carefully weighted silence allow every decay to function as orchestration and formal punctuation. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Rain Tree Sketch II* (1992) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

John Luther Adams, *The Farthest Place* (2001). from *piano work*. Layered resonance and spacious pacing make the piano suggest an environment while retaining the physical detail of hammer and string. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *The Farthest Place* (2001) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Nils Frahm, *Says* (2013). from *Spaces*. A repeating synthesizer figure grows into a wide live field, providing a clear example of envelopment increasing while musical material remains economical. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Says* (2013); test whether the form remains legible without level automation.

A.14.13 Experimental Voice, Installation, And Feedback

Pauline Oliveros, Stuart Dempster, *and Panaiotis*, Lear (1989). from *Deep Listening* (1989). The Fort Worden cistern's extraordinary decay turns each performed sound into material for the next gesture. YouTube catalog search. Pauline Oliveros discography.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Lear* (1989) with original material, then preserve the exact render command and JSON report.

Meredith Monk, *Gotham Lullaby* (1981). from *Dolmen Music*. Layered vocal resonance demonstrates how timbre, vowel, and recording space can create depth without conventional orchestration. YouTube catalog search. Artist recording catalog.

> **Study prompt.** Match the apparent depth of *Gotham Lullaby* (1981) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Janet Cardiff, *The Forty Part Motet* (2001). from *sound installation*. Forty individually recorded voices on forty loudspeakers let the listener move physically through Tallis's counterpoint. YouTube catalog search. Artist installation record.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *The Forty Part Motet* (2001); identify which path carries its spatial identity.

Maryanne Amacher, *Head Rhythm 1 and Plaything 2* (1999). from *Sound Characters (Making the Third Ear)*. High-level spectral interactions challenge the boundary between reproduced sound, room response, and listener-generated auditory phenomena. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Head Rhythm 1 and Plaything 2* (1999) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Ellen Fullman and Okkyung Lee, *The Air Around Her, Part I* (2018). from *The Air Around Her*. Long strings and cello activate architectural scale through sustained partials, beating, and slowly changing excitation. YouTube catalog search. Long String Instrument documentation.

> **Study prompt.** Translate the spatial premise of *The Air Around Her, Part I* (2018) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Éliane Radigue, *Jetsun Mila* (1986). on *Jetsun Mila* (1986). Nearly static electronic layers reward close listening to minute spectral drift and the playback room's own contribution. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Jetsun Mila* (1986); test whether the form remains legible without level automation.

Steve Reich, *Come Out* (1966). from *tape work*. Phase divergence transforms one recorded voice from intelligible speech into rhythm, spectrum, and a virtual field of multiplying sources. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Come Out* (1966) with original material, then preserve the exact render command and JSON report.

Cathy Berberian, *Stripsody* (1966). from *solo vocal work*. Rapid shifts among vocal characters test whether acoustic support preserves theatrical articulation without homogenizing timbre. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Stripsody* (1966) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Joan La Barbara, *Circular Song* (1975). from *Voice Is the Original Instrument*. Continuous breath, multiphonics, and resonant vocal technique complicate the distinction between source sustain and added reverberation. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Circular Song* (1975); identify which path carries its spatial identity.

Robert Ashley, *Automatic Writing* (1979). on *Automatic Writing* (1979). Whispered speech, electronics, and close-room ambiguity create an unstable boundary between private utterance and environmental sound. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Automatic Writing* (1979) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Trevor Wishart, *Vox 5* (1986). from *VOX Cycle*. Continuous transformation between voice and synthetic resonance offers a reference for spaces whose apparent material changes with the source. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Vox 5* (1986) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Pamela Z, *An In* (1998). from *A Delay Is Better*. Live voice, looping, and delay build contrapuntal layers whose intelligibility depends on precise timing and selective spectral space. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *An In* (1998); test whether the form remains legible without level automation.

A.14.14 Soul, Gospel, Country, And Vocal Presence

Sam Cooke, *A Change Is Gonna Come* (1964). from *Ain't That Good News*. The lead remains intimate while strings and chamber-like depth expand the emotional horizon around it. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *A Change Is Gonna Come* (1964) with original material, then preserve the exact render command and JSON report.

Aretha Franklin, *Amazing Grace* (1972). on *Amazing Grace* (1972). The live church recording joins solo voice, congregation, choir, and room response into one continuously changing performance system. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Amazing Grace* (1972) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Daniel Lanois, *The Maker* (1989). from *Acadie*. Percussion, voice, guitar, and dark ambience occupy distinct layers that remain legible despite a strongly atmospheric mix. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *The Maker* (1989); identify which path carries its spatial identity.

Chris Isaak, *Wicked Game* (1989). from *Heart Shaped World*. A close vocal and highly spacious guitar show how two sources can imply different rooms without breaking the song's unity. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Wicked Game* (1989) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Emmylou Harris, *Where Will I Be* (1995). from *Wrecking Ball*. Diffuse production enlarges the voice and instruments while preserving consonants, pulse, and a stable emotional foreground. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Where Will I Be* (1995) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Weyes Blood, *Movies* (2019). from *Titanic Rising*. The arrangement moves from intimate electronic detail toward orchestral bloom, making depth growth part of the song's form. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Movies* (2019); test whether the form remains legible without level automation.

Elvis Presley, *Blue Moon* (1954). from *Elvis Presley*. A distant, wavering vocal return contrasts with sparse accompaniment, making early studio echo part of the singer's vulnerable persona. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Blue Moon* (1954) with original material, then preserve the exact render command and JSON report.

The Staple Singers, *I'll Take You There* (1972). from *Be Altitude: Respect Yourself*. Economical groove and vocal call-and-response show how a compact ambience can support communal energy without softening rhythmic edges. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *I'll Take You There* (1972) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Dolly Parton, *I Will Always Love You* (1974). from *Jolene*. A direct lead vocal and restrained instrumental field demonstrate how subtle depth can preserve narrative intimacy at emotional peaks. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *I Will Always Love You* (1974); identify which path carries its spatial identity.

Bruce Springsteen, *I'm on Fire* (1984). from *Born in the U.S.A.*. Quiet voice, short percussion, and nocturnal ambience create psychological distance through low-level detail rather than spectacular decay. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *I'm on Fire* (1984) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Cowboy Junkies, *Sweet Jane* (1988). from *The Trinity Session*. A single ambisonic microphone in a resonant church captures performers and architecture as one scene, making placement inseparable from balance. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Sweet Jane* (1988) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Lucinda Williams, *Right in Time* (1998). from *Car Wheels on a Gravel Road*. Voice and band combine dryness, slap-like depth, and controlled sustain to keep lyrical presence inside a textured roots production. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Right in Time* (1998); test whether the form remains legible without level automation.

A.14.15 Heavy Music, Post-Rock, And Overwhelming Scale

Swans, *The Sound* (1996). from *Soundtracks for the Blind*. Long crescendos and saturated layers test whether a large field can grow without losing its central pulse and harmonic direction. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *The Sound* (1996) with original material, then preserve the exact render command and JSON report.

Sigur Rós, *Svefn-g-englar* (1999). from *Ágætis byrjun*. Bowed guitar, voice, and spacious drums demonstrate how slow attacks and long tails can remain rhythmically articulate. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Svefn-g-englar* (1999) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Godspeed You! Black Emperor, *Storm* (2000). from *Lift Your Skinny Fists Like Antennas to Heaven*. Ensemble accumulation and changing recording perspective make apparent room size rise with musical intensity. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Storm* (2000); identify which path carries its spatial identity.

Boris, *Flood III* (2000). from *Flood*. Distortion and extended decay fuse into a dense continuum whose clarity depends on spectral allocation rather than dryness. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Flood III* (2000) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Sunn O))), *It Took the Night to Believe* (2005). from *Black One*. Low-frequency sustain and enormous apparent volume reveal how room modes, distortion, and feedback can become orchestration. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *It Took the Night to Believe* (2005) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Deafheaven, *Dream House* (2013). from *Sunbather*. Fast drums and wide guitar fields provide a difficult test of transient preservation inside a bright, high-density wash. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Dream House* (2013); test whether the form remains legible without level automation.

Black Sabbath, *Black Sabbath* (1970). on *Black Sabbath* (1970). Rain, tolling bell, sparse guitar, and long decay establish dread before the ensemble's weight arrives, making environment part of the riff's meaning. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Black Sabbath* (1970) with original material, then preserve the exact render command and JSON report.

Joy Division, *Atmosphere* (1980). from *Licht und Blindheit*. Distant drums, baritone voice, and synthetic haze turn production depth into the song's defining emotional architecture. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Atmosphere* (1980) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Talk Talk, *After the Flood* (1991). from *Laughing Stock*. Organ, guitar distortion, room tone, and dynamic restraint create scale through accumulated resonance rather than continuous loudness. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *After the Flood* (1991); identify which path carries its spatial identity.

Slowdive, *When the Sun Hits* (1993). from *Souvlaki*. Vocal intimacy and broad guitar wash demonstrate how midrange density can remain emotionally legible when attacks and tails overlap. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *When the Sun Hits* (1993) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Mogwai, *Mogwai Fear Satan* (1997). from *Mogwai Young Team*. A long crescendo moves from exposed instrumental room to saturated mass, revealing how decay strategy must change with level and density. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Mogwai Fear Satan* (1997) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Cult of Luna, *Vicarious Redemption* (2013). from *Vertikal*. Layered guitars, electronics, and drums construct industrial scale while alternating narrow pressure with wide, high-density release. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Vicarious Redemption* (2013); test whether the form remains legible without level automation.

A.14.16 Dub Techno And Electronic Depth

Basic Channel, *Quadrant Dub I* (1994). from *Quadrant Dub*. Filtered chords and delay create a room whose boundaries are defined by repetition, noise, and spectral erosion rather than early reflections. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *Quadrant Dub I* (1994) with original material, then preserve the exact render command and JSON report.

Rhythm & Sound with Cornel Campbell, *King in My Empire* (2001). on *King in My Empire* (2001). Voice and rhythm remain centered while echoes recede into a dark, slowly modulated field. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *King in My Empire* (2001) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Deepchord Presents Echospace, *First Point of Aries* (2007). from *The Coldest Season*. Layered delays, hiss, and diffuse chords make depth emerge from many low-contrast events rather than one obvious tail. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *First Point of Aries* (2007); identify which path carries its spatial identity.

Actress, *Hubble* (2010). from *Splazsh*. Unstable transients and murky perspective show how partial obscurity can function as groove and timbral identity. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Hubble* (2010) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Andy Stott, *Numb* (2012). from *Luxury Problems*. A distant voice and heavy low-frequency structure demonstrate how spectral separation can support radically different apparent distances. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Numb* (2012) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Kelly Lee Owens, *Arthur* (2017). from *Kelly Lee Owens*. Pulses, voice fragments, and expanding returns create movement by changing envelopment while the center remains restrained. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Arthur* (2017); test whether the form remains legible without level automation.

Maurizio, *M4* (1995). from *M-Series*. A restrained pulse and slowly changing chord return demonstrate how filtering and feedback can imply vast depth with very few events. YouTube catalog search.

> **Study prompt.** Rebuild one 20–30 second spatial gesture from *M4* (1995) with original material, then preserve the exact render command and JSON report.

Porter Ricks, *Port Gentil* (1996). from *Biokinetics*. Submerged transients and pressure-like modulation make the field feel physical even when ordinary source and room boundaries disappear. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Port Gentil* (1996) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Pole, *Fahren* (1998). from *CD 1*. Clicks, bass, and unstable delay reveal how faults and residual noise can articulate the edges of a synthetic acoustic. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *Fahren* (1998); identify which path carries its spatial identity.

Burial, *Archangel* (2007). from *Untrue*. Pitch-shifted voice, rain-like noise, and displaced percussion create an urban depth field from fragments rather than continuous reverberation. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Archangel* (2007) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Yagya, *Rigning One* (2009). from *Rigning*. Rain and soft chord pulses maintain multiple time scales, allowing listeners to compare environmental continuity with musical decay. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Rigning One* (2009) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Vladislav Delay, *Kuopio* (2012). on *Kuopio* (2012). Irregular echo, clipped events, and broad low-frequency motion challenge assumptions that depth requires smooth or naturalistic late fields. YouTube catalog search.

Study prompt. Automate only one spatial parameter across a short form inspired by *Kuopio* (2012); test whether the form remains legible without level automation.

A.14.17 Soundscape Composition And Environmental Scale

Luc Ferrari, *Presque rien No. 1: Le Lever du jour au bord de la mer* (1970). from *Presque Rien*. A recorded place unfolds as composition, making source distance, perspective, and time of day primary musical parameters. YouTube catalog search.

Study prompt. Rebuild one 20–30 second spatial gesture from *Presque rien No. 1: Le Lever du jour au bord de la mer* (1970) with original material, then preserve the exact render command and JSON report.

R. Murray Schafer, *Music for Wilderness Lake* (1979). from *environmental work*. Trombonists positioned around a lake use propagation delay and landscape reflection as literal parts of ensemble timing. YouTube catalog search.

Study prompt. Match the apparent depth of *Music for Wilderness Lake* (1979) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Annea Lockwood, *A Sound Map of the Hudson River* (1982). on *A Sound Map of the Hudson River* (1982). Changing water textures and recording perspectives construct a long-form portrait whose spaces are real yet editorially composed. YouTube catalog search. Composer work catalog.

Study prompt. Create dry, early-only, late-only, and complete reconstructions of one cue from *A Sound Map of the Hudson River* (1982); identify which path carries its spatial identity.

Hildegard Westerkamp, *Kits Beach Soundwalk* (1989). from *Transformations*. Voice, city, water, and microscopic shoreline sounds shift scale through recording perspective and studio transformation. YouTube catalog search. Composer work page.

Study prompt. Analyze a lawful excerpt or an original reconstruction of *Kits Beach Soundwalk* (1989) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Chris Watson, *Vatnajökull* (2003). from *Weather Report*. Time-compressed glacier recordings turn environmental change into an eighteen-minute spectral and spatial trajectory. YouTube catalog search. Artist release notes.

Study prompt. Translate the spatial premise of *Vatnajökull* (2003) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Jana Winderen, *Aquaculture* (2010). from *Energy Field*. Hydrophones reveal underwater activity whose unfamiliar source cues require the listener to infer scale from spectrum and motion. YouTube catalog search. Artist release notes.

Study prompt. Automate only one spatial parameter across a short form inspired by *Aquaculture* (2010); test whether the form remains legible without level automation.

Pierre Schaeffer, *Étude aux chemins de fer* (1948). from *Cinq études de bruits*. Recorded trains are reorganized so perspective, mechanical rhythm, and captured environment become compositional parameters independent of their original scene. YouTube catalog search.

Study prompt. Rebuild one 20–30 second spatial gesture from *Étude aux chemins de fer* (1948) with original material, then preserve the exact render command and JSON report.

Iannis Xenakis, *Bohor* (1962). from *electroacoustic work*. Dense metallic spectra and sustained energy create a virtual enclosure whose scale changes with playback level, speaker spacing, and room response. YouTube catalog search.

> **Study prompt.** Match the apparent depth of *Bohor* (1962) with convolution and algorithmic engines at equal loudness; compare pre-delay, direct-to-reverberant ratio, and high-band decay.

Francisco López, *La Selva* (1998). on *La Selva* (1998). Rainforest recordings remove visual explanation and foreground the listener's changing inference of distance, density, and biological activity. YouTube catalog search.

> **Study prompt.** Create dry, early-only, late-only, and complete reconstructions of one cue from *La Selva* (1998); identify which path carries its spatial identity.

Toshiya Tsunoda, *Scenery from a Lighthouse* (2003). on *Scenery from a Lighthouse* (2003). Vibration and environmental detail expose transmission through structures, making contact resonance as important as airborne perspective. YouTube catalog search.

> **Study prompt.** Analyze a lawful excerpt or an original reconstruction of *Scenery from a Lighthouse* (2003) with `verbx analyze`; compare audible decay with broadband and frequency-dependent measurements.

Peter Cusack, *Sounds from Dangerous Places* (2012). on *Sounds from Dangerous Places* (2012). Recordings from environmentally damaged sites ask how audible place, documentation, and political context alter the interpretation of acoustic space. YouTube catalog search.

> **Study prompt.** Translate the spatial premise of *Sounds from Dangerous Places* (2012) to mono, stereo, and surround; document which distance and envelopment cues survive each layout.

Lawrence English, *Cruel Optimism* (2017). on *Cruel Optimism* (2017). Field recordings, voice, and dense electronic layers move between recognizable environment and overwhelming abstraction without a fixed listening distance. YouTube catalog search.

> **Study prompt.** Automate only one spatial parameter across a short form inspired by *Cruel Optimism* (2017); test whether the form remains legible without level automation.

A.15 Selected Primary Documentation

These creator and institutional pages document works whose site, installation, release, or recording history is especially important to the listening note. The list is alphabetical by creator.

- Annea Lockwood. *A Sound Map of the Hudson River* (1982). Composer work catalog.
- Chris Watson. *Vatnajökull* (2003). Artist release notes.
- Ellen Fullman and Okkyung Lee. *The Air Around Her, Part I* (2018). Long String Instrument documentation.
- Hildegard Westerkamp. *Kits Beach Soundwalk* (1989). Composer work page.
- Jana Winderen. *Aquaculture* (2010). Artist release notes.
- Janet Cardiff. *The Forty Part Motet* (2001). Artist installation record.
- Meredith Monk. *Gotham Lullaby* (1981). Artist recording catalog.
- Pauline Oliveros, Stuart Dempster, and Panaiotis. *Lear* (1989). Pauline Oliveros discography.

A.16 Appendix A Listening Record

Use one record for each focused comparison. The aim is to preserve enough evidence that another reader can repeat the listening conditions, understand the inference, and rebuild the spatial principle without copying the recording.

- **Source and version.** Record the work, performer, release, mix or remaster, playback service or physical medium, and the exact passage studied. Note when a catalog search led to a different recording than the one described in the entry.
- **Playback conditions.** Record sample rate, channel layout, loudspeaker or headphone system, listening level, room, normalization state, and whether any host spatialization or room correction was active.
- **Spatial event map.** Mark direct attacks, first audible reflections, echo clusters, late-field blooms, gates, feedback gestures, movement, and silence. Use elapsed time in seconds so observations can be checked without reproducing lyrics or score text.
- **Measured evidence.** Save `verbx analyze` JSON for lawful source material or for the reconstruction. Separate measured quantities from perceptual descriptions, and do not present program-material RT60 estimates as calibrated room measurements.
- **Reconstruction.** Preserve the dry source, exact command or plug-in state, random seed where relevant, output format, and one deliberately contrasting render. State which property was held constant during the comparison.
- **Listening judgment.** Describe what the reverberation contributes to rhythm, form, harmony, orchestration, drama, or place. End by naming one parameter that mattered less than expected and one interaction that mattered more.

Two-pass reconstruction protocol.

1. **Establish the baseline.** Trim a short original source, set conservative monitor gain, and render a dry reference. Write a prediction for apparent size, distance, decay color, and movement before choosing an engine or preset.
2. **Build controlled layers.** Create early-only and late-only versions before the complete reconstruction. Change one parameter family at a time, level-match every comparison, and keep at least one intentionally wrong version that makes the role of the target cue easier to hear.
3. **Test without labels.** Randomize the dry reference, reconstruction, and contrasting render. Ask at least one listener to rank distance, clarity, envelopment, and musical fit without seeing the settings. Record disagreement rather than averaging it away.
4. **Archive the evidence.** Save audio hashes, commands or plug-in state, reports, routing notes, and a short conclusion. If the study uses a commercial recording for listening, archive only bibliographic and timing information unless distribution rights permit more.

A complete record should fit on this page. It is evidence for critical listening, not a claim that one preset or metric explains the artistic result. The strongest conclusion names the audible interaction, the controlled comparison that exposed it, and the limit beyond which the reconstruction stopped serving the music.

"Sweetest Echo, sweetest nymph, that liv'st
unseen within thy airy shell."

— John Milton, *Comus* (1637)

Appendix B

Educational Exercises and Project Assignments

These projects form a progressive laboratory course in reverberation, spatial audio, realtime systems, critical listening, musical composition, and reproducible audio engineering. Twenty-four projects are explicitly composition-centered, and every technical exercise may be adapted to original musical material. Each assignment is scoped for one book page; instructors may treat it as a weekly laboratory, combine adjacent projects, or select a focused sequence.

B.1 Assessment Framework

Evaluate every submission on prediction, method, evidence, listening judgment, reproducibility, and clarity. Unless a project states otherwise, students should retain source audio, exact commands, parameter or preset state, analysis JSON, plots, and a short reflection distinguishing observation from interpretation.

B.2 Laboratory Protocol

The technical projects share one experimental protocol so that their individual pages can concentrate on the question at hand. Begin with a dry baseline and one conservative reference render. Create at least five documented variants across the useful range, including one deliberately poor case when it can be produced without risking equipment or hearing. Change one principal variable at a time, loudness-match comparisons, retain deterministic seeds, and record effective settings rather than relying on shell history alone.

Unless a project specifies a different submission, provide the source and rendered excerpts, runnable commands, presets or automation, analysis JSON, a compact comparison table, and one labeled figure. Record listening conditions, sample rate, channel layout, block or partition size, software revision, and warnings. Another student should be able to reproduce the central result, and conclusions should cite both measurement and level-matched listening rather than screenshots alone.

Every technical report should answer the same five questions: Which prediction was confirmed? Which result contradicted the model? What changed perceptually before a standard metric changed? Where does the preferred setting lie relative to a failure boundary? Which confound deserves a follow-up experiment? Finish by repeating the decisive comparison with a contrasting source, room, sample rate, or reproduction layout and state which conclusions generalize.

B.3 Project 1: Calibrate a Natural Room

Central concepts: RT60, pre-delay, early/late balance.

Design brief. Create three believable rooms for speech, percussion, and strings from one dry recording. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-1 presents Project 1 laboratory cycle: Calibrate a Natural Room and establishes the visual reference for the discussion that follows.

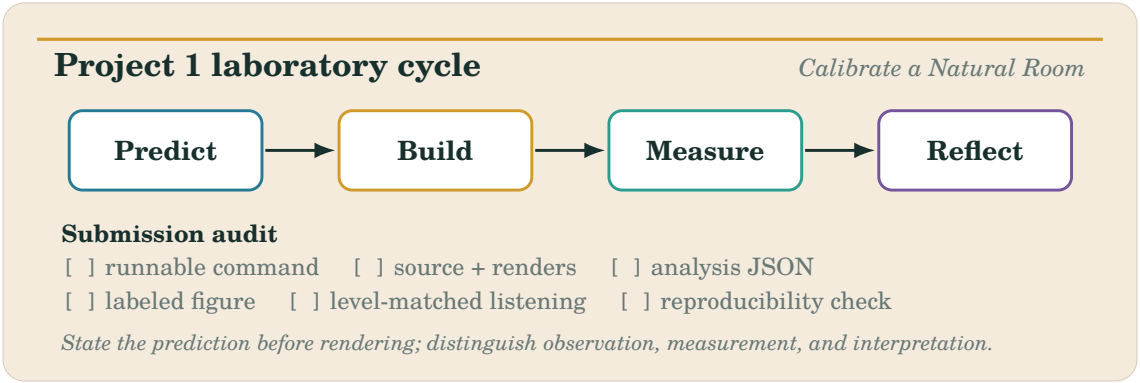


Figure B-1: Project 1 laboratory cycle: Calibrate a Natural Room

B.4 Project 2: Map the Reverb Event

Central concepts: direct sound, early reflections, late decay.

Design brief. Annotate arrival times and energy regions in an impulse response and in a rendered waveform. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-2 presents Project 2 laboratory cycle: Map the Reverb Event and establishes the visual reference for the discussion that follows.

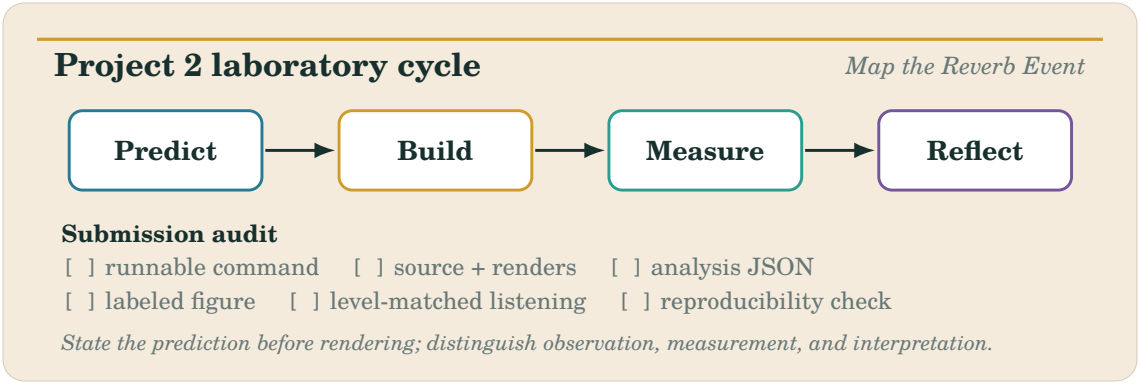


Figure B-2: Project 2 laboratory cycle: Map the Reverb Event

B.5 Project 3: Coarse and Fine RT60 Laboratory

Central concepts: logarithmic control and parameter resolution.

Design brief. Measure how coarse and fine controls distribute useful adjustment over 0.01 to 360 seconds. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-3 presents Project 3 laboratory cycle: Coarse and Fine RT60 Laboratory and establishes the visual reference for the discussion that follows.

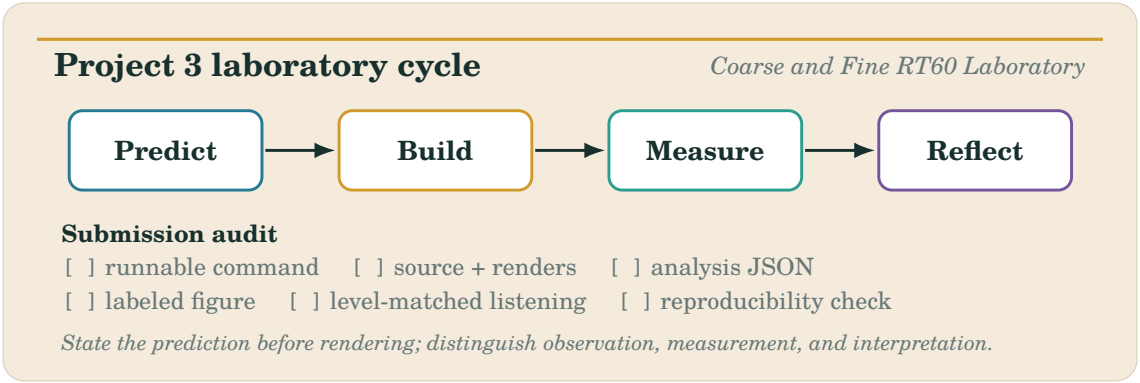


Figure B-3: Project 3 laboratory cycle: Coarse and Fine RT60 Laboratory

B.6 Project 4: Frequency-Dependent Decay

Central concepts: multiband RT60 and damping.

Design brief. Design warm, neutral, and bright tails with equal broadband RT60 but different band decay. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-4 presents Project 4 laboratory cycle: Frequency-Dependent Decay and establishes the visual reference for the discussion that follows.

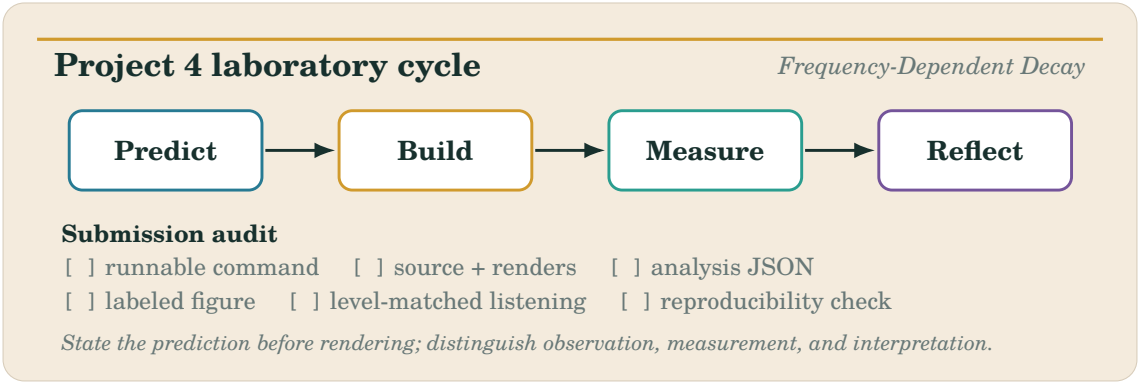


Figure B-4: Project 4 laboratory cycle: Frequency-Dependent Decay

B.7 Project 5: Pre-Delay and Apparent Distance

Central concepts: depth cues and source separation.

Design brief. Hold loudness constant while testing whether pre-delay changes foreground placement. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-5 presents Project 5 laboratory cycle: Pre-Delay and Apparent Distance and establishes the visual reference for the discussion that follows.

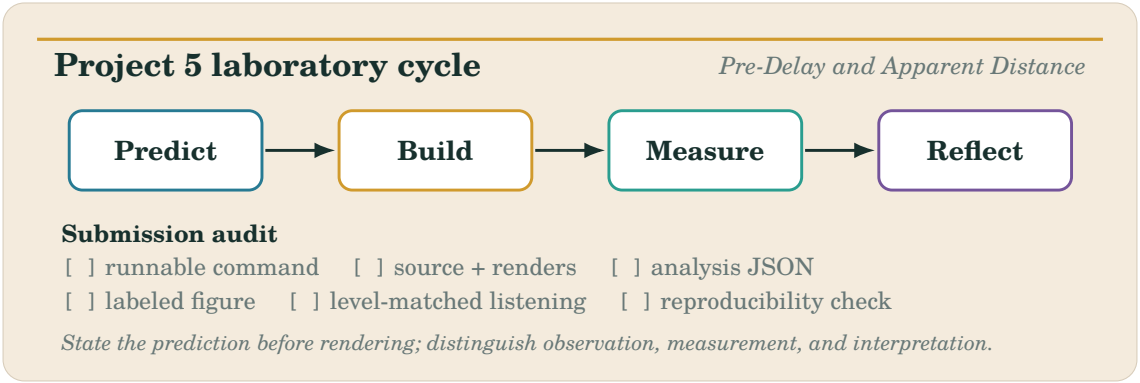


Figure B-5: Project 5 laboratory cycle: Pre-Delay and Apparent Distance

B.8 Project 6: Early-Reflection Geometry

Central concepts: room dimensions and directional cues.

Design brief. Compare compact, wide, and tall reflection patterns using the same late field. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-6 presents Project 6 laboratory cycle: Early-Reflection Geometry and establishes the visual reference for the discussion that follows.

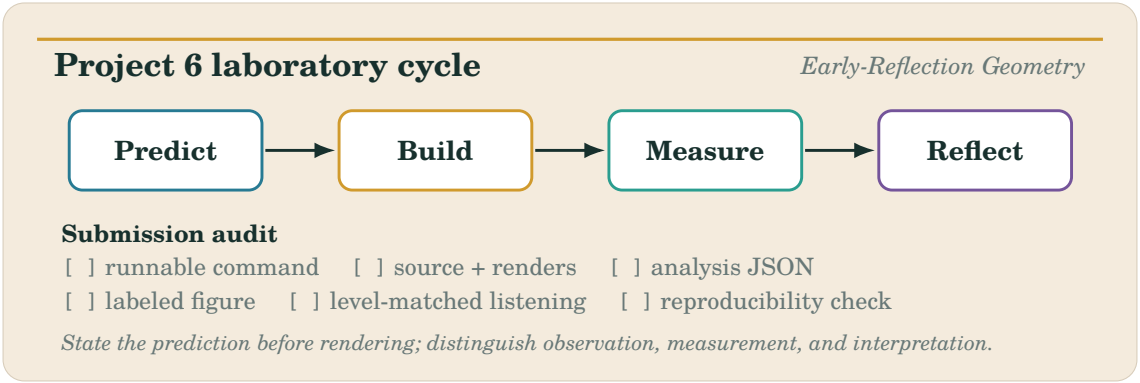


Figure B-6: Project 6 laboratory cycle: Early-Reflection Geometry

B.9 Project 7: Diffusion Build-Up

Central concepts: echo density and allpass structure.

Design brief. Find the transition from discrete echoes to a smooth field for three source types. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-7 presents Project 7 laboratory cycle: Diffusion Build-Up and establishes the visual reference for the discussion that follows.

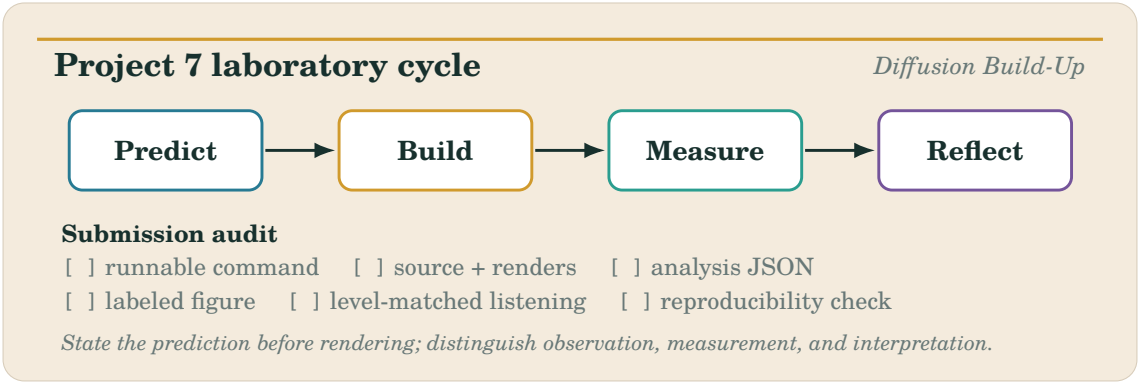


Figure B-7: Project 7 laboratory cycle: Diffusion Build-Up

B.10 Project 8: FDN Matrix Listening Test

Central concepts: feedback matrices and modal texture.

Design brief. Blind-compare matrix families under matched delay lengths, RT60, and output level. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-8 presents Project 8 laboratory cycle: FDN Matrix Listening Test and establishes the visual reference for the discussion that follows.

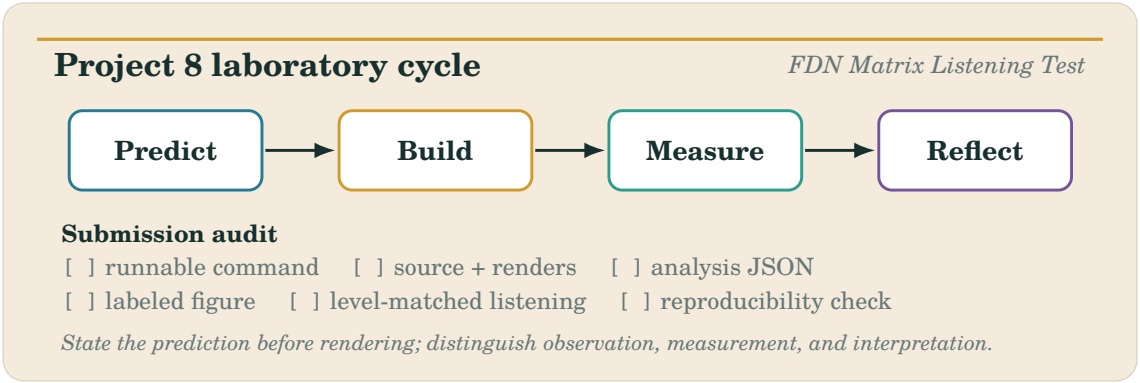


Figure B-8: Project 8 laboratory cycle: FDN Matrix Listening Test

B.11 Project 9: Delay-Line Distribution

Central concepts: mode spacing and metallic coloration.

Design brief. Construct prime, randomized, and clustered delay sets and relate spectra to listening notes. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-9 presents Project 9 laboratory cycle: Delay-Line Distribution and establishes the visual reference for the discussion that follows.

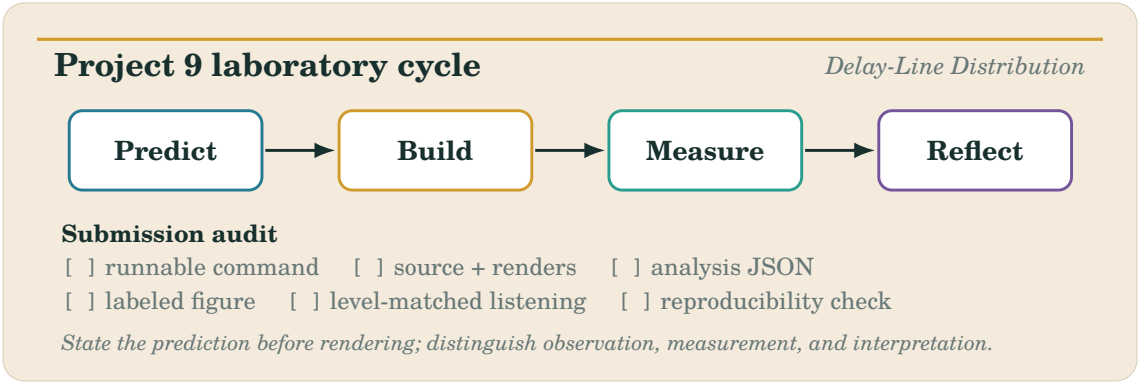


Figure B-9: Project 9 laboratory cycle: Delay-Line Distribution

B.12 Project 10: Compose with Infinite Sustain

Project mode: Musical composition and production.

Central concepts: freeze as harmonic memory, voice-leading, and density control.

Design brief. Compose a miniature in which captured tails become sustained harmonic material rather than background ambience. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write a two- to three-minute piece in three connected sections. Introduce a compact motif or chord field, capture at least three freeze states at structurally important moments, and release or filter them so each frozen layer changes the harmony or orchestration. Print separate dry, ordinary-wet, and frozen-return stems; leave deliberate rests so listeners can hear how the room memory develops. Test every sustained state for peak growth and spectral crowding before the final render.

Evidence package. Submit a stereo or multichannel master, dry and return stems, a one-page formal or score map marking every capture and release, exact commands or plug-in automation, presets, analysis JSON, and a labeled spectrogram or energy plot. Identify the pitch content retained by each frozen state and document the safety settings that keep it stable.

Questions for the report. When does a frozen tail become a musical voice rather than an effect? How does each capture alter the perceived harmony, density, or meter? Which register accumulates fastest, and what compositional decision solves that buildup without merely turning the return down?

Completion standard. The piece must make at least three clearly intentional freeze gestures that participate in voice-leading or form. A listener should be able to describe the harmonic function of the sustained material, and the full-duration render must remain stable and unclipped.

Extension. Adapt the piece for realtime performance, assigning capture, release, damping, and clear actions to playable controls; document a recovery gesture for an accidental or unstable freeze.

Figure B-10 presents Project 10 laboratory cycle: Compose with Infinite Sustain and establishes the visual reference for the discussion that follows.

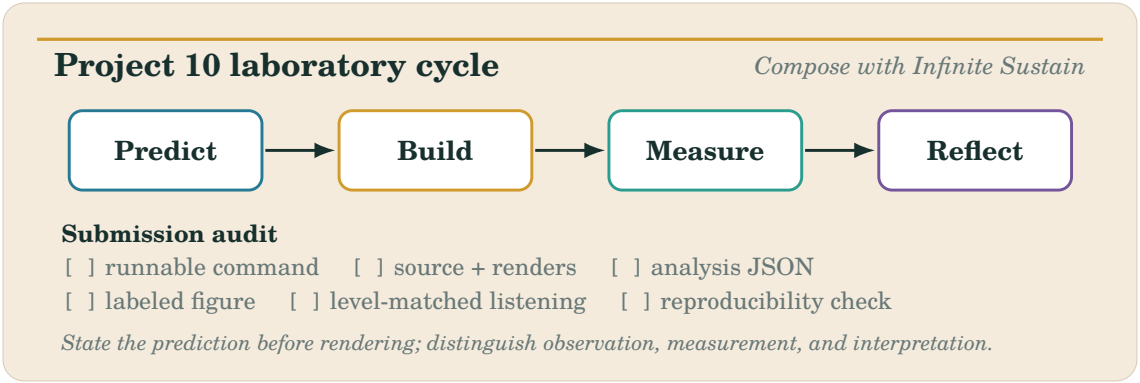


Figure B-10: Project 10 laboratory cycle: Compose with Infinite Sustain

B.13 Project 11: Reverse-Reverb Phrase Study

Project mode: Musical composition and production.

Central concepts: anacrusis, envelope reversal, and transition timing.

Design brief. Compose phrase-leading reverse tails whose arrivals articulate pickups, cadences, and formal boundaries. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Choose a tempo and compose a 60- to 90-second solo, vocal, or chamber etude with at least four reverse-reverb anticipations. Place one before an opening entrance, one across an internal phrase boundary, one at a cadence, and one that deliberately misdirects the listener. Align each swell to the destination attack in beats or milliseconds, then vary duration, spectral emphasis, and wet level without moving the performed note.

Evidence package. Submit the score or annotated timeline, dry and processed masters, isolated reverse-return stems, exact commands or automation, and a table relating tempo subdivisions to swell durations. Include analysis JSON and one labeled waveform showing the destination transient and at least two reverse envelopes.

Questions for the report. Which swells sound like pickups, crescendos, breaths, or causal impossibilities? How early can the destination be predicted without weakening surprise? Does the reverse gesture reinforce the meter, create a cross-rhythm, or suspend pulse?

Completion standard. At least four reverse events must have distinct phrase functions, land accurately, and preserve the destination transient. The musical result must still read as a coherent composition when the reverse-return stem is muted, while the processed version must reveal a stronger formal design.

Extension. Recompose one passage so the reverse tail supplies the only audible preparation for a silent or omitted destination note.

Figure B-11 presents Project 11 laboratory cycle: Reverse-Reverb Phrase Study and establishes the visual reference for the discussion that follows.

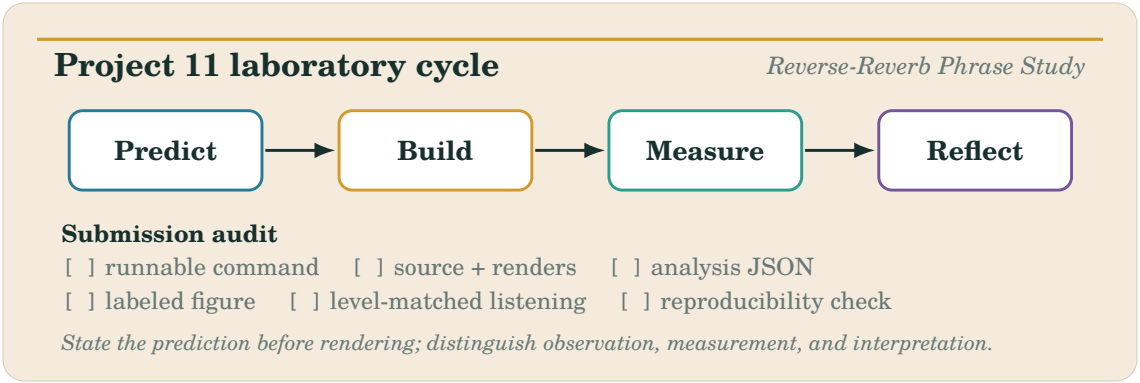


Figure B-11: Project 11 laboratory cycle: Reverse-Reverb Phrase Study

B.14 Project 12: Shimmer Canon and Harmonic Field

Project mode: Musical composition and production.

Central concepts: pitch-shifted feedback, canonic imitation, and spectral register.

Design brief. Write a canonic miniature in which shimmer intervals extend a limited pitch collection into a controlled harmonic field. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Limit the source to a five- to seven-note pitch collection and write a two-minute canon for two or more voices. Choose two shimmer intervals, such as an octave and a fifth, and assign each a formal role. Introduce voices one at a time, automate feedback and damping to move from sparse imitation to a shared harmonic cloud, and reserve the brightest register for a single structural climax.

Evidence package. Submit a notated score or piano-roll reduction, dry voices, shimmer returns, final master, pitch-interval map, presets or automation, analysis JSON, and a labeled time-frequency plot. Mark every point at which the feedback path creates a pitch not present in the performed material.

Questions for the report. Does the shimmer field preserve the canon or dissolve it? Which generated partials function as harmony, orchestration, or noise? How do feedback, damping, and source register change the perceived consonance of the same transposition interval?

Completion standard. The piece must demonstrate intelligible imitation, purposeful harmonic expansion, and a controlled spectral climax without runaway high-frequency energy. Both shimmer intervals must be identifiable by ear or convincingly justified in the analysis.

Extension. Create a second realization using a non-octave interval and revise the source pitch collection so the feedback field remains harmonically coherent.

Figure B-12 presents Project 12 laboratory cycle: Shimmer Canon and Harmonic Field and establishes the visual reference for the discussion that follows.

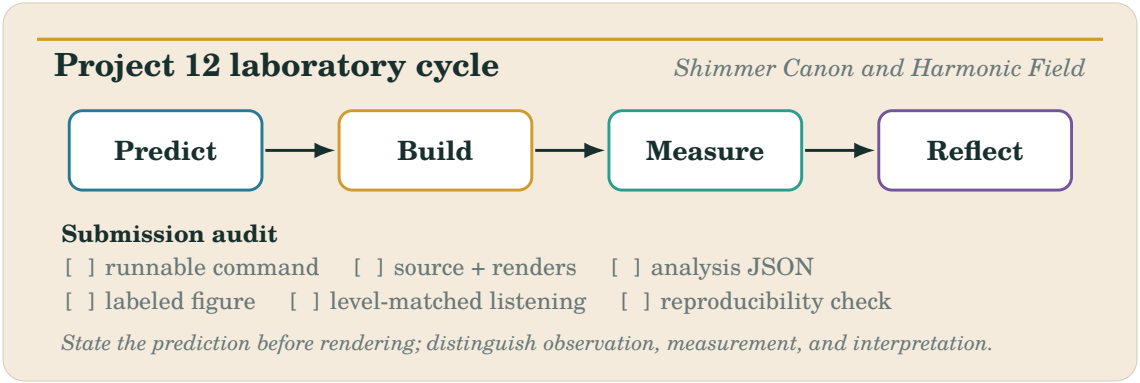


Figure B-12: Project 12 laboratory cycle: Shimmer Canon and Harmonic Field

B.15 Project 13: Ducked-Reverb Rhythmic Counterpoint

Project mode: Musical composition and production.

Central concepts: sidechain rhythm, negative space, and call-and-response.

Design brief. Compose a groove in which the wet return answers the dry material with clearly perceived rhythmic counterpoint. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose a 24- to 32-bar groove with a dry foreground line and a reverb return designed as its answering voice. Build three sections using release times related to contrasting subdivisions, such as eighth notes, dotted eighths, and quarter-note triplets. Shape the source rhythm and its rests so the return articulates syncopations rather than merely filling gaps; retain one passage where ducking is bypassed for comparison.

Evidence package. Submit a tempo map, rhythmic reduction of dry attacks and wet responses, dry and return stems, bypass comparison, final master, exact ducking and reverb settings, analysis JSON, and a labeled envelope plot covering at least two bars.

Questions for the report. Where does the return behave like echo, sustain, or a separate contrapuntal layer? Which release relationship strengthens the groove, and which creates metric ambiguity? How much source silence is required before the wet rhythm becomes independently legible?

Completion standard. The wet return must produce three audibly different rhythmic roles while the foreground remains intelligible. At least one section must create a convincing call-and-response pattern that disappears when ducking is bypassed.

Extension. Arrange the return as a second percussion part for live players, then compare the performed and signal-processed counterpoints.

Figure B-13 presents Project 13 laboratory cycle: Ducked-Reverb Rhythmic Counterpoint and establishes the visual reference for the discussion that follows.

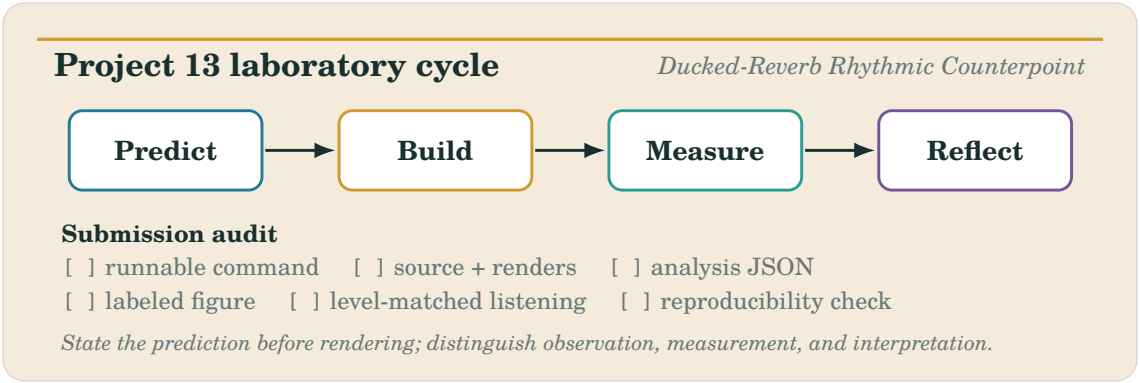


Figure B-13: Project 13 laboratory cycle: Ducked-Reverb Rhythmic Counterpoint

B.16 Project 14: Limiter Transparency

Central concepts: lookahead, knee, release, and true peak.

Design brief. Protect an extreme wet mix while minimizing transient flattening and pumping. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-14 presents Project 14 laboratory cycle: Limiter Transparency and establishes the visual reference for the discussion that follows.

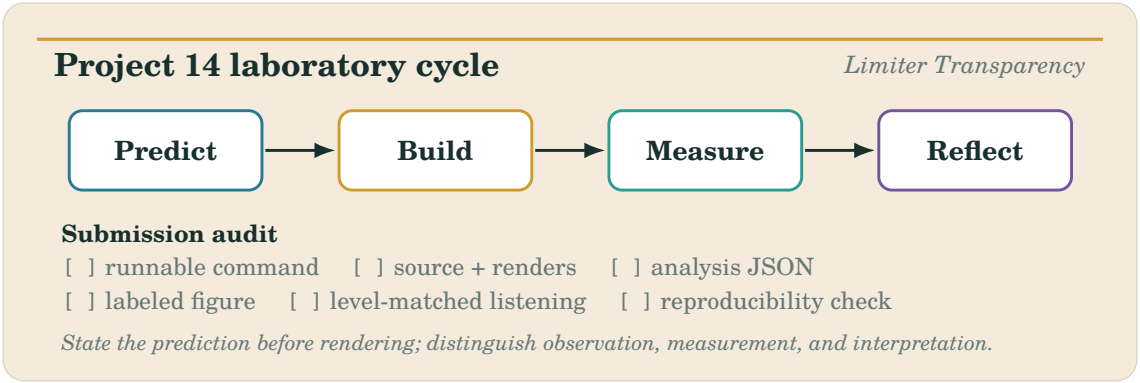


Figure B-14: Project 14 laboratory cycle: Limiter Transparency

B.17 Project 15: Convolution Partition Plan

Central concepts: latency versus throughput.

Design brief. Select partition sizes for live monitoring, music production, and offline archival rendering. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-15 presents Project 15 laboratory cycle: Convolution Partition Plan and establishes the visual reference for the discussion that follows.

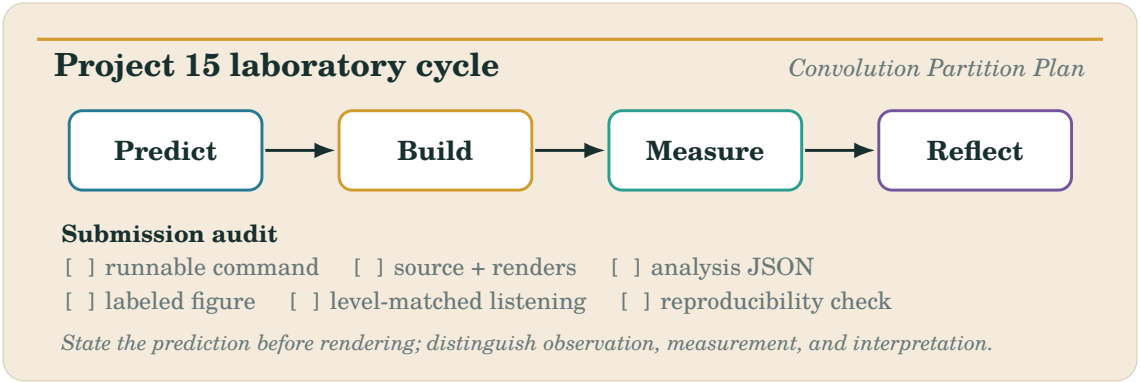


Figure B-15: Project 15 laboratory cycle: Convolution Partition Plan

B.18 Project 16: Impulse-Response Capture

Central concepts: sweeps, deconvolution, and noise.

Design brief. Plan and document a repeatable room capture with calibration and failure checks. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-16 presents Project 16 laboratory cycle: Impulse-Response Capture and establishes the visual reference for the discussion that follows.

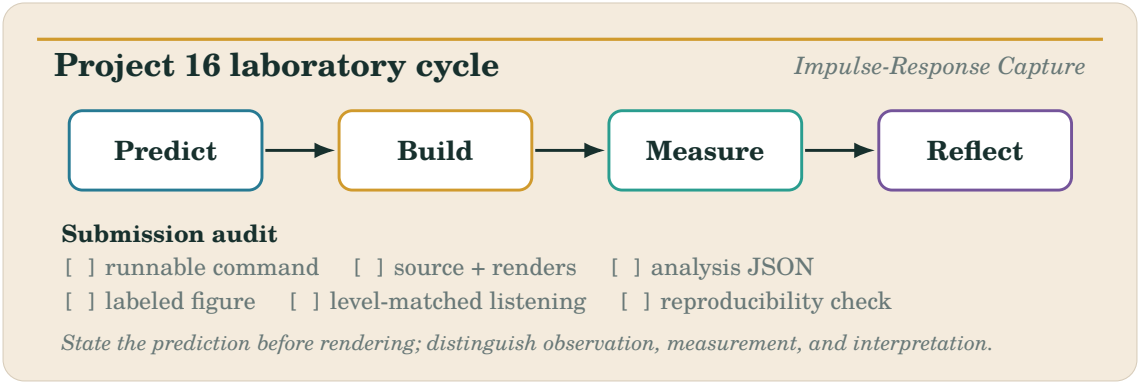


Figure B-16: Project 16 laboratory cycle: Impulse-Response Capture

B.19 Project 17: IR Editing and Provenance

Central concepts: trim, normalize, resample, and hash.

Design brief. Prepare an IR library while preserving source identity and reversible processing metadata. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-17 presents Project 17 laboratory cycle: IR Editing and Provenance and establishes the visual reference for the discussion that follows.

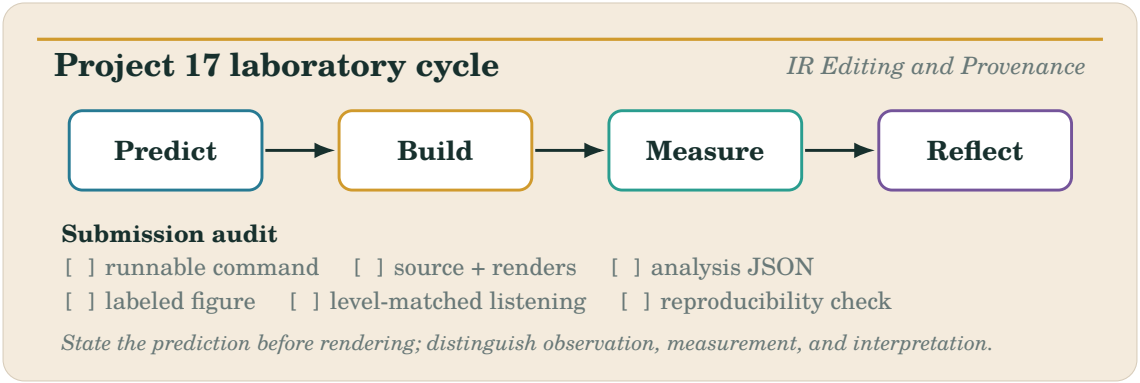


Figure B-17: Project 17 laboratory cycle: IR Editing and Provenance

B.20 Project 18: Morphing-Space Miniature

Project mode: Musical composition and production.

Central concepts: IR interpolation, formal trajectory, and perceptual continuity.

Design brief. Compose a miniature whose formal sections are defined by a continuous journey between contrasting spaces. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Select three contrasting impulse responses that suggest intimate, public, and impossible spaces. Compose a two- to four-minute miniature with an opening, transition, and arrival whose spatial trajectory is as deliberate as its harmonic trajectory. Build at least five reproducible morph states, include one continuous transition lasting a complete phrase, and avoid changing orchestration during one decisive morph so the space itself carries the form.

Evidence package. Submit the score or formal timeline, source and final audio, isolated wet stem, all IRs with provenance and hashes, morph settings, analysis JSON, and a labeled plot of morph position against musical time. Provide short listening annotations for every state and discontinuity check.

Questions for the report. Can listeners identify formal change before they identify the new room? Which acoustic attributes interpolate smoothly, and which appear to switch categories? Does the same harmony acquire a different function when its apparent architecture changes?

Completion standard. The spatial trajectory must create a perceptible beginning, transformation, and destination without relying only on level fades. IR provenance must be complete, all five states reproducible, and no unintended click or timbral discontinuity may remain.

Extension. Reverse the spatial journey while preserving the notes and durations, then explain which aspects of the musical form also seem reversed.

Figure B-18 presents Project 18 laboratory cycle: Morphing-Space Miniature and establishes the visual reference for the discussion that follows.

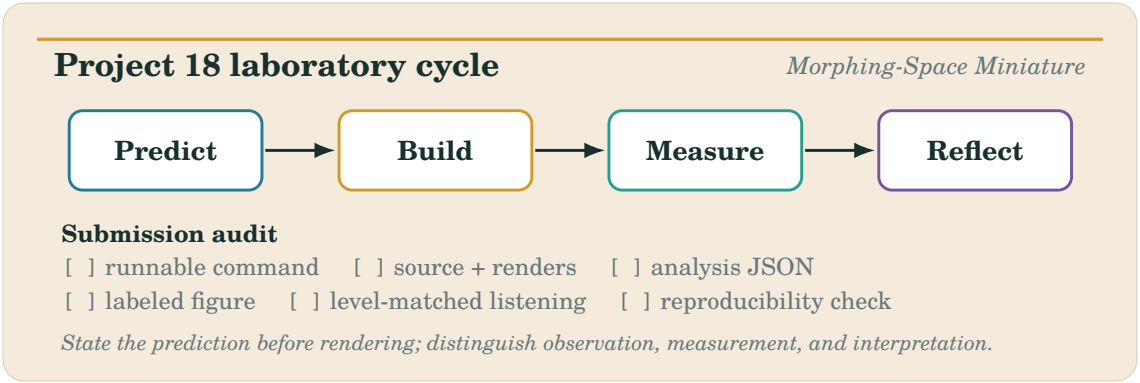


Figure B-18: Project 18 laboratory cycle: Morphing-Space Miniature

B.21 Project 19: Dereverberation Tradeoff

Central concepts: late-energy suppression and artifacts.

Design brief. Build a strength sweep and select an operating point from measured and blind-listening evidence. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-19 presents Project 19 laboratory cycle: Dereverberation Tradeoff and establishes the visual reference for the discussion that follows.

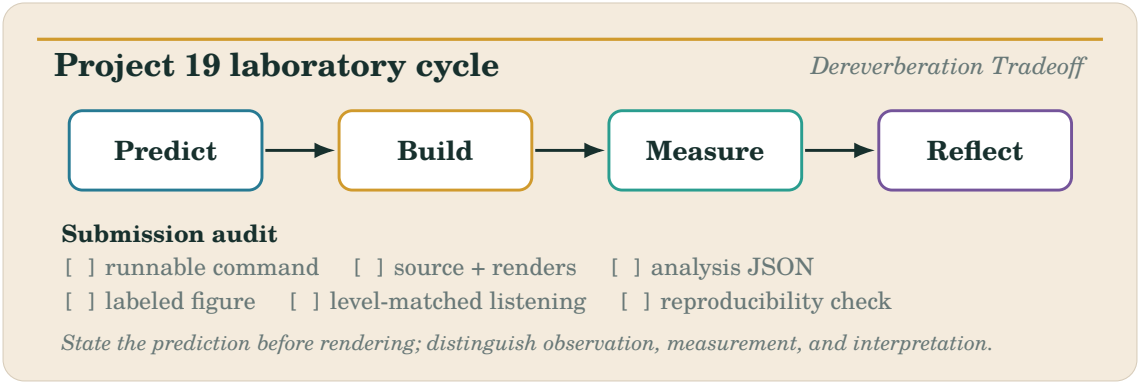


Figure B-19: Project 19 laboratory cycle: Dereverberation Tradeoff

B.22 Project 20: Speech Intelligibility Study

Central concepts: DRR, clarity, and spectral masking.

Design brief. Compare natural reverb, dereverberation, and re-reverberation for spoken material. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-20 presents Project 20 laboratory cycle: Speech Intelligibility Study and establishes the visual reference for the discussion that follows.

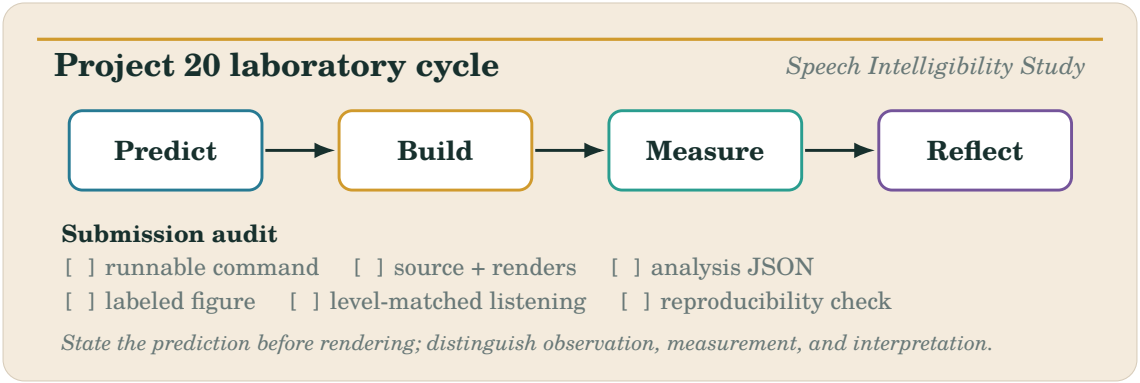


Figure B-20: Project 20 laboratory cycle: Speech Intelligibility Study

B.23 Project 21: Realtime Latency Audit

Central concepts: device, host, DSP, and acoustic delay.

Design brief. Measure round-trip latency and reconcile it with the predicted buffer stack. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-21 presents Project 21 laboratory cycle: Realtime Latency Audit and establishes the visual reference for the discussion that follows.

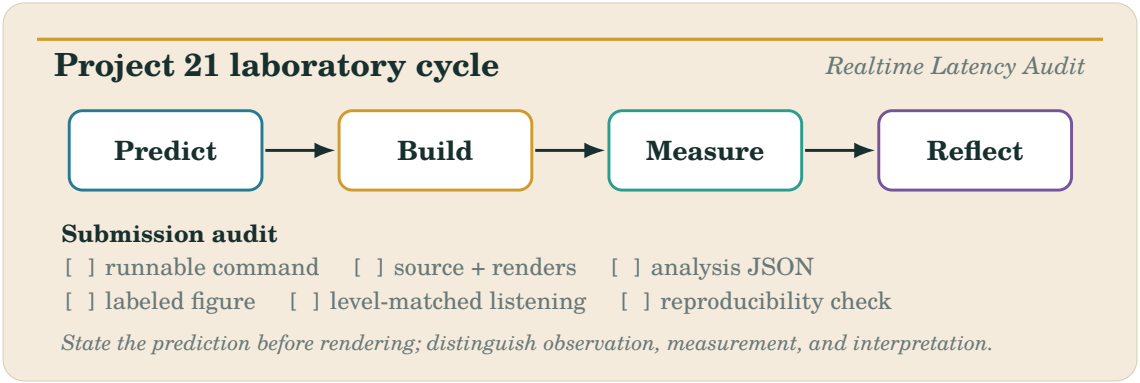


Figure B-21: Project 21 laboratory cycle: Realtime Latency Audit

B.24 Project 22: Realtime Failure Injection

Central concepts: xruns, overload, and recovery.

Design brief. Stress block size and quality modes while recording dropout thresholds and recovery behavior. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-22 presents Project 22 laboratory cycle: Realtime Failure Injection and establishes the visual reference for the discussion that follows.

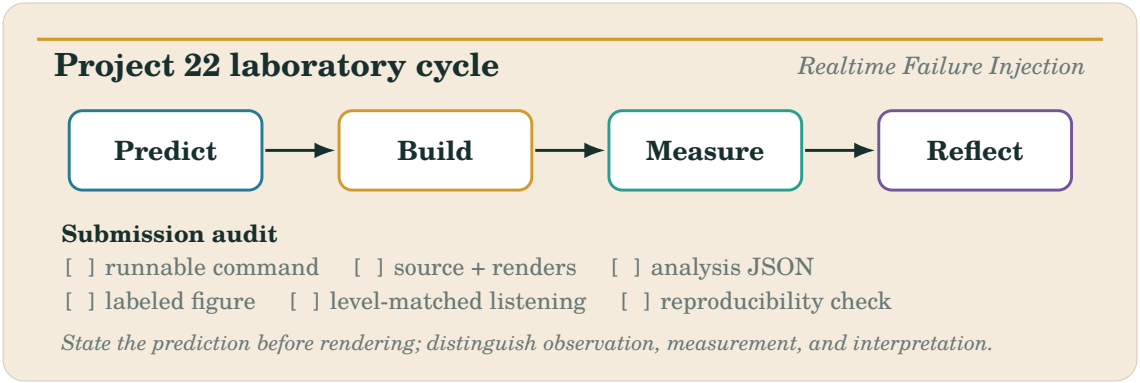


Figure B-22: Project 22 laboratory cycle: Realtime Failure Injection

B.25 Project 23: Spatial Automation as Musical Form

Project mode: Musical composition and production.

Central concepts: timeline interpolation, depth dramaturgy, and deterministic state.

Design brief. Compose a work in which automated depth, width, and decay create the principal large-scale form. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose a three- to five-minute work whose primary form is a trajectory through foreground, middle distance, enclosure, and release. Automate at least four spatial variables, including RT60, pre-delay, wet/dry balance, and width or position, but introduce them in stages so their musical functions remain audible. Render the same automation at multiple block sizes and preserve a fixed seed wherever the engine permits.

Evidence package. Submit the master, dry and wet stems, score or formal map, automation data, preset state, exact renders at two or more block sizes, analysis JSON, and a labeled multivariable timeline. Include null or difference measurements and listening notes for any block-boundary discrepancy.

Questions for the report. Which spatial parameter acts most like orchestration, harmony, dynamics, or tempo? Can a listener hear the formal return when the notes do not recur? Where does smooth automation obscure a boundary that a discrete change would articulate more effectively?

Completion standard. The composition must contain at least four clearly perceived spatial states and one convincing formal return created principally by automation. Repeated renders must satisfy the documented reproducibility tolerance or explain every divergence.

Extension. Produce a second version in which performers, rather than a fixed timeline, trigger the same spatial form from musical cues.

Figure B-23 presents Project 23 laboratory cycle: Spatial Automation as Musical Form and establishes the visual reference for the discussion that follows.

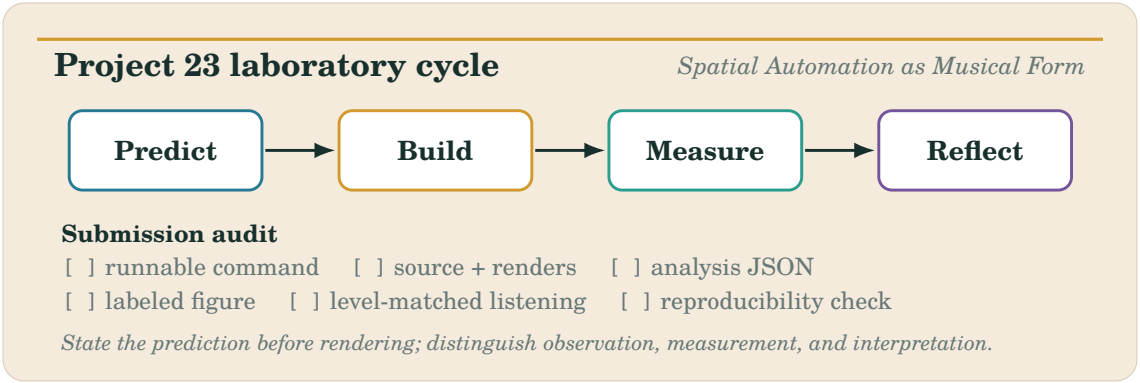


Figure B-23: Project 23 laboratory cycle: Spatial Automation as Musical Form

B.26 Project 24: Compose a Reverb-Preset Etude

Project mode: Musical composition and production.

Central concepts: semantic preset design, variation, and timbral orchestration.

Design brief. Design a related preset family and compose an etude whose sections reveal each preset's distinct musical role. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Design a family of six related reverb presets organized around musical functions such as pulse, veil, chamber, bloom, fracture, and horizon. Compose a 90-second etude in six short sections using the same source ensemble throughout; each section must make one preset necessary to its articulation, harmony, or texture. Constrain shared parameters so the family sounds related, and vary only the dimensions needed for each role.

Evidence package. Submit the six presets with semantic descriptions, the score or session timeline, dry and wet stems, final master, a parameter-difference table, recall-test results, analysis JSON, and one labeled comparison figure. Include a blind naming trial showing whether listeners can connect sound and intended role.

Questions for the report. Which preset names predict musical behavior rather than generic mood? What invariant makes the collection sound like one family? Can a preset remain distinctive when source register, density, or tempo changes?

Completion standard. All six presets must recall deterministically, remain distinguishable under loudness-matched listening, and perform a different compositional function in the etude. Names and descriptions must allow another musician to choose an appropriate preset without seeing its parameter values.

Extension. Orchestrate the etude for a contrasting ensemble and revise no more than two parameters per preset to preserve the family identity.

Figure B-24 presents Project 24 laboratory cycle: Compose a Reverb-Preset Etude and establishes the visual reference for the discussion that follows.

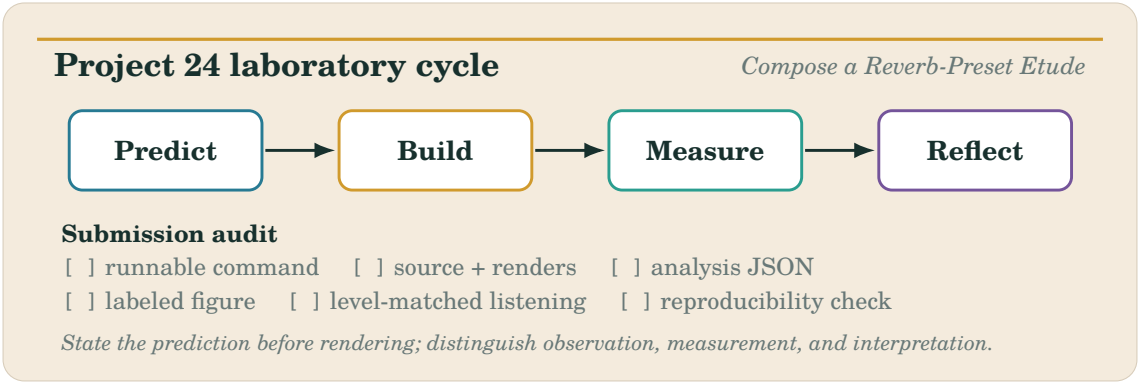


Figure B-24: Project 24 laboratory cycle: Compose a Reverb-Preset Etude

B.27 Project 25: Loudness-Matched Evaluation

Central concepts: LUFS, peak, and perceptual bias.

Design brief. Compare six reverb designs only after integrated loudness and peak constraints are matched. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-25 presents Project 25 laboratory cycle: Loudness-Matched Evaluation and establishes the visual reference for the discussion that follows.

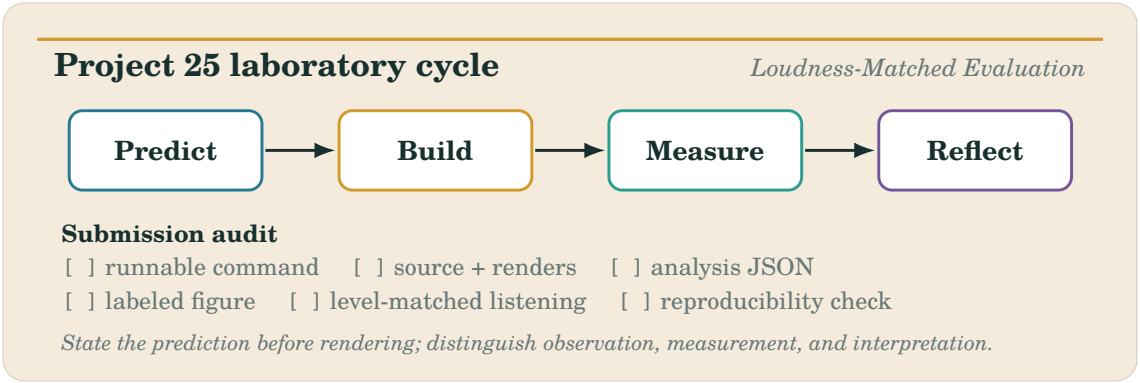


Figure B-25: Project 25 laboratory cycle: Loudness-Matched Evaluation

B.28 Project 26: Stereo Width Without Phase Damage

Central concepts: correlation and mono compatibility.

Design brief. Increase envelopment while maintaining a documented mono fold-down target. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-26 presents Project 26 laboratory cycle: Stereo Width Without Phase Damage and establishes the visual reference for the discussion that follows.

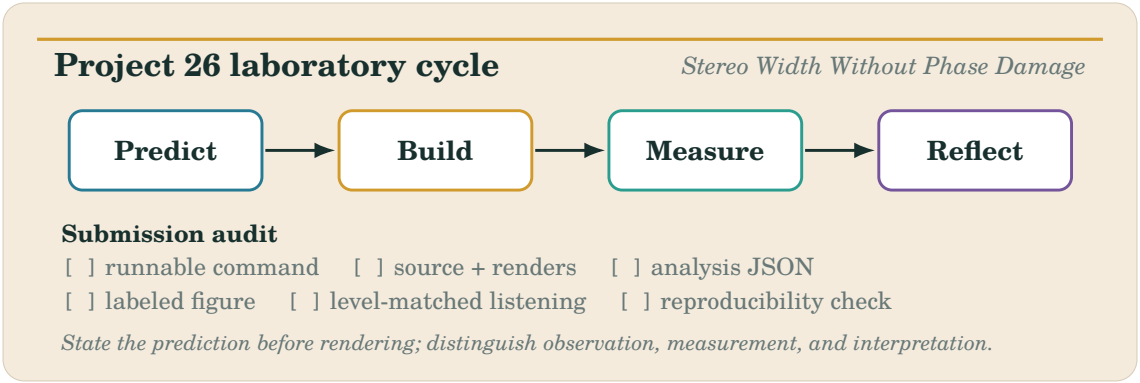


Figure B-26: Project 26 laboratory cycle: Stereo Width Without Phase Damage

B.29 Project 27: Multichannel Bus Verification

Central concepts: channel identity and matrix routing.

Design brief. Use labeled impulses to prove every input-to-output route in a surround configuration. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-27 presents Project 27 laboratory cycle: Multichannel Bus Verification and establishes the visual reference for the discussion that follows.

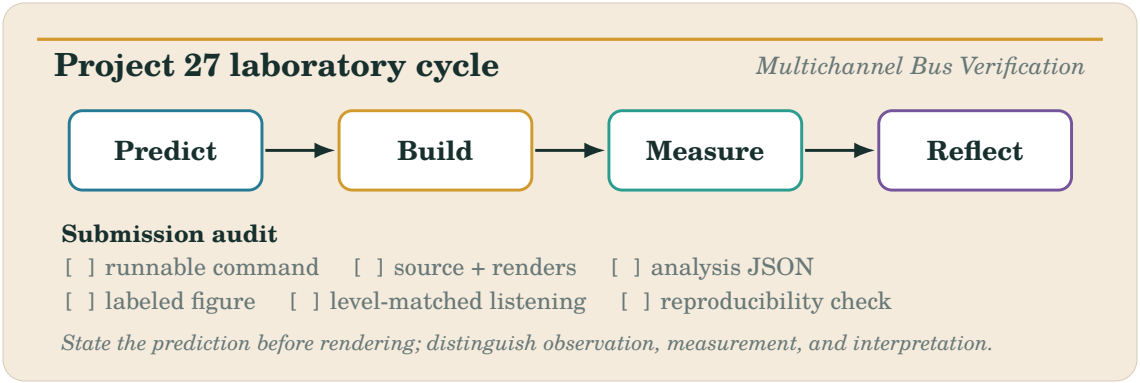


Figure B-27: Project 27 laboratory cycle: Multichannel Bus Verification

B.30 Project 28: Ambisonic Rotational Counterpoint

Project mode: Musical composition and production.

Central concepts: scene representation, spatial imitation, and decoding.

Design brief. Compose a multi-voice texture whose contrapuntal identities remain clear while their spatial trajectories rotate and exchange positions. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write a two- to four-minute three- or four-voice contrapuntal texture and encode it as a first-order Ambisonic scene. Give each voice a recurring spatial motif, then compose one imitative rotation, one exchange of positions, one stationary pedal against moving voices, and one global scene rotation. Decode to at least two playback formats and retain a binaural or stereo reference.

Evidence package. Submit the score, FOA master, decoded versions, isolated voices, trajectory diagram, channel-order and normalization settings, exact commands, analysis JSON, and labeled level or localization checks. Document every intended spatial invariant before comparing decoders.

Questions for the report. Which contrapuntal identities survive rotation because of timbre, rhythm, register, or location? When does motion clarify imitation, and when does it compete with voice-leading? What remains invariant across decoders, and what is reproduction-dependent?

Completion standard. The spatial motifs and at least three contrapuntal events must remain intelligible in both decodes, channel conventions must be verified, and global rotation must preserve the documented level relationships within tolerance.

Extension. Create a performance version in which one spatial voice follows a live conductor or performer while the remaining trajectories stay deterministic.

Figure B-28 presents Project 28 laboratory cycle: Ambisonic Rotational Counterpoint and establishes the visual reference for the discussion that follows.

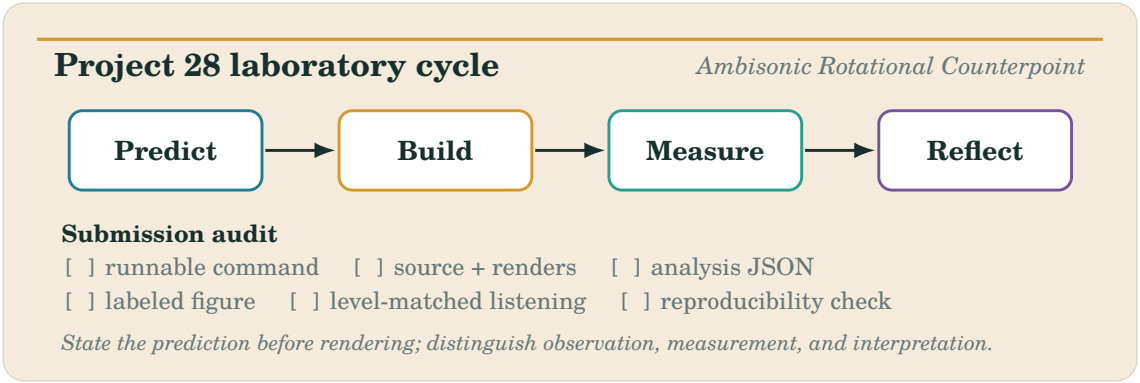


Figure B-28: Project 28 laboratory cycle: Ambisonic Rotational Counterpoint

B.31 Project 29: Room-Model Inversion

Central concepts: Sabine estimates and geometric constraints.

Design brief. Infer plausible dimensions and absorption from RT60, then explain non-unique solutions. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-29 presents Project 29 laboratory cycle: Room-Model Inversion and establishes the visual reference for the discussion that follows.

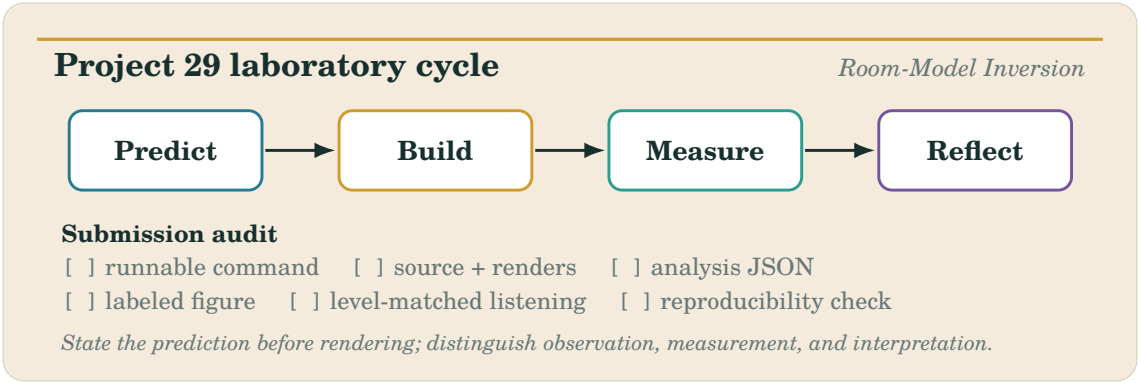


Figure B-29: Project 29 laboratory cycle: Room-Model Inversion

B.32 Project 30: Ray-Traced Early Field Proposal

Central concepts: CAD geometry, materials, and validation.

Design brief. Specify a DXF-to-reflection prototype with assumptions, error bounds, and listening tests. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-30 presents Project 30 laboratory cycle: Ray-Traced Early Field Proposal and establishes the visual reference for the discussion that follows.

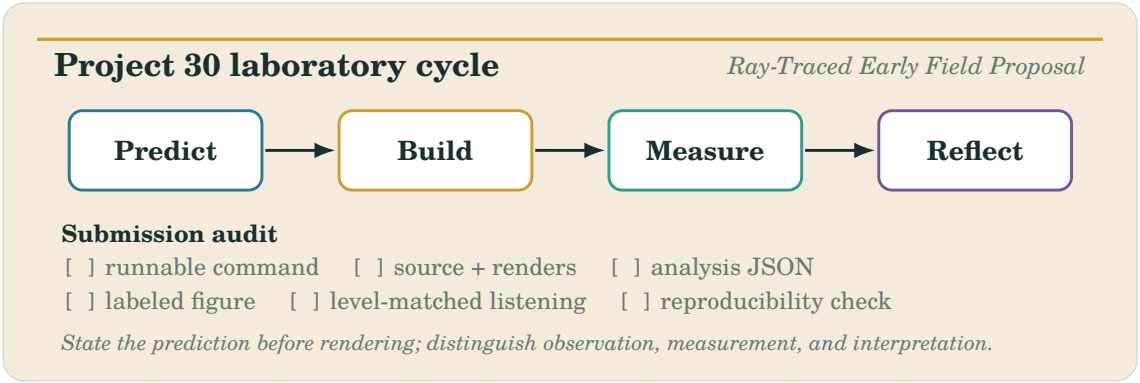


Figure B-30: Project 30 laboratory cycle: Ray-Traced Early Field Proposal

B.33 Project 31: Machine-Readable Evidence

Central concepts: analysis JSON and schema stability.

Design brief. Design a report consumed by a regression script without parsing human progress text. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-31 presents Project 31 laboratory cycle: Machine-Readable Evidence and establishes the visual reference for the discussion that follows.

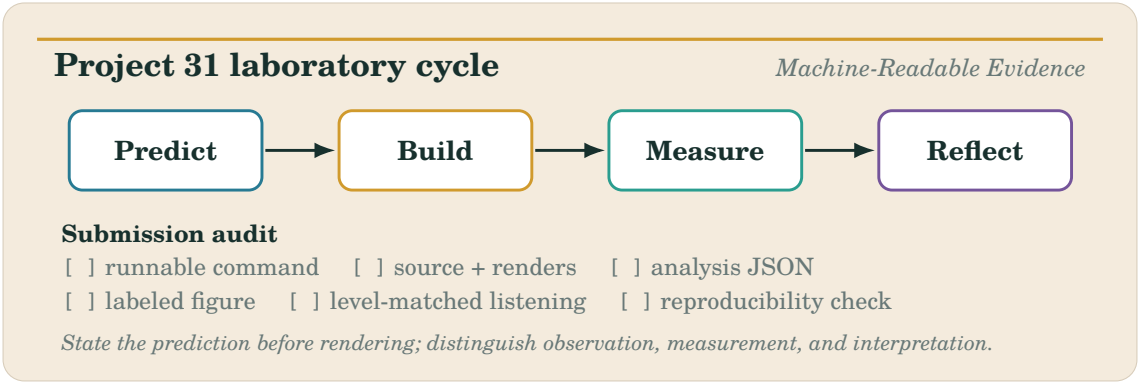


Figure B-31: Project 31 laboratory cycle: Machine-Readable Evidence

B.34 Project 32: Native Plug-in Parity Slice

Central concepts: C++ DSP, host state, and deterministic comparison.

Design brief. Port one narrow feature and prove parity against the Python reference. Begin with a written prediction that names the expected audible and measurable changes. Use controlled source material and alter one principal variable at a time before combining controls.

Method. Apply the shared laboratory protocol above, choosing the comparison range and deliberately poor case to expose this project's central mechanism rather than an unrelated failure.

Figure B-32 presents Project 32 laboratory cycle: Native Plug-in Parity Slice and establishes the visual reference for the discussion that follows.

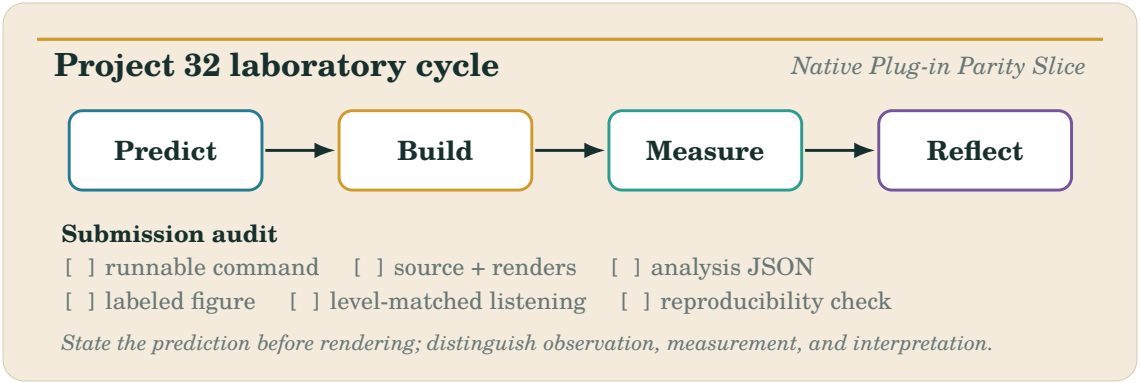


Figure B-32: Project 32 laboratory cycle: Native Plug-in Parity Slice

B.35 Project 33: Original Spatial Composition

Project mode: Musical composition and production.

Central concepts: original composition, spatial form, and integrated engineering practice.

Design brief. Create an original multichannel work in which reverberation is indispensable to the composition's harmony, rhythm, gesture, or form. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose and produce an original five- to eight-minute work for stereo, binaural, Ambisonic, or loudspeaker-array presentation. State a compositional thesis explaining how reverberation governs at least two domains among harmony, rhythm, gesture, orchestration, memory, and large-scale form. Develop the work through sketches, controlled listening comparisons, and a final production that integrates algorithmic or convolution reverb, automation, dynamics, and spatial routing without treating the room as a last-stage decoration.

Evidence package. Submit the final master and required playback derivatives, score or detailed formal map, dry and processed stems, presets, automation, impulse responses with provenance, exact commands, analysis JSON, plots, technical rider, and a reproducibility manifest with software revision and hashes. Include a five- to eight-minute presentation or written commentary that connects specific audible moments to the compositional thesis.

Questions for the report. What musical information exists only because of the designed acoustic? Where does the room act as instrument, ensemble, transition, or memory? Which technical constraint produced the strongest creative decision, and which apparently impressive effect was removed because it weakened the piece?

Completion standard. The work must be compositionally complete, technically deliverable, and reproducible by another student. Independent listeners must be able to identify at least two intended spatial-form events, and the submitted evidence must demonstrate stable levels, correct routing, and deliberate control of accumulated energy.

Extension. Prepare a concert or installation realization for a different reproduction system and write an adaptation note distinguishing musical invariants from venue-specific decisions.

Figure B-33 presents Project 33 laboratory cycle: Original Spatial Composition and establishes the visual reference for the discussion that follows.

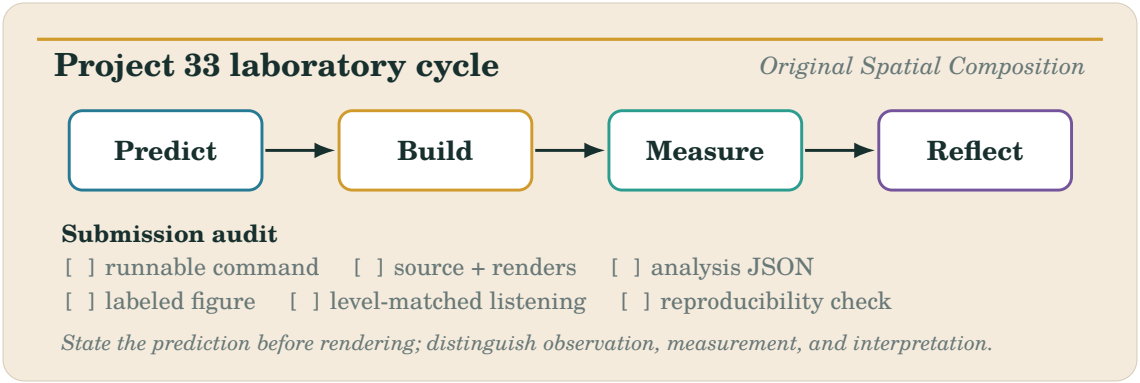


Figure B-33: Project 33 laboratory cycle: Original Spatial Composition

B.36 Project 34: Harmonic Pedals from Decay

Project mode: Musical composition and production.

Central concepts: crossfaded tails, harmonic memory, and voice-leading.

Design brief. Compose a harmonic sequence in which each decaying room becomes a pedal tone beneath the next sonority. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write a two- to three-minute progression of at least six sonorities. Render or automate each chord so its tail crosses into the next harmony, then decide whether the retained tones act as common tones, suspensions, dissonant pedals, or harmonic foreshadowing. Include one passage where the dry attack disappears and the listener hears only overlapping decays; control register and damping so the accumulated harmony remains intentional.

Evidence package. Submit the score or harmonic reduction, dry attacks, isolated tails, final master, exact settings, analysis JSON, and a labeled pitch-versus-time map showing the six transitions. Mark every retained pitch and explain whether it belongs to the departing chord, arriving chord, or neither.

Questions for the report. When does a decaying chord stop belonging to its source? Which crossings strengthen voice-leading, and which create unwanted harmonic mud? Can the room anticipate a harmony before any dry instrument plays it?

Completion standard. All six transitions must have a stated harmonic function, and at least three must be distinguishable in blind listening. The tails must remain stable, pitched enough to support the analysis, and free of accidental low-mid accumulation.

Extension. Reharmonize the piece while preserving every reverb print, treating the original tails as fixed contrapuntal material.

Figure B-34 presents Project 34 laboratory cycle: Harmonic Pedals from Decay and establishes the visual reference for the discussion that follows.

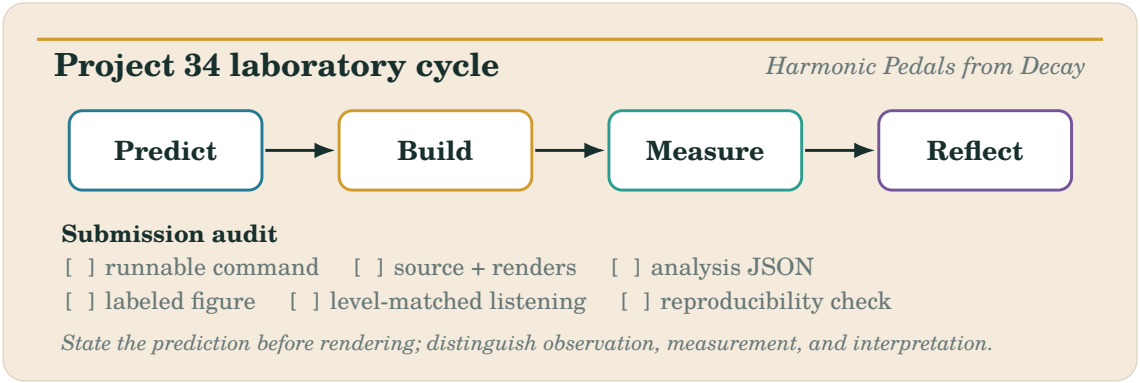


Figure B-34: Project 34 laboratory cycle: Harmonic Pedals from Decay

B.37 Project 35: Tempo-Synchronized Decay Canon

Project mode: Musical composition and production.

Central concepts: tempo ratios, RT60, meter, and imitation.

Design brief. Write a canon whose tail lengths articulate changing subdivisions and metric relationships. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Choose a tempo and derive three decay targets from beat subdivisions or multiples, such as an eighth note, dotted quarter, and two measures. Compose a two-minute canon that moves through the three time scales, then introduce one metric modulation while preserving an audible decay ratio. Compare mathematically exact RT60 values with nearby values selected by ear and keep level constant.

Evidence package. Submit the score, tempo map, decay calculations, dry and processed stems, exact automation or commands, analysis JSON, and a labeled timeline connecting entries, beat values, and measured decay. Include one click-referenced excerpt for timing verification and one musical master without click.

Questions for the report. Do listeners hear tail duration as meter, articulation, or texture? Which ratios remain perceptible without a click? When does mathematically synchronized decay sound less musical than a deliberately offset value?

Completion standard. The canon must present three clearly contrasting time scales and one convincing metric transition. Calculated and measured values must be documented, and the chosen musical deviations must be justified by listening rather than convenience.

Extension. Repeat the canon at a new tempo while preserving proportional decay relationships, then compare whether the same ratios retain their character.

Figure B-35 presents Project 35 laboratory cycle: Tempo-Synchronized Decay Canon and establishes the visual reference for the discussion that follows.

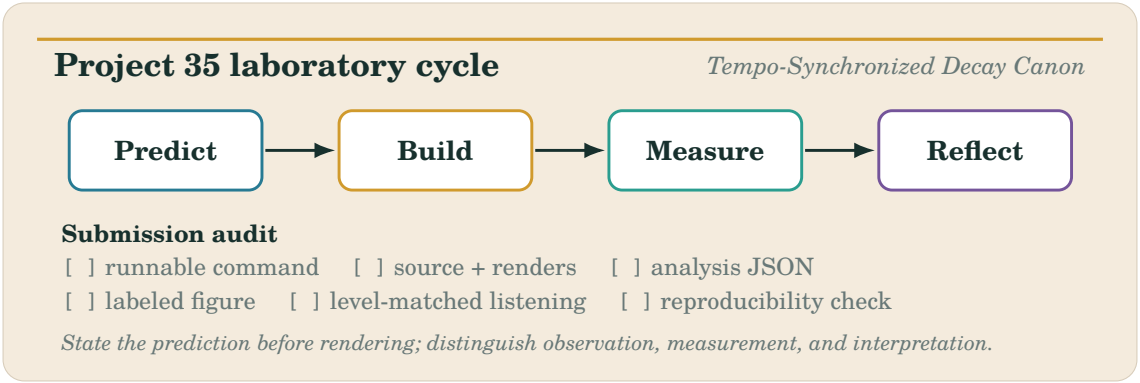


Figure B-35: Project 35 laboratory cycle: Tempo-Synchronized Decay Canon

B.38 Project 36: Antiphonal Virtual Architecture

Project mode: Musical composition and production.

Central concepts: call-and-response, early reflections, and contrasting rooms.

Design brief. Compose for two virtual ensembles whose phrases and rooms answer one another across an imagined architecture. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose a three-minute antiphonal work for two source groups. Give each group a contrasting virtual room, early-reflection pattern, and distance, then write six exchanges that evolve from literal call-and-response to overlap and interruption. Include one moment when the rooms exchange ensembles and one shared cadence in a third, unifying space.

Evidence package. Submit the score, isolated ensemble stems, isolated room returns, final master, room diagrams, presets or commands, analysis JSON, and a labeled formal map of the six exchanges. Document apparent direction, pre-delay, and early-to-late balance for every architectural state.

Questions for the report. Does architecture clarify ensemble identity as strongly as orchestration? What changes when a phrase moves into the other group's room? How does the shared final space alter the perceived social or dramatic relationship?

Completion standard. Listeners must identify the two virtual ensembles and the room exchange without seeing the session. The shared cadence must sound like a meaningful architectural convergence, not a global wetness increase.

Extension. Adapt the work for two live groups placed in a real venue and redesign the virtual rooms around the venue's measured response.

Figure B-36 presents Project 36 laboratory cycle: Antiphonal Virtual Architecture and establishes the visual reference for the discussion that follows.

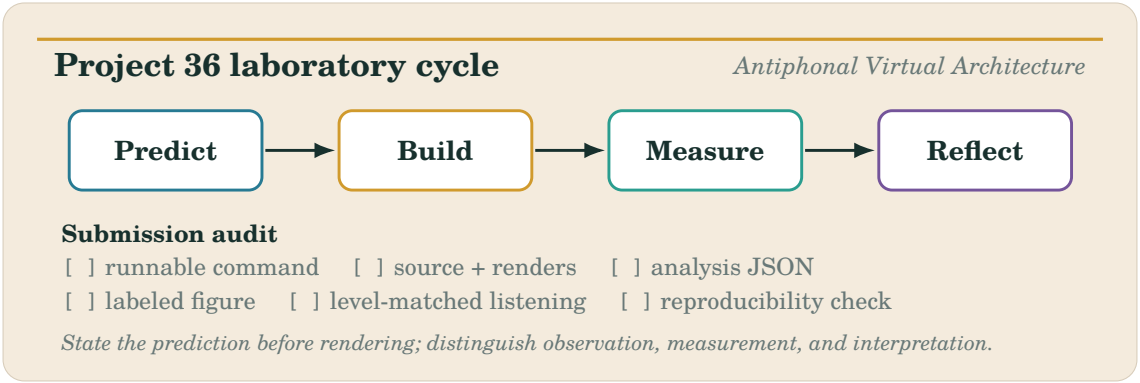


Figure B-36: Project 36 laboratory cycle: Antiphonal Virtual Architecture

B.39 Project 37: Multiband Tail Orchestration

Project mode: Musical composition and production.

Central concepts: register, frequency-dependent decay, and instrumental balance.

Design brief. Orchestrate low, middle, and high registers with different decay profiles that function as independent musical layers. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose for three registral layers using contrasting low-, mid-, and high-frequency decay. Begin with solo introductions that establish each layer, then combine them in a dense central section and redistribute the tails during a coda. Keep broadband RT60 or overall loudness sufficiently controlled that the form is created by spectral decay rather than simple level growth.

Evidence package. Submit a score or register chart, dry and band-isolated wet stems, final master, multiband settings, analysis JSON, and labeled decay curves for all three regions. Include a short orchestration note explaining which instruments leave spectral room for which tails.

Questions for the report. Which register reads as sustain, halo, or masking? Can a high-frequency tail articulate form without sounding merely bright? How does source orchestration need to change when the room occupies part of the spectrum?

Completion standard. All three decay layers must perform distinct musical roles and remain audible in the combined section. Measured curves must support the intended hierarchy, and the final mix must avoid uncontrolled low-frequency carryover.

Extension. Invert the decay orchestration so the shortest and longest bands exchange roles while the written notes remain unchanged.

Figure B-37 presents Project 37 laboratory cycle: Multiband Tail Orchestration and establishes the visual reference for the discussion that follows.

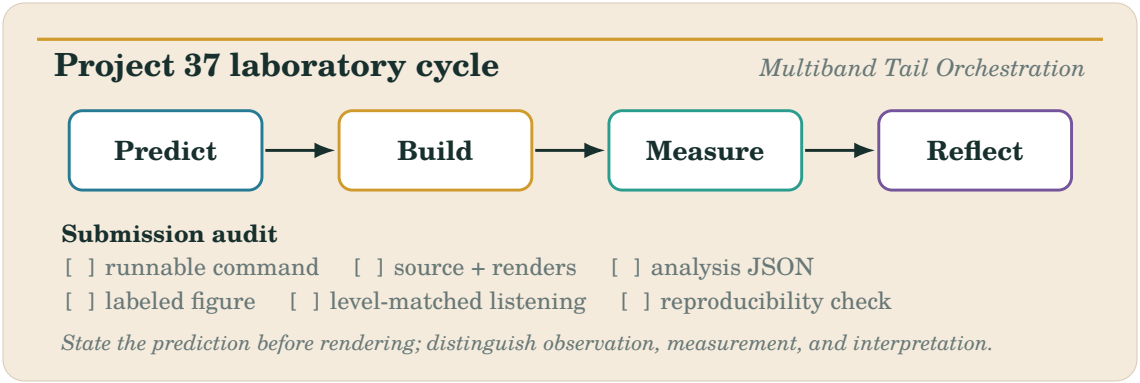


Figure B-37: Project 37 laboratory cycle: Multiband Tail Orchestration

B.40 Project 38: Percussive Reflection Groove

Project mode: Musical composition and production.

Central concepts: early-reflection timing, transient rhythm, and groove.

Design brief. Build a percussion study in which discrete early reflections supply notated offbeats, flams, and polyrhythms. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Record or synthesize a dry kit of at least five transient sounds. Configure early reflections to create flams, offbeats, tuplets, or short polyrhythms, then compose a 32-bar study in which those reflections provide an audible secondary percussion part. Use four sections: establishment, displacement, densification, and release; reserve the late tail for color rather than rhythmic definition.

Evidence package. Submit the dry one-shots, notated attack-and-reflection grid, dry and early-field stems, final master, exact settings, analysis JSON, and a labeled waveform over two representative bars. Distinguish performed attacks from generated reflections in the notation.

Questions for the report. Which reflections fuse into timbre, and which separate into rhythm? How does source articulation change perceived reflection timing? At what density does the virtual percussion part lose groove and become diffuse ambience?

Completion standard. The reflection part must create at least three reproducible rhythmic figures and remain intelligible without excessive late reverb. A listener must be able to tap or transcribe one generated pattern from the wet stem alone.

Extension. Arrange the generated reflection part for a live percussionist and compare timing, groove, and timbre with the signal-processed version.

Figure B-38 presents Project 38 laboratory cycle: Percussive Reflection Groove and establishes the visual reference for the discussion that follows.

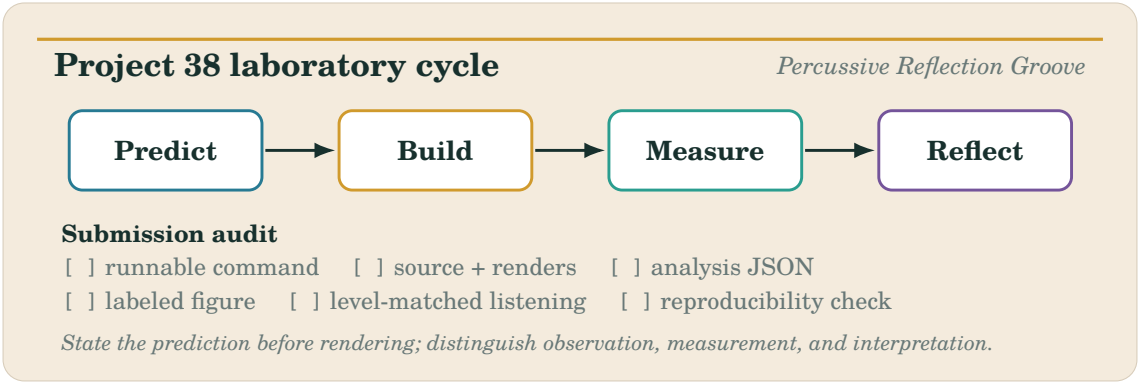


Figure B-38: Project 38 laboratory cycle: Percussive Reflection Groove

B.41 Project 39: Vocal Space Dramaturgy

Project mode: Musical composition and production.

Central concepts: narrative distance, intimacy, and spatial automation.

Design brief. Compose a song or spoken scene whose emotional arc is carried by changing vocal distance and enclosure. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write and record a three- to four-minute song, monologue, or dramatic scene with at least five vocal distance states. Map those states to a narrative arc that includes intimacy, withdrawal, confrontation, memory, and return. Coordinate pre-delay, early reflections, damping, width, and wet level with performance and arrangement rather than automating every parameter together.

Evidence package. Submit the text or score, dry vocal, wet return, final mix, five-state cue sheet, automation, presets, analysis JSON, and a labeled distance timeline. Include intelligibility notes from at least two listeners and identify every word intentionally obscured or exposed.

Questions for the report. Which acoustic cue most strongly changes the speaker's apparent psychological distance? When does a large room imply power rather than remoteness? Can a return to intimacy be heard without reducing overall loudness?

Completion standard. All five narrative states must be perceptually distinct while essential text remains intelligible. Spatial changes must coincide with dramatic decisions, and the final return must recall the opening state without simply copying it.

Extension. Create a binaural version in which apparent location and head-relative perspective reinforce the same dramatic arc.

Figure B-39 presents Project 39 laboratory cycle: Vocal Space Dramaturgy and establishes the visual reference for the discussion that follows.

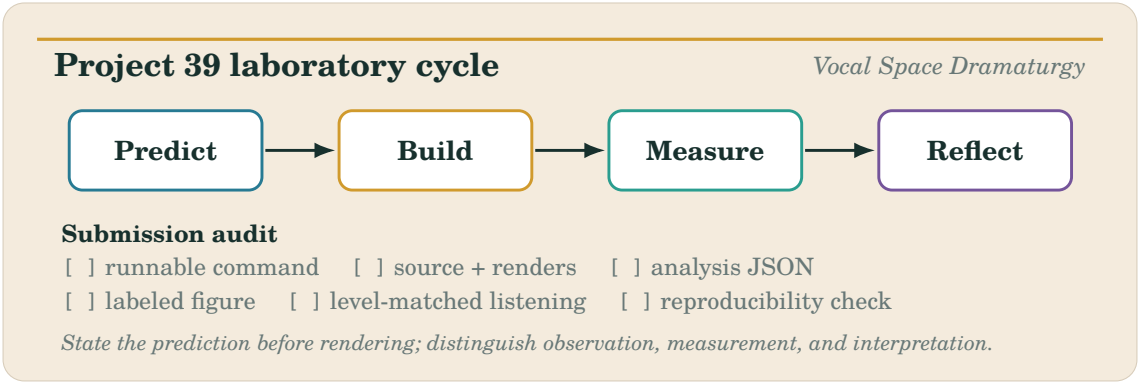


Figure B-39: Project 39 laboratory cycle: Vocal Space Dramaturgy

B.42 Project 40: Dub Send Performance

Project mode: Musical composition and production.

Central concepts: reverb throws, feedback, mute gestures, and live form.

Design brief. Perform a mix in which manually played sends and returns become an improvising instrumental part. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Prepare four contrasting stems and perform a three- to five-minute live dub mix using reverb sends, mutes, feedback, damping, freeze, and return level as playable gestures. Define a cue structure but leave room for improvisation. Rehearse safe maximum feedback, create at least eight intentional throws, and include one recovery from a deliberately over-dense texture.

Evidence package. Submit the source stems, unedited performance, final edited version if used, control map, cue sheet, presets, analysis JSON, and a labeled gesture timeline. Document peak and spectral behavior around the densest throw and explain the recovery strategy.

Questions for the report. Which controls behave most like instrumental articulation? How does performance timing differ from drawing automation? What makes a throw feel responsive to the source rather than pasted onto it?

Completion standard. The performance must contain eight identifiable send gestures, one controlled feedback build, and one musical recovery without clipping or panic muting. The cue structure must be repeatable while allowing meaningful variation between takes.

Extension. Perform the piece with another musician controlling source arrangement while you control the spatial instrument; document the shared cue vocabulary.

Figure B-40 presents Project 40 laboratory cycle: Dub Send Performance and establishes the visual reference for the discussion that follows.

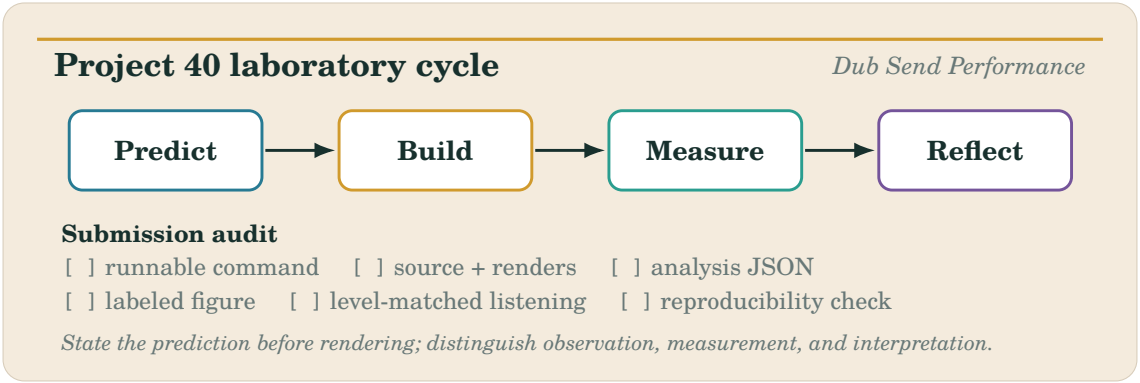


Figure B-40: Project 40 laboratory cycle: Dub Send Performance

B.43 Project 41: Silence and Cadential Decay

Project mode: Musical composition and production.

Central concepts: rests, phrase endings, and composed resonance.

Design brief. Write a sequence of cadences in which measured silence and audible decay complete, delay, or contradict closure. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose twelve short phrase endings for one ensemble. Vary harmonic closure, articulation, RT60, damping, and the notated silence that follows so the room completes, delays, questions, or erases each cadence. Sequence the endings into a two- to three-minute form in which silence has measured duration and at least one decay is interrupted before its natural end.

Evidence package. Submit the score with notated rests and tail cues, dry phrases, wet returns, final sequence, decay measurements, analysis JSON, and a labeled comparison of all twelve endings. Include listener judgments of closure strength without revealing the intended categories.

Questions for the report. Where does the cadence end: at the attack, release, threshold crossing, or next entrance? Can an unresolved harmony feel closed because its room disappears? How much silence is needed for a long decay to become structural rather than decorative?

Completion standard. At least four distinct closure functions must be recognized by listeners, and every rest must be compositionally justified. The interrupted decay must sound intentional and alter the formal reading of its phrase.

Extension. Perform the sequence in a reverberant real room and rewrite the notated silences to incorporate its natural decay.

Figure B-41 presents Project 41 laboratory cycle: Silence and Cadential Decay and establishes the visual reference for the discussion that follows.

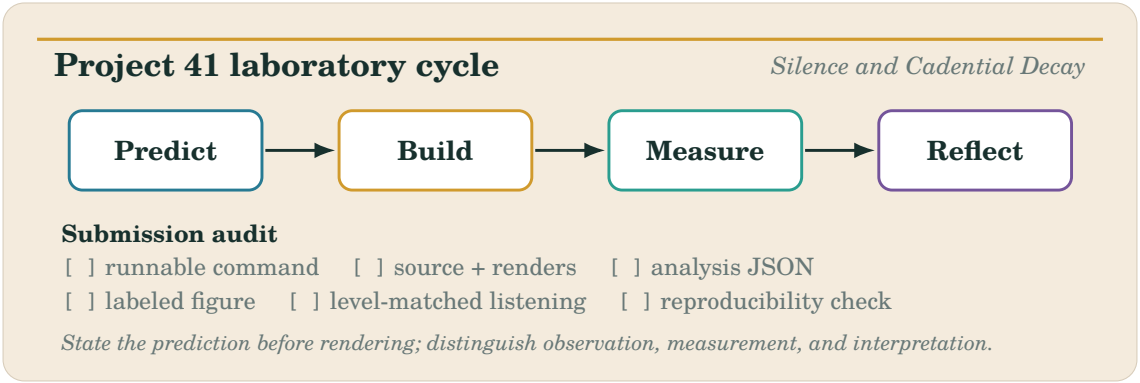


Figure B-41: Project 41 laboratory cycle: Silence and Cadential Decay

B.44 Project 42: Room-Swap Theme and Variations

Project mode: Musical composition and production.

Central concepts: variation form, orchestration, and acoustic identity.

Design brief. Create a theme and four variations whose musical characters emerge from contrasting room designs. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Compose a concise theme and four variations for the same core ensemble. Assign each variation a room identity such as chamber, hall, plate-like studio, or impossible synthetic space. Preserve the theme's notes in the first pass, then reorchestrate articulation, register, density, and tempo only where the room demands a musical response.

Evidence package. Submit the theme and four scores or timelines, dry common-source comparison, final variations, presets or IRs, analysis JSON, and a parameter-and-orchestration matrix. Include a blind test asking listeners to match each room with its intended variation character.

Questions for the report. How much musical character comes from room identity before notes change? Which reorchestrations are genuinely required by decay or clarity? Does an impossible room suggest formal possibilities unavailable to an acoustic model?

Completion standard. All four variations must remain recognizably related while presenting distinct acoustic characters. At least three room-to-character matches must exceed chance in the listening test, and every orchestration change must cite an acoustic reason.

Extension. Create a fifth variation by morphing continuously through the four spaces while holding orchestration fixed.

Figure B-42 presents Project 42 laboratory cycle: Room-Swap Theme and Variations and establishes the visual reference for the discussion that follows.

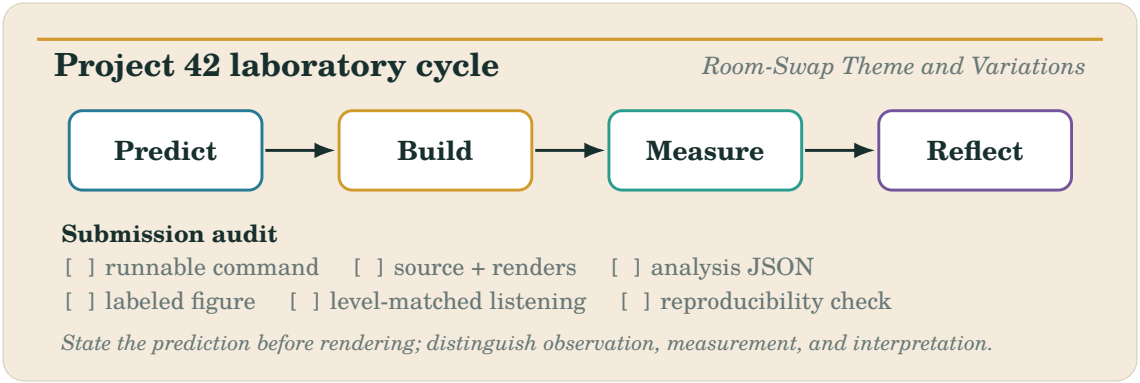


Figure B-42: Project 42 laboratory cycle: Room-Swap Theme and Variations

B.45 Project 43: Site-Specific IR Portrait

Project mode: Musical composition and production.

Central concepts: impulse-response capture, resonance, and place.

Design brief. Capture or document a real acoustic and compose a portrait that responds to its measured modes, clarity, and decay. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Capture a room impulse response or select a thoroughly documented local IR, then analyze its resonances, clarity, noise floor, and decay. Compose a three- to five-minute portrait that features at least two measured traits: one may be embraced as material and one may be countered through orchestration or timing. Include an unprocessed field recording or contextual sound if appropriate and ethically permissible.

Evidence package. Submit the raw and prepared IR, provenance, photographs or diagram when permitted, hashes, analysis JSON, measured plots, score or timeline, dry and convolved stems, and final master. Explain every edit made to the IR and retain a reversible preparation chain.

Questions for the report. Which audible traits belong uniquely to this place? Does convolution preserve identity when the source is unrelated to the site? What ethical, practical, or historical context should accompany an acoustic portrait?

Completion standard. The composition must make two measured site traits musically perceptible, and the IR provenance must permit another student to reproduce the result. Preparation may improve usability but must not erase the documented identity of the capture.

Extension. Present the piece in the captured location or simulate that return, then analyze the interaction between real and convolved acoustics.

Figure B-43 presents Project 43 laboratory cycle: Site-Specific IR Portrait and establishes the visual reference for the discussion that follows.

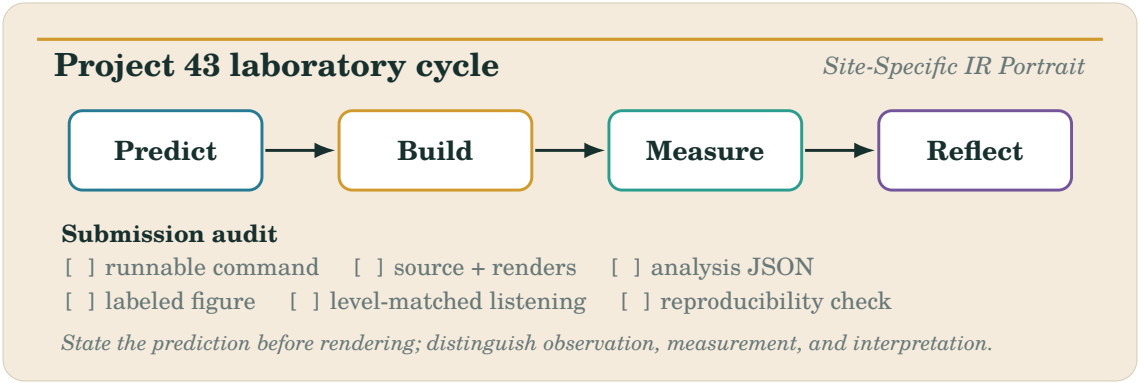


Figure B-43: Project 43 laboratory cycle: Site-Specific IR Portrait

B.46 Project 44: Modulated-Reverb Timbre Study

Project mode: Musical composition and production.

Central concepts: modulation rate, depth, pitch stability, and color.

Design brief. Compose a timbral etude in which reverb modulation changes orchestral color without obscuring pitch or pulse. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Choose two sustained and two transient sources, then compose a two-minute timbral etude with three modulation regimes: nearly static, gently animated, and deliberately unstable. Control modulation rate, depth, diffusion, and damping independently; introduce each regime in isolation before layering them. Preserve one pitched passage as a reference for detecting unwanted chorusing or intonation drift.

Evidence package. Submit source stems, wet-only stems, final master, modulation settings, analysis JSON, and labeled spectrogram and stereo-correlation plots. Include listening notes that distinguish motion, pitch change, density, and width rather than grouping them as a single impression.

Questions for the report. When does modulation read as room movement, ensemble size, chorus, or pitch instability? Which source envelopes reveal modulation most clearly? Can motion increase without widening the image?

Completion standard. The three regimes must be perceptually distinct and technically stable, and the reference passage must retain acceptable pitch clarity. The composition must use modulation to articulate form rather than cycling continuously without purpose.

Extension. Map modulation depth to a performer-controlled gesture and compare deterministic automation with embodied timing.

Figure B-44 presents Project 44 laboratory cycle: Modulated-Reverb Timbre Study and establishes the visual reference for the discussion that follows.

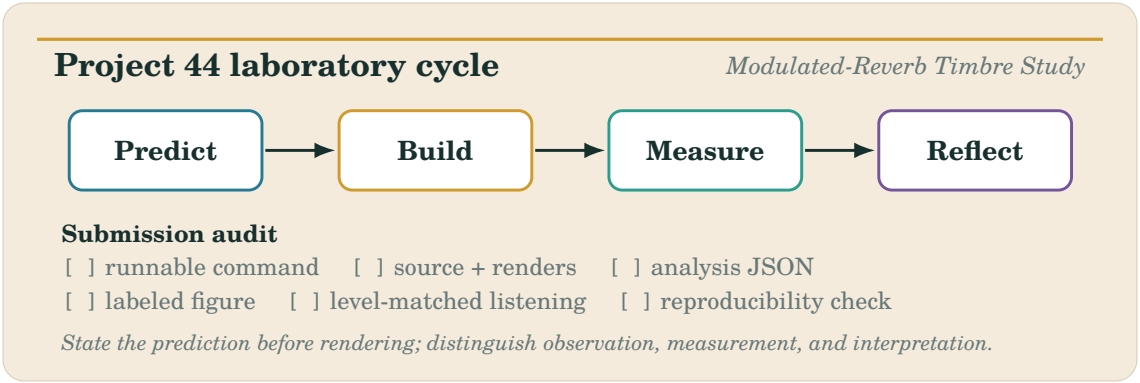


Figure B-44: Project 44 laboratory cycle: Modulated-Reverb Timbre Study

B.47 Project 45: Spatial Fugue Across Rooms

Project mode: Musical composition and production.

Central concepts: fugal process, room identity, and moving perspective.

Design brief. Write a fugue whose entries retain contrapuntal identity while migrating among distinct acoustic spaces. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write a three- or four-voice fugue or fugal exposition in which each subject entry begins in a recognizable room. As episodes develop, migrate voices between rooms, exchange apparent foreground and background, and include one stretto whose density is clarified by spatial separation. End with either convergence into one room or permanent architectural divergence.

Evidence package. Submit the score, isolated voices and room returns, final master, room-assignment map, routing or automation, analysis JSON, and a labeled trajectory diagram. Mark subject, answer, countersubject, episodes, migrations, and the spatial strategy used in the stretto.

Questions for the report. Can room identity function like instrumentation in a fugue? Which migrations preserve voice identity, and which deliberately destabilize it? Does spatial separation clarify counterpoint better than level or register changes?

Completion standard. Listeners must follow at least three subject entries and identify the major room migration. The stretto must remain contrapuntally intelligible, and the ending's architectural decision must support the harmonic and formal outcome.

Extension. Decode the work for a contrasting playback format and revise only spatial routing, not notes, to restore the intended contrapuntal clarity.

Figure B-45 presents Project 45 laboratory cycle: Spatial Fugue Across Rooms and establishes the visual reference for the discussion that follows.

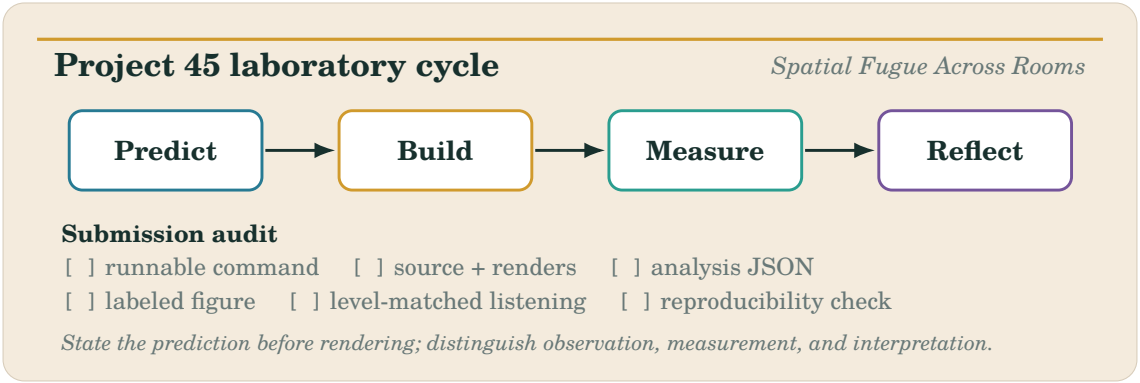


Figure B-45: Project 45 laboratory cycle: Spatial Fugue Across Rooms

B.48 Project 46: Adaptive Reverb Cue

Project mode: Musical composition and production.

Central concepts: interactive states, seamless transitions, and narrative space.

Design brief. Compose a modular cue whose reverb responds coherently to three imagined game or installation states. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Design three interactive states for an imagined game, installation, or performance: exploration, tension, and transformation. Compose modular stems that can recombine vertically and horizontally, then create reverb states and transition rules that preserve tails across state changes without exposing clicks or implausible room resets. Demonstrate at least six non-linear transitions, including a rapid reversal.

Evidence package. Submit all stems, state diagram, transition rules, preset and automation data, six transition renders, final demonstration, analysis JSON, and a labeled latency and tail-continuity report. Identify which parameters interpolate, switch, freeze, or wait for a musical boundary.

Questions for the report. What acoustic information must persist for a world to feel continuous? When should a narrative state override physical plausibility? Which transition sounds convincing in isolation but fails after repeated interactive use?

Completion standard. All six transitions must be click-free, musically coherent, and reproducible in arbitrary order. The rapid reversal must preserve intelligible state identity, and no transition may depend on a single fixed timeline.

Extension. Build a simple live controller or scripted simulator that chooses states unpredictably and record a five-minute emergent performance.

Figure B-46 presents Project 46 laboratory cycle: Adaptive Reverb Cue and establishes the visual reference for the discussion that follows.

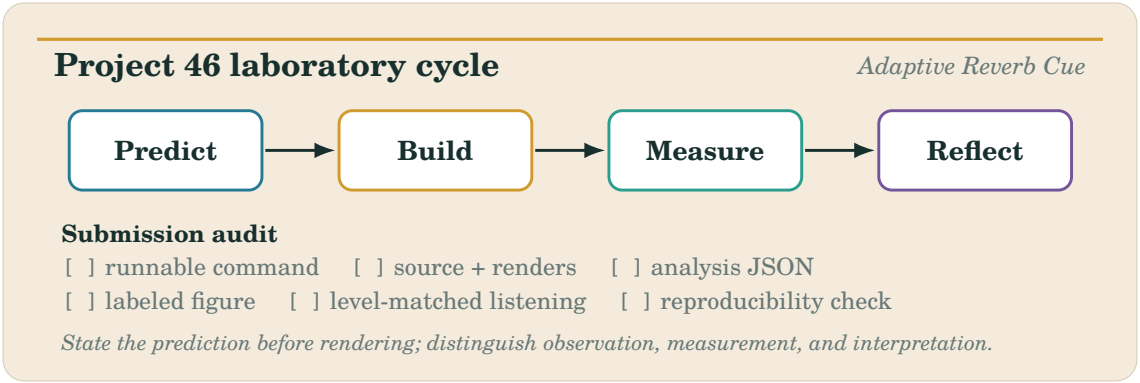


Figure B-46: Project 46 laboratory cycle: Adaptive Reverb Cue

B.49 Project 47: Reverb-Orchestrated Song

Project mode: Musical composition and production.

Central concepts: sectional contrast, vocal clarity, and send orchestration.

Design brief. Produce an original song whose verse, chorus, and bridge use distinct but related spatial orchestrations. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Write and produce an original song with verse, chorus, bridge, and final return. Design a related spatial orchestration for each section using send hierarchy, pre-delay, decay, width, ducking, and selective effects rather than one global preset. Keep the lead intelligible, give at least two supporting instruments independent depth roles, and make the final return transform an earlier space.

Evidence package. Submit the song chart or score, dry multitrack or grouped stems, wet returns, final mix, section-by-section send map, presets and automation, analysis JSON, and labeled loudness, clarity, and correlation comparisons. Include a vocal-up and instrumental reference for translation checks.

Questions for the report. Which section becomes larger because of arrangement, and which because of room design? How does pre-delay preserve lyrical focus at the largest moment? What makes the final return feel remembered rather than repeated?

Completion standard. Verse, chorus, bridge, and final return must have distinct but related spatial identities. The vocal must remain intelligible, mono compatibility must meet the stated target, and the spatial orchestration must survive at least two playback systems.

Extension. Create an alternate mix that reverses the expected scale hierarchy, making the verse enormous and chorus intimate without weakening the song's form.

Figure B-47 presents Project 47 laboratory cycle: Reverb-Orchestrated Song and establishes the visual reference for the discussion that follows.

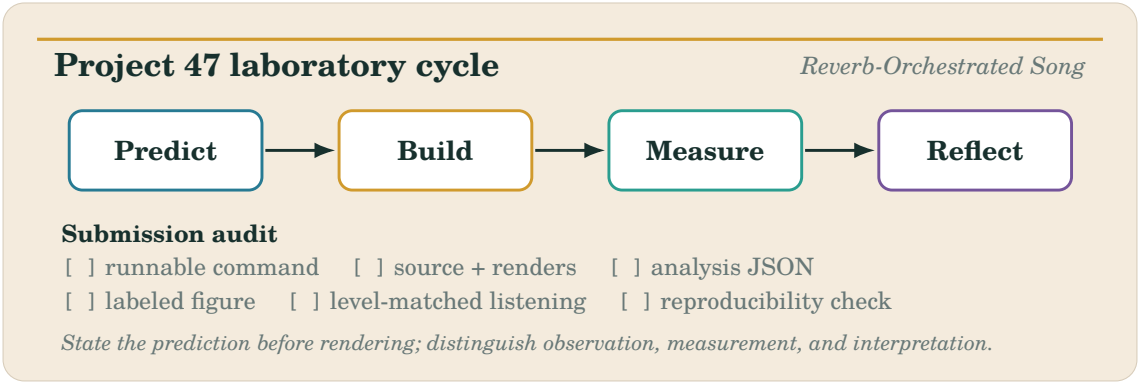


Figure B-47: Project 47 laboratory cycle: Reverb-Orchestrated Song

B.50 Project 48: Portfolio Capstone: Spatial Composition Cycle

Project mode: Musical composition and production.

Central concepts: portfolio curation, extended form, and production delivery.

Design brief. Create a multi-movement spatial composition cycle that synthesizes at least three earlier compositional techniques. State the intended musical function before processing, make an initial sketch, and preserve an unprocessed version so every spatial decision can be evaluated as part of the composition.

Procedure. Create an eight- to twelve-minute cycle of at least three connected movements. Synthesize three or more earlier techniques, including one harmonic or rhythmic process, one spatial-form process, and one realtime or interactive process. Establish recurring musical material and a recurring acoustic identity, then transform both across the cycle. Workshop sketches, conduct blind listening checks, and prepare a final concert, installation, or release-ready master.

Evidence package. Submit the complete cycle, movement stems, score or detailed formal document, presets, automation, IR provenance, exact commands, analysis JSON, technical rider, playback derivatives, and a reproducibility manifest with hashes. Include a portfolio commentary connecting specific audible moments to earlier exercises and explaining revisions made after listening tests.

Questions for the report. What unifies the cycle when rooms, media, or techniques change? Which spatial process became genuinely compositional, and which remained production support? How does the final movement reinterpret acoustic material introduced earlier?

Completion standard. The cycle must be compositionally complete, technically deliverable, and reproducible by another student. Independent listeners must recognize at least two recurring spatial ideas, all playback formats must pass routing and level checks, and the commentary must demonstrate substantive revision rather than feature accumulation.

Extension. Adapt the cycle to a new venue or medium and document which musical-spatial relationships are invariant and which require recomposition.

Figure B-48 presents Project 48 laboratory cycle: Portfolio Capstone: Spatial Composition Cycle and establishes the visual reference for the discussion that follows.

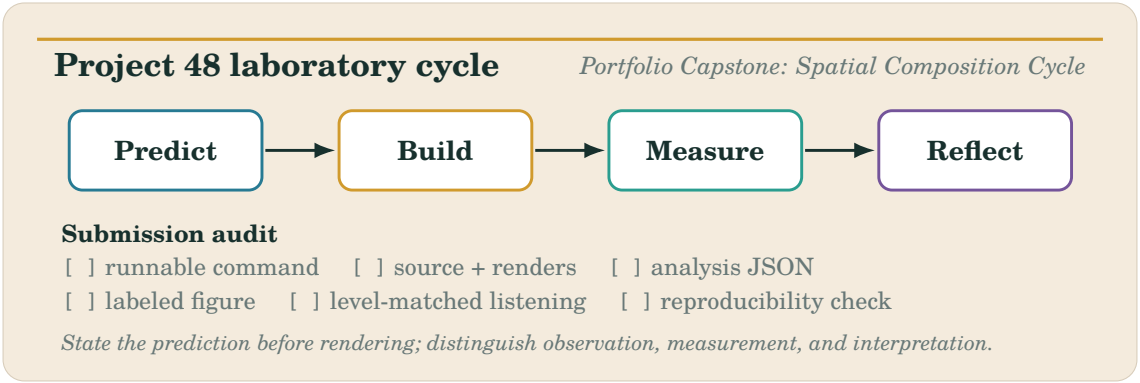


Figure B-48: Project 48 laboratory cycle: Portfolio Capstone: Spatial Composition Cycle

"How sweet the answer Echo makes to music at night."

— Thomas Moore, *Echo* (1821)

Appendix C

Finite-Element Modeling for Reverb and Resonant Systems

Finite-element modeling (FEM) provides a bridge between a physical description and an audible response. Instead of beginning with a desired RT60 or a familiar reverb topology, an FE model begins with geometry, material properties, constraints, sources, and receivers. The continuous wave or vibration problem is divided into a finite set of connected elements, and the resulting system of equations predicts how energy moves through the model. That prediction can be sampled as an impulse response and used directly for convolution, reduced to a modal model, or combined with a statistical late field.

This chapter distinguishes three related tasks that are easy to conflate:

- **Structural FEM** predicts vibration in solids such as plates, springs, shells, instrument bodies, and loudspeaker components.
- **Acoustic FEM** predicts pressure in a fluid volume such as air inside a room, duct, enclosure, or cavity.
- **Vibroacoustic FEM** couples structural motion to the surrounding fluid, as when a plate drives air or a flexible wall absorbs and reradiates energy.

verbx currently implements a bounded structural modal approximation for spring tanks and plates. It does not yet claim to be a general CAD-to-room acoustic FE solver. The distinction matters: a plate mesh predicts the plate's resonances, whereas a room mesh predicts the pressure field around sources and listeners.

C.1 From the Wave Equation to a Matrix System

For a homogeneous, lossless fluid, acoustic pressure $p(\mathbf{x}, t)$ obeys the wave equation

$$ \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}

- \nabla^2 p = s(\mathbf{x},t), $$

where c is the speed of sound, $\mathbf{x}$ is position, t is time, and s represents a source. Spatially varying density and compressibility require a more general form, but the modeling idea is unchanged. FEM multiplies the differential equation by spatial test functions, integrates over the domain, and uses integration by parts to obtain a weak form. The pressure field is then approximated as a weighted sum of local basis functions:

$$p(\mathbf{x}, t) \approx \sum_{i=1}^{N} N_i(\mathbf{x}) p_i(t).$$

Here $N_i(\mathbf{x})$ is the basis function associated with degree of freedom i, and $p_i(t)$ is its time-varying coefficient. Element contributions are assembled into the semidiscrete system

$$ \mathbf{M}\ddot{\mathbf{p}}(t)

- \mathbf{C}\dot{\mathbf{p}}(t)

- \mathbf{K}\mathbf{p}(t) = \mathbf{f}(t), $$

where **M** is the mass matrix, **C** represents damping and impedance losses, **K** is the stiffness matrix, and $\mathbf{f}(t)$ is the source vector. The same matrix form describes many structural systems when pressure coefficients are replaced by displacements.

The matrices are sparse because each basis function interacts only with nearby elements. This locality makes large models tractable, but three-dimensional audio-band room meshes can still contain millions of unknowns. Sparse storage, iterative solvers, domain decomposition, and parallel execution are therefore not optional refinements at realistic scale; they are the practical machinery that makes the calculation possible.

C.2 Boundary Conditions Are Part of the Instrument

Geometry alone does not define a room response. Every boundary must state what happens when energy reaches it.

- A rigid wall is commonly represented by a Neumann condition with zero normal particle velocity.
- A pressure-release boundary uses a Dirichlet condition, often $p = 0$.
- A locally reacting absorber uses an impedance or Robin condition relating pressure to normal velocity.
- An interface between a structure and air requires continuity of normal velocity and a force balance between pressure and structural stress.

Real absorbers are frequency dependent. Replacing a carpet, audience, curtain, or microperforated panel with one broadband absorption coefficient can produce a plausible scalar RT60 while getting modal damping, phase, and early decay wrong. For auralization, complex impedance is often more useful than a single absorption number because it preserves both magnitude and phase behavior.

Source and receiver definitions are equally consequential. A mathematical point source may overexcite mesh-scale energy; a distributed monopole or a measured directivity pattern can be more stable and more realistic. A receiver should interpolate pressure from the containing element rather than simply use the nearest node. Moving either endpoint by a fraction of a wavelength can change low-frequency modal balance substantially.

C.3 Mesh Resolution and Numerical Dispersion

An FE mesh must resolve the shortest wavelength of interest. If $h_{\max}$ is the largest relevant element dimension and n_λ is the chosen number of nodes or elements per wavelength, a useful planning estimate is

$$f_{\max} \approx \frac{c}{n_\lambda h_{\max}}.$$

Six to ten spatial samples per wavelength is a common starting range for low-order elements, not a guarantee. Element order, shape quality, formulation, solver, and acceptable phase error all affect the requirement. A mesh that is adequate for a 250 Hz room-mode study is nowhere near adequate for a 20 kHz full-band simulation. Halving the element size in three dimensions can increase the number of volume elements by roughly a factor of eight before time-step or solver costs are considered.

Under-resolution produces numerical dispersion: simulated waves travel at a frequency- and direction-dependent speed. The result can be a shifted mode, blurred arrival, or synthetic afterglow that looks like reverberation but is a numerical artifact. Good practice includes a mesh-convergence study. Repeat the calculation with a finer mesh and compare eigenfrequencies, receiver response, decay curves, and

arrival times. A result that changes materially under modest refinement is not yet a stable acoustical prediction.

Mesh quality matters as much as nominal spacing. Highly skewed or flattened elements can degrade conditioning and interpolation. Local refinement should follow geometric detail, impedance transitions, source and receiver regions, and zones with high pressure gradients. Refining an entire cathedral to the same scale as a small ornament is usually wasteful; simplifying irrelevant geometry is part of model design.

C.4 Frequency-Domain, Time-Domain, and Modal Solutions

Three solution strategies answer different questions.

C.4.1 Frequency-domain FEM

Assuming sinusoidal steady state at angular frequency ω gives

$$\left(\mathbf{K} + j\omega\mathbf{C} - \omega^2\mathbf{M}\right)\mathbf{P}(\omega) = \mathbf{F}(\omega).$$

Solving this system over a frequency grid yields transfer functions, pressure maps, and resonance detail. It is efficient when only a limited low-frequency band or a small set of frequencies is required. A broadband impulse response requires enough frequency samples, consistent phase, and an inverse transform.

C.4.2 Time-domain FEM

Time integration advances the matrix wave equation after an impulse, swept source, or other excitation. It produces arrivals and decays directly, but the time step must satisfy the formulation's stability and accuracy conditions. Long RT60 values make this expensive because the solver must continue until the field decays sufficiently. Absorbing boundaries and frequency-dependent losses also require causal time-domain realizations.

C.4.3 Modal reduction

For a linear system, free modes solve the generalized eigenproblem

$$\mathbf{K}\phi_r = \omega_r^2\mathbf{M}\phi_r,$$

where ω_r and ϕ_r are the angular frequency and shape of mode r. Retaining a subset of modes transforms a large spatial problem into a compact resonator bank. With source vector $\mathbf{b}$ and receiver vector $\mathbf{g}$, the contribution of mode r is weighted by

$$a_r = \left(\phi_r^{\mathsf{T}}\mathbf{b}\right)\left(\mathbf{g}^{\mathsf{T}}\phi_r\right).$$

A damped modal impulse response can then be written

$$h(t) = \sum_{r=1}^{R} \frac{a_r}{\omega_{d,r}} e^{-\sigma_r t}\sin(\omega_{d,r}t),$$

with damping rate σ_r and damped angular frequency $\omega_{d,r}$. Modal reduction is especially effective below the Schroeder frequency, where individual room or structure modes remain perceptually and numerically important. At higher frequencies the mode count grows rapidly, and a geometric or statistical late-field method may be more efficient.

C.5 From an FE Model to an Impulse Response

A practical room-acoustics pipeline is more than one matrix solve:

1. Clean and simplify geometry while retaining acoustically important volumes, openings, surfaces, and flexible structures.
2. Assign fluid and material parameters with units and frequency dependence.
3. Select boundary conditions, source directivity, receiver position, mesh order, spatial resolution, and target frequency range.
4. Solve in the frequency domain, time domain, or a reduced modal basis.
5. Convert receiver pressure to a causal impulse response with a documented normalization and reference distance.
6. Add high-frequency geometric or statistical energy if the FE band does not cover the complete audible range.
7. Validate modes, arrival times, energy decay, spectra, and spatial behavior against analytical cases or measurements.
8. Export the IR with geometry, mesh, solver, material, source, receiver, and software-version provenance.

Hybridization deserves special care. A low-frequency FE response and a high-frequency image-source, ray-tracing, or FDN response should overlap over a transition band. Match level, phase or group delay where meaningful, decay slope, and spatial convention before crossfading. A hard splice can create a spectral shelf or two unrelated rooms occupying adjacent bands.

C.6 Structural FEM in verbx

The current verbx `modal-fe` path applies the same reduction logic to bounded spring and plate structures. Spring tanks use lumped masses, chain stiffness, optional inter-spring coupling, drive and pickup vectors, and mode-dependent loss. Plates use a structured clamped grid, a mass-lumped matrix, a discrete thin-plate bending operator, optional tension, and a bilinearly interpolated pickup. Both systems solve normal modes offline and synthesize a causal modal IR for the regular render pipeline.

```
verbx render guitar.wav spring_fe.wav \
  --engine algo --algo-model spring \
  --electromechanical-solver modal-fe \
  --spring-count 3 --spring-fe-nodes 36 \
  --spring-fe-modes 48 --spring-fe-coupling 0.14 \
  --spring-fe-loss 0.42 --rt60 2.0 --wet 0.55 --dry 0.75

verbx render vocal.wav plate_fe.wav \
  --engine algo --algo-model plate \
  --electromechanical-solver modal-fe \
  --plate-fe-nx 20 --plate-fe-ny 14 --plate-fe-modes 72 \
  --plate-fe-loss 0.18 --plate-pickup-x 0.18 \
  --plate-pickup-y 0.76 --rt60 3.4 --wet 1 --dry 0
```

These commands are deterministic sound-design and research tools. Their node counts, mode limits, and simplified damping keep resource use bounded. They do not include a three-dimensional room

fluid mesh, detailed fixture geometry, measured transducers, nonlinear springs, or calibrated hardware losses.

C.7 Validation and Listening Tests

An FE render should be tested as both a numerical model and an audio object.

Table C-1: Validation and Listening Tests: reference table

Question	Numerical check	Listening check
Are modes in the right places?	Compare eigenfrequencies with analytical or measured peaks	Sweep a sine slowly and listen for shifted resonances
Is damping credible?	Fit per-band EDT, T20, and T30	Compare low and high decay independently
Is the mesh converged?	Repeat at finer resolution	Level-match and null or difference the two IR renders
Are source and receiver positions meaningful?	Plot mode participation and transfer magnitude	Move pickup or listener and listen for expected nodes
Is the hybrid crossover coherent?	Inspect magnitude, phase, and energy through overlap	Listen to impulses, speech, and sustained broadband material
Is the model stable?	Check finite samples, passive decay, and bounded energy	Listen for growing tones, clicks, or artificial afterglow

Use `verbx analyze` to preserve measurements beside the command and output:

```
verbx analyze plate_fe.wav --json-out plate_fe.analysis.json
```

The JSON report does not prove physical accuracy, but it makes comparisons repeatable and exposes accidental changes in decay, level, or spectrum.

C.8 Choosing FEM, Geometric Acoustics, or Statistical Reverb

FEM is strongest when wavelength-scale wave behavior matters: low-frequency room modes, diffraction, complex boundary impedance, small cavities, and coupled structural resonances. Image-source and ray methods are strongest when specular path geometry dominates and the room is many wavelengths across. FDNs and statistical tails are strongest when perceptual control, long decay, and realtime efficiency matter more than reconstructing every boundary interaction.

The methods are complementary. A musically useful room model may use FEM for low-frequency modes, image sources for early reflections, and an FDN for the dense late field. The engineering goal is not to choose the most prestigious solver; it is to choose the simplest combination that preserves the behavior the listener and the experiment require.

C.9 Further Reading

The bibliography includes Albert G. Prinn's 2023 review of FEM for room acoustics, the room-IR method of Papadakis and Stavroulakis (2015), the dispersion-reduced formulation of Okuzono and colleagues (2014), explicit time-domain studies by Okuzono and colleagues (2015–2023), and work on impedance boundaries, scalable wave simulation, and FEM-based room qualification. These sources are useful companions because they show where mesh dispersion, boundary modeling, solver choice, and validation dominate the quality of an apparently straightforward simulation.

C.10 Suggested Exercises

1. Model a simple rectangular cavity with stated boundary conditions and compare its lowest predicted modes with a measured or synthesized response.
2. Change one material or boundary parameter at a time and relate the simulated change to absorption, damping, or modal spacing.
3. Estimate the computational cost of refining a mesh in time and space, then explain why a faster method may be preferable for late reverberation.
4. Use a finite-element result only for the early or low-frequency part of a hybrid design, then choose a statistical late-field model for the remainder.
5. Write a validation plan that distinguishes agreement with one microphone position from agreement across a meaningful listening region.
6. Compare a finite-element prediction with a modal or waveguide approximation and state the domain in which each is most useful.
7. Perform a mesh-convergence thought experiment for one cavity and identify the evidence needed before trusting a high-frequency result.
8. Map one simulated mode onto a musical note range and propose an audible experiment to test the prediction.
9. Estimate memory and computational complexity for a three-dimensional simulation as resolution increases, using stated assumptions.
10. Write a short report that separates model inputs, numerical assumptions, validation evidence, and artistic interpretation.

“And more than echoes talk along the walls.”

— Alexander Pope, *Eloisa to Abelard* (1717)

Appendix D

Microtonal Workflows, Scala Import, and Scale-Tuned Reverberation

A conventional reverb is usually designed to avoid obvious pitch. Its modes are dense, irregular, or modulated so the tail supports many notes without advertising a key. Scale-tuned reverberation takes a different position: the decay field may have a harmonic vocabulary of its own. Resonances can reinforce the tuning of a work, create a stable spectral halo around selected degrees, or make out-of-scale material sound deliberately tense.

verbx imports Scala `.scl` files during synthetic IR generation. It expands the scale over a bounded frequency range, uses the resolved frequencies as modal targets, and applies a controlled constant-Q emphasis bank. The resulting WAV is an ordinary impulse response. It can therefore be used offline, in verbx realtime convolution, in a DAW convolution plug-in, or in a reproducible audio dataset without parsing Scala data in the audio callback.

This is **scale-conditioned reverberation**, not pitch correction. It does not detect each incoming note, retune the source, or change the IR as harmony moves. The fixed IR behaves more like a sympathetic resonator whose preferred frequencies were derived from a tuning system.

D.1 What a Scala File Describes

The Scala scale format is a compact text representation of one repeating pitch collection. Blank lines and text beginning with `!` are ignored. The first content line is a description, the second is the number of following pitch entries, and the remaining lines define strictly increasing degrees above an implicit ratio of 1/1.

An entry containing a decimal point is interpreted as cents. An integer is a whole-number ratio, and a fraction is an explicit ratio. The final entry is the period of repetition. It is often 2/1, or 1200 cents, but it can instead be a tritave, stretched octave, or another interval.

```
! 19edo.scl
19-tone equal temperament
19
63.1578947
126.3157895
189.4736842
252.6315789
315.7894737
378.9473684
442.1052632
505.2631579
568.4210526
631.5789474
694.7368421
757.8947368
```

```
821.0526316
884.2105263
947.3684211
1010.5263158
1073.6842105
1136.8421053
2/1
```

Scala keyboard mappings use the separate `.kbm` format. verbx currently reads `.scl`, not `.kbm`. Root mapping is supplied explicitly with `--scala-root-hz` and `--scala-root-degree`, which keeps the generated IR independent of MIDI note numbering.

D.2 How verbx Maps Degrees to Frequencies

Let r_d be the ratio for degree d, r_p the repeat-period ratio, f_r the frequency assigned to the selected root, and d_r the selected root degree. For register index k, verbx resolves

$$f_{d,k} = f_r \frac{r_d}{r_{d_r}} r_p^k.$$

Only targets between `--scala-low-hz` and `--scala-high-hz` are retained, and targets too close to Nyquist are rejected. If a high-division scale creates

more targets than `--scala-max-targets`, verbx samples evenly through the logarithmically ordered list. This preserves low-to-high coverage while bounding the number of filters and modal attractors.

`--scala-root-degree` is zero based and refers to the implicit unison plus the listed degrees before the final period. Degree zero is therefore the implicit $1/1$. Assigning 220 Hz to degree zero places the unison lattice on 220 Hz; assigning 220 Hz to another degree rotates the same scale around that reference.

D.3 Tuning Mathematics for Reverb Designers

Microtonal work becomes easier when ratios, cents, and frequencies are treated as different views of the same interval. A ratio r converts to cents by

$$c(r) = 1200 \log_2(r),$$

and a cent value c converts back to a ratio by

$$r(c) = 2^{c/1200}.$$

Equal divisions of a period use one repeated step. If a period ratio r_p is divided into N equal parts, degree d has ratio

$$r_d = r_p^{d/N}.$$

For ordinary N-EDO, $r_p = 2$; for 13 equal divisions of the Bohlen–Pierce tritave, $r_p = 3$. The distinction is structural. An octave-period scale repeats its pitch classes when frequency doubles, while a tritave-period scale repeats when frequency triples. Convolution exposes that difference because every source partial encounters a fixed lattice extending through many registers.

Just-intonation scales specify small-integer ratios instead of equal steps. A $3/2$ perfect fifth above 220 Hz resolves to 330 Hz, while a $5/4$ major third resolves to 275 Hz. Those exact values matter when tuned

resonances are narrow. If a source partial lies at frequency f_s and the corresponding IR resonance lies at f_r, their beating rate is approximately

$$f_b = |f_s - f_r|.$$

A 1 Hz difference produces a slow pulse; a 12 Hz difference can produce obvious roughness; a much larger difference may be heard as separate spectral components. The perceptual boundary is frequency dependent and affected by level, duration, masking, and the spectra of both source and return. This is why one fixed cent tolerance cannot describe every register equally well.

D.3.1 Bandwidth, quality factor, and selectivity

verbx expresses resonance width in cents because that is musically portable across frequency. If the full bandwidth is b cents around center frequency f_c, the approximate linear bandwidth is

$$\Delta f = f_c \left(2^{b/2400} - 2^{-b/2400}\right),$$

and the corresponding quality factor is

$$Q = \frac{f_c}{\Delta f}.$$

The same cent width therefore occupies more hertz in the treble than in the bass while preserving approximately equal pitch selectivity. Narrow bands produce long, identifiable ringing and reveal small tuning differences. Broad bands tolerate vibrato, bends, ensemble intonation, and detuned partials, but they also reduce the identity of individual scale degrees.

Bandwidth should be chosen with the source, not in isolation. A fixed-pitch synthesizer can support narrow bands; a choir, string ensemble, or analog oscillator often benefits from a wider field. Percussive noise may excite every band strongly enough that even moderate widths create an unmistakably pitched tail.

D.4 A Signal Model for Scale-Conditioned Decay

For input $x[n]$ and IR $h[n]$, convolution produces

$$y[n] = \sum_m x[m]h[n-m].$$

A useful conceptual decomposition writes the tuned IR as

$$h[n] = (1-\alpha)h_u[n] + \alpha h_s[n],$$

where $h_u[n]$ is the underlying untuned or diffuse response, $h_s[n]$ is the scale-conditioned component, and α represents the effective tuning strength. This equation is conceptual rather than an exact statement of every internal normalization stage, but it explains the audible continuum. Near $\alpha = 0$, the result behaves like an ordinary room with slight color. Near $\alpha = 1$, the tail behaves more like a resonator bank.

The tuned component may be approximated as a sum of damped modes:

$$h_s[n] = \sum_{k=1}^{K} A_k e^{-n/\tau_k} \sin\left(2\pi f_k n/f_s + \phi_k\right),$$

where each f_k is attracted toward a resolved Scala target. The input does not activate every mode equally. In the frequency domain,

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}),$$

so a resonance becomes prominent only when the source supplies energy near it. This explains why the same IR can sound gently colored on one instrument and strongly pitched on another.

The tail also acts as a memory. If note i ends at time t_i with decay constant τ_i, its residual amplitude at later time t is proportional to

$$a_i(t) = a_i(t_i)e^{-(t-t_i)/\tau_i}.$$

The harmony heard at time t therefore includes the present source plus weighted remnants of earlier events. Tuning and RT60 cannot be separated musically: the scale selects what persists, while decay time determines how many prior events remain available to interact.

D.5 Psychoacoustics of Tuned Reverberation

D.5.1 Fusion, beating, and roughness

Two nearby components may fuse into one fluctuating tone, produce audible beats, or separate into distinct pitches. Tuned reverb can place the dry source and wet resonance on either side of those perceptual regimes. Narrow bands and high wet level emphasize discrepancies; broad bands and diffuse energy promote fusion. Slow beating can sound like life or motion, while dense beating across many partials can make a tail grainy, unstable, or harsh.

The relationship changes over time. During the attack, the direct source often masks the return. After the source stops, the resonance is exposed and its pitch may appear to “resolve” even though the system is linear and time invariant. This temporal unmasking is one reason tuned reverberation can affect phrase endings more strongly than note beginnings.

D.5.2 Auditory masking and register

Strong source energy can hide nearby wet resonances during a phrase. A sparse arrangement or a rest reveals them. Low resonances need little gain to imply a pedal because the bass contains fewer competing modes and anchors pitch strongly. Midrange resonances interact with fundamentals and low partials, so they most readily alter perceived harmony. High resonances often read as sheen, metal, or air unless they form a sparse and sustained pattern.

Masking also explains why soloing the wet return is necessary but insufficient. The solo reveals the IR’s vocabulary; the full mix reveals whether that vocabulary survives, supports, or interferes with the source. Decisions should be made in both conditions at matched playback level.

D.5.3 Pitch salience versus spatial plausibility

Ordinary room reverberation contains modes, but a dense, irregular distribution usually prevents one scale from dominating. Pulling modes toward a pitch lattice increases pitch salience and can reduce the impression of a neutral physical room. There is no universal optimum. Production work may favor

a plausible space with subtle affinity, while installation, electroacoustic, or experimental work may intentionally cross into an audible instrument.

Three controls move the result along this continuum:

1. Increase `--scala-strength` to align and blend more strongly.
2. Increase `--scala-gain-db` to expose the conditioned bands.
3. Decrease `--scala-bandwidth-cents` to make each band more selective.

Changing all three at once makes diagnosis difficult. Move one coordinate at a time, level-match, and preserve the previous IR for comparison.

D.6 How Each Synthesis Mode Expresses a Scale

The same `.scl` file does not produce the same musical behavior in every mode.

Table D-1: How Each Synthesis Mode Expresses a Scale: reference table

Mode	Scale mechanism	Typical impression	Best first use
`fdn`	Conditioned bands color a recursively mixed late field	Smooth space with a persistent spectral center	General musical reverb that must remain cohesive
`stochastic`	Constant-Q emphasis shapes a noise-derived decay	Diffuse, breath-like spectrum with weak note identity	Vocals, speech, pads, and subtle alternate-tuning color
`modal`	Resonator frequencies are attracted toward scale targets, then emphasized	Clearly pitched object or sympathetic resonator	Percussion, impulses, drones, and exposed sound design
`hybrid`	Early field and multiple late families combine before conditioning	Balanced room cues plus audible harmonic affinity	Default composition and production workflow

`modal` is the most revealing diagnostic mode because individual resonances are easy to hear. It is also the easiest mode to overstate. `stochastic` can hide scale identity when bands are broad or gain is low, but that restraint may be exactly right for a vocal mix. `hybrid` is usually the safest starting point because untuned energy fills gaps between targets and maintains a continuous tail.

The selected mode should match the artistic metaphor. If the desired result is "a room sympathetic to this scale," begin with `hybrid` or `fdn`. If it is "an instrument made from this scale," begin with `modal`. If it is "air colored by this scale," begin with `stochastic`.

D.7 Designing and Auditing a Scala File

A scale file is small enough to inspect manually, and it should be. Before generation, verify:

1. The description identifies the scale and source clearly.
2. The declared count equals the number of pitch rows.
3. Every degree lies strictly above the implicit 1/1 and below or at the final period.
4. Degrees are strictly increasing.
5. Decimal cents and ratios express the intended values.
6. The last entry is the intended repeat period, not accidentally the last pitch class below it.
7. The file is UTF-8 text and comments begin with `!`.

Do not infer tuning from a filename alone. A file called `just_major.scl` may use a different seventh, omit a degree, or repeat at a nonstandard period. The

SHA-256 stored by verbx identifies the actual bytes used in synthesis.

For a new scale, begin with a narrow frequency range and print or inspect the resolved metadata. Confirm that the selected root appears where expected and that the highest targets remain below Nyquist. Then expand the range. This staged procedure catches root-degree and period mistakes before a long IR is rendered.

The bundled examples provide useful parser contrasts:

- `examples/scales/19edo.scl` uses cents and an octave period.
- `examples/scales/5_limit_major.scl` uses integer ratios and an octave period.
- `examples/scales/bohlen_pierce_13edo.scl` uses cents with a 3/1 period.

They are demonstrations, not canonical definitions of every performance practice associated with those tuning families.

D.8 First Complete Workflow

Generate the tuned IR once, inspect it, and then use it in as many renders as needed:

```
verbx ir gen irs/19edo_hybrid.wav \
  --mode hybrid --length 18 --rt60 7 --seed 19 \
  --scala-file examples/scales/19edo.scl \
  --scala-root-hz 220 --scala-root-degree 0 \
  --scala-low-hz 90 --scala-high-hz 10000 \
  --scala-strength 0.65 --scala-bandwidth-cents 22 \
  --scala-gain-db 5 --scala-max-targets 128

verbx ir analyze irs/19edo_hybrid.wav \
  --json-out irs/19edo_hybrid.analysis.json

verbx render source.wav tuned_space.wav \
  --engine conv --ir irs/19edo_hybrid.wav --wet 0.35 --dry 1
```

For low-latency auditioning, load the generated IR into realtime convolution:

```
verbx realtime \
  --engine conv --ir irs/19edo_hybrid.wav --block-size 128
```

Scala parsing, target expansion, and filter construction happen during `ir gen`, not in the realtime callback. Realtime latency is therefore determined by the audio device, host buffers, block size, safety buffering, and convolution partitioning rather than the number of degrees in the source scale.

D.9 The Controls as Musical Decisions

Table D-2: The Controls as Musical Decisions: reference table

Control	Musical question	Conservative start	More audible effect
`--scala-root-hz`	Where is the tuning lattice anchored?	Match the work's tuning reference	Shift by a scale degree or structural bass frequency
`--scala-root-degree`	Which degree receives the reference frequency?	`0`	Rotate the modal hierarchy around another degree
`--scala-low-hz`	Should bass modes participate?	100 Hz	35–70 Hz for drones and installation work
`--scala-high-hz`	How much brightness carries pitch identity?	6–10 kHz	12–16 kHz for bright, exposed resonances
`--scala-strength`	How strongly do modes approach the scale?	0.45–0.70	0.85–1.00 for an unmistakably tuned object
`--scala-bandwidth-cents`	Are resonances selective or forgiving?	20–35 cents	5–15 cents for narrow ringing bands

Control	Musical question	Conservative start	More audible effect
`--scala-gain-db`	How far does the lattice emerge from the diffuse tail?	3–6 dB	8–12 dB for special effects
`--scala-max-targets`	How dense may the target lattice become?	96–128	More targets for high-division scales and wide ranges

Strength and gain are related but not identical. Strength controls modal attraction and the blend of the emphasis layer. Gain controls the level of the filtered bank before final normalization. Bandwidth controls how much nearby pitch motion is accepted. A narrow, strong setting sounds like a bank of sympathetic strings; a broad, moderate setting sounds more like a room whose color gently favors the scale.

D.10 Musical Implications of a Tuned Decay Field

D.10.1 Consonance becomes time dependent

A dry pitch can end while its scale-related energy remains in the tail. The next note is therefore heard against a memory of previous notes. A melodic interval that is locally consonant may produce roughness when the older decay overlaps it, while a dissonant attack may resolve into a stable resonant field. Scale-tuned reverb makes harmonic rhythm partly a function of RT60.

Longer decay is not merely "more" of the same tuning. It increases the number of prior events simultaneously represented by the room. Slow music can expose individual degrees; fast music can accumulate the complete scale into a spectral aggregate. When testing a design, vary tempo before changing the scale or declaring the tuning ineffective.

D.10.2 The reverb can confirm or contradict the ensemble

When performers and IR share a tuning, resonant energy tends to reinforce stable scale positions. Near misses, expressive intonation, vibrato, and pitch bends pass through and may beat against narrow bands. This can make intonation audible in a productive way, but it can also make a flexible vocal or string line feel constrained.

Using a different scale in the IR creates a second harmonic layer. A 12-tone-equal-tempered source through a just-intonation IR may produce slow beating around nominally equivalent intervals. A diatonic source through a 19-EDO IR can make chromatic inflections leave different spectral residues. The result is not automatically dissonant; bandwidth, root, register, source spectrum, and decay determine whether the difference reads as color, chorus, or conflict.

D.10.3 Register controls whether tuning is heard as harmony or timbre

Low-frequency targets are sparse and individually audible. They can imply a fundamental, pedal, or room mode even when the source does not sustain that pitch. Midrange targets interact strongly with musical fundamentals and lower partials. High-frequency targets are more likely to be heard as sheen, brightness, or metallic identity than as named pitch.

For a mix that must remain harmonically agile, begin the tuned range above the bass and use broader bands. For an installation, drone, or resonant percussion piece, extending the lattice downward can make the reverb function as an instrumental voice.

D.10.4 Scale cardinality changes texture

A five- or seven-degree scale produces a relatively sparse lattice and leaves audible gaps. Equal divisions with 19, 31, or 53 degrees create increasing spectral density. A larger pitch count does not guarantee

a smoother tail: narrow filters can still expose beating and local clusters, while the target budget may omit some expanded degrees. Consider cardinality together with bandwidth and range.

Non-octave scales are especially distinctive because spectral relationships do not repeat at powers of two. A tritave-period scale can cause familiar octave equivalence to drift across registers. This may be musically compelling on bells, percussion, synthetic tones, and inharmonic sources, but it can feel unstable on octave-doubled orchestration unless that instability is intended.

D.10.5 The source spectrum determines which degrees awaken

An IR cannot emphasize energy that the source does not excite. A flute-like tone may illuminate only a few bands; noise, cymbals, consonants, and distorted sources can excite nearly the entire lattice. Transients reveal the IR itself, while sustained tones reveal the interaction between source partials and tuned resonances.

Test at least four source classes: an impulse or click, a chromatic sine sweep, a sustained harmonic sound, and broadband musical material. This separates the design of the resonator from the accident of one orchestration.

D.10.6 Spatialization can distribute harmonic function

A stereo or multichannel design need not use one identical tuned field in every channel. Related IRs can place complementary degree sets, roots, or bandwidths around the listener. Keep enough common energy to maintain one space, and avoid hard channel-specific peaks that disappear under downmixing. Treat each IR as part of a spatial harmony, then check mono, stereo, binaural, and speaker-array translations at matched level.

D.11 Source-Aware Orchestration

The scale file defines available resonances, but orchestration determines which ones receive energy. Designing the IR without considering the source is like specifying sympathetic strings without asking what will excite them.

D.11.1 Voice and choir

Vowels contain moving formants and a harmonic series whose exact frequencies follow the sung fundamental. Consonants provide broadband transients that can excite the entire tuned field. A narrow IR may reinforce stable vowels while making consonant tails unexpectedly pitched. For solo voice, begin above the lowest expected fundamental and use 25–45-cent bandwidth. For choir, broaden further unless exposing intonation differences is part of the piece.

Test vibrato at the slowest and fastest expected rates. If the dry pitch moves outside a narrow resonance while the tail remains fixed, the result can alternate between reinforcement and attenuation. That may sound luminous, unstable, or simply out of tune depending on context. A broad hybrid field generally follows human pitch variation more gracefully than a high-Q modal bank.

D.11.2 Strings and winds

String spectra change with bow position, pressure, mute, register, and articulation. Sul ponticello playing supplies abundant upper-partial energy and can reveal high Scala targets dramatically; a soft fundamental-rich tone may activate only a few low modes. Glissandi create a useful diagnostic because resonances brighten as partials cross the fixed lattice.

Wind instruments differ in harmonic structure and noise content. Clarinet-like odd-partial emphasis, flute-like breath energy, and brass brightness illuminate different degree subsets even on the same

written pitch. Test representative registers independently. A tuning that flatters a low horn line may turn a high reed texture into a dense whistle bank.

D.11.3 Piano, harp, and fixed-pitch instruments

Fixed-pitch instruments make tuning differences easy to reproduce. They also carry long natural decays, so instrument resonance and room resonance overlap. Before increasing RT60, compare the dry decay with the wet-only decay. If the source already sustains for several seconds, a shorter tuned IR may create more clarity than a long one.

Pedal behavior matters. A sustained piano resonance can feed the reverb with partials from many earlier notes, producing a two-stage memory: strings retain the harmony, then the IR retains the strings. Sparse textures expose this beautifully; dense pedal can turn even a seven-degree IR into broadband accumulation.

D.11.4 Percussion and impulsive excitation

Clicks, woodblocks, snare attacks, and mallet transients approximate broad excitation and make the IR's pitch set audible. A modal Scala IR can transform an unpitched attack into a pitched gesture. This is one of the clearest ways to treat reverberation as orchestration rather than ambience.

The apparent pitch may depend more on modal amplitude and damping than on the formal root. If one low target dominates, adjust `--scala-low-hz`, use a higher root, or reduce modal emphasis before changing the scale itself. Preserve peak headroom because simultaneous excitation of many resonances can create a large wet transient followed by a deceptively quiet tail.

D.11.5 Electronics, distortion, and noise

An ideal sine excites only a narrow part of the IR. Saturation or distortion adds harmonics that can awaken many more degrees. If tuning seems absent on a pure oscillator, compare the same line after gentle saturation. Conversely, if a distorted source makes the field overwhelming, generate a version with a lower high-frequency limit rather than merely reducing global wet level.

Noise reveals the transfer function almost directly. Filtered noise can audit one register at a time, while full-band noise exposes the complete lattice. Because noise is not harmonically committed, the tuned return may become the primary pitch-bearing layer.

D.12 Composition and Production Strategies

D.12.1 Stable halo

Use the composition's tuning, moderate strength, 25–40-cent bandwidth, and a high-pass-limited target range. The reverb supports pitch identity while the dry source remains dominant. This is a practical starting point for vocals, chamber textures, and tuned percussion.

D.12.2 Harmonic shadow

Root the IR on a structural pitch that is absent from the current passage. As the tail accumulates, it implies a latent pedal or tonal region. Automate the wet level rather than switching IRs abruptly so the shadow enters as memory.

D.12.3 Controlled contradiction

Generate two IRs from related but nonidentical scales. Render identical source material through each and crossfade between the wet returns. Because an IR is time invariant, the transition remains reproducible and can be shaped without changing the dry performance.

```
verbx ir gen irs/scale_a.wav \
  --mode hybrid --length 14 --rt60 6 --seed 41 \
  --scala-file scales/scale_a.scl --scala-root-hz 220 \
  --scala-strength 0.6 --scala-bandwidth-cents 28

verbx ir gen irs/scale_b.wav \
  --mode hybrid --length 14 --rt60 6 --seed 41 \
  --scala-file scales/scale_b.scl --scala-root-hz 220 \
  --scala-strength 0.6 --scala-bandwidth-cents 28

verbx ir morph irs/scale_a.wav irs/scale_b.wav irs/transition.wav \
  --mode equal-power --alpha 0.5
```

Using the same seed and synthesis parameters isolates the scale as the main experimental variable. Morphing the two final IRs produces a fixed intermediate response; it does not create time-varying tuning inside one render.

D.12.4 Percussive resonator

Use `modal`, strong alignment, narrow bands, and a lower target ceiling. Short impulses then excite pitched decays much like struck bars or strings. Preserve headroom because many resonances can sum during dense attacks.

D.12.5 Diffuse microtonal room

Use `hybrid` or `stochastic`, strength below 0.6, 30–60-cent bandwidth, and a moderate gain. This leaves enough untuned energy to bind the tail together and reduces the impression of a parallel synthesizer.

D.13 Worked Harmonic-Time Scenarios

D.13.1 One chord, several decay windows

Begin with one sustained sonority followed by silence. Render it through three IRs sharing scale, root, seed, and spectral limits but using RT60 values of 1.5, 5, and 15 seconds. The short response behaves primarily as color. The middle response lets individual degrees become legible after release. The long response turns the chord into a formal background for whatever follows.

This comparison demonstrates why scale-tuned reverb cannot be specified by tuning alone. The same frequency lattice may function as timbre, cadence, pedal, or independent layer solely because decay changes the listening window.

D.13.2 Diatonic phrase against a just field

Use a 5-limit scale rooted on the phrase's structural center, then play or render a 12-EDO phrase through it. Stable notes may align closely with IR targets, while tempered thirds and sevenths create slow discrepancies. Use broad bands first. Narrow the bands only after deciding whether the resulting beats articulate tension meaningfully.

Then transpose the dry phrase without changing the IR. This converts the field from confirmation to contradiction. Finally, transpose both source and root by the same ratio. The three conditions distinguish absolute resonance placement from interval structure.

D.13.3 Modulation between two roots

Prepare two IRs from the same scale, seed, mode, RT60, and bandwidth, but assign different root frequencies. Put each on a separate wet return. Before the harmonic modulation, fade the destination return in while the source still occupies the old region. Let both tails coexist briefly, then fade the old return after its important decay has passed.

The overlap is not merely technical concealment. It creates a harmonic corridor in which both resonant memories coexist. A short crossfade reads as a spatial switch; a long crossfade can become a composed modulation. Keep the dry signal stable so the listener can attribute the transformation to the field rather than to level movement.

D.13.4 Scale mutation with a fixed root

Generate matched IRs from two scales sharing the same root and period but differing in selected degrees. The common degrees preserve continuity while changed degrees alter the decay's internal harmony. Equal-power IR morphing creates one static midpoint; parallel returns with automated gains create a time-varying transition.

Use an impulse to confirm that the two endpoint IRs have comparable onset and energy. Then test sustained material. A difference that is subtle on the impulse may become obvious after repeated harmonic excitation.

D.13.5 Non-octave form

With a tritave-period IR, orchestrate successive registers rather than treating octave duplication as neutral. A bass note and its octave may excite different relations to the field, while a frequency tripling returns to the scale's periodic structure. Alternate octave and tritave displacement to make the reverb reveal the difference after each attack.

Non-octave design is especially effective when the source itself is inharmonic or when register functions structurally. Bells, metallic percussion, granular spectra, and synthesized partial sets can cooperate with the field without implying conventional octave-based harmony.

D.13.6 Silence as harmonic exposure

Compose rests long enough to hear the tuned return without the direct source. The attack may conceal differences that become clear only after 300 ms or one second. A rest can therefore operate as a harmonic microscope. Compare early silence, where room cues dominate, with late silence, where the narrowest and longest modes survive.

When a phrase ending sounds wrong, do not inspect only its final chord. The tail contains weighted remnants of earlier notes. Shorten RT60, reduce low-frequency participation, or alter harmonic rhythm before changing the final sonority.

D.14 Transposition, Modulation, and Changing Harmony

A scale-conditioned IR has a fixed root. To transpose the resonant field, regenerate it with a different `--scala-root-hz`. For a frequency ratio r, the transposed root is

$$f_r' = r f_r.$$

For n equal-tempered semitones, $r = 2^{n/12}$. For a Scala degree, use the degree's ratio directly. Regeneration is deterministic and cached, so a small library of roots can be prepared before a session.

Do not switch long IRs at arbitrary note boundaries and expect an inaudible transition. The old convolution state contains the previous tuning. Safer strategies include overlapping wet returns, equal-power crossfades longer than the important early field, or rendering sections separately with complete tails. A future dynamic resonator could track harmony continuously, but that is a different DSP architecture from static convolution.

D.15 Stereo, Surround, and Immersive Deployment

Scale tuning adds a harmonic dimension to spatial routing. A multichannel field can repeat one lattice everywhere, distribute related lattices, or separate diffuse and pitched responsibilities between layers.

D.15.1 Shared-field strategy

Use one multichannel IR or closely matched IRs with the same scale, root, and conditioning. This produces the strongest sense of one coherent room. Vary early reflections and decorrelation while keeping the late spectral center shared. It is the safest strategy for film, vocal, and ensemble work where spatial continuity matters more than hearing individual degrees move.

D.15.2 Complementary-degree strategy

Divide a scale into overlapping subsets and assign them to different returns. For example, front channels can carry stable structural degrees while side and rear returns emphasize tensions or upper extensions. Always retain common diffuse energy; completely disjoint narrow lattices can make head movement or downmixing expose holes.

This strategy is compositional, not a claim of physical-room realism. Document which degrees belong to each layer and test every fold-down. A stereo or mono sum may combine subsets into a denser spectrum than any individual speaker reveals.

D.15.3 Height as spectral perspective

Height returns often tolerate brighter target ranges because listeners can associate elevated energy with air or halo. Rather than sending the complete low-frequency lattice overhead, use a higher `--scala-low-hz`, moderate gain, and enough diffuse content to avoid isolated whistles. Keep bass and structural root information in the bed unless a deliberately disembodied image is wanted.

D.15.4 Moving objects and fixed tuned fields

An object can move through a fixed bed or field return while retaining one scale-conditioned decay. If several location-specific IRs use different roots, movement also becomes harmonic transformation. Crossfade states rather than hard-switching them, and distinguish panning automation from IR-state automation in the session record.

D.15.5 Translation checks

Evaluate at least speaker-array, binaural, stereo, and mono versions. Listen for degree cancellation, exaggerated shared bands, center buildup, and a height layer that disappears when folded down. Use matched loudness and include both sustained and transient material. A spatial design is not complete because its native layout sounds impressive.

D.16 Measurement and Analytical Listening

Standard room metrics do not fully describe scale conditioning. RT60, EDT, and clarity remain useful, but they should be paired with frequency-target evidence.

D.16.1 Target-to-neighbor contrast

For each resolved frequency f_k, compare energy in a narrow target band with energy in adjacent off-target bands. A simple contrast measure is

$$C_k = 10 \log_{10} \left(\frac{E_{\text{target},k}}{E_{\text{neighbor},k} + \epsilon} \right),$$

where ϵ prevents division by zero. Report the distribution of C_k rather than only its mean. One dominant low mode can otherwise hide weak upper conditioning.

D.16.2 Decay by target degree

Filter the IR around selected degrees and fit decay slopes separately. Two targets can have similar peak magnitude but different persistence. Degree-wise decay is musically important because late harmony is governed by the modes that survive, not merely those that begin loudly.

D.16.3 Matched-seed controls

Generate an untuned and tuned IR with identical mode, seed, length, RT60, channel count, and broad spectral shaping. Render the same source through both and loudness-match the results. This isolates conditioning from random topology and gross level. Repeat across several seeds before making a general claim; one favorable random realization is an anecdote, not a robust result.

D.16.4 A listening vocabulary

Use consistent descriptors so notes can be compared across sessions:

- **Pitch salience:** how clearly the return suggests stable pitch.
- **Roughness:** how strongly close components produce rapid fluctuation.
- **Fusion:** whether source and return form one object.
- **Harmonic drag:** whether old tail energy resists a new harmony.
- **Spectral halo:** broad affinity without distinct notes.
- **Resonator identity:** perception of the return as an instrument.
- **Register bias:** concentration of the effect in bass, middle, or treble.
- **Decay revoicing:** change in apparent harmony as modes disappear.

Collect notes during full-mix, wet-only, and post-release listening. The three conditions answer different questions and should not be collapsed into one preference score.

D.17 Reproducible Scale Libraries

Treat the `.scl` file as source data, not merely a preset name. Two files with

the same filename can contain different degrees. verbx records the scale description, content SHA-256, root mapping, resolved frequencies, frequency limits, strength, bandwidth, gain, and target budget in the IR metadata. The content hash also participates in cache identity.

For a production or research corpus, retain:

- the original `.scl` file and its license or source note;
- the exact `verbx ir gen` command or configuration;
- the generated IR and metadata sidecar;
- an analysis JSON report;
- the source audio identity and render command;
- listening notes describing tuning, root, register, and perceived interaction.

When generating machine-learning data, split scale families before expanding roots, seeds, RT60 values, and source files. Variants from one scale should not leak across training, validation, and test sets if the experiment claims generalization to unseen tunings. Include untuned controls and level-matched ablation renders so a model cannot solve the task from loudness alone.

D.18 Scale-Tuned Reverb for Audio AI and Data Augmentation

Scale-conditioned IRs can test whether an audio model is robust to structured, musically meaningful coloration rather than only generic room decay. They are useful for source separation, pitch estimation, transcription, instrument recognition, dereverberation, acoustic-scene classification, and generative audio evaluation, but only when the data split prevents tuning leakage.

D.18.1 Define the experimental factor

Decide what "tuning variation" means before generating files. Possible factors include scale family, degree count, period ratio, root mapping, bandwidth, strength, target range, or the relation between source tuning and IR tuning. Changing all of them together produces variety but weak scientific evidence.

A controlled factorial design might vary:

Table D-3: Define the experimental factor: reference table

Factor	Example levels
Scale family	19-EDO, 5-limit just collection, 13-EDT Bohlen–Pierce
Root relation	matched, one degree displaced, unrelated
Strength	0, 0.4, 0.7, 1.0
Bandwidth	12, 30, 60 cents
RT60	1.5, 5, 12 seconds
Source class	speech, monophonic music, polyphonic music, percussion, noise

The strength-zero condition is essential. It reveals whether performance changes come from reverberation itself or from the scale-conditioned component.

D.18.2 Split by lineage, not rendered filename

One `.scl` file can generate hundreds of roots, seeds, RT60 values, and source combinations. Randomly splitting those WAV files allows nearly identical tuning lattices into training and evaluation sets. Instead, assign the parent scale or scale family to a split first, then generate descendants inside that split.

For tests of unseen roots but known interval structure, keep scale files shared and split root mappings. For tests of unseen tuning systems, keep complete scale families out of training. State which generalization claim the split supports.

D.18.3 Preserve dry and wet evidence

For supervised dereverberation or augmentation, retain the dry source, IR, wet-only convolution, final mixture, and exact gain relation. A useful linear mixture is

$$y[n] = g_d x[n] + g_w (x * h)[n],$$

where g_d and g_w are documented dry and wet gains. If normalization runs after mixing, record it because it changes the direct-to-reverberant relation.

Do not let filenames encode the target class if a data loader or model can see them. Store scale identity and parameters in a manifest. Hash the source, Scala file, IR, and output so regenerated corpora can be audited.

D.18.4 Avoid shortcut learning

A classifier may appear to recognize a tuning while actually recognizing loudness, spectral tilt, IR length, one source instrument, or one random seed. Use matched-seed controls, loudness matching, balanced source classes, and multiple roots. Include adversarial controls where scale is held constant but gain and RT60 change, and where room statistics remain constant while the scale changes.

For dereverberation, evaluate whether the model removes the tail without flattening legitimate source harmonics near Scala targets. A model trained only on untuned rooms may interpret stable tuned decay as part of the instrument; a model trained too aggressively on tuned IRs may suppress sustained musical partials.

D.18.5 Evaluation beyond one aggregate score

Report task metrics separately by scale family, root relation, source class, strength, bandwidth, and RT60. Add target-band residual measurements for dereverberation and pitch error for transcription. A single mean can hide a model that performs well on diffuse fields and fails on sparse modal ones.

Listen to representative errors. Scale-conditioned tails can create musically important artifacts that generic perceptual scores underweight, including a wrong residual pedal, beating around a held note, or late energy that changes apparent chord quality.

D.19 Failure Modes and Corrections

Table D-4: Failure Modes and Corrections: reference table

Symptom	Likely cause	Correction
Tail sounds like unrelated sine tones	Strength or gain too high; bandwidth too narrow	Lower strength, broaden bands, or use `hybrid`
Bass implies the wrong harmony	Root mapping or low limit conflicts with the work	Verify root degree and raise `--scala-low-hz`
Effect disappears in a dense mix	Too little source energy near targets or bands too broad	Solo the return, inspect spectrum, then increase gain modestly
Vocal vibrato sounds trapped	Resonances are too selective	Broaden bandwidth or move tuning emphasis above the vocal fundamentals
High-division scale sounds incomplete	Target budget is truncating the lattice	Increase `--scala-max-targets` or narrow the frequency range
Realtime launch seems slow	IR is being generated in the session path	Pre-generate and cache the IR before performance
A scale edit seems ignored	An old IR file was loaded	Regenerate and verify the recorded SHA-256

Downmix becomes phasey or hollow	Channel variants are too spectrally independent	Increase shared diffuse energy and validate mono/stereo folds

D.20 A Focused Listening Exercise

Choose one dry sustained chord and one short percussion phrase. Generate four IRs with the same mode, seed, length, RT60, root, and scale:

1. An untuned control.
2. A broad setting at 50 cents and strength 0.4.
3. A moderate setting at 25 cents and strength 0.65.
4. A narrow setting at 10 cents and strength 0.9.

Loudness-match the wet returns. For each source, describe pitch stability, roughness, decay continuity, apparent register, and whether the reverb is heard as space, timbre, harmony, or a separate instrument. Repeat after transposing the root by one scale degree. The exercise reveals a central principle: scale-tuned reverberation is not one effect but a continuum between diffuse space and resonant composition.

D.21 Extended Laboratory and Composition Studies

D.21.1 Study 1: Cents, ratios, and audible beating

Create three two-note source files whose upper frequencies differ slightly around one target ratio. Render them through a narrow modal IR. Measure the frequency difference, predict the beat rate, and compare prediction with the wet-only decay. Repeat one octave higher while preserving the same cent offset. Explain why the hertz difference changes.

D.21.2 Study 2: Bandwidth as performance tolerance

Generate IRs at 10, 25, 50, and 100 cents with all other parameters fixed. Process a stable oscillator, a singer or string tone with vibrato, and a glissando. Identify the point at which tuning changes from selective resonance to broad coloration for each source.

D.21.3 Study 3: Scale cardinality and modal density

Compare five-, seven-, 12-, 19-, and 31-degree octave-period scales over the same frequency range and target budget. Keep seed, mode, RT60, strength, and gain fixed. Count resolved targets, inspect spectral spacing, and describe when individual pitch identity gives way to a continuous field.

D.21.4 Study 4: Octave and non-octave periodicity

Process the same sequence through an octave-period IR and a tritave-period IR. Transpose the source by 2/1, then by 3/1. Determine which transposition preserves the relation to each field. Compose a short passage in which the reverb, rather than the dry line, reveals the periodic structure.

D.21.5 Study 5: Harmonic memory and tempo

Use one chord progression at three tempi and three RT60 settings. Do not change the scale or wet gain. Mark where prior harmonies remain audible under later ones. Find a tempo-decay combination that supports continuity and another that creates deliberate harmonic drag.

D.21.6 Study 6: Root-degree rotation

Hold `--scala-root-hz` fixed while changing `--scala-root-degree`. Compare the resolved target lists before listening. Render one source through each IR and explain why rotation is not equivalent to an ordinary equal-tempered transposition.

D.21.7 Study 7: Orchestration as excitation

Render click, speech, bowed string, piano, cymbal, sine, and noise through one IR. Use wet-only outputs and matched levels. Create a table of which registers and degrees each source excites most strongly, then orchestrate a one-minute study that reveals the lattice gradually without changing the IR.

D.21.8 Study 8: Two-root modulation

Build two rooted IRs with identical seeds. Place them on parallel returns and compose a transition using a short crossfade, a long crossfade, and a period of intentional overlap. Compare whether each version sounds like a key change, room change, or emergence of a second instrument.

D.21.9 Study 9: Spatial harmony

Assign a shared diffuse field to all channels and complementary tuned layers to front, side, rear, and height groups. Produce speaker-array, binaural, stereo, and mono renders. Document which harmonic relationships survive each translation and revise the layout to reduce cancellation or buildup.

D.21.10 Study 10: Matched-seed perceptual test

Generate an untuned control and three conditioned IRs from the same seed. Loudness-match randomized renders and conduct a blinded listening test. Ask participants to rate pitch salience, roughness, fusion, and spatial plausibility. Report individual responses as well as means; tuned reverb can divide listeners according to attention and musical experience.

D.21.11 Study 11: Dereverberation stress test

Create matched dry, ordinary-reverb, and scale-tuned-reverb examples. Run the same dereverberation configuration on both wet conditions. Compare target-band residuals, source-harmonic damage, RT60 reduction, and listening quality. Explain whether the estimator treats tuned decay as room energy or musical sustain.

D.21.12 Study 12: A composed resonant architecture

Design a five-minute work in which at least three formal sections use the same dry instrumental palette but different relationships to a fixed Scala IR. Section one should confirm the field, section two should contradict it, and section three should use silence to expose accumulated decay. Submit the scale, commands, IR metadata, analysis JSON, score or timeline, and a commentary on how reverberation carries form.

These studies progress from isolated variables to complete musical design. In every case, retain an untuned or matched control and distinguish what the measurement demonstrates from what the listening interpretation suggests.

D.22 Suggested Exercises

1. Import one Scala scale and create two scale-tuned reverb designs with different root frequencies. Compare their interaction with the same sustained chord.
2. Use an impulse, a chromatic sweep, a harmonic sound, and a musical excerpt to test whether a tuned resonance design generalizes across sources.
3. Compare a five-degree, twelve-degree, and thirty-one-degree scale with matched target budgets. Describe changes in spectral density and beating.
4. Compose a short phrase in which reverb tuning changes at a harmonic boundary without changing the dry instrument's tuning.
5. Document the Scala file, root mapping, frequency range, target count, and any transposition used so another listener can reproduce the scale-tuned field.
6. Use the same Scala file with two different tonal centers and identify which resonant bands move and which remain fixed.
7. Compare a just-intonation scale and an equal-division scale in a scale-tuned reverb, using a sustained drone and a melodic line.
8. Design a scale-aware preset for an ensemble with two tunings, then state how the reverb should behave at their shared and conflicting degrees.
9. Create a sweep visualization showing the emphasized bands from a Scala import, with frequency in hertz and cents clearly labeled.
10. Compose a brief transition between two scale maps and document how reverberant tuning supports, obscures, or reframes the modulation.

“The reverb was so long that a trainwreck would sound good in there.”

— Frank Speller (1938–2017), American organist and composer,
on his recital in Westminster Abbey (1992)

Appendix E

Research Papers and References

Focused bibliography for extreme reverberation DSP and reverberation research in general (algorithmic reverb, FDN, convolution/IR, late-field modeling, and room reverberation metrics).

Pruned from the broader bibliography on 2026-03-01 to keep only reverb-centric papers.

Total entries: 1,066 (166 curated and primary-source entries + 900 extended Crossref entries)

Within each topical section, entries are alphabetized by first-author surname. Records without credited authors sort under "Unknown authors."

E.1 Sources for Chapter Epigraphs

The chapter-opening quotations are brief literary and musical interludes rather than technical evidence. Public-domain echo passages by Addison, Longfellow, Milton, Moore, Pope, Saxe, Shakespeare, and Shelley are collected with work attributions on the Wikiquote Echo page. Pauline Oliveros's account of using cardboard tubes, contact amplification, and a bathtub for reverberation appears in her American Mavericks interview. The separate Oliveros observation about vertical space and hearing a sound's tail is linked through the quotation record supplied for this edition. The Johnny Thunders epigraph is included as an attributed popular quotation supplied for this edition; it is not used to support a technical claim. Frank Speller's 1992 observation about the long reverberation during his Westminster Abbey recital is likewise included as an attributed recollection supplied for this edition.

E.2 Where to Start

If you are new to reverb DSP and want to understand the intellectual lineage of this tool, start with these seven papers, listed alphabetically by first author. The notes suggest a useful reading sequence; everything else in the bibliography builds on, extends, or argues with what these establish.

1. Jot and Chaigne (1997) – Maximally diffusive yet efficient FDN (entry 95)

The pivot point between classic Schroeder-style structures and modern FDN design. Jot formalized the feedback delay network as a matrix + delay system, gave a clean stability condition, and showed how to design the feedback matrix for maximum diffusion. This paper is the reason every serious reverb plugin since the late 1990s has an "FDN" somewhere inside it.

2. Rocchesso and Smith (1997) – Circulant and elliptic FDN families (entry 96)

Published the same year as Jot-Chaigne and essential reading alongside it. Rocchesso and Smith worked out the eigenvalue structure of feedback matrices, explained why some matrix designs produce audible resonances, and gave concrete construction rules for matrices that avoid them. The circulant family they describe remains a go-to for large-order FDNs.

3. Schlecht and Habets (2015) – Time-varying feedback matrices (entry 39)

Identifies the modal ringing problem that plagues fixed-matrix FDNs (you hear it as metallic "flutter" on transients) and proposes time-varying rotation of the feedback matrix as the solution. This is the theoretical grounding for verbx's reactive-control stream. Dense math, but the payoff is real.

4. Schlecht and Habets (2019) – Dense Reverberation with Delay Feedback Matrices (entry 29)

The most recent of the mandatory reads. Introduces the delay-feedback matrix concept, which lets you pack feedback directly into the delay topology rather than separating the two. Yields higher echo density for a given compute budget. The paper that shapes verbx's next-generation topology expansion.

5. Schroeder (1961) – Further Progress with Colorless Artificial Reverberation (entry 97)

The immediate follow-up, published the same year. Refines the allpass topology and introduces the parallel-comb / series-allpass architecture that became the canonical "Schroeder reverberator." The coloration problems it leaves unsolved are exactly what motivated everything in the FDN section below.

6. Schroeder and Logan (1961) – "Colorless" artificial reverberation (entry 98)

This is the paper that launched a thousand reverb plugins. Schroeder identified the two fundamental primitives – the comb filter and the allpass filter – and showed how to combine them to produce a plausible diffuse tail without coloring the sound. Almost every algorithmic reverb written in the six decades since has either implemented these ideas directly or consciously departed from them. Read this before anything else.

7. Valimaki et al. (2012) – Fifty Years of Artificial Reverberation (entry 59)

A historical survey connecting mechanical, algorithmic, convolution, and perceptual reverberation.

E.3 Key Results Reference

Quick-lookup table of the equations you will cite most often during development. These are not derivations – follow the source links for those.

Table E-1: Key Results Reference: reference table

Result	Formula	Source	Notes
Sabine equation	$T_{60} = 0.161V/A$, where $A = \sum_i S_i \alpha_i$	Sabine (1900), summarized in RA2, RA3, and RA5	Assumes perfectly diffuse field. Breaks down in rooms with non-uniform absorption or very low average absorption coefficient. Over-predicts T_{60} in dead rooms.
Eyring correction	$T_{60} = 0.161V/[-S\ln(1-\bar{\alpha})]$	Eyring (1930), see RA2 and RA5	More accurate when average absorption is high ($\bar{\alpha} > 0.3$). Reduces to Sabine in the limit of low absorption.
FDN gain calibration	$g_i = 10^{-3d_i/T_{60}}$ per delay line, where d_i is delay length in seconds	Jot and Chaigne (1997), entry 95; Schlecht and Habets (2015), entry 39	Applied per-band when using frequency-dependent absorption filters on the delay outputs. This is the central calibration formula for matching a target RT60.
EDT definition	Early Decay Time = time for first 10 dB of decay on the energy decay curve, extrapolated to 60 dB	ISO 3382-1; summarized in entry 80	EDT correlates better with perceived liveness than RT60 in spaces with non-exponential decay.
C80 (Clarity)	$C_{80} = 10\log_{10}\left(\frac{\int_0^{80\,\mathrm{ms}} h^2(t)\,dt}{\int_{80\,\mathrm{ms}}^{\infty} h^2(t)\,dt}\right)$ dB	ISO 3382-1; see entry 80	Ratio of early to late energy, 80 ms threshold. Positive values indicate clear/direct sound; negative values indicate reverberant/muddy.

Result	Formula	Source	Notes
D50 (Definition)	$D_{50} = \frac{\int_0^{50\,\mathrm{ms}} h^2(t)\,dt}{\int_0^{\infty} h^2(t)\,dt}$	ISO 3382-1; see entry 80	Fraction of total energy arriving in first 50 ms. Ranges 0–1; higher values correlate with better speech intelligibility. Uses 50 ms threshold versus C80's 80 ms.

E.4 Roadmap Citation Index (v0.7.0 Completion Program)

E.4.1 By stream

- `Stream R1 (reactive control hardening)`: Schlecht and Habets (2015): time-varying FDN matrices, Valimaki et al. (2012): fifty years of artificial reverberation
- `Stream R2 (perceptual/Jot-inspired calibration)`: Jot and Chaigne (1997): maximally diffusive FDN, Valimaki et al. (2012): fifty years of artificial reverberation
- `Stream R3 (IR morph productionization)`: Valimaki et al. (2012): fifty years of artificial reverberation
- `Stream R4 (next-generation topology expansion)`: Rocchesso and Smith (1997): circulant/elliptic FDN families, Schlecht and Habets (2019): delay feedback matrices

E.4.2 Core canonical links

- Rocchesso and Smith (1997): circulant/elliptic FDN families
- Schlecht and Habets (2015): time-varying FDN matrices
- Schlecht and Habets (2019): delay feedback matrices
- Valimaki et al. (2012): fifty years of artificial reverberation

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E.5 Section 1: Foundational Works

The papers every reverb developer must read. These define the vocabulary, the fundamental structures, and the problems that all subsequent work attempts to solve.

Audfray, R.; Jot, J. "Reverberation Loudness Model for Mixed-Reality Audio." *Audio Eng. Soc. Conference on Headphone Technology*. Source: Primary source, 2019. [F48]

Dahl, L.; Jot, J. "A Reverberator Based on Absorbent All-Pass Filters." *Proceedings of the COST G-6 Conference on Digital Audio Effects*. Source: DAFx paper, 2000. [F11]

> Annotation: Extends the all-pass primitive with frequency-dependent absorption, yielding a compact route to spectrally shaped decay that is directly relevant to multiband RT60 control.

Funkhouser, T.; Jot, J.; Tsingos, N. "Survey of Methods for Modeling Sound Propagation in Interactive Virtual Environment Systems." *Presence: Teleoperators and Virtual Environments*. Source: Author manuscript, 2003. [F12]

> Annotation: Places algorithmic reverberation within the broader problem of interactive sound propagation, including geometry, visibility, diffraction, and real-time rendering budgets.

Jot, J.; Chaigne, A. “Digital Delay Networks for Designing Artificial Reverberators.” *90th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 1991. [F13]

Annotation: Introduces the digital delay network design viewpoint that precedes the formal FDN treatment and establishes practical relationships among delay structure, modal density, stability, and decay.

Jot, J. “An Analysis/Synthesis Approach to Real-Time Artificial Reverberation.” *1992 IEEE International Conference on Acoustics, Speech, and Signal Processing* 221–224 vol.2. DOI: 10.1109/icassp.1992.226080, 1992. [F14]

Annotation: Develops the analysis and resynthesis framework behind perceptually meaningful, frequency-dependent reverberation control and the energy-decay-relief representation.

Jot, J. “Étude et réalisation d’un spatialisateur de sons par modèles physiques et perceptifs.” *Télécom Paris Doctoral Dissertation*. Source: Télécom Paris record, 1992. [F15]

Annotation: Provides the broad physical and perceptual foundation for Jot’s later work on artificial reverberation, distance rendering, and spatialization.

Jot, J. “Étude et réalisation d’un spatialisateur de sons par modèles physiques et perceptifs.” *Telecom Paris doctoral dissertation*. Source: Primary source, 1992. [F35]

Jot, J.; Warusfel, O.; Kahle, E.; Mein, M. “Binaural Concert Hall Simulation in Real Time.” *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*. Source: IRCAM paper, 1993. [F16]

Annotation: Demonstrates an early real-time union of room response modeling and binaural reproduction, anticipating modern headphone-based architectural auralization.

Jot, J.; Larcher, V.; Warusfel, O. “Digital Signal Processing Issues in the Context of Binaural and Transaural Stereophony.” *98th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 1995. [F17]

Annotation: Examines the filtering, equalization, and rendering constraints that arise when reverberant spatial scenes must survive binaural or transaural playback.

Jot, J.; Warusfel, O. “Le Spatialisateur.” *GRAME Conf. ”Le Son et L’Espace”*. Source: Primary source, 1995. [F39]

Jot, J.; et al. “Method to simulate the acoustical quality of a room and associated audio-digital processor.” *Issued patent family: US 5,812,674*. Source: Patent record, 1995. [F55]

Jot, J.; Warusfel, O. “Spat~: A Spatial Processor for Musicians and Sound Engineers.” *CIARM 1995*. Source: IRCAM paper, 1995. [F18]

Annotation: Documents the production-facing spatial processor that translated perceptual room and source descriptors into controls usable by musicians and engineers.

Jot, J.; Caulkins, T.; Gottfried, R. “Spat: Introduction.” *IRCAM technical report*. Source: Primary source, 1995. [F40]

Jot, J.; Caulkins, T.; Gottfried, R. “Spat: Reference Manual.” *IRCAM technical report*. Source: Primary source, 1995. [F41]

Jot, J.; Vandernoot, G.; Warusfel, O. “Analysis and Synthesis of Room Reverberation in the Time and Frequency Domains: Application to the Restoration of Room Impulse Responses Corrupted by Measurement Noise.” *The Journal of the Acoustical Society of America* 99(4_Supplement): 2530–2574. DOI: 10.1121/1.415793, 1996. [F19]

Annotation: Connects time-frequency decay analysis to restoration of noisy measured impulse responses, a key bridge between room analysis, dereverberation, and synthetic tail reconstruction.

Jot, J. “Synthesizing 3-D Sound Scenes in Audio or Multimedia Production and Interactive Human-Computer Interfaces.” *Fifth Conference on Interface to Real and Virtual Worlds*. Source: IRCAM paper, 1996. [F20]

Annotation: Frames room effects as part of an interactive sound-scene model rather than as an isolated insert effect, clarifying the role of source distance and environment.

Jot, J.; Cerveau, L.; Warusfel, O. “Analysis and Synthesis of Room Reverberation Based on a Statistical Time-Frequency Model.” *103rd Convention of the Audio Engineering Society*. Source: AES E-Library paper 4629, 1997. [F22]

Annotation: Formalizes statistical time-frequency room-decay modeling and shows how measured decay behavior can drive a controllable synthetic reverberator.

Jot, J. “Efficient Models for Reverberation and Distance Rendering in Computer Music and Virtual Audio Reality.” *International Computer Music Conference*. Source: ICMC paper, 1997. [F21]

Annotation: Presents efficient perceptual models for coupled reverberation and distance, emphasizing control dimensions that remain musically useful under real-time constraints.

Jot, J.; Chaigne, A. “Maximally diffusive yet efficient feedback delay networks for artificial reverberation.” *IEEE Signal Processing Letters* 4(9): 252–255. DOI: 10.1109/97.623041, 1997. [F4]

Annotation: The paper that unified the FDN framework and gave practitioners the tools to actually design good reverbs rather than accidentally stumbling into them. Provides the stability condition, the diffuseness criterion, and the RT60 calibration formula that verbx uses directly.

Jot, J.; Wardle, S.; Larcher, V. “Approaches to Binaural Synthesis.” *105th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 1998. [F23]

Annotation: Surveys practical binaural synthesis strategies and their tradeoffs, supplying essential context for headphone delivery of reverberant and spatial audio.

Jot, J.; et al. “Environmental reverberation processor.” *Issued patent family: US 6,188,769; US 6,917,686; US 7,561,699*. Source: Patent record, 1998. [F56]

Jot, J.; Larcher, V.; Pernaux, J. “A Comparative Study of 3-D Audio Encoding and Rendering Techniques.” *16th International Conference of the Audio Engineering Society*. Source: AES E-Library paper, 1999. [F25]

Annotation: Compares spatial encoding and rendering families, helping distinguish scene representation from the loudspeaker or headphone renderer that ultimately presents the reverberant field.

Jot, J. “Real-Time Spatial Processing of Sounds for Music, Multimedia and Interactive Human-Computer Interfaces.” *Multimedia Systems*. DOI: 10.1007/s005300050111, 1999. [F24]

Annotation: Synthesizes the perceptual and computational architecture of a general-purpose spatial audio processor, including reverberation as an environmental cue.

Jot, J. “Advanced Audio BIFS: Environmental Spatialization of Audio in MPEG-4 Version 2.” *108th Convention of the Audio Engineering Society*. Source: MIT presentation, 2000. [F26]

Annotation: Shows how environmental audio parameters can be embedded in a standardized interactive scene description, an early precedent for portable object-based reverb metadata.

Jot, J. “Efficient Models for Interactive Environmental Audio Spatialization.” *DAFx International Conf. Digital Audio Effects*. Source: Primary source, 2000. [F42]

Jot, J.; et al. “Reverberation processor based on absorbent all-pass filters.” *Issued patent family: US 7,149,314*. Source: Patent record, 2000. [F58]

Jot, J.; et al. “Reverberation processor for interactive audio applications.” *Issued patent family: US 6,978,027*. Source: Patent record, 2000. [F57]

Jot, J.; et al. “Method and apparatus for the simulation of complex audio environments.” *Issued patent family: US 7,099,482*. Source: Patent record, 2001. [F59]

Jot, J. “Perceptual and Statistical Models for Virtual Audio Environments.” *ACM Campfire: Acoustic Rendering for Virtual Environments*. Source: Official bibliography, 2001. [F43]

Jot, J.; Walsh, M.; Philp, A. “Binaural Simulation of Complex Acoustic Scenes for Interactive Audio.” *121st Convention of the Audio Engineering Society*. Source: AES E-Library paper, 2006. [F28]

Annotation: Applies binaural rendering to complex interactive scenes where source directivity, room response, and listener motion must remain coherent in real time.

Jot, J.; Trivi, J. “Scene Description Model and Rendering Engine for Interactive Virtual Acoustics.” *120th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 2006. [F27]

Annotation: Separates the acoustic scene model from its renderer, a useful architectural pattern for verbx presets, automation, game-audio integration, and future room-geometry engines.

Jot, J.; Walsh, M. “Center-Channel Processing in Virtual 3-D Audio Reproduction over Headphones or Loudspeakers.” *128th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 2010. [F29]

Annotation: Addresses center-image stability across headphone and loudspeaker virtualizers, a practical concern when reverberation and spatial widening alter localization cues.

Jot, J.; et al. “Spatial audio encoding and reproduction of diffuse sound.” *Issued patent family: US 8,908,874; US 9,042,565; US 9,728,181*. Source: Patent record, 2010. [F60]

Jot, J.; et al. “Direct-diffuse decomposition.” *Issued patent family: US 9,253,574*. Source: Patent record, 2011. [F61]

Jot, J.; et al. “System and method for estimating a reverberation time.” *Issued patent family: US 9,386,373*. Source: Patent record, 2012. [F62]

Jot, J.; Smith, B.; Thompson, J. “Dialog Control and Enhancement in Object-Based Audio Systems.” *139th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 2015. [F30]

Annotation: Explains how object-based rendering can preserve user control over dialog level and clarity, including interactions with ambient and reverberant components.

Jot, J. “Proportional Parametric Equalizers: Application to Digital Reverberation and Environmental Audio Processing.” *139th Convention of the Audio Engineering Society*. Source: AES E-Library paper, 2015. [F31]

Annotation: Develops efficient proportional parametric equalization structures for controllable spectral decay and environmental filtering in real-time reverberators.

Jot, J.; Lee, K. S. “Augmented Reality Headphone Environment Rendering.” *AES International Conference on Audio for Virtual and Augmented Reality*. Source: AES E-Library paper, 2016. [F32]

Annotation: Treats reverberation as a congruence problem between real and virtual environments, with implications for room adaptation, externalization, and mixed-reality headphone rendering.

Jot, J. "Environmental Audio Effects in Games, VR and AR." *Audio Eng. Soc. 61st Conf.: Audio for Games*. Source: Official bibliography, 2016. [F44]

Jot, J. "Augmented Reality Headphone Reverberation." *Seminar at IRCAM, Paris*. Source: Primary source, 2017. [F45]

Jot, J. "Augmented Reality Headphone Reverberation." *DSP Seminar at CCRMA, Stanford Univ*. Source: Primary source, 2017. [F46]

Jot, J. "Efficient Reverberation Rendering for Complex Interactive Audio Scenes." *DAFx International Conf. Digital Audio Effects*. Source: Primary source, 2017. [F36]

Jot, J. "Room-Adaptive Headphone 3D Audio." *143rd Conv. Audio Engineering Society*. Source: Primary source, 2017. [F47]

Jot, J.; et al. "Reverberation gain normalization." *Issued patent family: US 10,810,992; US 11,250,834; US 12,008,982*. Source: Patent record, 2018. [F64]

Jot, J.; et al. "Spatial audio for interactive audio environments." *Issued patent family: US 10,735,884; US 10,863,300; US 10,952,010; US 11,770,671; US 12,294,852*. Source: Patent record, 2018. [F63]

Jot, J.; et al. "Environment acoustics persistence." *Published patent application: US 2021/0176588, WO 2021/113781*. Source: Patent record, 2019. [F67]

Jot, J.; et al. "Mixed reality virtual reverberation." *Published patent application: US 2021/0084357*. Source: Patent record, 2019. [F66]

Jot, J.; et al. "Reverberation fingerprint estimation." *Issued patent family: US 11,304,017, US 11,540,072, US 12,149,896*. Source: Patent record, 2019. [F65]

Jot, J.; Audfray, R.; Hertensteiner, M.; Schmidt, B. "Rendering Spatial Sound for Interoperable Experiences in the Audio Metaverse." *International Conference on Immersive and 3D Audio*. DOI: 10.48550/arXiv.2109.12471, 2021. [F33]

Annotation: Proposes interoperable six-degree-of-freedom scene and rendering abstractions that combine object audio with room reverberation, reflectors, obstacles, and acoustic propagation solvers.

Jot, J.; Audfray, R.; Hertensteiner, M. "Spatial Audio Rendering for Interoperable XR Applications in the Audio Metaverse." *151st Conv. Audio Engineering Society*. Source: Primary source, 2021. [F49]

Jot, J. "6-DoF Spatial Audio Rendering of Complex Scenes." *154th Conv. Audio Engineering Society*. Source: Primary source, 2023. [F50]

Jot, J.; Carpentier, T.; Warusfel, O. "Perceptually Motivated Spatial Audio Scene Description and Rendering for 6-DoF Immersive Music Experiences." *2023 Immersive and 3D Audio: From Architecture to Automotive*. Source: IEEE Xplore record, 2023. [F34]

Annotation: Returns Jot's perceptual-control framework to immersive music, specifying scene and renderer concepts for navigable six-degree-of-freedom listening experiences.

Jot, J.; et al. "Spatial audio scene description and rendering." *Published patent application: WO 2025/054331*. Source: Patent record, 2023. [F68]

Jot, J. "On the Spatial Perception of Concurrent Free-Field and Diffuse-Field Components of a Sound Event." *NYU Immersive Audio Group meeting*. Source: Primary source, 2024. [F38]

Jot, J.; Carpentier, T.; Warusfel, O. “Creating and distributing immersive audio: from IRCAM Spat to Acoustic Objects.” *158th Conv. Audio Engineering Society*. Source: Primary source, 2025. [F51]

Jot, J.; Treanor, G.; Carpentier, T.; Warusfel, O. “Creating and distributing immersive audio: from IRCAM Spat to Acoustic Objects.” *Stanford CCRMA guest lecture*. Source: Primary source, 2025. [F52]

Jot, J.; Carpentier, T.; Warusfel, O. “Creating and distributing immersive audio experiences.” *in Immersive Sound, Volume II: The Design and Practice of Binaural and Multi-Channel Experiences*. Source: Primary source, 2026. [F54]

Jot, J.; Cohen, M. “Spatial Sound Rendering for Audio Augmented Reality.” *in Augmented Reality: Situated Spatial Synergy*. Source: Primary source, 2026. [F53]

Murgai, P.; Rau, M.; Jot, J. “Blind Estimation of the Reverberation Fingerprint of Unknown Acoustic Environments.” *143rd Conv. Audio Engineering Society*. Source: Primary source, 2017. [F37]

Polack, J. “Revisiting reverberation.” *Acta Acustica* 9: 30. DOI: 10.1051/aacus/2025010, 2025. [F8]

Annotation: A senior researcher’s reassessment of the theoretical assumptions behind classical reverberation theory. Worth reading for the perspective it offers on what the Sabine/Eyring/FDN frameworks do and do not capture.

Reiss, J. D.; McPherson, A. P. “Reverberation.” *Audio Effects* 229–253. DOI: 10.1201/9781003593942-10, 2026. [F7]

Annotation: A current textbook chapter that surveys the state of practical reverb algorithm design. Useful as a pedagogical introduction before diving into the primary literature above.

Rocchesso, D.; Smith, J. O. “Circulant and elliptic feedback delay networks for artificial reverberation.” *IEEE Transactions on Speech and Audio Processing* 5(1): 51–63. DOI: 10.1109/89.554269, 1997.[F3]

Annotation: Works out the eigenvalue structure of feedback matrices in detail and explains exactly why naive matrix choices cause audible modal ringing. Introduces the circulant family of matrices, which remain a practical default for large FDN orders. Indispensable alongside Jot-Chaigne.

Schroeder, M. R.; Logan, B. F. “Colorless” artificial reverberation.” *IRE Transactions on Audio* AU-9(6): 209–214. DOI: 10.1109/tau.1961.1166351, 1961. [F1]

Annotation: The founding document of algorithmic reverberation. Introduces parallel comb filters followed by series allpass sections as a way to produce high echo density without coloring the spectrum. Every subsequent algorithmic reverb either implements this architecture or explicitly departs from it. Read this first.

Schroeder, M. R. “Further Progress with Colorless Artificial Reverberation.” *The Journal of the Acoustical Society of America* 33(6_Supplement): 852–853. DOI: 10.1121/1.1936895, 1961. [F2]

Annotation: The companion paper published the same year, refining the allpass topology. Together with the IRE paper above, this establishes the canonical Schroeder reverberator that hardware units of the 1970s and virtually every software reverb through the 1990s was built on.

Theodore, M. “Altiverb Reverberation Software.” *Computer Music Journal* 28(2): 101–103. DOI: 10.1162/comj.2004.28.2.101, 2004. [F9]

Annotation: A practitioner’s account of the first commercially successful high-quality convolution reverb plugin. Important context for understanding where the convolution approach succeeded over algorithmic approaches, and what trade-offs it brought.

Unknown authors. "Artificial reverberation." *Journal of the Institution of Electrical Engineers* 1(6): 368–371. DOI: 10.1049/jiee-3.1955.0135, 1955. [F6]

Annotation: Pre-Schroeder documentation of early tape and mechanical approaches to artificial reverberation. Useful historical context for understanding what Schroeder's 1961 papers were responding to.

Unknown authors. "The Role of Modal Excitation in Colorless Reverberation." *2021 24th International Conference on Digital Audio Effects (DAFx)* 206–213. DOI: 10.23919/dafx51585.2021.9768217, 2021. [F10]

Annotation: Returns to Schroeder's original "colorless" criterion and examines it through the lens of modal analysis. Explains the precise mechanism by which poorly designed delay networks produce coloration, which is the problem verbx's time-varying matrix stream addresses.

Valimaki, V.; Parker, J. D.; Savioja, L.; et al. "Fifty Years of Artificial Reverberation." *IEEE Transactions on Audio, Speech, and Language Processing* 20(5): 1421–1448. DOI: 10.1109/tasl.2012.2189567, 2012. [F5]

Annotation: The definitive survey of the field from Schroeder's original papers through convolution and perceptual approaches. The single most useful reference document for a reverb developer. If you understand everything in this paper you understand the intellectual foundation of the entire discipline.

E.6 Section 2: Feedback Delay Networks

FDN theory, stability conditions, matrix design, and time-varying extensions. This is the core algorithmic section for verbx.

Koontz, W. L. "Artificial reverberation using multi-port acoustic elements." *Proceedings of Meetings on Acoustics* 055002. DOI: 10.1121/1.4854696, 2013. [FDN5]

Annotation: Proposes multi-port acoustic element networks as an alternative FDN-like architecture. Useful as a conceptual reference for exploring topologies beyond the standard scattering-matrix formulation.

Schlecht, S. J.; Habets, E. A. P. "Simulation of Room Reverberation Using a Feedback Delay Network." *Acoustics Australia* 43(1): 83–86. DOI: 10.1007/s40857-015-0005-8, 2015. [FDN2]

Annotation: Companion paper to [FDN1], covering implementation details of the full FDN simulation pipeline including frequency-dependent absorption and the gain calibration workflow. More accessible than the JASA paper; read this alongside it.

Schlecht, S. J.; Habets, E. A. P. "Time-varying feedback matrices in feedback delay networks and their application in artificial reverberation." *The Journal of the Acoustical Society of America* 138(3): 1389–1398. DOI: 10.1121/1.4928394, 2015. [FDN1]

Annotation: Proves that slowly rotating the feedback matrix breaks up the fixed modal structure that causes metallic flutter in fixed-matrix FDNs, without destabilizing the network or significantly changing the statistical RT60. The theoretical basis for Stream R1.

Schlecht, S. J.; Habets, E. A. P. "Dense Reverberation with Delay Feedback Matrices." *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)* 150–154. DOI: 10.1109/waspaa.2019.8937284, 2019. [FDN3]

Annotation: Introduces delay-feedback matrices, which embed the feedback network directly into the delay topology rather than treating the delay and feedback stages as separate. Achieves higher echo density per unit of computation. This is the architectural foundation for verbx's Stream R4 topology expansion.

Unknown authors. "Active Acoustics with a Phase Cancelling Modal Reverberator." *Journal of the Audio Engineering Society* 72(10): 705–715. DOI: 10.17743/jaes.2022.0171, 2024. [FDN4]

Annotation: Applies modal reverberator theory to active acoustic systems. Useful for understanding how FDN structures can be used to synthesize or suppress specific room modes, which informs edge cases in verbx's reactive control design.

E.7 Section 3: Convolution and IR Processing

Partitioned FFT convolution, impulse response measurement, and IR morphing. The theoretical backbone for verbx's Stream R3.

Arcas, K.; Chaigne, A. "On the quality of plate reverberation." *Applied Acoustics* 71(2): 147–156. DOI: 10.1016/j.apacoust.2009.07.013, 2010. [CV3]

Annotation: Analyzes plate reverb physically and identifies the spectral characteristics that distinguish high-quality plate IRs from poor ones. Useful as a reference when evaluating morphed or synthesized IRs against physical ground truth.

Ducasse, É.; Rodriguez, S.; Bonnet, M. "Early-reverberation imaging functions for bounded elastic domains." *Acta Acustica* 10: 2. DOI: 10.1051/aacus/2025069, 2026. [CV1]

Annotation: Develops imaging functions for early reflections from bounded elastic surfaces. Relevant to physically-grounded IR construction and to understanding what the early-reflection part of a measured IR actually encodes about room geometry.

Farina, A. "Simultaneous Measurement of Impulse Response and Distortion with a Swept-Sine Technique." *108th AES Convention*. Source: AES E-Library record, 2000. [CV6]

Annotation: Establishes the exponential swept-sine method used to recover a high-signal-to-noise linear impulse response while separating harmonic distortion responses in time. It is foundational for reproducible room, loudspeaker, and electroacoustic-system capture.

Mechel, F. "Reverberation with Mirror Sources." *Room Acoustical Fields* 347–390. DOI: 10.1007/978-3-642-22356-3_17, 2012. [CV4]

Annotation: Textbook chapter on image-source methods for computing room impulse responses. The image-source approach is the classical alternative to FDN for early reflections; understanding it illuminates the trade-offs in hybrid reverb architectures.

Schneiderwind, C.; De Sena, E.; Neidhardt, A. "Perceptual effects of modified late reverberation and reverberation time in auditory augmented reality in two rooms." *Acta Acustica* 10: 17. DOI: 10.1051/aacus/2026012, 2026. [CV5]

Annotation: Examines what happens perceptually when you modify the late tail of a measured IR independently of the early part. Directly relevant to IR morphing workflows – establishes the perceptual independence of early and late portions and the threshold above which manipulation becomes detectable.

Unknown authors. "Room Impulse Responses, Decay Curves and Reverberation Times." *Formulas of Acoustics* 939–940. DOI: 10.1007/978-3-540-76833-3_262, 2008. [CV2]

Annotation: Reference-book treatment of the mathematical relationship between the room impulse response and the energy decay curve. Essential grounding for anyone implementing EDC analysis or RT60 estimation from measured IRs.

E.8 Section 4: Room Acoustics Fundamentals

RT60, energy decay curves, Sabine/Eyring theory, and the physics of reverberant fields. These papers establish the physical quantities that verbx's algorithms attempt to match, measure, and explain to users. They are the backbone references for verbx's analysis JSON, room-model warnings, preset calibration, and documentation around when a single reverberation-time value is or is not enough.

Berger, R. "From Sports Arena to Sanctuary: Taming a Texas-Sized Reverberation Time." *Acoustics Today* 7(1): 24. DOI: 10.1121/1.3576192, 2011. [RA14]

Annotation: Practitioner case study of a space with exceptionally long RT60. Context entry useful for anchoring intuition about what multi-second reverberation actually sounds like in a physical space. It helps translate abstract values such as 6, 10, or more seconds of decay into practical consequences for speech, music, crowd noise, and treatment strategy, which is useful when documenting verbx's extreme-tail and infinite-style modes.

Clow, S. "Sound Insulation Testing: Sensitivity to Reverberation Time: T30 v T20." *Autumn Conference Acoustics 2003*. DOI: 10.25144/18162, 2023. [RA20]

Context entry. Compares T20 and T30 evaluation windows for building acoustics measurements. Peripheral to reverb synthesis but relevant to RT60 measurement methodology. It supports exposing multiple decay-window estimates in analysis because different windows can disagree when the noise floor, curvature, or early decay behavior contaminates a simple full-tail fit.

Diaz, C.; Pedrero, A. "The reverberation time of furnished rooms in dwellings." *Applied Acoustics* 66(8): 945–956. DOI: 10.1016/j.apacoust.2004.12.002, 2005. [RA15]

Annotation: Measurement study of RT60 in typical furnished domestic rooms. Useful reference for the short-RT60 end of verbx's operating range. This anchors small-room presets in measured domestic conditions rather than only studio or concert-hall ideals, and it is helpful for explaining why living-room reverberation usually needs early-reflection realism and spectral coloration more than a long, obvious tail.

Dunham, J. "Comparing Measured Sound Strength to Theory as a Function of Reverberation Time and Room Volume." *Auditorium Acoustics 2023*. DOI: 10.25144/15977, 2023. [RA18]

Context entry. Empirical validation of the relationship between room volume, RT60, and sound strength. Background context. This is relevant to gain staging and apparent distance because a longer or larger room is not only a longer tail; it also changes perceived loudness, source support, and the balance between direct and reverberant energy.

Fay, M. W. "Reverberation time slope ratio." *The Journal of the Acoustical Society of America* 146(4_Supplement): 2853–2853. DOI: 10.1121/1.5136901, 2019. [RA9]

Annotation: Proposes the slope ratio as a diagnostic for non-exponential decay curves, which arise whenever a room has coupled sub-spaces or highly non-uniform absorption. Relevant to detecting cases where a single RT60 number is insufficient. This is useful for future analysis reports because it suggests a compact way to flag curved or multi-stage decay behavior instead of silently fitting one straight line to a tail that does not behave exponentially.

Granzotto, N.; Caniato, M. “Influence of not homogeneous absorption on reverberation time.” *2023 Immersive and 3D Audio: from Architecture to Automotive (I3DA)* 1–5. DOI: 10.1109/i3da57090.2023.10289246, 2023. [RA12]

Annotation: Quantifies the error in RT60 prediction when absorption is spatially non-uniform. This is the regime where Sabine and Eyring both fail. Relevant to understanding the boundaries of the physical models verbx targets, especially for rooms with heavy treatment on one surface, audience absorption, curtains, or irregular furnishing. It motivates warnings around overconfident physical estimates when absorption cannot reasonably be treated as evenly distributed.

Jones, D. R. “Reverberation and Time Response.” *Sound of Worship* 233–249. DOI: 10.1016/b978-0-240-81339-4.00021-1, 2011. [RA13]

Annotation: Applied treatment of reverberation in large worship spaces. Context entry for understanding design targets at extreme RT60 values that verbx is built to reproduce. It provides a real architectural setting for long-decay presets, where the goal is not merely “more tail” but a musically and verbally usable balance among warmth, clarity, spaciousness, and intelligibility.

Kuttruff, H.; Vorländer, M. “Reverberation and steady-state energy density.” *Room Acoustics* 109–132. DOI: 10.1201/9781003389873-5, 2024. [RA2]

Annotation: Current edition textbook chapter on statistical room acoustics theory. The definitive modern treatment of Sabine and Eyring theory, with careful attention to the diffuse-field assumption and where it fails. This is the main reference for deciding when verbx should present a physical room estimate confidently and when it should warn that geometry, absorption distribution, or source/listener placement makes the model only approximate.

Nowoswiat, A.; Olechowska, M. “Investigation Studies on the Application of Reverberation Time.” *Archives of Acoustics* 41(1): 15–26. DOI: 10.1515/aoa-2016-0002, 2016. [RA11]

Annotation: Empirical study comparing different RT60 measurement protocols and their agreement across room types. Useful calibration context. It supports treating RT60 as a measured quantity with method-dependent uncertainty, which matters for verbx when comparing T20, T30, EDT, and fitted full-tail estimates in automated analysis output.

Pan, J. “Invited Review Paper: The Physics of Reverberation.” *Building Acoustics* 11(2): 81–95. DOI: 10.1260/1351010041494746, 2004. [RA7]

Annotation: A clear physics-first treatment of how reverberant fields form, covering both the modal and geometric views. Useful for developers who want to understand why the simple RT60 model is an approximation. It is a good conceptual anchor for the split between early reflections, modal coloration, and late diffuse decay, which maps directly onto verbx controls such as early level, density, modulation, and tail shaping.

Prato, A.; Casassa, F.; Schiavi, A. “Reverberation time measurements in non-diffuse acoustic field by the modal reverberation time.” *Applied Acoustics* 110: 160–169. DOI: 10.1016/j.apacoust.2016.03.041, 2016. [RA10]

Annotation: Addresses RT60 measurement in non-diffuse fields by separating individual modal contributions. Relevant context for understanding why blind estimation algorithms struggle in small rooms. The modal framing also helps explain low-frequency artifacts in synthesized or measured IRs: a short room can have individual resonances that dominate perceived decay even when the broadband RT60 looks plausible.

Schnitta, B. “Achieving optimal reverberation time in a room, using newly patented tuning tubes.” *Proceedings of Meetings on Acoustics* 015105–015105. DOI: 10.1121/1.4801071, 2013. [RA16]

Context entry. Describes physical treatment of reverberation time; tangential to algorithmic approaches. It remains useful as a reminder that algorithmic controls model only part of a larger acoustic design problem: real spaces use absorbers, resonators, diffusers, and tunable devices to change decay behavior in ways that can be uneven across frequency and position.

Smith, J. O. *Artificial Reverberation*. Physical Audio Signal Processing; W3K Publishing. Source: Artificial Reverberation, 2010. [BOOK3]

Annotation: A systems-level treatment of artificial reverberation spanning exact transfer-function and physical room models, perceptual requirements, early reflections, Schroeder and Freeverb structures, feedback delay networks, lossless feedback matrices, delay-length selection, frequency-dependent damping, digital waveguide reverberators, time variation, and scattering delay networks. The expanded verbx glossary uses this chapter as a primary terminology audit while retaining compact definitions suited to production and implementation work.

Smith, J. O. I. *Physical Audio Signal Processing*. W3K Publishing; ISBN 978-0-9745607-2-4. Source: Energy Decay Relief, 2010. [BOOK2]

Annotation: Defines the energy decay relief as reverse cumulative STFT-bin energy and explains its relationship to the broadband energy decay curve. This is the primary source for verbx's `--edr` analysis, including the distinction between a complete time-frequency decay surface and the compact per-band RT60 summaries written to analysis reports.

Stephenson, U. "On the Influence of Ceiling and Audience Profile on the Reverberation Time and Other Room Acoustical Parameters." *Auditorium Acoustics 2008*. DOI: 10.25144/17526, 2023. [RA17]

Context entry. Geometric acoustics study of audience and ceiling geometry effects on RT60. Background context for physical room modeling. It reinforces that audience layout and architectural geometry can change decay and clarity in ways that are not captured by volume and average absorption alone, which matters for any future room-parameter import or venue-style preset work.

Tohyama, M. "Reverberation Sound in Rooms." *Signals and Communication Technology* 191–218. DOI: 10.1007/978-3-642-20122-6_11, 2011. [RA8]

Annotation: Signal-processing perspective on room reverberation, bridging acoustic theory and DSP formulations. Useful for practitioners who are more comfortable with z-domain analysis than with acoustic physics. This helps connect room-acoustic quantities to filters, feedback structures, and impulse-response manipulation, making it a natural reference for the implementation side of algorithmic reverb and dereverberation.

Unknown authors. "Reverberation and steady state energy density." *Room Acoustics* 127–158. DOI: 10.1201/9781482286632-9, 2002. [RA3]

Annotation: Earlier edition of the Kuttruff treatment. Included because some derivations presented here are clearer than in the updated edition. It is useful as a companion reading source for implementation notes because the older presentation makes several intermediate assumptions more explicit, which helps when translating textbook acoustics into deterministic code paths and test fixtures.

Unknown authors. "Room Impulse Responses, Decay Curves and Reverberation Times." *Formulas of Acoustics* 939–940. DOI: 10.1007/978-3-540-76833-3_262, 2008. [RA1]

Annotation: The primary reference for the EDT, C80, and D50 definitions given in the Key Results Reference table above. The formulas here are the ones used in verbx's metric computation layer. It also provides the bridge between an impulse response as raw samples

and an energy decay curve as an interpretable acoustic object, which is why it is the practical starting point for validating analysis JSON, preset reports, and user-facing room clarity summaries.

Unknown authors. "Deriving the Reverberation Time Equation." *Acoustics and Psychoacoustics* 451–454. DOI: 10.1016/b978-0-240-52175-6.00014-4, 2010. [RA5]

Annotation: Companion derivation to [RA4], covering the wideband case. Read before [RA4]. It gives the simplest path from room volume and absorption to a single reverberation-time estimate, making it a useful baseline for docs, examples, and tests before adding frequency-dependent materials or per-band correction curves.

Unknown authors. "Deriving the Reverberation Time Equation for Different Frequencies and Surfaces." *Acoustics and Psychoacoustics* 455–457. DOI: 10.1016/b978-0-240-52175-6.00015-6, 2010. [RA4]

Annotation: Pedagogical derivation of frequency-dependent RT60, showing explicitly how absorption coefficients vary with frequency and how this produces frequency-dependent decay. The basis for verbx's per-band RT60 calibration. This reference is especially important for explaining why low, mid, and high bands should not be forced to share the same decay target, and why a preset can be technically correct only if its spectral tail shape matches the intended room material behavior.

Walker, R. "Microprocessor Controlled Equipment for the Measurement of Reverberation Time." *Acoustics '82*. DOI: 10.25144/23148, 2024. [RA19]

Context entry. Historical document on early automated RT60 measurement hardware. Included for completeness. It gives useful historical perspective on why reverberation-time measurement became standardized around repeatable excitation, decay capture, and fitted windows, which parallels verbx's emphasis on deterministic analysis output rather than subjective labeling alone.

Xiang, N. "Generalization of Sabine's reverberation theory." *The Journal of the Acoustical Society of America* 148(3): R5–R6. DOI: 10.1121/10.0001806, 2020. [RA6]

Annotation: Extends the Sabine framework to non-diffuse rooms using statistical energy analysis. Useful for understanding where the simple T60 prediction fails and what a more complete physical model needs to account for. For verbx, this supports the design choice to expose diagnostics rather than just a number: non-diffuse behavior can make the decay curve bend, split, or depend strongly on source and receiver position.

E.9 Section 4A: Archaeoacoustics and Chichén Itzá

Declercq, N. F.; Degrieck, J.; Briers, R.; Leroy, O. "A theoretical study of special acoustic effects caused by the staircase of the El Castillo pyramid at the Maya ruins of Chichen-Itza in Mexico." *The Journal of the Acoustical Society of America 116(6): 3328–3335*. DOI: 10.1121/1.1764833, 2004. [AA1]

Annotation: Compares measured chirped echoes with diffraction simulations and demonstrates the strong influence of the incident pulse. The paper provides the main physical account of why a handclap returned by El Castillo's periodic stairway can resemble a quetzal call.

Garza, C.; Medina, A.; Padilla, P.; Ramos, A.; Zalaquett, F. "Arqueoacústica maya. La necesidad del estudio sistemático de efectos acústicos en sitios arqueológicos." *Estudios de Cultura Maya 32*. Source: SciELO full text, 2008. [AA2]

Annotation: Places the Chichén Itzá chirped echo inside a systematic framework joining physical acoustics, mathematical modeling, perception, and archaeological interpretation. Especially valuable for separating measurable effects from still-unresolved claims about ancient design intent.

E.10 Section 5: Spatial Audio and Ambisonics

Betlehem, T.; Poletti, M. A. "Sound field reproduction around a scatterer in reverberation." *2009 IEEE International Conference on Acoustics, Speech and Signal Processing* 89–92. DOI: 10.1109/icassp.2009.4959527, 2009. [SP2]

Annotation: Addresses the problem of reproducing a reverberant sound field in the presence of a scattering object. Relevant context for spatial rendering of reverb in immersive systems. It highlights that the listener, playback setup, and objects in the field can alter the reproduced reverberant image, which is important background for Ambisonic, binaural, or multichannel render modes.

Corey, J. "Spatial Attributes and Reverberation." *Audio Production and Critical Listening* 51–75. DOI: 10.1016/b978-0-240-81295-3.00003-4, 2010. [SP1]

Annotation: Textbook chapter on how reverberation interacts with spatial audio perception, covering apparent source width, listener envelopment, and their relationship to early vs. late energy. Grounding for verbx's spatial output design. It is especially useful for separating "longer" from "larger": a reverb can increase depth, width, and immersion without simply increasing RT60, so spatial presets need independent control over early reflections, late-field decorrelation, and wet image spread.

Ren Gang; Shivaswamy, S. H.; Roessner, S.; et al. "An additional "Depth" of reverberation helps content stand out: Media content emphasis using audio reverberation effect." *2013 IEEE International Conference on Consumer Electronics (ICCE)* 98–99. DOI: 10.1109/icce.2013.6486811, 2013. [SP3]

Annotation: Explores the use of selective reverberation as a production tool for creating depth and emphasis in consumer audio. Useful framing for understanding how reverb functions as a mixing tool rather than purely a simulation. For verbx documentation, this supports creative features such as ducking, depth automation, and source-specific wetness because those controls are production decisions as much as acoustic-model decisions.

Zhan Haoke; Cai Zhiming; Yuan Bingcheng. "Space-time ODN suppressing reverberation method." *2008 9th International Conference on Signal Processing* 2559–2562. DOI: 10.1109/icosp.2008.4697671, 2008. [SP4]

Context entry. Dereverberation from a spatial signal processing perspective. Included for conceptual contrast with synthesis approaches. It gives a useful inverse-problem counterpoint to verbx's render path: the same spatial cues that help a reverb feel enveloping can be exploited or suppressed when the goal is clarity, separation, or reduced late energy.

E.11 Section 6: Perceptual Studies

How humans perceive reverberation: thresholds of detection, preference curves, and the perceptual dimensions of reverberant spaces. These papers inform which parameters verbx needs to control precisely and which it can treat as second-order. They are also the references behind documentation language

that distinguishes measured correctness from perceptual usefulness, since a numerically accurate RT60 does not always guarantee the listener hears the intended space.

Blasinski, L.; Felcyn, J.; Kocinski, J. “Perception of reverberation length in rooms with reverberation enhancement systems.” *The Journal of the Acoustical Society of America* 156(1): 450–462. DOI: 10.1121/10.0026485, 2024. [PE6]

Annotation: Studies the just-noticeable difference for RT60 changes in rooms with active enhancement systems. Directly relevant to calibrating the resolution of verbx's RT60 control. It helps determine sensible CLI step sizes, UI increments, and preset spacing so the tool does not imply precision that listeners cannot reliably perceive, while still preserving fine control for measurement and regression testing.

Erkelens, J. S.; Heusdens, R. “A statistical room impulse response model with frequency dependent reverberation time for single-microphone late reverberation suppression.” *Interspeech 2011* 201–204. DOI: 10.21437/interspeech.2011-82, 2011. [PE9]

Annotation: Derives a statistical model of the RIR late field that explicitly captures frequency-dependent decay. The model's assumptions are directly relevant to how verbx represents the decay filter bank. It is also a useful cross-reference between perceptual and statistical modeling: the late field can be compactly parameterized, but the usefulness of that parameterization depends on how well it predicts audible late reverberation.

Javed, H. A.; Naylor, P. A. “An extended reverberation decay tail metric as a measure of perceived late reverberation.” *2015 23rd European Signal Processing Conference (EUSIPCO)* 1063–1067. DOI: 10.1109/eusipco.2015.7362546, 2015. [PE2]

Annotation: Proposes a metric that correlates more closely with the perceived extent of late reverberation than RT60 alone. Useful for evaluating whether a verbx preset sounds as long as its nominal RT60 suggests it should. This matters for long-tail and infinite-style modes, where the perceptual impression of lingering energy can diverge from the fitted decay time because spectral balance, masking, and late-field texture all influence audibility.

Ninounakis, T.; Davies, W. “The Perception of Small Changes in Reverberation Time within Recording Studio Control Rooms.” *Reproduced Sound 1998*. DOI: 10.25144/18947, 2024. [PE7]

Annotation: Measures perceptual thresholds for RT60 change in a professional monitoring environment. Complements [PE6] with studio-context data. This is valuable because control-room listening emphasizes small changes, translation, and repeatable monitoring conditions, so it gives a stricter perceptual reference than casual playback or large-room listening.

Strother, T.; Davies, W. “Is There a Relationship Between Pitch Perception and Individual Preference of Concert Hall Reverberation Time?” *Acoustics 2023*. DOI: 10.25144/16625, 2023. [PE8]

Context entry. Investigates individual differences in preferred RT60. Peripheral to algorithm design; relevant to preset design philosophy. It supports keeping verbx presets adjustable rather than presenting one “correct” concert-hall or studio value, because preference can depend on listener, source material, pitch range, musical style, and listening task.

Tsilfidis, A.; Mourjopoulos, J. “Perceptually-motivated selective suppression of late reverberation.” *2009 16th International Conference on Digital Signal Processing* 1–6. DOI: 10.1109/icdsp.2009.5201165, 2009. [PE4]

Annotation: Earlier version of the work above; contains some derivations that the 2011 paper abbreviates. It is useful as a supplementary implementation reference because conference versions often expose the signal model and algorithmic tradeoffs more directly, making it easier to compare perceptual late-tail suppression with verbx's deterministic render and realtime controls.

Tsilfidis, A.; Mourjopoulos, J. “Blind single-channel suppression of late reverberation based on perceptual reverberation modeling.” *The Journal of the Acoustical Society of America*. DOI: 10.1121/1.3533690, 2011. [PE3]

Annotation: Develops a perceptual model of late reverberation and uses it to drive a suppression algorithm. The perceptual model itself, especially how the ear weights late energy, is the relevant contribution for verbx's design. It helps frame dereverberation and reverb reduction as selective control of perceptually harmful energy rather than blanket tail removal, which is important for low-latency real-time behavior.

Watkins, A. J. “Listening in real-room reverberation: Effects of extrinsic context.” *Auditory Signal Processing* 422–427. DOI: 10.1007/0-387-27045-0_52, 2005. [PE5]

Annotation: Psychoacoustic study on how listeners adapt to reverberation over time and how prior context affects the perception of a reverberant signal. Explains why very long reverb tails can become perceptually transparent under sustained exposure. This is important for preset evaluation because an effect that sounds dramatic in isolation may feel natural after adaptation, while a subtle change may be obvious in an A/B comparison.

Zahorik, P. “Perceptual scaling of room reverberation.” *The Journal of the Acoustical Society of America* 115(5_Supplement): 2598–2598. DOI: 10.1121/1.4784535, 2004. [PE1]

Annotation: Maps the perceptual space of reverberation using multidimensional scaling, identifying the primary perceptual dimensions, roughly decay time and reverberance character. Fundamental to understanding which physical parameters predict subjective quality. It is a key reason verbx should expose grouped controls and presets around perceptual outcomes such as clarity, bloom, distance, and room size rather than only offering raw acoustic variables.

E.12 Section 7: Machine Learning and Blind Estimation

CRNN-based RT60 estimation, neural dereverberation, and related machine learning approaches. Relevant to verbx's analysis pipeline, especially future work around blind room inference, automatic preset suggestion, dereverb assistance, and confidence scoring. These entries also provide baseline methods that can be compared against deterministic estimators before verbx grows heavier neural dependencies.

Ai, Y.; Wang, X.; Yamagishi, J.; et al. “Reverberation Modeling for Source-Filter-Based Neural Vocoder.” *Interspeech 2020* 3560–3564. DOI: 10.21437/interspeech.2020-1613, 2020. [ML6]

Annotation: Embeds a reverberation model inside a neural vocoder. Demonstrates how differentiable reverb models can be combined with neural speech synthesis. Relevant to any future verbx neural extensions. It is not part of the current deterministic core, but it frames a possible future where reverb parameters remain interpretable while being optimized or predicted inside a learned speech pipeline.

Andrijasevic, A. “Effect of phoneme variations on blind reverberation time estimation.” *Acta Acustica* 4(1): 3. DOI: 10.1051/aacus/2020001, 2020. [ML8]

Annotation: Examines how the phonetic content of speech affects the accuracy of blind RT60 estimators. Useful for understanding failure modes and the range of conditions over which blind estimators can be trusted. It supports surfacing caveats in analysis JSON, since voiced vowels, fricatives, pauses, and transient-rich material may provide very different evidence for decay fitting.

Choi, Y.; Xie, C.; Toda, T. “Reverberation-Controllable Voice Conversion Using Reverberation Time Estimator.” *INTERSPEECH 2023* 2103–2107. DOI: 10.21437/interspeech.2023-1356, 2023. [ML5]

Annotation: Uses an RT60 estimator as a conditioning signal inside a voice conversion network to control the reverberation of synthesized speech. Demonstrates an end-to-end neural approach to RT60-conditioned synthesis that is architecturally adjacent to verbx’s generative modes. The key design lesson is that estimated acoustics can become controllable latent variables, opening a path from analysis to automatic matching, transfer, or preset generation.

Deng, S.; Mack, W.; Habets, E. A. P. “Online Blind Reverberation Time Estimation Using CRNNs.” *Interspeech 2020* 5061–5065. DOI: 10.21437/interspeech.2020-2156, 2020. [ML1]

Annotation: Demonstrates that a convolutional-recurrent neural network can estimate RT60 reliably from a single speech channel in real time. Sets the state of the art for the blind estimation problem that verbx’s analysis mode needs to solve. It is a useful benchmark for future low-latency analysis because it treats estimation as an online problem, where window size, model state, and update cadence all affect end-to-end responsiveness.

Kodrasi, I.; Bourlard, H. “Single-channel Late Reverberation Power Spectral Density Estimation Using Denoising Autoencoders.” *Interspeech 2018* 1319–1323. DOI: 10.21437/interspeech.2018-1660, 2018. [ML7]

Annotation: Uses a denoising autoencoder to estimate the power spectral density of the late reverberant field. The estimated late-field PSD is then usable for suppression or analysis. Methodologically relevant to verbx’s analysis pipeline because late-field energy is exactly the component that must be separated from direct sound, noise, and early reflections when reporting dereverb strength or residual tail.

Lollmann, H. W.; Brendel, A.; Kellermann, W. “Efficient ML-Estimator for Blind Reverberation Time Estimation.” *2018 26th European Signal Processing Conference (EUSIPCO)* 2195–2199. DOI: 10.23919/eusipco.2018.8553001, 2018. [ML4]

Annotation: Maximum-likelihood formulation for blind RT60 estimation with a focus on computational efficiency. A strong non-neural baseline to compare CRNN approaches against. This is important for the 0.7.x stabilization track because deterministic, lightweight estimators are easier to ship, test, and run in CLI or realtime contexts than model-backed systems with larger dependencies.

Ma, C.; Shi, G. “Reverberation time estimation based on multidelay acoustic echo cancellation.” *2012 International Conference on Audio, Language and Image Processing* 230–234. DOI: 10.1109/icalip.2012.6376617, 2012. [ML11]

Context entry. Exploits an existing echo cancellation framework to derive RT60 estimates. Alternative estimation strategy for completeness. It is useful as a reminder that reverberation estimates can emerge from adjacent real-time systems, not only from offline IR analysis, which may matter if verbx later integrates with capture, monitoring, or conferencing workflows.

Mack, W.; Deng, S.; Habets, E. A. P. “Single-Channel Blind Direct-to-Reverberation Ratio Estimation Using Masking.” *Interspeech 2020* 5066–5070. DOI: 10.21437/interspeech.2020-2171, 2020. [ML2]

Annotation: Companion paper to [ML1], estimating the direct-to-reverberant ratio rather than RT60. DRR is the perceptually critical parameter for how “wet” a signal sounds, distinct from how long the tail is. For verbx, this points toward richer analysis output: RT60 can describe decay duration, but DRR and early-to-late balance describe distance, clarity, and apparent source presence.

Malik, H.; Farid, H. “Audio forensics from acoustic reverberation.” *2010 IEEE International Conference on Acoustics, Speech and Signal Processing* 1710–1713. DOI: 10.1109/icassp.2010.5495479, 2010. [ML13]

Context entry. Forensic identification of recording environments from reverberation. Companion study to [ML12]. This supports the broader idea that reverberation carries scene information, so analysis features in verbx should preserve details useful for matching, comparison, and provenance rather than collapsing everything into a single wet/dry description.

Malik, H.; Zhao, H. “Recording environment identification using acoustic reverberation.” *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 1833–1836. DOI: 10.1109/icassp.2012.6288258, 2012. [ML12]

Context entry. Uses reverberation signatures to fingerprint recording environments. Tangential but provides perspective on what makes each room’s impulse response distinctive. It helps explain why room matching is more than matching RT60: spectral coloration, early reflection pattern, and decay structure can identify a space even when broad statistics look similar.

Prodeus, A. “Parameter optimization of the single channel late reverberation suppression technique.” *2015 IEEE 35th International Conference on Electronics and Nanotechnology (ELNANO)* 269–274. DOI: 10.1109/elnano.2015.7146889, 2015. [ML10]

Context entry. Optimization of suppression filter parameters in a statistical dereverberation framework. Earlier work by the same author as [ML9]. It provides background for tuning dereverb options such as reduction amount, smoothing, and artifact control, where overly aggressive settings may improve measured tail reduction while hurting speech naturalness.

Prodeus, A. “Late reverberation reduction and blind reverberation time measurement for automatic speech recognition.” *2017 IEEE First Ukraine Conference on Electrical and Computer Engineering (UKRCON)* 634–639. DOI: 10.1109/ukrcon.2017.8100319, 2017. [ML9]

Context entry. Combines blind RT60 estimation with late-reverb suppression for ASR. Context for understanding the estimation-then-suppression pipeline architecture. It is relevant to verbx’s real-time dereverberation path because it treats measurement and processing as a coupled loop: estimation quality affects suppression strength, and suppression artifacts affect downstream speech recognition or intelligibility.

Xiong, F.; Goetze, S.; Kollmeier, B.; et al. “Joint Estimation of Reverberation Time and Early-To-Late Reverberation Ratio From Single-Channel Speech Signals.” *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 27(2): 255–267. DOI: 10.1109/taslp.2018.2877894, 2019. [ML3]

Annotation: Estimates RT60 and early-to-late ratio simultaneously from a single microphone. The joint approach avoids the error propagation that plagues sequential estimators and is worth evaluating for verbx’s blind analysis mode. It also suggests a product-level direction: when verbx reports room character, it should estimate related acoustic quantities together and expose confidence or consistency checks instead of presenting each metric as isolated truth.

Zheng, C.; Chan, W. “Late reverberation suppression using MMSE modulation spectral estimation.” *Interspeech 2013* 3239–3243. DOI: 10.21437/interspeech.2013-718, 2013. [ML14]

Context entry. MMSE estimator operating in the modulation spectrum domain for late reverb suppression. Useful for understanding the modulation-domain perspective on reverberation. It adds another lens for dereverberation design because late reverb smears temporal modulation, so suppression can be evaluated not only by decay curves but also by how well it restores speech envelope contrast.

E.13 Section 8: Speech and Room Acoustics

Papers on speech intelligibility in reverberant environments, dereverberation for ASR, and vocal effects in reverberant spaces. These are more peripherally related to verbx's core synthesis and analysis functions, but they matter for judging whether a reverb is useful, harmful, or stylistically appropriate. They support future intelligibility-aware modes, speech-oriented presets, QA metrics, and dereverb settings that prioritize clarity instead of only reducing measured tail energy.

Arai, T. "Preprocessing speech against reverberation." *The Journal of the Acoustical Society of America* 120(5_Supplement): 3323–3323. DOI: 10.1121/1.4781210, 2006. [SR8]

> Context entry. Signal processing approaches to pre-filtering speech before transmission through reverberant environments. Inverse problem perspective on the synthesis work. It is conceptually useful for verbx because adding reverb and preparing speech to survive reverb are mirror-image tasks, both concerned with how temporal envelopes and spectral cues survive a room response.

Barnett, P. "Study of Word Score Test Results to Determine the Robust Components of Speech Subject to Noise and Reverberation." *Reproduced Sound 1998*. DOI: 10.25144/18937, 2024. [SR13]

> Context entry. Word intelligibility scoring under combined noise and reverberation. Included for completeness. It is relevant to future test design because real-world speech problems often involve both noise and late reflection, so an evaluation corpus that varies only reverb may miss important interaction effects.

Bottalico, P.; Graetzer, S.; Hunter, E. "Effect of reverberation time on vocal fatigue." *Speech Prosody 2016* 494–497. DOI: 10.21437/speechprosody.2016-101, 2016. [SR14]

> Context entry. Studies how room RT60 affects talker vocal effort and fatigue. Peripheral to reverb synthesis; human factors context. It reminds us that room acoustics affect performers and speakers as well as listeners; for simulated environments, this can matter when designing presets for narration, voiceover, classroom, worship, or rehearsal contexts.

Choi, J.; Kim, J.; Kang, S. J.; et al. "Reverberation-robust acoustic indoor localization." *Interspeech 2015* 3422–3425. DOI: 10.21437/interspeech.2015-678, 2015. [SR19]

> Context entry. Room localization from reverberant acoustic signals. Tangential to synthesis; relevant to analysis. It reinforces that reverberation contains spatial and environmental information, so analysis tools can potentially infer room properties, device position, or capture context when enough cues survive in the signal.

Habets, E. A. P. "Speech Dereverberation Using Statistical Reverberation Models." *Signals and Communication Technology* 57–93. DOI: 10.1007/978-1-84996-056-4_3, 2010. [SR1]

> Annotation: Comprehensive chapter on statistical approaches to single-channel dereverberation. The statistical models of the late reverberant field described here inform verbx's internal representations of the reverberant tail. It is the central speech-dereverb reference in this bibliography because it connects room impulse response assumptions, late-tail statistics, suppression filters, and the practical artifacts that appear when trying to remove reverberation from a single channel.

Hodoshima, N.; Arai, T.; Kurisu, K. "Intelligibility of speech spoken in noise/reverberation for older adults in reverberant environments." *Interspeech 2012* 1464–1467. DOI: 10.21437/interspeech.2012-415, 2012. [SR12]

> Context entry. Earlier Hodoshima et al. study on older listener intelligibility. Part of a research series. It helps establish continuity across population-specific speech studies and supports documenting that dereverb or clarity presets may be especially valuable for accessibility, education, and public-address style material.

Hodoshima, N. “Effects of Urgent Speech and Congruent/Incongruent Text on Speech Intelligibility in Noise and Reverberation.” *Interspeech 2019* 3113–3117. DOI: 10.21437/interspeech.2019-1902, 2019. [SR11]

Context entry. Cognitive factors in speech intelligibility under reverberation. Peripheral to DSP design. It broadens the interpretation of “clarity” beyond acoustics alone, showing that urgency, text congruence, attention, and linguistic context can interact with room effects in ways that purely signal-level metrics may miss.

Hodoshima, N. “Effects of talker and playback rate of reverberation-induced speech on speech intelligibility of older adults.” *Interspeech 2024* 4229–4232. DOI: 10.21437/interspeech.2024-1721, 2024. [SR10]

Context entry. Speech intelligibility study focusing on older listeners in reverberant conditions. Peripheral to DSP design; population-specific psychoacoustic data. It is useful for accessibility-oriented thinking because reverberation that feels acceptable to one listener group may disproportionately reduce intelligibility for another, especially when playback rate and talker characteristics vary.

Kocinski, J.; Ozimek, E. “Speech Recognition in an Enclosure with a Long Reverberation Time.” *Archives of Acoustics* 41(2): 255–264. DOI: 10.1515/aoa-2016-0025, 2016. [SR6]

Context entry. Empirical study of intelligibility at long RT60 values. Relevant to understanding perceptual consequences of the extreme reverberation times verbx targets. It provides speech-focused grounding for why enormous tails should often be paired with ducking, predelay, spectral shaping, or clarity-preserving options when the source is spoken word rather than sustained music.

Leclere, T.; Lavandier, M.; Culling, J. F. “Speech intelligibility prediction in reverberation: Towards an integrated model of speech transmission, spatial unmasking, and binaural de-reverberation.” *The Journal of the Acoustical Society of America* 137(6): 3335–3345. DOI: 10.1121/1.4921028, 2015. [SR5]

Context entry. Integrated model of speech intelligibility in reverberation incorporating binaural effects. Background context for understanding the perceptual impact of long RT60 on speech. It shows that spatial hearing and binaural unmasking can change intelligibility outcomes, which matters when comparing mono, stereo, binaural, and multichannel reverberation or dereverberation modes.

Li, X.; Ma, X.; Hou, C. “Broadband DOA Estimation by Beamforming with Suppression of Reverberation.” *2008 Congress on Image and Signal Processing* 18–21. DOI: 10.1109/cisp.2008.550, 2008. [SR22]

Context entry. Broadband DOA estimation compensating for reverberation. Spatial processing context. It supports the broader analysis roadmap around multichannel and spatial features, where reverberation is not just a nuisance but a structured part of the observed sound field.

Nishiura, T.; Fukumori, T. “Suitable Reverberation Criteria for Distant-talking Speech Recognition.” *Speech Technologies*. DOI: 10.5772/18937, 2011. [SR4]

Context entry. Identifies which room acoustics metrics best predict ASR accuracy. Establishes C80 and DRR as more predictive than RT60 alone for intelligibility in ASR contexts. This is a strong reason for verbx analysis reports to include clarity and direct/reverberant metrics alongside RT60, especially when users are processing dialogue, conferencing audio, lectures, or datasets for recognition systems.

Pang, C.; Zhang, J.; Liu, H. “Direction of arrival estimation based on reverberation weighting and noise error estimator.” *Interspeech 2015* 3436–3440. DOI: 10.21437/interspeech.2015-681, 2015. [SR20]

Context entry. DOA estimation using reverberation structure. Spatial analysis context; peripheral to verbx. It is useful background for multichannel analysis because direction estimates can be biased or stabilized by reverberant energy depending on the method, array geometry, and late-field assumptions.

Peddinti, V.; Chen, G.; Povey, D.; et al. "Reverberation robust acoustic modeling using i-vectors with time delay neural networks." *Interspeech 2015*. DOI: 10.21437/interspeech.2015-527, 2015. [SR17]

Context entry. ASR robustness to reverberation using TDNN-based acoustic models. Neural ASR context; tangential to synthesis. It supports data-generation and benchmarking use cases where verbx could produce controlled reverberant examples for testing whether recognition models remain stable across room conditions.

Peng, J. "Chinese Syllable and Phoneme Identification in Noise and Reverberation." *Archives of Acoustics* 39(4): 483–488. DOI: 10.2478/aoa-2014-0052, 2015. [SR23]

Context entry. Language-specific phoneme intelligibility under reverberation. Peripheral to DSP design. It is included because reverberation can affect phoneme classes differently, and language-specific corpora may reveal clarity failures that broad acoustic metrics or English-only tests do not capture.

Petkov, P. N.; Braunschweiler, N.; Stylianou, Y. "Automated Pause Insertion for Improved Intelligibility Under Reverberation." *Interspeech 2016* 145–149. DOI: 10.21437/interspeech.2016-960, 2016.[SR16]

Context entry. Modifies speech rhythm to improve intelligibility in reverberant conditions. Peripheral to synthesis. It is useful for product thinking because not every reverb-related clarity problem must be solved by tail suppression; spacing, pauses, articulation, and source preprocessing can also improve intelligibility in reverberant playback.

Petkov, P. N.; Stylianou, Y. "Generalizing Steady State Suppression for Enhanced Intelligibility Under Reverberation." *Interspeech 2016* 2880–2884. DOI: 10.21437/interspeech.2016-1026, 2016. [SR15]

Context entry. Signal processing method for intelligibility enhancement in reverberant conditions. Included for completeness. It provides another example of improving intelligibility without necessarily reconstructing a perfectly dry signal, which is an important distinction for practical dereverberation controls that trade naturalness against clarity.

Raut, C. K.; Nishimoto, T.; Sagayama, S. "Model adaptation by state splitting of HMM for long reverberation." *Interspeech 2005* 277–280. DOI: 10.21437/interspeech.2005-157, 2005. [SR18]

Context entry. HMM adaptation for reverberant ASR. Historical ASR context. It shows the long-running nature of the reverberant speech problem and gives older-model context for why modern neural systems still benefit from controlled room simulation and reverberation-aware training data.

Rennies, J.; Brand, T.; Kollmeier, B. "Prediction of binaural intelligibility level differences in reverberation." *Interspeech 2011* 37–40. DOI: 10.21437/interspeech.2011-10, 2011. [SR7]

Context entry. Binaural intelligibility model predictions in reverberant conditions. Background reference for spatial dereverberation contexts. It supports the idea that a spatial reverb or dereverb mode should preserve useful interaural cues where possible, because apparent clarity can depend on binaural separation and not only on mono energy decay.

Saijo, K.; Wichern, G.; Germain, F. G.; et al. "Enhanced Reverberation as Supervision for Unsupervised Speech Separation." *Interspeech 2024* 607–611. DOI: 10.21437/interspeech.2024-1241, 2024. [SR2]

Context entry. Uses added reverberation as a data augmentation strategy for training speech separation models. Relevant for understanding how verbx might be used as a data synthesis tool. It supports a secondary use case where verbx is not only an effects processor but also a controllable corpus generator for robust speech, separation, or ASR experiments.

Saric, Z.; Jovicic, S. “Adaptive beamforming in room with reverberation.” *8th European Conference on Speech Communication and Technology (Eurospeech 2003)* 529–532. DOI: 10.21437/eurospeech.2003-218, 2003. [SR21]

Context entry. Beamforming in reverberant rooms. Spatial processing context. It provides a classic multichannel counterpoint to single-channel dereverberation, reminding readers that spatial filtering can reject some reverberant energy but is constrained by room reflections, array aperture, and source movement.

Sehr, A.; Kellermann, W. “New Results for Feature-Domain Reverberation Modeling.” *2008 Hands-Free Speech Communication and Microphone Arrays* 168–171. DOI: 10.1109/hscma.2008.4538713, 2008. [SR3]

Context entry. Feature-domain reverberation model for robust ASR. Peripheral to synthesis but relevant to understanding how reverb distorts feature representations. It helps explain why dereverberation quality cannot be judged only by waveform similarity: downstream feature spaces may amplify or ignore different kinds of smearing, coloration, and late-energy bias.

Tsilfidis, A.; Mourjopoulos, J. “Blind single-channel suppression of late reverberation based on perceptual reverberation modeling.” *The Journal of the Acoustical Society of America*. (Also listed as [PE3]), 2011. [SR9]

Annotation: Cross-listed perceptual dereverberation reference. In the speech section, its main value is showing that late-tail suppression can be guided by audibility and speech clarity rather than by raw energy reduction alone. This keeps the bibliography connected across perceptual modeling, dereverb design, and real-time speech use cases.

Unknown authors. “Fichera, I (2020). the Accuracy of Reverberation Time Predictions for General Teaching Spaces.” *Acoustics 2020*. DOI: 10.25144/13333, 1995. [SR25]

Context entry. Validation of RT60 prediction formulas in classroom environments. Context for applied room acoustics. It connects reverberation prediction to teaching spaces, where clarity and learning outcomes make acoustic estimates socially consequential rather than merely technical.

Unknown authors. “Reverberation Assessment in Audioband Speech Signals for Telepresence Systems.” *Proceedings of the International Conference on Signal Processing and Multimedia Applications* 257–262. DOI: 10.5220/0001936602570262, 2008. [SR24]

Context entry. Reverberation metrics for telepresence audio quality assessment. Peripheral context. It is relevant to practical deployments where users care about communication quality, echoic coloration, fatigue, and perceived professionalism rather than architectural realism alone.

E.14 Section 9: Underwater and Non-Room Reverberation

Papers whose “reverberation” subject matter refers to underwater acoustics, seabed scattering, structural media, musicology, or other domains unrelated to architectural room acoustics. Included because they were in the source bibliography; they have no direct bearing on verbx’s current design. Retained for bibliographic completeness, search disambiguation, and to make clear which references should not be used as evidence for room-reverb DSP decisions.

Burns, K. H., ed. *Women and Music in America Since 1900: An Encyclopedia*. Greenwood Press, Westport, CT; 2 vols.; ISBN 978-1-57356-267-6. Source: American Library Association, 2002. [BOOK1]

Annotation: This wide-ranging reference documents women as composers, performers, educators, producers, technologists, and institutional leaders across twentieth-century American music. It provides historical context for the guide's listening catalog and helps connect production techniques with the people and professional communities that developed and used them.

Chotiros, N. "Non-Rayleigh reverberation statistics." *The Journal of the Acoustical Society of America* 131(4_Supplement): 3448–3448. DOI: 10.1121/1.4708991, 2012. [UW2]

Context entry. Statistical characterization of underwater backscatter. Retained for completeness. It is useful mostly as a vocabulary boundary marker: similar statistical language appears in room acoustics, but the physical scattering mechanisms and operating assumptions are different from architectural reverberation.

Guo, Y.; Che, W. "Reverberation-Ray Matrix Analysis of Acoustic Waves in Multilayered Anisotropic Structures." *Acoustic Waves*. DOI: 10.5772/9788, 2010. [UW9]

Context entry. Reverberation-ray matrix method for layered elastic media. Structural acoustics domain; retained for completeness. It is another example where "reverberation" describes repeated wave interaction in a medium, but the target domain is not listener-room perception or audio-effect synthesis.

Harrison, C. "Reverberation versus time or reverberation versus range? A definitive relationship." *The Journal of the Acoustical Society of America* 126(4_Supplement): 2209–2209. DOI: 10.1121/1.3248705, 2009. [UW5]

Context entry. Underwater reverberation decay modeling. Retained for completeness. Its time-versus-range framing belongs to sonar and propagation analysis, not room impulse response synthesis, so it is cataloged here as an adjacent but non-authoritative use of the word reverberation.

Katsnelson, B.; Petnikov, V.; Lynch, J. "Low-Frequency Bottom Reverberation in Shallow Water." *Fundamentals of Shallow Water Acoustics* 239–266. DOI: 10.1007/978-1-4419-9777-7_6, 2011. [UW3]

Context entry. Shallow water bottom reverberation. Domain is underwater acoustics; retained for completeness only. It documents a non-room meaning of late returned acoustic energy and helps prevent accidental overextension of underwater decay or scattering models into verbx's room-oriented DSP.

LePage, K. D. "High-frequency broadband coherent reverberation predictions for the reverberation modeling workshop." *The Journal of the Acoustical Society of America* 126(4_Supplement): 2209–2209. DOI: 10.1121/1.3248711, 2009. [UW6]

Context entry. Underwater high-frequency reverberation modeling workshop proceedings. Retained for completeness. It is useful only for bibliography hygiene and terminology separation; verbx should continue to rely on room-acoustic and speech-acoustic references for implementation decisions.

Oh. "Low-Frequency Normal Mode Reverberation Model." *The Journal of the Acoustical Society of Korea* 34(3): 184. DOI: 10.7776/ask.2015.34.3.184, 2015. [UW4]

Context entry. Normal mode model for low-frequency underwater reverberation. Domain is underwater acoustics; retained for completeness only. The normal-mode framing is physically interesting but not directly applicable to ordinary room reverberation controls, except as a broad reminder that low-frequency propagation can be strongly modal in many acoustic systems.

Paul, J. "Amplification, Reverberation, and Distortion." *Music as Classical Reception* 236–256. DOI: 10.1093/9780198958024.003.0012, 2025. [UW10]

Context entry. Musicological analysis of reverberation as aesthetic phenomenon. Not DSP; retained for completeness. It may be useful for cultural or aesthetic framing, but it should not be cited for algorithm design, measurement formulas, or real-time processing behavior.

Preston, J. R.; Abraham, D. A.; Yang, J. "Non-stationary reverberation observations from the shallow water TREX13 reverberation experiments using the FORA triplet array." *The Journal of the Acoustical Society of America* 136(4_Supplement): 2225–2225. DOI: 10.1121/1.4900079, 2014. [UW1]

Context entry. Shallow-water acoustic reverberation measurements. Domain is underwater acoustics; retained for completeness only. The term "reverberation" here refers to environmental backscatter and propagation effects rather than enclosed-room decay, so it should not be used to justify verbx room modeling choices.

Tang, D. "Issues in reverberation modeling." *The Journal of the Acoustical Society of America* 134(5_Supplement): 4208–4208. DOI: 10.1121/1.4831447, 2013. [UW8]

Context entry. Underwater reverberation modeling issues review. Retained for completeness. It remains in the bibliography to avoid silently dropping source material, but it should be treated as out of scope for room-size, RT60, dereverb, and algorithmic reverb documentation.

Thorsos, E. I.; Perkins, J. S. "Reverberation modeling issues highlighted by the first Reverberation Modeling Workshop." *The Journal of the Acoustical Society of America* 122(5_Supplement): 3091–3091. DOI: 10.1121/1.2943047, 2007. [UW7]

Context entry. Summary of the first underwater acoustic reverberation modeling workshop. Retained for completeness. It helps identify a separate modeling community with different validation tasks, datasets, and physical assumptions than the architectural and production-audio problems verbx addresses.

E.15 Section 10: Extended Crossref Literature Index

This unannotated discovery index adds 900 Crossref-derived references to the 166 curated and primary-source entries above, bringing the guide bibliography to 1,066 total entries.

The entries below are intentionally not treated as vetted design authority. They are included to make the PDF a much broader literature map for reverberation, dereverberation, spatial audio, room acoustics, and related acoustic measurement work. Use the annotated sections above for canonical implementation guidance.

Source: Crossref Works API metadata, generated May 22, 2026.

Discovery queries:

- Room reverberation and room acoustics: `room acoustics reverberation time RT60`
- Artificial reverberation and FDNs: `artificial reverberation feedback delay network`
- Impulse responses and convolution reverb: `room impulse response convolution reverberation`
- Speech dereverberation and clarity: `speech dereverberation reverberation direct-to-reverberant ratio`
- Spatial audio and Ambisonics: `spatial audio ambisonics reverberation auralization`
- Perceptual reverberation: `perception reverberation listener envelopment clarity`
- Auralization and virtual acoustics: `auralization virtual acoustics reverberation`
- Acoustic measurement and decay metrics: `energy decay curve reverberation time acoustic measurement`

- Acoustic simulation and image-source models: `image source method room acoustic simulation reverberation`
- Late reverberation modeling: `late reverberation statistical model acoustic signal processing`

E.15.1 Extended Bibliography Entries

E.15.1.1 Room reverberation and room acoustics

Abd-Elbasseer, M.; Kh Mohamed, H. "Performance Evaluation and Effectiveness of the Reverberation Room." *Sound&Vibration* 55(1): 43–55. DOI: 10.32604/sv.2021.09417, 2021. [XREF0459]

Abolhasannejad, V.; Golmohammadi, R.; Aliabadi, M.; et al. "An image-based method for non-contact and dynamic room acoustics analysis." *Applied Acoustics* 140: 83–91. DOI: 10.1016/j.apacoust.2018.05.018, 2018. [XREF0696]

Accolti, E.; di Sciascio, F. "Room acoustics: Idealized field and real field considerations." *Proceedings of Meetings on Acoustics* 015003. DOI: 10.1121/2.0000795, 2017. [XREF0462]

Accolti, E.; di Sciascio, F. "On the use of time windows for the determination of sound strength parameter G from uncalibrated room impulse responses measurements." *Applied Acoustics* 178: 108023. DOI: 10.1016/j.apacoust.2021.108023, 2021. [XREF0251]

Accolti, E.; Gimenez, J.; Vorländer, M. "Uncertainties of room acoustics simulation due to directivity data of musical instruments." *Unspecified venue*. DOI: 10.36227/techrxiv.20858596.v3, 2023. [XREF0871]

Accolti, E.; Gimenez, J.; Vorländer, M. "Uncertainties of room acoustics simulation due to directivity data of musical instruments." *Unspecified venue*. DOI: 10.36227/techrxiv.20858596.v2, 2023. [XREF0873]

Accolti, E.; Gimenez, J.; Vorländer, M. "Uncertainties of room acoustics simulation due to directivity data of musical instruments." *Unspecified venue*. DOI: 10.36227/techrxiv.20858596, 2023.[XREF0900]

Achdjian, H.; Bustillo, J.; Arciniegas, A.; et al. "Contact surface fraction evaluation between aluminium and polymer using acoustic reverberation." *Applied Acoustics* 141: 208–212. DOI: 10.1016/j.apacoust.2018.07.016, 2018. [XREF0314]

Adelman-Larsen, N. W. "Amplified music and variable acoustics—Shorter reverberation times at low frequencies." *The Journal of the Acoustical Society of America* 141(5_Supplement): 3853–3853. DOI: 10.1121/1.4988599, 2017. [XREF0788]

Adelman-Larsen, N. W.; Jeong, C.; Støfringsdal, B. "Investigation on acceptable reverberation time at various frequency bands in halls that present amplified music." *Applied Acoustics* 129: 104–107. DOI: 10.1016/j.apacoust.2017.07.005, 2018. [XREF0087]

Ågren, A. "The design and evaluation of a hemi-anechoic engine test room." *Applied Acoustics*. DOI: 10.1016/0003-682x(92)90024-m, 1992. [XREF0579]

Ahnert, W.; Noy, D. "Room Acoustics and Sound System Design." *Sound Reinforcement for Audio Engineers* 20–67. DOI: 10.4324/9781003220268-2, 2022. [XREF0890]

Aitchison, G. J. "Measurement of Reverberation Time." *American Journal of Physics* 33(6): 493–495. DOI: 10.1119/1.1971726, 1965. [XREF0876]

Aknin, A.; Badeau, R. "Stochastic Reverberation Model with a Frequency Dependent Attenuation." *2021 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)* 351–355. DOI: 10.1109/waspaa52581.2021.9632792, 2021. [XREF0564]

Al-bayyar, Z.; Kitapci, K. “Effects of reverberation time and sound source composition on sense of place constructs.” *Applied Acoustics* 185: 108427. DOI: 10.1016/j.apacoust.2021.108427, 2022. [XREF0057]

Alarcão, D.; Coelho, J. L. B. “Lambertian Enclosures – A First Step towards Fast Room Acoustics Simulation.” *Building Acoustics* 10(1): 33–54. DOI: 10.1260/135101003765184807, 2003. [XREF0515]

Alarcão, D.; Inácio, O. “Determination of room acoustic parameters using spherical beamforming – The example of Lisbon’s Garrett Hall.” *Applied Acoustics* 192: 108734. DOI: 10.1016/j.apacoust.2022.108734, 2022. [XREF0701]

Alary, B.; Politis, A. “A Dataset for Location- and Direction-Dependent Reverberation Analysis.” *Proceedings of the 10th Convention of the European Acoustics Association Forum Acusticum 2023* 1663–1670. DOI: 10.61782/fa.2023.0833, 2022. [XREF0635]

Aletta, F.; Kang, J. “Historical Acoustics: Relationships between People and Sound over Time.” *Acoustics*. DOI: 10.3390/acoustics2010009, 2020. [XREF0770]

Algargoosh, A.; Navvab, M.; Granzow, J. “A method for analyzing room modal response using auralization.” *Applied Acoustics* 195: 108859. DOI: 10.1016/j.apacoust.2022.108859, 2022. [XREF0573]

Allen, J. B. “Effects of small room reverberation on subjective preference.” *The Journal of the Acoustical Society of America* 71(S1): S5–S5. DOI: 10.1121/1.2019478, 1982. [XREF0678]

Allen, J. B. “Method and apparatus for cancelling room reverberation and noise pickup.” *The Journal of the Acoustical Society of America* 65(4): 1097–1097. DOI: 10.1121/1.382584, 1979. [XREF0773]

Alpkocak, A.; Sis, M. “Computing Impulse Response of Room Acoustics Using the Ray-Tracing Method in Time Domain.” *Archives of Acoustics* 35(4): 505–519. DOI: 10.2478/v10168-010-0039-8, 2010. [XREF0150]

Álvarez, J.; Graus, R.; Martín-Nieva, H. “A historical overview of Higini Arau’s proposal for calculating reverberation time (1988).” *Proceedings of the 11th Convention of the European Acoustics Association Forum Acusticum / EuroNoise 2025* 925–931. DOI: 10.61782/fa.2025.0012, 2025. [XREF0239]

Amador, E.; Miry, C.; Bouyge, N. “Compatible susceptibility measurements in fully anechoic room and reverberation chamber.” *2014 International Symposium on Electromagnetic Compatibility* 860–865. DOI: 10.1109/emceurope.2014.6931024, 2014. [XREF0472]

Andrieu, G. “Frequency and time-domain performance assessment of vibrating intrinsic reverberation chambers.” *Electromagnetic Reverberation Chambers: Recent advances and innovative applications* 81–100. DOI: 10.1049/sbew544e_ch3, 2020. [XREF0347]

Aralikatti, R.; Boeddeker, C.; Wichern, G.; et al. “Reverberation as Supervision For Speech Separation.” *ICASSP 2023: 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 1–5. DOI: 10.1109/icassp49357.2023.10095022, 2023. [XREF0802]

Arau, H. “Variation of the Reverberation Time of Places of Public Assembly with Audience Size.” *Building Acoustics* 4(2): 87–98. DOI: 10.1177/1351010x9700400202, 1997. [XREF0062]

Arau-Puchades, H. “Sound Pressure Levels in Rooms: A Study of Steady State Intensity, Total Sound Level, Reverberation Distance, a New Discussion of Steady State Intensity and other Experimental Formulae.” *Building Acoustics* 19(3): 205–220. DOI: 10.1260/1351-010x.19.3.205, 2012. [XREF0628]

Arau-Puchades, H. “The Refurbishment of the Orchestra Rehearsal Room of the Great Theater of Liceu.” *Building Acoustics* 19(1): 45–56. DOI: 10.1260/1351-010x.19.1.45, 2012. [XREF0803]

Arau-Puchades, H.; Berardi, U. “The reverberation radius in an enclosure with asymmetrical absorption distribution.” *Proceedings of Meetings on Acoustics* 015141–015141. DOI: 10.1121/1.4800909, 2013. [XREF0427]

Aretz, M.; Orlowski, R. “Sound strength and reverberation time in small concert halls.” *Applied Acoustics* 70(8): 1099–1110. DOI: 10.1016/j.apacoust.2009.02.001, 2009. [XREF0042]

Aretz, M.; Orlowski, R. “Balancing Sound Strength and Reverberation Time in Small Concert Halls by Means of Variable Acoustics.” *Noise & Vibration Worldwide* 41(8): 11–24. DOI: 10.1260/0957-4565.41.8.11, 2010. [XREF0253]

Aretz, M.; Orlowski, R. “Balancing sound strength and reverberation time in small concert halls by means of variable acoustics.” *Noise Notes* 10(1): 3–19. DOI: 10.1260/1475-4738.10.1.3, 2011. [XREF0157]

Arnela, M.; Burbano-Escolà, R.; Scoczynski Ribeiro, R.; et al. “Reverberation time and random-incidence sound absorption measured in the audible and ultrasonic ranges with an omnidirectional parametric loudspeaker.” *Applied Acoustics* 229: 110414. DOI: 10.1016/j.apacoust.2024.110414, 2025. [XREF0126]

Atkins, J.; West, J. E. “Spatial Audio and Room Acoustics.” *Acoustics, Information, and Communication* 3–13. DOI: 10.1007/978-3-319-05660-9_1, 2015. [XREF0721]

Austin, B.; Cheer, J. “Experimental implementation of an actively controlled tuned vibration absorber using piezoelectric actuators.” *Proceedings of Meetings on Acoustics* 58: 065001. DOI: 10.1121/2.0002056, 2025. [XREF0879]

Avis, M.; Darlington, P. “Modifying Low Frequency Room Acoustics 1: Local Active De-Reverberation.” *RS 11*. DOI: 10.25144/20135, 2024. [XREF0008]

Aydin, A.; Baran, V.; Zhang, K. Y.; et al. “A real-world Lexicon 960L reverberation chamber: Simulating a hardware reverberation unit in virtual acoustics.” *The Journal of the Acoustical Society of America* 155(3_Supplement): A128–A128. DOI: 10.1121/10.0027047, 2024. [XREF0487]

Ayr, U.; Martellotta, F.; Rospi, G. “A method for the low frequency qualification of reverberation test rooms using a validated finite element model.” *Applied Acoustics* 116: 33–42. DOI: 10.1016/j.apacoust.2016.09.001, 2017. [XREF0358]

Azad, H. “Ali Qapu: Persian Historical Music Room.” *Auditorium Acoustics 2008*. DOI: 10.25144/17495, 2023. [XREF0448]

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“Repeating your ultimate word.”

— John Godfrey Saxe, *The Story of Echo* (1865)

Appendix F

Frequently Asked Questions

This appendix answers the questions most likely to arise while installing, running, measuring, and integrating verbx. Commands assume a shell in which `verbx` is on the path. Run `verbx COMMAND --help` whenever the installed version is the final authority for an option or default.

Table F-1: Frequently Asked Questions: reference table

If you need to…	Start with…
Verify the installation	`verbx doctor --strict`
Render a first example	`verbx render in.wav out.wav --preset warm-hall`
Inspect an audio file	`verbx analyze out.wav --lufs --edr --room --json-out analysis.json`
Find audio devices	`verbx realtime --list-devices`
List the preset library	`verbx presets`
Diagnose a failed render	`verbx render ... --failure-report-out failure.json`

F.1 Installation and First Launch

Q1. How do I verify that verbx is installed correctly?

Run `verbx version`, followed by `verbx doctor --strict`. For a stronger end-to-end check, use `verbx doctor --render-smoke-test`; it creates and processes a small test signal so the command path, audio dependencies, and render pipeline are exercised together.

Q2. What should I do if the shell says `verbx: command not found`?

First, open a new terminal and run `command -v verbx`. If the command prints no path, add the directory containing the `verbx` executable to your shell's `PATH`. The repository installer uses `~/.local/bin` by default, so add the following line to `~/.zshrc` on macOS or to the startup file used by your shell:

```
export PATH="$HOME/.local/bin:$PATH"
```

Open another terminal and run `verbx version`. If verbx was installed in a virtual environment, activate that environment instead. Run `python3 -m verbx version` to check whether the package is importable even when the `verbx` console command is not visible.

Q3. Should I install with Homebrew or from the repository?

Use Homebrew for the simplest supported macOS command-line installation. Use a source checkout when developing verbx, running the complete test suite, building plug-ins, or changing documentation. Do not mix two active installations while diagnosing version or plug-in discovery problems.

Q4. What does `install.sh` install?

The repository installer builds the available command-line and native targets and places supported plug-in bundles in the standard per-user macOS locations. It does not control a DAW's validation cache, so the host may still require a rescan or restart. Read the installer summary carefully because unavailable SDKs or formats are reported rather than silently fabricated.

Q5. Why does a newly installed plug-in not appear immediately in my DAW?

Quit and reopen the DAW, request a full plug-in rescan, and verify the bundle with the host's plug-in manager. On macOS, also confirm that the architecture matches the host and that quarantine, signing, or validation did not reject the bundle. The manufacturer name is **Colby Leider**.

Q6. Can installation diagnostics be saved for support or automation?

Yes. Use `verbx doctor --json-out doctor.json`. Add `--strict` when a warning should produce a failing exit status in continuous integration, and add `--render-smoke-test --smoke-out-dir smoke` when a reproducible audio artifact is useful.

F.2 Choosing and Shaping Reverb

Q7. When should I choose algorithmic reverb instead of convolution?

Choose algorithmic reverb for continuously variable decay, modulation, freeze, unusual feedback structures, and low-memory realtime operation. Choose convolution when a specific measured or designed impulse response is the sound. A hybrid workflow can use convolution for early reflections and an algorithmic network for the late field.

Q8. What does `--engine auto` do?

Auto selects an engine from the supplied inputs and requested controls. An explicit IR normally favors convolution; algorithm-only controls favor the algorithmic engine. Use `--engine algo` or `--engine conv` when reproducibility requires an unambiguous choice.

Q9. What does RT60 mean?

RT60 is the estimated time, in seconds, for reverberant energy to decay by 60 dB under the model's assumptions. It describes a decay slope, not necessarily the exact file length. Noise floors, frequency dependence, modulation, gates, and freeze can make the perceived tail differ from the nominal value.

Q10. Why is RT60 controlled logarithmically?

Useful decay times span several orders of magnitude, from tight rooms to minute-long sound-design fields. A logarithmic control gives short decays enough precision while still reaching extreme values. The plug-in combines coarse and fine controls so broad changes and exact adjustments are both practical.

Q11. What is the difference between pre-delay and early reflections?

Pre-delay is the initial dry-to-reverberant gap. Early reflections are discrete arrivals that follow and convey room size, source position, and surface geometry. Increasing pre-delay can preserve articulation without changing the late decay, while changing the early-reflection pattern changes spatial character.

Q12. How do damping, filtering, and diffusion differ?

Damping changes decay as a function of frequency. Input or output filters shape what enters or leaves the reverberator. Diffusion changes echo density and the audibility of individual repetitions. A dark but sparse tail and a bright but highly diffuse tail are therefore both possible.

F.3 Rendering and Output

Q13. What is a good first render command?

Start with:

```
verbx render in.wav out.wav --engine algo --rt60 2.4 --wet 0.35 --dry 0.8
```

Then change one parameter at a time. Add `--json-out out.report.json` when the settings and processing summary should travel with the audio.

Q14. Why can a render with an extremely long RT60 appear to hang?

The renderer may be generating and writing a tail proportional to the requested decay. An RT60 of 3,600 seconds is an hour-scale process, not an ordinary room render. Use freeze for intentional indefinite sustain, or choose a bounded output duration and watch the current processing status rather than treating elapsed time as a completion estimate.

Q15. What is the difference between a very long decay and freeze?

A very long RT60 still decays. Freeze deliberately approaches lossless recirculation and is intended for sustained textures. Freeze output must still be bounded during offline rendering; otherwise there is no natural endpoint at which a complete file can be written.

Q16. Can verbx write 32-bit and 64-bit floating-point WAV files?

Yes, where the selected backend supports the requested subtype. Float32 is the normal interchange choice; float64 preserves additional numerical precision at the cost of larger files and reduced compatibility with some software. File precision does not by itself guarantee that every internal stage or host uses the same precision.

Q17. How should I prevent clipping in wet renders?

Leave headroom in both dry and wet paths, especially when long tails accumulate energy. Use the limiter deliberately rather than as a substitute for gain staging, and inspect peak, true-peak, and loudness measurements after rendering. A normalized impulse response can still produce overload when dense material is convolved with it.

Q18. Which machine-readable render outputs are available?

Use `--json-out` for the processing report, `--analysis-out` for analysis results, `--frames-out` for frame-level data, `--repro-bundle` for reproducibility material, and `--failure-report-out` for structured failure context. `--dry-run` is useful before a large render because it validates and reports without committing to the full process.

F.4 Realtime Audio and Latency

Q19. How do I list realtime input and output devices?

Run `verbx realtime --list-devices`. Then pass the displayed device name or identifier with `--input-device` and `--output-device`. If a device disappears, close applications that may hold it exclusively and list devices again.

Q20. What does end-to-end realtime latency include?

It includes input conversion and driver buffering, verbx input buffering, algorithm or partitioned-convolution delay, output buffering, driver scheduling, and output conversion. The audio block size is only one component. Measure round-trip latency with a physical loopback when sample-accurate compensation matters.

Q21. How can I reduce realtime latency?

Choose a smaller `--block-size`, use an algorithmic engine or a smaller convolution partition, avoid unnecessary channel expansion, and select a low-latency device driver. Reduce one setting at a time while checking for xruns. The smallest accepted value is not necessarily the most reliable value under a real session load.

Q22. Why do I hear clicks or dropouts at a small block size?

The processor or driver is missing deadlines. Increase the block size, close competing audio applications, simplify the routing, reduce expensive analysis displays, or use a less demanding engine. Stable uninterrupted audio is more important than a nominally lower buffer that repeatedly underruns.

Q23. Can realtime mode dereverberate and add reverb in one pass?

Yes. Select `--live-mode dereverb-reverb` and configure both stages. This is more computationally demanding than either stage alone, so establish a stable block size with each stage separately before combining them.

Q24. How do I save a realtime session report?

Add `--json-out realtime.json`. The report records the resolved devices, stream format, buffer choices, mode, and available runtime counters so a session can be compared with a later run or attached to a bug report.

F.5 AUv3 and VST3 Plug-ins

Q25. Which sample rate and bit depth does the plug-in use?

The DAW negotiates the plug-in's sample rate and processing format. The project's 192 kHz, 32-bit float default describes its preferred high-resolution offline policy, not a request that overrides the host session. A 48 kHz DAW session therefore runs the plug-in at 48 kHz.

Q26. Why can a plug-in be discovered but fail to load?

Discovery only proves that the host found a bundle. Loading can still fail because of architecture, signing, deployment-target, factory-registration, bus-layout, or state initialization errors. Check the DAW validation log and run the repository's plug-in validation commands before changing the GUI.

Q27. Why does the plug-in appear under Colby Leider rather than verbx?

DAWs commonly group effects by manufacturer. **Colby Leider** is the manufacturer and author metadata; **verbx** is the product name. Both fields are intentional and should remain stable so saved sessions and host catalogs do not split into duplicate entries.

Q28. What is on the Expert page?

The Expert page exposes the detailed network, damping, modulation, spatial, dynamics, and quality controls that would overwhelm the primary performance page. Start from a preset, make broad changes on the main page, and enter Expert only when a specific technical adjustment is needed.

Q29. Does the realtime spectrum analyzer change the sound?

No. It is a metering overlay fed from the audio path, not an audio processor. Disabling or reducing its refresh rate should leave the rendered signal unchanged while reducing GUI work on constrained systems.

Q30. How should RT60 automation behave in the plug-in?

The coarse control moves logarithmically across the full 0.01–360 second performance range, while the fine control trims the current value. Hosts should automate the normalized parameter smoothly; abrupt jumps in a feedback network may still require internal smoothing to avoid clicks or unstable energy changes.

F.6 Analysis and Dereverberation

Q31. How do I extract reverb metrics from an audio file?

Run:

```
verbx analyze recording.wav --reverb --lufs --edr --room --json-out analysis.json
```

Use `--input-kind ir` for a measured impulse response, `--input-kind program` for music

or speech, or `--input-kind auto` when verbx should infer the interpretation.

Q32. Why do RT60 estimates differ between an impulse response and music?

An impulse response exposes a direct decay curve. Music and speech contain overlapping events, changing spectra, dynamics processing, and background noise, so their estimates are inferential. Treat program-material measurements as descriptive evidence rather than the same laboratory quantity obtained from a clean excitation and response.

Q33. What do EDR and frame-level outputs add?

An energy decay relief is a time-frequency generalization of the energy decay curve. It backward-integrates the squared magnitude of every STFT bin, so a narrow resonance can remain visible even when a broadband RT60 averages it away. `--edr` currently reports median, low-, middle-, and high-band RT60 summaries plus the number of usable fitted bins; it does not write the complete EDR matrix. Frame output preserves separate local descriptors for plotting, quality control, or downstream research. See Julius O. Smith III's Energy Decay Relief derivation for the underlying definition.

Q34. Can verbx estimate room properties from an ordinary recording?

It can report model-based room indicators with `--room`, but these are estimates, not a replacement for geometry, calibrated excitation, microphone positions, and atmospheric conditions. Confidence should decrease when the source is dense, dynamically processed, or recorded in a noisy environment.

Q35. What can dereverberation realistically remove?

Dereverberation can reduce diffuse late energy and improve intelligibility, but it cannot perfectly recover information masked by reflections or noise. Aggressive settings may create pumping, spectral holes, or speech artifacts. Compare bypassed and processed signals at matched loudness and preserve the original recording.

Q36. How do I start realtime dereverberation safely?

First verify devices with `verbx realtime --list-devices`. Then run realtime with `--live-mode dereverb`, conservative reduction, and a stable block size. Monitor speech transients, noise pumping, CPU deadline misses, and the saved JSON report before adding a reverb stage.

F.7 Impulse Responses, Geometry, and Spatial Audio

Q37. How do I discover the impulse-response tools?

Run `verbx ir --help`. The group includes generation, analysis, SOFA inspection and extraction, geometry tracing, processing, morphing, sweep creation, and fitting. Run the selected subcommand with `--help` because its required files and coordinate conventions are more specific than the top-level summary.

Q38. Can verbx generate reverberation from a DXF room model?

The geometry workflow can trace supported DXF room descriptions for early-reflection and IR work, subject to the documented parser and material limitations. Validate scale, closed surfaces, source position, receiver position, normals, and units before trusting the acoustical result. A CAD drawing is not automatically a complete acoustic model.

Q39. What is a hybrid geometry-plus-algorithmic reverb?

It uses traced or measured early arrivals for spatial identity and an algorithmic late field for dense, controllable decay. The handoff must match level, spectrum, direction, and time so the two parts sound like one room rather than an echo pattern followed by a separate tail.

Q40. What is SOFA used for?

SOFA stores spatially indexed acoustic measurements such as head-related or room impulse responses. Use `verbx ir sofa-info` to inspect a file and `verbx ir sofa-extract` to select responses. Confirm coordinate conventions and sample rate before convolution.

Q41. Does verbx support surround and immersive workflows?

Yes. The CLI includes multichannel routing, ambisonic analysis, matrix convolution, and immersive template, handoff, quality-control, and queue workflows. Preserve explicit channel labels and ordering throughout; a valid channel count does not prove that every speaker is mapped to the intended position.

Q42. Does verbx create a finished Dolby Atmos master?

verbx can prepare, render, inspect, and hand off spatial assets, but final Dolby Atmos authoring and delivery still belong in a licensed renderer and DAW workflow that meets

the destination specification. Treat verbx reports as technical companions, not as a replacement for required Dolby validation or deliverables.

F.8 Presets, Reproducibility, and Troubleshooting

Q43. How do I browse the large preset library?

Run `verbx presets` to list presets, `verbx presets --show NAME` to inspect one, and `verbx presets --validate NAME` to validate its resolved data. Start from a descriptive family rather than auditioning hundreds in arbitrary order, then compare at matched loudness.

Q44. How do I make a render reproducible?

Pin the verbx version, engine, preset or complete option set, random seed where used, input-file checksum, and output format. Save `--json-out` and `--repro-bundle` artifacts with the audio. Reproducibility includes the resolved defaults, not merely the switches typed on the command line.

Q45. How can I compare two processing choices objectively?

Use `verbx compare` for supported measurements, then perform a level-matched listening comparison. Keep the source, trim, output format, and analysis settings identical. A lower error metric or longer RT60 is not automatically the musically preferred result.

Q46. What information belongs in a useful bug report?

Include `verbx version`, operating system and architecture, the exact command, the smallest shareable input, expected and observed behavior, exit status, doctor JSON, processing or failure JSON, and relevant DAW validation logs. Remove private paths or audio only after preserving enough structure to reproduce the defect.

Q47. When should I clear or inspect the cache?

Use `verbx cache --help` when stale generated assets, unexpectedly high disk use, or a version transition points to cached data. Inspect before deleting whenever possible. Clearing a cache may make the next run slower, but it should not be required for ordinary parameter changes.

Q48. What is the safest way to explore extreme settings?

Lower monitor gain, keep a limiter after the experimental path, render a short excerpt, use a bounded duration, and save dry-run or failure reports. Feedback, freeze, modulation, and long decay can accumulate energy unexpectedly. Preserve the dry original and move from conservative values toward the extreme rather than starting at the maximum.

F.9 Measurement and Impulse-Response Capture

Q49. Can a handclap be used as an impulse response?

A handclap is useful for listening and rough scouting, but it is neither spectrally flat nor precisely repeatable. For an archival IR, use a known loudspeaker excitation and retain the raw capture, geometry, gain, sample rate, and environmental metadata.

Q50. When is a balloon pop appropriate?

A balloon pop is portable and can reveal gross arrival and decay behavior in a large space. Its directivity, spectrum, peak level, and source position vary from burst to burst, so compare repeated captures and do not treat it as an absolutely calibrated source without additional measurement.

Q51. Why is an exponential sine sweep usually preferred?

It places sustained energy throughout the measurement band without requiring one extreme impulse peak. With a matched inverse filter, it provides high signal-to-noise ratio and separates harmonic distortion responses from the principal linear response.

Q52. What is deconvolution?

Deconvolution estimates the unknown room response after a known excitation has passed through it. It is not simply reversing the recording: it applies the inverse spectral and phase behavior of the excitation, usually with regularization to avoid amplifying noise where the sweep has little energy.

Q53. What does the Farina method add to sweep measurement?

Farina's exponential swept-sine method maps harmonic distortion products to earlier, separate time regions after deconvolution. This lets the linear room response be windowed independently and also provides evidence about loudspeaker or device nonlinearity.

Q54. How long should the sweep be?

Longer sweeps improve energy and low-frequency signal-to-noise ratio, but they also increase sensitivity to motion and time variance. Choose a duration that covers the required band at safe level, then record silence after the sweep for at least the full expected decay plus a noise-floor margin.

Q55. Should I record a loopback channel?

Yes, when the interface permits it. A simultaneous electrical loopback documents the actual excitation timing and can expose latency, drift, dropout, or polarity errors that would otherwise be mistaken for room behavior.

Q56. How many source and receiver positions are enough?

One pair describes one transfer path, not the entire room. The necessary grid depends on whether the goal is one listening position, statistical room characterization, spatial rendering, or machine-learning coverage. Document every coordinate and avoid averaging positions that serve different scientific questions.

Q57. Should the direct-arrival delay remain in the IR?

Keep it when physical source distance and synchronization are part of the measurement. Trim or separate it when a production processor supplies independent pre-delay. Record the original direct-arrival sample either way so the operation can be reversed or audited.

Q58. Should a measured IR be peak normalized?

Peak normalization is convenient for production libraries but destroys absolute gain calibration. Preserve an untouched calibrated master and create a normalized derivative with the policy and target recorded in metadata.

Q59. How do I know whether the measured tail is noise?

Compare repeated captures and a background-noise recording, inspect band-limited energy decay, and identify where the decay becomes statistically indistinguishable from the noise floor. A long waveform is not evidence of a long reverberation time if its final seconds contain only ventilation or preamplifier noise.

Q60. Can verbx record and deconvolve the sweep itself?

Not currently. Use a measurement application, DAW, or validated scientific script for playback, synchronized capture, and deconvolution, then use `verbx ir analyze`, `verbx ir process`, and convolution rendering for inspection and production.

F.10 Filters, Poles, and Stability

Q61. What is a digital filter?

It is a difference equation that maps an input sample sequence to an output sequence. FIR filters use a finite input history; IIR filters include feedback and can sustain natural modes after the input stops.

Q62. What does a pole represent in a reverb?

A pole represents one natural exponential mode. Its angle corresponds to oscillation frequency and its radius to decay rate. A field of well-distributed stable poles usually sounds smoother than a few exposed or nearly coincident modes.

Q63. What does a zero represent?

A zero is a frequency-domain cancellation of the observed transfer function. In an FDN, transmission zeros depend on input and output projection as well as internal topology, so changing stereo projection can change color without moving the internal poles.

Q64. Why is the unit circle important?

For a causal discrete-time IIR system, poles strictly inside the unit circle decay. Poles on it persist ideally, and poles outside it grow. Finite precision and changing coefficients require practical stability margin rather than equality at radius one.

Q65. Why do high-order filters use second-order sections?

Direct high-order polynomial coefficients can be numerically ill conditioned. Pairing real roots and complex-conjugate roots into first- and second-order sections improves coefficient scaling, implementation stability, and diagnostic clarity.

Q66. Can an allpass have zeros outside the unit circle?

Yes. A stable allpass keeps its poles inside while placing reciprocal-conjugate zeros outside. The arrangement preserves ideal magnitude on the unit circle while changing phase and transient distribution.

Q67. Why can a stable reverb still sound metallic?

Stability only guarantees that modes do not grow. Sparse modal angles, repeated delay relationships, insufficient diffusion, static symmetry, or uneven loop damping can leave audible tones even when every pole is safely inside the unit circle.

Q68. How should a 360-second decay be validated?

Check loop gain and filter response in double precision, render a bounded impulse test, inspect energy by band, and verify that limiter activity is not hiding growth. Also test automation and sample-rate changes because tiny coefficient errors persist for many loops.

F.11 Rendering, Formats, and Long Tails

Q69. Why can a short source create a very large output?

Output duration includes the source plus the retained tail, and multichannel floating- point formats multiply storage quickly. Use a dry run and explicit tail policy before starting an extreme render.

Q70. When should I use W64 rather than WAV?

Use W64 when predicted output may exceed ordinary RIFF WAV size limits. Container choice does not change DSP quality; it changes how safely the file can represent long or highly multichannel data.

Q71. Does float32 output reduce internal precision?

No. verbx processes internally in float64 and converts at export. Float32 is usually adequate for production interchange, while float64 is useful for research, repeated offline transforms, and preserving very low-level tails.

Q72. Why does a render appear to pause during processing?

Some stages operate on long blocks or complete tails and cannot update a progress bar on every inner operation. Use current versions with dry-run estimates and status reporting; if elapsed time exceeds the estimate materially, retain the command and diagnostics for a reproducible report.

Q73. How should I render a one-hour RT60 safely?

Begin with a short source and bounded tail, estimate size, select a large-file container, and monitor peak growth and disk space. A nominal one-hour RT60 does not require writing the full theoretical decay if the project only needs a shorter musical window.

Q74. What is the difference between RT60 and output tail length?

RT60 is a decay-rate target; tail length is a file boundary. Several RT60 intervals may be required to reach the chosen noise floor, while a creative render may deliberately truncate or gate much earlier.

Q75. Can I process files at 192 kHz?

Yes, subject to engine, memory, and output-format constraints. Delay buffers, FFT work, and file size scale with sample rate, so test throughput and resample IRs carefully rather than assuming a higher rate is free.

Q76. Why should I retain JSON beside the WAV?

The WAV preserves audio but not the complete resolved configuration, estimates, warnings, or analysis. JSON makes the render auditable and lets batch systems distinguish completed, failed, and policy-rejected outputs without parsing human progress text.

F.12 Plug-ins and DAW Diagnosis

Q77. Why does the plug-in show under Colby Leider?

The manufacturer metadata is Colby Leider, so compatible hosts group the AU, AUv3, and VST3 there. A host may instead sort by category or product name; rescan after replacing a binary and remove only the host's documented plug-in cache when necessary.

Q78. Why might Logic reject an Audio Unit that another host sees?

Logic validates component metadata, architecture, code signing, bus layouts, and runtime behavior. Inspect `auval` and Logic validation output rather than assuming that appearance in a component folder proves loadability.

Q79. Why might Audacity reject a VST3?

The bundle may be built for an unsupported architecture, installed in the wrong scope, missing runtime dependencies, or failing initialization. Verify the universal or native binary, bundle structure, signing, and Audacity's plug-in scan log.

Q80. Why do plug-in controls sometimes appear to do nothing?

Confirm the host is sending audio, the wet path is audible, the control is not overridden by automation, and the parameter attachment reaches DSP state. Test a dramatic setting such as fully wet with a short decay before diagnosing subtler controls.

Q81. What belongs on the Expert page?

Expert controls should expose topology, multiband decay, modulation, diffusion, projection, limiting, and quality without changing parameter identifiers used by automation. The page is an alternate view of stable parameters, not a second incompatible processor.

Q82. Who decides the plug-in sample rate and output bit depth?

The DAW supplies the processing sample rate and buffer format. The plug-in can use higher internal precision or oversampling, but final file subtype and delivery sample rate are selected during the host's bounce or export.

Q83. How is plug-in latency different from CLI render time?

Plug-in latency is the sample delay from buffering and algorithms that the host may compensate. Render time is wall-clock computation. A processor can report low latency yet consume substantial CPU, or render faster than realtime offline while using a long FFT partition that would be unsuitable for live monitoring.

Q84. Why should the analyzer not allocate on the audio thread?

Dynamic allocation, locks, and GUI calls can miss realtime deadlines. The audio thread should copy bounded analysis data into a lock-free or carefully synchronized queue, while the editor performs FFT display work outside the callback.

F.13 Immersive, Surround, and Atmos Workflows

Q85. Should every surround channel receive the same reverb?

Usually no. One coherent late field can be projected differently to each output, while early reflections preserve plausible direction. Identical copies collapse spatially; unrelated reverbs lose one-room coherence.

Q86. What should go to LFE?

LFE is an effects channel, not the automatic bass remainder of every bed channel. Keep reverb out of LFE unless the production deliberately requires it, and audition through the intended bass-management path before drawing conclusions.

Q87. How should height channels be used for reverb?

Height returns can carry overhead early energy, diffuse late energy, or a deliberate creative layer. They should support enclosure rather than pulling every source upward; verify binaural, speaker, and stereo fold-down presentations.

Q88. What is the difference between a bed and an object?

A bed has a fixed channel layout, while an object carries audio plus position metadata for a renderer. verbx can prepare reverberant assets for either path, but final object authoring and Dolby delivery remain in the authorized production environment.

Q89. How do I test channel order?

Render or play one labeled impulse per input channel and verify arrival, level, polarity, and label at every output. A file that reports twelve channels can still have a dangerous ordering error.

Q90. Why does a wide reverb become weak in mono?

Signed projections or decorrelation can create cancellations when channels sum. Measure and listen to fold-down during design, and adjust projection, low-frequency correlation, or wet level rather than treating mono as an afterthought.

Q91. Can a stereo IR be turned into convincing Atmos reverb automatically?

It can be distributed creatively, but it does not contain measured height or rear transfer paths. A convincing expansion requires explicit assumptions, decorrelation, energy management, and fold-down tests; it should not be described as the original room capture.

F.14 Research, Education, and Dataset Practice

Q92. Should augmentation happen before or after splitting a dataset?

Always split clean source groups first. Augmenting first can place the same performance, speaker, room, or generated-IR family in multiple splits and produce deceptively strong metrics.

Q93. Should validation augmentation change every epoch?

No. Freeze validation and test waveforms or at least their exact seeds and parameters. Changing them on every evaluation adds measurement noise and makes checkpoint comparisons ambiguous.

Q94. When is on-the-fly augmentation appropriate?

It is useful for training when storage is constrained and deterministic worker behavior has been engineered. Avoid subprocess creation per example, cache IRs per worker, and record enough seed state to reproduce any sample.

Q95. How should TensorFlow users ingest a verbx corpus?

Read the JSONL manifest, create `tf.data` tensors from explicit paths and labels, use deterministic mapping for evaluation, and preserve the manifest beside any TFRecord shards. Directory names should not become the only source of truth.

Q96. How should PyTorch users ingest a verbx corpus?

Use a manifest-backed `Dataset`, shard with a distributed sampler, and define padding or cropping explicitly. Keep validation order fixed and derive any training randomness from the source identity, epoch, worker, rank, and global seed.

Q97. What should an unseen-room test contain?

Hold out complete measured rooms or synthesized room families before augmentation. Report performance separately for clean, seen-room, unseen-room, and severity buckets so one aggregate does not hide the generalization failure.

Q98. How can students verify a pole-zero lesson by ear?

Render impulses and sustained chords while moving one pole radius or one delay relation at a time. Compare waveform decay, spectrum, and the unit-circle plot, then explain which audible change follows from angle, radius, or cancellation.

Q99. How should a homework render be submitted reproducibly?

Submit the source citation or permitted source file, exact command or config, verbx version, output audio, JSON report, and a short listening analysis. A screenshot alone is not enough to reproduce the result.

Q100. What is the single best habit for serious verbx work?

Treat every render as an evidence-producing transformation: preserve the source, specify the operation, inspect the plan, retain machine-readable results, and listen at matched level. That habit scales from one musical experiment to a distributed research corpus.

Q101. Can verbx use a Scala file for microtonal reverb?

Yes. Pass `--scala-file SCALE.scl` to `verbx ir gen`, set the reference with `--scala-root-hz` and `--scala-root-degree`, and control the effect with `--scala-strength`, `--scala-bandwidth-cents`, and `--scala-gain-db`. verbx expands the scale across the requested frequency range, tunes modal targets, and emphasizes those bands in every synthetic IR mode. Generate the IR once, then use it with `verbx render --engine conv --ir ...`, `verbx realtime --engine conv --ir ...`, a convolution plug-in, or a dataset pipeline. The realtime callback never parses the Scala file, so this feature adds no callback latency.

Q102. Does verbx perform full finite-element room simulation?

Not yet. The current `--electromechanical-solver modal-fe` path is a bounded structural modal solver for spring tanks and plates. It assembles mass and stiffness matrices, solves retained normal modes, and synthesizes a causal IR. It does not mesh a three-dimensional air volume or claim calibrated CAD-to-room prediction. See the Finite-Element Modeling chapter for acoustic FEM, boundary conditions, mesh convergence, modal reduction, hybrid room simulation, and the exact boundary of the current implementation.

Q103. Does Scala import retune incoming notes in realtime?

No. Scala import conditions a fixed synthetic IR. The generated decay favors the scale's resolved frequencies, but the dry signal is not pitch-corrected and the IR does not follow chord changes. Prepare multiple rooted IRs and crossfade wet returns when harmony changes, or use one moderate, broad tuning field as a stable spectral halo. The Microtonal Workflows chapter explains the musical tradeoffs and reproducible workflow.

"As if a double hunt were heard at once."

— William Shakespeare, *Titus Andronicus* (1594)

Public Alpha Notes (v0.9.9)

Updated: 2026-08-05

`verbx` v0.9.9 is a public alpha focused on robust offline rendering, reproducibility, advanced DSP control, structural cleanup, and initial realtime duplex auditioning.

G.1 Scope

The supported alpha surface is summarized below. Each item describes shipped behavior, not a promise of parity across every command, host, or spatial format.

- Dual-engine render path (`algo`, `conv`, `auto`) with deterministic control plane behavior.
- Track C/D automation and IR morph workflows.
- Immersive handoff/QC tooling and AI augmentation workflows.
- Initial realtime duplex monitoring (`verbx realtime`) with selectable input and output devices, using direct convolution or an algorithmic proxy IR.

G.2 Known Limitations

The limits below define where testing and support remain narrower than the complete design vocabulary described elsewhere in the book.

- Realtime support is intentionally narrow: `verbx realtime` is for live auditioning, not for full offline feature parity or plugin-host workflows.
- Very long tails are intentionally compute-heavy; render time depends strongly on hardware and block/partition settings.
- GPU acceleration is most effective for convolution-heavy workloads.
- FOA (`--ambi-order 1`) is the validated Ambisonics path for this alpha.
- Cross-platform floating-point and library differences can introduce tiny low-level output deltas for long renders; use reproducibility bundles when comparing runs.
- Realtime device support depends on the optional `sounddevice` backend and the host audio stack. Confirm device visibility with `verbx realtime --list-devices`.

G.3 Support Channels

Use the channel that matches the report so release discussion, security information, and ordinary defects do not become mixed together.

- Bugs/regressions: GitHub Issues
- Security reports: GitHub Security Advisory submission
- Release artifacts and announcements: GitHub Releases

G.4 What To Include In Bug Reports

A useful report should let another person reproduce the failure without guessing at the environment or processing state. Include the following evidence when it is available and lawful to share.

- Exact CLI command and full stderr/stdout text.
- `verbx doctor --json-out doctor.json` output.
- `--repro-bundle` output from the failing render.
- For realtime issues: selected `--input-device`, `--output-device`, sample rate, block size, and whether `--engine algo` or `--engine conv` was used.
- OS, Python version, and commit/tag.

“Lost Echo sits amid the voiceless mountains, and feeds her grief.”

— Percy Bysshe Shelley, *Adonais* (1821)

Glossary

This colossal glossary defines 709 terms used throughout verbx and the wider literature of acoustics, reverberation, dereverberation, spatial audio, recording, music production, plug-in engineering, measurement, and Audio AI. Definitions are intentionally compact: each establishes the book's working meaning without pretending to replace the cited standards, textbooks, or research papers.

A term may have narrower meanings in a particular standard or discipline. Read units, channel conventions, measurement windows, reference levels, and algorithm settings with the surrounding chapter before comparing results. Acronyms are cross-defined where they are common enough to be encountered independently.

A

A-weighting. A frequency weighting that approximates human sensitivity at moderate levels and is commonly used for environmental-noise measurements.

AB microphone technique. A spaced-pair recording method in which arrival-time and level differences create stereo width and room impression.

Absolute threshold of hearing. The lowest sound pressure level detectable under specified conditions, varying strongly with frequency and listener.

Absorber. A material or structure that converts incident acoustic energy into heat rather than returning it as reflection.

Absorption. The loss of sound energy when a boundary, object, medium, or treatment prevents a reflection of equal energy.

Absorption coefficient. The frequency-dependent fraction of incident acoustic energy absorbed by a surface, usually expressed from zero to one.

Acoustic camera. A microphone-array system that estimates and visualizes the direction or location of sound radiation.

Acoustic clarity. The perceptual separation of successive sounds, influenced by direct sound, early energy, late decay, and source articulation.

Acoustic impedance. The ratio of acoustic pressure to volume velocity or particle velocity at a specified point and frequency.

Acoustic labyrinth. A folded passage that lengthens an acoustic path for delay, damping, loudspeaker loading, or artificial reverberation.

Acoustic mode. A resonant pressure pattern determined by enclosure geometry, boundary conditions, and the speed of sound.

Acoustic power. The rate at which a source emits sound energy, measured in watts and independent of measurement distance in free field.

Acoustic shadow. A region receiving reduced direct sound because an obstacle blocks or diffracts propagation.

Acoustic treatment. Deliberate use of absorbers, diffusers, resonators, and construction to shape a room's response.

Acoustics. The science of sound generation, propagation, interaction, perception, measurement, and control.

Active acoustics. Electroacoustic systems that use microphones, processing, and loudspeakers to alter a venue's apparent reverberation or spatial response.

Active noise control. Reduction of unwanted sound through a controlled secondary field designed to interfere destructively with the disturbance.

Adaptive filter. A filter whose coefficients update from an error criterion, often used for echo cancellation, system identification, or dereverberation.

Additive synthesis. Construction of a signal by summing sinusoidal or otherwise elementary components with controlled amplitudes and phases.

Air absorption. Frequency-dependent attenuation during propagation through air, strongest at high frequencies and dependent on humidity and temperature.

Air-loss filter. A frequency-dependent attenuation filter placed in a propagation or feedback path to model energy lost while sound travels through air.

Algorithmic latency. Delay caused by an algorithm's required lookahead, block size, transform, buffering, or internal state rather than hardware transport.

Algorithmic reverb. Artificial reverberation produced by delay networks, filters, modulation, and feedback rather than direct convolution with a measured IR.

Aliasing. Misrepresentation of frequencies above the Nyquist limit as lower frequencies after sampling or nonlinear processing.

Allpass diffuser. An allpass filter used to increase echo density and phase dispersion while approximately preserving magnitude response.

Allpass filter. A filter with nominally flat magnitude response and frequency-dependent phase response.

Ambience. The audible environmental field surrounding a source, including room tone, reflections, diffuse energy, and distant activity.

Ambience extraction. Estimation of diffuse or spatially incoherent content from a recording for remixing, upmixing, or analysis.

Ambisonics. A scene-based spatial-audio representation using spherical-harmonic components rather than fixed loudspeaker feeds.

Amplitude. The instantaneous or measured magnitude of a signal, wave, coefficient, or oscillation.

Amplitude envelope. The time-varying outline of signal magnitude, commonly described by attack, decay, sustain, and release.

Amplitude modulation. Multiplication of a signal by a varying gain, producing level motion and potentially upper and lower sidebands.

Analog-to-digital converter. Hardware that samples an analog voltage and represents it as digital numbers at a selected rate and resolution.

Analyzer. A tool that estimates signal, room, spectral, loudness, spatial, or decay metrics from audio.

Angular frequency. Frequency measured in radians per second, written $\omega = 2\pi f$.

Anisotropy. Directional dependence in a material, field, room, radiation pattern, or spatial decay.

Anti-aliasing filter. A low-pass filter applied before sampling or rate reduction to suppress content above the new Nyquist frequency.

Antiphony. Alternation or exchange between separated performers or groups, often using architecture as a compositional spatial parameter.

Apparent source width. The perceived lateral extent of a sound source, strongly influenced by early lateral reflections and interaural decorrelation.

Archaeoacoustics. Interdisciplinary study of sound, listening, architecture, and human activity at archaeological sites using acoustic measurement, modeling, and cultural evidence.

Artifact. An unintended audible or measurable product of processing, encoding, estimation, clipping, modulation, or numerical error.

Artificial reverberation. Deliberate creation or modification of reflected and decaying sound by architectural, mechanical, electromechanical, electronic, or digital means.

Attack. The onset portion of a sound or the time constant with which a processor responds to increasing level.

Attenuation. Reduction in amplitude, power, or level caused by distance, absorption, filtering, gain control, or cancellation.

Audio buffer. A block of samples held temporarily for device transfer or processing, contributing directly to realtime latency.

Audio callback. A host- or driver-invoked routine that must process a block before the playback deadline without blocking.

Audio unit. Apple's plug-in architecture for audio effects, instruments, generators, and music-processing components.

Auditory masking. Reduced audibility of one sound because another sound overlaps it in time, frequency, or both.

Auditory scene analysis. Perceptual organization of mixed sound into sources, events, streams, and environments.

Autocorrelation. Similarity of a signal to delayed copies of itself, useful for detecting periodicity, repetition, and modal structure.

Automatic gain control. A system that varies gain to maintain a target level or operating range over time.

Automation. Recorded or generated time variation of parameters, routing, gain, states, or events.

Aux send. A mixer path that routes a controllable copy of a channel to a shared processor such as reverb.

Azimuth. Horizontal angle around a listener or coordinate origin, usually measured in degrees.

B

B-format. A conventional name for an Ambisonic component signal set before loudspeaker or binaural decoding.

Back wall. The room boundary behind the listener, often responsible for strong delayed reflections and low-frequency modal behavior.

Backward energy integration. Reverse cumulative summation of squared impulse-response samples used to form an energy decay curve from the response tail toward its onset.

Backward prediction. Estimation of a sample from future samples, used in some linear-prediction and dereverberation formulations.

Band-pass filter. A filter that passes a selected frequency region while attenuating frequencies below and above it.

Bandlimited signal. A signal whose spectrum is zero or negligible above a specified maximum frequency.

Bandwidth. The frequency span occupied or passed by a signal, filter, channel, mode, or process.

Bass ratio. Room-acoustic measure comparing low-frequency reverberation times with mid-frequency values to describe perceived warmth or bass persistence.

Batch processing. Noninteractive execution of the same or parameterized operation across many files or manifest entries.

Beamforming. Combining microphone or loudspeaker array elements with delays and weights to emphasize selected directions.

Beat frequency. The difference frequency produced when two nearby sinusoids interfere.

Bell filter. A parametric equalizer band that boosts or cuts around a center frequency with controlled width.

Binaural audio. Two-channel audio designed to reproduce ear-input cues for headphone listening.

Binaural quality index. Spatial metric derived from interaural cross-correlation and used to characterize apparent source width or listener envelopment.

Binaural room impulse response. An impulse response measured or modeled from a source to the listener's two ears in a room.

Bit depth. The number or format of bits used to represent sample amplitude, affecting quantization and dynamic range.

Block convolution. Convolution performed on signal blocks, commonly accelerated with FFT methods.

Block size. The number of samples processed or transferred at once, trading scheduling margin and efficiency against latency.

Boundary. A surface or interface at which an acoustic wave reflects, transmits, absorbs, scatters, or changes medium.

Boundary element method. A numerical wave-simulation method that discretizes room or object surfaces rather than the full volume.

Brightness. Perceived emphasis of high-frequency energy, influenced by spectrum, transients, decay, and level.

Brilliance ratio. Room-acoustic ratio comparing high-frequency reverberation with mid-frequency reverberation to characterize treble liveliness or damping.

Brown noise. Noise with power decreasing approximately 6 dB per octave, produced by integrating white noise.

Brute-force convolution. Direct application of a long impulse response without structural approximation, often accurate but computationally expensive for realtime multichannel reverberation.

Buffer underrun. Failure to supply playback samples before the device deadline, typically heard as a click, dropout, or gap.

Bus. A mixer signal path that combines, distributes, or processes multiple channels.

Butterworth filter. A filter family designed for maximally flat passband magnitude response.

C

C-weighting. A relatively broad frequency weighting used for high-level sound measurements and peak-oriented assessment.

Cadence. A musical arrival or punctuation whose perceived duration can be extended or clarified by reverberation.

Calibration. Establishing a known relationship between digital values, electrical signals, acoustic level, distance, or measurement units.

Cardioid pattern. A first-order directional pattern with maximum sensitivity forward and a nominal null to the rear.

Causality. The requirement that a realtime system's output at a given instant cannot depend on future input.

Center time. Energy-weighted mean arrival time of an impulse response, commonly written T_s, that summarizes the balance between early and late energy.

Cepstrum. A transform-domain representation useful for separating source and filter periodicities, detecting echoes, or estimating pitch.

Channel. One discrete stream or component of audio within a file, bus, device, spatial format, or processor.

Channel decorrelation. Reduction of similarity between output channels so a reverberant field spreads spatially instead of collapsing toward a phantom source.

Channel layout. The ordered set of channel roles, labels, and positions associated with a multichannel signal.

Chirp. A signal whose instantaneous frequency changes over time, often used for system identification and impulse-response measurement.

Chirped echo. An echo whose dominant frequency changes over time because geometry, diffraction, dispersion, or a sequence of reflections reorganizes an impulsive source.

Chorus. A modulation effect that combines a signal with slowly varying delayed copies to create width and ensemble-like motion.

Chorus artifact. Audible pitch wandering or ensemble-like motion caused when delay modulation in a reverberator becomes too deep, fast, or coherent.

Circulant feedback matrix. FDN matrix whose rows are cyclic shifts of one another, enabling structured eigenanalysis and efficient implementations.

Clarity index. A room metric comparing early to late energy, such as C_{50} for speech or C_{80} for music.

Clipping. Nonlinear truncation when a signal exceeds the representable or permitted range.

Close microphone. A microphone placed near a source to emphasize direct sound and reduce room contribution.

Cloud reverb. A dense, diffuse texture built from many delayed, filtered, modulated, or granular components.

Cochlear filterbank. A model of frequency analysis in the inner ear using overlapping auditory bands.

Codec. An encoder-decoder system that represents audio for storage or transmission, often with compression.

Coefficient. A numerical multiplier controlling a filter, matrix, predictor, transform, or model term.

Coloration-duration factorization. Reverberator design strategy that separates control of spectral character from control of decay time as nearly independently as possible.

Colorless reverberation. Artificial reverberation designed to avoid conspicuous periodic resonances, metallic ringing, and comb-filter spectral coloration.

Comb filter. A feedforward or feedback delay filter with regularly spaced spectral peaks and notches.

Comb-filter coloration. Audible pitch or hollowness caused by interference between a signal and one or more delayed copies.

Common delay divisor. Integer factor shared by several delay lengths that can align recurrences and reduce the effective modal richness of a delay network.

Common-mode signal. A component shared by multiple channels, often contrasted with differential or side information.

Complex spectrum. A frequency-domain representation containing both magnitude and phase information.

Compression. Dynamic-range reduction in which gain decreases as signal level rises above a defined region.

Compressor knee. The transition shape between uncompressed and compressed gain behavior around threshold.

Computational room model. Numerical representation of acoustic propagation and boundary interaction used to estimate impulse responses or sound fields in an enclosure.

Concert hall. A performance room designed to balance clarity, blend, envelopment, loudness, and musical decay.

Condenser microphone. A microphone using a charged capacitive element and impedance-conversion electronics.

Conformal damping map. Interpretation of frequency-dependent delay-line loss as a mapping that moves lossless poles from the unit circle to frequency-dependent radii inside it.

Contractive matrix. Matrix whose induced gain does not exceed one, useful for ensuring that a feedback network does not increase signal energy.

Convolution. The operation that applies a linear time-invariant system's impulse response to an input signal.

Convolution matrix. A set of impulse responses mapping multiple input channels to multiple output channels.

Convolution reverb. Reverberation created by convolving audio with measured, synthesized, or designed impulse responses.

Coprime delay lengths. Delay lengths whose greatest common divisor is one, chosen to reduce coincident recurrences and improve modal distribution.

Correlation. Statistical similarity between signals, channels, time regions, or variables.

Correlation coefficient. A normalized measure of linear relationship, typically ranging from minus one to one.

Courant stability condition. Time-step and grid-spacing constraint that keeps an explicit finite-difference wave simulation numerically stable.

CPU load. The fraction of available processor time consumed by computation, scheduling, and data movement.

Crest factor. The difference in decibels between a signal's peak level and RMS level.

Critical band. A frequency region within which auditory interactions such as masking are especially strong.

Critical distance. The source distance at which direct and reverberant sound energies are approximately equal.

Cross-correlation. Similarity between two signals as a function of relative delay.

Crossfade. A smooth transition created by decreasing one signal while increasing another.

Crossover. A set of complementary filters that divides audio into frequency bands for separate processing.

Crosstalk. Unwanted leakage or coupling between channels, paths, transducers, or spatial components.

Cumulative spectral decay. A three-dimensional display of magnitude versus frequency and time, useful for observing resonant ringing.

Cutoff frequency. The reference frequency at which a filter transitions between passband and stopband behavior.

D

Damping. Frequency-dependent dissipation that reduces resonance amplitude or shortens decay.

Damping filter. Filter inside a reverberant feedback path that imposes frequency-dependent loss and therefore frequency-dependent decay time.

Damping ratio. A dimensionless measure relating decay to oscillation in a second-order resonant system.

Damping substitution. Replacement of each unit delay by a delay combined with a propagation-loss filter to convert a lossless reverberator into a decaying one.

DAW. Digital audio workstation software used to record, edit, route, process, automate, and mix audio.

dBFS. Decibels relative to digital full scale, where zero denotes the maximum nominal digital peak.

Decay. Reduction of sound energy or amplitude after excitation stops.

Decay curvature. Departure of a decay trace from a straight line in decibels, indicating changing decay rate, coupled spaces, modes, noise, or time variance.

Decay curve. Level or energy plotted against time to characterize reverberation, damping, or release behavior.

Decay eigenvalue. Eigenvalue of a recursive network whose magnitude and angle determine a modal decay rate and oscillation frequency.

Decay intercept. Level at which a fitted decay line crosses a chosen time origin, used with slope when estimating reverberation time.

Decay ridge. Persistent narrowband feature in an energy decay relief that reveals a mode or resonance lasting longer than neighboring frequencies.

Decay slope. Rate of level reduction over time on a decibel decay curve, usually expressed in decibels per second.

Decay time. The duration required for a specified drop in level or energy under defined measurement conditions.

Decibel. A logarithmic unit expressing a ratio of powers or amplitudes with an appropriate reference.

Decimation. Sample-rate reduction after low-pass filtering to prevent aliasing.

Decoder. A system that converts an encoded, scene-based, object-based, or compressed representation into playback channels or samples.

Deconvolution. Estimation or recovery of an impulse response or source by mathematically undoing convolution under stated assumptions.

Decorrelation. Reduction of similarity between channels or signal paths, often used to increase spaciousness and diffusion.

Definition index. Early-to-total energy ratio, commonly D_{50} for speech, expressing how much impulse-response energy arrives during the first 50 milliseconds.

Delay. A time shift measured in samples, milliseconds, seconds, or musical duration.

Delay density. Number and distribution of distinct delay events available to a reverberator over a specified interval.

Delay distribution. Statistical or designed arrangement of delay lengths that controls mode spacing, recurrence, density buildup, and temporal texture.

Delay line. Memory that stores samples so they can be read after a controlled time interval.

Delay modulation. Time variation of delay length, producing pitch change, phase movement, or decorrelation.

Delay network. An interconnected system of delays, filters, gains, and matrices used for reverberation or physical modeling.

Delay-line damping filter. Loss filter associated with a particular reverberator delay line and designed from a target frequency-dependent reverberation time.

Delay-line scaling. Multiplication of nominal delay lengths by room-size, sample-rate, or tuning factors while preserving required ordering and stability constraints.

Delta function. An idealized unit impulse with unit area and zero duration, used to define system response.

Density. The number or perceptual concentration of reflections, grains, echoes, or events per unit time.

Density buildup. Increase in the number of audible reflections per unit time as energy recirculates through a room or delay network.

Dereverberation. Estimation or recovery of a less reverberant signal from audio containing room reflections and decay.

Determinism. Property that identical inputs, parameters, seeds, and environment produce identical outputs.

Diffuse field. A statistical sound field with energy arriving approximately uniformly from many directions.

Diffuse reflection. Boundary interaction that redistributes incident sound across many outgoing directions rather than preserving a single mirror-like ray.

Diffuse-field assumption. Approximation that reverberant energy is statistically uniform in position and direction, underlying many classical room-acoustic formulas.

Diffuse-field equalization. Spectral correction referenced to the average response of sound arriving from many directions.

Diffuser. A surface or processor that redistributes reflected energy across directions or time.

Diffusion. The spreading of acoustic or signal energy across directions, delays, channels, or modes.

Diffusion coefficient. A measure of how evenly a surface distributes reflected energy over direction relative to a reference.

Diffusion time. Time required for a reverberator to develop a sufficiently dense, spatially distributed response after excitation.

Digital signal processing. Numerical representation, analysis, transformation, generation, and control of sampled signals.

Digital-to-analog converter. Hardware that converts digital samples into a continuous electrical signal for playback.

Direct sound. Energy traveling from source to receiver without an intervening reflection.

Direct-to-reverberant ratio. The energy or level ratio between direct sound and reverberant sound at a receiver.

Directional microphone. A microphone whose sensitivity varies with arrival angle.

Dispersion. Frequency-dependent propagation speed or spreading that causes components to arrive at different times.

Distance attenuation. Reduction in direct level as source-receiver distance increases, modified by geometry and environment.

Distance law. Relationship between propagation distance and sound level, such as inverse-distance pressure decay in an ideal free field.

Dither. Low-level noise added before quantization to decorrelate error and preserve small-signal linearity.

Dolby Atmos. An immersive-audio ecosystem combining beds, objects, metadata, rendering, and defined delivery formats.

Doppler shift. Apparent frequency change caused by relative motion between source and receiver.

Doppler-free modulation. Delay or network modulation designed to reduce audible pitch shift while still breaking up static resonances.

Double-slope decay. Energy decay containing two approximately linear decibel regions with different slopes, often caused by coupled spaces or layered processing.

Downmix. Reduction of a multichannel mix to fewer output channels using specified gains and routing.

Downsampling. Conversion to a lower sample rate, normally including anti-alias filtering.

DRR. Standard abbreviation for direct-to-reverberant ratio in room-acoustic measurement, analysis, and spatial rendering.

Dry signal. The source signal before the reverberation or other parallel effect under discussion.

Dual-mono. Two channels processed independently without stereo linking or cross-channel interaction.

Duck. To reduce one signal automatically when a key signal is active.

Ducking reverb. Reverb whose return level is reduced during source activity and allowed to bloom in gaps.

Dynamic convolution. Convolution whose response changes with level, time, state, or an interpolated set of impulse responses.

Dynamic range. The span between the lowest useful signal and the highest undistorted or permitted level.

E

Early decay time. A reverberation estimate derived from the first 10 dB of decay and extrapolated to a 60 dB rate.

Early reflections. The first individually structured room reflections arriving after the direct sound.

Early resonance. Low-frequency room or reverberator mode that remains individually perceptible before modal density becomes statistically high.

Early-to-late ratio. Ratio of impulse-response energy before a selected boundary to energy after it, used to quantify clarity and distance.

Echo. A delayed sound perceived as a distinguishable repetition rather than fused reverberation.

Echo cancellation. Estimation and subtraction of a known echo path, especially in communication systems.

Echo density. The number of significant reflections or delayed events per unit time.

Echo return loss enhancement. Additional echo reduction achieved by nonlinear post-processing after linear cancellation.

Echo-density growth. Time evolution of reflection count or normalized echo density as a reverberant response transitions from sparse echoes to a dense tail.

Eigenfrequency. A natural resonant frequency associated with a system mode.

Eigenmode. Characteristic oscillation associated with an eigenvector and eigenvalue of a linear acoustic or feedback system.

Eigenvalue. A scalar describing how a matrix transforms a corresponding eigenvector, central to feedback-network stability and decay.

Eigenvector. A direction in a vector space whose orientation is preserved by a matrix transformation.

Elevation. Vertical angle of a source or loudspeaker relative to the listener or coordinate origin.

Elliptic feedback delay network. Structured FDN family using elliptic or related matrix constructions to shape mode distribution and computational behavior.

Energy decay curve. Integrated impulse-response energy plotted backward in time, commonly used to estimate reverberation time.

Energy decay relief. Time-frequency surface formed by backward-integrating energy within spectral bins, revealing frequency-dependent decay and resonant ridges.

Energy preservation. Property of a lossless structure whose output or state energy equals its input or previous-state energy under the chosen norm.

Energy reflection coefficient. Fraction of incident acoustic energy returned by a boundary, equal to one minus absorbed and transmitted fractions when accounting is complete.

Energy-time curve. Squared impulse-response magnitude plotted against time to reveal direct sound and reflection structure.

Ensemble effect. Perceived multiplication or widening produced by small timing, pitch, and level variations among similar signals.

Envelopment. Perception of being surrounded by sound, strongly associated with late lateral and diffuse energy.

Equal-loudness contour. A curve showing frequency-dependent sound pressure levels judged equally loud.

Equal-power crossfade. A crossfade law intended to maintain approximately constant power for uncorrelated signals.

Equalization. Frequency-selective gain adjustment used to correct, balance, color, or prepare audio.

Equivalent continuous level. The steady level containing the same energy as a varying sound over a stated interval.

ERB. Equivalent rectangular bandwidth, a psychoacoustic measure approximating auditory-filter width.

Exact transfer-function model. Point-to-point room model that represents every relevant source-receiver impulse response directly rather than approximating its perceptual structure.

Excitation signal. A known signal used to stimulate a system for measurement, modeling, or synthesis.

Exponential decay. A decay whose amplitude changes by a constant ratio per unit time and appears linear in decibels.

External sidechain. A detector input derived from a signal other than the processor's main input.

Extrapolated RT60. Sixty-decibel decay time inferred from a shorter fitted interval such as EDT, T20, or T30 rather than observed over the full range.

F

Fabry-Pérot resonance. Resonance created by repeated reflections between approximately parallel boundaries.

Fade-in. A gradual increase from silence or a lower level to the intended signal level.

Fade-out. A gradual reduction from the current level toward silence or a lower level.

Far field. A region sufficiently distant from a source that radiation pattern and inverse-distance behavior are approximately established.

Fast Fourier transform. An efficient algorithm for computing the discrete Fourier transform and its inverse.

FDN reverberation. Late-reverberation synthesis using multiple delays coupled through a feedback matrix, with losses and output mixing controlling decay and spatial character.

Feedback. Routing a portion of a system's output back to its input.

Feedback comb bank. Parallel or coupled collection of feedback comb filters used to create many decaying modes and a dense reverberant tail.

Feedback delay network. A reverberator architecture containing multiple delays coupled by a feedback matrix and associated filters.

Feedback gain. The multiplier applied in a feedback path, controlling decay rate and stability.

Feedback matrix. The matrix that mixes delay outputs back into delay inputs in an FDN.

FFT bin. One discrete frequency sample in a discrete Fourier transform.

FFT convolution. Efficient convolution using multiplication of spectra from transformed signal blocks and impulse-response partitions.

Filter. A system that changes a signal according to frequency, time, phase, level, or statistical structure.

Filterbank. A collection of filters that decomposes or reconstructs a signal across frequency bands.

Finite impulse response filter. A filter with a response that becomes exactly zero after a finite number of samples.

Finite-difference time-domain method. Grid-based numerical method that advances discretized wave-equation variables through time while respecting stability and boundary conditions.

First-order delay filter. One-pole or one-zero loss filter designed for a reverberator delay line to approximate a target decay at selected frequencies.

Flanging. A modulation effect created by mixing a signal with a very short varying delay, producing moving comb filtering.

Flat response. A response whose magnitude remains approximately constant over a specified frequency range.

Flutter echo. Rapid, repetitive reflections between surfaces, heard as buzzing or pitched decay.

FOA. First-order Ambisonics, containing four spherical-harmonic components in a complete three-dimensional representation.

Folded delay line. A long acoustic or electronic path arranged compactly by folding its geometry or memory access.

Formant. A spectral resonance that helps define vowel identity, instrument character, or resonant coloration.

Forward masking. Reduced audibility of a sound because a preceding sound temporarily raises its perceptual threshold.

Fractional delay. A delay containing a noninteger number of samples, implemented by interpolation or phase methods.

Frame. A finite group of samples analyzed or processed together, often with overlap and a window.

Free field. An ideal or approximated environment without significant reflections.

Freeverb. Widely implemented Schroeder-Moorer-style reverberator using parallel lowpass-feedback comb filters followed by serial allpass-like diffusers.

Freeverb allpass approximation. Freeverb diffusion section conventionally called allpass although its implementation and coefficient choices only approximate ideal allpass behavior.

Freeze. A reverb state that sustains circulating energy by approaching lossless feedback while controlling stability.

Frequency. Repetition rate measured in hertz, equal to cycles per second.

Frequency modulation. Variation of instantaneous frequency by another signal, producing pitch motion and sidebands.

Frequency response. Magnitude and phase behavior of a system as a function of frequency.

Frequency-dependent decay. Reverberation in which different frequency bands decay at different rates.

Frequency-dependent energy decay curve. Family of decay curves computed in bands or spectral bins instead of after broadband energy summation.

Frequency-dependent reverberation time. Reverberation-time function over frequency, commonly specified in octave or fractional-octave bands and realized with feedback-path damping filters.

Frequency-domain processing. Signal processing performed on spectral representations rather than directly on time samples.

Full scale. The maximum nominal magnitude representable by a digital format or signal convention.

G

Gain. A multiplier or logarithmic level change applied to a signal.

Gain computer. The part of a dynamics processor that calculates desired gain from detector level and control parameters.

Gain reduction. The amount by which a dynamics processor attenuates a signal relative to unity gain.

Gated reverb. Reverberation shaped by a gate or envelope so the decay terminates sooner or more abruptly than naturally.

Gated reverse reverb. A reverse or reverse-style wet field restricted to a controlled pre-event or transition window and closed with a gate or authored gain envelope.

Gaussian noise. Random noise whose sample amplitudes follow a normal probability distribution.

Geometric acoustics. High-frequency room modeling using rays, image sources, or beams rather than full wave equations.

Geometric spreading. Reduction of wave intensity as energy expands over a growing area.

Grain. A short windowed segment of sound used as an elementary unit in granular processing.

Granular reverb. Reverberant texture assembled from overlapping short grains with controlled delay, pitch, envelope, and spatial distribution.

Grid dispersion. Frequency- and direction-dependent wave-speed error introduced by spatial discretization in finite-difference or waveguide-mesh simulation.

Grid point. Discrete spatial sample at which a numerical acoustic model stores and updates pressure, velocity, or traveling-wave variables.

Group delay. Negative derivative of phase with respect to angular frequency, interpreted as frequency-dependent envelope delay.

GUI. Graphical user interface containing visual controls, meters, displays, navigation, and interaction feedback.

H

Haas effect. Perceptual localization dominated by the first arrival when a similar delayed sound follows within a short interval.

Hadamard matrix. An orthogonal matrix of plus and minus entries often used for efficient energy-preserving diffusion in FDNs.

Hann window. A raised-cosine analysis window with zero-valued endpoints and useful spectral-leakage behavior.

Harmonic distortion. Added spectral components at integer multiples of input frequencies due to nonlinearity.

Head tracking. Measurement of listener orientation or position used to update binaural or immersive rendering.

Head-related impulse response. The time-domain response from a spatial source to an ear, including head, torso, and pinna effects.

Head-related transfer function. The frequency-domain counterpart of a head-related impulse response.

Headroom. Level margin between normal operating peaks and clipping or another hard limit.

Helmholtz resonator. A cavity-and-neck system that resonates over a relatively narrow low-frequency region.

High-pass filter. A filter that attenuates low frequencies while passing higher frequencies.

Higher-order Ambisonics. Ambisonics using spherical-harmonic orders above first order for greater spatial resolution.

Hilbert transform. A phase-shifting transform used to form analytic signals, estimate envelopes, or construct quadrature components.

Histogram. A count or probability display showing how values are distributed across ranges.

Homogeneous feedback delay network. FDN whose delay paths or attenuation structure follow a uniform design pattern rather than path-specific heterogeneous models.

Hop size. The sample advance between successive overlapping analysis or synthesis frames.

Householder matrix. An orthogonal reflection matrix useful for efficient mixing and diffusion in feedback networks.

HRTF personalization. Adapting head-related transfer functions to an individual listener's anatomy or measured responses.

Hybrid reverb. A reverberator combining distinct methods, commonly convolutional early reflections with an algorithmic late field.

I

IACC. Interaural cross-correlation coefficient, a room and spatial metric related to similarity between ear signals.

IEC. International Electrotechnical Commission, a standards body responsible for many audio and electroacoustic measurement standards.

Image-source method. A geometric room model that replaces specular reflection paths with mirrored virtual sources.

Image-source order. Number of boundary reflections represented by a virtual source in the image-source method.

Immersive audio. Audio designed to create enveloping three-dimensional experience using channels, objects, scenes, or binaural rendering.

Impulse. A brief excitation approximating a unit sample or Dirac delta for testing system response.

Impulse response. The output of a system excited by an impulse, fully describing a linear time-invariant system.

Impulse-response smoothness. Degree to which a late impulse response avoids isolated spikes, periodic gaps, and abrupt statistical changes.

In situ measurement. Measurement performed in the actual operating environment rather than a laboratory approximation.

Infinite impulse response filter. A recursive filter whose theoretical impulse response continues indefinitely.

Infinite reverb. A sustained reverb mode in which decay is halted or made extremely long under explicit stability controls.

Inhomogeneous feedback delay network. FDN with path-dependent delays, filters, gains, or routing intended to model nonuniform propagation and decay.

Initial time-delay gap. The interval between direct sound and the first significant reflection.

Input gain. Gain applied before the main processing network.

Input matrix. A matrix that maps external channels into an internal multichannel or delay-network representation.

Input-output room model. Representation of a room by transfer functions or impulse responses between chosen source and receiver points without explicitly simulating the full field.

Integrated loudness. Program loudness accumulated over an interval using gating and frequency weighting defined by a standard.

Interaural cross-correlation coefficient. Normalized similarity of left- and right-ear signals over a specified window and frequency band, used in spatial-acoustic assessment.

Interaural level difference. Difference in level between the ears, important for horizontal localization at higher frequencies.

Interaural time difference. Difference in arrival time between the ears, important for horizontal localization at lower frequencies.

Interchannel coherence. Frequency-dependent similarity between channels, often used to characterize diffuseness or spatial image stability.

Interleaved audio. Multichannel sample storage in which channel samples alternate within each frame.

Interpolation. Estimation of values between known samples, parameters, measurements, or responses.

Inverse filter. A filter designed to undo or compensate for another system's response within practical limits.

Inverse square law. Free-field intensity decreases with the square of distance from a point source, corresponding to about 6 dB per distance doubling.

IR. Common abbreviation for impulse response in acoustics, convolution, measurement, and digital signal processing.

ISO 3382. A family of standards defining room-acoustic measurement procedures and parameters.

Isotropic field. A field whose statistical properties are the same in every direction.

J

Jackknife estimate. A resampling method that estimates bias or variance by systematically omitting observations.

Jitter. Unwanted timing variation in sampling, clocks, events, callbacks, or packet arrival.

JSON report. A machine-readable structured report using JavaScript Object Notation for metrics, settings, diagnostics, or provenance.

Junction scattering. Redistribution of incoming traveling waves into outgoing branches according to impedance and conservation constraints at a waveguide junction.

Just intonation. Tuning based on simple frequency ratios rather than equal division of the octave.

Just-noticeable difference. The smallest change in a parameter or stimulus reliably perceived under defined conditions.

K

K-metering. A monitoring and metering system relating average level, peak headroom, and calibrated playback.

Kaiser window. A parameterized analysis window whose beta value trades main-lobe width against sidelobe suppression.

Key signal. The detector signal controlling a sidechain-dependent processor such as a ducker, gate, or compressor.

Knee. The region around a dynamics threshold where the input-output curve changes slope.

Kurtosis. A statistical measure of distribution tail weight or impulsiveness, sometimes used in audio-event analysis.

L

Lagrange interpolation. Polynomial interpolation commonly used for fractional-delay filters with controlled time-domain behavior.

Late field. The dense reverberant region after early reflections are no longer perceived as isolated events.

Late lateral energy fraction. The fraction of late sound energy arriving laterally, associated with listener envelopment.

Late-field isotropy. Condition in which late reverberant energy arrives with approximately equal statistics from all directions.

Late-reverberation approximation. Efficient statistical or recursive model that replaces explicit computation of every high-order reflection after the early response.

Late-tail onset. Time at which a reverberant response becomes dense enough to treat as a statistical late field rather than isolated reflections.

Latency. Elapsed time from an input event or sample to the corresponding output becoming available or audible.

Latency compensation. Host or system alignment that delays other paths to match a processor's reported latency.

Lateral energy fraction. Ratio of early lateral energy to total early energy, used in concert-hall assessment.

Leakage. Unwanted energy escaping between signals, channels, bands, frames, rooms, or spectral bins.

Least squares. Optimization that minimizes the sum of squared errors between observations and a model.

Level. A logarithmic or linear measure of signal magnitude, power, loudness proxy, or acoustic pressure.

Limiter. A dynamics processor intended to prevent peaks from exceeding a ceiling or controlled range.

Limiter ceiling. The maximum target output level enforced by a limiter, usually stated in dBFS or dBTP.

Linear phase. A phase response proportional to frequency, corresponding to constant group delay.

Linear prediction. Modeling a sample as a weighted combination of other samples, widely used in coding, analysis, and dereverberation.

Linear time-invariant system. A system satisfying superposition whose behavior does not change over time.

Linkwitz-Riley crossover. A crossover family formed from cascaded Butterworth sections so adjacent bands sum flat in phase-aligned conditions.

Listener envelopment. Perception of being surrounded by reverberant sound, strongly influenced by late lateral energy and interaural decorrelation.

Listening position. The specified receiver location from which spatial balance, calibration, or room response is evaluated.

Localization. Perceptual estimation of a sound source's direction, distance, or position.

Lookahead. Intentional delay that lets a processor inspect upcoming samples before applying gain or another decision.

Loop gain. Product of gains around a feedback path, determining stability and decay.

Lossless feedback matrix. Feedback matrix that preserves state energy, typically orthogonal in real-valued FDNs or unitary in complex-valued FDNs.

Lossless matrix. A matrix that preserves signal energy, commonly orthogonal or unitary in an FDN.

Lossless prototype reverberator. Recursive network configured without attenuation so its modes do not decay, used as a starting point before damping is introduced.

Low-frequency extension. The lowest frequency region a transducer or system reproduces within stated limits.

Low-pass filter. A filter that passes low frequencies while attenuating higher frequencies.

Lowpass-feedback comb filter. Feedback comb filter containing a low-pass loss filter in its loop so high frequencies decay faster than low frequencies.

LU. Loudness unit, representing a relative loudness difference of one decibel under the applicable standard.

LUFS. Loudness units relative to full scale, used for standardized program-loudness measurement.

M

Machine learning. Methods that infer mappings, representations, decisions, or generative behavior from data rather than only explicit rules.

Magnitude response. Frequency-dependent amplitude gain of a system, excluding phase information.

Manifest. A structured list describing batch inputs, outputs, parameters, labels, seeds, and provenance.

Masking threshold. Level below which a sound becomes inaudible because of competing spectral or temporal energy.

Matrix. A rectangular numerical array representing channel routing, mixing, transforms, or linear systems.

Matrix convolution. Multichannel convolution in which each input-output pair can have its own impulse response.

Matrix modulation. Time variation of a feedback or mixing matrix used to move resonances, alter diffusion, or decorrelate channels.

Maximum length sequence. A deterministic pseudorandom binary sequence used for system identification and impulse-response measurement.

Mean absorption coefficient. Surface-area-weighted average of frequency-dependent absorption coefficients across the boundaries of a room.

Mean free path. Average distance traveled by a sound ray between reflections in an enclosure.

Mean scattering coefficient. Surface-area-weighted average of boundary scattering coefficients used in geometric-acoustic estimates.

Measurement microphone. A microphone designed for known, stable, and often approximately flat response.

Median. The middle value of an ordered dataset, robust against isolated extreme values.

Mel scale. A perceptual frequency scale designed to approximate equal pitch-distance judgments.

Metadata. Descriptive information about audio, channels, objects, parameters, rights, measurements, or processing history.

Metallic ringing. Audible pitched or bell-like persistence caused by sparse, regularly spaced, or insufficiently damped reverberator modes.

Metering. Visual or machine-readable measurement of levels, loudness, peaks, dynamics, spectra, or processing state.

Mid-side processing. Stereo processing in the sum-and-difference domain rather than directly on left and right channels.

Minimum phase. A causal stable response whose energy is concentrated as early as possible for a given magnitude response.

Mix. The combination of multiple sources, channels, buses, and processing paths into a program.

Modal decay time. Time constant or RT60 associated with one resonant mode rather than a broadband or band-averaged response.

Modal degeneracy. Coincidence of two or more theoretically distinct modes at the same frequency, often increasing resonance strength.

Modal density. Number of acoustic modes per unit frequency, generally increasing with frequency and room volume.

Modal frequency. Resonant frequency associated with a room or system mode.

Modal overlap. Degree to which neighboring resonances overlap in bandwidth, contributing to a smoother statistical response.

Mode-density threshold. Frequency or time region beyond which modes or echoes are numerous enough that statistical description becomes more useful than individual tracking.

Mode-frequency distribution. Arrangement of resonant frequencies in a room or recursive network, including spacing, degeneracy, clustering, and irregularity.

Modulation. Time variation of amplitude, frequency, phase, delay, filter, matrix, or another parameter.

Modulation depth. Maximum excursion of a modulated parameter around its center value.

Modulation rate. Frequency or speed at which a modulation source evolves.

Mono compatibility. Degree to which a stereo or multichannel signal retains intended balance and timbre when combined to mono.

Monophonic. Containing one audio channel or one musical line, depending on context.

Moorer reverberator. Artificial-reverberation architecture extending Schroeder structures with explicit early reflections and frequency-dependent damping in feedback comb filters.

Moving average. A filter or statistic formed by averaging values over a sliding interval.

Multiband delay filter. Feedback-path filter designed from several target decay bands to approximate a detailed frequency-dependent RT60 curve.

Multiband processing. Independent or linked processing of frequency regions separated by a filterbank or crossover.

Multichannel audio. Audio containing more than two coordinated channels.

Multirate processing. Signal processing that uses more than one sample rate within a system.

Multitap delay. A delay line read at multiple times, with independent gains, filters, or positions for each tap.

Music Information Retrieval. Computational analysis of musical audio, scores, metadata, structure, similarity, or semantics.

Musical acoustics. Study of sound production, propagation, perception, and design in musical instruments, voices, rooms, and practices.

Musical time. Organization of duration, pulse, meter, phrase, and form, which often governs useful delay and decay choices.

Mute. Complete or near-complete attenuation of a signal path.

N

N-channel. Describing a signal or system with an arbitrary stated number N of channels.

Natural reverb. Reverberation generated by a physical acoustic environment rather than an electronic processor.

Near field. Region close to a source where pressure and particle velocity behavior is complex and simple far-field laws may not apply.

Nested allpass filter. Diffusion structure in which an allpass or delay network is embedded inside another allpass loop to increase echo complexity efficiently.

Neural audio codec. A learned encoder-decoder that represents audio with compact latent variables and neural synthesis.

Neural dereverberation. Dereverberation performed partly or wholly by a learned model trained on reverberant and target audio.

Neural impulse response. An impulse response represented, generated, estimated, or conditioned by a neural model.

Neural network. A parameterized layered function trained to model relationships from examples or objectives.

Noise floor. Residual background level below which measurements or low-level signal details become unreliable.

Noise gate. A dynamics processor that attenuates signals below a threshold according to timing and range controls.

Noise shaping. Spectral redistribution of quantization noise, usually moving energy toward less audible regions.

Noise-floor bend. Point where a measured decay curve departs from its reverberant slope because background noise begins to dominate backward-integrated energy.

Nonexponential decay. Reverberant decay that cannot be represented adequately by one constant exponential slope over the interval of interest.

Nonlinear distortion. New frequency components or waveform changes produced by a system that does not obey superposition.

Nonstationarity. Time variation in a signal's statistical properties, spectrum, level, source, or environment.

Normalization. Gain adjustment or numerical scaling to meet a peak, loudness, energy, or representational target.

Normalized echo density. Echo-density measure normalized against a Gaussian or reference process so responses with different levels and durations can be compared.

Normalized frequency. Frequency expressed relative to sample rate, Nyquist frequency, or another reference.

Notch filter. A narrow stop filter used to attenuate a specific frequency or resonance.

Nyquist frequency. Half the sample rate, the highest sinusoidal frequency uniquely representable under ideal sampling assumptions.

O

Object-based audio. Spatial audio represented as audio objects plus time-varying metadata interpreted by a renderer.

Octave. A frequency interval with a ratio of two to one.

Octave band. A standardized or practical band whose upper edge is twice its lower edge.

Offline rendering. Processing completed without realtime deadlines, allowing arbitrary lookahead, block sizes, and computation time.

Omnidirectional microphone. A microphone intended to have approximately equal sensitivity in every direction.

Onset. The beginning of an audible event, often detected from rapid changes in energy or spectrum.

Open Sound Control. A network-oriented protocol for exchanging structured control messages among musical and media systems.

Orthogonal matrix. A real square matrix whose transpose equals its inverse and that preserves Euclidean energy.

Orthogonal mixing. Energy-preserving mixing of real-valued channels or delay states using an orthogonal matrix.

Orthogonalized delay filter. Delay-line damping design transformed to preserve a desired lossless or orthogonal network relationship while imposing frequency-dependent decay.

Oscillation. Repeated variation around an equilibrium, periodic or quasi-periodic in time.

Output decorrelation delay. Short channel-specific delay used at reverberator outputs to reduce interchannel correlation and broaden spatial imaging.

Output gain. Gain applied after the principal processing network.

Output matrix. A matrix that maps internal channels or delay states to external output channels.

Overlap-add. Block-processing method that reconstructs linear convolution by adding overlapping transformed output blocks.

Overlap-save. Block-convolution method that discards corrupted circular-convolution samples and retains valid output regions.

Overload. Operation beyond a system's level, computational, thermal, bandwidth, or scheduling capacity.

Oversampling. Processing at a multiple of the base sample rate to reduce aliasing or improve numerical behavior.

P

Panning. Distribution of a source among output channels or spatial directions.

Parallel comb bank. Set of comb filters driven in parallel and summed, producing a superposition of modal families with different delay periods.

Parallel compression. Mixing a compressed path with a less compressed path to combine density and transient definition.

Parallel reverb. Reverb placed on a return path and blended with the dry signal rather than inserted destructively.

Parameter. A controllable numerical, categorical, or boolean value that changes system behavior.

Parameter smoothing. Gradual interpolation of parameter changes to avoid clicks, zipper noise, and unstable transitions.

Paraunitary matrix. Frequency-dependent matrix whose conjugate-transpose product is identity on the unit circle, preserving energy across frequency.

Partitioned convolution. FFT convolution that divides a long impulse response into partitions to manage latency and computation.

Passivity. Property that a system cannot generate net energy, providing a strong sufficient condition for stable physical and feedback-network models.

Peak hold. A meter function that retains the maximum observed value for a defined time or until reset.

Peak level. Maximum instantaneous or sample magnitude over an interval.

Peak normalization. Gain scaling so the largest sample or measured peak reaches a target.

Perceptual loss. A training objective based on features or distances intended to correlate with human hearing.

Perceptual model. A computational approximation of auditory sensitivity, masking, localization, quality, or preference.

Perceptual reverberator. Reverberator optimized to reproduce salient auditory cues rather than every geometrical reflection or exact point-to-point transfer function.

Phase. Position within a periodic cycle or the angular component of a complex spectrum.

Phase cancellation. Reduction caused by summing signals with opposing phase at some frequencies.

Phase response. System-induced phase shift as a function of frequency.

Phaser. An effect combining a signal with phase-shifted versions to create moving spectral notches.

Pink noise. Noise with approximately equal energy per octave and power decreasing about 3 dB per octave.

Pitch. Perceptual attribute ordering sounds from low to high, related primarily but not exclusively to fundamental frequency.

Plugin. A loadable software component that processes, generates, analyzes, or controls audio inside a host.

Plugin delay compensation. Host alignment based on latency reported by plug-ins so parallel paths remain synchronized.

Point-to-point transfer function. Acoustic transfer function from one specified source position to one specified receiver position.

Polar pattern. Directional sensitivity or radiation shown as magnitude versus angle.

Pole. A complex-frequency location where a transfer function becomes unbounded, governing resonance and decay.

Pole angle. Angular position of a pole in the complex plane, determining its oscillation frequency in a discrete-time resonant system.

Pole radius. Distance of a pole from the origin, determining modal decay rate and stability for a discrete-time recursive system.

Pre-delay. Time between dry sound and the onset of a reverberant response.

Pre-echo. Energy heard before a transient, caused by symmetric filtering, transform coding, or noncausal processing.

Prediction error. Difference between an observed sample or feature and its model prediction.

Preset. A named stored configuration of parameters and states.

Pressure reflection coefficient. Complex ratio of reflected to incident acoustic pressure at a boundary, carrying both magnitude and phase information.

Pressure zone microphone. A microphone designed for placement at a boundary to reduce direct-reflection comb filtering.

Prime delay length. Delay length chosen as a prime integer to reduce shared periodicities with other paths in a delay network.

Prime-power delay length. Delay length selected from powers of distinct primes to structure recurrence and mode distribution in an FDN.

Propagation-loss filter. Filter representing distance- and frequency-dependent energy loss accumulated while sound travels through a medium.

Psychoacoustics. Scientific study of relationships between physical sound and auditory perception.

Pumping. Audible gain movement in dynamics processing caused by detector behavior or excessive gain reduction.

Q

Q factor. Ratio of center frequency to bandwidth for a resonance or filter, indicating selectivity.

Quantization. Mapping continuous or high-resolution values to a finite set of representable levels.

Quantization noise. Error introduced by quantization, often modeled as noise under suitable dithering assumptions.

Quarter-inch jack. A common audio connector format used for balanced, unbalanced, insert, instrument, and headphone signals.

Quaternion rotation. A numerically robust representation of three-dimensional orientation used in spatial-audio head tracking.

Quiet zone. A region designed or controlled to have reduced sound level or improved isolation.

R

Ray tracing. Geometric simulation that follows many sound paths through reflections, transmission, scattering, and attenuation.

Ray-tracing order. Maximum number of reflections followed for each acoustic ray in a geometric room simulation.

Real-time factor. Processing time divided by signal duration; values below one indicate faster-than-realtime offline performance.

Real-time processing. Processing that must produce each output before a fixed playback or interaction deadline.

Receiver. The microphone, listener, ear, probe, or modeled point at which an acoustic response is evaluated.

Receiver directivity. Direction-dependent sensitivity of a microphone, ear model, or virtual receiver used during acoustic rendering or measurement.

Rectilinear waveguide mesh. Digital waveguide mesh arranged on an orthogonal grid, simple to implement but subject to direction-dependent dispersion.

Recursive filter. A filter that uses previous outputs, giving an impulse response that may continue indefinitely.

Reference level. A defined electrical, digital, acoustic, or perceptual level used for calibration and comparison.

Reflection. Sound energy returned from a boundary or impedance discontinuity.

Reflection coefficient. Complex or energy ratio describing how much incident sound a boundary reflects.

Reflection density. Number of reflections arriving per unit time within an impulse response.

Reflection order. Number of boundary interactions along a geometric propagation path.

Reflection path. Geometric route from source to receiver containing one or more boundary interactions and associated delay, loss, and filtering.

Regression interval. Selected decibel or time range over which a line is fitted to a decay curve for RT estimation.

Release. The time behavior with which a processor returns toward unity gain or a sound decays after excitation.

Render. To calculate and write processed audio, analysis, or an intermediate artifact.

Reproducibility. Ability to recreate a result from documented inputs, versions, parameters, seeds, and environment.

Resampling. Conversion between sample rates using interpolation and anti-alias filtering.

Resonance. Enhanced response near a natural frequency due to energy storage and repeated reinforcement.

Resonant frequency. Frequency at which a system exhibits a strong modal response.

Return channel. Mixer channel carrying the output of a shared send effect.

Reverb. Persistence and blending of sound after excitation due to many reflections or an artificial equivalent.

Reverb chamber. A reflective room used to create acoustic reverberation or perform diffuse-field measurements.

Reverb tail. The decaying late portion of a reverberant response.

Reverberance. Perceived prominence and duration of reverberant energy.

Reverberant power gain. Frequency-dependent ratio of reverberator output power to input power under specified stationary excitation and routing.

Reverberation coloration. Audible spectral shaping caused by uneven modal amplitudes, delays, feedback, damping, or output mixing in a reverberant system.

Reverberation diffusion. Temporal and spatial spreading that converts sparse reflections into a dense, less localized decay field.

Reverberation problem. Engineering task of reproducing the perceptually important behavior of a space under finite computation, memory, latency, and control constraints.

Reverberation radius. Another name for critical distance in some room-acoustic contexts.

Reverberation time. Time required for sound energy to decay by a specified amount, conventionally 60 dB.

Reverse cumulative integration. Summation from the end of a sequence toward its beginning, used in Schroeder energy-decay analysis.

Reverse reverb. Reverb whose envelope swells toward an event, commonly created by reversing a source or impulse-response workflow.

Ringing. Sustained oscillation after excitation, often caused by high-Q resonance, filtering, or unstable feedback.

Room constant. Absorption-related room quantity used in steady-state level and critical-distance formulas, often written from area and mean absorption.

Room correction. Measurement-guided equalization and sometimes timing control intended to improve playback at selected positions.

Room impulse response. Impulse response from a source position to a receiver position within an acoustic environment.

Room mode. Standing-wave resonance determined by room dimensions and boundary conditions.

Room tone. Background sound of a location without the foreground performance.

Room-mode transition frequency. Approximate frequency separating sparse individually resolvable room modes from an increasingly overlapping statistical field.

Root mean square. Square root of mean squared amplitude, used as an energy-related level measure.

Round-trip latency. Total delay from input conversion through processing to output conversion and acoustic playback.

Routing. Assignment and movement of signals among channels, buses, processors, devices, and outputs.

RT20. Reverberation estimate extrapolated from a 20 dB decay segment, often from minus 5 to minus 25 dB.

RT30. Reverberation estimate extrapolated from a 30 dB decay segment, often from minus 5 to minus 35 dB.

RT60. Time in seconds associated with a 60 dB reverberant decay, measured directly or extrapolated from a shorter reliable slope.

S

Sabine absorption area. Equivalent perfectly absorbing area obtained by summing each surface area multiplied by its absorption coefficient.

Sabine equation. Statistical relation among room volume, equivalent absorption area, and reverberation time under diffuse-field assumptions.

Sample. One discrete-time amplitude value for one channel.

Sample rate. Number of samples represented per second, measured in hertz.

Sample-rate conversion. Resampling between different rates with appropriate reconstruction and anti-alias filtering.

Sample-rate delay scaling. Adjustment of delay lengths in proportion to sampling rate so physical delay times remain approximately constant.

Saturation. Gradual nonlinear limiting that adds harmonics and compresses peaks rather than clipping abruptly.

Scattering. Redistribution of reflected sound caused by surface irregularity, geometry, or wavelength-scale structures.

Scattering coefficient. Fraction of reflected energy redirected away from the specular direction under a defined measurement method.

Scattering delay network. Room-reverberation structure connecting delay lines through scattering nodes derived from room geometry and boundary properties.

Scattering junction. Network node that maps incoming traveling-wave components to outgoing components while satisfying continuity and conservation relations.

Schroeder allpass section. Recursive delay structure with nominally flat magnitude response used to increase echo density and phase dispersion.

Schroeder frequency. Approximate transition above which room modes overlap sufficiently for statistical treatment.

Schroeder integration. Backward integration of squared impulse-response energy to form an energy decay curve.

Schroeder reverberator. Classic architecture using parallel feedback comb filters followed by serial allpass diffusers.

Score. Symbolic representation of musical instructions, events, structure, and often spatial or performance information.

Send. A routed copy of a channel feeding an auxiliary bus or processor.

Send level. Gain determining how much source signal enters a send path.

Sensitivity. Change in output, metric, or perception produced by a change in input or parameter.

Serial processing. Processing in which one stage's output feeds the next stage's input.

Series allpass chain. Cascade of allpass diffusers used to build echo density before or after a recursive reverberation stage.

Shelf filter. A filter that boosts or cuts frequencies above or below a transition region by an approximately constant amount.

Shimmer reverb. Reverb containing pitch-shifted feedback or parallel energy, often emphasizing intervals above the source.

Short-term loudness. Loudness measured over a moving interval of about three seconds under common broadcast standards.

Sidechain. A signal path used to control a processor's behavior without necessarily appearing in its audio output.

Signal. A time-varying quantity carrying audio, control, measurement, or encoded information.

Signal flow. Ordered routing and transformation of signals through a system.

Signal-to-noise ratio. Ratio of desired signal power to noise power under defined measurement conditions.

Silence. Absence of intended sound, not necessarily zero digital samples or zero acoustic energy.

Sine sweep. A sinusoid whose frequency changes over time, used for response measurement and system identification.

Slapback delay. A single or sparse short echo, usually long enough to be distinct but rhythmically attached to the source.

Slew rate. Maximum rate at which a value or signal can change.

Smoothing. Reduction of abrupt variation across time, frequency, space, or parameter values.

SOFA. Spatially Oriented Format for Acoustics, a standardized container for spatial impulse-response data and metadata.

Soft clipper. A nonlinear limiter with a gradual transfer curve near maximum level.

Sound absorption average. A single-number average of absorption coefficients over defined octave bands.

Sound field. Spatial distribution of acoustic pressure, particle velocity, phase, and energy over time.

Sound pressure. Local deviation from ambient pressure caused by a sound wave.

Sound pressure level. Logarithmic measure of RMS sound pressure relative to 20 micropascals in air.

Sound speed. Propagation speed of an acoustic disturbance, dependent on medium and environmental conditions.

Source. A physical, recorded, synthesized, or modeled origin of sound energy.

Source directivity. Direction-dependent radiation pattern of a source as a function of frequency.

Source separation. Estimation of individual sources or stems from a mixture.

Spatial aliasing. Directional artifacts caused by insufficient array sampling, Ambisonic order, or loudspeaker density at a frequency.

Spatial audio. Audio in which direction, distance, extent, environment, or motion is represented or rendered.

Spatial coherence. Consistency of phase or waveform relationships across positions, channels, or frequency.

Spatial impression. Combined percept of apparent source width, envelopment, distance, and room scale.

Spatialization. Placement, motion, extent, or environmental rendering of sound in a spatial scene.

Specific echo density. Echo count per unit time under a stated amplitude or statistical criterion, used to assess reverberant texture.

Spectral centroid. Energy-weighted mean frequency, often correlated with perceived brightness.

Spectral coloration equalizer. Filter used outside or around a reverberant network to correct its steady-state magnitude response independently of decay design.

Spectral convergence. Degree to which an estimated spectrum approaches a target spectrum over iterations or time.

Spectral decay. Reduction of energy over time as a function of frequency.

Spectral envelope. Smooth curve describing broad spectral shape independent of fine harmonic detail.

Spectral flux. Frame-to-frame spectral change, often used for onset detection and activity measurement.

Spectral leakage. Spread of sinusoidal energy across DFT bins because the analysis frame does not contain an integer number of periods.

Spectral subtraction. Noise- or reverberation-reduction method that subtracts an estimated unwanted magnitude spectrum.

Spectrogram. Time-frequency image showing spectral magnitude or power across successive frames.

Spectrum. Distribution of signal magnitude, power, phase, or energy over frequency.

Specular reflection. Mirror-like reflection in which outgoing angle follows the geometric law of reflection.

Speech clarity. Intelligibility-related perception influenced by articulation, direct sound, noise, and early-to-late energy balance.

Speech transmission index. Standardized estimate of speech intelligibility based on modulation transfer through noise and reverberation.

Spherical harmonics. Orthogonal angular basis functions used to represent three-dimensional sound fields in Ambisonics.

Spring reverb. Electromechanical reverberation produced by waves propagating and dispersing along metal springs.

Standing wave. Stationary interference pattern formed by opposing waves, producing fixed nodes and antinodes.

State. Stored information required for a processor's future output, such as delay contents, filter memories, envelopes, and random generators.

Statistical late field. Dense reverberant tail characterized through distributions, correlation, decay, and spectrum rather than individually modeled reflections.

Steady-state energy density. Average acoustic energy per unit volume after a continuous source and room losses reach statistical equilibrium.

Stem. A grouped audio submix intended for independent routing, processing, delivery, or reconstruction.

Stereo. Two-channel reproduction or representation conventionally associated with left and right.

Stereo width. Perceived or measured lateral spread of a stereo image.

STFT. Short-time Fourier transform, a sequence of windowed spectra representing time-varying frequency content.

Subwoofer. Loudspeaker designed primarily for low-frequency reproduction.

Summing. Combining signals by addition, with gains and routing determining the result.

Surface scattering coefficient. Frequency-dependent fraction of reflected energy redistributed away from the specular direction by boundary roughness or geometry.

Surround sound. Playback using loudspeakers around the listener to represent direction and envelopment beyond frontal stereo.

Sustain. Continued portion of a sound or envelope after onset, or maintenance of reverberant energy in a freeze system.

Sweet spot. Listening region in which localization, tonality, timing, and spatial balance meet the intended design.

System identification. Estimation of a system model or impulse response from known excitation and observed output.

T

T20. Common shorthand for a reverberation estimate based on a measured 20 dB decay range.

T30. Common shorthand for a reverberation estimate based on a measured 30 dB decay range.

Tail length. Rendered or effective duration of a reverberant decay after the source ends.

Tap. One read location and gain from a delay line.

Tapped-delay early-reflection model. Finite set of delayed, scaled, filtered, and spatialized taps approximating the perceptually important first room reflections.

Target curve. Desired frequency, loudness, decay, or spatial response used to guide correction or optimization.

Temporal masking. Reduced audibility of one sound because another occurs shortly before or after it.

Threshold. Input level or criterion at which a processor changes state or behavior.

Timbre. Perceptual quality distinguishing sounds with similar pitch and loudness through spectrum, envelope, modulation, and noise.

Time constant. Parameter describing the rate of exponential response, smoothing, attack, release, or decay.

Time domain. Representation of signal amplitude as a function of time or sample index.

Time stretching. Changing signal duration without proportionally changing pitch.

Time variance. Property of a system whose response changes with absolute time or internal modulation state.

Time-frequency mask. A matrix of gains applied to spectral bins over time for separation, enhancement, or dereverberation.

Time-varying delay line. Delay whose read position changes over time, requiring interpolation and potentially producing Doppler shift or decorrelation.

Time-varying reverberator. Reverberator whose delays, filters, matrices, or gains change over time to suppress static modes or create intentional motion.

Toeplitz matrix. A matrix constant along each diagonal, arising naturally in convolution and linear prediction.

Tonal correction filter. Equalizer that compensates the average spectral coloration of a reverberator without redefining its modal decay times.

Tonality. Degree to which a sound contains stable pitch or narrowband structure rather than noise-like energy.

Total harmonic distortion. Ratio of harmonic-distortion energy to fundamental energy under a specified test.

Transfer-function matrix. Matrix of point-to-point transfer functions mapping multiple acoustic or electrical inputs to multiple outputs.

Transient. A short, rapidly changing event such as an attack, click, or percussion hit.

Transient preservation. Retention of onset definition and peak structure through processing.

Transmission loss. Reduction in sound power transmitted through a partition, usually frequency dependent.

Triangular feedback matrix. Upper- or lower-triangular FDN matrix whose eigenvalues are visible on its diagonal and whose coupling has a directed structure.

Triangular waveguide mesh. Mesh using triangular spatial cells to improve angular isotropy relative to a simple rectilinear grid.

True peak. Estimated maximum of the reconstructed continuous waveform, which can exceed individual sample peaks.

Truncation. Cutting a signal, impulse response, numerical representation, or series to a finite duration or order.

Tukey window. A tapered-cosine window with an adjustable flat center region.

Two-port network. A system described by input-output variables at two interfaces, useful for electroacoustic and transmission modeling.

U

Uncorrelated signals. Signals with negligible linear correlation over the measurement interval.

Underflow. Numerical result too small for normal representation, potentially becoming denormalized or zero.

Uniform partitioning. Partitioned convolution using equal-size impulse-response blocks.

Unit impulse. A discrete signal equal to one at one sample and zero elsewhere.

Unit-circle pole. Pole lying exactly on the complex unit circle, corresponding to an undamped discrete-time mode in an ideal lossless system.

Unitary matrix. A complex matrix whose conjugate transpose equals its inverse and that preserves energy.

Unitary mixing. Energy-preserving mixing of complex-valued states or channels using a unitary matrix.

Unity gain. Gain of one, corresponding to zero decibels of level change.

Upmix. Creation of a higher-channel-count or more spatial representation from fewer input channels.

Upsampling. Conversion to a higher sample rate by interpolation and reconstruction filtering.

User preset. A parameter state saved and named by the user rather than shipped as factory content.

V

Validation. Checking that inputs, parameters, outputs, formats, and invariants meet defined requirements.

Variance. Mean squared deviation from a mean, describing statistical spread.

Vector base amplitude panning. Loudspeaker panning based on vector gains for a selected speaker basis.

Velocity microphone. A pressure-gradient microphone whose output relates to acoustic particle velocity.

Velvet noise. Sparse randomized impulse sequence used for efficient decorrelation, diffusion, and reverberation structures.

Venue. A physical performance, recording, or playback space with characteristic geometry and acoustics.

Vibrato. Periodic or quasi-periodic pitch modulation used expressively in voices, instruments, or processing.

Virtual acoustics. Simulation or reproduction of acoustic environments through signal processing and spatial rendering.

Virtual source. A perceived or modeled source location not occupied by a physical emitter.

Visualization. Graphical representation of signals, spectra, decay, geometry, routing, metrics, or model state.

Vocoder. A system that applies spectral-envelope or band-energy information from one signal to another.

Volume. Physical enclosure size or a colloquial level control; technical writing should state which meaning is intended.

VST3. Steinberg's cross-platform plug-in interface supporting audio processing, instruments, automation, and host integration.

W

WAV. A RIFF-based audio file container commonly storing PCM or floating-point samples and metadata chunks.

Wave equation. Partial differential equation governing propagation of acoustic pressure or related wave quantities.

Wave field synthesis. Spatial reproduction method using dense loudspeaker arrays to approximate desired wavefronts over an area.

Waveguide. A physical or digital structure that constrains wave propagation along one or more dimensions.

Waveguide numerical dispersion. Frequency- and direction-dependent propagation-speed error caused by discrete mesh geometry and sampling.

Waveguide-mesh reverberation. Physical-modeling approach that propagates traveling waves across an interconnected spatial mesh to approximate room acoustics.

Wavelength. Distance traveled by one wave cycle, equal to sound speed divided by frequency.

Wavelet transform. Multiresolution analysis using localized basis functions with scale-dependent time-frequency resolution.

Weighted prediction error. Multichannel linear-prediction method widely used for blind speech dereverberation.

Wet signal. The reverberated or otherwise effect-processed component in a wet-dry mixture.

Wet-dry mix. Relative blend of processed and unprocessed signal paths.

White noise. Noise with approximately equal power per hertz over the band of interest.

Window. A finite weighting function applied to a frame before spectral or statistical analysis.

Window function. A named taper such as Hann, Kaiser, or Tukey used to manage frame-edge discontinuities.

Word clock. A digital timing reference used to synchronize audio converters and devices at sample rate.

World-space coordinate. Position or direction expressed in a shared scene coordinate system rather than a source- or listener-local frame.

WPE. Standard abbreviation for weighted prediction error, a widely used multichannel dereverberation method.

X

X-curve. A cinema monitoring target response defined for calibrated dubbing stages and theaters.

XLR. A locking balanced connector family widely used for microphones, line-level audio, and digital interconnection.

XML. Extensible Markup Language, a structured text format used by some audio metadata and interchange standards.

XY microphone technique. A coincident stereo method using angled directional microphones to encode level differences with strong mono compatibility.

Y

Yaw. Rotation around the vertical axis, corresponding approximately to turning the head left or right.

YIN. A fundamental-frequency estimation algorithm based on a modified difference function and error control.

Yule-Walker equations. Linear equations relating autocorrelation to autoregressive model coefficients.

Z

Z-transform. Complex-frequency representation of discrete-time signals and systems used to analyze poles, zeros, and stability.

Zero crossing. A point where a waveform changes sign.

Zero padding. Appending zeros to a signal or frame to prevent circular wraparound or increase spectral sampling density.

Zero-latency convolution. A convolution design whose first partition is processed with no added block delay beyond the host buffer, while later partitions use FFTs.

Zero-phase filter. A noncausal offline filter with no phase shift, often implemented by forward-backward processing.

Zipper noise. Audible stepping caused by abrupt or insufficiently smoothed parameter changes.

Zita-Rev1. Open-source FDN reverberator by Fons Adriaensen with eight delay lines, frequency-dependent decay controls, and stereo output processing.

Zobel network. An impedance-equalization network used in loudspeaker and analog circuit design.

Zone. A defined spatial region assigned distinct playback, measurement, routing, or acoustic behavior.

Afterword

Reverberation begins as persistence: energy leaves a source, encounters a world, and returns changed. Every technique in this book chooses what part of that encounter to preserve. A measured impulse response preserves one source-receiver relationship. An algorithm preserves selected temporal and spectral statistics. A physical model preserves geometry and boundary behavior at a chosen resolution. A learned model preserves patterns represented in data. A musician may preserve none of these literally and instead use decay, resonance, reversal, gating, or spatial motion as independent compositional materials.

The command line and the plug-in embody complementary ways of working. The CLI favors reproducibility, inspection, batch rendering, datasets, reports, and unusual parameter ranges. The plug-in favors touch, automation, immediate comparison, and performance inside a DAW. Neither interface is the definition of the instrument. The durable center is a set of audible transformations whose assumptions can be measured, documented, repeated, and deliberately broken.

Future reverberation systems will likely combine measured spaces, parametric networks, geometry, scene metadata, differentiable models, and listener-aware rendering. Their success should not be judged by novelty alone. The useful questions remain old ones: Does the sound retain articulation? Does the room support the phrase? Does the decay reveal or conceal structure? Does the spatial field remain convincing when the listener moves? Most importantly, does the result invite another kind of music?

About the Author

Colby Leider, PhD, is the author and principal developer of verbx. His work on the project joins musical production, acoustics, digital signal processing, spatial audio, software design, technical writing, and education. He created this guide as both an operational manual and a durable reference for musicians, engineers, researchers, developers, and students who want to understand not only which controls to use, but why reverberant systems behave as they do.

verbx and its documentation are developed in Miami, Florida, USA. Current source, releases, examples, corrections, and generated editions are published at https://github.com/TheColby/verbx.

Errata and Corrigenda

Corrections belong to the edition in which an error appears. Before reporting one, record the edition date from the copyright page, the PDF page number, the printed page number when different, the chapter or appendix heading, and the shortest quotation that identifies the passage. For equations, figures, tables, commands, and citations, include the corrected form and enough surrounding context to distinguish a typographical error from a change in software behavior.

The live correction record is maintained with the source repository at https://github.com/TheColby/verbx/issues. Use an issue title beginning with `Book erratum:` and include the verbx version or commit when the correction concerns generated CLI output. Confirmed corrections are applied to the source files, protected by tests when practical, and incorporated into the next generated edition. Earlier PDFs remain historical artifacts; the edition date and repository history make the correction trail auditable.

A *corrigendum* records a substantive correction to reasoning, data, attribution, or technical instruction. An *erratum* records a smaller production error such as a misspelling, malformed symbol, incorrect page reference, or broken link. Suggestions, feature requests, and changes caused by later software releases should be filed separately so the correction record remains precise.

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Colophon

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The text face is TeX Gyre Schola, an open typeface in the New Century Schoolbook tradition. Its OpenType common ligatures are enabled throughout proportional prose, headings, captions, tables, front matter, and the index. Menlo is used without ligatures for commands and code so technical literals remain exact. The page is U.S. Letter with a book hierarchy, roman-numbered front matter, arabic-numbered main matter, lettered appendices, and two-column index. Tables use alternating neutral fills; diagrams use the verbx teal, gold, rust, blue, plum, and sage palette.

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